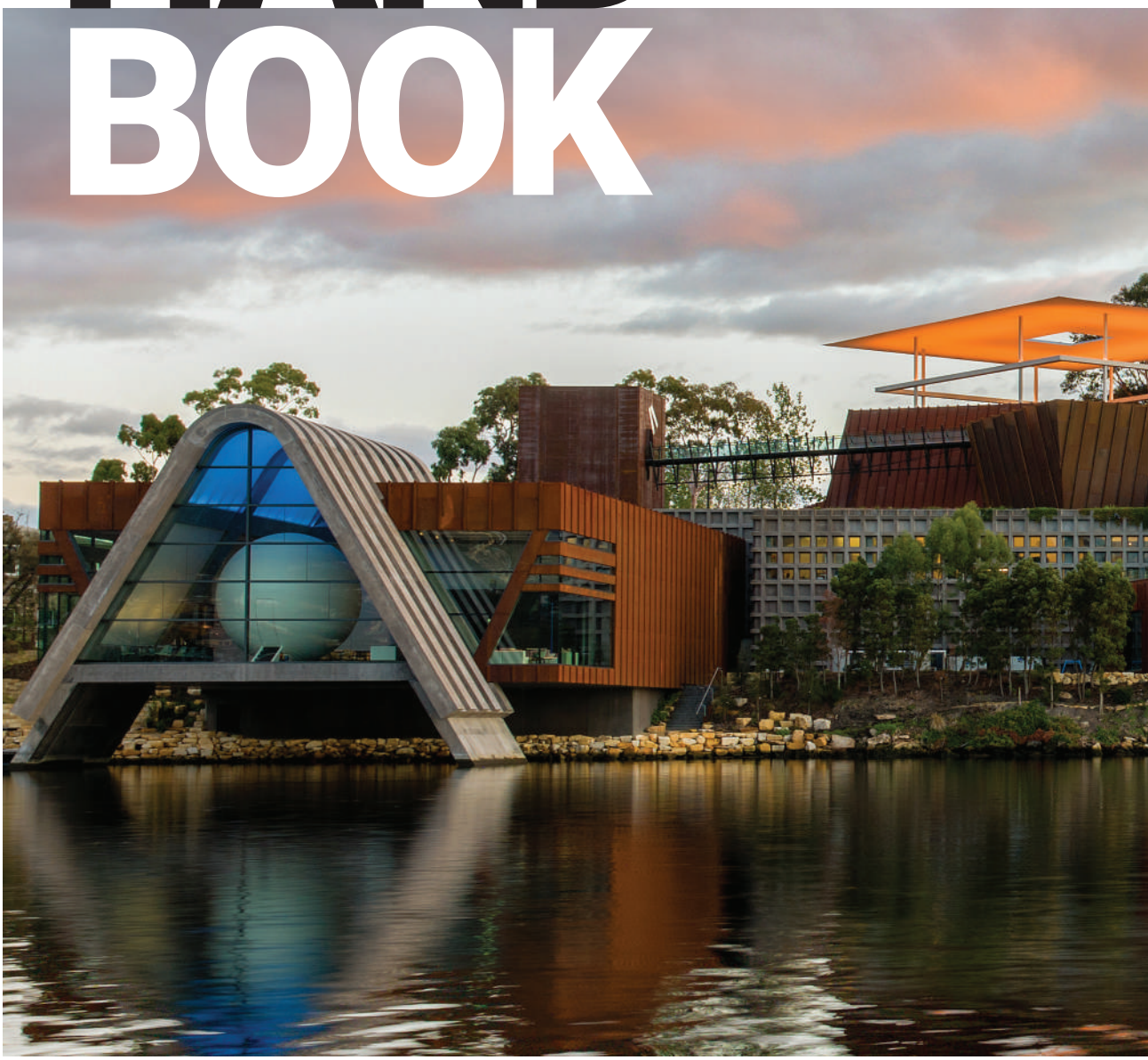
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THE HAND BOOK



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AUSTRALIAN CONSTRUCTION HANDBOOK 2021

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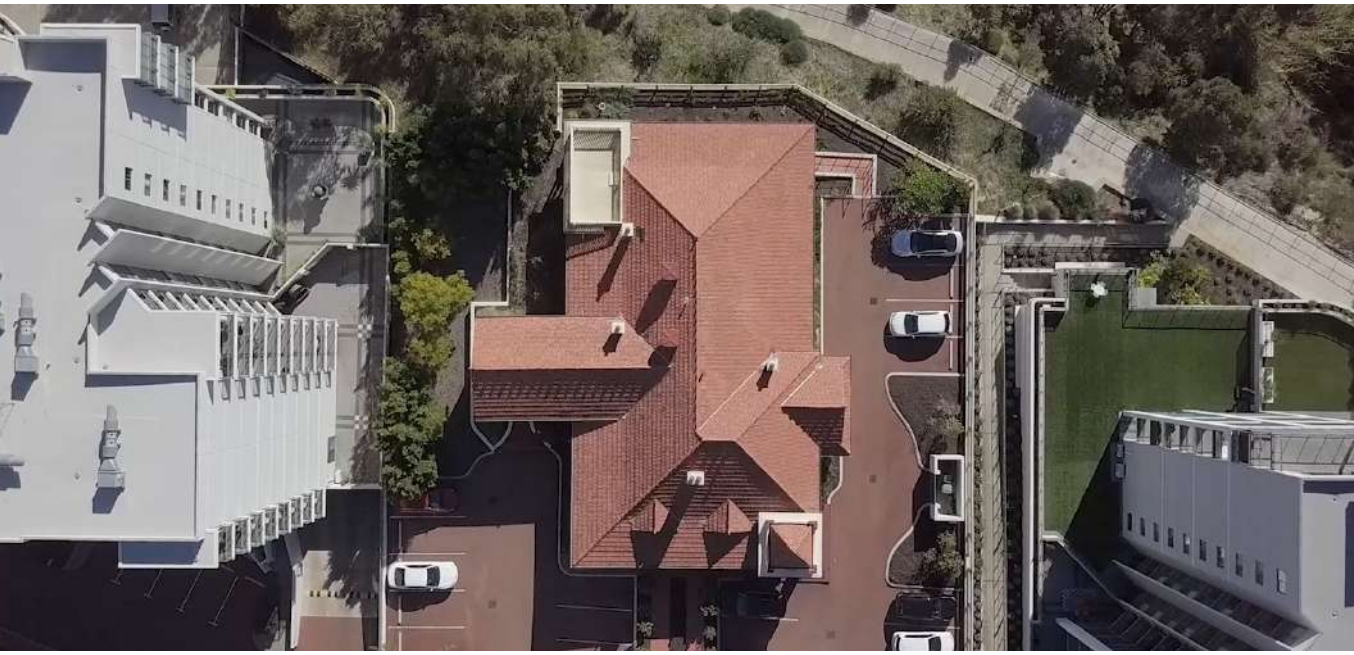
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AUSTRALIAN CONSTRUCTION HANDBOOK 2021

Edited by Rawlinsons
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THE

CONSTRUCTION COST EXPERTS



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Rawlinsons Cost Guide has been an essential reference book for the smaller builder, architect/consultant and many others in the construction industry since 1993



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HOW TO USE THIS HANDBOOK

BUILDING PRICE INDICES
These indices are a guide to the percentage variation in building costs for each year since 1978. This facilitates the updating of known costs to approximate current levels.

The indices are not valid for housing, small projects or remote country work, nor do they reflect the relative cost differential of the various cities.

The indices generally relate to all construction types, and as such, can only be a guide to specific project types.

REGIONAL INDICES
Regional indices give an indication of the locality adjustment factor to be applied to the figures of the base city in each state in order to arrive at a figure for country towns.

The respective index figures are a broad indication only of the cost variation and are applicable to the total project cost.

Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

ESTIMATING - BUILDING COSTS PER SQUARE METRE
This provides the average cost range for a wide selection of typical buildings. Whilst square metre costs are recommended for use in initial feasibility studies only, they also provide the base figure in the valuation of buildings, however when so used they should be adjusted as described in the following Elemental Costs of Buildings. It should also be noted that, as the figures given are for a typical building on a flat site, adjustment must be made for such factors as sloping sites, ground conditions, unusual shape and other design considerations.

The costs given are based on the total floor area of all levels, measured between the inner faces of external walls; refer pages 876 to 878 for measurement of areas.

It is important to refer to page 20 for further notes on the use of these costs.

ESTIMATING - ELEMENTAL COSTS OF BUILDINGS
This assists in the compilation of a more accurate cost per square metre through evaluation and adjustment of the mean figures of the various elements of the foregoing typical buildings, to reflect the particular building being examined. Examples of the major variations from the typical are the likes of special facade treatments, the extent of floor finishes and whether or not the building services include air-conditioning, lifts, etc.

It is important to refer to pages 20, 50 and 51 for further notes on the allowances and use of these costs.

ESTIMATING - COMPARATIVE COSTS
The more detailed rates given in this section facilitate a finer adjustment of the foregoing elemental figures through the application of unit costs to a particular element i.e. different design treatments such as facades, interior finishes, etc.

The costs given are all-up estimating costs, i.e. they include allowance for labour and minor items. For items not covered in this section it will be necessary to refer to the Detailed Prices section.

HOW TO USE THIS HANDBOOK

DETAILED PRICES
These provide a wide range of costs for particular items in all trades and services.

Due care and attention must be given to the explanatory notes for the complete section, page 172, and all subsection bullet points and notes.

All rates are supply and install rates unless otherwise noted.

Detailed prices can include specific items of work which, unless stated to the contrary, do not include for other related minor items and labours. When using to compliment the rates of Estimating - Comparative Costs, it will be necessary to make due allowance for such related work. The allowance can vary with particular item and project, but a range of between 5% and 10% would normally suffice.

GENERALLY
The rates within Rawlinsons Australian Construction Handbook allow for Builders' head office overheads and profit at the following rates (unless otherwise noted) :

Preliminaries, Transportation Systems and Base Material Prices - 0.0%
Electrical and Mechanical Services - 2.5%
All other trades and services - 5.0%.

• • • • •

When using any of the foregoing figures, at all times the nature and specific requirements of the particular project must be borne in mind.

INTRODUCTION

Since its inception in 1983, Rawlinsons Australian Construction Handbook has achieved a reputation as the leading authority on the various aspects of Australian construction costs. This Handbook, now in its thirty-ninth edition, is used by those involved in the various disciplines of the construction industry and is recognised as an authoritative text in the courts for cases relating to disputes regarding construction costs.

.

As in previous editions, this thirty-ninth edition is committed to the dissemination of information necessary to implement cost control, cost management and cost benefit studies effectively and at all stages of planning and construction.

It should be stressed that attention must be given to the particular circumstances and conditions of the project being reviewed, when using the data contained in this Handbook.

The editors wish to record their appreciation of the indispensable assistance received from the many individuals and firms around the country, in the production of this edition.

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All costs given exclude the Goods and Services Tax (G.S.T.), unless otherwise noted.

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PUBLISHER'S NOTE

While every effort has been made to ensure the accuracy of the information given in this publication, neither the editors nor the publishers in any way accept liability for loss of any kind resulting from the use of such information.

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EDITOR'S COMMENT

Adelaide - The unprecedented impacts of COVID-19 have resulted in a massive deterioration of the state's budget position, with significant challenges ahead for the economy. The federal budget in terms of personal tax cuts and investment incentives has been designed to encourage recovery and activity in the economy. The state government has responded by providing significant assistance to stimulate the economy, as infrastructure, e-commerce and defence are critical for growth in South Australia and will support the construction sector going forward. Developing the manufacturing sector is being emphasised with the view of becoming less reliant on imports from overseas. This strategy will be of significant benefit and in part will be driven by the defence sector, as the Naval Group commences the local manufacturing procurement process for the Frigate and Submarine programme. Concern has been expressed that industry (not just construction) does not have the capacity and skills to deliver efficiently the significant stimulus that will occur, due to reduced skilled immigration and the lack of productivity growth. The state government has committed $12.9 billion to the infrastructure plan, which includes commitment to school and hospital upgrades as well as local government funding. This is starting to come to fruition. Major projects are still lacking traction with industry groups calling on state government to develop a centrally coordinated capital works programme with the view to prioritise and fast track future projects. To attempt to accurately forecast the state of the construction sector over the next twelve months is difficult given the uncertainty and scale of the COVID-19 ongoing impacts, in particular, public and private sector future investment strategies, the material supply chain from within Australia and overseas, consumer and investor confidence and the fluctuation of the Australian dollar. All sectors are experiencing reduced activity, price discounting and very competitive tendering, which will be a constant for the 2021/2022 period.

Brisbane - Following the recent state election, the sitting Government has been returned for a further term in office. The outlook for 2021 is positive with investment in infrastructure emphasised a key factor in the economic recovery from COVID-19 by all levels of Government. The commitments of fast-tracked major infrastructure projects, coupled with significant volume of works already underway, are likely to maintain solid levels of activity across the sector for 2021 and beyond. Ongoing trade disputes, international border closures and uncertainty across international markets is anticipated to have a slowing effect on the resources sector in the year ahead. These international factors will also have an impact on construction sectors and has the potential to create volatile conditions from both a time and cost perspective. Tender activity and Building Approvals have continued to strengthen through the second half of 2020 and is anticipated to continue in 2021. While pricing currently remains competitive, upward pricing pressure is anticipated to return through 2021. Strong levels of activity are anticipated in the domestic housing sector with levels of activity across commercial and higher density residential sectors improving as border restrictions are relaxed and the economy recovers.

Canberra - Canberra's core group of developers/builders remain well-resourced and are actively looking for sites to develop, particularly townhouse sites which continue to feature heavily in Canberra. The Gungahlin area which is the staging point for the light rail has featured many high-rise apartment developments and continues to lead the way in future high-rise developments which are currently going through the development approval process. Sales in these developments continue to be relatively strong. Non-bank lenders continue to have a significant share of the development finance as developers turn to an alternative to traditional banks, stating that although interest rates are higher the processes and time saved makes it worthwhile.

Darwin - Market has been stable in 2019 with Northern Territory elections mid-year and COVID-19 which did not really affect the construction market as the Northern Territory had no case of local transmission enabling the industry to function within supplies and distancing restrictions. A number of projects have been and still are in the planning phase and should be ready to roll out as economic conditions improve. There is still very low level of private investment with most large projects receiving substantial federal funding. The state government, whilst keen to see more economic activity is itself constrained by high debt levels especially after significant investment in the market to mitigate the impact of COVID -19 on various sectors of the economy. We expect a number of large government and defence projects to keep the industry busy in 2021, but not to the point that it will affect prices markedly due to market spare capacity. Projects are still few and far between maintaining a lid on cost escalation.

Hobart - The Tasmanian construction industry appears to be weathering the storm of 2020 with evidence of increased tender pricing and reduced contractor availability towards the end of the year. Although some projects reliant on cross border visitation or associated investment were stopped or postponed in some cases, the ability for the industry to continue largely uninterrupted during the pandemic generated lockdowns experienced in other industries and states has reduced the impact 2020 may have had on the construction sector. Confidence appears to be returning to the industry creating a positive outlook for 2021. A contributing factor to the surge in activity towards the end of 2020 is the continued focus on alleviating the under supply of housing stock in the State. The increase in available government grants that stipulated substantial commencement prior to the end of 2020 has led to a spate of new house builds that will continue construction well into 2021 helping to ensure high levels of activity into the foreseeable future. In addition to the incentives offered in the residential market the State government has also announced an aggressive infrastructure spending policy to kick start the economy and create jobs in its response to the impact of COVID-19. Investment has been announced for major road and bridge developments, schools, public building maintenance, irrigation schemes, hospitals and the progression of the Macquarie Point development precinct. These announcements provide a confidence boost to a State that had achieved so much in recent years on the back of the visitor industry and associated activity and developments, all of which had the potential to regress with the closure of state and international borders. At this point the Tasmanian construction industry appears to be well placed to rebound strongly from the trials of 2020 and with a continued improvement in the global pandemic situation would be expected to return to pre-COVID levels quickly in 2021.

EDITOR'S COMMENT

Melbourne - The past year has been very different from any other year, not only in Victoria, but indeed around the country and the world. COVID-19 has changed our lives and is still changing so many things around us, including the way that people interact, work, how businesses operate, etc. The various sectors of the construction industry and property market have not been and will not be exempt from this. The COVID-19 effects of 2020 and the uncertain times, are set to continue into 2021, with there being some level of cautious optimism anticipated, by many, at some point in time. In Victoria, a number of construction projects, which stalled during the height of the pandemic, are set to take off again. This will be aided by current low interest rates, which are now at their lowest in living memory. Activity in the civil engineering sector will continue to remain strong and be a driving force within the industry, in the year ahead. Activity in the health sector will also be strong. The industrial sector should capitalise on factors which include low interest rates and the increase in on-line shopping activity. The change in immigration levels, international tourism and reduced number of overseas students, will continue to have their negative effects on the residential, retail, hotel and education sectors. Some sectors of the construction industry, as well as some associated suppliers will see new enterprise bargaining agreements coming into effect and this, in turn, will place additional pressure on construction costs. Activity generally across the construction industry, is anticipated to remain at reasonable levels, albeit that activity in some sectors will continue to soften. Tenders should remain competitive, but with a level of uncertainty continuing to be evident, as a result of COVID-19.

Perth - After a very unexpected year, Western Australia proved itself to be resilient due to hard boarder measures introduced by the State government. The State government reduction of 'red-tape' and construction stimulus via major infrastructure projects have in the short term negated some of the negative impacts of the downturn from COVID-19. Locally sourced materials have stabilised in price, whilst some uncertainty remains on internationally sourced materials, components and plant as greater lead and delivery times are subject to each manufacturing countries work place restrictions. Office developments remain on hold as weaker demand has risen from the recent consolidation of office space, especially in the CBD. The advent of remote working has led to a re-assessment of modern working practices. Shopping centre retail during 2020 has seen the on-site progress of refurbishment and expansion continue. Large format retail has endured any downturn, as the sector is predominately-home improvement, furniture or electronic good, which all became favourable during lock down. Online retail increased as the COVID-19 restriction commenced and this trend continues. Residentially, Western Australia is experiencing a falling vacancy rate; this coupled with the RBA cash rate of 0.10 per cent will entice developers to recommence some projects.

Sydney - New South Wales commenced 2020 heading into its third year of severe drought, with large portions of Australia's eastern seaboard on fire and whispers of an overseas virus that was gathering momentum. By the end of February, most of the fires had been extinguished, but before their embers had cooled, the coronavirus presented a new, fast-evolving and significant challenge to the Australian economy and health system. States also pushed through temporary planning acceleration programmes and brought forward infrastructure, health and education projects, in a bid to keep construction workers in jobs and the broader economy running. As a result of these initiatives, up to the June quarter, the construction industry has survived. This is mainly due to underlying demand and the closing out of projects that were already 'locked in'. Certain sectors, which were already in decline before COVID-19, have been hit hard. The retail sector was already repositioning itself to accommodate consumer trends towards an online marketplace. This shift in strategy has been fast tracked by COVID, as retailers look to reduce their brick-and-mortar presence and digitize their businesses. Residential house approvals have been on a downward trend since 2015. With no inward migration and the unemployment rate forecast to rise, this trend is set to continue in the short to medium term. In mid-August, Master Builders Australia announced its forecast for a twenty-seven per cent decline in new house construction between 2019-2020 and 2020-2021, with Treasury also projecting a drop to 140,000 new homes in the 2020-2021 period from 170,000. As international borders remain closed, the tourism and hospitality sector are relying on domestic demand, and this has been further disrupted by the states and territories closing their internal borders. All of this uncertainty has general contractors seeking to secure future work, and we have witnessed competitive tender returns during the first half of 2020. The sectors that are still relatively busy, such as industrial, healthcare, education and data centres are in a good position to avail of these favourable market conditions. With the effects of COVID having triggered a recession, from which the recovery is expected to be prolonged given the resurgence of the virus, the Australian economy, and by extension the construction industry, faces its greatest challenge in a generation.

ANTICIPATED BUILDING PRICE ADJUSTMENT, 2021

Prices given in the book are those ruling at the commencement of 2021 with an index of 100.00. To update prices refer to the following, which is a prediction of tender levels taking into account anticipated market conditions.

	31-Dec-20	31-Mar-21	30-Jun-21	30-Sep-21	31-Dec-21
Adelaide	100.00	100.50	100.75	102.00	102.50
Brisbane	100.00	101.00	101.50	102.50	103.00
Canberra	100.00	100.50	101.25	101.75	102.50
Darwin	100.00	100.00	100.20	100.30	100.40
Hobart	100.00	100.75	101.50	102.00	102.50
Melbourne	100.00	101.00	101.50	102.00	102.75
Perth	100.00	100.60	101.10	101.90	102.60
Sydney	100.00	100.30	100.60	100.90	101.20

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ESTIMATING BUILDING COSTS PER SQUARE METRE

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ESTIMATING

BUILDING COSTS PER SQUARE METRE

Costs given hereafter should be used for initial feasibility studies only.

Costs given are average prices for typical buildings at December 31, 2020 within the metropolitan areas including allowance for Preliminaries and Builders profit and overheads. ADD additional allowances for increased costs (refer Anticipated Building Price Adjustment page 15) and when referring to country areas (refer Regional Indices pages 834 to 844).

Costs given should be adjusted to take account of such factors as sloping sites, foundation problems, high wall to floor ratios, unusual shape and any special design factors where costs can vary considerably from the range given.

Costs are based on the total floor area of all levels measured between the inner faces of external walls.

Costs can provide no more than a rough guide to the probable cost of a building, and whilst in many instances a single rate is sufficient to indicate this ruling average cost, there are many other instances such as banks, function centres, theatres, churches, residential buildings, where costs can vary considerably from the range given.

Costs EXCLUDE parking areas. For basement parking to office buildings, etc. refer to Section 9.3, for open parking areas refer to Section 10.2.

Costs EXCLUDE land, demolitions, balconies, covered ways, external services outside 3.0m from the outside face of the building, external works other than those immediately adjacent to the building, loose or special equipment, furniture, furnishings, legal and professional fees.

For a more detailed breakdown, and to assist in the compilation of a more accurate estimate, refer to Elemental Costs of Buildings pages 49 to 105. For ease of reference identical item numbers have been used for each section.

Costs exclude the Goods and Services Tax (G.S.T.).

1.0	**ADMINISTRATION, CIVIC**		PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
1.1	ADMINISTRATIVE							
1.1.1	ADMINISTRATION OFFICE, single storey, standard finishes, subdivisional partitioning to offices, reception, amenities areas, air-conditioning	…	sqm	2445-2635	2095-2260	2395-2580	2445-2635	2485-2680
1.1.2	ADMINISTRATION OFFICE, 2-3 STOREY - standard finishes, subdivisional partitioning to offices, reception, amenities areas, air-conditioning, lift	…	sqm	2770-2985	2640-2845	2680-2885	2800-3015	2825-3050
1.1.3	ADMINISTRATION OFFICE, 3-10 STOREY - standard finishes and facilities to upper levels, good standard to ground level/ reception, sub-divisional partitioning to offices, reception, amenities areas, air-conditioning, lifts, fire sprinklers	…	sqm	3415-3685	3320-3580	3240-3495	3220-3470	3305-3560
1.2	CIVIC							
1.2.1	CITY HALL, 2000 CAPACITY - main hall, lesser hall, library, etc., medium standard finishes, air-conditioning	…	sqm	5585-6020	4885-5265	5555-5985	5705-6150	5920-6385
1.2.2	TOWN HALL, 1000 CAPACITY - main hall, lesser hall, library etc., medium standard finishes, air-conditioning	…	sqm	4380-4725	3875-4175	4250-4580	4480-4830	4510-4860

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGES 52 AND 53 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

1.0	**ADMINISTRATION, CIVIC**		PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
1.2	CIVIC (cont'd)							
1.2.3	CIVIC CENTRE, 300-500 CAPACITY - main hall suitable for cabarets, conventions, etc., ante-rooms, small kitchen, standard finishes, no air-conditioning	...	sqm	2120-2285	1980-2130	2115-2280	2160-2330	2155-2320
1.2.4	CIVIC CENTRE, 500-750 CAPACITY - main hall, lesser hall, ante-rooms, high standard finishes, kitchen and bar facilities, air-conditioning	...	sqm	3265-3520	2965-3200	3280-3535	3265-3520	3370-3635
1.2.5	LIBRARY, SUBURBAN - single storey, good standard finishes, air-conditioning, excluding loose fittings	...	sqm	2650-2860	2270-2445	2635-2840	2610-2815	2655-2865
1.2.6	LIBRARY, SUBURBAN - two to three storeys, good standard finishes, air-conditioning, lift, excluding loose fittings	...	sqm	2770-2985	2605-2810	2890-3115	2920-3150	2950-3180
1.2.7	LIBRARY, CAPITAL CITY - up to five storeys, high standard finishes, air-conditioned, slow lifts, fire sprinklers, excluding loose fittings	...	sqm	5125-5520	4260-4590	4750-5125	5105-5505	5850-6305
1.2.8	ART GALLERY, MUSEUM, CAPITAL CITY - up to six storeys, galleries and work areas, high standard finishes, air-conditioned, slow lifts, security provisions, excluding loose fittings	...	sqm	5960-6425	4900-5280	5540-5970	5920-6380	6805-7335
1.2.9	PUBLIC TOILETS - single storey, standard construction and basic finishes, male, female and handicapped areas	...	sqm	2735-2945	2545-2745	2670-2875	2800-3015	2800-3020
1.2.10	CHILD CARE CENTRE, refer 6.2.1	...	Note					
1.3	AUTHORITIES							
1.3.1	LAW COURTS, CAPITAL CITY - up to fifteen storeys for high and lower courts, administration, high standard finishes, air-conditioning, court fit-out, excluding loose fittings	...	sqm	4900-5285	4345-4685	4545-4895	5455-5875	5945-6410
1.3.2	MAGISTRATES COURT, SUBURBAN - single or two storey, standard finishes, air-conditioning, court fit-out, office partitions	...	sqm	2625-2830	2400-2585	2605-2810	2660-2865	2725-2935
1.3.3	FIRE STATION, SUBURBAN - single storey, 24 hour manned, drive through garage for 2 appliances, office, tunic room, watch room & stores, air-conditioned, overnight accommodation for up to 10, common/training room and common kitchen/dining	...	sqm	2400-2590	2260-2435	2340-2525	2460-2650	2415-2605

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 53 TO 56 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
1.0	**ADMINISTRATION, CIVIC**	PER	$	$	$	$	$
1.3	AUTHORITIES (cont'd)						
1.3.4	POLICE STATION, SUBURBAN - single storey, 24 hour manned, fully air-conditioned, public reception, general duties offices, charge room and maximum 2 holding cells, operational support facilities, mixed staff change room facilities, staff amenities ...	sqm	2625-2830	2405-2590	2615-2820	2765-2980	2760-2975
1.4	DIPLOMATIC						
1.4.1	EMBASSY CHANCERY -)						
1.4.2	EMBASSY RESIDENCE -) Refer page 1027 ...	Note					
2.0	**AGRICULTURAL**						
2.1	FARM BUILDINGS						
2.1.1	BULK STORAGE SHED - steel framed with metal deck roof and metal clad walls to one end and two sides; no ground slab ...	sqm	275-295	215-230	225-240	270-290	255-275
2.1.2	BULK SUPERPHOSPHATE SHED - steel framed with sliding metal roof, plywood or cellulose fibre cement wall lining to one end and two side walls, reinforced concrete ground slab ...	sqm	480-515	375-405	400-430	450-485	445-475
2.1.3	HAY SHED, steel framed with metal deck roof; no ground slab :						
2.1.3.1	Open all round ...	sqm	195-210	155-165	155-170	200-215	185-200
2.1.3.2	50% metal clad enclosure walls ...	sqm	215-230	165-175	185-200	210-225	200-215
2.1.4	MACHINERY SHED, steel framed with metal deck roof and metal clad walls, no ground slab :						
2.1.4.1	Open lean-to against adjoining building, with end walls only ...	sqm	145-155	125-130	125-130	145-160	145-155
2.1.4.2	Lean-to fully enclosed, sliding doors to long wall, windows to end walls ...	sqm	230-245	185-200	200-215	225-240	215-230
2.1.4.3	Freestanding, one long wall open ...	sqm	230-245	185-200	200-215	225-240	215-230
2.1.4.4	Freestanding, sliding doors to long wall, windows to end walls ...	sqm	260-280	205-220	215-230	255-275	240-260
	ADD extra for :						
2.1.4.5	Concrete ground slab ...	sqm	82.00	83.50	91.50	66.50	97.50
2.1.4.6	Electrical services ...	Note					
2.1.5	SHEARING SHED, steel framed with metal deck roof and metal clad walls, ventilated ridge, louvred windows, raised timber floor, internal yards and doors, ramps and loading platforms, sliding external doors ...	sqm	545-590	440-475	450-485	490-530	480-520
	ADD extra for :						
2.1.5.1	Electrical services ...	Note					
2.1.5.2	Fit-out in addition to internal yards ...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 56 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE POLICE STATION COST

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
3.0	**BANKS**							
3.1	SUBURBAN, SINGLE STOREY - standard finishes, air-conditioning, bank fittings	...	sqm	2755-2965	2680-2890	2960-3190	2755-2970	3065-3305
3.2	SUBURBAN, TWO TO THREE STOREY - standard finishes, air-conditioning, bank fittings, lift	...	sqm	2845-3070	2820-3040	3020-3255	2825-3045	3160-3405
3.3	CITY CENTRE, FOUR TO TEN STOREY - high standard finishes, air-conditioning, lifts, fire sprinklers, bank fittings to lower levels and office fit-out to upper levels, considerable strong-rooms and security	...	sqm	5595-6030	6010-6475	5705-6150	6040-6510	6525-7030
3.4	FIT-OUT ONLY :							
3.4.1	of lettable space (i.e. shell) to provide single level suburban bank including standard finishes, air-conditioning, bank fittings, amenities fittings	...	sqm	1930-2080	2010-2165	1825-1965	2050-2210	2100-2260
3.4.2	of lettable space in inner city office tower to provide bank with high standard finishes, upgraded air-conditioning, bank fittings, considerable strong rooms and security; excluding access floors and work stations	...	sqm	3500-3775	4205-4535	3235-3485	3495-3765	3600-3880
4.0	**EDUCATIONAL**							
4.1	SCHOOLS (TEACHING)							
4.1.1	PRIMARY - single storey, standard finishes, buildings only, no air-conditioning	...	sqm	1775-1915	1555-1675	1620-1750	1470-1585	1660-1790
4.1.2	SECONDARY - maximum three storeys, standard finishes, built-in cupboards and fittings, buildings only, no air-conditioning :							
4.1.2.1	Overall	...	sqm	2145-2310	1885-2035	2065-2230	1795-1935	2170-2340
4.1.2.2	Classrooms	...	sqm	1920-2070	1655-1785	1910-2060	1635-1760	1955-2110
4.1.2.3	Laboratories	...	sqm	2330-2515	2280-2460	2400-2585	2400-2585	2520-2715
4.1.2.4	Covered ways and balconies, refer 4.5.5 and 4.5.6	...	Note					
4.1.3	TECHNICAL - maximum two storeys, standard finishes, built-in cupboards and fittings, buildings only, no general air-conditioning :							
4.1.3.1	Overall	...	sqm	2215-2385	2030-2190	2185-2355	2000-2155	2290-2465
4.1.3.2	Classrooms	...	sqm	2065-2225	1885-2030	2030-2190	1850-1995	2120-2290
4.1.3.3	Arts and crafts	...	sqm	2115-2280	1905-2055	2070-2235	1880-2025	2145-2310
4.1.3.4	Catering	...	sqm	2260-2435	2070-2235	2240-2415	2040-2200	2330-2510
4.1.3.5	Engineering workshops	...	sqm	2375-2560	2155-2325	2340-2525	2105-2270	2440-2630
4.1.3.6	Covered ways and balconies, refer 4.5.5 and 4.5.6	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 57 TO 60 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
4.0 **EDUCATIONAL**	PER		$	$	$	$	$
4.2 SCHOOLS (SUPPORT)							
4.2.1 SECONDARY - maximum three storeys, standard finishes, built-in cupboards and fittings, buildings only, no general air-conditioning :							
4.2.1.1 Administration ...	sqm		1890-2035	1775-1915	1975-2125	2000-2155	2080-2240
4.2.1.2 Gymnasium ...	sqm		1905-2055	1770-1905	2010-2165	1795-1935	2105-2265
4.2.1.3 Residential ...	sqm		2190-2360	2010-2165	2175-2345	2105-2270	2260-2435
4.2.1.4 Covered ways and balconies, refer 4.5.5 and 4.5.6 ...	Note						
ADD extra for :							
4.3 External works and services to secondary schools at approximately 20-30% ...	Note						
4.4 UNIVERSITIES (TEACHING)							
4.4.1 ARTS - maximum three storeys, tutorial and lecture rooms, standard finishes, built-in fittings, air-conditioning ...	sqm		3675-3965	3005-3240	3340-3605	3245-3500	3555-3835
4.4.2 LECTURE THEATRE - single or two storey, standard finishes, built-in fittings and seating, acoustic treatment, air-conditioning ...	sqm		3945-4255	3315-3575	3645-3925	3540-3815	3855-4155
4.4.3 SCIENCE - laboratory block, maximum three storeys, standard finishes, built-in fittings, air-conditioning ...	sqm		4690-5055	4025-4340	4330-4670	4255-4590	4575-4935
4.5 UNIVERSITIES (SUPPORT)							
4.5.1 ADMINISTRATION - single or two storey, standard finishes, partitions and built-in fittings, air-conditioning ...	sqm		2795-3010	2325-2505	2580-2780	2465-2660	2750-2965
4.5.2 ANIMAL RESEARCH FACILITY - two storey, with holding rooms for large animals, laboratories, standard finishes, fully air-conditioned ...	sqm		5195-5600	4405-4745	4780-5155	4680-5040	5140-5540
4.5.3 LIBRARY - maximum three storeys, standard finishes, built-in fittings, air-conditioning, excluding movable furniture ...	sqm		3630-3910	3015-3250	3335-3595	3220-3470	3575-3855
4.5.4 RESIDENTIAL COLLEGE - maximum three storeys, standard finishes, built-in fittings, motel type accommodation and facilities, partial air-conditioning ...	sqm		2955-3185	2500-2695	2710-2925	2605-2810	2815-3035
ADD extra for :							
4.5.5 BALCONIES, walkway type, reinforced concrete with balustrade to one side, no roof over :							
4.5.5.1 Basic standard finish ...	sqm		545-680	510-665	560-720	495-580	575-755
4.5.5.2 Medium/high standard finish ...	sqm		670-865	655-820	685-860	615-805	705-870
4.5.6 COVERED WAYS, open sided :							
4.5.6.1 Basic standard finish ...	sqm		480-515	540-580	495-535	470-510	580-625
4.5.6.2 Medium/high standard finish ...	sqm		580-625	720-775	640-690	625-670	740-795

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 61 TO 63 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE EDUCATIONAL COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
5.0 **ENTERTAINMENT**	PER		$	$	$	$	$
5.1 CINEMA							
5.1.1 SINGLE, SUBURBAN - seating capacity 300/500, including air-conditioning, ancillary facilities, seats, projectors, etc.	...	seat	8370-9020	8110-8740	8220-8860	8520-9190	8720-9400
5.1.2 GROUP COMPLEX, INNER CITY - two to four cinemas, seating capacity 200/800 each, including air-conditioning, ancillary facilities, seats, projectors, etc.	...	seat	12010-12950	10830-11670	11410-12300	13160-14190	12640-13630
5.2 THEATRE - seating capacity 1000/1500, including air-conditioning, ancillary facilities, bars, seats, basic stage equipment	...	seat	15450-16650	13650-14700	14800-15950	16500-17800	17000-18350
5.3 INDOOR ARENA - seating capacity 6000/8000, standard finishes, including air-conditioning, ancillary facilities, bars, seats, basic stage equipment	...	seat	14550-15700	12800-13800	13750-14850	15250-16400	15900-17100
5.4 STUDIO - television or film of standard construction, including administration offices, partial air-conditioning	...	sqm	3240-3490	2955-3185	3135-3380	3230-3480	3385-3650
6.0 **HOSPITALS, HEALTH**							
6.1 HOSPITALS							
6.1.1 DISTRICT - single storey, 60 bed, operating theatre, partial air-conditioning	...	sqm	3975-4285	3795-4090	3765-4060	3920-4230	4165-4490
	...	bed	214700-231400	204900-220900	203300-219200	211800-228300	224800-242300
6.1.2 GENERAL - multi storey, 200 bed, all facilities, lifts, air-conditioning	...	sqm	6315-6805	6890-7425	5355-5775	6120-6595	6760-7285
	...	bed	568500-612500	620000-668000	482000-519500	551000-593500	608500-656000
6.1.3 MATERNITY - multi storey, 100 bed, all facilities, lifts, air-conditioning	...	sqm	4625-4985	4620-4980	3995-4305	4910-5295	5265-5675
	...	bed	231200-249300	231000-249000	199700-215200	245600-264700	263300-283800
6.1.4 PRIVATE - single storey, 40 bed, operating theatre, partial air-conditioning	...	sqm	3220-3475	3100-3345	3120-3365	3155-3400	3365-3630
	...	bed	138500-149300	133400-143800	134100-144600	135600-146100	144800-156000
6.1.5 NURSING HOME - aged person, refer 13.5.4	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 64 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE HOSPITALS COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
6.0	**HOSPITALS, HEALTH**							
6.2	ANCILLARY							
6.2.1	CHILD CARE CENTRE - single storey, standard finishes, no air-conditioning, one playroom per ten children	...	sqm	1905-2055	1925-2075	1845-1985	1850-1995	1925-2075
		...	child	17150-18490	17330-18680	16590-17880	16660-17960	17340-18690
	ADD extra for :							
	External play areas or courtyards with enclosure walls	...	Note					
	Air-conditioning	...	Note					
6.2.2	FAMILY CENTRE - single storey, standard finishes, no air-conditioning, activity areas, small kitchen, etc.	...	sqm	1630-1755	1625-1750	1600-1720	1630-1755	1660-1790
6.2.3	GROUP PRACTICE SURGERY - single storey, standard finishes with consulting rooms, surgery, partial air-conditioning	...	sqm	2070-2230	2000-2155	1995-2150	2015-2170	2115-2280
6.2.4	MULTI PURPOSE HEALTH CENTRE - standard finishes, with accident and emergency treatment facilities, labour suite, consulting rooms, occupational therapy facilities, limited aged persons rooms, partial air-conditioning	...	sqm	3240-3490	3130-3375	3120-3365	3200-3450	3310-3570
7.0	**HOTELS, MOTELS, CLUBS**							
7.1	SUBURBAN							
7.1.1	TAVERN - single storey, standard construction and finishes, partial air-conditioning, excluding loose furniture	...	sqm	2815-3035	2595-2800	2765-2980	2775-2990	2880-3105
7.1.2	CLUB - social/sporting club, single or two storey, standard construction and finishes, large bar, lounge, dining, recreation facilities, squash courts, club offices, air-conditioning	...	sqm	2660-2870	2360-2545	2705-2920	2665-2870	2765-2980
7.2	CITY HOTELS							
7.2.1	MEDIUM TO HIGH RISE, INTERMEDIATE STANDARD - three star, excluding fitting out and car parking :							
7.2.1.1	Overall (Two examples are given hereafter, the first with small bedrooms and a very low ratio of public areas to bedroom floors and the second with slightly larger bedrooms and a medium ratio of public areas to bedroom floors)							
7.2.1.1.1	Hotel with 30.00 to 32.00 sqm total floor area per bedroom	...	sqm	4185-4510	3600-3880	3685-3975	4295-4625	4220-4550
		...	b/room	129700-139800	111600-120300	114300-123200	133100-143400	130800-141000

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 65 TO 67 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

7.0 HOTELS, MOTELS, CLUBS

		PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
7.2	CITY HOTELS (cont'd)						
7.2.1.1.2	Hotel with 45.00 sqm total floor area per bedroom	... sqm	4290-4625	3680-3970	3780-4070	4390-4730	4310-4645
		... b/room	193000-208000	165500-178500	170000-183500	197500-213000	194000-209000
	ADD extra for :						
7.2.1.2	Fitting out of total building to make operational	... b/room	44000-47000	42000-45000	42500-45500	49000-52000	41500-44500
7.2.1.3	Balconies, individual, refer 13.3.1.6 and 13.3.1.7	... Note					
7.2.2	HIGH RISE, INTERNATIONAL STANDARD - five star, excluding fitting out and car parking :						
7.2.2.1	Overall (Two examples are given hereafter, the first with a low/medium ratio and the second with a medium/ high ratio of public areas to bedroom floors)						
7.2.2.1.1	Hotel with 67.50 sqm total floor area per bedroom, comprising :	... sqm	5730-6175	4805-5180	4820-5195	5420-5845	6080-6555
		... b/room	386500-416500	324500-349500	325000-350500	366000-394500	410500-442500
	- Bedroom floors						
	Bedrooms - 32.0sqm						
	Remainder of floor - 15.5sqm 47.5sqm						
	- Other areas/floors						
	Public floors - reception, bars, restaurants, club, convention, etc., areas - 10.0sqm						
	Service area - 6.5sqm						
	Plant areas - 3.5sqm 20.0sqm						
	67.5sqm						
7.2.2.1.2	Hotel with 77.50 sqm total floor area per bedroom, comprising :	... sqm	5760-6210	4920-5305	4940-5325	5605-6045	6240-6725
		... b/room	446500-481500	381500-411000	382500-412500	434500-468500	483500-521000
	Bedroom floors						
	Bedrooms - 32.0sqm						
	Remainder of floor - 15.5sqm 47.5sqm						
	Other areas/floors						
	Public floors - reception, bars, restaurants, club, convention, etc., areas - 18.5sqm						
	Service area - 7.5sqm						
	Plant areas - 4.0sqm 30.0sqm						
	77.5sqm						

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 67 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
7.0	**HOTELS, MOTELS, CLUBS**							
7.2	CITY HOTELS (cont'd)							
7.2.2	HIGH RISE, INTERNATIONAL STANDARD (cont'd)							
7.2.2.2	Bedroom floors	...	sqm	5605-6045	4875-5250	4805-5175	5475-5905	6075-6550
7.2.2.3	Other areas/floors :							
7.2.2.3.1	Public floors - reception, bars, restaurants, club, convention, etc., areas	...	sqm	7480-8065	6280-6765	6390-6885	7280-7845	8055-8685
7.2.2.3.2	Service areas - staff, stores, goods receival etc. areas	...	sqm	3810-4105	3230-3480	3275-3530	3720-4010	4135-4460
7.2.2.3.3	Plant rooms	...	sqm	2970-3205	2595-2795	2500-2695	2905-3135	3200-3450
	ADD extra for :							
7.2.2.4	Fitting out of total building to make operational	...	b/room	66500-78500	65000-77000	66500-78500	66000-78000	66000-78000
7.2.2.5	Car parking - cost per sqm or per car, refer 10.0	...	Note					
7.2.2.6	Balconies, individual, refer 13.3.1.6 and 13.3.1.7	...	Note					
7.3	COUNTRY MOTELS							
	Refer pages 834 to 844 for Regional Indices	...	Note					
7.3.1	STANDARD - travellers type, single or two storey, standard finishes, individual facilities, unit air-conditioning, excluding fitting out :							
7.3.1.1	Overall	...	sqm	2445-2635	2320-2500	2315-2495	2425-2615	2455-2650
		...	unit	82600	78350	78250	81850	82950
7.3.1.2	Accommodation units	...	sqm	2545-2745	2425-2615	2395-2580	2530-2730	2610-2815
		...	unit	60800	57950	57150	60500	62350
7.3.1.3	Reception, dining, kitchen	...	sqm	2210-2380	2070-2235	2085-2250	2190-2360	2260-2435
		...	unit	21800	20450	20600	21600	22300
	ADD extra for :							
7.3.1.4	Fitting out of total building to make operational	...	unit	23700-29700	23500-29500	23700-29700	23600-29600	23800-29800
7.3.2	HIGH STANDARD - single or two storey, individual facilities, well finished restaurant, bar and lounge facilities, amenities, fully air-conditioned, excluding fitting out :							
7.3.2.1	Overall	...	sqm	2935-3165	2750-2965	2750-2965	2920-3150	3005-3235
		...	unit	111300	104300	104250	110750	113850
7.3.2.2	Accommodation units	...	sqm	2940-3170	2795-3015	2750-2965	2920-3150	3010-3240
		...	unit	76400	72600	71400	75850	78150
7.3.2.3	Reception, restaurant, bar, lounges, etc.	...	sqm	2760-2975	2595-2800	2580-2780	2735-2945	2835-3060
		...	unit	34400	32400	32200	34100	35350
	ADD extra for :							
7.3.2.4	Fitting out of total building to make operational	...	unit	30300-36300	30000-36000	30200-36200	30100-36100	30400-36400

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 68 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
7.0	**HOTELS, MOTELS, CLUBS**							
7.4	SERVICED APARTMENTS							
7.4.1	Multi storey, one or two bedroom units, lifts, room air-conditioning units, high standard electrical services, fire sprinklers and alarms, ground floor with reception and provision for restaurant, excluding balconies :							
7.4.1.1	Two star	...	sqm	3230-3485	2925-3150	2890-3115	3260-3515	3395-3660
		...	unit	178000-191500	161000-173500	159000-171500	179500-193500	187000-201500
7.4.1.2	Three star	...	sqm	3480-3755	3190-3440	3105-3350	3435-3700	3450-3720
		...	unit	233500-251500	214000-230500	208000-224500	230000-248000	231000-249000
	ADD extra for : Fitting out of total building to make operational :							
7.4.1.3	Two star	...	unit	15600-17600	15500-17500	15600-17600	15600-17600	15700-17700
7.4.1.4	Three star	...	unit	18700-21700	18500-21500	18600-21600	18600-21600	18800-21800
7.4.1.5	Balconies, individual, refer 13.3.1.5 to 13.3.1.7	...	Note					
7.4.1.6	Parking, refer 10.0	...	Note					
7.4.1.7	Swimming pool, recreational facilities	...	Note					
7.5	PARKING							
	Refer PARKING page 35	...	Note					
8.0	**INDUSTRIAL**							
8.1	WAREHOUSES							
	Low bay provides 3500/4500mm and high bay 6500/9000mm headroom respectively	...	Note					
8.1.1	SINGLE STOREY for LETTING, standard shell construction, metal roof, roller shutters, electrical service to board, plumbing service only; no ventilation, fire sprinklers or fit-out :\ LOW BAY							
8.1.1.1	Brick external walls	...	sqm	690-745	685-740	660-710	655-705	665-715
8.1.1.2	Precast or tilt-up concrete external walls	...	sqm	635-685	665-715	610-660	650-700	625-675
	HIGH BAY							
8.1.1.3	Brick external walls	...	sqm	810-870	790-850	780-840	770-830	795-860
8.1.1.4	Precast or tilt-up concrete external walls	...	sqm	750-805	780-840	725-780	760-820	780-845
8.1.2	SINGLE STOREY for OWNER OCCUPATION, metal roof, roller shutters, small offices and amenities, all services; no ventilation or fire sprinklers :							
	LOW BAY							
8.1.2.1	Brick external walls	...	sqm	815-875	795-855	780-840	780-840	800-865
8.1.2.2	Precast or tilt-up concrete external walls	...	sqm	760-815	785-845	720-775	760-820	780-845

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 68 TO 70 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
8.0 **INDUSTRIAL**	PER		$	$	$	$	$
8.1 WAREHOUSES (cont'd)							
8.1.2 SINGLE STOREY for OWNER OCCUPATION, metal roof, roller shutters, small offices and amenities, all services; no ventilation or fire sprinklers (cont'd) :							
HIGH BAY							
8.1.2.3 Brick external walls ...	sqm		965-1040	950-1025	910-980	920-995	960-1035
8.1.2.4 Precast or tilt-up concrete external walls ...	sqm		865-930	930-1000	890-960	920-995	915-985
8.1.3 MULTI-STOREY, maximum six storeys, reinforced concrete construction with brick walls and metal roof; office, toilet and amenities areas, loading dock, goods lift, standard lighting, fire sprinklers, mechanical ventilation/filtered air ...	sqm		1665-1795	1500-1620	1600-1720	1670-1800	1675-1810
8.1.4 WAREHOUSE/OFFICE for OWNER OCCUPATION, warehouse with metal roof, roller shutters, all services; no ventilation or fire sprinklers. Offices and showroom, to front elevation, faced brick and glazed external walls, standard finishes, air-conditioning, no lift and excluding owner's fit-out. Amenities and change room areas to rear of office/ showroom including standard fittings and partial air-conditioning :							
Warehouse, low bay -							
8.1.4.1 Brick external walls ...	sqm		780-840	760-815	720-775	735-790	780-845
8.1.4.2 Precast or tilt-up concrete external walls ...	sqm		750-805	750-805	710-765	730-785	725-780
Warehouse, high bay -							
8.1.4.3 Brick external walls ...	sqm		915-990	900-970	875-940	875-945	930-1005
8.1.4.4 Precast or tilt-up concrete external walls ...	sqm		845-910	895-965	835-900	875-945	885-950
Offices and showrooms -							
8.1.4.5 Single storey ...	sqm		1930-2080	1770-1905	1880-2025	1905-2050	1960-2115
8.1.4.6 Two storey ...	sqm		2065-2225	1885-2030	1995-2150	2055-2215	2100-2260
Amenities -							
8.1.4.7 Single storey ...	sqm		2545-2745	2200-2370	2465-2655	2560-2755	2570-2770
8.2 FACTORIES							
Small span is up to 10.0m and large span is 20.0m to 30.0m ...	Note						
8.2.1 SINGLE STOREY SMALL SPAN for LETTING, standard shell construction, metal roof, roller shutters, electrical service to board, plumbing service only; no ventilation, fire sprinklers or fit-out :							
8.2.1.1 Framed and metal clad external walls ...	sqm		680-730	640-690	630-675	630-680	645-695
8.2.1.2 Brick external walls ...	sqm		690-745	685-740	665-715	660-710	675-730
8.2.1.3 Precast or tilt-up concrete external walls ...	sqm		640-690	665-715	625-670	660-710	645-695

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 71 TO 74 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
8.0 **INDUSTRIAL**	PER		$	$	$	$	$
8.2 FACTORIES (cont'd)							
8.2.2 SINGLE STOREY SMALL SPAN for OWNER OCCUPATION, metal roof, sliding doors, small office, toilet and amenities, all services; no ventilation or fire sprinklers :							
8.2.2.1 Brick external walls ...	sqm		860-925	845-915	810-875	840-905	860-925
8.2.2.2 Precast or tilt-up concrete external walls ...	sqm		810-870	840-905	795-855	810-875	815-875
8.2.3 LARGE SPAN, heavy industry for OWNER OCCUPATION, portal framed, metal roof, sliding doors. Two storey offices and showrooms to front elevation, faced brick and glazed external walls, standard finishes, air-conditioning, no lift and excluding owner's fit-out. Rear amenities/change room area with floor/ deck over, including standard amenities and change room fittings and partial air-conditioning :							
Factory, large span :							
8.2.3.1 Framed and metal clad external walls ...	sqm		1035-1115	1040-1120	1035-1115	1050-1135	1020-1100
8.2.3.2 Brick external walls ...	sqm		1115-1200	1075-1160	1050-1130	1075-1160	1080-1165
8.2.3.3 Precast or tilt-up concrete external walls ...	sqm		1000-1075	1050-1130	1025-1105	1060-1145	1015-1095
8.2.3.4 Offices, part ground floor and full first floor ...	sqm		1990-2145	1835-1980	1915-2065	2000-2155	1980-2135
8.2.3.5 Showrooms at ground floor ...	sqm		1875-2020	1700-1830	1815-1960	1850-1995	1850-1995
8.2.3.6 Amenities at ground floor ...	sqm		2605-2810	2365-2550	2510-2705	2585-2785	2625-2830
8.3 SUNDRIES							
8.3.1 COLD STORE, -30° storage, steel framed with metal clad insulated panels to walls and ceiling, additional roof cladding; NO facilities, amenities or special equipment such as racking, cranes, etc., internal height - 10.00m ...	sqm		2390-2580	2270-2445	2150-2320	2600-2805	2220-2395
ADD extra for :							
8.3.1.1 Racking, materials handling, etc., cranes, etc. to -30° storage area ...	Note						
8.3.1.2 Office, Toilets, Amenities, small ancillary building fully serviced ...	sqm		2030-2190	1870-2015	1930-2080	2015-2170	2050-2210
8.3.2 LABORATORY, WORKSHOP, OFFICE, single storey, brick walls, metal roof with equal areas of laboratory, workshop and offices together with toilets and amenities, all serviced and including built-in fittings, partially air-conditioning/ventilated; no fire sprinklers or special equipment ...	sqm		3255-3510	3020-3255	3140-3385	3205-3455	3280-3535
8.3.3 MAINTENANCE WORKSHOP, large span for heavy use, brick walls, metal roof, service pits, lubrication areas, small office, toilet and amenities, all services; no ventilation, fire sprinklers or special equipment ...	sqm		1365-1470	1280-1380	1310-1415	1390-1495	1385-1490

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 75 TO 79 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
8.0	**INDUSTRIAL**							
8.4	GARAGE, SERVICE STATION, simple single storey of brick walls, metal roof, including lubrication area, workshop, toilets; all serviced; excluding :							
8.4.1	Special equipment and fuel tanks	...	sqm	1685-1815	1575-1700	1610-1735	1670-1800	1675-1810
	ADD extra for :							
8.4.2	External metal clad awnings	...	sqm	675-730	720-775	700-750	690-745	760-820
8.4.3	Concrete driveways	...	sqm	164	161	166	149	175
8.4.4	Fuel tanks, refer page 702	...	Note					

9.0 OFFICES

9.1 LETTABLE

Costs exclude parking, tenants subdivisional partitions and any other special owners' or tenants' requirements ... Note
Buildings are all taken as above ground level ... Note
Parking costs, refer 9.3 and 10.2 ... Note

Costs given hereafter are for either :
FINISHED FLOOR, being where the building is handed over in a fully finished condition; this excludes the fitting out given in 9.1.5.
SHELL AND CORE, being where the typical office floors and lift lobbies of the building are handed over in a bare state whilst the core components are handed over in a finished state

The prices given hereafter for shell and core lettable assume the following, where applicable :
Typical office floors and lift lobbies -

Wall finishes - Plaster or plasterboard with no surface finish

Floor finishes - Monolithic finish only to receive carpet or other

Ceiling finishes - Suspension grid installed, tiles supplied only

Air-conditioning - All plant and rigid ducting installed, flexible ducting and registers supplied only

Electrical - Wired to distribution board in core, light fittings supplied only

Fire fire sprinklers - Generally fully installed, some plugged connection points provided.

Core -

Toilets, tea and cleaners - Fully finished and serviced

Fire stairs - Fully finished and serviced including fire doors

Lifts - Fully installed including doors

9.1.1	LOW RISE (BASIC SERVICES)							
	SINGLE STOREY - standard finishes, no air-conditioning :							
9.1.1.1	Finished floor lettable type	...	sqm	1445-1560	1340-1440	1400-1510	1460-1575	1430-1540
9.1.1.2	Shell and core lettable type	...	sqm	1300-1400	1205-1295	1235-1330	1300-1400	1285-1385

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 79 TO 81 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			Adel.	Bris.	Mel.	Per.	Syd.
9.0	**OFFICES**	PER	$	$	$	$	$
9.1	LETTABLE (cont'd)						
9.1.1	LOW RISE (BASIC SERVICES) (cont'd)						
	TWO STOREY - standard finishes, no air-conditioning or lifts :						
9.1.1.3	Finished floor lettable type ...	sqm	1630-1755	1540-1660	1605-1730	1635-1760	1650-1780
9.1.1.4	Shell and core lettable type ...	sqm	1490-1605	1365-1470	1450-1560	1470-1585	1500-1615
	THREE STOREY - standard finishes, no air-conditioning or lifts :						
9.1.1.5	Finished floor lettable type ...	sqm	1680-1810	1600-1720	1630-1760	1705-1840	1725-1860
9.1.1.6	Shell and core lettable type ...	sqm	1520-1640	1425-1535	1505-1620	1525-1640	1560-1680
9.1.2	LOW RISE (FULLY SERVICED)						
	SINGLE STOREY - standard finishes, air-conditioning :						
9.1.2.1	Finished floor lettable type ...	sqm	1800-1940	1645-1775	1760-1895	1850-1995	1815-1955
9.1.2.2	Shell and core lettable type ...	sqm	1645-1770	1500-1620	1600-1720	1705-1840	1660-1790
	TWO STOREY - standard finishes, air-conditioned, no lift :						
9.1.2.3	Finished floor lettable type ...	sqm	2040-2200	1915-2065	1995-2150	2060-2220	2085-2245
9.1.2.4	Shell and core lettable type ...	sqm	1860-2005	1735-1870	1795-1935	1860-2005	1900-2050
	THREE STOREY - standard finishes, air-conditioned, lift :						
9.1.2.5	Finished floor lettable type ...	sqm	2310-2495	2155-2325	2260-2435	2305-2485	2370-2555
9.1.2.6	Shell and core lettable type ...	sqm	2125-2290	1975-2125	2075-2240	2130-2295	2205-2375
	4 - 7 STOREY - standard finishes and facilities, air-conditioning, slow lifts, no fire sprinklers :						
9.1.2.7	Finished floor lettable type ...	sqm	2525-2720	2365-2550	2460-2650	2585-2785	2650-2860
9.1.2.8	Shell and core lettable type ...	sqm	2305-2485	2185-2355	2255-2430	2380-2565	2460-2655
9.1.3	HIGH RISE (FULLY SERVICED)						
	7 - 20 STOREY - standard finishes and facilities, air-conditioning, medium speed lifts, fire sprinklers :						
9.1.3.1	Finished floor lettable type ...	sqm	3745-4035	3050-3290	3325-3585	3520-3790	3775-4070
9.1.3.2	Shell and core lettable type ...	sqm	3485-3760	2850-3070	3100-3340	3320-3580	3515-3790
	21 - 35 STOREY - standard finishes and facilities, air-conditioning, multiple high speed lifts, fire sprinklers :						
9.1.3.3	Finished floor lettable type ...	sqm	4690-5055	3705-3990	4215-4545	4670-5030	5605-6040
9.1.3.4	Shell and core lettable type ...	sqm	4415-4760	3475-3745	3970-4280	4435-4780	5305-5715
	36 - 50 STOREY - standard finishes and facilities, air-conditioning, multiple high speed lifts, fire sprinklers :						
9.1.3.5	Finished floor lettable type ...	sqm	-	4105-4425	4635-5000	5160-5560	6185-6665
9.1.3.6	Shell and core lettable type ...	sqm	-	3835-4135	4405-4750	4870-5250	5875-6335
9.1.4	PRESTIGE NATURE - for projects with a high standard of finishes and services, increase the foregoing costs by approximately ...	+ 10%					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 80 TO 85 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
9.0	**OFFICES**							
9.1	LETTABLE (cont'd)							
9.1.5	FITTING OUT OFFICES							
	The following is a guide only of the additional cost over the foregoing Finished Floor Lettable costs for the fitting out of offices in NEW buildings including such work as partitions, amenities, reception, built-in furniture and only minor alterations to services	...	Note					
	No allowance is included for loose furniture or access floors	...	Note					
9.1.5.1	STANDARD QUALITY FIT-OUT - to provide average quality general offices with a small proportion of executive							
9.1.5.1.1	offices and boardroom areas	...	sqm	540-580	540-580	510-545	620-670	520-565
	ADD extra for :							
9.1.5.1.2	Standard work stations (one per 15sqm)	...	sqm	150-170	150-170	150-170	150-170	150-170
9.1.5.2	MEDIUM QUALITY FIT-OUT - to provide average quality general offices with a higher proportion of executive offices and boardroom areas, reception :							
9.1.5.2.1	Area and some staff amenities	...	sqm	1100-1175	1100-1175	1075-1150	1260-1350	1080-1145
	ADD extra for :							
9.1.5.2.2	Medium quality work stations (one per 15 sqm)	...	sqm	215-270	215-270	215-270	215-270	215-270
9.1.5.3	HIGH QUALITY FIT-OUT - to provide high standard offices with a high proportion of executive offices, board and meeting rooms. Also reception							
9.1.5.3.1	and additional facilities including kitchen and dining	...	sqm	2440-2600	2460-2645	2315-2485	2690-2900	2370-2545
	ADD extra for :							
9.1.5.3.2	High standard work stations (one per 15 sqm)	...	sqm	330-395	330-395	330-395	330-395	330-395
	ADD extra for :							
	Loose furniture	...	sqm	85-160	85-160	85-160	85-160	85-160
	Access floors, refer pages 295 and 296	...	Note					
	Work in existing buildings	...	Note					
9.2	ADMINISTRATIVE - Refer 1.1	...	Note					
9.3	PARKING							
9.3.1	UNDERCROFT PARKING TO 1-3 STOREY OFFICES, open at ground or slightly reduced ground level including additional excavation and structural costs	...	sqm	690-745	665-715	665-715	680-735	685-740
9.3.2	PARTIALLY UNDERGROUND, one level including additional excavation and structural costs, ramps, fire sprinklers, minimal mechanical ventilation. No roof over, as within building	...	sqm	1430-1540	1365-1470	1365-1475	1495-1610	1465-1580

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 86 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
9.0	**OFFICES**							
9.3	PARKING (cont'd)							
9.3.3	FULLY UNDERGROUND, two levels including additional excavation and structural costs, ramps, fire sprinklers, mechanical ventilation, lift service. No roof over, as within building	...	sqm	1620-1745	1540-1660	1545-1665	1780-1915	1810-1950
9.4	ARCADES - ground level arcades, shops and lift entry lobbies, fully enclosed with high standard finishes and services, excluding shop fit-out	...	sqm	3605-3885	3090-3330	3475-3745	3950-4260	4060-4375
9.5	RENOVATIONS							
	Refer Refurbishing and Recycling, pages 745 to 774	...	Note					
10.0	**PARKING**							
10.1	UNDERGROUND							
10.1.1	- Below offices, etc. refer 9.3	...	Note					
10.1.2	- Reinforced concrete construction including deck over, mechanical ventilation, fire sprinklers, landscaping to top of deck; minimal facilities, no lifts :							
10.1.2.1	One level	...	sqm	1820-1960	1625-1750	1745-1880	1960-2110	1910-2060
		...	car	58200-62700	52000-56000	55800-60100	62700-67600	61200-66000
10.1.2.2	Two levels	...	sqm	1820-1960	1580-1705	1730-1865	1960-2110	1910-2060
		...	car	58200-62700	50600-54500	55300-59600	62700-67600	61200-66000
10.2	OPEN PARKING AREAS, including bitumen paving, stormwater drainage, minimal lighting, and some landscaping	...	sqm	85-91	83-90	85-98	82-88	82-88
		...	car	3060-3275	2990-3240	3060-3530	2950-3170	2950-3170
10.3	PARKING STATIONS							
10.3.1	2 - 3 STOREY - reinforced concrete construction open sides, minimal toilet facilities; no lifts, ventilation or fire sprinklers							
10.3.1.1	Ground + one level	...	sqm	565-610	630-675	565-605	500-540	540-585
		...	car	16400-17700	18200-19600	16300-17600	14600-15700	15700-16900
10.3.1.2	Ground + two levels	...	sqm	735-790	770-830	705-760	680-735	655-705
		...	car	20500-22100	21600-23200	19800-21300	19100-20600	18300-19700

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 86 TO 88 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE PARKING COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
10.0	**PARKING**							
10.3	PARKING STATIONS							
10.3.2	MULTI-STOREY - reinforced concrete construction with open sides, slow lifts, average facilities; no ventilation or fire sprinklers	...	sqm	865-935	850-920	900-970	910-980	855-920
		...	car	24200-26100	23900-25700	25200-27100	25500-27500	23900-25800
	ADD extra for :							
10.3.2.1	Mechanical ventilation	...	sqm	89.00	92.00	94.00	94.00	97.00
10.3.2.2	Fire sprinklers	...	sqm	82.00	88.00	83.00	87.00	83.00
11.0	**RECREATIONAL**							
11.1	GRANDSTANDS							
	Costs given are for the base cost per seat including ONLY these costs directly associated with this provision. An indication of the ADDITIONAL allowance to be made for club areas, change rooms, toilets, etc. whether below the seating deck or in an attached structure, is given 11.1.3/11.1.5	...	Note					
11.1.1	SINGLE TIER - stepped deck of reinforced concrete construction, steel framed and metal clad roof over, rear walls, aisles, guardrails, fair face and off-form finish, minimal lighting, fixed bench seating :							
	Part on fill/part suspended	...	seat	915-990	885-955	890-960	915-985	990-1070
	Fully suspended	...	seat	1030-1110	1035-1115	1020-1100	1055-1140	1110-1195
11.1.2	MULTI-TIER - reinforced concrete construction, lower level part on fill/ suspended, steel framed and metal clad roof over, rear walls, staircases, concourse areas, aisles, guardrails, fair face and off-form finish, minimal lighting and fixed bench seating :							
	Two levels	...	seat	1035-1115	1040-1120	1025-1105	1070-1155	1145-1230
	Three levels	...	seat	1300-1400	1265-1360	1250-1345	1340-1445	1410-1520
	ADD extra for :							
	ANCILLARY AREAS :							
11.1.3	CHANGE ROOMS - medium standard, NO air-conditioning :							
11.1.3.1	- Under seating deck	...	sqm	2300-2480	2080-2245	2235-2410	2205-2380	2425-2615
11.1.3.2	- Attached building at rear	...	sqm	2770-2985	2490-2685	2665-2870	2775-2990	2895-3120
11.1.4	PUBLIC TOILETS - medium standard :							
11.1.4.1	Under seating deck	...	sqm	2595-2795	2380-2565	2485-2675	2530-2730	2685-2895
11.1.4.2	Attached building at rear	...	sqm	3060-3300	2860-3085	2940-3165	3055-3290	3160-3405
11.1.5	CLUB AREAS - two storey, medium standard construction and finishes, large bar, lounge, dining and kitchen areas, recreational facilities, club offices, air-conditioning :							
11.1.5.1	Attached building at rear	...	sqm	2740-2950	2470-2660	2635-2840	2720-2930	2895-3120

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS

- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 88 TO 90 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS (EXCLUDING 11.1.1 AND 11.1.2)

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
11.0	**RECREATIONAL**							
11.1	GRANDSTANDS (cont'd)							
11.1.6	LIFTS/ESCALATORS - refer pages 664 to 666	...	Note					
11.2	CLUBHOUSES etc.							
11.2.1	CHANGE ROOM/TOILET BUILDING - single storey, standard construction and basic finishes, large change area and adjoining showers/toilets	...	sqm	2790-3010	2730-2945	2785-3005	2805-3025	2930-3155
11.2.2	CLUBHOUSE AND CHANGE ROOMS - single storey, standard construction and finishes, large bar and lounge, small kitchen, dining area, large basic finished change rooms and shower/toilets; no air-conditioning	...	sqm	2580-2780	2385-2570	2505-2700	2520-2720	2690-2900
11.3	BASKETBALL CENTRE - standard construction, brick walls, metal roof, timber floor to playing area, change rooms	...	sqm	1365-1470	1270-1370	1340-1440	1365-1475	1460-1575
11.4	BOWLING ALLEY - single storey, standard construction and finishes including plumbing, electrical and air-conditioning services, excluding fitting out (refer 11.4.1). (Costs based on a 2000 sqm building with 24 alleys) - building only	...	sqm	1115-1200	1020-1100	1050-1130	1100-1185	1155-1245
		...	alley	97000-104600	88800-95700	91400-98500	95600-103000	100300-108100
	ADD extra for :							
11.4.1	Fitting out including installation of lane and machine package, automatic scoring system and all other ancillary items, cafeteria fittings :	...	alley	120000-140000	120000-140000	120000-140000	120000-140000	120000-140000
11.5	COMMUNITY RECREATION CENTRE							
11.5.1	BASIC standard with minimum amenities	...	sqm	1115-1200	980-1055	1045-1125	1055-1140	1115-1205
11.5.2	MEDIUM standard with moderate amenities	...	sqm	1365-1470	1235-1330	1335-1435	1325-1425	1425-1535
11.6	GYMNASIUM - single storey, standard construction, vinyl sports floor finish; excludes change rooms/toilets (refer separate cost)	...	sqm	1930-2080	1845-1990	1970-2120	1875-2020	2155-2320
	ADD extra for :							
11.6.1	Extra for sprung timber floor	...	sqm	117.00	121.00	112.00	123.00	124.00
11.7	SQUASH COURTS							
11.7.1	BASIC developer standard including change areas, small toilet, no air-conditioning	...	sqm	1490-1605	1390-1500	1470-1585	1470-1585	1555-1675
		...	court	134100-144500	125200-134900	132300-142600	132200-142500	139800-150700

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 91 TO 93 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
11.0	**RECREATIONAL**							
11.7	SQUASH COURTS (cont'd)							
11.7.2	HIGH standard private club courts including toilets, showers, change area and viewing gallery, no air-conditioning	...	sqm	1995-2150	1870-2015	1940-2090	1985-2140	2090-2250
		...	court	239000-258000	224500-242000	232500-251000	238000-256500	250500-270000
	ADD extra for :							
11.7.3	Glass rear wall	...	court	31000-32000	31000-32000	31000-32000	31000-32000	31000-32000
11.8	SWIMMING POOLS							
	Open in-ground pools of fully formed reinforced concrete construction unless otherwise described and including all plant and filtration equipment; refer 11.8.8 for additional works	...	Note					
11.8.1	Domestic	...	No	38200-41200	36600-39500	35700-38500	37600-40500	40800-44000
11.8.2	25.0 METRE x 18.5m wide x 2.2m deep	...	No	1078500-1162500	1036000-1117000	986500-1063500	1122000-1209500	1139500-1228500
11.8.3	33.0 METRE x 25.0m wide x 2.2m deep	...	No	1681000-1812000	1669000-1799000	1525000-1644000	1763000-1900000	1783000-1922000
11.8.4	50.0 METRE x 14.0m wide x 1.0/1.3m deep	...	No	1280500-1380000	1239500-1336000	1163500-1254000	1338500-1443000	1344000-1449000
11.8.5	50.0 METRE x 21.0m wide x 1.0/2.4m deep	...	No	1614500-1740500	1562500-1684500	1477500-1592500	1694000-1826000	1720500-1854500
11.8.6	FREEFORM 'leisure', 500 square metres	...	No	574500-619500	595500-642000	541500-583500	618000-666500	614500-662500
11.8.7	9.0 METRE 'DOMESTIC' x 4.5m wide x 1.0/2.0 deep - fibreglass construction	...	No	22500-27000	22500-27000	22500-27000	22500-27000	22500-27000
	ADD extra for :							
11.8.8	Wading pool, diving pool, ancillary buildings, heating, adjoining landscaping and floodlighting	...	Note					
11.8.9	Safety fencing	...	Note					
11.9	MARINAS							
11.9.1	DREDGING, STONE REVETMENTS - Refer DETAILED PRICES, page 713	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 93 FOR DETAILED ELEMENTAL BREAKDOWN OF SQUASH COURT COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
11.0	**RECREATIONAL**							
11.9	MARINAS (cont'd)							
11.9.2	WET PENS							
11.9.2.1	FLOATING							
11.9.2.1.1	Precast concrete on polystyrene for sheltered situations and lesser berthing loads, comprising main arm with finger units both sides, catwalk from land and average number of piles	...	berth	25000	20800	21400	28800	24400
11.9.2.1.2	Precast concrete for river and some open water situations and high berthing loads, comprising main arm with finger units both sides, catwalk from land and average number of piles	...	berth	30100	25500	25900	35500	29500
11.9.2.1.3	ADD extra for : Electrical and water services	...	berth	4750	4750	4750	4750	4750
11.9.2.2	FIXED							
11.9.2.2.1	Precast concrete plank deck 1.5m wide including fibre cement pipe piles, 5.0m long timber catwalks at 10.0m centres, mooring piles; electrical and water services	...	m	6150-6650	5400-5800	5900-6350	5800-6250	6850-7400
		...	berth	14800-15950	12950-13950	14150-15250	13900-15000	16450-17750
11.9.3	DRY PEN - building of mesh screen and gates, metal roof, electrical services	...	sqm	635-685	545-585	555-600	575-620	605-655
		...	pen	17800-19100	15200-16400	15600-16800	16100-17400	17000-18300
11.9.4	WHARVES							
11.9.4.1	FLOATING - precast concrete x 3.0m wide in 9.0m lengths with piles at 18.0m centres and with catwalks to land at 30.0m centres	...	m	6000	4900	5100	7200	5900
11.9.4.2	ADD extra for : Electrical and water services	...	m	670	650	640	670	630
11.9.4.3	LAND BACKED - comprising precast concrete interlocking piles 100mm x 4.0/5.0m long, top edging, reinforced concrete ground slab x 900mm wide with edge beam, electrical and water services	...	m	3630-3915	3410-3675	3275-3530	3630-3915	4015-4330
11.9.5	ADD extra for : Office and repair facility buildings, roads and parking, water and fire services on land, landing ramps, T.A.M.I. lift, landscaping	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS

11.0 **RECREATIONAL**		PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
11.10 GOLF COURSES							
Refer DETAILED PRICES, pages 705 and 706...		Note					
11.11 TENNIS COURTS - including minimal cut or fill, preparation, drainage, 3.6m chain-wire fence, nets and posts, surface finish :							
11.11.1 Grass	...	court	22600-24300	22900-24500	22600-24100	26100-28000	22900-24500
11.11.2 Bituminous concrete	...	court	31500-33600	30400-32600	29700-31800	35900-38300	29100-30800
11.11.3 Rebound Synpave	...	court	44800-50100	44400-49500	42500-47600	50400-56600	42900-48100
11.11.4 Sportgrass	...	court	57900-61800	58300-62300	55100-58100	66700-71300	55900-60200
11.12 BOWLING GREENS - six rink green including minimal cut or fill, preparation, drainage and edge treatment, excluding associated facilities :							
11.12.1 Grass	...	No	114400-123200	102400-112000	102500-111300	115900-126100	112600-113100
11.12.1.1 ADD extra for : Reticulation	...	No	14500	13900	12700	15700	13600
11.12.2 Sportgrass	...	No	185000-195000	185000-195000	185000-195000	185000-195000	185000-195000
11.13 LANDSCAPING - planting of shrubs, excluding reticulation :							
11.13.1 Around buildings, including top soil	...	sqm	46.70-50.20	44.20-47.50	45.00-48.50	49.70-54.10	48.60-51.90
11.13.2 Large areas, excluding top soil	...	sqm	6.70-7.10	6.80-7.30	6.35-6.75	7.10-7.60	6.60-7.10
11.14 CARAVAN PARKS							
11.14.1 BASIC STANDARD, communal facilities, electric power to all bays, small office, roads and parking	...	bay	22200-23900	20800-22400	21600-23300	21600-23300	23200-25000
11.14.2 MEDIUM STANDARD, communal facilities, electric power to all bays, office and retail, laundromat, barbecue areas, small swimming pool, roads and parking	...	bay	30300-32600	29900-32300	29800-32100	30500-32900	31900-34300
11.14.3 HIGH STANDARD, individual facilities and full services to each bay, office and retail, laundromat, barbecue areas, small swimming pool, roads and parking	...	bay	40800-44000	39900-43000	39600-42600	41600-44900	42500-45900

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS

12.0 RELIGIOUS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
12.1	CHURCH HALL - single storey, standard construction, basic finishes, small kitchen, toilets, no air-conditioning	...	sqm	1130-1215	1170-1260	1095-1185	1140-1230	1165-1255
12.2	CHAPEL, CHURCH, SYNAGOGUE							
12.2.1	BASIC STANDARD construction and finishes, minimal fit-out, including stackable seating	...	sqm	1520-1640	1610-1735	1500-1615	1580-1700	1575-1700
12.2.2	MEDIUM STANDARD construction and finishes, including seating	...	sqm	2345-2530	2475-2665	2285-2465	2375-2560	2430-2620
12.2.3	HIGH STANDARD construction and finishes, including seating	...	sqm	3360-3620	3500-3775	3245-3500	3365-3630	3515-3790

13.0 RESIDENTIAL

	Costs exclude the Goods and Services Tax which took effect from July 1, 2000	...	Note					
13.1	HOUSES - SINGLE UNIT							
	The costs given hereafter are based on the total floor area between inner faces of external walls. To adjust for measurement based on outer face of external walls REDUCE the following "Houses - single unit/costs per square metre" by approximately 10%	...	Note					
13.1.1	PROJECT HOUSE - tiled roof and built on a flat site :							
	Basic standard finish -							
	90/110 sqm :							
13.1.1.1	Brick veneer	...	sqm	950-1025	1150-1240	1185-1280	960-1030	1265-1365
13.1.1.2	Full brick	...	sqm	1025-1105	1250-1345	1260-1360	975-1055	1295-1395
	120/140 sqm :							
13.1.1.3	Brick veneer	...	sqm	890-960	1095-1180	1160-1250	920-995	1215-1310
13.1.1.4	Full brick	...	sqm	950-1025	1215-1305	1250-1350	945-1015	1270-1370
	Medium standard finish -							
	120/140 sqm :							
13.1.1.5	Brick veneer	...	sqm	1035-1115	1310-1410	1340-1445	1020-1100	1395-1500
13.1.1.6	Full brick	...	sqm	1100-1190	1405-1515	1430-1540	1060-1140	1430-1540
13.1.1.7	160/190 sqm :							
	Brick veneer	...	sqm	1000-1075	1265-1360	1295-1395	975-1055	1365-1470
13.1.1.8	Full brick	...	sqm	1075-1160	1395-1505	1410-1520	1015-1090	1385-1490
13.1.2	INDIVIDUAL HOUSE - tiled roof, approximately 150/350 sqm :							
	Medium standard finish -							
13.1.2.1	Framed	...	sqm	1130-1220	1560-1680	1465-1580	1495-1610	1815-1955
13.1.2.2	Brick veneer	...	sqm	1225-1325	1715-1845	1640-1765	1650-1775	1965-2120
13.1.2.3	Full brick	...	sqm	1275-1375	1870-2015	1730-1865	1745-1885	2050-2210

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 94 AND 95 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

13.0	**RESIDENTIAL**		PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
13.1	HOUSES - SINGLE UNIT (cont'd)							
	High standard finish, no air-conditioning -							
13.1.2.4	Framed	...	sqm	1635-1765	2265-2440	2120-2285	-	2505-2700
13.1.2.5	Brick veneer	...	sqm	1760-1895	2535-2730	2305-2485	-	2745-2960
13.1.2.6	Full brick	...	sqm	1830-1975	2680-2885	2435-2620	2500-2695	2815-3035
	Prestige standard, partial air-conditioning -							
13.1.2.7	Full brick	...	sqm	2395-2580	3485-3755	3060-3300	3350-3615	3625-3905
13.1.2.8	Prestige standard, top of the range	...	up to +100%					
13.1.2.9	Verandah - ground floor level comprising, ground floor timber framing, decking, timber balustrade, timber posts, roof tile of exposed timber rafters	...	sqm	435-470	545-590	480-515	520-560	565-610
13.1.3	HOUSE ADDITIONS - Ground floor level additions to project or individual house, tiled roof, medium standard finish, NO sanitary fixtures :							
	50/ 75 sqm -							
13.1.3.1	Framed	...	sqm	1070-1155	1400-1510	1430-1540	1395-1505	1690-1825
13.1.3.2	Brick veneer	...	sqm	1195-1285	1660-1790	1525-1645	1490-1605	1825-1965
13.1.3.3	Full brick	...	sqm	1195-1285	1775-1910	1585-1710	1545-1665	1910-2060
	75/125 sqm -							
13.1.3.4	Framed	...	sqm	1030-1110	1375-1480	1395-1500	1360-1465	1645-1775
13.1.3.5	Brick veneer	...	sqm	1140-1230	1585-1710	1470-1585	1460-1570	1835-1980
13.1.3.6	Full brick	...	sqm	1165-1255	1705-1835	1570-1690	1485-1600	1835-1980
	ADD extra for :							
13.1.3.7	Bathroom fit-out and services	...	each	19000-22000	24500-28000	21500-24500	24000-27500	23500-26500
13.1.3.8	Kitchen fit-out and services	...	each	16500-19000	20500-23500	18500-21500	20500-23000	19500-22000
	Upper floor addition to single storey house, tiled roof, medium standard finish, NO sanitary fixtures :							
	Insert (part roof area only utilised) -							
13.1.3.9	50/ 75 sqm	...	sqm	1685-1815	2150-2320	2115-2280	2125-2290	2640-2850
13.1.3.10	75/125 sqm	...	sqm	1615-1740	2085-2250	2060-2220	2070-2230	2550-2745
	Full first floor -							
13.1.3.11	75/100 sqm	...	sqm	1735-1870	2445-2640	2200-2375	2240-2415	2740-2950
13.1.3.12	100/150 sqm	...	sqm	1685-1815	2315-2495	2115-2280	2125-2290	2640-2850
	A frame or Mansard -							
13.1.3.13	75/100 sqm	...	sqm	1770-1905	2455-2650	2230-2405	2275-2450	2785-3000
13.1.3.14	100/125 sqm	...	sqm	1705-1835	2340-2520	2120-2285	2130-2295	2650-2860
	ADD extra for :							
13.1.3.15	Bathroom fit-out and services	...	each	19000-22000	24500-28000	21500-24500	24000-27500	23500-26500

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 96 AND 97 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
13.0	**RESIDENTIAL**							
13.2	MULTI UNIT - LOW DENSITY							
13.2.1	TOWN HOUSE - two storey, full brick with tiled roof :							
13.2.1.1	Medium standard finish	...	sqm	1565-1690	1785-1925	2000-2155	1915-2060	2115-2280
13.2.1.2	High standard finish	...	sqm	1755-1890	1950-2100	2265-2440	2155-2325	2390-2580
13.2.2	APARTMENTS - maximum three stories, one or two bedroom units, excluding balconies, no lift :							
13.2.2.1	Basic standard finish	...	sqm	1640-1770	1650-1775	1965-2115	1735-1870	2015-2170
		...	apart.	98500-106200	98900-106600	117800-127000	104200-112300	120800-130200
13.2.2.2	Medium standard finish	...	sqm	1815-1960	1800-1940	2180-2350	1900-2045	2190-2360
		...	apart.	109000-117500	108000-116400	130800-141000	113900-122700	131300-141600
	ADD extra for :							
13.2.2.3	Balconies, individual, refer 13.3.1.5 to 13.3.1.7	...	Note					
13.3	MULTI UNIT - HIGH DENSITY							
13.3.1	APARTMENTS - multi storey, one or two bedroom units, lift, excluding balconies :							
13.3.1.1	Basic standard finish	...	sqm	2125-2290	2245-2420	2250-2425	2395-2580	2280-2455
		...	apart.	110400-119000	116800-125900	117100-126200	124500-134200	118500-127700
13.3.1.2	Medium standard finish	...	sqm	2315-2495	2410-2600	2435-2625	2560-2760	2445-2635
		...	apart.	156000-168500	163000-175500	164000-177000	172500-186000	165000-178000
13.3.1.3	High standard finish, air-conditioning	...	sqm	2890-3115	2985-3220	3035-3270	3245-3500	3035-3270
		...	apart.	332500-358500	343500-370000	349000-376000	373500-402500	349000-376500
	Multi storey two or three bedroom units, large public areas, lift, air-conditioned, excluding balconies -							
13.3.1.4	- Prestige standard	...	sqm	3215-3465	3320-3580	3380-3645	3615-3895	3405-3670
		...	apart.	611000-658500	631000-680000	642000-692000	686500-740000	647000-697500
	ADD extra for :							
	Individual unit balcony, reinforced concrete -							
13.3.1.5	Basic standard	...	sqm	555-945	535-915	545-920	510-885	570-930
13.3.1.6	Medium standard	...	sqm	590-985	570-955	580-960	545-925	605-970
13.3.1.7	High standard	...	sqm	710-1170	705-1145	685-1125	655-1095	710-1130
	Walkway type balconies, refer 4.5.5	...	Note					
13.3.2	SERVICED APARTMENTS, refer 7.4	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 97 AND 99 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
13.0	**RESIDENTIAL**							
13.4	HOLIDAY UNITS							
13.4.1	SINGLE UNIT							
13.4.1.1	Brick walls, basic standard finish approximately 60/80 sqm	...	sqm	1325-1430	1620-1745	1580-1705	1615-1740	1700-1835
13.4.1.2	Transportable type, basic standard finish, approximately 70/90 sqm	...	sqm	1005-1080	1000-1075	1120-1210	1135-1225	1245-1340
13.4.1.3	Chalet type, basic standard finish, approximately 30/40 sqm	...	sqm	1620-1750	1885-2030	1870-2015	1875-2020	2090-2250
13.4.2	MULTI UNIT One or two storey, brick walls, approximately 60/80 sqm :							
13.4.2.1	Basic standard finish	...	sqm	1450-1560	1705-1835	1685-1820	1740-1875	1765-1900
13.4.2.2	Medium standard finish	...	sqm	1615-1740	1915-2060	1865-2010	1955-2105	1915-2065
13.4.2.3	High standard finish	...	sqm	2045-2205	2415-2600	2370-2555	2500-2695	2450-2640
13.5	AGED PERSONS HOMES							
13.5.1	SINGLE STOREY self-care housing units with medium standard finish, air-conditioning, communal laundries	...	sqm	1845-1990	2100-2265	1920-2070	2190-2360	2335-2520
		...	unit	110700-119400	126000-135800	115200-124200	131500-141700	140200-151100
	Support buildings :							
13.5.1.1	Single storey amenities and social centre	...	sqm	1615-1740	1830-1970	1735-1870	1860-2005	2125-2290
13.5.2	2 - 3 STOREY self-care and central care units, medium standard finish, climate control air-conditioning, lift and communal dining, amenities and social areas	...	sqm	2110-2275	2240-2415	2240-2415	2450-2640	2580-2780
13.5.3	MULTI STOREY self-care and central care units, medium standard finish, air-conditioning, lift and communal dining, amenities and social areas	...	sqm	2525-2720	2900-3125	2565-2760	2845-3065	2920-3150
		...	unit	116100-125100	133400-143800	117900-127100	130800-141000	134400-144800
	ADD extra for :							
13.5.3.1	Balconies, individual, refer 13.3.1.5 to 13.3.1.7	...	Note					
13.5.4	NURSING HOME, built as homes, single storey, self-care and central care units, medium standard finish, air-conditioning, communal dining, amenities and social areas	...	sqm	1930-2080	2055-2215	2050-2210	2240-2415	2370-2555
		...	unit	120600-130000	128500-138500	128300-138300	140000-151000	148000-159600
13.5.4.1	ADD extra for : High care units	...	sqm	120-130	127-137	114-123	150-161	134-145

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 100 AND 101 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
13.0	**RESIDENTIAL**							
13.6	HOSTELS							
13.6.1	1 - 3 STOREY hostel or boarding house, central toilet and dining facilities, minimum finish, no lift	...	sqm	1885-2030	2020-2175	2070-2235	2155-2325	2325-2505
		...	bed	32000-34500	34300-37000	35200-38000	36700-39500	39500-42600
13.6.2	4 - 10 STOREY hostel, central toilet and dining facilities, amenities area, minimum finish, partial air-conditioning	...	sqm	2250-2425	2685-2890	2320-2505	2460-2650	2600-2800
		...	bed	38300-41200	45600-49200	39500-42500	41800-45100	44200-47600
13.7	CARPORTS, GARAGES							
13.7.1	Garage, concrete floor, metal clad walls and roof on timber framing, roller shutter door	...	sqm	595-645	615-665	695-750	640-690	745-805
13.7.2	Garage, concrete floor, cavity brick walls, metal deck roof, roller shutter doors	...	sqm	700-755	690-745	785-845	725-780	860-925
13.7.3	Carport, concrete floor, metal deck roof on steel framing	...	sqm	295-320	315-340	330-355	315-335	325-350
13.8	PARKING							
13.8.1	OPEN parking area including bitumen paving, stormwater drainage, minimal lighting and some landscaping	...	sqm	85-91	83-90	85-98	82-88	82-88
		...	car	3060-3275	2990-3240	3060-3530	2950-3170	2950-3170
13.8.2	COVERED - Refer PARKING section, pages 34 and 35	...	Note					
14.0	**RETAIL**							
14.1	SUBURBAN							
14.1.1	NEIGHBOURHOOD SHOPS - standard shell construction including shop-fronts, plasterboard ceilings, electrical service to board, cold water supply to fixture point only and drainage. No fittings, hot water, air-conditioning, fire sprinklers or malls -							
14.1.1.1	Single storey	...	sqm	815-875	755-810	740-795	790-850	795-860
14.1.1.2	Two storey	...	sqm	920-995	855-925	880-945	940-1010	940-1015
	ADD extra for :							
14.1.1.3	FIT-OUT shell including finishes, shop fittings and services, but excluding air-conditioning	...	sqm	227	209	210	239	221
14.1.1.4	Air-conditioning (Package Unit)	...	sqm	315-335	280-300	300-320	335-355	305-325
14.1.1.5	Toilets	...	Note					
14.1.1.6	Awnings	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 102 AND 103 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE RETAIL COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
14.0	**RETAIL**							
14.1	SUBURBAN (cont'd)							
14.1.2	SUPERMARKETS - standard construction including shop fronts, standard finishes, all services, air-conditioning and fire sprinklers, standard amenities, offices, etc.; excluding							
14.1.2.1	cold rooms etc. and shop fittings	...	sqm	1730-1865	1570-1695	1625-1755	1665-1795	1760-1895
	ADD extra for :							
14.1.2.2	Shop fittings and installation of cold and freezer rooms	...	sqm	360-385	315-340	315-340	380-410	350-375
14.1.3	SHOPPING CENTRE - standard shell construction including shop fronts, plasterboard ceilings, electrical service to board, plumbing service only, air-conditioning, excluding fit-out,							
14.1.3.1	fire sprinklers and mall/arcade areas	...	sqm	1610-1735	1605-1730	1690-1820	1635-1760	1825-1970
	ADD extra for :							
14.1.3.2	Fully enclosed mall/arcade, good standard finishes, air-conditioning and landscaping etc.	...	sqm	2565-2760	2580-2780	2580-2780	2575-2775	2785-3000
14.1.3.3	Trading area fit-out including wall and floor finishes, additional electrical, plumbing services	...	sqm	735-795	715-775	680-735	735-795	745-805
14.2	REGIONAL							
14.2.1	DEPARTMENT STORE - single storey, large open planned sales area, standard construction and finishes, fully serviced, air-conditioning, fire sprinklers, all facilities							
14.2.1.1	and amenities, excluding shop fittings	...	sqm	1950-2100	1815-1955	1975-2130	1950-2100	2145-2310
	ADD extra for :							
14.2.1.2	Shop fittings	...	sqm	200-215	200-215	180-195	200-215	175-190

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 103 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

			PER	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
14.0	**RETAIL**							
14.2	REGIONAL (cont'd)							
14.2.2	SHOPPING CENTRE Integrated concept combining specialty shops, supermarkets and department stores with enclosed malls and public and staff toilets. Fully air-conditioned, electrical reticulation and light fittings, fire sprinklers, ceilings, painted block walls, floor finishes, shop fronts, moderate standard finishes to malls. Excludes - shop fitting work, e.g. check-outs, internal security systems, gondolas, counters, cold rooms and refrigeration cabinets. Car parking, refer separate section							
14.2.2.1	MEDIUM STANDARD :							
14.2.2.1.1	Single storey	...	sqm	2115-2280	1980-2130	2150-2320	2195-2365	2350-2535
14.2.2.1.2	Two storey	...	sqm	2385-2575	2240-2415	2470-2660	2460-2650	2650-2860
14.2.2.2	HIGH STANDARD :							
14.2.2.2.1	Single storey	...	sqm	2380-2565	2195-2370	2455-2645	2385-2570	2665-2875
14.2.2.2.2	Two storey	...	sqm	2670-2880	2515-2710	2755-2970	2725-2935	2980-3215
14.3	CITY							
14.3.1	DEPARTMENT STORE - three or four storey, standard construction and finishes, fully serviced, air-conditioning, fire sprinklers, all facilities and							
14.3.1.1	amenities, excluding shop fittings	...	sqm	2715-2930	2970-3205	2720-2930	2845-3065	2910-3135
	ADD extra for :							
14.3.1.2	Shop fittings	...	sqm	210-225	190-200	180-190	230-245	190-200
14.4	PARKING							
14.4.1	OPEN parking areas including bitumen paving, stormwater drainage, minimal lighting and some landscaping	...	sqm	85-91	83-90	85-98	82-88	82-88
		...	car	3060-3275	2990-3240	3060-3530	2950-3170	2950-3170
14.4.2	COVERED basement or multi-storey - refer PARKING section, pages 34 and 35	...	Note					

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 20 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 104 AND 105 FOR DETAILED ELEMENTAL BREAKDOWN OF ABOVE COSTS

ESTIMATING ELEMENTAL COSTS OF BUILDINGS

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This section is intended as a guide to the value of the various elements. The Total cost given is the approximate mean of the cost range for the respective buildings in the previous section BUILDING COSTS PER SQUARE METRE. Identical item numbers have been used for each section.

Refer page 20 for notes on what the costs given in BUILDING COSTS PER SQUARE METRE include and exclude.

The estimated costs within this section are those pertaining to SYDNEY, however the cost range of the other cities is given to FACILITATE ADJUSTMENT.

The breakdown into approximate elemental costs enables evaluation and adjustment of the figure used for initial feasibility studies.

For a more detailed assessment of the Elemental Costs of Buildings, it will be necessary to refer to the section COMPARATIVE COSTS, where it is possible to more accurately evaluate the various elements.

Costs exclude the Goods and Services Tax (G.S.T.).

DEFINITION OF ELEMENTS

The following is a general and abridged description of the various components of the elements listed hereafter.

PRELIMINARIES

Normal Builder's preliminaries i.e. plant and scaffolding, offices and sheds, hoardings and fences, temporary services, insurances, notices and fees.

SUBSTRUCTURE

The structurally sound and watertight base upon which to build, including excavation for foundations, piers, piles, column foundations, strip footings, foundation beams, foundation walls, hardcore filling; damp-proofing membranes; ground floor slab structures; subsoil drainage; ducts, pits and service tunnels; entrance steps, ramps and their finishes; structural screeds and toppings; all other work up to but excluding the lowest floor finish.

SUPERSTRUCTURE

COLUMNS (Framed buildings) - The upright supports to the upper floors and roof forming part of a framed structure, including internal and external columns from tops of columns to bases; column casings; all protective non-decorative coatings.Exclusions are portal frames (refer Roof); columns to non-framed (load bearing) structures (refer External or Internal Walls).

UPPER FLOORS - Floor structures above that at the lowest level.All beams; concrete, precast and in-situ floors; timber framed floors; structural screeds and toppings; overhangs and sunhoods integral with floors; steps and ramps in the one floor level; all protective non-decorative coatings.

STAIRS - The structural connection between two or more floor levels together with associated finishes, including landings; ramps between floor levels; fire escapes; supporting framework; access ladders; spiral staircases; tread, riser, string and soffit finishes; balustrades and handrails.

ROOF - To provide a structurally sound and watertight covering over the building including portal frames; roof construction; gable and other walls in roof spaces; parapet walls and roof balustrades; thermal insulation; rooflights; eaves, verges and fascias; rainwater goods; internal stormwater drainage runs; all protective non-decorative coatings.

EXTERNAL WALLS - The vertical enclosure around the building other than 'Windows' and 'External Doors' from Substructure to Roof including structural walls; spandrel, curtain and window walls; external shop fronts; glazed screen walls; columns and isolated piers to non-framed (load bearing) structures; solar screen walls; plant room air flow screens; insulation to external walls; external finishes to columns, slab edges, beams, projecting overhangs and walls.Exclusions are internal finishes to external walls and columns; walls in Roof; doors; sun protection to windows; beams integral with slabs (refer Upper Floors or Roof).

WINDOWS - Openings in External Walls to provide light and ventilation including flyscreens; louvres; guard grilles; remote control gear; sun protection to windows; curtains, blinds, track and pelmets; window sills and linings; hardware; decoration.Exclusions are rooflights; window walls and glazed screens; solar screen walls; window cleaning equipment.

EXTERNAL DOORS - The access ways into the building both for pedestrians and vehicles, including frames; linings; glazing; architraves; hardware; panels and highlights over; flydoors;roller shutters; garage doors; fire doors; grille and chain wire doors; gates; service cupboard doors and thresholds; decoration.

INTERNAL WALLS - Permanent division of internal spaces into separate rooms or to enclose ducts and other non-usable areas, including walls; internal columns and isolated piers to non-framed (load bearing) structures; part height solid walls glazed over to ceiling. Exclusions are Internal Screens and Borrowed Lights; wall finishes; works in Roof.

INTERNAL SCREENS - To screen off or temporarily divide internal spaces into separate compartments and to allow the transfer of light through Internal Walls including office partitioning; glazed screens; internal shop fronts; fold away and operable walls; chain wire and grille screens; toilet partitions and screen walls; borrowed lights; balustrades and rails not associated with staircases; all finishes and decorations.

INTERNAL DOORS - Passage ways through Internal Walls, internal screens and partitions and to provide access to other rooms, service cupboards and ducts, including frames, linings; glazing; architraves; pelmets; hardware and door grilles; chain wire and grille doors; toilet doors; strong room doors; fire doors; roller shutters; service cupboard doors; duct access panels; fanlights and panels over and linings to blank openings; decoration.

DEFINITION OF ELEMENTS (cont'd)

FINISHES	WALL - To finish and decorate all interior faces of Columns, External Walls, and Internal Walls, including finishes to internal faces of external walls and columns; acoustic wall linings; extra costs involved for face bricks and off form concrete; splashbacks and dados. Exclusions are finishes to Internal Screens; skirtings and cornices; all external finishes to external walls and columns. FLOOR - To provide a satisfactory finish to Upper Floors and Substructure for walking on, including all preparatory work and finishing; skirtings; screeds; timber floor finishes; dividing strips; mats and matwells; duct and pit covers; carpeting used as a permanent floor finish; timber and other finishes to concrete floors. Exclusions are structural screeds and toppings; landing and stair finishes; trafficable roof covering. CEILING - To finish and decorate all internal soffits of Upper Floors and Roof over rooms, including preparatory work; suspended false ceilings; proprietary suspended ceiling systems; acoustic ceiling linings; extra costs involved for off form concrete; ceiling manholes; framing to bulkheads and cornices.Exclusions are eaves soffits; stair and landing soffits.
FITTINGS	FITMENTS - To fit out the building with built-up fitments and fixed items included in the main contract, including benches; cupboards; shelving; racks; seats; counters; chalk-boards; notice boards; signs and name plates; coat rails and hooks; mirrors; wall hatches; daises and stages.Exclusions are loose furniture and furnishings; curtains and blinds; Special Equipment. SPECIAL EQUIPMENT - To provide items of equipment of unitary, commercially available type and/or a type not covered by other elements, including the likes of window cleaning, gymnasium, mortuary and photographic equipment; audio-visual aids; laboratory, laundry, kitchen and central sterile services department (C.S.S.D.) type equipment; dental and workshop equipment; boiling water units; laboratory stills; special black-out facilities; bed pan washers; linen and refuse disposal equipment; refrigerators and refrigerated drinking water coolers; incinerators; sanitary incinerators; circulating fans; specified Builders work in connection with this equipment. Exclusions are cool rooms and process cooling, fire fighting equipment; Sanitary Fixtures.
SERVICES	PLUMBING - To fit out the building with normal sanitary fixtures together with hot and cold water services and soil and waste plumbing systems together with all associated ancillaries.Exclusions are rainwater disposal systems and external services from the outside face of the building. MECHANICAL - Comprises air conditioning, evaporative cooling, mechanical ventilation, specialist hospital services and the like, reticulated steam and hot water systems. FIRE - To detect and/or extinguish fires, including sprinklers and other automatic extinguishing systems; fire indicator board; manual and automatic fire alarm installations; fire fighting equipment; hydrant installations and hose reels and cupboards; hand appliances. Exclusions are fire doors and fire proofing. ELECTRICAL - To provide all light and power and emergency light and power, power outlets and light fittings, including main distribution board; sub-mains and distribution boards; emergency lighting systems; power sub-mains to mechanical equipment, etc.; systems such as telephone, internal telephone, public address, call, emergency warning and intercommunication, personal paging, clock and/or bell, TV antenna and closed circuit TV. TRANSPORTATION - To transport personnel and/or goods from floor to floor or area to area including all lifts, hoists and conveyor systems; escalators; all associated equipment and work other than structural building work. SPECIAL SERVICES - To provide services or installations not covered by other elements.
EXTERNAL SERVICES	Covering sewer and stormwater drainage, water supply, etc. between and up to 3.0 metres from the outside face of the building.
CONTINGENCY	Being the contract contingency.

1.0 ADMINISTRATION, CIVIC

1.1 ADMINISTRATIVE

ELEMENT	1.1.1 Administration Office Single Storey		1.1.2 Administration Office 2-3 Storey		1.1.3 Administration Office 3-10 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	283.00	10.9	346.00	11.8	475.00	13.8
SUBSTRUCTURE	129.50	5.0	94.25	3.2	75.75	2.2
SUPERSTRUCTURE						
Columns	-	-	49.25	1.7	60.50	1.8
Upper Floors	-	-	221.00	7.5	296.75	8.6
Staircase	-	-	40.25	1.4	38.75	1.1
Roof	287.25	11.1	145.25	4.9	70.00	2.0
External Walls)	359.75	13.9	377.25	12.8	438.50	12.8
Windows)						
External Doors	34.50	1.3	34.00	1.2	10.25	0.3
Internal Walls	74.75	2.9	92.25	3.1	92.25	2.7
Internal Screens	117.75	4.6	119.75	4.1	103.50	3.0
Internal Doors	29.50	1.1	33.50	1.1	32.25	0.9
FINISHES						
Wall	62.00	2.4	69.00	2.4	69.75	2.0
Floor	111.25	4.3	113.00	3.8	108.50	3.2
Ceiling	88.75	3.4	89.25	3.0	90.00	2.6
FITTINGS						
Fitments	78.50	3.1	78.50	2.7	71.75	2.1
Special	-	-	-	-	-	-
SERVICES						
Plumbing	174.75	6.8	185.50	6.3	173.00	5.0
Mechanical	437.00	16.9	444.75	15.1	562.00	16.4
Fire	44.25	1.7	44.25	1.5	122.75	3.6
Electrical	198.00	7.7	226.50	7.7	237.25	6.9
Transportation	-	-	54.50	1.9	211.50	6.2
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.75	0.4	8.50	0.3	9.00	0.3
CONTINGENCY	64.75	2.5	73.50	2.5	86.00	2.5
TOTAL (Mean cost - SYDNEY)	2585.00	100.0	2940.00	100.0	3435.00	100.0
Cost Range : Adelaide	2445-2635		2770-2985		3415-3685	
Brisbane	2095-2260		2640-2845		3320-3580	
Canberra	2625-2825		2970-3205		-	
Darwin	2460-2660		2830-3060		-	
Hobart	2600-2800		2930-3155		-	
Melbourne	2395-2580		2680-2885		3240-3495	
Perth	2445-2635		2800-3015		3220-3470	
Sydney	2485-2680		2825-3050		3305-3560	

1.0 ADMINISTRATION, CIVIC

1.2 CIVIC

ELEMENT	1.2.1 City Hall 2000 Capacity		1.2.2 Town Hall 1000 Capacity		1.2.3 Civic Centre 300-500 Capacity		1.2.4 Civic Centre 500-750 Capacity	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	937.75	15.2	658.25	14.0	235.00	10.5	377.25	10.8
SUBSTRUCTURE	211.25	3.4	144.50	3.1	144.00	6.4	162.50	4.6
SUPERSTRUCTURE								
Columns	203.50	3.3	146.50	3.1	18.00	0.8	49.75	1.4
Upper Floors	413.50	6.7	349.50	7.5	-	-	47.50	1.4
Staircase	69.75	1.1	64.50	1.4	-	-	5.00	0.1
Roof	279.75	4.5	255.75	5.5	260.25	11.6	277.75	7.9
External Walls)	755.00	12.3	611.50	13.1	445.50	19.9	546.50	15.6
Windows)								
External Doors	19.25	0.3	26.25	0.6	30.00	1.3	41.00	1.2
Internal Walls	228.75	3.7	182.25	3.9	100.50	4.4	112.75	3.2
Internal Screens	102.00	1.7	52.50	1.1	6.75	0.3	5.00	0.1
Internal Doors	36.50	0.6	43.25	0.9	44.25	2.0	41.75	1.2
FINISHES								
Wall	126.25	2.1	109.75	2.3	69.25	3.1	123.75	3.6
Floor	174.75	2.9	141.00	3.0	134.75	6.0	155.50	4.4
Ceiling	195.50	3.2	158.25	3.4	115.75	5.2	175.50	5.0
FITTINGS								
Fitments	99.25	1.6	72.25	1.5	73.75	3.3	154.25	4.4
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	314.50	5.1	231.75	4.9	171.50	7.7	277.75	7.9
Mechanical	1108.75	18.0	719.75	15.4	81.75	3.6	583.00	16.6
Fire	154.00	2.5	124.25	2.6	39.50	1.8	47.75	1.4
Electrical	284.50	4.6	276.50	5.9	207.25	9.3	227.00	6.5
Transportation	277.50	4.5	196.00	4.2	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	9.00	0.2	3.50	0.1	6.25	0.3	6.00	0.2
CONTINGENCY	154.00	2.5	117.25	2.5	56.00	2.5	87.75	2.5
TOTAL (Mean cost - SYDNEY)	6155.00	100.0	4685.00	100.0	2240.00	100.0	3505.00	100.0
Cost Range : Adelaide	5585-6020		4380-4725		2120-2285		3265-3520	
Brisbane	4885-5265		3875-4175		1980-2130		2965-3200	
Canberra	-		-		2605-2805		-	
Darwin	-		-		2400-2580		-	
Hobart	-		-		2205-2375		-	
Melbourne	5555-5985		4250-4580		2115-2280		3280-3535	
Perth	5705-6150		4480-4830		2160-2330		3265-3520	
Sydney	5920-6385		4510-4860		2155-2320		3370-3635	

1.0 ADMINISTRATION, CIVIC

1.2 CIVIC

ELEMENT	1.2.5 Library Suburban Single Storey		1.2.6 Library Suburban 2-3 Storey		1.2.7 Library Capital City or State	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	282.25	10.2	319.50	10.4	1193.25	19.6
SUBSTRUCTURE	129.50	4.7	103.00	3.4	116.00	1.9
SUPERSTRUCTURE						
Columns	-	-	60.50	2.0	108.75	1.8
Upper Floors	-	-	232.25	7.6	424.75	7.0
Staircase	-	-	41.75	1.4	73.00	1.2
Roof	295.75	10.7	119.25	3.9	154.75	2.6
External Walls)	425.00	15.4	413.75	13.5	918.50	15.1
Windows)						
External Doors	38.75	1.4	37.75	1.2	37.50	0.6
Internal Walls	70.00	2.5	83.00	2.7	115.25	1.9
Internal Screens	94.25	3.4	94.00	3.1	55.75	0.9
Internal Doors	23.75	0.9	26.00	0.8	45.25	0.7
FINISHES						
Wall	52.00	1.9	50.75	1.6	146.00	2.4
Floor	114.00	4.1	110.00	3.6	161.50	2.7
Ceiling	114.75	4.2	114.50	3.7	165.00	2.7
FITTINGS						
Fitments	155.75	5.6	150.75	4.9	128.00	2.1
Special	-	-	-	-	-	-
SERVICES						
Plumbing	119.75	4.4	122.00	4.0	164.50	2.7
Mechanical	486.00	17.6	523.00	17.1	1024.25	16.8
Fire	43.75	1.6	44.00	1.4	192.00	3.2
Electrical	239.50	8.7	278.00	9.1	419.75	6.9
Transportation	-	-	58.25	1.9	211.25	3.5
Special	-	-	-	-	69.25	1.1
EXTERNAL SERVICES	6.25	0.2	6.25	0.2	3.75	0.1
CONTINGENCY	69.00	2.5	76.75	2.5	152.00	2.5
TOTAL (Mean cost - SYDNEY)	2760.00	100.0	3065.00	100.0	6080.00	100.0
Cost Range : Adelaide	2650-2860		2770-2985		5125-5520	
Brisbane	2270-2445		2605-2810		4260-4590	
Canberra	3010-3245		3065-3305		-	
Darwin	2725-2985		3075-3320		-	
Hobart	2860-3080		2985-3220		-	
Melbourne	2635-2840		2890-3115		4750-5125	
Perth	2610-2815		2920-3150		5105-5505	
Sydney	2655-2865		2950-3180		5850-6305	

1.0 ADMINISTRATION, CIVIC

1.2 CIVIC

ELEMENT	1.2.8 Art Gallery Museum Capital City		1.2.9 Public Toilets	
	$/sqm	%	$/sqm	%
PRELIMINARIES	1372.00	19.4	304.50	10.5
SUBSTRUCTURE	151.75	2.1	124.25	4.3
SUPERSTRUCTURE				
Columns	100.00	1.4	-	-
Upper Floors	466.50	6.6	-	-
Staircase	80.75	1.1	-	-
Roof	278.00	3.9	212.25	7.3
External Walls)	857.00	12.1	473.00	16.3
Windows)				
External Doors	18.25	0.3	59.00	2.0
Internal Walls	180.25	2.5	130.50	4.5
Internal Screens	18.25	0.3	156.75	5.4
Internal Doors	54.75	0.8	58.75	2.0
FINISHES				
Wall	76.50	1.1	68.75	2.4
Floor	215.75	3.1	82.75	2.8
Ceiling	423.50	6.0	77.25	2.6
FITTINGS				
Fitments	85.25	1.2	77.25	2.6
Special	43.00	0.6	-	-
SERVICES				
Plumbing	90.75	1.3	903.50	31.0
Mechanical	1400.25	19.8	-	-
Fire	160.50	2.3	-	-
Electrical	602.75	8.5	75.00	2.6
Transportation	77.50	1.1	-	-
Special	136.25	1.9	-	-
EXTERNAL SERVICES	3.75	0.1	33.75	1.2
CONTINGENCY	176.75	2.5	72.75	2.5
TOTAL (Mean cost - SYDNEY)	7070.00	100.0	2910.00	100.0
Cost Range : Adelaide	5960-6425		2735-2945	
Brisbane	4900-5280		2545-2745	
Canberra	-		-	
Darwin	-		-	
Hobart	-		2880-3105	
Melbourne	5540-5970		2670-2875	
Perth	5920-6380		2800-3015	
Sydney	6805-7335		2800-3020	

1.0 ADMINISTRATION, CIVIC

1.3 AUTHORITIES

ELEMENT	1.3.1 Law Courts Capital City		1.3.2 Magistrates Court Suburban		1.3.3 Fire Station Suburban		1.3.4 Police Station Suburban	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	1231.25	19.9	301.75	10.7	242.75	9.7	308.25	10.7
SUBSTRUCTURE	165.00	2.7	142.00	5.0	145.25	5.8	106.75	3.7
SUPERSTRUCTURE								
Columns	147.50	2.4	30.25	1.1	35.00	1.4	47.00	1.6
Upper Floors	386.50	6.3	78.25	2.8	-	-	143.00	5.0
Staircase	57.75	0.9	23.75	0.8	-	-	17.50	0.6
Roof	130.00	2.1	306.75	10.8	252.75	10.1	234.00	8.2
External Walls)	723.00	11.7	435.50	15.4	300.50	12.0	342.75	11.9
Windows)								
External Doors	8.75	0.1	13.75	0.5	194.50	7.8	18.50	0.6
Internal Walls	220.50	3.6	131.75	4.7	108.00	4.3	109.50	3.8
Internal Screens	67.50	1.1	12.00	0.4	5.00	0.2	36.25	1.3
Internal Doors	101.25	1.6	92.25	3.3	66.25	2.6	76.25	2.6
FINISHES								
Wall	144.50	2.3	76.25	2.7	62.00	2.5	80.75	2.8
Floor	187.50	3.0	98.50	3.5	72.75	2.9	96.75	3.4
Ceiling	194.75	3.2	105.75	3.7	130.75	5.2	96.75	3.4
FITTINGS								
Fitments	181.00	2.9	121.25	4.3	273.75	10.9	56.50	2.0
Special	-	-	-	-	-	-	14.25	0.5
SERVICES								
Plumbing	112.75	1.8	77.00	2.7	126.50	5.0	176.75	6.2
Mechanical	1018.75	16.5	468.50	16.5	175.00	7.0	497.00	17.3
Fire	142.00	2.3	21.00	0.7	15.00	0.6	9.00	0.3
Electrical	529.25	8.6	175.25	6.2	212.00	8.4	314.50	11.0
Transportation	214.25	3.5	-	-	-	-	-	-
Special	53.75	0.9	41.75	1.5	23.25	0.9	6.00	0.2
EXTERNAL SERVICES	8.00	0.1	6.00	0.2	6.25	0.2	10.25	0.4
CONTINGENCY	154.50	2.5	70.75	2.5	62.75	2.5	71.75	2.5
TOTAL (Mean cost - SYDNEY)	6180.00	100.0	2830.00	100.0	2510.00	100.0	2870.00	100.0
Cost Range : Adelaide	4900-5285		2625-2830		2400-2590		2625-2830	
Brisbane	4345-4685		2400-2585		2260-2435		2405-2590	
Canberra	-		-		-		-	
Darwin	-		-		-		-	
Hobart	-		-		-		-	
Melbourne	4545-4895		2605-2810		2340-2525		2615-2820	
Perth	5455-5875		2660-2865		2460-2650		2765-2980	
Sydney	5945-6410		2725-2935		2415-2605		2760-2975	

3.0 BANKS

ELEMENT	3.1 Bank Suburban Single Storey		3.2 Bank Suburban 2-3 Storey		3.3 Bank City Centre 4-10 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	329.75	10.4	373.00	11.4	1282.75	18.9
SUBSTRUCTURE	144.25	4.5	106.75	3.2	78.00	1.1
SUPERSTRUCTURE						
Columns	-	-	36.25	1.1	91.75	1.3
Upper Floors	-	-	235.75	7.2	441.75	6.5
Staircase	-	-	41.75	1.3	46.50	0.7
Roof	251.00	7.9	117.25	3.6	126.75	1.9
External Walls)	435.75	13.7	494.75	15.1	1148.25	16.9
Windows)						
External Doors	57.00	1.8	29.50	0.9	58.25	0.9
Internal Walls	92.75	2.9	123.00	3.8	185.25	2.7
Internal Screens	68.50	2.1	47.00	1.4	148.00	2.2
Internal Doors	53.25	1.7	33.75	1.0	44.25	0.7
FINISHES						
Wall	68.00	2.1	69.75	2.1	94.00	1.4
Floor	104.00	3.3	103.00	3.1	138.00	2.0
Ceiling	108.25	3.4	110.25	3.4	134.25	2.0
FITTINGS						
Fitments	456.00	14.3	191.25	5.8	262.25	3.9
Special	-	-	-	-	-	-
SERVICES						
Plumbing	168.25	5.3	163.75	5.0	192.25	2.8
Mechanical	504.00	15.8	546.50	16.6	1116.75	16.5
Fire	24.50	0.8	53.75	1.6	171.00	2.5
Electrical	233.75	7.3	257.00	7.8	496.50	7.3
Transportation	-	-	65.50	2.0	275.50	4.1
Special	-	-	-	-	74.00	1.1
EXTERNAL SERVICES	6.25	0.2	3.25	0.1	4.50	0.1
CONTINGENCY	79.75	2.5	82.25	2.5	169.50	2.5
TOTAL (Mean cost - SYDNEY)	3185.00	100.0	3285.00	100.0	6780.00	100.0
Cost Range : Adelaide	2755-2965		2845-3070		5595-6030	
Brisbane	2680-2890		2820-3040		6010-6475	
Canberra	3015-3255		3150-3400		-	
Darwin	3030-3250		3070-3320		-	
Hobart	3090-3330		3155-3400		-	
Melbourne	2960-3190		3020-3255		5705-6150	
Perth	2445-2635		2825-3045		6040-6510	
Sydney	3065-3305		3160-3405		6525-7030	

4.0 EDUCATIONAL

4.1 SCHOOLS (TEACHING)

ELEMENT	4.1.1 Primary School Single Storey		4.1.2.1 Secondary School Overall Max. 3 Storeys		4.1.2.2 Secondary School Classrooms Max. 3 Storeys		4.1.2.3 Secondary School Laboratories Max. 3 Storeys	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	182.00	10.6	283.50	12.6	254.00	12.5	325.75	12.4
SUBSTRUCTURE	109.25	6.3	86.75	3.8	86.25	4.2	83.75	3.2
SUPERSTRUCTURE								
Columns	-	-	38.75	1.7	39.75	2.0	39.75	1.5
Upper Floors	-	-	197.75	8.8	201.00	9.9	199.75	7.6
Staircase	-	-	35.75	1.6	36.25	1.8	35.75	1.4
Roof	306.75	17.8	168.25	7.5	168.25	8.3	165.25	6.3
External Walls)	215.75	12.5	260.50	11.6	275.75	13.5	277.75	10.6
Windows)								
External Doors	11.25	0.7	13.50	0.6	12.00	0.6	12.25	0.5
Internal Walls	84.50	4.9	113.75	5.0	86.00	4.2	97.25	3.7
Internal Screens	52.50	3.1	51.50	2.3	53.00	2.6	53.25	2.0
Internal Doors	36.50	2.1	38.75	1.7	29.50	1.4	53.50	2.0
FINISHES								
Wall	28.50	1.6	34.50	1.5	32.00	1.6	71.25	2.7
Floor	67.00	3.9	77.75	3.4	72.75	3.6	94.00	3.6
Ceiling	88.25	5.1	89.75	4.0	89.00	4.4	85.50	3.3
FITTINGS								
Fitments	179.00	10.4	279.25	12.4	222.25	10.9	331.25	12.7
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	102.25	5.9	128.00	5.7	44.50	2.2	205.00	7.8
Mechanical	12.50	0.7	23.25	1.0	13.75	0.7	80.25	3.1
Fire	10.50	0.6	19.75	0.9	20.75	1.0	20.75	0.8
Electrical	191.25	11.1	254.50	11.3	244.25	12.0	319.25	12.2
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	4.00	0.2	3.00	0.1	3.00	0.1	3.25	0.1
CONTINGENCY	43.25	2.5	56.50	2.5	51.00	2.5	65.50	2.5
TOTAL (Mean cost - SYDNEY)	1725.00	100.0	2255.00	100.0	2035.00	100.0	2620.00	100.0
Cost Range : Adelaide	1775-1915		2145-2310		1920-2070		2330-2515	
Brisbane	1555-1675		1885-2035		1655-1785		2280-2460	
Canberra	2200-2375		2350-2535		2255-2430		3100-3340	
Darwin	2385-2550		2530-2730		2350-2540		4400-4750	
Hobart	1855-2000		2375-2560		2305-2485		3025-3260	
Melbourne	1620-1750		2065-2230		1910-2060		2400-2585	
Perth	1470-1585		1795-1935		1635-1760		2400-2585	
Sydney	1660-1790		2170-2340		1955-2110		2520-2715	

4.0 EDUCATIONAL

4.1 SCHOOLS (TEACHING)

ELEMENT	4.1.3.1 Technical School Overall Max. 2 Storeys		4.1.3.2 Technical School Classrooms Max. 2 Storeys	
	$/sqm	%	$/sqm	%
PRELIMINARIES	292.50	12.3	277.75	12.6
SUBSTRUCTURE	129.00	5.4	131.75	6.0
SUPERSTRUCTURE				
Columns	52.00	2.2	51.25	2.3
Upper Floors	167.75	7.1	166.25	7.5
Staircase	31.00	1.3	30.50	1.4
Roof	216.25	9.1	213.25	9.7
External Walls)	342.00	14.4	333.00	15.1
Windows)				
External Doors	35.25	1.5	23.50	1.1
Internal Walls	83.50	3.5	95.75	4.3
Internal Screens	31.00	1.3	44.50	2.0
Internal Doors	38.50	1.6	34.50	1.6
FINISHES				
Wall	53.75	2.3	36.50	1.6
Floor	57.75	2.4	70.50	3.2
Ceiling	92.50	3.9	92.00	4.2
FITTINGS				
Fitments	187.75	7.9	223.25	10.1
Special	16.00	0.7	-	-
SERVICES				
Plumbing	110.25	4.6	63.75	2.9
Mechanical	100.75	4.2	-	-
Fire	20.00	0.8	19.50	0.9
Electrical	240.00	10.1	239.25	10.9
Transportation	-	-	-	-
Special	19.50	0.8	-	-
EXTERNAL SERVICES	3.50	0.1	3.00	0.1
CONTINGENCY	59.50	2.5	55.25	2.5
TOTAL (Mean cost - SYDNEY)	2380.00	100.0	2205.00	100.0
Cost Range : Adelaide	2215-2385		2065-2225	
Brisbane	2030-2190		1885-2030	
Canberra	2435-2620		2335-2520	
Darwin	2750-2960		2530-2730	
Hobart	2490-2685		2355-2540	
Melbourne	2185-2355		2030-2190	
Perth	2000-2155		1850-1995	
Sydney	2290-2465		2120-2290	

4.0 EDUCATIONAL

4.1 SCHOOLS (TEACHING)

ELEMENT	4.1.3.3 Technical School Arts and Crafts Max. 2 Storeys		4.1.3.4 Technical School Catering Max. 2 Storeys		4.1.3.5 Technical School Engineering Workshops Max. 2 Storeys	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	280.25	12.6	299.50	12.4	315.50	12.4
SUBSTRUCTURE	141.00	6.3	123.50	5.1	126.75	5.0
SUPERSTRUCTURE						
Columns	51.50	2.3	50.75	2.1	29.25	1.1
Upper Floors	167.25	7.5	175.00	7.2	93.50	3.7
Staircase	29.50	1.3	28.50	1.2	13.25	0.5
Roof	197.75	8.9	202.50	8.4	284.00	11.2
External Walls)	305.75	13.7	300.75	12.5	316.75	12.5
Windows)						
External Doors	25.00	1.1	20.25	0.8	47.00	1.9
Internal Walls	79.50	3.6	73.75	3.0	67.50	2.7
Internal Screens	35.00	1.6	12.25	0.5	10.50	0.4
Internal Doors	37.50	1.7	74.75	3.1	50.50	2.0
FINISHES						
Wall	64.75	2.9	69.25	2.9	44.00	1.7
Floor	59.75	2.7	78.50	3.2	58.00	2.3
Ceiling	92.75	4.2	75.50	3.1	67.25	2.6
FITTINGS						
Fitments	117.50	5.2	177.50	7.3	367.75	14.5
Special	36.75	1.6	40.25	1.7	-	-
SERVICES						
Plumbing	114.00	5.1	133.00	5.5	69.75	2.8
Mechanical	91.00	4.1	167.50	6.9	106.00	4.2
Fire	8.75	0.4	11.00	0.5	21.75	0.9
Electrical	236.25	10.6	242.50	10.0	281.50	11.1
Transportation	-	-	-	-	-	-
Special	-	-	-	-	96.50	3.8
EXTERNAL SERVICES	2.75	0.1	3.00	0.1	4.50	0.2
CONTINGENCY	55.75	2.5	60.50	2.5	63.50	2.5
TOTAL (Mean cost - SYDNEY)	2230.00	100.0	2420.00	100.0	2535.00	100.0
Cost Range : Adelaide	2115-2280		2260-2435		2375-2560	
Brisbane	1905-2055		2070-2235		2155-2325	
Canberra	2310-2490		2585-2785		2940-3170	
Darwin	2530-2730		2740-2960		2700-2920	
Hobart	2375-2560		2570-2770		2915-3140	
Melbourne	2070-2235		2240-2415		2340-2525	
Perth	1880-2025		2040-2200		2105-2270	
Sydney	2145-2310		2330-2510		2440-2630	

4.0 EDUCATIONAL

4.2 SCHOOLS (SUPPORT)

ELEMENT	4.2.1.1 Secondary School Administration Max. 3 Storeys		4.2.1.2 Secondary School Gymnasium Single Storey		4.2.1.3 Secondary School Residential Max. 3 Storeys	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	269.00	12.5	270.50	12.4	292.75	12.5
SUBSTRUCTURE	72.75	3.4	127.75	5.8	79.25	3.4
SUPERSTRUCTURE						
Columns	38.25	1.8	-	-	40.50	1.7
Upper Floors	203.25	9.4	-	-	214.25	9.1
Staircase	30.25	1.4	-	-	33.75	1.4
Roof	122.25	5.7	273.25	12.5	120.75	5.2
External Walls)	306.00	14.2	512.00	23.4	351.00	14.9
Windows)						
External Doors	8.50	0.4	37.00	1.7	8.50	0.4
Internal Walls	75.75	3.5	49.50	2.3	92.75	3.9
Internal Screens	35.75	1.7	37.50	1.7	38.25	1.6
Internal Doors	33.25	1.6	21.50	1.0	36.75	1.6
FINISHES						
Wall	61.50	2.8	61.75	2.8	45.75	2.0
Floor	91.50	4.2	175.25	8.0	94.00	4.0
Ceiling	82.75	3.8	65.00	3.0	42.25	1.8
FITTINGS						
Fitments	84.50	3.9	128.75	5.9	275.50	11.7
Special	-	-	-	-	-	-
SERVICES						
Plumbing	106.75	4.9	67.00	3.1	342.25	14.6
Mechanical	250.75	11.6	64.50	3.0	30.25	1.3
Fire	17.50	0.8	20.00	0.9	17.00	0.7
Electrical	213.00	9.8	216.00	9.9	132.50	5.6
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	2.75	0.1	3.00	0.1	3.25	0.1
CONTINGENCY	54.00	2.5	54.75	2.5	58.75	2.5
TOTAL (Mean cost - SYDNEY)	2160.00	100.0	2185.00	100.0	2350.00	100.0
Cost Range : Adelaide	1890-2035		1905-2055		2190-2360	
Brisbane	1775-1915		1770-1905		2010-2165	
Canberra	2255-2430		2175-2340		2375-2565	
Darwin	2430-2640		2630-2840		2425-2650	
Hobart	2175-2345		2220-2390		2385-2570	
Melbourne	1975-2125		2010-2165		2175-2345	
Perth	2000-2155		1795-1935		2105-2270	
Sydney	2080-2240		2105-2265		2260-2435	

4.0 EDUCATIONAL

4.4 UNIVERSITIES (TEACHING)

ELEMENT	4.4.1 University Arts Max. 3 Storeys		4.4.2 University Lecture Theatre 1-2 Storey		4.4.3 University Science Laboratory Max. 3 Storeys	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	522.50	14.2	580.75	14.5	695.25	14.6
SUBSTRUCTURE	90.00	2.4	94.25	2.4	92.00	1.9
SUPERSTRUCTURE						
Columns	49.75	1.4	64.50	1.6	50.50	1.1
Upper Floors	282.00	7.6	178.75	4.5	276.00	5.8
Staircase	33.75	0.9	40.25	1.0	34.25	0.7
Roof	139.50	3.8	190.00	4.7	136.00	2.9
External Walls)	532.25	14.4	324.50	8.1	520.25	10.9
Windows)						
External Doors	15.25	0.4	29.50	0.7	14.25	0.3
Internal Walls	96.75	2.6	110.75	2.8	91.50	1.9
Internal Screens	8.00	0.2	11.75	0.3	7.25	0.2
Internal Doors	56.50	1.5	41.00	1.0	50.75	1.1
FINISHES						
Wall	60.00	1.6	59.25	1.5	78.25	1.6
Floor	109.00	3.0	107.50	2.7	111.25	2.3
Ceiling	87.50	2.4	97.25	2.4	89.50	1.9
FITTINGS						
Fitments	186.00	5.0	365.75	9.1	388.50	8.2
Special	-	-	-	-	-	-
SERVICES						
Plumbing	107.25	2.9	177.75	4.4	288.00	6.1
Mechanical	576.00	15.6	951.25	23.8	989.25	20.8
Fire	20.00	0.5	22.50	0.6	17.50	0.4
Electrical	461.00	12.5	451.25	11.3	520.75	10.9
Transportation	164.50	4.5	-	-	169.25	3.6
Special	-	-	-	-	11.00	0.2
EXTERNAL SERVICES	5.00	0.1	6.25	0.1	4.75	0.1
CONTINGENCY	92.50	2.5	100.25	2.5	119.00	2.5
TOTAL (Mean cost - SYDNEY)	3695.00	100.0	4005.00	100.0	4755.00	100.0
Cost Range : Adelaide	3675-3965		3945-4255		4690-5055	
Brisbane	3005-3240		3315-3575		4025-4340	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	3340-3605		3645-3925		4330-4670	
Perth	3245-3500		3540-3815		4255-4590	
Sydney	3555-3835		3855-4155		4575-4935	

4.0 EDUCATIONAL

4.5 UNIVERSITIES (SUPPORT)

ELEMENT	4.5.1 University Administration 1-2 Storey		4.5.2 University Animal Research 2 Storey		4.5.3 University Library Max. 3 Storeys		4.5.4 University Residential Max. 3 Storeys	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	402.00	14.0	763.75	14.3	543.00	14.6	431.25	14.7
SUBSTRUCTURE	89.50	3.1	74.75	1.4	94.50	2.5	91.50	3.1
SUPERSTRUCTURE								
Columns	46.25	1.6	45.25	0.8	73.50	2.0	41.50	1.4
Upper Floors	190.75	6.7	271.00	5.1	289.25	7.8	203.25	7.0
Staircase	36.25	1.3	28.75	0.5	34.00	0.9	29.00	1.0
Roof	157.00	5.5	278.75	5.2	141.50	3.8	130.25	4.5
External Walls)	419.25	14.6	284.50	5.3	520.50	14.0	513.00	17.5
Windows)								
External Doors	17.25	0.6	30.00	0.6	16.25	0.4	28.75	1.0
Internal Walls	73.50	2.6	181.00	3.4	72.25	1.9	116.00	4.0
Internal Screens	36.25	1.3	7.25	0.1	83.00	2.2	17.00	0.6
Internal Doors	38.00	1.3	104.00	2.0	35.00	1.0	90.00	3.1
FINISHES								
Wall	64.75	2.3	178.00	3.3	69.50	1.9	93.25	3.2
Floor	96.25	3.4	115.75	2.2	104.25	2.8	113.50	3.9
Ceiling	87.50	3.1	120.25	2.3	123.75	3.3	60.25	2.1
FITTINGS								
Fitments	97.75	3.4	284.50	5.3	107.75	2.9	219.75	7.5
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	126.50	4.4	397.75	7.5	106.50	2.9	316.50	10.8
Mechanical	542.00	18.9	1457.75	27.3	675.25	18.2	109.00	3.7
Fire	19.00	0.7	21.00	0.4	18.75	0.5	18.00	0.6
Electrical	238.25	8.3	396.25	7.4	423.00	11.4	223.75	7.6
Transportation	-	-	114.00	2.1	40.75	1.1	-	-
Special	-	-	41.50	0.8	43.25	1.2	-	-
EXTERNAL SERVICES	10.50	0.4	10.75	0.2	6.50	0.2	6.25	0.2
CONTINGENCY	71.50	2.5	133.50	2.5	93.00	2.5	73.25	2.5
TOTAL (Mean cost - SYDNEY)	2860.00	100.0	5340.00	100.0	3715.00	100.0	2925.00	100.0
Cost Range : Adelaide	2795-3010		5195-5600		3630-3910		2955-3185	
Brisbane	2325-2505		4405-4745		3015-3250		2500-2695	
Canberra	-		-		-		-	
Darwin	-		-		-		-	
Hobart	-		-		-		-	
Melbourne	2580-2780		4780-5155		3335-3595		2710-2925	
Perth	2465-2660		4680-5040		3220-3470		2605-2810	
Sydney	2750-2965		5140-5540		3575-3855		2815-3035	

6.0 HOSPITALS, HEALTH

6.1 HOSPITALS

ELEMENT	6.1.1 Hospital District Single Storey		6.1.2 Hospital General Multi-Storey		6.1.3 Hospital Maternity Multi-Storey		6.1.4 Hospital Private Single Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	480.50	11.1	1294.50	18.4	1008.75	18.4	378.75	10.8
SUBSTRUCTURE	123.50	2.8	85.25	1.2	69.50	1.3	122.00	3.5
SUPERSTRUCTURE								
Columns	-	-	50.25	0.7	47.25	0.9	-	-
Upper Floors	-	-	333.00	4.7	314.00	5.7	-	-
Staircase	-	-	20.50	0.3	20.25	0.4	-	-
Roof	237.25	5.5	80.25	1.1	77.50	1.4	248.25	7.1
External Walls)	379.25	8.8	445.00	6.3	418.50	7.7	342.00	9.8
Windows)								
External Doors	30.75	0.7	16.25	0.2	15.50	0.3	16.00	0.5
Internal Walls	139.25	3.2	167.00	2.4	142.75	2.6	141.75	4.0
Internal Screens	14.00	0.3	17.75	0.3	15.00	0.3	11.75	0.3
Internal Doors	101.75	2.3	114.50	1.6	95.50	1.7	98.50	2.8
FINISHES								
Wall	82.25	1.9	104.25	1.5	88.75	1.6	69.50	2.0
Floor	98.25	2.3	103.25	1.5	98.25	1.8	87.75	2.5
Ceiling	104.25	2.4	112.75	1.6	110.75	2.0	94.75	2.7
FITTINGS								
Fitments	190.25	4.4	218.00	3.1	132.75	2.4	184.50	5.3
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	262.25	6.1	261.25	3.7	213.75	3.9	244.00	7.0
Mechanical	1511.25	34.9	2379.00	33.9	1610.25	29.4	977.00	27.9
Fire	22.50	0.5	153.50	2.2	140.25	2.6	18.25	0.5
Electrical	435.50	10.1	671.50	9.6	557.75	10.2	367.75	10.5
Transportation	-	-	220.00	3.1	154.50	2.8	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	9.00	0.2	1.50	0.1	1.75	0.1	10.00	0.3
CONTINGENCY	108.25	2.5	175.75	2.5	136.75	2.5	87.50	2.5
TOTAL (Mean cost - SYDNEY)	4330.00	100.0	7025.00	100.0	5470.00	100.0	3500.00	100.0
Cost Range : Adelaide	3975-4285		6315-6805		4625-4985		3220-3475	
Brisbane	3795-4090		6890-7425		4620-4980		3100-3345	
Canberra	4320-4660		-		-		3370-3630	
Darwin	4550-4900		-		-		4350-4730	
Hobart	4040-4355		-		-		3245-3500	
Melbourne	3765-4060		5355-5775		3995-4305		3120-3365	
Perth	3920-4230		6120-6595		4910-5295		3155-3400	
Sydney	4165-4490		6760-7285		5265-5675		3365-3630	

6.0 HOSPITALS, HEALTH

6.2 ANCILLARY

ELEMENT	6.2.1 Child Care Centre Single Storey		6.2.2 Family Centre Single Storey		6.2.3 Group Surgery Single Storey		6.2.4 Multi-Purpose Health Centre Single Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	186.00	9.3	160.75	9.3	197.00	9.0	405.00	11.8
SUBSTRUCTURE	99.75	5.0	107.25	6.2	109.75	5.0	135.50	3.9
SUPERSTRUCTURE								
Columns	-	-	25.00	1.4	-	-	-	-
Upper Floors	-	-	-	-	-	-	-	-
Staircase	-	-	-	-	-	-	-	-
Roof	225.50	11.2	259.75	15.1	228.75	10.4	329.00	9.6
External Walls)	309.75	15.5	258.00	15.0	316.50	14.4	211.50	6.1
Windows)								
External Doors	18.25	0.9	26.50	1.5	9.50	0.4	44.25	1.3
Internal Walls	84.00	4.2	80.00	4.6	106.50	4.8	115.25	3.4
Internal Screens	32.50	1.6	66.25	3.8	8.75	0.4	28.75	0.8
Internal Doors	71.50	3.6	44.00	2.6	53.00	2.4	104.25	3.0
FINISHES								
Wall	46.00	2.3	52.50	3.1	59.25	2.7	85.50	2.5
Floor	87.25	4.3	95.25	5.5	89.25	4.1	120.50	3.5
Ceiling	144.25	7.2	107.00	6.2	70.50	3.2	129.25	3.8
FITTINGS								
Fitments	147.25	7.4	65.50	3.8	109.50	5.0	211.50	6.1
Special	-	-	-	-	-	-	21.25	0.6
SERVICES								
Plumbing	221.25	11.0	124.25	7.2	199.00	9.0	206.50	6.0
Mechanical	13.25	0.7	51.75	3.0	370.00	16.8	507.00	14.7
Fire	13.00	0.7	9.50	0.6	12.25	0.6	52.00	1.5
Electrical	239.50	12.0	137.00	7.9	193.75	8.8	487.75	14.2
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	149.75	4.4
EXTERNAL SERVICES	11.00	0.6	11.50	0.7	11.75	0.5	9.50	0.3
CONTINGENCY	50.00	2.5	43.25	2.5	55.00	2.5	86.00	2.5
TOTAL (Mean cost - SYDNEY)	2000.00	100.0	1725.00	100.0	2200.00	100.0	3440.00	100.0
Cost Range : Adelaide	1905-2055		1630-1755		2070-2230		3240-3490	
Brisbane	1925-2075		1625-1750		2000-2155		3130-3375	
Canberra	-		-		2200-2375		-	
Darwin	-		-		2520-2730		-	
Hobart	2075-2240		1835-1980		2220-2390		-	
Melbourne	1845-1985		1600-1720		1995-2150		3120-3365	
Perth	1850-1995		1630-1755		2015-2170		3200-3450	
Sydney	1925-2075		1660-1790		2115-2280		3310-3570	

7.0 HOTELS, MOTELS, CLUBS

7.1 SUBURBAN

ELEMENT	7.1.1 Tavern Single Storey		7.1.2 Club Social/Sporting 1-2 Storey	
	$/sqm	%	$/sqm	%
PRELIMINARIES	279.75	9.3	304.00	10.6
SUBSTRUCTURE	108.00	3.6	94.75	3.3
SUPERSTRUCTURE				
Columns	7.75	0.3	25.00	0.9
Upper Floors	-	-	122.50	4.3
Staircase	-	-	24.50	0.8
Roof	243.50	8.1	196.75	6.8
External Walls)	332.00	11.1	378.00	13.1
Windows)				
External Doors	10.00	0.3	10.50	0.4
Internal Walls	58.25	1.9	56.00	1.9
Internal Screens	3.00	0.1	10.25	0.4
Internal Doors	35.00	1.2	34.75	1.2
FINISHES				
Wall	62.75	2.1	61.50	2.1
Floor	107.75	3.6	108.25	3.8
Ceiling	91.25	3.1	90.00	3.1
FITTINGS				
Fitments	452.75	15.1	335.50	11.7
Special	-	-	-	-
SERVICES				
Plumbing	273.25	9.1	244.00	8.5
Mechanical	630.75	21.1	476.50	16.6
Fire	11.50	0.4	9.75	0.3
Electrical	203.50	6.8	214.50	7.5
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	9.25	0.3	6.00	0.2
CONTINGENCY	75.00	2.5	72.00	2.5
TOTAL (Mean cost - SYDNEY)	2995.00	100.0	2875.00	100.0
Cost Range : Adelaide	2815-3035		2660-2870	
Brisbane	2595-2800		2360-2545	
Canberra	3080-3320		-	
Darwin	3150-3400		-	
Hobart	3035-3270		2930-3155	
Melbourne	2765-2980		2705-2920	
Perth	2775-2990		2665-2870	
Sydney	2880-3105		2765-2980	

7.0 HOTELS, MOTELS, CLUBS

7.2 CITY HOTELS

ELEMENT	7.2.1.1.1 City Hotel Three Star Med/High Rise Overall (0/32sqm)		7.2.1.1.2 City Hotel Three Star Med/High Rise Overall (45sqm)		7.2.2.1.1 City Hotel Internat. Std. High Rise Overall (67.5sqm)		7.2.2.1.2 City Hotel Internat. Std. High Rise Overall (77.5sqm)	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	875.25	20.0	897.00	20.0	1389.00	22.0	1428.00	22.0
SUBSTRUCTURE	53.75	1.2	53.50	1.2	57.00	0.9	56.50	0.9
SUPERSTRUCTURE								
Columns	48.50	1.1	50.00	1.1	71.75	1.1	73.25	1.1
Upper Floors	323.25	7.4	329.75	7.4	371.50	5.9	375.00	5.8
Staircase	37.75	0.9	40.50	0.9	44.50	0.7	43.75	0.7
Roof	47.25	1.1	48.75	1.1	63.50	1.0	66.50	1.0
External Walls)	485.75	11.0	500.25	11.2	760.50	12.0	792.75	12.2
Windows)								
External Doors	9.00	0.2	9.00	0.2	14.00	0.2	14.00	0.2
Internal Walls	216.25	4.9	213.75	4.8	214.00	3.4	211.25	3.3
Internal Screens	10.75	0.2	9.50	0.2	9.00	0.1	9.75	0.1
Internal Doors	117.75	2.7	118.25	2.6	116.50	1.9	115.75	1.8
FINISHES								
Wall	151.50	3.5	153.50	3.4	206.00	3.3	212.00	3.3
Floor	122.00	2.8	130.25	2.9	165.50	2.6	173.50	2.7
Ceiling	91.25	2.1	92.25	2.1	134.50	2.1	142.00	2.2
FITTINGS								
Fitments	140.00	3.2	148.00	3.3	209.75	3.3	220.75	3.4
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	372.50	8.5	369.25	8.2	505.25	8.0	505.75	7.8
Mechanical	542.75	12.3	552.50	12.3	1034.75	16.4	1054.75	16.3
Fire	157.25	3.6	160.75	3.6	196.75	3.1	200.25	3.1
Electrical	280.25	6.4	287.75	6.4	376.75	6.0	399.00	6.1
Transportation	190.00	4.3	201.00	4.5	215.50	3.4	222.25	3.4
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	2.50	0.1	2.50	0.1	6.00	0.1	6.00	0.1
CONTINGENCY	109.75	2.5	112.00	2.5	158.00	2.5	162.25	2.5
TOTAL (Mean cost - SYDNEY)	4385.00	100.0	4480.00	100.0	6320.00	100.0	6485.00	100.0
Cost Range : Adelaide	4185-4510		4290-4625		5730-6175		5760-6210	
Brisbane	3600-3880		3680-3970		4805-5180		4920-5305	
Canberra	3600-3880		3600-3880		-		-	
Darwin	3500-3750		3720-4050		-		-	
Hobart	3795-4090		3900-4205		-		-	
Melbourne	3685-3975		3780-4070		4820-5195		4940-5325	
Perth	4295-4625		4390-4730		5420-5845		5605-6045	
Sydney	4220-4550		4310-4645		6080-6555		6240-6725	

7.0 HOTELS, MOTELS, CLUBS

7.3 COUNTRY MOTELS
7.4 SERVICED APARTMENTS

ELEMENT	7.3.1.1 Country Motel Std. Travellers 1-2 Storey Overall		7.3.2.1 Country Motel High Std. 1-2 Storey Overall		7.4.1.1 Serviced Apartments Multi-Storey Two Star		7.4.1.2 Serviced Apartments Multi-Storey Three Star	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	256.75	10.1	316.50	10.1	600.75	17.0	614.50	17.1
SUBSTRUCTURE	107.50	4.2	69.25	2.2	44.00	1.2	45.00	1.3
SUPERSTRUCTURE								
Columns	11.25	0.4	39.75	1.3	46.00	1.3	47.75	1.3
Upper Floors	42.25	1.7	151.75	4.9	316.50	9.0	309.75	8.6
Staircase	6.25	0.2	14.00	0.4	23.00	0.7	25.75	0.7
Roof	238.50	9.3	141.50	4.5	46.25	1.3	50.75	1.4
External Walls)	338.00	13.2	395.75	12.7	329.50	9.3	331.75	9.3
Windows)								
External Doors	35.50	1.4	43.00	1.4	6.75	0.2	8.00	0.2
Internal Walls	75.25	3.0	84.25	2.7	259.50	7.3	242.75	6.8
Internal Screens	10.75	0.4	10.75	0.3	9.75	0.3	9.50	0.3
Internal Doors	71.75	2.8	79.00	2.5	111.75	3.1	101.75	2.8
FINISHES								
Wall	57.50	2.3	64.50	2.1	109.50	3.1	113.00	3.2
Floor	99.00	3.9	103.50	3.3	118.50	3.4	122.50	3.4
Ceiling	83.75	3.3	87.25	2.8	86.50	2.5	88.25	2.5
FITTINGS								
Fitments	301.50	11.8	389.75	12.5	104.50	3.0	112.00	3.1
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	300.75	11.8	348.50	11.2	317.00	9.0	305.50	8.5
Mechanical	263.75	10.3	459.50	14.7	313.00	8.9	329.75	9.2
Fire	13.75	0.5	14.75	0.5	152.00	4.3	143.50	4.0
Electrical	171.50	6.7	223.25	7.2	266.25	7.5	290.25	8.1
Transportation	-	-	-	-	146.75	4.1	160.75	4.5
Special	-	-	-	-	32.25	0.9	41.00	1.1
EXTERNAL SERVICES	5.75	0.2	5.50	0.2	1.75	0.1	1.50	0.1
CONTINGENCY	64.00	2.5	78.00	2.5	88.25	2.5	89.75	2.5
TOTAL (Mean cost - SYDNEY)	2555.00	100.0	3120.00	100.0	3530.00	100.0	3585.00	100.0
Cost Range : Adelaide	2445-2635		2935-3165		3230-3485		3480-3755	
Brisbane	2320-2500		2750-2965		2925-3150		3190-3440	
Canberra	2540-2740		-		-		-	
Darwin	2500-2690		-		-		-	
Hobart	2645-2850		3130-3375		-		-	
Melbourne	2315-2495		2750-2965		2890-3115		3105-3350	
Perth	2425-2615		2920-3150		3260-3515		3435-3700	
Sydney	2455-2650		3005-3235		3395-3660		3450-3720	

8.0 INDUSTRIAL

8.1 WAREHOUSES

ELEMENT	8.1.1.1 Warehouse Low Bay Letting Brick Walls		8.1.1.2 Warehouse Low Bay Letting Precast or Tilt-up Walls		8.1.1.3 Warehouse High Bay Letting Brick Walls		8.1.1.4 Warehouse High Bay Letting Precast or Tilt-up Walls	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	65.25	9.4	60.00	9.2	79.50	9.6	75.75	9.3
SUBSTRUCTURE	110.50	16.0	98.75	15.2	115.75	13.9	110.75	13.6
SUPERSTRUCTURE								
Columns	37.00	5.4	34.50	5.3	47.25	5.7	46.00	5.6
Upper Floors	-	-	-	-	-	-	-	-
Staircase	-	-	-	-	-	-	-	-
Roof	180.75	26.2	176.00	27.1	197.50	23.8	202.25	24.8
External Walls)	130.50	18.9	116.75	18.0	200.50	24.2	187.50	23.0
Windows)								
External Doors	14.50	2.1	14.25	2.2	29.00	3.5	29.50	3.6
Internal Walls	-	-	-	-	-	-	-	-
Internal Screens	-	-	-	-	-	-	-	-
Internal Doors	-	-	-	-	-	-	-	-
FINISHES								
Wall	7.00	1.0	6.75	1.0	6.00	0.7	6.50	0.8
Floor	10.25	1.5	9.75	1.5	9.75	1.2	10.00	1.3
Ceiling	-	-	-	-	-	-	-	-
FITTINGS								
Fitments	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	36.25	5.3	36.25	5.6	36.50	4.4	37.75	4.6
Mechanical	-	-	-	-	-	-	-	-
Fire	9.50	1.4	9.25	1.4	10.75	1.3	10.75	1.3
Electrical	61.50	8.9	62.00	9.5	67.50	8.1	68.00	8.4
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	9.75	1.4	9.50	1.5	9.25	1.1	9.75	1.2
CONTINGENCY	17.25	2.5	16.25	2.5	20.75	2.5	20.50	2.5
TOTAL (Mean cost - SYDNEY)	690.00	100.0	650.00	100.0	830.00	100.0	815.00	100.0
Cost Range : Adelaide		690-745		635-685		810-870		750-805
Brisbane		685-740		665-715		790-850		780-840
Canberra		775-840		830-890		940-1010		925-1000
Darwin		820-880		830-890		930-1010		965-1050
Hobart		725-780		715-770		865-935		855-925
Melbourne		660-710		610-660		780-840		725-780
Perth		655-705		650-700		770-830		760-820
Sydney		665-715		625-675		795-860		780-845

8.0 INDUSTRIAL

8.1 WAREHOUSES

ELEMENT	8.1.2.1 Warehouse Low Bay Owner Occupation Brick Walls		8.1.2.2 Warehouse Low Bay Owner Occupation Precast or Tilt-up Walls	
	$/sqm	%	$/sqm	%
PRELIMINARIES	75.75	9.1	71.75	8.8
SUBSTRUCTURE	110.75	13.3	102.00	12.5
SUPERSTRUCTURE				
Columns	38.00	4.6	36.25	4.4
Upper Floors	-	-	-	-
Staircase	-	-	-	-
Roof	187.25	22.4	187.00	22.9
External Walls)	130.50	15.6	122.75	15.1
Windows)				
External Doors	15.75	1.9	16.25	2.0
Internal Walls	22.00	2.6	21.25	2.6
Internal Screens	1.00	0.1	1.00	0.1
Internal Doors	10.25	1.2	10.25	1.3
FINISHES				
Wall	6.25	0.7	6.50	0.8
Floor	18.00	2.2	17.75	2.2
Ceiling	12.25	1.5	13.00	1.6
FITTINGS				
Fitments	10.25	1.2	10.25	1.3
Special	-	-	-	-
SERVICES				
Plumbing	47.25	5.7	48.00	5.9
Mechanical	1.00	0.1	1.00	0.1
Fire	7.75	0.9	7.75	1.0
Electrical	111.00	13.3	112.75	13.8
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	9.00	1.1	9.00	1.1
CONTINGENCY	21.00	2.5	20.50	2.5
TOTAL (Mean cost - SYDNEY)	835.00	100.0	815.00	100.0
Cost Range : Adelaide	815-875		760-815	
Brisbane	795-855		785-845	
Canberra	925-1000		925-1000	
Darwin	970-1040		1060-1140	
Hobart	865-935		865-935	
Melbourne	780-840		720-775	
Perth	780-840		760-820	
Sydney	800-865		780-845	

8.0 INDUSTRIAL

8.1 WAREHOUSES

ELEMENT	8.1.2.3 Warehouse High Bay Owner Occupation Brick Walls		8.1.2.4 Warehouse High Bay Owner Occupation Precast or Tilt-up Walls		8.1.3 Warehouse Multi-Storey Max. 6 storeys	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	91.00	9.1	85.25	9.0	237.75	13.6
SUBSTRUCTURE	120.00	12.0	111.00	11.7	78.50	4.5
SUPERSTRUCTURE						
Columns	48.50	4.8	43.75	4.6	58.00	3.3
Upper Floors	-	-	-	-	263.50	15.1
Staircase	-	-	-	-	25.25	1.4
Roof	200.75	20.1	199.00	20.9	64.75	3.7
External Walls)	209.50	21.0	185.50	19.5	191.75	11.0
Windows)						
External Doors	29.75	3.0	29.50	3.1	17.25	1.0
Internal Walls	22.00	2.2	21.25	2.2	35.25	2.0
Internal Screens	1.50	0.2	1.50	0.2	1.00	0.1
Internal Doors	13.25	1.3	13.25	1.4	9.00	0.5
FINISHES						
Wall	8.00	0.8	7.25	0.8	16.50	0.9
Floor	18.50	1.8	18.00	1.9	28.25	1.6
Ceiling	16.75	1.7	16.50	1.7	17.25	1.0
FITTINGS						
Fitments	9.25	0.9	9.00	0.9	10.00	0.6
Special	-	-	-	-	-	-
SERVICES						
Plumbing	50.75	5.1	50.50	5.3	65.25	3.8
Mechanical	1.25	0.1	1.25	0.2	190.25	10.9
Fire	8.50	0.9	8.50	0.9	123.75	7.1
Electrical	116.50	11.6	116.25	12.2	141.50	8.1
Transportation	-	-	-	-	123.75	7.1
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.25	0.9	9.00	1.0	2.75	0.2
CONTINGENCY	25.00	2.5	23.75	2.5	43.75	2.5
TOTAL (Mean cost - SYDNEY)	1000.00	100.0	950.00	100.0	1745.00	100.0
Cost Range : Adelaide	965-1040		865-930		1665-1795	
Brisbane	950-1025		930-1000		1500-1620	
Canberra	1025-1105		1025-1105		-	
Darwin	1070-1170		1150-1250		-	
Hobart	1025-1105		1000-1075		-	
Melbourne	910-980		890-960		1600-1720	
Perth	920-995		920-995		1670-1800	
Sydney	960-1035		915-985		1675-1810	

8.0 INDUSTRIAL

8.1 WAREHOUSES

ELEMENT	8.1.4.1 Warehouse Low Bay Owner Occupation Brick Walls		8.1.4.2 Warehouse Low Bay Owner Occupation Precast or Tilt-up Walls		8.1.4.3 Warehouse High Bay Owner Occupation Brick Walls		8.1.4.4 Warehouse High Bay Owner Occupation Precast or Tilt-up Walls	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	75.00	9.2	66.00	8.7	91.00	9.4	82.25	8.9
SUBSTRUCTURE	108.00	13.3	95.50	12.6	119.50	12.3	109.25	11.9
SUPERSTRUCTURE								
Columns	37.00	4.5	33.00	4.4	46.50	4.8	45.25	4.9
Upper Floors	-	-	-	-	-	-	-	-
Staircase	-	-	-	-	-	-	-	-
Roof	179.25	22.0	171.50	22.7	196.50	20.3	191.50	20.8
External Walls)	121.25	14.9	101.00	13.4	193.25	19.9	172.75	18.8
Windows)								
External Doors	15.75	1.9	15.75	2.1	30.00	3.1	29.25	3.2
Internal Walls	19.00	2.3	18.75	2.5	21.25	2.2	20.50	2.2
Internal Screens	1.25	0.2	1.00	0.1	1.75	0.2	1.75	0.2
Internal Doors	8.75	1.1	8.75	1.2	12.50	1.3	12.25	1.3
FINISHES								
Wall	7.50	0.9	6.75	0.9	8.50	0.9	8.00	0.9
Floor	17.00	2.1	16.75	2.2	18.50	1.9	18.75	2.0
Ceiling	12.25	1.5	11.50	1.5	16.00	1.6	16.00	1.8
FITTINGS								
Fitments	9.25	1.1	9.50	1.3	9.50	1.0	9.25	1.0
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	52.00	6.4	50.75	6.8	48.75	5.0	48.25	5.3
Mechanical	1.75	0.2	1.50	0.2	1.00	0.1	1.00	0.1
Fire	7.75	1.0	7.75	1.0	8.75	0.9	8.50	0.9
Electrical	112.75	13.8	111.25	14.7	113.00	11.6	113.25	12.3
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	9.00	1.1	9.00	1.2	9.50	1.0	9.25	1.0
CONTINGENCY	20.50	2.5	19.00	2.5	24.25	2.5	23.00	2.5
TOTAL (Mean cost - SYDNEY)	815.00	100.0	755.00	100.0	970.00	100.0	920.00	100.0
Cost Range : Adelaide		780-840		750-805		915-990		845-910
Brisbane		760-815		750-805		900-970		895-965
Canberra		-		-		-		-
Darwin		-		-		-		-
Hobart		-		-		-		-
Melbourne		720-775		710-765		875-940		835-900
Perth		735-790		730-785		875-945		875-945
Sydney		780-845		725-780		930-1005		885-950

8.0 INDUSTRIAL

8.1 WAREHOUSES

ELEMENT	8.1.4.5 Offices, Showrooms Owner Occupation Single Storey		8.1.4.6 Offices, Showrooms Owner Occupation Two Storey		8.1.4.7 Amenities Owner Occupation Single Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	195.00	9.5	209.25	9.6	255.50	9.6
SUBSTRUCTURE	122.25	6.0	65.75	3.0	145.00	5.4
SUPERSTRUCTURE						
Columns	-	-	48.50	2.2	45.00	1.7
Upper Floors	-	-	162.75	7.5	-	-
Staircase	-	-	20.50	0.9	-	-
Roof	243.50	11.9	137.25	6.3	303.25	11.4
External Walls)	391.50	19.2	399.50	18.3	244.75	9.2
Windows)						
External Doors	22.25	1.1	10.75	0.5	19.25	0.7
Internal Walls	52.75	2.6	67.25	3.1	101.75	3.8
Internal Screens	9.50	0.5	9.00	0.4	37.75	1.4
Internal Doors	13.75	0.7	23.00	1.0	48.00	1.8
FINISHES						
Wall	42.00	2.0	56.50	2.6	81.25	3.0
Floor	92.50	4.5	88.50	4.1	112.75	4.2
Ceiling	87.25	4.3	80.75	3.7	71.50	2.7
FITTINGS						
Fitments	22.00	1.1	21.25	1.0	234.50	8.8
Special	-	-	-	-	-	-
SERVICES						
Plumbing	101.50	5.0	118.25	5.4	390.50	14.6
Mechanical	389.00	19.1	396.50	18.2	299.00	11.2
Fire	15.50	0.7	14.75	0.7	7.75	0.3
Electrical	177.25	8.7	186.75	8.6	197.25	7.4
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	11.50	0.6	8.75	0.4	8.50	0.3
CONTINGENCY	51.00	2.5	54.50	2.5	66.75	2.5
TOTAL (Mean cost - SYDNEY)	2040.00	100.0	2180.00	100.0	2670.00	100.0
Cost Range : Adelaide	1930-2080		2065-2225		2545-2745	
Brisbane	1770-1905		1885-2030		2200-2370	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	1880-2025		1995-2150		2465-2655	
Perth	1905-2050		2055-2215		2560-2755	
Sydney	1960-2115		2100-2260		2570-2770	

8.0 INDUSTRIAL

8.2 FACTORIES

ELEMENT	8.2.1.1 Factory Small Span Letting Framed Walls		8.2.1.2 Factory Small Span Letting Brick Walls		8.2.1.3 Factory Small Span Letting Precast or Tilt-up Walls	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	60.50	9.0	67.25	9.5	61.00	9.1
SUBSTRUCTURE	111.00	16.6	115.25	16.3	104.50	15.5
SUPERSTRUCTURE						
Columns	42.50	6.3	37.25	5.3	37.75	5.6
Upper Floors	-	-	-	-	-	-
Staircase	-	-	-	-	-	-
Roof	180.50	26.9	182.00	25.8	178.50	26.7
External Walls)	103.25	15.4	128.75	18.3	116.75	17.4
Windows)						
External Doors	14.75	2.2	14.75	2.1	14.75	2.2
Internal Walls	-	-	-	-	-	-
Internal Screens	-	-	-	-	-	-
Internal Doors	-	-	-	-	-	-
FINISHES						
Wall	7.50	1.1	7.75	1.1	7.25	1.1
Floor	11.25	1.7	11.00	1.6	11.00	1.6
Ceiling	-	-	-	-	-	-
FITTINGS						
Fitments	-	-	-	-	-	-
Special	-	-	-	-	-	-
SERVICES						
Plumbing	35.50	5.3	36.00	5.1	35.50	5.3
Mechanical	-	-	-	-	-	-
Fire	10.00	1.5	10.00	1.4	9.75	1.5
Electrical	66.75	10.0	67.50	9.6	66.75	10.0
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.75	1.5	9.75	1.4	9.75	1.5
CONTINGENCY	16.75	2.5	17.75	2.5	16.75	2.5
TOTAL (Mean cost - SYDNEY)	670.00	100.0	705.00	100.0	670.00	100.0
Cost Range : Adelaide		680-730		690-745		640-690
Brisbane		640-690		685-740		665-715
Canberra		775-840		775-840		775-840
Darwin		800-860		800-860		820-895
Hobart		720-775		730-785		720-775
Melbourne		630-675		665-715		625-670
Perth		630-680		660-710		660-710
Sydney		645-695		675-730		645-695

8.0 INDUSTRIAL

8.2 FACTORIES

ELEMENT	8.2.2.1 Factory Small Span Owner Occupation Brick Walls		8.2.2.2 Factory Small Span Owner Occupation Precast or Tilt-up Walls		8.2.3.1 Factory Large Span Owner Occupation Framed Walls	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	85.50	9.5	74.25	8.8	96.75	9.1
SUBSTRUCTURE	118.75	13.3	101.50	12.0	121.00	11.4
SUPERSTRUCTURE						
Columns	39.25	4.4	39.50	4.7	51.25	4.8
Upper Floors	-	-	-	-	-	-
Staircase	-	-	-	-	-	-
Roof	184.75	20.6	180.25	21.3	233.00	22.0
External Walls)	126.25	14.1	111.50	13.2	172.50	16.3
Windows)						
External Doors	18.50	2.1	18.50	2.2	28.75	2.7
Internal Walls	36.75	4.1	36.25	4.3	34.25	3.2
Internal Screens	1.00	0.1	1.00	0.1	-	-
Internal Doors	9.50	1.1	9.75	1.2	12.00	1.1
FINISHES						
Wall	8.00	0.9	8.00	0.9	8.75	0.8
Floor	18.00	2.0	18.25	2.2	18.00	1.7
Ceiling	11.00	1.2	11.00	1.3	14.50	1.4
FITTINGS						
Fitments	10.25	1.2	10.25	1.2	9.50	0.9
Special	-	-	-	-	-	-
SERVICES						
Plumbing	50.00	5.6	49.25	5.8	67.50	6.4
Mechanical	1.50	0.2	1.50	0.2	6.25	0.6
Fire	8.25	0.9	8.25	1.0	10.75	1.0
Electrical	136.00	15.2	135.50	16.0	139.50	13.2
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.25	1.0	9.25	1.1	9.25	0.9
CONTINGENCY	22.50	2.5	21.25	2.5	26.50	2.5
TOTAL (Mean cost - SYDNEY)	895.00	100.0	845.00	100.0	1060.00	100.0
Cost Range : Adelaide		860-925		810-870		1035-1115
Brisbane		845-915		840-905		1040-1120
Canberra		900-975		925-1000		-
Darwin		980-1060		1075-1160		-
Hobart		925-995		910-980		-
Melbourne		810-875		795-855		1035-1115
Perth		840-905		810-875		1050-1135
Sydney		860-925		815-875		1020-1100

8.0 INDUSTRIAL

8.2 FACTORIES

ELEMENT	8.2.3.2 Factory Large Span Owner Occupation Brick Walls		8.2.3.3 Factory Large Span Owner Occupation Precast or Tilt-up Walls	
	$/sqm	%	$/sqm	%
PRELIMINARIES	105.50	9.4	93.00	8.8
SUBSTRUCTURE	122.50	10.9	107.50	10.2
SUPERSTRUCTURE				
Columns	54.00	4.8	49.75	4.7
Upper Floors	-	-	-	-
Staircase	-	-	-	-
Roof	231.25	20.6	228.75	21.7
External Walls)	216.00	19.2	187.00	17.7
Windows)				
External Doors	27.75	2.5	28.75	2.7
Internal Walls	37.50	3.3	37.25	3.5
Internal Screens	-	-	-	-
Internal Doors	13.00	1.2	12.00	1.2
FINISHES				
Wall	8.75	0.8	8.75	0.8
Floor	18.50	1.6	18.25	1.7
Ceiling	15.00	1.3	14.50	1.4
FITTINGS				
Fitments	9.75	0.9	9.75	0.9
Special	-	-	-	-
SERVICES				
Plumbing	68.25	6.1	67.25	6.4
Mechanical	6.00	0.5	6.00	0.6
Fire	11.50	1.0	10.75	1.0
Electrical	142.00	12.6	140.00	13.3
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	9.50	0.8	9.25	0.9
CONTINGENCY	28.25	2.5	26.50	2.5
TOTAL (Mean cost - SYDNEY)	1125.00	100.0	1055.00	100.0
Cost Range : Adelaide	1115-1200		1000-1075	
Brisbane	1075-1160		1050-1130	
Canberra	-		-	
Darwin	-		-	
Hobart	-		-	
Melbourne	1050-1130		1025-1105	
Perth	1075-1160		1060-1145	
Sydney	1080-1165		1015-1095	

8.0 INDUSTRIAL

8.2 FACTORIES

ELEMENT	8.2.3.4 Factory Large Span Offices Two Storey		8.2.3.5 Factory Large Span Showroom Ground Floor		8.2.3.6 Factory Large Span Amenities Ground Floor	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	206.50	10.0	183.75	9.5	260.75	9.5
SUBSTRUCTURE	83.50	4.0	116.75	6.1	119.25	4.4
SUPERSTRUCTURE						
Columns	33.75	1.6	-	-	35.75	1.3
Upper Floors	121.50	5.9	-	-	-	-
Staircase	16.50	0.8	-	-	-	-
Roof	181.00	8.8	234.00	12.2	282.00	10.3
External Walls)	331.00	16.1	369.00	19.2	230.00	8.4
Windows)						
External Doors	14.25	0.7	21.75	1.1	28.00	1.0
Internal Walls	76.00	3.7	39.25	2.0	125.00	4.6
Internal Screens	8.75	0.4	7.25	0.4	76.25	2.8
Internal Doors	22.75	1.1	14.00	0.7	56.50	2.1
FINISHES						
Wall	53.00	2.6	43.50	2.3	101.25	3.7
Floor	92.25	4.5	90.25	4.7	97.25	3.6
Ceiling	84.75	4.1	87.25	4.5	70.50	2.6
FITTINGS						
Fitments	19.75	1.0	21.25	1.1	229.00	8.4
Special	-	-	-	-	-	-
SERVICES						
Plumbing	99.75	4.8	82.00	4.3	424.00	15.5
Mechanical	366.25	17.8	370.75	19.3	304.50	11.2
Fire	14.00	0.7	14.25	0.7	8.50	0.3
Electrical	174.25	8.5	172.25	8.9	205.50	7.5
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.00	0.4	9.50	0.5	7.75	0.3
CONTINGENCY	51.50	2.5	48.25	2.5	68.25	2.5
TOTAL (Mean cost - SYDNEY)	2060.00	100.0	1925.00	100.0	2730.00	100.0
Cost Range : Adelaide	1990-2145		1875-2020		2605-2810	
Brisbane	1835-1980		1700-1830		2365-2550	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	1915-2065		1815-1960		2510-2705	
Perth	2000-2155		1850-1995		2585-2785	
Sydney	1980-2135		1850-1995		2625-2830	

8.0 INDUSTRIAL

8.3 SUNDRIES

ELEMENT	8.3.1 Cold Store		8.3.1.2 Cold Store Ancillary Building Single Storey	
	$/sqm	%	$/sqm	%
PRELIMINARIES	222.25	9.6	212.50	10.0
SUBSTRUCTURE	319.50	13.8	116.75	5.5
SUPERSTRUCTURE				
Columns	681.00	29.5	-	-
Upper Floors	-	-	-	-
Staircase	-	-	-	-
Roof	-	-	242.00	11.3
External Walls)	-	-	396.25	18.6
Windows)				
External Doors	-	-	37.00	1.7
Internal Walls	-	-	67.50	3.2
Internal Screens	-	-	59.50	2.8
Internal Doors	-	-	42.50	2.0
FINISHES				
Wall	-	-	65.75	3.1
Floor	-	-	99.00	4.6
Ceiling	-	-	63.75	3.0
FITTINGS				
Fitments	106.25	4.6	135.75	6.4
Special	-	-	-	-
SERVICES				
Plumbing	-	-	246.75	11.6
Mechanical	739.00	32.0	83.00	3.9
Fire	-	-	15.75	0.7
Electrical	168.75	7.3	185.00	8.7
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	15.50	0.7	8.00	0.4
CONTINGENCY	57.75	2.5	53.25	2.5
TOTAL (Mean cost - SYDNEY)	2310.00	100.0	2130.00	100.0
Cost Range : Adelaide	2390-2580		2030-2190	
Brisbane	2270-2445		1870-2015	
Canberra	-		-	
Darwin	-		-	
Hobart	-		-	
Melbourne	2150-2320		1930-2080	
Perth	2600-2805		2015-2170	
Sydney	2220-2395		2050-2210	

8.0 INDUSTRIAL

8.3 SUNDRIES
8.4 GARAGE, SERVICE STATION

ELEMENT	8.3.2 Laboratory Workshop Single Storey		8.3.3 Maintenance Workshop		8.4.1 Garage Service Station	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	325.00	9.5	137.50	9.6	161.75	9.3
SUBSTRUCTURE	136.25	4.0	218.50	15.2	288.50	16.6
SUPERSTRUCTURE						
Columns	-	-	59.25	4.1	54.00	3.1
Upper Floors	-	-	-	-	-	-
Staircase	-	-	-	-	-	-
Roof	215.75	6.3	203.50	14.1	198.00	11.3
External Walls)	248.00	7.3	254.50	17.7	274.75	15.7
Windows)						
External Doors	29.50	0.9	54.50	3.8	106.25	6.1
Internal Walls	83.25	2.4	53.75	3.7	76.50	4.4
Internal Screens	15.25	0.4	7.25	0.5	3.75	0.2
Internal Doors	61.00	1.8	19.25	1.3	10.25	0.6
FINISHES						
Wall	107.75	3.2	22.00	1.5	35.00	2.0
Floor	127.75	3.7	38.75	2.7	44.75	2.6
Ceiling	78.25	2.3	24.00	1.7	28.25	1.6
FITTINGS						
Fitments	370.50	10.9	15.75	1.1	23.25	1.3
Special	-	-	-	-	-	-
SERVICES						
Plumbing	230.50	6.8	97.25	6.8	157.50	9.0
Mechanical	1042.25	30.6	10.00	0.7	8.25	0.5
Fire	12.00	0.4	12.50	0.9	19.00	1.1
Electrical	233.25	6.8	167.75	11.6	204.50	11.7
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	8.50	0.2	8.00	0.5	7.00	0.4
CONTINGENCY	85.25	2.5	36.00	2.5	43.75	2.5
TOTAL (Mean cost - SYDNEY)	3410.00	100.0	1440.00	100.0	1745.00	100.0
Cost Range : Adelaide	3255-3510		1365-1470		1685-1815	
Brisbane	3020-3255		1280-1380		1575-1700	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		1795-1935	
Melbourne	3140-3385		1310-1415		1610-1735	
Perth	3205-3455		1390-1495		1670-1800	
Sydney	3280-3535		1385-1490		1675-1810	

9.0 OFFICES

9.1 LETTABLE LOW RISE (BASIC SERVICES) - Finished Floor Lettable

ELEMENT	9.1.1.1 Office Building No A/C Single Storey		9.1.1.3 Office Building No A/C or Lift 2 Storey		9.1.1.5 Office Building No A/C or Lift 3 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	162.50	10.9	196.75	11.5	207.00	11.5
SUBSTRUCTURE	115.75	7.8	66.00	3.8	56.75	3.2
SUPERSTRUCTURE						
Columns	-	-	47.50	2.8	58.00	3.2
Upper Floors	-	-	169.00	9.9	223.75	12.5
Staircase	-	-	21.00	1.2	22.50	1.2
Roof	232.25	15.6	135.25	7.9	115.25	6.4
External Walls)	283.50	19.1	300.50	17.5	307.00	17.1
Windows)						
External Doors	22.50	1.5	11.50	0.7	11.75	0.7
Internal Walls	64.75	4.4	84.25	4.9	85.75	4.8
Internal Screens	9.75	0.7	9.50	0.6	9.75	0.5
Internal Doors	15.00	1.0	28.50	1.6	31.25	1.7
FINISHES						
Wall	38.75	2.6	56.75	3.3	58.00	3.2
Floor	90.00	6.1	90.75	5.3	90.75	5.1
Ceiling	84.00	5.7	83.50	4.8	86.50	4.8
FITTINGS						
Fitments	20.25	1.4	20.00	1.2	19.50	1.1
Special	-	-	-	-	-	-
SERVICES						
Plumbing	102.75	6.9	122.00	7.1	127.25	7.1
Mechanical	9.25	0.6	10.50	0.6	9.50	0.5
Fire	17.50	1.2	17.00	1.0	20.75	1.2
Electrical	165.25	11.1	191.75	11.2	200.25	11.2
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	14.00	0.9	10.00	0.6	8.75	0.5
CONTINGENCY	37.25	2.5	43.00	2.5	45.00	2.5
TOTAL (Mean cost - SYDNEY)	1485.00	100.0	1715.00	100.0	1795.00	100.0
Cost Range : Adelaide	1445-1560		1630-1755		1680-1810	
Brisbane	1340-1440		1540-1660		1600-1720	
Canberra	1575-1700		1765-1905		-	
Darwin	1650-1760		1780-1920		-	
Hobart	1535-1655		1765-1900		-	
Melbourne	1400-1510		1605-1730		1630-1760	
Perth	1460-1575		1635-1760		1705-1840	
Sydney	1430-1540		1650-1780		1725-1860	

9.0 OFFICES

9.1 LETTABLE LOW RISE (BASIC SERVICED) - Shell and Core Lettable

ELEMENT	9.1.1.2 Office Building No A/C Single Storey		9.1.1.4 Office Building No A/C or Lift 2 Storey		9.1.1.6 Office Building No A/C or Lift 3 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	146.75	11.0	182.25	11.7	190.00	11.7
SUBSTRUCTURE	115.00	8.6	68.25	4.4	56.00	3.5
SUPERSTRUCTURE						
Columns	-	-	48.50	3.1	57.25	3.5
Upper Floors	-	-	169.00	10.8	225.50	13.9
Staircase	-	-	20.50	1.3	22.75	1.4
Roof	233.75	17.5	134.75	8.6	116.25	7.2
External Walls)	284.75	21.3	300.50	19.3	308.25	19.1
Windows)						
External Doors	22.50	1.7	11.25	0.8	13.50	0.8
Internal Walls	65.50	4.9	83.75	5.4	84.50	5.2
Internal Screens	10.00	0.8	9.75	0.6	10.00	0.6
Internal Doors	15.00	1.1	28.50	1.8	31.25	1.9
FINISHES						
Wall	20.25	1.5	36.00	2.3	36.25	2.2
Floor	30.25	2.3	30.50	2.0	27.75	1.7
Ceiling	69.50	5.2	63.75	4.1	64.50	4.0
FITTINGS						
Fitments	20.25	1.5	20.00	1.3	19.50	1.2
Special	-	-	-	-	-	-
SERVICES						
Plumbing	102.25	7.7	124.00	7.9	125.75	7.8
Mechanical	9.25	0.7	9.25	0.6	9.75	0.6
Fire	18.25	1.4	16.25	1.0	20.00	1.3
Electrical	124.25	9.3	153.50	9.8	152.25	9.4
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	14.00	1.0	10.75	0.7	8.50	0.5
CONTINGENCY	33.50	2.5	39.00	2.5	40.50	2.5
TOTAL (Mean cost - SYDNEY)	1335.00	100.0	1560.00	100.0	1620.00	100.0
Cost Range : Adelaide	1300-1400		1490-1605		1520-1640	
Brisbane	1205-1295		1365-1470		1425-1535	
Canberra	1225-1325		1605-1735		-	
Darwin	1400-1500		1680-1810		-	
Hobart	1340-1440		1595-1720		-	
Melbourne	1235-1330		1450-1560		1505-1620	
Perth	1300-1400		1470-1585		1525-1640	
Sydney	1285-1385		1500-1615		1560-1680	

9.0 OFFICES

9.1 LETTABLE LOW RISE (FULLY SERVICED) - Finished Floor Lettable

ELEMENT	9.1.2.1 Office Building Single Storey		9.1.2.3 Office Building 2 Storey		9.1.2.5 Office Building 3 Storey		9.1.2.7 Office Building 4-7 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	209.50	11.1	243.75	11.3	287.00	11.6	359.25	13.0
SUBSTRUCTURE	113.50	6.0	68.25	3.1	56.00	2.3	51.75	1.9
SUPERSTRUCTURE								
Columns	-	-	47.25	2.2	55.50	2.3	56.50	2.1
Upper Floors	-	-	171.25	7.9	224.50	9.1	275.00	10.0
Staircase	-	-	21.00	1.0	22.50	0.9	20.00	0.7
Roof	222.75	11.8	133.00	6.1	113.75	4.6	74.00	2.7
External Walls)	282.00	15.0	310.50	14.3	317.00	12.9	391.50	14.2
Windows)								
External Doors	21.50	1.1	12.00	0.6	13.00	0.5	9.25	0.3
Internal Walls	66.75	3.5	88.50	4.1	94.25	3.8	90.00	3.3
Internal Screens	9.50	0.5	9.25	0.4	9.75	0.4	8.00	0.3
Internal Doors	14.25	0.8	28.50	1.3	30.50	1.2	30.50	1.1
FINISHES								
Wall	37.00	2.0	55.75	2.6	60.25	2.4	60.25	2.2
Floor	87.75	4.7	91.75	4.3	91.75	3.7	102.00	3.7
Ceiling	84.25	4.5	83.75	3.9	86.00	3.5	89.25	3.2
FITTINGS								
Fitments	20.00	1.1	19.75	0.9	19.75	0.8	22.25	0.8
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	102.75	5.4	126.25	5.8	126.25	5.1	130.00	4.7
Mechanical	378.50	20.1	386.25	17.8	403.75	16.4	478.25	17.4
Fire	17.75	0.9	16.00	0.7	19.00	0.8	22.50	0.8
Electrical	158.00	8.4	188.50	8.7	198.50	8.1	209.75	7.6
Transportation	-	-	-	-	167.25	6.8	202.50	7.4
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	12.00	0.6	9.50	0.5	7.00	0.3	3.50	0.1
CONTINGENCY	47.25	2.5	54.25	2.5	61.75	2.5	69.00	2.5
TOTAL (Mean cost - SYDNEY)	1885.00	100.0	2165.00	100.0	2465.00	100.0	2755.00	100.0
Cost Range : Adelaide	1800-1940		2040-2200		2310-2495		2525-2720	
Brisbane	1645-1775		1915-2065		2155-2325		2365-2550	
Canberra	1865-2015		2200-2370		2355-2535		2580-2780	
Darwin	2010-2160		2410-2500		2620-2830		2740-2960	
Hobart	1885-2035		2185-2355		2445-2635		2640-2845	
Melbourne	1760-1895		1995-2150		2260-2435		2460-2650	
Perth	1850-1995		2060-2220		2305-2485		2585-2785	
Sydney	1815-1955		2085-2245		2370-2555		2650-2860	

9.0 OFFICES

9.1 LETTABLE LOW RISE (FULLY SERVICED) - Shell and Core Lettable

ELEMENT	9.1.2.2 Office Building Single Storey		9.1.2.4 Office Building 2 Storey		9.1.2.6 Office Building 3 Storey		9.1.2.8 Office Building 4-7 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	195.25	11.3	228.25	11.6	282.75	12.3	342.25	13.4
SUBSTRUCTURE	114.50	6.6	67.25	3.4	55.50	2.4	52.50	2.0
SUPERSTRUCTURE								
Columns	-	-	46.75	2.4	55.50	2.4	57.25	2.2
Upper Floors	-	-	169.00	8.6	223.25	9.7	275.75	10.8
Staircase	-	-	21.25	1.0	22.25	1.0	20.25	0.8
Roof	223.25	12.9	131.50	6.7	113.75	5.0	74.00	2.9
External Walls)	282.75	16.4	307.75	15.6	316.00	13.8	392.50	15.3
Windows)								
External Doors	21.25	1.2	12.00	0.6	13.00	0.6	9.75	0.4
Internal Walls	66.25	3.9	88.25	4.5	93.50	4.1	91.00	3.5
Internal Screens	9.25	0.6	9.25	0.5	10.00	0.4	8.75	0.3
Internal Doors	13.75	0.8	28.50	1.4	29.50	1.3	30.50	1.2
FINISHES								
Wall	19.25	1.1	33.00	1.7	37.50	1.6	37.25	1.5
Floor	29.75	1.7	28.75	1.4	31.25	1.4	36.00	1.4
Ceiling	63.75	3.7	59.50	3.0	61.75	2.7	68.00	2.7
FITTINGS								
Fitments	19.75	1.1	19.75	1.0	20.00	0.9	22.25	0.9
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	102.00	5.9	125.75	6.3	125.75	5.5	129.50	5.1
Mechanical	361.75	21.0	365.25	18.5	377.25	16.5	444.00	17.3
Fire	17.25	1.0	17.00	0.8	19.25	0.8	22.75	0.9
Electrical	130.50	7.6	157.50	8.0	169.25	7.4	175.50	6.9
Transportation	-	-	-	-	168.75	7.4	202.75	7.9
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	11.50	0.7	9.25	0.5	7.00	0.3	3.50	0.1
CONTINGENCY	43.25	2.5	49.50	2.5	57.25	2.5	64.00	2.5
TOTAL (Mean cost - SYDNEY)	1725.00	100.0	1975.00	100.0	2290.00	100.0	2560.00	100.0
Cost Range : Adelaide	1645-1770		1860-2005		2125-2290		2305-2485	
Brisbane	1500-1620		1735-1870		1975-2125		2185-2355	
Canberra	1710-1840		2060-2225		2205-2380		2410-2600	
Darwin	1850-1990		2130-2300		2420-2600		2565-2775	
Hobart	1715-1850		1985-2140		2255-2430		2445-2635	
Melbourne	1600-1720		1795-1935		2075-2240		2255-2430	
Perth	1705-1840		1860-2005		2130-2295		2380-2565	
Sydney	1660-1790		1900-2050		2205-2375		2460-2655	

9.0 OFFICES

9.1 LETTABLE HIGH RISE (FULLY SERVICED) - Finished Floor Lettable

ELEMENT	9.1.3.1 Office Building 7-20 Storey		9.1.3.3 Office Building 21-35 Storey		9.1.3.5 Office Building 36-50 Storey		9.1.4 Office Building Prestige 36-50 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	719.00	18.3	1317.75	22.6	1574.50	24.5	1733.00	24.5
SUBSTRUCTURE	52.00	1.3	55.75	1.0	51.75	0.8	51.25	0.7
SUPERSTRUCTURE								
Columns	78.00	2.0	124.00	2.1	140.50	2.2	144.75	2.1
Upper Floors	345.00	8.8	437.25	7.5	440.75	6.9	452.75	6.4
Staircase	22.25	0.6	36.25	0.6	35.25	0.6	42.00	0.6
Roof	41.25	1.0	36.25	0.6	29.75	0.5	33.75	0.5
External Walls)	562.75	14.3	1021.50	17.5	1104.00	17.2	1218.00	17.2
Windows)								
External Doors	6.75	0.2	6.75	0.1	7.25	0.1	7.50	0.1
Internal Walls	102.50	2.6	185.75	3.2	194.25	3.0	203.00	2.9
Internal Screens	8.75	0.2	17.00	0.3	19.25	0.3	20.50	0.3
Internal Doors	33.75	0.9	38.50	0.7	41.00	0.6	44.00	0.6
FINISHES								
Wall	72.00	1.8	100.50	1.7	118.00	1.8	141.75	2.0
Floor	117.25	3.0	138.25	2.4	148.00	2.3	176.00	2.5
Ceiling	104.50	2.7	133.00	2.3	135.00	2.1	171.75	2.4
FITTINGS								
Fitments	31.25	0.8	34.25	0.6	36.75	0.6	42.00	0.6
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	144.75	3.7	178.25	3.1	193.00	3.0	213.00	3.0
Mechanical	678.25	17.3	871.75	14.9	973.25	15.2	1037.25	14.7
Fire	134.00	3.4	162.50	2.8	170.25	2.6	178.00	2.5
Electrical	293.50	7.5	377.25	6.5	392.50	6.1	480.00	6.8
Transportation	273.50	7.0	399.75	6.9	452.00	7.0	481.75	6.8
Special	-	-	-	-	-	-	13.75	0.2
EXTERNAL SERVICES	5.75	0.1	7.00	0.1	7.25	0.1	7.50	0.1
CONTINGENCY	98.25	2.5	145.75	2.5	160.75	2.5	176.75	2.5
TOTAL (Mean cost - SYDNEY)	3925.00	100.0	5825.00	100.0	6425.00	100.0	7070.00	100.0
Cost Range : Adelaide	3745-4035		4690-5055		-		-	
Brisbane	3050-3290		3705-3990		4105-4425		-	
Canberra	3345-3605		-		-		-	
Darwin	3380-3670		-		-		-	
Hobart	3055-3295		-		-		-	
Melbourne	3325-3585		4215-4545		4635-5000		-	
Perth	3520-3790		4670-5030		5160-5560		-	
Sydney	3775-4070		5605-6040		6185-6665		-	

9.0 OFFICES

9.1 LETTABLE HIGH RISE (FULLY SERVICED) - Shell and Core Lettable

ELEMENT	9.1.3.2 Office Building 7-20 Storey		9.1.3.4 Office Building 21-35 Storey		9.1.3.6 Office Building 36-50 Storey		9.1.4 Office Building Prestige 36-50 Storey	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	658.75	18.0	1252.00	22.7	1522.00	24.9	1668.00	24.8
SUBSTRUCTURE	52.50	1.4	56.25	1.0	52.00	0.9	51.50	0.8
SUPERSTRUCTURE								
Columns	77.25	2.1	123.50	2.2	140.50	2.3	144.25	2.2
Upper Floors	342.25	9.4	436.25	7.9	439.75	7.2	449.75	6.7
Staircase	23.00	0.6	35.00	0.7	35.50	0.6	41.00	0.6
Roof	43.00	1.2	35.00	0.7	30.25	0.5	33.50	0.5
External Walls)	558.25	15.3	1019.25	18.5	1097.50	18.0	1213.50	18.1
Windows)								
External Doors	6.75	0.2	6.75	0.1	7.25	0.1	7.75	0.1
Internal Walls	104.00	2.9	185.75	3.4	194.00	3.2	202.00	3.0
Internal Screens	8.75	0.2	16.75	0.3	19.75	0.3	20.50	0.3
Internal Doors	33.75	0.9	38.50	0.7	39.75	0.7	43.00	0.6
FINISHES								
Wall	45.50	1.2	62.75	1.1	63.75	1.0	83.25	1.2
Floor	45.50	1.2	54.75	1.0	60.00	1.0	70.75	1.1
Ceiling	88.25	2.4	103.00	1.9	107.00	1.7	142.25	2.1
FITTINGS								
Fitments	31.75	0.9	34.50	0.6	36.50	0.6	42.75	0.6
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	146.50	4.0	177.50	3.2	186.00	3.0	214.00	3.2
Mechanical	640.00	17.5	828.00	15.0	933.25	15.3	1012.75	15.1
Fire	134.25	3.7	162.00	3.0	168.75	2.8	162.75	2.4
Electrical	246.25	6.8	337.50	6.1	359.50	5.9	443.25	6.6
Transportation	271.25	7.4	399.75	7.3	451.75	7.4	484.25	7.2
Special	-	-	-	-	-	-	13.75	0.2
EXTERNAL SERVICES	6.00	0.2	7.50	0.1	7.50	0.1	7.50	0.1
CONTINGENCY	91.50	2.5	137.75	2.5	152.75	2.5	168.00	2.5
TOTAL (Mean cost - SYDNEY)	3655.00	100.0	5510.00	100.0	6105.00	100.0	6720.00	100.0
Cost Range : Adelaide	3485-3760		4415-4760		-		-	
Brisbane	2850-3070		3475-3745		3835-4135		-	
Canberra	2960-3195		-		-		-	
Darwin	3205-3440		-		-		-	
Hobart	2870-3095		-		-		-	
Melbourne	3100-3340		3970-4280		4405-4750		-	
Perth	3320-3580		4435-4780		4870-5250		-	
Sydney	3515-3790		5305-5715		5875-6335		-	

9.0 OFFICES

9.3 PARKING

ELEMENT	9.3.1 Office Building Parking Undercroft		9.3.2 Office Building Parking Partially Underground		9.3.3 Office Building Parking Fully Underground	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	88.25	12.4	193.25	12.7	401.75	21.4
SUBSTRUCTURE	36.25	5.1	263.25	17.3	365.25	19.4
SUPERSTRUCTURE						
Columns	35.00	4.9	77.00	5.1	87.25	4.6
Upper Floors	305.50	42.7	333.75	21.9	179.50	9.5
Staircase	11.25	1.6	17.00	1.1	21.75	1.2
Roof	-	-	-	-	-	-
External Walls)	54.25	7.6	145.25	9.5	182.50	9.7
Windows)						
External Doors	-	-	10.50	0.7	9.75	0.5
Internal Walls	44.75	6.3	95.25	6.3	102.00	5.4
Internal Screens	-	-	-	-	-	-
Internal Doors	8.50	1.2	14.00	0.9	16.00	0.9
FINISHES						
Wall	4.00	0.6	11.25	0.7	16.25	0.9
Floor	11.75	1.6	17.00	1.1	18.25	1.0
Ceiling	12.75	1.8	20.00	1.3	21.00	1.1
FITTINGS						
Fitments	-	-	1.75	0.1	1.75	0.1
Special	-	-	-	-	-	-
SERVICES						
Plumbing	9.50	1.3	31.75	2.1	37.25	2.0
Mechanical	-	-	51.00	3.3	121.75	6.5
Fire	20.25	2.8	132.75	8.7	131.50	7.0
Electrical	35.25	4.9	48.75	3.2	81.50	4.3
Transportation	17.25	2.4	21.75	1.4	38.00	2.0
Special	-	-	-	-	-	-
EXTERNAL SERVICES	2.50	0.3	1.50	0.1	-	-
CONTINGENCY	18.00	2.5	38.25	2.5	47.00	2.5
TOTAL (Mean cost - SYDNEY)	715.00	100.0	1525.00	100.0	1880.00	100.0
Cost Range : Adelaide		690-745		1430-1540		1620-1745
Brisbane		665-715		1365-1470		1540-1660
Canberra		700-750		840-910		-
Darwin		870-950		1350-1450		-
Hobart		765-825		1555-1675		-
Melbourne		665-715		1365-1475		1545-1665
Perth		680-735		1495-1610		1780-1915
Sydney		685-740		1465-1580		1810-1950

10.0 PARKING

10.1 UNDERGROUND

ELEMENT	10.1.2.1 Parking Underground 1 Level		10.1.2.2 Parking Underground 2 Levels	
	$/sqm	%	$/sqm	%
PRELIMINARIES	338.00	17.0	344.75	17.4
SUBSTRUCTURE	300.25	15.1	371.25	18.7
SUPERSTRUCTURE				
Columns	77.50	3.9	79.75	4.0
Upper Floors	-	-	176.75	8.9
Staircase	23.00	1.2	19.75	1.0
Roof	513.50	25.9	264.75	13.3
External Walls)	179.50	9.0	178.25	9.0
Windows)				
External Doors	35.75	1.8	20.50	1.0
Internal Walls	43.25	2.2	42.75	2.2
Internal Screens	-	-	-	-
Internal Doors	11.00	0.5	10.50	0.5
FINISHES				
Wall	20.00	1.0	19.75	1.0
Floor	18.00	0.9	17.75	0.9
Ceiling	18.00	0.9	17.75	0.9
FITTINGS				
Fitments	1.25	0.1	1.25	0.1
Special	-	-	-	-
SERVICES				
Plumbing	47.25	2.4	46.50	2.3
Mechanical	108.25	5.5	122.75	6.2
Fire	129.50	6.5	127.75	6.4
Electrical	69.75	3.5	71.25	3.6
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	1.50	0.1	1.50	0.1
CONTINGENCY	49.75	2.5	49.75	2.5
TOTAL (Mean cost - SYDNEY)	1985.00	100.0	1985.00	100.0
Cost Range : Adelaide	1820-1960		1820-1960	
Brisbane	1625-1750		1580-1705	
Canberra	-		-	
Darwin	-		-	
Hobart	-		-	
Melbourne	1745-1880		1730-1865	
Perth	1960-2110		1960-2110	
Sydney	1910-2060		1910-2060	

10.0 PARKING

10.3 PARKING STATIONS

ELEMENT	10.3.1.1 Parking Station Ground Plus 1 Level		10.3.1.2 Parking Station Ground Plus 2 Levels		10.3.2 Parking Station Multi-Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	73.50	13.0	88.25	13.0	148.00	16.6
SUBSTRUCTURE	58.25	10.3	50.50	7.4	52.75	5.9
SUPERSTRUCTURE						
Columns	30.00	5.3	34.00	5.0	50.50	5.7
Upper Floors	168.25	29.8	240.75	35.4	317.50	35.6
Staircase	4.75	0.8	6.00	0.9	8.25	0.9
Roof	12.25	2.2	14.25	2.1	8.25	0.9
External Walls)	85.75	15.2	104.50	15.4	135.75	15.3
Windows)						
External Doors	7.25	1.3	4.50	0.6	1.25	0.2
Internal Walls	4.75	0.8	14.50	2.1	12.00	1.3
Internal Screens	1.00	0.2	1.00	0.1	1.25	0.2
Internal Doors	1.00	0.2	1.50	0.2	1.75	0.2
FINISHES						
Wall	3.50	0.6	5.25	0.8	1.50	0.2
Floor	8.75	1.5	9.25	1.4	8.50	1.0
Ceiling	8.25	1.5	8.50	1.2	8.50	1.0
FITTINGS						
Fitments	17.25	3.1	15.50	2.3	14.75	1.6
Special	-	-	-	-	-	-
SERVICES						
Plumbing	11.75	2.1	13.75	2.0	16.00	1.8
Mechanical	1.50	0.3	1.50	0.2	1.75	0.2
Fire	0.75	0.1	1.25	0.2	2.50	0.3
Electrical	49.25	8.7	45.75	6.8	52.00	5.8
Transportation	-	-	-	-	22.00	2.5
Special	-	-	-	-	-	-
EXTERNAL SERVICES	3.00	0.5	2.50	0.4	3.00	0.3
CONTINGENCY	14.25	2.5	17.00	2.5	22.25	2.5
TOTAL (Mean cost - SYDNEY)	565.00	100.0	680.00	100.0	890.00	100.0
Cost Range : Adelaide		565-610		735-790		865-935
Brisbane		630-675		770-830		850-920
Canberra		625-675		675-725		-
Darwin		840-900		830-890		-
Hobart		620-670		785-845		-
Melbourne		565-605		705-760		900-970
Perth		500-540		680-735		910-980
Sydney		540-585		655-705		855-920

11.0 RECREATIONAL

11.1 GRANDSTANDS -

ELEMENT	11.1.3.1 Change Room Under Deck		11.1.3.2 Change Room Attached Building	
	$/sqm	%	$/sqm	%
PRELIMINARIES	285.00	11.3	338.00	11.3
SUBSTRUCTURE	119.00	4.7	132.75	4.4
SUPERSTRUCTURE				
Columns	-	-	-	-
Upper Floors	-	-	-	-
Staircase	-	-	-	-
Roof	10.00	0.4	266.50	8.9
External Walls)	264.00	10.5	410.50	13.6
Windows)				
External Doors	14.00	0.5	17.50	0.6
Internal Walls	75.00	3.0	87.50	2.9
Internal Screens	205.00	8.1	204.25	6.8
Internal Doors	67.75	2.7	67.00	2.2
FINISHES				
Wall	83.25	3.3	85.25	2.8
Floor	123.75	4.9	124.00	4.1
Ceiling	80.50	3.2	64.25	2.1
FITTINGS				
Fitments	173.50	6.9	178.00	5.9
Special	-	-	-	-
SERVICES				
Plumbing	608.75	24.1	633.00	21.0
Mechanical	160.25	6.4	135.00	4.5
Fire	9.50	0.4	9.25	0.3
Electrical	166.25	6.6	170.50	5.7
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	11.50	0.5	11.50	0.4
CONTINGENCY	63.00	2.5	75.25	2.5
TOTAL (Mean cost - SYDNEY)	2520.00	100.0	3010.00	100.0
Cost Range : Adelaide	2300-2480		2770-2985	
Brisbane	2080-2245		2490-2685	
Canberra	-		-	
Darwin	-		-	
Hobart	-		-	
Melbourne	2235-2410		2665-2870	
Perth	2205-2380		2775-2990	
Sydney	2425-2615		2895-3120	

11.0 RECREATIONAL

11.1 GRANDSTANDS -

ELEMENT	11.1.4.1 Public Toilets Under Deck		11.1.4.2 Public Toilets Attached Building		11.1.5.1 Club Area Attached Building	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	316.25	11.3	370.25	11.3	333.25	11.1
SUBSTRUCTURE	120.25	4.3	133.25	4.1	92.25	3.1
SUPERSTRUCTURE						
Columns	-	-	-	-	43.00	1.4
Upper Floors	-	-	-	-	208.75	6.9
Staircase	-	-	-	-	32.25	1.1
Roof	10.50	0.4	277.25	8.4	168.25	5.6
External Walls)	261.75	9.4	362.00	11.0	418.25	13.9
Windows)						
External Doors	19.00	0.7	20.25	0.6	14.25	0.5
Internal Walls	76.25	2.7	94.00	2.9	73.25	2.4
Internal Screens	227.25	8.1	234.75	7.1	20.75	0.7
Internal Doors	85.50	3.1	84.75	2.6	43.00	1.4
FINISHES						
Wall	114.25	4.1	96.00	2.9	67.25	2.2
Floor	130.00	4.7	129.00	3.9	123.75	4.1
Ceiling	78.00	2.8	76.00	2.3	92.75	3.1
FITTINGS						
Fitments	131.75	4.7	130.50	4.0	294.25	9.8
Special	-	-	-	-	-	-
SERVICES						
Plumbing	826.00	29.6	904.75	27.5	234.75	7.8
Mechanical	135.00	4.8	103.50	3.2	445.00	14.8
Fire	7.00	0.3	3.50	0.1	8.50	0.3
Electrical	171.25	6.1	171.00	5.2	213.75	7.1
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	10.25	0.4	12.00	0.4	7.50	0.2
CONTINGENCY	69.75	2.5	82.25	2.5	75.25	2.5
TOTAL (Mean cost - SYDNEY)	2790.00	100.0	3285.00	100.0	3010.00	100.0
Cost Range : Adelaide	2595-2795		3060-3300		2740-2950	
Brisbane	2380-2565		2860-3085		2470-2660	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	2485-2675		2940-3165		2635-2840	
Perth	2530-2730		3055-3290		2720-2930	
Sydney	2685-2895		3160-3405		2895-3120	

11.0 RECREATIONAL

11.2 CLUBHOUSES
11.3 BASKETBALL CENTRES

ELEMENT				11.2.1 Change Room/ Toilet Building		11.2.2 Clubhouse and Change Rooms		11.3 Basketball Centre	
				$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	...	...	...	288.25	9.5	264.25	9.5	159.50	10.5
SUBSTRUCTURE	...	...	...	125.75	4.1	125.25	4.5	112.25	7.4
SUPERSTRUCTURE									
Columns	...	...	...	-	-	-	-	-	-
Upper Floors	...	...	...	-	-	-	-	-	-
Staircase	...	...	...	-	-	-	-	-	-
Roof	...	...	...	272.50	9.0	271.75	9.7	269.00	17.7
External Walls)	...	...	...	471.25	15.5	530.75	19.0	341.50	22.5
Windows)	...	...	...						
External Doors	...	...	...	24.25	0.8	33.75	1.2	9.00	0.6
Internal Walls	...	...	...	64.50	2.1	64.75	2.3	31.50	2.1
Internal Screens	...	...	...	204.50	6.7	111.25	4.0	4.00	0.3
Internal Doors	...	...	...	69.50	2.3	62.00	2.2	12.75	0.8
FINISHES									
Wall	...	...	...	112.75	3.7	121.50	4.3	23.50	1.5
Floor	...	...	...	126.75	4.2	121.75	4.4	234.00	15.4
Ceiling	...	...	...	43.75	1.4	75.75	2.7	26.00	1.7
FITTINGS									
Fitments	...	...	...	197.00	6.5	218.50	7.8	14.75	1.0
Special	...	...	...	-	-	-	-	-	-
SERVICES									
Plumbing	...	...	...	664.50	21.8	459.50	16.4	59.75	3.9
Mechanical	...	...	...	120.50	4.0	63.00	2.3	44.25	2.9
Fire	...	...	...	7.25	0.2	7.00	0.3	9.75	0.6
Electrical	...	...	...	165.50	5.4	185.25	6.6	123.50	8.1
Transportation	...	...	...	-	-	-	-	-	-
Special	...	...	...	-	-	-	-	-	-
EXTERNAL SERVICES	...	...	...	10.25	0.3	9.00	0.3	7.00	0.5
CONTINGENCY	...	...	...	76.25	2.5	70.00	2.5	38.00	2.5
TOTAL (Mean cost - SYDNEY)	...	...	...	3045.00	100.0	2795.00	100.0	1520.00	100.0
Cost Range : Adelaide	...	...	...	2790-3010		2580-2780		1365-1470	
Brisbane	...	...	...	2730-2945		2385-2570		1235-1330	
Canberra	...	...	...	2660-2870		2630-2835		1545-1665	
Darwin	...	...	...	3100-3350		2900-3130		1530-1660	
Hobart	...	...	...	3100-3340		2835-3055		1535-1655	
Melbourne	...	...	...	2785-3005		2505-2700		1335-1435	
Perth	...	...	...	2805-3025		2520-2720		1325-1425	
Sydney	...	...	...	2930-3155		2690-2900		1425-1535	

11.0 RECREATIONAL

11.5 COMMUNITY RECREATION CENTRES

ELEMENT	11.5.1 Community Recreation Centre Basic Std.		11.5.2 Community Recreation Centre Medium Std.	
	$/sqm	%	$/sqm	%
PRELIMINARIES	111.75	9.6	142.75	9.7
SUBSTRUCTURE	102.00	8.8	104.50	7.1
SUPERSTRUCTURE				
Columns	12.25	1.1	14.75	1.0
Upper Floors	44.00	3.8	60.25	4.1
Staircase	3.50	0.3	6.75	0.4
Roof	206.00	17.8	223.50	15.1
External Walls)	146.50	12.6	224.25	15.2
Windows)				
External Doors	10.50	0.9	14.75	1.0
Internal Walls	35.00	3.0	49.25	3.3
Internal Screens	9.00	0.8	10.25	0.7
Internal Doors	5.75	0.5	13.00	0.9
FINISHES				
Wall	27.25	2.3	44.75	3.0
Floor	128.75	11.1	138.75	9.4
Ceiling	31.75	2.7	55.25	3.7
FITTINGS				
Fitments	13.50	1.2	25.50	1.7
Special	-	-	-	-
SERVICES				
Plumbing	69.25	6.0	89.25	6.0
Mechanical	40.00	3.4	73.50	5.0
Fire	5.25	0.4	5.50	0.3
Electrical	120.25	10.4	137.75	9.3
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	8.75	0.8	8.75	0.6
CONTINGENCY	29.00	2.5	37.00	2.5
TOTAL (Mean cost - SYDNEY)	1160.00	100.0	1480.00	100.0
Cost Range : Adelaide	1115-1200		1365-1470	
Brisbane	980-1055		1235-1330	
Canberra	1320-1420		1545-1665	
Darwin	1360-1470		1530-1660	
Hobart	1225-1320		1535-1655	
Melbourne	1045-1125		1335-1435	
Perth	1055-1140		1325-1425	
Sydney	1115-1205		1425-1535	

11.0 RECREATIONAL

11.6 GYMNASTICS
11.7 SQUASH COURTS

ELEMENT				11.6 Gymnasium		11.7.1 Squash Courts Basic Std.		11.7.2 Squash Courts High Std.	
				$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	...	...	...	247.75	11.1	151.75	9.4	201.50	9.3
SUBSTRUCTURE	...	...	...	128.75	5.7	118.25	7.3	123.00	5.7
SUPERSTRUCTURE									
Columns	...	...	...	-	-	-	-	20.00	0.9
Upper Floors	...	...	...	-	-	-	-	89.25	4.1
Staircase	...	...	...	-	-	-	-	49.00	2.3
Roof	...	...	...	268.00	12.0	234.00	14.4	292.75	13.5
External Walls)	...	...	...	526.75	23.5	330.75	20.4	458.50	21.1
Windows)	...	...	...						
External Doors	...	...	...	40.75	1.8	19.50	1.2	19.75	0.9
Internal Walls	...	...	...	54.75	2.4	123.50	7.6	141.00	6.5
Internal Screens	...	...	...	42.00	1.9	-	-	12.75	0.6
Internal Doors	...	...	...	22.25	1.0	16.50	1.0	20.75	1.0
FINISHES									
Wall	...	...	...	64.00	2.9	74.75	4.6	111.00	5.1
Floor	...	...	...	192.50	8.6	235.25	14.5	223.25	10.3
Ceiling	...	...	...	76.00	3.4	77.00	4.8	77.50	3.6
FITTINGS									
Fitments	...	...	...	124.00	5.5	19.50	1.2	35.50	1.6
Special	...	...	...	-	-	-	-	-	-
SERVICES									
Plumbing	...	...	...	76.50	3.4	65.75	4.1	95.50	4.4
Mechanical	...	...	...	80.00	3.6	12.50	0.8	23.00	1.0
Fire	...	...	...	17.25	0.8	-	-	-	-
Electrical	...	...	...	213.25	9.5	90.00	5.6	111.25	5.1
Transportation	...	...	...	-	-	-	-	-	-
Special	...	...	...	-	-	-	-	-	-
EXTERNAL SERVICES	...	...	...	9.50	0.4	10.50	0.6	10.50	0.5
CONTINGENCY	...	...	...	56.00	2.5	40.50	2.5	54.25	2.5
TOTAL (Mean cost - SYDNEY)	...	...	...	2240.00	100.0	1620.00	100.0	2170.00	100.0
Cost Range : Adelaide	...	...	...	1930-2080		1490-1605		1995-2150	
Brisbane	...	...	...	1845-1990		1390-1500		1870-2015	
Canberra	...	...	...	2200-2370		1610-1740		2130-2295	
Darwin	...	...	...	2280-2470		1850-2000		2330-2520	
Hobart	...	...	...	2305-2485		1665-1795		2170-2340	
Melbourne	...	...	...	1970-2120		1470-1585		1940-2090	
Perth	...	...	...	1875-2020		1470-1585		1985-2140	
Sydney	...	...	...	2155-2320		1555-1675		2090-2250	

12.0 RELIGIOUS

12.1 CHURCH HALL
12.2 CHAPEL, CHURCH

ELEMENT	12.1 Church Hall		12.2.1 Chapel/Church Basic Std.		12.2.2 Chapel/Church Medium Std.		12.2.3 Chapel/Church High Std.	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	115.50	9.6	154.25	9.4	242.00	9.6	390.50	10.7
SUBSTRUCTURE	102.25	8.5	118.00	7.2	134.00	5.3	145.75	4.0
SUPERSTRUCTURE								
Columns	-	-	45.00	2.7	79.00	3.1	85.75	2.3
Upper Floors	-	-	-	-	74.00	2.9	81.50	2.2
Staircase	-	-	-	-	41.00	1.6	46.25	1.3
Roof	200.25	16.5	246.75	15.0	338.75	13.4	867.25	23.7
External Walls)	230.50	19.0	389.00	23.7	623.25	24.7	781.00	21.4
Windows)								
External Doors	16.75	1.4	22.50	1.4	36.25	1.5	45.50	1.2
Internal Walls	59.00	4.9	32.00	2.0	34.50	1.4	36.50	1.0
Internal Screens	6.25	0.5	-	-	-	-	7.50	0.2
Internal Doors	26.75	2.2	14.50	0.9	22.25	0.9	21.00	0.6
FINISHES								
Wall	37.00	3.1	39.25	2.4	45.25	1.8	63.00	1.7
Floor	61.50	5.1	90.75	5.5	133.50	5.3	153.75	4.2
Ceiling	68.25	5.6	77.50	4.7	142.00	5.6	183.50	5.0
FITTINGS								
Fitments	42.75	3.5	163.25	10.0	230.50	9.1	298.00	8.2
Special	-	-	-	-	-		-	-
SERVICES								
Plumbing	105.00	8.7	58.75	3.6	86.50	3.4	111.75	3.1
Mechanical	-	-	-	-	13.75	0.6	19.00	0.5
Fire	13.25	1.1	5.75	0.4	6.00	0.2	10.75	0.3
Electrical	86.00	7.1	133.25	8.1	170.75	6.8	206.75	5.7
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	8.75	0.7	8.50	0.5	8.50	0.3	8.50	0.2
CONTINGENCY	30.25	2.5	41.00	2.5	63.25	2.5	91.50	2.5
TOTAL (Mean cost - SYDNEY)	1210.00	100.0	1640.00	100.0	2525.00	100.0	3655.00	100.0
Cost Range : Adelaide	1130-1215		1520-1640		2345-2530		3360-3620	
Brisbane	1170-1260		1610-1735		2475-2665		3500-3775	
Canberra	1495-1615		-		2350-2530		-	
Darwin	1550-1710		-		2650-2890		-	
Hobart	1275-1375		1600-1725		2455-2645		-	
Melbourne	1095-1185		1500-1615		2285-2465		3245-3500	
Perth	1140-1230		1580-1700		2375-2560		3365-3630	
Sydney	1165-1255		1575-1700		2430-2620		3515-3790	

13.0 RESIDENTIAL

13.1 HOUSES - SINGLE UNIT

ELEMENT	13.1.2.1 Individual House Medium Std. Framed		13.1.2.2 Individual House Medium Std. Brick Veneer		13.1.2.3 Individual House Medium Std. Full Brick	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	191.00	10.1	207.25	10.1	219.25	10.3
SUBSTRUCTURE	149.25	7.9	167.50	8.2	177.75	8.3
SUPERSTRUCTURE						
Columns	-	-	-	-	-	-
Upper Floors	-	-	-	-	-	-
Staircase	-	-	-	-	-	-
Roof	283.00	15.0	283.25	13.9	282.25	13.3
External Walls)	331.25	17.6	435.75	21.3	456.50	21.4
Windows)						
External Doors	35.50	1.9	36.00	1.8	35.75	1.7
Internal Walls	111.75	5.9	123.25	6.0	124.75	5.9
Internal Screens	12.25	0.7	15.50	0.8	15.50	0.7
Internal Doors	55.00	2.9	55.25	2.7	54.75	2.6
FINISHES						
Wall	61.50	3.3	60.75	3.0	105.25	4.9
Floor	45.75	2.4	45.75	2.2	46.00	2.2
Ceiling	88.75	4.7	89.00	4.4	87.00	4.1
FITTINGS						
Fitments	145.00	7.7	146.00	7.1	144.75	6.8
Special	-	-	-	-	-	-
SERVICES						
Plumbing	233.25	12.4	234.00	11.4	232.75	10.9
Mechanical	-	-	-	-	-	-
Fire	-	-	-	-	-	-
Electrical	79.75	4.2	79.75	3.9	79.75	3.7
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	14.75	0.8	14.75	0.7	14.75	0.7
CONTINGENCY	47.25	2.5	51.25	2.5	53.25	2.5
TOTAL (Mean cost - SYDNEY)	1885.00	100.0	2045.00	100.0	2130.00	100.0
Cost Range : Adelaide	1130-1220		1225-1325		1275-1375	
Brisbane	1560-1680		1715-1845		1870-2015	
Canberra	1610-1740		1780-1920		1805-1945	
Darwin	2060-2230		-		2050-2220	
Hobart	1825-1970		1950-2100		2085-2250	
Melbourne	1465-1580		1640-1765		1730-1865	
Perth	1495-1610		1650-1775		1745-1885	
Sydney	1815-1955		1965-2120		2050-2210	

13.0 RESIDENTIAL

13.1 HOUSES - SINGLE UNIT - HIGH STANDARD

ELEMENT	13.1.2.4 Individual House High Std. Framed		13.1.2.5 Individual House High Std. Brick Veneer		13.1.2.6 Individual House High Std. Full Brick	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	268.00	10.3	291.25	10.2	293.75	10.0
SUBSTRUCTURE	156.00	6.0	183.50	6.4	203.00	6.9
SUPERSTRUCTURE						
Columns	-	-	-	-	-	-
Upper Floors	-	-	-	-	-	-
Staircase	-	-	-	-	-	-
Roof	351.25	13.5	353.75	12.4	352.25	12.0
External Walls)	402.75	15.5	571.25	20.0	585.75	20.0
Windows)						
External Doors	46.25	1.8	46.25	1.6	48.25	1.7
Internal Walls	147.50	5.6	165.25	5.8	158.50	5.4
Internal Screens	21.00	0.8	23.00	0.8	22.25	0.8
Internal Doors	71.25	2.7	73.00	2.6	71.25	2.5
FINISHES						
Wall	83.50	3.2	83.25	2.9	123.25	4.2
Floor	145.00	5.6	145.00	5.1	145.50	5.0
Ceiling	123.00	4.7	123.25	4.3	124.25	4.2
FITTINGS						
Fitments	294.00	11.3	295.25	10.4	293.75	10.1
Special	-	-	-	-	-	-
SERVICES						
Plumbing	289.00	11.1	289.25	10.1	290.00	9.9
Mechanical	-	-	-	-	-	-
Fire	-	-	-	-	-	-
Electrical	126.50	4.8	125.75	4.4	125.25	4.3
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	14.75	0.6	14.50	0.5	14.75	0.5
CONTINGENCY	65.25	2.5	71.50	2.5	73.25	2.5
TOTAL (Mean cost - SYDNEY)	2605.00	100.0	2855.00	100.0	2925.00	100.0
Cost Range : Adelaide	1635-1765		1760-1895		1830-1975	
Brisbane	2265-2440		2535-2730		2680-2885	
Canberra	2215-2385		2360-2540		2455-2645	
Darwin	2520-2740		-		2450-2650	
Hobart	2365-2550		2525-2720		2635-2840	
Melbourne	2120-2285		2305-2485		2435-2620	
Perth	-		-		2500-2695	
Sydney	2505-2700		2745-2960		2815-3035	

13.0 RESIDENTIAL

13.1 HOUSES - SINGLE UNIT - PRESTIGE STANDARD
13.2 MULTI UNIT - LOW DENSITY

ELEMENT	13.1.2.7 Individual House Prestige Std. Full Brick		13.2.1.1 Town House 2 Storey Medium Std.		13.2.1.2 Town House 2 Storey High Std.	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	386.25	10.3	226.25	10.3	257.75	10.4
SUBSTRUCTURE	142.25	3.8	109.00	4.9	109.00	4.4
SUPERSTRUCTURE						
Columns	-	-	-	-	-	-
Upper Floors	173.50	4.6	165.50	7.5	176.00	7.1
Staircase	58.00	1.6	72.00	3.3	78.75	3.2
Roof	235.50	6.3	141.25	6.4	157.75	6.3
External Walls)	641.00	17.0	426.25	19.4	471.00	19.0
Windows)						
External Doors	47.25	1.3	32.50	1.5	34.50	1.4
Internal Walls	162.25	4.3	128.25	5.8	154.25	6.2
Internal Screens	24.00	0.6	16.75	0.8	21.00	0.8
Internal Doors	73.25	1.9	36.25	1.6	35.00	1.4
FINISHES						
Wall	148.00	3.9	85.25	3.9	111.50	4.5
Floor	223.50	5.9	96.50	4.4	107.00	4.3
Ceiling	135.75	3.6	62.50	2.8	70.00	2.8
FITTINGS						
Fitments	345.50	9.2	135.75	6.2	182.00	7.3
Special	-	-	-	-	-	-
SERVICES						
Plumbing	302.25	8.0	259.25	11.8	275.75	11.1
Mechanical	381.75	10.1	35.00	1.6	46.50	1.9
Fire	-	-	-	-	-	-
Electrical	176.00	4.7	102.75	4.7	121.00	4.8
Transportation	-	-	-	-	-	-
Special	-	-	-	-	-	-
EXTERNAL SERVICES	14.75	0.4	14.00	0.6	14.00	0.6
CONTINGENCY	94.25	2.5	55.00	2.5	62.25	2.5
TOTAL (Mean cost - SYDNEY)	3765.00	100.0	2200.00	100.0	2485.00	100.0
Cost Range : Adelaide	2395-2580		1565-1690		1755-1890	
Brisbane	3485-3755		1785-1925		1950-2100	
Canberra	-		2215-2385		2430-2620	
Darwin	-		1960-2110		2180-2350	
Hobart	3495-3765		2430-2620		2645-2850	
Melbourne	3060-3300		2000-2155		2265-2440	
Perth	3350-3615		1915-2060		2155-2325	
Sydney	3625-3905		2115-2280		2390-2580	

13.0 RESIDENTIAL

13.2 MULTI UNIT - LOW DENSITY

ELEMENT	13.2.2.1 Apartments Max. 3 Storeys Basic Std.		13.2.2.2 Apartments Max. 3 Storeys Medium Std.	
	$/sqm	%	$/sqm	%
PRELIMINARIES	225.25	10.8	247.50	10.9
SUBSTRUCTURE	98.75	4.7	100.50	4.4
SUPERSTRUCTURE				
Columns	-	-	-	-
Upper Floors	227.75	10.9	235.25	10.3
Staircase	40.00	1.9	44.75	2.0
Roof	106.50	5.1	118.75	5.2
External Walls)	326.75	15.6	357.25	15.7
Windows)				
External Doors	30.00	1.4	34.25	1.5
Internal Walls	126.75	6.0	135.50	6.0
Internal Screens	8.25	0.4	10.00	0.4
Internal Doors	34.25	1.6	37.25	1.6
FINISHES				
Wall	71.00	3.4	76.25	3.4
Floor	74.50	3.6	84.75	3.7
Ceiling	71.75	3.4	78.50	3.5
FITTINGS				
Fitments	122.50	5.8	143.50	6.3
Special	-	-	-	-
SERVICES				
Plumbing	321.75	15.4	334.50	14.7
Mechanical	38.00	1.8	45.00	2.0
Fire	10.00	0.5	10.00	0.4
Electrical	98.75	4.7	114.25	5.0
Transportation	-	-	-	-
Special	-	-	-	-
EXTERNAL SERVICES	10.00	0.5	10.25	0.5
CONTINGENCY	52.50	2.5	57.00	2.5
TOTAL (Mean cost - SYDNEY)	2095.00	100.0	2275.00	100.0
Cost Range : Adelaide	1640-1770		1815-1960	
Brisbane	1650-1775		1800-1940	
Canberra	2550-2750		2840-3060	
Darwin	2080-2240		2150-2320	
Hobart	2430-2620		2550-2750	
Melbourne	1965-2115		2180-2350	
Perth	1735-1870		1900-2045	
Sydney	2015-2170		2190-2360	

13.0 RESIDENTIAL

13.3 MULTI UNIT - HIGH DENSITY

ELEMENT	13.3.1.1 Apartments Multi-Storey Basic Std.		13.3.1.2 Apartments Multi-Storey Medium Std.		13.3.1.3 Apartments Multi-Storey High Std.		13.3.1.4 Apartments Multi-Storey Prestige Std.	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	352.50	14.5	376.00	14.4	463.75	14.3	521.50	14.3
SUBSTRUCTURE	43.75	1.8	44.25	1.7	44.00	1.4	46.00	1.3
SUPERSTRUCTURE								
Columns	59.25	2.4	59.25	2.3	59.50	1.8	61.75	1.7
Upper Floors	297.50	12.2	304.75	11.6	308.25	9.5	315.25	8.6
Staircase	32.00	1.3	35.25	1.3	39.00	1.2	42.50	1.2
Roof	39.75	1.6	40.00	1.5	43.50	1.3	45.75	1.3
External Walls)	307.75	12.7	358.25	13.7	375.75	11.6	432.25	11.8
Windows)								
External Doors	30.00	1.2	32.75	1.3	35.25	1.1	46.00	1.3
Internal Walls	112.75	4.6	117.00	4.5	118.75	3.7	123.25	3.4
Internal Screens	7.75	0.3	9.75	0.4	11.75	0.4	13.50	0.4
Internal Doors	30.00	1.2	35.75	1.4	43.25	1.3	55.50	1.5
FINISHES								
Wall	68.00	2.8	72.00	2.7	90.00	2.8	109.75	3.0
Floor	84.00	3.4	93.75	3.6	104.25	3.2	130.00	3.6
Ceiling	40.75	1.7	47.00	1.8	59.00	1.8	78.50	2.1
FITTINGS								
Fitments	136.50	5.6	154.50	5.9	196.50	6.1	279.00	7.7
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	367.50	15.1	381.50	14.5	385.75	11.9	418.00	11.5
Mechanical	37.25	1.6	44.50	1.7	388.25	11.9	424.75	11.6
Fire	56.50	2.3	56.25	2.2	57.50	1.8	61.25	1.7
Electrical	101.25	4.2	112.25	4.3	132.75	4.1	138.75	3.8
Transportation	167.75	6.9	173.50	6.6	173.25	5.3	177.25	4.9
Special	-	-	-	-	27.75	0.9	27.25	0.7
EXTERNAL SERVICES	1.50	0.1	1.25	0.1	1.25	0.1	1.25	0.1
CONTINGENCY	61.00	2.5	65.50	2.5	81.00	2.5	91.00	2.5
TOTAL (Mean cost - SYDNEY)	2435.00	100.0	2615.00	100.0	3240.00	100.0	3640.00	100.0
Cost Range : Adelaide	2165-2335		2360-2545		2950-3175		3280-3535	
Brisbane	2290-2470		2460-2650		3045-3285		3385-3650	
Canberra	-		-		-		-	
Darwin	-		-		-		-	
Hobart	-		-		-		-	
Melbourne	2295-2475		2485-2675		3095-3335		3450-3715	
Perth	2425-2615		2590-2795		3285-3545		3660-3945	
Sydney	2345-2525		2515-2710		3120-3360		3500-3775	

13.0 RESIDENTIAL

13.4 HOLIDAY UNITS

ELEMENT	13.4.1.1 Holiday Unit Single Unit Basic Std.		13.4.2.1 Holiday Unit Multi-Unit Basic Std.		13.4.2.2 Holiday Unit Multi-Unit Medium Std.		13.4.2.3 Holiday Unit Multi-Unit High Std.	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	164.75	9.3	188.00	10.3	206.50	10.4	255.25	10.0
SUBSTRUCTURE	122.75	6.9	73.75	4.0	73.75	3.7	72.75	2.9
SUPERSTRUCTURE								
Columns	-	-	-	-	-	-	-	-
Upper Floors	-	-	138.00	7.5	148.25	7.4	151.50	6.0
Staircase	-	-	37.00	2.0	37.25	1.9	43.50	1.7
Roof	205.25	11.6	136.25	7.4	146.75	7.4	158.50	6.2
External Walls)	363.25	20.5	317.25	17.3	345.50	17.4	375.50	14.8
Windows)								
External Doors	37.25	2.1	35.75	1.9	36.75	1.8	41.00	1.6
Internal Walls	147.00	8.3	138.25	7.5	141.25	7.1	143.50	5.6
Internal Screens	6.25	0.4	5.75	0.3	7.75	0.4	9.25	0.4
Internal Doors	45.00	2.5	44.25	2.4	46.75	2.3	58.25	2.3
FINISHES								
Wall	55.00	3.1	59.75	3.3	67.50	3.4	82.25	3.2
Floor	58.75	3.3	70.50	3.8	81.00	4.1	95.75	3.8
Ceiling	63.25	3.6	53.25	2.9	56.00	2.8	65.00	2.5
FITTINGS								
Fitments	141.00	8.0	141.75	7.8	157.25	7.9	173.50	6.8
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	220.75	12.5	238.25	13.0	267.75	13.5	292.50	11.5
Mechanical	-	-	-	-	-	-	320.50	12.6
Fire	-	-	-	-	-	-	-	-
Electrical	83.25	4.7	102.75	5.6	112.00	5.6	134.25	5.3
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	12.25	0.7	8.50	0.5	8.25	0.4	8.50	0.3
CONTINGENCY	44.25	2.5	46.00	2.5	49.75	2.5	63.75	2.5
TOTAL (Mean cost - SYDNEY)	1770.00	100.0	1835.00	100.0	1990.00	100.0	2545.00	100.0
Cost Range : Adelaide	1325-1430		1450-1560		1615-1740		2045-2205	
Brisbane	1620-1745		1705-1835		1915-2060		2415-2600	
Canberra	-		-		-		-	
Darwin	-		-		-		-	
Hobart	1670-1800		1825-1970		1975-2125		2510-2705	
Melbourne	1580-1705		1685-1820		1865-2010		2370-2555	
Perth	1615-1740		1740-1875		1955-2105		2500-2695	
Sydney	1700-1835		1765-1900		1915-2065		2450-2640	

13.0 RESIDENTIAL

13.5 AGED PERSONS HOME

ELEMENT	13.5.1 Aged Persons Homes Single Storey		13.5.2 Aged Persons Homes 2-3 Storey		13.5.3 Aged Persons Homes Multi-Storey		13.5.4 Aged Persons Homes Nursing Home	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	248.75	10.0	304.25	11.0	459.25	14.7	255.75	10.1
SUBSTRUCTURE	155.00	6.2	84.25	3.1	41.75	1.3	117.00	4.6
SUPERSTRUCTURE								
Columns	-	-	41.00	1.5	51.00	1.6	-	-
Upper Floors	-	-	179.00	6.5	294.75	9.5	-	-
Staircase	-	-	37.75	1.4	34.25	1.1	-	-
Roof	249.00	10.0	92.75	3.4	42.00	1.3	213.00	8.4
External Walls)	303.50	12.2	311.50	11.3	309.25	9.9	325.50	12.8
Windows)								
External Doors	40.00	1.6	30.25	1.1	29.25	1.0	40.00	1.6
Internal Walls	119.00	4.8	130.00	4.7	126.25	4.1	121.25	4.8
Internal Screens	9.75	0.4	10.00	0.4	9.75	0.3	12.00	0.5
Internal Doors	41.50	1.7	39.50	1.4	50.25	1.6	54.75	2.2
FINISHES								
Wall	72.00	2.9	68.50	2.5	72.00	2.3	69.50	2.7
Floor	95.00	3.8	86.00	3.1	89.25	2.9	88.25	3.5
Ceiling	66.25	2.6	70.75	2.6	55.25	1.8	72.25	2.9
FITTINGS								
Fitments	165.50	6.6	156.75	5.7	199.75	6.4	174.25	6.9
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	315.25	12.6	320.75	11.6	440.00	14.1	346.50	13.7
Mechanical	342.00	13.7	282.75	10.3	285.00	9.1	325.50	12.9
Fire	18.50	0.7	66.25	2.4	68.25	2.2	70.00	2.8
Electrical	184.50	7.4	199.00	7.2	196.75	6.3	174.75	6.9
Transportation	-	-	165.50	6.0	182.00	5.8	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	7.00	0.3	9.50	0.3	6.00	0.2	6.50	0.2
CONTINGENCY	62.50	2.5	69.00	2.5	78.00	2.5	63.25	2.5
TOTAL (Mean cost - SYDNEY)	2495.00	100.0	2755.00	100.0	3120.00	100.0	2530.00	100.0
Cost Range : Adelaide	1885-2030		2150-2320		2575-2775		1970-2120	
Brisbane	2100-2265		2240-2415		2960-3190		2055-2215	
Canberra	2695-2905		-		-		2580-2780	
Darwin	2460-2680		-		-		2700-2925	
Hobart	2835-3055		-		-		2575-2775	
Melbourne	1940-2090		2285-2465		2615-2820		2090-2255	
Perth	2210-2385		2480-2675		2880-3105		2270-2445	
Sydney	2400-2590		2650-2860		3005-3235		2435-2625	

13.0 RESIDENTIAL

13.6 HOSTELS

ELEMENT	13.6.1 Hostel 1-3 Storey		13.6.2 Hostel 4-10 Storey	
	$/sqm	%	$/sqm	%
PRELIMINARIES	288.00	11.9	393.25	14.6
SUBSTRUCTURE	81.50	3.4	65.50	2.4
SUPERSTRUCTURE				
Columns	40.00	1.7	56.75	2.1
Upper Floors	211.25	8.7	273.50	10.1
Staircase	33.75	1.4	39.00	1.5
Roof	124.50	5.2	61.50	2.3
External Walls)	340.25	14.1	334.25	12.4
Windows)				
External Doors	9.00	0.4	8.50	0.3
Internal Walls	104.50	4.3	101.75	3.8
Internal Screens	40.00	1.7	35.50	1.3
Internal Doors	39.75	1.6	38.25	1.4
FINISHES				
Wall	46.25	1.9	46.50	1.7
Floor	95.50	4.0	95.75	3.5
Ceiling	42.50	1.7	52.50	1.9
FITTINGS				
Fitments	277.25	11.5	287.75	10.7
Special	-	-	-	-
SERVICES				
Plumbing	372.25	15.4	357.50	13.2
Mechanical	37.00	1.5	52.75	2.0
Fire	23.00	1.0	20.50	0.8
Electrical	142.75	5.9	160.50	5.9
Transportation	-	-	145.75	5.4
Special	-	-	-	-
EXTERNAL SERVICES	5.50	0.2	5.25	0.2
CONTINGENCY	60.50	2.5	67.50	2.5
TOTAL (Mean cost - SYDNEY)	2415.00	100.0	2700.00	100.0
Cost Range : Adelaide	1885-2030		2250-2425	
Brisbane	2020-2175		2685-2890	
Canberra	-		-	
Darwin	-		-	
Hobart	-		-	
Melbourne	2070-2235		2320-2505	
Perth	2155-2325		2460-2650	
Sydney	2325-2505		2600-2800	

14.0 RETAIL

14.1 SUBURBAN

ELEMENT	14.1.1.1 Suburban Neighbourhood Shell Single Storey		14.1.1.2 Suburban Neighbourhood Shell 2 Storey		14.1.2.1 Suburban Supermarket		14.1.3.1 Suburban Shopping Centre Shell Excl. Mall/Arcade	
	$/sqm	%	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	71.00	8.5	88.75	9.1	166.75	9.1	186.50	9.8
SUBSTRUCTURE	93.25	11.2	74.50	7.6	110.75	6.1	117.75	6.2
SUPERSTRUCTURE								
Columns	27.00	3.3	33.00	3.4	54.75	3.0	47.00	2.5
Upper Floors	-	-	170.50	17.4	18.25	1.0	-	-
Staircase	-	-	13.50	1.4	7.00	0.4	-	-
Roof	186.25	22.4	110.25	11.3	200.25	10.9	269.50	14.2
External Walls)								
Windows)	174.50	21.0	188.25	19.2	316.50	17.3	340.25	17.9
External Doors								
Internal Walls	43.50	5.3	48.25	4.9	33.50	1.8	49.25	2.6
Internal Screens	-	-	-	-	2.50	0.1	4.00	0.2
Internal Doors	2.75	0.4	6.25	0.6	16.00	0.9	12.00	0.6
FINISHES								
Wall	9.00	1.1	9.25	0.9	36.75	2.0	19.00	1.0
Floor	17.25	2.1	18.00	1.8	55.25	3.0	26.50	1.4
Ceiling	71.75	8.6	66.25	6.8	72.00	3.9	82.75	4.3
FITTINGS								
Fitments	-	-	-	-	30.25	1.7	3.00	0.2
Special	-	-	-	-	-	-	-	-
SERVICES								
Plumbing	48.00	5.8	55.75	5.7	91.00	5.0	81.75	4.3
Mechanical	-	-	-	-	296.25	16.2	494.50	26.0
Fire	6.50	0.8	5.50	0.5	127.00	6.9	16.75	0.9
Electrical	47.50	5.7	59.00	6.0	141.00	7.7	90.25	4.8
Transportation	-	-	-	-	-	-	-	-
Special	-	-	-	-	-	-	-	-
EXTERNAL SERVICES	11.00	1.3	8.50	0.9	8.50	0.5	11.75	0.6
CONTINGENCY	20.75	2.5	24.50	2.5	45.75	2.5	47.50	2.5
TOTAL (Mean cost - SYDNEY)	830.00	100.0	980.00	100.0	1830.00	100.0	1900.00	100.0
Cost Range : Adelaide	815-875		920-995		1730-1865		1610-1735	
Brisbane	755-810		855-925		1570-1695		1605-1730	
Canberra	1095-1180		1155-1245		1925-2070		1830-1975	
Darwin	1125-1215		1300-1410		1930-2090		1820-1950	
Hobart	970-1045		1065-1150		1885-2035		1810-1950	
Melbourne	740-795		880-945		1625-1755		1690-1820	
Perth	790-850		940-1010		1665-1795		1635-1760	
Sydney	795-860		940-1015		1760-1895		1825-1970	

14.0 RETAIL

14.2 REGIONAL

ELEMENT	14.2.1.1 Regional Department Store		14.2.2.1.1 Regional Shopping Centre Incl. Malls Medium Standard		14.2.2.1.2 Regional Shopping Centre Incl. Malls Medium Standard Two Storey	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	284.50	12.8	308.50	12.6	375.75	13.6
SUBSTRUCTURE	120.25	5.4	103.50	4.2	62.50	2.3
SUPERSTRUCTURE						
Columns	67.00	3.0	46.25	1.9	61.00	2.2
Upper Floors	18.00	0.8	10.00	0.4	215.50	7.8
Staircase	12.00	0.5	3.25	0.2	20.50	0.8
Roof	223.25	10.0	263.75	10.8	157.00	5.7
External Walls)						
Windows)	429.50	19.3	434.75	17.8	454.00	16.5
External Doors						
Internal Walls	34.25	1.5	52.25	2.1	58.75	2.1
Internal Screens	3.25	0.1	4.25	0.2	5.25	0.2
Internal Doors	14.50	0.7	13.25	0.5	29.50	1.1
FINISHES						
Wall	22.25	1.0	42.00	1.7	49.50	1.8
Floor	84.25	3.8	81.00	3.3	83.25	3.0
Ceiling	84.25	3.8	86.75	3.6	89.25	3.2
FITTINGS						
Fitments	3.00	0.1	9.50	0.4	9.50	0.3
Special	-	-	21.25	0.9	21.50	0.8
SERVICES						
Plumbing	84.50	3.8	83.50	3.4	93.75	3.4
Mechanical	383.25	17.2	515.75	21.1	527.75	19.2
Fire	148.50	6.7	145.25	5.9	146.75	5.3
Electrical	148.25	6.6	151.25	6.2	161.50	5.9
Transportation	-	-	-	-	60.50	2.2
Special	-	-	-	-	-	-
EXTERNAL SERVICES	9.50	0.4	7.75	0.3	3.00	0.1
CONTINGENCY	55.75	2.5	61.25	2.5	69.00	2.5
TOTAL (Mean cost - SYDNEY)	2230.00	100.0	2445.00	100.0	2755.00	100.0
Cost Range : Adelaide	1950-2100		2115-2280		2385-2575	
Brisbane	1815-1955		1980-2130		2240-2415	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	1975-2130		2150-2320		2470-2660	
Perth	1950-2100		2195-2365		2460-2650	
Sydney	2145-2310		2350-2535		2650-2860	

14.0 RETAIL

14.2 REGIONAL
14.3 CITY

ELEMENT	14.2.2.2.1 Regional Shopping Centre Incl. Malls High Standard Single Storey		14.2.2.2.2 Regional Shopping Centre Incl. Malls High Standard Two Storey		14.3.1.1 City Department Store	
	$/sqm	%	$/sqm	%	$/sqm	%
PRELIMINARIES	363.50	13.1	420.00	13.6	498.75	16.5
SUBSTRUCTURE	113.50	4.1	69.75	2.3	68.00	2.2
SUPERSTRUCTURE						
Columns	53.50	1.9	62.75	2.0	66.00	2.2
Upper Floors	13.25	0.5	230.25	7.4	311.25	10.3
Staircase	3.75	0.1	22.00	0.7	16.00	0.5
Roof	286.00	10.3	170.50	5.5	86.75	2.9
External Walls)						
Windows)	500.25	18.1	531.00	17.1	327.00	10.8
External Doors						
Internal Walls	60.50	2.2	63.25	2.0	19.00	0.6
Internal Screens	7.00	0.3	5.50	0.2	3.00	0.1
Internal Doors	14.75	0.5	33.75	1.1	15.50	0.5
FINISHES						
Wall	65.00	2.3	74.50	2.4	28.75	0.9
Floor	96.00	3.5	107.75	3.5	94.00	3.1
Ceiling	93.50	3.4	100.25	3.2	89.50	3.0
FITTINGS						
Fitments	14.25	0.5	14.25	0.5	10.75	0.4
Special	28.75	1.0	29.00	0.9	-	-
SERVICES						
Plumbing	93.50	3.4	104.00	3.4	72.00	2.4
Mechanical	570.75	20.6	583.25	18.8	617.00	20.4
Fire	148.50	5.4	146.50	4.7	158.25	5.2
Electrical	168.25	6.1	177.25	5.7	205.00	6.8
Transportation	-	-	74.00	2.4	261.50	8.6
Special	-	-	-	-	-	-
EXTERNAL SERVICES	6.25	0.2	3.00	0.1	1.25	0.1
CONTINGENCY	69.25	2.5	77.50	2.5	75.75	2.5
TOTAL (Mean cost - SYDNEY)	2770.00	100.0	3100.00	100.0	3025.00	100.0
Cost Range : Adelaide	2380-2565		2670-2880		2715-2930	
Brisbane	2195-2370		2515-2710		2970-3205	
Canberra	-		-		-	
Darwin	-		-		-	
Hobart	-		-		-	
Melbourne	2455-2645		2755-2970		2720-2930	
Perth	2385-2570		2725-2935		2845-3065	
Sydney	2665-2875		2980-3215		2910-3135	

ESTIMATING COMPARATIVE COSTS

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EXPLANATORY NOTES :

This section is intended as a guide to the comparative unit costs of different design treatments within the various building elements so that, by selection or interpolation, elemental building costs can be obtained which specifically reflect the given sizes, proportions, etc. of any project.

The costs given are all-up estimating costs, i.e. they include allowance for labours and other minor items. Builder's head office overheads and profits have been included at approximately 5% (except for electrical and mechanical services at 2.5%).

Costs exclude the Goods and Services Tax (G.S.T.).

PRELIMINARIES

The following percentage allowances should be considered as an approximate guide only, to be used when compiling elemental cost estimates of a project.

The percentage allowances include for all usual Builder's preliminaries, i.e. insurance, setting out, plant and scaffolding, supervision, temporary services and facilities, cleaning and handover, maintenance, etc., assuming normal site conditions and average contract period relative to the value of the works.

ADD to the following percentage allowances where considered necessary on account of current market conditions, when the value of the works is below $500,000, or when the contract period is longer than that usually allowed, particular attention being paid to supervision, temporary services and plant hire.

			Approximate Preliminaries Adel. Bris. Melb. Per. Syd. %		
EDUCATIONAL					
School	- primary, single storey	...	8	-	14
	- secondary, maximum three stories	...	9	-	16
University	- teaching building, maximum three stories	...	11	-	18
	- support building, maximum three stories	...	11	-	18
HOSPITALS					
District/Private, single storey		...	9	-	14
General/Maternity, multi-storey		...	15	-	25
Aged Person's Home, single storey		...	9	-	15
HOTELS, ETC.					
City hotel, medium/high rise		...	15	-	25
Country motel, single or double storey		...	9	-	15
Club, sporting or social, maximum three stories		...	9	-	15
INDUSTRIAL					
Warehouse/Store	- single storey	...	8	-	14
	- multi-storey	...	11	-	19
Factory	- single storey, small span	...	8	-	14
	- single storey, large span	...	8	-	14
Laboratory/Workshop	- single storey	...	9	-	14
OFFICES					
Single/two storey		...	9	-	15
Three/seven storey		...	10	-	19
Seven/twenty storey		...	15	-	25
Twenty/fifty storey		...	18	-	28
RESIDENTIAL					
Individual/town house		...	8	-	14
Flats	- maximum three stories	...	10	-	15
	- multi-storey up to twenty stories	...	12	-	20
RETAIL					
Suburban	- neighbourhood, shell	...	7	-	12
	- supermarket	...	7	-	13
	- shopping centre, shell	...	7	-	13
Regional	- department store/shopping centre	...	10	-	18
City	- department store	...	14	-	22
	- multi-level arcade	...	14	-	24

SUBSTRUCTURE

		UNIT	Adel. $	Bris. $	Mel. $	Per $	Syd. $
The following site soil conditions have been assumed :							
Adelaide - Clay							
Brisbane - Clay							
Melbourne - Clay							
Perth - Sand							
Sydney - Clay							
REDUCED LEVEL EXCAVATION)							
BASEMENT EXCAVATION) Refer Detailed							
DEWATERING) Prices pages							
PILES AND PIERS) 211 to 221							
SHEET PILING, ETC.)	...	Note					

FOUNDATION BEAMS

25 MPa concrete in foundation beam including reinforcement (75 kg/cum), formwork, excavation of equal depth, planking and strutting :							
Foundation beams generally	...	cum	1060.00	1020.00	1130.00	870.00	1140.00
Foundation beam size -							
450mm wide x 600mm deep	...	m	288.00	277.00	307.00	236.00	309.00
450mm wide x 750mm deep	...	m	359.00	345.00	382.00	294.00	385.00
600mm wide x 750mm deep	...	m	423.00	407.00	448.00	351.00	457.00

COLUMN FOUNDATIONS

25 MPa concrete in column foundation including reinforcement (40 kg/cum), formwork, excavation of equal depth, planking and strutting	...	cum	638.00	624.00	685.00	538.00	708.00
ADD extra for :							
Excavation twice depth of foundation	...	cum	125.50	106.00	120.50	77.20	118.50

STRIP FOOTINGS

25 MPa concrete in strip footing including reinforcement (20 kg/cum), formwork, excavation of equal depth, planking and strutting	...	cum	567.00	534.00	574.00	453.00	612.00
ADD extra for :							
Excavation twice depth of footing	...	cum	162.50	138.00	138.50	420.00	148.00

RAFT FOUNDATIONS

25 MPa concrete in raft foundation including reinforcement (100 kg/cum) and formwork	...	cum	573.00	582.00	606.00	552.00	664.00

FILLING

Clean sand filling compacted in layers, under ground slab :							
150mm deep	...	sqm	11.50	10.70	9.05	3.95	11.05
250mm deep	...	sqm	19.20	17.90	15.15	6.60	18.45
500mm deep	...	sqm	38.40	35.80	30.30	13.20	36.90
1000mm deep	...	sqm	76.80	71.50	60.50	26.40	73.70

SUBSTRUCTURE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FILLING (cont'd)							
Crushed rock filling compacted in layers under ground slab :							
150mm deep	...	sqm	17.25	13.40	13.40	10.70	11.80
250mm deep	...	sqm	28.70	22.30	22.30	17.85	19.70
500mm deep	...	sqm	57.50	44.60	44.60	35.70	39.40
GROUND SLABS							
25 MPa concrete in ground slab including F72 fabric reinforcement, formwork to edges, joints :							
100mm thick	...	sqm	82.10	83.70	91.60	66.60	97.70
125mm thick	...	sqm	89.10	90.00	100.00	74.50	108.00
150mm thick	...	sqm	96.10	96.30	108.50	82.50	118.00
ADD extra for :							
Termite treatment refer pages 220 and 221	...	Note					
Orange polythene underlay	...	sqm	4.80	5.55	6.05	5.40	7.25
TIMBER GROUND FLOOR FRAMING							
Framing comprising 100 x 50mm joists at 450mm centres, 100 x 75mm bearers at 1800mm centres, stumps and foundations	...	sqm	68.70	106.50	78.90	89.80	110.50
UNDERFLOOR DUCTS							
25 MPa concrete in 100mm thick sides and base of underfloor duct, including reinforcement (65 kg/cum), formwork (Class 4) to sides, duct internal size :							
450mm wide x 300mm deep	...	m	277.00	267.00	294.00	230.00	312.00
450mm wide x 450mm deep	...	m	381.00	370.00	407.00	317.00	432.00
600mm wide x 600mm deep	...	m	495.00	481.00	528.00	411.00	561.00
ADD extra for :							
Excavation and planking and strutting to duct size :							
450mm wide x 300mm deep	...	m	32.50	26.30	26.00	26.00	23.80
450mm wide x 450mm deep	...	m	44.70	36.10	35.80	35.80	32.80
600mm wide x 600mm deep	...	m	65.80	53.20	52.70	48.60	48.30
Grating or cover to top	...	Note					
UNDERPINNING							
Underpinning existing 600mm wide footing in mass concrete including preliminary excavation not exceeding 1000mm deep, depth below underside of existing footing :							
1000mm	...	m	1215.00	1295.00	1355.00	1080.00	1420.00
2000mm	...	m	2325.00	2445.00	2625.00	2095.00	2695.00
3000mm	...	m	3590.00	3685.00	4100.00	3260.00	4335.00
WALLS BELOW GROUND SLAB							
Refer EXTERNAL WALLS, pages 125 to 129	...	Note					

COLUMNS, UPPER FLOORS

REINFORCED IN SITU CONCRETE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Unless otherwise described, costs are based on 25 MPa concrete, Class 3 formwork, bar reinforcement and assume standard loadings and spans	...	Note					
For the additional cost of higher strength concrete, pumping, additives, formwork alternatives, refer Detailed Prices, pages 237 to 248	...	Note					
COLUMNS							
(Reinforcement 190 kg/cum)							
300mm x 300mm	...	cum	3240.00	2765.00	3215.00	2770.00	3320.00
	...	m	291.00	249.00	289.00	249.00	299.00
400mm x 400mm	...	cum	2650.00	2265.00	2625.00	2280.00	2725.00
	...	m	424.00	363.00	420.00	365.00	436.00
500mm x 500mm	...	cum	2315.00	1980.00	2285.00	1995.00	2380.00
	...	m	579.00	495.00	571.00	499.00	594.00
600mm x 600mm	...	cum	2075.00	1775.00	2040.00	1795.00	2130.00
	...	m	746.00	639.00	735.00	646.00	768.00
750mm x 750mm	...	cum	1830.00	1570.00	1795.00	1590.00	1885.00
	...	m	1040.00	894.00	1025.00	906.00	1075.00
Tube formed :							
200mm dia.	...	m	128.00	140.00	138.00	138.00	161.00
300mm dia.	...	m	178.00	187.00	189.00	187.00	217.00
400mm dia.	...	m	250.00	254.00	260.00	255.00	294.00
500mm dia.	...	m	341.00	339.00	350.00	342.00	393.00
600mm dia.	...	m	450.00	430.00	449.00	437.00	501.00
ATTACHED BEAMS							
Sizes given are overall dimensions below soffit of slab	...	Note					
(Reinforcement 150 kg/cum)							
200mm x 300mm deep	...	cum	2965.00	2600.00	3060.00	2680.00	3150.00
	...	m	178.00	156.00	183.00	161.00	189.00
300mm x 300mm deep	...	cum	2405.00	2105.00	2470.00	2175.00	2555.00
	...	m	217.00	189.00	222.00	196.00	230.00
300mm x 450mm deep	...	cum	2220.00	1940.00	2275.00	2010.00	2360.00
	...	m	300.00	262.00	307.00	271.00	319.00
400mm x 600mm deep	...	cum	1850.00	1610.00	1885.00	1675.00	1965.00
	...	m	444.00	386.00	452.00	402.00	471.00
450mm x 750mm deep	...	cum	1700.00	1475.00	1725.00	1540.00	1800.00
	...	m	574.00	498.00	583.00	520.00	609.00
ADD extra for :							
Additional reinforcement for that part of the beam within the depth of the suspended slab (@150 kg/cum)	...	cum	413.00	380.00	394.00	374.00	394.00

COLUMNS, UPPER FLOORS

REINFORCED IN SITU CONCRETE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SUSPENDED SLABS							
Suspended slab in load bearing structure (reinforcement 80 kg/cum) :							
100mm thick	...	sqm	195.00	179.00	198.00	155.00	210.00
125mm thick	...	sqm	207.00	190.00	211.00	168.00	225.00
150mm thick	...	sqm	220.00	201.00	225.00	181.00	240.00
Suspended flat plate slab (reinforcement 100 kg/cum) :							
200mm thick	...	sqm	261.00	238.00	269.00	221.00	285.00
225mm thick	...	sqm	282.00	257.00	291.00	241.00	308.00
250mm thick	...	sqm	289.00	263.00	299.00	249.00	318.00
275mm thick	...	sqm	303.00	276.00	314.00	263.00	334.00
300mm thick	...	sqm	316.00	288.00	328.00	276.00	351.00
SPEEDFLOOR							
Composite suspended slab comprising cold form steel joist and plywood formwork :							
200 series joist	...	sqm	115.00-125.00	115.00-125.00	115.00-125.00	115.00-125.00	115.00-125.00
ADD extra for :							
Insitu concrete topping	...	Note					
Reinforcement	...	Note					
Surface finish	...	Note					
PRESTRESSING IN SITU CONCRETE							
Average allowance for prestressing and reinforcement only assuming a typical slab situation i.e. 7500 to 8000 spans and 30.0m tendons, slab thickness :							
200mm - post tensioning	...	sqm	40.50	41.00	39.00	40.00	41.00
- reinforcement (25 kg/cum)	...	sqm	15.00	13.80	14.30	13.60	14.30

LIFT SLABS

Whilst in the past the cost of lift slab construction might, at the very best, have been comparable or only marginally ahead of the all-up cost of conventional methods the situation has now changed and in most capital cities the overall cost of a post-tensioned lift slab is now on average some 15 to 20% less than a conventionally formed reinforced or prestressed flat slab. (The major factor in this marked improvement in the cost of lift slab construction has been generally the high level of conventional formwork costs in all capital cities).

Apart from the cost savings occasioned through the various works being carried out with greater convenience at ground level than at heights, another advantage of lift slab construction is the likely saving in construction period and, depending upon plant requirements, a possible reduction in the use or deletion of a tower crane.

COLUMNS, UPPER FLOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PRECAST CONCRETE							
Costs are for supply and erection complete	...	Note					
COLUMNS							
300mm x 300mm	...	m	365.00	379.00	405.00	534.00	395.00
400mm x 400mm	...	m	400.00	505.00	530.00	626.00	515.00
500mm x 500mm	...	m	760.00	722.00	470.00	745.00	715.00
600mm x 600mm	...	m	950.00	949.00	905.00	920.00	895.00
750mm x 750mm	...	m	1130.00	1150.00	1200.00	1140.00	1175.00
BEAMS							
450mm x 450mm deep	...	m	690.00	641.00	645.00	534.00	633.00
450mm x 600mm deep	...	m	742.00	700.00	730.00	607.00	721.00
450mm x 750mm deep	...	m	800.00	888.00	845.00	684.00	832.00
450mm x 800mm deep	...	m	910.00	937.00	945.00	742.00	942.00
450mm x 1050mm deep	...	m	970.00	1090.00	1070.00	882.00	1050.00
450mm x 1200mm deep	...	m	1080.00	1150.00	150.00	949.00	1125.00
FLOORS							
Hollowcore pretensioned 1200mm wide units :							
150mm thick	...	sqm	130.00-140.00	140.00-150.00	-	123.00	126.00-142.00
200mm thick	...	sqm	150.00-160.00	175.00-190.00	180.00-195.00	126.00	181.00-195.00
250mm thick	...	sqm	180.00-190.00	185.00-200.00	188.00-195.00	145.00	188.00-195.00
300mm thick	...	sqm	190.00-210.00	200.00-215.00	195.00-210.00	155.00	195.00-212.00
Floor systems, refer pages 252 and 253	...	Note					
STRUCTURAL STEEL							
Costs are for supply and erection complete, including shop treatment with zinc phosphate primer	...	Note					
COLUMNS							
Plain steel :							
150 UC 37.2	...	m	229.00	242.00	233.00	241.00	234.00
200 UC 59.5	...	m	363.00	381.00	369.00	379.00	371.00
250 UC 89.5	...	m	546.00	573.00	555.00	570.00	558.00
310 UC 158.0	...	m	965.00	1010.00	980.00	1005.00	985.00
100 x 100 x 6 .3 square hollow section	...	m	140.00	142.00	137.00	141.00	144.00
125 x 125 x 6 .3 square hollow section	...	m	183.00	184.00	178.00	181.00	188.00
ADD extra for :							
Zinc silicate treatment	...	t	340.00	420.00	345.00	395.00	315.00
Hot dip galvanising	...	t	1055.00	630.00	1155.00	920.00	1155.00

COLUMNS, UPPER FLOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STRUCTURAL STEEL (cont'd)							
COLUMNS (cont'd)							
Concrete encasing (25 MPa) and including Class 3 formwork :							
150 UC 37.2	...	m	215.00	186.00	218.00	187.00	229.00
200 UC 59.5	...	m	257.00	223.00	262.00	225.00	276.00
250 UC 89.5	...	m	311.00	269.00	317.00	273.00	335.00
310 UC 158.0	...	m	398.00	346.00	407.00	353.00	434.00
Sprayed fireproofing (P.C. $62/sqm applied) :							
150 UC 37.2	...	m	57.00	57.00	57.00	57.00	57.00
200 UC 59.5	...	m	74.00	74.00	74.00	74.00	74.00
250 UC 89.5	...	m	93.00	93.00	93.00	93.00	93.00
310 UC 158.0	...	m	114.00	114.00	114.00	114.00	114.00
ADD or DEDUCT for each variation of $5.00 per sqm for type or rating of sprayed fireproofing	...	+/-12.5%					
Refer pages 671 to 673 for alternative fireproofing materials	...	Note					
BEAMS							
Plain steel :							
200 UB 29.8	...	m	184.00	194.00	186.00	193.00	187.00
250 UB 37.3	...	m	230.00	243.00	233.00	242.00	235.00
310 UB 46.2	...	m	285.00	301.00	289.00	299.00	291.00
360 UB 56.7	...	m	346.00	363.00	352.00	361.00	354.00
410 UB 59.7	...	m	365.00	382.00	370.00	380.00	372.00
460 UB 82.1	...	m	501.00	526.00	509.00	523.00	512.00
530 UB 92.4	...	m	564.00	592.00	573.00	589.00	576.00
610 UB 125.0	...	m	763.00	801.00	776.00	797.00	780.00
ADD extra for :							
Zinc silicate treatment	...	t	340.00	420.00	345.00	395.00	315.00
Hot dip galvanising	...	t	1055.00	630.00	1155.00	920.00	1155.00
Concrete encasing (25 MPa) and including Class 3 formwork :							
200 UB 29.8	...	m	211.00	188.00	224.00	196.00	234.00
250 UB 37.3	...	m	236.00	211.00	251.00	221.00	263.00
310 UB 46.2	...	m	267.00	239.00	284.00	250.00	299.00
360 UB 56.7	...	m	293.00	262.00	312.00	275.00	329.00
410 UB 59.7	...	m	315.00	281.00	336.00	296.00	354.00
460 UB 82.1	...	m	351.00	314.00	375.00	330.00	396.00
530 UB 92.4	...	m	387.00	346.00	414.00	365.00	438.00
610 UB 125.0	...	m	434.00	388.00	464.00	410.00	493.00

COLUMNS, UPPER FLOORS

STRUCTURAL STEEL (cont'd)

BEAMS (cont'd)

Plain steel (cont'd) :

ADD extra for (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Sprayed fireproofing (to surface area less top of upper flange) (P.C. $43/sqm applied) :							
200 UB 29.8	...	m	34.00	34.00	34.00	34.00	34.00
250 UB 37.3	...	m	40.00	40.00	40.00	40.00	40.00
310 UB 46.2	...	m	47.00	47.00	47.00	47.00	47.00
360 UB 56.7	...	m	52.00	52.00	52.00	52.00	52.00
410 UB 59.7	...	m	56.00	56.00	56.00	56.00	56.00
460 UB 82.1	...	m	63.00	63.00	63.00	63.00	63.00
530 UB 92.4	...	m	71.00	71.00	71.00	71.00	71.00
610 UB 125.0	...	m	80.00	80.00	80.00	80.00	80.00
ADD or DEDUCT for each variation of $5.00 per sqm for type or rating of sprayed fireproofing	...	+/-12.5%					
Refer pages 671 to 673 for alternative fireproofing materials	...	Note					

TIMBER

Posts 100mm x 100mm, dressed :							
Hardwood	...	m	41.90	-	-	-	-
Oregon	...	m	-	-	41.20	-	46.80
Upper floor framing comprising joists, plates and blocking :							
150mm x 50mm joists at 450mm centres	...	sqm	75.40	123.50	89.10	110.50	130.50
200mm x 50mm joists at 600mm centres	...	sqm	82.50	139.00	98.80	103.50	145.50
ADD extra for :							
Decking, refer page 340	...	Note					

BRICKWORK

Piers in standard clay common brickwork :							
350mm x 230mm	...	m	103.50	77.90	88.90	79.00	101.00
350mm x 350mm	...	m	155.00	117.00	133.50	118.50	151.50
470mm x 470mm	...	m	277.00	209.00	239.00	212.00	271.00
ADD extra for :							
Galvanised structural grade bar reinforcement -							
4/6mm dia.	...	m	3.80	4.00	4.00	4.00	4.40
4/10mm dia.	...	m	8.00	8.40	8.20	8.20	8.60

STAIRCASES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Costs of staircases are given per metre of Vertical Rise	...	Note					

IN SITU CONCRETE

BASIC staircase rising 3.00 metres between levels in 1000mm wide flights with intermediate landing; including monolithic finish to walking surface, Class 3 off-form formwork soffit and edges, M.S. tubular balustrade to inner leg	...	M/Rise	2680.00	2775.00	3320.00	2430.00	3085.00
ADD to BASIC staircase extra for :							
WALKING SURFACE -							
Granolithic (including edges)	...	M/Rise	468.00	275.00	436.00	374.00	352.00
Vinyl tiles	...	M/Rise	126.00	129.00	127.00	130.00	137.00
Carpet	...	M/Rise	220.00	222.00	220.00	223.00	230.00
Terrazzo	...	M/Rise	4085.00	3615.00	3075.00	-	3115.00
Marble	...	M/Rise	2325.00	2430.00	2140.00	2325.00	2485.00
Aluminium safety nosing	...	M/Rise	473.00	472.00	477.00	481.00	561.00
SOFFIT -							
Boarded off-form	...	M/Rise	274.00	-	-	341.00	281.00
Hardwall plaster and seal and two coats acrylic paint	...	M/Rise	279.00	212.00	158.00	196.00	195.00
Seal and two coats acrylic paint	...	M/Rise	76.00	69.50	73.00	71.00	70.00
ADDITIONAL WIDTH -							
For each additional 100mm in flight width add to the foregoing BASIC staircase and associated additional finishing costs	...	+ 7%					
BALUSTRADES -							
200mm x 38mm timber handrail	...	M/Rise	164.00	181.00	186.00	150.00	179.00
Aluminium section balustrade with laminated glass and posts at 1200mm centres	...	M/Rise	454.00	380.00	405.00	399.00	369.00
Stainless steel section balustrade with laminated glass and posts at 1200mm centres	...	M/Rise	921.00	816.00	892.00	834.00	825.00
UNENCLOSED STAIRCASES -							
Additional cost of balustrade to outer perimeter :							
M.S. painted balustrade	...	M/Rise	1155.00	1055.00	1155.00	1215.00	1160.00
Ditto with 200 x 38mm timber handrail	...	M/Rise	1435.00	1365.00	1470.00	1475.00	1470.00
Aluminium section balustrade with laminated glass and posts at 1200mm centres	...	M/Rise	1935.00	1705.00	1845.00	1900.00	1795.00
Stainless steel section balustrade with laminated glass and posts at 1200mm centres	...	M/Rise	2735.00	2450.00	2680.00	2645.00	2575.00

STAIRCASES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STEEL FRAMED							
BASIC steel framed heavy duty staircase rising 3.00 metres between levels in 1000mm wide flights with intermediate landing, including expanded steel panels to treads and landings, M.S. tubular balustrade to inner leg, all visible steel painted	...	M/Rise	3620.00	3640.00	3660.00	3840.00	3735.00
ADD to BASIC staircase extra for :							
TIMBER TREADS -							
Polished hardwood open treads and landings	...	M/Rise	1230.00	1155.00	765.00	785.00	1120.00
NOSING -							
Aluminium safety nosing	...	M/Rise	475.00	470.00	475.00	480.00	560.00
ADDITIONAL WIDTH -							
For each additional 100mm in flight width add to the foregoing BASIC staircase and associated additional finishing costs	...	+ 6%					
BALUSTRADES -							
200 x 38mm timber handrail	...	M/Rise	164.00	181.00	186.00	150.00	179.00
UNENCLOSED STAIRCASES -							
Additional cost of balustrade to outer perimeter :							
M.S. painted balustrade	...	M/Rise	1155.00	1055.00	1155.00	1215.00	1160.00
Ditto with 200 x 38mm timber handrail	...	M/Rise	1435.00	1365.00	1470.00	1475.00	1470.00
STEEL ACCESS LADDER							
Steel access ladder 450mm wide with 16mm bar rungs :							
Primed and paint finish	...	M/Rise	63.00	66.10	62.90	67.00	64.40
Galvanised and paint finish	...	M/Rise	71.10	70.90	72.20	74.70	72.70
TIMBER							
TRADITIONAL							
Traditional pattern 1000mm wide dog-leg stair in hardwood with closed treads and risers, simple pattern balustrade with moulded rails, landing between floor levels and with clear finish to all visible surfaces	...	M/Rise	1590.00	1870.00	1670.00	1735.00	1720.00
ADD extra for :							
Carpet to treads and risers, landings	...	M/Rise	298.00	294.00	299.00	294.00	315.00
Each additional 100mm in width	...	+ 7%					
Similar balustrade to outer perimeter	...	M/Rise	950.00	1020.00	940.00	930.00	1050.00

STAIRCASES

TIMBER (cont'd)

CONTEMPORARY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Contemporary pattern 1000mm wide dog-leg stair in hardwood with open treads, plain strings, simple pattern balustrade, landing between floor levels and with clear finish to all visible surfaces	...	M/Rise	1655.00	1970.00	1725.00	1825.00	1865.00
ADD extra for :							
Carpet to treads and landings	...	M/Rise	148.00	147.00	149.00	146.00	157.00
Each additional 100mm in width	...	+ 6%					
Similar balustrade to outer perimeter	...	M/Rise	860.00	920.00	850.00	840.00	945.00

SPIRAL

Steel spiral staircase, standard pattern with plate treads, centre newel, simple balustrade with plastic capping, all painted :							
1300mm overall diameter	...	M/Rise	2165.00	2120.00	2155.00	2110.00	2160.00
1500mm overall diameter	...	M/Rise	2355.00	2245.00	2260.00	2210.00	2270.00
Timber spiral staircase, standard pattern, with open treads, steel centre newel, simple balustrade with plastic capping, all polished or painted :							
1200mm overall diameter	...	M/Rise	1360.00	1375.00	2015.00	1720.00	1450.00
1500mm overall diameter	...	M/Rise	1430.00	1630.00	2115.00	1915.00	1645.00

ROOFS		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Costs are based on flat plan area to overall dimensions with allowance for average pitches	...	Note					
For roof pitches of 20° and over ADD to roofing finishes :							

Pitch		% Increase
20°	-	6.41
25°	-	10.34
30°	-	15.47
35°	-	22.08
40°	-	30.54
45°	-	41.42
50°	-	55.57
55°	-	74.34
60°	-	100.00

CONCRETE ROOF CONSTRUCTION

Reinforced concrete flat plate slab average 200mm thick finished with :							
Three layer bituminous felt and loose quartz chip finish	...	sqm	392.00	368.00	396.00	349.00	412.00
Ditto with concrete tile finish	...	sqm	472.00	447.00	477.00	437.00	494.00
ADD extra for :							
Cement and sand screed, refer page 428	...	Note					
Insulation, refer page 403	...	Note					
Rainwater disposal, refer page 121	...	Note					
Parapet walls, refer page 121	...	Note					
ALTERNATIVES :							
Suspended slabs, comparative costs, refer page 112	...	Note					
Roofing materials and surface finishes, refer Detailed Prices pages 400 to 403	...	Note					

STEEL FRAMED ROOF CONSTRUCTION

Steelwork in roof comprising trusses/beams up to 10.0m clear span, purlins, insulation batts on mesh and finishing with :							
French pattern terracotta tiles	...	sqm	219.00	248.00	228.00	228.00	260.00
Swiss pattern terracotta tiles	...	sqm	223.00	259.00	244.00	236.00	270.00
Standard pattern concrete tiles	...	sqm	200.00	209.00	214.00	214.00	229.00
Steel ribbed decking -							
Zincalume® Steel	...	sqm	185.00	193.00	193.00	191.00	205.00
Colour coated	...	sqm	197.00	206.00	206.00	204.00	217.00
Aluminium ribbed decking	...	sqm	200.00	208.00	208.00	206.00	220.00
ADD extra for :							
Span 10.0m to 15.0m	...	sqm	18.40	18.40	18.00	18.80	18.00
Rainwater disposal, refer page 121	...	Note					
Eaves treatment, refer page 120	...	Note					
Deep fascias, refer page 120	...	Note					
ALTERNATIVES :							
Roofing materials, refer Detailed Prices, pages 397 to 400	...	Note					

ROOFS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TIMBER FRAMED ROOF CONSTRUCTION							
Timber framed roof up to 10.0m overall span comprising trusses, rafters, ceiling joists, etc., insulation batts on mesh, standard fascia and finishing with :							
French pattern terracotta tiles	…	sqm	176.00	259.00	186.00	230.00	237.00
Swiss pattern terracotta tiles	…	sqm	180.00	270.00	202.00	239.00	247.00
Standard pattern concrete tiles	…	sqm	158.00	220.00	172.00	217.00	206.00
Steel ribbed decking -							
Zincalume® Steel	…	sqm	151.00	205.00	160.00	196.00	188.00
Colour coated	…	sqm	163.00	217.00	173.00	208.00	201.00
Aluminium ribbed decking	…	sqm	166.00	220.00	175.00	211.00	203.00
ADD extra for :							
Rainwater disposal, refer page 121	...	Note					
Eaves treatment, refer page 120	...	Note					
Deep fascias, refer page 120	...	Note					
ALTERNATIVES :							
Roofing materials, refer Detailed Prices, pages 397 to 400	...	Note					
FASCIA TREATMENT							
Cost per square metre fascia area	...	Note					
Standard timber fascias included in foregoing roof costs	...	Note					
Steel truss framing clad with :							
Steel ribbed decking, colour coated, both sides	…	sqm	332.00	351.00	357.00	347.00	382.00
Fibre cement sheeting, painted, both sides	…	sqm	333.00	360.00	364.00	355.00	396.00
Precast G.R.C. panels with textured paint finish one side, colour coated steel decking the other side	…	sqm	757.00	756.00	783.00	763.00	798.00
EAVES SOFFIT TREATMENT							
Costs per square metre eaves soffit area	...	Note					
Timber framing sheeted on underside with :							
Colour coated steel ribbed decking	…	sqm	110.00	133.00	116.50	112.00	139.50
Fibre cement sheeting with paint finish	…	sqm	107.50	137.50	120.00	116.00	149.00
Timber battens	…	sqm	146.50	220.00	225.00	165.50	194.00
Water resistant plasterboard with paint finish	…	sqm	92.40	116.50	87.40	93.00	107.50
Precast G.R.C. panels with textured paint finish	…	sqm	529.00	532.00	537.00	523.00	548.00

ROOFS

RAINWATER DISPOSAL

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cost per square metre roof area, giving an indicative range only	...	Note					
Attention is drawn to the necessity to make additional allowance when the gutter/roof area ratio increases	...	Note					
For a more accurate assessment refer to ROOFING AND ROOF PLUMBING pages 410 to 413	...	Note					
Gutters, sumps, downpipes and internal stormwater drainage up to external face of building, all in standard materials, for building :							
1 - 2 storeys high - (Minimum)	...	sqm	30.40	32.80	33.30	32.30	41.10
- (Maximum)	...	sqm	39.50	42.30	42.80	41.70	52.00
3 - 5 storeys high - (Minimum)	...	sqm	47.00	50.40	51.50	49.70	62.70
- (Maximum)	...	sqm	63.80	67.90	68.90	67.10	82.50
6 - 10 storeys high - (Minimum)	...	sqm	74.70	79.90	81.70	78.80	98.70
- (Maximum)	...	sqm	104.30	110.50	112.50	109.50	133.00
11 - 20 storeys high - (Minimum)	...	sqm	130.00	138.50	142.00	137.00	171.00
- (Maximum)	...	sqm	185.00	196.00	199.00	194.00	235.00
21 - 30 storeys high - (Minimum)	...	sqm	185.50	197.50	202.50	195.50	243.00
- (Maximum)	...	sqm	266.00	281.00	286.00	278.00	336.00
ADD extra for :							
Rainwater disposal system in -							
Copper	...	+100/130%					
Stainless steel	...	+100/140%					
Rainwater disposal to hoods, balconies (not included in roof area)	...	Note					

PARAPET WALLS

Cost per square metre of parapet or balustrade	...	Note					
Brickwork in 270mm thick cavity parapet wall faced both sides and including capping (average height 1000mm)	...	sqm	383.00	297.00	321.00	273.00	369.00
Blockwork in 250mm thick cavity parapet wall in hollow blocks, including solid top course, and pointing both faces (average height 1000mm) :							
Natural colour blocks	...	sqm	265.00	234.00	260.00	227.00	325.00
Coloured blocks	...	sqm	293.00	263.00	268.00	242.00	-
Reinforced concrete in 150mm thick parapet wall	...	sqm	387.00	380.00	416.00	326.00	439.00
ADD extra for :							
Surface or applied finishes, cladding, refer page 125	...	Note					

ROOFS

AWNINGS/COVERED WAY ROOFS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Steel roof framing and purlins together with gutters and downpipes, painted finish to visible surfaces of soffit, framing etc., roof of :							
Steel ribbed decking -							
Zincalume® Steel	…	sqm	245.00	252.00	216.00	229.00	283.00
Colour coated one side	…	sqm	275.00	284.00	248.00	261.00	317.00
Timber roof framing and rafters together with gutters and downpipes, painted finish to visible surfaces of soffit, framing etc., roof of :							
Steel ribbed decking -							
Zincalume® Steel	…	sqm	216.00	257.00	225.00	249.00	267.00
Colour coated one side	…	sqm	247.00	289.00	257.00	281.00	300.00
French pattern terracotta tiles	…	sqm	227.00	302.00	237.00	275.00	298.00
Swiss pattern terracotta tiles	…	sqm	231.00	313.00	254.00	283.00	308.00
Standard pattern concrete tiles	…	sqm	209.00	263.00	223.00	261.00	267.00
ADD extra for :							
Columns, piers	...	Note					
Fascia, trusses and linings	...	Note					
Insulation and soffit linings	...	Note					

SPACEFRAMES

Cost per square metre flat surface area. This cost should be investigated in detail due to considerable cost variations of differing materials or systems	...	Note					
DECORATIVE TYPE							
Spaceframe clad with polycarbonate sheeting in either barrel vaults or domes, including all accessories, gutters, etc., framed in :							
Steel tube section	...	sqm	800.00-870.00	780.00-850.00	815.00-885.00	800.00-870.00	820.00-890.00
Aluminium tube section	...	sqm	920.00-970.00	900.00-950.00	940.00-1090.00	920.00-970.00	945.00-1095.00
ADD extra for :							
Columns, supporting structures	...	Note					
Spaceframe to a 1500mm module x 750mm deep, framed up in steel tubing and with a powdercoat polyester paint finish (cost range depending upon structural support system)	...	sqm	295.00-430.00	290.00-425.00	295.00-430.00	295.00-430.00	295.00-430.00
ADD extra for :							
Columns, supporting structures	...	Note					
In situ assembly on scaffold when crane and lift not feasible	...	Note					
Modules less than 1500mm	...	Note					
Cladding	...	Note					

ROOFS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SPACEFRAMES (cont'd)							
STRUCTURAL TYPE							
Spaceframe to a 2400mm module x 1300mm deep, framed up in cold formed channel chords with purlins at 2400mm centres, all galvanized finish (cost range depending upon span)	...	sqm	125.00-150.00	120.00-145.00	125.00-150.00	125.00-150.00	125.00-150.00
ADD extra for :							
Columns, supporting structures	...	Note					
In situ assembly on scaffold when crane lift not feasible	...	Note					
Modules less than 2400mm	...	Note					
Cladding	...	Note					
Cover plates	...	Note					
Paint finish	...	Note					
TENSION MEMBRANES							
The smaller the area the higher the cost	...	Note					
Barrel vaults, domed, conical or folded plate profile glazed with Teflon coated fibreglass membrane :							
Plan area	...	sqm	300.00-400.00	300.00-400.00	300.00-400.00	300.00-400.00	300.00-400.00
ADD extra for :							
Structural supports, wires, etc.	...	Note					
PERGOLAS							
Pergola comprising 150 x 50mm sawn joists at 900mm centres and 200 x 50mm sawn beams at 3600mm centres	...	sqm	39.00	68.90	44.40	46.40	49.80
ADD extra for :							
50 x 38mm battens at 75mm centres	…	sqm	72.20	122.50	67.30	80.80	137.00
Safety mesh	…	sqm	12.05	13.95	12.80	13.35	14.25
Shadecloth (70%)	…	sqm	7.10	8.05	8.15	7.90	9.50
Two coats acrylic timber finish on -							
Joists and beams	…	sqm	6.60	7.05	6.60	8.55	7.10
Battens at 75mm centres	…	sqm	27.80	28.90	28.40	32.40	24.30
BARREL VAULT DOMES AND CANOPIES							
Cost per square metre surface area	...	Note					
Barrel vault profile domed roof or canopy of steel framing glazed with polycarbonate sheeting :							
Up to 4000mm diameter (average)	…	sqm	780.00	765.00	780.00	775.00	780.00
4000/8000mm diameter (average)	…	sqm	705.00	690.00	710.00	705.00	710.00

ROOFS

CLEAR SPAN ROOFING SYSTEMS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Installed costs per square metre floor plan area	...	Note					
This cost should be investigated in detail due to the considerable cost variations of differing spans, heights, load conditions, systems or cladding materials	...	Note					
Clear span roofing system comprising proprietary stressed-arch framed roof structure including supporting columns and foundations, roof purlins, roof cladding, gutters and downpipes and engineering design by manufacturer/fabricator	...	Note					
LOW RISE STRUCTURE							
Suitable for clear spans from 60.0m to 120.0m	...	sqm	370.00-475.00	370.00-475.00	370.00-475.00	370.00-475.00	370.00-475.00
ADD extra for :							
Insulation	...	Note					
Rooflights	...	Note					
Roof ventilation	...	Note					
Transport to locations remote from capital cities or major provincial centres	...	Note					
All other elements, i.e. ground slab, external walls and windows, internal fit-out, finishes, services, etc.	...	Note					
Cost variations could be considerable due to span, wall height, roof rise, wind loadings, cladding materials and systems	...	Note					

EXTERNAL WALLS

IN SITU CONCRETE WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
25 MPa reinforced concrete wall formed in Class 4 formwork and reinforced at the rate of 100 kg/cum :							
150mm thick	...	sqm	381.00	374.00	410.00	321.00	434.00
200mm thick	...	sqm	413.00	402.00	440.00	350.00	468.00
250mm thick	...	sqm	445.00	430.00	470.00	378.00	501.00
300mm thick	...	sqm	478.00	458.00	501.00	407.00	535.00
ADD extra for :							
CONCRETE -							
32 MPa per 25mm wall thickness	...	sqm	0.27	0.15	0.30	0.21	0.24
40 MPa per 25mm wall thickness	...	sqm	0.56	0.44	0.59	0.50	0.56
White or off-white cement per 25mm wall thickness	...	sqm	-	5.05	2.50	1.40	-
Brightonlite cement per 25mm wall thickness	...	sqm	5.30	-	2.50	-	-
FORMWORK -							
To one face :							
Class 1	...	sqm	94.80-174.00	96.80-177.00	106.50-195.00	78.70-144.00	111.00-203.00
Class 2	...	sqm	31.60	32.20	35.50	26.20	37.00
Class 3	...	sqm	15.80	16.10	17.75	13.10	18.50
Boarded off-form	...	sqm	112.50	94.10	82.20	118.00	102.50
REINFORCEMENT -							
For every 10 kg/cum increase per 25mm wall thickness	...	sqm	0.68	0.63	0.65	0.62	0.65
SURFACE FINISHES -							
To one face of plain vertical surface :							
Bush hammered	...	sqm	69.00	59.60	69.20	64.00	75.80
Sand blasted	...	sqm	43.50	38.90	46.90	50.70	50.30
Water washed to expose aggregate including additional cost of better aggregate	...	sqm	32.30	26.00	33.20	35.20	31.70
Acid etched	...	sqm	49.90	37.90	45.20	48.50	46.50
APPLIED FINISHES -							
To one face :							
Cement render, two coats wood float	...	sqm	56.50	49.30	49.10	44.60	48.20
Hardwall plaster	...	sqm	58.60	57.70	54.80	28.40	52.60
Textured finishes -							
Preparatory treatment	...	sqm	4.20-21.00	4.20-21.00	4.20-26.30	4.20-21.00	4.20-26.30
Sprayed 2mm application with satin impact top coat (GranoSpray)	...	sqm	43.50	45.90	48.30	48.30	48.30
Refer Detailed Prices pages 419 to 421 for range of textured finishes	...	Note					
Seal and two coats acrylic paint	...	sqm	12.90	12.20	12.55	12.00	12.35

EXTERNAL WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
IN SITU CONCRETE WALLS (cont'd)							
ADD extra for :							
CLADDING -							
CLADDING SYSTEMS, METAL - Refer Detailed Prices pages 297 to 300	...	Note					
MARBLE, GRANITE - Refer Detailed Prices pages 272 and 274	...	Note					
PRECAST G.R.C. PANELS - Refer Detailed Prices page 255	...	Note					
PRECAST CONCRETE WALLS							
Wall panels with smooth finish both sides :							
150mm thick	...	sqm	320.00	320.00	335.00	320.00	316.00
175mm thick	...	sqm	345.00	353.00	370.00	343.00	367.00
200mm thick	...	sqm	365.00	386.00	405.00	366.00	398.00
ADD extra for :							
SURFACE FINISHES) As foregoing IN SITU) CONCRETE WALLS APPLIED FINISHES) where applicable	...	Note					
Hollowcore wall units, refer page 252	...	Note					
PRECAST AERATED CONCRETE WALL PANELS							
Refer Detailed Prices, page 254	...	Note					
GYPCRETE WALL PANELS							
Refer Detailed Prices, page 266	...	Note					
TILT-UP CONCRETE WALLS							
Refer Detailed Prices, page 254	...	Note					
SLIP FORM CONCRETE WALLS							
As a general rule, slip-forming is not usually considered for repetitive walls for heights below 15.0m (4 storeys). Above this level and up to 30.0m similar costs to in situ concrete walls would apply, whilst above 30.0m there may be some saving in forming costs.	...	Note					

EXTERNAL WALLS

BRICK WALLS

CLAY BRICKWORK

STANDARD BRICKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Wall :							
110mm thick	...	sqm	154.60	116.60	133.10	118.30	151.30
230mm thick	...	sqm	309.00	233.00	266.00	237.00	303.00
270mm cavity wall in two 110mm skins	...	sqm	312.00	236.00	268.00	239.00	305.00
390mm ditto in 110mm and 230mm skins	...	sqm	466.00	352.00	401.00	357.00	456.00
110mm thick (Jumbo brick)	...	sqm	-	-	115.00	-	136.40
230mm thick (Jumbo brick)	...	sqm	-	-	229.90	-	272.80
270mm cavity wall in two 110mm skins (Jumbo brick)	...	sqm	-	-	232.10	-	275.00
110mm thick (double brick)	...	sqm	-	81.40	-	-	128.70
230mm thick (double brick)	...	sqm	-	162.80	-	-	257.40
270mm cavity wall in two 110mm skins (double brick)	...	sqm	-	165.20	-	-	259.60
ADD extra for :							
FAIR FACE AND FACINGS -							
To one face :							
Bagged finish (single application)	...	sqm	9.45	6.10	8.85	6.50	9.25
Fair face and clean down to common wall	...	sqm	11.70	8.05	11.80	8.35	8.00
Facing bricks and pointing (from)	...	sqm	35.70	27.60	23.20	14.30	31.50
APPLIED FINISHES -							
To one face :							
Cement render, two coats wood float	...	sqm	56.50	49.30	49.10	44.60	48.20
Hardwall plaster	...	sqm	58.60	57.70	54.80	28.40	52.60
Textured finishes -							
Preparatory treatment	...	sqm	4.20-21.00	4.20-21.00	4.20-26.30	4.20-21.00	4.20-26.30
Sprayed 2mm application with satin impact top coat (GranoSpray)	...	sqm	43.50	45.90	48.30	48.30	48.30
Refer Detailed Prices pages 419 to 421 for range of textured finishes	...	Note					
Seal and two coats acrylic paint	...	sqm	12.90	12.20	12.55	12.00	12.35
CLADDING -							
CLADDING SYSTEMS, METAL - Refer Detailed Prices pages 297 to 300	...	Note					
MARBLE, GRANITE - Refer Detailed Prices pages 272 to 274	...	Note					
PRECAST G.R.C. PANELS - Refer Detailed Prices page 255	...	Note					

EXTERNAL WALLS

BRICK WALLS (cont'd)

CLAY BRICKWORK (cont'd)

MODULAR BRICKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Wall :							
90mm thick	...	sqm	-	-	-	90.20	124.30
230mm thick cavity wall in two 90mm skins	...	sqm	-	-	-	182.80	251.00
90mm thick (Longreach)	...	sqm	-	-	-	82.10	153.40
305mm thick (Longreach)	...	sqm	-	-	-	301.00	563.00
230mm cavity wall in two 90mm skins (Longreach)	...	sqm	-	-	-	166.50	309.00
90mm thick (Maxibrick)	...	sqm	-	-	-	55.10	99.90
305mm thick (Maxibrick)	...	sqm	-	-	-	173.30	314.00
230mm cavity wall in two 90mm skins (Maxibrick)	...	sqm	-	-	-	112.60	202.00
ADD extra for :							
FAIR FACE AND FACINGS							
To one face -							
Bagged finish (single application)	...	sqm	-	-	-	6.50	9.25
Fair face and clean down to common wall	...	sqm	-	-	-	8.35	8.00
Facing bricks and pointing (from)	...	sqm	-	-	-	-	23.10
APPLIED FINISHES) As foregoing							
CLADDING) CLAY BRICKWORK	...	Note					

CONCRETE MASONRY BRICKWORK

STANDARD BRICKS

Wall :							
110mm thick	...	sqm	151.30	-	-	-	153.40
230mm thick	...	sqm	303.00	-	-	-	307.00
270mm thick cavity wall in two 110mm skins	...	sqm	305.00	-	-	-	309.00
ADD extra for :							
FAIR FACE AND FACINGS -							
To one face :							
Fair face and clean down to common wall	...	sqm	11.70	-	-	-	8.00
Facing bricks and pointing (from)	...	sqm	-	-	-	-	-
APPLIED FINISHES) As foregoing							
CLADDING) CLAY BRICKWORK	...	Note					

MODULAR BRICKS

Wall :							
90mm thick	...	sqm	-	-	-	-	-
230mm thick cavity wall in two 90mm skins	...	sqm	-	-	-	-	-
ADD extra for :							
FAIR FACE AND FACINGS -							
To one face :							
Fair face and clean down to common wall	...	sqm	11.70	-	-	-	-
Facing bricks and pointing (from)	...	sqm	-	-	-	-	-
APPLIED FINISHES) As foregoing							
CLADDING) CLAY BRICKWORK	...	Note					

EXTERNAL WALLS

BLOCK WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Plain face concrete blocks, 200mm x 400mm in wall/skin :							
100mm thick -							
Hollow blocks :							
Natural colour	...	sqm	119.50	104.50	116.50	98.00	145.50
Coloured	...	sqm	134.00	118.50	121.00	105.50	-
Solid blocks :							
Natural colour	...	sqm	131.50	112.00	140.50	117.00	169.50
Coloured	...	sqm	150.50	127.00	150.50	133.50	-
150mm thick -							
Hollow blocks :							
Natural colour	...	sqm	145.50	118.00	131.50	122.50	167.50
Coloured	...	sqm	165.00	140.00	137.50	129.00	-
Solid blocks :							
Natural colour	...	sqm	-	-	-	159.50	-
Coloured	...	sqm	-	-	-	-	-
200mm thick -							
Hollow blocks :							
Natural colour	...	sqm	177.00	132.50	147.00	138.50	193.00
Coloured	...	sqm	194.30	166.90	154.90	152.30	-
250mm thick cavity wall, with two 100mm thick skins in 200mm high blocks -							
Hollow blocks :							
Each skin natural colour	...	sqm	-	-	-	-	-
Each skin coloured	...	sqm	270.00	240.00	244.00	213.00	-
One skin natural, one skin coloured	...	sqm	256.00	225.00	240.00	206.00	-
ADD extra for :							
Alternative block types, refer page 263	...	Note					
Pointing and cleaning down one face	...	sqm	8.25	10.20	9.80	9.90	12.45
APPLIED FINISHES -							
To one face :							
Cement render, two coats wood float	...	sqm	54.90	48.30	47.70	43.40	46.40
Hardwall plaster	...	sqm	57.60	56.70	53.00	26.70	52.60
Textured finishes -							
Preparatory treatment	...	sqm	4.20-21.00	4.20-21.00	4.20-26.30	4.20-21.00	4.20-26.30
Sprayed 2mm application with satin impact top coat (GranoSpray)	...	sqm	43.50	45.90	48.30	48.30	48.30
Refer Detailed Prices pages 419 to 421 for range of textured finishes	...	Note					
Seal and two coats acrylic paint	...	sqm	12.90	12.20	12.55	12.00	12.35

STUD FRAMED WALLS

FRAMING

Timber framed wall in :							
100 x 38mm framing, studs @ 450mm ctrs	...	sqm	54.50	65.20	56.80	56.40	76.90
75 x 50mm framing, studs @ 600mm ctrs	...	sqm	46.40	55.40	51.80	49.50	61.60
100 x 50mm framing, studs @ 600mm ctrs	...	sqm	56.30	68.00	59.40	62.40	74.30
Steel framed wall in :							
75 x 32mm studs @ 450mm ctrs	...	sqm	59.10	64.30	66.40	63.70	73.70
75 x 32mm studs @ 600mm ctrs	...	sqm	54.70	59.40	61.30	59.00	68.10

EXTERNAL WALLS

STUD FRAMED WALLS (cont'd)

FRAMING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for :							
INSULATION -							
430/530 double sided aluminium foil insulation	...	sqm	5.85	6.75	6.90	6.60	8.10
Glasswool batts :							
R1.5	...	sqm	13.10	15.25	15.10	14.85	17.85
R2.0	...	sqm	13.60	15.75	15.60	15.40	18.40
R2.5	...	sqm	14.10	16.25	16.10	15.90	18.85
Additional insulation materials - refer Detailed Prices pages 335 to 338	...	Note					
CLADDING (EXTERNAL) -							
To one face :							
FIBRE CEMENT -							
4.5mm thick flexible sheeting	...	sqm	34.20	39.80	40.90	38.90	48.60
6.0mm thick Stucco	...	sqm	59.00	66.40	68.70	65.10	79.10
7.5mm thick 230mm Woodgrain finish plank	...	sqm	60.10	68.40	70.40	67.00	82.00
Additional sheeting materials refer Detailed Prices pages 341 to 346	...	Note					
STEEL -							
0.42mm corrugated	...	sqm	43.90	45.60	49.00	44.80	54.70
0.42mm Monoclad, Trimdek®	...	sqm	43.90	45.60	49.00	44.80	54.70
0.42mm Longspan, Spandek®	...	sqm	49.50	51.30	54.60	50.40	60.30
0.55mm Mini Orb®	...	sqm	73.60	75.30	78.70	74.50	84.40
ADD extra for :							
Colour coating	...	sqm	11.70	11.70	11.70	11.70	11.70
Additional steel cladding materials refer Detailed Prices page 414	...	Note					
TIMBER, ETC. -							
25mm thick hardwood horizontal weather-boarding	...	sqm	94.60	88.00	91.20	107.50	118.50
25mm thick Western Red Cedar horizontal Shiplap panelling	...	sqm	158.00	147.50	171.00	128.00	204.50
ADD extra for :							
Acrylic, timber finish	...	sqm	9.15	9.70	9.15	11.80	9.80
APPLIED FINISHES -							
Applied finishes and painting of foregoing cladding materials, where applicable, refer relevant sections of Detailed Prices	...	Note					

EXTERNAL WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STUD FRAMED WALLS (cont'd)							
FRAMING (cont'd)							
ADD extra for (cont'd) :							
LININGS (INTERNAL)							
To one face :							
PLASTERBOARD -							
Plain plasterboard :							
1 x 10mm thick layer	…	sqm	36.80	42.00	32.00	34.60	35.40
1 x 13mm thick layer	…	sqm	38.90	42.00	33.10	36.20	36.50
1 x 10mm thick water resistant	…	sqm	41.00	43.00	34.10	38.90	37.80
Fire rated plasterboard :							
1 x 13mm thick layer	…	sqm	42.00	42.50	37.80	42.00	40.00
2 x 13mm thick layer	…	sqm	75.60	75.60	64.40	68.30	66.20
1 x 16mm thick layer	…	sqm	47.80	48.30	41.00	47.80	55.50
2 x 16mm thick layer	…	sqm	85.60	86.10	76.70	77.20	92.90
TIMBER, ETC. -							
Fibre cement, hardboard, particle board, plywood and other timber linings, refer Detailed Prices, pages 341 to 346	...	Note					
TILING -							
Mosaic tiles fixed with adhesive (P.C. $40.00/sqm supplied)	...	sqm	128.00	146.50	127.00	116.00	134.50
APPLIED FINISHES -							
Applied finishes and painting of foregoing cladding materials, where applicable, refer relevant sections of Detailed Prices	...	Note					
CLADDING SYSTEMS							
Cladding systems refer Detailed Prices pages 297 to 300	...	Note					
CURTAIN WALLING							
Curtain walling in clear anodised aluminium framing with 50% coloured glass spandrils and 50% clear glazed windows :							
Windows -							
Single glazed, fixed	…	sqm	780.00	815.00	795.00	745.00	805.00
Single glazed, 25% opening	…	sqm	815.00	840.00	825.00	780.00	840.00
Double glazed, fixed	…	sqm	885.00	920.00	895.00	855.00	900.00
Double glazed, 25% opening	…	sqm	910.00	945.00	925.00	880.00	935.00
ADD/ADJUST foregoing for :							
Spandril infill, alternatives, opening light %, refer Detailed Prices page 366	...	Note					
Glass alternatives, refer Detailed Prices pages 443 to 446	...	Note					

EXTERNAL WALLS

FRAMELESS GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer Detailed Prices page 367	...	Note					
SHOP FRONTS							
Clear anodised aluminium shop fronts glazed with 6mm toughened safety glass -							
Budget standard	...	sqm	395.00	420.00	430.00	410.00	453.00
Commercial standard	...	sqm	660.00	650.00	670.00	665.00	725.00
STALLBOARD							
110mm brick backing wall finished one face with :							
Face brick	...	sqm	204.00	155.00	168.50	143.50	196.50
Render and paint	...	sqm	265.00	198.50	224.00	213.00	249.00
Ceramic tiles	...	sqm	312.00	293.00	295.00	270.00	334.00
Polished marble, Travertine	...	sqm	850.00	875.00	820.00	950.00	915.00
COMPOSITE CONCRETE WALL CONSTRUCTION							
Composite concrete wall construction comprising plasterboard internal lining, 60mm eps panel, concrete with reinforcement, 60mm eps panel and reinforced textured render finish :							
100mm thick concrete, over all thickness 220mm	...	sqm	190.00	206.00	191.00	180.00	191.00
150mm thick concrete, over all thickness 270mm	...	sqm	210.00	226.00	211.00	200.00	211.00
200mm thick concrete, over all thickness 320mm	...	sqm	218.00	233.00	218.00	207.00	219.00
Composite concrete wall construction comprising plasterboard internal lining, 60mm eps panel, concrete with reinforcement, 6mm cfc board :							
112mm thick concrete, over all thickness 178mm	...	sqm	212.00	228.00	213.00	202.00	214.00
136mm thick concrete, over all thickness 202mm	...	sqm	231.00	246.00	231.00	220.00	232.00
262mm thick concrete, over all thickness 228mm	...	sqm	248.00	264.00	249.00	238.00	249.00

WINDOWS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
The following composite costs are for the installation complete including all accessories. NO adjustment has been made for the cost of the wall displaced by the window. All aluminium windows are standard commercial section - for domestic windows refer Detailed Prices, page 363	...	Note					

ALUMINIUM, COMMERCIAL

Clear anodised aluminium framing and clear float glazing in windows :							
Single glazed -							
Fixed	...	sqm	505.00	555.00	525.00	500.00	565.00
50% sliding	...	sqm	590.00	625.00	580.00	505.00	610.00
50% pivoted	...	sqm	685.00	735.00	665.00	650.00	695.00
Double glazed -							
Fixed	...	sqm	690.00	740.00	635.00	650.00	670.00
Opening	...	sqm	800.00	835.00	750.00	750.00	775.00
ADD/ADJUST foregoing for :							
Opening light alternatives, refer Detailed Prices, page 364	...	Note					
Glass alternatives, refer Detailed Prices, pages 443 to 446	...	Note					
Colour anodising	...	sqm	17.70	16.30	17.35	14.40	18.25
Insect screens, sliding	...	sqm	61.00	61.50	64.00	60.00	65.00
Venetian blinds -							
Micro	...	sqm	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00
Mini, Slimline	...	sqm	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00
Integral venetian blinds	...	sqm	178.00	180.00	183.00	177.00	187.00
Vertical blinds (range)	...	sqm	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00
Curtains and blinds generally, refer Detailed Prices, pages 384 to 386	...	Note					

TIMBER, DOMESTIC

Timber windows with clear float glazing and paint finish :							
Fixed lights	...	sqm	345.00	370.00	325.00	430.00	325.00
Casement or awning lights -							
25% opening	...	sqm	465.00	505.00	440.00	380.00	445.00
50% opening	...	sqm	490.00	550.00	505.00	450.00	535.00
Double hung sashes	...	sqm	630.00	670.00	595.00	695.00	650.00
ADD/ADJUST foregoing for :							
Opening light, alternatives, refer Detailed Prices, page 365	...	Note					
Glass alternatives, refer Detailed Prices, pages 443 to 446	...	Note					
Insect screens, hinged	...	sqm	55.00	55.50	59.00	55.00	60.00
Venetian blinds -							
Micro	...	sqm	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00
Mini, Slimline	...	sqm	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00
Roller blinds (range)	...	sqm	90.00-145.00	90.00-145.00	90.00-145.00	90.00-145.00	90.00-145.00
Vertical blinds (range)	...	sqm	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00
Curtain and blinds generally, refer Detailed Prices pages 384 to 386	...	Note					

WINDOWS

SUN CONTROL

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Blinds :							
Roller -							
Standard Holland	...	sqm	90.00-110.00	90.00-110.00	90.00-110.00	90.00-110.00	90.00-110.00
Felt heavy duty	...	sqm	115.00-145.00	115.00-145.00	115.00-145.00	115.00-145.00	115.00-145.00
Verosol reflective fabric	...	sqm	165.00-185.00	165.00-185.00	165.00-185.00	165.00-185.00	165.00-185.00
Venetian -							
Aluminium :							
Micro	...	sqm	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00
Mini, Slimline	...	sqm	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00
Timber, lacquer finish	...	sqm	350.00-450.00	350.00-450.00	350.00-450.00	350.00-450.00	350.00-450.00
Vertical, free hanging	...	sqm	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00
Cantilevered slab, 100mm thick reinforced in situ concrete slab with smooth off-form finish	...	sqm	225.00	209.00	228.00	185.00	240.00
ADD extra for :							
Surface finish	...	Note					
Curtains including track :							
Standard curtains -							
Cotton print and standard lining	...	sqm	118.50	118.50	120.00	118.50	121.00
Wool scrim unlined	...	sqm	88.00	86.00	89.50	88.00	90.00
Velvet, cotton pile, lined, mostly blackout	...	sqm	224.00	224.00	226.00	224.00	227.00
Additional types, refer pages 384 to 386	...	Note					
Solar control blinds :							
Manually operated	...	sqm	310.00-330.00	310.00-330.00	310.00-330.00	310.00-330.00	310.00-330.00
Motorised pleated shades	...	sqm	P.O.A.	P.O.A.	P.O.A.	P.O.A.	P.O.A.
Sunscreens, external and fully retractable, refer page 387	...	Note					
Louvres, colour finished fixed aluminium excluding supporting framing	...	sqm	360.00	360.00	360.00	360.00	360.00
Solar control film, refer page 447	...	Note					

INTERNAL WALLS

This section comprises the average costs of the various types of wall in general use followed by composite items of similar walls and standard finishes

IN SITU CONCRETE WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer External Walls page 125	...	Note					

PRECAST CONCRETE WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer External Walls page 126	...	Note					

GYPCRETE WALL PANELS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer Detailed Prices page 266	...	Note					

BRICK WALLS

CLAY BRICKWORK

Standard brickwork in wall :							
110mm thick	...	sqm	154.60	116.60	133.10	118.30	151.30
230mm thick	...	sqm	309.00	233.00	266.00	237.00	303.00
270mm cavity wall in two 110mm skins	...	sqm	312.00	236.00	268.00	239.00	305.00
110mm thick (Jumbo brick)	...	sqm	-	-	115.00	-	136.40
230mm thick (Jumbo brick)	...	sqm	-	-	229.90	-	272.80
270mm cavity wall in two 110mm skins (Jumbo brick)	...	sqm	-	-	232.10	-	275.00
110mm thick (double brick)	...	sqm	-	81.40	-	-	128.70
230mm thick (double brick)	...	sqm	-	162.80	-	-	257.40
270mm cavity wall in two 110mm skins (double brick)	...	sqm	-	165.20	-	-	259.60
Modular brickwork in wall :							
90mm thick	...	sqm	-	-	-	90.20	124.30
230mm thick cavity wall in two 90mm skins	...	sqm	-	-	-	182.80	251.00
90mm thick (Longreach)	...	sqm	-	-	-	82.10	153.40
305mm thick (Longreach)	...	sqm	-	-	-	301.00	563.00
230mm cavity wall in two 90mm skins (Longreach)	...	sqm	-	-	-	166.50	309.00
90mm thick (Maxibrick)	...	sqm	-	-	-	55.10	99.90
305mm thick (Maxibrick)	...	sqm	-	-	-	173.30	314.00
230mm cavity wall in two 90mm skins (Maxibrick)	...	sqm	-	-	-	112.60	202.00

CONCRETE MASONRY BRICKWORK

Standard brickwork in wall :							
110mm thick	...	sqm	151.30	-	-	-	153.40
230mm thick	...	sqm	303.00	-	-	-	307.00
270mm thick cavity wall in two 110mm skins	...	sqm	305.00	-	-	-	309.00
Modular brickwork in wall :							
90mm thick	...	sqm	-	-	-	-	-
230mm thick cavity wall in two 90mm skins	...	sqm	-	-	-	-	-

INTERNAL WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BLOCK WALLS							
Plain face natural colour hollow concrete blocks, 200mm high coursing, in wall :							
100mm thick	...	sqm	119.50	104.50	116.50	98.00	145.50
150mm thick	...	sqm	145.50	118.00	131.50	122.50	167.50
200mm thick	...	sqm	177.00	132.50	147.00	138.50	193.00
250mm thick cavity wall in two 100mm skins	...	sqm	242.00	211.00	235.00	198.50	293.00
Plain face natural colour solid concrete blocks, 200mm high coursing in wall :							
100mm thick	...	sqm	131.50	112.00	140.50	117.00	169.50
150mm thick	...	sqm	-	-	-	159.50	-
STUD FRAMED WALLS							
Refer pages 129 to 131	...	Note					
Framed partitions refer page 143	...	Note					
AERATED CONCRETE BLOCK WALLS							
Refer page 266	...	Note					
COMPOSITE COSTS							
CONCRETE WALLS							
150mm thick in situ reinforced concrete (25 MPa) wall formed in Class 4 formwork and reinforced at the rate of 100 kg/cum	...	sqm	381.00	374.00	410.00	321.00	434.00
ADD extra for :							
Each additional 10mm thickness	...	sqm	6.40	5.60	6.00	5.70	6.70
Class 2 formwork to one side	...	sqm	31.60	32.20	35.50	26.20	37.00
Class 3 formwork to one side	...	sqm	15.80	16.10	17.75	13.10	18.50
Hardwall plaster to one side	...	sqm	57.60	56.70	53.80	27.60	52.60
Render for tiling to one side	...	sqm	36.10	31.00	20.80	21.60	28.60
Tiles, 150 x 150mm, white glazed to one side	...	sqm	106.00	121.50	105.00	95.30	111.00
Seal and two coats acrylic paint to one side	...	sqm	12.90	12.20	12.55	12.00	12.35
Vinyl fabric, heavy duty commercial standard, to one side	...	sqm	67.40	60.30	66.30	68.00	62.70
PLASTERBOARD WALLS							
Lightweight intertenancy wall systems for apartment buildings providing a minimum FRL -/60/60 and Rw+Ctr=50dB (Boral IntRWall) :							
13mm plasterboard lining both sides, cavity one side, nominal width 177mm (System 50IW13S13)	...	sqm	206.00	203.00	200.00	200.00	204.00
13mm soundstop plasterboard lining both sides, cavity one side, nominal width 161mm (System 50IW13AS13A)	...	sqm	220.00	218.00	215.00	217.00	218.00
13mm soundstop plasterboard lining both sides, cavity both sides, nominal width 245mm (System 50IWS13AS13A)	...	sqm	273.00	269.00	266.00	269.00	268.00
13mm soundstop plasterboard lining both sides, cavity both sides, nominal width 251mm (System 25IWS13AS13A)	...	sqm	253.00	252.00	248.00	251.00	251.00

INTERNAL WALLS

COMPOSITE COSTS (cont'd)

PLASTERBOARD WALLS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Lightweight intertenancy wall systems for attached dwellings providing a minimum FRL 60/60/60 and Rw+Ctr=50dB (Boral PartiWALL) :							
10mm soundstop plasterboard lining both sides, R2.5 glasswool insulation both sides (System 25TP1010A)	...	sqm	251.00	251.00	246.00	250.00	248.00
13mm soundstop plasterboard lining both sides, R2.0 glasswool insulation both sides (System 25TP1313A)	...	sqm	257.00	257.00	251.00	255.00	254.00
2 x 10mm plasterboard lining both sides, R1.5 glasswool insulation both sides (System 25TP2020)	...	sqm	264.00	260.00	257.00	256.00	259.00

STANDARD BRICK WALLS

Clay common brick wall 110mm thick	...	sqm	154.60	116.60	133.10	118.30	151.30
ADD extra for :							
Each additional 110mm thickness	...	sqm	154.60	116.60	133.10	118.30	151.30
Fair face and facings to one side -							
Fair face and clean down	...	sqm	11.70	8.05	11.80	8.35	8.00
Facing bricks and pointing (from)	...	sqm	35.70	27.60	23.20	14.30	31.50
Applied finishes to one side -							
Hardwall plaster	...	sqm	56.60	55.70	52.00	25.70	51.60
Render for tiling	...	sqm	35.00	29.60	20.80	21.60	27.80
Tiles, 150 x 150mm white glazed	...	sqm	106.00	121.50	105.00	95.30	111.00
Seal and two coats acrylic paint	...	sqm	13.15	12.45	12.75	12.25	12.55
Vinyl fabric, commercial standard	...	sqm	67.40	60.30	66.30	68.00	62.70
Concrete masonry common brick wall 110mm thick	...	sqm	151.30	-	-	-	153.40
ADD extra for :							
Each additional 110mm thickness	...	sqm	151.30	-	-	-	153.40
165mm thick special performance brick	...	sqm	-	-	-	-	-
Fair face and facings to one side -							
Fair face and clean down	...	sqm	11.70	-	-	-	8.00
Facing bricks and pointing (from)	...	sqm	-	-	-	-	-
Applied finishes to one side -							
As foregoing Clay common brick wall 110mm thick	...	Note					

MODULAR BRICK WALLS

Clay common brick wall 90mm thick	...	sqm	-	-	-	90.20	124.30
ADD extra for :							
Fair face and facings to one side -							
Fair face and clean down	...	sqm	-	-	-	8.35	8.00
Facing bricks and pointing (from)	...	sqm	-	-	-	-	23.10
Applied finishes to one side -							
As foregoing Clay common brick wall 110mm thick	...	Note					

INTERNAL WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
COMPOSITE COSTS (cont'd)							
MODULAR BRICK WALLS (cont'd)							
Concrete masonry common brick wall							
90mm thick	...	sqm	-	-	-	-	-
ADD extra for :							
Fair face and facings to one side -							
Fair face and clean down	...	sqm	11.70	-	-	-	8.00
Facing bricks and pointing (from)	...	sqm	-	-	-	-	-
Applied finishes to one side -							
As foregoing Clay common brick wall							
110mm thick	...	Note					
BLOCK WALLS							
Plain face natural colour hollow concrete blocks 200 x 400mm in wall :							
100mm thick	…	sqm	119.50	104.50	116.50	98.00	145.50
150mm thick	…	sqm	145.50	118.00	131.50	122.50	167.50
200mm thick	…	sqm	177.00	132.50	147.00	138.50	193.00
ADD extra for :							
Pointing and cleaning down one side	...	sqm	8.25	10.20	9.80	9.90	12.45
Applied finishes to one side -							
As foregoing Clay common brick wall							
110mm thick	...	Note					
SCORIA BLEND BLOCKWORK							
Refer page 264	...	Note					

FIRE RESISTANCE LEVELS

For infomation on the fire resistance levels of walls refer to the structural adequacy charts and design manuals issued by the various manufacturers and the specific test results for the particular material. Once satisfied as to the particular wall required refer the foregoing INTERNAL WALLS and the subsection Composite Costs therein.	...	Note					

DOORS		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TIMBER DOORS							
Costs of doors and frames include for paint or clear polyurethene finish	...	Note					
HOLLOW CORE FLUSH DOORS							
2040 x 820mm standard hollow core door, prime coated hardboard faced, including fully welded zinc annealed frame, average quality mortice lock and furniture and paint finish	...	No	500.00	538.00	542.00	551.00	598.00
ADD extra for :							
Rebated timber frame and architraves	...	No	510.00	389.00	295.00	334.00	585.00
Each additional 100mm in height of door -							
Non-standard size, increase base cost of door	...	No	27.60	34.00	36.00	33.60	42.40
ADD per 100mm additional height	...	No	17.80	20.10	20.50	20.60	23.60
Edge strip all round	...	No	80.50	113.00	107.00	105.00	111.00
600 x 820mm fixed panel over	...	No	139.00	154.50	156.50	159.00	179.00
Top 750mm of door glazed	...	No	96.60	97.70	101.00	99.80	101.00
Waterproof plywood	...	No	23.00	23.00	44.00	(0.50)	44.00
Sliced veneer - Pacific maple	...	No	30.50	30.00	28.00	62.00	46.00
Hardware - High quality lock	...	No	150.00	150.00	150.00	150.00	150.00
For additional hardware, etc., refer Detailed Prices, pages 388 to 392	...	Note					
600 x 300mm air relief grille	...	No	178.50	187.50	196.50	189.50	207.00
2040 x 1640mm standard hollow core double door, prime coated hardboard faced and with rebated meeting stiles and edge strips all round, including fully welded zinc annealed frame, average quality lock and furniture and paint finish	...	No	1030.00	1180.00	1170.00	1180.00	1280.00
SOLID CORE FLUSH DOORS							
2040 x 820mm standard solid core door, prime coated hardboard faced, including fully welded zinc annealed frame, average quality mortice lock and furniture and paint finish	...	No	562.00	586.00	604.00	613.00	654.00
ADD extra for :							
Rebated timber frame and architraves	...	No	510.00	389.00	295.00	334.00	585.00
Each additional 100mm in height of door -							
Non standard size, increase base cost of door	...	No	40.00	43.60	48.40	46.00	53.60
ADD per 100mm additional height	...	No	21.90	23.20	24.60	24.60	27.20

DOORS

TIMBER DOORS (cont'd)

SOLID CORE FLUSH DOORS (cont'd)

ADD extra for (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Edge strip all round	...	No	76.30	112.90	106.90	105.10	111.20
600 x 820mm fixed panel over	...	No	164.00	174.00	182.00	184.00	201.00
Plywood for paint	...	No	39.00	31.00	19.00	31.00	38.00
Sliced veneer - Pacific maple	...	No	50.00	42.00	29.00	45.00	49.00
Hardware - High quality lock	...	No	150.00	150.00	150.00	150.00	150.00
For additional hardware, etc., refer Detailed Prices, pages 388 to 392	...	Note					
600 x 300mm air relief grille	...	No	180.50	189.50	198.50	192.00	209.00
2040 x 1640mm standard solid core double door, prime coated hardboard faced and with rebated meeting stiles, including fully welded zinc annealed frame, average quality lock and furniture and paint finish	...	No	1145.00	1280.00	1290.00	1305.00	1390.00

ACOUSTIC DOORS

Refer pages 374 and 375 ... Note

W.C. CUBICLE DOORS

1800 x 600mm standard hollow core door including terrazzo set and paint finish	...	No	357.00	412.00	419.00	410.00	486.00

PANELLED DOORS

2040 x 820mm four panel door with applied mouldings including rebated timber frame and architraves, average quality mortice lock and furniture and paint finish	...	No	1210.00	1165.00	1045.00	1105.00	1420.00

FRAMED, LEDGED AND BRACED DOORS

2040 x 820mm framed, ledged and braced door with T & G V-jointed vertical boarding including timber frame, average quality mortice lock and furniture and paint finish	...	No	1335.00	1240.00	1230.00	1255.00	1435.00

LOUVRED DOORS

Fully louvred door including timber frame and architraves, average quality mortice latch and furniture and paint finish, door size :

2040 x 720mm	...	No	610.00	665.00	645.00	655.00	765.00
2040 x 820mm	...	No	635.00	705.00	665.00	680.00	790.00

GLAZED TIMBER DOORS

2040 x 820mm glazed door including timber frame and architraves, average quality mortice lock and furniture and paint finish :

Single panel door glazed with clear safety glass	...	No	1155.00	1120.00	1030.00	1070.00	1340.00
Two panel door glazed with clear safety glass	...	No	1195.00	1060.00	870.00	1025.00	1355.00

DOORS

METAL DOORS

GLAZED ALUMINIUM ENTRY DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Clear anodised aluminium glazed entry doors with 6mm toughened safety glass :							
2100 x 900mm single, hinged	…	No	2650.00	2600.00	2660.00	2760.00	2785.00
ADD for automatic opening	…	No	8800.00	8800.00	8800.00	8800.00	8800.00
2100 x 1200mm single, sliding	…	No	3450.00	3500.00	3400.00	3500.00	3535.00
ADD for automatic opening	…	No	7000.00	7000.00	7000.00	7000.00	7000.00
2100 x 1800mm double, hinged	…	No	4200.00	4200.00	4070.00	3940.00	4405.00
ADD for automatic opening	…	No	9300.00	9300.00	9300.00	9300.00	9300.00
2100 x 1800mm double, sliding	…	No	4500.00	4600.00	4690.00	4440.00	4915.00
ADD for automatic opening	…	No	7650.00	7650.00	7650.00	7650.00	7650.00

REVOLVING DOORS

Refer pages 376 and 377	...	Note					

ROLLER DOORS AND GRILLES

DOMESTIC/COMMERCIAL

Ribbed roller door (continuous curtain) :							
Zincanneal® Steel	…	sqm	267.00	274.00	272.00	273.00	280.00
Pre-enamelled finish	…	sqm	280.00	286.00	284.00	285.00	292.00
ADD extra for :							
Electrically operated opening mechanism - refer page 378	...	Note					

INDUSTRIAL

Continuous curtain roller door :							
Zincanneal® Steel	…	sqm	300.00-420.00	310.00-430.00	310.00-430.00	310.00-430.00	320.00-445.00
Pre-enamelled or in situ paint finish	…	sqm	330.00-450.00	340.00-465.00	335.00-460.00	340.00-460.00	350.00-475.00
ADD extra for :							
Electrically operated opening mechanism	...	Sum	1390.00	1475.00	1455.00	1460.00	1560.00
Interlocking slat roller door :							
Zincanneal® Steel	…	sqm	325.00-415.00	335.00-425.00	335.00-425.00	335.00-425.00	350.00-440.00
Pre-enamelled or in situ paint finish	…	sqm	375.00-465.00	390.00-480.00	385.00-475.00	385.00-475.00	400.00-490.00
ADD extra for :							
Electrically operated opening mechanism	...	Sum	1230.00	1275.00	1260.00	1265.00	1315.00

DECORATIVE

Roller grille of natural anodised aluminium, including locks and manual control :							
3 .0sqm grille	…	sqm	520.00-625.00	520.00-625.00	520.00-625.00	520.00-625.00	520.00-625.00
6 .0sqm grille	…	sqm	435.00-540.00	435.00-540.00	435.00-540.00	435.00-540.00	435.00-540.00
12 .0sqm grille	…	sqm	420.00-525.00	420.00-525.00	420.00-525.00	420.00-525.00	420.00-525.00

DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
METAL DOORS (cont'd)							
SECTIONAL OVERHEAD DOOR							
Ribbed sectional panel door :							
Colour coated finish	…	sqm	325.00	340.00	335.00	340.00	360.00
Translucent fibreglass	…	sqm	700.00	720.00	715.00	715.00	735.00
ADD extra for :							
Electrically operated opening mechanism -							
Domestic	…	Sum	895.00	955.00	940.00	945.00	1015.00
Industrial	…	Sum	2105.00	2165.00	2150.00	2155.00	2225.00
TRANSPARENT INDUSTRIAL SWING DOORS							
See through PVC sheet door, self closing and sealing including steel frame :							
Medium duty -							
Single door	…	sqm	830.00	865.00	855.00	860.00	900.00
Double doors	…	sqm	815.00	845.00	835.00	840.00	875.00
Heavy duty -							
Single door	…	sqm	1170.00	1220.00	1205.00	1210.00	1270.00
Double doors	…	sqm	565.00	585.00	580.00	585.00	605.00
ADD extra for :							
Mounting plates to frame	...	m	70.00	70.00	70.00	70.00	70.00
DISABLED PERSONS DOOR							
2000 x 920mm steel flush finish door, including frame, hardware and paint finish	...	No	3500.00	3500.00	3600.00	3550.00	3500.00
ADD extra for :							
Automatic release button fully installed	...	No	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00
DOMESTIC SECURITY DOORS							
Anodised aluminium security door 2040 x 820mm with patterned security grille, insect screen mesh, hinges and lock	...	No	385.00	390.00	390.00	390.00	390.00
FIRE DOORS							
2040 x 820mm fire rated door including steel frame, average quality hardware and paint finish :							
One hour rated	…	No	1400.00	1415.00	1350.00	1380.00	1415.00
Two hour rated	…	No	1540.00	1555.00	1575.00	1505.00	1550.00
Three hour rated	…	No	1655.00	1675.00	1720.00	1655.00	1680.00
2040 x 1640mm double ditto :							
One hour rated	…	No	3720.00	3070.00	3530.00	3770.00	3740.00
Two hour rated	…	No	4020.00	2350.00	4050.00	4200.00	4090.00
Three hour rated	…	No	4360.00	3550.00	4170.00	4390.00	4200.00
Sliding fire doors, refer page 382	...	Note					
ACCORDION DOORS							
Refer page 383	...	Note					
OPERABLE WALLS							
Refer pages 144 and 383	...	Note					

INTERNAL PARTITIONS AND SCREENS		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FRAMED PARTITIONS							
Partitions in 64mm steel stud framing lined both sides with :							
10mm plasterboard and painted	…	sqm	145.00	154.00	131.00	135.00	146.00
10mm plasterboard and vinyl fabric	…	sqm	185.00	190.00	171.00	177.00	182.00
One hour fire rated plasterboard and painted	…	sqm	168.00	167.00	149.00	159.00	188.00
One hour fire rated plasterboard and vinyl fabric	…	sqm	208.00	203.00	189.00	201.00	224.00
Two hour fire rated plasterboard (2x13mm layers to both sides) and painted	…	sqm	227.00	225.00	199.00	202.00	210.00
Two hour fire rated plasterboard (2x13mm layers to both sides) and vinyl fabric	…	sqm	266.00	261.00	239.00	244.00	246.00
MODULAR SEMI-DEMOUNTABLE PARTITIONS							
Costs based on overall area including openings	...	Note					
Partition in 1200mm wide panels, full height 2400/2700mm, framed and lined both sides with :							
13mm plasterboard and painted	…	sqm	137.00	141.00	142.00	133.00	141.00
13mm plasterboard and Flexiclad vinyl	…	sqm	134.00	138.00	140.00	132.00	138.00
One hour fire rated plasterboard and painted	…	sqm	160.00	154.00	161.00	157.00	183.00
One hour fire rated plasterboard and Flexiclad vinyl	…	sqm	157.00	151.00	158.00	156.00	180.00
ADD extra for :							
Partition to door height with capping	...	sqm	19.00	13.50	18.50	27.00	5.50
Standard door with lock and furniture -							
Hollow core (average), painted	…	No	595.00	635.00	630.00	655.00	695.00
Solid core (average), painted	…	No	655.00	685.00	695.00	715.00	750.00
MODULAR DEMOUNTABLE PARTITIONS							
Costs based on overall area including openings	...	Note					
Partition in 1200mm wide panels, full height 2400/2700mm, solid infill, faced both sides with :							
'Flexiclad' vinyl	…	sqm	158.00	155.00	167.00	166.00	154.00
Vinyl fabric	…	sqm	221.00	214.00	230.00	230.00	214.00
Veneered plywood	…	sqm	224.00	223.00	231.00	243.00	223.00
ADD extra for :							
Partition to door height with capping	...	sqm	5.00	8.50	4.00	22.50	9.50
Standard door with lock and furniture -							
Hollow core (average), painted	…	No	595.00	635.00	630.00	655.00	695.00
Solid core (average), painted	…	No	655.00	685.00	695.00	715.00	750.00
MODULAR SCREEN PARTITIONS							
Rigid screen panels 70mm thick x 1350 to 1850mm high faced both sides with front runner fabric :							
Straight panel	…	sqm	245.00	220.00	235.00	235.00	250.00
Radiused panel	…	sqm	615.00	560.00	600.00	600.00	625.00

INTERNAL PARTITIONS AND SCREENS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GLAZED SCREENS							
Clear anodised aluminium screens glazed with :							
6mm toughened glass	…	sqm	565.00	575.00	580.00	605.00	630.00
10mm laminated glass	…	sqm	650.00	600.00	630.00	640.00	660.00
Timber framed screens, clear finish, glazed with :							
6mm toughened glass	…	sqm	605.00	595.00	595.00	665.00	710.00
10mm laminated glass	…	sqm	630.00	620.00	620.00	690.00	655.00
OPERABLE WALLS							
Acoustically insulated (STC-45) operable screen manually operated with linen backed vinyl finish, complete with jambs, overhead track, hardware, etc. :							
10 to 25sqm wall	...	sqm	1650.00	1650.00	1650.00	1650.00	1650.00
25 to 50sqm wall	...	sqm	1355.00	1355.00	1355.00	1355.00	1355.00
Exceeding 50sqm wall	...	sqm	985.00	985.00	985.00	985.00	985.00
Glazed operable wall, refer page 383	...	Note					
CHAIN WIRE PARTITIONS							
Galvanised chain wire partitions excluding gates	...	sqm	38.70	39.40	36.10	33.90	38.00
TOILET SCREENS							
COMPRESSED CELLULAR FIBREBOARD 18mm thick including paint finish, aluminium channels and fixings in :							
Screen	…	sqm	340.00	375.00	375.00	345.00	380.00
Jamb or end x 300mm wide	…	m	211.00	213.00	206.00	198.00	199.00
HMR PARTICLE BOARD 25mm thick faced both sides with plastic laminate and including aluminium channels and fixings in :							
Screen	…	sqm	270.00	270.00	260.00	265.00	275.00
Pilaster or end post x 300mm wide	…	m	273.00	297.00	276.00	273.00	291.00
PRECAST TERRAZZO 35mm thick including anodised aluminium plated capping and fixings in :							
Screen	…	sqm	800.00	610.00	-	-	600.00
Jamb or end x 300mm wide	…	m	504.00	204.00	-	-	212.00
BRICK wall, 110mm thick finished both sides with :							
Plaster and paint	…	sqm	295.00	254.00	264.00	195.00	281.00
Plaster and paint 50%, tiled 50%	…	sqm	367.00	337.00	325.00	275.00	356.00
Fully tiled	…	sqm	439.00	421.00	387.00	354.00	431.00

WALL FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CONCRETE SURFACE FINISHES							
Bush hammer	…	sqm	69.00	59.60	69.20	64.00	75.80
Sand blast	…	sqm	43.50	38.90	46.90	50.70	50.30
Water wash to expose aggregate including additional cost of better aggregate	...	sqm	32.30	26.00	33.20	35.20	31.70
Water wash only (no special finish)	...	sqm	11.80	12.50	11.00	10.10	10.80
Acid etch	...	sqm	49.90	37.90	45.20	48.50	46.50
MARBLE AND GRANITE							
GRANITE							
South Australian imperial black, 30mm thick :							
Sawn face	…	sqm	625.00	615.00	550.00	620.00	565.00
Polished finish	…	sqm	780.00	735.00	690.00	750.00	710.00
MARBLE							
Wombeyan facing, 19mm thick :							
Sawn face	…	sqm	415.00	395.00	395.00	-	420.00
Filled and polished finish	…	sqm	495.00	475.00	475.00	-	510.00
Sicilian pearl facing, 19mm thick :							
Sawn face	…	sqm	520.00	560.00	480.00	560.00	560.00
Filled and polished finish	…	sqm	600.00	640.00	565.00	655.00	650.00
Travertine facing, 19mm thick :							
Sawn face	…	sqm	465.00	520.00	455.00	560.00	510.00
Filled and polished finish	…	sqm	545.00	600.00	540.00	655.00	600.00
Travertine tiles, 300 x 300/600 x 300 x 10mm thick, pre-polished	...	sqm	270.00	275.00	270.00	270.00	285.00
TIMBER							
FIBREBOARD							
16mm thick medium density fibreboard with :							
Gloss acrylic paint finish	…	sqm	51.10	57.80	62.30	64.30	73.30
Melamine, white both sides	…	sqm	56.40	64.00	65.80	64.50	77.70
Melamine, plain colours both sides	…	sqm	50.90	58.50	60.30	59.00	72.20
PARTICLE BOARD							
16mm thick standard grade faced with :							
Melamine, satin white	...	sqm	50.50	59.20	61.10	57.80	73.10
Timber veneer and clear finish	...	sqm	72.00-145.00	82.00-151.00	82.00-151.00	82.00-159.00	94.00-163.00
PLYWOOD							
6mm thick and gloss acrylic paint finish	...	sqm	47.70	54.20	58.90	59.80	69.10
6mm thick veneered and clear finish	...	sqm	75.60	82.90	83.00	83.10	91.60
CANE-ITE HARDBOARD PLYWOOD TIMBER BOARDING ETC.	) Refer Detailed Prices)) pages 341 to 346))	...	Note				

WALL FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PLASTER AND RENDER							
Costs include average allowance for narrow widths and labours	...	Note					
CEMENT RENDER							
Wood float finish -							
One coat	...	sqm	41.60	42.90	32.40	33.50	29.70
Two coats	...	sqm	55.70	48.80	48.40	44.00	47.30
Steel float finish -							
One coat	...	sqm	42.90	43.70	34.00	36.00	30.60
Two coats	...	sqm	57.20	50.10	53.90	41.80	47.30
HARDWALL PLASTER							
Two coats	...	sqm	58.10	57.20	53.90	27.60	52.60
ADD extra for :							
Painting -							
Seal and two coats acrylic latex paint	...	sqm	12.15	11.80	11.80	11.30	12.20
Seal and two coats acrylic semi-gloss paint	...	sqm	12.90	12.20	12.55	12.00	12.35
Seal and two coats alkyd enamel paint	...	sqm	13.15	14.20	13.05	12.50	12.55
PLASTERBOARD							
Plasterboard fixed to steel or timber framing - refer page 131	...	Note					
10mm thick plasterboard fixed to solid wall with -							
Metal furring	...	sqm	66.70	72.30	55.00	64.80	58.40
Alignment strips and plaster daubs	...	sqm	52.50	57.80	46.60	50.10	53.30
50 x 38mm battens	...	sqm	63.50	61.80	49.00	53.00	53.30
ADD extra for :							
Painting -							
Seal and two coats acrylic latex paint	...	sqm	11.85	11.50	11.55	11.05	12.00
Seal and two coats acrylic semi gloss paint	...	sqm	12.60	11.90	12.30	11.75	12.15
Seal and two coats alkyd enamel paint	...	sqm	12.75	13.65	12.75	12.10	12.15
TEXTURED FINISHES							
Render (pre-blended) skimcoat to masonry substrates	...	sqm	15.80-21.00	15.80-21.00	21.00-26.30	15.80-21.00	21.00-26.30
Skimcoat with mesh to polystyrene base sheet (GranoRender PM and GranoSonge Fine Acrylic)	...	sqm	28.40	29.90	31.50	31.50	31.50
Acrylic texture coating, lambswool mitt or brush applied on prepared substrate, GranoSahara	...	sqm	15.10	16.00	16.80	16.00	16.80
Acrylic texture coating, lambswool mitt or brush applied on prepared substrate, GranoSorrento	...	sqm	20.80	22.00	23.10	23.10	23.10
Acrylic texture coating on prepared substrate with impact topcoat, roller applied, GranoRoll-on Medium Impact	...	sqm	23.60	24.90	26.30	26.30	26.30

WALL FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TEXTURED FINISHES (cont'd)							
Acrylic texture coating on prepared substrate with impact topcoat, roller applied, GranoRoll-on Smooth Impact	…	sqm	20.80	22.00	23.10	23.10	23.10
Acrylic texture coating on prepared substrate with elastomeric top coat, trowel applied, GranoMarble Membrane	…	sqm	33.20	35.00	36.90	36.90	36.90
Acrylic texture coating, 2mm aggregate, sprayed on prepared substrate with impact topcoat, GranoSpray-on 2mm Impact	…	sqm	43.50	45.90	48.30	48.30	48.30
TILE AND SHEET							
VINYL							
Vinyl sheet 1.50mm thick (P.C. $20.00/sqm supplied)	...	sqm	57.40	59.70	58.20	60.40	66.10
CERAMIC							
Tiling in main areas, fixed with adhesive :							
150 x 150mm white glazed (P.C. $29.00/sqm supplied)	…	sqm	106.00	121.50	105.00	95.30	111.00
150 x 150mm plain colour glazed (P.C. $30.00/sqm supplied)	…	sqm	107.00	122.50	106.00	96.50	112.00
200 x 100mm white and plain colour glazed (P.C. $40.00/sqm supplied)	…	sqm	118.00	133.50	117.00	107.50	123.00
200 x 200mm white and plain colour glazed (P.C. $40.00/sqm supplied)	…	sqm	118.00	133.50	117.00	107.50	123.00
MOSAIC							
61 x 61mm (P.C. $40.00/sqm supplied) in main areas, fixed with adhesive	...	sqm	128.00	146.50	127.00	116.00	134.50
ADD extra for :							
Render for tiling	...	sqm	35.50	30.30	20.80	21.60	28.20
PAPER HANGING							
WALLPAPER							
Prepare and hang paper (P.C. $7.00/sqm supplied)	...	sqm	22.40	20.50	21.40	20.90	20.20
FABRIC							
Prepare and hang :							
Vinyl, paper backed (P.C. $18.00/sqm supplied)	...	sqm	31.70	29.60	31.60	31.80	30.00
Vinyl fabric -							
Heavy duty (commercial standard) (P.C. $25.00/sqm supplied)	...	sqm	61.30	54.80	60.30	61.80	57.00
Extra heavy duty (P.C. $33.00/sqm supplied)	...	sqm	74.00	67.80	73.70	74.40	69.60

WALL FINISHES

PAINTING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Seal and two coats acrylic paint on :							
Plaster or similar smooth surface	...	sqm	12.60	11.90	12.30	11.75	12.15
Fair face masonry or similar rough surface	...	sqm	13.15	12.45	12.75	12.25	12.55
Seal and two coats acrylic latex paint on :							
Plaster or similar smooth surface	...	sqm	11.85	11.50	11.55	11.05	12.00
Fair face masonry or similar rough surface	...	sqm	12.40	12.05	12.00	11.55	12.40
Seal and two coats alkyd enamel paint on :							
Plaster or similar smooth surface	...	sqm	12.75	13.65	12.75	12.10	12.15
Fair face masonry or similar rough surface	...	sqm	13.50	14.70	13.35	12.85	12.90
Seal and three coats alkyd enamel paint on :							
Plaster or similar smooth surface	...	sqm	16.80	17.60	16.40	15.70	16.15
Fair face masonry or similar rough surface	...	sqm	17.55	18.65	17.00	16.45	16.90
Two coats clear polyurethane on plywood or similar surface	...	sqm	9.45	10.95	9.35	11.25	9.15

FLOOR FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
IN SITU CONCRETE, GRANOLITHIC							
SURFACE FINISHES							
Broomed	...	sqm	6.00	7.55	7.40	4.85	6.40
Power float monolithic	...	sqm	8.15	7.55	8.25	6.15	6.60
Steel trowel	...	sqm	8.25	7.55	8.25	7.20	6.80
Wood float	...	sqm	8.25	7.00	7.90	6.55	6.65
Water wash to expose aggregate including additional cost of better aggregate	...	sqm	32.30	26.00	33.20	35.20	31.70
Water wash only (no special finish)	...	sqm	11.80	12.50	11.00	10.10	10.80
GRANOLITHIC							
Granolithic with steel trowel finish, large areas, including labours :							
19mm thick	...	sqm	37.10	36.00	37.30	35.30	36.80
25mm thick	...	sqm	41.00	46.40	36.10	39.40	46.40
38mm thick	...	sqm	44.80	59.40	59.90	43.80	48.90
50mm thick	...	sqm	47.90	62.60	64.00	47.50	56.40
MARBLE AND GRANITE							
MARBLE							
Wombeyan tiles, 450 x 300 x 10mm :							
Filled and honed finish	...	sqm	324.00	276.00	260.00	-	265.00
Filled and polished finish	...	sqm	354.00	306.00	294.00	-	279.00
Sicilian pearl paving, 19mm thick :							
Sawn finish	...	sqm	495.00	520.00	460.00	490.00	530.00
Filled and polished finish	...	sqm	575.00	600.00	545.00	585.00	620.00
Travertine paving, 19mm thick :							
Sawn finish	...	sqm	440.00	460.00	405.00	440.00	470.00
Filled and polished finish	...	sqm	520.00	540.00	490.00	535.00	560.00
Travertine tiles, 300 x 300/600 x 300 x 10mm thick, pre-polished	...	sqm	270.00	275.00	270.00	270.00	285.00
GRANITE							
South Australian imperial black, 30mm thick :							
Sawn face	...	sqm	560.00	465.00	425.00	550.00	455.00
Polished finish	...	sqm	715.00	585.00	565.00	680.00	600.00
SLATE							
Flag paving, 10mm thick, sawn finish :							
Chinese or Indian	...	sqm	90.00-110.00	95.00-115.00	95.00-115.00	100.00-120.00	95.00-115.00
BRICK AND BLOCK							
Refer page 223	...	Note					
TERRAZZO							
IN SITU, 37mm thick terrazzo paving	...	sqm	605.00	535.00	460.00	-	460.00
PRECAST, 300 x 300 x 25mm thick tiles	...	sqm	310.00	300.00	290.00	300.00	295.00

FLOOR FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TIMBER							
HARDBOARD - 5.5mm underlay nailed and glued	...	sqm	36.60	40.10	40.80	39.50	45.60
PARTICLE BOARD - 22mm panels with T and G edges	...	sqm	35.10	38.90	39.60	38.20	44.70
PARQUETRY, BLOCK - Wood block parquetry including two or three coats of finish (Average)	...	sqm	226.00	264.00	252.00	236.00	244.00
PARQUETRY, MOSAIC - Mosaic parquetry including two or three coats of finish (Average)	...	sqm	352.00	383.00	-	378.00	394.00
PLYWOOD - 18.5mm panels with T and G edges	...	sqm	39.70	40.30	41.10	42.10	46.10
TONGUED AND GROOVED							
25mm T and G flooring (standard grade) :							
Hardwood	...	sqm	132.00	105.00	-	103.50	146.50
Radiata pine	...	sqm	69.50	76.90	80.20	77.20	86.10
ADD extra for :							
Specially selected prime quality flooring (range)	...	sqm	30.00-40.00	30.00-40.00	30.00-40.00	35.00-45.00	35.00-45.00
Three coats clear polyurethane finish	...	sqm	33.30	27.80	40.40	32.20	29.30
Seal and wax polish	...	sqm	33.30	27.80	40.40	32.20	29.30
CARPET							
Commercial grade carpet (average standard) :							
Nylon -							
Medium use ($17.00/sqm supplied)	...	sqm	30.30	31.10	30.60	31.40	33.40
Heavy use ($30.00/sqm supplied)	...	sqm	44.00	44.80	44.20	45.00	47.10
Polypropylene, direct stick, general range (P.C. $10.00 to $20.00/sqm supplied)	...	sqm	23.00-33.50	23.80-34.30	23.20-33.70	24.00-34.50	26.10-36.60
Wool tufted -							
Medium use ($30.00/sqm supplied)	...	sqm	44.00	44.80	44.20	45.00	47.10
Heavy use ($43.00/sqm supplied)	...	sqm	57.60	58.40	57.90	58.70	60.70
Wool woven carpet -							
Medium use ($70.00/sqm supplied)	...	sqm	86.00	86.80	86.20	87.00	89.10
Heavy use ($90.00/sqm supplied)	...	sqm	107.00	108.00	107.00	108.00	110.00
ADD extra for :							
Rubber underlay	...	sqm	24.60	25.40	25.00	25.80	26.90
Commercial grade modular carpet :							
Refer pages 435 and 436 for a range of modular carpet	...	Note					
Cost range	...	sqm	63.50-84.50	64.20-85.20	63.70-84.70	64.40-85.40	66.20-87.20

FLOOR FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TILES, NON-RESILIENT							
Refer pages 433 to 434 for additional tiles	...	Note					
MOSAIC							
61 x 61mm tiles (P.C. $40.00/sqm supplied)	...	sqm	128.00	146.50	127.00	116.00	134.50
100 x 100mm tiles (P.C. $40.00/sqm supplied)	...	sqm	125.00	142.00	123.50	113.50	130.50
QUARRY							
250 x 250mm tiles (P.C. $33.00/sqm supplied)	...	sqm	110.50	126.00	109.00	99.80	115.50
300 x 300mm tiles (P.C. $38.00/sqm supplied)	...	sqm	114.00	129.50	113.00	104.00	119.00
STEEL							
305 x 305 x 3mm steel tiles :							
Black	...	sqm	271.00	285.00	266.00	262.00	274.00
Galvanised	...	sqm	251.00	265.00	246.00	242.00	254.00
TILE AND SHEET, RESILIENT							
Refer pages 436 to 441 for additional floor finishes	...	Note					
CORK TILES							
6.0mm thick tiles with polyurethene finish (P.C. $35.00/sqm supplied)	...	sqm	102.50	107.00	104.00	108.00	119.00
LINOLEUM SHEET							
Heavy duty 2.5mm thick sheet (P.C. $40.00/sqm supplied)	...	sqm	87.90	90.80	88.80	91.80	99.10
RUBBER TILES							
Studded surface, standard colours :							
Medium traffic -							
2.7mm thick (P.C. $50.00 sqm/supplied)	...	sqm	95.80	98.50	96.70	99.40	106.00
Heavy duty -							
3.0mm thick (P.C. $58.00 sqm/supplied)	...	sqm	107.00	109.50	108.00	110.50	118.00
VINYL TILES							
Semi rigid :							
2.0mm thick (P.C. $10.00sqm/supplied)	...	sqm	23.00	23.70	23.20	24.00	26.00
3.0mm thick (P.C. $19.00sqm/supplied)	...	sqm	32.40	33.20	32.70	33.50	35.40
VINYL SHEET							
Flexible :							
Medium duty -							
2.0mm thick (P.C. range $16.50 to $18.00 sqm/supplied)	...	sqm	43.50-45.00	45.00-46.50	44.00-45.50	45.50-47.00	49.50-51.00
Heavy duty -							
2.0mm thick (P.C. range $20.00 to $47.00 sqm/supplied)	...	sqm	47.00-75.50	48.50-77.00	47.50-76.00	49.00-77.50	53.00-81.50

FLOOR FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TILE AND SHEET, RESILIENT (cont'd)							
VINYL SHEET (cont'd)							
Semi flexible heterogeneous :							
Heavy duty -							
2.0mm thick (P.C. $51.00 sqm/supplied)	...	sqm	79.50	81.00	80.00	81.50	85.50
Acoustic, foam backed :							
2.0 to 4.0mm thick (P.C. range $44.00 to $54.00 sqm/supplied)	...	sqm	72.00-82.50	74.00-84.50	72.50-83.00	74.50-85.00	78.50-89.00
Conductive :							
2.0mm thick (P.C. range $42.00 to $72.00 sqm/supplied)	...	sqm	80.00-111.50	82.00-113.50	80.50-112.00	83.00-114.50	88.50-120.00
Slip resistant :							
2.0mm thick (P.C. range $40.00 to $65.00 sqm/supplied)	...	sqm	78.00-104.00	80.00-106.50	78.50-105.00	81.00-107.00	86.50-112.50
Sports :							
6.5mm thick (P.C.$55.00 sqm/supplied)	...	sqm	118.50	122.50	120.00	123.50	133.00
SKIRTINGS							
Granolithic 75mm high coved	…	m	28.30	21.80	30.30	25.70	30.80
Terrazzo 150mm high coved	…	m	153.00	112.00	117.00	-	124.00
Timber (MDF) 100 x 25mm, painted	…	m	20.30	23.70	24.80	23.30	31.00
Vitreous tile 75mm high	…	m	89.30	93.10	93.50	93.90	101.00
Vinyl 100mm high, feather edged	…	m	11.70	12.10	12.00	12.40	13.00
Ducted 130mm high	…	m	34.00-84.00	34.00-84.00	34.00-84.00	34.00-84.00	34.00-84.00
MATWELLS, MATS							
Matwell, 2000 x 1000mm with brass angle surround	...	No	437.00	462.00	451.00	461.00	481.00
Mat, 2000 x 1000mm :							
Coir	...	No	262.00	277.00	272.00	279.00	289.00
PVC link	...	No	192.00	205.00	202.00	207.00	219.00
Closed construction mat with alternate aluminium and carpet inserts	...	No	1055.00	1090.00	1045.00	1110.00	1085.00

CEILING FINISHES		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ON CONCRETE SOFFIT							
PLASTER							
Costs include average allowance for narrow widths and labours	...	Note					
HARDWALL PLASTER							
Skim coat	...	sqm	39.70	33.90	25.40	29.60	28.70
Two coats	...	sqm	69.80	62.60	62.50	32.90	57.00
ADD extra for :							
Seal and two coats acrylic latex paint	...	sqm	12.40	11.95	11.80	11.50	12.25
Seal and two coats alkyd enamel paint	...	sqm	13.55	14.30	13.75	12.80	12.25
VERMICULITE PLASTER							
Sprayed finish, 10mm thick	...	sqm	30.20	35.70	28.50	30.00	28.70
ON FRAMED SOFFIT							
BOARD AND TIMBER							
FIBRE CEMENT							
4.5mm thick sheeting and seal and two coats acrylic latex paint	...	sqm	57.30	65.30	67.30	63.50	79.70
PLASTERBOARD							
Plasterboard :							
10mm thick	...	sqm	39.80	44.60	34.40	38.20	36.50
13mm thick	...	sqm	41.90	44.60	35.50	39.20	35.50
Fire rated plasterboard :							
1 x 13mm thick layer	...	sqm	46.50	47.30	40.30	45.70	41.90
2 x 13mm thick layer	...	sqm	80.90	81.20	67.50	72.60	69.80
1 x 16mm thick layer	...	sqm	53.20	53.80	46.80	52.70	57.00
2 x 16mm thick layer	...	sqm	91.90	92.50	83.30	82.80	93.80
ADD extra for :							
Seal and two coats acrylic latex paint	...	sqm	12.40	11.95	11.80	11.50	12.25
TIMBER BOARDING							
25mm thick T & G pine boarding with clear finish	...	sqm	90.50	102.50	105.00	102.50	115.00
25mm thick hardwood shiplap panelling with clear finish	...	sqm	170.00	178.50	180.50	168.50	234.00
SUSPENDED CEILINGS							
PLASTERBOARD							
FLUSH plasterboard with concealed suspension :							
10mm thick	...	sqm	74.00	77.40	74.00	77.50	87.00
13mm thick	...	sqm	76.30	80.40	76.00	80.50	92.00
FLUSH fire rated plasterboard with concealed suspension :							
2 x 13mm thick (one hour)	...	sqm	115.00	118.00	115.00	117.50	124.00
2 x 16mm thick (one hour)	...	sqm	125.00	129.00	127.00	131.50	141.00

CEILING FINISHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SUSPENDED CEILINGS (cont'd)							
PLASTERBOARD (cont'd)							
FLUSH fire rated plasterboard with concealed suspension (cont'd) :							
Two skins each of 2 x 16mm thick with furring channels between	...	sqm	228.00	247.00	262.00	252.00	288.00
PLAIN tiles, 1200 x 600mm in prefinished exposed suspension system	...	sqm	68.00	67.90	66.00	56.50	69.00
ACOUSTIC tiles, 600 x 600mm in prefinished exposed suspension system	...	sqm	83.50	77.40	78.00	81.50	83.00
TIMBER BOARD, ETC.							
All in exposed suspension system	...	Note					
MINERAL FIBRE prefinished tiles	...	sqm	64.50-134.00	68.00-141.00	65.00-140.00	70.00-139.00	72.00-147.00
PLYWOOD veneered finish	...	sqm	126.00	131.00	131.00	153.00	133.00
FIBRE CEMENT prefinished panels	...	sqm	67.00	65.50	63.50	72.00	69.50
METAL, ETC.							
All with suspension system	...	Note					
ALUMINIUM, 75 x 75mm open cell 'U' blades	...	sqm	177.00	182.00	180.00	185.00	183.00
ALUMINIUM strip with baked enamel finish	...	sqm	161.50	164.50	161.50	166.50	164.50
STEEL ACOUSTIC tile, prepainted finish :							
1200 x 400mm in one way suspension system - range	...	sqm	87.50-167.50	87.50-168.50	85.50-163.50	90.50-174.50	90.50-169.50
1200 x 600mm in two way suspension system, lay in	...	sqm	71.50	70.50	69.50	72.50	72.50
METAL STRIP 75mm module with baked enamel finish	...	sqm	150.00	158.00	150.00	160.00	150.00
PAINTING							
ADD to foregoing where applicable :							
Seal and two coats acrylic latex paint on plaster or similar smooth surface	...	sqm	12.40	11.95	11.80	11.50	12.25
Seal and two coats alkyd enamel paint on plaster or similar smooth surface	...	sqm	13.55	14.30	13.75	12.80	12.25

FITMENTS	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $

CHECK LIST

METALWORK :

Lockers, refer page 307 ... Note

Shelving and storage systems, refer pages 311 to 313 ... Note

Toilet accessories, refer pages 315 to 319 ... Note

JOINERY :

Cupboards and fittings, refer pages 354 to 358 ... Note

Notice boards, refer page 358 ... Note

Shelving, refer page 353 ... Note

Signs and letters, refer pages 361 and 362 ... Note

GLAZING :

Mirrors, refer page 445 ... Note

MISCELLANEOUS :

Bank fit-out, refer pages 682 and 683 ... Note

Dormitory fit-out, refer page 683 ... Note

Ecclesiastical fit-out, refer page 683 ... Note

Hospital fit-out, refer page 684 ... Note

Hotel fit-out, refer pages 685 to 687 ... Note

Laboratory fit-out, refer pages 688 to 692 ... Note

Library fit-out, refer page 693 ... Note

Retail fit-out, refer pages 694 to 696 ... Note

Office furniture, refer pages 696 to 698 ... Note

Seating, refer page 699 ... Note

PLUMBING AND DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $

The following composite prices include good quality sanitary fixtures/tapware and typical water/drainage pipework to a main/riser or to the outside of the building. PC Amounts show the included costs of the fixture (and cistern) and tapware.

Refer pages 161 and 162 for ADDITIONAL COSTS

SINGLE STOREY, COMMERCIAL / INSTITUTIONAL (Inside Building)

WC Suite, with :							
Plastic integrated cistern	(PC $ 435.00)	No	2450.00	2550.00	2550.00	2600.00	3050.00
Vitreous china close coupled suite	(PC $ 900.00)	No	3000.00	3100.00	3100.00	3150.00	3600.00
Duct cistern	(PC $ 865.00)	No	3000.00	3100.00	3100.00	3100.00	3600.00
An-wall cistern	(PC $1040.00)	No	3300.00	3400.00	3400.00	3400.00	3900.00
Junior size, connector suite	(PC $ 740.00)	No	2550.00	2600.00	2650.00	2650.00	3100.00
Care style & integrated cistern	(PC $1535.00)	No	3900.00	4000.00	4000.00	4000.00	4500.00
Anti vandal style & in-wall cistern	(PC $1910.00)	No	4300.00	4400.00	4400.00	4400.00	4900.00
ADD extra for :							
Longer pvc pipe run (over 3m)		m	240.00	245.00	250.00	245.00	300.00
Cast iron pipe (in lieu of pvc)		m	360.00	370.00	370.00	380.00	440.00
Wall Basin - type :							
Small	(PC $ 675.00)	No	2050.00	2100.00	2100.00	2100.00	2400.00
Modest	(PC $ 935.00)	No	2350.00	2400.00	2400.00	2450.00	2750.00
Prestigious	(PC $1220.00)	No	2750.00	2800.00	2800.00	2850.00	3150.00
Health	(PC $2245.00)	No	4100.00	4100.00	4100.00	4100.00	4500.00
Care	(PC $1265.00)	No	2900.00	2950.00	2950.00	3000.00	3300.00
Anti vandal	(PC $ 860.00)	No	2150.00	2200.00	2250.00	2250.00	2550.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	82.00	80.00	84.00	78.00	98.00
Chrome plated exposed trap & waste	(PC $ 130.00)	No	198.00	200.00	202.00	204.00	214.00
Chrome plated bottle trap & waste	(PC $ 350.00)	No	460.00	460.00	460.00	460.00	470.00
FWG Floor Waste Gully - type :							
Basic, pvc gully		No	620.00	610.00	630.00	600.00	740.00
Stainless steel, pvc gully		No	750.00	740.00	760.00	730.00	880.00
100 dia stainless steel & strainer, hdpe gully		No	1020.00	1030.00	1040.00	1030.00	1130.00
Urinal - type :							
Single, wall, in-wall cistern	(PC $2225.00)	No	4000.00	4000.00	4000.00	4000.00	4300.00
Single, SS in-duct cistern	(PC $2465.00)	No	4300.00	4300.00	4300.00	4300.00	4600.00
Single, auto-flush	(PC $3040.00)	No	4800.00	4900.00	4900.00	4900.00	5200.00
Single, waterless	(PC $1540.00)	No	2400.00	2450.00	2450.00	2450.00	2650.00
1800mm, auto-flush	(PC $3950.00)	No	5600.00	5600.00	5800.00	5800.00	6000.00
Stainless Steel Sink / Trough - type :							
Sink, 1230mm long, single bowl	(PC $ 880.00)	No	2550.00	2600.00	2650.00	2650.00	3000.00
Sink, 1800mm long, double bowl	(PC $1370.00)	No	3200.00	3300.00	3300.00	3300.00	3700.00
Sink, 2500mm long, double bowl	(PC $3635.00)	No	6200.00	6200.00	6400.00	6400.00	6800.00
Cleaner's sink	(PC $1105.00)	No	3200.00	3300.00	3300.00	3300.00	3700.00
Laundry, double bowl inset	(PC $1755.00)	No	3600.00	3600.00	3600.00	3700.00	4000.00
Drinking trough, with 4 outlets	(PC $2550.00)	No	4600.00	4600.00	4700.00	4700.00	5100.00
Ablution trough, 1200mm, 3 taps	(PC $1650.00)	No	3500.00	3600.00	3600.00	3600.00	4000.00
Activity trough, 1800mm, 4 taps	(PC $2095.00)	No	4200.00	4200.00	4300.00	4300.00	4700.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	320.00	315.00	325.00	305.00	395.00

PLUMBING AND DRAINAGE

SINGLE STOREY, COMMERCIAL / INSTITUTIONAL (Inside Building) (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Bath, porcelain enamel - type							
1675mm bath, in built	(PC $1145.00)	No	2800.00	2800.00	2850.00	2850.00	3200.00
Prestige / health	(PC $4285.00)	No	7400.00	7400.00	7400.00	7400.00	7800.00
Shower (floor waste, tapware & rose) - type							
General	(PC $ 585.00)	No	2050.00	2100.00	2150.00	2150.00	2450.00
Hand shower and rail	(PC $ 830.00)	No	2350.00	2400.00	2450.00	2450.00	2750.00
Care	(PC $1205.00)	No	2900.00	2950.00	2950.00	3000.00	3300.00
Care, with flushing rim and cistern	(PC $3425.00)	No	6800.00	7000.00	7000.00	7000.00	7800.00
ADD extra for :							
TMV Thermostatic Mixing Valve		No	1700.00	1700.00	1750.00	1750.00	2050.00

PLUMBING AND DRAINAGE

Refer pages 161 and 162 for ADDITIONAL COSTS

MULTI STOREY, COMMERCIAL / INSTITUTIONAL (Inside Building)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
WC Suite, with							
Plastic integrated cistern	(PC $ 435.00)	No	2600.00	2700.00	2700.00	2750.00	3200.00
Vitreous china close coupled suite	(PC $ 900.00)	No	3150.00	3300.00	3300.00	3350.00	3800.00
Duct cistern	(PC $ 865.00)	No	3000.00	3100.00	3100.00	3200.00	3600.00
In-wall cistern	(PC $1040.00)	No	3400.00	3500.00	3500.00	3600.00	4100.00
Junior size, connector suite	(PC $ 740.00)	No	2650.00	2800.00	2800.00	2850.00	3300.00
Care style & integrated cistern	(PC $1535.00)	No	4100.00	4200.00	4200.00	4200.00	4700.00
Anti vandal style & in-wall cistern	(PC $1910.00)	No	4500.00	4600.00	4600.00	4700.00	5100.00
ADD extra for :							
Longer pvc pipe run (over 3m)		m	370.00	390.00	390.00	390.00	470.00
Cast iron pipe (in lieu of pvc)		m	430.00	440.00	440.00	510.00	530.00
Wall Basin - type							
Small	(PC $ 675.00)	No	2000.00	2050.00	2100.00	2100.00	2400.00
Modest	(PC $ 935.00)	No	2350.00	2400.00	2400.00	2450.00	2700.00
Prestigious	(PC $1220.00)	No	2750.00	2800.00	2800.00	2850.00	3150.00
Health	(PC $2245.00)	No	4100.00	4200.00	4200.00	4200.00	4500.00
Care	(PC $1265.00)	No	2900.00	2950.00	2950.00	3000.00	3300.00
Anti vandal	(PC $ 860.00)	No	2100.00	2200.00	2200.00	2250.00	2500.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	102.00	108.00	108.00	108.00	134.00
Chrome plated exposed trap & waste	(PC $ 130.00)	No	206.00	208.00	210.00	210.00	222.00
Chrome plated bottle trap & waste	(PC $ 350.00)	No	470.00	480.00	480.00	480.00	490.00
FWG Floor Waste Gully - type							
Basic, pvc gully		No	680.00	720.00	720.00	730.00	890.00
Stainless steel, pvc gully		No	825.00	870.00	870.00	885.00	1035.00
100 dia stainless steel & strainer, hdpe gully		No	1215.00	1290.00	1305.00	1305.00	1440.00
Urinal - type							
Single, wall, in-wall cistern	(PC $2225.00)	No	4000.00	4100.00	4100.00	4100.00	4400.00
Single, SS in-duct cistern	(PC $2465.00)	No	4300.00	4400.00	4400.00	4400.00	4700.00
Single, auto-flush	(PC $3040.00)	No	4900.00	5000.00	5000.00	5000.00	5300.00
Single, waterless	(PC $1540.00)	No	2400.00	2450.00	2450.00	2450.00	2650.00
1800mm, auto-flush	(PC $3950.00)	No	5800.00	5800.00	5800.00	5800.00	6200.00
Stainless Steel Sink / Trough - type							
Sink, 1230mm long, single bowl	(PC $ 880.00)	No	2750.00	2850.00	2850.00	2900.00	3300.00
Sink, 1800mm long, double bowl	(PC $1370.00)	No	3400.00	3500.00	3500.00	3600.00	4000.00
Sink, 2500mm long, double bowl	(PC $3635.00)	No	6600.00	6600.00	6600.00	6600.00	7200.00
Cleaner's Sink	(PC $1105.00)	No	3400.00	3500.00	3500.00	3500.00	3900.00
Laundry, double bowl inset	(PC $1755.00)	No	3700.00	3800.00	3800.00	3800.00	4200.00
Drinking trough, with 4 outlets	(PC $2550.00)	No	4800.00	4800.00	5000.00	5000.00	5400.00
Ablution trough, 1200mm, 3 taps	(PC $1650.00)	No	3700.00	3800.00	3800.00	3900.00	4300.00
Activity trough, 1800mm, 4 taps	(PC $2095.00)	No	4300.00	4400.00	4500.00	4500.00	4900.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	98.00	104.00	104.00	104.00	128.00
Bath, porcelain enamel - type							
1675mm bath, in built	(PC $1145.00)	No	2750.00	2850.00	2850.00	2900.00	3200.00
Prestige / health	(PC $4285.00)	No	7400.00	7600.00	7600.00	7600.00	8000.00
Shower (floor waste, tapware & rose) - type							
general	(PC $ 585.00)	No	2150.00	2200.00	2200.00	2250.00	2550.00
Hand shower and rail	(PC $ 830.00)	No	2450.00	2500.00	2500.00	2550.00	2850.00
Care	(PC $1205.00)	No	3000.00	3050.00	3100.00	3100.00	3400.00
Care, with flushing rim and cistern	(PC $3425.00)	No	7200.00	7200.00	7200.00	7400.00	8000.00
ADD extra for :							
TMV Thermostatic Mixing Valve		No	1625.00	1675.00	1700.00	1725.00	1975.00

PLUMBING AND DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer pages 161 and 162 for ADDITIONAL COSTS							
SINGLE STOREY, RESIDENTIAL / DOMESTIC (Inside Building)							
WC Suite, with							
Plastic integrated cistern	(PC $ 435.00)	No	2100.00	2150.00	2200.00	2200.00	2600.00
Vitreous china close coupled suite	(PC $ 900.00)	No	2650.00	2700.00	2700.00	2700.00	3100.00
Duct cistern	(PC $ 865.00)	No	2600.00	2650.00	2700.00	2700.00	3100.00
In-wall cistern	(PC $1040.00)	No	2850.00	2950.00	2950.00	2950.00	3350.00
Junior size, connector suite	(PC $ 740.00)	No	2150.00	2250.00	2250.00	2250.00	2650.00
Care style & integrated cistern	(PC $1535.00)	No	3500.00	3500.00	3500.00	3600.00	4000.00
Anti vandal style & in-wall cistern	(PC $1910.00)	No	3900.00	3900.00	4000.00	4000.00	4400.00
ADD extra for :							
Longer pvc pipe run (over 3m)		m	185.00	190.00	195.00	190.00	235.00
Cast iron pipe (in lieu of pvc)		m	310.00	310.00	315.00	320.00	370.00
Wall Basin - type							
Small	(PC $ 675.00)	No	1800.00	1850.00	1850.00	1850.00	2100.00
Modest	(PC $ 935.00)	No	2100.00	2150.00	2150.00	2150.00	2400.00
Prestigious	(PC $1220.00)	No	2450.00	2500.00	2550.00	2550.00	2800.00
Health	(PC $2245.00)	No	3700.00	3800.00	3800.00	3800.00	4100.00
Care	(PC $1265.00)	No	2600.00	2650.00	2650.00	2700.00	2950.00
Anti vandal	(PC $ 860.00)	No	1900.00	1950.00	1950.00	1950.00	2250.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	70.00	68.00	72.00	66.00	84.00
Chrome plated exposed trap & waste	(PC $ 130.00)	No	188.00	192.00	192.00	194.00	206.00
Chrome plated bottle trap & waste	(PC $ 350.00)	No	435.00	435.00	440.00	440.00	450.00
FWG Floor Waste Gully - type							
Basic, pvc gully			540.00	540.00	550.00	520.00	650.00
Stainless steel, pvc gully			660.00	660.00	680.00	650.00	780.00
100 dia stainless steel & strainer, hdpe gully			930.00	940.00	950.00	940.00	1030.00
Urinal - type							
Single, wall, in-wall cistern	(PC $2225.00)	No	3650.00	3700.00	3700.00	3700.00	4000.00
Single, SS in-duct cistern	(PC $2465.00)	No	3900.00	4000.00	4000.00	4000.00	4200.00
Single, auto-flush	(PC $3040.00)	No	4500.00	4500.00	4500.00	4500.00	4800.00
Single, waterless	(PC $1540.00)	No	2200.00	2200.00	2250.00	2200.00	2400.00
1800mm, auto-flush	(PC $3950.00)	No	5200.00	5200.00	5400.00	5400.00	5600.00
Stainless Steel Sink / Trough - type							
Sink, 1230mm long, single bowl	(PC $ 880.00)	No	2200.00	2200.00	2250.00	2250.00	2550.00
Sink, 1800mm long, double bowl	(PC $1370.00)	No	2800.00	2800.00	2850.00	2850.00	3150.00
Sink, 2500mm long, double bowl	(PC $3635.00)	No	5600.00	5600.00	5600.00	5600.00	6000.00
Cleaner's sink	(PC $1105.00)	No	2950.00	3000.00	3000.00	3000.00	3300.00
Laundry, double bowl inset	(PC $1755.00)	No	2950.00	2950.00	3000.00	3000.00	3300.00
Drinking trough, with 4 outlets	(PC $2550.00)	No	4100.00	4200.00	4200.00	4200.00	4500.00
Ablution trough,1200mm, 3 taps	(PC $1650.00)	No	3100.00	3200.00	3200.00	3200.00	3500.00
Activity trough, 1800mm, 4 taps	(PC $2095.00)	No	3700.00	3800.00	3800.00	3800.00	4100.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	65.00	63.00	66.00	60.00	78.00
Bath, porcelain enamel - type							
1675mm bath, in built	(PC $1145.00)	No	2450.00	2500.00	2500.00	2500.00	2800.00
Prestige / health	(PC $4285.00)	No	6800.00	6800.00	6800.00	6800.00	7100.00
Shower (floor waste, tapware & rose) - type							
General	(PC $ 585.00)	No	1850.00	1900.00	1950.00	1950.00	2200.00
Hand shower and rail	(PC $ 830.00)	No	2150.00	2200.00	2200.00	2200.00	2500.00
Care	(PC $1205.00)	No	2650.00	2700.00	2750.00	2750.00	3000.00
Care, with Flushing Rim and cistern	(PC $3425.00)	No	6400.00	6600.00	6600.00	6600.00	7200.00
ADD extra for :							
TMV Thermostatic Mixing Valve		No	1475.00	1500.00	1525.00	1550.00	1800.00

PLUMBING AND DRAINAGE

Refer pages 161 and 162 for ADDITIONAL COSTS

MULTI STOREY, RESIDENTIAL / DOMESTIC (Inside Building)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
WC Suite, with							
Plastic integrated cistern	(PC $ 435.00)	No	2150.00	2250.00	2250.00	2250.00	2650.00
Vitreous china close coupled suite	(PC $ 900.00)	No	2700.00	2800.00	2800.00	2800.00	3200.00
Duct cistern	(PC $ 865.00)	No	2550.00	2650.00	2650.00	2700.00	3050.00
In-wall cistern	(PC $1040.00)	No	2900.00	3000.00	3000.00	3050.00	3450.00
Junior size, connector suite	(PC $ 740.00)	No	2200.00	2300.00	2300.00	2350.00	2700.00
Care style & integrated cistern	(PC $1535.00)	No	3500.00	3600.00	3600.00	3700.00	4100.00
Anti vandal style & in-wall cistern	(PC $1910.00)	No	4000.00	4100.00	4100.00	4100.00	4500.00
ADD extra for :							
Longer pvc pipe run (over 3m)		m	220.00	230.00	230.00	235.00	285.00
Cast iron pipe (in lieu of pvc)		m	290.00	300.00	300.00	365.00	355.00
Wall Basin - type							
Small	(PC $ 675.00)	No	1700.00	1750.00	1800.00	1800.00	2050.00
Modest	(PC $ 935.00)	No	2050.00	2100.00	2100.00	2100.00	2350.00
Prestigious	(PC $1220.00)	No	2400.00	2450.00	2500.00	2500.00	2750.00
Health	(PC $2245.00)	No	3700.00	3700.00	3800.00	3800.00	4000.00
Care	(PC $1265.00)	No	2550.00	2600.00	2600.00	2650.00	2900.00
Anti vandal	(PC $ 860.00)	No	1850.00	1900.00	1900.00	1950.00	2150.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	86.00	90.00	90.00	90.00	112.00
Chrome plated exposed trap & waste	(PC $ 130.00)	No	196.00	198.00	198.00	200.00	212.00
Chrome plated bottle trap & waste	(PC $ 350.00)	No	450.00	450.00	450.00	450.00	470.00
FWG Floor Waste Gully - type							
Basic, pvc gully		No	550.00	580.00	590.00	590.00	720.00
Stainless steel, pvc gully		No	690.00	720.00	730.00	730.00	860.00
100 dia stainless steel & strainer, hdpe gully		No	1020.00	1080.00	1095.00	1095.00	1200.00
Urinal - type							
Single, wall, in-wall cistern	(PC $2225.00)	No	3700.00	3750.00	3750.00	3750.00	4050.00
Single, SS in-duct cistern	(PC $2465.00)	No	4000.00	4000.00	4000.00	4000.00	4300.00
Single, auto-flush	(PC $3040.00)	No	4500.00	4600.00	4600.00	4600.00	4900.00
Single, waterless	(PC $1540.00)	No	2200.00	2250.00	2250.00	2250.00	2450.00
1800mm, auto-flush	(PC $3950.00)	No	5400.00	5400.00	5400.00	5400.00	5800.00
Stainless Steel Sink / Trough - type							
Sink, 1230mm long, single bowl	(PC $ 880.00)	No	2250.00	2300.00	2350.00	2350.00	2650.00
Sink, 1800mm long, double bowl	(PC $1370.00)	No	2850.00	2950.00	2950.00	3000.00	3250.00
Sink, 2500mm long, double bowl	(PC $3635.00)	No	5600.00	5800.00	5800.00	5800.00	6200.00
Cleaner's sink	(PC $1105.00)	No	3000.00	3100.00	3100.00	3150.00	3450.00
Laundry, double bowl inset	(PC $1755.00)	No	3000.00	3100.00	3100.00	3150.00	3450.00
Drinking trough, with 4 outlets	(PC $2550.00)	No	4200.00	4300.00	4300.00	4400.00	4700.00
Ablution trough,1200mm, 3 taps	(PC $1650.00)	No	3200.00	3300.00	3300.00	3300.00	3600.00
Activity trough, 1800mm, 4 taps	(PC $2095.00)	No	3800.00	3900.00	3900.00	3900.00	4300.00
ADD extra for :							
Longer pvc pipe run (over 5m)		m	80.00	86.00	86.00	86.00	106.00
Bath, porcelain enamel - type							
- 1675mm bath, in built	(PC $1145.00)	No	2450.00	2500.00	2500.00	2550.00	2800.00
- prestige / health	(PC $4285.00)	No	6900.00	6900.00	6900.00	7000.00	7200.00
Shower (floor waste, tapware & rose) - type							
General	(PC $ 585.00)	No	1850.00	1900.00	1900.00	1950.00	2200.00
Hand shower and rail	(PC $ 830.00)	No	2150.00	2200.00	2200.00	2250.00	2500.00
Care	(PC $1205.00)	No	2650.00	2700.00	2750.00	2750.00	3000.00
Care, with flushing rim and cistern	(PC $3425.00)	No	6600.00	6800.00	6800.00	7000.00	7600.00
ADD extra for :							
TMV Thermostatic Mixing Valve		No	1350.00	1400.00	1400.00	1425.00	1650.00

PLUMBING AND DRAINAGE

OTHER SERVICES AND COSTS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD to the foregoing for :							
Light-use hot water heater (for one or two fixtures)		No	1500.00	1525.00	1525.00	1550.00	1725.00
General-use gas hot water heater (for four or five fixtures)		No	2600.00	2650.00	2650.00	2700.00	2900.00
Heavy-duty gas hot water heater (commercial 265L)		No	7250.00	7250.00	7250.00	7250.00	7500.00
Corner Bath	(PC $1430.00)		5200.00	5200.00	5200.00	5200.00	5600.00
Bidette -							
With manual mixer	(PC $1235.00)	No	3300.00	3400.00	3400.00	3400.00	3900.00
With luxury remote control	(PC $1825.00)	No	3800.00	4000.00	4000.00	4000.00	4400.00
Boiling water unit -							
30 cup, 5 litre	(PC $ 990.00)	No	1750.00	1750.00	1750.00	1800.00	1950.00
150 cup, 25 litre	(PC $1370.00)	No	2200.00	2200.00	2200.00	2250.00	2400.00
Boiling/chilled water tap unit	(PC $3810.00)	No	5400.00	5600.00	5600.00	5600.00	5800.00
Chilled water tap unit	(PC $2460.00)	No	3900.00	4000.00	4000.00	4000.00	4300.00
Boiling/chilled/sparkling water tap unit	(PC $5165.00)	No	7000.00	7000.00	7000.00	7000.00	7400.00
Refrigerated drinking trough with disabled access and 4 bubblers	(PC $6405.00)	No	8000.00	8000.00	8000.00	8200.00	8400.00
Service distribution pipe (water / fire / fuel gas), with fittings, buried in light / sand / clay ground 600mm to 1000mm deep :							
25mm dia Copper		m	64.00	67.00	66.00	65.00	79.00
25mm dia Copper, insulated		m	92.00	96.00	96.00	94.00	114.00
32mm dia Copper		m	62.00	65.00	64.00	63.00	75.00
32mm dia Copper, insulated		m	98.00	100.00	102.00	100.00	118.00
50mm dia Copper		m	88.00	92.00	92.00	90.00	106.00
50mm dia Copper, insulated		m	134.00	140.00	142.00	140.00	164.00
100mm dia Copper		m	185.00	195.00	190.00	180.00	215.00
100mm dia Galvanised Steel		m	140.00	150.00	145.00	150.00	170.00
100mm dia pvc drain		m	72.00	80.00	78.00	82.00	102.00
100mm dia cast iron drain		m	155.00	160.00	160.00	215.00	190.00
150mm dia Copper		m	320.00	335.00	325.00	305.00	360.00
150mm dia cast iron drain		m	220.00	230.00	230.00	235.00	265.00
Service distribution pipe (water / fire / fuel gas), with fittings, fixed in building :							
25mm dia copper		m	59.00	57.00	56.00	50.00	64.00
25mm dia pvc		m	47.00	46.00	46.00	42.00	54.00
32mm dia Copper		m	62.00	60.00	59.00	53.00	66.00
32mm dia pvc		m	51.00	48.00	49.00	44.00	54.00
50mm dia Copper		m	88.00	88.00	88.00	78.00	98.00
63mm dia Polyethylene		m	63.00	61.00	61.00	57.00	73.00
100mm dia Copper		m	164.00	166.00	162.00	148.00	178.00
110mm dia Polyethylene		m	84.00	82.00	82.00	78.00	98.00
150mm dia Copper		m	290.00	300.00	290.00	260.00	310.00
160mm dia Polyethylene		m	138.00	140.00	136.00	138.00	162.00
Isolating valve and valve pit in ground :							
25mm dia		m	650.00	630.00	650.00	670.00	710.00
50mm dia		m	900.00	885.00	900.00	930.00	990.00
100mm dia		m	1450.00	1450.00	1450.00	1500.00	1650.00
150mm dia		m	3350.00	3300.00	3350.00	3350.00	3550.00
Sewer / Stormwater drain pipe, with fittings, buried in light soil / clay ground :							
100mm dia pipe up to 1000mm deep		m	86.00	84.00	80.00	68.00	96.00
150mm dia pipe up to 1000mm deep		m	106.00	104.00	100.00	88.00	120.00
225mm dia pipe up to 1000mm deep		m	156.00	154.00	150.00	140.00	176.00
100mm dia pipe 1000mm-2000mm deep		m	154.00	138.00	136.00	108.00	156.00
150mm dia pipe 1000mm-2000mm deep		m	174.00	160.00	156.00	130.00	180.00
225mm dia pipe 1000mm-2000mm deep		m	210.00	210.00	206.00	182.00	236.00

PLUMBING AND DRAINAGE

OTHER SERVICES AND COSTS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD to the foregoing for (cont'd) :							
Manhole - 1100mm dia precast with heavy slab cover in light soil / clay ground :							
1000mm deep	…	No	3400.00	3000.00	3200.00	3600.00	3800.00
2000mm deep	…	No	3800.00	3800.00	4200.00	4000.00	5000.00
Surface drainage channel in light soil / clay ground :							
100mm wide Light Duty with ss grate	…	m	590.00	570.00	570.00	620.00	590.00
100mm wide Medium Duty with DI grate	…	m	280.00	270.00	270.00	285.00	290.00
200mm wide Heavy Duty with DI grate	…	m	570.00	550.00	550.00	580.00	580.00
Surface drainage sump in light soil / clay ground with galvanised steel grate :							
Light Duty 600 x 600 x 900mm deep	…	No	910.00	880.00	910.00	940.00	970.00
Medium Duty 600 x 600 x 900mm deep	…	No	1035.00	990.00	1020.00	1050.00	1095.00
Medium Duty 900 x 900 x 500mm deep	…	No	1800.00	1725.00	1775.00	1875.00	1850.00

CHECK LIST

Project specific costs may apply including specialist and facility equipment And the associated service and installation requisites :

Mains / ring mains - water / hot water / tepid water / fuel gas, etc	…	Note
Service stacks - soil / waste / vent / water / stormwater, etc	…	Note
Hot water boilers / central hot water systems	…	Note
Fire service - mains, tanks, pumps, hydrants, hose reels, etc	…	Note
Fuel gas service - mains, valves, equipment, branch runs, etc	…	Note
Recycled water / grey water / non potable water systems	…	Note
Stormwater and rain water drainage - Rwp connections, drains, collectors, Sumps, manholes,	…	Note
Channels, basins, separators, large underground storage,	…	Note
Gross pollutant traps, stormwater harvesting and reuse, etc	…	Note
Metering and controls	…	Note
Electrical service - supply, mains and submains, switchboards, subcircuits, etc	…	Note
Controls / data systems / bms - supply / connect / interface etc	…	Note
Storage tanks and pumps	…	Note
Sewer and stormwater pumps	…	Note
Medical and specialist gasses, suction, compressed air, etc	…	Note
Authority, connection and service fees (refer to notices and fees in Preliminaries section)	…	Note

ELECTRICAL

COSTS PER SQUARE METRE

The following are average costs for typical capital city buildings. They do not include for significant site conditions and specialised services and equipment including high voltage cabling, substation, main telephone, data and IT services.

	$/sqm
EDUCATIONAL	
Primary Schools	181.00-248.00
Secondary Schools	210.00-332.00
Technical Schools	205.00-311.00
University Arts	398.00-564.00
University Lecture Theatre	386.00-555.00
HOSPITAL	
District, Private	301.00-479.00
General	539.00-848.00
HOTELS, MOTELS	
City, three star	253.00-365.00
City, five star	334.00-443.00
Motel, standard	150.00-213.00
INDUSTRIAL	
Warehouse with small office	75.00-160.00
Factory with small office	83.00-211.00
Laboratory, workshop	193.00-303.00
OFFICES	
Basic - 1/3 storey	178.00-233.00
Serviced - 1/3 storey	166.00-256.00
- 4/7 storey	172.00-275.00
- 7/20 storey	235.00-382.00
- 20/50 storey	332.00-524.00

	$/sqm
PARKING	
Underground, two level	61.00-92.00
Multi-storey	42.00-69.00
RECREATIONAL	
Clubhouse and changerooms	170.00-224.00
Community recreation centre - medium std	117.00-178.00
Squash courts - high std	100.00-137.00
RESIDENTIAL	
Individual house - medium std	65.00-99.00
Flats, 1/3 storey - medium std	99.00-143.00
Multi-storey - basic std	82.00-130.00
- prestige std	104.00-184.00
Hostel or boarding house	118.00-182.00
RETAIL	
Supermarket, suburban	130.00-171.00
Regional centre, prestige	140.00-199.00
Department store, city	172.00-245.00
RELIGIOUS	
Church, medium std.	145.00-220.00
ADMINISTRATION, CIVIC	
Administration office, single storey	179.00-243.00
Civic Centre, 300-500 capacity	190.00-252.00
Library, suburban	227.00-282.00

GUIDE TO TYPICAL BREAK-UP OF FOREGOING CAPITAL CITIES COSTS/SQM :

	EDUCATIONAL				HOSPITALS	
CONTENT	Primary Schools	Secondary Schools	University Arts	University Lecture Theatre	District Private	General
	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
H.V. Cable/H.V. Substation	-	-	-	-	-	-
Mains and submains	41.25	50.00	52.25	45.25	71.00	82.25
Switchgear	29.00	50.25	56.00	61.25	39.25	73.25
Standby generated power	-	-	-	-	36.25	85.00
Lighting - Circuits	22.50	27.75	41.25	46.75	36.75	75.75
- Fittings	59.50	75.75	190.75	203.25	103.75	111.75
Power	34.25	48.25	45.75	37.75	39.00	65.50
Telephone	-	6.50	24.50	11.25	10.25	5.00
PBX	-	-	-	-	-	-
Security system	-	-	6.75	6.50	-	-
Automatic fire alarm	-	-	20.25	21.25	30.50	27.00
Emergency evacuation and intercommunication	-	-	7.00	6.75	13.00	-
General communications	11.50	12.50	10.25	19.50	40.50	90.50
	198.00	**271.00**	**454.75**	**459.50**	**420.25**	**616.00**

ELECTRICAL

GUIDE TO TYPICAL BREAK-UP OF FOREGOING CAPITAL CITIES COSTS/SQM :

	HOTELS, MOTELS			INDUSTRIAL		
	City Three Star	City Five Star	Motel Standard	Warehouse Small Office	Factory Small Office	Laboratory Workshop
CONTENT	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
H.V. Cable/H.V. Substation	-	-	-	-	-	-
Mains and submains	21.75	23.75	20.00	8.75	17.25	32.75
Switchgear	43.75	45.00	33.50	13.50	13.50	39.75
Standby generated power	33.50	35.25	-	-	-	-
Lighting - Circuits	34.00	37.00	26.00	21.50	24.50	17.50
- Fittings	71.00	88.75	45.75	40.50	40.25	66.50
Power	29.75	30.75	24.50	19.25	25.75	41.75
Telephone	5.25	7.25	3.00	-	-	14.75
PBX	-	-	-	-	-	-
Security system	-	-	-	-	-	-
Automatic fire alarm	21.50	21.25	-	-	-	19.75
Emergency evacuation and intercommunication	-	-	-	-	-	-
General communications	27.50	36.75	31.75	1.75	4.75	24.25
	288.00	**325.75**	**184.50**	**105.25**	**126.00**	**257.00**

	OFFICES					PARKING
	Basic 1/3 Storey	Serviced 1/3 Storey	Serviced 4/7 Storey	Serviced 7/20 Storey	Serviced 20/50 Storey	Under-ground Two Level
CONTENT	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
H.V. Cable/H.V. Substation	-	-	-	-	-	-
Mains and submains	31.75	34.25	37.00	36.75	39.50	17.75
Switchgear	19.00	18.50	20.00	27.50	40.50	4.00
Standby generated power	-	-	-	14.75	17.00	-
Lighting - Circuits	20.50	20.75	23.00	18.50	25.50	5.00
- Fittings	38.50	43.25	50.50	67.50	76.25	17.00
Power	18.25	23.50	25.50	26.25	28.25	3.25
Telephone and Data Wiring	23.50	25.50	27.00	30.25	32.50	-
PBX	-	-	-	-	-	-
Security system	-	1.75	2.00	2.75	-	-
Automatic fire alarm	21.75	20.75	23.50	25.75	27.75	18.25
Emergency evacuation and intercommunication	-	-	-	22.25	26.75	-
General communications	3.00	4.75	5.00	5.25	12.50	-
	176.25	**193.00**	**213.50**	**277.50**	**326.50**	**65.25**

ELECTRICAL

GUIDE TO TYPICAL BREAK-UP OF FOREGOING CAPITAL CITIES COSTS/SQM :

	PARKING	RECREATIONAL			RESIDENTIAL	
	Multi-Storey	Clubhouse and Change-rooms	Community Recreation Centre	Squash Courts High Std.	Individual House Med. Std.	Flats 1-3 Storey Med. Std.
CONTENT	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
H.V. Cable/H.V. Substation	-	-	-	-	-	-
Mains and submains	13.25	46.75	27.50	25.75	3.75	24.50
Switchgear	4.25	22.50	13.25	12.50	5.25	6.50
Standby generated power	-	-	-	-	-	-
Lighting - Circuits	5.00	18.50	21.50	17.75	11.50	23.50
- Fittings	14.75	39.25	52.75	32.75	19.50	27.00
Power	3.50	17.25	11.75	14.75	18.75	27.50
Telephone	-	5.75	2.50	-	-	-
PBX	-	-	-	-	-	-
Security system	-	9.25	9.00	-	-	-
Automatic fire alarm	10.00	-	-	-	-	-
Emergency evacuation and intercommunication	-	-	-	-	-	-
General communications	-	14.50	-	-	-	-
	50.75	**173.75**	**138.25**	**103.50**	**58.75**	**109.00**

	RESIDENTIAL			RETAIL		
	Multi-Storey Basic Std.	Multi-Storey Prestige Std.	Hostel or Boarding House	Super-market Suburban	Regional Centre, Prestige	Department Store, City
CONTENT	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
H.V. Cable/H.V. Substation	-	-	-	-	-	-
Mains and submains	28.50	29.00	16.50	27.75	34.25	30.75
Switchgear	15.50	17.00	12.50	14.50	16.50	16.50
Standby generated power	-	-	-	-	-	-
Lighting - Circuits	16.25	21.50	13.25	21.00	24.00	26.75
- Fittings	9.00	9.25	26.75	47.00	54.00	60.75
Power	19.50	23.75	56.75	7.25	5.75	15.75
Telephone	-	-	-	-	-	-
PBX	-	-	-	-	-	-
Security system	-	-	-	3.00	3.25	4.50
Automatic fire alarm	14.00	15.50	-	12.00	12.50	16.25
Emergency evacuation and intercommunication	-	-	-	-	-	-
General communications	3.50	21.75	5.25	7.00	6.00	8.75
	106.25	**137.75**	**131.00**	**139.50**	**156.25**	**180.00**

MECHANICAL

AIR-CONDITIONING

Costs given hereafter indicate the average costs for typical buildings and exclude any significant site conditions. Refer to Design Data section pages 648 to 660 to establish typical cooling loads.

				Adel. Bris. Mel. Per. Syd $ AIR-CONDITIONED AREA (RANGE)
CINEMAS AND THEATRES				
Seating area	- standard (0.9 sqm/seat)	...	sqm	1005.00-1245.00
	- premium seating	...	sqm	870.00-1080.00
Foyer	- standard	...	sqm	570.00-660.00
HOTELS AND MOTELS				
Public areas		...	sqm	590.00-910.00
Bedrooms (per 30sqm room)				
Central plant system				
	- medium standard	...	room	9400.00-11750.00
	- high standard	...	room	12250.00-15250.00
Package unit				
	- medium standard	...	room	3300.00-4700.00
	- high standard	...	room	5800.00-7600.00
LABORATORIES				
Physical Containment Level -				
PC1		...	sqm	810.00-920.00
PC2		...	sqm	870.00-1350.00
PC3		...	sqm	1080.00-1850.00
OFFICES AND ADMINISTRATION				
1/7 storey	- packaged plant ducted system	...	sqm	315.00-400.00
	- medium standard	...	sqm	375.00-460.00
	- high standard	...	sqm	450.00-530.00
8/14 storey	- medium standard	...	sqm	450.00-560.00
	- high standard	...	sqm	490.00-620.00
15/28 storey	- medium standard	...	sqm	630.00-780.00
	- high standard	...	sqm	670.00-850.00
29/60 storey	- medium standard	...	sqm	680.00-870.00
	- high standard	...	sqm	820.00-1095.00
RESIDENTIAL				
House - ducted	- medium standard	...	sqm	178.00-235.00
	- high standard	...	sqm	260.00-325.00
Apartment	- wall mounted	...	sqm	182.00-260.00
	- ducted	...	sqm	265.00-370.00
RETAIL (Excluding smoke exhaust)				
Supermarket, suburban				
	- packaged plant ducted	...	sqm	240.00-315.00
	- chilled water system	...	sqm	270.00-380.00
Shopping arcade, shops				
	- split system AC units	...	sqm	215.00-325.00
	- rooftop packaged AC units	...	sqm	300.00-380.00
	- chilled water fixed volume AHU	...	sqm	270.00-380.00
	- chilled water VAV AHU	...	sqm	325.00-430.00
Shopping arcade, malls				
	- split system AC units	...	sqm	215.00-325.00
	- rooftop packaged AC units	...	sqm	300.00-380.00
	- chilled water AHU	...	sqm	325.00-430.00
Department stores				
	- rooftop packaged AC units	...	sqm	270.00-380.00
	- chilled water fixed volume AHU	...	sqm	300.00-380.00
	- chilled water VAV AHU	...	sqm	325.00-430.00

MECHANICAL

MECHANICAL VENTILATION

Costs per square metre given hereafter, apply to ADELAIDE, BRISBANE, MELBOURNE, PERTH AND SYDNEY and indicate the range of average costs for typical buildings and exclude any significant site conditions.

		COST/SQM VENTILATED AREA $
CHANGE ROOMS AND TOILETS		
Industrial or recreational buildings :		
medium standard	...	58.00-86.00
high standard	...	100.00-128.00
KITCHENS		
Including exhaust hoods, evaporative cooling, etc. :		
medium standard	...	450.00-580.00
high standard	...	610.00-900.00
TOILETS, ETC.		
Low rise office buildings	...	52.00-88.00
Medium/high rise office building	...	66.00-94.00
PARKING		
Basement fully underground parking	...	72.00-102.00

STAIR PRESSURISATION

Average cost per stair (up to 20 storeys)	...	31000.00-41000.00

LIFTS AND ESCALATORS

Refer pages 664 to 666

FIRE PROTECTION

Refer pages 667 to 670

SITEWORKS	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PAVINGS						
SPRAYED BITUMEN surfacing on and including 100mm thick cement treated crushed rock basecourse ...	sqm	35.00	38.30	26.60	40.00	23.20
BITUMINOUS CONCRETE 25mm thick on and including 100mm thick cement treated crushed rock basecourse ...	sqm	39.10	35.80	36.90	35.40	33.60
ADD extra for :						
150mm crushed rock basecourse in addition ...	sqm	16.00	12.70	17.00	13.40	17.45
Kerbing, stormwater drainage ...	Note					
IN SITU CONCRETE paving with broomed finish :						
100mm thick, unreinforced …	sqm	35.60	34.30	43.20	37.70	48.70
100mm thick, reinforced with SL72 fabric …	sqm	47.50	45.90	56.80	48.70	61.50
BRICK paving :						
230 x 110/115 x 50mm thick …	sqm	74.80	75.90	69.80	50.90	65.90
230 x 110/150 x 75mm thick …	sqm	79.10	70.60	-	67.60	78.60
INTERLOCKING BLOCK paving :						
60mm thick …	sqm	66.90	77.80	78.10	51.20	90.70
80mm thick …	sqm	73.20	83.40	82.40	58.60	97.30
ADD extra for :						
Laying paving to patterns ...	sqm	6.00-11.00	6.50-11.50	6.00-11.00	6.00-11.00	6.00-11.00
Compacted sand bed 75mm thick ...	sqm	6.50	6.55	7.55	5.35	6.55
Polythene underlay ...	sqm	2.55	3.00	3.40	2.90	4.10
Mortar bed 25mm thick ...	sqm	8.15	6.40	6.60	5.95	6.55
RETAINING WALLS						
In situ reinforced concrete walls, refer page 125 ...	Note					
Brickwork, standard common, faced one side, in wall :						
230mm thick …	sqm	345.00	264.00	291.00	254.00	334.00
350mm thick …	sqm	492.00	375.00	418.00	367.00	479.00
470mm thick …	sqm	640.00	487.00	545.00	480.00	623.00
Blockwork, plain faced hollow concrete block, 400 x 200 x 200mm thick, including core filling and bar reinforcement ...	sqm	265.00	212.00	249.00	236.00	313.00
Composite, 260mm thick comprising a 150mm thick in situ reinforced concrete wall with a veneer skin of 110mm standard brickwork in facings ...	sqm	564.00	513.00	560.00	448.00	610.00
Composite, 320mm thick comprising two 110mm skins of standard common brickwork faced one side and a 100mm cavity filled with concrete and reinforced with a layer of F82 fabric reinforcement ...	sqm	381.00	297.00	333.00	292.00	385.00
Precast concrete crib block wall :						
Heavy duty (Pincrib) …	sqm	520.00	515.00	535.00	520.00	545.00
Light duty (Minicrib) …	sqm	300.00	300.00	305.00	310.00	330.00
ADD extra for :						
Reinforced concrete footing including excavation ...	cum	567.00	534.00	574.00	453.00	612.00

SITEWORKS

SCREEN WALLS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Brickwork, standard common in facings with coping, including reinforced concrete footing and excavation, wall :							
110mm thick -							
x 1200mm high	...	m	298.00	237.00	259.00	213.00	290.00
x 1500mm high	...	m	355.00	280.00	306.00	253.00	344.00
230mm thick -							
x 1500mm high	...	m	676.00	544.00	584.00	494.00	664.00
x 2000mm high	...	m	869.00	693.00	745.00	631.00	850.00
Blockwork, coloured, fair face both sides with coping, including reinforced concrete footing and excavation, wall :							
100mm thick -							
x 1200mm high	...	m	251.00	232.00	239.00	208.00	-
x 1600mm high	...	m	311.00	288.00	295.00	258.00	-
200mm thick -							
x 1600mm high	...	m	444.00	398.00	386.00	363.00	-
x 2000mm high	...	m	528.00	473.00	456.00	432.00	-
Brickwork cement rendered and painted both sides, including reinforced concrete footing and excavation, wall :							
100mm thick -							
x 1200mm high	...	m	381.00	333.00	332.00	301.00	349.00
x 1500mm high	...	m	458.00	399.00	397.00	362.00	417.00
230mm thick -							
x 1500mm high	...	m	721.00	605.00	627.00	565.00	674.00
x 2000mm high	...	m	923.00	770.00	798.00	722.00	859.00
Blockwork cement rendered and painted both sides, including reinforced concrete footing and excavation, wall :							
100mm thick -							
x 1200mm high	...	m	346.00	323.00	318.00	282.00	348.00
x 1600mm high	...	m	437.00	408.00	400.00	356.00	439.00
200mm thick -							
x 1600mm high	...	m	571.00	492.00	489.00	455.00	557.00
x 2000mm high	...	m	684.00	588.00	583.00	546.00	667.00

DETAILED PRICES

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The Detailed Prices are indicative average prices for reasonable quantities of work and would apply to projects in excess $1,500,000 and having average site conditions. For works in the CBD it will be necessary to make due allowance to cover the particular requirements of the project especially in respect of site restrictions, productivity and allowances applicable to major works.

Prices, unless otherwise described, are for the completely installed item of work including Builders' overheads and profit normally applying at the commencement of 2021. Whilst including for site allowance at the average rate per hour, they do not include any other special allowances such as structural and height allowances etc.

The prices in the Detailed Prices have been set to reflect current sensible pricing levels with, where appropriate, an adjustment that reflects reasonable discounting of materials.

No allowance has been included in the prices of the respective trade sections for Builder's Preliminaries items such as site establishment, supervision, large plant, scaffolding, temporary services, notices and fees, insurances, etc.; whilst the prices for the various trades and services sections include allowance for their own minor Preliminaries items (i.e. small tools and hand plant and supervision), separate provision must be made for Builder's Preliminaries items as previously described in order to arrive at a realistic total cost.

As prices can differ appreciably because of the nature and specific requirements of each particular contract, it is not recommended that they be used for quotations without checking in detail the requirements of the contract.

Costs exclude the Goods and Services Tax (G.S.T.).

In these Detailed Prices the allowance for Builders' head office overheads and profit that has been included is (unless otherwise noted) :

Preliminaries, Transportation Systems and Base Material Prices	**-**	**0.0%**
Electrical and Mechanical Services	**-**	**2.5%**
All other trades and services	**-**	**5.0%**

PRELIMINARIES

PLANT AND SCAFFOLDING

GENERAL AND SMALL PLANT

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Refer PLANT HIRE RATES pages 737 to 741	...	Note					

HOISTING

CRANES - MOBILE CRANES (CASUAL USE)

BASIC HIRE RATES

Prices include operators and fuel

6 tonne	...	Hour	125.00-140.00	130.00-150.00	145.00-175.00	120.00-140.00	145.00-175.00
9 tonne	...	Hour	130.00-140.00	130.00-150.00	145.00-175.00	125.00-145.00	145.00-175.00
15 tonne	...	Hour	135.00-145.00	160.00-180.00	145.00-175.00	135.00-165.00	145.00-175.00
20 tonne	...	Hour	145.00-165.00	170.00-190.00	165.00-195.00	145.00-175.00	155.00-195.00
30 tonne	...	Hour	170.00-190.00	190.00-220.00	165.00-195.00	155.00-185.00	165.00-195.00
40 tonne	...	Hour	200.00-210.00	210.00-240.00	175.00-215.00	165.00-195.00	175.00-225.00
50 tonne	...	Hour	210.00-230.00	270.00-310.00	195.00-245.00	190.00-230.00	195.00-265.00
60 tonne	...	Hour	240.00-270.00	290.00-340.00	235.00-305.00	185.00-225.00	245.00-335.00
75 tonne	...	Hour	300.00-350.00	345.00-395.00	280.00-355.00	220.00-300.00	355.00-470.00
100 tonne	...	Hour	350.00-420.00	420.00-470.00	405.00-480.00	290.00-400.00	450.00-550.00

ADD extra for :

Dogmen if required	...	Note					

ADDITIONAL TO BASE HIRE RATES :

Cartage to and from site, including pilot cars, low loaders, semi-trailers, police escort, overtime, rates; permit fee; special on-site and living away allowances; statutory authority charges; supervision; travelling time of operators	...	Note					

CRANES - TOWER

Prices exclude operators, fuel and servicing	...	Note					

Prices include ten tower sections :

Favco 350	...	Week	2500.00-3500.00	3600.00-4000.00	3000.00-3500.00	2100.00-2600.00	2950.00-3600.00
Favco 500	...	Week	3500.00-4500.00	4500.00-5200.00	-	2600.00-3100.00	3250.00-4150.00
Favco 750	...	Week	4500.00-5500.00	4500.00-5200.00	-	3200.00-3700.00	3600.00-4250.00
Favco 1000	...	Week	5500.00-6000.00	6000.00	5000.00-6000.00	4200.00-4700.00	5150.00-5800.00

ADD extra for :

Transport, setting up, maintenance, jumping and returning to yard	...	Note					
Operators, fuel and servicing	...	Note					
For all other charges, minimum hire rates etc., refer to hiring company	...	Note					

PRELIMINARIES

PLANT AND SCAFFOLDING (cont'd)

HOISTING (cont'd)

HOISTS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Barrow hoist, double barrow (3 storey)	...	Week	350.00-500.00	320.00-475.00	270.00-450.00	300.00-500.00	290.00-390.00
ADD extra for :							
Transport, setting up and removal	...	Item	1500.00-2000.00	1500.00-2000.00	1150.00-2150.00	1200.00-2200.00	1140.00-2290.00
Platform hoist, 0.5 tonne two barrow (9 storey)	...	Week	450.00-620.00	310.00-580.00	350.00-600.00	400.00-600.00	350.00-640.00
ADD extra for :							
Transport, installation and removal	...	Item	4000.00-5000.00	2500.00-4000.00	2100.00-4200.00	1800.00-2800.00	2380.00-4500.00
Man and material hoist, 1.0 tonne/16 person to 12 floors :							
Hoist	...	Week	2000.00-2500.00	2000.00-2650.00	2000.00-2600.00	2100.00-2500.00	2050.00-2520.00
Transport, erect, dismantle and remove, including all associated costs	...	Item	25000.00-32000.00	25000.00-45000.00	23000.00-33000.00	23000.00-33000.00	29000.00-35000.00

SCAFFOLDING

TUBULAR

Tubular steel scaffolding including erecting, dismantling, cartage, etc. (First month)	...	sqm	37.00-42.00	35.00-45.00	35.00-45.00	37.00-44.00	52.00
ADD extra for :							
Hire charge for scaffolding (per month after first month)	...	sqm/Mth	12.00-16.00	12.00-16.00	13.00-16.00	18.50-23.50	17.00
Timber or aluminium planks (hire only) (per month)	...	m/Mth	4.00	3.50-8.50	3.50	4.00	4.00
Safety netting (hire only) (total period)	...	sqm	-	Incl.	Incl.	-	Incl.
Safety screen panels (hire only) (per week)	...	sqm/week	Incl.	Incl.	Incl.	-	1.00

SUNDRY

Mobile scaffolds -							
1.80 x 1.20 x 4.00m high	...	Week	130.00-160.00	200.00-250.00	150.00-250.00	160.00-210.00	280.00
Scissor platforms -							
5.80m	...	Week	550.00-750.00	700.00-900.00	600.00-800.00	600.00-800.00	620.00-930.00
7.62m	...	Week	700.00-800.00	800.00-1100.00	750.00-1100.00	750.00-1000.00	880.00-1360.00
9.15m	...	Week	800.00-1100.00	900.00-1200.00	1000.00-1250.00	1050.00-1250.00	1010.00-1960.00
12.20m	...	Week	1150.00-1450.00	1200.00-1500.00	1200.00-1700.00	1450.00-1500.00-	1150.00-2430.00

PRELIMINARIES

PLANT AND SCAFFOLDING (cont'd)

SCAFFOLDING (cont'd)

SUNDRY (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cherry pickers -							
12.00m excluding operator	...	Day	300.00-400.00	300.00-350.00	300.00-400.00	220.00-320.00	270.00-440.00
15.00m excluding operator	...	Day	350.00-400.00	350.00-450.00	350.00-450.00	250.00-350.00	380.00-570.00
22.00m excluding operator	...	Day	380.00-500.00	450.00-650.00	500.00-700.00	350.00-500.00	620.00-720.00
27.40m excluding operator	...	Day	500.00-800.00	575.00-800.00	600.00-800.00	500.00-700.00	720.00-880.00
33.50m excluding operator	...	Day	850.00-1000.00	950.00-1150.00	850.00-1100.00	900.00-1000.00	960.00-1150.00
ADD extra for :							
Cartage	...	Note					

ROOF GUARDRAIL

Tubular guardrail including erection, dismantling and removal to :							
Metal deck roof	...	m	25.00	24.00	26.00	28.50	27.00
Tiled roof	...	m	22.00	24.00	26.00	28.50	21.00

HOARDINGS AND FENCES

Prices given hereafter are based on the use of second hand material and include the cost of erecting, painting, maintaining and removing	...	Note					
STUD FRAMED :							
Chainwire clad x 1800mm high	...	m	40.50	40.00	42.00	38.00	40.00
Tempered hardboard clad x 1800mm high	...	m	46.00	45.00	55.00	43.00	60.00
Waterproof plywood clad x 1800mm high	...	m	78.00-89.00	80.00-100.00	80.00-100.00	78.00	80.00-110.00
Hire rate for foregoing hoardings/ fences, including erection and dismantling, six month minimum period	...	m/Mth	7.00-9.00	6.00-9.00	7.00-10.00	6.00-9.00	Incl.
CHAINWIRE :							
Galvanised security fence x 2100mm high	...	m	40.00	40.00	42.00	40.00	32.00
GANTRY :							
3500mm wide deck including electrical illumination, handrails and kickboards and maintaining for twelve months	...	m	1300.00-1750.00	1250.00-1700.00	1250.00-1700.00	1250.00-1750.00	1500.00
OVERHEAD PROTECTION :							
Non-trafficable gantry over sheds, including erection, hire and removal, twelve months period	...	m	78.00-89.00	-	80.00-90.00	-	105.00
ADD extra for :							
Maintaining gantry for period in excess of twelve months	...	m/week	8.50-10.50	7.00-9.00	8.00-12.00	-	8.00-12.00
Prestige appearance, allowance for additional height lettering or graphics	...	Note					
Builders sheds to gantries	...	Note					

PRELIMINARIES

HOARDINGS AND FENCES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
POLYETHYLENE BARRIERS, SCREENS							
Prices of the following are based on new polyethylene mesh and second hand star pickets :							
Light duty Site Mesh x 930mm high with pickets at 1500/2000mm centres	...	m	7.50-9.50	6.00-9.00	7.00-10.00	6.00-8.00	9.00
Site Screen x 915mm high having rigid vertical strands and pickets at 1500/2000mm centres	...	m	8.50-10.50	7.00 10.00	8.00-14.00	7.00-9.50	15.50
Barrier Mesh x 1000mm high for pedestrians and motorist control, pickets at 3000mm centres	...	m	10.50-13.00	10.00-12.00	10.00-14.00	9.50-11.50	14.75

OFFICES AND SHEDS

ADD extra for :							
Delivery to site and return -							
Up to 6000mm long	...	Item	350.00	130.00-230.00	200.00-250.00	200.00-250.00	235.00
Over 6000mm long (by quotation)	...	Note					
Erection and dismantling (Included)	...	Note					
Electrical connections (by electrical contractor)	...	Note					
Maintenance (per hire period)	...	Item	-	55.00	70.00	50.00	70.00
SITE OFFICE							
Insulated panel construction :							
4880 x 2400mm	...	Week	55.00-70.00	55.00-65.00	60.00-75.00	50.00-70.00	60.00-80.00
6000 x 3000mm	...	Week	65.00-80.00	65.00-75.00	60.00-75.00	70.00-80.00	80.00-100.00
9000 x 3000mm	...	Week	90.00-110.00	75.00-85.00	90.00-110.00	90.00-110.00	100.00-120.00
12000 x 3000mm	...	Week	110.00-130.00	85.00-140.00	100.00-120.00	110.00-130.00	120.00-150.00
ADD extra for :							
Tables, chairs, benches, office equipment, etc.	...	Unit/Mth	20.00-30.00	10.00-50.00	20.00-25.00	20.00	20.00-24.00
LUNCHROOM							
Insulated panel construction, to comply with building regulations, including tables, benches, exhaust fan, etc. :							
4880 x 2400mm	...	Week	65.00-75.00	60.00-70.00	70.00-90.00	65.00-80.00	70.00-90.00
6000 x 3000mm	...	Week	75.00-90.00	70.00-80.00	80.00-100.00	80.00-100.00	100.00-120.00
9000 x 3000mm	...	Week	90.00-120.00	80.00-90.00	100.00-130.00	125.00-155.00	110.00-140.00
12000 x 3000mm	...	Week	120.00-135.00	95.00-145.00	115.00-140.00	180.00-220.00	125.00-155.00
ADD extra for :							
Cold water supply, sink, refrigerator, pie warmer, urns, heaters, bins, air conditioning, etc.	...	Unit/Mth	25.00-30.00	20.00-40.00	20.00-30.00	50.00	29.00

PRELIMINARIES

OFFICES AND SHEDS (cont'd)

TOILETS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Insulated panel construction with W.C's, urinal, basin, water inlet and soil/waste outlet ready for mains connection :							
20 man unit	…	Week	65.00-80.00	70.00-90.00	70.00-100.00	70.00-85.00	95.00-125.00
30 man unit	…	Week	80.00-90.00	80.00-100.00	85.00-120.00	85.00-115.00	115.00-145.00
50 man unit	…	Week	90.00-100.00	90.00-120.00	95.00-125.00	110.00-160.00	150.00-190.00
100 man unit	…	Week	140.00-160.00	130.00-180.00	130.00-190.00	160.00-210.00	200.00-260.00
ADD extra for :							
Water supply and sewer connections, soap and paper supplies	...	Note					
Single chemical pan toilet including pumping out and cleaning	...	Week	65.00	50.00	60.00-70.00	55.00	60.00-70.00

STORAGE SHEDS

Metal clad and lockable :							
3000 x 2400mm	...	Week	30.00-40.00	27.00-37.00	40.00-50.00	40.00-45.00	40.00-50.00
4880 x 2400mm	...	Week	40.00-50.00	37.00-47.00	40.00-50.00	40.00-45.00	45.00-55.00
6000 x 2400mm	...	Week	40.00-50.00	43.00-58.00	40.00-60.00	40.00-45.00	45.00-65.00

WASTE MANAGEMENT COSTS

For Disposal of Waste During the Construction of a Building :
Assumes removal bins of size utilised in the industry in that city.
All prices include G.S.T. and EPA levies (taxes) where levied.

House	…	sqm*	6.20	7.55	4.95	5.55	8.50
Townhouse	…	sqm*	7.95	9.10	6.25	7.05	10.45
Block of Flats (3 storey)	…	sqm*	6.20	7.55	4.95	5.60	8.40
Factory	…	sqm*	2.05	2.80	1.50	1.70	3.00
High Rise Office	…	sqm*	9.65	10.60	7.70	8.45	12.10

* The cost of waste disposal is expressed as a cost per square metre of floor area of the type of building being constructed.

PRELIMINARIES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TEMPORARY SERVICES							
TELEPHONES - AREAS WITHOUT NBN							
Costs include G.S.T.							
As a new service - connection fee per first line	...	Each	99.00	99.00	99.00	99.00	99.00
Rental per line - per month	...	Month	25.00-55.00	25.00-55.00	25.00-55.00	25.00-55.00	25.00-55.00
Calls - standard rates apply	...	Note					
TELEPHONES - AREAS WITH NBN							
Costs include G.S.T.							
As a new service - connect fee per NBN connection	...	Each	300.00	300.00	300.00	300.00	300.00
Calls - rates as per service provider plan	...	Note					
ELECTRICITY							
Charge for connecting three phase temporary service for commercial building and later removal of service and meters :							
ADELAIDE By quotation only							
BRISBANE By quotation only							
MELBOURNE By quotation only							
PERTH By quotation only							
SYDNEY By quotation only							
ADD extra for :							
Power consumed, including for testing of lifts and mechanical services	...	Note					
Mains and submains, main board, sub boards, GPO's, welding points, light points and fittings, night lighting and leads	...	Note					
WATER							
Refer NOTICES AND FEES	...	Note					

SUPERVISION AND LABOUR

In costing this item it is necessary to establish the number of staff needed to administer the project, the length of time each member of staff will be required on the project, and the gross weekly cost to the employer of each grade of staff. The following list is given as a broad indication of the grades of staff likely to be applicable to medium to large projects :

Project Manager	Timekeeper	L.H. Carpenter
Project Engineer	Typist	Foreman Labourer
General Foreman	Telephonist	Labourers
Administrator/Quantity Surveyor	Sub Foreman	Apprentices
Site Clerk	Foreman Carpenter	

MISCELLANEOUS COSTS

Refer check list on page 204

PRELIMINARIES

INSURANCE

CONTRACT WORKS INSURANCE

Provides cover to the works for loss or damage from :

- fire, lightning, explosion, crashing aircraft
- water (or other fire-fighting measures) used to extinguish a fire
- flood, heavy rain, snow
- windstorm, cyclone
- earthquake, subsidence, land or rock-slide
- theft, burglary
- impact, collapse
- malicious acts, riots, strikes
- the consequences of :
  - bad workmanship, lack of skill, faulty materials,
  - error or omission in design or specifications.

The premiums and excesses vary greatly depending upon :

- construction techniques
- physical exposures on individual sites
- geological and meteorological features at the construction site
- builders experiences
- construction and maintenance periods
- width of insurance cover required

The following table is for indicative purposes only and does not include fire services levies or stamp duty charges :

PROJECT TYPE	RATE x TCV	DEDUCTIBLE (EXCESS)
Bridges (wet) over a known water course	0.275% to 0.40%	0.25% of TCV (min $100,000)
Bridge (dry) e.g. : overpasses	0.20% to 0.30%	0.20% of TCV (min $25,000)
Residential (TCV less than $1,000,000 - house)	0.10% to 0.17%	$500.00 to $2,500.00
Residential (TCV greater than $1,000,000 - apartments)	0.12% to 0.22%	$1,000.00 to $25,000.00
Light engineering	0.10% to 0.22%	$1,000.00 to $10,000.00
Commercial construction, say small office blocks, units, schools less than $10,000,000 TCV	0.10% to 0.18%	$1,000.00 to $25,000.00

Current fire service levies and stamp duty charges to be added to any contract works premium :

STATE	FIRE SERVICES LEVY	STAMP DUTY
New South Wales	37**% x basic premium	9% x addition of premium and F.S.L. + G.S.T
Victoria*	0%	10% x (as above)
South Australia	0%	11% x (as above)
Western Australia	0%	10% x (as above)
Queensland	0%	9% x (as above)
Northern Territory	0%	10% x (as above)
A.C.T. (Canberra)	0%	0% x (as above)
Tasmania	28% x basic premium	10% x (as above)

* Fire service levy in Victoria was phased out during 2013.
** May vary between insurers.

MATERIAL DAMAGE ATTRACTS :

Premium (including Terrorism Premium*)
F.S.L. on Premium
G.S.T. on (Premium + F.S.L.)
Stamp duty on (Premium + F.S.L. + G.S.T.)

*** Refer page 181 for explanation**

PRELIMINARIES

INSURANCE (cont'd)

CONTRACT WORKS PUBLIC LIABILITY

Provides cover for the insured's legal obligation to pay compensation to third parties for property damage and bodily injury arising during the construction and subsequent visits during defect liability/maintenance periods and associated with the performance of the contract works.

The premium and excess vary greatly depending on :

- contractual arrangements between principals and contractors
- contractual arrangements between sub-contractors and contractors
- method of construction
- particular hazards to the individual risk e.g.: underpinning, excavation, piling activities
- safety programmes on site
- protection provided to the public
- state in which the risk is situated
- limit of liability required

The following table indicates an average range and does not include Stamp Duty charges. The indicative rates allow for worker to worker exposure and the operation of normal construction plant and equipment on site, but does not allow for professional indemnity cover, pollution exposure or completed operations exposure :

PROJECT TYPE	RATE x TCV	LIMIT OF LIABILITY	DEDUCTIBLE (EXCESS)
Bridges (wet)	0.12% to 0.25%	$20,000,000 Any one occurrence	$5,000 + All losses
Dry bridges (not rail)	0.12% to 0.25%	$20,000,000 Any one occurrence	$2,500 + All losses
Residential TCV < $1,000,000	0.15% to 0.25%	$20,000,000 Any one occurrence	$1,000 + All losses
Residential TCV $1,000,000 to $10,000,000	0.08% to 0.18%	$20,000,000 Any one occurrence	$1,000 + All losses
Light Engineering	0.12% to 0.20%	$20,000,000 Any one occurrence	$1,000 + All losses
Commercial construction say small office blocks, units, schools	0.11% to 0.20%	$20,000,000 Any one occurrence	$1,000 + All losses

NOTE : T.C.V. - Total Contract Value
T.P.P.D. - Third Party Property Damage
T.P.P.I. - Third Party Property Insurance

Worker to Worker is a major exposure - deductibles range from $10,000.00 up to $100,000.00

PRELIMINARIES

INSURANCE (cont'd)

CONTRACT WORKS PUBLIC LIABILITY (cont'd)

Current stamp duty charges to be added to any liability premium :

STATE	STAMP DUTY	STATE	STAMP DUTY	STATE	STAMP DUTY
New South Wales	9% of premium	South Australia	11% of premium	A.C.T.	0% of premium
Victoria	10% of premium	Western Australia	10% of premium	Tasmania	10% of premium
Queensland	9% of premium	Northern Territory	10% of premium		

It is recommended that anyone requiring insurance cover for contract works have their insurance company or insurance broker advise terms before tendering for any project.

CONSTRUCTION LIABILITY ATTRACTS : Premium
G.S.T. on Premium
Stamp duty on (Premium + G.S.T.)

TERRORISM LEVY

The Federal Government established the Terrorism Insurance Act, (2003) to provide terrorism cover on certain policies for terrorist events in Australia. Only policies that exclude terrorism and the insurance company has purchased Terrorism reinsurance from the Australian Reinsurance Pool Corporation will be covered by the Act. If the Minister declares a terrorist event then the Act will operate to write back the terrorist exclusion and provide cover.

TERRORISM PREMIUM : C.B.D. - 16% URBAN - 5.3% COUNTRY - 2.6%

The reinsurance premium due to ARPC is calculated as a percentage of the gross base premium. Hence the base premium may need to be grossed up for brokerage (for example).
See http://arpc.gov.au/qanda/#Premiums

PRELIMINARIES

NOTICES AND FEES, ETC. - ADELAIDE

BUILDING FEE — G.S.T. INCLUSIVE FEE

CALCULATION OF FEES PAYABLE

The fee payable for approval of building work must be calculated from the following formula -

Generally- Fee (in dollars) = 0.0023 x CI x A x CF
Change of Classification -
- Fee (in dollars) = 0.00184 x CI x A x CF

where CI = Construction Index from Minister's Schedule
A = Floor Area as computed according to regulation 4.1
CF = Complexity Factor as detailed below
Where a building is composed of parts having different Construction Indices the fee shall be the sum of the fees for each part.

Construction Indices

The appended List of Construction Indices illustrates the basis of the proposal. Note that these figures are not final, but will enable meaningful comment.

Building Classes		Construction Indices	APPROXIMATE Fee per sqm
Class 1, 2, 4 (Dwelling)	...	1395	$3.29
Class 3, 5, 6 (Offices, shops)	...	1858	$4.38
Class 7, 8 (Warehouses, factories)	...	1231	$2.91
Class 9a, 9c (Hospitals, churches)	...	2105	$4.97
Class 9b (Clubs, etc.)	...	1849	$4.36
Class 10 (Outbuilding e.g. garage if not attached to dwelling or shop)	...	416	$0.98

Complexity Factor

The appropriate Complexity Factor tabulated hereunder will be used :

(a) except as below	...	1.0
(b) for buildings exceeding 6 storeys	...	1.3
(c) for buildings containing an atrium	...	1.3
(d) for buildings containing a mall exceeding 40m in length	...	1.3
(e) for demolition of buildings (based on area of existing building)	...	0.2
(f) for approval of change of classification where no building work is proposed	...	0.8

Building Classes

Class 1a - A single dwelling including detatched house, town houses and villa units.
Class 1b - Guest house, boarding house or the like accommodating not more than 12 persons and not exceeding 300sqm.
Class 2 - A building containing 2 or more separate dwellings, excluding buildings of Class 1.
Class 3 - A residential building, other than Class 1 or Class 2, which is a common place of living for a number of unrelated persons.
Class 4 - A dwelling in a building that is Class 5, 6, 7, 8 or 9 if it is the only dwelling in the building.
Class 5 - An office building for professional or commercial purposes, excluding buildings of Class 6, 7, 8 or 9.
Class 6 - A shop or other building for the sale of goods by retail or the supply of services direct to the public.
Class 7 - A building which is :
Class 7a - A carpark.
Class 7b - For storage or display of goods or produce for sale by wholesale.
Class 8 - A factory or laboratory.

PRELIMINARIES

NOTICES AND FEES, ETC. - ADELAIDE (cont'd)

BUILDING FEE (cont'd)

CALCULATION OF FEES PAYABLE (cont'd)

Building Classes (cont'd)

Class 9 - A building of a public nature :
Class 9a - A health care building.
Class 9b - An assembly building.
Class 9c - An aged care building.

Class 10 - A non-habitable building or structure :
Class 10a - An open or private garage, shed, or the like.
Class 10b - A fence, mast, antenna, retaining or free-standing wall, swimming pool or the like.

DEPARTMENT OF WORKPLACE SERVICES

		Design Fee	Registration Fee
Notification of commencement of certain work (previously Construction Safety Notification Advice Fee)	...	N/A	Nil
Lifts and escalators :			
			(five year term)
Passenger or goods lift	...	$132.00	$383.00
Escalator or moving walk	...	$132.00	$383.00
Window cleaning lift	...	$132.00	$383.00
Service lift	...	$132.00	$383.00
Chair lift	...	$132.00	$383.00
Special lift	...	$132.00	$383.00
Hand operated lift	...	$132.00	$383.00
Cranes and hoists :			
Tower cranes	...	$132.00	$383.00
Work boxes suspended from cranes	...	$132.00	$383.00
Gantry cranes (refer Department's schedule)	...	$132.00	$383.00
Bridge cranes (refer Department's schedule)	...	$132.00	$383.00
Mobile cranes (refer Department's schedule)	...	$132.00	$383.00
Hoists (to lift people) (refer Department's schedule)	...	$132.00	$383.00
Vehicle hoists	...	$132.00	$383.00
Any modification	...	$132.00	$383.00

DEVELOPMENT AND PLANNING FEES

Lodgement fee		
Not exceeding $5,000.00	...	$68.50
Over $5,000.00	...	$77.00
Additional lodgement fee	...	$109.00
Planning fee		
Not exceeding $10,000.00	...	$42.50
$10,000 - $100,000	...	$116.00
Over $100,000	...	0.125% of cost
State heritage referral	...	$243.00

PRELIMINARIES

NOTICES AND FEES, ETC. - ADELAIDE (cont'd)

DEVELOPMENT AND PLANNING FEES (cont'd) — G.S.T. INCLUSIVE FEE

Minimum building fee	...	$74.50
Adjoining Owner notification fee	...	$116.00
Non-complying notification fee	...	$140.00
Non-complying advertising fee	...	$515.90

HOARDING FEE

Fees for hoarding licences to cover occupation of streets in the City of Adelaide :

ALL DEVELOPMENT

$2.45 per square metre of footpath occupied per week or part thereof for all distances up to and including 2.5 metres from the site building alignment for periods greater than eight weeks.

$2.85 per square metre of footpath occupied per week or part thereof for all areas in excess of 2.5 metres from the site building alignment for periods greater than eight weeks.

An application fee of $60.00 per week is payable in all cases.

CRANES

Mobile in streets and park lands	...	$108.00 per day

CONCRETE PUMP

Operation of concrete pump	...	$104.00 per day

NEW JERSEY KERBING

New Jersey kerbing on roadway (concrete or water filled barriers)	...	included in road/footpath occupation fees

LOCAL AUTHORITY FEES

Check with the relevant local authority to ascertain fees chargeable and any statutory deposits. Refer also to MISCELLANEOUS FEES.

S.A. WATER CORPORATION FEES

SEWER CONNECTION

To a junction in the sewer line (prelaid) :

100mm dia.	...	$54.50
150mm dia.	...	$54.50

When junction to be inserted (direct) :

100mm dia. up to 12m	...	$4987.00
150mm dia. up to 12m	...	$5944.00
Disconnection fee (100mm and 150mm)	...	$1176.00

PRELIMINARIES

NOTICES AND FEES, ETC. - ADELAIDE (cont'd)

S.A. WATER CORPORATION FEES (cont'd)

G.S.T. INCLUSIVE FEE

WATER CONNECTION

		Water Meter (where connection is existing)	Metered Connection (Connection and Meter)
20mm dia. up to 12m	...	$268.00	$2705.00
25mm dia. up to 12m	...	$470.00	$3080.00
40mm dia. up to 12m	...	$837.00	$4168.00
50mm dia. up to 12m	...	$1941.00	$5637.00
75 and 100mm dia.	...	Quotation	Quotation
Disconnection fee	...		$725.00
Metal underground box (20mm or 25mm service)	...		$708.00
Shorten a water service 20mm or 25mm	...		$1394.00
Raise/lower water service :			
15mm or 20mm dia.	...		$1204.00
25mm - 50mm dia.	...		$1364.00

METER RELOCATION (based on 20mm service)

Up to 0.5 metre	...	$828.00
0.5 to 1.0 metre	...	$851.00
1.0 to 2.0 metres	...	$851.00
2.0 to 3.0 metres	...	$1134.00
3.0 to 4.0 metres	...	$1134.00

FIRE SERVICES

100mm dia.	...	Quotation
150mm dia.	...	Quotation
Disconnect	...	Quotation

OTHER CHARGES

Refer scale of charges issued by S.A. Water

STAMP DUTY

Normal Agreement	...	Exempt
Under Seal	...	N/A

ENVIRONMENTAL FEES

GOVERNMENT OF SOUTH AUSTRALIA

Application to erect, construct, enlarge or modify a dam or other structure that will collect or divert water	...	each	$60.00
Drain or discharge water into a watercourse or lake	...	each	$60.00
Application for a permit to drain or discharge water into a well	...	each	$60.00
Application for a water licence in a prescribed wells area	...	each	$254.00
Application to clear native vegetation	...	each	$162.75-$651.00

EPA SOUTH AUSTRALIA

EPA licence application require by some businesses for prescribed activities that pollute or have the potential to pollute the environment	...	each	$219.00

MISCELLANEOUS FEES

CHECK LIST

Alignment fee
Inspection fees
Restoration
Road opening and crossover fee

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE

DEVELOPMENT ASSESSMENT AND COMPLIANCE FEES

DEVELOPMENT ASSESSMENT ENQUIRIES

Standard prelodgement meeting (miniumum one hour fee)	...	per hour	$590.00
AdviceSMART prelodgement meeting (miniumum one hour fee)	...	per hour	$1534.00
Discussion and calculation of infrastructure charge (minimum one hour fee)	...		$590.00
Request for written information on development related matters	...		$590.00
File scanning, preparation and storage of hardcopy lodgements	...		$236.00

DEVELOPMENT ASSESSMENT FEES

Minor development :			
Flat fee	...		$590.00
Domestic development :			
Flat fee	...		$1888.00
Demolition development :			
Demolition type A :			
Flat fee	...		$2832.00
Demolition type B :			
Flat fee	...		$944.00
Reconfiguration development :			
Base fee (upto 4 lots)	...		$3540.00
Fee per lot	...	per lot	$590.00
Multiple dwellings :			
Base fee	...		$7080.00
Fee per unit, 3-70 units	...	per unit	$472.00
Fee per unit, more than 71 units	...	per unit	$236.00
Development assessment area change :			
Base fee	...		$4248.00
Fee per 100sqm or part thereof, between 51sqm and 2050sqm	...	per 100sqm	$354.00
Fee per 100sqm or part thereof, greater than 2050sqm	...	per 100sqm	$236.00

COMPLIANCE AND INSPECTION FEES

Reconfiguration :			
Boundary realignment	...		$826.00
Access easement	...		$1298.00
1-3 lots (including boundary realignment)	...		$1888.00
4-7 lots - one project component	...		$4720.00
4-7 lots - two project components	...		$6136.00
4-7 lots - three project components	...		$7080.00
4-7 lots - four project components	...		$8614.00
8-15 lots - up to two project components	...		$9440.00
8-15 lots - three project components	...		$11210.00
8-15 lots - four project components	...		$13216.00
16-30 lots - up to three project components	...		$15340.00
16-30 lots - four project components	...		$17110.00
31 or more lots - up to three project components	...		$19116.00
31 more or lots - four project components	...		$25016.00

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE (cont'd)

DEVELOPMENT ASSESSMENT AND COMPLIANCE FEES (cont'd)

COMPLIANCE AND INSPECTION FEES (cont'd)

Additional compliance fees :			
Major infrastructure items	...	per item	$8024.00
Additional inspection fee	...	per inspection	$354.00
Approval for building over or near stormwater infrastructure/overland flow paths where assessed by Council	...		$944.00

PLAN SEALING FEES

Compliance assessment of subdivision plan or endorsement of survey plan			
Compliance assessment/survey plan endorsement	...	$415 plus $200 per lot on survey plans	
Resubmission/re-endorsement of survey plan	...	per survey plan	$472.00
Request for Council to accept bond for incomplete work			
Where the value of the bond is up to and including $200000	...	per request	$2124.00
Where the value of the bond is between $200000-500000	...	per request	$2714.00
Where the value of the bond is more than $500000	...	per request	$3186.00
Resubmission of bond request if initial request is inadequate or incomplete	...	per resubmission	$354.00
Request for approval of street naming where not part of survey plan endorsement	...	per request	$472.00
Change/written response with respect of official house numbering	...	per request	$354.00
Request to change the official house number	...	per request	$354.00

BUILDING APPLICATION FEES

Siting variations :			
New single storey dwelling including removal dwelling, two inspections, frame and final (footing and slab done by engineer)	...		$1546.86
New two storey dwelling including removal dwelling, two inspections, frame and final (footing and slab done by engineer)	...		$1811.50
Major additions/alterations (> 50 square metres), two inspections, frame and final (footing and slab done by engineer)	...		$1430.59
Outbuildings single storey with engineer's certificate	...	per inspection	$935.36
Outbuildings single storey without engineer's certificate	...	per inspection	$1027.55
Swimming pools	...	per inspection	$1186.95
Application for pool safety certificate inspection. Requires inspection, assessment and decision	...	per application	$411.05
Request for assessment against the Queensland Development Code (QDC) design and siting requirements (inspection, assessment and decision by Council)	...	per request	$826.00

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE (cont'd)

PLUMBING AND DRAINAGE FEES

Plumbing and drainage fees

Single detached dwellings and additions/alterations - sewered

Compliance assessment and inspections		
Application fee (each application) - online lodgement	...	$56.00
Application fee (each application) - hardcopy lodgement	...	$123.00
Assessment fee	...	$56.50
Inspection of plumbing and drainage - new dwelling		
Each application (up to five inspections including cancellations)	...	$816.00
Inspection of plumbing and drainage - additions and/or alterations to a single detached dwelling		
Each application (up to three inspections including cancellations)	...	$468.50

All other classes except single detached dwelling - sewered

Assessment of hydraulic plans		
Standard permit application - online lodgement	...	$56.00
Standard permit application - hardcopy lodgement	...	$123.00
Plan assessment for all classes except a single detached dwelling		
Each application	...	$175.50
Each fixture or water meter		
In addition to application and assessment fee	...	$20.50
Each testable backflow prevention device		
In addition to application and assessment fee	...	$65.50
Plans less than 10 fixtures, excluding assessments which attract a development application. The plan will be assessed within five working days of receipt of payment by Council, provided all necessary information is provided.		
In addition to standard application and fixture fees	...	$218.50
Plans up to and including 50 fixtures. The plan will be assessed within 10 working days of receipt of payment by Council, provided all necessary information is provided.		
In addition to standard application and fixture fees	...	$434.50
Plans more than 50 fixtures and up to and including 250 fixtures. The plan will be assessed within 10 working days of receipt of payment by Council, provided all necessary information is provided.		
In addition to standard application and fixture fees	...	$1002.50
Request to stage a previously submitted application for assessment of plans for all classes except a single detached dwelling		
Each stage	...	$612.00

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE (cont'd)

PLUMBING AND DRAINAGE FEES (cont'd)

Plumbing and drainage fees (cont'd)

Inspections

Inspection of plumbing and drainage installations resulting from the approval of hydraulic plans for all classes except a single detached dwelling

Each application	...	$68.00
Plus each fixture or water meter	...	$45.00
Minimum fee	...	$246.00

Inspection of common drainage and water or fire services (civil works only)

Minimum fee	...	$246.00
Per metre	...	$5.10

Minor commercial

Inspection and approval of up to four fixtures for all classes except a single detached dwelling not involving in-ground drainage or trade waste

Each inspection	...	$582.00

Other inspections and related charges - all classes

Outside business hours inspection (for up to two hours)

Minimum fee	...	$244.00
Per hour after two hours	...	$122.00

Issue notice for defective or incomplete plumbing and drainage works for all classes

Each notice	...	$103.50

Minor works - inspection only

Inspection of minor plumbing works (one inspection only) e.g. seal-off

Each inspection	...	$204.00

Inspection of pre-fabricated units (pods)

Up to five units (pods)	...	$204.00
Plus each additional unit (pod) up to 10 per application	...	$41.00

Inspection of tank	...	$113.00
Plus each additional tank up to 4 per application	...	$51.50

Backflow prevention devices

Backflow prevention devices may be required where there is potential for the potable water supply to become contaminated by the backflow of water. Testable devices must be registered and require an annual inspection by a licensed plumber who will send the results to Council. Council is required to maintain a register of all testable backflow prevention devices.

Annual registration fee - per device	...	$65.50

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE (cont'd)

PLUMBING AND DRAINAGE FEES (cont'd)

On-site sewage and greywater facilities

New dwelling and additions / alterations

Application fee - online lodgement	...	$56.00
Application fee - hardcopy lodgement	...	$123.00
Assessment fee - each application	...	$56.50
Wastewater assessment fee	...	$207.00
Inspection fee - new dwelling (up to 5 inspections)	...	$816.00
Inspection fee - additions / alterations (up to 3 inspections)	...	$468.50
Inspection fee - relocations of trenches / replacement of plant	...	$204.00

Concurrence agency (referral) advice

Request for concurrence agency (referral) advice - application	...	$384.50

All other classes except single detached dwelling

Greywater installation

Greywater systems enable reuse of water. Compliance assessment of these systems is performed to ensure the water is used appropriately and safely.

Assessment of proposed greywater treatment installation (all classes except a single detached dwelling)

Each application (in addition to the hydraulic plan assessment fees)	...	$450.00

Inspection of greywater installation for all classes except a single detached dwelling

Each application	...	$450.00

Note: For all classes except a single detached dwelling, greywater fees are charged in addition to the applicable assessment and inspection fees.

OTHER FEES

Refer list of charges issued by the Brisbane City Council (or other local authority), Brisbane Certification Group, Department of Water Supply and Sewerage and any other relevant authority/department.

LOCAL AUTHORITY FEES

Check with the relevant local authority to ascertain fees chargeable and any statutory deposits.
Refer also to MISCELLANEOUS FEES

WORKPLACE HEALTH AND SAFETY REGULATIONS 1990, FEES :

ON NOTIFIABLE SCHEDULE OF FEES

The notification of works fee is now incorporated into a single fee with the Long Service Leave Levy and the Construction Industry Training Levy. The combined fee is currently 0.475% (as at July, 2014) of the cost of construction on projects over $150000 (excl. GST).Reduced fee applies to projects exceeding $1.169 billion

PRELIMINARIES

NOTICES AND FEES, ETC. - BRISBANE (cont'd)

STAMP DUTY

Refer www.qld.gov.au for transfer duty rates

ENVIRONMENTAL FEES

G.S.T. INCLUSIVE FEE

DEPARTMENT OF NATURAL RESOURCES AND MINES

Application for permit to take water	...	No fee

DEPARTMENT OF ENVIRONMENT AND HERITAGE PROTECTION

Application for contaminated soil disposal permit	...	No fee
Obtaining extract from the environment management register or contaminated land registerv(one search per lot)		
Online search	...	$53.00

MISCELLANEOUS FEES

CHECK LIST
- Alignment fee
- Inspection fees
- Restoration
- Land tax
- Search fees

GOVERNMENT FEES

DEPARTMENT OF NATURAL RESOURCES, MINES AND ENERGY

Refer www.dmrme.qld.gov.au regarding the following :
- Land and property
- Water management
- Mining and exploration
- Energy
- Mapping and data

DEPARTMENT OF ENERGY AND SCIENCE

Refer www.des.qld.gov.au regarding the following :
- Spatial data
- Licences, permits, approvals and resource use (incl waste management)
- Searches (environmental and heritage)
- Recreation and tourism services

PRELIMINARIES

NOTICES AND FEES, ETC. - MELBOURNE

VICTORIAN BUILDING AUTHORITY - FEES FOR BUILDING SURVEYING SERVICES

Pursuant to section 188 (1)(b) of the Building Act 1993, the following Guideline has been issued concerning the charges to be made by the Building Control Commission for building surveying services including applications for permits and approvals under Parts 5 and 12 of the Act and the building regulations including applications for approval for prescribed temporary structures and places of public entertainment and applications for amendments to occupancy permits. Note that under sections 188 (6) and 199 (7) the Building Control Commission must have regard to this Guideline in fixing charges.

Building Permit or Temporary Approval

This is now based on quotation, being assessed on the amount of work involved and the professional services rendered.

Note :	A fee of 0.128% of the cost of the works is applicable to commercial works over $10000 in value (Government Duty)	...	0.128%
	Projects of $10000 and under in value	...	No levy

SUNDRY FEES, PERMITS, etc.

Check with the various authorities for all other sundry fees, permits, services, etc. such as :

Preliminary examination, structural computation
Relocation of buildings
Demolition of buildings
Sundry permits etc., i.e. exterior illumination and all other relevant fees etc.

HOARDING AND SPACE RENTAL FEES

The following fees are payable to the City of Melbourne :

Hoarding permit fee	...	$158.80
Hoarding space at ground level, together with ground level area of all office or occupied space at higher levels	...	$1.30/sqm per day (incl. G.S.T.)

Additional fee to be charged when Engineer required to check structural computations for special overhead protective awnings, etc.

Refer to the City of Melbourne Schedule of Fees for details of sundry fees.

WORK ON ROADS FEES

Refer City of Melbourne Schedule of fees

PRELIMINARIES

NOTICES AND FEES, ETC. - MELBOURNE (cont'd)

LIFT AND ESCALATOR INSPECTION FEES

This is a fee payable every three years after lifts or escalators go into service. No fees are payable during the installation period

Refer Schedule (2) of the Occupational Health and Safety (Plant) Regulations 1995

LOCAL AUTHORITY FEES

Check with the relevant local authority to ascertain fees chargeable and any statutory deposits, etc. Refer also Sundry Permits and MISCELLANEOUS FEES

CITY WEST WATER FEES

Refer other authorities depending upon location

FEES FOR PLUMBING SERVICES

Application for sewerage connection + water connection		
Standard	...	$88.15
Complex (extra cost if assessment/offer required)	...	$415.05
Single residential sewer branch design	...	$117.25
Customer requested water/sewage consent variation	...	$117.25
Property service plans	...	$31.80
Asset plots - water and sewer	...	$50.45
Asset plots - water only	...	$25.40
Sewer details easement	...	$67.90
Build over assessment application	...	$278.90
Pressure and flow information -		
Standard	...	$286.05
Urgent	...	$572.10

SERVICE CONNECTION

Tapping fee :		
20mm service - short side installation	...	$1823.65
25mm service - short side installation	...	$1951.20
32mm service - short side installation	...	$2682.70
40mm service - short side installation	...	$2838.85
50mm service - short side installation	...	$3559.05
20mm service - long side installation	...	$2217.25
25mm service - long side installation	...	$2361.90
32mm service - long side installation	...	$3554.80
40mm service - long side installation	...	$3710.95
50mm service - long side installation	...	$4442.80
20mm-50mm service - extended length installation	...	Actual cost
Plugging fee :		
20mm service - all mains	...	$653.25
25mm service - all mains	...	$653.25
32mm service - all mains	...	$670.50
40mm service - all mains	...	$674.50
50mm service - all mains	...	$685.10
80mm service and above	...	Actual cost

PRELIMINARIES

NOTICES AND FEES, ETC. - MELBOURNE (cont'd)

CITY WEST WATER FEES (cont'd)

SERVICE CONNECTION (cont'd)

Meter connection fee :

20	...	$78.25
20 recycled	...	$72.85
25	...	$253.75
25 recycled	...	$259.15
32	...	$648.60
40	...	$805.85
50	...	$987.80
80 and above	...	Actual cost

Remote meter device

Remote meter device plus 20mm meter plus installation	...	$221.55
Remote meter device plus 25mm meter plus installation	...	$342.00

Recycled water by-pass connector	...	$42.35

Removal and testing of water meter :

20mm	...	$246.10
25mm	...	$353.10
32mm	...	$426.95
40mm	...	$581.10
50mm	...	$662.15
80mm	...	Actual cost
100mm	...	Actual cost
150mm	...	Actual cost

Trade waste compliance

Non-compliance charge	...	$128.50-$334.50 per hour
Sampling charge	...	$125.00-$334.50 per hour
Laboratory charge	...	Actual cost

CONTRIBUTION TOWARDS WATER SUPPLY DISTRIBUTION WORKS

For information refer to the current schedule of contribution fees obtainable from the authority.

WATER USED DURING CONSTRUCTION

Refer to Contract Specification for details.

DEPOSIT FOR ACTUAL COST TEE INSERTION

Refer to scale of deposits issued by the authority.

TEE INSERTION. PLUMBER'S RESPONSIBILITY INCLUDES

Cost of excavation, refilling and road repairs
Cost of providing assistance to the Board's personnel for the full duration of the operation.

PRELIMINARIES

NOTICES AND FEES, ETC. - MELBOURNE (cont'd)

METROPOLITAN FIRE BRIGADE'S BOARD FEES

BUILDING APPROVALS

Applications for consent by the Chief Fire Officer to issue a building approval subject to the provisions of Section 20 of the Building Control Act.

The Metropolitan Fire Brigade's Board now operates on a price on application consultancy fee.

Dangerous site report fee (No set fee, refer to Board)

ENVIRONMENTAL FEES	G.S.T. INCLUSIVE FEE
DEPARTMENT OF ENVIRONMENT AND PRIMARY INDUSTRIES	
Application for a permit to remove native vegetation ...	No Fee

MISCELLANEOUS FEES

Consent for works (Excavations) Road Management Act 2004
Space occupation fee for each device ... $33.80-$388.50 per day

Flood Level - Land Information Certificate $41.71 through rates dept.

Restoration Fees - Check with the relevant authority.

PRELIMINARIES

NOTICES AND FEES, ETC. - PERTH

The Development Fee and Building Application Fee charges given hereafter are those of the City of Perth. As fees levied by other authorities may vary considerably, all such fees and charges should be checked in detail with the relevant authority.

DEVELOPMENT/PLANNING FEE

The following fees are payable upon the submission of applications to commence development. The fees are assessed according to the estimated cost of development :

Estimated Cost of Development (excluding G.S.T.) (other than Extractive Industry)		Fee
Less than $50,000	...	$147.00
$50,000 - $500,000	...	0.32% of estimated cost of development
$500,000 - $2,500,000	...	$1700 + 0.25% per $1 over $500,000
$2,500,000 - $5,000,000	...	$7161 + 0.206% per $1 over $2,500,000
$5,000,000 - $21,500,000	...	$12633 + 0.123% per $1 over $5,000,000
Greater than $21,500,000	...	$34196.00
Subdivision Clearance		
Not less than 5 lots	...	$73 per lot
5 lots - 195 lots	...	$73 for first 5 lots then $35 per lot
Greater than 195 lots	...	$7393.00
Approval of Home Occupation		
Initial fee	...	$222.00
Renewal fee	...	$73.00
Change of Use	...	$295.00
Zoning Certificate	...	$73.00
Written Planning Advice	...	$73.00
Development Assessment Panel Fees :		
$10.0 million to $12.5 million	...	$9411.00
$12.5 million to $15.0 million	...	$9680.00
$15.0 million to $17.5 million	...	$9948.00
$17.5 million to $20.0 million	...	$10218.00
Greater than $20 million	...	$10486.00

BUILDING APPLICATION FEE

Based on estimated cost of development (including G.S.T.) :		
Dwellings (Class 1 and 10) - Certified	...	0.19% Minimum $105.00
Other developments (Classes 2 to 9 incl.)	...	0.09% Minimum $105.00
Construction Training Levy	...	0.20%

WORKS BOND

Refer table of fees issued by the City of Perth

PRELIMINARIES

NOTICES AND FEES, ETC. - PERTH (cont'd)

CONSTRUCTION SAFETY FEE

No longer any fee payable

LIFT AND ESCALATOR INSTALLATION FEES

No longer any fee payable

LOCAL AUTHORITY AND SUNDRY LICENCE FEES, etc.

Check with the relevant local authority to ascertain fees chargeable and any statutory deposits. Refer also to MISCELLANEOUS FEES.

WATER CORPORATION FEES AND CHARGES

BUILDING APPLICATION FEE

FULL CONSTRUCTION COST	FEE FOR OTHER THAN SINGLE RESIDENTIAL BUILDING	FEE FOR SINGLE RESIDENTIAL
Up to $22,500	$31.00	$125.99
Over $22,500 but less than $200,000	$106.00	$125.99
Over $200,000 but less than $500,000	$420.00	+ Service Activation Fee $141.87
Over $500,000 but less than $1,000,000	$680.00	+ Sewer Connection Fee $22.20
Over $1,000,000 but less than $10,000,000	$1.10 for each $1000	
Over $10,000,000	$11000 plus $0.30 for each $1000	

PLUMBING INSPECTION FEES

Refer 'SEWER CONNECTION FEES'

WATER METERS AND FIRE SERVICES FEES

Water mains and fire services connection :

20mm dia. (Domestic service only)	...	$1267.47
25mm dia.	...	$1284.19
40mm dia.	...	$1869.49
50mm dia.	...	$2310.17
80mm dia.	...	$4473.43
100mm dia.	...	$4473.43
150mm dia.	...	$5339.94

Relocations :		Under 500mm	Over 500mm
20mm dia.	...	$478.50	$1407.68
25mm dia.	...	$545.81	$1424.37
40mm dia.	...	$778.96	$2010.81
50mm dia.	...	$913.03	$2450.93
80mm dia.	...	Quotation	$4615.29
100mm dia.	...	Quotation	$4615.29
150mm dia.	...	Quotation	$5481.04

Water mains and fire service connections - C.B.D.	...	By quotation only

WATER USED DURING CONSTRUCTION

Refer to Water Corporation for residential and commercial sliding scale of water usage charges

PRELIMINARIES

NOTICES AND FEES, ETC. - PERTH (cont'd)

WATER CORPORATION FEES AND CHARGES (cont'd)

CONTRIBUTION TO HEADWORKS SYSTEM

Applies in the first instance to all subdivisions and then at the building stage to all buildings other than single residential.

Refer to schedule of payments issued by the Water Corporation or country local authority.

SEWER CONNECTION FEES

New sewer connection fee, applies to first connection only	...	$22.20

SEWER JUNCTIONS

New junction to sewer main :		
100mm junction		
- Dry or Wet	...	$545.04
150mm junction		
- Dry or Wet	...	$679.45

ADD extra to above for :
Where main is concrete encased by quotation
Other jobs and weekend work by quotation

STORMWATER BOUNDARY CONNECTION

Refer Water Corporation for quotation

UTILITY SERVICES EXTENSION COSTS

Refer Water Corporation for quotation

OTHER CHARGES

Refer Water Corporation fixed fees and charges

ENVIRONMENTAL FEES

G.S.T. INCLUSIVE FEE

DEPARTMENT OF WATER AND ENVIRONMENT REGULATION

Application for native vegetation clearing		Intensive land-use zone	Extensive land-use zone
Area < 1ha	...	$400.00	$400.00
Area 1 to 5 ha	...	$600.00	$600.00
Area 5 to 10 ha	...	$1500.00	$750.00
Area 10 to 50 ha	...	$2000.00	$1000.00
Area 50 to 100 ha	...	$3000.00	$1500.00
Area 100 to 500 ha	...	$4000.00	$2000.00
Area 500 to 1000 ha	...	$5000.00	$2500.00
Area > 1000 ha	...	$10000.00	$5000.00

Application for a license as a controlled waste carrier required for transporting wastes including waste waters, contaminated soils, tyres, grease wastes etc.		
1 year	...	$225.00
3 year	...	$555.00
5 year	...	$885.00

PRELIMINARIES

NOTICES AND FEES, ETC. - PERTH (cont'd)

ENVIRONMENTAL FEES (cont'd) — G.S.T. INCLUSIVE FEE

DEPARTMENT OF WATER

Application for licence to take water :		
Low risk	…	$5357.00
Medium risk	…	$7143.00
High risk	…	$8929.00

MISCELLANEOUS FEES

CHECK LIST

Alignment fees
Development fees
Inspection fees
Restorations
Road opening and crossover fees
Local authority services contribution fees

PRELIMINARIES

NOTICES AND FEES, ETC. - SYDNEY

For a full list of fees refer to Sydney City Council's
schedules of fees & charges from 1 July 2020 to 30 June 2021

DEVELOPMENT APPLICATION FEES		G.S.T. INCLUSIVE FEE
ESTIMATED VALUE OF CONTRACT		
Up to $5000	...	$110.00 per application
$5,001 but not exceeding $50,000	...	$170.00 + $3.00 per $1,000
$50,001 but not exceeding $250,000	...	$352.00 + $3.64 per $1,000 above $50,000
$250,001 but not exceeding $500,000	...	$1,160.00 + $2.34 per $1,000 above $250,000
$500,001 but not exceeding $1,000,000	...	$1,745.00 + $1.64 per $1,000 above $500,000
$1,000,001 but not exceeding $10,000,000	...	$2,615.00 + $1.44 per $1,000 above $1,000,000
More than $10,000,000	...	$15,875.00 + $1.19 per $1,000 above $10,000,000
ADVERTISING/NOTIFICATION FEES FOR COMPLYING DEVELOPMENT APPLICATIONS		
Application that involves a new dwelling or addition to an existing dwelling or land	...	$285.00 per application
SUBDIVISIONS		
STRATA TITLE SUBDIVISION (STRATA SCHEMES ACT)		
Application Fee plus additional fees listed below	...	$575.00 per application
Additional Fees for each proposed lot -		
1-10 lots	...	$585.00 per lot
11-50 lots	...	$5850.00 + $420.00 per lot over 10 lots
Exceeding 50 lots	...	$22650.00 + $205.00 per lot over 50 lots
Certificate of conversion	...	$480.00 per application
LAND SUBDIVISION (EP&A Act)		
Land Subdivision - Only involving road dedication to council	...	$0.00 per application
Land Subdivision - Other		
Application Fee plus additional fees listed below	...	$595.00 per application
Additional Fees -		
For each proposed lot	...	$595.00 per lot
Re-execution of Strata Plan (minimum fee $350)	...	$74.00 per sheet
BUILDING APPLICATION FEES		
Construction Certificate and Complying Development Certificate under the EP&A Act		
ESTIMATED VALUE OF CONTRACT		
Class 1a and 1b residential buildings :		
Up to $5,000	...	$412.00 per application
$5,000 - $10,000	...	$486.00 per application
$10,000 - $100,000	...	$655.00 per application
$100,000 - $250,000	...	$1154.00 per application
Exceeding $250,000	...	$1154.00 + $2.36 per $1,000 above $250,000
Class 2 - 9 buildings :		
Less than $10,000	...	$655.00 per application
$10,000 - $100,000	...	$655 + $5.55 per $1,000 above $10,000
$100,000 - $500,000	...	$1190 + $3.15 per $1,000 above $100,000
$500,000 - $1,000,000	...	$2490 + $2.34 per $1,000 above $500,000
Exceeding $1,000,000	...	$3687 + $1.64 per $1,000 above $1,000,000

PRELIMINARIES

NOTICES AND FEES, ETC. - SYDNEY (cont'd)

BUILDING APPLICATION FEES (cont'd)		G.S.T. INCLUSIVE FEE
Class 10a and 10b buildings :		
Up to $10,000	...	$418.00 per application
$10,000 - $100,000	...	$490.00 per application
Exceeding $100,000	...	$654.00 per application
INSPECTION FEES (For council acting as principal certifying authority)		
Residential Dwellings (Class 1a & 1b)	...	$241.00 per inspection
Class 10 & 10b Buildings & Signs	...	$241.00 per inspection
Class 2-9	...	$357.00 per inspection
Re-inspection due to incomplete work	...	$357.00 per inspection
Multiple Unit Inspection Fee (In addition to critical stage inspection) 6 items or more	...	$62.22 per item
Multiple Unit Inspection Fee (In addition to critical stage inspection) less than 6 items	...	$357.00 per hour
HOARDING AND RELATED ACTIVITIES FEES		
(Charged per month or part thereof)		
Application Fee	...	$367.00 per application
A Class hoardings without traffic barrier -		
City East, City South, Ultimo, Pyrmont	...	$7.89 lineal mt/week
A Class hoardings without traffic barrier -		
General CBD	...	$10.82 lineal mt/week
A Class hoardings with traffic barrier -		
City East, City South, Ultimo, Pyrmont	...	$9.42 lineal mt/week
A Class hoardings with traffic barrier -		
General CBD	...	$13.42 lineal mt/week
B Class hoardings without sheds -		
City East, City South, Ultimo, Pyrmont	...	$10.82 lineal mt/week
B Class hoardings without sheds -		
General CBD	...	$15.20 lineal mt/week
B Class hoardings with single storey sheds -		
City East, City South, Ultimo, Pyrmont	...	$15.86 lineal mt/week
B Class hoardings with single storey sheds -		
General CBD	...	$29.11 lineal mt/week
B Class hoardings with double row sheds -		
City East, City South, Ultimo, Pyrmont	...	$31.72 lineal mt/week
B Class hoardings with double row sheds -		
General CBD	...	$58.23 lineal mt/week
B Class hoardings with double storey sheds -		
City East, City South, Ultimo, Pyrmont	...	$30.80 lineal mt/week
B Class hoardings with double storey sheds -		
General CBD	...	$43.34 lineal mt/week
B Class hoardings with double storey double row sheds -		
City East, City South, Ultimo, Pyrmont	...	$61.71 lineal mt/week
B Class hoardings with double storey double row sheds -		
General CBD	...	$86.68 lineal mt/week
Form compound area Zone 1 - Circular Quay, City Centre	...	$14.41 per sqm/week
Form compound area Zone 2 - All other areas	...	$7.20 per sqm/week

PRELIMINARIES

NOTICES AND FEES, ETC. - SYDNEY (cont'd)

SECURITY DEPOSITS OR BANK GUARANTEES FOR HOARDINGS		G.S.T. INCLUSIVE FEE
Minimum of $10,000 or the greater of the following to a maximum of $100,000		
A Class on normal/old paving	...	$0.00 per sqm
A Class on new granite paving	...	$671.00 per sqm
B Class on normal/old paving	...	$1337.00 per lineal m
B Class on new granite paving	...	$671.00 per sqm
Street trees impacted by the erection of the Hoarding	...	$2000-$20000 per tree

BARRICADES AND RELATED ACTIVITIES FEES		
Approval for placement of barricades on footway	...	$95.00 per week or part of
Road opening permit only	...	$128.00 per opening
Road opening deposits - per permit	...	$2800 per permit
Road restorations vary between	...	$344.00 to $711.00 per square metre
Kerb deposits - concrete kerb and gutter	...	$1445.00 per lineal metre
Kerb & Gutter restorations	...	$646.00 to $1665 per lineal metre
Emergency barricades and lights - Initial attendance (Call out)	...	$334.00 each
Time charge on site	...	$66.30 per hour

For a full list refer to Sydney City Council's schedules of fees & charges

LOCAL AUTHORITY FEES

Check with the relevant local authority to ascertain fees chargeable and any statutory deposits. Refer also to MISCELLANEOUS FEES.

City of Sydney - Sundry Fees :

Crane Operations - approval to stand & operate a mobile crane, concrete boom pump etc.	...	$190.00 per day
Hoisting activity	...	$367.00 per application
Application to close a roadway	...	$367.00 per application
Temporary partial minor road lane closure - 24 hours	...	$617.00 per lane/day
Temporary partial major road lane closure - 24 hours	...	$1235 per lane/day
Hoist operations - approval to stand and operate a hoist on the footway or B Class hoarding	...	$190.00 per week
Temporary footway crossing - standard footway crossing or conduit ramp	...	$95.00 per week

PRELIMINARIES

NOTICES AND FEES, ETC. - SYDNEY (cont'd)

SYDNEY WATER FEES AND CHARGES

For a full list of fees refer to Sydney Water's ancillary charges from 1 July 2020 to 30 June 2021

Hourly rate to administer a service not listed in the price sheet	...		$165.52 per hour
Building over or adjacent asset letter	...	each	$47.02
Note that these fees are Sydney Water's administration charges and do not include the cost of carrying out the physical work described			
Conveyancing Certificate			
Electronic	...	each	$7.16
Property Sewerage Services Diagram (Up to A4)			
Over the counter	...	each	$24.56
Electronic	...	each	$13.67
Section 88g certificate	...	each	$7.88
Service Location Diagram - Location of sewer and or water main in relation to a properties boundaries			
Over the counter	...	each	$16.55
Electronic	...	each	$7.80
Building plan approval			
Application per approval of council approved building plans	...	each	$17.63
Request for Asset Construction Details			
Detailed Plans of Sydney Water's assets showing water sewer and drainage	...	each	$51.54
Sydney Water Supply System Diagram	...	each	$138.43
Applications for water service connections - administration fees only			
Up to 25mm	...	each	$0.00
32mm or greater	...	each	$334.18
Disconnection	...	each	$0.00

MISCELLANEOUS FEES

CHECK LIST

Builder's Licensing Board fees
Inspection fees
Restoration
Road opening and crossover fees

PRELIMINARIES

MISCELLANEOUS COSTS

- The following additional items are noted as a check list

ADVERTISING

AUDITS
- Audit of compliance with the
  - Construction Environmental Management Plan
  - Environmental Management System

CLEANING
- Contaminated waste bags
- Mobile spill kit
- vehicle spill kit

- Interim
  - Labour
  - Truck hire
  - Tip fees

- Rubbish chute
  - Material
  - Erect and dismantle

- Final
  - Labour
  - Truck hire
  - Tip fees

COMPOUND
- For material storage

CONTRACT DOCUMENTS

COST ADJUSTMENT
- Rise and Fall or Fixed Price

CROSSOVERS

DEFECTS LIABILITY/MAINTENANCE

EMPLOYEE BENEFITS
- Long service leave
- Severance and redundancy pay
- Site allowance
- Superannuation

ENVIRONMENTAL MONITORING
- Dust gauges
- Vibration meters
- Weather stations

ENVIRONMENTAL REMEDIATION
- Nest boxes

EXISTING SERVICES
- Disconnect or divert :
  - Water
  - Gas
  - Sewer
  - Stormwater
  - Electrical
  - Telephone

FINANCE
- Performance Bond costs
- Security Deposit charges
- Bank Guarantee charges
- Interest on Retention
- Turnover Tax

FIRE SERVICE, TEMPORARY
- Extinguishers
- Hydrants

FIRST AID

INSPECTIONS
- Environmental site inspections
- Pre-works inspections
  - Habitat clearing inspection
  - Tree clearing inspection
  - Weed clearing inspection

MAKE GOOD
- External :
  - Road
  - Fences
  - Footpaths
  - Facade
- Internal :
  - Dust protection

MANAGEMENT PLANS
- Aboriginal Heritage
- Air Quality
- Bushfire
- Chemical and Dangerous Goods
- Clearing and Grubbing
- Climate Change
- Construction Environmental
- Contaminated Land
- Erosion and Sediment Control
- Flora/Fauna
- Ground Vibration and Air Blast
- Noise
- Non-Aboriginal Heritage
- Pest
- Safety
- Site Restoration
- Soil and Water
- Traffic
- Waste
- Weed

NOTICE BOARD
- Including signwriting

OFFICE ADMINISTRATION

OFFICE FURNITURE AND EQUIPMENT

P.A. SYSTEM

PETTY CASH

PHOTOGRAPHS

PLANT
- Depreciation
- Maintenance

PROTECTION
- Adjoining buildings
- Services
- External facade
- Windows
- Internal
- Carpet

PROTECTIVE CLOTHING

QUALITY ASSURANCE

RESOURCES
- Environmental Management Representative
- Environmental Site Representative
- Soil Conservation Consultant

SCHEDULING
- Computer or otherwise

SECURITY SERVICES/WATCHMAN

SETTING OUT
- Licensed surveyor, refer Scale of Fees for Surveyors laid down by the Institute of Surveyors, Australia

SHOP DRAWINGS

SIGNAGE
- Contaminated area signage
- Environmentally sensitive area signage
- Stockpile signage

SITE ACCESS

SITE MEETINGS

SITE VISITS

SPECIAL ITEMS
- Refer Specification

STAFF VEHICLES

TAXES
- Goods and Services Tax
- Fringe Benefit Tax
- Payroll Tax

TEMPORARY ROADS

TESTING AND REPORTING
- Air quality monitoring
- Dilapidation reports
- Noise monitoring
- Pre-clearing ecological assessments
- Soil testing
- Surface and ground water testing
- Vibration monitoring

TESTS AND SAMPLES

NOTE :
- Check that they are not covered in other trades :
  - Tests, generally
  - Tests, concrete
  - Samples
  - Prototypes
  - Guarantee

DEMOLITIONS AND ALTERATIONS

DEMOLITIONS

WHOLE STRUCTURES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices assume normal demolition periods, no site access problems, and include for removal of all debris	...	Note					
ADD extra for work executed outside normal working hours	...	Note					
No allowance has been made for removal of asbestos	...	Note					
Refer to page 213 for tip fees	...	Note					
No allowance included for credit value of demolished material	...	Note					
Demolition including grubbing up foundations, sealing off services and removing debris of :							
FACTORIES, WAREHOUSES :							
Single storey, light industry/warehouse, reinforced concrete ground slab, framed walls, metal roof	...	sqm	40.90	45.30	50.80	42.80	60.50
Ditto, with brick walls	...	sqm	53.70	62.00	72.30	54.60	91.20
Single storey, heavy industry, reinforced concrete ground slab, framed walls, metal roof	...	sqm	53.10	57.10	65.60	55.00	76.80
Ditto, with brick walls	...	sqm	63.80	76.60	85.80	66.00	102.30
HOUSES :							
Single storey, timber framed with timber cladding, metal roof	...	sqm	36.90	48.50	43.40	37.00	53.10
Single/two storey, brick and tile	...	sqm	43.70	58.10	58.10	43.00	63.70
OFFICE BUILDINGS :							
Two storey, reinforced concrete slab, brick walls, metal roof	...	sqm	94.30	96.90	101.30	86.50	113.20
Six storey ditto	...	sqm	105.10	126.00	106.60	98.20	124.10
Fifteen storey, reinforced concrete frame and slabs, brick/stone external walls	...	sqm	116.10	137.00	128.70	119.00	137.30
Fifteen storey ditto but with structural steel frame	...	sqm	145.20	157.80	145.70	130.00	149.20
ADD extra for : Prestressed concrete structure	...	+ 50%					
RETAIL PREMISES :							
Single storey, reinforced concrete ground slab, brick walls, metal roof	...	sqm	61.30	73.90	82.60	62.20	99.60
Two storey ditto with reinforced concrete slabs	...	sqm	94.30	101.80	101.30	86.20	115.10
Three storey ditto but with structural steel frame and timber floors	...	sqm	72.70	90.00	82.60	74.70	102.20

DEMOLITIONS AND ALTERATIONS

DEMOLITIONS (cont'd)

PARTIAL DEMOLITION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices allow for the removal of debris to a loading point within 50 metres	...	Note					
ADD extra for :							
Work executed outside normal working hours	...	Note					
Removal of debris from upper floor levels	...	Note					
Disposal of debris and tip fees, refer page 213	...	Note					
Cut away concrete ground slab :							
100mm thick unreinforced	...	sqm	36.00	35.40	46.60	33.10	50.70
150mm thick ditto	...	sqm	47.80	49.60	61.10	43.70	62.50
100mm thick reinforced	...	sqm	47.80	49.60	61.10	43.70	62.50
150mm thick ditto	...	sqm	58.30	61.90	76.00	55.90	74.70
Cut away reinforced concrete suspended slab :							
125mm thick	...	sqm	69.70	67.60	97.20	67.40	78.10
200mm thick	...	sqm	111.70	117.50	155.70	107.40	128.70
250mm thick	...	sqm	137.70	153.00	192.00	130.50	162.50
Cut away reinforced concrete :							
Walls	...	cum	427.00	405.00	600.00	386.00	561.00
Columns	...	cum	323.00	281.00	566.00	289.00	572.00
Beams	...	cum	398.00	385.00	566.00	358.00	572.00
Plinths, 100mm thick	...	sqm	41.90	39.80	83.00	37.20	78.10
Demolish external brick walls :							
110mm thick	...	sqm	23.90	23.80	30.60	22.80	39.10
230mm thick	...	sqm	43.40	39.60	50.00	42.90	39.10
280mm thick cavity wall	...	sqm	43.40	44.10	50.00	46.00	80.20
Demolish external stone walls :							
300mm thick	...	sqm	59.30	60.80	83.00	62.70	104.40
500mm thick	...	sqm	87.00	94.00	130.50	97.10	160.40
Demolish internal walls :							
75mm or 110mm brick walls	...	sqm	28.30	29.00	30.40	30.10	36.20
Ditto plastered both sides	...	sqm	30.00	30.60	30.40	32.30	36.20
Stud partition with plasterboard both sides	...	sqm	14.40	13.70	14.35	15.45	14.00
Stud partition with metal lathing and render both sides	...	sqm	19.70	18.15	17.70	17.50	17.10
Strip roof covering to sloping roof :							
Slates including battens and sarking	...	sqm	8.10	10.15	8.40	8.70	9.20
Tiles ditto	...	sqm	7.90	10.15	8.40	8.20	9.20
Asbestos including battens and safety mesh	...	sqm	36.00	31.90	62.70	25.10	37.60
Metal roofing	...	sqm	9.95	14.15	9.85	10.25	11.00
Strip roof covering to flat roof :							
Sheet metal decking including insulation	...	sqm	11.05	12.20	11.65	11.85	11.00
Felt or composition roofing	...	sqm	18.45	21.10	13.80	18.55	18.80
Take down :							
Timber roof framing	...	sqm	12.00	14.15	13.80	11.75	9.20
Eaves lining	...	sqm	12.00	13.55	9.85	10.25	12.60
Eaves gutter	...	m	6.15	7.65	5.80	7.50	8.50
Downpipes	...	m	6.15	7.65	5.80	7.50	8.50

DEMOLITIONS AND ALTERATIONS

DEMOLITIONS (cont'd)

PARTIAL DEMOLITION (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Take up or strip floors :							
Timber floor framing	...	sqm	8.10	10.00	9.85	8.55	8.75
Floor boards including preparing joists for new	...	sqm	13.10	13.70	19.70	13.40	10.60
Carpet including fixing strips	...	sqm	2.80	3.40	2.80	2.50	3.50
Carpet, adhesive laid	...	sqm	7.90	10.00	7.15	8.30	9.20
Vinyl tiles	...	sqm	12.30	14.15	11.65	13.45	15.10
Vinyl sheet	...	sqm	8.55	10.55	9.85	9.20	10.30
Parquetry flooring	...	sqm	8.55	10.55	13.00	9.20	10.30
Ceramic/mosaic tiles	...	sqm	18.45	20.20	17.80	19.35	21.70
Granolithic	...	sqm	19.75	23.80	22.00	22.50	21.70
Hack finishes from walls (no preparation for new work included) :							
Plaster	...	sqm	13.30	15.75	16.20	14.05	18.10
Cement render	...	sqm	14.30	17.65	22.00	14.85	18.90
Ceramic tiles and backing screed	...	sqm	19.75	28.50	23.80	21.50	23.00
Take down :							
Suspended ceiling and metal suspension grid	...	sqm	6.75	9.65	10.70	7.50	10.40
Timber framed and sheet lined ceiling	...	sqm	12.80	16.75	13.80	14.35	13.60
Lath and plaster ceiling	...	sqm	21.20	29.00	32.70	22.30	29.40
Asbestos sheet ceiling lining	...	sqm	39.80	52.90	70.90	49.10	41.60
ADD extra for :							
Special requirements such as vacuum air extraction, screens if occupied	...	Note					
Take out :							
Single door and timber frame	...	No	33.10	42.10	41.80	34.50	37.50
Window and frame (up to 2000 x 1000mm)	...	No	64.80	75.00	76.10	70.80	75.20
Room air conditioner units built in	...	No	39.80	42.10	55.50	41.90	46.00
Counter/bench unit	...	m	24.00	35.40	25.90	25.70	29.50

ALTERATIONS, ETC.

OPENINGS

Cut 2100 x 900mm opening for door, provide and build in lintel and face up jambs in :							
110mm brick wall	...	No	266.00	325.00	360.00	283.00	339.00
230mm brick wall	...	No	524.00	633.00	572.00	530.00	633.00
280mm cavity brick wall	...	No	541.00	633.00	593.00	530.00	680.00
150mm reinforced concrete wall	...	No	375.00	474.00	502.00	393.00	477.00
ADD extra for :							
Making good plaster or render one side	...	No	74.50	82.80	97.50	72.40	100.00
Making good white glazed wall tiling one side	...	No	223.00	270.00	287.00	232.00	275.00
Making good boarded floor including bearers	...	No	53.50	61.90	67.10	56.00	62.90
Making good vinyl floor including screed	...	No	58.30	77.00	76.20	63.60	69.70
Cut 2100 x 1600mm opening for door, provide and build in lintel and face up jambs in :							
110mm brick wall	...	No	423.00	529.00	538.00	442.00	537.00
230mm brick wall	...	No	833.00	1013.00	856.00	817.00	993.00
280mm cavity brick wall	...	No	849.00	1031.00	856.00	834.00	1022.00
150mm reinforced concrete wall	...	No	659.00	818.00	733.00	657.00	819.00

DEMOLITIONS AND ALTERATIONS

ALTERATIONS, ETC. (cont'd)

OPENINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cut 2100 x 1600mm opening for door, provide and build in lintel and face up jambs in (cont'd) :							
ADD extra for :							
Making good plaster or render one side	...	No	91.00	109.30	118.80	93.30	124.30
Making good white glazed wall tiling one side	...	No	307.00	385.00	380.00	335.00	375.00
Making good boarded floor including bearers	...	No	97.00	107.80	121.10	101.60	113.70
Making good vinyl floor including screed	...	No	97.00	121.60	132.40	101.60	124.30

ADDITIONS AND NEW WORKS WITHIN EXISTING BUILDING

Refer page 210	...	Note					

PREPARATORY WORK/REPAIRS

WALLS

Prepare existing walls for new finishes :							
Hack concrete for key	...	sqm	15.65	19.55	17.80	15.50	24.10
Rake out joints and hack face of brickwork	...	sqm	12.25	13.55	18.10	12.75	15.90
Dubbing in cement and sand (1:3) average 13mm thick	...	sqm	8.75	11.90	11.90	9.70	18.70

FLOORS

Prepare existing concrete floors for new finishes :							
Hack concrete for key	...	sqm	17.65	20.40	20.70	16.20	20.30
Grind to provide a true, smooth surface	...	sqm	13.50	16.75	16.45	13.70	16.25

WINDOWS AND DOORS

Overhaul ease and adjust :							
Single door	...	No	51.30	55.50	79.20	51.00	58.20
Casement window (1.0 sqm)	...	No	75.60	101.30	87.30	80.00	105.80
Double-hung sash window (1.0 sqm)	...	No	75.60	101.30	121.90	80.00	105.80
Reglazing windows :							
Hack out, clean rebates and prepare for reglazing	...	sqm	62.00	83.60	84.60	67.30	82.60

PREPARATION FOR REDECORATION

Refer page 453	...	Note					

REMOVAL OF ASBESTOS FIREPROOFING

Remove asbestos fireproofing from structural steel framing in office buildings and the like, including all usual protective measures and approved method of disposal (per sqm of floor area)	...	sqm	320.00-440.00	320.00-440.00	320.00-440.00	320.00-440.00	320.00-440.00
ADD extra for :							
Small areas	...	Note					
Removal and relocation of furniture and the like	...	Note					

DEMOLITIONS AND ALTERATIONS

ALTERATIONS, ETC. (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
REMOVAL OF ASBESTOS FIREPROOFING (cont'd)							
ADD extra for (cont'd) :							
Rectification of flush ceilings including new areas where required	...	Note					
Repairs to existing finishes and services	...	Note					
New fireproofing of steelwork, refer pages 671 to 673	...	Note					
WORK IN CONNECTION WITH PLUMBING							
Take out sanitary fittings complete with brackets, overflow, services, waste and soil pipe connections and make good floor and wall surfaces :							
Sink or basin	...	No	201.00	254.00	241.00	197.00	231.00
W.C. suite	...	No	210.00	264.00	248.00	203.00	239.00
Bath	...	No	740.00	909.00	842.00	738.00	805.00
Cut into existing 100mm dia. cast iron stack and provide and insert branch	...	No	201.00	254.00	239.00	197.00	231.00
Ditto with inspection opening	...	No	217.00	274.00	259.00	217.00	231.00
Cut chase in brick wall for :							
Small pipe	...	m	14.35	16.75	19.55	13.85	19.15
Large pipe	...	m	18.80	23.30	23.40	18.25	24.10
Cut hole for new small pipe through :							
110mm brick wall	...	No	14.35	16.75	18.45	13.10	15.70
230mm brick wall	...	No	22.90	30.30	29.60	23.00	24.90
Cut hole for new large pipe through :							
110mm brick wall	...	No	26.60	30.30	31.80	24.40	29.40
230mm brick wall	...	No	37.00	44.00	44.00	36.30	42.20
Concrete Cutting refer page 243	...	Note					
TEMPORARY SCREENS							
Temporary screen comprising timber framing covered one side with :							
Building paper taped at joints and edges	...	sqm	20.10	19.15	23.30	19.45	28.70
Polythene sheet ditto	...	sqm	24.30	21.10	23.60	23.30	30.30
3mm hardboard	...	sqm	33.00	30.60	34.00	33.30	38.50
13mm insulation board	...	sqm	38.30	42.40	42.20	38.50	41.80
PROTECTION							
Protect floors with polythene, hessian or sawdust	...	sqm	2.70	3.00	3.20	2.70	2.85
Protect handrails with felt tied on	...	m	6.75	7.30	8.40	7.50	8.05
Protect timber stair treads with temporary softwood treads	...	m	8.85	9.15	10.15	9.15	10.05

DEMOLITIONS AND ALTERATIONS

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $

ADDITIONS AND NEW WORKS WITHIN EXISTING BUILDING

On the assumption that the additions and new works within the existing building include reasonable quantities of the respective items then the prices given in the Detailed Prices of the respective sections will apply with the following provisos :

EXCAVATION

Will preclude the use of mechanical plant, therefore the prices for hand excavation will apply with the addition of 30/50% for additional disposal costs ... Note

CONCRETE WORK

Due to additional handling costs the prices should be increased by 20/30% ... Note

BRICKWORK

Due to the nature of the work and additional handling costs, prices should be increased by 50/75% ... Note

PLUMBING AND DRAINAGE

Due to short runs and connections to existing, each scheme will have to be examined in detail and a suitable percentage added ... Note

ALL OTHER TRADES

Likely to be affected to a lesser extent but examine and where necessary add a suitable percentage ... Note

DAYWORKS

No allowance has been included for sundry dayworks ... Note

EXCAVATION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices of excavation and subsequent disposal relate to volume before excavation	...	Note					
Prices include allowance for bulking factors as follows :							
Sand 10% increase							
Light soil 20-30% increase							
Clay 30-50% increase							
Rock 50-75% increase	...	Note					
All excavation is by machine unless otherwise stated	...	Note					
Prices for excavation include for backfilling of trenches and pits and for removal of surplus material from site, a distance of approximately 10km, including payment of dumping fees of up to $2.50/cum. Refer page 213 for a guide to tip fees	...	Note					

SITE PREPARATION

Clear site of vegetation and cart away :							
Light vegetation	...	sqm	0.71	0.33	0.41	0.38	0.39
Medium vegetation	...	sqm	1.19	0.51	0.56	0.59	0.55
Clear bush with bulldozer, ball and chain, grub up roots and burn on site :							
Light bush	...	ha	875.00	1030.00	1045.00	1050.00	1150.00
Medium bush	...	ha	2470.00	2980.00	3115.00	3080.00	3200.00
Excavate to remove vegetable soil average 150mm deep and spread and level on site	...	sqm	1.75	2.25	2.25	1.90	1.85
Cut down tree, grub up stump, roots and cart away or burn on site :							
500mm girth - 1 off	...	No	211.00	207.00	180.00	217.00	187.00
500mm girth - 10 off	...	No	184.00	127.00	148.00	189.00	166.00
500/1000mm girth - 1 off	...	No	291.00	307.00	239.00	298.00	195.00
500/1000mm girth - 10 off	...	No	253.00	187.00	204.00	266.00	172.00
1000/2000mm girth - 1 off	...	No	538.00	565.00	513.00	535.00	530.00
1000/2000mm girth - 10 off	...	No	435.00	367.00	435.00	449.00	441.00
2000/3000mm girth - 1 off	...	No	939.00	1265.00	958.00	975.00	1123.00
2000/3000mm girth - 10 off	...	No	645.00	800.00	855.00	832.00	963.00
Cut existing turf into sods size 600 x 300mm stack and keep watered for re-use	...	sqm	5.95	5.10	6.55	6.45	8.10
Break up and remove reinforced concrete in :							
Open excavations	...	cum	213.00	187.00	210.00	256.00	317.00
Trenches	...	cum	282.00	231.00	273.00	316.00	392.00
Break up and remove brickwork in :							
Open excavations	...	cum	113.00	80.00	117.00	128.00	223.00
Trenches	...	cum	163.00	96.00	168.00	190.00	276.00
Break up and remove reinforced concrete ground slab - refer page 206	...	Note					
Break up and remove bitumen paving with basecourse under	...	sqm	3.95	3.80	3.65	4.15	3.55
Take down and remove :							
Timber fence 1800mm high	...	m	5.50	4.70	4.30	5.45	6.15
Chain link fence 1800mm high	...	m	4.95	4.35	3.95	5.05	6.15
Silt fence, refer page 713	...	Note					

EXCAVATION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DEWATERING							
Prices are a guide only and are for the total area to be dewatered and include for 150mm dia. header pipe, well points, pump, etc.	...	Note					
SHALLOW system (to reduce water level by 1000mm), comprising header with well points at 1500mm centres, including installation, daily running costs for average duration and removal	...	sqm	46.00-58.00	46.00-58.00	45.00-57.00	45.50-57.50	47.00-59.00
DEEP system (to reduce water level by over 1000mm), comprising header with well points at 1000mm centres, including installation, daily running costs for average duration and removal	...	sqm	57.00-69.00	57.00-69.00	56.00-68.00	56.50-68.50	56.00-68.00
ADD extra for :							
Turbidity meter, monitoring and flocculent agents if required	...	Note					
EXCAVATION							
Prices for pit and trench excavation not exceeding 1.00m deep including planking and strutting and deposit in spoil heaps within 500m	...	Note					
Based on minimum quantity of 500 cum	...	Note					
Excavate over site to reduce levels :							
in sand	...	cum	17.35	13.80	15.95	11.85	21.10
in light soil	...	cum	17.95	15.15	20.80	13.65	22.50
in clay	...	cum	19.35	16.55	21.45	20.70	31.70
in soft rock	...	cum	35.40	30.20	39.80	39.60	75.40
in hard rock	...	cum	95.00	96.10	168.00	116.50	118.50
Excavate for basement or similar :							
N.E. 3.00m deep -							
in sand	...	cum	17.35	16.55	18.85	15.85	21.20
in light soil	...	cum	17.95	17.65	23.90	17.80	22.50
in clay	...	cum	19.35	19.45	25.70	25.40	31.70
in soft rock	...	cum	35.40	62.60	51.70	47.80	75.40
in hard rock	...	cum	95.00	134.50	174.00	126.50	118.50
3.00/6.00m deep -							
in sand	...	cum	17.95	17.60	19.75	22.95	22.50
in light soil	...	cum	18.75	23.90	24.50	29.40	24.30
in clay	...	cum	20.15	21.20	27.30	31.90	33.60
in soft rock	...	cum	36.80	62.70	51.50	57.20	75.40
in hard rock	...	cum	95.60	134.50	174.00	203.50	117.50
Excavate trenches :							
N.E. 1.00m deep -							
in sand	...	cum	75.40	37.30	50.00	29.60	51.90
in light soil	...	cum	75.40	44.80	64.70	32.30	51.90
in clay	...	cum	85.50	69.10	68.50	46.00	62.70
in soft rock	...	cum	96.50	90.00	103.40	95.30	225.00
in hard rock	...	cum	208.00	138.00	329.00	211.00	339.00
1.00/2.00m deep -							
in sand	...	cum	75.40	38.90	51.40	31.00	59.20
in light soil	...	cum	75.40	46.30	66.10	34.10	59.20
in clay	...	cum	85.60	69.00	69.20	48.00	72.10
in soft rock	...	cum	85.60	90.00	110.50	98.00	225.00
in hard rock	...	cum	211.00	138.00	334.00	221.00	339.00

EXCAVATION

EXCAVATION (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Excavate pits :							
N.E. 1.00m deep -							
in sand	...	cum	67.40	38.90	59.30	31.00	52.10
in light soil	...	cum	67.40	46.30	75.40	34.10	52.10
in clay	...	cum	80.70	69.00	78.40	48.00	66.60
in soft rock	...	cum	97.50	90.00	103.80	98.00	225.00
in hard rock	...	cum	198.50	137.50	320.00	221.00	339.00
1.00/2.00m deep -							
in sand	...	cum	70.60	40.00	60.70	32.40	59.20
in light soil	...	cum	70.60	49.30	76.50	35.60	59.20
in clay	...	cum	84.50	69.10	79.70	49.30	72.20
in soft rock	...	cum	106.00	90.00	110.50	104.00	225.00
in hard rock	...	cum	208.00	137.50	334.00	233.00	339.00
Excavate trenches or pits in confined spaces, by hand :							
N.E. 1.00m deep -							
in sand	...	cum	131.50	86.20	173.00	66.40	188.00
in light soil	...	cum	131.50	97.30	173.00	84.50	188.00
in clay	...	cum	156.50	127.50	193.00	112.00	219.00
in soft rock	...	cum	224.00	380.00	381.00	205.00	408.00
in hard rock	...	cum	321.00	501.00	462.00	343.00	526.00
Excavate for thickening to ground slabs by hand including forming both sides to slope :							
in sand	...	cum	131.50	86.20	173.00	66.40	188.00
in light soil	...	cum	131.50	97.30	173.00	84.50	188.00
in clay	...	cum	156.50	127.50	193.00	112.00	219.00
in soft rock	...	cum	224.00	381.00	381.00	205.00	408.00
in hard rock	...	cum	321.00	501.00	462.00	343.00	526.00
DISPOSAL							
Additional cost of carting excavated material each additional 5km over 10km	...	cum	3.15	3.00	3.50	2.95	3.10
ADD extra for :							
Waste Management Centre and tip fees in excess of $2.50 per cum							
Tip fees vary with tip location and sometimes with type of material, the following is a guide only -							
Recyclable material	...	t	40.00-70.00	5.00-105.00	55.00-130.00	-	22.00-190.00
Materials/generally	...	t	40.00-60.00	130.00-160.00	55.00-185.00	95.00-125.00	370.00-380.00
Demolition material - mixed	...	t	150.00-210.00	130.00-160.00	105.00-180.00	140.00-160.00	360.00-370.00
Mixture of concrete, soil, etc.	...	t	90.00-130.00	15.00-105.00	90.00-180.00	-	30.00-200.00
Trees	...	t	60.00-80.00	80.00-170.00	120.00-180.00	-	225.00-355.00
Uncontaminated clean fill	...	t	25.00-30.00	20.00-40.00	50.00-170.00	8.00-16.00	200.00-375.00
Contaminated soil to approved land fill :							
Low level contamination	...	t	190.00-220.00	70.00-145.00	120.00-500.00	125.00-225.00	280.00-400.00
High level contamination (and possibly mix with cement)	...	t	450.00-500.00	P.O.A.	650.00-1400.00	150.00-250.00	770.00-790.00

Note : Preparation fee and EPA levies may apply

EXCAVATION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CUT AND FILL							
Balanced cut and fill over site average 0.50/1.00m deep :							
in sand	...	cum	15.35	13.20	11.00	9.85	10.90
in light soil	...	cum	15.35	14.20	11.90	11.70	11.55
in clay	...	cum	18.80	22.10	14.45	19.30	13.80
FILLING							
Backfilling of trenches and pits is included in excavation prices	...	Note					
Prices for filling are for consolidated quantities in place	...	Note					
Excavated material as filling (on site)	...	cum	23.50	15.00	17.00	7.00	8.45
Clean sand filling	...	cum	69.80	65.00	55.00	24.00	67.00
Crushed rock filling laid and consolidated in 150mm layers	...	cum	109.50	85.00	85.00	68.00	75.00
Crushed recycled concrete, ditto	...	cum	104.00	85.00	85.00	47.00	83.00
General filling, ditto (= CBR 15)	...	cum	-	45.00	-	-	-
Decomposed granite, ditto	...	cum	-	80.00	95.00	-	-
River gravel filling, ditto	...	cum	-	90.00	90.00	-	147.00
Labour forming sloping edge to filling :							
N.E. 0.50m wide	...	m	3.20	3.10	2.75	2.40	2.75
0.50/1.00m wide	...	m	3.20	3.10	3.05	2.90	2.75
50mm sand bed laid in trenches or pits under footings and under slabs	...	sqm	5.40	5.95	5.70	3.50	3.75
Backfillling :							
Cement stabilised sand (1:20)	...	cum	158.50	130.00	150.00	110.00	135.00
Flowable (self levelling) material	...	cum	125.00-165.00	125.00-165.00	125.00-165.00	125.00-165.00	125.00-165.00
SURFACE TREATMENTS							
Compaction only of :							
Foundation trenches	...	sqm	2.10	3.60	2.75	3.35	3.55
Ground under slabs	...	sqm	2.10	2.70	2.10	2.65	3.65
Level ground surface under floor slabs, paving and the like, including compaction	...	sqm	2.70	3.60	3.15	3.25	3.65
Level and grade to falls ditto	...	sqm	3.10	3.60	3.70	3.55	3.65
Trim surfaces of cuttings and embankments in other than rock to slope	...	sqm	3.70	3.65	3.40	3.20	3.65
Scabble rock in levelling bottoms of excavations	...	sqm	28.50	26.20	31.50	21.50	27.30
Scabble and trim vertical or sloping rock sides of excavations	...	sqm	31.60	29.80	37.90	28.70	34.10

EXCAVATION

PLANKING AND STRUTTING, SHORING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Sides of basement excavation (sand and light soils only) :							
1.00 to 3.00m deep	...	sqm	20.50	21.50	25.30	29.50	30.00
ADD extra for :							
Site boundaries abutting adjacent roadway or building - refer Sheet Piling	...	Note					
Sides of trench excavation :							
N.E. 1.00m deep	...	sqm	Incl.	Incl.	Incl.	Incl.	Incl.
1.00-3.00m deep -							
Sand and light soil	...	sqm	19.90	19.95	20.90	18.45	30.70
Clay	...	sqm	10.20	10.35	10.40	11.20	11.95
Sides of pit excavation :							
N.E. 1.00m deep	...	sqm	Incl.	Incl.	Incl.	Incl.	Incl.
1.00-3.00m deep -							
Sand and light soil	...	sqm	21.30	21.10	21.90	20.40	32.80
Clay	...	sqm	10.80	10.85	11.45	11.70	15.65
Shoring comprising R.S.J. soldier piles in bored pier hole including concrete base, infilling R.S.J.'s with timber or precast waling and filling at back with weak concrete average 100mm thick (average depth 3.00m)	...	sqm	280.00	265.00	275.00	270.00	325.00
Shoring/retaining wall system comprising precast soldier piles/columns at approximately 2100mm centres and 250mm thick sprayed reinforced concrete infill walls, including rock anchors, strip footing, tie beams, stabilised sand backfill and strip drainage (based on three basement levels)	...	sqm	-	800.00-900.00	770.00-880.00	-	-
ADD extra for :							
Two basement levels	...	sqm	-	35.00	41.00	-	-
Precast concrete wall panels in lieu sprayed	...	sqm	-	25.00-35.00	29.00-41.00	-	-
Internal tanking or wall lining	...	Note					

SHEET STEEL PILING

Sheet steel piling including all supports :							
Permanent -							
43.2kg/m	...	sqm	660.00	480.00	500.00	480.00	485.00
58.4kg/m	...	sqm	705.00	550.00	605.00	580.00	640.00
74.0kg/m	...	sqm	775.00	620.00	685.00	670.00	700.00
ADD extra for :							
Toe, allow x 1/3 of height retained plus 300mm at bottom and 700mm at top	...	Note					
Removal and disposal for temporary locations, costed on a job by job basis	...	Note					
Waler with ground anchors at 2400/3200mm centres :							
Temporary	...	m	840.00-940.00	850.00-950.00	660.00-715.00	750.00-850.00	640.00-660.00
Left in	...	m	1050.00-1150.00	1150.00-1300.00	815.00-915.00	950.00-1050.00	745.00-835.00

EXCAVATION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SHEET STEEL PILING (cont'd)							
Sheet steel piling including all supports (cont'd) :							
ADD extra for (cont'd) :							
Typical allowance for walers and ground anchors :							
Excavation -							
up to 3000mm deep - no walers (but increase toe to equal IN and OUT)							
3.0/6.0m deep - 1 waler							
6.0/9.0m deep - 2 walers							
Lightweight steel trench sheeting including all supports	...	sqm	335.00	275.00	225.00	250.00	215.00
DIAPHRAGM WALLS							
Prices give a broad indication only for work in typical conditions of the respective cities	...	Note					
Mobilisation and setting up, dismantling and removal :							
Minor work	...	Sum	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00
Major works	...	Sum	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00	155000.00-185000.00
Set up guide walls, average 1.00/1.20m deep	...	m	750.00-850.00	400.00-500.00	800.00-900.00	750.00-850.00	750.00-850.00
500mm thick wall including excavating trench under Bentonite support, reinforcement, concrete and disposal of surplus material, including all blockouts and sundries (sqm of wall and toe)	...	sqm	625.00-750.00	900.00-1200.00	650.00-775.00	625.00-750.00	625.00-750.00
ADD extra for :							
Ground anchors	...	Note					
SECANT PILE WALLS							
Prices are a broad indication only for work in typical conditions	...	Note					
Mobilisation and setting up, dismantling and removal (range)	...	Sum	24000.00-26000.00	20000.00-40000.00	25000.00-27000.00	24500.00-27000.00	24500.00-27000.00
Soft secant pile wall in 500mm dia. reinforced concrete piles including all necessary boring, displacement, temporary steel casing, reinforcement and testing on completion (sqm of wall and toe)	...	sqm	625.00	750.00	680.00	640.00	675.00
ADD extra for :							
Ground anchors, typical allowance 1 anchor per 5sqm of secant wall, refer immediately following prices	...	Note					
Site conditions	...	Note					
Water	...	Note					

EXCAVATION

GROUND ANCHORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices give a broad indication only for work in typical conditions	...	Note					
Prices are based on a minimum of fifty anchors in kilonewtons working load	...	Note					
Prices are based on an assumed 15.2mm diameter superstand	...	Note					
Temporary anchor in :							
Rock -							
Up to 330kN WL (3 strand or bar)	...	Each	800.00	800.00	800.00	800.00	800.00
340 to 770kN WL (3 to 7 strand or bar)	...	Each	1100.00	1100.00	1100.00	1100.00	1100.00
780 to 1430kN WL (7 to 12 strand or high strength bar)	...	Each	3200.00	3200.00	3200.00	3200.00	3200.00
1440 to 2090kN WL (12 to 17 strand or large diameter high strength bar)	...	Each	7000.00	7000.00	7000.00	7000.00	7000.00
2100 to 2970kN WL (17 to 24 strand or very large diameter high strength bar)	...	Each	10000.00	10000.00	10000.00	10000.00	10000.00
2980 and above (price on application)	...	Note					
Sand and soil -							
Up to 220kN WL (up to 2 strand)	...	Each	2600.00	2600.00	2600.00	2600.00	2600.00
230 to 440kN WL (2 to 4 strand)	...	Each	3000.00	3000.00	3000.00	3000.00	3000.00
Larger anchors not generally available	...	Note					
ADD extra for :							
25 to 50 anchors	...	+ 15%					
10 to 20 anchors	...	+ 25%					
Corrosion protection to permanent anchors	...	+ 15%					
Casing or difficult drilling conditions	...	+25/50%					
Waterproof grouting/redrilling in rock with cavaties	...	+25/50%					
Marine anchors	...	+200%					
Craneage	...	Note					
Depth	...	Note					
Permanent anchors :							
Price on application, could be 100 to 200% higher than temporary	...	Note					

BORED PIERS

Prices are for mechanically bored pier holes in ground other than rock and not exceeding 4.0m in depth	...	Note					
Prices include for all site establishment costs, setting out, setting up rigs and removal of excavated material	...	Note					
Bored piers including 40 MPa concrete (based on 100m) :							
300mm dia.	...	m	92.00	100.00	92.00	86.00	100.00
450mm dia.	...	m	126.00	150.00	125.00	112.00	145.00
600mm dia.	...	m	210.00	270.00	210.00	205.00	220.00
750mm dia.	...	m	325.00	425.00	285.00	260.00	265.00
900mm dia.	...	m	462.00	610.00	355.00	345.00	355.00

EXCAVATION

BORED PIERS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Bored piers including 40 MPa concrete (based on 100m) (cont'd) :							
ADD extra for :							
Belling to base -							
450 - 750mm dia.	...	No	185.00	100.00	107.00	190.00	99.00
600 - 900mm dia.	...	No	189.00	120.00	122.00	240.00	131.00
750 - 1050mm dia.	...	No	210.00	135.00	158.00	265.00	149.00
900 - 1200mm dia.	...	No	240.00	150.00	178.00	320.00	160.00
Reinforcement							
Allow 50-100kg/cum of pier	...	Note					
If in water, costs could increase by 50 to 100%	...	Note					
Restricted sites	...	Note					
Temporary or permanent liners if required	...	Note					

PILES

Site establishment costs, setting out and setting up rigs	...	Sum	7800.00	7500.00	8100.00	7800.00	7800.00
Timber piles, pressure treated hardwood, including driving and labour (based on 500m) :							
30t/40t bearing capacity	...	m	-	119.00	150.00	182.00	138.00
60t bearing capacity	...	m	-	134.00	206.00	233.00	194.00
Precast reinforced concrete piles, including driving and all labour (based on 500m) :							
235mm square	...	m	140.00	114.00	124.00	153.00	134.00
270mm square	...	m	167.00	132.00	142.00	182.00	158.00
300mm square	...	m	217.00	166.00	182.00	240.00	185.00
350mm square	...	m	283.00	188.00	219.00	308.00	237.00
400mm square	...	m	360.00	257.00	296.00	375.00	313.00
450mm square	...	m	454.00	342.00	402.00	444.00	382.00
Reinforced concrete driven cast in place displacement pile with enlarged base including temporary steel tube casings, 25 MPa concrete, reinforcement (@ 0.5%) (based on 500m) :							
450mm dia.	...	m	283.00	250.00	282.00	275.00	292.00
600mm dia.	...	m	350.00	287.00	352.00	331.00	365.00
ADD extra for :							
Cut off and trim head of pile	...	No	149.00	135.00	156.00	145.00	163.00
Removal of spoil and restricted sites	...	Note					

STEEL SCREW PILES

Prices give a broad indication assuming founding in stiff clay or medium density sands	...	Note					
Pile length assumed as 3m long							
Compression piles (based on No 30+) :							
10t bearing capacity	...	m	91.00	91.00	91.00	91.00	91.00
20t bearing capacity	...	m	123.00	123.00	123.00	123.00	123.00
30t bearing capacity	...	m	155.00	155.00	155.00	155.00	155.00
40t bearing capacity	...	m	176.00	176.00	176.00	176.00	176.00
60t bearing capacity	...	m	209.00	209.00	209.00	209.00	209.00
80t bearing capacity	...	m	263.00	263.00	263.00	263.00	263.00

EXCAVATION

STEEL SCREW PILES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Tension piles (based on No 30+) :							
10t bearing capacity	...	m	91.00	91.00	91.00	91.00	91.00
20t bearing capacity	...	m	123.00	123.00	123.00	123.00	123.00
30t bearing capacity	...	m	155.00	155.00	155.00	155.00	155.00
40t bearing capacity	...	m	176.00	176.00	176.00	176.00	176.00
60t bearing capacity	...	m	209.00	209.00	209.00	209.00	209.00
80t bearing capacity	...	m	263.00	263.00	263.00	263.00	263.00
Founding or bearing screw piles in rock strata	...	No	27.00	27.00	27.00	27.00	27.00
Restricted site	...	Note					
Screw pile design is site specific and based on accurate geotechnical information. Pile lengths and costs may vary accordingly	...	Note					

SOIL STABILISATION

Mobilisation and setting up	...	Sum	-	750.00	-	750.00	-
In situ ultrafine cement grout installation to stabilise and solidify soil under foundations (per cum volume of soil to be stabilised)	...	cum	-	700.00-750.00	-	700.00-750.00	-
ADD extra for :							
Any access problems	...	Note					

WATERPROOFING, TANKING

WATERPROOFING

Three coats liquid application :							
Vertically	...	sqm	45.50	39.20	42.20	41.80	42.30
Prepare and apply single layer :							
Bituminous felt horizontally	...	sqm	29.40	27.40	29.40	28.40	29.70
Butyl rubber horizontally	...	sqm	24.60	36.80	42.60	39.20	43.90

TANKING

Two coats bituminous paint :							
Horizontally	...	sqm	15.90	12.95	13.30	13.15	14.55
Vertically	...	sqm	16.55	13.45	14.15	13.90	15.25
Prepare and apply single layer bituminous felt :							
Horizontally	...	sqm	30.50	28.40	30.30	29.50	30.80
Vertically	...	sqm	33.50	31.20	33.40	32.70	33.90
Prepare and apply two layers bituminous felt :							
Horizontally	...	sqm	52.30	48.20	51.90	49.80	52.70
Vertically	...	sqm	55.20	50.80	53.80	53.80	54.70

EXCAVATION

WATERPROOFING, TANKING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TANKING (cont'd)							
Corflute protection board :							
Vertically -							
3.0mm thick	...	sqm	8.15	8.95	9.05	8.80	10.20
4.0mm thick	...	sqm	8.75	9.55	9.70	9.40	10.85
5.0mm thick	...	sqm	11.15	12.05	12.20	11.90	13.50
Styrofoam tongued and grooved protection board :							
Vertically -							
25mm thick	...	sqm	21.60	23.10	22.60	23.50	24.70
30mm thick	...	sqm	29.20	30.80	30.00	31.50	32.20
50mm thick	...	sqm	45.60	47.60	46.20	48.90	49.20
75mm thick	...	sqm	64.50	66.80	64.60	68.90	68.00
ADDITIONAL TANKING AND MEMBRANE TYPE APPLICATIONS							
Refer pages 404 and 405	...	Note					
TANK LININGS							
Refer page 405	...	Note					

PAPER AND PLASTIC MEMBRANES

Prices include for all laps and jointing	...	Note					
Building paper	...	sqm	3.90	4.30	4.15	3.95	4.20
Polythene film, 200µm :							
Black	...	sqm	2.55	3.00	3.40	2.90	4.10
Orange	...	sqm	4.35	5.05	5.50	4.90	6.60

WEED AND TERMITE CONTROL

Weed control, (average) :							
Sub grade areas	...	sqm	0.95	0.95	0.90	0.90	0.90
Paved areas	...	sqm	0.95	0.95	0.90	0.90	0.90
Soil sterilisation, fumigation, (average)	...	sqm	2.25	2.20	2.20	2.25	2.20
ADD extra for :							
Temporary fencing or security guards to site	...	Note					
Termite control :							
Crushed stone granite (Granitgard) -							
Perimeter treatment (in cavity of perimeter	...	m	16.00-19.00	14.00-17.00	15.00-18.00	16.00-19.00	12.00-16.00
Construction joints and joints at extensions	...	m	15.00-18.00	15.00-18.00	15.00-18.00	15.00-18.00	15.00-18.00
Penetrations	...	No	16.00-17.00	12.00-16.00	14.00-17.00	16.00-17.00	11.00-14.00
Granitgard Blockaid modified rubberised termite resistant compound :							
Retaining walls	...	sqm	70.00	65.00	70.00	70.00	60.00

EXCAVATION

WEED AND TERMITE CONTROL (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Termite control (cont'd) :							
Full barrier treatment under slab and perimeter with Biflex	...	sqm	5.30	4.20	5.30	4.80	5.30
Anti termite irrigation system with Biflex	...	sqm	-	12.20	-	13.20	-
ADD extra for :							
Perimeter treatment	...	m	-	15.80	-	16.80	-
Laminated sheet barrier (Kordon) -							
Full barrier	...	sqm	32.00-37.00	32.00-37.00	32.00-37.00	32.00-37.00	32.00-37.00
Perimeter barrier to edge of slab and turned into cavity wall	...	m	18.00	18.00	18.00	18.00	18.00
Penetrations	...	No	18.00	18.00	18.00	18.00	18.00
Stainless steel mesh barrier (Termimesh) -							
Perimeter barrier to edge of slab and turned into cavity wall	...	m	22.00-28.00	22.00-28.00	22.00-28.00	22.00-28.00	22.00-28.00
Penetrations	...	No	16.00-22.00	16.00-22.00	16.00-22.00	16.00-22.00	16.00-22.00
Pest control, vermin proofing or eradication of cockroaches, flies, etc. (average)	...	sqm	1.80	1.75	1.65	1.80	1.65

EXTERNAL WORKS

PAVINGS

BITUMEN PAVING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices given hereafter are for 500 to 1000sqm. For smaller quantities refer page 724 and for in excess of 5000sqm refer page 716	...	Note					
Crushed rock/blue metal base course including grading, rolling and consolidating to receive paving :							
100mm thick	...	sqm	10.90	9.65	12.00	10.05	13.75
150mm thick	...	sqm	16.00	12.70	17.00	13.40	17.45
200mm thick	...	sqm	21.00	16.55	22.00	16.80	22.90
250mm thick	...	sqm	27.60	24.30	28.70	20.10	29.70
Cement treated crushed rock including grading, rolling and consolidating to receive paving :							
50mm thick	...	sqm	9.60	8.55	9.60	10.95	8.00
100mm thick	...	sqm	15.95	13.35	16.85	18.25	14.40
Crushed limestone base course including grading, rolling and consolidating to receive paving :							
150mm thick	...	sqm	-	-	-	12.60	-
200mm thick	...	sqm	-	-	-	15.75	-
250mm thick	...	sqm	-	-	-	18.80	-
Prime and two coats sprayed bitumen seal	...	sqm	19.00	24.90	9.70	21.80	8.80
Ditto in repairs to service trenches including making good to existing	...	sqm	23.90	32.00	23.20	24.00	15.60
Hot bituminous concrete including tack coat :							
20mm thick	...	sqm	20.80	20.60	17.50	14.60	17.10
25mm thick	...	sqm	23.20	22.40	20.00	17.10	19.20
40mm thick	...	sqm	27.00	26.90	25.30	23.10	26.60
50mm thick	...	sqm	33.60	33.20	29.80	30.50	29.70
ADD extra for :							
Laterite and 2% red oxide	...	+40%					

IN SITU CONCRETE PAVING

In situ concrete paving including F72 fabric reinforcement, expansion joints, edge forms and broomed finish :							
100mm thick	...	sqm	80.30	82.00	89.50	64.70	95.30
150mm thick	...	sqm	71.20	69.60	83.30	68.60	91.70
ADD extra for :							
Steel trowel finish	...	sqm	2.25	-	0.85	2.35	0.40

EXTERNAL WORKS

PAVINGS (cont'd)

BRICK, BLOCK AND SLAB PAVING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Clay brick paving laid on prepared foundation :							
180 x 180 x 50mm	...	sqm	63.50	-	-	-	-
230 x 110/115 x 50mm	...	sqm	71.20	72.30	66.50	48.50	62.80
230 x 110/150 x 65mm	...	sqm	-	72.30	67.50	-	83.40
230 x 110/150 x 70mm	...	sqm	72.90	-	-	57.70	-
230 x 110/150 x 75mm	...	sqm	75.30	67.20	-	64.40	74.90
230 x 150 x 50mm	...	sqm	-	68.80	-	-	74.90
230 x 230 x 50/60mm	...	sqm	-	78.60	-	60.60	-
230 x 230 x 65mm	...	sqm	72.90	-	-	-	-
Concrete brick or block paving laid on prepared foundation :							
Cobblestone pavers -							
190 x 190 x 50mm	...	sqm	-	64.70	60.20	-	73.80
190 x 190 x 60mm	...	sqm	-	65.30	62.50	48.10	76.40
Interlocking pavers -							
225 x 112 x 60mm	...	sqm	63.70	74.10	74.40	48.80	86.40
225 x 112 x 80mm	...	sqm	69.70	79.40	78.50	55.80	92.70
Paving bricks -							
230 x 115 x 50mm	...	sqm	58.50	73.90	-	-	73.80
230 x 115 x 60mm	...	sqm	-	74.80	70.80	47.50	84.40
Urbanstone paving slabs laid on prepared foundation :							
Shotblast finish, colours -							
Category A, B, C and D -							
400 x 400 x 40mm	...	sqm	126.50	132.00	130.00	121.50	129.50
400 x 400 x 60mm	...	sqm	139.00	147.00	148.50	134.50	143.50
Category E -							
400 x 400 x 40mm	...	sqm	131.50	141.00	134.50	127.00	136.00
400 x 400 x 60mm	...	sqm	143.50	152.50	150.50	136.50	146.50
Milled finish, colours -							
Category A, B, C and D -							
400 x 400 x 40mm	...	sqm	161.00	164.50	146.00	161.50	155.50
400 x 400 x 60mm	...	sqm	170.50	176.50	174.00	171.00	169.50
Category E -							
400 x 400 x 40mm	...	sqm	169.50	176.50	166.50	161.50	168.00
400 x 400 x 60mm	...	sqm	181.00	188.00	182.50	171.00	179.00
ADD extra for :							
Laying paving to patterns	...	sqm	6.00-11.00	6.50-11.50	6.00-11.00	6.00-11.00	6.00-11.00
75mm thick compacted sand bed	...	sqm	6.50	6.55	7.55	5.35	6.55
Black polythene underlay	...	sqm	2.55	3.00	3.40	2.90	4.10
25mm thick mortar bed	...	sqm	8.15	6.40	6.60	5.95	6.55
Two coats acrylic total surface treatment	...	sqm	-	7.85	7.80	7.20	7.65

TREADS AND RISERS

Brick paving 230 x 115 x 76mm or similar to treads and risers, laid :							
Flat	...	sqm	110.50	95.30	101.50	90.70	127.50
On edge	...	sqm	152.00	115.00	130.50	121.50	150.00

EXTERNAL WORKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
KERBS AND DRAINS							
Prices include for excavation, backfilling and disposal of surplus.	...	Note					
Precast concrete kerb :							
150 x 75mm to 50mm	...	m	32.20	-	-	-	-
200 x 125mm to 100mm	...	m	38.70	-	-	38.60	-
250 x 150mm to 100mm	...	m	49.40	-	-	48.20	-
Cast in situ concrete including reinforcement, formwork and surface finishes in :							
150 x 150mm kerb	...	m	80.40	85.20	82.70	47.10	86.30
250 x 150mm kerb	...	m	87.60	91.60	91.00	54.60	96.40
600 x 225mm kerb and gutter	...	m	150.00	153.00	160.00	97.90	170.50
Box drain 300mm wide x 250mm deep internally with 100mm thick bottom and sides	...	m	178.00	168.00	206.50	151.00	204.00
150 x 75mm edging strip	...	m	57.40	61.20	58.80	32.10	61.20
Extruded in situ concrete kerb :							
150 x 350mm high, 150mm in ground (x 500 metres)	...	m	37.60	41.50	40.90	-	41.90
250 x 150mm high (x 500 metres)	...	m	27.50	-	-	16.70	-
300 x 150mm high including 150 x 150mm trenching (x 500 metres)	...	m	39.10	-	-	25.50	-
600 x 225mm high kerb and gutter (x 500 metres)	...	m	40.20	53.60	-	-	-
CULVERTS							
Prices include for excavation, backfilling, and disposal of surplus.	...	Note					
Precast concrete box culvert :							
375 x 225mm	...	m	214.00	312.00	255.00	230.00	258.00
600 x 450mm	...	m	414.00	456.00	409.00	454.00	491.00
750 x 600mm	...	m	559.00	609.00	525.00	631.00	682.00
Larger sized culverts - refer pages 718 and 719	...	Note					
SUNDRIES							
ROAD MARKING							
One coat road marking paint on bitumen paving :							
Line -							
75mm wide	...	m	1.80	1.60	1.50	1.40	1.60
100mm wide	...	m	1.90	1.90	1.80	1.65	1.75
Number or letter -							
200mm high	...	No	4.10	6.30	5.10	3.80	5.05
300mm high	...	No	5.45	7.65	6.50	4.25	6.10
Arrow, not exceeding -							
500 x 500mm	...	No	25.60	25.10	24.80	20.00	22.50
Thermoplastic marking of bitumen paving :							
Prices are based on a minimum of 500 metres	...	Note					
Establishment cost	...	Sum	800.00-1000.00	800.00-1000.00	800.00-1000.00	800.00-1000.00	800.00-1000.00
100mm wide line	...	m	1.60-1.90	1.80-2.05	1.70-2.00	1.65-1.95	1.70-2.00

EXTERNAL WORKS

SUNDRIES (cont'd)

BOLLARDS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Standard styled 115mm dia. Aluminium powder coated bollard 850-965mm high :							
Surface mounted	...	No	455.00	440.00	440.00	460.00	450.00
Subsurface mounted	...	No	465.00	455.00	455.00	470.00	465.00
Standard styled 115mm dia. Stainless steel (304) bollard 850-965mm high :							
Surface mounted	...	No	760.00	730.00	730.00	760.00	740.00
Subsurface mounted	...	No	745.00	720.00	720.00	750.00	730.00
Standard styled 165mm dia. Aluminium powder coated bollard 790-920mm high :							
Surface mounted	...	No	590.00	570.00	570.00	595.00	580.00
Subsurface mounted	...	No	590.00	570.00	570.00	595.00	580.00
Removable surface mounted	...	No	695.00	670.00	670.00	700.00	680.00
Removable subsurface mouted	...	No	1355.00	1300.00	1300.00	1360.00	1315.00
Standard styled 165mm dia. Stainless steel (304) bollard 790-920mm high :							
Surface mounted	...	No	785.00	755.00	755.00	790.00	765.00
Subsurface mounted	...	No	785.00	755.00	755.00	790.00	765.00
Removable surface mounted	...	No	945.00	910.00	910.00	950.00	915.00
Removable subsurface mouted	...	No	1565.00	1500.00	1500.00	1570.00	1515.00
Lighting bollards - refer pages 509 and 595	...	Note					
Crash and guard rails - refer page 304	...	Note					

RETAINING WALLS

Comparative costs of various retaining walls - refer page 190	...	Note					

FENCING AND GATES

TIMBER

Standard butt jointed paling fence (excluding Perth) or butt jointed picket fence (Perth only) :							
1500mm high	...	m	95.00	84.00	75.00	97.00	78.00
1800mm high	...	m	105.00	94.00	80.00	115.00	83.00
ADD extra for :							
Lapping palings or pickets -							
1500mm high	...	m	10.00	11.55	8.75	8.50	8.40
1800mm high	...	m	11.00	12.60	10.00	10.00	9.40
100mm wide splayed capping	...	m	12.00	10.50	8.40	10.00	8.40
1800mm high with horizontal boards both sides alternately	...	m	145.00	-	-	153.00	150.00
Standard brush fence on steel supports, with capping and set in concrete :							
11500mm high	...	m	210.00	-	205.00	225.00	244.00
1800mm high	...	m	225.00	231.00	220.00	235.00	283.00

EXTERNAL WORKS

FENCING AND GATES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FIBRE CEMENT							
1800mm high corrugated fibre cement set into ground and with metal capping	...	m	-	-	-	78.00	-
METAL							
RIBBED SHEET							
Square profile fence of ribbed steel decking on steel posts and rails, with capping and set in concrete :							
Colorbond® Steel -							
1500mm high	...	m	85.00	110.00	87.00	92.00	90.00
1800mm high	...	m	97.00	121.00	95.00	96.00	100.00
MESH							
1200mm high fence of galvanised welded mesh roll top panels and tubular posts	...	m	89.00	80.30	92.00	104.00	98.00
ADD extra for :							
900mm wide single gate with two posts	...	No	345.00	330.00	290.00	395.00	302.00
2750mm wide double ditto	...	No	650.00	570.00	650.00	580.00	690.00
1525 or 1575mm high swimming pool safety fence of galvanised welded mesh panels with top and bottom rails and tubular posts	...	m	132.00	143.00	133.00	125.00	141.00
ADD extra for :							
1000mm wide single gate with two posts	...	No	459.00	495.00	450.00	450.00	535.00
Pre-painted finish to fence	...	m	21.50	22.50	22.00	23.00	25.00
1800mm high galvanised chain mesh rail-less fence with tubular posts	...	m	58.00	77.00	56.00	48.00	66.00
ADD extra for :							
Timber slat infill to link mesh	...	m	-	-	112.00	-	135.00
900mm wide single gate with two posts, braces and fittings	...	No	500.00	550.00	435.00	510.00	424.00
6000mm wide, double ditto	...	No	1300.00	1320.00	1305.00	1220.00	1400.00
Green plastic coating to link mesh	...	m	15.00	14.50	14.00	14.50	13.00
1800mm high security fence of galvanised chain mesh plus three rows of barbed wire to top	...	m	64.00	85.00	66.00	56.00	70.00
ADD extra for :							
900mm wide single gate with two posts, braces and fittings	...	No	500.00	550.00	510.00	510.00	544.00
3000mm wide double gate with two posts, braces and fittings	...	No	900.00	935.00	890.00	950.00	913.00
5000mm wide, ditto	...	No	1200.00	1265.00	1225.00	1190.00	1335.00
Barbed tape (TIGERTAPE) to top of fence (range)	...	m	25.00-28.00	33.00-38.00	30.00-35.00	29.00-31.00	41.00-44.00
Tubular top or bottom rail	...	m	20.00	22.00	18.25	17.20	18.00

EXTERNAL WORKS

FENCING AND GATES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
METAL (cont'd)							
MESH (cont'd)							
2100mm high security fence of galvanised chain mesh plus three rows barbed wire to top	...	m	85.00	92.00	74.00	72.00	75.00
ADD extra for :							
900mm wide single gate with two posts, braces and fittings	...	No	600.00	600.00	610.00	695.00	615.00
3000mm wide double gate with two posts, braces and fittings	...	No	1050.00	950.00	990.00	1180.00	1025.00
5000mm wide, ditto	...	No	1250.00	1300.00	1275.00	1300.00	1340.00
Green plastic coating to link mesh	...	m	16.00	16.50	14.00	18.00	13.00
Tubular top or bottom rail	...	m	20.00	22.00	18.25	16.70	17.00
3600mm high tennis court enclosure fence of galvanised chain mesh with tubular posts	...	m	133.00	126.00	133.00	121.00	145.00
ADD extra for :							
1000mm wide entry gate	...	No	330.00	440.00	380.00	385.00	400.00
Green plastic coating to link mesh	...	m	22.00	25.00	26.00	30.00	14.00
PALISADE							
Steel palisade high security fence 'D' pale with galvanised finish and RSJ posts set in concrete :							
1800mm high	...	m	440.00	440.00	440.00	440.00	440.00
2100mm high	...	m	450.00	450.00	450.00	450.00	450.00
2400mm high	...	m	465.00	465.00	465.00	465.00	465.00
up to 4200mm high	...	m	P.O.A.	P.O.A.	P.O.A.	P.O.A.	P.O.A.
ADD extra for :							
Personnel single gate with posts -							
1000mm wide :							
x 1800mm high	...	No	2650.00	2650.00	2650.00	2650.00	2650.00
x 2400mm high	...	No	2975.00	2975.00	2975.00	2975.00	2975.00
Double leaf gate with posts -							
4000mm wide :							
x 1800mm high	...	No	4850.00	4850.00	4850.00	4850.00	4850.00
x 2400mm high	...	No	5475.00	5475.00	5475.00	5475.00	5475.00
6000mm wide :							
x 1800mm high	...	No	6475.00	6475.00	6475.00	6475.00	6475.00
x 2400mm high	...	No	7275.00	7275.00	7275.00	7275.00	7275.00
Powder coated finish to fencing and gates	...	-10% - 20%					
TUBULAR							
Tubular steel modular pre-painted landscape fence with posts set in concrete :							
950mm high -							
Rounded top	...	m	118.00	125.00	118.00	118.00	121.00
Fleur-de-lis top	...	m	133.00	140.00	133.00	133.00	135.00
1400mm high -							
Rounded top	...	m	124.00	130.00	124.00	124.00	128.00
Fleur-de-lis top	...	m	144.00	150.00	144.00	144.00	149.00

EXTERNAL WORKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FENCING AND GATES (cont'd)							
METAL (cont'd)							
TUBULAR (cont'd)							
Tubular steel modular pre-painted security fence with posts set in concrete :							
1700mm high -							
Pointed head	...	m	178.00	185.00	178.00	178.00	182.00
Spearhead	...	m	190.00	195.00	190.00	190.00	195.00
2000mm high -							
Pointed head	...	m	194.00	200.00	194.00	194.00	199.00
Spearhead	...	m	198.00	205.00	198.00	198.00	203.00
ADD extra for :							
986mm wide gate with 64 x 64mm posts -							
1700mm high	...	No	980.00	990.00	980.00	980.00	980.00
2000mm high	...	No	990.00	1000.00	990.00	990.00	990.00
4000mm wide double gate with 76 x 76mm posts -							
1700mm high	...	No	1925.00	1950.00	1925.00	1925.00	1925.00
2000mm high	...	No	1975.00	2000.00	1975.00	1975.00	1975.00
SOILING AND GRASSING							
Top soil filled into planting boxes, spread and levelled	...	cum	70.00	68.50	63.20	68.50	67.80
Top soil filled into beds, spread and levelled	...	cum	62.00	60.50	52.50	54.50	54.30
Top soil spread in layers over the ground and raked and levelled :							
150mm thick	...	sqm	8.40	10.50	9.70	11.25	11.60
300mm thick	...	sqm	15.80	19.60	17.40	19.40	19.80
150mm thick exceeding 45° from the horizontal	...	sqm	9.60	11.00	10.80	12.35	12.65
Prepare the ground surface to receive top soil by rotary hoeing (large areas)	...	sqm	0.48	0.53	0.50	0.41	0.51
50mm layer of loam spread and levelled and sown with approved couch grass seed applied at the rate of 0.70 kilogrammes per 10 square metres including maintaining for six months	...	sqm	8.50	9.20	9.20	9.35	9.65
Level, grade, prepare and grass seed including preparatory fertilizing and watering and maintaining for six months :							
Bowling greens and the like	...	sqm	7.90	8.50	8.30	7.60	9.00
Playing fields, golf courses and the like	...	ha	8150.00	8500.00	8350.00	7850.00	8700.00
Native grasses	...	ha	10400.00-17150.00	10500.00-17500.00	10700.00-17600.00	10500.00-17300.00	11000.00-17850.00
Cut 600 x 325mm turfs, stockpile and water	...	sqm	5.50	6.20	6.50	5.45	7.40
Turfs, laid, rolled and watered for two weeks, 400mm wide roll turf :							
Couch	...	sqm	10.50	8.70	9.00	10.40	9.80
Broad leaf	...	sqm	15.90	13.20	14.70	15.40	16.15
Synthetic turf - refer page 234	...	Note					
Reticulation - refer page 706	...	Note					

EXTERNAL WORKS

LANDSCAPING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GROUND FINISHES							
Coarse pine bark chips spread in beds	...	cum	86.00	76.50	73.50	87.50	80.00
Wood chips spread in beds	...	cum	-	65.00	-	93.50	-
Lightweight planting mixture (vermiculite/ polystyrene/sawdust, etc.) spread in planters	...	cum	102.00	97.00	107.00	87.00	120.00
River pebbles average 50mm thick including plastic underlay	...	sqm	20.00	13.70	15.90	54.00	24.60
Blue metal chips average 50mm thick including plastic underlay	...	sqm	12.90	10.60	11.20	11.50	12.50
Filter fabric - refer page 715	...	Note					
Hydro mulch - refer page 715	...	Note					
Stablilisation - refer page 715	...	Note					
PLANTING							
Ground cover including planting (average cost)	...	sqm	8.00	8.00	8.00	8.00	8.00
Shrubs including planting (average cost) :							
Small (tube)	...	No	4.50-10.00	4.50-10.00	4.50-10.00	4.50-10.00	4.50-10.00
Average (150mm pot)	...	No	8.00-13.00	8.00-13.00	8.00-13.00	8.00-13.00	8.00-13.00
Large (200mm pot)	...	No	11.00-20.00	11.00-20.00	11.00-20.00	11.00-20.00	11.00-20.00
Trees including planting (average cost) :							
Small (5 litre)	...	No	12.00-18.00	12.00-18.00	12.00-18.00	12.00-18.00	12.00-18.00
Average (45 litre)	...	No	80.00-150.00	80.00-150.00	80.00-150.00	80.00-150.00	80.00-150.00
Large (100 litre)	...	No	225.00-325.00	225.00-325.00	225.00-325.00	225.00-325.00	225.00-325.00
PROTECTION							
BOLLARDS - refer page 225	...	Note					
LOG BARRIERS							
Treated pine log barrier comprising 125/150mm dia. rail in approximately 3 metre lengths supported on posts set into ground	...	m	31.50	32.00	31.50	31.50	32.00

EXTERNAL WORKS

LANDSCAPING (cont'd)

PROTECTION (cont'd)

TREE GRILLES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cast aluminium tree grille and frame set in position :							
Square - 1200 x 1200mm (approx.)	...	No	1490.00	1525.00	1490.00	1525.00	1490.00
Polyethylene HDPE recycled plastic tree grille and frame set in position :							
Round - 1200mm (approx.) dia.	...	No	915.00	935.00	915.00	935.00	915.00
Square - 1200 x 1200mm (approx.)	...	No	915.00	935.00	915.00	935.00	915.00

TREE GUARDS

Cast aluminium tree guard set in position :							
Round - 725mm dia.	...	No	1365.00	1395.00	1365.00	1395.00	1365.00
Cast aluminium tree guard and grille combined, and frame set in position :							
1200 x 1200mm (approx.) dia. grille with 725mm dia. tree guard	...	No	2850.00	2920.00	2850.00	2920.00	2850.00
Timber framed tree guard 1800mm high comprising three timber posts with galvanised chicken wire or similar wrapping, up to 300mm dia.	...	No	42.00	40.10	41.10	32.90	40.50
Shadecloth wrapping 1800mm high on three timber posts	...	No	37.00	31.90	34.20	34.50	35.00

TREE ROOT MANAGEMENT

Polypropylene tree root management modular cell, including filling with crushed rock :							
360mm deep, 65 tons per sqm crushing strength, trafficable	...	sqm	223.00	206.00	206.00	208.00	204.00
440mm deep, 18 tons per sqm crushing strength	...	sqm	167.00	151.00	151.00	149.00	148.00

SHADECLOTH

Shadecloth covering to pergolas and the like :							
Knitted type, sun resistance -							
50%	...	sqm	6.85	7.80	7.90	7.60	9.25
70%	...	sqm	7.10	8.05	8.15	7.90	9.50
ADD extra for :							
Pergola framing, battens	...	Note					

WHEELSTOP

Precast concrete wheelstop, 200 x 150mm thick with dowels to paving	...	m	46.50	46.50	46.00	45.50	46.50

EXTERNAL WORKS

LANDSCAPING (cont'd)

OUTDOOR FURNITURE

SEATING

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cast aluminium framed with powdercoat finish and with bases set into paving :						
Seat -						
1800 x 560mm deep :						
Aluminium slats ...	No	1040.00	1040.00	1040.00	1040.00	1040.00
Aluminium slats; Disability Discrimination Act (DDA) Compliant ...	No	1040.00	1040.00	1040.00	1040.00	1040.00
Timber slats ...	No	1040.00	1040.00	1040.00	1040.00	1040.00
Timber slats; Disability Discrimination Act (DDA) Compliant ...	No	1040.00	1040.00	1040.00	1040.00	1040.00
Bench seat -						
1800 x 295mm deep :						
Aluminium slats ...	No	550.00	560.00	560.00	560.00	570.00
Timber slats ...	No	550.00	560.00	560.00	560.00	570.00
1800 x 485mm deep :						
Aluminium slats ...	No	810.00	810.00	810.00	820.00	820.00
Aluminium slats; Disability Discrimination Act (DDA) Compliant ...	No	810.00	810.00	810.00	820.00	820.00
Timber slats ...	No	810.00	810.00	810.00	820.00	820.00
Timber slats; Disability Discrimination Act (DDA) Compliant ...	No	810.00	810.00	810.00	820.00	820.00
Galvanised steel framed with bases set into paving :						
Seat -						
1500 x 595mm wide, recycled wood plastic composite battens ...	No	1450.00	1450.00	1450.00	1460.00	1450.00
Bench seat -						
1500 x 410mm wide, recycled wood plastic composite battens ...	No	1230.00	1230.00	1230.00	1240.00	1230.00
Aluminium frame with powdercoat finish fixed to substrate :						
Leaning rail -						
3000 x 180mm deep, ribbed aluminium leaning rail :						
Wall mounted ...	No	1160.00	1160.00	1160.00	1170.00	1170.00
Surface mounted ...	No	1150.00	1150.00	1150.00	1150.00	1150.00
Subsurface mounted ...	No	1170.00	1170.00	1170.00	1180.00	1180.00

CAFÉ SETTINGS

Cast aluminium framed with powdercoat finish						
Café table, free standing including base :						
600mm dia. -						
480mm high ...	No	500.00	500.00	500.00	500.00	510.00
720mm high ...	No	530.00	530.00	530.00	530.00	530.00
1100mm high ...	No	580.00	580.00	580.00	580.00	590.00
830mm dia. ...	No	630.00	630.00	630.00	640.00	640.00
830mm dia. heavy duty ...	No	900.00	900.00	900.00	900.00	900.00
ADD extra for :						
Matching stool ...	No	450.00	450.00	450.00	450.00	460.00

EXTERNAL WORKS

LANDSCAPING (cont'd)

OUTDOOR FURNITURE (cont'd)

TABLES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cast aluminium framed with powdercoat finish and with bases set into paving :							
1800 x 670mm wide -							
Aluminium slats	...	No	1060.00	1060.00	1060.00	1070.00	1070.00
Timber slats	...	No	1060.00	1060.00	1060.00	1070.00	1070.00
1800 x 860mm wide -							
Aluminium slats; Disability Discrimination Act (DDA) Compliant	...	No	1410.00	1410.00	1410.00	1420.00	1410.00
Timber slats; Disability Discrimination Act (DDA) Compliant	...	No	1410.00	1410.00	1410.00	1420.00	1410.00
Galvanised steel framed with bases set into paving :							
1500 x 620mm wide, recycled wood plastic composite battens	...	No	1470.00	1460.00	1460.00	1470.00	1460.00

LITTER SOLUTIONS

Single 80L surface mounted bin with polyliner :							
Rolled sheet - Aluminium - solid	...	No	1300.00	1300.00	1300.00	1310.00	1300.00
Rolled sheet - Stainless steel (316) - solid	...	No	1680.00	1670.00	1670.00	1690.00	1670.00
Fluted - Aluminium	...	No	1350.00	1350.00	1350.00	1360.00	1350.00
Single frame bin enclosure with curved roof, signage, frame and perforated metal body, receive 120L wheeled bin :							
Aluminium	...	No	2240.00	2230.00	2230.00	2250.00	2220.00
Stainless steel (316)	...	No	3230.00	3200.00	3200.00	3230.00	3190.00
Single frame bin enclosure with curved roof, signage, frame and perforated metal body, receive 240L wheeled bin :							
Aluminium	...	No	2460.00	2440.00	2450.00	2470.00	2440.00
Stainless steel (316)	...	No	3690.00	3660.00	3660.00	3700.00	3640.00
Double framed bin enclosure with curved roof, signage, frame and perforated metal body, receive 120L wheeld bin :							
Aluminium	...	No	3780.00	3760.00	3760.00	3790.00	3740.00
Stainless steel (316)	...	No	5520.00	5490.00	5490.00	5540.00	5470.00
Double framed bin enclosure with curved roof, signage, frame and perforated metal body, receive 240L wheeld bin :							
Aluminium	...	No	4160.00	4130.00	4140.00	4170.00	4110.00
Stainless steel (316)	...	No	6250.00	6200.00	6210.00	6260.00	6160.00
ADD extra for :							
Ash box 1.5 litre	...	No	740.00	740.00	740.00	740.00	730.00

EXTERNAL WORKS

LANDSCAPING (cont'd)

OUTDOOR FURNITURE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DRINKING FOUNTAIN							
Cast aluminium with integral ceramic bowl water fountain, 1025mm high with bubbler and mouthguard :							
Two pack finish	...	No	2220.00	2210.00	2210.00	2230.00	2210.00
Water fountain 1125mm high, Disability Discrimination Act (DDA) Compliant :							
Powder coated aluminium	...	No	4260.00	4230.00	4230.00	4270.00	4200.00
Stainless steel (316)	...	No	6870.00	6810.00	6810.00	6880.00	6760.00
ADD extra for :							
Dog bubbler	...	No	460.00	450.00	450.00	460.00	450.00
Water pole 1495mm high water bottle refill station :							
Powder coated aluminium	...	No	2320.00	2310.00	2320.00	2330.00	2310.00
Stainless steel (316)	...	No	4070.00	4040.00	4040.00	4080.00	4020.00
ADD extra for :							
Filter	...	No	260.00	260.00	260.00	260.00	250.00
PLANTERS							
Powder coated aluminium sheet planter box, including polyethylene liner :							
100 x 500mm wide	...	No	1320.00	1310.00	1310.00	1320.00	1310.00
ASH CYLINDER							
Cast aluminium ash cylinder, 3 litre	...	No	435.00	435.00	435.00	440.00	440.00
LIGHTING							
Exterior and sports - refer pages 509, 510 and 594 to 598	...	Note					

ARTIFICIAL SURFACES/FINISHES

Laid on prepared surface :							
Plexiflor surface	...	sqm	13.00	13.30	13.00	13.30	13.00
Plexipave surface	...	sqm	12.50	12.80	12.50	12.80	12.50
Plexicushion prestige surface	...	sqm	65.10	66.70	65.10	66.70	65.10
Plexicushion tournament surface	...	sqm	58.80	59.90	58.80	59.90	58.80
Har-Tru clay court	...	sqm	P.O.A.	P.O.A.	P.O.A.	P.O.A.	P.O.A.
Sportgrass surfaces -							
7400 denier (adhesive fixed)	...	sqm	40.80	41.90	40.80	41.90	40.80
9600 denier (sand filled)	...	sqm	46.10	47.20	46.10	47.20	46.10
Rebound Synpave surface	...	sqm	18.40	18.60	18.40	18.60	18.40
Rebound Pro surface	...	sqm	59.90	60.90	59.90	60.90	59.90
Rebound Ace (5mm) surface	...	sqm	113.50	114.50	113.50	114.50	113.50
Rebound Impact surface	...	sqm	168.00	171.00	168.00	171.00	168.00

EXTERNAL WORKS

ARTIFICIAL SURFACES/FINISHES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Laid on prepared surface (cont'd) :							
Running track surfacing (full range)	...	sqm	100.00-220.00	100.00-220.00	100.00-220.00	100.00-220.00	100.00-220.00
ADD extra for :							
Special pavement requirements to running track	...	Note					
Synthetic ice	...	sqm	400.00	400.00	400.00	400.00	400.00
Golf course sundries -							
Walkways of synthetic grass or recycled rubber	...	sqm	84.00-101.50	86.50-105.00	84.00-102.00	85.50-104.00	84.00-102.00
Tees and greens, synthetic turf, Sport Grass	...	sqm	120.00-160.00	125.00-165.00	120.00-160.00	125.00-165.00	120.00-160.00

SUBSOIL DRAINAGE

Refer pages 497 and 499 ... Note

STORMWATER DRAINAGE

Refer pages 494 to 497 ... Note

MARINAS

Refer page 39 ... Note

SWIMMING POOLS

Refer page 38 ... Note

TENNIS COURTS

Refer page 40 ... Note

GOLF COURSES

Refer page 705 ... Note

UNDERPINNING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices assume the work to be carried out in lengths not exceeding 1200mm	...	Note					
Additional allowance must be made for shoring and supporting the structure to be underpinned, this allowance will depend on the nature and extent of the work	...	Note					
All excavation is assumed to be in other than rock in typical city ground conditions	...	Note					

EXCAVATION

Excavation alongside existing structure to expose footings :							
Not exceeding -							
1.00m deep	...	cum	99.90	90.90	68.70	43.10	71.50
2.00m deep	...	cum	107.50	92.00	79.50	47.00	72.10
3.00m deep	...	cum	115.00	96.70	96.60	52.70	104.50
4.00m deep	...	cum	122.00	103.00	106.00	55.80	105.50
Excavate below level of existing structure for underpinning commencing at bottom of preparatory excavation and going down a further depth :							
Not exceeding -							
1.00m deep	...	cum	166.00	251.00	234.50	103.50	289.50
2.00m deep	...	cum	182.50	264.00	257.00	115.50	296.00
3.00m deep	...	cum	200.00	270.00	289.00	139.50	356.00
4.00m deep	...	cum	221.00	305.00	346.00	168.50	407.00

DISPOSAL

Remove surplus excavated material from site to tip not exceeding 10km distance including payment of dumping fees of up to $10.00/cum	...	cum	27.00	26.30	28.90	26.00	26.80
ADD extra for :							
Additional tip fees, refer page 213 for a guide to tip fees	...	Note					

FILLING

Excavated material as backfilling	...	cum	23.50	23.50	25.45	13.45	15.60
Imported sand filling as backfilling	...	cum	90.80	90.00	88.00	44.40	82.00

PLANKING AND STRUTTING

Planking and strutting to sides of preparatory trench :							
Not exceeding -							
1.00m deep	...	sqm	20.30	Incl.	17.10	18.85	Incl.
2.00m deep	...	sqm	24.80	14.25	20.80	24.00	Incl.
3.00m deep	...	sqm	32.20	21.30	25.60	30.10	Incl.
4.00m deep	...	sqm	38.40	31.70	37.90	37.10	Incl.
Planking and strutting to sides of excavation for underpinning commencing at base of preparatory excavation :							
Not exceeding -							
1.00m deep	...	sqm	37.40	21.80	32.10	35.50	38.20
2.00m deep	...	sqm	45.10	21.80	35.50	45.60	38.20
3.00m deep	...	sqm	59.30	31.70	46.80	54.80	48.30
4.00m deep	...	sqm	72.30	42.10	60.50	68.80	64.60

UNDERPINNING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CONCRETE 25 MPa							
Footing to underpinning :							
Unreinforced	...	cum	300.00	287.00	328.00	311.00	364.00
Reinforced	...	cum	302.00	290.00	330.00	311.00	356.00
Underpinning	...	cum	337.00	321.00	353.00	326.00	382.00
FORMWORK							
Sides of footings	...	sqm	87.00	88.00	100.00	85.00	100.00
Ditto - left-in	...	sqm	101.00	102.00	122.00	95.00	110.00
Sides of underpinning	...	sqm	92.00	93.00	111.00	108.00	100.00
Ditto - left-in	...	sqm	113.00	114.00	133.00	125.00	115.00
REINFORCEMENT							
Steel bar reinforcement	...	t	2910.00	3305.00	3015.00	2790.00	2970.00
BRICKWORK							
Reduced brickwork	...	cum	1260.00	1300.00	1280.00	1090.00	1300.00
SUNDRIES							
Cut away existing concrete footings :							
Unreinforced	...	cum	365.00	475.00	555.00	474.00	538.00
Reinforced	...	cum	402.00	600.00	641.00	531.00	584.00
Cut away existing brick footings	...	cum	300.00	254.00	362.00	273.00	380.00
100mm thick expanding grout to gap between top of underpinning and underside of footing including all pinning up and jacking for the full length prior to packing	...	sqm	145.00	153.00	144.00	144.00	147.50
COMPOSITE PRICES							
Refer page 110	...	Note					
SOIL STABILISATION							
Soil stabilisation under foundations, refer to page 219	...	Note					

CONCRETE WORKS

BASE PRICES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Concrete, delivered to site :							
20 MPa	...	cum	162.00	130.00	180.00	170.00	220.00
25 MPa	...	cum	169.00	135.00	190.00	175.00	226.00
32 MPa	...	cum	178.00	140.00	200.00	182.00	234.00
40 MPa	...	cum	188.00	150.00	210.00	192.00	245.00
50 MPa	...	cum	203.00	170.00	230.00	212.00	270.00
60 MPa	...	cum	221.00	190.00	255.00	232.00	288.00
ADD extra for :							
Cartage in excess of 20km	...	km/cum	3.00	2.80	3.00	2.75	3.00
Loads less than 3.5 cum (average)	...	cum	40.00-45.00	75.00	40.00-60.00	45.00	95.00
Reinforcement :							
Deformed bar reinforcement, cut to size and delivered and including rolling margin, chairs and tie wire	...	t	1750.00	1400.00	1475.00	1475.00	1560.00
ADD extra for :							
Placing and fixing in position	...	t	850.00	990.00	1000.00	875.00	910.00
SL92 fabric supply only, full sheets	...	sqm	7.40	5.76	8.10	6.06	8.19
SL72 ditto	...	sqm	4.70	3.82	5.25	3.92	5.10
SL62 ditto	...	sqm	3.70	3.06	4.20	3.10	4.03

CONCRETE

Prices for concrete are based on ready-mixed concrete prices and include for delivery to site, handling and placing in position, as well as an allowance for wastage and loss during handling and placing. Refer page 238 for additional allowances to be made for pumping, samples and testing ... Note

UNREINFORCED CONCRETE 25 MPa

Blinding layer, 50mm thick	...	cum	248.00	222.00	360.00	336.00	433.00
Column or pier foundation (in ground)	...	cum	245.00	255.00	281.00	261.00	336.00
Strip footing (in ground)	...	cum	245.00	255.00	281.00	261.00	336.00
Slabs and thickening on fill :							
N.E. 150mm thick	...	cum	250.00	226.00	306.00	285.00	367.00
150/300mm thick	...	cum	250.00	222.00	302.00	281.00	363.00
Machine bases and the like	...	cum	253.00	230.00	340.00	317.00	409.00
Plinths, kerbs and steps	...	cum	253.00	230.00	340.00	317.00	409.00
Cavity wall filling	...	cum	368.00	340.00	422.00	395.00	500.00

REINFORCED CONCRETE 25 MPa

Foundation beams	...	cum	244.00	256.00	283.00	263.00	340.00
Column or pier foundation (in ground)	...	cum	248.00	259.00	285.00	264.00	341.00
Strip footing (in ground)	...	cum	261.00	259.00	285.00	264.00	341.00
Raft foundation	...	cum	231.00	256.00	272.00	253.00	326.00

CONCRETE WORKS

CONCRETE (cont'd)

REINFORCED CONCRETE 25 MPa (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Slabs and thickening on fill :							
N.E. 150mm thick	…	cum	253.00	230.00	310.00	288.00	372.00
150/300mm thick	…	cum	253.00	222.00	306.00	285.00	367.00
Suspended slabs :							
N.E. 150mm thick	…	cum	253.00	226.00	306.00	285.00	367.00
150/300mm thick	…	cum	250.00	222.00	302.00	281.00	363.00
ADD extra for :							
Control strips, delayed pour	...	cum	12.40	33.10	18.75	17.85	23.00
Ribbed or waffle slabs :							
Suspended slab top -							
N.E. 150mm thick	...	cum	253.00	237.00	306.00	285.00	367.00
150/300mm thick	...	cum	253.00	237.00	306.00	285.00	367.00
Ribs	...	cum	306.00	237.00	306.00	285.00	367.00
Walls :							
N.E. 150mm thick	…	cum	368.00	281.00	317.00	295.00	381.00
150/300mm thick	…	cum	337.00	281.00	313.00	292.00	376.00
Columns	...	cum	368.00	292.00	355.00	331.00	427.00
Attached beams	...	cum	306.00	226.00	302.00	281.00	363.00
Lintels	...	cum	498.00	281.00	430.00	403.00	519.00
Stairs and landings	...	cum	337.00	278.00	355.00	331.00	427.00
String courses, copings and the like	...	cum	362.00	347.00	430.00	403.00	519.00
Casings to structural steel :							
Columns	…	cum	368.00	329.00	403.00	378.00	487.00
Beams	…	cum	306.00	263.00	355.00	331.00	427.00
Lintels	…	cum	498.00	384.00	486.00	456.00	588.00
Filling :							
Piers	...	cum	572.00	660.00	655.00	617.00	794.00
Cores of hollow blocks	...	cum	585.00	660.00	711.00	670.00	863.00
Lintel blocks	...	cum	585.00	660.00	655.00	617.00	794.00
ADD extra for :							
Concrete strengths -							
32 MPa	…	cum	9.70	5.40	10.75	7.55	8.60
40 MPa	…	cum	20.40	16.15	21.50	18.30	20.40
50 Mpa	…	cum	36.60	37.70	43.00	39.80	47.40
60 MPa	…	cum	56.00	59.20	70.00	61.30	66.70
Pumping -							
Minimum charge	...	Item	500.00	-	300.00-400.00	600.00	500.00
Hourly rate	...	Hour	-	-	80.00	200.00	220.00
Quantity charge	...	cum	-	-	9.00	16.00	12.50
Allowance for travelling one way * at hourly rate	...	Item	-	-	*	-	-
Approximate additional costs in lieu of foregoing	...	cum	20.00	25.00-40.00	-	-	-
Pneumatically placing, (spraying) additional cost for quantities in excess of 22.50 cum per pour (based on minimum 3 hours with 7.50 cum/hour output)	...	cum	32.00	33.00-50.00	25.00	26.00	33.00
Samples and testing -							
Site sample and three cylinder tests	...	Set	236.00	189.00	236.00	173.00	252.00
Ditto average cost per cum	...	cum	15.75	12.60	15.75	11.55	16.80

CONCRETE WORKS

CONCRETE (cont'd)

REINFORCED CONCRETE 25 MPa (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for (cont'd) :							
Cements and aggregates -							
White cement	...	cum	-	183.00	91.50	51.70	-
Brightonlite or equivalent cement	...	cum	192.50	-	91.50	-	-
Coloured cements generally	...	cum	-	50.00-90.00	-	-	-
Rapid hardening cement (minimum rate)	...	cum	15.05	17.50	19.35	9.70	-
White quartz aggregate in lieu of blue metal	...	cum	61.30	215.50	78.60	48.40	-
Special coloured aggregate	...	cum	-	40.00-100.00	-	-	-
Micro reinforcing systems -							
Average additional cost for incorporating :							
Polypropylene fibres @ 4.0kg per cum of concrete (FIBERMESH M.D.)	...	cum	40.00	40.00	40.00	40.00	40.00
Steel Fibre Reinforced Concrete (SFRC) - @30kg per cum :							
Dramix 3D steel fibre	...	cum	87.00	87.00	87.00	87.00	87.00
Dramix 4D steel fibre	...	cum	81.00	81.00	81.00	81.00	81.00
Dramix 5D steel fibre	...	cum	99.00	99.00	99.00	99.00	99.00
Twisted Steel Micro Reinforcement (TSMR) -							
Helix 5-25 Micro-Rebar fibre reinforcement @10kg per cum (HELIX STEEL)	…	cum	78.00	78.00	78.00	78.00	78.00
Helix 5-25 Micro-Rebar typical application range (5kg to 35kg) per cum (HELIX STEEL)	…	kg	7.80	7.80	7.80	7.80	7.80
Bonding agents -							
Megapoxyh bonding agent	...	cum	45.00	45.00	45.00	45.00	45.00
Waterproofing agents -							
Aquapel/3cc (maximum rate)	...	cum	70.00	110.00	105.00	70.00	80.00
Caltite	...	cum	140.00	125.00	105.00	150.00	102.00
Xypex Admix at 2.6 to 3.4kg/cum	...	cum	65.00-85.00	65.00-85.00	65.00-85.00	65.00-85.00	65.00-85.00
ADD extra Xypex (by volume) for :							
Any special chemical treatment	...	Note					
Increased MPa	...	Note					
Air entraining agents -							
Plastet	...	cum	9.00	-	Incl.	-	Incl.
Micro-Air	...	cum	7.00	-	Incl.	-	Incl.
Rel-Air	...	cum	7.00	-	Incl.	-	Incl.
Silica fume technology -							
General purpose	...	cum	10.00	10.00	10.00	10.00	10.00
Special	...	cum	10.00-40.00	10.00-40.00	10.00-40.00	10.00-40.00	10.00-40.00

LIGHTWEIGHT CONCRETE

BST lightweight concrete, **supply only** cost delivered to site with cartage not exceeding 15km :							
600kg/cum (lightweight infill, roof insulation)	…	cum	725.00	705.00	705.00	745.00	650.00
1200kg/cum (toppings, walls)	…	cum	650.00	625.00	625.00	665.00	580.00
1800kg/cum (lightweight structural uses)	…	cum	525.00	515.00	515.00	535.00	485.00
ADD extra for :							
Placing in position	...	Note					

CONCRETE WORKS

LABOURS AND SUNDRIES ON CONCRETE

CURING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Cure concrete for at least 7 days with :							
Building paper kept damp	…	sqm	2.10	1.80	2.15	1.75	2.20
50mm layer of sand kept damp	…	sqm	2.70	3.05	2.95	2.30	2.70
Plastic membrane	…	sqm	2.40	2.60	2.70	2.45	2.50
Cure concrete with :							
Ashford curing membrane	...	sqm	5.40	5.45	5.40	5.50	5.40
Calcure curing agent	...	sqm	1.35	1.50	1.65	1.40	1.60

JOINTS

Keyed construction joint in :							
150mm slab	…	m	24.60	32.60	24.60	24.00	24.10
200mm slab	…	m	25.60	34.70	25.80	25.60	25.90
ADD extra for :							
25 x 25mm groove filled with non-shrink grout	...	m	11.25	12.95	12.00	11.20	11.35
Expansion joint 13mm wide filled with bitumen impregnated fibreboard and pointed with polysulphide sealant 25mm deep, in :							
150mm slab	...	m	42.40	41.00	44.20	41.10	44.60
200mm slab	...	m	44.90	43.20	46.70	42.80	47.30
ADD extra for :							
20mm dia. dowel rods at 300mm centres including expansion caps (per metre of expansion joint)	...	m	12.65	13.90	15.20	13.00	14.45
Joints and sealants, refer page 265	...	Note					

WATERSTOPS

PVC double bulb waterstop cast into concrete including formwork :							
150mm wide	…	m	34.80	37.60	38.00	37.10	42.20
200mm wide	…	m	40.60	43.80	44.30	43.20	48.90
ADD extra for :							
Mitre on flat	…	No	107.00	109.50	110.00	109.00	113.00
Three way intersection on flat (average)	…	No	113.00	117.00	117.50	116.00	123.50
T-piece (average)	…	No	116.50	120.50	121.00	120.00	127.00
Self expanding waterstop joint fixed to face of construction joint, including all angles and intersections	...	m	49.70	52.00	52.30	51.60	55.60

SEPARATION STRIPS

Heavy duty building paper bond breaker between face of concrete and sheet piling (flat area, add for profile)	...	sqm	4.15	4.40	4.40	4.35	4.75
Polystyrene sheeting between face of concrete and existing building to :							
10mm (nominal) gap	…	sqm	7.30	7.85	7.95	7.75	8.80
50mm ditto	…	sqm	24.70	25.40	25.50	25.30	26.50

CONCRETE WORKS

LABOURS AND SUNDRIES ON CONCRETE (cont'd)

SEPARATION STRIPS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Bitumen impregnated fibreboard strip 12.0mm thick between edge of slab and wall :							
100mm high	…	m	9.10	9.45	9.45	9.35	10.00
150mm high	…	m	13.15	13.55	13.60	13.50	14.25
200mm high	…	m	16.90	17.40	17.45	17.30	18.10
Self expanding cork strip 15mm thick :							
100mm high	…	m	32.60	33.30	33.30	33.10	34.30
200mm high	…	m	62.10	62.80	62.90	62.70	63.90
300mm high	…	m	91.30	92.00	92.10	91.90	93.10
Preformed bitumen impregnated polyurethene foam sealing strip, Compriband :							
15 x 15mm	…	m	5.05	5.65	5.70	5.55	6.55
25 x 25mm	…	m	7.10	7.80	7.90	7.70	8.90
40 x 20mm	…	m	7.25	8.10	8.20	7.95	9.35
50 x 25mm	…	m	12.65	13.80	13.95	13.60	15.65
Joints and sealants, refer page 265	...	Note					

SUNDRIES "CAST-IN"

Continuity strip starter bar system comprising steel bar reinforcement embedded in a steel casing cast in, steel cover later removed and bars bent to line :							
150mm wide x 35mm thick with three pairs 'u' shaped Y12 bars per metre	...	m	113.00	117.00	121.00	113.00	122.00
175mm wide x 40mm thick with six pairs of Y16 bars per 1.20 metres	...	m	129.00	133.00	139.00	130.00	142.00
Galvanised steel tie, 25 x 4 x 300mm long cast in	...	No	1.00	1.10	1.05	1.05	1.00
Polystyrene blockout in 200mm wall including later removal, filling with epoxy mortar and all necessary formwork, blockout to suit pipe size :							
100mm dia.	…	No	47.00	51.00	52.00	47.00	49.00
150mm dia.	…	No	51.00	56.00	58.00	52.00	54.00
200mm dia.	…	No	56.00	61.00	65.00	57.00	61.00
250mm dia.	…	No	62.00	67.00	70.00	62.00	65.00
Ditto in 300mm wall, blockout size :							
100mm dia.	…	No	50.00	53.00	56.00	50.00	51.00
150mm dia.	…	No	53.00	58.00	62.00	55.00	56.00
200mm dia.	…	No	61.00	65.00	68.00	61.00	63.00
250mm dia.	…	No	66.00	72.00	75.00	67.00	69.00

CONCRETE WORKS

CONCRETE FINISHES

SURFACE HARDENERS AND SEALERS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Proprietary surface hardener and dustproofer applied :							
Ashford Formula	...	sqm	6.50	6.55	6.50	6.60	6.50
Diamite	...	sqm	3.55	3.50	3.50	3.55	3.65
Mastertop 200 (Masterplate 200)	...	sqm	31.00	31.00	31.00	31.00	31.00
Mastertop 230 (Anvil Top 3000)	...	sqm	130.00-430.00	130.00-430.00	130.00-430.00	130.00-430.00	130.00-430.00
Masterseal 333 (Silaseal)	...	sqm	1.35	1.35	1.35	1.35	1.35
Rokite	...	sqm	4.85	4.95	4.80	4.95	4.90

SURFACE FINISHES

Labour only :							
Grading to falls	...	sqm	3.50	7.05	4.65	3.85	4.20
Grading to crossfalls or camber	...	sqm	4.05	7.20	4.85	4.40	5.10
Broomed finish	...	sqm	6.00	7.55	7.40	4.85	6.40
Power float monolithic finish	...	sqm	8.15	7.55	8.25	6.15	6.60
Steel trowel finish by hand with driers	...	sqm	8.25	7.55	8.25	7.20	6.80
Wood float finish	...	sqm	8.25	7.00	7.90	6.55	6.65
Water wash only (no special finish)	...	sqm	11.80	12.50	11.00	10.10	10.80

ARCHITECTURAL SURFACE FINISHES

Bush hammered finish to :							
Plain vertical surfaces	…	sqm	69.00	59.60	69.20	64.00	75.80
Ribbed vertical surfaces	…	sqm	84.90	76.60	92.10	88.40	92.50
Sand blasted finish to :							
Plain vertical surfaces	…	sqm	43.50	38.90	46.90	50.70	50.30
Ribbed vertical surfaces	…	sqm	51.00	47.20	56.20	58.70	58.10
Water wash and scrub to expose aggregate and finish to a non-slip surface including additional cost of better aggregate	...	sqm	32.30	26.00	33.20	35.20	31.70
Acid etch plain vertical surfaces	...	sqm	49.90	37.90	45.20	48.50	46.50
Polished finish to concrete, including surface hardeners and sealing	...	sqm	121.00	124.00	130.00	126.00	125.00

SURFACE TREATMENTS

Anti-Graffiti shield, refer page 453	...	Note					

WATERPROOFING

Bituminous paint, two coats	...	sqm	11.85	11.35	11.60	11.35	11.80
Driwal surface application	…	sqm	17.85	16.90	17.90	17.40	18.25
Krystol surface application	…	sqm	16.35	15.45	16.55	16.85	16.95
Xypex concentrate brushed on :							
One coat	…	sqm	15.90	15.85	15.90	15.90	15.90
Two coats	…	sqm	31.80	31.70	31.80	31.80	31.80
ADD extra for :							
Preparation of surfaces	...	Note					

CONCRETE WORKS

CONCRETE CUTTING AND DRILLING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DRILLING							
Average minimum charge $150.00 / $300.00	...	Note					
ADD site establishment fee (average)	...	Sum	90.00	100.00	100.00	95.00	100.00
Drilling through reinforced concrete to form hole :							
Not exceeding 10 holes -							
25mm dia. - not exceeding 10 holes	...	10mm	2.00	2.00	2.65	3.35	3.00
50mm dia. - not exceeding 10 holes	...	10mm	2.50	2.00	3.20	3.45	3.00
100mm dia. - not exceeding 10 holes	...	10mm	3.50	2.75	4.10	3.80	5.80
150mm dia. - not exceeding 10 holes	...	10mm	4.50	4.10	6.15	5.20	6.60
200mm dia. - not exceeding 10 holes	...	10mm	6.00	6.60	9.20	6.85	9.70
300mm dia. - not exceeding 10 holes	...	10mm	12.00	15.40	18.35	12.65	16.10
450mm dia. - not exceeding 10 holes	...	10mm	25.00	27.50	26.00	24.70	18.50
600mm dia. - not exceeding 10 holes	...	10mm	30.00	39.50	35.00	28.40	-
800mm dia. - not exceeding 10 holes	...	10mm	-	-	43.80	32.50	-
SLAB SAWING							
Average minimum charge $150.00 / $300.00	...	Note					
ADD site establishment fee (average)	...	Sum	95.00	100.00	100.00	95.00	100.00
Straight cutting reinforced concrete :							
50mm deep	...	m	5.00	5.00	5.50	6.20	5.10
75mm deep	...	m	7.50	7.50	7.75	8.75	7.70
100mm deep	...	m	10.00	10.00	11.25	11.60	11.80
150mm deep	...	m	15.00	15.00	15.80	16.10	15.30
200mm deep	...	m	25.00	20.00	22.00	24.10	19.90
250mm deep	...	m	30.00	26.00	29.50	36.40	28.50
450mm deep	...	m	68.00	63.00	75.00	78.00	-
WALL SAWING							
Minimum charges range from $250.00 / $1000.00	...	Note					
ADD site establishment fee (average)	...	Sum	95.00	300.00	150.00	135.00	200.00
Straight cutting reinforced concrete :							
50mm deep	...	m	17.50	14.50	21.00	47.00	21.00
75mm deep	...	m	21.00	22.00	28.30	52.00	26.60
100mm deep	...	m	30.90	29.50	41.00	62.00	41.10
150mm deep	...	m	42.00	42.00	66.00	88.00	71.20
200mm deep	...	m	62.90	100.00	92.00	113.00	117.00
250mm deep	...	m	119.00	130.00	143.00	139.00	155.50
300mm deep	...	m	223.00	237.00	205.00	165.00	183.00
400mm deep	...	m	333.00	-	-	216.00	286.00
600mm deep	...	m	-	-	-	268.00	508.00
WATER CUTTING							
Average minimum charge $600.00	...	Note					
Cut only 50mm to 100mm deep to expose reinforcement in slab :							
On flat	...	sqm	40.00-125.00	40.00-125.00	40.00-130.00	40.00-125.00	40.00-125.00
Overhead	...	sqm	160.00-220.00	160.00-220.00	170.00-230.00	160.00-220.00	175.00-225.00

CONCRETE WORKS

FORMWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices assume moderate quantities and five uses	...	Note					
Prices exclude perimeter screens and handrails	...	Note					

CLASSES OF FORMWORK

CLASS 1

Surfaces of high quality, fine tolerances, subject to close scrutiny, (visual quality).

CLASS 2

Surfaces of uniform texture, close tolerances, (visual quality).

CLASS 3

Surfaces with limited imperfections, not subject to close scrutiny, (good visual quality when viewed as a whole).

CLASS 4

Surfaces to be rendered or concealed, (visual quality not significant).

CLASS 5

Surfaces permanently concealed, i.e. footings, rear of retaining walls and the like.

CLASS 5 FORMWORK

ADJUST THE PRICES GIVEN HEREAFTER FOR :								
THREE USES	+ 15%	...	Note					
ONE USE	+ 50%	...	Note					
ONE USE LEFT IN	+ 75%	...	Note					
Sides of column or pier foundations		...	sqm	92.00	90.00	100.00	65.00	86.00
Sides of foundation beams		...	sqm	92.00	90.00	105.00	71.00	96.00
Sides of foundations etc. :								
100/200mm high		...	m	32.00	35.00	32.00	16.00	33.00
200/300mm high		...	m	36.00	40.00	42.00	17.00	43.00

CLASS 3 FORMWORK

ADJUST THE PRICES GIVEN HEREAFTER FOR :								
CLASS 1	+ 50% to + 100%	...	Note					
CLASS 2	+ 10%	...	Note					
CLASS 4	- 10%	...	Note					
CLASS 5	- 20%	...	Note					
THREE USES	+ 15%	...	Note					
ONE USE	+ 50%	...	Note					
ONE USE LEFT IN	+ 75%	...	Note					
STRUTTING, FOR EVERY ADDITIONAL 1.00 METRE OVER 3.50 METRES		...	sqm	15.00	15.00	16.00	12.00	16.00

CONCRETE WORKS

FORMWORK (cont'd)

CLASS 3 FORMWORK (cont'd)

ADJUST THE PRICES HEREAFTER FOR (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Soffit of suspended slab :							
100/200mm thick	...	sqm	130.00	120.00	128.00	93.00	133.00
200/300mm thick	...	sqm	136.00	125.00	135.00	98.00	137.00
Soffit of suspended sloping slab not exceeding 15° from the horizontal :							
100/200mm thick	...	sqm	136.00	140.00	180.00	99.00	190.00
200/300mm thick	...	sqm	141.00	150.00	210.00	100.00	195.00
Soffit of drop panels	...	sqm	136.00	140.00	168.00	100.00	156.00
Table form to support rib or waffle pans	...	sqm	124.00	125.00	140.00	93.00	137.00
Soffit of suspended slab between ribs	...	sqm	147.00	140.00	168.00	110.00	162.00
Sides and soffit of ribs in suspended ribbed floor	...	sqm	147.00	150.00	140.00	110.00	162.00
Sloping soffit of stair flights	...	sqm	152.00	240.00	250.00	116.00	186.00
Sides of core walls	...	sqm	147.00	180.00	150.00	131.00	168.00
Sides of walls	...	sqm	147.00	150.00	165.00	122.00	172.00
Sides of walls curved to :							
2000mm radius	...	sqm	227.00	220.00	315.00	167.00	240.00
6000mm radius	...	sqm	204.00	220.00	230.00	165.00	240.00
Sides of wall cast against existing (one side only measured)	...	sqm	175.00	180.00	205.00	143.00	221.00
Sides of attached pier	...	sqm	159.00	150.00	174.00	138.00	162.00
Sides of isolated column or pier :							
Square and/or rectangular	...	sqm	159.00	135.00	160.00	133.00	162.00
Hexagonal	...	sqm	175.00	150.00	200.00	137.00	188.00
L-shaped	...	sqm	175.00	150.00	220.00	137.00	188.00
U-shaped	...	sqm	175.00	150.00	220.00	143.00	188.00
Circular	...	sqm	159.00	150.00	126.00	142.00	172.00
Tube formers, refer page 248	...	Note					
Sides and soffit of attached beams	...	sqm	152.00	135.00	160.00	137.00	162.00
Sides and soffit of isolated beam or lintel	...	sqm	159.00	150.00	175.00	140.00	188.00
ADD extra for :							
Coating with a retarding agent	...	sqm	3.60	3.50	3.50	2.55	3.00

CONCRETE WORKS

FORMWORK (cont'd)

SUNDRY CLASS 3 FORMWORK (cont'd)

ADJUST THE PRICES HEREAFTER FOR (cont'd) :

			UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CLASS 1	+ 45%	...	Note					
CLASS 2	+ 10%	...	Note					
CLASS 4	- 10%	...	Note					
CLASS 5	- 20%	...	Note					
SINGLE USE	+ 75%	...	Note					
Edges, sides or risers :								
N.E. 100mm high		...	m	22.50	30.00	60.00	21.00	36.00
100/200mm high		...	m	29.00	35.00	42.00	23.00	49.00
200/300mm high		...	m	34.00	40.00	48.00	31.00	57.00
Receding riser :								
100/200mm high		...	m	39.00	40.00	60.00	32.00	54.00
200/300mm high		...	m	45.50	45.00	48.00	32.00	54.00
Stair strings not exceeding 300mm wide (extreme) including cutting and fitting to treads and risers		...	m	68.00	65.00	100.00	54.00	77.00
Form groove :								
12 x 12mm		...	m	10.50	17.50	12.00	9.00	19.00
25 x 25mm		...	m	11.50	20.00	15.00	10.00	23.00
32 x 32mm		...	m	12.50	22.50	17.00	11.10	31.00
Form rebate :								
75 x 50mm		...	m	14.50	25.00	27.00	11.00	19.00
100 x 50mm		...	m	18.50	30.00	37.00	14.50	25.00
150 x 75mm		...	m	29.00	40.00	48.00	22.90	48.00
Rounded edge not exceeding 50mm girth		...	m	20.50	20.00	37.00	16.00	31.00
Moulded edge not exceeding 50mm girth		...	m	22.50	25.00	42.00	21.90	42.00
Chamfered edge not exceeding 50mm wide		...	m	13.50	15.00	13.00	11.20	40.00
V-joint		...	m	13.50	15.00	13.00	11.20	15.00
Splayed internal angle not exceeding 50mm wide		...	m	17.50	20.00	21.00	13.00	20.00
Raking cutting		...	m	31.00	15.00	21.00	11.10	17.00
Circular cutting		...	m	36.00	20.00	27.00	17.20	17.00
Form hole/opening in concrete :								
100/200mm thick, size -								
100 x 100mm		...	No	29.00	35.00	32.00	25.00	40.00
150 x 150mm		...	No	34.00	40.00	37.00	26.50	40.00
200 x 200mm		...	No	39.00	45.00	42.00	31.00	40.00
250 x 250mm		...	No	45.00	50.00	52.00	35.50	48.00
300 x 300mm		...	No	48.00	55.00	54.00	38.00	55.00
350 x 350mm		...	No	50.00	60.00	60.00	42.00	64.00
400 x 400mm		...	No	55.00	65.00	70.00	45.00	75.00
450 x 450mm		...	No	57.00	70.00	75.00	50.50	80.00
500 x 500mm		...	No	62.00	75.00	80.00	55.00	89.00

CONCRETE WORKS

FORMWORK (cont'd)

SUNDRY CLASS 3 FORMWORK (cont'd)

ADJUST THE PRICES HEREAFTER FOR (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Form hole/opening in concrete :							
200/300mm thick, size -							
100 x 100mm	…	No	34.00	35.00	34.00	25.50	43.00
150 x 150mm	…	No	37.00	40.00	40.00	29.00	48.00
200 x 200mm	…	No	40.00	45.00	45.00	34.00	49.00
250 x 250mm	…	No	45.00	50.00	62.00	38.00	57.00
300 x 300mm	…	No	48.00	55.00	64.00	41.00	72.00
350 x 350mm	…	No	52.00	60.00	70.00	45.50	79.00
400 x 400mm	…	No	57.00	65.00	80.00	49.00	89.00
450 x 450mm	…	No	62.00	70.00	85.00	53.00	102.00
500 x 500mm	…	No	68.00	75.00	90.00	59.00	115.00

BOARDED FORMWORK

Formwork of sawn timber boarding to provide an off-form finish to :							
Soffit of suspended slab 200/300mm thick -							
(3) Uses	…	sqm	198.00	195.00	200.00	188.00	224.00
(1) Use	…	sqm	260.00	250.00	250.00	230.00	257.00
Sides of walls -							
(3) Uses	…	sqm	206.00	195.00	200.00	197.00	233.00
(1) Use	…	sqm	268.00	250.00	250.00	242.00	267.00
Sides of column -							
(3) Uses	…	sqm	227.00	240.00	241.00	215.00	245.00
(1) Use	…	sqm	273.00	270.00	267.00	245.00	279.00
Sides and soffit of attached beam -							
(3) Uses	…	sqm	212.00	240.00	220.00	204.00	239.00
(1) Use	…	sqm	273.00	270.00	250.00	249.00	270.00

METAL TRAY FORMWORK

Bondek or Condek permanent formwork to suspended slab :							
0.75mm thick	…	sqm	97.00	130.00	120.00	94.00	116.00
1.00mm thick	…	sqm	103.00	135.00	125.00	105.00	124.00
Kingflor deep deck permanent formwork to suspended slab :							
1.25mm thick	...	sqm	135.00	135.00	135.00	135.00	138.00
ADD extra for :							
Edge form -							
100mm	…	m	23.00	24.00	28.00	19.00	33.00
125mm	…	m	28.00	27.50	32.00	21.00	40.00
150mm	…	m	33.00	39.50	34.00	22.00	49.00
Shear studs welded on	…	No	9.00	10.00	9.00	9.00	10.00
Fireproofing soffit	...	Note					
Coating in excess of Z 200	...	Note					
Special paint treatment on exposed surfaces	...	Note					
Propping when required	...	Note					

CONCRETE WORKS

FORMWORK (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
C.G.I. FORMWORK							
Corrugated galvanised iron permanent formwork to sides of retaining walls and the like against ground	...	sqm	80.00	91.00	78.00	74.00	101.00
SLIP-FORM FORMWORK							
Refer page 126	...	Note					
TUBE FORMERS							
Prices are based on minimum 100 metres	...	Note					
Fibre tube form, including stripping :							
300mm I.D.	...	m	120.00	139.50	137.50	141.50	162.50
350mm I.D.	...	m	131.00	151.50	150.00	154.00	176.00
400mm I.D.	...	m	146.00	167.00	166.50	171.50	194.00
450mm I.D.	...	m	160.50	182.50	182.50	188.00	211.00
500mm I.D.	...	m	171.50	194.50	194.50	200.00	224.00
600mm I.D.	...	m	196.50	221.50	222.00	229.00	254.50
F.R.C. pipe tube form, left in :							
Classic (premium) -							
200mm dia.	...	m	142.00	162.00	156.50	184.50	183.00
260mm dia.	...	m	167.50	190.00	184.00	196.00	214.00
345mm dia.	...	m	215.00	239.00	232.00	254.00	265.00
425mm dia.	...	m	271.00	296.00	289.00	315.00	326.00
585mm dia.	...	m	370.00	393.00	387.00	389.00	422.00
Spirally wound plastic tube form Profile 98, including stripping :							
200mm I.D.	...	m	98.00	114.50	110.00	111.50	130.50
300mm I.D.	...	m	111.50	129.50	125.00	126.50	147.50
400mm I.D.	...	m	132.50	152.00	147.00	149.00	171.50
500mm I.D.	...	m	156.50	179.00	173.00	175.00	201.00
600mm I.D.	...	m	184.50	200.00	194.00	196.50	224.00
Spirally wound steel tube form Duraduct, left in :							
200mm dia.	...	m	87.50	104.00	99.50	101.00	120.00
300mm dia.	...	m	103.50	120.50	116.00	117.50	137.50
400mm dia.	...	m	122.50	141.50	136.50	138.00	160.00
500mm dia.	...	m	140.50	161.00	156.00	157.50	181.50
600mm dia.	...	m	175.50	198.00	192.00	194.00	220.00
VOID FORMERS							
Spirally wound steel tube void form, left in :							
500mm dia.	...	m	76.40	79.60	79.00	79.00	83.10
750mm dia.	...	m	156.50	160.50	159.50	159.50	164.50
1000mm dia.	...	m	278.00	283.00	282.00	282.00	288.50
1250mm dia.	...	m	326.00	332.00	331.00	331.00	339.00

CONCRETE WORKS

REINFORCEMENT

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for reinforcement include for bending, tying and placing, rolling margin and bending schedules	...	Note					
Prices for fabric reinforcement include 15% for laps	...	Note					
BAR							
Structural grade round bar reinforcement	...	t	2730.00	2510.00	2600.00	2470.00	2600.00
Ditto in ligatures	...	t	2990.00	2510.00	2600.00	2520.00	2820.00
Tempcore, deformed bar reinforcement :							
Y 12mm	...	t	2680.00	2510.00	2600.00	2470.00	2600.00
Y 16mm	...	t	2680.00	2510.00	2600.00	2470.00	2600.00
Y 20 - 36mm	...	t	2680.00	2510.00	2600.00	2470.00	2600.00
Threaded reinforcing bar (Reidbar)	...	t	2990.00	2510.00	2600.00	2925.00	3080.00
ADD extra for :							
Bar reinforcement in prestressed situations	...	+10%					
Galvanising	...	t	1500.00	1800.00	1375.00	1300.00	1800.00
FABRIC							
RECTANGULAR MESH							
RL 1218	...	sqm	34.90	32.60	39.50	35.80	51.50
RL 1118	...	sqm	28.90	26.90	33.30	30.20	40.40
RL 1018	...	sqm	24.30	21.50	27.60	24.80	32.40
RL 918	...	sqm	20.70	19.45	23.50	21.50	27.40
RL 818	...	sqm	17.60	16.60	19.40	18.30	20.90
RL 718	...	sqm	15.00	14.20	17.45	15.65	20.00
ADD extra for galvanising :							
RL 1118	...	sqm	16.50	20.00	15.00	14.50	20.00
RL 918	...	sqm	12.00	14.00	11.00	10.00	14.00
RL 718	...	sqm	8.50	10.00	8.00	7.50	10.00
SQUARE MESH							
SL 81	...	sqm	23.40	21.50	27.30	22.20	26.80
SL 102	...	sqm	17.80	16.55	20.90	16.95	20.80
SL 92	...	sqm	14.60	13.30	16.40	13.00	16.00
SL 82	...	sqm	12.60	11.80	12.65	11.35	13.65
SL 72	...	sqm	11.10	10.75	12.65	10.20	11.95
SL 62	...	sqm	9.15	9.00	10.55	8.50	9.90
SL 52	...	sqm	7.45	7.65	8.80	7.70	7.75
SL 42	...	sqm	-	-	-	6.60	-
ADD extra for galvanising :							
SL 92	...	sqm	8.50	10.50	8.00	7.50	10.50
SL 72	...	sqm	5.50	6.50	5.00	4.50	6.50

CONCRETE WORKS

REINFORCEMENT (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FIBRE							
Refer below for FIBERMESH M.D. as a replacement for wire mesh temperature and shrinkage reinforcement	...	Note					
Steel fibres at :							
20kg/cum of concrete	...	cum	40.00	40.00	40.00	40.00	40.00
40kg/cum of concrete	...	cum	80.00	80.00	80.00	80.00	80.00
TRENCH MESH							
L 8TM :							
200mm wide	...	m	4.00	4.55	5.15	4.90	6.35
300mm wide	...	m	5.25	6.15	6.50	6.30	7.25
400mm wide	...	m	6.50	7.00	8.40	7.75	9.05
L11TM :							
200mm wide	...	m	7.35	7.10	9.05	8.00	10.20
300mm wide	...	m	9.60	9.30	11.45	10.45	13.15
400mm wide	...	m	11.85	13.65	14.55	13.55	16.20
ROD CONNECTIONS							
Butt weld column rods, rod size :							
16mm dia.	...	No	15.90	18.45	18.75	19.05	23.90
20mm dia.	...	No	18.50	21.50	21.90	22.20	27.90
24mm dia.	...	No	21.50	25.00	25.40	25.80	32.40
28mm dia.	...	No	24.80	28.80	29.30	29.70	37.40
32mm dia.	...	No	28.10	32.70	33.20	33.70	42.40
36mm dia.	...	No	33.10	38.50	39.10	39.60	49.90
Mechanical threaded connection (Lenton System), rod size :							
16mm dia.	...	No	78.40	83.90	84.60	85.10	95.60
20mm dia.	...	No	88.50	94.10	94.70	95.30	106.00
24mm dia.	...	No	105.00	110.50	111.00	112.00	122.50
28mm dia.	...	No	125.00	131.00	131.50	132.00	143.50
32mm dia.	...	No	133.00	139.00	140.00	140.50	151.50
36mm dia.	...	No	156.00	162.00	162.50	163.50	175.00
MICRO REINFORCING SYSTEM USING SYNTHETIC FIBRES							
Synthetic fibres incorporated in concrete as a replacement for wire mesh temperature and shrinkage reinforcement, additional cost for :							
Fibrillated polypropylene fibres @ 0.9kg per cum of concrete (FIBERMESH M.D.) in -							
100mm thick slab	...	sqm	2.75	2.85	2.75	2.85	2.75
150mm thick slab	...	sqm	4.15	4.30	4.15	4.30	4.15

CONCRETE WORKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CFRP STRENGTHENING							
Carbon fibre reinforced polymer strengthening strip bonded to concrete surface (SIKA CARBO DUR) :							
Single strip -							
80mm wide - x 1.2mm thick	…	m	130.00-135.00	130.00-135.00	140.00-145.00	130.00-135.00	140.00-145.00
90mm wide - x 1.4mm thick	…	m	145.00-155.00	145.00-155.00	150.00-160.00	145.00-155.00	150.00-160.00
120mm wide - x 1.2mm thick	…	m	205.00-230.00	205.00-230.00	210.00-235.00	205.00-230.00	210.00-235.00
120mm wide - x 1.4mm thick	…	m	230.00-250.00	230.00-250.00	235.00-255.00	230.00-250.00	235.00-255.00
ADD extra for :							
Substrate preparation	...	Note					
Additional strips at similar rates	...	Note					
Overcoating and protection of strips	...	Note					
Scaffolding, working platform, etc.	...	Note					
PRESTRESSING IN SITU CONCRETE							
Refer page 112 for indicative costs per sqm slab area	...	Note					
Prestressing including tendons, tendon ducts, anchorages, tensioning, grouting, etc., complete, based on spans of 7.50m to 8.00m and tendon length :							
10.0m	…	t	8700.00-9100.00	8800.00-9200.00	8700.00-9100.00	8700.00-9100.00	8700.00-9100.00
20.0m	…	t	7800.00-8100.00	7900.00-8200.00	7900.00-8200.00	7800.00-8100.00	7900.00-8200.00
30.0m	…	t	7600.00-7900.00	7800.00-8100.00	7700.00-8000.00	7800.00-8000.00	7700.00-8000.00
40.0m	…	t	7200.00-7500.00	7500.00-7800.00	7300.00-7600.00	7200.00-7500.00	7300.00-7600.00
PRECAST CONCRETE							
Prices for precast concrete are based on reasonable quantities of a repetitive nature, when numerous moulds and small quantities are involved, prices could increase appreciably	...	Note					
Prices include supply, delivery, mobile craneage, erection and fixing in position	...	Note					
WALLING							
Non load bearing wall panel typical size 15 to 20sqm with four fixing points using cast in ferrules or grouted dowel connections and a smooth off form grey external face and a floated finish internal face :							
150mm thick	…	sqm	310.00	310.00	325.00	310.00	306.00
175mm thick	…	sqm	335.00	343.00	360.00	333.00	357.00
200mm thick	…	sqm	355.00	376.00	395.00	356.00	388.00

CONCRETE WORKS

PRECAST CONCRETE (cont'd)

WALLING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Load bearing panels, typical size 20 to 25 sqm for industrial buildings with 2 bottom grouted dowel connections and 2 welded bracket top fixing and a class 2 off form grey external face and a floated finish internal face :							
150mm thick	...	sqm	335.00	345.00	365.00	342.00	357.00
175mm thick	...	sqm	365.00	376.00	395.00	367.00	383.00
200mm thick	...	sqm	385.00	407.00	425.00	394.00	393.00
ADD extra for :							
Galvanised fixing brackets	...	sqm	26.00	31.00	51.00	-	51.00
Stainless steel fixing brackets	...	sqm	80.00	160.00	-	-	70.00
White cement - two layer construction	...	sqm	33.00	40.00	25.00	-	25.00
Off white cement for full thickness	...	sqm	33.00	30.00	18.00	16.00	18.00
Quartz or coloured aggregate	...	sqm	67.00	60.00	39.00	80.00	40.00
Oxide colouring - 1% for full thickness	...	sqm	33.00	45.00	33.00	20.00	50.00
Surface finish -							
Profiled and fenestrated panels	...	sqm	52.00	48.00	52.00	-	52.00
Acid etched	...	sqm	68.00	50.00	69.00	-	66.00
Honed	...	sqm	137.00	205.00	132.00	148.00	128.00
Polished	...	sqm	159.00	240.00	155.00	197.00	167.00
Sand blasted	...	sqm	68.00	65.00	68.00	65.00	52.00
Water washed	...	sqm	62.00	65.00	53.00	65.00	52.00
Mould Liner (based on 40 uses)	...	sqm	15.00-45.00	15.00-45.00	15.00-45.00	15.00-45.00	13.00-44.00
Stained	...	Note					
Tower craneage, additional cost if required, refer Preliminaries	...	Note					
Non load bearing pretensioned 1200mm wide hollowcore units :							
150mm thick	...	sqm	-	-	-	148.00	176.00
200mm thick	...	sqm	-	175.00	200.00	154.00	189.00
250mm thick	...	sqm	-	-	210.00	168.00	199.00
ADD extra for :							
Painted finish	...	sqm	-	15.00	-	-	11.00
Integral exposed aggregate finish	...	sqm	-	-	21.00	-	11.00-20.00
Load bearing units	...	Note					

FLOORING

Hollowcore pretensioned 1200mm wide units :							
150mm thick	...	sqm	130.00-140.00	140.00-150.00	-	123.00	126.00-142.00
200mm thick	...	sqm	150.00-160.00	175.00-190.00	180.00-195.00	126.00	181.00-195.00
250mm thick	...	sqm	180.00-190.00	185.00-200.00	188.00-195.00	145.00	188.00-195.00
300mm thick	...	sqm	190.00-210.00	200.00-215.00	195.00-210.00	155.00	195.00-212.00
350mm thick	...	sqm	205.00-225.00	-	-	162.00	-
400mm thick	...	sqm	225.00-250.00	220.00-240.00	205.00-225.00	181.00	-
ADD extra for :							
65mm thick in situ concrete topping including mesh reinforcement	...	sqm	50.00-60.00	70.00-80.00	75.00-85.00	58.00	81.00
Penetrations	...	Note					

CONCRETE WORKS

PRECAST CONCRETE (cont'd)

FLOORING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Permanent formwork flooring system. Precast reinforced concrete permanent formwork panels, typical size 6000mm long x 2500mm wide x 65mm thick with smooth off-form soffit, including temporary propping (average cost range)	...	sqm	199.00	197.00	205.00	135.00	205.00
ADD extra for :							
In situ concrete structural topping and reinforcement as required	...	Note					
Penetrations	...	Note					
Beam and infill system. Prestressed concrete inverted T beams with fibre cement formboard (average cost range) :							
130mm deep	...	sqm	110.00-130.00	110.00-130.00	110.00-130.00	-	105.00-125.00
150mm deep	...	sqm	120.00-140.00	120.00-140.00	120.00-140.00	125.00-145.00	110.00-130.00
200mm deep	...	sqm	125.00-145.00	125.00-145.00	125.00-145.00	130.00-150.00	115.00-135.00
250mm deep	...	sqm	135.00-155.00	135.00-155.00	135.00-155.00	140.00-160.00	120.00-140.00
ADD extra for :							
In situ concrete structural topping and reinforcement as required	...	Note					
Penetrations	...	Note					

COLUMNS

Columns in lengths of 3000mm with reinforcement @150kg/cum, grouted dowel connections and a smooth off-form finish :							
300 x 300mm	...	m	365.00	379.00	405.00	534.00	395.00
400 x 400mm	...	m	400.00	505.00	530.00	626.00	515.00
500 x 500mm	...	m	760.00	722.00	470.00	745.00	715.00
600 x 600mm	...	m	950.00	949.00	905.00	920.00	895.00
750 x 750mm	...	m	1130.00	1150.00	1200.00	1140.00	1175.00
ADD extra for :							
Steel baseplate connection	...	Note					

BEAMS

Beams in lengths of 8000mm with reinforcement @ 150kg/cum, stressing strand @ 30kg/cum, grouted dowel connections and a smooth off-form finish :							
450 x 450mm deep	...	m	690.00	641.00	645.00	534.00	633.00
450 x 600mm deep	...	m	742.00	700.00	730.00	607.00	721.00
450 x 750mm deep	...	m	800.00	888.00	845.00	684.00	832.00
450 x 800mm deep	...	m	910.00	937.00	945.00	742.00	942.00
450 x 1050mm deep	...	m	970.00	1090.00	1070.00	882.00	1050.00
450 x 1200mm deep	...	m	1080.00	1150.00	1150.00	949.00	1125.00

CONCRETE WORKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TILT-UP CONCRETE WALL PANELS							
Prices include mobile craneage, erection and fixing in position	...	Note					
The following are site cast industrial standard wall panels	...	Note					
Openings up to 10sqm not deducted	...	Note					
Wall panel in 30 MPa concrete with smooth off-form finish, fixed in position, in panels of approximately 25.0 sqm :							
100mm thick	...	sqm	155.00	155.00	158.00	150.00	155.00
125mm thick	...	sqm	165.00	164.00	163.00	155.00	160.00
150mm thick	...	sqm	167.00	169.00	173.00	170.00	172.00
170mm thick	...	sqm	184.00	182.00	183.00	186.00	176.00
200mm thick	...	sqm	220.00	194.00	203.00	219.00	195.00
ADD extra for :							
Higher standard finish and better tolerances	...	sqm	20.00-30.00	19.00-29.00	18.00-30.00	17.50-27.50	19.00-31.00
Surface finish one face -							
Ribbed or grooved	...	sqm	29.00	28.00	29.00	25.00	34.00
Vertical timber grain	...	sqm	18.00	20.00	20.00	25.00	22.00
Exposed aggregate	...	sqm	23.00	23.00	26.00	22.00	26.00
Cream coloured cement in -							
100mm thick panel	...	sqm	-	8.50	9.50	9.00	9.00
125mm thick panel	...	sqm	12.00	10.50	11.50	11.00	11.00
White cement and aggregate in -							
100mm thick panel	...	sqm	-	22.00	25.00	25.00	24.00
125mm thick panel	...	sqm	29.00	27.00	30.00	30.00	29.00
Joints and sealants -							
Normal situation	...	sqm	15.00-25.00	15.00-20.00	15.00-24.00	14.00-19.00	15.00-24.00
Fire rated situation	...	sqm	20.00-30.00	20.00-28.00	20.00-30.00	19.00-27.00	23.00-32.00
PRECAST AERATED CONCRETE							
Floor panels :							
150mm thick (4000mm long)	...	sqm	223.00	229.00	223.00	219.00	195.00
200mm thick (5000mm long)	...	sqm	301.00	310.00	301.00	306.00	261.00
250mm thick (6000mm long)	...	sqm	362.00	372.00	362.00	374.00	317.00
Wall panels, non load bearing :							
125mm thick	...	sqm	213.00	216.00	213.00	210.00	193.00
150mm thick	...	sqm	244.00	250.00	244.00	242.00	217.00
200mm thick	...	sqm	350.00	372.00	362.00	374.00	317.00
ADD extra for :							
Surface hardener	...	sqm	6.75	7.00	6.80	6.80	6.80
Surface hardener and waterproofer	...	sqm	13.30	13.70	13.45	13.45	13.35
Cladding panels 75mm thick and up to 3000mm long, fixed to wall framing	...	sqm	108.00	110.00	108.00	108.00	99.00
ADD extra for :							
Wall framing	...	Note					
Finishes, external and internal	...	Note					

CONCRETE WORKS

PRECAST GLASS REINFORCED CEMENT PANELS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices of G.R.C. panels are based on reasonable quantities of panels of a repetitive nature, when numerous moulds and small quantities are involved rates could increase by up to 100%.	...	Note					
Prices include supply, delivery, erection and fixing	...	Note					
Refer Preliminaries page 173 for tower or other crane rates	...	Note					
10mm thick fascia or spandril panel, standard grey colour with smooth finish, plain butt joints, all fixings and standard finishes, sealants and pointing :							
Flat panels	...	sqm	390.00	380.00	405.00	390.00	400.00
Curved panels	...	sqm	445.00	430.00	455.00	440.00	450.00
ADD extra for :							
Brass ferrules in lieu of galvanised	...	sqm	50.00	50.00	50.00	50.00	50.00
Integral stud framing	...	sqm	150.00	150.00	150.00	155.00	155.00
Bone colour using Brightonlite cement and white sand	...	sqm	65.00	55.00	66.00	61.00	56.00
White colour using white cement and white sand	...	sqm	67.00	70.00	66.00	76.00	66.00
Integral aggregate and sand blast finish	...	sqm	58.00	55.00	65.00	58.00	61.00
Integral aggregate polished finish	...	sqm	183.00	175.00	195.00	170.00	185.00
Profiled finish formed in the mould	...	sqm	33.00	32.00	34.00	32.00	34.00
Roller applied textured paint finish	...	sqm	28.00	28.00	28.00	28.00	28.00
Two pack ceramic type finish	...	sqm	85.00	85.00	85.00	85.00	85.00
Trowel applied textured paint finish 2/3mm thick	...	sqm	37.00	37.00	37.00	37.00	37.00
Silicone protection treatment	...	sqm	15.00	15.00	15.00	15.00	15.00

BRICKWORK AND BLOCKWORK

			UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for brickwork are based on the cheapest general purpose common or facing brick available		...	Note					
A distance of 30 kilometres has been assumed for cartage of bricks and blocks		...	Note					
Prices for brickwork and blockwork include for plumbing angles, normal straight cutting, forming reveals, openings, weepholes, etc., raking out or leaving joints for plastering, all normal bedding, building in ends of lintels and all other sundry labours; blockwork prices include for fractional corner and end blocks		...	Note					

BASE PRICES

Bricks delivered to site :

Clay bricks, standard -								
230 x 100 x 76mm	- common	...	1000	824.00	640.00	750.00	850.00	813.00
	- facing	...	1000	1187.00	960.00	900.00	940.00	1120.00
	- glazed	...	1000	5900.00	6020.00	5860.00	5310.00	5710.00
230 x 110 x 119mm	- common	...	1000	-	-	975.00	1290.00	1183.00
230 x 110 x 162mm	- common	...	1000	-	920.00	-	-	1625.00
Clay bricks, modular -								
290 x 90 x 90mm	- common	...	1000	-	-	-	980.00	954.00
	- facing	...	1000	-	-	-	-	1364.00
290 x 90 x 162mm	- facing	...	1000	-	-	-	1950.00	1979.00
305 x 90 x 76mm	- common	...	1000	-	-	-	780.00	954.00
305 x 90 x 162mm	- common vert. cored	...	1000	-	-	-	1180.00	1548.00
	- common horz. cored	...	1000	-	-	-	1000.00	1548.00
Concrete masonry bricks -								
230 x 110 x 76mm standard	- common	...	1000	770.00	-	-	-	856.00
	- facing	...	1000	-	-	-	-	-
290 x 90 x 90mm modular	- common	...	1000	-	-	-	-	-
	- facing	...	1000	-	-	-	-	-

Blocks delivered to site :

Concrete masonry blocks -								
100 x 100 x 400mm, hollow	- natural colour	...	100	371.00	162.00	199.00	260.00	204.00
	- colour	...	100	485.00	210.00	259.00	296.00	-
100 x 200 x 400mm, hollow	- natural colour	...	100	346.00	180.00	186.00	302.00	215.00
	- colour	...	100	448.00	283.00	216.00	356.00	-
150 x 200 x 400mm, hollow	- natural colour	...	100	466.00	195.00	224.00	410.00	257.00
	- colour	...	100	602.00	351.00	269.00	459.00	-
200 x 200 x 400mm, hollow	- natural colour	...	100	662.00	239.00	279.00	475.00	301.00
	- colour	...	100	785.00	485.00	334.00	573.00	-
100 x 100 x 400mm, solid	- natural colour	...	100	-	-	183.00	265.00	189.00
	- colour	...	100	-	-	183.00	306.00	-
100 x 200 x 400mm, solid	- natural colour	...	100	398.00	193.00	218.00	388.00	286.00
	- colour	...	100	538.00	303.00	289.00	504.00	-
150 x 200 x 400mm, solid	- natural colour	...	100	-	-	-	583.00	-
	- colour	...	100	-	-	-	750.00	-
100 x 100 x 400mm, split face solid	- natural colour	...	100	-	-	-	291.00	-
	- colour	...	100	-	278.00	290.00	342.00	-
100 x 200 x 400mm, split face hollow	- natural colour	...	100	-	-	-	406.00	-
	- colour	...	100	-	394.00	324.00	563.00	-
100 x 200 x 400mm, ribbed split face solid	- natural colour	...	100	-	-	-	413.00	-
	- colour	...	100	-	-	-	559.00	438.00

BRICKWORK AND BLOCKWORK

BASE PRICES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Blocks delivered to site (cont'd) :							
Scoria blend blocks -							
100 x 200 x 400mm :							
2 hour (10-201)	…	100	-	-	-	-	309.00
3 hour (10-331)	…	100	-	-	-	-	345.00
150 x 200 x 400mm :							
2 hour (15-201)	…	100	-	-	-	-	333.00
3 hour (15-301)	…	100	-	-	-	-	358.00
4 hour (15-401)	…	100	-	-	-	-	441.00
200 x 200 x 400mm :							
2 hour (20-201)	…	100	-	-	-	-	405.00
3 hour (20-301)	…	100	-	-	-	-	437.00
4 hour (20-401)	…	100	-	-	-	-	519.00
Cement Mortar	…	cum	190.00	185.00	215.00	180.00	305.00
per 1000 standard bricks	…	1000	95.00	92.50	107.50	90.00	152.50
per 1000 modular bricks	…	1000	-	98.05	113.95	95.40	161.65
Labour, subcontract rate (excluding mortar) for :							
Standard -							
230 x 100 x 76mm :							
Common	…	1000	1820.00	1330.00	1500.00	1150.00	1710.00
Facing	…	1000	2030.00	1440.00	1700.00	1250.00	1900.00
230 x 110 x 119mm :							
Common	…	1000	-	-	1950.00	-	2270.00
230 x 110 x 162mm :							
Common	…	1000	-	1850.00	-	-	2750.00
Modular -							
290 x 90 x 90mm :							
Common	…	1000	-	-	-	1250.00	2100.00
Facing	…	1000	-	-	-	-	2280.00
290 x 90 x 162mm :							
Facing	…	1000	-	-	-	1430.00	2805.00
305 x 90 x 76mm :							
Common	…	1000	-	-	-	1350.00	3060.00
305 x 90 x 162mm :							
Common vertically cored	…	1000	-	-	-	1450.00	1880.00
Common horizontally cored	…	1000	-	-	-	1450.00	2910.00
Masonry blocks, hollow -							
100 x 100 x 400mm	…	100	305.00	545.00	570.00	250.00	715.00
100 x 200 x 400mm	…	100	490.00	550.00	630.00	380.00	800.00
150 x 200 x 400mm	…	100	540.00	620.00	680.00	430.00	890.00
200 x 200 x 400mm	…	100	560.00	670.00	730.00	475.00	1020.00
Masonry blocks, solid -							
100 x 100 x 400mm	…	100	-	590.00	680.00	250.00	860.00
100 x 200 x 400mm	…	100	520.00	590.00	770.00	430.00	900.00
150 x 200 x 400mm	…	100	-	-	-	525.00	-

BRICKWORK AND BLOCKWORK

BRICKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on the supply and laying costs given on pages 256 and 257	...	Note					
ADD or DEDUCT for each variation of $5.00 per 1000 in supply cost of bricks :							
Standard bricks, 110mm thick wall	...	sqm	0.26	0.26	0.26	0.26	0.26
Modular bricks, 90mm thick wall	...	sqm	0.18	0.18	0.18	0.18	0.18
Prices for work in facings include fair face and pointing one side	...	Note					
CLAY BRICKWORK							
STANDARD BRICKS, 230 x 110 x 76mm							
Common Brickwork :							
Base rate	...	1000	2900.00	2180.00	2495.00	2215.00	2830.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	140.50	106.00	121.00	107.50	137.50
230mm thick	...	sqm	281.00	212.00	242.00	215.00	275.00
Pier, attached or isolated	...	cum	1170.00	882.00	1010.00	896.00	1145.00
Face Brickwork :							
Base rate	...	1000	3510.00	2640.00	2865.00	2420.00	3360.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	170.00	128.00	139.00	117.50	163.00
Glazed Brickwork :							
Base rate	...	1000	8790.00	8240.00	8380.00	7250.00	8500.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	427.00	400.00	407.00	352.00	412.00
STANDARD BRICKS, 230 x 110 x 119mm							
Common Brickwork :							
Base rate	...	1000	-	-	3230.00	-	3840.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	-	-	104.50	-	124.00
230mm thick	...	sqm	-	-	209.00	-	248.00
Pier, attached or isolated	...	cum	-	-	869.00	-	1035.00
STANDARD BRICKS, 230 x 110 x 162mm							
Common Brickwork :							
Base rate	...	1000	-	3060.00	-	-	4845.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	-	74.00	-	-	117.00
230mm thick	...	sqm	-	148.00	-	-	234.00
Pier, attached or isolated	...	cum	-	618.00	-	-	978.00
MODULAR BRICKS, 290 x 90 x 90mm							
Common Brickwork :							
Base rate	...	1000	-	-	-	2465.00	3395.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	82.00	113.00
290mm thick	...	sqm	-	-	-	246.00	340.00
Pier, attached or isolated	...	cum	-	-	-	821.00	1130.00
Face Brickwork :							
Base rate	...	1000	-	-	-	-	4025.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	-	134.00

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BRICKWORK (cont'd)							
CLAY BRICKWORK (cont'd)							
MODULAR BRICKS, 290 x 90 x 162mm							
Face Brickwork :							
Base rate	...	1000	-	-	-	3715.00	5270.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	72.10	102.00
MODULAR BRICKS, 305 x 90 x 76mm							
Common Brickwork :							
Base rate	...	1000	-	-	-	2355.00	4405.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	74.60	139.50
305mm thick	...	sqm	-	-	-	273.50	512.00
Pier, attached or isolated	...	cum	-	-	-	869.00	1625.00
MODULAR BRICKS, 305 x 90 x 162mm							
Common Brickwork - vertically cored :							
Base rate	...	1000	-	-	-	2910.00	3840.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	53.90	71.10
305mm thick	...	sqm	-	-	-	169.00	223.00
Pier, attached or isolated	...	cum	-	-	-	537.00	709.00
Common Brickwork - horizontally cored :							
Base rate	...	1000	-	-	-	2710.00	4910.00
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	50.10	90.80
305mm thick	...	sqm	-	-	-	157.50	285.50
Pier, attached or isolated	...	cum	-	-	-	500.00	906.00
CONCRETE MASONRY BRICKWORK							
STANDARD BRICKS 230 x 110 x 76mm							
Common Brickwork :							
Base rate	...	1000	2840.00	-	-	-	2875.00
Wall or skin of hollow wall -							
110mm thick	...	sqm	137.50	-	-	-	139.50
230mm thick	...	sqm	275.50	-	-	-	279.00
Pier, attached or isolated	...	cum	1160.00	-	-	-	1175.00
Face Brickwork :							
Base rate	...	1000	-	-	-	-	-
Wall or skin of hollow wall -							
110mm thick	...	sqm	-	-	-	-	-
MODULAR BRICKS, 290 x 90 x 90mm							
Common Brickwork :							
Base rate	...	1000	-	-	-	-	-
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	-	-
Pier, attached or isolated	...	cum	-	-	-	-	-
Face Brickwork :							
Base rate	...	1000	-	-	-	-	-
Wall or skin of hollow wall -							
90mm thick	...	sqm	-	-	-	-	-

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BRICKWORK (cont'd)							
CONCRETE MASONRY BRICKWORK (cont'd)							
SPECIAL PERFORMANCE BRICKS, 230 x 165 x 119mm*							
Common Brickwork :							
Base rate	...	1000	-	-	-	-	-
Wall x 165mm thick	...	sqm	-	-	-	-	-
* 230 x 165 x 76mm in Melbourne							
LOAD BEARING BRICKWORK							
ADD extra to foregoing common brickwork prices (average)	...	+ 5%					
BRICKWORK - LABOURS, ETC.							
BUILDING UP EXISTING OPENINGS							
Extra labour for :							
90 or 110mm wall	...	sqm	13.95	13.70	26.10	13.85	13.85
190 or 230mm wall	...	sqm	27.50	27.40	52.30	27.70	27.90
CAVITIES							
Mortar cavity filling	...	cum	375.00	408.00	433.00	397.00	586.00
Form 50mm cavity including 4 ties per square metre :							
Galvanised wire ties	...	sqm	2.70	2.40	2.20	2.40	2.20
Stainless steel, 200 x 1mm ties	...	sqm	6.30	6.35	5.85	6.30	5.40
Close cavity with brickwork 110mm thick, cavity :							
50mm wide	...	m	13.25	13.45	14.00	14.75	16.90
100mm wide	...	m	14.60	15.10	15.60	16.35	18.90
CIRCULAR WALLING							
Extra labour for :							
90 or 100mm wall to 2 metre radius -							
Common	...	sqm	11.90	11.05	22.10	11.40	11.85
Face	...	sqm	12.80	12.25	24.00	12.45	12.65
190 or 230mm wall to 2 metre radius -							
Common	...	sqm	23.80	22.30	43.90	22.80	23.60
Face	...	sqm	25.20	24.30	47.30	24.50	25.10
CUTTINGS							
Circular cutting :							
Common -							
90 or 110mm wide	...	m	13.05	18.95	35.00	14.95	13.40
190 or 230mm wide	...	m	35.20	24.90	69.70	30.10	26.80
Face -							
90 or 110mm wide	...	m	20.40	22.80	37.00	21.60	21.80
190 or 230mm wide	...	m	40.50	42.00	74.30	43.00	43.80
Raking cutting :							
Common -							
90 or 110mm wide	...	m	13.10	16.75	26.80	13.55	10.60
190 or 230mm wide	...	m	26.20	30.90	53.30	27.10	21.10
Face -							
90 or 110mm wide	...	m	17.65	20.90	29.60	17.95	15.00
190 or 230mm wide	...	m	35.20	38.60	59.00	36.00	29.90

BRICKWORK AND BLOCKWORK

BRICKWORK - LABOURS, ETC. (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CUTTINGS (cont'd)							
Chase 50mm wide and 25mm deep :							
Horizontal	...	m	12.65	20.80	35.40	13.85	13.55
Raking or vertical	...	m	14.25	26.10	42.20	18.50	18.10
Cut toothings and bond new work to old :							
110mm wide	...	m	30.00	40.20	59.30	30.50	33.00
230mm wide	...	m	60.20	80.40	119.00	60.80	66.20
FACEWORK							
Cleaning down face brickwork on completion with acid	...	sqm	3.25	3.35	3.35	3.25	3.40
Pointing only reverse face	...	sqm	8.15	5.00	8.20	5.05	4.45
FINISHES							
Bagged finish :							
Single application	...	sqm	9.45	6.10	8.85	6.50	9.25
Two applications	...	sqm	18.85	12.20	17.80	13.10	18.50
Fair face to common wall	...	sqm	8.45	4.70	8.45	5.10	4.60
PREPARATION OF EXISTING WORK							
Level and prepare old work for raising :							
110mm wide	...	m	8.45	9.80	17.15	9.60	8.45
230mm wide	...	m	12.85	14.70	25.60	14.45	12.65
Pin up new work to old soffit :							
110mm wide	...	m	16.90	16.10	27.10	15.75	17.85
230mm wide	...	m	33.80	32.30	53.80	31.50	35.70
PROJECTIONS AND RECESSES							
Set back two courses of brickwork 25mm including cutting :							
Common	...	m	8.45	10.05	14.45	7.80	8.75
Face	...	m	10.35	12.65	18.00	11.35	10.25
Set forward two courses of brickwork 25mm :							
Common	...	m	6.40	7.25	11.75	6.50	6.65
Face	...	m	8.70	10.75	14.70	9.20	8.85
Oversailing two courses a total projection of 110mm as corbel including cutting :							
Common	...	m	16.20	17.55	17.35	15.10	15.75
Face	...	m	17.60	21.10	20.60	18.20	20.00
ARCHES							
Two ring arch in 230mm wall, 2 metres wide and rising 600mm from spring line, including all necessary centreing and supports and all bricks and cutting :							
Common bricks	...	No	455.00	370.00	410.00	355.00	450.00
Face bricks	...	No	550.00	450.00	500.00	405.00	555.00

BRICKWORK AND BLOCKWORK

BRICKWORK SUNDRIES

AIR BRICKS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Terracotta air brick built in :							
230 x 76mm	...	No	21.20	22.60	23.60	21.00	26.30
230 x 160mm	...	No	23.40	25.30	25.20	25.10	28.20

ANT CAPPING

Galvanised sheet steel ant capping to :							
110mm wall	...	m	6.55	6.95	6.85	6.90	7.40
230mm wall	...	m	12.30	12.65	12.60	12.60	13.10
230 x 230mm pier	...	No	4.80	4.95	4.95	4.95	5.15

LINTELS

Steel flat arch bars :							
Zinc silicate treated	...	t	3595.00	4005.00	3690.00	3810.00	3995.00
75 x 8mm flat	...	m	17.50	19.55	18.05	18.60	19.65
75 x 10mm flat	...	m	20.60	22.90	21.10	21.80	22.70
Galvanised or similarly treated	...	t	4325.00	4225.00	4520.00	4350.00	4855.00
75 x 8mm flat	...	m	20.90	20.60	21.90	21.20	23.70
75 x 10mm flat	...	m	24.90	24.20	26.00	25.00	27.80
Steel angle lintel :							
Zinc silicate treated	...	t	3165.00	3480.00	3185.00	3300.00	3350.00
75 x 75 x 8mm angle	...	m	29.80	33.10	30.40	31.40	32.50
150 x 90 x 10mm angle	...	m	52.60	57.50	52.50	54.50	54.70
Galvanised or similarly treated	...	t	3895.00	3695.00	4010.00	3835.00	4210.00
75 x 75 x 8mm angle	...	m	36.20	34.90	37.60	36.10	40.00
150 x 90 x 10mm angle	...	m	65.30	61.20	66.80	63.80	69.60
Ribbed galvanised steel lintel :							
85 x 8mm flat	...	m	23.20	24.30	24.00	24.10	25.50
100 x 100 x 6mm angle	...	m	39.60	41.20	40.90	41.00	43.00
150 x 100 x 6mm angle	...	m	48.50	50.00	49.70	49.80	51.80
200 x 200 x 7mm T bar	...	m	125.50	128.50	128.00	128.00	132.00
200 x 200 x 9mm T bar	...	m	150.00	153.00	152.50	152.50	156.50

SILLS, COPINGS

Brick-on-edge sill set sloping including pointing to top and edge	...	m	32.80	30.90	32.60	28.40	33.60
Brick-on-edge coping to 230mm wall including pointing to top and both sides	...	m	32.80	30.90	32.60	28.40	33.60
Terracotta tile sill 152 x 152mm set sloping including pointing	...	m	31.30	29.80	29.00	28.90	29.10
Quarry tile sill 152 x 152mm, ditto	...	m	31.10	30.60	29.00	30.40	29.50
Glazed tile sill 152 x 152mm, ditto	...	m	30.20	28.60	27.60	28.30	28.10

TIES/CRAMPS

Wall ties built-in :							
8 gauge galvanised wire	...	No	0.48	0.50	0.49	0.48	0.45
200 x 1mm galvanised	...	No	0.55	0.56	0.55	0.53	0.49
200 x 1mm stainless steel	...	No	1.39	1.42	1.30	1.31	1.24
37mm x 16 gauge galvanised hoop iron cramp built-in	...	No	1.75	1.80	1.90	1.70	1.80
Flexible and acoustic anchors, large range available	...	Note					

BRICKWORK AND BLOCKWORK

BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on the supply and laying costs given on pages 256 and 257	...	Note					
PLAIN FACE BLOCKWORK							
Natural colour hollow concrete blocks :							
100mm high coursing in -							
100mm thick wall	...	sqm	185.00	191.50	209.00	140.50	250.50
200mm high coursing in -							
100mm thick wall	...	sqm	114.00	99.30	111.00	93.30	138.50
150mm ditto	...	sqm	138.50	112.50	125.00	116.50	159.50
200mm ditto	...	sqm	168.50	126.00	140.00	132.00	184.00
Natural colour solid concrete blocks :							
100mm high coursing in -							
100mm thick wall	...	sqm	-	-	233.50	141.50	284.50
200mm high coursing in -							
100mm thick wall	...	sqm	125.00	106.50	134.00	111.50	161.50
150mm ditto	...	sqm	-	-	-	152.00	-
Standard coloured hollow concrete blocks :							
100mm high coursing in -							
100mm thick wall	...	sqm	215.50	204.50	225.00	150.00	-
200mm high coursing in -							
100mm thick wall	...	sqm	127.50	113.00	115.00	100.50	-
150mm ditto	...	sqm	157.00	133.50	131.00	123.00	-
200mm ditto	...	sqm	185.00	159.00	147.50	145.00	-
Standard coloured solid concrete blocks :							
100mm high coursing in -							
100mm thick wall	...	sqm	-	-	233.50	152.50	-
200mm high coursing in -							
100mm thick wall	...	sqm	143.50	121.00	143.50	127.00	-
150mm ditto	...	sqm	-	-	-	-	-
RIBBED FACE BLOCKWORK							
Solid ribbed “off the mould” face concrete blocks, 200mm high coursing in 100mm thick wall :							
Natural colour	...	sqm	-	-	-	-	-
Coloured	...	sqm	-	-	-	-	-
SPLIT FACE BLOCKWORK							
Solid split face concrete blocks, 100mm high coursing in 100mm thick wall :							
Natural colour	...	sqm	-	-	-	148.50	-
Coloured	...	sqm	-	234.50	262.00	162.50	-
Hollow split face concrete blocks, 200mm high coursing in 100mm thick wall :							
Natural colour	...	sqm	-	-	-	107.50	-
Coloured	...	sqm	-	128.00	129.50	128.50	-

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BLOCKWORK (cont'd)							
RIBBED/SPLIT FACE BLOCKWORK							
Solid ribbed split face blocks, 200mm high coursing in 100mm thick wall :							
Natural colour	...	sqm	-	-	-	115.00	-
Coloured	...	sqm	-	-	-	134.50	182.00
SCORIA BLEND BLOCKWORK							
Two hour insulation, 400 x 200mm high coursing in :							
100mm thick wall	...	sqm	-	-	-	-	151.50
150mm thick wall	...	sqm	-	-	-	-	169.50
200mm thick wall	...	sqm	-	-	-	-	182.00
Three hour insulation, 400 x 200mm high coursing in :							
100mm thick wall	...	sqm	-	-	-	-	156.00
150mm thick wall	...	sqm	-	-	-	-	173.00
200mm thick wall	...	sqm	-	-	-	-	202.50
Four hour insulation, 400 x 200mm high coursing in :							
150mm thick wall	...	sqm	-	-	-	-	184.00
200mm thick wall	...	sqm	-	-	-	-	213.50
BLOCKWORK - LABOURS, ETC.							
CAVITIES							
Form 50mm cavity including 4 wire ties per square metre	...	sqm	2.55	2.35	2.30	2.40	2.30
COPINGS							
Standard natural colour coping blocks 40mm thick x 200mm wide x 400mm long including pointing	...	m	15.25	15.05	15.10	16.95	19.55
CUTTINGS							
Fair raking cutting, hollow or solid blocks :							
100mm wide	...	m	25.90	22.40	24.50	18.45	16.65
150mm wide	...	m	33.40	30.30	30.90	21.80	21.30
200mm wide	...	m	40.80	31.70	35.90	24.60	26.40
Fair vertical splay cutting solid or hollow blocks :							
100mm wide	...	m	26.10	22.40	24.50	18.45	16.65
150mm wide	...	m	33.40	30.30	30.90	21.80	21.30
200mm wide	...	m	40.80	31.70	35.90	24.60	26.40
FINISHES							
Pointing and cleaning down	...	sqm	8.25	10.20	9.80	9.90	12.45

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BLOCKWORK - LABOURS, ETC. (cont'd)							
LINTELS							
Lintel block, natural colour including concrete filling and temporary supports :							
150mm wide x 200mm high	...	m	67.80	59.10	60.40	81.70	74.00
200mm wide x 200mm high	...	m	81.90	63.90	71.20	93.20	86.30
200mm wide x 400mm high	...	m	107.00	94.90	102.50	150.00	124.00
300mm wide x 400mm high	...	m	160.00	143.50	151.50	192.50	174.00
Steel lintels, refer page 262	...	Note					
SOLID TOP COURSE							
Extra over plain face natural colour hollow blocks for solid top course in :							
100mm wall with 200mm high top course	...	m	2.45	1.25	1.90	3.35	2.95
THREADING BLOCKS							
Extra for threading 400mm long block over vertical bar reinforcement	...	No	1.40	Incl.	1.50	1.40	1.55
CORE FILLING							
Weak concrete filling to cores of hollow blocks :							
150mm thick block	...	sqm	35.00	39.50	42.60	40.10	51.70
200mm thick block	...	sqm	52.30	59.00	63.50	59.80	77.10
JOINTS AND SEALANTS							
Form 9mm wide vertical control joint in 110mm wall including 25mm dia. polyethylene rope and mastic pointing	...	m	18.45	21.30	20.60	20.80	24.50
Two layers of 2-ply bituminous felt laid on top of 110mm wall	...	m	1.75	1.90	1.85	1.85	2.05
Caulk and seal with :							
Non-fire rated -							
Mastic sealant :							
6 x 6mm	...	m	10.90	13.05	12.60	12.75	15.60
12 x 12mm	...	m	17.10	19.60	19.05	19.20	22.50
Polysulphide sealant :							
6 x 6mm	...	m	11.85	14.05	13.55	13.70	16.55
12 x 12mm	...	m	21.00	23.50	22.90	23.10	26.40
Polyurethane sealant :							
6 x 6mm	...	m	10.60	12.75	12.30	12.45	15.30
12 x 12mm	...	m	15.95	18.45	17.90	18.05	21.30
Silicone sealant :							
6 x 6mm	...	m	10.75	12.95	12.45	12.60	15.45
12 x 12mm	...	m	16.60	19.10	18.55	18.70	22.00

BRICKWORK AND BLOCKWORK

JOINTS AND SEALANTS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Caulk and seal with (cont'd) :							
Fire rated (in two hour rated walls) -							
Polyurethane sealant, two or three hour rated, to :							
10mm wide gap x 10mm deep	...	m	14.70	17.15	16.60	16.80	20.10
15mm wide gap x 10mm deep	...	m	16.75	19.20	18.70	18.85	22.10
20mm wide gap x 10mm deep	...	m	20.10	22.90	22.30	22.50	26.20
Silicone sealant, two hour rated, to :							
10mm wide gap x 15mm deep	...	m	16.85	19.35	18.80	18.95	22.20
15mm wide gap x 15mm deep	...	m	20.00	22.50	21.90	22.10	25.40
20mm wide gap x 20mm deep	...	m	30.00	33.10	32.40	32.60	36.70
ADD extra for :							
Backing rod -							
10mm	...	m	5.75	7.00	6.70	6.80	8.45
12mm	...	m	6.45	7.65	7.40	7.50	9.10
25mm	...	m	8.20	9.60	9.25	9.35	11.20
Sealant to both sides when necessary	...	Note					

AERATED CONCRETE BLOCK WALLS

Aerated concrete blocks, non load bearing, including all labours :							
600 x 200mm high blocks in wall -							
100mm thick	...	sqm	113.00	112.00	113.00	112.00	113.00
150mm thick	...	sqm	154.00	153.00	154.00	153.00	155.00
200mm thick	...	sqm	166.00	166.00	166.00	166.00	168.00
250mm thick	...	sqm	183.00	182.00	183.00	182.00	185.00
300mm thick	...	sqm	204.00	203.00	204.00	203.00	206.00
Aerated concrete wall panels, refer page 254	...	Note					
Aerated concrete partition wall system, refer page 393	...	Note					

GYPCRETE WALL PANELS

Prices are based on gross area and include concrete filling and cut-outs for windows and doors							
Prefinished gypcrete panels, fire rated (minimum height 2700mm) :							
Load bearing -							
Internal	...	sqm	-	-	170.00	-	170.00
External	...	sqm	-	-	190.00	-	190.00
ADD extra for :							
Craneage	...	Note					
Door frames	...	Note					
Paint	...	Note					

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DAMP PROOF COURSES							
Prices are for net lengths as laid. Allowance has been made for laps, angles, etc.	...	Note					
Mortar incorporating Dampco :							
110mm wide horizontal d.p.c.	...	m	0.26	0.26	0.26	0.26	0.26
230mm wide ditto	...	m	0.36	0.36	0.36	0.36	0.36
350mm wide ditto	...	m	0.52	0.52	0.52	0.52	0.52
Bituminous two-ply felt :							
110mm wide horizontal d.p.c.	...	m	0.35	0.40	0.40	0.40	0.50
230mm wide ditto	...	m	0.55	0.65	0.65	0.65	0.80
300mm wide ditto	...	m	0.65	0.80	0.80	0.80	1.00
Aluminium cored bitumen coated :							
110mm wide horizontal d.p.c.	...	m	1.50	1.60	1.55	1.55	1.65
230mm wide ditto	...	m	3.20	3.35	3.30	3.30	3.50
350mm wide ditto	...	m	4.35	4.50	4.50	4.50	4.70
300mm girth cavity gutter and flashing	...	m	4.15	4.40	4.35	4.35	4.70
150mm wide vertical d.p.c. one edge fixed to timber or metal frame	...	m	3.60	4.05	3.95	3.95	4.60
150mm wide sill flashing	...	m	2.90	3.20	3.15	3.20	3.40
Pure soft zinc :							
100mm wide horizontal d.p.c.	...	m	11.45	11.50	11.50	11.50	11.60
230mm wide ditto	...	m	25.30	25.40	25.30	25.40	25.50
380mm wide ditto	...	m	42.40	42.50	42.50	42.50	42.70
300mm girth cavity gutter and flashing	...	m	34.20	34.40	34.40	34.40	34.70
150mm wide vertical d.p.c. one edge fixed to timber or metal frame	...	m	18.50	19.00	18.90	18.90	19.50
150mm wide sill flashing	...	m	17.85	18.15	18.10	18.10	18.55
Flashings - for flashings in connection with roofing refer pages 408 to 410	...	Note					
REINFORCEMENT							
Galvanised structural grade round bar reinforcement built in :							
6mm dia.	...	m	0.95	1.00	1.00	1.00	1.10
10mm dia.	...	m	2.00	2.10	2.05	2.05	2.15
Galvanised wire mesh reinforcement built in :							
50mm wide	...	m	1.25	1.35	1.35	1.35	1.50
90mm wide	...	m	1.35	1.50	1.45	1.45	1.65
140mm wide	...	m	1.50	1.65	1.60	1.60	1.80
ACID RESISTANT BRICKWORK							
115mm acid resistant brickwork skin	...	sqm	410.00	430.00	425.00	425.00	455.00
FIREBRICK LININGS							
115mm firebrick lining to chimney shaft bedded and pointed in special compound and block bonded to brickwork (average)	...	sqm	410.00	430.00	425.00	425.00	455.00

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BRICK PAVING							
Refer page 223	...	Note					
GLASS BLOCK PANELS							
190 x 190 x 80mm plain clear glass blocks in straight panels, built in Ezylay Glass Block Installation silicone system including aluminium framing, expansion materials, connectors and reinforcing :							
1000 x 1000mm panel size	...	No	595.00	625.00	1085.00	675.00	1085.00
1000 x 2000mm panel size	...	No	1020.00	1050.00	1655.00	1125.00	1655.00
1000 x 3000mm panel size	...	No	1435.00	1520.00	2260.00	1615.00	2260.00
2400 x 2400mm panel size	...	No	2425.00	2725.00	3835.00	2885.00	3835.00
190 x 190 x 80mm plain clear glass blocks in straight panels, built in Ezylay Glass Block Installation mortar system including aluminium framing, expansion materials, connectors and reinforcing :							
1000 x 1000mm panel size	...	No	765.00	645.00	1140.00	685.00	1140.00
1000 x 2000mm panel size	...	No	1360.00	1175.00	1710.00	1235.00	1710.00
1000 x 3000mm panel size	...	No	1945.00	1705.00	2335.00	1780.00	2335.00
2400 x 2400mm panel size	...	No	3410.00	3085.00	3945.00	3225.00	3945.00
ADD extra for :							
Seves design range Q19 0 glass block -							
1000 x 1000mm panel size	...	No	45.00	45.00	120.00	55.00	120.00
1000 x 2000mm panel size	...	No	90.00	90.00	240.00	110.00	240.00
1000 x 3000mm panel size	...	No	135.00	135.00	360.00	170.00	360.00
2400 x 2400mm panel size	...	No	260.00	260.00	695.00	320.00	695.00
Seves design range Q19 0 coloured glass block -							
1000 x 1000mm panel size	...	No	170.00	170.00	225.00	200.00	225.00
1000 x 2000mm panel size	...	No	335.00	335.00	455.00	400.00	455.00
1000 x 3000mm panel size	...	No	500.00	500.00	680.00	605.00	680.00
2400 x 2400mm panel size	...	No	965.00	965.00	1305.00	1155.00	1305.00
190 x 190 x 80mm plain clear glass blocks in straight panels, fire rated to one hour, built in Ezylay Glass Block Installation mortar system including aluminium framing, expansion materials, connectors and reinforcing :							
1000 x 1000mm panel size	...	No	1035.00	805.00	1430.00	1150.00	1430.00
1000 x 2000mm panel size	...	No	1700.00	1335.00	2185.00	1860.00	2185.00
1000 x 3000mm panel size	...	No	2360.00	1930.00	2995.00	2550.00	2995.00
2400 x 2400mm panel size	...	No	4005.00	3345.00	4990.00	4330.00	4990.00
190 x 190 x 80mm plain clear glass blocks in paving panels, built in Ezylay Glass Block Installation system, concrete and includes expansion materials, connectors and reinforcing :							
1000 x 1000mm panel size	...	No	2030.00	2030.00	3115.00	2990.00	3115.00
1000 x 2000mm panel size	...	No	2925.00	2925.00	4835.00	4280.00	4835.00
1000 x 3000mm panel size	...	No	4255.00	4255.00	6725.00	6635.00	6725.00
2400 x 2400mm panel size	...	No	7760.00	7760.00	12240.00	10830.00	12240.00

BRICKWORK AND BLOCKWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PROTECTIVE TREATMENTS							
ANTI-GRAFFITI							
Refer page 453	...	Note					
FUNGUS PROTECTION							
Fungishield 211 treatment on :							
Porous surfaces -							
Horizontal	...	sqm	10.45	11.00	11.00	10.75	10.60
Vertical	...	sqm	13.10	13.70	13.80	13.45	13.30
Non-porous surfaces -							
Horizontal	...	sqm	8.75	9.40	9.25	9.10	8.90
Vertical	...	sqm	11.15	11.75	11.75	11.50	11.30
ADD extra for :							
Pre-treatment cleaning with high pressure jet if required	...	sqm	5.60	5.80	5.60	5.75	5.65
SILICONE SOLUTION							
Driwall or similar silicone solution on fair face brick/block	...	sqm	21.00	21.80	21.40	21.00	21.20
MONOLITHIC STABILISED RAMMED EARTH WALLING							
Prices include all temporary shutters, plant, etc.	...	Note					
Prices include for blockouts up to 1000 x 1000mm and for standard 900mm wide door openings	...	Note					
Prices are indicative only as work is generally priced on a project by project basis	...	Note					
300mm thick local earth wall with an off-form finish :							
Walls -							
to 3000mm high	...	sqm	395.00-460.00	-	400.00-475.00	280.00-335.00	-
3000 to 6000mm high	...	sqm	480.00-545.00	-	450.00-545.00	395.00-450.00	-
over 6000mm high	...	sqm	610.00	-	500.00-535.00	505.00-560.00	-
curved to 3000mm high	...	sqm	490.00-550.00	-	445.00-515.00	395.00-490.00	-
overfills and underfills	...	sqm	-	-	580.00	560.00	-

BRICKWORK AND BLOCKWORK

MONOLITHIC STABILISED RAMMED EARTH WALLING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
400mm thick insulated earth wall with an off-form finish :							
Walls -							
to 3000mm high	...	sqm	-	-	630.00-705.00	595.00-675.00	-
ADD extra for :							
Lintels -							
Rod and dowel bar	...	m	-	-	103.00	94.00	-
Mesh and dowel bar	...	m	-	-	-	-	-
T-bar	...	m	-	-	-	-	-
Raking and curved tops (per metre of rake or curve) -							
Raking top to :							
Straight wall	...	m	112.00	-	59.00	-	-
Curved on plan wall	...	m	-	-	-	225.00	-
Curved top to :							
Straight wall	...	m	-	-	59.00	-	-
Curved on plan wall	...	m	-	-	-	225.00	-
Blockouts larger than above (include in area of wall)	...	Note					

MASONRY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for fixing and all bedding and pointing, cramps, dowels, ties and the like. Prices for facings and pavings include for cutting to size and setting out	...	Note					
The approximate weights of stone per cubic metre are as follows -							
Sandstone - 2.323 t							
Limestone - 2.403 t							
Marble and Granite - 2.691 t							
Slate - 2.915 t							
Bluestone - 2.650 t	...	Note					

SANDSTONE

WALLING

300mm thick random rubble :							
Faced one side	...	sqm	-	340.00	495.00	-	-
Faced both sides	...	sqm	-	495.00	540.00	-	-
300mm thick regular coursed squared rubble :							
Faced one side	...	sqm	740.00	700.00	730.00	-	855.00
Faced both sides	...	sqm	870.00	810.00	800.00	-	970.00

FACING

75mm thick sawn facing bedded in mortar	...	sqm	410.00	295.00	375.00	-	350.00

PAVING

Flagstones in regular squares including sand bed and pointing :							
12 to 15mm thick	...	sqm	-	-	-	100.00-135.00	-
50mm thick	...	sqm	-	140.00	-	-	190.00
75mm thick	...	sqm	-	170.00	-	-	-
140mm thick	...	sqm	-	340.00	-	-	450.00

COPINGS, SILLS

400 x 150mm thick dressed coping	...	m	-	195.00	-	-	290.00
225 x 150mm dressed sill or threshold	...	m	-	110.00	-	-	170.00

LIMESTONE

WALLING

350mm thick random rubble :							
Sawn faced one side	...	sqm	245.00	-	-	240.00	-
Sawn faced both sides	...	sqm	255.00	-	-	245.00	-
350mm thick regular coursed squared rubble :							
Sawn faced one side	...	sqm	255.00	-	-	250.00	-
Sawn faced both sides	...	sqm	270.00	-	-	265.00	-

MASONRY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LIMESTONE (cont'd)							
PAVING							
Flagstones in regular squares including sand bed and pointing :							
70mm thick	...	sqm	-	-	-	100.00	-
REVETMENT WALLING							
Refer page 713	...	Note					
GRANITE							
FACING							
Wall facing, 30mm thick, sawn face :							
Harcourt grey	...	sqm	610.00	605.00	530.00	-	555.00
Riverina grey	...	sqm	580.00	605.00	505.00	-	525.00
South Australian imperial black	...	sqm	625.00	615.00	550.00	620.00	565.00
PAVING							
Slab paving, 30mm thick, sawn face laid in mortar bed :							
Harcourt grey	...	sqm	550.00	425.00	415.00	-	430.00
Riverina grey	...	sqm	510.00	425.00	385.00	-	390.00
South Australian imperial black	...	sqm	560.00	465.00	425.00	550.00	455.00
ADD extra for :							
Exfoliated face	...	sqm	84.00	65.00	71.00	75.00	77.00
Rubbed face	...	sqm	127.00	106.00	109.00	106.00	123.00
Polished face	...	sqm	154.00	122.00	142.00	129.00	145.00
Patterned work and cutting	...	Note					
Divisional strips and inserts	...	Note					
RECONSTRUCTED GRANITE							
FACING							
Wall panels, 90/100mm thick with thickened edges and ribs, including fixing in position	...	sqm	650.00	590.00	540.00	805.00	640.00
MARBLE							
AUSTRALIAN MARBLE							
FACING							
Wall facing, 19mm thick, sawn face :							
Wombeyan	...	sqm	415.00	395.00	395.00	-	420.00
Wall facing, 25mm thick, sawn face :							
Wombeyan	...	sqm	460.00	410.00	420.00	-	445.00
ADD extra for :							
Unfilled and honed finish	...	sqm	57.00	55.00	63.00	-	69.00
Filled and honed finish	...	sqm	65.00	63.00	72.00	-	78.00
Filled and polished finish	...	sqm	79.00	80.00	82.00	-	92.00

MASONRY

MARBLE (cont'd)

AUSTRALIAN MARBLE (cont'd)

PAVING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Wombeyan tiles, 450 x 300 x 10mm thick, unfilled and honed finish, laid in mortar bed	...	sqm	305.00	257.00	240.00	-	247.00
ADD extra for :							
Filled and honed finish	...	sqm	19.10	19.45	20.15	-	18.25
Filled and polished finish	...	sqm	48.50	49.00	54.00	-	32.00

IMPORTED MARBLE

FACING

Wall facing, 19mm thick, sawn face :							
Carrara white	...	sqm	515.00	565.00	475.00	560.00	560.00
Sicilian pearl	...	sqm	520.00	560.00	480.00	560.00	560.00
Travertine	...	sqm	465.00	520.00	455.00	560.00	510.00
ADD extra for :							
Unfilled and honed finish	...	sqm	58.00	55.00	63.00	67.00	67.00
Filled and honed finish	...	sqm	69.00	63.00	73.00	95.00	75.00
Filled and polished finish	...	sqm	81.00	80.00	83.00	95.00	90.00
Tiles, 300 x 300/600 x 300 x 10mm thick, pre-polished :							
Carrara white	...	sqm	295.00	310.00	295.00	295.00	315.00
Sicilian pearl	...	sqm	300.00	315.00	300.00	295.00	320.00
Travertine	...	sqm	270.00	275.00	270.00	270.00	285.00

PAVING

Slab paving, 19mm thick, sawn face, laid in mortar bed :							
Carrara white	...	sqm	495.00	520.00	455.00	490.00	530.00
Sicilian pearl	...	sqm	495.00	520.00	460.00	490.00	530.00
Travertine	...	sqm	440.00	460.00	405.00	440.00	470.00
ADD extra for :							
Unfilled and honed finish	...	sqm	58.00	55.00	63.00	67.00	67.00
Filled and honed finish	...	sqm	69.00	63.00	73.00	95.00	75.00
Filled and polished finish	...	sqm	81.00	80.00	83.00	95.00	90.00
Tiles, 300 x 300/600 x 300 x 10mm thick, pre-polished, laid in mortar bed :							
Carrara white	...	sqm	295.00	300.00	290.00	295.00	315.00
Sicilian pearl	...	sqm	300.00	305.00	290.00	295.00	315.00
Travertine	...	sqm	270.00	260.00	265.00	255.00	285.00
ADD extra for :							
Honed finish to pre-polished surface	...	sqm	42.00	44.00	41.00	67.00	42.00
Tiles 900 x 450 x 10mm thick, sawn face, laid in mortar bed :							
Carrara white	...	sqm	410.00	390.00	365.00	430.00	430.00
Sicilian pearl	...	sqm	410.00	390.00	370.00	430.00	430.00
Travertine	...	sqm	350.00	325.00	310.00	380.00	375.00

MASONRY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
MARBLE (cont'd)							
IMPORTED MARBLE (cont'd)							
PAVING (cont'd)							
ADD extra for :							
Unfilled and honed finish	...	sqm	58.00	55.00	61.00	67.00	67.00
Filled and honed finish	...	sqm	69.00	63.00	70.00	95.00	75.00
Filled and polished finish	...	sqm	81.00	80.00	80.00	95.00	90.00
EDGE TREATMENT							
Polished edge to :							
10mm marble	...	m	24.70	21.00	20.40	40.00	27.60
19mm marble	...	m	25.70	26.10	24.60	60.00	28.70
SLATE							
AUSTRALIAN SLATE							
Mintaro wall facing, sawn finish :							
19mm thick	...	sqm	278.00	148.00	152.00	285.00	168.00
25mm thick	...	sqm	407.00	163.00	164.00	380.00	179.00
Willunga wall facing, sawn finish :							
10/12mm thick	...	sqm	278.00	163.00	160.00	-	173.00
Mintaro flag paving, 25mm thick, sawn finish, to regular pattern, laid in sand bed and pointed	...	sqm	375.00	115.00	124.00	333.00	131.00
IMPORTED SLATE							
Flag paving, average 10mm thick, sawn finish, to regular pattern, laid in mortar bed :							
Chinese or Indian	...	sqm	90.00-110.00	95.00-115.00	95.00-115.00	100.00-120.00	95.00-115.00
BLUESTONE							
Wall facing, 40mm thick, diamond cut	...	sqm	492.00	500.00	465.00	523.00	504.00
Slab paving, 40mm thick, diamond cut, laid in mortar bedding	...	sqm	311.00	317.00	283.00	380.00	310.00
Slab paving, 60mm thick, diamond cut, laid in mortar bedding	...	sqm	349.00	351.00	323.00	428.00	352.00

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

UNIVERSAL BEAMS

GRADES 300 and 350

Size		M/t	Surface Area Per m. sqm
150UB	14.0	71.4	0.576
	18.0	55.6	0.584
180UB	16.1	62.1	0.682
	18.1	55.2	0.685
	22.2	45.0	0.691
200UB	18.2	54.9	0.764
	22.3	44.8	0.910
	25.4	39.4	0.912
	29.8	33.6	0.922
250UB	25.7	38.9	0.961
	31.4	31.8	1.060
	37.3	26.8	1.070
310UB	32.0	31.2	1.160
	40.4	24.8	1.240
	46.2	21.6	1.250
360UB	44.7	22.4	1.350
	50.7	19.7	1.360
	56.7	17.6	1.370
410UB	53.7	18.6	1.480
	59.7	16.8	1.490
460UB	67.1	14.9	1.630
	74.6	13.4	1.640
	82.1	12.2	1.650
530UB	82.0	12.2	1.850
	92.4	10.8	1.860
610UB	101	9.9	2.070
	113	8.8	2.080
	125	8.0	2.090

UNIVERSAL COLUMNS

GRADES 300 and 350

Size		M/t	Surface Area Per m. sqm
100UC	14.8	67.6	0.563
150UC	23.4	42.7	0.885
	30.0	33.3	0.899
	37.2	26.9	0.908
200UC	46.2	21.6	1.180
	42.3	19.1	1.190
	59.5	16.8	1.220
250UC	72.9	13.7	1.480
	89.5	11.2	1.500
310UC	96.8	10.3	1.790
	118	8.5	1.810
	137	7.3	1.820
	158	6.3	1.840

PARALLEL FLANGE CHANNELS

GRADES 300 and 350

Size		M/t	Surface Area Per m. sqm
74x40x	5.9	168.9	0.286
100x50x	8.3	120.0	0.374
125x65x	11.9	84.0	0.480
150x75x	17.7	56.5	0.579
180x75x	20.9	47.8	0.638
200x75x	22.9	43.7	0.678
230x75x	25.1	39.8	0.737
250x90x	35.5	28.2	0.834
300x90x	40.1	24.9	0.932
380x100x	55.2	18.1	1.130

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

TEES CUT FROM UNIVERSAL BEAMS

GRADES 300 and 350

Size		M/t	Surface Area Per m. sqm
75BT	7.0	142.9	0.291
	9.0	111.1	0.296
90BT	8.1	123.5	0.343
	9.1	109.9	0.345
	11.1	90.1	0.349
100BT	9.1	109.9	0.385
	11.2	89.3	0.458
	12.7	78.7	0.460
	14.9	67.1	0.465
125BT	12.9	77.5	0.484
	15.7	63.7	0.534
	18.7	53.5	0.539
155BT	16.0	62.5	0.583
	20.2	49.5	0.622
	23.1	43.3	0.627
180BT	22.4	44.6	0.682
	25.4	39.4	0.686
	28.4	35.2	0.691
205BT	26.9	37.2	0.747
	29.9	33.4	0.751
230BT	33.6	29.8	0.822
	37.3	26.8	0.826
	41.1	24.3	0.831
265BT	41.0	24.4	0.932
	46.3	21.6	0.937
305BT	50.5	19.8	1.040
	56.5	17.7	1.050
	62.5	16.0	1.060

TEES CUT FROM UNIVERSAL COLUMNS

GRADES 300 and 350

Size		M/t	Surface Area Per m. sqm
50CT	7.4	135.1	0.284
75CT	11.7	85.5	0.447
	15.0	66.7	0.454
	18.6	53.8	0.460
100CT	23.1	43.3	0.598
	26.1	38.3	0.603
	29.8	33.6	0.608
125CT	36.5	27.4	0.748
	44.8	22.3	0.758
155CT	48.4	20.7	0.902
	59.0	16.9	0.912
	68.5	14.6	0.922
	79.0	12.7	0.933

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

TAPER FLANGE BEAMS

GRADES 250 and 350

Size	M/t	Surface Area Per m. sqm
100 x 45 x 7.2	138.9	0.349
125 x 65 x 13.1	76.3	0.470

TAPER FLANGE CHANNELS

GRADES 250 and 350

Size	M/t	Surface Area Per m. sqm
75 x 40 x 6.65	150.4	0.286
100 x 50 x 9.34	107.1	0.374
125 x 65 x 13.4	74.6	0.480

WELDED BEAMS

GRADES 300 and 400

Size	M/t	Surface Area Per m. sqm
700 WB 115	8.7	2.36
130	7.7	2.38
150	6.7	2.40
173	5.8	2.51
800 WB 122	8.2	2.56
146	6.8	2.68
168	6.0	2.70
192	5.2	2.81
900 WB 175	5.7	2.98
218	4.6	3.20
257	3.9	3.41
282	3.5	3.42
1000 WB 215	4.7	3.17
258	3.9	3.39
296	3.4	3.60
322	3.1	3.62
1200 WB 249	4.0	3.41
278	3.6	3.71
317	3.2	3.92
342	2.9	3.94
392	2.6	4.34
423	2.4	4.35
455	2.2	4.37

WELDED COLUMNS

GRADES 300 and 400

Size	M/t	Surface Area Per m. sqm
350 WC 197	5.1	2.02
230	4.3	2.03
258	3.9	2.04
280	3.6	2.05
400 WC 144	6.9	2.33
181	5.5	2.34
212	4.7	2.36
270	3.7	2.38
303	3.3	2.39
328	3.0	2.40
361	2.8	2.38
500 WC 228	4.4	2.94
267	3.7	2.96
290	3.4	2.97
340	2.9	2.98
383	2.6	2.88
414	2.4	2.90
440	2.3	2.88

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

FLAT PLATE SECTIONS

THICKNESS	WIDTH/mm							
	10	20	30	40	50	60	70	80
mm	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m
3	0.236	0.471	0.707	0.942	1.18	1.41	1.65	1.88
4	0.314	0.628	0.942	1.26	1.57	1.88	2.20	2.51
5	0.393	0.785	1.18	1.57	1.96	2.36	2.75	3.14
6	0.471	0.942	1.41	1.88	2.36	2.82	3.30	3.77
8	0.628	1.26	1.88	2.51	3.14	3.77	4.40	5.02
10	0.785	1.57	2.36	3.14	3.93	4.71	5.50	6.28
12	0.942	1.88	2.83	3.77	4.71	5.65	6.59	7.54
16	1.260	2.51	3.77	5.02	6.28	7.54	8.79	10.05
20	1.570	3.14	4.71	6.28	7.85	9.42	10.99	12.56
25	1.960	3.93	5.89	7.85	9.81	11.78	13.74	15.70
28	2.200	4.40	6.59	8.79	10.99	13.19	15.39	17.58
32	2.510	5.02	7.54	10.05	12.56	15.07	17.58	20.10
36	2.830	5.65	8.48	11.30	14.13	16.96	19.78	22.61
40	3.140	6.28	9.42	12.56	15.70	18.84	21.98	25.12
45	3.530	7.07	10.60	14.13	17.66	21.20	24.73	28.26
50	3.930	7.86	11.78	15.70	19.63	23.55	27.48	31.40
60	4.710	9.42	14.13	18.84	23.55	28.26	32.97	37.68
70	5.500	10.99	16.49	21.98	27.48	32.97	38.47	43.96
80	6.280	12.56	18.84	25.12	31.40	37.68	43.96	50.24
90	7.070	14.13	21.20	28.26	35.33	42.39	49.46	56.52
100	7.850	15.70	23.55	31.40	39.25	47.10	54.95	62.80

THICKNESS	WIDTH/mm							
	90	100	200	300	400	500	1000	2000
mm	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m	Kg/m
3	2.12	2.36	4.71	7.07	9.42	11.78	23.55	47.10
4	2.83	3.14	6.28	9.42	12.56	15.70	31.40	62.80
5	3.53	3.93	7.85	11.78	15.70	19.63	39.25	78.50
6	4.24	4.71	9.42	14.13	18.84	23.55	47.10	94.20
8	5.65	6.28	12.56	18.84	25.12	31.40	62.80	125.60
10	7.07	7.85	15.70	23.55	31.40	39.25	78.50	157.00
12	8.48	9.42	18.84	28.26	37.68	47.10	94.20	188.40
16	11.30	12.56	25.12	37.68	50.24	62.80	125.60	251.20
20	14.13	15.70	31.40	47.10	62.80	78.50	157.00	314.00
25	17.66	19.63	39.25	58.88	78.50	98.13	196.25	392.50
28	19.78	21.98	43.96	65.94	87.92	109.90	219.80	439.60
32	22.61	25.12	50.24	75.36	100.48	125.60	251.20	502.40
36	25.43	28.26	56.52	84.78	113.04	141.30	282.60	565.20
40	28.26	31.40	62.80	94.20	125.60	157.00	314.00	628.00
45	31.79	35.33	70.65	105.98	141.30	176.63	353.25	706.50
50	35.33	39.25	78.50	117.75	157.00	196.25	392.50	785.00
60	42.39	47.10	94.20	141.30	188.40	235.50	471.00	942.00
70	49.46	54.95	109.90	164.85	219.80	274.75	549.50	1099.00
80	56.52	62.80	125.60	188.40	251.20	314.00	628.00	1256.00
90	63.59	70.65	141.30	211.95	282.60	353.25	706.50	1413.00
100	70.65	78.50	157.00	235.50	314.00	392.50	785.00	1570.00

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

EQUAL ANGLES

GRADES 250 and 350

Size mm	Kg/m	M/t	Surface Area Per m. sqm
25 x 25 x 3	1.12	892.9	0.0953
x 5	1.65	606.1	0.0953
x 6	2.08	480.8	0.0953
30 x 30 x 3	1.35	740.7	0.115
5	2.01	497.5	0.115
6	2.56	390.6	0.115
40 x 40 x 3	1.83	546.4	0.155
5	2.73	366.3	0.155
6	3.50	285.7	0.155
45 x 45 x 3	2.06	485.4	0.175
5	3.10	322.6	0.175
6	3.97	251.9	0.175
50 x 50 x 3	2.31	432.9	0.195
x 5	3.48	287.4	0.195
x 6	4.46	224.2	0.195
x 8	5.68	176.1	0.195
55 x 55 x 5	3.84	260.4	0.215
x 6	4.93	202.8	0.215
65 x 65 x 5	4.56	219.3	0.255
x 6	5.87	170.4	0.255
x 8	7.51	133.2	0.255
x 10	9.02	110.9	0.255
75 x 75 x 5	5.27	189.8	0.292
x 6	6.81	146.8	0.292
x 8	8.73	114.5	0.292
x 10	10.50	95.2	0.292
90 x 90 x 6	8.22	121.7	0.352
x 8	10.60	94.3	0.352
x 10	12.70	78.7	0.352
100 x 100 x 6	9.16	109.2	0.392
x 8	11.80	84.7	0.392
x 10	14.20	70.4	0.392
x 12	17.70	56.5	0.392

GRADES 300 and 350

Size mm	Kg/m	M/t	Surface Area Per m. sqm
125 x 125 x 8	14.90	67.1	0.491
x 10	18.00	55.6	0.491
x 12	22.50	44.4	0.491
x 16	29.10	34.4	0.491
x 10	21.90	45.7	0.590
x 12	27.30	36.6	0.590
x 16	35.40	28.2	0.590
x 19	42.10	23.7	0.590
200 x 200 x 13	40.00	25.0	0.788
x 16	48.70	20.5	0.788
x 18	54.40	18.4	0.788
x 20	60.10	16.6	0.788
x 26	76.80	13.0	0.788

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

UNEQUAL ANGLES

GRADES 250 and 350

Size mm	Kg/m	M/t	Surface Area Per m. sqm
65 x 50 x 5	4.02	248.8	0.225
x 6	5.16	193.8	0.225
x 8	6.59	151.7	0.225
75 x 50 x 5	4.36	229.4	0.244
x 6	5.62	177.9	0.244
x 8	7.19	139.1	0.244
100 x 75 x 6	7.98	125.3	0.342
x 8	10.30	97.1	0.342
x 10	12.40	80.6	0.342
125 x 75 x 6	9.16	109.2	0.392
x 8	11.80	84.7	0.392
x 10	14.20	70.4	0.392
x 12	17.70	56.6	0.392

GRADES 300 and 350

Size mm	Kg/m	M/t	Surface Area Per m. sqm
150 x 90 x 8	14.30	69.9	0.471
x 10	17.30	57.8	0.471
x 12	21.60	46.3	0.471
x 16	27.90	35.8	0.471
150 x 100 x 10	18.00	55.6	0.491
x 12	22.50	44.4	0.491

ROUND BARS

Size/mm	Kg/m	M/t
10	0.616	1623
12	0.887	1127
13	1.040	962
14	1.210	826
15	1.390	719
16	1.580	633
17	1.780	562
18	1.990	503
19	2.230	448
20	2.460	406
21	2.720	368
22	2.980	336
23	3.260	307
24	3.550	282
26	4.170	240
27	4.490	223
28	4.834	207
30	5.550	180
32	6.310	158
33	6.710	149
36	7.990	125
39	9.380	107
42	10.900	92
45	12.500	80
48	14.200	70
50	15.400	65
56	19.300	52
60	22.200	45
65	26.000	38
75	34.700	29
80	39.500	25
90	49.900	20
100	61.700	16

SQUARE BARS

Size/mm	Kg/m	M/t
10	0.790	1266
12	1.130	885
16	2.010	498
20	3.140	318
25	4.910	204
40	12.500	80

FLOOR PLATE

Size/mm	Kg/Sqm	Sqm/t
3	25.6	39.06
5	41.3	24.21
6	49.1	20.37
8	64.8	15.43
10	80.5	12.42
12	96.2	10.40

PRESTRESSING

	Size/mm	Kg/m	M/t
Wire	7	0.302	3310.0
Strand	12.7	0.787	1270.0

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

SQUARE HOLLOW SECTIONS

GRADE C350

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
13 x 13 x 1.8	0.568	0.0458
15 x 15 x 1.8	0.681	0.0538
20 x 20 x 1.6	0.873	0.0745
25 x 25 x 1.6	1.12	0.0945
x 2.0	1.36	0.0931
x 2.5	1.64	0.0914
x 3.0	1.89	0.0897
30 x 30 x 1.6	1.38	0.115
x 2.0	1.68	0.113
35 x 35 x 1.6	1.63	0.135
x 2.0	1.99	0.133
x 2.5	2.42	0.131
x 3.0	2.83	0.130
40 x 40 x 1.6	1.88	0.155
x 2.0	2.31	0.153
x 2.5	2.82	0.151
x 3.0	3.30	0.150
x 4.0	4.09	0.143
50 x 50 x 1.6	2.38	0.195
x 2.0	2.93	0.193
x 2.5	3.60	0.191
x 3.0	4.25	0.190
x 4.0	5.35	0.183
x 5.0	6.39	0.179
65 x 65 x 2.0	3.88	0.253
x 2.5	4.78	0.251
x 3.0	5.66	0.250
x 4.0	7.23	0.243
x 5.0	8.75	0.239
x 6.0	10.10	0.234

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
75 x 75 x 2.5	5.56	0.291
x 3.0	6.60	0.290
x 3.5	7.53	0.285
x 4.0	8.49	0.283
x 5.0	10.30	0.279
x 6.0	12.00	0.274
89 x 89 x 3.5	9.06	0.341
x 5.0	12.50	0.334
x 6.0	14.70	0.330
100 x 100 x 3.0	8.96	0.390
x 4.0	11.60	0.383
x 5.0	14.20	0.379
x 6.0	16.70	0.374
x 9.0	23.50	0.361
125 x 125 x 4.0	14.80	0.483
x 5.0	18.20	0.479
x 6.0	21.40	0.474
x 9.0	30.60	0.461
150 x 150 x 5.0	22.10	0.579
x 6.0	26.20	0.574
x 9.0	37.70	0.561
200 x 200 x 5.0	29.90	0.779
x 6.0	35.60	0.774
x 9.0	51.80	0.761
250 x 250 x 6.0	45.00	0.974
x 9.0	65.90	0.961
300 x 300 x 10.0	88.40	1.160
x 12.5	109.00	1.150
350 x 350 x 10.0	104.00	1.360
x 12.5	128.00	1.350
400 x 400 x 12.5	148.00	1.550
x 16.0	186.00	1.530

GRADE C450

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
25 x 25 x 2.3	1.53	0.0921
35 x 35 x 2.3	2.25	0.132
x 2.8	2.67	0.130
40 x 40 x 2.3	2.62	0.152
x 2.8	3.11	0.150
50 x 50 x 2.3	3.34	0.192
x 2.8	3.99	0.190

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
65 x 65 x 2.3	4.42	0.252
75 x 75 x 2.3	5.14	0.292
x 2.8	6.19	0.290
x 3.3	7.14	0.286
100 x 100 x 2.8	8.39	0.390
x 3.3	9.73	0.386
x 3.8	11.10	0.384

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

RECTANGULAR HOLLOW SECTIONS

GRADE C350

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
50 x 20 x 1.6	1.63	0.135
x 2.0	1.99	0.133
x 2.5	2.42	0.131
x 3.0	2.83	0.130
50 x 25 x 1.6	1.75	0.145
x 2.0	2.15	0.143
x 2.5	2.62	0.141
x 3.0	3.07	0.140
65 x 35 x 2.0	2.93	0.193
x 2.5	3.60	0.191
x 3.0	4.25	0.190
75 x 25 x 1.6	2.38	0.195
x 2.0	2.93	0.193
x 2.5	3.60	0.191
75 x 50 x 2.0	3.72	0.243
x 2.5	4.58	0.241
x 3.0	5.42	0.240
x 4.0	6.92	0.233
x 5.0	8.35	0.229
100 x 50 x 2.0	4.50	0.293
x 2.5	5.56	0.291
x 3.0	6.60	0.290
100 x 50 x 3.5	7.53	0.285
x 4.0	8.49	0.283
x 5.0	10.30	0.279
x 6.0	12.00	0.274
125 x 75 x 3.0	8.96	0.390
x 4.0	11.60	0.383
x 5.0	14.20	0.379
x 6.0	16.70	0.374
150 x 50 x 3.0	8.96	0.390
x 4.0	11.60	0.383
x 5.0	14.20	0.379
150 x 100 x 4.0	14.80	0.483
x 5.0	18.20	0.479
x 6.0	21.40	0.474
200 x 100 x 4.0	17.90	0.583
x 5.0	22.10	0.579
x 6.0	26.20	0.574
x 9.0	37.70	0.561
250 x 150 x 5.0	29.90	0.779
x 6.0	35.60	0.774
x 9.0	51.80	0.761
300 x 200 x 8.0	59.10	0.966
x 10.0	72.70	0.957
400 x 200 x 10.0	88.40	1.160
x 12.5	109.00	1.150

GRADE C450

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
50 x 20 x 2.3	2.25	0.132
x 2.8	2.67	0.130
50 x 25 x 2.3	2.44	0.142
x 2.8	2.89	0.140
65 x 35 x 2.3	3.34	0.192
x 2.8	3.99	0.190
75 x 50 x 2.3	4.24	0.242
x 2.8	5.09	0.240
100 x 50 x 2.3	5.14	0.292
x 2.8	6.19	0.290
x 3.3	7.14	0.286
125 x 75 x 2.8	8.39	0.390
x 3.3	9.73	0.386
x 3.8	11.10	0.384

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

CIRCULAR HOLLOW SECTIONS

GRADE C250

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
21.3 x 3.6	1.57	0.0669
26.9 x 2.6	1.56	0.0845
x 3.2	1.87	0.0845
x 4.0	2.26	0.0845
33.7 x 3.2	2.41	0.1060
x 4.0	2.93	0.1060
x 4.5	3.24	0.1060
42.4 x 3.2	3.09	0.1330
x 4.0	3.79	0.1330
x 4.9	4.53	0.1330
48.3 x 3.2	3.50	0.152
x 4.0	4.37	0.152
x 4.9	5.71	0.152
60.3 x 3.6	5.03	0.189
x 4.5	6.19	0.189
x 5.4	7.31	0.189
76.1 x 3.6	6.44	0.239
x 4.5	7.95	0.239
x 5.9	10.20	0.239
88.9 x 4.0	8.38	0.279
x 5.0	10.30	0.279
x 5.9	12.10	0.279
101.6 x 4.0	9.63	0.319
x 5.0	11.90	0.319
114.3 x 4.5	12.20	0.359
x 5.4	14.50	0.359
139.7 x 5.0	16.60	0.439
x 5.4	17.90	0.439
165.1 x 5.0	19.70	0.519
x 5.4	21.30	0.519
508.0 x 6.4	79.20	1.600
x 9.5	117.00	1.600
x 12.7	155.00	1.600
610.0 x 6.4	95.30	1.920
x 9.5	141.00	1.920
x 12.7	187.00	1.920

GRADE C350

Designation mm	Kg/m	Ext. Surface Area Per m. sqm
26.9 x 2.0	1.23	0.0845
x 2.3	1.40	0.0845
33.7 x 2.0	1.56	0.106
x 2.6	1.99	0.106
42.4 x 2.0	1.99	0.133
x 2.6	2.55	0.133
48.3 x 2.3	2.61	0.152
x 2.9	3.25	0.152
60.3 x 2.3	3.29	0.189
x 2.9	4.11	0.189
76.1 x 2.3	4.19	0.239
x 3.2	5.75	0.239
88.9 x 2.6	5.53	0.279
x 3.2	6.76	0.279
x 4.8	9.96	0.279
x 5.5	11.30	0.279
101.6 x 2.6	6.35	0.319
x 3.2	7.77	0.319
114.3 x 3.2	8.77	0.359
x 3.6	9.83	0.359
x 4.8	13.00	0.359
x 6.0	16.00	0.359
139.7 x 3.0	10.10	0.439
x 3.5	11.80	0.439
165.1 x 3.0	12.00	0.519
x 3.5	13.90	0.519
168.3 x 4.8	19.40	0.529
x 6.4	25.60	0.529
x 7.1	28.20	0.529
219.1 x 4.8	25.40	0.688
x 6.4	33.60	0.688
x 8.2	42.60	0.688
273.1 x 4.8	31.80	0.858
x 6.4	42.10	0.858
x 9.3	60.50	0.858
323.9 x 6.4	50.10	1.020
x 9.5	73.70	1.020
x 12.7	97.50	1.020
355.6 x 6.4	55.10	1.120
x 9.5	81.10	1.120
x 12.7	107.00	1.120
406.4 x 6.4	63.10	1.28
x 9.5	93.00	1.28
x 12.7	123.00	1.28
457.0 x 6.4	71.10	1.44
x 9.5	105.00	1.44
x 12.7	139.00	1.44

STRUCTURAL STEELWORK

TABLES OF WEIGHTS OF STEEL SECTIONS

PIPE-SCREWED ENDS

Nominal Bore	O.D.	Wall Thickness	BLACK		GALVANISED	
mm	mm	mm	Kg/m	M/t	Kg/m	M/t
EXTRA LIGHT						
20	26.9	2.00	1.23	814.0	1.29	774.0
25	33.7	2.00	1.56	640.0	1.64	608.0
32	42.4	2.00	1.99	502.0	2.10	477.0
40	48.3	2.30	2.61	383.0	2.73	367.0
50	60.3	2.30	3.29	304.0	3.44	291.0
LIGHT						
15	21.3	2.00	0.95	1050.0	1.00	999.0
20	26.9	2.30	1.40	717.0	1.46	686.0
25	33.7	2.60	1.99	501.0	2.07	482.0
32	42.4	2.60	2.55	392.0	2.65	377.0
40	48.3	2.90	3.25	308.0	3.36	297.0
50	60.3	2.90	4.11	244.0	4.25	235.0
65	76.1	3.20	5.75	174.0	5.94	168.0
80	88.9	3.20	6.76	148.0	6.98	143.0
MEDIUM						
8	13.5	2.30	0.64	1560.0	0.67	1490.0
10	17.2	2.30	0.84	1190.0	0.87	1140.0
15	21.3	2.60	1.21	830.0	1.25	798.0
20	26.9	2.60	1.56	642.0	1.62	617.0
25	33.7	3.20	2.41	414.0	2.49	401.0
32	42.4	3.20	3.10	322.0	3.20	312.0
40	48.3	3.20	3.57	280.0	3.68	272.0
50	60.3	3.60	5.03	199.0	5.18	193.0
65	76.1	3.60	6.43	156.0	6.61	151.0
80	88.9	4.00	8.37	120.0	8.58	117.0
90	101.6	4.00	9.63	104.0	9.88	101.0
100	114.3	4.50	12.20	82.2	12.40	80.4
125	139.7	5.00	16.60	60.2	16.90	59.0
150	165.1	5.00	19.70	50.7	20.10	49.6
HEAVY						
8	13.5	2.90	0.76	1310.0	0.79	1260.0
10	17.2	2.90	1.02	985.0	1.05	951.0
15	21.3	3.20	1.44	696.0	1.48	674.0
20	26.9	3.20	1.87	535.0	1.93	518.0
25	33.7	4.00	2.94	340.0	3.02	332.0
32	42.4	4.00	3.80	263.0	3.90	257.0
40	48.3	4.00	4.38	228.0	4.49	223.0
50	60.3	4.50	6.19	161.0	6.33	158.0
65	76.1	4.50	7.93	126.0	8.12	123.0
80	88.9	5.00	10.30	96.8	10.50	94.8
90	101.6	5.00	11.90	84.0	12.20	82.3
100	114.3	5.40	14.50	69.1	14.80	67.8
125	139.7	5.40	17.90	55.9	18.20	54.9
150	165.1	5.40	21.30	47.0	21.70	46.1
EXTRA HEAVY						
15	21.3	3.60	1.57	636.0	-	-
20	26.9	4.00	2.26	443.0	-	-
25	33.7	4.50	3.24	309.0	-	-
32	42.4	4.90	4.53	221.0	-	-
40	48.3	5.40	5.71	175.0	-	-
50	60.3	5.40	7.31	137.0	-	-
65	76.1	5.90	10.20	97.9	-	-
80	88.9	5.90	12.10	82.8	-	-

STRUCTURAL STEELWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BASE PRICES							
The base prices given hereafter are applicable to a structural steel content of 20 to 50 tonnes. For smaller tonnages an approximate adjustment percentage is given at the beginning of the supply and erect section on this page	...	Note					
Average supply only price for the base grade of the respective sections :							
Universal beams	...	t	1325.00	1375.00	1300.00	1325.00	1325.00
Universal columns	...	t	1350.00	1375.00	1300.00	1325.00	1325.00
Welded beams	...	t	1900.00	2050.00	1975.00	2050.00	2025.00
Taper flange sections	...	t	1850.00	1850.00	1900.00	2100.00	2050.00
Angles	...	t	1500.00	1500.00	1425.00	1500.00	1450.00
Plates	...	t	1400.00	1400.00	1325.00	1400.00	1350.00
Square and rectangular hollow sections	...	t	1700.00	1725.00	1650.00	1700.00	1675.00
Circular hollow sections	...	t	1750.00	1825.00	1750.00	1800.00	1775.00
Bar	...	t	1350.00	1400.00	1375.00	1425.00	1575.00
ADD extra for :							
Higher grade steelwork (average)	...	+ 6%					
Average price of shop drawings, fabrication, delivery to site and erection of steelwork :							
Universal beams and columns	...	t	4000.00	4250.00	4100.00	4300.00	4050.00
Built-up sections	...	t	3950.00	4100.00	3950.00	4150.00	3900.00
Taper flange sections	...	t	3950.00	4100.00	3950.00	4150.00	3900.00
Angle sections	...	t	4300.00	4500.00	4350.00	4550.00	4300.00
Hollow sections	...	t	4850.00	4850.00	4700.00	4900.00	4650.00
STRUCTURAL STEELWORK SUPPLY AND ERECT							
Prices include for shop fabrication, preparation of shop drawings, marking, delivery, unloading, hoisting, erecting and fixing	...	Note					
Prices are based on a total structural steelwork content of 20 to 50 tonnes							
ADD extra for quantities :							
Below 1 tonne	...	+20%					
1 - 5 tonnes	...	+15%					
5 - 10 tonnes	...	+10%					
10 - 20 tonnes	...	+ 5%					
Prices are for the base grade of the respective sections, for higher grades adjust supply cost by (average)	...	+ 6%					
Prices include an allowance for shop drawings at the average rates ruling in the respective cities	...	Note					
ADD extra to the steelwork items given hereafter for :							
Architectural structural steelwork (cost varies with degree of complexity)	...	t	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00
Surface treatments, refer page 290	...	Note					
Universal beams and columns with :							
Simple bolted connections -							
Up to 50kg/m	...	t	5250.00	5550.00	5350.00	5550.00	5300.00
Over 50kg/m	...	t	5200.00	5450.00	5250.00	5500.00	5250.00

STRUCTURAL STEELWORK

STRUCTURAL STEELWORK SUPPLY AND ERECT (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Universal beams and columns with (cont'd) :							
Welded fitments, bolted connections -							
Up to 50kg/m	...	t	5450.00	5750.00	5500.00	5750.00	5500.00
Over 50kg/m	...	t	5400.00	5650.00	5450.00	5650.00	5450.00
Fully rigid connections -							
Up to 50kg/m	...	t	5650.00	5950.00	5700.00	5950.00	5700.00
Over 50kg/m	...	t	5600.00	5850.00	5650.00	5850.00	5600.00
Built-up plate beams and columns :							
Up to 50kg/m	...	t	5250.00	5450.00	5200.00	5500.00	5200.00
Over 50kg/m	...	t	5200.00	5400.00	5150.00	5400.00	5150.00
Portal frames, industrial type, fabricated from universal sections	...	t	5650.00	5950.00	5700.00	5950.00	5700.00
Taper flange beams and channels with :							
Simple bolted connections	...	t	6200.00	6450.00	6350.00	6800.00	6450.00
Fully rigid connections	...	t	6350.00	6600.00	6500.00	6950.00	6600.00
Hollow Sections :							
Black Steel -							
Square hollow sections :							
up to 100 x 100mm	...	t	6850.00	6850.00	6650.00	6900.00	7050.00
over 100 x 100mm	...	t	6800.00	6750.00	6600.00	6750.00	7000.00
Rectangular hollow sections :							
up to 125 x 75mm	...	t	6850.00	6850.00	6650.00	6900.00	7050.00
over 125 x 75mm	...	t	6800.00	6750.00	6600.00	6750.00	7000.00
Circular hollow sections :							
up to 89mm dia.	...	t	7150.00	7250.00	7000.00	7400.00	7000.00
over 89mm dia.	...	t	6900.00	7000.00	6750.00	7000.00	6700.00
Pre-galvanised -							
Square hollow sections :							
up to 100 x 100mm	...	t	6900.00	6950.00	6700.00	6950.00	6700.00
Rectangular hollow sections :							
up to 150 x 50mm	...	t	6900.00	6950.00	6700.00	6950.00	6700.00
Purlins and Girts :							
Channels	...	t	6550.00	6950.00	6700.00	7000.00	6700.00
Angles	...	t	6400.00	6550.00	6500.00	6750.00	6150.00
Bracing :							
Angles	...	t	6150.00	6350.00	6100.00	6400.00	6100.00
Flats	...	t	5900.00	5900.00	6100.00	5900.00	5650.00
Roof Trusses :							
Fabricated from -							
Universal sections	...	t	6300.00	6400.00	6100.00	6400.00	6050.00
Angle sections	...	t	6150.00	6350.00	6100.00	6400.00	6100.00
Circular hollow sections	...	t	7000.00	7050.00	6800.00	7100.00	6800.00
Trusses -							
up to 10.00m span	...	m span	183.00	188.00	181.00	189.00	180.00
10.00 - 15.00m span	...	m span	229.00	234.00	226.00	236.00	225.00
Staircases :							
Fabricated from taper flange and plate sections	...	t	6250.00	6500.00	6300.00	6650.00	6350.00

STRUCTURAL STEELWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STRUCTURAL STEELWORK SUPPLY AND ERECT (cont'd)							
PLATE CONNECTIONS							
Base and cap plates	...	t	5900.00	5950.00	5700.00	5850.00	5700.00
Plate and angle in :							
Attached connections	...	t	6050.00	6250.00	6050.00	6300.00	6000.00
Loose connections	...	t	5450.00	6250.00	6000.00	6300.00	6000.00
ADD extra when connections weigh on average :							
1.0 kg each	...	t	860.00	940.00	905.00	950.00	900.00
1.5 kg each	...	t	575.00	625.00	600.00	630.00	600.00
Average % weight of plate connections per tonne of structural steel, say 2.5% - 10.0%	...+ 2.5/10.0%						
WALL FRAMING							
Fully welded wall framing to standard height walls of cold formed steel sections in top and bottom plates, studs at 600mm centres and one row on noggings, including angles and intersections, in :							
76mm framing	...	sqm	79.50	81.50	72.00	69.00	86.00
92mm framing	...	sqm	81.50	86.00	76.00	75.00	90.00
DECKING							
Chequer plate :							
6mm thick	...	sqm	247.00	273.00	269.00	277.00	308.00
10mm thick	...	sqm	351.00	381.00	375.00	386.00	418.00
12mm thick	...	sqm	394.00	427.00	420.00	433.00	467.00
Grid decking :							
General floor areas -							
60 x 50mm openings :							
Black bitumen painted -							
25mm thick x 22.3 kg/sqm	...	sqm	166.00	162.00	166.00	162.00	166.00
32mm thick x 27.0 kg/sqm	...	sqm	185.00	181.00	185.00	181.00	185.00
Galvanised -							
25mm thick x 22.3 kg/sqm	...	sqm	186.00	182.00	186.00	182.00	186.00
32mm thick x 27.0 kg/sqm	...	sqm	199.00	194.00	199.00	194.00	199.00
40 x 50mm openings :							
Black bitumen painted -							
20mm thick x 22.5 kg/sqm	...	sqm	179.00	174.00	179.00	174.00	179.00
Galvanised -							
20mm thick x 22.5 kg/sqm	...	sqm	199.00	194.00	199.00	194.00	199.00
30 x 100mm openings :							
Black bitumen painted -							
25mm thick x 36.2 kg/sqm	...	sqm	207.00	202.00	207.00	202.00	207.00
32mm thick x 45.5 kg/sqm	...	sqm	240.00	234.00	240.00	234.00	240.00
Galvanised -							
25mm thick x 36.2 kg/sqm	...	sqm	229.00	223.00	229.00	223.00	229.00
32mm thick x 45.5 kg/sqm	...	sqm	267.00	260.00	267.00	260.00	267.00
Conveyors and walkways -							
Allow as foregoing General Floor Areas with the addition of :							
Black bitumen painted decking (ADD)	...	sqm	7.00	7.50	7.00	7.50	7.00
Galvanised decking (ADD)	...	sqm	8.00	8.50	8.00	8.50	8.00
Aluminium grid decking refer pages 305 and 306	...	Note					

STRUCTURAL STEELWORK

PROPRIETARY PURLINS, GIRTS

All galvanised unless otherwise described.

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for :							
Laps to Z sections	...	Note					
Standard purlin bolted to steel, excluding bolts :							
Z or C10010	...	m	16.25	16.95	17.00	17.10	18.40
Z or C10012	...	m	19.05	19.75	19.85	19.90	21.20
Z or C10015	...	m	19.15	19.85	19.95	20.00	21.30
Z or C10019	...	m	22.20	22.90	23.00	23.00	24.30
Z or C15012	...	m	23.90	24.70	24.70	24.80	26.20
Z or C15015	...	m	24.30	25.00	25.10	25.20	26.60
Z or C15019	...	m	29.80	30.50	30.60	30.70	32.10
Z or C15024	...	m	35.20	35.90	36.00	36.10	37.50
Z or C20015	...	m	29.90	30.70	30.80	30.90	32.50
Z or C20019	...	m	35.70	36.50	36.60	36.70	38.30
Z or C20024	...	m	43.00	43.80	43.90	44.00	45.60
C 25019	...	m	42.70	43.60	43.70	43.80	45.50
C 25024	...	m	51.20	52.10	52.20	52.30	54.00
Fascia purlin bolted to steel, excluding bolts :							
230F	...	m	80.60	81.50	81.60	81.70	83.50
Bridging member bolted on, excluding bolts :							
B100	...	No	24.50	25.80	25.90	26.00	28.40
B150	...	No	25.20	26.60	26.70	26.90	29.50
B200	...	No	25.90	27.40	27.60	27.70	30.50
Purlins, composite rates :							
Purlins at 1200 centres including tie rods, bridging, bolting, etc. -							
C10019	...	sqm	30.30	31.40	31.50	31.60	33.60
C15019	...	sqm	38.10	39.20	39.40	39.50	41.70
C20019	...	sqm	44.20	45.50	45.70	45.80	48.20
Girts zinc coated and bolted to steel excluding bolts :							
LC07630	...	m	23.60	24.40	24.50	24.50	26.00
LC08330	...	m	23.60	24.40	24.50	24.50	26.00
LC09530	...	m	23.10	24.00	24.10	24.10	25.80
LC10330	...	m	23.10	24.00	24.10	24.10	25.80
LC10230	...	m	28.50	29.40	29.60	29.60	31.40
LC15230	...	m	34.00	35.00	35.20	35.30	37.20
ADD extra for :							
Downturned lip	...	Note					
Heavy galvanising	...	Note					

STRUCTURAL STEELWORK

BOLTS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include nuts and washers	...	Note					
Black steel bolts :							
12mm dia. x 100mm	...	No	0.85	0.85	0.85	0.85	0.85
x 150mm	...	No	1.20	1.20	1.20	1.20	1.20
16mm dia. x 100mm	...	No	1.50	1.50	1.50	1.50	1.50
x 150mm	...	No	2.30	2.30	2.30	2.30	2.30
20mm dia. x 100mm	...	No	2.60	2.60	2.60	2.60	2.60
x 150mm	...	No	3.55	3.55	3.55	3.55	3.55
24mm dia. x 100mm	...	No	3.80	3.80	3.80	3.80	3.80
x 150mm	...	No	5.25	5.25	5.25	5.25	5.25
Galvanised steel bolts :							
12mm dia. x 100mm	...	No	0.85	0.85	0.85	0.85	0.85
x 150mm	...	No	1.10	1.10	1.10	1.10	1.10
16mm dia. x 100mm	...	No	1.45	1.45	1.45	1.45	1.45
x 150mm	...	No	1.95	1.95	1.95	1.95	1.95
20mm dia. x 100mm	...	No	2.45	2.45	2.45	2.45	2.45
x 150mm	...	No	3.40	3.40	3.40	3.40	3.40
24mm dia. x 100mm	...	No	4.10	4.10	4.10	4.10	4.10
x 150mm	...	No	5.20	5.20	5.20	5.20	5.20
High strength steel friction grip bolts, galvanised :							
16mm dia. x 100mm	...	No	2.25	2.25	2.25	2.25	2.25
20mm dia. x 100mm	...	No	2.60	2.60	2.60	2.60	2.60
x 150mm	...	No	3.90	3.90	3.90	3.90	3.90
25mm dia. x 100mm	...	No	4.50	4.50	4.50	4.50	4.50
x 150mm	...	No	6.40	6.40	6.40	6.40	6.40
Stainless steel, grade 316 bolts :							
12mm dia. x 100mm	...	No	2.50	2.50	2.50	2.50	2.50
16mm dia. x 100mm	...	No	4.20	4.20	4.20	4.20	4.20
20mm dia. x 100mm	...	No	7.40	7.40	7.40	7.40	7.40
24mm dia. x 100mm	...	No	13.10	13.10	13.10	13.10	13.10
Purlin bolts :							
12mm dia. x 30mm, flanged	...	No	0.60	0.60	0.60	0.60	0.60
Steel holding down bolt including setting in concrete :							
16mm dia. x 300mm long	...	No	26.20	29.00	29.30	29.60	35.00
x 400mm long	...	No	28.20	31.30	31.70	32.00	37.90
x 500mm long	...	No	30.80	34.20	34.60	34.90	41.40
20mm dia. x 300mm long	...	No	34.00	37.40	37.80	38.10	44.60
x 400mm long	...	No	36.60	40.30	40.70	41.10	48.10
x 500mm long	...	No	39.80	43.80	44.20	44.60	52.10
24mm dia. x 300mm long	...	No	40.20	44.10	44.60	45.00	52.50
x 400mm long	...	No	41.90	46.20	46.70	47.10	55.10
x 500mm long	...	No	43.70	48.20	48.70	49.20	57.70
Holding down bolt, U shaped, including setting in concrete :							
16mm dia. x 500mm girth	...	No	51.60	55.80	56.40	56.80	64.80
x 700mm girth	...	No	55.70	60.20	60.80	61.20	69.80
x 900mm girth	...	No	61.80	66.90	67.50	68.00	77.60
20mm dia. x 500mm girth	...	No	61.50	66.20	66.80	67.30	76.40
x 700mm girth	...	No	68.30	73.40	74.00	74.50	84.10
x 900mm girth	...	No	77.20	82.80	83.50	84.00	94.80
24mm dia. x 500mm girth	...	No	71.10	76.40	77.10	77.60	87.80
x 700mm girth	...	No	79.10	84.70	85.40	86.00	96.70
x 900mm girth	...	No	86.70	92.90	93.70	94.30	106.10

STRUCTURAL STEELWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SUNDRIES							
Fixings :							
Sleeve anchor including drilling concrete -							
10mm dia. x 110mm	...	No	11.95	12.55	11.95	12.55	12.55
12mm dia. x 125mm	...	No	13.60	14.25	13.60	14.25	14.25
16mm dia. x 140mm	...	No	15.10	15.80	15.10	15.80	15.80
Stud anchor including drilling concrete -							
16mm dia. x 100mm	...	No	18.30	19.15	18.30	19.15	19.15
16mm dia. x 175mm	...	No	34.30	35.90	34.30	35.90	35.90
20mm dia. x 215mm	...	No	46.90	48.70	46.90	48.70	48.70
Chemical anchor including drilling concrete -							
8mm dia. x 110mm long	...	No	19.10	20.10	19.10	20.10	20.10
12mm dia. x 160mm long	...	No	30.90	32.50	30.90	32.50	32.50
16mm dia. x 190mm long	...	No	38.60	40.50	38.60	40.50	40.50
20mm dia. x 260mm long	...	No	65.00	68.10	65.00	68.10	68.10
Steel wedge and grout up under steel with cement and sand 1:2 well rammed in :							
25mm thick	...	sqm	175.50	181.00	186.50	179.00	201.00
50mm thick	...	sqm	197.00	210.00	215.00	208.00	233.00
Site drill or cut hole through 10mm thick steel :							
6mm dia. hole	...	No	5.70	6.25	5.75	5.35	7.05
9mm dia. hole	...	No	8.50	9.00	8.60	7.85	10.45
13mm dia. hole	...	No	11.25	12.15	11.30	10.20	13.75
150mm dia. hole	...	No	62.00	64.70	60.40	56.70	73.40
200 x 100mm hole	...	No	82.30	83.10	80.30	74.10	97.90
Site fillet weld (minimum 20 metres) :							
6mm	...	m	31.50	32.60	33.70	31.10	38.80
12mm	...	m	115.50	113.00	122.50	109.50	136.00
Machined end to 310 U.C.	...	No	51.00	49.50	55.40	46.60	54.20
SURFACE TREATMENTS							
Prices given are an indication only and depend on the section and mass of steel to be treated.	...	Note					
Lightly sandblast and paint one coat of zinc phosphate primer :							
Columns, beams, channels, etc.	...	t	415.00	450.00	460.00	420.00	490.00
Small angles, tubular sections, etc.	...	t	495.00	600.00	500.00	475.00	500.00
Class 3 sandblast and zinc silicate treatment :							
Columns, beams, channels, etc.	...	t	695.00	800.00	760.00	650.00	720.00
Small angles, tubular sections, etc.	...	t	820.00	1000.00	830.00	850.00	800.00
Hot dip galvanize (single dip) :							
Columns, beams, channels, etc.	...	t	1190.00	900.00	1250.00	900.00	1250.00
Small angles, tubular sections, etc.	...	t	1500.00	1200.00	1600.00	1350.00	1600.00
SPACEFRAMES							
Refer pages 122 and 123	...	Note					

METALWORK

BASE MATERIAL PRICES

ALUMINIUM

Prices are based on the following quantities :
Sheet - 150/250 kg
Extrusions - 250 kg

EQUAL ANGLES

	$/m
20 x 20 x 3.0mm	2.16
25 x 25 x 3.0mm	2.67
32 x 32 x 3.0mm	3.54
40 x 40 x 3.0mm	4.48
50 x 50 x 3.0mm	5.64

UNEQUAL ANGLES

	$/m
40 x 25 x 3.0mm	3.61
50 x 25 x 3.0mm	4.19

SQUARE TUBE

	$/m
25.4 x 25.4 x 3.25mm	4.96
40.0 x 40.0 x 3.00mm	8.60
50.8 x 50.8 x 3.18mm	11.49

ROUND TUBE

	$/m
25.0 O.D. x 3.0mm	3.74
32.0 O.D. x 3.0mm	4.93
40.0 O.D. x 3.0mm	6.32
44.0 O.D. x 3.0mm	7.64

UNEQUAL TEES

	$/m
38.1 x 32.2 x 2.6mm	3.68
40.0 x 40.0 x 1.6mm	2.29

EQUAL TEES

	$/m
20.0 x 20.0 x 1.6mm	1.14
25.4 x 25.4 x 1.6mm	1.41
38.1 x 38.1 x 3.1mm	4.48

RECTANGULAR BAR

	$/m
40 x 5mm	3.62
25 x 6mm	2.91
50 x 6mm	5.96
100 x 6mm	11.66
50 x 10mm	9.69
80 x 10mm	15.50

SQUARE BARS

	$/m
10mm	2.47
25mm	13.18

ROUND BAR

	$/m
12.7mm	2.20
20.0mm	5.72
25.4mm	9.19

CHANNELS

	$/m
25.0 x 25.0 x 3.0mm	4.01
31.7 x 25.4 x 3.1mm	5.17
38.1 x 19.0 x 3.1mm	4.44
50.0 x 25.0 x 3.1mm	5.46

SHEET

ALLOY 5005 - (2400 x 1200mm sheets)
(Sheets 2.88 sqm)

	$/sheet
1.6mm thick	72.44
2.0mm thick	90.54
2.5mm thick	113.15
3.0mm thick	135.81

METALWORK

BASE MATERIAL PRICES (cont'd)

BRASS

RECTANGULAR BAR	
	$/m
25.40 x 3.18mm	11.84
38.10 x 3.18mm	17.54
50.80 x 3.18mm	23.68
25.40 x 6.35mm	27.05
38.10 x 6.35mm	37.93
50.80 x 6.35mm	46.62

EQUAL ANGLES	
	$/m
19.0 x 19.0 x 3.00mm	18.48
25.0 x 25.0 x 3.00mm	25.25
38.0 x 38.0 x 3.00mm	39.04
50.0 x 50.0 x 6.35mm	109.15

1800 x 600mm SHEET		
	(Sheets 1.08sqm)	$/sheet
0.6mm thick		142.00

COPPER

1800 x 900mm SHEET		
	(Sheets 1.62sqm)	$/sheet
0.55mm thick		193.50
0.70mm thick		227.10
0.90mm thick		286.70

STAINLESS STEEL

TYPE 304 FLAT	
	$/m
25.0 x 6.0mm	6.60
50.0 x 10.0mm	23.70
75.0 x 10.0mm	35.00
100.0 x 10.0mm	92.00

TYPE 304 ANGLE	
	$/m
50.0 x 50.0 x 6.0mm	26.95
65.0 x 65.0 x 6.0mm	34.09

TYPE 316 FLAT	
	$/m
25.0 x 6.0mm	9.50
40.0 x 10.0mm	26.50
75.0 x 10.0mm	51.00
100.0 x 10.0mm	68.00

TYPE 316 ANGLE	
	$/m
25.0 x 25.0 x 3.0mm	10.15
40.0 x 40.0 x 5.0mm	25.20
65.0 x 65.0 x 6.0mm	51.00
75.0 x 75.0 x 6.0mm	58.25

TYPE 304 TUBE, POLISHED	
	$/m
25.4 dia. x 1.2mm	3.60
31.8 dia. x 1.2mm	5.40
38.1 dia. x 1.6mm	10.15

TYPE 304 SQUARE TUBE, POLISHED	
	$/m
19.05 x 19.05 x 1.2mm	4.00
25.40 x 25.40 x 1.2mm	6.00
31.80 x 31.80 x 1.2mm	6.45
38.10 x 38.10 x 1.6mm	10.20

METALWORK

BASE MATERIAL PRICES (cont'd)

STAINLESS STEEL (cont'd)

302/304 SHEET - (2400 x 1200mm sheets)	
(Sheets 2.88sqm) (No 4PC)	$/sheet
0.7mm thick	87.95
0.9mm thick	108.40
1.2mm thick	137.95
1.5mm thick	170.20

316 SHEET - (2400 x 1200mm sheets)	
(Sheets 2.88sqm) (No 4PC)	$/sheet
1.2mm thick	193.00

STEEL

STEEL SECTIONS

Refer STRUCTURAL STEELWORK page 285 for average supply only prices and pages 275 to 284 for weights.

ADD cost of Protective Treatment when required.

GALVANISED STEEL PIPE

GALVANISED STEEL PIPE - UNSOCKETTED	
MEDIUM	$/m
Nom. Bore	
15mm	8.21
20mm	9.56
25mm	13.51
32mm	16.35
40mm	18.68
50mm	21.58
65mm	27.64
80mm	41.59
100mm	64.55
150mm	99.72

STEEL SHEET

GALVABOND® STEEL	
	$/sheet
0.40mm thick (sheets 2400 x 1200 - 2.88sqm)	23.60
0.55mm thick (sheets 2400 x 1200 - 2.88sqm)	29.22
0.75mm thick (sheets 2400 x 1200 - 2.88sqm)	38.23
0.95mm thick (sheets 2400 x 1200 - 2.88sqm)	46.90
1.15mm thick (sheets 2400 x 1200 - 2.88sqm)	52.73

ZINCANNEAL® STEEL	
	$/sheet
0.55mm thick (sheets 2400 x 1200 - 2.88sqm)	29.15
0.80mm thick (sheets 2400 x 1200 - 2.88sqm)	41.94
1.00mm thick (sheets 2400 x 1200 - 2.88sqm)	51.27
1.20mm thick (sheets 2400 x 900 - 2.16sqm)	61.54

ZINCALUME® STEEL	
	$/sheet
0.40mm thick (sheets 2400 x 1200 - 2.88sqm)	22.05
0.55mm thick (sheets 2400 x 1200 - 2.88sqm)	29.74
0.75mm thick (sheets 2400 x 1200 - 2.88sqm)	39.77
1.00mm thick (sheets 2400 x 1200 - 2.88sqm)	53.03
1.20mm thick (sheets 2400 x 1200 - 2.88sqm)	63.64

METALWORK

PROTECTIVE TREATMENTS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for preparation. Square metre prices are for treated area unless otherwise described	...	Note					
Anodising :							
Clear -							
Extrusions :							
x 0.010mm thick	...	sqm	13.80	10.80	9.75	9.90	10.70
x 0.025mm thick	...	sqm	22.60	16.60-17.60	15.80	16.15	17.10
Coloured -							
Extrusions :							
x 0.010mm thick	...	sqm	16.10	12.60	11.40	11.50	12.20
x 0.025mm thick	...	sqm	25.20	18.80-19.80	17.60	18.00	19.10
Chrome plating, bright or satin finish :							
Tubing -							
25mm dia.	...	m	13.90	10.10	10.05	10.30	10.65
50mm dia.	...	m	18.60	18.70	17.75	18.10	18.90
Hot dip galvanising :							
General surfaces	...	sqm	29.50	26.00	26.10	25.80	26.40
Tubing -							
25mm dia.	...	m	2.30	2.90	2.90	2.80	2.85
50mm dia.	...	m	5.95	5.70	5.50	5.35	5.40
Zinc phosphate primer, one coat :							
General surfaces -	...	sqm	8.50	8.15	8.10	7.50	8.20
Tubing :							
25mm dia.	...	m	0.72	0.85	0.70	0.65	1.10
50mm dia.	...	m	1.75	2.05	1.60	1.60	2.00
Zinc silicate treatment -							
General surfaces :	...	sqm	17.25	18.95	19.05	19.55	18.75
Tubing -							
25mm dia.	...	m	2.80	2.90	2.80	2.90	2.85
50mm dia.	...	m	4.55	4.80	4.40	4.60	4.55
Powder coat finish -							
General surfaces :	...	sqm	15.40	12.15	10.90	11.20	11.75
Tubing -							
25mm dia.	...	m	3.65	2.60	2.30	2.35	2.60
50mm dia.	...	m	4.40	4.00	3.70	3.70	4.00
Baked finishes -							
Acrylic :							
General surfaces -	...	sqm	14.10	9.05	9.00	8.95	9.50
Tubing :							
25mm dia.	...	m	3.50	1.75	1.75	1.75	1.95
50mm dia.	...	m	4.10	2.65	2.65	2.65	2.75
Enamel :							
General surfaces -	...	sqm	9.30	9.40	9.05	9.05	9.50
Tubing :							
25mm dia.	...	m	3.45	1.80	1.75	1.80	1.90
50mm dia.	...	m	4.10	2.90	2.70	2.75	2.80
Painting generally :							
Refer Painting page 449	...	Note					

METALWORK

ACCESS FLOORING

RAISED PANEL TYPE

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Access floor system comprising **unfinished** floor panels on a 600 x 600mm grid together with understructure 200mm high above slab, including minor labours, edge trims, cut-outs etc., panels of :						
Steel construction/cementitious core suitable for static and dynamic loads :						
Medium grade (general offices) - (100 sqm) ...	sqm	97.00	97.00	95.00	99.00	90.00
(cornerlock stringerless system) - (1000 sqm) ...	sqm	82.00	82.00	81.00	83.00	76.00
(bare for carpet) - (5000 sqm) ...	sqm	76.00	76.00	75.00	79.00	72.00
- (10000 sqm) ...	sqm	75.00	75.00	73.00	76.00	71.00
- (50000 sqm) ...	sqm	74.00	74.00	73.00	75.00	70.00
Extra heavy grade including high pressure laminate finishing panel (information processing, heavy duty computer areas)						
(bolted stringer system) - (1000 sqm) ...	sqm	141.00	141.00	139.00	148.00	137.00
- (5000 sqm) ...	sqm	137.00	137.00	135.00	144.00	132.00
- (10000 sqm) ...	sqm	135.00	135.00	133.00	143.00	131.00
Steel encapsulated woodcore construction including high pressure laminate finishing panel :						
Heavy grade (computer areas)						
(bolt-on stringer system) - (100 sqm) ...	sqm	145.00	145.00	143.00	152.00	141.00
- (1000 sqm) ...	sqm	140.00	140.00	138.00	147.00	137.00
- (5000 sqm) ...	sqm	139.00	139.00	137.00	146.00	135.00
- (10000 sqm) ...	sqm	138.00	138.00	136.00	145.00	134.00
Low finish floor height systems :						
Steel construction/cementitious core						
(cornerlock - medium grade) - (100 sqm) ...	sqm	113.00	113.00	110.00	113.00	109.00
(bare for carpet) - (1000 sqm) ...	sqm	108.00	108.00	97.00	100.00	94.00
- (10000 sqm) ...	sqm	92.00	92.00	91.00	94.00	88.00
Wood core construction :						
(stringerless - medium grade) - (100 sqm) ...	sqm	97.00	97.00	95.00	104.00	93.00
(bare for carpet) - (1000 sqm) ...	sqm	92.00	92.00	91.00	100.00	89.00
- (10000 sqm) ...	sqm	81.00	81.00	88.00	97.00	86.00
ADD extra for :						
Each additional 100mm in height above slab ...	sqm	3.15	3.15	3.15	3.15	3.15
Finishing panel with -						
Carpet - modular tiles 457 x 457mm ...	sqm	64.00	64.00	64.00	64.00	64.00
- modular tiles 600 x 600mm (locating buttons) ...	sqm	75.00	75.00	75.00	75.00	75.00
Vinyl, tiles or sheet ...	sqm	64.00	64.00	64.00	64.00	64.00
Air flow floor grilles -						
Perforated panel 25% free area ...	No	125.00	125.00	125.00	125.00	125.00
600 x 300mm air grille with damper ...	No	165.00	165.00	165.00	165.00	165.00
Electrical floor box 200 x 200mm ...	No	155.00	155.00	155.00	155.00	155.00
Vapour seal concrete slab ...	sqm	3.00	3.00	3.00	3.00	3.00
Handrails ...	m	200.00	200.00	200.00	200.00	200.00
Ramps (x 200mm rise) ...	sqm	260.00	260.00	260.00	260.00	260.00
Steps (x 200mm rise) ...	m	165.00	165.00	165.00	165.00	165.00

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ACCESS FLOORING (cont'd)							
CABLE MANAGEMENT FLOOR SYSTEM							
Accessible cable management floor system comprising fibre reinforced thermoplastic self levelling modules including direct cable access covers, for floor heights from 75mm to 95mm :							
(x 100 sqm)	...	sqm	141.00	141.00	146.00	141.00	138.00
(x 1000 sqm)	...	sqm	120.00	120.00	116.00	125.00	127.00
ADD extra for :							
Vinyl or carpet finish	...	Note					
AIR RELIEF GRILLES							
Refer page 373 for forming openings in doors	...	Note					
Clear anodised aluminium air relief grille with inverted V-shaped louvres screwed to timber :							
600 x 150mm	...	No	83.50	91.00	94.60	92.00	105.50
600 x 200mm	...	No	88.70	96.30	100.50	97.40	111.00
600 x 250mm	...	No	96.10	104.00	108.50	105.00	119.00
600 x 300mm	...	No	102.50	110.50	115.50	111.50	126.00
600 x 450mm	...	No	137.00	146.00	153.50	147.00	164.00
600 x 600mm	...	No	159.00	169.00	178.00	170.50	190.00
600 x 900mm	...	No	242.00	255.00	270.00	257.00	284.00
Powder coated aluminium air relief grille with inverted V-shaped louvres screwed to timber :							
600 x 150mm	...	No	92.90	100.50	105.00	101.50	115.50
600 x 200mm	...	No	101.50	109.00	114.00	110.50	125.00
600 x 250mm	...	No	112.00	120.00	125.50	121.00	136.50
600 x 300mm	...	No	121.50	129.50	136.00	130.50	147.00
600 x 450mm	...	No	166.50	176.00	186.00	177.00	196.50
600 x 600mm	...	No	198.00	209.00	221.00	210.00	233.00
600 x 900mm	...	No	299.00	313.00	333.00	315.00	347.00
ALUMINIUM TRIM							
Natural anodised aluminium angle trim including screwing to backing surface, angles and ends, etc. :							
12 x 12 x 3mm angle	...	m	10.05	11.40	11.25	11.60	13.40
20 x 20 x 3mm angle	...	m	10.60	11.95	11.80	12.20	13.95
25 x 25 x 3mm angle	...	m	11.55	12.95	12.80	13.15	14.95
32 x 32 x 3mm angle	...	m	13.15	14.55	14.35	14.75	16.50
40 x 25 x 3mm angle	...	m	13.90	15.45	15.25	15.70	17.60
ADD extra for :							
Colour anodising	...	+ 10%					
ANGLE GUARDS							
Steel angle guard with lugs cast into concrete or drilled and bolted to brickwork :							
75 x 75 x 8mm -							
Zinc phosphate primed	...	m	69.90	77.00	74.80	76.60	83.40
Galvanised	...	m	79.40	82.80	85.20	85.10	93.80
100 x 100 x 10mm -							
Zinc phosphate primed	...	m	94.20	103.50	100.50	102.50	111.00
Galvanised	...	m	109.00	112.50	116.50	116.00	127.00
Stainless steel angle guard with ditto :							
200mm girth x 6mm	...	m	270.00	284.00	279.00	286.00	295.00
300mm girth x 6mm	...	m	354.00	374.00	366.00	376.00	388.00

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ASH TRAYS							
Stainless steel, satin finished wall urn, 230mm wide x 300mm high x 130mm projection from wall, with lockable butt collector and mounting brackets fixed to wall	...	No	107.00	111.00	109.00	112.00	113.00
BENCH FRAMING							
Steel tube framing to benches including welded joints :							
25 x 25 x 2.5mm (1.64kg/m)	...	m	16.55	17.45	17.40	16.80	17.40
35 x 35 x 2.0mm (1.99kg/m)	...	m	17.15	17.25	17.90	17.40	18.00
Stainless steel tube in framing to benches including welded joints :							
25 x 25 x 1.2mm	...	m	50.00	53.00	50.00	50.50	51.00
32 x 32 x 1.2mm	...	m	62.00	66.50	62.50	63.00	63.50
38 x 38 x 1.2mm	...	m	68.00	74.00	71.00	72.00	71.50
BRACKETS							
Small section steel angle or flat in brackets plugged to wall (maximum 2kg each)	...	kg	15.60	16.40	17.10	16.10	16.95
CHAIN WIRE PARTITIONS							
Partition comprising 2.5mm wire in interwoven 40mm chain link mesh and framing of 40mm dia. steel tubing in posts at maximum 3000mm centres, top and bottom rails, including all fixings and galvanising	...	sqm	38.70	39.40	36.10	33.90	38.00
CLADDING SYSTEMS							
BUCHTAL CLADDING SYSTEM							
600 x 600mm ceramic tile cladding fixed to and including aluminium substructure :							
Vertical clip fixing system	...	sqm	305.00-485.00	310.00-495.00	305.00-485.00	305.00-490.00	305.00-485.00
Quadro invisible fixing system	...	sqm	305.00-485.00	310.00-495.00	305.00-485.00	305.00-490.00	305.00-485.00
Optional Hydrotect self-cleaning system	...	+5%					
KeraTwin 600 x 280mm ceramic tile cladding fixed to and including horizontal clip system (K1)	...	sqm	250.00-355.00	255.00-365.00	250.00-355.00	250.00-360.00	250.00-355.00
Ditto fixed to and including aluminium horizontal profiles (K3)	...	sqm	300.00-370.00	305.00-380.00	300.00-370.00	300.00-375.00	300.00-370.00

METALWORK

CLADDING SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FIRECLAD SYSTEM							
The costs given hereafter are for the fire rated element only which is integrated into the standard portal frame structure	...	Note					
Cladding comprising three layers of 13mm fire rated plasterboard together with vapour permeable membrane fixed to steel girt framing system, fixed/erected by :							
Tilt up method	...	sqm	78.00	77.00	81.00	74.00	77.00
In situ method	...	sqm	102.00	99.00	104.00	98.00	99.00
ADD extra for :							
External cladding, for various corrugated and ribbed roofing type materials, refer page 414	...	Note					
Internal finish to plasterboard	...	Note					
Building column and frame members together with girt framing system	...	Note					
ESL							
ESL certified and insurer approved cladding systems comprising 1000mm wide panels with CFC and HCFC free PIR, PUR and 'non combustible' biodegradable rockwool cores. Metal skins to both sides, including fixing to primary steel frame and all joints and sealants in :							
Wall panels - corrugated profile insulated metal panel with concealed fixings, external face PVDF finish, internal face colorcoat enamel -							
60mm thick	...	sqm	255.00	255.00	255.00	255.00	255.00
80mm thick	...	sqm	305.00	305.00	305.00	305.00	305.00
100mm thick	...	sqm	325.00	325.00	325.00	325.00	325.00
120mm thick	...	sqm	365.00	365.00	365.00	365.00	365.00
Wall panels - corrugated profile rockwool wall panel with concealed fixings, external face PVDF finish, internal face colorcoat enamel -							
60mm thick	...	sqm	195.00	195.00	195.00	195.00	195.00
80mm thick	...	sqm	235.00	235.00	235.00	235.00	235.00
100mm thick	...	sqm	280.00	280.00	280.00	280.00	280.00
120mm thick	...	sqm	320.00	320.00	320.00	320.00	320.00
140mm thick	...	sqm	365.00	365.00	365.00	365.00	365.00
160mm thick	...	sqm	410.00	410.00	410.00	410.00	410.00
180mm thick	...	sqm	450.00	450.00	450.00	450.00	450.00
200mm thick	...	sqm	500.00	500.00	500.00	500.00	500.00
Wall and roof panel - Enviroslips structural insulated polyurethane PUR panels, faced both sides with timber OSB board -							
80mm thick	...	sqm	85.00	85.00	85.00	85.00	85.00
100mm thick	...	sqm	90.00	90.00	90.00	90.00	90.00
125mm thick	...	sqm	96.00	96.00	96.00	96.00	96.00
150mm thick	...	sqm	118.00	118.00	118.00	118.00	118.00
180mm thick	...	sqm	139.00	139.00	139.00	139.00	139.00

METALWORK

CLADDING SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ESL (cont'd)							
Roof panels - galvalume trapezoidally profiled insulated metal panel, external face PVDF finish, internal face color coat enamel -							
75mm thick	...	sqm	95.00	95.00	95.00	95.00	95.00
95mm thick	...	sqm	115.00	115.00	115.00	115.00	115.00
115mm thick	...	sqm	135.00	135.00	135.00	135.00	135.00
Chill store panels - clad both sides with 0.5mm white liner -							
40mm thick	...	sqm	54.00	54.00	54.00	54.00	54.00
60mm thick	...	sqm	59.00	59.00	59.00	59.00	59.00
80mm thick	...	sqm	75.00	75.00	75.00	75.00	75.00
100mm thick	...	sqm	92.00	92.00	92.00	92.00	92.00
120mm thick	...	sqm	115.00	115.00	115.00	115.00	115.00
ADD extra for :							
Primary steel framing	...	Note					
LUXALON MULTIPANEL FACADE SYSTEM							
Panels with a thermosetting acrylic surface finish fixed to and including sub-framing :							
Aluminium -							
200mm module	...	sqm	283.00	283.00	283.00	293.00	273.00
300mm module	...	sqm	245.00	245.00	245.00	255.00	235.00
400mm module	...	sqm	232.00	232.00	232.00	242.00	222.00
LUXALON INVERTER-RIB CLADDING SYSTEM							
Panels with 'Color-Cote' polyester paint finish, 194mm panel module, fixed to and including sub-framing :							
Aluminium	...	sqm	154.00	154.00	154.00	164.00	144.00
Steel	...	sqm	148.00	148.00	148.00	158.00	138.00
FLAT PANELS AND FRAME SYSTEM							
Framing only to support cladding	...	sqm	330.00	350.00	340.00	335.00	345.00
Flat panel cladding fixed to framing including all joints and seals :							
Aluminium panels, 4mm thick sheet -							
Powder coat finish	...	sqm	315.00	325.00	330.00	325.00	335.00
Fluoropolymer finish	...	sqm	360.00	370.00	370.00	370.00	370.00
Alucobond panels, 4mm thick -							
Fluoropolymer finish	...	sqm	365.00	360.00	380.00	370.00	395.00
Externit fibre cement panels 7.5mm thick -							
Glasal	...	sqm	205.00	215.00	215.00	210.00	215.00
Miral	...	sqm	235.00	250.00	245.00	245.00	245.00
ADD extra for :							
Narrow width panels up to 600mm girth	...	+ 25%					
Curved panels	...	+ 50%					
50mm thick plain insulation	...	sqm	17.00	17.30	17.60	17.80	17.60
Windows, refer Curtain Walling page 366	...	Note					

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CLADDING SYSTEMS (cont'd)							
CURTAIN WALLING							
Refer page 366	...	Note					
WALL CLADDING							
Corrugated and ribbed roofing type materials - refer page 414	...	Note					
SHEET CLADDING							
Flat sheet cladding to ducts, enclosures, etc. - refer page 310	...	Note					
Perforated cladding - refer page 309	...	Note					
CLOTHES HOIST, LINE							
Rotary clothes hoist :							
4000mm dia.	...	No	605.00	610.00	605.00	610.00	605.00
5000mm dia.	...	No	610.00	620.00	610.00	620.00	610.00
6000mm dia.	...	No	645.00	655.00	645.00	655.00	645.00
Extendable clothes line with 50 metres of line including cabinet to wall at one end, the other end with line receiving post and socket in ground	...	No	475.00	480.00	475.00	480.00	475.00
Swinging frame clothes line with 23 metres of line :							
Ground mounted	...	No	425.00	430.00	425.00	430.00	425.00
Wall mounted	...	No	305.00	310.00	305.00	310.00	305.00
ACCESS COVERS AND GRATINGS							
Sizes stated are clear opening	...	Note					
DUCTILE IRON ACCESS COVERS							
Class B cover and frame including concrete filling :							
305 x 305mm	...	No	250.00	263.00	265.00	269.00	292.00
457 x 457mm	...	No	290.00	306.00	308.00	313.00	341.00
457 x 610mm	...	No	336.00	353.00	355.00	362.00	392.00
610 x 610mm	...	No	381.00	400.00	403.00	410.00	444.00
762 x 610mm	...	No	463.00	484.00	487.00	495.00	531.00
762 x 762mm	...	No	536.00	560.00	563.00	571.00	612.00
914 x 610mm	...	No	536.00	560.00	563.00	571.00	612.00
914 x 914mm	...	No	677.00	706.00	709.00	720.00	771.00
600mm dia.	...	No	336.00	353.00	355.00	362.00	392.00
Class D cover and frame including concrete filling :							
457 x 457mm	...	No	481.00	499.00	501.00	507.00	538.00
457 x 610mm	...	No	542.00	562.00	564.00	572.00	606.00
610 x 610mm	...	No	627.00	649.00	652.00	660.00	699.00
762 x 610mm	...	No	713.00	737.00	740.00	749.00	792.00
762 x 762mm	...	No	798.00	824.00	828.00	838.00	885.00
914 x 610mm	...	No	822.00	849.00	852.00	862.00	909.00
914 x 914mm	...	No	1060.00	1095.00	1100.00	1110.00	1175.00
600mm dia.	...	No	460.00	481.00	483.00	491.00	526.00

METALWORK

ACCESS COVERS AND GRATINGS (cont'd)

DUCTILE IRON ACCESS COVERS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Class D cover and frame including concrete filling :							
Two part cover -							
457 x 984mm	...	No	1035.00	1055.00	1060.00	1070.00	1110.00
457 x 1137mm	...	No	1100.00	1120.00	1125.00	1135.00	1175.00
457 x 1289mm	...	No	1160.00	1185.00	1190.00	1200.00	1240.00
610 x 984mm	...	No	1150.00	1175.00	1175.00	1185.00	1230.00
610 x 1137mm	...	No	1235.00	1260.00	1265.00	1275.00	1320.00
610 x 1289mm	...	No	1315.00	1345.00	1345.00	1355.00	1400.00
610 x 1441mm	...	No	1380.00	1405.00	1410.00	1420.00	1465.00
610 x 1594mm	...	No	1460.00	1485.00	1490.00	1500.00	1550.00
762 x 1289mm	...	No	1465.00	1495.00	1500.00	1510.00	1560.00
762 x 1441mm	...	No	1545.00	1575.00	1580.00	1590.00	1645.00
762 x 1594mm	...	No	1625.00	1660.00	1665.00	1675.00	1730.00
Class E cover and frame including concrete filling :							
457 x 457mm	...	No	949.00	967.00	970.00	976.00	1010.00
457 x 610mm	...	No	976.00	998.00	1000.00	1010.00	1045.00
610 x 610mm	...	No	1000.00	1025.00	1030.00	1035.00	1080.00
760mm dia.	...	No	992.00	1020.00	1025.00	1035.00	1085.00

GALVANISED STEEL ACCESS COVERS

Class B cover and frame, unfilled :							
Single unit cover -							
450 x 450mm	...	No	450.00	441.00	443.00	488.00	467.00
600 x 450mm	...	No	555.00	542.00	544.00	600.00	571.00
600 x 600mm	...	No	615.00	602.00	604.00	665.00	634.00
750 x 600mm	...	No	698.00	682.00	684.00	754.00	717.00
750 x 750mm	...	No	772.00	753.00	756.00	834.00	793.00
900 x 450mm	...	No	721.00	705.00	707.00	780.00	744.00
900 x 900mm	...	No	835.00	819.00	822.00	906.00	869.00
Two part cover -							
450 x 1020mm	...	No	1000.00	972.00	974.00	1075.00	1010.00
450 x 1170mm	...	No	1080.00	1050.00	1050.00	1160.00	1090.00
450 x 1320mm	...	No	1160.00	1125.00	1125.00	1245.00	1165.00
600 x 1020mm	...	No	1160.00	1125.00	1125.00	1245.00	1165.00
600 x 1170mm	...	No	1230.00	1190.00	1190.00	1315.00	1230.00
600 x 1320mm	...	No	1310.00	1265.00	1270.00	1400.00	1310.00
600 x 1470mm	...	No	1430.00	1380.00	1380.00	1530.00	1425.00
600 x 1620mm	...	No	1455.00	1405.00	1410.00	1560.00	1455.00
750 x 1320mm	...	No	1525.00	1470.00	1475.00	1630.00	1520.00
Class C cover and frame, unfilled :							
Single unit cover -							
450 x 450mm	...	No	492.00	482.00	483.00	533.00	509.00
600 x 450mm	...	No	577.00	564.00	566.00	624.00	596.00
600 x 600mm	...	No	636.00	623.00	625.00	689.00	658.00
750 x 600mm	...	No	746.00	729.00	731.00	806.00	768.00
750 x 750mm	...	No	818.00	800.00	802.00	884.00	843.00
900 x 600mm	...	No	818.00	800.00	802.00	884.00	843.00
900 x 900mm	...	No	849.00	831.00	833.00	919.00	877.00
Two part cover -							
450 x 1020mm	...	No	1200.00	1160.00	1160.00	1285.00	1205.00
450 x 1170mm	...	No	1135.00	1100.00	1105.00	1220.00	1145.00
450 x 1320mm	...	No	1235.00	1195.00	1200.00	1325.00	1240.00

METALWORK

ACCESS COVERS AND GRATINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GALVANISED STEEL ACCESS COVERS (cont'd)							
Class C cover and frame, unfilled (cont'd) :							
Two part cover (cont'd) -							
600 x 1020mm	...	No	1235.00	1195.00	1200.00	1325.00	1240.00
600 x 1170mm	...	No	1330.00	1285.00	1290.00	1425.00	1330.00
600 x 1320mm	...	No	1385.00	1340.00	1345.00	1485.00	1390.00
600 x 1470mm	...	No	1490.00	1440.00	1445.00	1595.00	1495.00
600 x 1620mm	...	No	1520.00	1470.00	1475.00	1630.00	1520.00
600 x 1770mm	...	No	1560.00	1510.00	1515.00	1675.00	1565.00
600 x 1920mm	...	No	1590.00	1540.00	1540.00	1705.00	1595.00
750 x 1320mm	...	No	1610.00	1560.00	1565.00	1725.00	1625.00
750 x 1470mm	...	No	1680.00	1630.00	1630.00	1805.00	1695.00
750 x 1620mm	...	No	1775.00	1720.00	1725.00	1905.00	1705.00
900 x 1320mm	...	No	1660.00	1615.00	1620.00	1790.00	1695.00
CAST IRON SUMP GRATINGS							
Class B cast iron sump or gully grating and frame (Type 2) :							
305 x 305mm	...	No	232.00	243.00	245.00	249.00	270.00
457 x 457mm	...	No	320.00	335.00	337.00	342.00	367.00
610 x 610mm	...	No	396.00	414.00	416.00	422.00	453.00
Class D cast iron sump or gully grating and frame (Type 2) :							
457 x 457mm	...	No	457.00	474.00	477.00	483.00	514.00
610 x 610mm	...	No	624.00	646.00	649.00	656.00	694.00
762 x 762mm	...	No	992.00	1020.00	1020.00	1030.00	1080.00
CAST IRON TRENCH GRATINGS							
All gratings are transverse bar type	...	Note					
Class B cast iron grating and frame :							
102mm wide	...	m	286.00	293.00	294.00	296.00	308.00
152mm wide	...	m	339.00	346.00	347.00	350.00	363.00
229mm wide	...	m	334.00	342.00	343.00	346.00	360.00
305mm wide	...	m	388.00	397.00	398.00	402.00	418.00
457mm wide	...	m	452.00	466.00	468.00	473.00	498.00
610mm wide	...	m	524.00	541.00	543.00	549.00	577.00
Class D cast iron grating :							
152mm wide	...	m	354.00	365.00	366.00	370.00	388.00
229mm wide	...	m	369.00	380.00	382.00	386.00	405.00
305mm wide	...	m	414.00	427.00	429.00	433.00	456.00
457mm wide	...	m	690.00	707.00	709.00	716.00	746.00
610mm wide	...	m	802.00	821.00	824.00	831.00	866.00
762mm wide	...	m	1035.00	1055.00	1060.00	1065.00	1105.00
PLATE COVERS AND FRAMES							
6mm thick galvanised plate cover to :							
Trench -							
150mm wide	...	m	58.40	63.20	63.80	64.20	73.50
300mm wide	...	m	93.60	99.60	100.50	100.50	112.00
450mm wide	...	m	129.00	136.00	137.00	137.50	151.00
Sump -							
450 x 450mm	...	No	85.00	92.50	93.40	94.00	108.50
600 x 600mm	...	No	135.00	145.50	147.00	147.50	168.50

METALWORK

ACCESS COVERS AND GRATINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PLATE COVERS AND FRAMES (cont'd)							
6mm thick galvanised floor plate cover to :							
Trench -							
150mm wide	...	m	60.60	65.50	66.10	66.40	75.70
300mm wide	...	m	94.70	100.00	101.00	101.00	111.50
450mm wide	...	m	132.00	138.50	139.50	140.00	152.50
Sump -							
450 x 450mm	...	No	91.20	99.30	100.50	101.00	116.50
600 x 600mm	...	No	140.00	151.00	152.00	153.00	174.00
Galvanised angle frame with lugs welded on, angle :							
30 x 30 x 6mm	...	m	38.50	42.80	43.30	43.70	51.90
40 x 40 x 6mm	...	m	46.20	51.10	51.60	52.00	61.30
50 x 50 x 6mm	...	m	60.70	67.10	67.90	68.40	80.80
65 x 65 x 6mm	...	m	80.50	89.10	90.20	90.80	107.50
LIFTING KEYS							
Gatic standard lifting key	...	Pair	59.50	59.50	59.50	59.50	59.50
HOSE TAP BOX							
Gatic standard hose tap box	...	No	95.00	99.00	99.50	100.00	108.50
STOP COCK BOX							
Gatic standard stop cock box	...	No	70.50	74.00	74.50	75.00	82.00
TREE GRILLES							
Refer page 230	...	Note					
BICYCLE STAND							
Stainless steel (304) bicyle parking hoop with bases set into paving :							
270mm wide	...	No	400.00	400.00	400.00	400.00	410.00
845mm wide	...	No	420.00	420.00	420.00	420.00	430.00
Stainless steel (304) bicyle parking stand with bases set into paving :							
1550mm long for four bicycles	...	No	1090.00	1090.00	1090.00	1100.00	1100.00
BBQ							
Outdoor electric barbeque, stand alone unit, stainless steel construction for and including :							
Single BBQ	...	No	5630.00	5590.00	5590.00	6540.00	5550.00
Double BBQ	...	No	6760.00	6710.00	6710.00	6780.00	6670.00
Single BBQ, Disability Discrimination Act (DDA) compliant	...	No	7870.00	7810.00	7810.00	7880.00	7750.00
Double BBQ, Disability Discrimination Act (DDA) compliant	...	No	13580.00	13460.00	13460.00	13590.00	13350.00
ADD extra for :							
Electrical connection	...	Note					
Concrete slab	...	Note					

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CRASH AND GUARD RAILS							
Galvanised steel double corrugated guard rail with bolts and splice plates	...	m	100.00	111.00	109.50	113.00	127.00
ADD extra for :							
Curved guardrail	...	m	8.60	8.80	8.60	8.80	8.60
1800mm long steel post set into and including concrete footing	...	No	208.00	229.00	228.00	233.00	264.00
Terminal section -							
“In” section or “Out” section	...	No	84.80	89.60	87.80	90.10	93.20
Bullnose terminal end	...	No	106.00	111.00	109.00	111.50	114.00
CURTAIN TRACKS							
BED SCREEN TRACK							
45 x 19 x 1.6mm anodised aluminium box section track together with 12mm dia. tubular hangers at 3000mm centres fixed to underside of ceiling suspension system including splice plates, bends, junctions, etc., as required (average)	...	m	63.50	66.50	64.00	65.00	65.50
WINDOW CURTAIN TRACK							
Refer page 386	...	Note					
DOORS							
Refer pages 375 to 381	...	Note					
DOOR FRAMES							
Refer page 369	...	Note					
DUCT ACCESS PANELS							
Access panel comprising Zincanneal® steel sheet panel with frame and sub-frame, budget lock, baked enamel finish, built-in :							
Standard panel -							
300 x 300mm wide	...	No	110.50	117.50	119.00	119.00	134.00
450 x 450mm wide	...	No	103.00	111.00	112.00	113.00	128.50
600 x 600mm wide	...	No	131.50	139.50	141.50	141.50	158.50
ADD extra for :							
Hinged door	...	No	7.90	8.05	7.90	8.05	7.90
One hour rated panel with approved infill -							
300 x 300mm wide	...	No	292.00	294.00	300.00	296.00	316.00
450 x 450mm wide	...	No	349.00	350.00	358.00	352.00	374.00
600 x 600mm wide	...	No	381.00	382.00	390.00	383.00	407.00
DUCT ACCESS DOORS							
Refer page 378	...	Note					
DUCTED SKIRTINGS							
Refer pages 543 and 544	...	Note					
Standard metal skirting, refer page 313	...	Note					

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DUCT ACCESS COVERS (FIRE RATED)							
Fire rated sound attenuated removable duct access cover comprising Zincanneal® steel sheet panel with frame and sub-frame, baked enamel finish, built-in, size :							
1500 x 800mm wide -							
One hour rated	...	No	815.00	815.00	830.00	815.00	865.00
Two hour rated	...	No	960.00	955.00	980.00	960.00	1010.00
Three hour rated	...	No	1020.00	1010.00	1035.00	1015.00	1070.00
2000 x 800mm wide							
One hour rated	...	No	1015.00	1005.00	1030.00	1010.00	1065.00
Two hour rated	...	No	1240.00	1225.00	1255.00	1230.00	1290.00
Three hour rated	...	No	1330.00	1315.00	1345.00	1315.00	1380.00
FENCING							
Refer pages 226 to 228	...	Note					
FLAGPOLES							
Galvanised steel flagpole, tabernacle type, complete with halyards, pulleys and chromium plated brass cleats, set 600mm into ground including 300 x 300 x 600mm concrete base, excavation, etc. :							
6000mm high	...	No	1370.00	1450.00	1425.00	1460.00	1510.00
9000mm high	...	No	1615.00	1695.00	1665.00	1705.00	1755.00
12000mm high	...	No	2760.00	2885.00	2825.00	2895.00	2935.00
FLOOR AND PLATFORM DECKING							
EXPANDED MESH							
General floor areas and platforms, including allowance for edge treatment, penetrations, bolting and welding :							
Alu-tread, in panels 6000 x 600mm wide -							
7.70 kg/sqm	...	sqm	146.00	156.00	153.00	157.00	164.00
9.70 kg/sqm	...	sqm	264.00	277.00	271.00	278.00	283.00
Steel, galvanised -							
14.00 kg/sqm	...	sqm	162.00	172.00	169.00	174.00	181.00
19.00 kg/sqm	...	sqm	237.00	250.00	245.00	251.00	258.00
22.00 kg/sqm	...	sqm	245.00	258.00	253.00	260.00	267.00
28.00 kg/sqm	...	sqm	317.00	332.00	325.00	333.00	339.00
EXPANDED METAL PANELS							
General floor areas and platforms, including allowance for edge treatment, penetrations, bolting and welding :							
Steel galvanised, in panels 6000 x 300mm wide -							
15.70 kg/sqm	...	sqm	246.00	256.00	251.00	257.00	259.00
25.30 kg/sqm	...	sqm	312.00	324.00	317.00	325.00	326.00
Aluminium, in panels 6000 x 300mm wide -							
10.70 kg/sqm	...	sqm	262.00	272.00	267.00	273.00	276.00

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FLOOR AND PLATFORM DECKING (cont'd)							
GRID DECKING							
General floor areas and platforms, including allowance for edge treatment, penetrations, bolting and welding :							
Aluminium -							
13mm thick x 7.64 kg/sqm	...	sqm	260.00	270.00	265.00	265.00	255.00
22mm thick x 9.58 kg/sqm	...	sqm	285.00	300.00	290.00	295.00	295.00
30mm thick x 12.60 kg/sqm	...	sqm	405.00	420.00	410.00	415.00	420.00
40mm thick x 14.90 kg/sqm	...	sqm	450.00	460.00	450.00	460.00	460.00
Steel -							
Refer page 287	...	Note					
General floor areas and platforms, including roof walkways x 600mm wide including end treatment :							
Aluminium -							
22mm thick x 9.58 kg/sqm	...	m	170.00	185.00	180.00	175.00	185.00
Steel -							
Refer page 287	...	Note					
RAISED STORAGE FLOOR							
Raised storage (mezzanine) floor comprising structural steel columns and beams, particle board decking and including access stairs, handrails, loading/sliding gate, deck approximately 2400mm above ground floor (Dexion)	...	sqm	550.00	570.00	555.00	560.00	570.00
INSULATED PANEL WALLS							
Insulated panels can be manufactured from 50mm - 300mm thick in 1200mm wide panels and in very long lengths	...	Note					
Prices exclude structural frames, girts, etc.	...	Note					
Insulated PIR panels in walls together with fixings, flashings, channels, etc., Colorbond® steel panel faced both sides :							
75mm thick	...	sqm	168.00	172.00	168.00	172.00	168.00
100mm thick	...	sqm	170.00	174.00	170.00	174.00	170.00
150mm thick	...	sqm	192.00	197.00	192.00	197.00	192.00
200mm thick	...	sqm	206.00	211.00	206.00	211.00	206.00
ADD extra for :							
Colorbond® steel external grade facing one side (Equitilt panels)	...	sqm	15.20	15.60	15.20	15.60	15.20
Fire resistant FM Rockwool panel in lieu of PIR core	...	+50%					
LADDERS							
Ladder 450mm wide comprising 50 x 6mm steel strings and 16mm dia. bar rungs :							
Zinc phosphate primed	...	m	55.90	58.70	56.40	59.10	56.80
	...	t	8000.00	8400.00	8050.00	8450.00	8100.00
Galvanised	...	m	62.90	62.90	64.10	65.20	64.50
	...	t	9000.00	9000.00	9150.00	9350.00	9200.00

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LOCKERS							
Single tier locker size 450mm wide x 300 x 1800mm high with baked enamel finish	...	No	256.00	267.00	262.00	268.00	273.00
Two tier ditto	...	No	282.00	297.00	291.00	298.00	306.00
Four tier ditto	...	No	325.00	344.00	337.00	346.00	359.00
LOUVRE PANELS							
Refer pages 368 and 369	...	Note					
MAIL BOXES							
Bank of anodised aluminium mail boxes fixed into prepared opening, bank of :							
5 boxes (av. P.C. $100 each)	...	No	665.00	680.00	670.00	680.00	670.00
15 boxes (av. P.C. $90 each)	...	No	1550.00	1570.00	1555.00	1570.00	1555.00
25 boxes (av. P.C. $85 each)	...	No	2365.00	2390.00	2370.00	2390.00	2370.00
35 boxes (av. P.C. $80 each)	...	No	3080.00	3110.00	3090.00	3110.00	3090.00
MATWELL SURROUNDS							
Brass angle matwell surround :							
12 x 25 x 3mm angle	...	m	51.20	54.60	53.60	55.10	57.90
20 x 25 x 3mm angle	...	m	56.80	60.40	59.20	60.80	63.50
25 x 25 x 3mm angle	...	m	62.40	66.10	64.80	66.50	69.10
Aluminium matwell surround :							
12 x 20 x 3mm angle	...	m	25.90	28.70	28.30	29.10	32.60
20 x 25 x 3mm angle	...	m	28.10	31.00	30.50	31.40	34.80
25 x 25 x 3mm angle	...	m	30.40	33.30	32.80	33.80	37.10
ADD extra for :							
Welded corners (per corner)	...	No	24.00	25.00	24.00	24.00	25.00
MATS							
Coir on flexible backing mat :							
17mm thick	...	sqm	116.50	123.50	121.50	124.50	130.00
23mm thick	...	sqm	131.00	138.50	136.00	139.50	144.50
30mm thick	...	sqm	153.00	161.00	157.50	162.00	166.50
PVC link mat	...	sqm	96.00	102.50	101.00	103.50	109.50
Cocamat closed construction mat comprising coir inserts in aluminium extrusions :							
x 10mm	...	sqm	453.00	468.00	447.00	479.00	464.00
x 16mm	...	sqm	537.00	554.00	531.00	565.00	548.00
x 10mm, slotted	...	sqm	537.00	554.00	531.00	565.00	548.00
Duramat closed construction heavy traffic area mat comprising alternate treads of aluminium and heavy duty ribbed rubber/vinyl inserts :							
x 10mm	...	sqm	432.00	446.00	426.00	457.00	443.00
x 16mm	...	sqm	516.00	532.00	510.00	543.00	527.00
x 10mm, slotted	...	sqm	516.00	532.00	510.00	543.00	527.00

METALWORK

MATS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Duragrit closed construction heavy traffic area mat with abrasive grip inserts :							
x 10mm	...	sqm	637.00	656.00	631.00	667.00	647.00
x 16mm	...	sqm	663.00	683.00	657.00	694.00	674.00
x 10mm, slotted	...	sqm	663.00	683.00	657.00	694.00	674.00
Pedimat surface mounted mat with abrasive, carpet or vinyl inserts and edge frame	...	sqm	450.00-550.00	450.00-550.00	450.00-550.00	450.00-550.00	450.00-550.00
ADD extra for :							
Recessed installation	...	sqm	50.00-75.00	50.00-75.00	50.00-75.00	50.00-75.00	50.00-75.00
Pedigrid mat system comprising treads with abrasive, carpet or vinyl inserts and supporting frame under	...	sqm	475.00-575.00	450.00-550.00	450.00-550.00	550.00-650.00	450.00-550.00
ADD extra for :							
Forming sinking or catch basin, gratings, traps and drainage	...	Note					
Ultramat closed construction internal entrance mat comprising alternate treads of aluminium and carpet inserts :							
x 10mm	...	sqm	443.00	459.00	438.00	470.00	459.00
x 16mm	...	sqm	527.00	545.00	522.00	556.00	543.00
x 10mm, slotted	...	sqm	527.00	545.00	522.00	556.00	543.00
DIVISION STRIPS, WEATHER BARS, refer page 431	...	Note					

MISCELLANEOUS METALWORK

Prices given hereafter are for the fabrication of miscellaneous and ornamental metalwork, NOT for structural work.	...	Note					
ALUMINIUM, small section or tubular, mill finish, based on supply cost of $15.00 to $19.00/kg and labour cost of $6.00 to $10.00/kg	...	kg	21.00-29.00	21.00-29.00	21.00-29.00	21.00-29.00	21.00-29.00
ADD extra for :							
Surface finish	...	Note					
Fixings	...	Note					
STAINLESS STEEL, small section or tubular, polished finish, based on supply cost of $14.50 to $25.50/kg and labour cost of $8.00 to $12.50/kg	...	kg	23.00-38.50	23.00-38.50	23.00-38.50	23.00-38.50	23.00-38.50
ADD extra for :							
Fixings	...	Note					
STEEL, small section angle or flat, unfinished, based on average supply cost of $2200/t and labour cost of $5.50 to $9.00/kg	...	kg	8.00-12.00	8.00-12.00	8.00-12.00	8.00-12.00	8.00-12.00
ADD extra for :							
Hollow section or tubular steel	...	kg	0.30	0.30	0.30	0.30	0.30
Surface finish	...	Note					
Fixings	...	Note					

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PAPER CUP DISPENSERS							
Stainless steel, satin finished paper cup dispenser, 597mm high, surface mounter, to suit cup 180-300ml (P.C. $140.00)	...	No	173.50	182.00	178.50	183.00	187.00
PERFORATED METAL CLADDING							
Perforated metal in flat panel cladding spot welded to support framing and including an allowance of 10% for waste :							
Mild steel sheet (2400 x 1200mm sheets) -							
x 1.00mm thick	...	sqm	116.50	123.50	120.00	124.50	128.50
x 1.60mm thick	...	sqm	135.50	143.50	140.50	144.00	149.50
x 3.00mm thick	...	sqm	248.00	259.00	253.00	260.00	263.00
ADD extra for :							
Waste in excess of 10%	...	Note					
Purpose made with non standard perforations, add to base supply cost	...	+60/80%					
Support framing	...	Note					
Electro-galvanising of panel and frame	...	Note					
Powder coat finish of panel and frame	...	Note					
Aluminium sheet - Price on application	...	Note					
SECURITY SCREENS							
Mesh security screen in anodised aluminium surround	...	sqm	133.00	131.00	134.00	131.00	134.00
SECURITY GRILLES							
Angle frame and welded mesh bolted in position, all galvanised	...	sqm	130.00	135.00	129.00	135.00	132.00
40 x 10mm steel flat surround with 16mm dia. bar infill at maximum 125mm centres, welded together and bolted in position, all galvanised	...	sqm	220.00	235.00	225.00	225.00	230.00
SHEET ETC. CLADDING							
EXTERNAL INSULATED CLADDING							
Proprietary metal clad panel, 100mm thick with a thermal rating of R2-6, comprising polystyrene core sandwiched between Colorbond® steel skins, fixed to steelwork (price based on 500/1000sqm)	...	sqm	103.00	103.00	102.00	103.00	105.00
ADD extra for :							
Areas of 1 to 100sqm	...	+/sqm	10.00	10.00	10.00	10.00	10.00
Areas of 100 to 500sqm	...	+/sqm	5.00	5.00	5.00	5.00	5.00
Steelwork frame	...	Note					

METALWORK

SHEET ETC. CLADDING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PRE-FINISHED CLADDING							
Chemically cured fluropolymer coating system (Vitraflow 700) on 9mm thick compressed fibre cement, fixed to framing or backing	...	sqm	250.00-310.00	250.00-310.00	250.00-310.00	250.00-310.00	250.00-310.00
ADD extra for :							
Sub framing	...	sqm	25.00	25.00	25.00	25.00	25.00
Unfinished 9mm thick fibre cement sheet proprietary cladding system with 10mm expressed vertical and horizontal joints screw fixed to proprietary framing including all trims, gaskets, sealing, etc.	...	sqm	160.00-195.00	160.00-195.00	160.00-195.00	160.00-195.00	160.00-195.00
SHEETMETAL CLADDING							
Sheet cladding to ducts, enclosures and the like, fixed to framing, in :							
0.80mm Zincanneal® steel sheet	...	sqm	109.00	125.00	123.50	127.50	149.50
1.00mm ditto	...	sqm	132.00	151.50	149.50	154.50	181.00
0.75mm galvanised sheet steel	...	sqm	127.50	146.00	144.50	149.00	174.50
0.95mm ditto	...	sqm	138.00	157.00	155.50	160.00	186.50
0.90mm, 302/304 grade stainless steel sheet	...	sqm	230.00	262.00	259.00	267.00	311.00
1.60mm sheet aluminium	...	sqm	236.00	263.00	260.00	268.00	303.00
ADD extra for :							
Baked enamel finish	...	sqm	9.30	9.40	9.05	9.05	9.50
CLADDING SYSTEMS							
Refer pages 297 to 300	...	Note					
EXTERNAL WALL CLADDING							
Refer page 414	...	Note					
PERFORATED METAL CLADDING							
Refer page 309	...	Note					
VANDAL RESISTANT BOARD							
Refer page 343	...	Note					

METALWORK

SHELVING AND STORAGE SYSTEMS

ADJUSTABLE SHELVING UNITS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fabricated from pre-coated sheet and having solid back and fixed top and bottom panels, all baked enamel finish, average prices	...	Note					
Unit with two adjustable shelves :							
300mm deep x 975mm high	...	m	306.00	319.00	339.00	320.00	350.00
500mm deep x 1775mm high	...	m	339.00	354.00	376.00	355.00	389.00
Unit with four adjustable shelves :							
300mm deep x 1775mm high	...	m	329.00	344.00	365.00	346.00	381.00
500mm deep x 1775mm high	...	m	393.00	411.00	437.00	413.00	454.00
ADD extra for :							
End panel -							
300mm deep x 975mm high	...	No	32.90	34.40	36.60	34.50	37.90
500mm deep x 975mm high	...	No	52.90	54.80	58.50	54.90	59.80
300mm deep x 1775mm high	...	No	52.20	54.20	57.80	54.40	59.30
500mm deep x 1775mm high	...	No	80.30	83.00	88.80	83.20	90.30
Vertical division panel -							
300mm deep x 975mm high	...	No	28.90	30.20	32.10	30.30	33.40
500mm deep x 975mm high	...	No	36.90	38.50	41.00	38.60	42.30
300mm deep x 1775mm high	...	No	44.60	46.50	49.50	46.60	51.00
500mm deep x 1775mm high	...	No	56.50	58.60	62.50	58.80	64.00
Additional adjustable shelf -							
300mm deep	...	m	24.40	25.70	27.20	25.80	28.50
500mm deep	...	m	32.20	33.60	35.80	33.80	37.10
Pair of doors -							
900mm wide x 975mm high	...	No	407.00	421.00	449.00	422.00	458.00
900mm wide x 1775mm high	...	No	481.00	498.00	532.00	499.00	542.00

LIBRARY SHELVING UNITS

Fabricated from mild steel sheet, all baked enamel finish	...	Note					
Single sided unit 250mm deep x 2135mm high with six intermediate shelves and canopy over :							
890mm long in one bay	...	No	590.00	609.00	651.00	610.00	662.00
1800mm long in two bays	...	No	848.00	876.00	936.00	878.00	953.00
Double sided unit 500mm deep x 2135mm high with six intermediate shelves and canopy over :							
890mm long in one bay	...	No	958.00	989.00	1060.00	991.00	1075.00
1800mm long in two bays	...	No	1505.00	1550.00	1660.00	1555.00	1680.00

STAINLESS STEEL SHELVES

Satin finished shelf 150mm wide x 20mm deep with brackets fixed to wall, shelf :							
450mm long	...	No	133.50	143.50	141.00	144.50	154.00
900mm long	...	No	198.00	210.00	205.00	211.00	218.00

METALWORK

SHELVING AND STORAGE SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
COMPACTUS FILE AND STORAGE SYSTEM							
Base and track system :							
1600 x 940mm	...	No	480.00	490.00	485.00	490.00	480.00
2560 x 940mm	...	No	635.00	635.00	630.00	645.00	640.00
Open shelf cabinets 900mm wide x 1800mm high with five intermediate shelves :							
300mm deep -							
Static	...	No	510.00	520.00	510.00	520.00	510.00
Mobile	...	No	540.00	550.00	540.00	550.00	540.00
400mm deep -							
Static	...	No	535.00	545.00	535.00	545.00	535.00
Mobile	...	No	565.00	575.00	565.00	575.00	565.00
ADD extra for :							
Mechanical operation	...	Per Bay	215.00	215.00	215.00	215.00	215.00
- (+ power)	...	Note					
DEXION SELECTA-FLO LIVE STORAGE CARTON PICKING SYSTEM							
Bays 2400mm deep x 2400mm high x 1530mm long with four shelf levels and five live storage lanes per level :							
Starter bay	...	Each	6100.00	6200.00	6100.00	6200.00	6100.00
Extension bay	...	Each	5800.00	5900.00	5800.00	5900.00	5800.00
ADD extra for :							
One additional shelf per bay with five live storage lanes	...	Each	1040.00	1060.00	1040.00	1060.00	1040.00
BROWNBUILT INDUSTRIAL STORAGE SHELVING							
Storage shelving unit 900 x 400 x 2175mm high with seven shelves per bay :							
Bay suitable for -							
Light applications	...	No	380.00	380.00	375.00	385.00	375.00
Moderate applications	...	No	550.00	555.00	555.00	560.00	550.00
PALLET RACKING							
Selection pallet rack unit overall size 2600mm wide x 1200mm deep x 4800mm high to accommodate eight one tonne pallets each 1100 x 1100 x up to 1200mm high	...	Each	900.00	920.00	900.00	920.00	900.00
STORLOK SHELVING							
Fabricated from steel frames and beams with powder coat finish	...	Note					
Unit with four MDF adjustable tiers/shelves :							
450mm deep -							
x 900 long x 1800 high	...	No	-	311.00	305.00	287.00	315.00
x 1200 long x 1800 high	...	No	-	360.00	352.00	332.00	363.00
x 1800 long x 1800 high	...	No	-	373.00	365.00	344.00	376.00
600mm deep -							
x 900 long x 1800 high	...	No	-	330.00	323.00	304.00	333.00
x 1200 long x 1800 high	...	No	-	398.00	389.00	366.00	400.00
x 1800 long x 1800 high	...	No	-	410.00	401.00	377.00	412.00

METALWORK

SHELVING AND STORAGE SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Unit with five MDF adjustable tiers/shelves :							
450mm deep -							
x 900 long x 2100 high	...	No	-	356.00	348.00	328.00	359.00
x 1200 long x 2100 high	...	No	-	417.00	408.00	384.00	419.00
x 1800 long x 2100 high	...	No	-	433.00	424.00	398.00	435.00
600mm deep -							
x 900 long x 2100 high	...	No	-	374.00	366.00	345.00	377.00
x 1200 long x 2100 high	...	No	-	463.00	453.00	426.00	464.00
x 1800 long x 2100 high	...	No	-	478.00	468.00	440.00	479.00
ADD extra for :							
Galvanised steel tiers/shelves (per each shelf) -							
450mm deep x 900 long	...	No	-	60.20	58.80	54.70	58.80
600mm deep x 900 long	...	No	-	65.40	63.90	59.50	63.90

SKIRTINGS

Versiclad finish steel skirting fixed to wall :							
100mm high	...	m	19.00	20.00	19.50	19.50	19.50
150mm high	...	m	21.00	22.00	21.50	21.50	21.50
Ducted skirtings, refer pages 543 and 544	...	Note					

STAIRCASES, BALUSTRADES, ETC.

STAIRCASES

Straight single flight of steel stairs 1000mm wide with open treads in diamond mesh on steel strings and supports to rise 3000mm, tubular balustrade to one side (medium duty)	...	No	5900.00	5500.00	5750.00	5600.00	5950.00
ADD extra for :							
Mid level landing	...	No	860.00	805.00	835.00	810.00	870.00
Spiral staircase to rise 3000mm with plate treads and simple balustrade -							
1300mm diameter	...	No	6100.00	5950.00	6100.00	5900.00	6050.00
1500mm diameter	...	No	6600.00	6250.00	6350.00	6100.00	6300.00
Concrete staircases, refer page 116	...	Note					
Steel framed staircases, heavy duty, refer pages 117	...	Note					
Timber staircases, refer pages 117, 118 and 352	...	Note					

METALWORK

STAIRCASES, BALUSTRADES, ETC. (cont'd)

BALUSTRADES AND HANDRAILS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for ramps and bends	...	Note					
GLAZED BALUSTRADES							
ADD extra to the following for crowd loading situations	...	Note					
The following balustrades are good standard architectural	...	Note					
Balustrade 1000mm high, post free comprising 6mm toughened glass set in channel at base and with approximately 60mm dia. handrail, in :							
Powder coated aluminium	...	m	850.00	770.00	805.00	870.00	830.00
Polished stainless steel	...	m	945.00	885.00	930.00	1000.00	935.00
ADD extra for :							
12mm tinted toughened glass	...	m	405.00	385.00	400.00	385.00	285.00
The following balustrades are good standard architectural	...	Note					
Balustrade 1000mm high comprising 6mm laminated glass with posts at 1200mm centres, in :							
Powder coated aluminium	...	m	545.00	480.00	520.00	535.00	505.00
Polished stainless steel	...	m	770.00	690.00	755.00	745.00	725.00
ADD extra for :							
10mm laminated glass	...	m	96.00	92.00	106.00	95.00	103.00
6mm tinted laminated glass	...	m	42.00	32.00	36.00	42.00	47.00
10mm tinted laminated glass	...	m	135.00	113.00	130.00	136.00	137.00

STEEL BALUSTRADES

Balustrade 900mm high comprising 32mm dia. tubular top and mid rails and standards at 1800mm centres with feet set in concrete -							
Zinc phosphate primed	...	m	310.00	280.00	310.00	325.00	310.00
Galvanised	...	m	320.00	290.00	330.00	335.00	320.00
Balustrade 900mm high comprising 50 x 9mm top and bottom rails filled in with 16 x 16mm balusters at 150mm centres and 25 x 25mm standards at 1800mm centres set in concrete -							
Zinc phosphate primed	...	m	455.00	400.00	450.00	450.00	450.00
Galvanised	...	m	480.00	430.00	485.00	470.00	480.00
Monowills industrial tubular balustrade comprising 25mm dia. knee-rail, 32mm dia. handrail and 40mm dia. stanchions at 1500mm centres and set in concrete, all galvanised	...	m	245.00	220.00	245.00	235.00	235.00
Stainless steel cable railing system 900mm high comprising 6mm stainless steel wires, 50mm dia. handrail, and balusters at 2000mm centres set in concrete	...	m	460.00-500.00	460.00-500.00	460.00-500.00	460.00-500.00	460.00-500.00

METALWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STAIRCASES, BALUSTRADES, ETC. (cont'd)							
BALUSTRADES AND HANDRAILS (cont'd)							
ALUMINIUM BALUSTRADES							
Powder coated aluminium framed balustrade 1000mm high comprising 60mm dia. handrail, RHS bottom rail, 50mm dia. stanchions at 1000mm centres set into concrete and filled in with 20mm dia. balustrades at 125mm centres	...	m	275.00	265.00	290.00	270.00	280.00
HANDRAILS							
Tubular handrail including brackets at 1200mm centres fixed to wall :							
Aluminium, clear anodised -							
60mm dia.	...	m	219.00	206.00	216.00	205.00	207.00
75mm dia.	...	m	233.00	219.00	233.00	220.00	221.00
Aluminium, colour anodised -							
60mm dia.	...	m	211.00	222.00	212.00	203.00	205.00
75mm dia.	...	m	223.00	233.00	226.00	215.00	217.00
Brass, polished finish -							
50mm dia.	...	m	295.00	295.00	295.00	280.00	285.00
63.5mm dia.	...	m	305.00	310.00	305.00	310.00	295.00
Stainless steel, polished finish -							
50.8mm dia.	...	m	225.00	230.00	235.00	220.00	225.00
63.6mm dia.	...	m	235.00	240.00	250.00	240.00	240.00
76.2mm dia.	...	m	275.00	285.00	290.00	280.00	280.00
101.6mm dia.	...	m	365.00	370.00	375.00	375.00	370.00
Steel, 32mm dia. -							
Zinc chromate primed	...	m	100.00	101.00	97.00	122.00	93.00
Galvanised	...	m	106.00	106.00	102.00	132.00	98.00
Flat handrail including brackets at 1200mm centres fixed to wall, steel flat 50 x 10mm :							
Zinc chromate primed	...	m	89.00	88.00	87.00	117.00	85.00
Galvanised	...	m	93.00	93.00	90.00	127.00	86.00
SUPPORT FRAMING GENERALLY							
Small section steel angle framing in duct enclosures, tank stands, bulkheads, etc., with bolted or welded joints	...	kg	11.50-17.00	12.50-18.00	12.00-17.50	11.00-16.50	12.00-17.50
TOILET ACCESSORIES							
TOWEL RAILS							
STAINLESS STEEL, satin finished 25mm dia. x 1mm tube with flanged ends fixed to brick wall :							
600mm long	...	No	98.20	106.00	104.00	107.00	115.00

METALWORK

TOILET ACCESSORIES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TOWEL RINGS							
STAINLESS STEEL, bright polished 6mm dia. bar in 150mm diameter round ring with mounting plate fixed to brick wall	...	No	82.00	88.50	87.00	89.50	95.50
GRAB RAILS							
STAINLESS STEEL, satin finished with flanged ends fixed to tile brick wall :							
38mm dia. x 1mm tube in straight grab rail -							
300mm long	...	No	71.00	78.10	76.90	79.10	87.70
600mm long	...	No	82.40	89.90	88.50	91.00	99.20
900mm long	...	No	88.70	96.30	94.80	97.40	105.50
1200mm long	...	No	105.50	113.50	111.50	114.50	122.50
STAINLESS STEEL, satin finished handicapped persons grab rail of varying configurations with average three wall or floor fixing points (average) :							
32mm dia. x 1mm tube	...	m	165.00	173.50	170.00	174.50	178.50
38mm dia. x 1mm tube	...	m	247.00	257.00	252.00	258.00	260.00
ADD extra for :							
Non-slip gripping surface to 32 or 38mm dia. tube	...	m	42.00	43.00	42.00	43.00	42.00
SHOWER CURTAIN RAILS							
STAINLESS STEEL, satin finished 32mm dia. x 1mm tube with flanged ends fixed to brick wall :							
1000mm wide	...	No	130.00	139.00	136.00	140.00	147.00
1200mm wide	...	No	146.50	156.00	153.00	157.00	165.00
2000mm girth	...	No	203.50	215.00	210.50	216.00	223.50
STAINLESS STEEL shower curtain hook to suit 32mm dia. rail	...	No	2.00	2.05	2.00	2.05	2.00
PAPER TOWEL DISPENSERS							
Built in models :							
320mm wide x 675mm high x 100mm deep, for interfold towels, stainless steel - (P.C. $465.00)	...	No	520.00	540.00	525.00	540.00	540.00
Surface mounted models :							
256mm wide x 360mm high x 270mm deep, for roll towels, plastic - (P.C. $ 85.00)	...	No	122.50	130.50	128.50	132.00	139.00
260mm wide x 393mm high x 95mm deep, for interleaved towels -							
Stainless steel - (P.C. $180.00)	...	No	222.00	233.00	228.00	234.00	239.00
White epoxy - (P.C. $ 47.50)	...	No	83.00	90.50	89.00	91.50	100.00
265mm wide x 390mm high x 72mm deep, for unfolded sheets -							
Plastic - (P.C. $ 28.00)	...	No	62.50	69.50	68.50	70.50	79.50
Stainless steel - (P.C. $170.00)	...	No	212.00	222.00	218.00	223.00	228.00
White epoxy - (P.C. $ 45.00)	...	No	80.50	87.50	86.50	89.00	97.00

METALWORK

TOILET ACCESSORIES (cont'd)

PAPER TOWEL DISPENSERS (cont'd)

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Surface mounted models (cont'd) :						
267mm wide x 285mm high x 148mm deep, for interfold towels, white enamel - (P.C. $ 150.00) ...	No	190.50	200.50	196.50	201.50	207.50
282mm wide x 514mm high x 104mm deep, for interleaved towels, plastic - (P.C. $ 99.00) ...	No	137.00	146.00	143.00	147.00	154.00

PAPER TOWEL/WASTE CABINETS

All combined units with concealed used towel receptacles ...	Note					
330mm wide x 1660mm high x 184mm deep, roll or interleaved dispenser, stainless steel - (P.C. $1095.00) ...	No	1225.00	1265.00	1235.00	1265.00	1260.00
Built in models :						
330mm wide x 1070mm high x 184mm deep, interleaved or roll dispenser, stainless steel - (P.C. $ 595.00) ...	No	700.00	725.00	710.00	730.00	735.00
435mm wide x 1420mm high x 208mm deep, interleaved dispenser, stainless steel - (P.C. $1440.00) ...	No	1585.00	1635.00	1600.00	1635.00	1620.00
Semi-recessed models :						
437mm wide x 1422mm high x 207mm deep, interleaved dispenser, stainless steel - (P.C. $1990.00) ...	No	2155.00	2220.00	2170.00	2220.00	2190.00
437mm wide x 1422mm high x 230mm deep, roll dispenser, stainless steel - (P.C. $2300.00) ...	No	2480.00	2550.00	2495.00	2555.00	2515.00
Set into bench top model :						
325mm wide x 150mm high, front to back x 435mm deep, interleaved dispenser, stainless steel - (P.C. $ 670.00) ...	No	755.00	785.00	765.00	785.00	785.00
ADD extra for :						
Housing below bench top to conceal dispenser and waste receptacle ...	Note					

TOILET TISSUE DISPENSERS

Built in models :						
189mm wide x 306mm high x 143mm deep, multi roll - (P.C. $ 185.00) ...	No	220.00	230.00	224.00	231.00	234.00
Surface mounted models :						
115mm wide x 315mm high x 128mm deep, multi roll dispenser -						
Stainless steel - (P.C. $ 115.00) ...	No	143.00	151.00	147.00	152.00	155.00
White epoxy - (P.C. $ 77.00) ...	No	104.00	110.00	107.00	111.00	116.00
115mm wide x 410mm high x 128mm deep, multi roll dispenser -						
Stainless steel - (P.C. $ 131.00) ...	No	160.00	168.00	164.00	169.00	172.00

METALWORK

TOILET ACCESSORIES (cont'd)

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TOILET TISSUE DISPENSERS (cont'd)						
Surface mounted models (cont'd) :						
130mm wide x 230mm high x 76mm deep, tissue cabinet, stainless steel - (P.C. $ 210.00) ...	No	244.00	253.00	248.00	254.00	255.00
145mm wide x 180mm high x 110mm deep, interleaved tissue, white plastic - (P.C. $ 46.00) ...	No	71.50	77.00	75.50	78.00	83.00
272mm wide x 280mm high x 122mm deep, jumbo tissue, plastic - (P.C. $ 54.00) ...	No	80.00	85.50	84.00	86.50	91.50
TOILET ROLL HOLDERS						
Surface mounted models :						
152mm wide x 310mm high x 145mm deep, two rolls, plastic - (P.C. $ 25.00) ...	No	44.50	52.50	51.50	49.50	54.50
156mm wide x 156mm high x 75mm deep, single roll, stainless steel with chrome plated roller - (P.C. $ 65.00) ...	No	94.00	98.50	100.00	97.00	100.50
Ditto 313mm wide for two rolls - (P.C. $ 95.00) ...	No	123.00	131.00	132.00	121.00	135.00
164mm wide x 114mm high x 95mm deep, single roll, plastic - (P.C. $ 17.00) ...	No	35.50	42.50	40.50	39.50	41.50
FACIAL TISSUE DISPENSERS						
Built in models :						
282mm wide x 130mm high x 60mm deep, interleaved tissues, with face plate, plastic - (P.C. $ 15.00) ...	No	31.00	36.00	34.00	34.00	36.50
Surface mounted models :						
260mm wide x 132mm high x 76mm deep, interleaved tissue, stainless steel - (P.C. $ 95.00) ...	No	124.00	129.00	132.00	131.00	135.00
266mm wide x 130mm high x 60mm deep, interleaved tissues, stainless steel - (P.C. $ 52.00) ...	No	76.50	83.00	83.50	80.50	82.50
HAND CLEANSERS						
Surface mounted models :						
Plastic dispenser -						
1.5 litre capacity - (P.C. $110.00) ...	No	135.00	142.00	139.00	143.00	145.00
800ml pod capacity - (P.C. $ 52.00) ...	No	74.50	79.50	78.00	80.00	84.50
SOAP DISPENSERS						
Glass container, liquid soap pump dispenser, satin chrome fixtures, surface mounted - (P.C. $105.00) ...	No	133.00	140.00	138.00	141.00	145.00
Stainless steel dispenser with push in valve, built in - (P.C. $ 95.00) ...	No	126.00	134.00	131.00	134.00	140.00

METALWORK

TOILET ACCESSORIES (cont'd)

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DISPOSAL BASKETS						
425mm x 337mm x 616mm high, freestanding, collapsible, white plastic coated wire - (P.C. $ 46.00) ...	No	55.00	55.50	53.00	54.50	54.50
DISPOSAL BINS						
340mm x 340mm x 750mm high, freestanding, top with self closing doors, stainless steel - (P.C. $790.00) ...	No	835.00	655.00	640.00	655.00	640.00
420mm x 330mm x 595mm high, freestanding, removable top, white plastic - (P.C. $ 28.00) ...	No	36.00	55.00	53.50	55.00	55.00
HAND DRYERS						
Hand dryer, 240 volt, enamel finish, surface mounted, including electrical connection :						
Push button model - (P.C. $500.00) ...	No	675.00	715.00	700.00	720.00	750.00
Automatic model - (P.C. $485.00) ...	No	660.00	700.00	685.00	705.00	735.00
Automatic ducted model -						
Polycarbonate - (P.C. $450.00) ...	No	620.00	660.00	650.00	665.00	695.00
Stainless steel - (P.C. $650.00) ...	No	830.00	875.00	860.00	880.00	905.00
Vitreous china - (P.C. $550.00) ...	No	725.00	770.00	755.00	775.00	800.00
BABY CHANGE STATION						
High impact plastic, pneumatic shock operation, nappy liner dispenser, surface mounted - (P.C. $650.00) ...	No	855.00	890.00	870.00	895.00	895.00

WALL PROTECTION

Prices are based on the Acrovyn system ...	Note					
Bumper guard (BG-10), fixed to wall ...	m	106.00	116.50	115.00	118.00	131.00
ADD extra for :						
End cap ...	No	14.60	16.30	16.10	16.60	18.80
Corner guard (SM-20), fixed to plastered wall ...	m	111.00	123.50	121.50	125.00	140.50
Handrail (HR-6C), fixed to plastered wall ...	m	93.00	101.50	99.50	102.50	111.50
ADD extra for :						
End cap ...	No	20.00	22.50	22.00	23.00	26.00
Corner cap ...	No	24.00	26.50	26.50	27.00	30.00
Wall covering, refer page 454 ...	Note					

METALWORK

WINDOW CLEANING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Unistrut P.1000 galvanised concrete insert channel with end stops cast into soffit of concrete beam	...	m	39.50	44.00	43.50	45.00	51.00
G.K.N. removable Saf-T-Trolley	...	No	222.00	228.00	223.00	228.00	225.00
Anchoring point for harness fixed to concrete :							
Galvanised	...	No	55.50	60.50	59.50	61.00	66.50
Plastic coated	...	No	61.50	67.00	65.50	67.50	73.00
Stainless steel	...	No	138.00	145.00	142.00	146.00	149.00

ROOF SAFETY SYSTEMS

Static line roof safety system comprising stanchions, together with flashings to roof, stainless steel life line, shock absorbers and complete with all necessary accessories, harnesses, etc. (Cost range - per metre of ridge)	...	m	360.00-470.00	360.00-470.00	350.00-460.00	360.00-470.00	350.00-460.00
Saferidge full arrest system fully installed (Cost range - per metre of ridge)	...	m	360.00-470.00	360.00-470.00	350.00-470.00	360.00-470.00	350.00-460.00

CARPENTRY

BASE MATERIAL PRICES

The following tables indicate the comparative supply prices of commonly used timber sizes in standard lengths and of general construction grade.

Sections not listed can be assessed by interpolation.

Average trade (SUPPLY ONLY) prices :

ADELAIDE

HARDWOOD		LOCAL PINE				OREGON		
				Untreated	Pressure Treated			
Nominal Size	Dry Sawn F17	Sawn Size	Nominal Green Dressed Size	Dry Dressed MGP10	Dry Sawn MGP10	Nominal Size	Merchant Sawn F5	Select Merchant Dressed F7
	$/m			$/m	$/m		$/m	$/m
50 x 25	2.44	50 x 26	42 x 19	1.50	2.08	50 x 25	-	-
75 x 25	3.66	81 x 26	70 x 19	1.80	2.98	75 x 25	-	-
100 x 25	4.88	100 x 26	90 x 19	2.35	3.70	100 x 25	-	-
50 x 38	3.71	50 x 42	42 x 35	1.35	2.50	50 x 38	-	-
100 x 38	7.41	100 x 42	90 x 35	2.00	4.00	100 x 38	-	-
150 x 38	11.12	153 x 42	140 x 35	3.75	6.10	150 x 38	-	-
75 x 50	7.31	81 x 53	70 x 45	2.60	4.00	75 x 50	-	-
100 x 50	9.75	100 x 53	90 x 45	2.60	5.10	100 x 50	-	-
150 x 50	14.63	153 x 53	140 x 45	5.00	7.60	150 x 50	-	-
200 x 50	19.50	206 x 53	190 x 45	7.30	10.70	200 x 50	-	-
100 x 75	14.63	100 x 81	90 x 70	5.25	9.90	100 x 75	-	-
150 x 75	24.94	132 x 81	120 x 70	-	12.08	150 x 75	-	-
100 x 100	19.50	100 x 100	90 x 90	12.60	14.70	100 x 100	-	-

BRISBANE

LOCAL HARDWOOD			LOCAL PINE						OREGON		
					Untreated		Pressure Treated				
Nom. Size	Green Sawn F14	Green Dressed	Nom Green Sawn Size	Dry Dressed Size	Sawn MGP10	Sawn MGP12	Dry Sawn	Dry Dressed	Nom. Size	Sawn Merch. F5	Sawn Merch. F7
	$/m				$/m	$/m	$/m	$/m		$/m	$/m
50 x 25	2.62	-	50 x 25	42 x 19	-	1.44	1.52	1.97	50 x 25	-	-
75 x 25	4.25	-	75 x 25	68 x 19	-	.	2.22	2.98	75 x 25	-	-
100 x 25	5.65	-	100 x 25	93 x 19	-	2.88	3.03	3.99	100 x 25	-	-
50 x 38	3.90	-	50 x 38	42 x 35	-	2.22	2.22	2.73	50 x 38	-	-
100 x 38	7.76	-	100 x 38	90 x 35	-	4.37	4.14	3.94	100 x 38	-	-
150 x 38	13.43	-	150 x 38	140 x 35	-	7.15	5.05	5.96	150 x 38	-	-
75 x 50	7.81	-	75 x 50	70 x 45	-	4.31	3.74	3.94	75 x 50	-	-
100 x 50	10.10	-	100 x 50	90 x 45	-	5.76	5.25	5.20	100 x 50	-	-
150 x 50	16.48	-	150 x 50	140 x 45	-	9.42	6.57	7.83	150 x 50	-	-
200 x 50	23.95	-	200 x 50	190 x 45	-	13.85	9.29	10.40	200 x 50	-	-
100 x 75	19.76	-	100 x 75	90 x 70	-	8.64	9.77	15.40	100 x 75	-	-
150 x 75	28.05	-	150 x 75	120 x 70	-	14.13	15.97	29.30	150 x 75	-	-
100 x 100	26.19	-	100 x 100	90 x 90	-	11.50	10.05	11.06	100 x 100	-	-

CARPENTRY

BASE MATERIAL PRICES (cont'd)

MELBOURNE

HARDWOOD			LOCAL PINE					OREGON		
					Untreated		Pressure Treated			
Nom. Size	Green Sawn F8	K/Dry Sawn F17	Sawn Size	Dry Dressed Size	Sawn Struct MGP10	Dry Sawn	Dry Sawn	Nom. Size	Sawn F7	Select Merch. Dressed F7
	$/m				$/m	$/m	$/m		$/m	$/m
50 x 25	1.32	-	50 x 26	42 x 19	-	1.10	-	50 x 25	1.43	1.86
75 x 25	2.35	-	81 x 26	70 x 19	-	1.65	-	75 x 25	2.09	2.75
100 x 25	2.65	-	100 x 26	90 x 19	-	2.20	-	100 x 25	2.86	3.67
50 x 38	2.04	-	50 x 42	42 x 35	1.94	-	-	50 x 38	2.16	2.75
100 x 38	4.00	5.10	100 x 42	90 x 35	2.32	-	-	100 x 38	4.34	5.50
150 x 38	5.90	8.15	153 x 42	140 x 35	6.42	5.40	-	150 x 38	6.50	8.16
75 x 50	3.90	7.00	81 x 53	70 x 45	3.06	3.80	-	75 x 50	4.28	5.50
100 x 50	5.30	7.15	100 x 53	90 x 45	3.16	4.85	-	100 x 50	5.40	7.05
150 x 50	7.80	10.70	153 x 53	140 x 45	6.83	7.15	-	150 x 50	7.40	7.65
200 x 50	11.50	14.75	206 x 53	190 x 45	9.59	9.50	-	200 x 50	11.73	14.80
100 x 75	8.15	-	100 x 81	90 x 70	5.81	-	-	100 x 75	10.52	13.05
150 x 75	11.80	-	132 x 81	120 x 70	8.72	-	-	150 x 75	12.85	16.05
100 x 100	10.70	-	100 x 100	90 x 90	10.20	-	-	100 x 100	12.65	15.81

PERTH

HARDWOOD		LOCAL PINE						OREGON		
				Untreated			Treated			
Nom. Size	Green Sawn F8	Nom Green Sawn Size	Dry Dressed Size	Dry Sawn Std. Grade F7	Dry Sawn MGP10	Dry Dressed Std. Grade	Dry Sawn MGP10	Nom. Size	Unseason Struct Grade Sawn	Seasoned C and B Dressed
	$/m				$/m	$/m	$/m	$/m		$/m
50 x 25	1.87	50 x 25	42 x 19	-	0.99	1.10	-	50 x 25	-	7.03
75 x 25	2.74	75 x 25	70 x 19	-	1.45	2.30	-	75 x 25	-	12.28
100 x 25	3.72	100 x 25	90 x 19	-	1.79	1.97	2.72	100 x 25	-	15.44
50 x 38	2.74	50 x 38	42 x 35	1.86	-	2.19	-	50 x 38	-	-
100 x 38	5.97	100 x 38	90 x 35	4.38	2.91	-	3.55	100 x 38	-	-
150 x 38	9.97	150 x 38	140 x 35	8.16	5.38	-	4.22	150 x 38	14.03	-
75 x 50	5.77	75 x 50	70 x 45	4.95	3.21	-	4.16	75 x 50	9.75	22.72
100 x 50	9.26	100 x 50	90 x 45	5.81	4.77	-	5.06	100 x 50	12.96	30.01
150 x 50	13.89	150 x 50	140 x 45	9.47	7.89	-	8.03	150 x 50	21.81	44.95
200 x 50	18.52	200 x 50	190 x 45	12.50	9.91	-	11.28	200 x 50	25.99	59.96
100 x 75	11.59	100 x 75	90 x 70	10.60	9.15	-	-	100 x 75	-	-
150 x 75	16.59	150 x 75	120 x 70	12.68	-	-	-	150 x 75	-	-
100 x 100	17.76	100 x 100	90 x 90	17.15	15.25	-	16.37	100 x 100	24.71	-

CARPENTRY

BASE MATERIAL PRICES (cont'd)

SYDNEY

HARDWOOD			LOCAL PINE						OREGON		
					Untreated		Pressure Treated				
Nom. Size	Green Sawn F11	Dry Sawn F17	Nom. Green Sawn Size	Dry Dressed Size	K/Dry Sawn F5/MGP10	Dry Dressed	K/Dry Sawn F7	Dry Dressed	Nom. Size	Merch. Sawn F5	Selected Merch. Dressed F7
	$/m	$/m			$/m	$/m	$/m	$/m		$/m	$/m
50 x 25	2.05	-	50 x 26	42 x 19	-	1.32	1.68	1.85	50 x 25	1.77	2.10
75 x 25	-	-	81 x 26	70 x 19	2.18	1.85	-	2.86	75 x 25	2.05	2.32
100 x 25	4.67	-	100 x 26	90 x 19	-	3.84	-	2.97	100 x 25	2.14	2.82
50 x 38	3.42	-	50 x 42	42 x 35	1.89	1.94	-	2.49	50 x 38	1.95	2.28
100 x 38	7.25	-	100 x 42	90 x 35	3.11	3.84	4.37	4.07	100 x 38	3.59	4.42
150 x 38	10.65	-	153 x 42	140 x 35	-	6.07	6.12	5.52	150 x 38	5.68	6.16
75 x 50	6.81	-	81 x 53	70 x 45	2.85	2.89	4.18	3.67	75 x 50	2.42	3.64
100 x 50	8.53	-	100 x 53	90 x 45	3.60	3.70	5.11	4.38	100 x 50	2.97	4.57
150 x 50	15.45	-	153 x 53	140 x 45	6.91	6.07	7.49	7.31	150 x 50	5.32	5.42
200 x 50	22.32	-	206 x 53	190 x 45	7.08	7.51	10.13	11.74	200 x 50	7.38	7.71
100 x 75	17.35	-	100 x 81	90 x 70	-	11.11	13.65	11.74	100 x 75	5.80	6.23
150 x 75	28.42	-	132 x 81	120 x 70	-	13.96	15.81	18.43	150 x 75	12.68	15.21
100 x 100	-	-	100 x 100	90 x 90	-	15.47	18.50	15.65	100 x 100	12.50	15.00

CARPENTRY

STRUCTURAL TIMBERS - UNIT PRICES

Prices for Carpentry timbers are based on :

	FLOOR FRAMING	WALL FRAMING	ROOF AND CEILING FRAMING
ADELAIDE	PINE	PINE	PINE
BRISBANE	LOCAL HARDWOOD	PINE	LOCAL HARDWOOD
MELBOURNE	LOCAL HARDWOOD	PINE	PINE
PERTH	LOCAL HARDWOOD	PINE	LOCAL HARDWOOD/PINE
SYDNEY	NORTHERN HARDWOOD/PINE	PINE	PINE

		UNIT	Adel. $ Treated Pine MGP10	Bris. $ Loc H'wd F14	Mel. $ H'wd F17	Per. $ Loc H'wd F8	Syd. $ Nth H'wd F11/14
GROUND FLOOR FRAMING							
Plates :							
100 x 25mm	...	m	8.40	11.70	8.45	9.35	12.25
100 x 38mm	...	m	9.40	14.90	11.15	12.70	16.60
100 x 50mm	...	m	11.95	19.20	14.15	18.00	20.10
Bearers, beams :							
100 x 75mm	...	m	21.60	35.40	22.80	25.80	36.90
125 x 75mm	...	m	-	43.50	27.60	30.60	46.70
150 x 75mm	...	m	27.00	48.50	30.90	35.00	54.40
Joists :							Pine Treated F7
100 x 50mm	...	m	12.60	20.00	14.95	18.80	20.10
125 x 50mm	...	m	15.70	25.50	18.45	22.40	25.90
150 x 50mm	...	m	18.40	30.90	21.70	27.60	32.30
200 x 50mm	...	m	24.80	43.00	29.70	31.20	44.50
250 x 50mm	...	m	-	55.20	35.40	39.60	23.40
UPPER FLOOR FRAMING							
Joists :							
150 x 50mm	...	m	19.35	32.20	22.80	28.80	33.80
200 x 50mm	...	m	26.10	44.70	31.30	32.80	46.50
250 x 50mm	...	m	-	57.30	37.30	41.60	25.90
Blocking :							
150 x 50mm	...	m	20.70	33.80	24.80	30.40	35.80
200 x 50mm	...	m	27.40	46.30	33.30	34.40	48.50
250 x 50mm	...	m	-	58.90	39.30	43.20	27.90
Timber engineered joist :							
200 x 47mm	...	m	22.70	23.40	23.70	25.80	26.50
240 x 57mm	...	m	26.40	27.00	27.60	29.80	30.70
300 x 63mm	...	m	27.10	27.90	28.40	30.80	31.80
360 x 63mm	...	m	34.40	34.70	35.20	38.70	38.80

CARPENTRY

STRUCTURAL TIMBERS - UNIT PRICES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
			Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10
WALL FRAMING							
Plates :							
75 x 38mm	...	m	9.75	11.95	10.80	10.75	14.20
75 x 50mm	...	m	13.35	16.00	15.15	14.35	18.20
100 x 38mm	...	m	13.35	16.05	14.35	14.05	19.30
100 x 50mm	...	m	16.55	20.10	18.00	18.55	22.60
Studs :							
75 x 38mm	...	m	9.10	11.15	10.00	9.95	13.20
75 x 50mm	...	m	11.70	13.95	12.80	12.40	15.20
100 x 38mm	...	m	11.70	14.00	12.00	12.05	16.30
100 x 50mm	...	m	13.95	16.80	14.50	15.35	18.10
100 x 75mm	...	m	21.60	22.90	20.60	23.00	30.40
100 x 100mm	...	m	29.60	29.50	28.70	33.10	39.30
Noggings :							
75 x 38mm	...	m	9.75	11.95	10.80	10.75	14.20
75 x 50mm	...	m	12.35	14.75	14.00	13.20	16.70
100 x 38mm	...	m	12.35	14.85	13.15	12.85	17.80
100 x 50mm	...	m	15.25	18.45	16.05	16.95	20.10
Lintels :							
150 x 50mm	...	m	24.30	30.40	28.40	28.00	33.30
200 x 50mm	...	m	31.70	40.40	36.60	35.10	41.40
150 x 75mm	...	m	33.90	41.50	-	-	49.60
Bracing :							
5mm thick fibre cement	...	sqm	32.90	38.50	39.40	37.60	47.10
6.4mm thick hardboard	...	sqm	40.90	46.50	47.40	45.60	55.20
Galvanised steel strap	...	m	6.15	7.55	7.95	7.30	9.90
ROOF AND CEILING FRAMING			Pine MGP10	Loc H'wd F14	Pine MGP10	Loc H'wd F8	Pine MGP10
Plates :							
75 x 38mm	...	m	8.80	14.50	9.60	12.20	12.70
75 x 50mm	...	m	11.40	18.30	12.45	15.65	14.70
100 x 38mm	...	m	11.40	18.25	11.60	15.90	15.80
100 x 50mm	...	m	13.95	22.90	14.50	21.60	18.10
Rafters :							
100 x 38mm	...	m	10.40	17.00	10.45	14.70	14.30
125 x 38mm	...	m	12.25	21.00	13.50	17.75	16.55
150 x 38mm	...	m	14.75	26.30	17.75	22.00	20.30
100 x 50mm	...	m	13.30	22.10	13.70	20.80	17.10
125 x 50mm	...	m	17.35	28.80	18.50	25.60	16.45
150 x 50mm	...	m	19.70	34.20	22.50	30.80	25.80
200 x 50mm	...	m	26.50	46.80	29.50	34.80	32.40
250 x 50mm	...	m	-	58.90	-	43.50	28.90
Purlins :							
100 x 50mm	...	m	14.60	23.80	15.30	22.40	19.10
150 x 50mm	...	m	20.40	35.10	23.30	31.60	26.80
200 x 50mm	...	m	25.10	45.10	28.00	33.20	30.40
100 x 75mm	...	m	22.90	38.70	22.20	29.00	32.40
150 x 75mm	...	m	29.60	53.80	30.90	40.20	42.60

CARPENTRY

STRUCTURAL TIMBERS - UNIT PRICES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
			Pine MGP10	Loc H'wd F14	Pine MGP10	Loc H'wd F8	Pine MGP10
ROOF AND CEILING FRAMING (cont'd)							
Ridges :							
150 x 25mm	...	m	11.70	19.85	14.65	17.15	11.95
200 x 25mm	...	m	16.10	26.70	18.25	21.70	14.45
150 x 38mm	...	m	16.05	27.90	19.35	23.60	22.30
200 x 38mm	...	m	-	38.60	24.70	31.40	27.90
Hips, valley boards :							
200 x 25mm	...	m	16.10	26.70	18.25	21.70	14.45
200 x 38mm	...	m	-	38.60	24.70	31.40	27.90
Struts and ties :							
75 x 50mm	...	m	11.60	18.50	12.75	15.90	15.15
100 x 50mm	...	m	15.25	24.60	16.05	23.20	20.10
100 x 75mm	...	m	24.20	40.40	23.70	30.60	34.40
						Pine MGP10	
Ceiling joists :							
100 x 38mm	...	m	11.40	18.25	11.60	11.65	15.80
125 x 38mm	...	m	12.90	21.80	14.30	13.45	17.55
100 x 50mm	...	m	13.30	22.10	13.70	14.55	17.10
125 x 50mm	...	m	16.70	28.00	17.75	22.30	15.45
150 x 50mm	...	m	19.05	33.40	21.80	21.60	24.80
200 x 50mm	...	m	25.80	45.90	28.80	27.90	31.40
Hangers :							
200 x 38mm	...	m	-	37.40	23.50	25.10	26.40
200 x 50mm	...	m	25.80	45.90	28.80	27.90	31.40
250 x 38mm	...	m	-	49.70	25.90	33.70	32.50
250 x 50mm	...	m	-	58.10	-	46.90	27.90
Gutter boarding :							
18mm waterproof plywood -							
Fixed to timber	...	sqm	77.10	83.50	85.20	86.20	97.00
On and including metal straps fixed to timber	...	sqm	85.50	91.90	89.60	94.60	100.40

PERGOLA FRAMING

Posts :	Refer page 326	...	Note
Beams :	Refer page 327	...	Note
Joists :	Refer page 329	...	Note
Battens :	Refer page 329	...	Note
Fixings :	Post anchor base and bolts, refer page 347	...	Note

POSTS

100 x 100 post -									
Oregon	- sawn	- F 5	...	m	-	-	37.70	49.30	44.00
	- dressed	- F 7	...	m	-	-	41.20	-	46.80
Hardwood	- sawn	- F 8	...	m	-	-	37.80	43.90	-
		- F11/14	...	m	36.20	-	-	-	32.90
		- F14	...	m	-	54.20	-	-	-
		- F17 (K/D)	...	m	-	-	-	-	-
	- dressed	- (green)	...	m	41.90	-	-	-	-

CARPENTRY

STRUCTURAL TIMBERS - UNIT PRICES (cont'd)

BEAMS

Laminated Timber Beams, refer pages 330 to 332 ... Note

			UNIT	125 x 50	150 x 50	200 x 50	250 x 50	300 x 50	200 x 75	300 x 75
				$	$	$	$	$	$	$
ADELAIDE										
Oregon :										
Sawn	- F 5 (unseasoned)	...	m	-	-	-	-	-	-	-
Dressed	- F 7 (unseasoned)	...	m	-	-	-	-	-	-	-
Hardwood :										
Sawn	- F 17	...	m	26.00	30.70	41.00	51.10	60.30	61.60	88.10
BRISBANE										
Oregon :										
Sawn	- F 7	...	m	30.60	34.00	44.90	56.60	-	67.80	-
Hardwood :										
Sawn	- F 14 (green)	...	m	31.60	37.10	52.80	73.70	84.80	82.70	124.30
Dressed	- F 14 (green)	...	m	33.50	39.50	55.80	80.10	85.90	87.30	133.70
MELBOURNE										
Oregon :										
Sawn	- F 7 (unseasoned)	...	m	20.30	23.80	32.40	41.30	49.80	-	-
Dressed	- F 7 (unseasoned)	...	m	22.50	26.30	37.60	48.90	59.50	-	-
Hardwood :										
Sawn	- F 17 (dry)	...	m	23.30	27.40	38.20	48.20	-	-	-
PERTH										
Oregon :										
Sawn	- Struct. Grade (unseasoned)	...	m	30.00	35.60	47.40	59.60	69.70	71.30	96.10
Dressed	- C and B (seasoned)	...	m	50.50	60.40	83.10	104.20	123.50	125.20	182.90
Hardwood :										
Sawn	- F 8 (green)	...	m	26.30	30.90	41.40	55.80	65.60	56.50	79.20
Dressed	- Paint grade (dry)	...	m	30.60	36.00	48.20	60.00	70.30	-	-
SYDNEY										
Oregon :										
Sawn	- F 7 (unseasoned)	...	m	25.00	28.40	37.50	48.60	58.00	66.20	91.80
Sawn	- Unseasoned select merch	...	m	25.80	29.20	38.60	50.40	60.10	69.00	97.00
Hardwood :										
Sawn	- F 11 (green)	...	m	32.40	34.50	49.60	65.50	71.00	72.30	102.70

CARPENTRY

STRUCTURAL TIMBERS - COMPOSITE PRICES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
			H'wd F17	Loc H'wd F14	H'wd F17	Loc H'wd F8	Nth H'wd F11/14
GROUND FLOOR FRAMING							
Bearers :							
100 x 75mm	@ 1800mm centres ...	sqm	11.90	19.50	12.55	14.20	20.30
	@ 2400mm centres ...	sqm	9.50	15.60	10.05	11.35	16.25
125 x 75mm	@ 1800mm centres ...	sqm	-	23.90	15.20	16.85	25.70
	@ 2400mm centres ...	sqm	-	19.10	12.15	13.50	20.50
150 x 75mm	@ 1800mm centres ...	sqm	14.85	26.70	16.95	19.25	29.90
	@ 2400mm centres ...	sqm	11.90	21.30	13.60	15.40	23.90
							Pine Treated F7
Joists :							
100 x 50mm	@ 450mm centres ...	sqm	34.50	54.70	40.80	51.30	54.80
	@ 600mm centres ...	sqm	26.50	42.10	31.40	39.40	42.10
150 x 50mm	@ 600mm centres ...	sqm	38.60	64.90	45.50	57.90	67.90
200 x 50mm	@ 600mm centres ...	sqm	52.10	90.40	62.50	65.50	93.50
Steel floor framing, refer page 332	...	Note					
UPPER FLOOR FRAMING							
Joists including blocking at 1800mm centres :							
150 x 50mm	@ 450mm centres ...	sqm	68.40	113.00	81.00	101.50	119.00
	@ 600mm centres ...	sqm	56.20	92.90	66.60	83.20	97.90
200 x 50mm	@ 450mm centres ...	sqm	91.90	156.50	110.50	115.00	163.50
	@ 600mm centres ...	sqm	75.50	128.50	90.70	94.60	134.00
250 x 50mm	@ 450mm centres ...	sqm	-	200.50	131.50	146.00	91.80
	@ 600mm centres ...	sqm	-	164.50	108.00	119.50	75.40
Timber engineered joists :							
200 x 47mm	@ 600mm centres ...	sqm	45.50	46.90	47.40	51.60	53.00
246 x 57mm	@ 600mm centres ...	sqm	52.70	53.90	55.10	59.70	61.40
300 x 63mm	@ 600mm centres ...	sqm	54.30	55.80	56.90	61.50	63.60
360 x 63mm	@ 600mm centres ...	sqm	68.80	69.40	70.40	77.50	77.50
ADD extra for :							
Bracing as required	...	Note					
WALL FRAMING			Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10
2400mm high with plates, studs and two rows of noggings :							
75 x 50mm studs	@ 450mm centres ...	sqm	54.80	65.30	60.90	58.20	72.50
	@ 600mm centres ...	sqm	47.20	56.40	52.60	50.30	62.80
100 x 38mm studs	@ 450mm centres ...	sqm	54.80	65.60	57.10	56.70	77.30
	@ 600mm centres ...	sqm	47.20	56.60	49.40	49.00	66.90
100 x 50mm studs	@ 450mm centres ...	sqm	66.20	80.10	69.90	73.30	87.40
	@ 600mm centres ...	sqm	57.30	69.30	60.50	63.50	75.80
2700mm high with plates, studs and two rows of noggings :							
75 x 50mm studs	@ 450mm centres ...	sqm	51.90	61.80	57.60	55.10	68.60
	@ 600mm centres ...	sqm	44.20	52.80	49.30	47.10	58.70
100 x 38mm studs	@ 450mm centres ...	sqm	51.90	62.10	54.10	53.70	73.20
	@ 600mm centres ...	sqm	44.20	53.00	46.30	45.90	62.60
100 x 50mm studs	@ 450mm centres ...	sqm	62.70	75.80	66.00	69.40	82.60
	@ 600mm centres ...	sqm	53.60	64.80	56.60	59.40	70.80
ADD extra for :							
Bracing, refer page 325	...	Note					
Holding down bolts, refer page 347	...	Note					
Steel stud wall framing, refer pages 332 and 333	...	Note					

CARPENTRY

STRUCTURAL TIMBERS - COMPOSITE PRICES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CEILING FRAMING			Pine MGP10	Loc H'wd F14	Pine MGP10	Pine MGP10	Pine MGP10
Hangers :							
200 x 38mm	@ 2400mm centres ...	sqm	-	16.45	10.35	11.05	11.60
250 x 38mm	@ 2400mm centres ...	sqm	-	21.90	11.40	14.85	14.30
Ceiling joists :							
100 x 50mm	@ 450mm centres ...	sqm	36.20	60.30	37.50	39.70	46.70
	@ 600mm centres ...	sqm	27.90	46.40	28.80	30.50	36.00
150 x 50mm	@ 450mm centres ...	sqm	52.00	91.20	59.40	59.00	67.60
	@ 600mm centres ...	sqm	40.00	70.10	45.70	45.40	52.00
200 x 50mm	@ 450mm centres ...	sqm	70.40	125.50	78.50	76.10	85.70
	@ 600mm centres ...	sqm	54.20	96.40	60.40	58.60	65.90
ROOF FRAMING						Loc H'wd F8	
Rafters :							
100 x 50mm	@ 450mm centres ...	sqm	36.20	60.30	37.50	56.70	46.70
125 x 50mm	@ 600mm centres ...	sqm	36.40	60.60	38.90	53.80	34.60
150 x 50mm	@ 600mm centres ...	sqm	41.40	71.90	47.30	64.60	54.10
200 x 50mm	@ 600mm centres ...	sqm	55.60	98.20	62.10	73.00	68.00
Purlins including bolting to cleats :							
150 x 50mm	@ 1200mm centres ...	sqm	21.10	34.80	24.50	31.60	27.90
150 x 75mm	@ 1200mm centres ...	sqm	29.60	52.10	31.40	39.60	42.40
FLAT ROOF FRAMING							
Rafters including blocking at 1800mm centres :							
150 x 50mm	@ 450mm centres ...	sqm	69.30	119.00	80.10	107.00	97.20
150 x 50mm	@ 600mm centres ...	sqm	56.90	97.20	65.90	87.40	81.00
200 x 50mm	@ 450mm centres ...	sqm	92.80	162.50	105.50	120.50	125.00
200 x 50mm	@ 600mm centres ...	sqm	76.10	133.00	87.00	98.80	104.50
EAVES FRAMING							
100 x 38mm framing :							
	@ 450mm centres ...	sqm	31.10	49.80	31.70	31.80	43.10
	@ 600mm centres ...	sqm	23.90	38.30	24.40	24.50	33.10
PERGOLA FRAMING			Treated Pine	Loc H'wd F14	Treated Pine	Treated Pine	Treated Pine
Joists :							
100 x 38mm	@ 900mm centres ...	sqm	18.75	26.40	19.80	18.50	25.20
100 x 50mm	@ 900mm centres ...	sqm	21.70	32.00	23.90	22.60	27.40
150 x 50mm	@ 900mm centres ...	sqm	31.00	48.40	37.30	32.80	38.70
Battens :							
50 x 38mm	@ 100mm centres ...	sqm	52.00	87.20	103.50	61.00	72.40
	@ 450mm centres ...	sqm	12.35	20.70	24.60	14.50	17.20
	@ 900mm centres ...	sqm	6.20	10.35	12.30	7.25	8.60
75 x 38mm	@ 100mm centres ...	sqm	75.60	151.00	114.00	88.30	96.60
	@ 450mm centres ...	sqm	18.00	35.90	27.00	21.00	22.90
	@ 900mm centres ...	sqm	9.00	17.95	13.50	10.50	11.45

CARPENTRY

ROOF TRUSSES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Standard prefabricated roof truss in local sawn softwood or hardwood for tiled roofs to 22° pitch and including 750mm overhang each side :							
6000mm span	...	No	169.50	238.00	207.00	253.00	229.00
7000mm span	...	No	187.00	264.00	229.00	269.00	252.00
8000mm span	...	No	209.00	295.00	254.00	305.00	281.00
9000mm span	...	No	250.00	358.00	302.00	342.00	331.00
10000mm span	...	No	270.00	386.00	326.00	378.00	358.00

LAMINATED TIMBER BEAMS

ADD extra for preservative treatment ... Note

ADELAIDE

Dressed pine, “B” grade, interior		231 x 60	297 x 60	330 x 60	264 x 80	330 x 80	363 x 80	396 x 80	429 x 80	297 x 110
	Price $/m	58.40	72.00	77.90	81.90	101.20	109.00	120.00	128.00	123.00
		330 x 110	363 x 110	396 x 110	429 x 110					
	Price $/m	134.00	145.00	161.50	173.00					
Dressed oregon, “B” grade, exterior		191 x 79	229 x 79	267 x 79	305 x 79	343 x 79	381 x 79	419 x 79	267 x 130	305 x 130
	Price $/m	85.00	101.70	116.00	133.00	149.50	163.00	177.00	182.50	201.00
		343 x 130	381 x 130	419 x 130	457 x 130					
	Price $/m	233.00	256.00	285.00	302.00					
Laminated veneered lumber, structural grade, interior		200 x 36	240 x 36	300 x 36	100 x 45	200 x 45	240 x 45	300 x 45	360 x 45	400 x 45
	Price $/m	23.20	25.90	30.30	17.70	26.70	33.50	41.90	47.30	53.70
		100 x 63	240 x 63	300 x 63	360 x 63	400 x 63	450 x 63			
	Price $/m	21.30	45.10	55.50	62.90	71.10	80.20			

BRISBANE

Dressed brush box, select grade, GL18		600 x 40	115 x 70	215 x 70	280 x 70	345 x 70	345 x 80	215 x 90	280 x 90	410 x 90
	Price $/m	151.50	61.80	101.50	129.00	153.00	170.50	128.00	158.00	223.00
Dressed oregon, “B” grade, exterior		191 x 80	229 x 80	267 x 80	305 x 80	343 x 80	381 x 80	419 x 80	267 x 130	305 x 130
	Price $/m	92.30	110.50	125.00	143.50	161.50	176.00	190.00	196.00	218.00
		343 x 130	381 x 130	419 x 130	457 x 130					
	Price $/m	250.00	273.00	305.00	328.00					
Laminated veneered lumber, structural grade, interior		200 x 36	240 x 36	300 x 36	100 x 45	200 x 45	240 x 45	300 x 45	360 x 45	400 x 45
	Price $/m	27.10	30.10	34.60	21.40	30.90	38.90	48.80	54.50	62.10
		100 x 63	240 x 63	300 x 63	360 x 63	400 x 63	450 x 63			
	Price $/m	25.20	52.20	63.90	71.90	81.50	91.90			

CARPENTRY

LAMINATED TIMBER BEAMS (cont'd)

MELBOURNE

Dressed pine, "B" grade, interior		231 x 60	297 x 60	330 x 60	264 x 80	330 x 80	363 x 80	396 x 80	429 x 80	297 x 110
	Price $/m	57.10	69.60	74.80	78.30	96.10	103.00	114.00	121.50	117.00
		330 x 110	363 x 110	396 x 110	429 x 110					
	Price $/m	126.50	136.10	153.30	162.90					
Dressed oregon, "B" grade, exterior		191 x 79	229 x 79	267 x 79	305 x 79	343 x 79	381 x 79	419 x 79	267 x 130	305 x 130
	Price $/m	72.10	86.50	97.20	111.00	126.00	136.50	147.50	152.00	169.00
		343 x 130	381 x 130	419 x 130	457 x 130					
	Price $/m	194.00	211.00	236.00	253.00					
Laminated veneered lumber, structural grade, interior		200 x 36	240 x 36	300 x 36	100 x 45	200 x 45	240 x 45	300 x 45	360 x 45	400 x 45
	Price $/m	24.70	27.30	31.40	19.50	28.10	35.10	43.90	49.00	56.00
		100 x 63	240 x 63	300 x 63	360 x 63	400 x 63	450 x 63			
	Price $/m	22.90	46.90	57.80	65.00	73.40	82.90			

PERTH

Dressed jarrah, appearance grade 4		210 x 50	240 x 50	240 x 75	300 x 75	360 x 75	420 x 75	450 x 100	510 x 100	570 x 100
	Price $/m	118.50	132.50	194.00	238.00	280.00	326.00	460.00	517.00	581.00
Dressed oregon, "B" grade, exterior		191 x 80	229 x 80	267 x 80	305 x 80	343 x 80	381 x 80	419 x 80	267 x 130	305 x 130
	Price $/m	64.10	76.80	85.70	98.80	111.00	120.00	143.50	-	-
Laminated veneered lumber, structural grade, interior		150 x 36	200 x 36	240 x 36	150 x 45	200 x 45	240 x 45	300 x 45	400 x 45	200 x 63
	Price $/m	20.70	23.90	26.40	23.10	27.10	34.40	43.10	54.80	41.50
		240 x 63	300 x 63	360 x 63	400 x 63					
	Price $/m	46.00	56.40	63.20	71.80					

CARPENTRY

LAMINATED TIMBER BEAMS (cont'd)

SYDNEY

Dressed pine, "B" grade, interior		231 x 60	297 x 60	330 x 60	264 x 80	330 x 80	363 x 80	396 x 80	429 x 80	297 x 110
	Price $/m	66.30	79.90	85.30	88.90	108.50	116.00	128.00	136.00	131.00
		330 x 110	363 x 110	396 x 110	429 x 110					
	Price $/m	141.00	151.50	171.50	181.50					
Dressed oregon, "B" grade, exterior		191 x 80	229 x 80	267 x 80	305 x 80	343 x 80	381 x 80	419 x 80	267 x 130	305 x 130
	Price $/m	83.80	100.50	112.00	129.00	145.00	157.50	169.00	173.00	192.00
		343 x 130	381 x 130	419 x 130	457 x 130					
	Price $/m	221.00	240.00	270.00	287.00					
Laminated veneered lumber, structural grade, interior		200 x 36	240 x 36	300 x 36	100 x 45	200 x 45	240 x 45	300 x 45	360 x 45	
	Price $/m	30.00	33.00	37.40	24.50	33.70	42.40	53.10	58.70	
		400 x 45	100 x 63	240 x 63	300 x 63	360 x 63	400 x 63	450 x 63		
	Price $/m	67.40	28.10	56.30	69.30	76.90	87.20	98.80		

PROPRIETARY STEEL WALL AND FLOOR FRAMING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GROUND FLOOR FRAMING							
Bearers :							
100 x 75 x 1.6mm, external	...	m	38.80	40.90	41.60	40.60	44.60
100 x 75 x 1.2mm, internal	...	m	31.20	33.20	33.90	32.90	36.80
Joists, 100 x 50 x 1.0mm	...	m	23.70	25.00	25.50	24.80	27.40
Composite rates :							
Bearers at 1800mm centres and joists -							
@ 450mm centres	...	sqm	85.90	90.90	92.50	90.10	99.70
@ 600mm centres	...	sqm	70.20	74.40	75.70	73.70	81.60
WALL FRAMING							
Bottom plate, 78 x 31 x 1.2mm	...	m	11.20	12.05	12.20	11.90	13.40
Top plate, 79 x 75 x 1.6mm	...	m	24.70	25.70	25.80	25.50	27.20
Stud :							
Standard, 75 x 32 x 1.2mm	...	m	12.50	13.70	14.20	13.50	15.90
Junction, 75 x 38 x 1.2mm	...	m	20.30	21.70	22.10	21.50	24.10
Corner, 75 x 38 x 1.2mm	...	m	20.30	21.70	22.10	21.50	24.10
Nogging, 72 x 34 x 1.2mm	...	m	13.50	14.95	15.35	14.70	17.40
Bracing :							
25 x 0.6mm strapping	...	m	5.60	6.80	-	-	-
25 x 1.2mm strapping	...	m	-	-	7.85	7.20	9.60
30 x 1.2mm strapping	...	m	-	-	-	-	-

CARPENTRY

PROPRIETARY STEEL WALL AND FLOOR FRAMING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
WALL FRAMING (cont'd)							
Composite rates :							
2400mm high with plates, studs and one row of noggings -							
75 x 32mm studs	@ 450mm centres ...	sqm	59.40	64.60	66.70	64.10	74.10
	@ 600mm centres ...	sqm	55.00	59.80	61.70	59.30	68.50
2700mm high with plates, studs and two rows of noggings -							
75 x 32mm studs	@ 450mm centres ...	sqm	56.30	61.20	63.20	60.70	70.20
	@ 600mm centres ...	sqm	52.10	56.60	58.40	56.20	64.90
ADD extra for :							
Bracing, refer pages 325 and 332 ...		Note					
PROPRIETARY STEEL FRAMING SYSTEMS							
Wall frames 2400mm high with studs at 600mm centres and one row noggings (per sqm wall area) :							
Computer designed system ...		sqm	33.00	33.50	-	-	-
Manual measure and cut system ...		sqm	41.30	39.80	47.60	42.50	42.50
Roof trusses at 600mm or 900mm centres depending on system (per sqm floor area) :							
Computer designed system ...		sqm	58.50-97.00	81.00-97.50	-	-	-
Manual measure and cut system ...		sqm	96.50-115.50	90.50-107.00	98.00-113.50	82.50-99.00	98.50-105.00
Composite wall frame and roof truss costs per sqm floor area :							
Computer designed system ...		sqm	131.00-152.00	127.50-145.50	-	-	-
Manual measure and cut system ...		sqm	148.00-167.50	143.00-157.50	150.00-164.50	148.00-165.00	154.50-173.50

GROUNDS, BATTENS

Grounds and Battens :							
25 x 19mm -							
Local hardwood	...	m	-	4.41	3.24	3.56	6.05
Pine	...	m	2.64	3.31	4.11	3.23	4.72
Treated pine	...	m	4.10	3.59	4.20	-	4.98
38 x 19mm -							
Local hardwood	...	m	-	4.85	3.32	3.72	6.35
Pine	...	m	2.64	3.31	4.11	3.23	4.72
Treated pine	...	m	4.18	3.73	4.47	-	5.25
38 x 25mm -							
Local hardwood	...	m	3.45	5.05	3.41	3.93	6.45
Pine	...	m	3.36	3.55	4.72	3.65	5.35
Treated pine	...	m	4.39	3.83	6.56	4.59	5.40
50 x 25mm -							
Local hardwood	...	m	4.02	6.54	4.06	4.56	7.32
Pine	...	m	3.51	4.23	5.69	4.14	5.85
Treated pine	...	m	5.09	4.61	7.38	4.99	6.16
50 x 38mm -							
Local hardwood	...	m	5.09	8.63	4.75	5.70	9.65
Pine	...	m	4.34	5.24	6.29	5.19	6.54
Treated pine	...	m	5.15	5.72	10.26	6.04	7.17

CARPENTRY

GROUNDS, BATTENS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Grounds and Battens (cont'd) :							
75 x 38mm -							
Local hardwood	...	m	7.55	14.95	6.62	8.40	11.58
Pine	...	m	5.59	7.55	9.38	7.52	7.94
Treated pine	...	m	7.49	8.29	11.27	8.74	9.56
ADD extra for :							
Plugging and screwing to brickwork or concrete	...	m	3.35	3.35	3.45	3.45	3.50
Battens (Composite)							
38 x 25mm @ 450mm centres :							
Local hardwood	...	sqm	8.30	12.15	8.20	9.45	15.50
Pine	...	sqm	8.05	8.50	11.35	8.75	12.85
Treated pine	...	sqm	10.55	9.20	15.75	11.05	12.95
38 x 25mm @ 600mm centres :							
Local hardwood	...	sqm	6.20	9.10	6.15	7.10	11.65
Pine	...	sqm	6.05	6.40	8.50	6.55	9.65
Treated pine	...	sqm	7.90	6.90	11.80	8.25	9.70
50 x 25mm @ 450mm centres :							
Local hardwood	...	sqm	9.65	15.70	9.75	10.95	17.60
Pine	...	sqm	8.45	10.15	13.65	9.95	14.05
Treated pine	...	sqm	12.20	11.05	17.70	12.00	14.80
50 x 25mm @ 600mm centres :							
Local hardwood	...	sqm	7.25	11.75	7.30	8.20	13.20
Pine	...	sqm	6.35	7.60	10.25	7.45	10.55
Treated pine	...	sqm	9.15	8.30	13.30	9.00	11.10
50 x 38mm @ 600mm centres :							
Local hardwood	...	sqm	9.15	15.55	8.55	10.25	17.40
Pine	...	sqm	7.80	9.45	11.30	9.35	11.75
Treated pine	...	sqm	9.25	10.30	18.45	10.85	12.90
50 x 38mm @ 900mm centres :							
Local hardwood	...	sqm	6.10	10.35	5.70	6.85	11.60
Pine	...	sqm	5.20	6.30	7.55	6.25	7.85
Treated pine	...	sqm	6.20	6.85	12.30	7.25	8.60
75 x 38mm @ 600mm centres :							
Local hardwood	...	sqm	13.60	26.90	11.90	15.15	20.90
Pine	...	sqm	10.05	13.60	16.90	13.55	14.30
Treated pine	...	sqm	13.45	14.90	20.30	15.75	17.20
75 x 38mm @ 900mm centres :							
Local hardwood	...	sqm	9.05	17.95	7.95	10.10	13.90
Pine	...	sqm	6.70	9.05	11.25	9.00	9.50
Treated pine	...	sqm	9.00	9.95	13.50	10.50	11.45

FASCIAS AND BARGE BOARDS

All in dry dressed timber							
175 x 31mm							
Local hardwood	...	m	35.90	39.20	41.90	42.90	52.50
Oregon	...	m	-	63.30	38.10	42.90	31.00
Western red cedar	...	m	51.40	39.70	43.20	50.80	59.50
175 x 38mm -							
Local hardwood	...	m	41.20	44.50	47.70	49.20	59.60
Oregon	...	m	-	74.40	43.20	49.30	33.50
Western red cedar	...	m	60.30	45.40	49.10	59.10	68.50

CARPENTRY

FASCIAS AND BARGE BOARDS (cont'd)

All in dry dressed timber (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
200 x 31mm							
Local hardwood	...	m	38.50	42.10	-	47.00	60.40
Oregon	...	m	-	69.30	44.10	47.10	32.50
Western red cedar	...	m	57.20	42.90	51.30	62.70	63.40
200 x 38mm -							
Local hardwood	...	m	44.30	48.00	-	54.00	69.20
Oregon	...	m	-	81.60	50.60	54.40	35.30
Western red cedar	...	m	67.40	49.20	59.30	73.60	73.20
250 x 31mm -							
Local hardwood	...	m	45.60	35.90	-	55.70	71.60
Oregon	...	m	-	83.70	52.00	56.00	36.50
Western red cedar	...	m	69.10	50.60	60.90	75.50	75.20
250 x 38mm -							
Local hardwood	...	m	52.80	56.70	-	64.50	82.50
Oregon	...	m	-	98.90	60.20	65.10	40.40
Western red cedar	...	m	81.80	58.40	71.10	89.00	87.70
ADD extra for :							
Grooving	...	m	0.35-0.40	0.35-0.40	0.35-0.40	0.35-0.40	0.35-0.40

INSULATION

SAFETY MESH SYSTEM

Galvanised roof safety mesh with cold formed galvanised insulation brackets, mechanically fixed to purlins	…	sqm	12.05	13.95	12.80	13.35	14.25

ALUMINIUM

SARKING

Aluminium foil sarking laid over and fixed to purlins :							
Double sided, fire resistant -							
430/530	…	sqm	5.85	6.75	6.90	6.60	8.10
433/533, anti glare	…	sqm	5.95	6.80	6.95	6.65	8.15
450/550	…	sqm	6.10	6.95	7.10	6.80	8.30
453/553, anti glare	…	sqm	6.15	7.00	7.15	6.85	8.35
ADD extra for :							
Galvanised roofing wire supporting mesh	…	sqm	3.75	4.25	4.10	4.15	4.75

REFLECTIVE AIR CELL

Thermo reflective insulation sheeting comprising a cross linked (XL) closed cell insulation core sandwiched by two layers of low emittance aluminium foil :							
Fixed on roof - over purlins (Insuliner)	…	sqm	14.00	13.90	13.75	15.25	15.40
Fixed to walls -							
Concrete (Permiwall)	…	sqm	15.25	15.95	15.90	16.95	17.95
Brick veneer (Permicav XV)	…	sqm	15.25	15.95	15.90	16.95	17.95
Steel framed, lightweight clad (Insulbreak 70)	…	sqm	17.00	17.55	18.00	17.25	21.10
Timber framed, lightweight clad (Permishield 70)	…	sqm	16.10	18.05	18.65	17.65	20.60
Double brick cavity (Permicav XV)	…	sqm	15.20	17.30	16.90	16.85	19.10
Fixed to underfloor - suspended framed (Permifloor)	…	sqm	12.05	13.90	13.95	14.85	15.35

CARPENTRY

INSULATION (cont'd)

ALUMINIUM (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CONCERTINA CORRUGATED FOIL							
Aluminium corrugated and perforated foil sheeting combined with 25 fibreglass insulation and polythene including support bar system, ceiling height 6.0 to 12.0 metres (Ripple Sound)	…	sqm	152.50	152.50	152.50	152.50	152.50
Aluminium corrugated and perforated foil sheeting, laminated with acoustic textile including support bar system, ceiling height 6.0 to 12.0 metres (Ripple Tone)	…	sqm	157.50	157.50	157.50	157.50	157.50
ADD extra for :							
Ceiling heights of 12.0 to 24.0 metre	...	sqm	20.00-40.00	20.00-40.00	20.00-40.00	20.00-40.00	20.00-40.00

GLASSWOOL

Glasswool batts laid between ceiling joists :							
R 1.5	…	sqm	8.50	9.45	9.60	9.25	10.85
R 2.0	…	sqm	9.00	9.95	10.10	9.80	11.40
R 2.5	…	sqm	9.50	10.45	10.60	10.30	11.85
R 3.0	…	sqm	10.85	11.80	11.95	11.65	13.25
R 3.5	…	sqm	11.60	12.55	12.70	12.40	14.00
R 4.0	…	sqm	12.20	13.15	13.30	13.00	14.60
R 5.0	…	sqm	15.05	16.00	16.15	15.80	17.40
ADD extra for :							
Fixing vertically	…	sqm	4.60	5.80	5.50	5.60	7.00
Glasswool blanket laid over purlins :							
R 1.25 (x 50mm)	…	sqm	8.95	9.80	9.95	9.65	11.15
R 1.75 (x 75mm)	…	sqm	9.35	10.20	10.40	10.05	11.55
R 2.00 (x 100mm)	…	sqm	10.85	11.70	11.85	11.55	13.05
Glasswool blanket with black fibreglass tissue facing one side (Sonomatt) laid over ceiling panels :							
25mm thick	…	sqm	11.10	11.95	12.15	11.80	13.30
50mm thick	…	sqm	13.45	14.30	14.50	14.15	15.65
Glasswool blanket with lightweight foil facing one side (Anticon or Permastop) laid over purlins :							
R 1.5	…	sqm	8.85	9.80	9.95	9.65	11.25
R 2.0	…	sqm	10.00	10.95	11.10	10.80	12.40
R 2.5	…	sqm	12.30	13.25	13.40	13.10	14.70
ADD extra for :							
Medium weight perforated foil facing	…	sqm	0.85	0.85	0.85	0.85	0.85
Heavy weight antiglare foil facing	…	sqm	1.85	1.85	1.85	1.85	1.85

CARPENTRY

INSULATION (cont'd)

GLASSWOOL (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Glasswool batts made from recycled glass bottles laid between ceiling joists :							
R 2.5	…	sqm	9.55	10.45	10.60	10.30	11.90
R 3.0	…	sqm	10.60	11.50	11.65	11.35	12.95
R 3.5	…	sqm	11.40	12.35	12.45	12.15	13.75
R 4.0	…	sqm	12.65	13.60	13.75	13.45	15.05
R 5.0	…	sqm	17.15	18.05	18.20	17.90	19.50
R 6.0	…	sqm	19.45	20.40	20.50	20.20	21.80
ADD extra for :							
Fixing vertically	…	sqm	4.60	5.80	5.50	5.60	7.00
Glasswool batts made from recycled glass bottles laid within wall frame :							
R 1.5	…	sqm	9.20	10.75	10.75	10.50	12.75
R 2.0	…	sqm	10.30	11.85	11.80	11.55	13.85
R 2.5	…	sqm	13.80	15.35	15.35	15.10	17.35

PHENOLIC BOARD

Rigid board insulation comprising a closed-cell thermoset phenolic insulation core sandwiched by layers of either low emittance foil or tissue based facings depending on application :							
Fixed to walls - concrete including clip and channel system (Kooltherm K12) -							
R 1.40 (x 30mm)	…	sqm	37.70	36.30	36.30	40.40	38.20
R 1.90 (x 40mm)	…	sqm	42.90	41.00	41.00	45.80	42.90
Fixed to walls - brick veneer (Kooltherm K12) -							
R 1.15 (x 25mm)	…	sqm	35.40	34.10	34.10	37.90	36.10
R 1.40 (x 30mm)	…	sqm	37.70	36.30	36.30	40.40	38.20
Fixed to walls - steel framed, lightweight clad (Kooltherm K12) -							
R 1.15 (x 25mm)	…	sqm	35.40	34.10	34.10	37.90	36.10
R 1.40 (x 30mm)	…	sqm	37.70	36.30	36.30	40.40	38.20
Fixed to walls - timber framed, lightweight clad (Kooltherm K12) -							
R 1.15 (x 25mm)	…	sqm	35.40	34.10	34.10	37.90	36.10
R 1.40 (x 30mm)	…	sqm	37.70	36.30	36.30	40.40	38.20
Fixed to walls - double brick cavity (Kooltherm K8) -							
R 1.15 (x 25mm)	…	sqm	35.40	34.10	34.10	37.90	36.10
Fixed to underfloor - concrete soffit (Kooltherm K10 G2) -							
R 1.15 (x 25mm)	…	sqm	32.10	30.30	33.20	36.90	35.20
R 1.90 (x 40mm)	…	sqm	40.30	37.80	40.70	45.40	42.60
R 2.50 (x 50mm)	…	sqm	49.20	45.90	46.10	54.80	47.30
R 3.00 (x 60mm)	…	sqm	56.60	52.60	55.50	62.50	57.40
R 3.50 (x 70mm)	…	sqm	66.70	61.80	64.70	73.10	66.60
R 4.00 (x 80mm)	…	sqm	71.00	65.70	68.60	77.50	70.50
R 4.50 (x 90mm)	…	sqm	82.40	76.10	79.00	89.50	80.90

CARPENTRY

INSULATION (cont'd)

POLYESTER

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Polyester batts laid between ceiling joists :							
R 2.0	…	sqm	12.30	12.25	12.40	12.85	13.70
R 2.5	…	sqm	14.00	13.80	13.90	14.55	15.20
R 3.0	…	sqm	16.85	16.30	16.45	17.30	17.75
R 3.5	…	sqm	18.75	18.00	18.15	19.15	19.40
R 4.0	…	sqm	21.50	20.50	20.60	21.90	21.90
ADD extra for :							
Fixing vertically	…	sqm	4.60	5.80	5.50	5.60	7.00
Polyester blanket laid over purlins :							
x 50mm thick	…	sqm	7.15	8.00	8.15	7.85	9.35
x 65mm thick	…	sqm	8.10	8.95	9.10	8.80	10.30
x 75mm thick	…	sqm	9.55	10.40	10.55	10.25	11.75
Polyester blanket with lightweight foil facing one side laid over purlins :							
R 1.0 (x 50mm)	…	sqm	13.75	14.65	14.80	14.50	16.10
R 1.3 (x 75mm)	…	sqm	15.45	16.35	16.50	16.20	17.80
R 1.7 (x 100mm)	…	sqm	16.40	17.35	17.50	17.20	18.80
ADD extra for :							
Medium weight perforated foil facing	…	sqm	2.15	2.15	2.15	2.15	2.15
Heavy weight antiglare foil facing	…	sqm	2.45	2.45	2.45	2.45	2.45

POLYSTYRENE

Expanded polystyrene sheet/batts (Isolite ClassSL) laid in roof space :							
25mm thick	…	sqm	10.55	10.60	10.75	11.10	11.95
50mm thick	…	sqm	15.30	14.80	14.95	15.70	16.15
75mm thick	…	sqm	26.80	25.10	25.30	27.00	26.60
100mm thick	…	sqm	27.80	26.10	26.30	28.10	27.70
Extruded polystyrene tongued and grooved boards (Styrofoam) laid in roof space :							
25mm thick	…	sqm	18.15	18.35	18.70	19.15	20.90
30mm thick	…	sqm	25.60	25.00	25.40	26.50	27.50
50mm thick	…	sqm	40.40	38.70	39.00	41.30	41.70
ADD extra for :							
Fixing vertically	…	sqm	4.60	5.80	5.50	5.60	7.00

WOOL

Fire retarded wool batts laid between ceiling joists :							
R 3.0	…	sqm	26.20	26.60	26.70	27.50	28.00
R 3.5	…	sqm	34.70	34.90	35.00	36.20	36.30
ADD extra for :							
Fixing vertically	…	sqm	4.60	5.80	5.50	5.60	7.00

CARPENTRY

FLOORING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FIBRE CEMENT							
Compressed fibre cement sheets fixed to timber :							
15mm thick	...	sqm	96.50	102.00	103.50	101.00	111.00
18mm thick	...	sqm	110.00	116.50	118.00	115.50	126.50
24mm thick	...	sqm	152.00	159.50	161.00	158.50	171.00
Fibre cement sheet underlay fixed to timber :							
6mm thick for ceramic tiles	...	sqm	29.30	32.50	33.20	31.90	37.70
HARDBOARD							
Hardboard fixed to timber :							
4.8mm thick	...	sqm	26.70	29.80	30.50	29.30	34.80
6.4mm thick	...	sqm	32.20	35.40	36.10	34.90	40.70
5.5mm thick underlay	...	sqm	34.90	38.20	38.90	37.60	43.40
PARTICLE BOARD							
Particle board panels with tongued and grooved edges, fixed to timber :							
Untreated -							
19mm thick standard (yellow)	...	sqm	27.60	31.00	31.70	30.50	36.40
22mm thick mid range (red)	...	sqm	33.40	37.00	37.70	36.40	42.60
25mm thick heavy duty (blue)	...	sqm	42.50	46.30	46.90	45.60	52.00
Fungicide or termicide treated -							
19mm thick	...	sqm	30.50	34.00	34.60	33.40	39.40
22mm thick	...	sqm	35.60	39.10	39.80	38.50	44.70
PLYWOOD							
Panels with tongued and grooved edges fixed to timber :							
15mm thick	...	sqm	34.60	35.10	35.90	36.60	40.20
19mm thick	...	sqm	37.80	38.40	39.10	40.10	43.90
21mm thick	...	sqm	46.60	46.60	47.20	48.90	52.20
TONGUED AND GROOVED							
25mm tongued and grooved flooring (standard grade) in 100mm widths fixed to timber :							
Brush box	...	sqm	85.70	101.50	-	-	168.00
Cypress pine	...	sqm	72.40	81.70	84.60	84.20	87.30
Jarrah	...	sqm	125.50	-	-	98.60	139.50
Mixed hardwood	...	sqm	-	100.00	-	-	148.50
Radiata pine	...	sqm	66.20	73.20	76.40	73.50	82.00
Tasmanian oak	...	sqm	98.20	95.60	90.80	120.50	152.00
ADD extra for :							
Secret nailing	...	sqm	12.50	15.75	14.90	15.20	19.00
Fine sand and apply -							
Clear polyurethene finish :							
Two coats	...	sqm	-	-	-	23.60	-
Three coats	...	sqm	33.30	27.80	40.40	32.20	29.30
Three coats seal and one coat wax polish	...	sqm	33.30	27.80	40.40	32.20	29.30
Specially selected or prime quality flooring of the foregoing where available (range)	...	sqm	30.00-40.00	30.00-40.00	30.00-40.00	35.00-45.00	35.00-45.00

CARPENTRY

FLOORING (cont'd)

TONGUED AND GROOVED (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Proprietary tongued and grooved veneered floating floor panel comprising marine ply core with resin impregnated timber veneer facing, fixed to battens (Parklex 2000) :							
188mm wide plank (P.C. $115.00)	...	sqm	262.00	311.00	288.00	318.00	266.00
290mm wide plank (P.C. $119.00)	...	sqm	241.00	286.00	266.00	298.00	250.00
590mm wide plank (P.C. $125.00)	...	sqm	221.00	262.00	244.00	274.00	229.00

TIMBER DECKING

Timber decking, close spaced and fixed to timber :							
Pencil rounded -							
75 x 25mm :							
Jarrah (prime grade)	...	sqm	201.00	204.50	188.00	139.00	218.50
Treated pine	...	sqm	81.30	93.00	99.70	98.40	103.00
100 x 25mm :							
Jarrah (prime grade)	...	sqm	191.50	193.50	177.00	128.50	204.00
Treated pine	...	sqm	77.40	89.40	97.40	83.10	97.40
Reeded -							
75 x 25mm :							
Jarrah (standard grade)	...	sqm	140.00	144.00	134.50	100.00	157.50
Treated pine	...	sqm	79.70	97.70	106.00	83.70	108.50
100 x 25mm :							
Jarrah (standard grade)	...	sqm	133.00	140.00	130.00	92.30	146.50
Treated pine	...	sqm	66.60	84.70	92.20	69.90	93.00
ADD extra for :							
Prime grade boards where available (range)	...	sqm	28.00-33.00	28.00-33.00	28.00-33.00	28.00-33.00	28.00-33.00
Recycled wood plastic composite decking, close spaced and fixed to timber :							
88 x 23mm	...	sqm	151.00	158.00	159.50	157.00	168.50
137 x 29mm	...	sqm	156.50	163.50	164.50	162.00	174.00

PARQUETRY FLOORING

19mm wood block parquetry flooring laid herring-bone pattern on cement base with adhesive, including all cutting, sanding, sealing and applying two or three coats of finish :							
Brush box	...	sqm	210.00	255.00	-	-	220.00
Blackbutt	...	sqm	210.00	235.00	240.00	230.00	235.00
Tallowood	...	sqm	225.00	265.00	-	-	235.00
Tasmanian Oak	...	sqm	-	-	240.00	220.00	240.00
Jarrah	...	sqm	-	-	240.00	225.00	-
ADD extra for :							
Special pattern and border work	...	Note					
9mm best quality mosaic parquetry flooring laid on cement base with adhesive and including all cutting, sanding, sealing and applying three coats of finish :							
Brush box	...	sqm	335.00	365.00	-	360.00	375.00
Bluegum	...	sqm	-	365.00	-	360.00	375.00
Tallowood	...	sqm	335.00	365.00	-	360.00	375.00
ADD extra for :							
Special pattern and border work	...	Note					

CARPENTRY

FLOORING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TONGUED AND GROOVED PARQUETRY FLOORING							
19mm acrylic impregnated tongued and grooved parquetry flooring (Permagrain)	...	sqm	260.00	260.00	265.00	260.00	270.00
19mm hardwood tongued and grooved parquetry flooring (4mm timber) (Kahrs)	...	sqm	145.00	150.00	155.00	150.00	160.00
ADD extra for :							
Resilient foam underlay	...	Note					
Moisture barrier	...	Note					
LAMINATED FLOORING							
8mm tongued and grooved flooring (96% timber) (Comfort)	...	sqm	46.50	47.00	52.00	47.50	54.00
Floating acoustic timber veneer strip flooring	...	sqm	-	-	110.00	-	-
TIMBER FLOORING SYSTEM							
Low level system 22mm thick overall and comprising a layer of 12mm plywood glued and nailed to concrete and finished on top with 10mm thick x 80mm wide square dressed blackbutt or jarrah flooring in random lengths from 0.9m to 1.80m and sanded, sealed and polished (average)	...	sqm	-	235.00	-	250.00	-
AIR THRUST FLOORS							
Air Thrust floor comprising two layers of exterior grade plywood, attached air thrust pads and finished with 9mm thick mosaic flooring laid to wood weave pattern and finished with three coats sealer	...	sqm	210.00	215.00	260.00	-	220.00
Parq-Air elastic flooring system comprising 12mm neoprene pads, two layers of 12mm structural grade plywood and finished with bluegum mosaic and three coats sealer	...	sqm	-	-	-	170.00	-

WALL LININGS

Prices are for fixing to timber framing	...	Note					
ADD extra for :							
Fixing to metal framing	...	sqm	1.55	1.65	1.70	1.70	1.80
50 x 25mm battens at 450mm centres fixed to brick or concrete wall	...	sqm	8.45	10.15	13.65	9.95	14.05
CANE-ITE							
Cane-ite sheeting fixed to timber framing :							
12.7mm thick natural finish	...	sqm	38.50	42.80	43.60	42.00	49.50
12.7mm thick ivory finish	...	sqm	40.40	44.70	45.50	44.00	51.50

CARPENTRY

WALL LININGS (cont'd)

MEDIUM DENSITY FIBREBOARD

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Medium density fibreboard sheet fixed to timber framing :							
Plain -							
4.5mm thick	...	sqm	26.00	31.00	32.70	31.00	40.40
6.0mm thick	...	sqm	26.60	31.60	33.30	31.70	41.00
9.0mm thick	...	sqm	30.40	35.50	37.10	36.00	45.30
12.0mm thick	...	sqm	33.00	38.30	39.80	38.90	48.40
16.0mm thick	...	sqm	36.20	41.90	43.30	42.70	52.50
18.0mm thick	...	sqm	38.10	43.80	46.00	44.80	55.70
25.0mm thick	...	sqm	51.10	56.80	58.70	59.50	69.50
ADD extra for :							
Melamine -							
White, both sides	...	sqm	12.25	13.80	14.10	13.50	16.25
Plain colours, both sides	...	sqm	17.50	19.05	19.35	18.75	21.50
Profiled tongued and grooved panelling 9mm thick fixed to timber framing :							
EasyPanel	...	sqm	42.30	42.70	42.10	47.30	49.30
EasyVJ	...	sqm	42.30	44.10	43.40	48.60	50.70
EasyRegency	...	sqm	42.70	44.10	43.40	48.60	50.70
EasyAscot	...	sqm	45.70	65.00	46.70	52.00	54.00

FIBRE CEMENT

Fibre cement flexible sheeting fixed to timber framing :							
4.5mm thick (exterior)	...	sqm	31.80	37.00	38.00	36.20	45.20
6.0mm thick (exterior)	...	sqm	39.90	45.60	46.90	44.60	54.80
4.5mm thick (interior)	...	sqm	31.80	37.00	38.00	36.20	45.20
6.0mm thick (interior)	...	sqm	39.90	45.60	46.90	44.60	54.80
Fibre cement decorative sheeting fixed to timber framing :							
6.0mm thick Stucco	...	sqm	54.90	61.80	63.90	60.60	73.60
Fibre cement decorative planks fixed to timber framing :							
230mm x 7.5mm Smooth finish	...	sqm	53.60	61.30	63.20	60.00	74.00
230mm x 7.5mm Woodgrain finish	...	sqm	55.90	63.60	65.50	62.30	76.30
300mm x 9.0mm Weatherboard	...	sqm	69.00	74.90	76.50	73.90	84.90
170mm x 9.0mm Weatherboard	...	sqm	98.00	108.50	109.50	106.50	123.50
Compressed fibre cement sheeting fixed to timber framing :							
Plain sheets -							
6.0mm thick	...	sqm	68.60	76.20	78.10	74.90	88.90
9.0mm thick	...	sqm	84.60	93.20	94.80	91.70	106.50
12.0mm thick	...	sqm	107.00	117.50	118.50	115.50	132.50
15.0mm thick	...	sqm	118.50	129.50	130.50	127.50	145.50
18.0mm thick	...	sqm	132.50	144.50	145.00	142.50	161.50
24.0mm thick	...	sqm	177.00	190.50	191.00	188.50	209.00
Wallboard fibre cement sheeting flush jointed and fixed to timber framing :							
6mm thick	...	sqm	43.00	50.20	51.80	49.00	61.90
9mm thick	...	sqm	61.40	69.90	70.80	68.40	82.40
12mm thick	...	sqm	72.40	82.10	83.10	80.50	96.20

CARPENTRY

WALL LININGS (cont'd)

FIBRE CEMENT (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Vandal resistant board comprising two layers of 6.0mm fibre cement sheeting with a 1.50mm thick steel sheet between, fixed to framing	...	sqm	258.00	273.00	277.00	270.00	298.00
7.5mm thick texture base sheet flush jointed and fixed to framing	...	sqm	41.90	48.30	49.70	47.20	58.70

HARDBOARD

Hardboard fixed to timber framing :							
Standard -							
3.2mm thick	...	sqm	26.00	30.60	31.70	29.80	38.20
4.8mm thick	...	sqm	33.30	38.00	39.10	37.20	45.80
6.4mm thick	...	sqm	40.70	46.20	47.90	45.20	55.60
Pegboard -							
4.8mm thick	...	sqm	52.60	57.70	58.80	56.90	65.90
5.5mm thick	...	sqm	58.60	63.70	64.70	62.80	71.80
Readi-fix -							
5.5mm thick	...	sqm	38.90	44.00	45.00	43.20	52.20
Tempered -							
3.2mm thick	...	sqm	34.10	38.70	39.80	37.90	46.30
4.8mm thick	...	sqm	38.60	43.30	44.40	42.50	51.10
6.4mm thick	...	sqm	47.00	52.40	54.10	51.50	61.90
Weathertex planks -							
300 x 9.5mm Millwood	...	sqm	71.00	77.80	79.20	76.70	88.70
200 x 9.5mm Smooth	...	sqm	64.90	72.60	74.50	71.30	85.30
200 x 9.5mm Ruff Sawn	...	sqm	68.10	75.80	77.70	74.50	88.40
200 x 9.5mm Rusticated	...	sqm	75.90	83.60	85.50	82.30	96.30

PARTICLE BOARD

Particle board, moisture resistant, fixed to timber framing :							
9/10mm thick	...	sqm	30.40	36.20	37.80	35.20	46.00
12/13mm thick	...	sqm	33.20	39.30	40.90	38.30	49.50
16mm thick	...	sqm	35.20	41.90	43.30	40.80	52.50
18mm thick	...	sqm	37.00	43.80	46.00	42.70	55.70
25mm thick	...	sqm	48.20	51.80	53.90	50.60	63.60
33mm thick	...	sqm	52.70	61.20	62.90	59.70	74.70
ADD extra for :							
Veneers -							
Melamine, satin white	...	sqm	12.85	14.45	14.85	14.20	17.10
Timber :							
Hoop pine	...	sqm	24.60	25.40	25.70	26.90	28.00
Brush box	...	sqm	39.90	39.90	40.30	42.80	42.50
Jarrah	...	sqm	94.00	91.40	91.80	99.50	94.00
Nyatoh	...	sqm	37.40	37.50	37.90	40.20	40.10
Pacific maple	...	sqm	35.00	35.20	35.60	37.70	37.90
Sapele (Rotary)	...	sqm	41.30	41.20	41.60	44.30	43.90
Silky oak	...	sqm	39.80	39.90	40.20	42.80	42.50
Teak	...	sqm	49.90	49.50	49.80	53.40	52.10
Victorian ash	...	sqm	46.00	45.70	46.10	49.20	48.30

CARPENTRY

WALL LININGS (cont'd)

TIMBER CLADDING

Prices exclude structural frames, girts, backings, etc.
All prices based on 500 sqm minimum
All prices are for standard range veneers
All prices based on a 2.4 x 1.2m panel format

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
High density timber panel comprising thermoset core with resin impregnated timber veneer facing, exterior grade, fixed to framing or backing with exposed face fixings (Parklex Façade) :							
6mm thick (P.C. $143.00) - soffits only	…	sqm	300.00	286.00	272.00	315.00	243.00
8mm thick (P.C. $161.00)	…	sqm	325.00	309.00	293.00	342.00	271.00
10mm thick (P.C. $179.00)	…	sqm	351.00	333.00	315.00	369.00	294.00
ADD extra for :							
Enhanced fire rating	…	sqm	26.60	24.70	22.80	28.50	19.00
Double faced panels	…	sqm	24.80	24.80	24.80	24.80	24.80
High abrasion coating	…	sqm	7.70	7.15	6.60	8.25	5.50
Concealed fixings	...	+ 25%					
High density timber panel comprising thermoset core with resin impregnated timber veneer facing, interior grade, fixed to framing or backing with exposed face fixings (Parklex 700) :							
6mm thick (P.C. $110.00) - soffits only	…	sqm	218.00	246.00	213.00	268.00	210.00
8mm thick (P.C. $130.00)	…	sqm	238.00	270.00	233.00	294.00	230.00
10mm thick (P.C. $165.00)	…	sqm	239.00	267.00	234.00	289.00	231.00
ADD extra for :							
Enhanced fire rating	…	sqm	26.60	24.70	22.80	28.50	19.00
Double faced panels	...	+14%					
High abrasion coating	…	sqm	7.70	7.15	6.60	8.25	5.50
Concealed fixings	...	+25%					
High density timber panel comprising marine ply core with resin impregnated timber veneer facing, interior grade, fixed to framing or backing with exposed face fixings (Parklex 500) :							
8mm thick (P.C. $100.00)	…	sqm	205.00	232.00	202.00	252.00	198.00
11mm thick (P.C. $105.00)	…	sqm	212.00	239.00	209.00	259.00	205.00
ADD extra for :							
Enhanced fire rating	…	sqm	26.60	24.70	22.80	28.50	19.00
Double faced panels	...	+14%					
High density timber panel comprising thermoset core with solid colour finish, interior grade, fixed to framing or backing with exposed face fixings (Parklex 500) :							
8mm thick (P.C. $125.00)	…	sqm	275.00	263.00	250.00	288.00	225.00
11mm thick (P.C. $130.00)	…	sqm	282.00	269.00	256.00	295.00	230.00
ADD extra for :							
Enhanced fire rating	…	sqm	26.60	24.70	22.80	28.50	19.00
Double faced panels	…	sqm	15.40	14.30	13.20	16.50	11.00

CARPENTRY

WALL LININGS (cont'd)

PLYWOOD

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Interior standard grade plywood for paint finish fixed to timber framing :							
3mm thick	...	sqm	28.40	33.50	34.40	33.20	41.90
4/5mm thick	...	sqm	30.70	35.90	36.80	35.80	44.50
6mm thick	...	sqm	33.00	38.50	40.10	38.40	48.50
9mm thick	...	sqm	36.80	42.90	44.30	42.90	53.60
12mm thick	...	sqm	41.20	47.90	49.10	47.90	59.20
15mm thick	...	sqm	46.80	54.00	55.80	54.20	67.00
18mm thick	...	sqm	52.30	59.90	61.60	60.30	73.40
25mm thick	...	sqm	64.30	71.90	73.50	73.20	85.90
38mm thick	...	sqm	83.90	92.00	93.30	94.50	107.00
Exterior grade waterproof plywood for paint finish fixed to timber framing :							
3mm thick	...	sqm	27.60	32.80	33.70	32.50	41.20
4/5mm thick	...	sqm	31.10	36.30	37.10	36.20	44.90
6mm thick	...	sqm	37.10	42.40	44.00	42.80	52.40
9mm thick	...	sqm	42.10	48.00	49.40	48.40	58.60
12mm thick	...	sqm	50.60	56.90	58.10	57.80	68.20
15mm thick	...	sqm	58.00	64.70	66.50	65.90	77.70
18mm thick	...	sqm	65.80	72.80	74.50	74.50	86.30
25mm thick	...	sqm	79.60	86.50	88.10	89.30	100.50
38mm thick	...	sqm	111.50	118.50	119.50	123.50	133.50
Marine plywood for paint finish fixed to timber framing :							
3/4mm thick	...	sqm	32.80	37.70	38.60	37.90	46.10
5/6mm thick	...	sqm	39.50	44.70	46.20	45.20	54.60
9mm thick	...	sqm	49.20	54.70	56.10	55.80	65.40
12mm thick	...	sqm	57.70	63.60	64.90	65.20	75.00
15mm thick	...	sqm	66.20	72.60	74.40	74.60	85.60
18/19mm thick	...	sqm	77.10	83.50	85.20	86.20	97.00
25mm thick	...	sqm	93.80	100.00	101.50	104.00	114.00
Formwork grade plywood plastic faced both sides fixed to timber framing :							
17mm thick	...	sqm	57.30	63.60	65.50	65.00	76.30
ADD extra for :							
Veneers -							
Hoop pine	...	sqm	31.00	31.50	31.80	33.50	34.10
Jarrah	...	sqm	81.80	79.80	80.20	86.80	82.40
Maple	...	sqm	25.60	26.30	26.70	27.90	29.00
Nyatoh	...	sqm	27.90	28.50	28.90	30.30	31.10
Pacific maple	...	sqm	25.60	26.30	26.70	27.90	28.90
Sapele (Rotary)	...	sqm	32.10	32.50	32.80	34.60	35.10
Silky oak	...	sqm	30.30	30.80	31.10	32.80	33.40
Teak	...	sqm	40.50	40.50	40.80	43.50	43.10
Victorian ash	...	sqm	36.70	36.80	37.20	39.50	39.40

TIMBER BOARDING

Weatherboarding fixed to timber studs :							
200 x 25mm sawn, splayed -							
Hardwood	...	sqm	88.00	81.90	84.80	100.00	110.00
Western red cedar	...	sqm	88.20	81.10	91.30	67.40	110.50
150 x 25mm dressed, rusticated -							
Treated Pine	...	sqm	103.00	106.00	124.00	101.00	124.00
Hardwood	...	sqm	111.50	99.40	116.50	116.00	147.00

CARPENTRY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
WALL LININGS (cont'd)							
TIMBER BOARDING (cont'd)							
Shiplap panelling fixed to timber studs :							
150 x 25mm -							
Oregon	...	sqm	-	138.50	128.00	128.50	192.00
Tasmanian oak	...	sqm	142.00	146.00	148.00	136.50	195.00
Western red cedar	...	sqm	147.00	137.00	159.00	119.00	190.00
Shakes and shingles, refer page 405	...	Note					
LATTICE							
Fibre cement lattice panels including treated pine surrounds and joiners between panels (typical range)	...	sqm	50.00-65.00	50.00-65.00	50.00-65.00	50.00-65.00	50.00-65.00
Treated pine lattice panels including surrounds and joiners between panels (typical range)	...	sqm	60.00-75.00	60.00-75.00	60.00-75.00	60.00-75.00	60.00-75.00
CEILING LININGS							
As foregoing WALL LININGS,							
ADD extra for fixing to soffits	...	sqm	11.50	14.45	15.65	13.95	19.95
EAVES LININGS							
As foregoing WALL LININGS,							
ADD extra for fixing to soffits, widths :							
N.E. 2000mm wide	...	sqm	14.75	18.60	19.55	17.95	24.90
2000/4000mm wide	...	sqm	13.15	16.55	17.60	15.95	22.40
Exceeding 4000mm wide	...	sqm	11.50	14.45	15.65	13.95	19.95
SUNDRY FIXINGS							
BOLTS							
Galvanised cup head bolt including drilling :							
6mm dia. x 75mm long	...	No	5.40	6.60	6.70	6.40	8.30
10mm dia. x 100mm long	...	No	7.65	9.35	9.70	9.05	12.05
10mm dia. x 150mm long	...	No	9.30	11.35	11.60	11.00	14.40
12mm dia. x 100mm long	...	No	8.90	10.75	11.05	10.45	13.65
12mm dia. x 150mm long	...	No	10.70	12.90	13.10	12.50	16.10
Galvanised hexagonal head bolt including drilling :							
6mm dia. x 75mm long	...	No	5.55	6.75	6.85	6.55	8.45
10mm dia. x 100mm long	...	No	7.90	9.60	9.95	9.30	12.30
10mm dia. x 150mm long	...	No	9.85	11.90	12.15	11.55	14.95
12mm dia. x 100mm long	...	No	9.05	10.95	11.20	10.60	13.80
12mm dia. x 150mm long	...	No	11.10	13.30	13.50	12.95	16.55
16mm dia. x 100mm long	...	No	12.10	14.35	14.55	13.95	17.55
16mm dia. x 150mm long	...	No	14.65	17.20	17.30	16.75	20.70

CARPENTRY

SUNDRY FIXINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BOLTS (cont'd)							
Zinc plated holding down bolt including drilling and casting into concrete :							
10mm dia. x 2400mm long	...	No	62.10	75.80	80.00	73.50	99.40
10mm dia. x 2700mm long	...	No	62.70	76.40	80.60	74.10	100.00
12mm dia. x 2400mm long	...	No	65.20	78.80	83.10	76.50	102.40
12mm dia. x 2700mm long	...	No	66.00	79.70	83.90	77.40	103.30
Additional holding down bolts, refer page 289	...	Note					
COACHSCREWS							
Zinc plated coachscrew including drilling :							
10mm dia. x 50mm long	...	No	5.25	6.45	6.50	6.25	8.15
10mm dia. x 100mm long	...	No	6.75	8.25	8.65	8.00	10.80
12mm dia. x 50mm long	...	No	6.70	8.00	8.05	7.80	9.80
12mm dia. x 100mm long	...	No	8.30	9.90	10.25	9.60	12.50
MASONRY ANCHORS							
Expanding masonry anchor and galvanised bolt fixing 50mm timber including drilling concrete, bolt dia. :							
6mm	...	No	9.80	11.85	12.10	11.50	14.85
8mm	...	No	10.30	12.35	12.55	12.00	15.35
10mm	...	No	10.80	12.95	13.55	12.60	16.55
12mm	...	No	11.40	13.65	14.25	13.25	17.35
16mm	...	No	12.85	15.15	15.75	14.80	18.95
Expanding masonry anchor and galvanised bolt fixing 50mm timber including drilling brickwork, bolt dia. :							
6mm	...	No	8.45	10.15	10.50	9.90	12.90
8mm	...	No	8.95	10.65	11.00	10.40	13.40
10mm	...	No	9.50	11.30	12.00	11.00	14.55
12mm	...	No	10.10	12.00	12.65	11.65	15.35
16mm	...	No	11.55	13.50	14.20	13.20	17.00
Plastic plug and screw fixing including drilling concrete or brickwork :							
5mm dia. hole, 25mm plug	...	No	5.40	6.75	6.40	6.50	8.10
6mm dia. hole, 30mm plug	...	No	5.45	6.80	6.45	6.55	8.15
8mm dia. hole, 40mm plug	...	No	6.45	8.10	7.65	7.80	9.70
POST ANCHOR BASE							
Post anchor base cast into concrete to suit :							
Solid post	...	No	48.40	58.70	59.90	56.90	73.90
Twin posts	...	No	51.50	61.70	62.90	60.00	76.90
SHOT FIXINGS							
Shot fixing 50mm timber to :							
Concrete	...	No	3.10	3.65	3.50	3.55	4.15
Brickwork	...	No	3.10	3.65	3.50	3.55	4.15
Steel	...	No	3.10	3.65	3.50	3.55	4.15

CARPENTRY

SUNDRY FIXINGS (cont'd)

TIMBER CONNECTORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Light gauge galvanised connectors :							
Standard framing anchor	...	No	2.75	2.90	3.10	2.76	3.14
Universal framing anchor	...	No	2.74	2.83	3.07	2.69	3.07
Joist/hanger connection, stirrup type	...	No	3.23	3.28	3.52	3.21	3.58
Joist/hanger, strip type	...	No	1.59	1.72	1.84	1.61	1.87
Nail on plate (average size)	...	No	3.23	3.28	3.52	3.21	3.57

STAINLESS STEEL NAILS

Stainless steel nails :							
Supply only cost of flat head grade 314 - 304 stainless steel nails -							
40mm x 9 (314 per kg)	...	Kg	90.00	90.00	90.00	90.00	90.00
50mm x 10 (290 per kg)	...	Kg	90.00	90.00	90.00	90.00	90.00
65mm x 9 (188 per kg)	...	Kg	90.00	90.00	90.00	90.00	90.00
75mm x 9 (154 per kg)	...	Kg	90.00	90.00	90.00	90.00	90.00
90mm x 8 (100 per kg)	...	Kg	90.00	90.00	90.00	90.00	90.00

JOINERY

BASE MATERIAL PRICES

The following table indicates the comparative supply price of timbers in general use. As prices per cubic metre vary considerably depending upon section, the following can only give a guide as to the likely price differential.

All timbers, unless otherwise described, are selected grade, dressed all round.

Average trade (SUPPLY ONLY) prices :

	Nominal size	Jarrah	Local hardwood	Oregon	Pacific maple (Meranti)	Radiata pine	Silky oak	Tasmanian oak	Western red cedar
		$/m	$/m	$/m	$/m	$/m	$/m	$/m	$/m
Adelaide	50 x 25mm	8.10	-	-	2.01	3.90	-	9.20	9.45
	100 x 25mm	15.10	-	-	5.01	6.90	-	16.30	17.50
	150 x 38mm	33.90	-	-	17.79	11.35	-	37.00	36.60
	200 x 50mm	58.10	-	-	26.70	20.75	-	70.20	62.60
Brisbane	50 x 25mm	5.10	5.53	6.16	-	2.17	-	7.71	5.18
	100 x 25mm	11.07	10.20	12.31	-	4.35	-	15.43	10.37
	150 x 38mm	25.25	23.26	28.07	-	9.91	-	35.19	23.65
	200 x 50mm	44.32	40.81	49.25	-	17.40	-	61.74	41.49
Melbourne	50 x 25mm	5.50	3.70	2.95	2.05	2.35	-	4.50	5.75
	100 x 25mm	11.25	7.65	8.50	3.87	4.70	-	9.75	12.60
	150 x 38mm	25.50	21.50	16.00	16.48	10.25	-	28.00	28.75
	200 x 50mm	46.95	49.25	22.60	29.50	19.50	-	40.00	52.50
Perth	50 x 25mm	4.11	-	5.88	3.26	2.38	-	5.63	6.88
	100 x 25mm	8.75	-	11.75	6.53	4.75	-	11.25	13.75
	150 x 38mm	23.57	-	26.79	14.88	10.83	-	25.65	31.35
	200 x 50mm	45.26	-	47.00	26.10	19.00	-	45.00	55.00
Sydney	50 x 25mm	6.29	5.45	3.24	1.87	2.96	-	6.23	8.56
	100 x 25mm	12.57	10.88	5.59	3.95	5.15	-	12.45	17.12
	150 x 38mm	28.67	24.81	-	17.18	11.49	-	28.40	39.03
	200 x 50mm	69.55	69.66	21.73	32.23	23.55	-	49.82	68.48

JOINERY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
COVER MOULDS							
Quadrant fillet fixed to timber :							
13mm	...	m	7.65	9.25	9.45	7.65	10.60
19mm	...	m	7.15	8.70	8.85	7.95	10.55
25mm	...	m	7.00	8.60	8.75	8.50	10.45
Scotia fixed to timber :							
25mm	...	m	7.50	9.20	9.25	9.70	11.05
38mm	...	m	11.25	13.45	14.00	12.00	16.80
50mm	...	m	19.35	22.60	23.50	21.00	25.00
DOOR LININGS, ARCHITRAVES, BEADS, ETC.							
Refer pages 370 to 372	...	Note					
HAT AND COAT RAILS							
100 x 25mm hat and coat rail plugged to brickwork or concrete :							
Oregon	...	m	-	30.10	25.80	28.60	26.20
Local hardwood	...	m	30.10	26.30	24.20	24.20	32.20
Tasmanian oak	...	m	30.10	32.20	26.60	27.00	33.90
Pacific maple	...	m	17.45	-	20.00	21.70	24.40
MD Fibreboard	...	m	14.75	17.80	18.55	17.30	22.90
PELMETS							
Fascia or top :							
100 x 25mm -							
Oregon	...	m	-	25.10	21.90	24.10	22.10
Local hardwood	...	m	25.80	22.80	16.80	20.80	28.00
Tasmanian oak	...	m	27.20	28.60	23.30	23.60	29.70
Pacific maple	...	m	14.70	-	16.80	18.35	20.30
MD Fibreboard	...	m	12.05	14.45	15.40	14.05	18.80
150 x 25mm -							
Oregon	...	m	-	33.60	29.40	32.20	25.10
Local hardwood	...	m	35.90	30.10	21.50	28.10	36.00
Tasmanian oak	...	m	36.20	38.80	31.60	31.40	38.50
Pacific maple	...	m	18.75	-	21.50	23.60	26.80
MD Fibreboard	...	m	14.80	17.50	18.35	17.05	22.20
200 x 25mm -							
Oregon	...	m	-	42.00	36.20	40.30	30.10
Local hardwood	...	m	44.90	37.40	27.70	36.40	43.90
Tasmanian oak	...	m	46.10	48.90	43.10	39.20	47.40
Pacific maple	...	m	23.60	-	27.70	28.80	34.20
ADD extra for :							
Returned end	...	No	20.70	24.50	25.10	23.90	30.20
Standard bracket (average) fixed to wall	...	No	19.50	22.50	23.70	22.00	28.00

JOINERY

SKIRTINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Rounded or splayed fixed to timber :							
75 x 25mm -							
Oregon	...	m	-	19.65	15.15	18.85	17.65
Local hardwood	...	m	20.40	17.85	13.30	15.85	21.80
Tasmanian oak	...	m	19.75	22.30	17.20	18.45	23.10
Pacific maple	...	m	11.60	-	13.30	14.45	15.45
Western red cedar	...	m	22.60	18.00	20.20	20.50	27.10
MD Fibreboard	...	m	9.55	11.45	12.15	11.15	14.80
100 x 25mm -							
Oregon	...	m	-	23.90	20.70	22.90	20.40
Local hardwood	...	m	25.00	21.60	15.45	19.60	26.40
Tasmanian oak	...	m	26.40	27.40	22.10	22.40	28.10
Pacific maple	...	m	13.65	-	15.45	17.10	18.55
Western red cedar	...	m	27.70	21.70	25.30	25.20	33.40
MD Fibreboard	...	m	10.95	13.00	14.05	12.65	17.05
150 x 25mm -							
Oregon	...	m	-	32.50	28.30	31.10	23.40
Local hardwood	...	m	35.20	28.90	20.30	26.90	34.50
Tasmanian oak	...	m	35.50	37.80	30.50	30.30	37.10
Pacific maple	...	m	17.80	-	20.30	22.30	25.10
Western red cedar	...	m	39.40	29.20	35.80	34.50	45.00
MD Fibreboard	...	m	13.75	16.10	17.05	15.70	20.50
Moulded fixed to timber :							
150 x 25mm -							
Oregon	...	m	-	34.80	30.50	33.50	25.90
Local hardwood	...	m	37.40	31.30	22.60	29.20	37.00
Tasmanian oak	...	m	37.70	40.10	32.80	32.60	39.60
Pacific maple	...	m	19.95	-	22.60	24.70	27.60
Western red cedar	...	m	41.60	31.60	38.10	36.80	47.50
MD Fibreboard	...	m	15.90	18.45	19.35	18.05	23.00
ADD extra for :							
Fixing to brickwork or concrete including plugging	...	m	5.90	7.45	7.80	7.20	9.95

SEAT BATTENS

Battens screwed to metal :							
75 x 38mm -							
Oregon	...	m	-	26.50	21.00	25.40	-
Local hardwood	...	m	27.60	23.80	19.25	22.60	28.90
100 x 38mm -							
Oregon	...	m	-	32.60	26.10	31.20	-
Local hardwood	...	m	34.50	29.00	23.50	27.50	34.50

JOINERY

STAIRCASES, BALUSTRADES AND HANDRAILS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STAIRCASES							
All in select grade hardwood	...	Note					
Traditional pattern 1000mm wide straight flight staircase with closed treads and risers, simple alustrade one side and handrail the other	...	M/rise	1545.00	1815.00	1625.00	1680.00	1675.00
Ditto but in two flights with half space landing	...	M/rise	1735.00	2075.00	1855.00	1910.00	1970.00
Contemporary pattern 1000mm wide straight flight staircase with open treads, plain strings, simple balustrade one side and handrail the other	...	M/rise	1590.00	1895.00	1665.00	1750.00	1805.00
Ditto but in two flights with half space landing	...	M/rise	1860.00	2060.00	1845.00	2035.00	2140.00
Spiral with open timber treads, steel centre newel, simple balustrade with plastic capping :							
1200mm overall dia.	...	M/rise	1285.00	1285.00	1940.00	1630.00	1375.00
1500mm overall dia.	...	M/rise	1335.00	1520.00	2020.00	1800.00	1555.00
1800mm overall dia.	...	M/rise	1810.00	1905.00	2130.00	2080.00	1915.00
BALUSTRADES							
Balustrade 900mm high comprising 65 x 40mm moulded handrail and floor capping with 40 x 40mm turned balusters at 200mm centres	...	m	228.00	244.00	226.00	221.00	253.00
HANDRAILS							
Plain handrail with rounded edges, fixed to balustrade :							
150 x 50mm -							
Oregon	...	m	-	71.30	51.30	68.30	56.60
Local hardwood	...	m	73.20	64.10	55.20	66.90	76.60
Tasmanian oak	...	m	79.80	81.80	61.90	66.60	81.90
Pacific maple	...	m	42.20	-	55.20	50.70	68.10
200 x 38mm -							
Oregon	...	m	-	71.80	62.60	68.90	-
Local hardwood	...	m	73.70	64.60	57.20	65.80	99.40
Tasmanian oak	...	m	74.70	82.50	85.40	67.20	82.40
Pacific maple	...	m	50.30	-	57.20	51.00	66.10
200 x 50mm -							
Oregon	...	m	-	92.50	64.50	88.70	74.30
Local hardwood	...	m	94.80	83.10	72.30	86.80	128.00
Tasmanian oak	...	m	108.50	106.50	84.10	86.50	106.00
Pacific maple	...	m	59.50	-	72.30	65.20	86.10
ADD extra for :							
Mitred joint (average)	...	No	59.10	74.40	78.20	71.80	99.80
Standard metal bracket fixed to wall	...	No	34.30	40.50	42.00	39.40	50.60

JOINERY

SHELVING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
50 x 25mm pine ladder framing	...	m	17.50	19.00	19.85	18.65	25.30
Slat shelving of 50 x 25mm battens at 75mm centres :							
Pine	...	sqm	131.00	123.50	129.00	123.50	149.00
Oregon	...	sqm	-	183.50	138.00	176.00	168.50
Local hardwood	...	sqm	194.00	174.00	149.50	149.50	202.00
Tasmanian oak	...	sqm	210.50	207.00	161.50	172.50	213.50
Pacific maple	...	sqm	102.50	-	124.50	137.00	148.00
150 x 25mm shelving :							
Pine	...	m	22.70	22.10	22.90	22.30	28.60
Oregon	...	m	-	35.40	31.10	34.00	27.10
Local hardwood	...	m	37.40	31.90	23.20	29.80	38.10
Pacific maple	...	m	20.20	-	23.20	25.30	28.90
Western red cedar	...	m	41.60	32.20	38.50	37.30	48.50
MD Fibreboard	...	m	16.15	19.20	20.00	18.70	24.30
200 x 25mm shelving :							
Pine	...	m	29.80	26.20	26.30	26.50	37.40
Oregon	...	m	-	44.00	38.30	42.10	32.70
Local hardwood	...	m	46.50	39.20	29.80	38.20	46.70
Pacific maple	...	m	25.00	-	29.80	30.50	36.90
Western red cedar	...	m	52.00	39.60	53.40	46.60	60.60
16mm particle board edge stripped one side :							
300mm wide	...	m	19.00	22.90	23.80	22.20	29.20
450mm wide	...	m	23.40	27.90	29.10	27.10	35.30
600mm wide	...	m	27.50	32.50	33.90	31.70	40.90
16mm melamine faced particle board :							
300mm wide	...	m	19.55	23.40	24.30	22.70	29.70
450mm wide	...	m	24.20	28.70	29.80	27.90	36.10
600mm wide	...	m	28.60	33.50	35.00	32.70	42.00
Adjustable wall shelving, slotted steel framing strip :							
600mm long	...	No	13.60	15.30	16.00	15.00	18.50
900mm long	...	No	17.40	19.50	20.10	19.15	23.10
1200mm long	...	No	21.40	23.90	24.80	23.50	28.40
1800mm long	...	No	27.60	30.70	31.90	30.20	36.30
2400mm long	...	No	33.70	37.10	38.20	36.50	43.00
Interlocking shelf brackets to suit shelving, width :							
150mm	...	No	4.95	5.30	5.20	5.20	5.60
250mm	...	No	6.35	6.70	6.60	6.65	7.00
400mm	...	No	12.35	13.30	12.90	13.10	13.05
500mm	...	No	14.25	14.60	14.50	14.55	14.95
Steel shelving and shelving units refer pages 311 to 313	...	Note					

JOINERY

CUPBOARDS AND FITTINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
All custom made quality units	...	Note					
BENCH TOP, including HMR substrate (as required), fixed to top of cupboard unit or steel framing :							
Solid surface (12mm Corian, Staron, Hi-Macs et al)	...	sqm	750.00	900.00	750.00	740.00	760.00
Built-up/downturn edge profile	...	m	180.00	200.00	160.00	180.00	190.00
Cutouts for sinks and hotplates, top-mounted	...	No	100.00	150.00	85.00	105.00	90.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	200.00	250.00	150.00	180.00	155.00
Quartz surface (20mm Ceasarstone, Essastone, et al) :	...	sqm	750.00-950.00	700.00-900.00	750.00-950.00	800.00-1000.00	840.00-1040.00
Built-up/downturn edge profile	...	m	135.00	30.00	135.00	145.00	135.00
Cutouts for sinks and hotplates, top-mounted	...	No	75.00	150.00	90.00	70.00	155.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	200.00	250.00	200.00	200.00	210.00
Granite (20mm thick natural stone) :	...	sqm	1450.00	1600.00	1400.00	1400.00	1570.00
Built-up/downturn edge profile	...	m	135.00	40.00	145.00	145.00	195.00
Cutouts for sinks and hotplates, top-mounted	...	No	75.00	150.00	90.00	70.00	155.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	200.00	250.00	200.00	200.00	310.00
Marble (20mm thick natural stone) :	...	sqm	1450.00	2150.00	1550.00	1400.00	1880.00
Built-up/downturn edge profile	...	m	135.00	40.00	150.00	145.00	180.00
Cutouts for sinks and hotplates, top-mounted	...	No	75.00	150.00	90.00	70.00	115.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	200.00	250.00	200.00	200.00	310.00
Postformed laminate (35-40mm Laminex, Polytec, Wilsonart et al) :	...	sqm	500.00	500.00	500.00	490.00	550.00
Cutouts for sinks and hotplates, top-mounted	...	No	80.00	100.00	85.00	75.00	75.00
Stainless steel (type 304) :	...	sqm	1000.00	1250.00	1100.00	850.00	1150.00
Built-up/downturn edge profile	...	m	80.00	50.00	125.00	50.00	100.00
Cutouts for sinks and hotplates, top-mounted	...	No	100.00	150.00	90.00	75.00	115.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	350.00	250.00	250.00	475.00	308.00
Laminated solid timber (40mm sealed with 2 pack clear) :	...	sqm	1030.00	1150.00	1050.00	1060.00	1095.00
Cutouts for sinks and hotplates, top-mounted	...	No	110.00	125.00	90.00	105.00	75.00
Cutouts for sinks and hotplates, underslung (finished internal edge)	...	No	300.00	300.00	180.00	275.00	155.00
Glassfibre reinforced concrete (GRC)	...	sqm	1300.00	1400.00	1350.00	1260.00	1395.00
ADD extra for :							
Integral sinkbowl -							
Solid surface (12mm Corian, Staron)	...	No	650.00	600.00	700.00	625.00	780.00
Stainless steel 200 x 300 x 200mm	...	No	450.00	500.00	425.00	600.00	415.00
Stainless steel 400 x 550 x 250mm	...	No	550.00	600.00	525.00	625.00	520.00
Glassfibre reinforced concrete (GRC)	...	No	750.00	800.00	760.00	710.00	780.00

JOINERY

CUPBOARDS AND FITTINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Wall upstand/splashback applied to rendered wall :							
Colour backed glass (6mm toughened)	...	sqm	350.00	450.00	390.00	335.00	365.00
Colour backed "Starfire" ultra clear Glass (6mm toughened)	...	sqm	450.00	500.00	480.00	385.00	470.00
Solid surface (12mm Corian, Staron)	...	sqm	680.00	900.00	725.00	660.00	760.00
Granite (20mm thick natural stone)	...	sqm	1450.00	1600.00	1200.00	1400.00	1570.00
Marble (20mm thick natural stone)	...	sqm	1450.00	2150.00	1450.00	1400.00	1780.00
Stainless steel (type 304)	...	sqm	1000.00	1250.00	1100.00	170.00	1150.00
Ceramic tile splash, refer page 432	...	Note					
Floor based four drawer units - best quality - between 500 and 900mm wide :							
Carcass of 16mm HMR particleboard with 1mm ABS edging to seen edges, 100mm levelled recessed plinth, soft-close good quality full extension metal drawer sides/runners and good quality average price metal handle -							
Coloured/patterned melamine faced HMR MDF, 2mm ABS edging	...	m	1100.00	1100.00	1100.00	1060.00	1200.00
Lacquered finish to plain un-routed MDF	...	m	1260.00	1250.00	1280.00	1290.00	1305.00
Lacquered finish to pattern-routed MDF	...	m	1250.00	1300.00	1350.00	1340.00	1360.00
High gloss lacquered finish to plain un-routed MDF	...	m	1350.00	1350.00	1375.00	1400.00	1410.00
Polished timber veneer, 2mm multi-layered veneer edging	...	m	1450.00	1525.00	1495.00	1475.00	1515.00
Polished timber, solid framed with timber insert panel	...	m	1700.00	1850.00	1800.00	1790.00	1830.00
Carcass of 16mm HMR particleboard with 1mm ABS edging to seen edges, 100mm levelled recessed plinth, white HMR melamine ABS edged drawer boxes and good quality full extension metal drawer runners and good quality average price metal handle -							
Coloured/patterned melamine faced HMR MDF, 2mm ABS edging	...	m	870.00	850.00	875.00	820.00	900.00
Lacquered finish to plain un-routed MDF	...	m	1000.00	950.00	1055.00	1050.00	980.00
Lacquered finish to pattern-routed MDF	...	m	980.00	1000.00	1125.00	1100.00	1020.00
High gloss lacquered finish to plain un-routed MDF	...	m	1025.00	1050.00	1150.00	1160.00	1060.00
Polished timber veneer, 2mm multi-layered veneer edging	...	m	1200.00	1300.00	1270.00	1235.00	1135.00
Polished timber, solid framed with timber insert panel	...	m	1500.00	1625.00	1570.00	1550.00	1370.00

JOINERY

CUPBOARDS AND FITTINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Floor based cabinets (measured to back edge), excluding doors :							
400-600mm deep x 800mm high carcass (measured to back edge) entirely of 16mm HMR board with 1mm ABS edging to all seen edges with one intermediate adjustable/fixed shelf, supported on 100mm levelled recessed plinth	...	m	350.00	350.00	360.00	345.00	365.00
400-600mm deep x 2100mm high carcass (measured to back edge) entirely of 16mm HMR board with 1mm ABS edging to all seen edges with five intermediate adjustable/fixed shelves, supported on 100mm levelled recessed plinth	...	m	450.00	450.00	465.00	450.00	470.00
Allowance for dishwasher alcove	...	No	200.00	200.00	205.00	200.00	205.00
Overhead wall-mounted cabinets (measured to back edge), excluding doors :							
300-400mm deep x 800mm high carcass of 16mm HMR board with 1mm ABS edging to all seen edges (measured to back edge) and applied laminate to visible underside, soft-close, self-close hinges, metal handle	...	m	350.00	350.00	375.00	340.00	365.00
Lights beneath overhead cabinets :							
Recessed LED, from cabinet hardware supplier (measure to complete back edge)	...	m	180.00	180.00	180.00	180.00	180.00
ADD extra for :							
Electrical service	...	Note					
Doors and endpanels to all cabinets and integral appliances including best quality cast cup all metal soft-close, self-close hinges and good quality average price metal handle :							
Coloured/patterned melamine faced HMR MDF, 2mm ABS Edging	...	sqm	175.00	180.00	185.00	170.00	180.00
Lacquered finish to plain un-routed MDF	...	sqm	200.00	200.00	205.00	210.00	200.00
Lacquered finish to pattern-routed MDF	...	sqm	220.00	220.00	230.00	240.00	220.00
High gloss lacquered finish to plain un-routed MDF	...	sqm	245.00	240.00	250.00	285.00	245.00
Polished timber veneer, 2mm multi-layered veneer edging	...	sqm	330.00	350.00	280.00	310.00	265.00
Polished timber, solid framed with timber insert panel	...	sqm	400.00	415.00	375.00	380.00	340.00
Aluminium framed glass (clear or frosted)	...	sqm	230.00	260.00	250.00	220.00	235.00
ADD extra for :							
Cutouts for appliances (dishwasher, wall oven, microwave, extract etc.) may all be offset by applying the sqm rate to the entire front of the cabinet in which are accomodated	...	sqm	50.00	50.00	53.00	50.00	55.00
Allowance for dishwasher with matching door applied, include appropriate door at relevant sqm rate and include the following allowance for the dishwasher alone	...	sqm	225.00	230.00	230.00	230.00	235.00
Allowance for dishwasher alcove alone	...	No	200.00	200.00	205.00	210.00	205.00

JOINERY

CUPBOARDS AND FITTINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Built-in Wardrobes 2.4m high :							
Floor to ceiling fitted with no back or end/s where fitted against fully finished wall surface, and with no base, scribed around cornice, including one fixed hanging rail shelf	...	m	750.00	750.00	600.00	770.00	780.00
All as above with allowance made for hinged doors and exposed end panels as below :							
Coloured/patterned melamine faced HMR particleboard, 2mm ABS edging	...	m	950.00	950.00	930.00	940.00	990.00
Coloured/patterned melamine faced HMR MDF, 2mm ABS edging	...	m	1010.00	1000.00	980.00	1025.00	1095.00
Lacquered finish to plain un-routed MDF	...	m	1120.00	1100.00	1080.00	1140.00	1150.00
Lacquered finish to pattern-routed MDF	...	m	1160.00	1150.00	1130.00	1170.00	1200.00
Polished timber veneer, 2mm multi-layered veneer edging	...	m	1350.00	1400.00	1320.00	1320.00	1360.00
Polished timber, solid framed with timber insert panel	...	m	1600.00	1660.00	1570.00	1580.00	1670.00
All as above with allowance made for sliding doors as below :							
Powdercoated aluminium framed mirror glass	...	m	940.00	900.00	900.00	985.00	1045.00
Powdercoated aluminium framed coloured glass	...	m	1080.00	1100.00	1025.00	1060.00	1150.00
Powdercoated aluminium framed coloured/ patterned melamine faced HMR MDF	...	m	1100.00	950.00	1100.00	1280.00	1360.00
ADD extra for :							
Drawers, wire baskets, shoe racks, pullout hanging rails etc	...	Note					
Built-in Wardrobes 2.7M high :							
Floor to ceiling fitted with no back or end/s where fitted against fully finished wall surface, and with no base, scribed around cornice, including one fixed hanging rail shelf	...	m	880.00	850.00	720.00	865.00	885.00
All as above with allowance made for hinged doors and exposed end panels as below							
Coloured/patterned melamine faced HMR particleboard, 2mm ABS edging	...	m	1100.00	1050.00	1075.00	1115.00	1120.00
Coloured/patterned melamine faced HMR MDF, 2mm ABS edging	...	m	1150.00	1100.00	1100.00	1225.00	1235.00
Lacquered finish to plain un-routed MDF	...	m	1260.00	1250.00	1200.00	1285.00	1295.00
Lacquered finish to pattern-routed MDF	...	m	1310.00	1300.00	1275.00	1315.00	1360.00
Polished timber veneer, 2mm multi-layered veneer edging	...	m	1430.00	1500.00	1425.00	1410.00	1530.00
Polished timber, solid framed with timber insert panel	...	m	1760.00	1800.00	1750.00	1745.00	1885.00
All as above with allowance made for sliding doors as below :							
Powdercoated aluminium framed mirror glass	...	m	1100.00	1000.00	1100.00	1160.00	1180.00
Powdercoated aluminium framed coloured glass	...	m	1275.00	1250.00	1200.00	1285.00	1295.00
Powdercoated aluminium framed coloured/ patterned melamine faced HMR MDF	...	m	1130.00	1050.00	1200.00	1290.00	1330.00
ADD extra for :							
Drawers, wire baskets, shoe racks, pullout hanging rails etc	...	Note					

JOINERY

CUPBOARDS AND FITTINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LABORATORY							
Refer pages 688 to 692	...	Note					
LIBRARY							
Refer page 693	...	Note					
RETAIL							
Refer pages 694 to 696	...	Note					
RECEPTION							
RECEPTION desk :							
Medium standard	...	m	1750.00-2800.00	2000.00-3000.00	1800.00-3000.00	1715.00-3015.00	2230.00-3240.00
High standard	...	m	3500.00-5300.00	4000.00-6000.00	3500.00-5500.00	3530.00-5530.00	4570.00-6250.00
Prestige standard	...	m	6000.00-7900.00	6000.00-8500.00	6000.00-8250.00	6040.00-8540.00	7500.00-9350.00

NOTICE BOARDS

All units to manufacturers standard sizes	...	Note					
PINBOARD							
Plain colour melded fabric faced dense Caneite board with clear anodised aluminium trim	...	sqm	110.00	120.00	115.00	105.00	115.00
Designer patterned fabric faced dense Caneite board with clear anodised aluminium trim	...	sqm	150.00	150.00	145.00	155.00	120.00
WHITEBOARD (TRADITIONAL BEST QUALITY)							
Baked ceramic faced sheet steel bonded to particleboard backing with clear anodised aluminium trim	...	sqm	220.00	220.00	225.00	220.00	235.00
Baked ceramic faced sheet steel bonded to particleboard backing with clear anodised aluminium trim with integral custom graphic e.g. oil rig diagram emergency plans etc.	...	sqm	1550.00	1500.00	1050.00	1500.00	1045.00
WHITEBOARDS (6mm TOUGHENED GLASS)							
Plain white backed glass	...	sqm	345.00	350.00	355.00	335.00	365.00
Plain graphic printed back	...	sqm	445.00	450.00	410.00	435.00	415.00
Custom design graphic printed back	...	sqm	515.00	500.00	465.00	500.00	470.00
WHITEBOARDS INTERACTIVE							
Interactive whiteboard, 2400mm wide including short throw projector, writable screen and stand	...	No	4000.00	4000.00	4000.00	4000.00	4000.00
Electronic whiteboard 1800 x 900mm	...	No	3000.00	3000.00	3000.00	3000.00	3000.00
Touchscreens :							
1200mm wide	...	No	3000.00-7000.00	3000.00-7000.00	3000.00-7000.00	3000.00-7000.00	3000.00-7000.00
1800mm wide	...	No	6000.00-12000.00	6000.00-12000.00	6000.00-12000.00	6000.00-12000.00	6000.00-12000.00
2400mm wide	...	No	12000.00-25000.00	12000.00-25000.00	12000.00-25000.00	12000.00-25000.00	12000.00-25000.00

JOINERY

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TOILET PARTITIONS							
FULLY COMPRESSED CELLULAR FIBREBOARD							
18mm thick including fixing channels :							
Main panels	...	sqm	305.00	340.00	340.00	315.00	350.00
Pilaster or end post -							
150mm wide	...	m	150.00	138.00	136.00	129.00	136.00
300mm wide	...	m	183.00	184.00	178.00	171.00	172.00
1650 x 600mm wide standard hollow core door plywood faced to receive paint including self closing butt hinges, W.C. bolt, hook and buffer	...	No	395.00	445.00	415.00	405.00	415.00
ADD extra for :							
Finish to partitions and doors	...	Note					
HMR PARTICLE BOARD CORE							
25mm thick faced both sides and edges with laminated plastic :							
Floor anchored system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	270.00	270.00	260.00	265.00	275.00
Pilaster or end post x 1770mm long :							
150mm wide	...	No	430.00	455.00	435.00	435.00	455.00
300mm wide	...	No	460.00	500.00	465.00	460.00	490.00
Ceiling hung system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	270.00	270.00	260.00	265.00	275.00
Pilaster or end posts x 1980mm long :							
150mm wide	...	No	520.00	550.00	520.00	490.00	550.00
300mm wide	...	No	555.00	600.00	560.00	550.00	590.00
Overhead braced system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	270.00	270.00	260.00	265.00	275.00
Pilaster or end post x 2150mm long :							
150mm wide	...	No	520.00	550.00	520.00	490.00	550.00
300mm wide	...	No	555.00	600.00	560.00	550.00	590.00
Overhead bracing	...	m	33.00	34.00	32.50	31.50	34.00
Doors -							
Door to match including self-closing rising butt hinges, W.C. bolt, hook and buffer :							
1470mm x 600mm wide	...	No	520.00	535.00	530.00	495.00	560.00
1470mm x 800mm wide	...	No	550.00	565.00	560.00	530.00	585.00
ADD extra for :							
Stainless steel channel edging to panels and doors	...	m	52.00	55.00	53.00	52.00	57.00
Post formed coved internal corners	...	m	224.00	236.00	220.00	241.00	239.00

JOINERY

TOILET PARTITIONS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PHENOLIC CORE							
18mm thick solid phenolic core faced both sides with laminated plastic :							
Floor anchored system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	355.00	380.00	360.00	360.00	380.00
Pilaster or end post x 1770mm long :							
150mm wide	...	No	460.00	490.00	465.00	465.00	485.00
300mm wide	...	No	505.00	540.00	510.00	510.00	535.00
Ceiling hung system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	355.00	380.00	360.00	360.00	380.00
Pilaster or end post x 1980mm long :							
150mm wide	...	No	515.00	550.00	520.00	520.00	535.00
300mm wide	...	No	555.00	590.00	560.00	556.00	585.00
Overhead braced system -							
Main panel (based on 1500mm x 1470mm high)	...	sqm	355.00	380.00	360.00	360.00	380.00
Pilaster or end post x 2150mm long :							
150mm wide	...	No	515.00	550.00	520.00	520.00	535.00
300mm wide	...	No	555.00	590.00	560.00	560.00	585.00
Overhead bracing	...	m	26.00	27.00	26.00	26.00	28.00
Doors							
Door to match including self closing rising butt hinges, w.c. bolt, hook and buffer -							
1470mm x 600mm wide	...	No	565.00	605.00	570.00	560.00	585.00
1470mm x 800mm wide	...	No	640.00	690.00	645.00	630.00	650.00
COMPACT LAMINATE							
13mm thick panel of solid compact laminated panels faced both sides with laminated plastic :							
Floor anchored system -							
Main panel (based on 1800mm x 1800mm high)	...	sqm	210.00	205.00	210.00	210.00	215.00
Nib panel x 1800mm high :							
100mm wide	...	No	185.00	180.00	185.00	185.00	190.00
200mm wide	...	No	270.00	265.00	270.00	270.00	275.00
Doors :							
1800 x 600mm wide	...	No	425.00	420.00	425.00	425.00	430.00
1800 x 800mm wide	...	No	585.00	580.00	585.00	585.00	590.00
ADD extra for :							
Head rail x 1000mm long	...	No	85.00	85.00	85.00	85.00	85.00

JOINERY

SIGNS AND LETTERS

PRELAMINATED PANELS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
16mm thick prelaminated particle board panel with radiused corners and PVC edge stripping fixed to wall :							
300 x 300mm	...	No	46.00	47.00	48.50	46.00	50.00
600 x 300mm	...	No	74.50	75.00	76.50	74.00	79.50
900 x 300mm	...	No	108.00	110.00	112.00	108.00	115.00
1200 x 300mm	...	No	126.00	128.00	131.00	127.00	135.00
3.0mm thick natural anodised aluminium plate with radiused corners and arrised edges fixed with adhesive to wall :							
100 x 100mm	...	No	26.00	26.00	28.00	26.50	28.50
200 x 50mm	...	No	27.00	27.00	29.00	27.50	29.00
200 x 100mm	...	No	42.00	42.00	44.00	42.00	44.00
300 x 100mm	...	No	56.00	57.00	60.00	57.00	60.00
400 x 100mm	...	No	65.00	67.00	69.00	67.00	69.00

LETTERING

PVC self adhesive lettering :							
25mm high	...	No	2.75	2.85	2.90	2.80	3.00
50mm high	...	No	3.25	3.35	3.40	3.35	3.60
75mm high	...	No	4.10	4.20	4.25	4.15	4.40
100mm high	...	No	5.50	5.60	5.70	5.60	5.95
150mm high	...	No	7.15	7.25	7.35	7.25	7.55
PVC self adhesive arrow :							
25 x 25mm	...	No	2.75	2.85	2.90	2.80	3.00
50 x 50mm	...	No	3.25	3.35	3.40	3.35	3.60
75 x 75mm	...	No	4.10	4.20	4.25	4.15	4.40
100 x 100mm	...	No	5.50	5.60	5.70	5.60	5.95
150 x 150mm	...	No	7.15	7.25	7.35	7.25	7.55
Bronze lettering, plugged and screwed to brickwork :							
50mm high	...	No	37.50	38.50	39.50	38.00	40.50
75mm high	...	No	44.50	46.50	47.50	46.00	49.00
100mm high	...	No	51.00	53.00	54.00	52.00	55.50
150mm high	...	No	76.00	77.00	78.00	77.00	82.00
200mm high	...	No	102.00	104.00	107.00	103.00	111.00
Cut out acrylic lettering, plugged and screwed to brickwork :							
50mm high	...	No	15.50	16.00	16.50	15.50	17.00
75mm high	...	No	19.00	19.50	20.00	19.00	20.50
100mm high	...	No	22.00	22.50	23.00	22.00	24.00
150mm high	...	No	32.00	32.50	33.00	32.00	34.50
200mm high	...	No	43.50	44.00	44.50	44.00	47.50

SIGNWRITING

Refer page 454	...	Note					

JOINERY

SIGNS AND LETTERS (cont'd)

SILK SCREENED SIGNS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Silk screened printing of sign on prepared/finished board or plate background, size 1000 x 150mm high and comprising 15 to 20 and 50mm high letters :							
Preparation of plate and first sign	...	No	155.00	160.00	165.00	160.00	165.00
Additional signs, identical apart from minor numerical changes	...	No	7.30	7.30	7.40	7.20	7.50
ADD extra for :							
Board or plate background	...	Note					

WINDOWS AND DOORS

WINDOWS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for windows are based on stock pattern sections single glazed with clear glass unless otherwise stated, fixed in position and complete with any hardware	...	Note					
PROTECTIVE TREATMENTS							
Prices are per square metre of window measured overall including glass	...	Note					
Anodising, clear :							
0.025µm thick	...	sqm	14.50	16.10	13.10	15.40	12.50
Anodising, coloured :							
0.025µm thick	...	sqm	17.70	18.10	16.30	17.20	16.50
Polyester powder coat finish	...	sqm	17.70	16.30	17.35	14.40	18.25
Fluro polymer coating :							
Standard colours	...	sqm	38.70	34.70	39.40	35.90	47.00
Metallic colours	...	sqm	88.50	81.90	92.00	83.80	107.50
ALUMINIUM WINDOWS							
DOMESTIC STANDARD							
All powder coat finished	...	Note					
Adjust for protective treatments as noted above	...	Note					
Prices are for architectural standard windows	...	Note					
Standard catalogue size windows, reduce prices by 10% to 30%	...	Note					
Upgrading with heavier sections, prices could increase by 20% to 30%	...	Note					
Single glazed units :							
Fixed light	...	sqm	265.00	260.00	268.00	270.00	273.00
Awning -							
25% opening	...	sqm	319.00	345.00	324.00	310.00	341.00
50% opening	...	sqm	360.00	390.00	360.00	370.00	390.00
Casement -							
25% opening	...	sqm	320.00	345.00	361.00	310.00	343.00
50% opening	...	sqm	360.00	390.00	400.00	370.00	391.00
Dual hung sash	...	sqm	490.00	500.00	450.00	525.00	524.00
Sliding -							
25% opening	...	sqm	270.00	280.00	285.00	299.00	309.00
50% opening	...	sqm	315.00	300.00	300.00	309.00	324.00
Sealed double glazed unit comprising two panes of 6mm clear float glass :							
Fixed light	...	sqm	455.00	450.00	445.00	440.00	459.00
Awning -							
50% opening	...	sqm	550.00	570.00	540.00	539.00	575.00
Casement -							
50% opening	...	sqm	550.00	570.00	575.00	539.00	573.00
Sliding -							
50% opening	...	sqm	500.00	490.00	480.00	480.00	513.00

WINDOWS AND DOORS

WINDOWS (cont'd)

ALUMINIUM WINDOWS (cont'd)

COMMERCIAL STANDARD

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
All clear anodised	...	Note					
Adjust for protective treatments as noted above	...	Note					
Prices are those of windows for general commercial use	...	Note					
Adjust for :							
Custom designed and extra heavy duty windows, increase prices by 15% to 20%	...	Note					
Severe location loadings	...	Note					
Fixed light	...	sqm	500.00	550.00	520.00	495.00	562.00
Awning :							
25% opening	...	sqm	586.00	640.00	585.00	577.00	607.00
50% opening	...	sqm	650.00	700.00	650.00	601.00	680.00
Casement :							
25% opening	...	sqm	586.00	640.00	585.00	605.00	589.00
50% opening	...	sqm	650.00	700.00	650.00	671.00	637.00
Pivoted :							
50% opening	...	sqm	678.00	730.00	660.00	647.00	690.00
Sliding :							
25% opening	...	sqm	560.00	600.00	555.00	561.00	573.00
50% opening	...	sqm	585.00	620.00	575.00	500.00	606.00
ADD extra for :							
Timber sub-frame (average)	...	m	47.90	55.00	48.50	55.90	54.00
Alternative glass costs, refer pages 443 to 446	...	Note					
Insect screens, refer page 369	...	Note					

COMMERCIAL DOUBLE GLAZED

Refer Commercial Standard on page 364 for Notes	...	Note					
Fixed light	...	sqm	680.00	730.00	625.00	638.00	658.00
Vertically pivoted	...	sqm	788.00	825.00	740.00	741.00	765.00
ADD extra for :							
Integral venetian blinds	...	sqm	180.00	182.00	180.00	175.00	185.00

WINDOWS AND DOORS

WINDOWS (cont'd)

TIMBER WINDOWS - DOMESTIC

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Window wall with awning sashes :							
25% opening	...	sqm	361.00	390.00	354.00	350.00	346.00
50% opening	...	sqm	387.00	420.00	390.00	360.00	378.00
Fixed light	...	sqm	316.00	340.00	295.00	400.00	295.00
Fixed light short glazed	...	sqm	321.00	350.00	310.00	305.00	324.00
Awning sash :							
25% opening	...	sqm	421.00	450.00	380.00	330.00	390.00
50% opening	...	sqm	452.00	500.00	450.00	430.00	479.00
Side hung casement :							
25% opening	...	sqm	444.00	500.00	440.00	370.00	436.00
50% opening	...	sqm	464.00	540.00	495.00	410.00	527.00
Double hung sash	...	sqm	602.00	640.00	565.00	665.00	621.00
Bi-fold external doors	...	sqm	685.00-1000.00	700.00-1000.00	665.00-975.00	660.00-950.00	680.00-750.00
ADD extra for :							
Alternative glass costs, refer pages 443 to 446	...	Note					
Insect screens, refer page 369	...	Note					

HIGHLIGHT WINDOWS

Aluminium highlight, clear anodised :							
Fixed	...	sqm	520.00	550.00	470.00	495.00	516.00
Opening	...	sqm	560.00	600.00	514.00	536.00	557.00
Horizontally pivoted, 25% opening	...	sqm	560.00	600.00	514.00	574.00	557.00
Overlapping glazing as fixed ventilation	...	sqm	535.00	560.00	500.00	519.00	521.00
Timber framed highlight :							
Fixed	...	sqm	321.00	370.00	309.00	322.00	327.00
Sliding, 25% opening	...	sqm	420.00	470.00	424.00	322.00	450.00
Horizontally pivoted, 25% opening	...	sqm	420.00	470.00	386.00	340.00	451.00
Overlapping glazing as fixed ventilation	...	sqm	339.00	380.00	309.00	340.00	345.00
ADD extra for :							
Window control gearing -							
Manually operated :							
Awning windows -							
1 operator to open :							
1 awning	...	Set	460.00	470.00	470.00	450.00	470.00
2 awnings	...	Set	630.00	660.00	660.00	620.00	660.00
3 awnings	...	Set	800.00	840.00	840.00	780.00	840.00
4 awnings	...	Set	970.00	1020.00	1020.00	950.00	1020.00
5 awnings	...	Set	1160.00	1200.00	1200.00	1120.00	1200.00
6 awnings	...	Set	1320.00	1365.00	1365.00	1270.00	1365.00
Sliding sashes -							
1 operator to open :							
1 sliding sash	...	Set	410.00	420.00	420.00	390.00	420.00
2 sliding sashes	...	Set	510.00	530.00	530.00	490.00	530.00
3 sliding sashes	...	Set	640.00	660.00	660.00	610.00	660.00
4 sliding sashes	...	Set	770.00	805.00	805.00	740.00	805.00

WINDOWS AND DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
WINDOWS (cont'd)							
HIGHLIGHT WINDOWS (cont'd)							
Timber framed highlight (cont'd) -							
ADD extra for (cont'd) :							
Window control gearing -							
Electrically operated :							
Awning windows -							
1 operator to open :							
1 awning	…	Set	490.00	505.00	505.00	465.00	505.00
2 awnings	…	Set	870.00	885.00	885.00	820.00	885.00
3 awnings	…	Set	1060.00	1165.00	1165.00	1020.00	1165.00
4 awnings	…	Set	1220.00	1270.00	1270.00	1175.00	1270.00
Electrically operated :							
Sliding sashes -							
1 operator to open :							
1 sliding sash	…	Set	600.00	625.00	625.00	580.00	625.00
2 sliding sashes	…	Set	750.00	785.00	785.00	725.00	785.00
3 sliding sashes	…	Set	860.00	885.00	885.00	820.00	885.00
4 sliding sashes	…	Set	960.00	1005.00	1005.00	930.00	1005.00

ALUMINIUM CURTAIN WALLING

All clear anodised	...	Note					
Prices are those of curtain walling for general medium/high rise	...	Note					
Adjust for :							
Custom designed and upgraded curtain walling for special high rise, increase prices by 20% to 50%	...	Note					
Severe location loadings	...	Note					
Curtain wall framing only	...	sqm	535.00	550.00	545.00	510.00	548.00
ADD SPANDREL PANEL (x net area) :							
Coloured 6mm thick float glass	…	sqm	183.00	200.00	184.00	166.00	193.00
6mm thick toughened safety glass	…	sqm	345.00	350.00	345.00	312.00	371.00
Anodised ribbed aluminium sheet	…	sqm	345.00	350.00	343.00	348.00	371.00
Porcelain enamelled sheet	…	sqm	120.00	130.00	125.00	112.00	132.00
50mm thick insulation -							
Plain	…	sqm	27.00	30.00	28.00	26.00	29.00
Compressed	…	sqm	45.00	50.00	45.00	43.50	45.00
Alucobond panels	…	sqm	520.00	500.00	510.00	510.00	545.00
Black-out panels	…	sqm	43.00	45.00	42.00	39.00	43.00
WINDOW SINGLE GLAZED (x net area) -							
Fixed light	…	sqm	189.00	200.00	194.00	190.00	198.00
Vertically pivoted	…	sqm	432.00	400.00	445.00	455.00	454.00
Horizontally pivoted	…	sqm	467.00	400.00	478.00	455.00	489.00
WINDOW DOUBLE GLAZED (x net area) -							
Fixed light	…	sqm	385.00	400.00	390.00	400.00	385.00
Vertically pivoted	…	sqm	600.00	595.00	600.00	600.00	621.00
ADD extra for :							
Integral venetian blinds	...	sqm	178.00	180.00	183.00	177.00	187.00
Alternative glass costs, refer pages 443 to 446	...	Note					
Special glass	...	Note					

WINDOWS AND DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SHOP FRONTS							
COMMERCIAL STANDARD							
Prices are those of standard shop fronts for general commercial use	...	Note					
Adjust for :							
Custom designed and upgraded shop fronts, increase prices by 15% to 20%	...	Note					
Severe location loadings	...	Note					
Clear anodised aluminium shop fronts, standard section, glazed with 6mm toughened safety glass	...	sqm	660.00	650.00	670.00	665.00	725.00
ADD extra for :							
Doors, refer pages 375 to 377	...	Note					
BUDGET STANDARD							
Clear anodised shop fronts in lighter weight sections, glazed with :							
6mm clear float glass, generally	...	sqm	320.00	360.00	340.00	320.00	339.00
6mm toughened or laminated glass	...	sqm	395.00	420.00	430.00	410.00	453.00
FRAMELESS GLAZING							
Clear anodised finished aluminium surrounds glazed with 12mm clear float glass with -							
Butt joints silicone sealed	...	sqm	1060.00	1080.00	1080.00	1000.00	1090.00
Glass mullions (at 1200mm centres)	...	sqm	1195.00	1205.00	1200.00	1130.00	1220.00
Stainless steel surrounds and brackets glazed with 12mm toughened safety glass with -							
Butt joints silicone sealed	...	sqm	1480.00	1600.00	1575.00	1345.00	1825.00
Glass mullions (at 1200mm centres)	...	sqm	1595.00	1700.00	1700.00	1515.00	1960.00
PATENT GLAZING							
Standard system with bars at 600mm centres and glazed with 6mm white wired glass (vertical situations), in :							
Galvanised steel	...	sqm	330.00	380.00	375.00	320.00	417.00
Aluminium clear anodised	...	sqm	345.00	380.00	335.00	318.00	361.00
ADD extra for :							
6.38mm laminated glass (sloping situations)	...	sqm	75.00	78.00	79.00	67.00	75.00
INTERNAL GLAZED SCREENS							
COMMERCIAL STANDARD							
Clear anodised aluminium screen, fully glazed with :							
10mm clear float glass	...	sqm	480.00	500.00	525.00	525.00	554.00
10mm laminated glass	...	sqm	650.00	600.00	630.00	640.00	662.00
6mm toughened glass	...	sqm	565.00	575.00	580.00	605.00	630.00
Timber framed screen in solid rebated timber fully glazed with :							
10mm clear float glass	...	sqm	500.00	500.00	500.00	570.00	538.00
10mm laminated glass	...	sqm	612.00	600.00	600.00	665.00	639.00
6mm toughened glass	...	sqm	586.00	575.00	575.00	640.00	694.00

WINDOWS AND DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
INTERNAL GLAZED SCREENS (cont'd)							
BUDGET STANDARD							
Glazed screen comprising 10mm clear float glass set in anodised aluminium top and bottom channels and with silicone sealed joints ...		sqm	330.00-360.00	350.00-430.00	330.00-370.00	355.00-410.00	315.00-360.00
GLASSWALL SYSTEM							
Prices include for installation together with all fixings and sealants. ...		Note					
Rear wall panels to suit regulation size squash court 6401mm wide :							
Championship wall 2134mm high ...	- two panel	No	32000.00	32000.00	32000.00	32000.00	32000.00
...	- four panel	No	35000.00	35000.00	35000.00	35000.00	35000.00
Club wall 2134mm high ...	- four panel	No	*	*	*	*	*
Extended height Championship or Club wall 3688mm high ...		No	*	*	*	*	*
* Price on application							
LOUVRED PANELS, WINDOWS							
LOUVRED PANELS							
SINGLE STAGE TYPE							
Fixed weatherproof louvre blades and frame :							
Steel -							
Galvanised ...		sqm	390.00	386.00	390.00	414.00	389.00
Colorbond finish ...		sqm	420.00	416.00	410.00	441.00	419.00
Aluminium -							
Mill finish ...		sqm	398.00	388.00	400.00	428.00	397.00
Clear anodised finish ...		sqm	562.00	563.00	560.00	600.00	554.00
Powder coat finish ...		sqm	620.00	603.00	603.00	699.00	602.00
ADD extra for :							
Birdproof backing (12.5 x 12.5 x 0.80mm galvanised Weldmesh) ...		sqm	35.00	35.00	35.00	35.00	36.00
TWO STAGE TYPE							
Fixed weatherproof louvre blades and frame :							
Steel -							
Galvanised ...		sqm	610.00	603.00	608.00	654.00	597.00
Colorbond finish ...		sqm	714.00	707.00	710.00	766.00	703.00
Aluminium -							
Mill finish ...		sqm	625.00	617.00	612.00	669.00	604.00
Clear anodised finish ...		sqm	995.00	912.00	954.00	1035.00	986.00
Powder coat finish ...		sqm	1030.00	1045.00	1015.00	1125.00	1035.00
ADD extra for :							
Birdproof backing (12.5 x 12.5 x 0.80mm galvanised Weldmesh) ...		sqm	35.00	35.00	35.00	35.00	36.00

WINDOWS AND DOORS

LOUVRED PANELS, WINDOWS (cont'd)

LOUVRED WINDOWS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Louvred window comprising clear glass blades, aluminium louvre frame and timber surround size :							
640mm high x 1000mm wide with -							
7 fixed blades	...	No	620.00	610.00	585.00	600.00	630.00
4 adjustable blades	...	No	505.00	445.00	430.00	470.00	435.00
920mm high x 1000mm wide with -							
11 fixed blades	...	No	870.00	805.00	785.00	835.00	795.00
1060mm high x 1000mm wide with -							
7 adjustable blades	...	No	775.00	695.00	680.00	730.00	685.00

INSECT SCREENS

Powder coat finished aluminium surround with fibreglass gauze :							
Hinged	...	sqm	55.00	55.50	59.00	55.00	60.00
Sliding	...	sqm	61.00	61.50	64.00	60.00	65.00

DOOR FRAMES, LININGS, ETC.

METAL FRAMES

1.25mm Zinc annealed knock-down door frame, twice rebated, built into 90/110mm brick or concrete wall and grouted up solid, to suit door size :							
2040 x 820mm	...	No	137.00	145.50	143.50	154.50	154.00
2040 x 870mm	...	No	137.00	145.50	143.50	154.50	154.00
2040 x 1240mm	...	No	155.00	164.50	162.00	174.50	174.00
2040 x 1640mm	...	No	158.50	168.50	166.00	178.50	179.00
1.25mm Zinc annealed door frame with welded joints, four times rebated, built into 90/110mm brick or concrete wall and grouted up solid, to suit door size :							
2040 x 820mm	...	No	107.50	117.50	115.00	122.50	128.00
2040 x 870mm	...	No	107.50	117.50	115.00	122.50	128.00
2040 x 1240mm	...	No	118.00	129.00	126.00	134.00	140.00
2040 x 1640mm	...	No	124.50	143.50	137.00	147.50	150.00
2340 x 820mm	...	No	124.50	136.50	133.50	142.00	148.50
2340 x 1640mm	...	No	158.50	182.00	173.50	187.50	189.00
2640 x 820mm	...	No	141.50	155.00	151.50	161.50	169.00
2640 x 1640mm	...	No	189.00	206.00	202.00	215.00	223.00
ADD extra for :							
Transom to form fanlight or the like	...	m	30.20	33.00	32.30	34.40	36.00
Frame in 190/230mm wall	...	+ 7.5%					

WINDOWS AND DOORS

DOOR FRAMES, LININGS, ETC. (cont'd)

TIMBER LININGS, FRAMES, ETC.

LININGS AND FRAMES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include fixing to brickwork	...	Note					
100 x 25mm plain lining :							
Oregon	...	m	-	34.10	31.10	32.80	33.10
Local hardwood	...	m	33.50	32.70	26.00	29.50	38.90
Pacific maple	...	m	22.40	-	26.00	27.10	31.30
Western red cedar	...	m	36.10	31.90	35.70	35.00	45.80
MD Fibreboard	...	m	19.70	23.40	24.60	22.70	29.80
150 x 25mm plain lining :							
Oregon	...	m	-	41.40	37.60	39.80	34.70
Local hardwood	...	m	42.70	39.40	29.80	35.70	45.60
Pacific maple	...	m	25.60	-	29.80	31.20	36.40
Western red cedar	...	m	46.80	38.20	44.90	43.20	55.90
MD Fibreboard	...	m	21.60	25.30	26.60	24.70	31.90
100 x 38mm plain lining :							
Oregon	...	m	-	41.70	35.60	40.10	-
Local hardwood	...	m	42.10	39.60	33.00	36.40	45.80
Pacific maple	...	m	30.40	-	33.00	31.40	41.40
Western red cedar	...	m	44.10	38.40	44.10	43.50	56.20
150 x 38mm plain lining :							
Oregon	...	m	-	52.50	41.00	50.50	-
Local hardwood	...	m	55.10	49.40	41.50	46.90	56.20
Pacific maple	...	m	37.40	-	41.50	37.30	47.80
Western red cedar	...	m	58.10	47.70	55.00	55.50	71.90
100 x 50mm window or door frame, twice rebated :							
Oregon	...	m	-	55.10	41.10	52.60	43.90
Local hardwood	...	m	55.20	52.40	43.30	50.40	59.30
Pacific maple	...	m	41.00	-	43.30	41.10	54.10
Western red cedar	...	m	57.60	50.80	56.10	57.00	73.00
100 x 75mm window or door frame, twice rebated :							
Oregon	...	m	-	70.20	50.80	67.00	58.00
Local hardwood	...	m	77.30	66.10	58.40	62.40	94.70
Pacific maple	...	m	56.50	-	58.40	49.70	70.60
Western red cedar	...	m	81.80	63.80	74.70	73.60	93.70
150 x 50mm window or door frame, twice rebated :							
Oregon	...	m	-	70.20	50.90	67.00	53.50
Local hardwood	...	m	72.80	66.10	54.70	65.60	73.00
Pacific maple	...	m	42.40	-	54.70	49.70	64.70
Western red cedar	...	m	76.60	63.80	70.70	73.60	93.70
150 x 75mm window or door frame, twice rebated :							
Oregon	...	m	-	94.40	-	90.20	78.90
Local hardwood	...	m	103.00	88.30	-	83.20	128.50
Pacific maple	...	m	72.90	-	-	64.20	-
Western red cedar	...	m	114.00	84.80	101.50	100.00	127.00

WINDOWS AND DOORS

DOOR FRAMES, LININGS, ETC. (cont'd)

TIMBER LININGS, FRAMES, ETC. (cont'd)

LININGS AND FRAMES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
100 x 50mm mullion or transom, twice rebated :							
Oregon	…	m	-	59.20	45.60	56.60	49.20
Local hardwood	…	m	58.60	56.50	47.70	54.40	64.60
Pacific maple	…	m	44.50	-	47.70	45.10	59.40
Western red cedar	…	m	61.10	54.90	60.60	61.00	78.30
150 x 50mm mullion or transom, twice rebated :							
Oregon	…	m	-	73.90	55.00	70.60	58.30
Local hardwood	…	m	76.00	69.80	58.80	69.10	77.80
Pacific maple	…	m	45.60	-	58.80	53.30	69.50
Western red cedar	…	m	79.70	67.40	74.80	77.20	98.50
150 x 75mm mullion or transom, twice rebated :							
Oregon	…	m	-	96.90	-	92.70	81.90
Local hardwood	…	m	105.00	90.80	-	85.80	131.50
Pacific maple	…	m	75.20	-	-	66.70	-
Western red cedar	…	m	116.50	87.30	104.50	102.50	130.00
100 x 75mm sill, once rebated, weathered and throated :							
Oregon	…	m	-	75.80	56.60	72.30	63.80
Local hardwood	…	m	82.20	71.80	64.20	67.60	100.50
Pacific maple	…	m	61.30	-	64.20	55.00	76.50
Western red cedar	…	m	86.60	69.40	80.50	78.90	99.50
150 x 75mm sill, once rebated, weathered and throated :							
Oregon	…	m	-	101.50	-	96.60	86.00
Local hardwood	…	m	109.00	95.10	-	89.70	135.50
Pacific maple	…	m	78.90	-	-	70.70	-
Western red cedar	…	m	120.00	91.60	108.50	106.50	134.00

ARCHITRAVES

50 x 25mm rounded or splayed architrave fixed to timber :							
Oregon	…	m	-	13.40	10.25	12.85	12.40
Local hardwood	…	m	14.30	12.70	9.25	10.90	14.90
Pacific maple	…	m	7.45	-	9.25	9.95	10.85
Western red cedar	…	m	15.80	12.30	13.40	14.00	18.40
MD Fibreboard	…	m	6.65	7.95	8.40	7.70	10.20
75 x 25mm rounded or splayed architrave fixed to timber :							
Oregon	…	m	-	17.25	12.50	16.55	14.30
Local hardwood	…	m	18.55	16.20	10.70	13.60	18.45
Pacific maple	…	m	9.75	-	10.70	12.15	12.10
Western red cedar	…	m	20.50	15.65	17.60	18.25	23.70
MD Fibreboard	…	m	7.70	9.10	9.50	8.85	11.45

WINDOWS AND DOORS

DOOR FRAMES, LININGS, ETC. (cont'd)

TIMBER LININGS, FRAMES, ETC. (cont'd)

ARCHITRAVES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
100 x 25mm rounded or splayed architrave fixed to timber :							
Oregon	...	m	-	21.10	17.25	20.30	16.05
Local hardwood	...	m	22.80	19.75	12.05	16.90	22.00
Pacific maple	...	m	11.50	-	12.05	14.40	14.20
Western red cedar	...	m	25.50	18.95	21.90	22.50	29.00
MD Fibreboard	...	m	8.75	10.20	10.65	10.00	12.65
ADD extra for :							
Fixing to brickwork or concrete including plugging	...	m	5.20	7.25	6.20	6.35	7.30

GLAZING BEADS, STOPS, FILLETS

19 x 13mm glazing bead mitred and bradded :							
Oregon	...	m	-	10.75	9.05	10.35	11.15
Local hardwood	...	m	10.70	10.35	8.40	9.15	12.65
Pacific maple	...	m	6.60	-	8.40	8.60	10.25
Western red cedar	...	m	11.60	10.10	10.90	11.00	14.75
25 x 13mm glazing bead mitred and bradded :							
Oregon	...	m	-	12.30	10.15	11.80	12.50
Local hardwood	...	m	12.30	11.80	9.45	10.40	14.25
Pacific maple	...	m	7.50	-	9.45	9.75	11.45
Western red cedar	...	m	13.35	11.50	12.35	12.60	16.70
ADD extra for :							
Fixing with brass screws and cups	...	m	3.70	3.75	3.75	3.65	3.95
19mm quadrant fillet :							
Oregon	...	m	-	11.05	8.95	10.60	11.05
Local hardwood	...	m	11.30	10.55	8.25	9.20	12.75
Pacific maple	...	m	6.50	-	8.25	8.55	9.95
Western red cedar	...	m	12.35	10.30	11.15	11.40	15.20
38 x 19mm door stop or fillet :							
Oregon	...	m	-	12.55	10.10	12.05	12.40
Local hardwood	...	m	12.85	12.00	9.25	10.50	14.35
Pacific maple	...	m	7.40	-	9.25	9.70	11.15
Western red cedar	...	m	14.05	11.70	12.60	12.95	17.15
50 x 25mm door stop or fillet :							
Oregon	...	m	-	14.35	11.15	13.80	13.60
Local hardwood	...	m	15.00	13.65	10.15	11.80	16.10
Pacific maple	...	m	8.15	-	10.15	10.85	12.10
Western red cedar	...	m	16.50	13.25	14.30	14.90	19.60

WINDOWS AND DOORS

DOORS

TIMBER

Prices include for hinges and hanging to frames but exclude supply and fixing of hardware and painting ... Note

HOLLOW CORE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Standard hollow core flush door size 2040 x 820 x 35mm thick, faced both sides with :							
Prime coated hardboard for paint finish	...	No	138.00	170.00	180.00	168.00	212.00
Waterproof plywood for paint finish	...	No	161.00	193.00	224.00	167.50	256.00
Sliced Pacific maple veneer	...	No	168.50	200.00	208.00	230.00	258.00
Non-standard size 35mm thick hollow core flush door faced both sides with :							
Prime coated hardboard for paint finish	...	sqm	110.50	136.00	144.00	134.50	169.50
Waterproof plywood for paint finish	...	sqm	128.50	154.00	179.00	134.00	204.00
Sliced Pacific maple veneer	...	sqm	134.50	160.50	166.50	183.50	206.50
ADD extra for :							
40mm thick in lieu of 35mm thick standard door	...	No	41.60	25.50	19.85	40.60	41.60
Edge strips to 2040 x 820mm door -							
2 edges	...	No	47.20	60.10	51.60	48.00	60.30
4 edges	...	No	80.40	112.90	106.90	105.10	111.20
Top 750mm of 2040 x 820mm door glazed with 4mm clear glass	...	No	96.60	97.70	101.00	99.80	101.00
Form 600 x 300mm air relief grille opening (refer page 296 for grilles)	...	No	56.90	57.90	60.30	59.20	59.90

SOLID CORE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Standard solid core flush door size 2040 x 820 x 35mm thick, faced both sides with :							
Prime coated tempered hardboard for paint finish	...	No	200.00	218.00	242.00	230.00	268.00
Plywood for paint finish	...	No	239.00	249.00	261.00	261.00	306.00
Sliced Pacific maple veneer	...	No	250.00	260.00	271.00	275.00	317.00
Non-standard size 35mm thick solid core flush door faced both sides with :							
Prime coated tempered hardboard for paint finish	...	sqm	160.50	174.00	194.00	184.00	214.00
Plywood for paint finish	...	sqm	191.00	199.00	209.00	209.00	245.00
Sliced Pacific maple veneer	...	sqm	200.00	208.00	217.00	220.00	254.00
ADD extra for :							
40mm thick in lieu of 35mm thick standard door	...	No	41.60	25.50	19.85	40.60	41.60
Edge strips to 2040 x 820mm door -							
2 edges	...	No	47.20	60.10	51.60	48.00	60.10
4 edges	...	No	76.30	112.90	106.90	105.10	111.20
Top 750mm of 2040 x 820mm door glazed with 4mm clear glass	...	No	98.70	99.80	103.00	102.00	103.00
Form 600 x 300mm air relief grille opening (refer page 296 for grilles)	...	No	59.00	60.00	62.40	61.30	62.00

WINDOWS AND DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DOORS (cont'd)							
TIMBER (cont'd)							
PANELLED							
Four panel door size 2040 x 820 x 40mm in sliced Pacific maple, of stile and rail construction with veneered fielded panels	...	No	329.00	395.00	376.00	377.00	439.00
ADD extra for :							
Rebated meeting stile to pair of doors	...	Pair	52.10	61.80	55.30	58.60	61.80
FRAMED AND LEDGED							
Framed and ledged door size 2040 x 820 x 40mm in hardwood with T & G v-jointed vertical boarding set in	...	No	625.00	630.00	635.00	620.00	660.00
FRAMED, LEDGED AND BRACED							
Framed, ledged and braced door size 2040 x 820 x 40mm in hardwood with T & G v-jointed vertical boarding set in	...	No	660.00	635.00	670.00	665.00	690.00
LOUVRED							
Fully louvred door in softwood :							
2040 x 720 x 33mm	...	No	177.00	211.00	194.00	206.00	244.00
2040 x 820 x 33mm	...	No	184.50	232.00	202.00	215.00	255.00
Fully louvred screen door in softwood :							
2040 x 520 x 29mm	...	No	141.50	172.50	170.00	164.00	196.00
2040 x 620 x 29mm	...	No	149.50	180.50	178.00	174.00	206.00
Fully louvred pair of batwing doors in softwood :							
915 x 820 x 29mm	...	No	191.50	245.00	249.00	215.00	259.00
GLAZED							
Single panel door size 2040 x 820mm, Meranti framed including glazing with clear safety glass	...	No	376.00	432.00	419.00	412.00	479.00
Two panel door size 2040 x 820mm, Meranti framed including glazing with clear safety glass	...	No	416.00	375.00	256.00	365.00	495.00
ACOUSTIC DOORS							
Acoustic rated door and frame comprising door faced both sides with veneered particle board and timber jambs with adjustable stops and seals, including fixing only hardware supplied by others :							
Standard single door 2040 x 820mm -							
STC 28 dB	...	No	3380.00	3390.00	3390.00	3330.00	3340.00
STC 30 dB	...	No	3400.00	3390.00	3390.00	3370.00	3350.00
STC 34 dB	...	No	4710.00	4690.00	4680.00	4640.00	4620.00
STC 37 dB	...	No	4660.00	4660.00	4660.00	4610.00	4580.00
STC 40 dB	...	No	5030.00	5010.00	5010.00	4950.00	4910.00
STC 43 dB	...	No	5310.00	5290.00	5290.00	5250.00	5200.00
STC 47 dB	...	No	6220.00	6190.00	6190.00	6160.00	6040.00

WINDOWS AND DOORS

DOORS (cont'd)

TIMBER (cont'd)

ACOUSTIC DOORS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Standard double door 2040 x 1640mm :							
STC 28 dB	...	No	3780.00	3820.00	3830.00	3740.00	3770.00
STC 30 dB	...	No	3750.00	3790.00	3800.00	3700.00	3720.00
STC 34 dB	...	No	6310.00	6350.00	6320.00	6240.00	6110.00
STC 37 dB	...	No	6380.00	6380.00	6350.00	6280.00	6260.00
STC 40 dB	...	No	9090.00	9050.00	9050.00	8960.00	8850.00
STC 43 dB	...	No	9590.00	9500.00	9500.00	9460.00	9330.00
STC 47 dB	...	No	11400.00	11300.00	11250.00	11250.00	9850.00
Non-standard size single doors are specials	...	Note					
ADD extra for :							
Supply of hardware	...	Note					
Sub-frame when in framed walls	...	Note					
Architraves to frames	...	Note					

METAL

GLAZED ENTRY DOORS

HINGED, SLIDING

All in clear anodised aluminium unless otherwise described	...	Note					
DOMESTIC							
Sliding, standard size, door unit complete :							
2100 x 1800mm wide with one 900mm wide sliding panel	...	No	640.00	650.00	659.00	600.00	744.00
2100 x 2400mm wide with one 1200mm wide sliding panel	...	No	800.00	850.00	850.00	800.00	854.00
2100 x 3600mm wide with one 1200mm wide sliding panel	...	No	1190.00	1250.00	1185.00	1175.00	1170.00
COMMERCIAL							
Single hinged door and frame, 2100 x 900mm in one panel and fitted with surface mounted closer, lock etc., glazed with 6mm toughened safety glass :							
Architectural standard	...	No	2650.00	2600.00	2660.00	2760.00	2785.00
Budget standard	...	No	2350.00	1900.00	1855.00	1780.00	1730.00
Double hinged door and frame, 2100 x 1800mm each leaf in one panel and fitted with surface mounted closer, lock etc., glazed with 6mm toughened safety glass :							
Architectural standard	...	No	4200.00	4200.00	4070.00	3940.00	4405.00
Budget standard	...	No	2700.00	2950.00	2730.00	2770.00	2745.00

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

GLAZED ENTRY DOORS (cont'd)

HINGED, SLIDING (cont'd)

COMMERCIAL (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Single sliding door in one panel installed in prepared opening and complete with sliding track, lock etc. :							
6mm toughened safety glass -							
2100 x 900mm	...	No	3200.00	3200.00	3200.00	3350.00	3370.00
2100 x 1200mm	...	No	3450.00	3500.00	3400.00	3500.00	3535.00
2100 x 1500mm	...	No	3600.00	3800.00	3545.00	3680.00	3755.00
Double sliding door 2100 x 1800mm each leaf in one panel installed in prepared opening and complete with sliding track, lock, etc. :							
6mm toughened safety glass	...	No	4500.00	4600.00	4690.00	4440.00	4915.00
ADD extra for :							
Mid rail to door -							
900mm wide	...	No	125.00	165.00	120.00	121.00	124.00
1200mm wide	...	No	180.00	197.00	177.00	198.00	183.00
1500mm wide	...	No	240.00	244.00	248.00	254.00	257.00
1800mm wide	...	No	275.00	266.00	281.00	291.00	291.00
Fixed side or intermediate panel including framing, glazed with :							
6mm toughened safety glass	...	sqm	435.00	475.00	450.00	481.00	462.00
Hyperwave/microwave sensor opening device to :							
Sliding door -							
Single	...	No	7000.00	7000.00	7000.00	7000.00	7000.00
Double	...	No	7650.00	7650.00	7650.00	7650.00	7650.00
Hinged door -							
Single	...	No	8800.00	8800.00	8800.00	8800.00	8800.00
Double	...	No	9300.00	9300.00	9300.00	9300.00	9300.00

REVOLVING DOORS

COMMERCIAL QUALITY

3 leaf manual revolving door, leaves of 6mm toughened safety glass, sidelights of 8.38mm curved laminated glass, unit 2100mm high with 325mm canopy over door :							
2100mm dia. -							
Aluminium	...	Set	50000.00	52500.00	50000.00	48000.00	54000.00
Brass	...	Set	74500.00	78500.00	75000.00	72500.00	80500.00
Stainless steel	...	Set	63000.00	67000.00	65000.00	62000.00	69000.00
Timber and brass	...	Set	71500.00	74000.00	70500.00	69000.00	76500.00
2400mm dia. -							
Aluminium	...	Set	53000.00	56000.00	56000.00	52000.00	57500.00
Brass	...	Set	77500.00	82000.00	80000.00	76000.00	84000.00
Stainless steel	...	Set	68500.00	72500.00	67500.00	67500.00	75000.00
Timber and brass	...	Set	74500.00	79000.00	80000.00	73000.00	82000.00

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

GLAZED ENTRY DOORS (cont'd)

REVOLVING DOORS (cont'd)

COMMERCIAL QUALITY (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
3 leaf manual revolving door, leaves of 6mm toughened safety glass, sidelights of 8.38mm curved laminated glass, unit 2100mm high with 325mm canopy over door (cont'd) :							
2700mm dia. -							
Aluminium	...	Set	61000.00	64500.00	64500.00	60000.00	67000.00
Brass	...	Set	88500.00	93000.00	93000.00	86500.00	95500.00
Stainless steel	...	Set	76500.00	81000.00	80000.00	75000.00	84000.00
Timber and brass	...	Set	84500.00	90000.00	88000.00	83000.00	92500.00
ADD extra for :							
Automatic controls	...	Set	16000.00	16000.00	16500.00	16000.00	16000.00
3 leaf automatic revolving door (Windlock), leaves of 8.38mm glass, sidelights of 10mm clear toughened segmented glass, unit 2100mm high with 400mm high canopy over, door :							
Stainless steel -							
2400mm dia.	...	Set	95000.00	100000.00	93500.00	94500.00	90000.00
2600mm dia.	...	Set	95000.00	100000.00	93500.00	94500.00	90000.00
2800mm dia.	...	Set	95000.00	100000.00	93500.00	94500.00	90000.00
3000mm dia.	...	Set	101000.00	106000.00	99000.00	101000.00	96000.00

HEAVY DUTY AUTOMATIC

4 leaf revolving door, leaves of 8.38mm clear laminated glass, curved 8.38mm laminated glass to centre carousel and 10mm clear toughened segmented glass to sidelights, unit 2100mm high with 400mm canopy over, door :							
Stainless steel -							
3600mm dia.	...	Set	155000.00	161000.00	150500.00	155500.00	146000.00
4200mm dia.	...	Set	169500.00	177000.00	162000.00	172500.00	157000.00
4800mm dia.	...	Set	192000.00	200000.00	183000.00	194500.00	177000.00
ADD extra for :							
Computer integration, custom requirements, etc.	...+ 15%-20%						

DISABLED PERSONS DOOR

Disabled persons door and frame unit to provide a clear opening of 2000 x 920mm wide comprising steel flush finish single action pivot door and steel sub-frame together with hinges, hardware, indicator bolts etc. complete	...	No	3400.00	3400.00	3500.00	3440.00	3400.00
ADD extra for :							
Automatic release button fully installed	...	No	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00	2000.00-3000.00

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

DUCT ACCESS DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fire rated door and frame comprising Zincanneal® Steel sheet clad door with frame, self closing mechanism, baked enamel finish, built-in, size :							
2040 x 820mm wide - one hour rated	...	No	1415.00	1410.00	1450.00	1380.00	1470.00

SHEET STEEL CLAD DOORS

Sheet steel clad door and frame unit comprising flush finished door with rigid and reinforced core and fully welded frame together with heavy duty hinges and deadlock :							
2040 x 820mm single	...	No	1400.00	1440.00	1420.00	1440.00	2000.00
2040 x 1640mm double	...	No	2500.00	2550.00	2510.00	2540.00	4000.00

ROLLER DOORS

DOMESTIC/COMMERCIAL

Ribbed roller door (continuous curtain) :							
Zincanneal® Steel -							
2400 x 2515mm wide	...	No	1550.00	1585.00	1575.00	1580.00	1620.00
2400 x 3000mm wide	...	No	1835.00	1880.00	1865.00	1870.00	1920.00
3000 x 3000mm wide	...	No	2050.00	2100.00	2085.00	2090.00	2150.00
Pre-enamelled finish one side -							
2400 x 2515mm wide	...	No	1625.00	1660.00	1650.00	1655.00	1695.00
2400 x 3000mm wide	...	No	1920.00	1960.00	1950.00	1955.00	2005.00
3000 x 3000mm wide	...	No	2145.00	2195.00	2180.00	2185.00	2245.00
ADD extra for :							
Centre mullion -							
2400mm high	...	No	281.00	293.00	290.00	291.00	305.00
3000mm high	...	No	301.00	313.00	310.00	311.00	325.00
Electrically operated opening mechanism	...	Sum	780.00	815.00	805.00	810.00	850.00

INDUSTRIAL

Continuous curtain roller door, chain operated :							
Zincanneal® Steel -							
2700 x 2700mm wide	...	No	2915.00	3000.00	2975.00	2985.00	3085.00
3000 x 3000mm wide	...	No	3055.00	3155.00	3130.00	3140.00	3260.00
3600 x 3600mm wide	...	No	3725.00	3840.00	3810.00	3820.00	3960.00
Pre-enamelled or in situ paint finish -							
2700 x 2700mm wide	...	No	3130.00	3215.00	3195.00	3200.00	3300.00
3000 x 3000mm wide	...	No	3290.00	3395.00	3365.00	3375.00	3495.00
3600 x 3600mm wide	...	No	4070.00	4190.00	4155.00	4170.00	4310.00
ADD extra for :							
Centre mullion -							
3000mm high	...	No	675.00	690.00	685.00	685.00	700.00
3600mm high	...	No	750.00	765.00	760.00	760.00	775.00
Electrically operated opening mechanism	...	Sum	1390.00	1475.00	1455.00	1460.00	1560.00

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

ROLLER DOORS (cont'd)

INDUSTRIAL (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Interlocking slat, 1mm thick, roller door, chain operated :							
Zincanneal® Steel -							
3000 x 3000mm wide	…	No	3570.00	3660.00	3635.00	3645.00	3755.00
3600 x 3600mm wide	…	No	4420.00	4565.00	4530.00	4540.00	4710.00
4000 x 4000mm wide	…	No	4950.00	5140.00	5090.00	5105.00	5325.00
Pre-enamelled or in situ paint finish -							
3000 x 3000mm wide	…	No	4005.00	4100.00	4075.00	4085.00	4195.00
3600 x 3600mm wide	…	No	5055.00	5200.00	5160.00	5175.00	5345.00
4000 x 4000mm wide	…	No	5730.00	5920.00	5870.00	5885.00	6105.00
ADD extra for :							
Electrically operated opening mechanism	…	Sum	1230.00	1275.00	1260.00	1265.00	1315.00
Wicket gate	…	No	2260.00	2310.00	2295.00	2300.00	2360.00

ROLLING GRILLES

Steel heavy duty grille including locks and manual control :							
3.0sqm grille	…	sqm	605.00	605.00	610.00	600.00	615.00
6.0sqm grille	…	sqm	640.00	540.00	545.00	535.00	550.00
12.0sqm grille	…	sqm	445.00	450.00	455.00	445.00	460.00
ADD extra for :							
Removable mullion -							
1000mm long	…	No	235.00	240.00	245.00	235.00	245.00
2000mm long	…	No	315.00	320.00	325.00	310.00	325.00
3000mm long	…	No	400.00	410.00	415.00	400.00	415.00
Natural anodised aluminium decorative pattern purpose made grille including locks and manual control :							
3.0sqm grille	…	sqm	520.00-625.00	520.00-625.00	520.00-625.00	520.00-625.00	520.00-625.00
6.0sqm grille	…	sqm	435.00-540.00	435.00-540.00	435.00-540.00	435.00-540.00	435.00-540.00
12.0sqm grille	…	sqm	420.00-525.00	420.00-525.00	420.00-525.00	420.00-525.00	420.00-525.00
ADD extra for :							
Removable mullion -							
1000mm long	…	No	300.00	300.00	300.00	300.00	300.00
2000mm long	…	No	375.00	375.00	375.00	375.00	375.00
3000mm long	…	No	465.00	465.00	465.00	465.00	465.00
Colour anodising	...	+ 20%					

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

SECTIONAL OVERHEAD DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Ribbed pattern sectional panel door complete with hardware, etc. :							
Colour coated finish one side -							
2700 x 4000mm wide	...	No	3345.00	3515.00	3470.00	3485.00	3685.00
3000 x 4800mm wide	...	No	3985.00	4190.00	4135.00	4155.00	4395.00
Translucent fibreglass -							
2700 x 4000mm wide	...	No	7220.00	7390.00	7345.00	7360.00	7560.00
3000 x 4800mm wide	...	No	7875.00	8080.00	8025.00	8045.00	8285.00
4000 x 5000mm wide	...	No	10900.00	11200.00	11100.00	11150.00	11500.00
ADD extra for -							
Electrically operated opening mechanism :							
Domestic	...	Sum	895.00	955.00	940.00	945.00	1015.00
Industrial	...	Sum	2105.00	2165.00	2150.00	2155.00	2225.00

LOUVRED DOORS

Heavy duty louvred door and frame comprising door with chevron louvre blades, steel frame together with stainless steel heavy duty butt hinges with mortice lock preparation, etch prime finish :							
2040 x 818mm single	...	No	1900.00	2010.00	1960.00	1935.00	1960.00
2040 x 1100mm single	...	No	1960.00	2120.00	2060.00	2070.00	2160.00

TRANSPARENT INDUSTRIAL SWING DOORS

Industrial safety swing door of see-through PVC sheeting, self-closing and sealing :							
Medium duty, aluminium frame -							
2100 x 1050mm single	...	No	1745.00	1815.00	1800.00	1805.00	1885.00
2100 x 2100mm double	...	No	3430.00	3550.00	3515.00	3530.00	3670.00
Heavy duty, steel frame -							
2100 x 1500mm double	...	No	3505.00	3660.00	3620.00	3630.00	3810.00
2700 x 3000mm double	...	No	4365.00	4520.00	4480.00	4495.00	4675.00
3600 x 3000mm double	...	No	5080.00	5230.00	5190.00	5205.00	5385.00
ADD extra for :							
Air curtains, refer page 383	...	Note					
Mounting plates to jambs and head	...	m	70.00	70.00	70.00	70.00	70.00
Heavy duty galvanised frame in coolstores and freezers -							
Up to 2700 x 3000mm	...	No	600.00	605.00	590.00	595.00	595.00
3600 x 3000mm	...	No	650.00	660.00	650.00	660.00	655.00
Full/half height opaque	...	No	50.30	50.30	50.30	50.30	50.30
Gantry frame	...	m	62.50	62.50	62.50	62.50	62.50

STRONGROOM DOORS

Refer page 701	...	Note					

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

COOL ROOM AND COLD STORE DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Single hinged cool room door and frame installed in prepared opening, clad with Zincanneal® Steel sheeting and complete with hardware :							
2000 x 900 x 50mm thick	...	No	1695.00	1705.00	1700.00	1695.00	1725.00
ADD extra for -							
Stainless steel cladding	...	No	580.00	580.00	580.00	580.00	580.00
Single hinged cold store door and frame installed in prepared opening clad with Zincanneal® Steel sheeting and complete with hardware, door size :							
1900 x 1200mm light duty	...	No	2780.00	2810.00	2790.00	2800.00	2820.00
2000 x 1200mm heavy duty	...	No	3070.00	3070.00	3070.00	3080.00	3090.00
2000 x 1500mm heavy duty	...	No	3315.00	3355.00	3345.00	3340.00	3355.00
ADD extra for -							
Stainless steel cladding :							
2000 x 1200mm door	...	No	835.00	835.00	835.00	835.00	835.00
2000 x 1500mm door	...	No	900.00	900.00	900.00	900.00	900.00
Single sliding cold store door and frame installed in prepared opening clad with Zincanneal® Steel sheeting and complete with sliding track and hardware, door size :							
2000 x 1500 x 100mm thick	...	No	3075.00	3080.00	3075.00	3080.00	3095.00
3000 x 2000 x 100mm thick	...	No	4185.00	4200.00	4185.00	4200.00	4215.00
3000 x 3000 x 100mm thick	...	No	4710.00	4720.00	4710.00	4710.00	4720.00
ADD extra for -							
Stainless steel cladding :							
2000 x 1500mm door	...	No	1025.00	1025.00	1025.00	1025.00	1025.00
3000 x 2000mm door	...	No	1255.00	1255.00	1255.00	1255.00	1255.00
3000 x 3000mm door	...	No	1520.00	1520.00	1520.00	1520.00	1520.00

SECURITY DOORS

Security door and frame comprising steel flush finish security door and fully welded frame together with security hinge bolts and locking system :							
Standard single door 2040 x 820mm	...	No	2220.00	2220.00	2225.00	2220.00	2220.00
Standard double door 2040 x 1640mm	...	No	4215.00	4215.00	4225.00	4215.00	4215.00
Heavy duty single door 2040 x 820mm	...	No	2785.00	2785.00	2790.00	2785.00	2785.00
Heavy duty double door 2040 x 1640mm	...	No	5410.00	5410.00	5420.00	5410.00	5410.00
Extra heavy duty single door 2040 x 820mm	...	No	2945.00	2945.00	2950.00	2945.00	2945.00
Extra heavy duty double door 2040 x 1640mm	...	No	5480.00	5480.00	5490.00	5480.00	5480.00

DOMESTIC SECURITY DOORS

Anodised aluminium security door 2040 x 840mm with patterned security grille, insect screen mesh and fitted with hinges and lock	...	No	385.00	390.00	390.00	390.00	390.00

WINDOWS AND DOORS

DOORS (cont'd)

METAL (cont'd)

STRONGROOM DOORS

Refer page 688

FIRE DOORS

HINGED FIRE DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fire rated door and frame comprising door faced both sides with plywood for paint finish, steel frame built in and fitted with butt hinges, surface mounted closer and latchset :							
Standard single door 2040 x 820mm -							
One hour rated	…	No	1290.00	1310.00	1250.00	1265.00	1305.00
Two hour rated	…	No	1430.00	1450.00	1475.00	1390.00	1440.00
Three hour rated	…	No	1545.00	1570.00	1620.00	1540.00	1570.00
Non-standard single door (cost per sqm) -							
One hour rated	…	sqm	920.00	935.00	895.00	900.00	930.00
Two hour rated	…	sqm	1015.00	1035.00	1050.00	985.00	1030.00
Three hour rated	…	sqm	1100.00	1115.00	1150.00	1100.00	1115.00
Standard double door 2040 x 1640mm -							
One hour rated	…	No	3530.00	2890.00	3355.00	3570.00	3545.00
Two hour rated	…	No	3835.00	2165.00	3875.00	4000.00	3895.00
Three hour rated	…	No	4170.00	3365.00	4000.00	4190.00	4010.00
Non-standard double door (cost per sqm) -							
One hour rated	…	sqm	1140.00	965.00	1120.00	1190.00	1185.00
Two hour rated	…	sqm	1285.00	1070.00	1295.00	1335.00	1305.00
Three hour rated	…	sqm	1390.00	1120.00	1330.00	1395.00	1330.00
ADD extra for -							
Veneer for clear finish (surface area excluding edges)	…	sqm	26.00	26.00	26.00	26.00	26.00
Melamine plastic laminate veneer (surface area excluding edges)	…	sqm	155.00	155.00	155.00	155.00	155.00
1mm Zincanneal® Steel sheeting (surface area excluding edges)	…	sqm	80.00	80.00	80.00	80.00	80.00

SLIDING FIRE DOORS

Fire rated sliding door comprising metal clad door complete with jamb assembly, automatic closing system, tracks, guides and hardware :							
Two or three hour rated door (area taken to nearest 0.25sqm up), door area -							
2.0sqm	…	sqm	2045.00	2095.00	2045.00	2010.00	1850.00
12.0sqm	…	sqm	1285.00	1265.00	1250.00	1260.00	1165.00
20.0sqm	…	sqm	1205.00	1240.00	1115.00	1180.00	1090.00
Four hour rated door (area taken to nearest 0.25sqm up), door area -							
2.0sqm	…	sqm	3150.00	3275.00	3265.00	3095.00	2905.00
12.0sqm	…	sqm	1900.00	2390.00	2080.00	1865.00	1790.00
20.0sqm	…	sqm	1700.00	2310.00	1865.00	1665.00	1580.00

WINDOWS AND DOORS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DOORS (cont'd)							
FOLDING DOORS AND OPERABLE WALLS							
ACCORDION DOORS							
Accordion door with linen-backed vinyl finish, hung on overhead track, including hardware and reveal linings :							
N/R	…	sqm	425.00	425.00	425.00	425.00	425.00
STC - 23	…	sqm	605.00	605.00	605.00	605.00	605.00
STC - 28	…	sqm	775.00	775.00	775.00	775.00	775.00
STC - 38	…	sqm	835.00	835.00	835.00	835.00	835.00
OPERABLE WALLS							
Acoustically insulated operable wall manually operated and with linen-backed vinyl finish to panels and pass door, hung on overhead track, including hardware and reveal linings (min. 50sqm) :							
STC - 40	…	sqm	775.00	775.00	775.00	775.00	775.00
STC - 42	…	sqm	825.00	825.00	825.00	825.00	825.00
STC - 45	…	sqm	985.00	985.00	985.00	985.00	985.00
STC - 50	…	sqm	1115.00	1115.00	1115.00	1115.00	1115.00
ADD extra for -							
Walls of :							
10 to 25sqm	...+60-75%						
25 to 50sqm	...+25-50%						
Heights of less than 3000mm	...	Note					
Pelmets	...	Note					
Supporting steelwork	...	Note					
Fully glazed operable wall with 10/12mm clear toughened glass in panels 800/1300mm wide x 3000mm maximum height, hung on overhead track, including pass door and hardware	...	sqm	1800.00-2100.00	1800.00-2100.00	1800.00-2100.00	1800.00-2100.00	1800.00-2100.00
AIR CURTAINS							
Variable speed unit (one unit per 1000mm width of opening) :							
Commercial	…	No	845.00	890.00	880.00	880.00	930.00
Industrial - stainless steel	…	No	2060.00	2145.00	2125.00	2130.00	2230.00

WINDOWS AND DOORS

CURTAINS AND BLINDS

BLINDS AND SHUTTERS

Prices are based on quantities of 200 sqm

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for :							
Lesser quantities	...+10-50%						
ROLLER BLINDS							
Standard Holland	...	sqm	90.00-110.00	90.00-110.00	90.00-110.00	90.00-110.00	90.00-110.00
Solar control to IEQ-4 rating	...	sqm	120.00-140.00	120.00-140.00	120.00-140.00	120.00-140.00	120.00-140.00
Felt heavy duty	...	sqm	115.00-145.00	115.00-145.00	115.00-145.00	115.00-145.00	115.00-145.00
Verosol reflective fabric :							
Standard type	...	sqm	165.00-185.00	165.00-185.00	165.00-185.00	165.00-185.00	165.00-185.00
Premium (spring assisted) type	...	sqm	300.00-350.00	300.00-350.00	300.00-350.00	300.00-350.00	300.00-350.00
VENETIAN BLINDS							
Aluminium :							
Micro (16mm)	...	sqm	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00	130.00-170.00
Mini, Slimline (25mm)	...	sqm	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00	110.00-160.00
Timber (50mm or 60mm) with lacquer finish	...	sqm	350.00-450.00	350.00-450.00	350.00-450.00	350.00-450.00	350.00-450.00
ADD extra for :							
Paint finish	...	+25%					
VERTICAL BLINDS							
Heavy duty cotton, vinyl impregnated, free hanging including headrail (range)	...	sqm	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00	200.00-230.00
TIMBER SHUTTERS							
Timber (68mm) with oiled finish	...	sqm	550.00-750.00	550.00-750.00	550.00-750.00	550.00-750.00	550.00-750.00
ADD extra for :							
Lacquer finish	...	sqm	30.00	30.00	30.00	30.00	30.00
Paint finish	...	sqm	70.00	70.00	70.00	70.00	70.00

WINDOWS AND DOORS

CURTAINS AND BLINDS (cont'd)

CURTAINS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on the curtained covered area and include an additional allowance of plus 140/150% for fullness	...	Note					
Prices exclude track	...	Note					
CURTAINS with pinch pleated top in :							
ACRYLIC SCRIM (P.C. $25.00/metre @ 1500mm wide) in drops :							
2000mm long	...	sqm	78.50	78.50	80.00	78.50	80.50
2500/3000mm long	...	sqm	68.50	68.50	70.00	68.50	70.50
ADD extra for :							
Cotton lining sewn in	...	sqm	22.00	22.00	22.00	22.00	22.00
Detachable coated lining -							
Soft light entry	...	sqm	28.50	28.50	28.50	28.50	28.50
Mostly blackout	...	sqm	37.50	37.50	37.50	37.50	37.50
WOOL SCRIM (P.C. $30.00/metre @ 1500mm wide) in drops :							
2000mm long	...	sqm	88.00	86.00	89.50	88.00	90.00
2500/3000mm long	...	sqm	79.00	79.00	79.50	79.00	80.00
ADD extra for :							
Cotton lining sewn in	...	sqm	22.00	22.00	22.00	22.00	22.00
Detachable coated lining -							
Soft light entry	...	sqm	28.50	28.50	28.50	28.50	28.50
Mostly blackout	...	sqm	37.50	37.50	37.50	37.50	37.50
COTTON PRINT (P.C. $25.00/metre @ 1200mm wide), including medium quality cotton lining, in drops :							
2000mm long	...	sqm	118.50	118.50	120.00	118.50	121.00
2500/3000mm long	...	sqm	106.00	106.00	107.00	106.00	107.50
COTTON PRINT (P.C. $23.00/metre @ 1200mm wide), including soft light entry lining, in drops :							
2000mm long	...	sqm	133.00	133.00	134.50	133.00	135.00
2500/3000mm long	...	sqm	118.50	118.50	119.50	118.50	120.00
COTTON PRINT (P.C. $23.00/metre @ 1200mm wide), including mostly blackout lining, in drops :							
2000mm long	...	sqm	143.00	143.00	145.00	143.00	145.50
2500/3000mm long	...	sqm	128.50	128.50	119.50	128.50	130.50
COTTON PRINT, COATED BACK (P.C. $26.00/metre @ 1200mm wide) in drops :							
2000mm long	...	sqm	99.50	99.50	101.50	99.50	101.00
2500/3000mm long	...	sqm	89.50	89.50	90.50	89.50	91.00

WINDOWS AND DOORS

CURTAINS AND BLINDS (cont'd)

CURTAINS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CURTAINS with pinch pleated top in (cont'd) :							
VELVET, ACRYLIC PILE (P.C. $44.00/metre @ 1370mm wide) :							
Unlined (against walls) in drops -							
2500/3000mm long	...	sqm	116.00	116.00	117.00	116.00	118.00
Lined with mostly blackout, in drops -							
2000mm long	...	sqm	160.00	160.00	162.00	160.00	163.00
2500/3000mm long	...	sqm	145.00	145.00	146.00	145.00	147.00
VELVET, COTTON PILE (P.C. $60.00/metre @ 1200mm wide) :							
Unlined (against walls) in drops -							
2500/3000mm long	...	sqm	174.00	174.00	175.00	174.00	176.00
Lined with mostly blackout, in drops -							
2000mm long	...	sqm	224.00	224.00	226.00	224.00	227.00
2500/3000mm long	...	sqm	208.00	208.00	209.00	208.00	210.00

TRACK

Extruded aluminium track :							
Hand draw type	...	m	35.00	35.00	35.00	35.00	35.00
Corded type	...	m	42.00	42.00	42.00	42.00	42.00

SOLAR CONTROL BLINDS

VEROSOL 321, 316 and 975 reflective fabric	...	Note					
Prices include tracks	...	Note					
Prices are based on a size of 2 metres x 2 metres. Smaller or larger sizes will vary this price.	...	Note					
ADD for electrical service and switches	...	Note					
HORIZONTAL PLEATED SHADES							
Manually operated	...	sqm	310.00-330.00	310.00-330.00	310.00-330.00	310.00-330.00	310.00-330.00
Motorised pleated shades	...	sqm	P.O.A.	P.O.A.	P.O.A.	P.O.A.	P.O.A.

SOLAR CONTROL FILM

Solar control film refer page 447	...	Note					

WINDOWS AND DOORS

SUNSCREENS

EXTERNAL

CANVAS AWNINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Canvas awnings, roll-up	...	sqm	250.00-310.00	250.00-310.00	250.00-310.00	250.00-310.00	250.00-310.00

FIXED LOUVRES

Horizontal or vertical louvres of 132mm wide extruded aluminium modules, powder coat finished, including supporting brackets and framing	...	sqm	340.00-400.00	340.00-400.00	340.00-400.00	340.00-400.00	340.00-400.00
Weatherproof louvred panels, refer pages 368 and 369	...	Note					

HORIZONTAL STRIPS

Horizontal strips with baked enamel finish, including brackets and framing :							
Roll-Up	...	sqm	290.00-360.00	290.00-360.00	290.00-360.00	290.00-360.00	290.00-360.00
Adjustable	...	sqm	350.00-460.00	350.00-460.00	350.00-460.00	350.00-460.00	350.00-460.00

RETRACTABLE SUNSCREENS

Prices are based on 50sqm. For quantities in excess of 100sqm the prices may reduce by up to 30%	...	Note					
Retractable horizontal aluminium louvres :							
Manually operated -							
80mm convex shape	...	sqm	700.00	700.00	650.00	700.00	710.00
90mm special profile	...	sqm	780.00	780.00	730.00	780.00	780.00
NS 100 shutter type	...	sqm	885.00	885.00	860.00	885.00	910.00
Electrically operated -							
80mm convex shape	...	sqm	920.00	920.00	870.00	920.00	940.00
90mm special profile	...	sqm	990.00	990.00	940.00	990.00	1000.00
NS 100 shutter type	...	sqm	1085.00	1085.00	1030.00	1085.00	1085.00
ADD extra for :							
Central command with wind and sun sensor	...	sqm	150.00	150.00	150.00	150.00	150.00
Pelmet	...	m	Incl.	Incl.	Incl.	Incl.	Incl.

VERGOLA ROOFING SYSTEM

Refer page 416	...	Note					

HARDWARE AND DOOR SUNDRIES

SUPPLY ONLY

HINGES

		HINGES - SUPPLY ONLY WITH FIXINGS - ALL CITIES				
		Bright Steel	Zinc Plated	Satin Chrome	Brass	Stainless Steel
		Each $	Each $	Each $	Each $	Each $
Light butt, loose pin :						
100 x 70 x 1.6mm	...	1.60	2.25	-	-	3.05
100 x 85 x 1.6mm	...	1.70	2.35	-	-	3.40
100 x 100 x 1.6mm	...	1.80	2.50	-	-	3.85
Light butt, fixed pin :						
100 x 70 x 1.6mm	...	1.60	2.15	-	-	3.20
100 x 85 x 1.6mm	...	1.70	2.35	-	-	3.65
100 x 100 x 1.6mm	...	-	-	-	-	4.65
Heavy butt, loose pin :						
Standard -						
100 x 75 x 2.5mm	...	5.30	3.30	-	10.90	-
100 x 100 x 2.5mm	...	-	3.50	-	12.00	-
Button tipped -						
100 x 75 x 2.5mm	...	5.30	6.50	3.50	-	5.30
100 x 100 x 2.5mm	...	-	6.85	3.65	-	6.35
Ball bearing, button tipped -						
100 x 75 x 2.5mm	...	-	10.10	5.30	-	16.00
100 x 100 x 2.5mm	...	-	10.10	5.45	-	19.00
Heavy butt, fixed pin :						
Button tipped -						
100 x 75 x 3.2mm	...	-	6.75	3.60	-	16.00
100 x 100 x 3.2mm	...	-	7.05	3.75	-	19.00
Security broad butt :						
100 x 75 x 2.5mm	...	-	-	9.70	-	26.80
100 x 100 x 2.5mm	...	-	-	10.10	-	28.50

SUPPLY AND FIX

HINGES

The supply and fixing of the hinges is included with the hanging of the door in the respective items ... Note

HARDWARE AND DOOR SUNDRIES

SUPPLY AND FIX (cont'd)

LOCKS AND LATCHES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for locks and latches include furniture	...	Note					
Cupboard or drawer projection lock :							
Basic	- (P.C. $ 23.50) ...	No	55.80	61.90	63.80	60.60	74.50
Cupboard or drawer rim lock :							
High quality	- (P.C. $120.00) ...	No	172.00	184.00	188.50	182.00	206.00
Mortice latch :							
Basic	- (P.C. $250.00) ...	No	315.00	329.00	333.00	326.00	352.00
High quality	- (P.C. $310.00) ...	No	378.00	392.00	396.00	389.00	415.00
Cylinder mortice lock :							
Basic	- (P.C. $295.00) ...	No	362.00	376.00	380.00	374.00	400.00
High quality	- (P.C. $385.00) ...	No	457.00	470.00	475.00	468.00	494.00
Cylindrical latch :							
Domestic	- (P.C. $ 76.00) ...	No	132.50	146.00	150.00	143.50	169.50
High quality	- (P.C. $200.00) ...	No	263.00	276.00	280.00	274.00	300.00
Cylindrical lock :							
Basic	- (P.C. $125.00) ...	No	184.00	197.50	201.50	195.00	221.00
High quality	- (P.C. $285.00) ...	No	352.00	365.00	370.00	363.00	389.00
Mortice deadlock :							
Domestic	- (P.C. $ 80.00) ...	No	133.00	146.00	150.50	144.00	169.00
High quality	- (P.C. $175.00) ...	No	233.00	246.00	250.00	244.00	269.00
Narrow stile	- (P.C. $160.00) ...	No	217.00	230.00	234.00	228.00	253.00
Rim night latch or deadlock :							
Basic	- (P.C. $ 67.50) ...	No	117.00	129.00	133.50	126.50	150.50
High quality (fire rated)	- (P.C. $135.00) ...	No	187.50	199.50	204.50	197.50	221.50

SECURITY LOCKS

Tubular deadbolt, double cylinder	- (P.C. $195.00) ...	No	254.00	267.00	271.00	265.00	290.00
Digital door lock :							
Domestic	- (P.C. $325.00) ...	No	397.00	412.00	416.00	409.00	436.00
Commercial	- (P.C. $550.00) ...	No	640.00	656.00	664.00	653.00	687.00
Patio door bolt	- (P.C. $ 38.50) ...	No	79.80	90.00	95.20	88.30	110.50
Window lock	- (P.C. $ 28.50) ...	No	46.30	50.60	49.50	49.90	54.90

ELECTRIC LOCKS AND MAGNETS

ADD extra to the following for electrical circuits and connections	...	Note					
Refer page 522 for access control	...	Note					
Monitoral Electrified mortice lock and furniture	- (P.C. $685.00) ...	No	778.00	794.00	797.00	791.00	819.00
Magnetic lock with monitor	- (P.C. $435.00) ...	No	509.00	523.00	527.00	521.00	547.00
Electric strike with monitor	- (P.C. $470.00) ...	No	553.00	568.00	572.00	565.00	593.00
Electric locking bolt	- (P.C. $525.00) ...	No	604.00	617.00	622.00	615.00	641.00
Power transfer hinge	- (P.C. $ 90.00) ...	No	134.00	144.00	149.50	142.50	164.50
Power transfer lead cover	- (P.C. $ 90.00) ...	No	134.00	144.00	149.50	142.50	164.50

HARDWARE AND DOOR SUNDRIES

SUPPLY AND FIX (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DOOR CLOSERS							
Surface mounted overhead type :							
Basic	- (P.C. $130.00) ...	No	215.00	236.00	238.00	232.00	266.00
High quality	- (P.C. $410.00) ...	No	509.00	530.00	532.00	526.00	560.00
Semi concealed ditto :							
Average	- (P.C. $415.00) ...	No	531.00	556.00	561.00	551.00	595.00
Concealed ditto :							
Average	- (P.C. $415.00) ...	No	537.00	564.00	569.00	559.00	605.00
Transom concealed ditto	- (P.C. $320.00) ...	No	438.00	464.00	469.00	460.00	506.00
FLOOR SPRINGS							
Double action, light duty, floor mounted	- (P.C. $360.00) ...	No	536.00	576.00	581.00	570.00	637.00
Double action, medium duty, recess mounted	- (P.C. $660.00) ...	No	870.00	916.00	920.00	908.00	982.00
Double action, heavy duty, recess mounted	- (P.C. $895.00) ...	No	1130.00	1180.00	1195.00	1170.00	1265.00
BOLTS							
Barrel bolt :							
75mm	- (P.C. $11.00) ...	No	28.00	32.20	33.10	31.50	39.00
100mm	- (P.C. $16.00) ...	No	33.20	37.50	38.30	36.80	44.20
150mm	- (P.C. $20.50) ...	No	39.60	44.30	45.00	43.50	51.50
200mm	- (P.C. $25.50) ...	No	44.80	49.50	50.20	48.70	56.70
Flush bolt :							
150mm	- (P.C. $23.00) ...	No	70.10	82.00	82.80	80.00	99.00
200mm	- (P.C. $27.00) ...	No	82.50	96.60	96.80	94.20	115.50
Strap/skeleton bolt :							
300mm	- (P.C. $41.50) ...	No	62.60	67.60	67.80	67.50	74.50
450mm	- (P.C. $53.50) ...	No	77.20	82.60	83.60	82.50	91.10
450mm, offset	- (P.C. $59.50) ...	No	83.50	88.90	89.90	88.80	97.40
PANIC BOLTS							
Kaba Type 22, silver finish :							
To single door -							
Rim and crossbar	- (P.C. $320.00) ...	Set	402.00	419.00	426.00	416.00	451.00
Vertical rod and crossbar	- (P.C. $485.00) ...	Set	575.00	592.00	599.00	589.00	624.00
To double door -							
One vertical rod and two crossbars	- (P.C. $795.00) ...	Set	982.00	1021.00	1030.00	1014.00	1084.00
Briton 3000 door selector (fire rated)	- (P.C. $115.00) ...	No	137.00	141.00	141.00	141.00	147.00

HARDWARE AND DOOR SUNDRIES

SUPPLY AND FIX (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TERRAZZO SET							
Comprising hinges, indicator bolt, bumper and coat hook, all s.c.p.	- (P.C. $ 89.50) ...	No	192.50	218.00	219.00	214.00	254.00
Indicator bolt	- (P.C. $ 39.00) ...	No	60.00	64.90	65.20	64.90	71.90
PLATES							
Pull handles and push plates :							
Stainless steel, push, 300 x 75mm	...	No	30.30	32.40	32.60	32.00	35.50
Stainless steel, pull handle on 300 x 100mm plate	...	No	53.40	55.50	55.70	55.10	58.60
Sheet vinyl, 225 x 2mm to full width of 820mm wide door	...	No	26.60	28.70	28.90	28.40	31.80
Kick plates, full width of 820mm door :							
Sheet vinyl, 225 x 2mm	...	No	31.50	34.90	35.60	34.30	40.30
Stainless steel, 225 x 1.2mm	...	No	88.20	91.60	92.30	91.00	97.00
Letter plate :							
S.c.p. brass, 280 x 90mm	...	No	140.50	145.50	146.50	144.50	153.50
CATCHES, FASTENERS, ETC.							
Casement spring catch	- (P.C. $19.50) ...	No	32.30	35.40	36.10	34.80	40.40
Casement fastener	- (P.C. $20.50) ...	No	33.30	36.40	37.20	35.90	41.50
Casement friction stays	- (P.C. $23.00) ...	Pair	57.00	65.50	67.20	64.00	79.00
Sash fastener	- (P.C. $19.50) ...	No	36.90	41.10	42.00	40.40	47.90
Sash lift	- (P.C. $12.50) ...	No	23.00	25.50	26.00	25.10	29.60
Cupboard ball catch	- (P.C. $ 6.00) ...	No	16.15	18.70	19.20	18.25	22.80
Magnetic cupboard catch	- (P.C. $15.50) ...	No	26.10	28.70	29.20	28.20	32.70
HASP AND STAPLE							
To suit padlock :							
Galvanised	- (P.C. $ 9.25) ...	No	19.60	22.10	22.60	21.70	26.20
Architectural	- (P.C. $52.00) ...	No	64.40	67.00	67.50	66.60	71.10
HOOKS, ETC.							
Cabin hook :							
75mm	- (P.C. $ 8.00) ...	No	14.95	16.65	17.00	16.40	19.35
150mm	- (P.C. $12.50) ...	No	19.70	21.40	21.70	21.10	24.10
Coat hook :							
Standard	- (P.C. $ 8.00) ...	No	14.95	16.65	17.00	16.40	19.35
Stainless steel	- (P.C. $20.50) ...	No	28.10	29.80	30.10	29.50	32.50
Bumper and hook	- (P.C. $ 8.00) ...	No	14.95	16.65	17.00	16.40	19.35
DOOR STOPS							
Floor mounted, light duty	- (P.C. $ 5.50) ...	No	12.35	14.05	14.40	13.75	16.75
Floor mounted, heavy duty	- (P.C. $11.00) ...	No	18.10	19.80	20.20	19.55	22.50
Roller cushioned stop 40 dia.	- (P.C. $ 7.00) ...	No	13.90	15.60	15.95	15.35	18.30
Air cushioned stop	- (P.C. $ 2.00) ...	No	11.95	14.50	15.00	14.05	18.55

HARDWARE AND DOOR SUNDRIES

SUPPLY AND FIX (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
KEYING							
Non-restricted masterkeying		...Per Door	21.50	21.50	21.50	21.50	21.50
Restricted masterkeying		...Per Door	84.00	84.00	84.00	84.00	84.00
SLIDING TRACK							
Overhead track including fittings (domestic) :							
820mm wide door	- (P.C. $ 67.00) ...	Set	129.50	145.00	152.50	142.00	175.00
1200mm wide door	- (P.C. $115.00) ...	Set	186.50	203.00	211.00	201.00	235.00
Overhead track including fittings (heavy duty) :							
820mm wide door	- (P.C. $155.00) ...	Set	245.00	266.00	272.00	263.00	302.00
1200mm wide door	- (P.C. $230.00) ...	Set	340.00	366.00	374.00	361.00	411.00
Cupboard track :							
450mm wide panel	- (P.C. $ 49.00) ...	Set	90.80	101.00	106.00	99.30	121.50
900mm wide panel	- (P.C. $ 57.00) ...	Set	106.00	117.50	122.50	115.50	139.50
DOOR SEALS							
To timber or metal clad butt hinged doors in metal or timber frames, internal and external	...	Note					
Type of seal :							
A - Acoustic							
L - Light							
S - Smoke							
W - Weather							
Threshold and door bottom protection :							
Raven RP4 storm proof seal (A, L, S, W)	...	m	91.60	100.00	102.00	98.70	113.50
Raven RP54 rain stop seal (L, S, W)	...	m	123.50	132.00	134.00	130.50	145.50
Raven RP4T and RP77 seal, handicapped standard (A, L, S, W)	...	m	134.00	143.50	145.00	142.00	158.00
Head and jamb protection :							
Raven RP23 to stops/rebated jambs (L, S, W)	...	m	43.50	46.90	47.60	46.30	52.30
Raven RP78 ditto (A, L, S, W)	...	m	35.70	39.10	39.80	38.50	44.50
Raven RP24 heavy duty door stop seal (A, L, S, W)	...	m	122.00	126.00	126.50	125.50	132.50
Raven RP47 ditto (A, L, S, W)	...	m	101.00	105.00	105.50	104.50	111.00
Raven RP10 commercial duty door stop seal (A, L, S, W)	...	m	47.80	51.20	51.80	50.60	56.60
Meeting stile protection :							
Raven RP16 to plain or rebated stiles (A, L, S, W)	...	m	43.80	47.20	47.90	46.60	52.60
Raven RP71 ditto (A, L, S)	...	m	32.20	35.70	36.30	35.10	41.10
Automatic door bottom seals :							
Raven RP8 concealed (A, L, S, W)	...	m	87.40	90.80	91.50	90.30	96.30
Raven RP38, face mounted, semi-mortised (A, L, S, W)	...	m	194.50	198.50	199.00	198.00	205.00
Sliding and double acting doors :							
Raven RP2A nylon brush strip seal, all round (S, W)	...	m	40.70	43.30	43.80	42.90	47.30
Raven RP15 ditto (S, W)	...	m	43.40	46.00	46.50	45.50	50.00
Raven RP75 ditto (S, W)	...	m	41.90	44.50	45.00	44.00	48.50

PARTITIONS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
AERATED CONCRETE PARTITION WALL SYSTEM							
158mm thick partition wall system comprising 75mm thick Hebel panel 3000mm high, S4 polyester insulation and finished both faces with 10mm plasterboard (Powerpanel 75)	...	sqm	211.00	213.00	211.00	211.00	202.00
ADD extra for :							
Fire rated sealants	...	Note					
FRAMED							
All based on height of 3000mm	...	Note					
STEEL							
51mm galvanised steel stud framing lined both sides with 10mm plasterboard	...	sqm	120.50	134.50	106.00	111.00	119.50
64mm ditto	...	sqm	121.50	131.00	107.50	112.50	122.00
76mm ditto	...	sqm	123.00	135.00	108.50	113.50	124.50
92mm ditto	...	sqm	124.00	137.50	111.00	117.00	126.50
ADD extra for :							
R1.5 insulation batts	...	sqm	13.10	15.25	15.10	14.85	17.85
Acoustic insulation (regular range)	...	sqm	26.00-36.00	26.00-36.00	26.00-36.00	26.00-36.00	26.00-36.00
Fire rated plasterboard -							
1 x 16mm layer (one hour) to both sides	...	sqm	23.00	13.20	18.70	24.00	42.00
2 x 13mm layers (two hours) to both sides	...	sqm	81.50	70.50	68.00	67.00	64.00
2 x 16mm layers (two hours) to both sides	...	sqm	102.50	92.50	93.50	86.00	116.50
Vinyl fabric (paper backed) facing to both sides	...	sqm	63.00	59.00	63.00	64.00	60.00
Skirtings and trims	...	Note					
TIMBER							
Timber framed and plasterboard lined walls, refer pages 129 and 131	...	Note					
MODULAR SEMI-DEMOUNTABLE							
Price based on overall area including doors	...	Note					
Partition in 1200mm wide panels of galvanised steel stud framing faced both sides with 13mm bevelled edge plasterboard sheets :							
Full height, 2400/2700mm	...	sqm	113.00	117.50	119.00	110.50	117.00
Door height, 2100/2400mm with capping	...	sqm	132.00	131.00	137.50	137.50	122.50
ADD extra for :							
R1.5 insulation batts	...	sqm	13.10	15.25	15.10	14.85	17.85
Single glazing with 10.38mm clear laminated glass	...	sqm	205.00	206.00	224.00	225.00	218.00
One hour fire rated plasterboard to both sides	...	sqm	93.00	84.00	116.00	85.00	78.00
Veneered (average standard) particleboard to both sides	...	sqm	80.00	80.50	103.50	98.00	117.00

PARTITIONS

MODULAR SEMI-DEMOUNTABLE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for (cont'd) :							
Flexiclad vinyl (unbacked) facing to both sides	...	sqm	20.50	20.00	20.50	21.00	20.50
Vinyl fabric facing to both sides	...	sqm	62.00	56.50	62.50	63.50	60.00
2040 x 820mm flush hollow core door including frame and hardware	...	No	500.00	540.00	540.00	550.00	600.00
2040 x 820mm solid core ditto	...	No	560.00	585.00	605.00	615.00	655.00
Vinyl skirting, 100mm high, feather edge	...	m	11.70	12.10	12.00	12.40	13.00

MODULAR DEMOUNTABLE

Price based on overall area including doors	...	Note					
Partition in 1200mm wide panels of aluminium framing infilled with 33mm particle board and faced with Flexiclad vinyl :							
Full height, 2400/2700mm	...	sqm	158.00	155.00	166.50	166.00	154.00
Door height, 2100/2400mm with capping	...	sqm	163.00	163.50	170.50	188.50	163.50
ADD extra for :							
Single glazing in clear glass	...	sqm	121.50	96.00	109.00	104.50	77.00
Double ditto	...	sqm	289.50	253.50	270.50	278.50	243.50
Veneered ply facing to both sides	...	sqm	47.50	46.00	45.50	54.50	51.00
Vinyl fabric facing to both sides	...	sqm	41.50	36.50	43.50	43.00	39.50
2040 x 820mm flush hollow core door including frame and hardware	...	No	500.00	540.00	540.00	550.00	600.00
2040 x 820mm solid core ditto	...	No	560.00	585.00	605.00	615.00	655.00
Vinyl skirting, 100mm high, feather edge	...	m	11.70	12.10	12.00	12.40	13.00

MODULAR SCREENS

Prices include for fixing the various module panels into a rigid screen	...	Note					
Straight panel, 70mm thick covered with front runner fabric, panel :							
600mm long -							
1350mm high	...	No	215.00	200.00	210.00	210.00	220.00
1500mm high	...	No	220.00	205.00	215.00	215.00	225.00
1650mm high	...	No	225.00	210.00	220.00	220.00	230.00
1800mm high	...	No	245.00	230.00	240.00	245.00	250.00
2100mm high	...	No	295.00	260.00	280.00	285.00	300.00
900mm long -							
1350mm high	...	No	320.00	275.00	295.00	300.00	325.00
1500mm high	...	No	330.00	295.00	320.00	320.00	335.00
1650mm high	...	No	335.00	320.00	345.00	345.00	340.00
1800mm high	...	No	365.00	330.00	355.00	355.00	370.00
2100mm high	...	No	425.00	380.00	410.00	410.00	435.00
1200mm long -							
1350mm high	...	No	330.00	295.00	320.00	320.00	335.00
1500mm high	...	No	360.00	325.00	350.00	350.00	365.00
1650mm high	...	No	375.00	335.00	365.00	365.00	380.00
1800mm high	...	No	395.00	350.00	380.00	380.00	400.00
2100mm high	...	No	450.00	410.00	435.00	435.00	460.00

PARTITIONS

MODULAR SCREENS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Radiused corner panel, 70mm thick covered with synthetic carpet fabric, panel :							
400mm radius -							
1350mm high	...	No	350.00	325.00	350.00	340.00	355.00
1500mm high	...	No	370.00	335.00	360.00	360.00	375.00
1650mm high	...	No	380.00	340.00	370.00	370.00	385.00
1800mm high	...	No	410.00	365.00	395.00	395.00	415.00
2100mm high	...	No	460.00	420.00	450.00	450.00	470.00
ADD extra for :							
Services ducting to both sides	...	m	69.00	66.00	69.00	69.00	69.00
Strengthened panels for shelving	...	No	28.00	27.00	28.00	28.00	28.00
Step down panels	...	No	32.00	30.00	32.00	32.00	32.00

RADIATION SHIELDED PARTITIONS

Based on a height of 3500mm	...	Note					
Partition framing comprising 92mm studs, SHS steel uprights @ 2.0m centres, bracing and fixing to concrete top and bottom	...	sqm	81.50	86.00	76.00	75.50	90.00
LININGS to one side :							
13mm thick interior grade plywood with lead core or bonded backing, using -							
10kg/sqm lead	...	sqm	260.00	255.00	260.00	265.00	270.00
20kg/sqm lead	...	sqm	265.00	350.00	345.00	350.00	355.00
30kg/sqm lead	...	sqm	425.00	430.00	425.00	435.00	445.00
Plasterboard -							
1 x 10mm thick	...	sqm	35.00	40.00	30.50	33.50	34.00
1 x 13mm thick	...	sqm	37.00	40.00	31.50	34.50	35.00
1 x 10mm thick 'water resistant'	...	sqm	39.00	41.00	32.50	37.50	36.00
1 x 13mm thick fire rated	...	sqm	40.00	40.50	36.00	39.00	39.00
1 x 16mm thick fire rated	...	sqm	45.50	46.00	39.00	44.50	53.00
DOORS							
Solid core door 2040 x 820 x 35mm thick faced both sides with select grade interior plywood and with lead core to centre of :							
10kg/sqm lead	...	No	885.00	905.00	915.00	945.00	1000.00
20kg/sqm lead	...	No	1040.00	1060.00	1065.00	1100.00	1150.00
30kg/sqm lead	...	No	1190.00	1210.00	1220.00	1250.00	1305.00
ADD extra for :							
Door frame and hardware	...	Note					
FLASHINGS							
Perimeter flashing 50mm wide :							
10kg/sqm lead	...	m	23.40	23.50	24.50	24.00	25.00
20kg/sqm lead	...	m	42.40	44.90	42.40	44.90	44.90
30kg/sqm lead	...	m	63.80	65.80	61.80	65.80	66.30
Flashing to :							
2040 x 820mm door frame -							
10kg/sqm lead	...	No	255.00	260.00	270.00	265.00	275.00
20kg/sqm lead	...	No	380.00	405.00	380.00	405.00	405.00
30kg/sqm lead	...	No	445.00	460.00	435.00	460.00	465.00

PARTITIONS

RADIATION SHIELDED PARTITIONS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FLASHINGS (cont'd)							
ADD extra for :							
Penetrations for electrical outlet, plumbing pipe -							
10kg/sqm lead	...	No	23.00	23.70	22.20	23.90	24.00
20kg/sqm lead	...	No	39.30	39.30	36.80	40.00	39.40
30kg/sqm lead	...	No	51.30	52.60	49.10	53.50	53.20
SUNDRIES							
Insulation	...	Note					
Paint or other finish	...	Note					

TOILET PARTITIONS

Laminate faced particle board or phenolic core, refer pages 359 and 360	...	Note					
Compressed cellular fibreboard, refer page 359	...	Note					
Terrazzo, refer page 431	...	Note					

OPERABLE WALLS

Refer page 383	...	Note					

SOUND BARRIERS

Sound insulation barrier in ceiling space fixed to concrete soffit :							
Vinyl surfaced barium compound with hessian back -							
Low	...	sqm	121.00	128.00	127.00	124.00	126.00
Medium range	...	sqm	142.00	147.00	146.00	143.00	144.00
High range	...	sqm	194.00	202.00	199.00	197.00	196.00
Sheet lead -							
0.9mm thick	...	sqm	122.50	130.00	124.00	121.50	132.00
1.3mm thick	...	sqm	159.50	167.00	164.00	169.50	174.00
Plasterboard, refer page 423	...	Note					
ADD extra for :							
Trimming to penetrations (range)	...	sqm	15.00-45.00	15.00-45.00	15.00-45.00	15.00-45.00	15.00-45.00

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TILE ROOFING							
CONCRETE							
Prices are for work up to two storeys	...	Note					
Standard pattern coloured tiles to slope not exceeding 30° and fixed to and including 38 x 25mm timber battens or standard steel battens	...	sqm	43.50	44.50	52.50	50.00	60.50
ADD extra for :							
50 x 38mm battens in lieu of 38 x 25mm	...	sqm	3.40	3.70	3.25	3.35	2.90
Laying to slope -							
30° to 40°(27°/35° - Perth)	...	sqm	3.50	4.45	-	3.65	-
40° to 45°(36°/50° - Perth)	...	sqm	4.30	8.90	7.40	10.60	-
Exceeding 50°	...	sqm	8.60	P.O.A	17.85	35.00	-
Laying 7.5 to 12.5 metres above ground level at eaves/or three storey	...	sqm	P.O.A.	3.70	6.10	3.95	8.70
Ridge or hip tile including angles and intersections, bedding and pointing	...	m	28.00	31.30	32.50	28.40	40.70
Apex tile ditto	...	No	-	25.10	22.40	-	21.70
Bedding and pointing at verge including fibre cement bedding strip	...	m	32.00	24.10	23.00	19.60	30.60
Tapered barge (dry fitting)	...	m	-	-	-	-	-
Aluminium foil sarking, refer page 335	...	Note					
Over two storeys	...	Note					
Standard coloured shingles on and including battens	...	sqm	50.00	70.00-80.00	65.00-85.00	70.00-75.00	-
ADD extra for :							
Sarking (* must be included in Adelaide and Melbourne)	...	sqm	10.70	8.60	8.00	11.00	7.90
TERRACOTTA							
Prices are for up to two storey work	...	Note					
French pattern standard colour terracotta tiles to slope not exceeding 30° (27°/Perth) and fixed to and including 38 x 25mm timber battens or standard steel battens	...	sqm	60.00	80.00	60.00-70.00	60.00-65.00	88.80
Swiss pattern standard colour terracotta tiles as above	...	sqm	64.00	90.00	80.00	70.00	98.00
ADD extra for :							
50 x 38mm battens in lieu of 38 x 25mm	...	sqm	3.60	3.70	3.25	3.35	2.90
Laying to slope -							
30° to 40°(27°/35° - Perth)	...	sqm	4.80	8.50	11.00	10.80	-
40° to 45°(36°/50° - Perth)	...	sqm	10.00	17.00	14.00	35.90	-
Exceeding 50°	...	sqm	P.O.A.	P.O.A.	P.O.A.	53.70	-
Laying 7.5 to 12.5 metres above ground level at eaves/or three storey	...	sqm	P.O.A.	3.70	6.00	4.00	10.20

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TILE ROOFING (cont'd)							
TERRACOTTA (cont'd)							
Prices are for up to two storey work (cont'd)	...	Note					
ADD extra for (cont'd) :							
Ridge or hip tile including angles and intersections, bedding and pointing	...	m	41.50	43.50	42.80	29.10	41.50
Apex/hip tile ditto	...	No	-	35.00	27.50	-	21.80
Bedding and pointing at verge including fibre cement bedding strip	...	m	36.00	27.50	23.00	20.40	30.60
Aluminium foil sarking, refer page 335	...	Note					
Over two storeys	...	Note					
SLATE							
Nu-lok system comprising 400 x 250mm slates on and including steel batten grid, ridges and hips	...	sqm	205.00-225.00	205.00-225.00	215.00-235.00	215.00-235.00	210.00-230.00
ADD extra for :							
Mitred hips	...	Note					
Lead ridge capping	...	Note					
Traditional system of 500 x 250mm slates with a 64% overlap fixed to and including roofing battens, including ridges and hips	...	sqm	300.00-450.00	300.00-450.00	300.00-450.00	300.00-450.00	300.00-450.00
ADD extra for :							
Mitred hips	...	Note					
Lead ridge capping	...	Note					
FIBREGLASS ROOFING							
Translucent fibreglass to AS/NZS4256, Part 3, fixed to steel purlins at 1200mm centres :							
76mm corrugation - 1800 g/sqm	...	sqm	32.60	35.90	36.20	35.30	37.40
- 2400 g/sqm	...	sqm	38.20	41.60	41.80	40.90	43.10
Gel coated - 2400 g/sqm	...	sqm	43.10	46.40	46.60	45.70	47.90
Ribbed profile - 1800 g/sqm	...	sqm	32.60	35.90	36.20	35.30	37.40
- 2400 g/sqm	...	sqm	38.20	41.60	41.80	40.90	43.10
POLYCARBONATE ROOFING							
Clear or tinted polycarbonate fixed to timber :							
Corrugated profile -							
0.8mm	...	sqm	40.10	43.70	43.50	43.00	45.30
Ribbed profile -							
0.8mm :							
Greca	...	sqm	40.10	43.70	43.50	43.00	45.30
Trimdek®	...	sqm	42.30	45.90	45.60	45.20	47.40
ADD extra for :							
Polycarbonate infill to eaves or ridge	...	m	7.85	8.50	8.35	8.40	9.45
Self adhesive foam sealant at fascias, gutters	...	m	7.20	7.70	7.60	7.60	8.40

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ALUMINIUM ROOFING							
All roofing fixed to steel purlins	...	Note					
ALSPAN®, profile C, ribbed decking :							
0.70mm	...	sqm	60.40	62.00	64.90	60.90	70.40
0.90mm	...	sqm	70.80	72.30	75.20	71.30	80.70
ADD extra for :							
Painted finish one side	...	sqm	7.15	7.15	7.15	7.15	7.15
Ridge capping	...	m	26.50	31.00	29.90	30.10	37.30
Barge or fascia capping	...	m	25.40	29.90	28.80	29.00	36.20
Stop ending	...	m	9.85	12.30	11.75	11.85	15.75
Closing ends and lipping	...	m	13.15	16.45	15.65	15.80	21.00
KALZIP® standing seam decking :							
1.00mm	...	sqm	140.50	143.50	149.50	141.50	160.50
ADD extra for :							
Curveable in two planes and tappered	...	sqm	56.40	59.70	65.30	57.50	76.30
STEEL ROOFING							
All roofing fixed to steel purlins	...	Note					
CORRUGATED, Zincalume® steel							
0.42mm	...	sqm	38.00	39.60	42.50	38.60	48.00
0.48mm	...	sqm	44.60	46.20	49.10	45.10	54.60
0.60mm	...	sqm	56.30	57.90	60.80	56.90	66.30
0.80mm	...	sqm	71.80	73.40	76.30	72.30	81.80
ADD extra for :							
Colour coating one side -							
XRW (average)	...	sqm	8.10	8.10	8.10	8.10	8.10
Ultra (average)	...	sqm	35.10	35.10	35.10	35.10	35.10
Barge capping -							
Zincalume® steel	...	m	27.20	31.30	30.30	30.50	37.00
Colour coated	...	m	37.30	41.40	40.40	40.60	47.10
Ridge or hip capping -							
Zincalume® steel	...	m	30.90	35.00	34.00	34.20	40.70
Colour coated	...	m	41.20	45.30	44.30	44.50	51.00
CURVEABLE CORRUGATED, Zincalume® coated 0.60mm	...	sqm	52.10	53.60	56.50	52.60	62.00
ADD extra for :							
Colour coating -							
XRW (average)	...	sqm	8.10	8.10	8.10	8.10	8.10
RIBBED, Zincalume® steel :							
LONGSPAN, SPANDEK -							
0.42mm	...	sqm	43.20	44.70	47.60	43.70	53.10
0.48mm	...	sqm	47.80	49.40	52.30	48.40	57.80
MONOCLAD, TRIMDEK® -							
0.42mm	...	sqm	38.00	39.60	42.50	38.60	48.00
0.48mm	...	sqm	44.60	46.20	49.10	45.10	54.60
KLIP-LOK®, SPEED DECK -							
0.42mm	...	sqm	49.10	50.70	53.50	49.60	59.10
0.48mm	...	sqm	50.80	52.30	55.20	51.30	60.70
0.60mm	...	sqm	62.90	64.50	67.40	63.50	72.90

ROOFING AND ROOF PLUMBING

STEEL ROOFING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
RIBBED, Zincalume® steel (cont'd)							
ADD extra for :							
Colour coating -							
XRW :							
Ribbed (average)	...	sqm	10.60	10.60	10.60	10.60	10.60
Klip-lok®	...	sqm	13.65	13.65	13.65	13.65	13.65
Ultra :							
Trimdek®	...	sqm	29.60	29.60	29.60	29.60	29.60
Barge capping -							
Zincalume® steel	...	m	40.90	45.00	44.00	44.20	50.70
Colour coated	...	m	51.00	55.10	54.10	54.30	60.80
Ridge or hip capping -							
Zincalume® steel	...	m	30.90	35.00	34.00	34.20	40.70
Colour coated	...	m	41.20	45.30	44.30	44.50	51.00
Stop ending	...	m	9.85	12.30	11.75	11.85	15.75
Closing ends and lipping	...	m	13.15	16.45	15.65	15.80	21.00

COPPER ROOFING SYSTEM

Profiled copper panel roofing system fixed to and including plywood diaphragm deck, all flashings and accessories (cost range)	...	sqm	400.00-500.00	400.00-500.00	400.00-500.00	450.00-550.00	450.00-550.00
Copper shingles refer page 405	...	Note					

BUILT-UP ROOFING

Prices are based on a minimum of 500sqm	...	Note					
BITUMINOUS FELT FIBREGLASS REINFORCED BONDED WITH HOT BITUMEN							
Three layer laid to falls on concrete	...	sqm	75.70	74.50	75.20	73.60	75.00
ADD extra for :							
Dressing to eaves gutter including 50mm down-turn and anchor strip	...	m	31.90	32.40	31.10	31.40	32.30
150mm high skirting dressed over angle fillet at bottom and turned into groove at top including cover flashing	...	m	68.40	70.70	64.80	69.00	72.80
Dress to roof outlet 100 to 150mm dia.	...	No	32.00	33.10	31.80	25.50	41.30
Dress to pipe up to 100mm dia. including overflashing	...	No	32.00	33.10	31.80	31.80	41.30

ROOFING AND ROOF PLUMBING

BUILT-UP ROOFING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BITUMAT APP MODIFIED WATERPROOF MEMBRANE							
Single layer polyester reinforced APP modified bitumen lining 4mm thick (BITUMAT TORCHSEAL) on concrete with welded seams	...	sqm	68.40	70.70	64.80	69.00	72.80
ADD extra for :							
Additional layer including glass reinforced APP modified bitumen 3mm thick laid on concrete slab	...	sqm	23.90	23.90	21.50	23.40	21.50
Bitumen SBS modified waterproof membrane	...	sqm	5.60	5.60	5.55	5.60	5.55
Granular surface for exposed finish	...	sqm	6.10	6.15	6.10	6.10	6.10
Ventilated base sheet	...	sqm	8.40	8.40	8.30	8.15	8.35
150mm high skirting including sealed aluminium trim	...	m	44.10	44.20	40.80	42.80	39.80
Dress to roof outlet 100 to 150mm dia.	...	No	51.60	51.70	47.00	51.00	47.00
Dress to pipe up to 100mm dia. including overflashing	...	No	29.60	29.60	26.70	28.50	26.70
BITUMEN APP MODIFIED WATERPROOF MEMBRANE UNDER ASPHALT TOPPING							
Single layer polyester reinforced APP modified bitumen lining 3mm thick (BITUMAT POLYFLAME 3mm) on concrete with torch welded seams	...	sqm	41.40	41.70	39.80	39.90	39.40
ADD extra for :							
Additional layer of reinforced APP modified bitumen 3mm thick laid on concrete (BITUMAT POLYFLAME 3mm)	...	sqm	27.10	27.10	22.00	27.50	22.00
150mm high skirting including sealed aluminium trim	...	m	28.70	38.80	42.70	42.80	42.80
Dress to roof outlet 100 to 150mm dia.	...	No	52.70	52.80	42.70	52.00	42.80
Dress to pipe up to 100mm dia. including overflashing	...	No	32.30	32.30	26.70	31.60	26.70
Asphalt topping average 50mm thick	...	sqm	35.30	35.30	28.60	33.60	28.60

ROOFING AND ROOF PLUMBING

BUILT-UP ROOFING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
POLYURETHANE ROOF MEMBRANE, VEHICULAR TRAFFIC							
Roofing membrane laid to falls on concrete, low VOC	...	sqm	69.90	70.00	68.50	72.30	68.50
ADD extra for :							
Turning areas, ramps and driveways	...	sqm	28.40	28.20	28.00	29.00	28.00
Aggregate non-slip finish	...	sqm	8.60	8.55	7.50	7.85	7.50
150mm high turn up and sealant fillet	...	m	19.30	19.15	17.95	21.05	17.95
Detail cracks with sawcut and membrane	...	m	21.50	21.40	19.30	20.00	19.30
Dress to roof outlet 100 to 150mm dia.	...	No	27.80	27.60	25.50	28.40	25.50
Dress to pipe penetration up to 100mm dia.	...	No	19.35	19.25	17.40	18.15	20.80
Decorative stencil pattern acrylic finish laid over membrane	...	sqm	90.00	92.00	90.00	90.00	88.00
Low odour and low VOC	...	sqm	3.20	3.15	3.15	3.15	3.15
WET AREA MEMBRANES							
Single component polyurethane membrane, low VOC :							
Wet area polyurethane membrane laid to floors and walls	...	sqm	45.10	44.60	41.00	44.90	43.10
Fibre reinforced polyurethane membrane :							
Wet area polyurethane membrane laid to floors and walls	....	sqm	40.30	40.00	37.00	41.30	36.90
ADD extra for :							
Sand seeding with 16/30 aggregate under render topping or tiles	...	sqm	10.40	10.40	9.50	10.40	9.50
Mixer tap proprietary flange system	...	each	26.80	23.30	23.70	26.10	23.70
Tap or pipe penetration flange or sealant	...	each	17.20	17.00	15.10	15.65	15.10
Fibreglass reinforcing and sealant bead to wall floor junction	...	m	21.50	21.20	19.40	23.00	19.40
Shower recess and threshold 'bund' angle	...	m	129.00	127.50	120.50	130.50	121.00
Single layer polyurethane roofing and balcony membrane (pedestrian traffic) laid to falls on concrete, low VOC	...	sqm	69.90	70.00	68.50	72.30	68.50
ADD extra for :							
Aggregate non-slip finish	...	sqm	8.60	8.55	7.50	7.85	7.50
150mm high turn up and sealant fillet	...	m	19.30	19.15	17.95	21.05	17.95
Dress to roof outlet 100 to150mm dia.	...	No	27.80	27.60	25.50	28.40	25.50
Dress to pipe penetration up to 100mm dia.	...	No	19.35	19.25	17.40	18.15	20.75
Detail cracks with saw-cut and membrane	...	m	21.50	21.40	19.30	20.00	19.30
Decorative stencil patterned acrylic finish laid over membrane	...	sqm	90.00	92.00	90.00	90.00	88.00
Low odour and low VOC	...	sqm	3.10	3.10	3.10	3.10	3.10

ROOFING AND ROOF PLUMBING

BUILT-UP ROOFING (cont'd)

WET AREA MEMBRANES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Single ply synthetic roof and balcony membrane laid to falls on concrete, 1.5mm thick with internal polyester reinforcing	...	sqm	64.90	65.50	67.70	64.00	68.90
ADD extra for :							
Dressing to eaves gutter including 50mm turndown	...	m	23.70	26.00	25.80	24.00	25.80
Corners, internal and external	...	No	35.40	33.90	34.40	36.60	32.30
150mm high skirting including sealed aluminium flat bar trim	...	m	50.40	48.80	49.40	50.10	48.40
Dress to roof outlet 100 to 150mm dia.	...	No	59.00	55.20	55.90	60.60	53.80
Dress to pipe including overflashing up to 100mm dia.	...	No	41.80	41.40	38.70	41.80	37.70
Trafficable "Trac walkway" 760mm wide	...	m	66.50	68.90	64.50	67.90	64.60

INSULATION

Expanded polystyrene for IRMA roof application :							
Class H -							
20mm thick	...	sqm	17.75	17.85	18.05	18.15	18.00
50mm thick	...	sqm	31.00	30.70	28.30	30.70	30.80
Class VH -							
20mm thick	...	sqm	18.45	18.15	17.80	18.40	18.25
50mm thick	...	sqm	35.90	34.60	31.10	34.70	34.70
ADD extra for :							
Prefabricating foregoing insulation to falls	...	+25%					
Extruded polystyrene for IRMA roof application :							
30mm thick	...	sqm	30.50	30.00	28.50	30.00	30.10
50mm thick	...	sqm	47.00	46.30	44.80	46.30	46.50
IRMA-MATE fabric (500gsm)	...	sqm	5.50	5.40	5.45	5.60	5.40

LIGHTWEIGHT CONCRETE

Refer page 239	...	Note					

FINISHES AND PROTECTION TO BUILT-UP ROOFING

Course washed river gravel, 50mm thick, laid on membrane	...	sqm	28.00	28.30	28.90	28.40	28.80
Loose quartz chips laid on membrane, 50mm thick	...	sqm	36.30	36.00	33.10	36.20	33.00
400 x 400 x 50mm plain finished concrete paving laid on santoprene pads	...	sqm	116.00	115.50	113.50	124.00	115.50
400 x 400 x 50mm exposed aggregate finished paving laid on santoprene pads	...	sqm	120.00	119.50	115.50	124.00	115.50

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LIQUID ACRYLIC MEMBRANE							
Prices are based on a minimum of 500sqm	...	Note					
Roofing (BITUMAT SYNROOF HI-BUILD) laid to falls on concrete	...	sqm	41.00	39.80	40.30	41.00	40.20
ADD extra for :							
Non-woven polyester fabric reinforcing including extra membrane	...	sqm	17.60	17.50	16.35	20.15	16.30
Internal corner to turn ups including fibreglass	...	m	13.90	13.70	12.10	13.25	12.10
TANKING AND MEMBRANE TYPE APPLICATIONS							
Prices are based on a minimum of 500sqm	...	Note					
BITUMAT APP MODIFIED WATERPROOF MEMBRANE							
Single layer polyester reinforced APP modified bitumen lining 4mm thick (BITUMAT TORCHSEAL) laid to falls on concrete and with welded seams	...	sqm	43.30	42.90	41.80	43.90	41.70
ADD extra for :							
Additional layer including glass reinforced APP modified bitumen 3mm thick laid on concrete slab	...	sqm	26.50	26.60	21.50	26.90	21.50
Bitumat anti-root garden membrane in lieu of TORCHSEAL to planter boxes and landscaping	...	sqm	7.20	7.05	7.15	7.05	7.15
Bitumat SBS modified waterproof membrane	...	sqm	6.65	6.50	6.60	6.50	6.60
150mm high skirting including sealed aluminium trim	...	m	44.40	43.50	41.90	38.00	38.40
Dress to roof outlet 100 to 150mm dia.	...	No	47.70	48.90	43.00	47.70	42.80
Dress to pipe including overflashing up to 100mm dia.	...	No	29.70	29.20	26.80	29.60	26.80
Single layer self adhesive bitumen membrane 1.8mm thick (BITUMAT BITUSEAL) laid to falls on concrete and with welded seams	...	sqm	32.40	31.80	31.20	32.90	31.10
ADD extra for :							
150mm high skirting including sealed top edge	...	m	20.70	18.50	18.60	22.70	18.50
Dress to roof outlet 100 to 150mm dia.	...	No	22.80	21.60	21.60	24.80	21.60
Dress to pipe including overflashing up to 100mm dia.	...	No	20.70	19.60	19.60	21.70	19.00
Prices are based on a minimum of 200sqm	...	Note					
Spray applied (2 component) polyurea membrane including primer applied to sound concrete, 2.2mm nominal thickness	...	sqm	220.00	214.00	216.00	210.00	214.00
ADD extra for :							
Surface preparation to provide 'clean sound Class 2 concrete'	...	sqm	15.70	15.60	15.70	17.60	15.50
Night work and out of hours	...	sqm	56.00	55.70	56.10	55.00	55.40
Dressing to edges masked with ‘wired double sided tape and plastic’	...	m	39.20	39.00	39.30	41.80	38.80

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TANKING AND MEMBRANE TYPE APPLICATIONS (cont'd)							
BITUMAT APP MODIFIED WATERPROOF MEMBRANE (cont'd)							
Prices are based on a minimum of 1000sqm	...	Note					
Spray applied (2 component) polyurea membrane including primer laid to falls on concrete at 1.8mm nominal thickness	...	sqm	176.50	171.50	176.00	172.00	174.00
ADD extra for :							
Additional 1mm thickness	...	sqm	58.30	57.95	58.30	60.50	57.60
Surface preparation to provide 'clean sound Class 2 concrete'	...	sqm	3.45	13.35	13.45	16.50	13.30
Dressing to roof edges masked with 'wired double sided tape and plastic'	...	m	37.60	37.30	37.60	39.00	37.10
150mm high skirting masked with 'wired double sided tape and plastic'	...	m	37.00	36.80	37.00	38.50	36.50
Dress to roof outlet 100 to 150mm dia.	...	No	39.20	39.00	39.30	41.80	38.80
Dress to pipes and penetrations up to 300mm dia.	...	No	56.00	55.70	56.10	60.50	55.40
Movement expansion/contraction joints up to 50mm wide	...	m	145.50	145.00	146.00	154.00	144.00
Dressing to edges masked with 'wired double sided tape and plastic'	...	m	39.20	39.00	39.30	41.80	38.80
MEMBRANE LINING TO WATER STORAGE TANKS							
Prices are based on a minimum of 300sqm	...	Note					
Potable synthetic single layer sheet membrane lining applied to prepared surface	...	sqm	188.50	188.00	184.00	186.50	183.50
ADD extra for :							
Dressing to -							
Inlet/outlet valves	...	No	239.00	248.00	232.00	265.00	232.00
Access ladder fixings	...	No	61.50	59.10	58.10	63.50	57.90
Penetrations	...	No	58.30	57.00	54.90	59.30	54.70
SHINGLES							
American shingles, either slate, wood or dimensional look, including roof deck, flashings, vents, underlay and accessories :							
Standard quality, wind loading to 110kmh, 25 year warranty	...	sqm	120.00-145.00	110.00-135.00	100.00-125.00	120.00-145.00	110.00-135.00
Premium quality, wind loading to 175kmh, 40-50 year warranty	...	sqm	145.00-170.00	135.00-160.00	125.00-150.00	145.00-170.00	135.00-160.00
Flat copper shingles fixed to and including plywood deck, all flashings and accessories	...	sqm	450.00-550.00	450.00-550.00	450.00-550.00	500.00-600.00	500.00-600.00
Porcelain enamelled shingle (imitation slate) fixed to and including battens	...	sqm	215.00	220.00	210.00	220.00	215.00

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SHINGLES (cont'd)							
Timber shingles fixed to slope not exceeding 35° and nailed to and including battens (not exceeding 7.5 metres above ground level) :							
Australian hardwood -							
Shakes	...	sqm	156.00	156.00	156.00	167.00	162.00
Shingles	...	sqm	150.00	145.00	140.00	166.00	150.00
Western Red Cedar -							
Shakes	...	sqm	150.00	150.00	150.00	155.00	150.00
Shingles	...	sqm	133.00	133.00	133.00	144.00	133.00
Concrete shingles, refer page 397	...	Note					
INSULATION, SARKING							
Refer pages 335 to 338 for insulation and sarking materials	...	Note					
ROOF PENETRATIONS							
FLEXIBLE FLASHINGS							
Prices include penetration, flashing cone, silicone sealant, bedding and fixings	...	Note					
Steel Roof							
Dektite® E.P.D.M. penetration flashing cone to suit pipe of approximately :							
to 55mm dia.	...	No	64.30	70.00	67.40	68.80	84.80
50/70mm dia.	...	No	69.50	75.20	72.60	74.00	90.00
70/110mm dia.	...	No	75.70	81.30	78.70	80.10	96.10
110/160mm dia.	...	No	81.90	87.50	85.00	86.30	102.50
160/220mm dia.	...	No	114.50	121.00	118.00	120.00	138.50
220/300mm dia.	...	No	145.50	152.00	149.00	150.50	170.00
290/380mm dia.	...	No	195.50	202.50	199.50	201.00	222.00
METAL FLASHINGS							
Prices include penetration, tray and/or collar flashing, no allowance for overflashing	...	Note					
Tiled Roof :							
Lead, 1.80mm (20.41kg/sqm) to -							
50mm dia. pipe, 600 x 600mm tray	...	No	115.00	123.00	119.50	121.50	143.50
100mm dia. pipe, 600 x 600mm tray	...	No	115.00	123.00	119.50	121.50	143.50
150mm dia. pipe, 900 x 900mm tray	...	No	183.50	192.50	188.50	190.50	215.00
Steel Roof :							
Zincalume® steel, 0.70mm including silicone seal bedding to -							
50mm dia. pipe, 600 x 600mm tray	...	No	108.50	116.00	112.50	114.50	137.00
100mm dia. pipe, 600 x 600mm tray	...	No	108.50	116.00	112.50	114.50	137.00
150mm dia. pipe, 900 x 900mm tray	...	No	158.00	166.50	162.50	165.00	189.50
Built-up Roofing							
Refer pages 400 to 403	...	Note					

ROOFING AND ROOF PLUMBING

ROOFLIGHTS, SKYLIGHTS

SOLA-TUBE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Toughened dome skylight complete with reflector, 1250mm of reflective tube and curved diffuser, fully installed, size :							
250mm dia.	...	No	485.00	500.00	495.00	495.00	500.00
400mm dia.	...	No	630.00	645.00	645.00	640.00	650.00

ROOF WINDOWS

Laminated glass fixed roof window including flashings to roof :							
500 x 700mm	...	No	530.00	555.00	550.00	550.00	590.00
780 x 980mm	...	No	750.00	785.00	775.00	780.00	830.00
1050 x 1050mm	...	No	960.00	1005.00	995.00	995.00	1060.00
Laminated glass opening roof window including flashings to roof :							
500 x 700mm	...	No	1000.00	1025.00	1020.00	1020.00	1060.00
780 x 980mm	...	No	1175.00	1210.00	1205.00	1205.00	1260.00
1050 x 1050mm	...	No	1430.00	1475.00	1465.00	1465.00	1535.00
Double glazed, toughened and laminated glass, fixed roof window including flashings to roof :							
500 x 700mm	...	No	630.00	655.00	650.00	650.00	690.00
780 x 980mm	...	No	865.00	900.00	890.00	895.00	945.00
1050 x 1050mm	...	No	1130.00	1170.00	1160.00	1165.00	1230.00
Double glazed, toughened and laminated glass, opening roof window including flashings to roof :							
500 x 700mm	...	No	1045.00	1070.00	1065.00	1065.00	1110.00
780 x 980mm	...	No	1255.00	1290.00	1285.00	1285.00	1340.00
1050 x 1050mm	...	No	1650.00	1695.00	1685.00	1685.00	1755.00
ADD extra for :							
Pleated Window blind to roof window -							
500 x 700mm	...	No	200.00	200.00	200.00	200.00	200.00
780 x 980mm	...	No	225.00	225.00	225.00	225.00	225.00
1050 x 1050mm	...	No	280.00	280.00	280.00	280.00	280.00
Electronic opening system	...	No	760.00	760.00	760.00	760.00	760.00

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FLASHINGS							
ALUMINIUM							
Straight :							
0.60mm x 250mm girth	...	m	36.50	40.00	42.30	39.20	54.40
0.60mm x 500mm girth	...	m	51.00	55.30	57.20	54.40	72.00
0.60mm x 750mm girth	...	m	65.40	70.60	72.20	69.50	89.60
0.70mm x 250mm girth	...	m	37.50	41.00	43.30	40.20	55.40
0.70mm x 500mm girth	...	m	53.00	57.30	59.20	56.40	74.00
0.70mm x 750mm girth	...	m	68.40	73.60	75.10	72.50	92.60
Stepped :							
0.60mm x 250mm girth	...	m	73.30	81.10	85.30	79.40	112.00
0.60mm x 500mm girth	...	m	87.70	96.40	100.50	94.50	129.50
0.60mm x 750mm girth	...	m	102.00	111.50	115.00	109.50	147.50
0.70mm x 250mm girth	...	m	74.30	82.00	82.40	80.40	108.00
0.70mm x 500mm girth	...	m	89.70	98.40	102.00	96.50	131.50
0.70mm x 750mm girth	...	m	105.00	114.50	118.00	112.50	150.50
COPPER							
Straight :							
0.50mm x 250mm girth	...	m	64.60	68.10	70.40	67.30	82.50
0.50mm x 500mm girth	...	m	107.00	111.50	113.50	110.50	128.00
0.50mm x 750mm girth	...	m	149.50	155.00	156.50	154.00	174.00
0.55mm x 250mm girth	...	m	66.80	70.30	72.60	69.60	84.70
0.55mm x 500mm girth	...	m	111.50	116.00	118.00	115.00	132.50
0.55mm x 750mm girth	...	m	156.50	161.50	163.00	160.50	180.50
Stepped :							
0.50mm x 250mm girth	...	m	101.50	109.00	109.50	107.50	135.00
0.50mm x 500mm girth	...	m	144.00	152.50	156.50	151.00	186.00
0.50mm x 750mm girth	...	m	186.50	196.00	199.50	194.00	231.50
0.55mm x 250mm girth	...	m	103.50	111.50	112.00	109.50	137.00
0.55mm x 500mm girth	...	m	148.50	157.00	161.00	155.00	190.50
0.55mm x 750mm girth	...	m	193.00	202.50	206.00	200.50	238.50
ZINCALUME® STEEL							
Straight :							
0.55mm x 250mm girth	...	m	32.20	35.20	37.70	34.60	48.40
0.55mm x 500mm girth	...	m	46.00	49.90	52.00	49.00	65.40
0.55mm x 750mm girth	...	m	59.80	64.50	66.30	63.50	82.30
0.80mm x 250mm girth	...	m	34.70	37.80	40.30	37.10	51.00
0.80mm x 500mm girth	...	m	51.10	55.00	57.10	54.20	70.50
0.80mm x 750mm girth	...	m	67.50	72.20	74.00	71.20	90.00
Stepped :							
0.55mm x 250mm girth	...	m	68.90	76.30	76.80	74.70	101.00
0.55mm x 500mm girth	...	m	82.70	90.90	95.00	89.20	123.00
0.55mm x 750mm girth	...	m	96.50	105.50	109.50	103.50	140.00
0.80mm x 250mm girth	...	m	71.50	78.90	83.30	77.30	109.00
0.80mm x 500mm girth	...	m	87.90	96.10	100.00	94.30	128.50
0.80mm x 750mm girth	...	m	104.00	113.50	117.00	111.50	148.00

ROOFING AND ROOF PLUMBING

FLASHINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ZINCALUME® STEEL, COLOUR COATED							
Straight :							
0.55mm x 250mm girth	...	m	43.40	47.30	49.40	46.50	62.80
0.55mm x 500mm girth	...	m	61.00	65.80	67.50	64.80	83.60
0.55mm x 750mm girth	...	m	78.70	84.30	85.70	83.10	104.50
0.80mm x 250mm girth	...	m	46.00	49.90	52.00	49.10	65.40
0.80mm x 500mm girth	...	m	66.30	71.00	72.80	70.00	88.80
0.80mm x 750mm girth	...	m	86.50	92.20	93.50	91.00	112.50
Stepped :							
0.55mm x 250mm girth	...	m	80.10	88.40	92.40	86.60	120.50
0.55mm x 500mm girth	...	m	97.80	107.00	110.50	105.00	141.50
0.55mm x 750mm girth	...	m	115.50	125.50	128.50	123.50	162.00
0.80mm x 250mm girth	...	m	82.70	91.00	95.10	89.20	123.00
0.80mm x 500mm girth	...	m	103.00	112.00	116.00	110.00	146.50
0.80mm x 750mm girth	...	m	123.50	133.00	136.50	136.00	170.00
LEAD							
Straight :							
0.90mm x 250mm girth	...	m	42.50	45.50	48.10	44.90	58.80
0.90mm x 500mm girth	...	m	66.60	70.50	72.70	69.70	86.10
0.90mm x 750mm girth	...	m	90.80	95.50	97.30	94.50	113.50
1.80mm x 250mm girth	...	m	59.30	62.30	64.80	61.70	75.60
1.80mm x 500mm girth	...	m	100.00	104.00	106.00	103.50	119.50
1.80mm x 750mm girth	...	m	141.00	146.00	147.50	145.00	163.50
Stepped :							
0.90mm x 250mm girth	...	m	79.30	86.60	91.10	85.10	116.50
0.90mm x 500mm girth	...	m	103.50	111.50	115.50	110.00	144.00
0.90mm x 750mm girth	...	m	127.50	136.50	140.50	134.50	171.00
1.80mm x 250mm girth	...	m	96.00	103.50	108.00	102.00	133.50
1.80mm x 500mm girth	...	m	137.00	145.00	149.50	143.50	177.50
1.80mm x 750mm girth	...	m	178.00	187.00	190.50	185.00	221.50
STAINLESS STEEL							
Straight :							
0.50mm x 250mm girth	...	m	60.30	65.50	67.00	64.40	84.40
0.50mm x 500mm girth	...	m	83.80	89.90	91.00	88.60	111.00
0.50mm x 750mm girth	...	m	107.50	114.50	115.00	113.00	138.00
Stepped :							
0.50mm x 250mm girth	...	m	97.00	106.50	110.00	104.50	142.00
0.50mm x 500mm girth	...	m	120.50	131.00	134.00	128.50	169.00
0.50mm x 750mm girth	...	m	144.00	155.50	158.00	153.00	195.50
ZINC :							
Straight :							
0.70mm x 250mm girth	...	m	46.40	49.80	52.20	49.10	64.20
0.70mm x 500mm girth	...	m	70.70	75.00	76.90	74.10	91.70
0.70mm x 750mm girth	...	m	95.00	100.00	101.50	99.10	119.00
0.80mm x 250mm girth	...	m	48.20	51.70	54.00	51.00	66.10
0.80mm x 500mm girth	...	m	74.40	78.70	80.70	77.80	95.40
0.80mm x 750mm girth	...	m	100.50	111.00	110.00	108.50	124.50

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FLASHINGS (cont'd)							
ZINC (cont'd) :							
Stepped :							
0.70mm x 250mm girth	...	m	83.10	90.90	95.20	89.30	122.00
0.70mm x 500mm girth	...	m	107.50	116.00	120.00	114.50	149.50
0.70mm x 750mm girth	...	m	131.50	141.50	145.00	139.00	177.00
0.80mm x 250mm girth	...	m	85.00	92.80	97.10	91.10	124.00
0.80mm x 500mm girth	...	m	111.00	120.00	123.50	118.00	153.00
0.80mm x 750mm girth	...	m	137.50	147.00	150.50	145.00	182.50
DOWNPIPES							
Prices include strap fixings	...	Note					
ALUMINIUM, 0.70mm :							
100 x 55mm	...	m	34.80	37.10	38.00	36.60	46.00
110 x 75mm	...	m	36.20	38.60	39.40	38.10	47.50
CAST IRON :							
Refer pages 469 and 470, allow as Soil Waste and Vent pipework	...	Note					
COPPER, 0.55mm :							
75mm dia.	...	m	79.80	82.20	83.10	81.70	91.10
100mm dia.	...	m	95.60	98.40	99.50	97.80	109.00
150mm dia.	...	m	125.00	129.00	130.50	128.00	142.50
100 x 75mm	...	m	89.20	91.90	93.00	91.30	102.50
150 x 100mm	...	m	106.00	109.50	110.00	108.50	120.50
ZINCALUME® STEEL, 0.55mm :							
75mm dia.	...	m	34.50	36.90	37.80	36.40	45.80
100mm dia.	...	m	41.70	44.40	45.50	43.80	54.90
150mm dia.	...	m	60.80	64.30	65.80	63.60	77.60
100 x 50mm	...	m	42.00	44.60	45.80	44.10	54.60
100 x 75mm	...	m	57.10	60.50	62.90	59.80	74.90
ZINCALUME® STEEL, COLOUR COATED, 0.61mm :							
100 x 50mm	...	m	54.60	57.80	58.70	57.20	69.40
100 x 75mm	...	m	61.40	64.70	65.90	64.00	76.90
150 x 100mm	...	m	80.50	84.60	86.70	83.80	100.50
STAINLESS STEEL, 0.50mm :							
75mm dia.	...	m	89.80	93.00	93.50	92.30	104.00
100mm dia.	...	m	119.50	123.00	123.50	122.50	136.00
150mm dia.	...	m	163.00	168.00	169.00	167.00	185.00
PVC RIGID :							
80mm dia.	...	m	31.90	34.30	34.80	33.80	42.80
100 x 50mm	...	m	34.90	37.30	37.70	36.80	45.80
ADD extra for :							
Bend, elbow or shoe to downpipe -							
Aluminium :							
100 x 50mm	...	No	40.40	45.20	46.90	44.20	63.00
100 x 75mm	...	No	44.10	49.30	50.80	48.20	68.30

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DOWNPIPES (cont'd)							
ADD extra for (cont'd) :							
Bend, elbow or shoe to downpipe (cont'd) -							
Copper :							
75mm dia.	...	No	58.80	65.70	66.50	64.30	89.30
100mm dia.	...	No	66.20	73.90	78.20	72.30	105.00
150mm dia.	...	No	91.90	102.50	105.50	100.50	142.00
100 x 50mm	...	No	58.80	65.70	66.50	64.30	89.30
100 x 75mm	...	No	62.50	69.80	70.40	68.30	94.50
Zincalume® steel :							
75mm dia.	...	No	33.10	37.00	39.10	36.10	52.50
100mm dia.	...	No	40.40	45.20	46.90	44.20	63.00
150mm dia.	...	No	55.10	61.60	62.60	60.20	84.00
100 x 75mm	...	No	36.80	41.10	43.00	40.20	57.80
150 x 100mm	...	No	58.80	65.70	66.50	64.30	89.30
Zincalume® steel, colour coated :							
100mm dia.	...	No	47.80	53.40	54.80	52.20	73.50
100 x 75mm	...	No	44.10	49.30	50.80	48.20	68.30
150 x 100mm	...	No	66.20	73.90	74.30	72.30	99.80
Stainless steel :							
75mm dia.	...	No	69.80	78.10	78.20	76.30	105.00
100mm dia.	...	No	84.50	94.50	93.90	92.40	126.00
150mm dia.	...	No	132.50	148.00	148.50	144.50	199.50
PVC rigid :							
80mm dia.	...	No	33.20	36.20	36.80	35.60	46.80
100 x 50mm	...	No	36.80	39.80	40.40	39.10	50.40
EAVES GUTTERS							
Prices include strap fixings	...	Note					
ALUMINIUM, 115 x 0.60mm, quadrant	...	m	43.20	46.00	46.70	45.40	56.10
COPPER, 115 x 0.56mm, quadrant	...	m	83.70	87.00	87.50	86.30	98.20
ZINCALUME® STEEL, 115 x 0.42mm, quadrant	...	m	35.80	38.70	39.30	38.10	48.70
ZINCALUME® STEEL, COLOUR COATED, 115 x 0.48mm, quadrant	...	m	43.00	46.30	46.80	45.60	57.50
STAINLESS STEEL, 115 x 0.50mm, quadrant	...	m	81.50	85.20	85.40	84.40	97.50
PVC rigid, 113mm quadrant	...	m	34.40	36.70	37.60	36.20	45.70
ADD extra for :							
Mitred angle to quadrant eaves gutter -							
Aluminium, 115 x 0.60mm	...	No	38.60	43.10	43.00	42.20	57.80
Copper, 115 x 0.56mm	...	No	45.90	51.40	50.80	50.20	68.30
Zincalume® steel, 115 x 0.42mm	...	No	36.80	41.10	41.10	40.20	55.10
Zincalume® steel, colour coated, 115 x 0.48mm	...	No	45.90	51.40	50.80	50.20	68.30
Stainless steel, 115 x 0.50mm	...	No	51.50	57.50	58.70	56.20	78.80
PVC, rigid, 113mm	...	No	49.50	53.60	53.70	52.70	67.10

ROOFING AND ROOF PLUMBING

EAVES GUTTERS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra for (cont'd) :							
Stopped end to quadrant eaves gutter -							
Aluminium, 115 x 0.60mm	...	No	38.60	43.00	41.00	42.10	54.40
Copper, 115 x 0.56mm	...	No	52.60	57.80	55.40	56.70	71.50
Zincalume® steel, 115 x 0.42mm	...	No	40.40	44.70	42.70	43.80	56.10
Zincalume® steel, colour coated, 115 x 0.48mm	...	No	47.10	52.30	49.90	51.20	66.00
Stainless steel, 115 x 0.50mm	...	No	50.50	55.70	53.40	54.60	69.40
PVC, rigid, 113mm	...	No	33.20	36.70	35.10	35.90	45.80
Expansion joint to quadrant eaves gutter -							
Aluminium, 115 x 0.60mm	...	No	93.30	103.50	103.00	101.50	136.50
Copper, 115 x 0.56mm	...	No	106.00	117.00	116.00	114.50	152.00
Zincalume® steel, 115 x 0.42mm	...	No	95.50	106.00	105.00	103.50	138.50
Zincalume® steel, colour coated, 115 x 0.48mm	...	No	116.50	129.00	127.50	126.50	167.50
Stainless steel, 115 x 0.50mm	...	No	119.00	132.00	130.00	129.00	171.50
Outlet and joint to 100 x 50mm or nearest size thereto downpipe together with wire mesh guard -							
Aluminium, 115 x 0.60mm	...	No	43.40	47.70	45.80	46.80	59.20
Copper, 115 x 0.56mm	...	No	65.40	70.50	68.20	69.40	84.30
Zincalume® steel, 115 x 0.42mm	...	No	51.10	55.50	53.50	54.50	66.90
Zincalume® steel, colour coated, 115 x 0.48mm	...	No	62.10	67.30	64.90	66.20	81.00
Stainless steel, 115 x 0.50mm	...	No	60.10	65.30	63.00	64.20	79.00
PVC, rigid, 113mm	...	No	45.50	48.90	47.40	48.20	58.10

BOX GUTTERS

Prices do not include gutter boarding	...	Note					
ALUMINIUM, 1000mm girth x 0.80mm	...	m	84.00	89.20	94.60	88.10	113.50
COPPER, 1000mm girth x 0.55mm	...	m	160.00	166.00	171.00	164.50	192.50
ZINCALUME® STEEL, 1000mm girth x 0.70mm	...	m	82.80	88.00	93.50	86.90	112.00
STAINLESS STEEL, 1000mm girth x 0.50mm	...	m	123.50	130.00	135.00	129.00	158.00
ZINC, 1000mm girth x 0.70mm	...	m	123.50	129.50	134.50	128.00	156.00
ADD extra for :							
Mitred angle to 1000mm girth box gutter -							
Aluminium x 0.80mm	...	No	44.10	49.30	50.80	48.20	68.30
Copper x 0.55mm	...	No	62.50	69.80	70.40	68.30	94.50
Zincalume® steel x 0.70mm	...	No	44.10	49.30	50.80	48.20	68.30
Stainless steel x 0.50mm	...	No	66.20	73.90	74.30	72.30	99.80
Zinc x 0.70mm	...	No	51.50	57.50	58.70	56.20	78.80
Stopped end to 1000mm girth box gutter -							
Aluminium x 0.80mm	...	No	63.50	70.00	70.90	68.60	92.30
Copper x 0.55mm	...	No	103.00	112.00	112.00	110.00	141.50
Zincalume® steel x 0.70mm	...	No	63.50	70.00	71.00	68.60	92.40
Stainless steel x 0.50mm	...	No	99.50	109.50	109.00	107.50	141.00
Zinc x 0.70mm	...	No	78.50	85.80	86.40	84.30	110.50

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
BOX GUTTERS (cont'd)							
ADD extra for (cont'd) :							
Expansion joint to 1000mm girth box gutter -							
Aluminium x 0.80mm	...	No	146.50	163.00	159.00	159.50	211.50
Copper x 0.55mm	...	No	194.00	214.00	213.00	210.00	278.00
Zincalume® steel x 0.70mm	...	No	139.50	155.00	152.00	152.00	201.50
Stainless steel x 0.50mm	...	No	218.00	242.00	239.00	237.00	317.00
Zinc x 0.70mm	...	No	167.00	185.50	185.00	181.50	244.00
Outlet and joint to 100 x 0.50mm or nearest size thereto downpipe together with wire mesh guard -							
Aluminium x 0.80mm	...	No	60.70	67.20	68.10	65.80	89.60
Copper x 0.55mm	...	No	93.60	102.50	102.50	100.50	132.00
Zincalume® steel x 0.70mm	...	No	62.30	68.80	69.80	67.40	91.20
Stainless steel x 0.50mm	...	No	97.00	107.00	106.50	105.00	138.50
Zinc x 0.70mm	...	No	75.80	83.20	83.70	81.60	108.00
VALLEY GUTTERS							
ALUMINIUM, 500mm girth x 0.80mm	...	m	50.40	54.30	56.40	53.40	69.80
COPPER, 500mm girth x 0.55mm	...	m	107.00	112.00	113.50	111.00	129.50
ZINCALUME® STEEL, 500mm girth x 0.60mm	...	m	50.60	54.50	56.60	53.60	70.00
ZINC, 500mm girth x 0.70mm	...	m	67.10	71.00	73.10	70.20	86.50
ROOF OUTLETS							
100mm dia. flat outlet with flanged joints :							
Brass	...	No	698.00	709.00	716.00	707.00	756.00
Cast iron	...	No	482.00	494.00	489.00	492.00	525.00
SUMP OUTLETS							
Prices include for jointing to gutter and to downpipe, outlet and wire mesh guard	...	Note					
ALUMINIUM, 0.80mm :							
300 x 300 x 150mm deep	...	No	177.50	196.00	195.50	192.00	256.00
450 x 450 x 150mm deep	...	No	221.00	243.00	245.00	238.00	318.00
COPPER, 0.55mm :							
300 x 300 x 150mm deep	...	No	241.00	263.00	265.00	259.00	339.00
450 x 450 x 150mm deep	...	No	304.00	330.00	334.00	325.00	421.00
ZINCALUME® STEEL, 0.80mm :							
300 x 300 x 150mm deep	...	No	177.50	196.00	195.50	192.00	256.00
450 x 450 x 150mm deep	...	No	220.00	243.00	244.00	238.00	318.00
STAINLESS STEEL, 0.50mm :							
300 x 300 x 150mm deep	...	No	224.00	247.00	248.00	242.00	322.00
450 x 450 x 150mm deep	...	No	275.00	302.00	305.00	296.00	392.00
ZINC, 0.70mm :							
300 x 300 x 150mm deep	...	No	193.00	212.00	211.00	208.00	271.00
450 x 450 x 150mm deep	...	No	245.00	268.00	269.00	263.00	343.00

ROOFING AND ROOF PLUMBING

WALL CLADDING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
All cladding fixed to steel	...	Note					
FIBREGLASS							
Corrugated translucent sheeting with 76mm corrugations :							
1800 g/sqm	...	sqm	36.50	38.10	41.10	37.30	46.30
2400 g/sqm	...	sqm	42.20	43.80	46.80	43.00	52.00
STEEL							
All zincalume® coated :							
0.42mm corrugated	...	sqm	39.90	41.50	44.50	40.70	49.70
0.42mm Longspan, Spandek®	...	sqm	45.00	46.60	49.60	45.80	54.80
0.42mm Monoclad, Trimdek®	...	sqm	39.90	41.50	44.50	40.70	49.70
0.42mm Klip-lok®, Speed Deck	...	sqm	50.90	52.50	55.50	51.70	60.70
0.42mm Panelrib®	...	sqm	51.60	53.20	56.20	52.40	61.40
0.35mm Multiclad®	...	sqm	38.00	39.60	42.60	38.80	47.80
0.42mm Multiclad®	...	sqm	42.90	44.50	47.50	43.70	52.70
0.35mm Wallclad®	...	sqm	38.00	39.60	42.70	38.80	47.90
0.55mm Mini Orb®	...	sqm	66.90	68.50	71.50	67.70	76.70
0.60mm Curveable corrugated	...	sqm	53.90	55.50	58.50	54.70	63.70
ADD extra for :							
Corner mould -							
Plain	...	m	25.00	29.10	28.20	28.40	34.90
Colour coated	...	m	25.80	29.90	28.90	29.10	35.60
Colour coating (XRW) one side -							
Corrugated	...	sqm	8.10	8.10	8.10	8.10	8.10
Ribbed generally	...	sqm	10.60	10.60	10.60	10.60	10.60
Klip-lok®	...	sqm	13.65	13.65	13.65	13.65	13.65
Colour coating (Ultra) one side -							
Refer page 399	...	Note					
ALUMINIUM							
0.70mm ribbed :							
Plain	...	sqm	57.50	59.10	62.10	58.30	67.30
Painted finish, one side	...	sqm	64.70	66.20	69.30	65.40	74.50
CLADDING SYSTEMS							
Refer pages 297 to 300 for various cladding systems and page 366 for curtain walling	...	Note					

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
RIDGE VENTILATORS							
CONTINUOUS RIDGE VENTILATOR							
Fully designed engineered model :							
300mm throat width -							
Zincalume® steel	...	m	327.00	344.00	340.00	340.00	366.00
Colorbond® steel	...	m	364.00	380.00	376.00	377.00	403.00
500mm throat width :							
Zincalume® steel	...	m	429.00	451.00	446.00	447.00	482.00
Colorbond® steel	...	m	496.00	518.00	513.00	514.00	549.00
ADD extra for :							
Operable dampers to ventilator -							
300mm throat width :							
Zincalume® steel	...	m	171.50	179.50	177.50	185.00	191.00
Colorbond® steel	...	m	227.00	235.00	233.00	243.00	246.00
500mm throat width :							
Zincalume® steel	...	m	228.00	237.00	235.00	245.00	250.00
Colorbond® steel	...	m	295.00	304.00	302.00	315.00	316.00
Birdmesh to ventilator -							
300mm throat width	...	m	39.20	39.20	39.20	39.20	39.20
500mm throat width	...	m	59.50	59.50	59.50	59.50	59.50
End to ventilator -							
300mm throat width :							
Zincalume® steel	...	No	181.00	194.50	191.00	192.00	213.00
Colorbond® steel	...	No	187.00	200.00	197.00	197.50	218.00
500mm throat width :							
Zincalume® steel	...	No	255.00	271.00	267.00	268.00	294.00
Colorbond® steel	...	No	260.00	277.00	273.00	274.00	300.00
Standard model :							
200mm throat width -							
Zincalume® steel	...	m	152.50	164.00	161.00	161.50	180.00
Colorbond® steel	...	m	157.50	169.50	166.50	167.00	185.50
300mm throat width -							
Zincalume® steel	...	m	204.00	219.00	216.00	216.00	240.00
Colorbond® steel	...	m	208.00	223.00	220.00	220.00	244.00
ADD extra for :							
End -							
200mm and 300mm throat :							
Zincalume® steel	...	No	40.70	44.80	43.80	44.00	50.50
Colorbond® steel	...	No	41.40	45.50	44.50	44.70	51.20
NATURAL AND POWER VENTILATION							
Refer pages 642 to 647	...	Note					

ROOFING AND ROOF PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
FABRIC STRUCTURES							
Prices are based on plan area	...	Note					
The smaller the structure the higher the cost	...	Note					
Custom designed tensioned membrane structure fabricated from PVC coated polyester material and having an expected life span of fifteen to twenty years and in an application where fire performance is non critical	...	sqm	300.00-400.00	300.00-400.00	300.00-400.00	300.00-400.00	300.00-400.00
Custom designed tensioned membrane structure fabricated from PTFE Teflon coated fibreglass material and having an expected life span of twenty five years and in an application where fire performance is critical	...	sqm	500.00-600.00	500.00-600.00	500.00-600.00	500.00-600.00	500.00-600.00
Custom designed tensioned membrane car park canopy to standard arched designs and having an expected life span of fifteen to twenty years, based on 100 car park spaces :							
PVC mesh fabric	...	sqm	170.00-220.00	170.00-220.00	170.00-220.00	170.00-220.00	170.00-220.00
HDPE mesh fabric	...	sqm	120.00-170.00	120.00-170.00	120.00-170.00	120.00-170.00	120.00-170.00
Custom designed air inflated structure fabricated from ETFE foil material and having an expected life span of twenty years, applications where lightweight transparent foil as an alternative to glazing	...	sqm	800.00-1000.00	800.00-1000.00	800.00-1000.00	800.00-1000.00	800.00-1000.00
ADD extra for :							
Support fans to air supported configurations	...	Note					
Custom designed tensioned membrane fabric façade from lightweight PVC fabric mesh and having an expected life span of fifteen to twenty years, fixed to support framework by others	...	sqm	350.00-500.00	350.00-500.00	350.00-500.00	350.00-500.00	350.00-500.00
VERGOLA ROOFING SYSTEM							
Louvred opening roof system comprising winder operated Colorbond® steel louvres together with all framings and posts	...	sqm	800.00-900.00	800.00-900.00	900.00-1000.00	800.00-900.00	900.00-1000.00
ADD extra for :							
Electrically operated control mechanism	...	P.O.A.					

PLASTERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for render and plaster include for all labours, protection and making good around pipes, etc.	...	Note					

PREPARATORY WORK

Bonding agent applied to surface	...	sqm	5.25	3.95	5.15	4.35	3.75
Preparing and dubbing out not exceeding 13mm thick on existing work before plastering	...	sqm	15.50	10.20	12.50	10.75	13.20
Pricking up coat on metal lathing	...	sqm	15.80	10.45	13.70	12.50	12.65

RENDER FOR TILING

CEMENT AND SAND RENDER IN ONE COAT 13MM THICK ON :

Brick walls	...	sqm	35.00	29.60	20.80	21.60	27.80
Concrete walls	...	sqm	36.10	31.00	20.80	21.60	28.60
Isolated columns	...	sqm	42.90	39.40	26.60	33.00	45.00
Narrow widths up to 250mm wide	...	sqm	50.20	43.30	37.80	44.50	47.80

CEMENT RENDER

WOOD FLOAT FINISH IN ONE COAT 13MM THICK ON :

Brick walls	...	sqm	40.40	41.40	31.30	31.60	28.60
Concrete walls	...	sqm	41.80	43.30	32.40	34.40	29.80
Concrete soffits	...	sqm	49.30	45.00	39.80	34.40	32.30
Sloping concrete soffits of stairs	...	sqm	51.60	45.00	39.80	46.10	33.90
Isolated columns	...	sqm	51.60	51.10	39.80	42.90	46.80
Sides and soffits of beams	...	sqm	66.30	51.10	45.30	49.30	46.80
Brick or concrete filling to existing openings	...	sqm	67.40	61.40	57.50	56.60	68.90
Narrow widths up to 250mm wide	...	sqm	70.50	57.50	46.70	58.10	49.40
ADD extra for :							
Waterproofing agent	...	sqm	1.00	1.00	1.00	1.00	1.00
Labours, included in sqm rate	...	Note					

WOOD FLOAT FINISH IN TWO COATS 19MM THICK ON :

Brick walls	...	sqm	53.90	47.30	46.70	42.40	45.40
Concrete walls	...	sqm	55.50	48.30	48.10	43.60	47.20
Concrete soffits	...	sqm	63.80	51.10	55.20	44.40	48.80
Sloping concrete soffits of stairs	...	sqm	68.30	51.10	55.20	60.50	51.60
Isolated columns	...	sqm	65.10	57.00	55.20	52.60	50.00
Sides and soffits of beams	...	sqm	73.40	57.00	62.40	63.50	50.00
Brick or concrete filling to existing openings	...	sqm	79.50	68.70	70.20	63.70	74.50
Narrow widths up to 250mm wide	...	sqm	83.40	59.00	61.20	70.90	71.70
ADD extra for :							
Waterproofing agent	...	sqm	1.00	1.00	1.00	1.00	1.00
Labours, included in sqm rate	...	Note					

PLASTERING

CEMENT RENDER (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
STEEL FLOAT FINISH IN ONE COAT 13MM THICK ON :							
Brick walls	…	sqm	41.80	42.40	32.50	34.20	29.80
Concrete walls	…	sqm	42.90	43.90	34.40	36.90	30.30
Concrete soffits	…	sqm	50.50	45.60	38.40	36.90	33.10
Sloping concrete soffits of stairs	…	sqm	53.00	45.60	38.40	50.20	34.40
Isolated columns	…	sqm	58.50	51.10	38.40	44.40	48.30
Sides and soffits of beams	…	sqm	66.00	51.10	46.70	52.90	48.30
Brick or concrete filling to existing openings	…	sqm	70.60	61.90	61.90	56.60	73.10
Narrow widths up to 250mm wide	…	sqm	71.90	57.50	52.00	59.00	51.20
ADD extra for :							
Waterproofing agent	...	sqm	1.00	1.00	1.00	1.00	1.00
Labours, included in sqm rate	...	Note					
STEEL FLOAT FINISH IN TWO COATS 19MM THICK ON :							
Brick walls	…	sqm	55.70	48.30	52.00	40.10	45.40
Concrete walls	…	sqm	56.60	49.90	53.80	41.50	47.20
Concrete soffits	…	sqm	65.10	52.00	59.50	40.10	49.40
Sloping concrete soffits of stairs	…	sqm	69.70	52.00	59.50	59.10	51.60
Isolated columns	…	sqm	66.00	58.00	59.50	50.10	50.00
Sides and soffits of beams	…	sqm	74.90	58.00	66.30	61.60	50.00
Brick or concrete filling to existing openings	…	sqm	79.80	69.30	73.50	63.90	74.50
Narrow widths up to 250mm wide	…	sqm	83.90	59.60	66.30	72.60	71.70
ADD extra for :							
Waterproofing agent	...	sqm	1.00	1.00	1.00	1.00	1.00
Labours, included in sqm rate	...	Note					

HARDWALL PLASTER

RENDER AND SET WITH HARDWALL PLASTER IN TWO COATS 16MM THICK ON :							
Brick walls	…	sqm	56.60	55.70	52.00	25.70	51.60
Concrete walls	…	sqm	57.60	56.70	53.80	27.40	51.60
Concrete soffits	…	sqm	66.50	59.60	59.50	31.30	54.30
Sloping concrete soffits of stairs	…	sqm	72.90	59.60	59.50	47.60	60.40
Isolated columns	…	sqm	70.10	65.10	59.50	48.80	59.80
Sides and soffits of beams	…	sqm	79.50	65.10	68.60	53.40	59.80
Brick or concrete filling to existing openings	…	sqm	83.50	74.20	80.30	56.80	75.70
Narrow widths up to 250mm wide	…	sqm	87.70	69.00	66.30	65.10	73.10
ADD extra for :							
Labours, included in sqm rate	...	Note					
SKIM COAT HARDWALL PLASTER ON :							
Concrete soffits	…	sqm	36.10	30.60	22.90	27.30	26.60
Sloping concrete soffits of stairs	…	sqm	37.80	32.30	24.20	28.20	27.30

PLASTERING

TEXTURED FINISHES

ACRATEX APPLICATIONS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Preparatory treatment :							
AcraTex patch preparatory treatment on -							
Class 2 finished concrete surfaces	...	sqm	8.50	8.10	10.90	9.35	10.85
Fair faced masonry block, brickwork or aerated concrete block surfaces	...	sqm	18.30	33.20	24.30	18.55	18.30
AcraTex texture coat / AcraShield top coat on prepared base including primer :							
Roll on 00 Low Profile -	...	sqm	35.20	27.30	34.10	34.80	31.70
3mm High Profile	...	sqm	37.60	34.70	34.20	35.00	34.20
9mm High Profile	...	sqm	40.10	39.70	41.30	37.40	36.60
9mm Trowel Finish	...	sqm	42.60	42.20	42.60	39.90	41.40
AcraTex 2mm Trowel on -	...	sqm	48.60	42.50	48.70	42.50	48.70
1mm Spray on	...	sqm	44.90	32.70	43.80	37.60	42.60
Wall Membrane 953-103 roll on	...	sqm	36.60	30.10	42.50	34.90	36.60
Roofing Membrane 105	...	sqm	36.60	43.40	43.70	-	34.20
AcraStone 1mm/2mm	...	sqm	85.20	74.00	61.30	74.90	84.60
AcraSand two coats	...	sqm	32.70	29.90	35.40	37.00	30.50
Tuscany trowel on	...	sqm	54.50	43.40	54.50	42.20	48.80
Coventry trowel on	...	sqm	54.50	43.40	54.50	42.20	48.80
Substrate system :							
Trowel on -							
Aerated concrete	...	sqm	54.50	49.50	58.40	-	72.00
Fibre cement sheeting	...	sqm	60.60	53.40	60.80	61.90	72.10

GRANOSITE APPLICATIONS

Preparation systems :							
Cleaning concrete to remove residual bondbreaker (GranoTilt Clean)	...	sqm	4.20-6.30	4.20-6.30	4.20-6.30	4.20-6.30	4.20-6.30
Fill holes and ridges in concrete using acrylic patching compounds (GranoPatch Smooth, GranoPatch Spackle)	...	sqm	1.00-10.50	1.00-10.50	1.00-10.50	1.00-10.50	1.00-10.50
Acrylic jointing compound and mesh with skimcoat of polymer render to "Blue-Board" base sheet (Granoflex to joints; GranoSponge Fine Acrylic as skimcoat)	...	sqm	18.90	18.90	21.00	21.00	21.00
Render (pre-blended) skimcoat to masonry substrates (GranoRender)	...	sqm	15.80-21.00	15.80-21.00	21.00-26.30	15.80-21.00	21.00-26.30
Skimcoat (pre-mixed) to masonry or previously painted substrates (GranoPatch Spackle, GranoReady Render, GranoSponge Fine Acrylic)	...	sqm	15.80-21.00	15.80-21.00	21.00-26.30	15.80-21.00	21.00-26.30
Render with mesh to polystyrene base sheet (GranoRender PM)	...	sqm	28.40	29.90	31.50	31.50	31.50

PLASTERING

TEXTURED FINISHES (cont'd)

GRANOSITE APPLICATIONS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Roller applied finishes :							
Acrylic texture coating on prepared substrate with GranoImpact top coat -							
GranoRoll-on Coarse Impact	...	sqm	26.50	27.90	29.40	29.40	29.40
GranoRoll-on Medium Impact	...	sqm	23.60	24.90	26.30	26.30	26.30
GranoRoll-on Smooth Impact	...	sqm	20.80	22.00	23.10	23.10	23.10
Acrylic "Sand Finish" texture coating on prepared substrate -							
GranoSahara Ripple (texture sleeve)	...	sqm	20.80	22.00	23.10	23.10	23.10
GranoSahara Roll-on (nap roller sleeve)	...	sqm	18.90	20.00	21.00	21.00	21.00
Elastomeric acrylic texture coating on prepared substrate -							
GranoShield Low Profile	...	sqm	20.80	22.00	23.10	23.10	23.10
GranoShield High Profile	...	sqm	22.70	23.90	25.20	25.20	25.20
GranoImpact Low Profile	...	sqm	18.90	20.00	21.00	21.00	21.00
GranoImpact High Profile	...	sqm	20.80	22.00	23.10	23.10	23.10
Elastomeric acrylic membrane coating on prepared substrate -							
GranoSkin Membrane Low Profile	...	sqm	22.70	23.90	25.20	25.20	25.20
GranoSkin Membrane High Profile	...	sqm	24.60	25.90	27.30	27.30	27.30
Bag and Paint Finishes :							
Acrylic "Sand Finish" texture coating, applied by lambswool mitt or brush onto prepared substrate -							
GranoSahara	...	sqm	15.10	16.00	16.80	16.00	16.80
GranoSorrento	...	sqm	20.80	22.00	23.10	23.10	23.10
Trowel applied finishes :							
Acrylic texture coating onto prepared substrate with elastomeric membrane top coat -							
GranoQuartz	...	sqm	33.20	35.00	36.90	36.90	36.90
GranoMarble	...	sqm	33.20	35.00	36.90	36.90	36.90
GranoMarble Fast (winter grade)	...	sqm	33.00	34.80	36.70	36.70	36.70
GranoTrowl Sand Coarse	...	sqm	38.00	40.10	42.20	42.20	42.20
Acrylic texture coating (travertine finish) onto prepared substrate with elastomeric membrane top coat -							
GranoTrowl 2mm	...	sqm	34.40	36.30	38.20	38.20	38.20
ADD extra for :							
E/O for elastomeric top coat or membrane -							
Elastomeric top coat or membrane to GranoSkin Membrane	...	sqm	9.45	10.00	10.50	10.50	10.50
Elastomeric top coat or membrane to GranoShield	...	sqm	7.55	8.00	8.40	8.40	8.40
Elastomeric top coat or membrane to GranoImpact	...	sqm	6.60	7.00	7.35	7.35	7.35
Spray applied finish :							
Acrylic texture coating, sprayed onto prepared substrate with impact topcoat -							
GranoSpray-on Fine Impact	...	sqm	41.60	43.90	46.20	46.20	46.20
GranoSpray-on 2mm Impact	...	sqm	43.50	45.90	48.30	48.30	48.30

PLASTERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TEXTURED FINISHES (cont'd)							
GRANOSITE APPLICATIONS (cont'd)							
Aggregate finishes :							
Aggregate finish onto prepared base and coloured baselayer, spray applied -							
GranoStone 1mm	...	sqm	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00
GranoStone 2mm	...	sqm	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00
Aggregate finish, trowel applied onto prepared base. Aggregate bound in clear resin :							
GranoGranit 2mm	...	sqm	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00
Aggregate finish, spray applied onto prepared base. Aggregate bound in clear resin :							
GranoElegance	...	sqm	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00	105.00-147.00
SPRAYED VERMICULITE, CELLULOSE FIBRE							
Prices include all labours and narrow widths	...	Note					
Sprayed vermiculite finish 10mm thick on :							
Concrete soffits	...	sqm	30.20	35.70	28.50	30.00	28.70
Sides and soffits of beams	...	sqm	34.00	40.50	35.30	36.10	32.60
Cellulose fibre surface spray nominally 30mm thick (Envirospray)	...	sqm	46.10	48.00	44.80	41.10	46.80
Cellulose fibre concealed wet spray nominally 75mm thick (Enviropanel)	...	sqm	28.90	30.80	28.30	26.10	29.60
EXPANDED METAL LATH, BEADS							
Expanded metal lath in small areas, fixed to :							
Timber studding	...	sqm	19.10	20.10	19.00	17.90	18.65
Metal studding	...	sqm	20.00	20.40	19.20	18.10	18.85
Soffit of timber joists	...	sqm	20.70	21.10	20.00	18.30	19.40
Stainless steel metal lath fixed to metal studding	...	sqm	215.00	211.00	205.00	206.00	205.00
P.G.I. : External corner bead	...	m	11.40	12.25	11.15	12.05	11.75
Bullnose corner bead	...	m	14.50	15.85	14.90	15.70	15.10
Casing or stopping bead	...	m	11.40	11.80	11.00	11.85	11.80
Acrovyn corner guards, refer page 319	...	Note					

PLASTERING

PLASTERBOARD

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Plasterboard, flush finished :							
10mm thick to -							
Timber wall framing	...	sqm	35.00	40.00	30.50	33.00	33.50
Steel wall framing	...	sqm	35.00	40.00	30.50	33.00	34.00
Timber ceiling framing	...	sqm	37.00	41.50	32.00	35.50	34.00
13mm thick to -							
Timber wall framing	...	sqm	37.00	40.00	31.50	34.50	34.50
Steel wall framing	...	sqm	37.00	40.00	31.50	34.50	35.00
Timber ceiling framing	...	sqm	39.00	41.50	33.00	36.50	33.00
10mm thick water resistant to -							
Timber wall framing	...	sqm	39.00	41.00	32.50	37.00	36.00
Steel wall framing	...	sqm	39.00	41.00	32.50	37.00	36.00
Fire rated plasterboard, flush finished :							
1x13mm thick to -							
Timber wall framing	...	sqm	40.00	40.50	36.00	40.00	37.10
Steel wall framing	...	sqm	40.00	40.50	36.00	40.00	39.00
Timber ceiling framing	...	sqm	43.30	44.00	37.50	42.50	39.00
2x13mm thick to -							
Timber wall framing	...	sqm	72.00	72.00	61.30	65.00	63.00
Steel wall framing	...	sqm	72.00	72.00	61.30	65.00	63.00
1x16mm thick to -							
Timber wall framing	...	sqm	45.50	46.00	39.00	45.50	52.70
Steel wall framing	...	sqm	45.50	46.00	39.00	45.50	53.00
Timber ceiling framing	...	sqm	49.50	50.00	43.50	49.00	53.00
2x16mm thick to -							
Timber wall framing	...	sqm	81.50	82.00	73.00	73.50	87.00
Steel wall framing	...	sqm	81.50	82.00	73.00	73.50	90.00
Foil backed plasterboard, flush finished :							
10mm thick to -							
Timber wall framing	...	sqm	37.00	41.00	35.00	37.00	37.50
Steel wall framing	...	sqm	37.00	41.00	35.00	37.00	38.00
13mm thick to -							
Timber wall framing	...	sqm	39.00	43.50	36.80	40.00	39.00
Steel wall framing	...	sqm	39.00	43.50	36.80	40.00	39.00
Impact resistant plasterboard, flush finished :							
10mm thick to -							
Timber wall framing	...	sqm	41.80	45.00	39.00	41.00	43.00
Steel wall framing	...	sqm	41.80	45.00	39.00	41.00	43.00
13mm thick to -							
Timber wall framing	...	sqm	51.00	56.50	47.00	48.00	53.00
Steel wall framing	...	sqm	51.00	56.50	47.00	49.00	53.00
ADD extra for :							
Fixing to walls 2700-4000mm high	...	sqm	5.30	5.55	6.85	6.80	3.00
Fixing to ceilings above 3000mm and up to 4500mm high	...	sqm	5.50	4.15	7.00	4.50	3.70
Metal furring at 600mm centres	...	sqm	29.90	30.30	23.00	30.20	23.00
50 x 38mm fixing battens at 600mm centres	...	sqm	26.80	19.80	17.00	18.30	17.90
Alignment strips and plaster daubs to brick or concrete	...	sqm	15.80	15.80	14.60	15.50	17.90
Standard 90mm plasterboard cornice	...	m	13.90	19.35	12.25	16.50	13.00
Setting perimeter angle where no cornice	...	m	19.00	17.35	15.25	16.50	12.00
External angles and corner beads, refer page 421	...	Note					

PLASTERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SOUND BARRIER							
Two layers 10mm thick plasterboard bonded together including galvanised steel furring channel framing and sealing all edges and intersections, between ceiling framing and slab soffit :							
600mm high	...	m	85.00	105.00	95.00	99.00	104.00
1000mm high	...	m	118.50	139.00	125.00	135.00	134.00
1500mm high	...	m	152.00	192.50	160.00	180.00	171.00
2000mm high	...	m	182.00	227.00	195.00	222.00	214.00
Vinyl surfaced barium, refer page 396	...	Note					
ADD extra for :							
Penetrations for ducts, etc.	...	Note					
WALL FRAMING							
Refer pages 129 to 131 for stud wall costs and pages 143 and 393 for framed partition costs	...	Note					
Wall framing to standard height walls of cold formed steel sections in top and bottom plates, studs at 600mm centres and one row of noggings, including angles and intersections, in :							
51mm framing	...	sqm	37.70	40.60	34.00	32.50	39.00
64mm framing	...	sqm	38.80	37.60	35.00	33.70	41.00
76mm framing	...	sqm	39.80	40.80	36.00	34.50	43.00
92mm framing	...	sqm	40.80	43.10	38.00	37.60	45.00
ADD extra for :							
Additional row noggings	...	m	11.70	8.00	12.25	8.60	15.00
Additional height	...	Note					
Door and window openings	...	Note					
SHAFTWALL SYSTEM							
96 or 98mm thick shaftwall, two hour fire rated both sides, comprising 64mm steel framing infilled with 25mm plasterboard shaft liner and 2/16mm fire rated plasterboard to one face or single layer to each face	...	sqm	200.00	250.00	205.00	198.00	260.00
ADD extra for :							
50mm thick glasswool insulation batts	...	sqm	17.00	17.50	16.75	16.70	17.30
134 or 136mm ditto but with 102mm steel framing	...	sqm	212.00	267.00	225.00	220.00	229.00

SUSPENDED CEILINGS

SUSPENSION SYSTEMS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on hangers 750/900mm long fixed to concrete soffit	...	Note					
Fully concealed suspension system	...	sqm	36.00	35.60	36.00	35.00	41.50
Exposed suspension system :							
Steel, baked enamel finish -							
One-way system	...	sqm	34.00	33.40	28.50	27.60	33.40
Two-way system	...	sqm	29.90	30.00	26.00	27.60	29.50
Anodised aluminium :							
One-way system	...	sqm	40.00	43.70	38.00	37.00	44.00
Two-way system	...	sqm	36.00	40.00	36.00	29.00	66.00
Anti-vibrating ceiling mounting against concrete soffit	...	No	37.00	43.00	41.00	41.00	48.50

SUSPENDED CEILINGS

Prices are based on a minimum of 500sqm and include for suspension system, minor labours and edge trim	...	Note					
PLASTERBOARD							
FLUSH CEILING with a concealed suspension system :							
Plasterboard -							
10mm thick	...	sqm	72.00	75.40	72.00	75.50	85.00
13mm thick	...	sqm	74.30	78.40	74.00	78.50	90.00
Fire rated plasterboard -							
2 x 13mm thick (one hour)	...	sqm	112.00	115.00	112.00	114.50	121.00
2 x 16mm thick (one hour)	...	sqm	122.00	126.00	124.00	128.50	138.00
Two hour fire rated comprising two skins each of two layers of fire rated plasterboard finishing 32mm thick with 30mm furring channels between (total thickness 94mm)	...	sqm	225.00	244.00	259.00	249.00	285.00
PLAIN PLASTERBOARD tiles in a pre-finished steel two-way exposed suspension system :							
600 x 600mm tiles	...	sqm	66.00	65.90	64.00	54.50	67.00
1200 x 600mm tiles	...	sqm	60.80	52.90	58.00	44.50	65.00
ACOUSTIC PLASTER tiles in a pre-finished two-way exposed steel suspension system :							
600 x 600mm tiles	...	sqm	81.50	75.40	76.00	79.50	81.00
1200 x 600mm tiles	...	sqm	-	60.40	64.00	58.50	78.00
MINERAL FIBRE							
MINERAL FIBRE, pre-finished face pattern panels with a white paint finish unless otherwise described :							
Exposed ceiling system on a 24mm steel T-bar suspension system :							
600 x 600mm module -							
Recessed edge panel :							
Stock line	...	sqm	79.00	78.00	75.00	76.00	82.00
Non stock line	...	sqm	93.00	89.00	88.00	89.00	98.00
Fabric covered panel	...	sqm	132.00	139.00	138.00	137.00	145.00

SUSPENDED CEILINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
MINERAL FIBRE (cont'd)							
MINERAL FIBRE, pre-finished face pattern panels with a white paint finish unless otherwise described (cont'd) :							
Exposed ceiling system on a 24mm steel T-bar suspension system (cont'd) -							
1200 x 600mm module :							
Board	...	sqm	62.50	66.00	63.00	68.00	70.00
Recessed edge panel -							
Stock line	...	sqm	73.00	73.00	69.50	74.50	79.50
Non tock line	...	sqm	82.50	83.00	79.00	87.00	87.50
Scored recessed panel	...	sqm	74.00	83.00	74.00	85.50	83.00
ADD extra for :							
Aluminium suspension system	...	sqm	16.80	18.10	18.30	21.60	21.10
Coloured panels	...	Note					
Exposed ceiling system on a 15mm steel T-bar suspension system -							
600 x 600mm module :							
Recessed edge panel -							
Stock line	...	sqm	76.00	82.00	78.00	82.50	82.00
Non stock line	...	sqm	93.50	91.00	87.00	94.50	91.50
Open cell panel	...	sqm	117.00	110.50	111.00	122.00	122.00
1200 x 600mm module :							
Scored recessed panel	...	sqm	91.50	85.50	81.00	91.50	94.00
Decorative scored recessed panel	...	sqm	99.50	97.50	92.50	99.50	99.50
ADD extra for :							
Aluminium suspension system	...	sqm	16.80	18.10	18.30	21.60	21.10
Coloured panels	...	Note					
TIMBER BOARD, ETC.							
PLYWOOD, 1800 x 900 x 4.7mm thick panels, veneered finish and hung in a two-way exposed aluminium suspension system	...	sqm	123.00	128.00	128.00	150.00	130.00
PLYWOOD, ACOUSTIC, ditto	...	sqm	132.00	134.00	137.00	159.00	133.00
STRIP ceiling of timber ribs fixed to suspended carrier rails and including insulation blanket over, ribs :							
Pine	…	sqm	266.00	265.00	266.00	297.00	265.00
Selected hardwood	…	sqm	372.00	371.00	371.00	406.00	372.00
Specially selected premium hardwood	…	sqm	443.00	447.00	442.00	471.00	443.00
FIBRE CEMENT, 1200 x 600 x 6.0mm thick panels pre-finished and hung in a two-way exposed aluminium suspension system	...	sqm	65.00	63.50	61.50	70.00	67.50

SUSPENDED CEILINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
METAL, ETC.							
ALUMINIUM, cell ceiling system comprising 'U' shaped blades 9mm wide 40mm thick and suspension system, in standard colours, blades at :							
50 x 50mm centres	...	sqm	239.00	244.00	242.00	247.00	245.00
75 x 75mm centres	...	sqm	172.00	177.00	175.00	180.00	178.00
86 x 86mm centres	...	sqm	162.00	167.00	165.00	170.00	168.00
100 x 100mm centres	...	sqm	153.00	158.00	156.00	161.00	159.00
120 x 120mm centres	...	sqm	135.00	140.00	138.00	143.00	141.00
150 x 150mm centres	...	sqm	128.00	133.00	131.00	136.00	134.00
ADD extra for :							
Custom colours	...	Note					
ALUMINIUM STRIP ceiling of ribs with baked enamel finish fixed to suspended carrier rails, 300mm wide module and including insulation blanket over	...	sqm	159.00	162.00	159.00	164.00	162.00
METAL ACOUSTIC, perforated tiles including 25mm insulation heat sealed in 25mm black polythene bag :							
Fully concealed suspension system -							
1200 x 300mm tiles :							
Steel with colorbond® steel	...	sqm	110.00	110.00	108.00	113.00	112.00
Steel with powder coat finish	...	sqm	110.00	110.00	108.00	113.00	112.00
Aluminium with painted finish	...	sqm	141.00	141.00	138.00	146.00	143.00
Stainless steel (unperforated)	...	sqm	181.00	181.00	176.00	187.00	182.00
One-way suspension system -							
1200 x 400mm tiles :							
Steel with colorbond® steel	...	sqm	85.00	85.00	83.00	88.00	88.00
Steel with powder coat finish	...	sqm	85.00	85.00	83.00	88.00	88.00
Aluminium with painted finish	...	sqm	126.00	126.00	122.00	130.00	128.00
Stainless steel (unperforated)	...	sqm	165.00	166.00	161.00	172.00	167.00
ADD extra for :							
Sound attenuation panel	...	sqm	21.00	21.50	20.50	22.00	22.00
Two-way suspension system, lay in -							
600 x 600mm tiles :							
Steel with colorbond® steel	...	sqm	70.00	70.00	68.00	72.00	71.00
Steel with powder coat finish	...	sqm	70.00	70.00	68.00	72.00	71.00
1200 x 600mm tiles :							
Steel with colorbond® steel	...	sqm	69.00	68.00	67.00	70.00	70.00
Steel with powder coat finish	...	sqm	69.00	68.00	67.00	70.00	70.00
ADD extra for :							
Sound attenuation panel	...	sqm	20.00	20.50	19.50	21.00	20.50
METAL PAN system comprising powder coated zincalume® pans with a concealed suspension system :							
600 x 600mm pans	...	sqm	147.00	152.00	150.00	155.00	153.00
1200 x 600mm pans	...	sqm	142.00	112.00	145.00	150.00	148.00

SUSPENDED CEILINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
METAL, ETC. (cont'd)							
FIBREGLASS 1200 x 600mm panels, pre-finished face and hung in a two-way exposed aluminium suspension system	...	sqm	49.50	52.00	52.00	56.50	51.00
ADD extra for :							
Sound attenuation system to steel or aluminium strip ceilings	...	sqm	21.00	21.00	20.00	21.00	21.00
Aluminium linear ceiling of panels with Hunter Douglas coil coated polyester finish, attached to suspended carrier rails :							
30mm wide module -							
Type 30B	...	sqm	198.00	206.00	198.00	208.00	198.00
Type 30BD	...	sqm	209.00	217.00	209.00	219.00	209.00
80mm wide module -							
Type 80B	...	sqm	149.00	157.00	149.00	159.00	149.00
84mm wide module -							
Type 84B	...	sqm	147.00	155.00	147.00	157.00	147.00
Type 84C	...	sqm	147.00	155.00	147.00	157.00	147.00
Type 84R	...	sqm	148.00	156.00	148.00	158.00	148.00
130mm wide module -							
Type 130B	...	sqm	149.00	157.00	149.00	159.00	149.00
180mm wide module -							
Type 180B	...	sqm	142.00	150.00	142.00	152.00	142.00
ADD extra for :							
Insulation blanket over	...	sqm	12.50	12.70	12.60	12.70	12.50
Aluminium screen ceiling of panels with Hunter Douglas coil coated polyester finish, in standard colours, attached to suspended carrier rails :							
Type V100 at 150mm centres	...	sqm	132.00	140.00	132.00	142.00	132.00
Type V200 at 150mm centres	...	sqm	146.00	154.00	146.00	156.00	146.00
Polyester honeycomb construction acousitc panels, pre-finished face and clipped on a standard steel T-bar suspension system :							
1200 x 600mm panels	...	sqm	143.00	148.00	143.00	148.00	149.00
1200 x 1200mm panels	...	sqm	143.00	148.00	143.00	148.00	149.00

BULKHEADS TO SUSPENDED CEILINGS

For bulkheads and narrow widths to suspended ceilings increase the foregoing prices by the following percentages :		
Up to 600mm deep/wide - +100 to 150%	...	Note
600/1200mm deep/wide - + 75 to 125%	...	Note
1200/2000mm deep/wide - + 50 to 100%	...	Note

PAVIOR AND TERRAZZO

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for minor labours, i.e. rounded angles, fair edges	...	Note					

CEMENT BASED PAVING

CONCRETE SURFACE FINISHES

Power float monolithic finish	...	sqm	8.15	7.55	8.25	6.15	6.60
Additional surface finishes, refer page 242	...	Note					

SCREEDS

Cement and sand (1:3) screed with steel trowel finish, laid on concrete, in large areas :							
13mm thick	...	sqm	23.80	21.30	23.00	22.70	29.40
19mm thick	...	sqm	26.00	26.30	30.10	23.40	32.50
25mm thick	...	sqm	28.00	32.20	35.20	28.10	36.90
32mm thick	...	sqm	31.10	35.80	39.80	30.70	39.30
ADD extra for :							
Small areas (up to 10sqm)	...	sqm	7.35	7.25	7.85	8.10	7.70
Laying to falls or crossfalls not exceeding 15° from horizontal	...	sqm	0.70	0.70	0.65	0.80	0.60
Surface hardener brushed on	...	sqm	5.00-12.00	5.00-12.00	5.00-12.00	5.00-12.00	5.00-12.00
Water repellant additive, per 13mm thickness of finish	...	sqm	1.50	1.50	1.50	1.50	1.55
Oil repellant additive, per 13mm thickness of finish	...	sqm	2.45	2.45	2.45	2.50	2.50
Applying bonding agent before laying	...	sqm	2.65	2.65	2.70	2.70	2.70

GRANOLITHIC

Granolithic (1:2:5) with steel trowel finish laid on concrete while still green, in large areas :							
19mm thick	...	sqm	29.70	28.80	29.80	28.20	29.40
25mm thick	...	sqm	32.80	37.10	28.90	31.50	37.10
38mm thick	...	sqm	35.80	47.50	47.90	35.00	39.10
50mm thick	...	sqm	38.30	50.10	51.20	38.00	45.10
ADD extra for :							
Small areas (up to 10 sqm)	...	sqm	7.35	7.05	7.50	8.00	7.70
Floor ducts, pits	...	sqm	8.20	7.55	7.10	9.00	7.75
Laying to falls or crossfalls not exceeding 15° from horizontal	...	sqm	1.05	0.80	1.15	1.15	0.90
Surface hardener brushed on	...	sqm	5.00-12.00	5.00-12.00	5.00-12.00	5.00-12.00	5.00-12.00
Carborundum grains trowelled in (approx. 1kg/sqm)	...	sqm	7.00	7.00	7.00	7.00	7.00
Water repellant additive, per 13mm thickness of finish	...	sqm	1.50	1.50	1.50	1.50	1.50
Colour additive, integral -							
Black	...	sqm	2.00	2.00	2.00	2.00	2.00
Green	...	sqm	6.20	6.20	6.20	6.20	6.20
All other colours	...	sqm	2.25	2.25	2.25	2.25	2.25

PAVIOR AND TERRAZZO

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
CEMENT BASED PAVING (cont'd)							
GRANOLITHIC (cont'd)							
19mm skirting 75mm high with cove at bottom and rounded top edge	...	m	28.30	21.80	30.30	25.70	30.80
19mm finish to tread exceeding 150mm and not exceeding 300mm wide	...	m	29.60	18.80	31.70	24.50	29.70
19mm finish to riser not exceeding 150mm wide with cove at bottom and rounded top edge	...	m	36.00	24.80	37.00	27.90	25.70
19mm finish to stair string maximum 300mm wide with rounded top edge and cove to treads	...	m	36.80	24.80	37.00	27.90	25.70
19mm finish to kerb 450mm girth with two rounded angles and two coves at junction with paving including angles	...	m	71.10	54.20	79.90	71.70	85.10
Coved internal angle 50mm girth	...	m	5.20	4.35	4.65	6.10	7.45
Extra over for spoon drain not exceeding 300mm wide	...	m	9.70	8.85	10.45	11.40	11.20
EPOXY FINISHES AND PAVING							
SELF LEVELLING TOPPINGS							
Cementitious liquid as underlay to carpet or tiles (Vibro K-15) :							
4.8mm thick	...	sqm	54.50	57.00	57.00	56.50	55.50
Epoxy, 3mm thick, abrasion resistant, industrial applications (Epoxy-Pave type ESL)	...	sqm	95.50	98.50	98.50	97.50	98.00
Acrylic, abrasion resistant, industrial applications (ACRYDUR M.M.A.) :							
4.0mm thick	...	sqm	125.00	132.00	132.00	121.00	129.00
Vinyl-ester, 4mm thick (Epoxy-Pave type VEM)	...	sqm	150.00	154.00	154.00	150.00	151.00
TRAFFIC PROTECTION TOPPINGS							
Light Traffic :							
Surface coating not exceeding 0.025mm thick of -							
Acrylic	...	sqm	9.90	10.00	10.00	10.20	10.00
Chlorinated rubber	...	sqm	9.30	9.50	9.40	9.50	9.40
Epoxy	...	sqm	44.50	45.50	45.00	45.50	45.00
Silicate (Ashford formula treatment)	...	sqm	5.85	6.00	5.90	6.00	5.90
Urethane	...	sqm	32.00	33.00	32.50	33.00	32.50

PAVIOR AND TERRAZZO

EPOXY FINISHES AND PAVING (cont'd)

TRAFFIC PROTECTION TOPPINGS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Medium Traffic :							
Prepare concrete to provide key by -							
Acid etching	...	sqm	6.30	6.50	6.40	6.50	6.40
Blastrac enclosed shotblasting treatment	...	sqm	8.50	8.75	8.60	8.75	8.60
Surface treatment on prepared surface of -							
Pigmented high build epoxy, two coats finishing approximately 200µm thick	...	sqm	48.50	49.50	49.00	49.50	49.00
Polyurethane, 100µm thick	...	sqm	48.50	50.50	49.00	50.50	49.00
Methylmethacrylate, 4.5mm thick (ACRYDUR M.M.A.)	...	sqm	157.00	162.00	161.00	162.00	159.00
Heavy Traffic :							
Prepare concrete to provide key by -							
Acid etching	...	sqm	6.40	6.40	6.40	6.50	6.40
Blastrac enclosed shotblasting treatment	...	sqm	8.50	8.75	8.60	8.75	8.60
Scabbling of old weak concrete	...	sqm	21.40	21.80	21.50	21.80	21.50
Surface treatments -							
Refer Epoxy, Composition, etc. Finishes	...	Note					

EPOXY, COMPOSITION, ETC. FINISHES

Vinyl-ester finish 5mm thick to mild chemical areas (Epoxy-Pave type VEM) :							
General surfaces (250sqm)	...	sqm	168.00	173.00	171.00	173.00	171.00
Small areas (up to 50sqm or in narrow widths)	...	sqm	215.00	217.00	214.00	217.00	214.00
Skirting 100/150mm high with cove	...	m	83.00	85.00	84.00	85.00	84.00
Methylmethacrylate (M.M.A.) 4.5mm thick trowel applied to strong chemical areas (ACRYDUR M.M.A.) :							
General surfaces (250sqm)	...	sqm	158.00	162.00	159.00	162.00	159.00
Small areas (up to 50sqm or in narrow widths)	...	sqm	201.00	207.00	203.00	207.00	203.00
Skirting 100/150mm high with cove	...	m	89.00	91.00	90.00	91.00	90.00
Epoxy finish, 3mm thick in one application to medium chemical areas :							
General surfaces (250sqm)	...	sqm	97.00	100.00	98.00	100.00	98.00
Small areas (up to 50sqm or in narrow widths)	...	sqm	137.00	142.00	139.00	142.00	139.00
Skirting 100/150mm high with cove	...	m	88.00	90.00	89.00	90.00	89.00
Epoxy paint including acid etching :							
One coat pigmented high-build	...	sqm	33.50	34.30	33.80	34.30	33.80
Two coats clear thin coating	...	sqm	42.50	43.40	42.80	43.40	42.80
Three coats pigmented thin coating	...	sqm	57.50	58.50	58.00	58.50	58.00
ADD extra for :							
Non-skid finish (carborundum or aluminium oxide)	...	sqm	1.60	1.65	1.65	1.65	1.65
Seamless finishes on general surfaces (250sqm) :							
Polyurethane with vinyl flake	...	sqm	112.00	115.00	113.00	115.00	113.00
ACRYDUR M.M.A. with acrylic flake	...	sqm	133.00	137.00	134.00	137.00	134.00
ADD extra for :							
Small areas (by quotation)	...	Note					

PAVIOR AND TERRAZZO

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TERRAZZO							
IN SITU PAVING							
15mm thick in situ terrazzo paving incorporating white marble chips on and including 22mm thick screed	...	sqm	550.00	485.00	420.00	-	420.00
ADD extra for :							
Each additional 10mm of screed	...	sqm	3.45	2.95	2.35	-	2.85
Laying to falls or crossfalls not exceeding 15° from horizontal	...	sqm	8.80	7.45	6.40	-	7.45
Division strips	...	Note					
Patterned work	...	Note					
6mm thick terrazzo in skirting 150mm high on and including 13mm screed and with cove at bottom and rounded at top edge	...	m	153.00	112.00	117.00	-	124.00
PRECAST TERRAZZO PAVING							
300 x 300mm tiles laid on and including 13mm cement and sand (1:3) screed to :							
Floors	…	sqm	260.00	245.00	245.00	300.00	240.00
Landings	…	sqm	360.00	355.00	330.00	300.00	345.00
ADD extra for :							
400 x 400mm tiles	...	sqm	21.50	21.50	22.00	40.00	22.00
Expansion strips (average allowance)	...	sqm	20.00-40.00	20.00-40.00	20.00-40.00	20.00-40.00	20.00-40.00
Laying to patterns	...	Note					
PRECAST TERRAZZO PARTITIONS							
Prices include for anodised aluminium capping, supports, etc.	...	Note					
38mm double faced partition polished on all exposed faces	...	sqm	800.00	610.00	-	-	600.00
38mm double faced jamb or end :							
150mm wide	...	m	370.00	168.00	-	-	187.00
300mm wide	...	m	480.00	194.00	-	-	202.00
DIVISION STRIPS, WEATHER BARS							
Brass strip set in :							
25 x 3.18mm	…	m	39.50	43.50	44.00	44.40	52.10
38 x 3.18mm	…	m	46.80	50.90	51.40	51.70	59.40
50 x 3.18mm	…	m	55.80	60.10	60.60	61.10	69.20
Brass angle set in :							
12 x 6 x 1.6mm	…	m	32.20	36.50	37.00	37.50	45.70
38 x 19 x 3mm	…	m	55.80	60.10	60.60	61.00	69.20
ADD extra for :							
Lugs brazed on	...	m	11.50	13.10	13.30	13.50	16.50
Latham joint	...	m	50.00-150.00	50.00-150.00	50.00-150.00	50.00-150.00	50.00-150.00

TILING, NON RESILIENT

WALL TILING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices unless otherwise described include for fixing with adhesive and pointing	...	Note					
Render for tiling refer page 417	...	Note					
CERAMIC							
Tiling in main areas :							
150 x 150mm white glazed (P.C. $29.00/sqm supplied)	...	sqm	103.00	118.50	102.00	92.30	108.00
150 x 150mm plain colour glazed (P.C. $30.00/sqm supplied)	...	sqm	104.00	119.50	103.00	93.50	109.00
200 x 100mm white and plain colour glazed (P.C. $40.00/sqm supplied)	...	sqm	115.00	130.50	114.00	104.50	120.00
200 x 200mm white and plain colour glazed (P.C. $40.00/sqm supplied)	...	sqm	115.00	130.50	114.00	104.50	120.00
200 x 400mm white and plain colour glazed (P.C. $45.00/sqm supplied)	...	sqm	119.00	134.00	118.00	108.50	124.00
300 x 400mm white and plain colour glazed (P.C. $45.00/sqm supplied)	...	sqm	118.00	133.00	116.50	107.50	123.00
300 x 600mm white and plain colour glazed (P.C. $45.00/sqm supplied)	...	sqm	117.00	132.00	116.00	107.00	122.00
ADD extra for :							
Splashbacks -							
150mm or 200mm high	...	sqm	59.50	69.20	70.40	71.30	89.80
300mm or 400mm high	...	sqm	39.70	46.10	46.90	47.60	59.90
Acid resistant adhesive fixing	...	sqm	4.50	4.50	4.50	4.50	4.50
Epoxy grouting	...	sqm	30.00-35.00	30.00-35.00	30.00-35.00	30.00-35.00	30.00-35.00
Round edge tile (150 x 150mm white tiles only)	...	m	5.50	5.50	5.50	5.50	5.50
Tile rounded two adjacent edges (150 x 150mm white tiles only)	...	No	1.10	1.10	1.10	1.10	1.10
Coved internal angle tile (150 x 75mm white tiles only)	...	m	38.60	38.60	38.60	38.60	38.60
FITTINGS							
Footrest 150 x 150mm	...	No	24.50	25.00	25.00	25.00	26.00
Soap holder	...	No	24.50	25.00	25.00	25.00	26.00
Toilet paper holder	...	No	24.50	25.00	25.00	25.00	26.00
MOSAIC							
GLAZED							
Tiling, in sheets, in main areas							
61 x 61mm - (P.C. $40.00/sqm supplied)	...	sqm	125.00	143.50	124.00	113.00	131.50
100 x 100mm - (P.C. $40.00/sqm supplied)	...	sqm	122.00	139.00	120.50	110.50	127.50
ADD extra for :							
Splashbacks -							
Up to 150mm high	...	sqm	59.50	69.20	70.40	71.30	89.80
150/300mm high	...	sqm	39.70	46.10	46.90	47.60	59.90

TILING, NON RESILIENT

FLOOR TILING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for bedding and pointing	...	Note					
CERAMIC							
Tiling in main areas, light duty commercial/ domestic :							
200 x 200mm (P.C. $40.00/sqm supplied)	...	sqm	115.00	130.50	114.00	104.50	120.00
300 x 300mm (P.C. $40.00/sqm supplied)	...	sqm	113.50	128.50	112.00	103.00	118.50
330 x 330mm (P.C. $35.00/sqm supplied)	...	sqm	108.00	123.00	106.50	97.50	113.00
450 x 450mm (P.C. $40.00/sqm supplied)	...	sqm	112.00	127.50	111.00	102.00	117.50
600 x 600mm (P.C. $55.00/sqm supplied)	...	sqm	128.00	143.00	127.00	118.00	133.00
ADD extra for :							
Small areas	...	sqm	49.60	57.70	58.70	59.50	74.80
MOSAIC							
GLAZED							
Tiling, in sheets, in main areas :							
61 x 61mm - (P.C. $40.00/sqm supplied)	...	sqm	125.00	143.50	124.00	113.00	131.50
100 x 100mm - (P.C. $40.00/sqm supplied)	...	sqm	122.00	139.00	120.50	110.50	127.50
ADD extra for :							
Small areas	...	sqm	49.60	57.70	58.70	59.50	74.80
Pointing with acid resisting grout	...	sqm	4.50	4.50	4.50	4.50	4.50
Patterned work	...	sqm	16.00	16.50	16.50	16.50	16.50
NON SLIP							
Unglazed tiling, 50 x 50mm in sheets (P.C. $40.00/sqm supplied), in main areas	...	sqm	125.00	143.50	124.00	113.00	131.50
Homogeneous tiling, 50 x 50mm in sheets (P.C. $40.00/sqm supplied), in main areas	...	sqm	125.00	143.50	124.00	113.00	131.50
QUARRY							
Australian manufactured tiles	...	Note					
Tiling in main areas :							
Light duty commercial/domestic - 250 x 250mm - (P.C. $33.00/sqm supplied)	...	sqm	107.50	123.00	106.00	96.80	112.50
Heavy duty commercial - 300 x 300mm - (P.C. $38.00/sqm supplied)	...	sqm	111.00	126.50	110.00	101.00	116.00
200 x 200mm unglazed acid resisting (P.C. $50.00/sqm supplied)	...	sqm	126.00	142.00	125.00	115.50	131.50
ADD extra for :							
Small areas	...	sqm	49.60	57.70	58.70	59.50	74.80
Pointing with acid resisting grout	...	sqm	4.50	4.50	4.50	4.50	4.50
Patterned work	...	sqm	16.00	16.50	16.50	16.50	16.50

TILING, NON RESILIENT

FLOOR TILING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TERRACOTTA							
Natural terracotta tiles, in main areas :							
300 x 300mm - (P.C. $42.00/sqm supplied)	...	sqm	115.50	131.00	114.50	105.00	120.50
ADD extra for -							
Acrylic surface sealer	...	sqm	14.50	15.00	15.60	15.10	15.60
Tactile strip x 300mm wide, 300 x 300mm or 300 x 150mm tiles with raised dots or ridges (P.C. $48.00/sqm supplied)	...	sqm	124.00	139.50	122.50	113.50	129.00
VITREOUS							
Unglazed porcelain china clay tiling in main areas :							
200 x 200mm - (P.C. $50.00/sqm supplied)	...	sqm	126.00	142.00	125.00	115.50	131.50
300 x 300mm - (P.C. $50.00/sqm supplied)	...	sqm	124.50	139.50	123.50	114.00	129.50
ADD extra for :							
Small areas	...	sqm	49.60	57.70	58.70	59.50	74.80
Patterned work	...	sqm	16.00	16.50	16.50	16.50	16.50
STEEL							
305 x 305 x 3mm steel tiles set into wet concrete or bedding :							
Black	...	sqm	266.00	280.00	261.00	257.00	269.00
Galvanised	...	sqm	246.00	260.00	241.00	237.00	249.00
ADD extra for :							
50mm thick bedding if required	...	sqm	31.00	43.50	43.50	33.00	44.00
TERRAZZO							
Terrazzo tile paving, refer page 431	...	Note					
SKIRTINGS							
VITREOUS, with rounded top edge and cove :							
75mm high	...	m	89.30	93.10	93.50	93.90	101.00
TREADS AND RISERS							
QUARRY tile to -							
300mm wide tread	...	m	44.00	49.40	50.00	50.60	60.80
150mm high riser	...	m	28.60	32.40	32.80	33.20	40.40
ANTI SLIP TREATMENT							
Gelseal SR treatment	...	sqm	20.50	21.50	21.00	21.00	21.00
Surestop treatment	...	sqm	16.00	17.00	16.75	16.75	17.00
BRICK AND BLOCK PAVING, refer page 223	...	Note					
MARBLE PAVING, refer pages 273 and 274	...	Note					
WOOD BLOCK PAVING, refer page 340	...	Note					

RESILIENT FINISHES

CARPET

CARPET

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Commercial grade laid on prepared surface (no underlay)	...	Note					
Nylon :							
Medium use (18 oz) (P.C. $17.00/sqm supplied)	...	sqm	30.30	31.10	30.60	31.40	33.40
Heavy use (20 oz) (P.C. $30.00/sqm supplied)	...	sqm	44.00	44.80	44.20	45.00	47.10
Extra heavy use (24 oz) (P.C. $39.00/sqm supplied)	...	sqm	53.40	54.20	53.70	54.50	56.50
Wool tufted :							
Medium use (35-45 oz) (P.C. $30.00/sqm supplied)	...	sqm	44.00	44.80	44.20	45.00	47.10
Heavy use (45-50 oz) (P.C. $43.00/sqm supplied)	...	sqm	57.60	58.40	57.90	58.70	60.70
Extra heavy use (50-55 oz) (P.C. $56.00/sqm supplied)	...	sqm	71.30	72.10	71.50	72.30	74.40
Wool woven :							
Medium use (35-45 oz) (P.C. $70.00/sqm supplied)	...	sqm	86.00	86.80	86.20	87.00	89.10
Heavy use (45-50 oz) (P.C. $90.00/sqm supplied)	...	sqm	107.00	108.00	107.00	108.00	110.00
Electrostatically flocked heavy duty carpet/vinyl, carpet or tiles :							
Light use (P.C. $49.00/sqm supplied)	...	sqm	63.90	64.70	64.20	65.00	67.00
Medium use (P.C. $52.00/sqm supplied)	...	sqm	67.10	67.90	67.30	68.10	70.20
Heavy use (P.C. $57.00/sqm supplied)	...	sqm	72.30	73.10	72.60	73.40	75.40
Nylon, direct stick, “healthcare” including slab sealing (P.C. $47.00/sqm supplied)	...	sqm	67.20	68.00	67.50	68.30	70.30
Polypropylene, direct stick, general range (P.C. $10.00 to $20.00/sqm supplied)	...	sqm	23.00-33.50	23.80-34.30	23.20-33.70	24.00-34.50	26.10-36.60
Indoor/outdoor (average standard $29.00/sqm supplied)	...	sqm	45.10	45.90	45.40	46.20	48.20
Synthetic turf, refer page 234	...	Note					
ADD extra for :							
Rubber underlay	...	sqm	24.60	25.40	25.00	25.80	26.90
Foam underlay	...	sqm	16.90	18.25	17.45	18.80	20.10
Lay only carpet including fixing strips, protection and cleaning but excluding waste :							
Commercial grade	...	sqm	12.50	13.30	12.70	13.50	15.60
Woven wool, heavy use	...	sqm	17.50	18.60	17.80	19.00	21.80

RESILIENT FINISHES

CARPET (cont'd)

MODULAR CARPET

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Commercial grade laid on prepared surface ...	Note					
ADD extra for :						
Printed pattern, laying to design effect, logos ...	Note					
Ontera :						
Nylon, anti-static, loop pile -						
Medium duty, (P.C. $50.00/sqm supplied) ...	sqm	63.50	64.20	63.70	64.40	66.20
Heavy duty, (P.C. $60.00/sqm supplied) ...	sqm	74.00	74.70	74.20	74.90	76.70
Nylon, anti-static, cut pile :						
Heavy duty, (P.C. $70.00/sqm supplied) ...	sqm	84.50	85.20	84.70	85.40	87.20
InterfaceFLOR :						
Nylon, anti-static, tufted loop pile or tufted cut and loop pile -						
Medium duty, (P.C. $50.00/sqm supplied) ...	sqm	63.50	64.20	63.70	64.40	66.20
Heavy duty, (P.C. $60.00/sqm supplied) ...	sqm	74.00	74.70	74.20	74.90	76.70

CARPET CLEANING

Vacuum and steam clean cleared areas :						
500sqm ...	sqm	2.00-2.65	1.75-2.30	2.35-3.15	1.80-2.40	2.30-3.00
1000sqm ...	sqm	1.65-2.30	1.55-2.00	1.95-2.75	1.60-2.05	2.00-2.60

CORK

Laid on prepared floor surface ...	Note					

TILES

Natural cork tiles 305 x 305 x 6.0mm thick including sanding and finishing with three coats polyurethane (P.C. $35.00/sqm supplied) ...	sqm	101.50	106.00	103.00	107.00	118.00

SHEET

3.0mm thick simulated cork sheet safety floor, laid on prepared surface :						
General use (Low Profile) (P.C. $63.00/sqm supplied) ...	sqm	124.50	128.50	126.00	129.50	139.00
Industrial use (Aquadeck) (P.C. $65.00/sqm supplied) ...	sqm	126.50	130.50	128.00	131.50	141.00

RESILIENT FINISHES

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LINOLEUM						
Laid on prepared floor surface ...	Note					
TILES						
Heavy duty :						
2.5mm thick (Marmoleum), anti-static (P.C. $50.00/sqm supplied) ...	sqm	93.40	96.00	94.30	96.90	103.50
SHEET						
Heavy duty :						
2.5mm thick (Linoleum XF Allegro) (P.C. $40.00/sqm supplied) ...	sqm	86.90	89.80	87.80	90.80	98.10
2.5mm thick (Linoleum XF Etrusco) (P.C. $50.00/sqm supplied) ...	sqm	97.40	100.50	98.30	101.50	108.50
Acoustic :						
3.8mm thick (Linoleum XF Etrusco) (P.C. $58.00/sqm supplied) ...	sqm	106.00	108.50	106.50	109.50	117.00
ADD extra for :						
5.5mm hardboard underlay including feather finish ...	sqm	38.00	40.50	39.00	41.00	42.00
VINYL						
Laid on prepared floor surface unless otherwise described ...	Note					
TILES						
Semi rigid :						
2.0mm thick (Excelon Designer) (P.C. $10.00/sqm supplied) ...	sqm	22.50	23.20	22.70	23.50	25.50
2.0mm thick (Excelon ECO) (P.C. $8.00/sqm supplied) ...	sqm	19.40	20.10	19.60	20.40	22.20
2.0mm thick (Excelon Terrazz) (P.C. $11.00/sqm supplied) ...	sqm	23.50	24.30	23.80	24.60	26.50
2.0mm thick (Excelon Stonetex) (P.C. $26.00/sqm supplied) ...	sqm	39.30	40.00	39.50	40.30	42.30
3.0mm thick (Excelon Designer) (P.C. $19.00/sqm supplied) ...	sqm	31.90	32.70	32.20	33.00	34.90
3.0mm thick (Excelon Stonetex) (P.C. $22.00/sqm supplied) ...	sqm	35.10	35.80	35.30	36.10	38.10
3.0mm thick (Polyflor VC Tile) (P.C. $19.00/sqm supplied) ...	sqm	31.90	32.70	32.20	33.00	34.90
SHEET						
Homogeneous :						
1.5mm thick (Polyflor XL) (P.C. $16.50/sqm supplied) ...	sqm	42.30	43.90	42.80	44.40	48.50
2.0mm thick (Accolade Plus) (P.C. $38.00/sqm supplied) ...	sqm	64.80	66.50	65.40	67.00	71.10
2.0mm thick (Armalon) (P.C. $18.00/sqm supplied) ...	sqm	43.80	45.50	44.40	46.00	50.10
2.0mm thick (Bravo) (P.C. $35.00/sqm supplied) ...	sqm	61.70	63.30	62.20	63.80	67.90

RESILIENT FINISHES

VINYL (cont'd)

SHEET (cont'd)

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Homogeneous (cont'd) :						
2.0mm thick (Eminent) (P.C. $47.00/sqm supplied) ...	sqm	74.30	75.90	74.80	76.40	80.50
2.0mm thick (Granit) (P.C. $38.00/sqm supplied) ...	sqm	64.80	66.50	65.40	67.00	71.10
2.0mm thick (Optima) (P.C. $42.00/sqm supplied) ...	sqm	69.00	70.70	69.60	71.20	75.30
2.0mm thick (Polyflor XL) (P.C. $20.00/sqm supplied) ...	sqm	45.90	47.60	46.50	48.10	52.20
2.0mm thick (Prestige) (P.C. $47.00/sqm supplied) ...	sqm	74.30	75.90	74.80	76.40	80.50
Hetrogeneous :						
2.0mm thick (Acczent Excellence 80 Wood) (P.C. $37.00/sqm supplied) ...	sqm	63.80	65.40	64.30	65.90	70.00
2.0mm thick (Contract Interior) (P.C. $50.00/sqm supplied) ...	sqm	77.40	79.10	78.00	79.60	83.70
2.0mm thick (Design FX) (P.C. $51.00/sqm supplied) ...	sqm	78.50	80.10	79.00	80.60	84.70
2.0mm thick (Forest FX) (P.C. $51.00/sqm supplied) ...	sqm	78.50	80.10	79.00	80.60	84.70
2.0mm thick (Optic) (P.C. $48.00/sqm supplied) ...	sqm	75.30	77.00	75.90	77.50	81.60
2.0mm thick (Stella) (P.C. $26.00/sqm supplied) ...	sqm	52.20	53.90	52.80	54.40	58.50
2.0mm thick (Timberline) (P.C. $50.00/sqm supplied) ...	sqm	77.40	79.10	78.00	79.60	83.70
Flexible, factory finished :						
2.0mm thick (Smaragd) (P.C. $46.00/sqm supplied) ...	sqm	73.20	74.90	73.80	75.40	79.50
Acoustic, foam backed :						
3.0mm thick (Polytred Acoustic) (P.C. $44.00/sqm supplied) ...	sqm	71.10	72.80	71.70	73.30	77.40
3.7mm thick (Forest FX Acoustic) (P.C. $54.00/sqm supplied) ...	sqm	81.60	83.30	82.20	83.80	87.90
4.0mm thick (Granit Acoustic) (P.C. $51.00/sqm supplied) ...	sqm	78.50	80.10	79.00	80.60	84.70
Conductive :						
2.0mm thick (Granit SD) (P.C. $51.00/sqm supplied) ...	sqm	87.50	89.70	88.20	90.40	95.90
2.0mm thick (Polyflor EC) (P.C. $72.00/sqm supplied) ...	sqm	109.50	111.50	110.00	112.50	118.00
2.0mm thick (Polyflor SD) (P.C. $42.00/sqm supplied) ...	sqm	78.00	80.20	78.70	81.00	86.50
2.0mm thick (Toro SC) (P.C. $61.00/sqm supplied) ...	sqm	98.00	100.00	98.70	101.00	106.50



Clears the Air.

On average people spend 90% of their time indoors. At Tarkett, we strive to contribute to people's health and wellbeing by improving indoor air quality.

DESSO AirMaster captures harmful fine dust from the air more effectively than any other flooring solution, and can help play a fundamental role in contributing to the health and wellbeing in today's work and living spaces.

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Tarkett

RESILIENT FINISHES

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
VINYL (cont'd)						
SHEET (cont'd)						
Slip resistant :						
2.0mm thick (Accolade Safe Plus R10) (P.C. $40.00/sqm supplied) ...	sqm	75.90	78.10	76.60	78.90	84.40
2.0mm thick (Armstrong Safeguard R10) (P.C. $45.00/sqm supplied) ...	sqm	77.30	79.40	78.00	80.10	85.40
2.0mm thick (Armstrong Safeguard R11) (P.C. $55.00/sqm supplied) ...	sqm	87.30	89.40	88.00	90.10	95.40
2.0mm thick (Armstrong Safeguard R12) (P.C. $65.00/sqm supplied) ...	sqm	102.00	104.50	103.00	105.00	110.50
2.0mm thick (Granit Safe T. R11) (P.C. $46.00/sqm supplied) ...	sqm	82.20	84.40	82.90	85.20	90.70
2.0mm thick (Polysafe Standard R10) (P.C. $46.00/sqm supplied) ...	sqm	82.20	84.40	82.90	85.20	90.70
2.5mm thick (Polysafe Standard R10) (P.C. $59.00/sqm supplied) ...	sqm	95.90	98.10	96.60	98.80	104.50
2.5mm thick (Polysafe Ultima R11) (P.C. $66.00/sqm supplied) ...	sqm	103.00	105.50	104.00	106.00	111.50
2.5mm thick (Safetred Universal Plus R12) (P.C. $52.00/sqm supplied) ...	sqm	88.50	90.70	89.20	91.50	97.00
Sports :						
3.5mm thick (Omnisports Speed) (P.C. $45.00/sqm supplied) ...	sqm	105.00	109.00	106.50	110.00	119.50
6.5mm thick (Omnisports Reference) (P.C. $55.00/sqm supplied) ...	sqm	115.50	119.50	117.00	120.50	130.00
6.7mm thick (Polytred Sports 67) (P.C. $88.00/sqm supplied) ...	sqm	150.00	154.00	151.50	155.50	164.50
8.0mm thick (Polytred Sports 80) (P.C. $97.00/sqm supplied) ...	sqm	159.50	163.50	161.00	164.50	174.00
8.3mm thick (Omnisports Excel) (P.C. $75.00/sqm supplied) ...	sqm	136.50	140.50	138.00	141.50	151.00
Luxury tile and plank :						
2.0mm thick (Camaro Studioflor) (P.C. $47.00/sqm supplied) ...	sqm	69.30	70.60	69.70	71.00	74.30
2.0mm thick (Expona Domestic) (P.C. $49.00/sqm supplied) ...	sqm	71.40	72.70	71.80	73.10	76.40
3.0mm thick (Expona Design) (P.C. $68.00/sqm supplied) ...	sqm	91.30	92.60	91.80	93.10	96.30
ADD extra for :						
Borders, corners and feature strips ...	Note					
Wall sheeting :						
1.0mm thick (Aquarelle) (P.C. $28.00/sqm supplied) ...	sqm	64.30	66.60	65.10	67.30	73.00
1.25mm thick (Wallflex) (P.C. $24.00/sqm supplied) ...	sqm	60.10	62.40	60.90	63.10	68.80
1.25mm thick (Polyclad) (P.C. $20.00/sqm supplied) ...	sqm	55.90	58.20	56.70	58.90	64.60
2.00mm thick (Wallguard) (P.C. $21.00/sqm supplied) ...	sqm	57.00	59.20	57.70	60.00	65.70
SKIRTINGS						
100mm high feather edge ...	m	11.70	12.10	12.00	12.40	13.00
100mm high coved sheeting including fillet (average) ...	m	23.20	23.90	23.50	25.00	25.50

RESILIENT FINISHES

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
RUBBER						
Laid on prepared floor surface ...	Note					
SHEET (VOC free)						
Regupol :						
3.0mm thick (Intensity Berlin)						
(P.C. $110.00/sqm supplied) ...	sqm	161.50	164.50	162.50	165.50	173.00
3.0mm thick (Intensity Nome)						
(P.C. $102.00/sqm supplied) ...	sqm	153.00	156.00	154.00	157.00	164.50
4.0mm thick (Core Mons)						
(P.C. $60.00/sqm supplied) ...	sqm	112.00	115.00	113.00	116.00	124.00
5.0mm thick (Shock Pad Underlay)						
(P.C. $20.00/sqm supplied) ...	sqm	82.80	86.90	84.20	88.20	98.30
Rephouse :						
3.0mm thick (Neoflex 600)						
(P.C. $114.00/sqm supplied) ...	sqm	165.50	168.50	166.50	169.50	177.00
4.0mm thick (Neoflex 600)						
(P.C. $138.00/sqm supplied) ...	sqm	194.00	197.00	195.00	198.00	206.00
6.0mm thick (Neoflex 600)						
(P.C. $198.00/sqm supplied) ...	sqm	270.00	274.00	271.00	275.00	285.00
TILES (VOC free)						
Studded, standard colours (Kinetics) :						
2.7mm thick, medium traffic (K2)						
(P.C. $50.00/sqm supplied) ...	sqm	93.80	96.50	94.70	97.40	104.00
4.0mm thick, heavy duty (K1)						
(P.C. $58.00/sqm supplied) ...	sqm	105.00	107.50	106.00	108.50	116.00
2.7mm thick, high performance (K4)						
(P.C. $55.00/sqm supplied) ...	sqm	99.00	101.50	99.90	102.50	109.50
3.0mm thick, heavy traffic						
(P.C. $60.00/sqm supplied) ...	sqm	104.50	107.00	105.00	108.00	114.50
Sports, standard colours (Kinetics) :						
6.0mm thick recreational/impact tile						
(P.C. $79.00/sqm supplied) ...	sqm	138.50	142.00	140.00	143.50	152.50
12.0mm thick gymtile						
(P.C. $116.00/sqm supplied) ...	sqm	183.50	187.50	185.00	189.00	199.00
DIMINISHING STRIPS						
25mm wide vinyl diminishing strip at junction of differing finishes, tapering from :						
2.0mm thick ...	m	9.80	9.80	10.00	10.30	10.40
3.0mm thick ...	m	11.70	11.60	11.90	12.30	12.40
TACTILE						
PVC tactile tiles 600 x 300 x 8mm total thickness						
(P.C. $30.00/each supplied) ...	No	51.40	52.70	51.90	53.20	56.40
Stainless steel tactile indicator single stud with concentric ridges to top ...	No	4.35	4.00	4.15	5.35	5.00

RESILIENT FINISHES

STAIR TREAD NOSINGS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Aluminium safety stair tread nosing for vinyl, together with silicon carbide slip resistant strip, mechanically fixed	...	m	72.60	72.40	73.20	73.80	86.10
Aluminium safety stair tread nosing for carpet together with silicon carbide slip resistant strip, mechanically fixed	...	m	72.60	72.40	73.20	73.80	86.10
Vinyl, standard ribbed stair tread nosing	...	m	10.10	10.00	10.25	10.50	10.55

GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on average pane sizes up to 2.4sqm and on glazing to metal or wood with beads and mastic or neoprene strips	...	Note					
Prices are for site glazing. For shop glazing further discounting of glass could reduce the prices by 10 to 35%	...	Note					
Glazing to metal or wood with putty :							
Deduct $3.00/sqm	...	Note					

CLEAR FLOAT GLASS

Glazing to metal or wood :							
3.00mm	...	sqm	97.00	97.00	99.00	99.00	101.00
4.00mm	...	sqm	107.00	107.00	108.00	111.00	110.00
5.00mm	...	sqm	120.00	119.00	125.00	122.00	123.00
6.00mm	...	sqm	125.00	124.00	129.00	129.00	127.00
8.00mm	...	sqm	166.00	167.00	169.00	169.00	170.00
10.00mm	...	sqm	188.00	182.00	186.00	192.00	189.00
12.00mm	...	sqm	218.00	212.00	218.00	222.00	222.00
Tinted, refer to Solar Control Glass	...	Note					

FIGURED ROLLED GLASS

Glazing to metal or wood :							
White -							
3.00mm	...	sqm	106.00	106.00	99.00	111.00	110.00
5.00mm	...	sqm	115.00	115.00	118.00	122.00	118.00
6.00mm	...	sqm	125.00	124.00	129.00	129.00	127.00
Amber, bronze, grey -							
3.00mm	...	sqm	124.00	123.00	129.00	129.00	127.00
5.00mm	...	sqm	143.00	142.00	145.00	145.00	146.00
6.00mm	...	sqm	155.00	151.00	155.00	153.00	156.00

WIRED GLASS

Glazing to metal or wood :							
White obscure, 6mm	...	sqm	159.00	156.00	160.00	162.00	162.00
Coloured obscure, 6mm	...	sqm	191.00	191.00	190.00	184.00	192.00
Clear plate, 6mm	...	sqm	236.00	230.00	235.00	242.00	239.00

LAMINATED GLASS

Glazing to metal or wood :							
Clear float -							
6.38mm	...	sqm	221.00	224.00	228.00	221.00	225.00
10.38mm	...	sqm	313.00	320.00	328.00	321.00	330.00
12.38mm	...	sqm	326.00	335.00	339.00	325.00	345.00
Cyclone, 14mm clear	...	sqm	608.00	627.00	622.00	628.00	626.00
Tinted, refer Solar Control Glass	...	Note					
Fritting, refer page 447	...	Note					

GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TOUGHENED GLASS							
Glazing to metal or wood :							
Clear float -							
4.00mm	...	sqm	237.00	228.00	237.00	242.00	238.00
5.00mm	...	sqm	255.00	247.00	255.00	259.00	257.00
6.00mm	...	sqm	275.00	277.00	279.00	280.00	279.00
10.00mm	...	sqm	534.00	534.00	541.00	550.00	549.00
12.00mm	...	sqm	635.00	634.00	644.00	654.00	649.00
Tinted, refer Solar Control Glass	...	Note					
SOLAR CONTROL GLASS							
Glazing to metal or wood :							
FLOAT GLASS							
Heat absorbing :							
4.00mm bronze and grey	...	sqm	130.00	126.00	129.00	132.00	129.00
5.00mm bronze and grey	...	sqm	136.00	133.00	136.00	137.00	139.00
6.00mm bronze and grey	...	sqm	147.00	141.00	148.00	144.00	148.00
6.00mm green	...	sqm	153.00	149.00	151.00	153.00	156.00
10.00mm green	...	sqm	224.00	216.00	220.00	227.00	224.00
10.00mm bronze and grey	...	sqm	204.00	199.00	203.00	209.00	206.00
12.00mm bronze and grey	...	sqm	240.00	233.00	236.00	240.00	239.00
Heat absorbing, reflective :							
3.00mm bronze (Solarcool)	...	sqm	150.00	145.00	148.00	150.00	154.00
6.00mm bronze (Solarcool)	...	sqm	202.00	200.00	199.00	209.00	204.00
LAMINATED GLASS							
Heat absorbing :							
6.38mm tinted with single interlayer (8 colour range)	...	sqm	264.00	255.00	262.00	266.00	273.00
6.76mm tinted with two interlayers (36 colour range)	...	sqm	310.00	301.00	307.00	315.00	316.00
7.14mm tinted with three interlayers (120 colour range)	...	sqm	351.00	338.00	349.00	360.00	364.00
7.52mm tinted with four interlayers (342 colour range)	...	sqm	404.00	392.00	405.00	415.00	417.00
10.38mm bronze and grey	...	sqm	353.00	338.00	349.00	363.00	361.00
Cyclone, tinted, 14mm	...	sqm	661.00	638.00	664.00	682.00	681.00
TOUGHENED GLASS							
Heat absorbing :							
4.00mm tinted	...	sqm	272.00	265.00	269.00	276.00	279.00
5.00mm tinted	...	sqm	292.00	283.00	290.00	296.00	297.00
6.00mm tinted	...	sqm	317.00	298.00	311.00	321.00	326.00
10.00mm tinted	...	sqm	577.00	564.00	571.00	582.00	582.00
12.00mm tinted	...	sqm	677.00	660.00	666.00	680.00	679.00
Heat absorbing, reflective :							
6.00mm bronze (Solarcool)	...	sqm	210.00	203.00	207.00	212.00	215.00
Reflective :							
6.00mm clear (Suncool)	...	sqm	227.00	224.00	226.00	228.00	230.00

GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
LOUVRES							
5mm clear float glass with ground and arrised edges :							
100mm wide	…	m	36.70	37.00	36.60	37.40	36.90
150mm wide	…	m	45.00	43.40	42.00	44.50	42.90
DOUBLE GLAZING							
Sealed units :							
6.00mm clear float/6mm clear float	…	sqm	284.00	273.00	282.00	288.00	285.00
6.00mm clear float/6.38mm clear laminated	…	sqm	390.00	389.00	389.00	396.00	395.00
6.00mm clear float/6.38mm tinted laminated	…	sqm	421.00	414.00	424.00	432.00	430.00
6.00mm clear float/6mm clear toughened	…	sqm	434.00	427.00	435.00	440.00	438.00
6.00mm clear float/10mm clear toughened	…	sqm	696.00	692.00	698.00	698.00	707.00
6.00mm clear float/6mm bronze toughened (Solarcool)	…	sqm	357.00	361.00	366.00	370.00	376.00
MIRRORS							
6mm clear float glass with silvered and coppered back, fixed with mirror screws	...	sqm	235.00	227.00	236.00	236.00	230.00
One way observation mirror, 900 x 600mm, built into prepared opening	...	No	965.00	925.00	985.00	980.00	1055.00
POLYCARBONATE SHEET							
STANDARD grade and glazing to metal :							
Clear -							
3.00mm	…	sqm	111.00	124.50	125.00	125.50	148.50
4.50mm	…	sqm	134.00	148.50	148.50	149.50	173.50
6.00mm	…	sqm	154.50	169.00	169.50	170.50	195.50
Tinted -							
3.00mm	…	sqm	112.00	125.50	125.50	126.50	149.50
4.50mm	…	sqm	138.00	152.00	152.50	153.50	177.50
6.00mm	…	sqm	184.50	199.00	199.50	200.50	225.50
ABRASION RESISTANT grade and glazing to metal :							
Clear -							
4.70mm	…	sqm	294.00	308.00	308.00	309.00	333.00
6.30mm	…	sqm	329.00	344.00	344.00	345.00	370.00
ACRYLIC SHEET							
STANDARD grade and glazing to metal :							
Clear -							
3.00mm	…	sqm	111.00	125.00	125.00	126.00	148.50
4.50mm	…	sqm	120.50	134.50	135.00	136.00	159.50
6.00mm	…	sqm	145.50	160.00	160.50	161.50	186.50
10.00mm	…	sqm	205.00	220.50	220.50	221.50	247.50
Tinted -							
3.00mm	…	sqm	98.90	112.50	112.50	113.50	136.50
4.50mm	…	sqm	114.00	128.00	128.50	129.50	153.00
6.00mm	…	sqm	134.50	149.00	149.50	150.50	175.50
10.00mm	…	sqm	197.00	212.50	212.50	213.50	239.50

GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ATTACK RESISTANT GLASS							
Attack resistant glass thermally toughened both sides and glazing to metal :							
12.00mm (Attack guard)	...	sqm	865.00	855.00	865.00	885.00	880.00
14.00mm	...	sqm	940.00	925.00	925.00	955.00	950.00
14.00mm (Prison guard)	...	sqm	1455.00	1435.00	1430.00	1470.00	1490.00
17.50mm	...	sqm	1805.00	1775.00	1770.00	1820.00	1845.00
BULLET RESISTANT GLASS							
Bullet resistant glass and glazing to metal :							
38mm (G1 rated)	...	sqm	1515.00	1540.00	1505.00	1565.00	1590.00
45mm (G2, S0 rated)	...	sqm	2165.00	2205.00	2175.00	2230.00	2285.00
51mm (R1 rated)	...	sqm	2230.00	2285.00	2260.00	2320.00	2370.00
No spall bullet resistant glass and glazing to metal :							
24mm (G1 rated)	...	sqm	1580.00	1605.00	1585.00	1580.00	1655.00
34mm (G2, S0 rated)	...	sqm	2220.00	2255.00	2210.00	2290.00	2340.00
38mm (R1, S1 rated)	...	sqm	2475.00	2525.00	2460.00	2570.00	2615.00
51mm (R2 rated)	...	sqm	2840.00	2885.00	2830.00	2945.00	2990.00
EDGE PROCESSING							
TOUGHENED GLASS							
Polished edge :							
Straight -							
4 - 6mm glass	...	m	9.90	10.50	11.50	10.50	11.20
8 - 12mm glass	...	m	21.70	22.20	24.40	22.60	24.10
Shaped -							
4 - 6mm glass	...	m	18.60	18.90	22.00	19.30	20.40
8 - 12mm glass	...	m	42.70	44.00	47.70	43.90	46.80
Holes up to 85mm dia. :							
Plain -							
4 - 6mm glass	...	No	18.60	18.80	20.90	19.20	20.30
8 - 12mm glass	...	No	44.00	26.10	28.80	25.80	27.80
Countersunk -							
4 - 6mm glass	...	No	26.40	27.10	29.60	27.30	29.00
8 - 12mm glass	...	No	39.90	30.30	44.10	40.90	43.20
Finger slot	...	No	23.80	24.50	26.90	24.20	26.30
OTHER GLASS TYPES							
Polished edge :							
Up to 6mm glass	...	m	14.30	14.60	15.90	14.50	15.50
8 and 10mm glass	...	m	19.90	20.40	22.50	20.40	21.80
12mm glass	...	m	25.30	26.10	28.10	25.00	27.60
Bevelled and polished edge on 5 and 6mm glass :							
17mm bevel	...	m	31.90	32.50	34.90	32.50	34.30
25mm bevel	...	m	42.30	43.70	47.60	43.60	46.70

GLAZING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
EDGE PROCESSING (cont'd)							
OTHER GLASS TYPES (cont'd)							
Drill hole :							
6mm dia. in -							
6mm glass	...	No	3.60	3.70	3.85	3.60	3.85
12mm glass	...	No	8.10	8.30	8.85	8.25	8.60
25mm dia. in -							
6mm glass	...	No	6.65	6.80	7.35	6.70	7.05
12mm glass	...	No	11.00	11.35	12.00	11.05	11.85
Fan hole	...	No	55.00	55.30	59.40	55.00	56.20
Finger slot	...	No	6.30	6.45	6.85	7.00	6.90
Polished speak or pay hole	...	No	86.70	89.00	95.50	88.20	93.80
FRITTING							
Fritting of glass	...	sqm	60.00-80.00	60.00-80.00	60.00-80.00	60.00-80.00	60.00-80.00
SHOWER SCREENS							
Clear toughened glass, semi frameless :							
Front panel -							
900 x 1870mm high	...	No	750.00	765.00	780.00	750.00	770.00
Two sided -							
900 x 900 x 1870mm high with pivoted door	...	No	1050.00	1080.00	1100.00	1035.00	1080.00
Three sided -							
1800mm girth x 1870mm high comprising two 450mm end panels with 900mm wide pivoted door on splay between	...	No	1195.00	1240.00	1250.00	1190.00	1230.00
Frameless laminated glass :							
Cost range	...	No	2050.00-3200.00	2050.00-3200.00	2050.00-3200.00	2050.00-3200.00	2050.00-3200.00
SOLAR CONTROL FILM							
Prices include fixing to glass and are based on quantities of 5 to 50sqm	...	Note					
Reflective high performance film	...	sqm	86.00	86.50	87.00	86.50	87.00
Non-reflective film	...	sqm	81.50	82.00	82.50	82.00	82.50
Decorator film	...	sqm	81.50	82.00	82.50	82.00	82.50
Safety film	...	sqm	81.50	82.00	82.50	82.00	82.50
Security film	...	sqm	127.00	127.50	128.00	127.50	128.00
Anti-graffiti film	...	sqm	81.50	82.00	82.50	82.00	82.50
ADD extra for :							
Areas of in excess of 50sqm	...	-5%					

PAINTING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for preparation	...	Note					

INTERNAL PAINTING

WOODWORK

ACRYLIC

Prime, one sealer undercoat and two coats semi-gloss or gloss acrylic on :							
General surfaces	...	sqm	13.05	13.80	16.80	19.45	18.15
Surfaces -							
N.E.100mm girth	...	m	2.70	2.45	3.05	3.35	3.70
100/200mm girth	...	m	3.65	3.45	4.05	4.65	4.80
200/300mm girth	...	m	5.75	6.70	6.55	7.25	7.80
Two coats timber finish acrylic on :							
General surfaces	...	sqm	8.70	9.25	8.70	11.25	9.35
Surfaces -							
N.E.100mm girth	...	m	2.20	2.20	2.15	2.65	2.10
100/200mm girth	...	m	2.75	2.85	2.80	3.20	2.40
200/300mm girth	...	m	3.30	3.35	3.40	3.80	4.00

ALKYD ENAMEL

Prime, one undercoat and two coats semi-gloss or gloss alkyd enamel on :							
General surfaces	...	sqm	17.85	18.00	16.80	19.75	18.15
Surfaces -							
N.E.100mm girth	...	m	3.40	3.25	2.90	3.40	4.00
100/200mm girth	...	m	4.60	4.90	4.35	4.80	4.75
200/300mm girth	...	m	6.90	6.70	6.75	7.70	7.80
ADD extra for :							
Additional coat undercoat or enamel	...	sqm	4.50	4.15	3.90	4.15	4.45

CLEAR FINISH

Two coats clear polyurethane on :							
General surfaces	...	sqm	9.45	10.95	9.35	11.25	9.15
Surfaces -							
N.E.100mm girth	...	m	2.40	2.20	2.15	2.70	2.10
100/200mm girth	...	m	2.75	2.85	2.65	3.20	2.40
200/300mm girth	...	m	3.90	3.70	3.70	3.85	4.00
ADD extra for :							
Additional coat clear polyurethane	...	sqm	4.20	4.45	4.20	4.60	4.45

STAIN

Two coats stain on :							
General surfaces	...	sqm	8.20	8.25	8.05	9.70	7.90
Surfaces -							
N.E.100mm girth	...	m	2.35	2.00	2.15	2.50	1.90
100/200mm girth	...	m	2.75	2.20	2.80	2.80	2.35
200/300mm girth	...	m	3.50	3.15	3.35	3.75	3.45
ADD extra for :							
Additional coat stain	...	sqm	3.90	3.80	3.70	3.95	4.00

FRENCH POLISH

French polish general surfaces	...	sqm	139.00	132.50	140.50	136.00	127.00

WAX POLISH

Two coats wax polish on floors	...	sqm	9.80	9.60	9.30	8.80	9.10

PAINTING

INTERNAL PAINTING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
METALWORK							
ACRYLIC							
Touch up primer and two coats semi-gloss or gloss acrylic on :							
General surfaces	...	sqm	10.10	11.60	10.15	12.65	14.55
Surfaces -							
N.E.100mm girth	...	m	2.45	2.30	2.35	2.75	2.35
100/200mm girth	...	m	2.85	3.00	3.00	3.20	3.15
200/300mm girth	...	m	4.15	4.60	4.20	4.90	5.25
Pipes -							
N.E. 50mm dia.	...	m	3.50	3.40	3.25	3.65	3.20
50/100mm dia.	...	m	4.45	4.65	4.35	5.25	4.25
ADD extra for :							
Additional coat acrylic	...	sqm	3.90	3.85	3.65	4.05	5.30
ALKYD ENAMEL							
One coat zinc phosphate primer on general surfaces	...	sqm	4.60	3.90	4.25	5.05	5.30
One coat of two pack etch primer on general surfaces	...	sqm	4.85	4.60	5.30	5.80	5.80
Touch up primer, one undercoat and two coats semi-gloss or gloss alkyd enamel on :							
General surfaces	...	sqm	15.70	16.40	14.50	17.45	16.95
Surfaces -							
N.E.100mm girth	...	m	3.65	3.25	3.35	3.65	3.40
100/200mm girth	...	m	4.15	3.70	3.75	4.10	4.35
200/300mm girth	...	m	6.05	5.75	5.70	6.35	7.10
Pipes -							
N.E. 50mm dia.	...	m	4.95	4.85	4.25	4.70	4.35
50/100mm dia.	...	m	6.90	6.30	6.00	6.80	7.10
Prime, one undercoat and two coats semi-gloss or gloss alkyd enamel on :							
General surfaces	...	sqm	18.30	17.75	17.95	21.10	18.20
Surfaces -							
N.E.100mm girth	...	m	3.50	3.55	3.40	3.55	3.80
100/200mm girth	...	m	4.90	4.95	4.65	4.75	4.75
200/300mm girth	...	m	7.60	6.60	6.75	7.10	7.85
Pipes -							
N.E. 50mm dia.	...	m	6.05	5.20	5.75	5.60	5.30
50/100mm dia.	...	m	7.75	6.90	7.50	7.10	7.90
EPOXY MASTIC							
Two coats epoxy mastic coating on :							
General surfaces	...	sqm	30.50	30.40	29.50	37.20	30.00
Surfaces -							
N.E.100mm girth	...	m	6.45	5.15	5.00	5.30	5.95
100/200mm girth	...	m	7.85	6.60	6.75	7.05	7.90
200/300mm girth	...	m	10.55	10.95	11.00	11.90	10.30
Pipes -							
N.E. 50mm dia.	...	m	8.80	6.80	6.75	7.10	8.65
50/100mm dia.	...	m	10.60	10.95	10.85	11.65	10.30
BAKED FINISHES							
Refer Metalwork, page 294	...	Note					

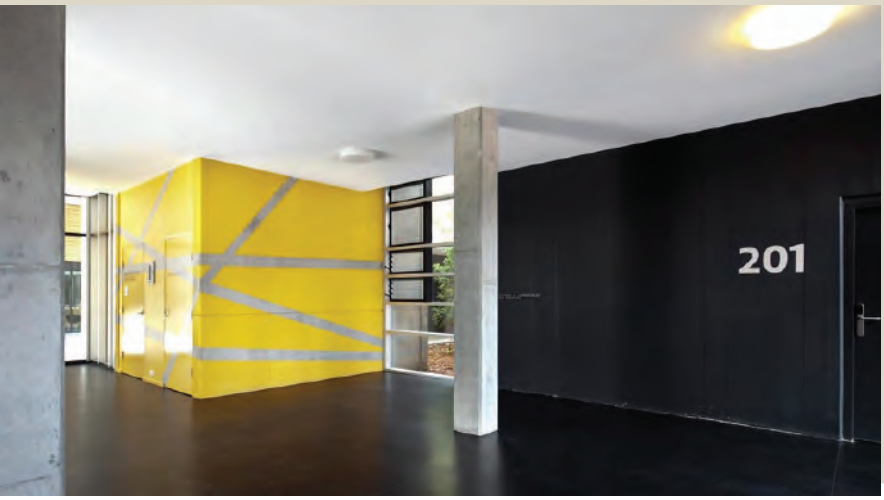
PAINTING

INTERNAL PAINTING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
MASONRY, CEMENT AND PLASTER							
ACRYLIC							
Sealer undercoat and two coats flat, low sheen or semi-gloss acrylic on :							
Plaster or similar smooth surface of -							
Walls	...	sqm	12.60	11.90	12.30	11.75	12.15
Soffits	...	sqm	13.30	12.15	12.85	12.45	12.35
ADD extra for :							
Rough surface	...	sqm	0.55	0.55	0.45	0.50	0.40
Additional coat acrylic	...	sqm	3.90	3.90	3.60	3.55	4.15
ACRYLIC LATEX							
Sealer undercoat and two coats flat acrylic latex on :							
Plaster or similar smooth surface of -							
Walls	...	sqm	11.85	11.50	11.55	11.05	12.00
Soffits	...	sqm	12.40	11.95	11.80	11.50	12.25
ADD extra for :							
Rough surface	...	sqm	0.55	0.55	0.45	0.50	0.40
Additional coat acrylic latex	...	sqm	3.90	3.55	3.55	3.60	4.15
ALKYD ENAMEL							
One coat sealer, one undercoat and one coat semi-gloss or gloss alkyd enamel on :							
Plaster or similar smooth surface of -							
Walls	...	sqm	12.75	13.65	12.75	12.10	12.15
Soffits	...	sqm	13.55	14.30	13.75	12.80	12.25
ADD extra for :							
Rough surface	...	sqm	0.75	1.05	0.60	0.75	0.75
Additional coat undercoat or enamel	...	sqm	4.05	3.95	3.65	3.60	4.00
CHLORINATED RUBBER ENAMEL							
Two coats chlorinated rubber enamel on :							
Plaster or similar smooth surface of -							
Walls	...	sqm	11.85	12.15	11.00	10.75	9.55
Soffits	...	sqm	13.30	12.75	11.95	11.50	9.95
ADD extra for :							
Rough surface	...	sqm	0.95	1.05	0.95	0.90	0.80
Additional coat enamel	...	sqm	5.50	5.30	4.95	4.90	4.65
PVA							
Two coats flat ceiling white pva copolymer on :							
Soffits - smooth surface	...	sqm	7.25	8.00	6.85	6.70	7.50
- rough surface	...	sqm	8.10	8.50	7.40	7.20	8.10
WATER BASED EPOXY							
One coat epoxy primer and two coats water based epoxy enamel on :							
Smooth wall surface	...	sqm	30.30	27.40	28.40	27.70	25.00
Rough wall surface	...	sqm	32.80	28.70	29.90	29.50	26.40









Environmentally ECO Friendly Coating Systems For Concrete and Masonry

Keim Mineral Paints originated in Bavaria in 1878 and are now used extensively around the world, with buildings painted well over 100 years, still intact!

EXTREMELY LOW VOCs **COLOURS OF NATURE** **BREATHABLE**

The unique longevity of KEIM coating systems not only provides a low maintenance, colour fast and breathable paint finish, but a truly environmentally sustainable coating system.

With a range of paint finishes and concrete stains, Keim Mineral Paints are a simple choice for your next project.

KEIM

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PAINTING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
INTERNAL PAINTING (cont'd)							
MASONRY, CEMENT AND PLASTER (cont'd)							
SEALER							
One coat sealer on fair faced masonry	...	sqm	4.20	4.05	3.65	3.75	3.75
SILICONE WATER REPELLANT							
Two coats silicone water repellant on fair faced masonry	...	sqm	18.55	19.30	18.10	17.25	13.85
GranoRepel UV resistant silicone based water repellant on fair faced masonry	...	sqm	14.85	12.15	13.05	12.85	10.05
EXTERNAL PAINTING							
ADD to the foregoing INTERNAL PAINTING prices the following :							
Working off ladders -							
N.E. 6 metres above ground	...	sqm	1.25	1.30	1.60	1.30	1.30
N.E. 9 metres above ground	...	sqm	2.00	1.95	2.05	2.10	1.75
Working off swinging stage	...	sqm	2.30	2.50	2.65	2.50	2.20
MINERAL PAINT							
Prices are based on a minimum of 500 sqm							
Two coat application of KEIM ROYALAN potassium silicate on external smooth concrete surfaces :							
Matt finish, whites and pastels	...	sqm	39.00	38.50	28.50	38.00	38.50
ADD extra for :							
Dark colours	...	sqm	5.35	5.35	5.35	5.30	5.35
Two coat application of KEIM ECOSIL potassium silicate paint on internal smooth surfaces :							
Matt finish, whites and pastels	...	sqm	19.50	18.00	18.00	19.00	19.50
ADD extra for :							
Dark colours	...	sqm	4.80	4.30	4.30	4.25	4.30
MINERAL STAIN							
Prices are based on a minimum of 500 sqm							
Two coat application of KEIM CONCRETAL LASUR potassium silicate mineral stain on smooth concrete surfaces :							
Matt finish, whites and pastels	...	sqm	32.00	32.00	32.00	28.50	29.00
ADD extra for :							
Dark colours	...	sqm	3.20	3.20	3.20	3.15	3.20
Allow for additional consumption of KEIM on absorbent surfaces	...	Note					

PAINTING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ANTI-GRAFFITI COATINGS							
Sacrifical coating system using Polysaccharide technology PSS-20 on smooth surfaces	...	sqm	29.00	28.90	27.90	29.60	30.00
Nano technology impregnation glaze coating, Faceal Oleo HD on smooth surfaces	...	sqm	32.20	30.50	30.00	29.60	30.00
ADD extra for :							
Porous or course surfaces	...	Note					
SURFACE PROTECTION COATINGS							
Nano technology impregnation glaze coating, Faceal Oleo HD on smooth surfaces	...	sqm	31.10	30.00	30.00	29.60	30.00
Nano technology impregnation glaze coating, with anti-graffiti properties, Faceal Oleo HD on smooth surfaces	...	sqm	35.40	35.30	35.30	36.00	36.40
KEIM SILIAN-100 water repellent for mineral construction materials on smooth surfaces	...	sqm	32.20	30.50	30.00	29.60	30.00
KEIM CONCRETAL BETONSIEGEL sealer to concrete, polyurethane based, solvent free coating to smooth concrete surfaces	...	sqm	42.30	40.40	40.70	40.20	40.70
ADD extra for :							
Porous or course surfaces	...	Note					
ASBESTOS ENCAPSULATION							
Apply one coat primer and one coat acrylic coating on top surface of corrugated asbestos roofing	...	sqm	23.00-28.00	23.50-28.50	24.50-29.50	22.50-27.50	22.00-27.00
LEAD ENCAPSULATION							
Apply one coat primer and one coat acrylic coating to surface of existing paint	...	sqm	15.00-19.00	15.00-19.00	15.50-19.50	13.50-17.50	15.00-20.00
PREPARATION OF EXISTING SURFACES							
Prepare painted woodwork including washing and rubbing down, priming bare patches :							
General surfaces	...	sqm	6.45	6.35	6.00	5.95	5.15
Surfaces not exceeding 300mm girth	...	m	2.90	3.10	3.10	3.00	2.40
ADD extra for :							
Burning off	...	sqm	23.40	22.70	23.40	20.40	18.35
Prepare painted metalwork including washing and wire brushing, priming bare patches :							
General surfaces	...	sqm	6.70	6.60	6.55	6.15	5.45
Surfaces not exceeding 300mm girth	...	m	2.90	3.15	3.10	3.00	2.40
ADD extra for :							
Burning off	...	sqm	23.20	21.90	21.10	21.50	18.25
Prepare decorated plaster including washing surfaces, cutting out cracks and making good :							
Walls	...	sqm	6.70	6.80	6.85	6.70	5.65
Ceilings	...	sqm	8.60	8.80	8.55	8.35	6.60
ADD extra for :							
Stripping off one layer wallpaper	...	sqm	5.30	4.70	5.15	5.20	3.90

PAINTING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SIGNWRITING							
Letters or numerals in two coats enamel :							
Up to 50mm high	...	No	6.40	6.60	6.95	6.45	6.45
51mm to 75mm high	...	No	7.05	7.40	7.55	7.65	7.50
76mm to 100mm high	...	No	8.35	8.35	8.35	8.55	8.45
WALLPAPER							
Prepare, size and hang lining paper on :							
Walls	...	sqm	6.90	5.85	6.80	6.70	5.80
Ceilings	...	sqm	7.25	6.15	7.25	7.05	6.15
Prepare, size and hang pattern paper (P.C. $7.00/sqm) (average commercial standard) on :							
Walls	...	sqm	22.40	20.50	21.40	20.90	20.20
Ceilings	...	sqm	23.80	21.50	22.50	22.20	21.20
ADD extra for :							
One coat acrylic water based glaze	...	sqm	4.20	3.75	3.95	3.75	3.30
FABRICS							
Prepare and hang on walls :							
Acrovyn semi-rigid covering, 1.60mm thick (P.C. $80.00/sqm)	...	sqm	133.00	125.50	131.50	131.50	127.50
Vinyl, paper backed (P.C. $18.00/sqm)	...	sqm	31.70	29.60	31.60	31.80	30.00
Vinyl fabric -							
Heavy duty (P.C. $25.00/sqm) (commercial standard)	...	sqm	61.30	54.80	60.30	61.80	57.00
Extra heavy duty (P.C. $33.00/sqm) (lift cars and heavy use areas)	...	sqm	74.00	67.80	73.70	74.40	69.60
Vinyl, suede, cloth backed (heavy use areas) (P.C. $38.00/sqm)	...	sqm	85.20	78.50	84.70	83.60	79.90

SANITARY PLUMBING

SANITARY FIXTURES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for sanitary fixtures include for installation complete with plugs, grated waste outlets, brackets and all fixings including jointing to pipes but do not include for pipework, traps, cocks or outlet fittings ...		Note					
BASINS							
Stainless steel basin:							
Vanity - 520 x 440mm	(P.C. $ 440.00) ...	No	640.00	660.00	660.00	660.00	710.00
Wall - 500 x 420mm	(P.C. $ 220.00) ...	No	305.00	310.00	310.00	310.00	330.00
Vitreous china basin:							
Wall, hand rinse, 4L	(P.C. $ 160.00) ...	No	340.00	360.00	360.00	360.00	410.00
Wall, 5.5L, with large shelf	(P.C. $ 640.00) ...	No	860.00	870.00	870.00	880.00	920.00
Wall, 10L	(P.C. $ 420.00) ...	No	620.00	630.00	640.00	640.00	690.00
Wall, with integrated shroud for trap, 10L	(P.C. $ 705.00) ...	No	970.00	990.00	990.00	990.00	1050.00
Wall, with pedestal, 10L	(P.C. $ 725.00) ...	No	990.00	1005.00	1005.00	1020.00	1080.00
Vanity, 12L	(P.C. $ 355.00) ...	No	450.00	450.00	450.00	450.00	470.00
Vanity, semi recessed, 9L	(P.C. $ 415.00) ...	No	510.00	520.00	520.00	520.00	540.00
Above counter, 20L	(P.C. $ 750.00) ...	No	1020.00	1030.00	1040.00	1040.00	1100.00
Vitreous china, wall basin, for accessible toilet to AS1428 :							
Basic, 8L	(P.C. $ 465.00) ...	No	750.00	760.00	770.00	770.00	840.00
With integrated shroud for trap, 8L	(P.C. $ 750.00) ...	No	1095.00	1110.00	1125.00	1125.00	1200.00
With integrated shelf, 9.5L	(P.C. $ 545.00) ...	No	850.00	870.00	870.00	880.00	950.00
With integrated trap, for infection control, 15L	(P.C. $1435.00) ...	No	1800.00	1825.00	1825.00	1825.00	1900.00
Corner style, 6.5L	(P.C. $ 390.00) ...	No	680.00	700.00	700.00	710.00	790.00
BATHS							
Shower baths, refer page 457	...	Note					
Spa and air baths, refer page 458	...	Note					
Acrylic :							
1530 x 745mm	(P.C. $ 385.00) ...	No	590.00	600.00	600.00	610.00	650.00
1680 x 745mm	(P.C. $ 490.00) ...	No	700.00	710.00	710.00	720.00	770.00
Corner type -							
1300 x 1300mm	(P.C. $ 815.00) ...	No	970.00	970.00	970.00	970.00	1000.00
1500 x 1500mm	(P.C. $ 995.00) ...	No	1155.00	1170.00	1170.00	1170.00	1185.00
Porcelain enamelled steel bath :							
1525 x 760mm	(P.C. $ 660.00) ...	No	880.00	900.00	900.00	900.00	950.00
1675 x 760mm	(P.C. $ 740.00) ...	No	970.00	980.00	980.00	990.00	1030.00
915 x 915mm	(P.C. $ 810.00) ...	No	960.00	970.00	970.00	970.00	990.00
BIDETTE, MANUAL							
Vitreous china pedestal (with over rim mixer)	(P.C. $1235.00) ...	No	1575.00	1575.00	1575.00	1600.00	1650.00
BIDETTE, ELECTRONIC							
On-pan type :							
Twin nozzle, heated	(P.C. $ 745.00) ...	No	930.00	945.00	945.00	945.00	975.00
Luxury, remote control	(P.C. $1825.00) ...	No	2100.00	2100.00	2100.00	2100.00	2100.00
ADD extra for :							
Pan, backflow prevention and power outlet ...		Note					

SANITARY PLUMBING

SANITARY FIXTURES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
W.C. SUITES							
Vitreous china pan with integrated plastic cistern, lid and seat	(P.C. $ 435.00) ...	No	710.00	720.00	730.00	730.00	800.00
Vitreous china cistern and pan with plastic lid and seat :							
Close coupled suite	(P.C. $ 900.00) ...	No	1200.00	1215.00	1230.00	1230.00	1290.00
Close coupled suite with integrated hand basin	(P.C. $1040.00) ...	No	1350.00	1380.00	1380.00	1380.00	1440.00
Junior size, connector suite	(P.C. $ 740.00) ...	No	1030.00	1050.00	1050.00	1060.00	1120.00
Care connector cistern and back rest, for accessible toilet to AS 1428	(P.C. $1535.00) ...	No	1975.00	2000.00	2000.00	2000.00	2100.00
Vitreous china wall hung pan with plastic lid and seat, inwall cistern, fascia and buttons	(P.C. $1050.00) ...	No	1395.00	1410.00	1410.00	1410.00	1485.00
W.C. PANS							
Stainless steel :							
Floor mounted pan	(P.C. $1340.00) ...	No	1575.00	1575.00	1575.00	1575.00	1625.00
Wall mounted pan	(P.C. $1355.00) ...	No	1725.00	1725.00	1725.00	1750.00	1800.00
Vitreous china :							
Connector floor mounted	(P.C. $ 155.00) ...	No	300.00	310.00	310.00	315.00	350.00
Skew trap floor mounted	(P.C. $ 565.00) ...	No	760.00	770.00	770.00	780.00	820.00
Close coupled floor mounted	(P.C. $ 450.00) ...	No	620.00	630.00	630.00	630.00	670.00
Wall faced	(P.C. $ 590.00) ...	No	770.00	780.00	780.00	780.00	820.00
Wall hung	(P.C. $ 720.00) ...	No	910.00	920.00	920.00	920.00	960.00
CISTERNS							
Prices include flushpipes	...	Note					
Plastic :							
Wall cistern	(P.C. $ 155.00) ...	No	260.00	265.00	265.00	270.00	290.00
Duct cistern							
Urinal	(P.C. $ 605.00) ...	No	740.00	750.00	750.00	750.00	770.00
Dual flush	(P.C. $ 305.00) ...	No	420.00	430.00	430.00	430.00	450.00
Concealed, in wall cistern	(P.C. $ 340.00) ...	No	480.00	490.00	490.00	490.00	520.00
ADD extra for :							
Push buttons & flush plate	(P.C. $ 140.00) ...	No	180.00	182.00	182.00	182.00	188.00
Stainless steel :							
Wall cistern, for W.C.	(P.C. $ 570.00) ...	No	710.00	710.00	710.00	710.00	740.00
Duct cistern, for W.C.	(P.C. $ 755.00) ...	No	900.00	910.00	910.00	910.00	930.00
Wall cistern for urinal to suit up to 2m length	(P.C. $ 570.00) ...	No	750.00	760.00	760.00	760.00	800.00
Duct cistern for urinal to suit up to 2m length	(P.C. $ 720.00) ...	No	910.00	920.00	920.00	920.00	960.00
Vitreous china :							
Wall cistern	(P.C. $ 540.00) ...	No	710.00	720.00	720.00	730.00	760.00
Flusher Devices etc. :							
Refer Valves and Controls, page 466	...	Note					

SANITARY PLUMBING

SANITARY FIXTURES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TOILET SEATS							
Hard plastic :							
Medium quality -							
Double flap	(P.C. $ 90.00) ...	No	122.00	124.00	124.00	124.00	130.00
Single flap	(P.C. $ 75.00) ...	No	106.00	108.00	108.00	108.00	114.00
Top quality -							
Double flap	(P.C. $ 405.00) ...	No	460.00	460.00	460.00	460.00	470.00
Single flap	(P.C. $ 400.00) ...	No	460.00	460.00	460.00	460.00	460.00
Open front single flap	(P.C. $ 340.00) ...	No	390.00	390.00	395.00	390.00	400.00
SHOWER BASES							
Porcelain enamelled steel, 900 x 900mm	(P.C. $ 540.00) ...	No	760.00	770.00	770.00	770.00	820.00
Stainless steel, 900 x 916mm	(P.C. $1110.00) ...	No	1365.00	1380.00	1380.00	1380.00	1425.00
Acrylic :							
900 x 900mm	(P.C. $ 425.00) ...	No	630.00	640.00	650.00	650.00	700.00
1200 x 900mm	(P.C. $ 565.00) ...	No	780.00	800.00	800.00	800.00	850.00
SHOWER BATH							
Porcelain enamelled steel -							
1540 x 750 x 2175mm high	(P.C. $1150.00) ...	No	1500.00	1515.00	1515.00	1515.00	1590.00
SHOWER ENCLOSURE							
Reinforced polyester, 900 x 765 x 1875mm high	(P.C. $1250.00) ...	No	1775.00	1800.00	1800.00	1825.00	1925.00
SINKS							
Bar Sink, 380mm dia bowl	(P.C. $ 170.00) ...	No	230.00	230.00	230.00	235.00	245.00
Cleaner's Sink :							
Stainless steel with gallows brackets, 500 x 400 x 180mm	(P.C. $ 520.00) ...	No	730.00	750.00	750.00	750.00	800.00
White vitreous china, 580 x 435 x 280mm with brackets	(P.C. $1255.00) ...	No	1530.00	1530.00	1545.00	1545.00	1590.00
Drip Sink	(P.C. $ 225.00) ...	No	290.00	290.00	290.00	290.00	305.00
Kitchen Sink :							
Stainless steel, standard grade -							
Under mount :							
Single bowl	(P.C. $ 340.00) ...	No	430.00	440.00	440.00	440.00	460.00
Double bowl	(P.C. $ 620.00) ...	No	730.00	740.00	740.00	740.00	760.00
Single bowl, with drainer -							
930mm long	(P.C. $ 500.00) ...	No	600.00	610.00	610.00	610.00	630.00
1234mm long	(P.C. $ 540.00) ...	No	650.00	650.00	650.00	650.00	670.00
1386mm long	(P.C. $ 745.00) ...	No	870.00	870.00	870.00	870.00	890.00
Double bowl, with drainer -							
1386mm long	(P.C. $ 905.00) ...	No	1040.00	1040.00	1050.00	1040.00	1060.00
1538mm long	(P.C. $ 925.00) ...	No	1060.00	1070.00	1070.00	1060.00	1080.00
1842mm long	(P.C. $1030.00) ...	No	1185.00	1185.00	1185.00	1185.00	1200.00

SANITARY PLUMBING

SANITARY FIXTURES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SINKS (cont'd)							
Stainless steel, commercial grade, fabricated -							
Single bowl (400 x 400 x 300), with drainer :							
1500mm long, 304 grade	(P.C. $1255.00) ...	No	1440.00	1440.00	1440.00	1440.00	1470.00
1500mm long, 316 grade	(P.C. $1780.00) ...	No	2000.00	2000.00	2000.00	2000.00	2025.00
2000mm long, 304 grade	(P.C. $1475.00) ...	No	1675.00	1700.00	1700.00	1675.00	1725.00
2000mm long, 316 grade	(P.C. $2105.00) ...	No	2350.00	2350.00	2350.00	2350.00	2400.00
Double bowl (400 x 400 x 300), with drainer :							
1500mm long, 304 grade	(P.C. $2035.00) ...	No	2300.00	2300.00	2300.00	2300.00	2300.00
1500mm long, 316 grade	(P.C. $3100.00) ...	No	3400.00	3450.00	3450.00	3400.00	3450.00
2000mm long, 304 grade	(P.C. $2385.00) ...	No	2650.00	2650.00	2650.00	2650.00	2700.00
2000mm long, 316 grade	(P.C. $3600.00) ...	No	3950.00	3950.00	4000.00	3950.00	4000.00
2500mm long, 304 grade	(P.C. $2980.00) ...	No	3300.00	3300.00	3300.00	3300.00	3350.00
2500mm long, 316 grade	(P.C. $4450.00) ...	No	4900.00	4900.00	4900.00	4900.00	4900.00
Laboratory Sink :							
Stainless steel, acid resisting -							
300 x 300 x 250mm	(P.C. $ 950.00) ...	No	1070.00	1070.00	1070.00	1070.00	1080.00
450 x 350 x 350mm	(P.C. $1160.00) ...	No	1290.00	1290.00	1290.00	1290.00	1305.00
Vitreous china -							
375 x 375 x 265mm	(P.C. $ 880.00) ...	No	1010.00	1020.00	1020.00	1020.00	1030.00
480 x 405 x 265mm	(P.C. $1200.00) ...	No	1365.00	1365.00	1365.00	1365.00	1380.00
SPA BATHS							
Acrylic :							
1650mm long with 10 jets	(P.C. $1420.00) ...	No	1775.00	1800.00	1800.00	1800.00	1875.00
1790mm long with 15 jets	(P.C. $3850.00) ...	No	4400.00	4400.00	4400.00	4400.00	4500.00
1570 round, free standing, 14 jets	(P.C. $6470.00) ...	No	7200.00	7250.00	7250.00	7200.00	7300.00
1490 x 1490mm corner, 12 jets	(P.C. $2110.00) ...	No	2600.00	2650.00	2650.00	2650.00	2750.00
1490 x 1490mm corner, 22 jets	(P.C. $3855.00) ...	No	4500.00	4500.00	4500.00	4500.00	4600.00
1400 x 1400mm square, 24 jets	(P.C. $5415.00) ...	No	6150.00	6200.00	6200.00	6200.00	6300.00
ADD extra for :							
Tiled enclosure, raised deck or floor sinking	...	Note					
SLOP HOPPERS							
Vitreous china, pedestal with grate	(P.C. $3175.00) ...	No	3650.00	3650.00	3650.00	3650.00	3750.00
ADD extra for :							
Cistern	...	Note					
TROUGHS							
Ablution Trough :							
Stainless steel, 304 grade with 370mm x 200mm deep bowl and brackets, length :							
1200mm with :							
Tile lip	(P.C. $ 995.00) ...	No	1245.00	1260.00	1260.00	1260.00	1305.00
150mm upstand	(P.C. $1045.00) ...	No	1305.00	1320.00	1320.00	1320.00	1365.00
300mm upstand	(P.C. $1155.00) ...	No	1425.00	1450.00	1450.00	1450.00	1500.00

SANITARY PLUMBING

SANITARY FIXTURES (cont'd)

TROUGHS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Ablution Trough :							
Stainless steel, 304 grade with 370mm							
x 200mm deep bowl and brackets, length :							
1800mm with :							
Tile lip	(P.C. $1195.00) ...	No	1500.00	1515.00	1515.00	1515.00	1575.00
150mm upstand	(P.C. $1300.00) ...	No	1625.00	1625.00	1650.00	1650.00	1700.00
300mm upstand	(P.C. $1435.00) ...	No	1775.00	1775.00	1800.00	1800.00	1850.00
2400mm with :							
Tile lip	(P.C. $1435.00) ...	No	1800.00	1815.00	1815.00	1830.00	1890.00
150mm upstand	(P.C. $1550.00) ...	No	1925.00	1950.00	1950.00	1950.00	2025.00
300mm upstand	(P.C. $1720.00) ...	No	2125.00	2150.00	2150.00	2150.00	2225.00
Drinking Trough :							
Stainless steel, 304 grade with 300mm							
x 160mm deep bowl and brackets, length -							
900mm with :							
150 upstand	(P.C. $1195.00) ...	No	1455.00	1470.00	1470.00	1470.00	1515.00
2 push bubblers	(P.C. $1285.00) ...	No	1575.00	1575.00	1575.00	1575.00	1625.00
1200mm with :							
150 upstand	(P.C. $1830.00) ...	No	2150.00	2150.00	2150.00	2150.00	2200.00
3 push bubblers	(P.C. $2155.00) ...	No	2500.00	2500.00	2500.00	2500.00	2550.00
1800mm with							
4 push bubblers	(P.C. $2550.00) ...	No	2950.00	3000.00	3000.00	3000.00	3050.00
Refrigerated, stainless steel,							
with 4 push bubblers and disabled							
access station -							
2000mm	(P.C. $6405.00) ...	No	7200.00	7200.00	7200.00	7200.00	7200.00
2000mm with bottle filler							
and auto flush	(P.C. $7980.00) ...	No	8800.00	8800.00	8800.00	8800.00	9000.00
Laundry Trough, stainless steel :							
Single compartment 45L laundry trough -							
Tub, flush into benchtop	(P.C. $ 425.00) ...	No	520.00	530.00	530.00	530.00	550.00
Standard tub with cabinet	(P.C. $ 425.00) ...	No	630.00	640.00	640.00	650.00	690.00
Single compartment 70L laundry trough -							
Tub, flush into benchtop	(P.C. $ 685.00) ...	No	800.00	810.00	810.00	810.00	830.00
Standard tub with cabinet	(P.C. $ 850.00) ...	No	1080.00	1095.00	1095.00	1095.00	1155.00
Double compartment 2 x 45L laundry trough -							
Tub, flush into benchtop	(P.C. $1170.00) ...	No	1350.00	1350.00	1350.00	1350.00	1380.00
Scrub up trough, stainless steel, grade 316,							
fabricated with double skin and brackets :							
1200mm long (2 person)	(P.C. $ 5130.00) ...	No	5600.00	5800.00	5800.00	5600.00	5800.00
1800mm long (3 person)	(P.C. $ 7180.00) ...	No	8000.00	8000.00	8000.00	8000.00	8000.00

SANITARY PLUMBING

SANITARY FIXTURES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
URINALS							
Refer page 456 for cisterns	...	Note					
Stainless steel, floor style urinals, with integral sparge :							
330mm wide, length -							
900mm (1 water inlet)	(P.C. $ 1620.00) ...	No	2050.00	2050.00	2050.00	2050.00	2150.00
1200mm (1 water inlet)	(P.C. $ 1955.00) ...	No	2400.00	2450.00	2450.00	2450.00	2550.00
1500mm (1 water inlet)	(P.C. $ 2260.00) ...	No	2750.00	2800.00	2800.00	2800.00	2900.00
1800mm (1 water inlet)	(P.C. $ 2650.00) ...	No	3200.00	3200.00	3200.00	3200.00	3300.00
2100mm (2 water inlets)	(P.C. $ 3010.00) ...	No	3600.00	3600.00	3700.00	3700.00	3800.00
2400mm (2 water inlets)	(P.C. $ 3230.00) ...	No	3900.00	3900.00	3900.00	3900.00	4100.00
2700mm (2 water inlets)	(P.C. $ 3485.00) ...	No	4200.00	4200.00	4300.00	4300.00	4400.00
3000mm (2 water inlets)	(P.C. $ 3940.00) ...	No	4700.00	4800.00	4800.00	4800.00	4900.00
ADD extra for :							
Side panel exposed, per side	(P.C. $ 265.00) ...	No	330.00	335.00	335.00	335.00	345.00
610mm wide, with 300 wide step on grate, length -							
900mm (1 water inlet)	(P.C. $ 2650.00) ...	No	3150.00	3200.00	3200.00	3200.00	3300.00
2100mm (2 water inlets)	(P.C. $ 4815.00) ...	No	5600.00	5600.00	5600.00	5600.00	5700.00
3000mm (2 water inlets)	(P.C. $ 6065.00) ...	No	7000.00	7200.00	7200.00	7200.00	7200.00
Other lengths, pro-rata	...	Note					
900mm wide, with 600mm wide step on grate, length -							
900mm (1 water inlet)	(P.C. $ 4805.00) ...	No	5600.00	5600.00	5600.00	5600.00	5600.00
2100mm (2 water inlets)	(P.C. $ 8555.00) ...	No	9600.00	9600.00	9600.00	9600.00	9800.00
3000mm (2 water inlets)	(P.C. $12205.00) ...	No	13750.00	13750.00	13750.00	13750.00	13750.00
Other lengths, pro-rata	...	Note					
Single Stall Urinals, wall hung, with sparge pipe :							
Stainless steel	(P.C. $ 520.00) ...	No	730.00	750.00	750.00	750.00	800.00
Vitreous china	(P.C. $ 1745.00) ...	No	2025.00	2050.00	2050.00	2050.00	2100.00
Vitreous china, including electronic flush	(P.C. $ 3230.00) ...	No	3700.00	3725.00	3725.00	3725.00	3775.00
Stainless steel Water Saving urinal with integral sparge and Hygiene Clean flushing device, up to 1400mm long	(P.C. $ 3845.00) ...	No	4400.00	4500.00	4500.00	4500.00	4500.00
Waterless Single Stall Urinals, wall hung :							
Stainless steel	(P.C. $ 1310.00) ...	No	1575.00	1600.00	1600.00	1600.00	1650.00
Vitreous china	(P.C. $ 1540.00) ...	No	1825.00	1850.00	1850.00	1850.00	1900.00
Glazed titanium steel	(P.C. $ 1915.00) ...	No	2225.00	2250.00	2250.00	2250.00	2300.00
Glazed titanium steel, with hybrid technology (drain rinsing flush)	(P.C. $ 2575.00) ...	No	2950.00	2950.00	2950.00	2950.00	3000.00
WATER COOLERS							
Mains connected :							
Under bench in line -							
8L per hour chilling	(P.C. $ 1030.00) ...	No	1365.00	1380.00	1380.00	1395.00	1455.00
15L per hour chilling	(P.C. $ 1215.00) ...	No	1550.00	1575.00	1575.00	1575.00	1650.00
Floor mounted, bubbler							
26L per hour chilling	(P.C. $ 1760.00) ...	No	2150.00	2150.00	2150.00	2150.00	2250.00
Heavy duty weatherproof, twin bubbler	(P.C. $ 4345.00) ...	No	5000.00	5000.00	5000.00	5000.00	5100.00
TOILET ACCESSORIES							
Dispensers, dryers, refer pages 315 to 319	...	Note					

SANITARY PLUMBING

HOT WATER UNITS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for Hot Water Units (HWU) include for installation, connection to HW&CW service and CW isolation valve ...		Note					
ADD extra for :							
Gas / electrical service ...		Note					
Safety, pressure, temperature and tempering valves as applicable, pages 472 and 473 ...		Note					
Thermostatic Mixing Valves, pages 471 and 472 ...		Note					
ELECTRIC							
Instantaneous 3 phase multi point	(P.C. $ 640.00) ...	No	870.00	880.00	880.00	890.00	930.00
Mains pressure, vitreous glass lining :							
Domestic type -							
Internal/external :							
25 litre	(P.C. $ 430.00) ...	No	630.00	640.00	640.00	650.00	700.00
50 litre	(P.C. $ 430.00) ...	No	630.00	640.00	640.00	650.00	700.00
80 litre	(P.C. $ 625.00) ...	No	915.00	945.00	945.00	945.00	1020.00
125 litre	(P.C. $ 720.00) ...	No	1110.00	1140.00	1140.00	1155.00	1245.00
160 litre	(P.C. $1030.00) ...	No	1450.00	1475.00	1475.00	1475.00	1575.00
250 litre	(P.C. $1310.00) ...	No	1750.00	1775.00	1775.00	1775.00	1875.00
315 litre	(P.C. $1450.00) ...	No	1900.00	1900.00	1900.00	1900.00	2000.00
Commercial type -							
315 litre, 3 element	(P.C. $2850.00) ...	No	3400.00	3400.00	3400.00	3400.00	3500.00
315 litre, 6 element	(P.C. $4000.00) ...	No	4600.00	4600.00	4600.00	4600.00	4700.00
Sink Heaters :							
Over sink 5 litres	(P.C. $ 330.00) ...	No	530.00	550.00	550.00	550.00	600.00
Under sink 10 litres	(P.C. $ 365.00) ...	No	570.00	580.00	590.00	590.00	640.00
ELECTRIC HEAT PUMPS							
STC's - Small-scale Technology Certificates can apply to heat-pump HWU's - may offset part of a purchase cost. Refer to HWU seller / Clean Energy Regulator. ...		Note					
310 litre (cooler climates)	(P.C. $4190.00) ...	No	5000.00	5000.00	5000.00	5000.00	5200.00
325 litre (warmer climates)	(P.C. $3530.00) ...	No	4300.00	4400.00	4400.00	4400.00	4400.00
GAS							
Continuous flow, 6 star :							
Internal (flued) :							
27 litre/m -							
Factory set at 50°C	(P.C. $2340.00) ...	No	2700.00	2700.00	2700.00	2700.00	2750.00
Up to 82°C - commercial	(P.C. $2880.00) ...	No	3400.00	3400.00	3400.00	3400.00	3500.00
ADD extra for :							
Flue (varies with design and length) nominal per metre ...		m	265.00	270.00	270.00	275.00	300.00
External, factory set at 50°C or 60°C :							
12 litres/m	(P.C. $ 605.00) ...	No	860.00	880.00	880.00	890.00	950.00
16 litres/m	(P.C. $ 605.00) ...	No	860.00	880.00	880.00	890.00	950.00
20 litres/m	(P.C. $ 670.00) ...	No	930.00	945.00	945.00	960.00	1020.00
26 litres/m	(P.C. $ 735.00) ...	No	1005.00	1020.00	1020.00	1020.00	1080.00
27 litres/m up to 82°C - commercial	(P.C. $2260.00) ...	No	2750.00	2800.00	2800.00	2800.00	2900.00
ADD extra for :							
Standard Flue Diverter (12/16/26L) ...		No	280.00	285.00	285.00	285.00	295.00

SANITARY PLUMBING

HOT WATER UNITS (cont'd)

GAS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Commercial multipack continuous flow heater bank - factory manifolded, with staging valves, CW/HW/gas unit isolation valves and delivering up to 82°C (given capacities are peak flow rates at 50°C) :							
Internal (flued) :							
Two unit 27 litres/m	(P.C. $ 7710.00) ...	No	8800.00	8800.00	8800.00	8800.00	8800.00
Three unit 41 litres/m	(P.C. $11140.00) ...	No	12750.00	12750.00	12750.00	12750.00	13000.00
Four unit 54 litres/m	(P.C. $14560.00) ...	No	16500.00	16500.00	16500.00	16500.00	16750.00
Five unit 54 litres/m	(P.C. $18440.00) ...	No	21000.00	21000.00	21000.00	21000.00	21250.00
Six unit 81 litres/m	(P.C. $21860.00) ...	No	25000.00	25000.00	25000.00	25000.00	25500.00
ADD extra for :							
Flue (for each unit, varies with design and length)	...	Note					
External:							
Two unit 27 litres/m	(P.C. $ 6350.00) ...	No	7200.00	7200.00	7200.00	7200.00	7400.00
Three unit 41 litres/m	(P.C. $ 9090.00) ...	No	10250.00	10500.00	10500.00	10500.00	10500.00
Four unit 54 litres/m	(P.C. $11830.00) ...	No	13500.00	13500.00	13500.00	13500.00	13750.00
Five unit 54 litres/m	(P.C. $15020.00) ...	No	17000.00	17250.00	17250.00	17250.00	17500.00
Six unit 81 litres/m	(P.C. $17770.00) ...	No	20250.00	20250.00	20250.00	20250.00	20750.00
Storage type, mains pressure :							
Domestic type with vitreous glass lining :							
Internal (with basic flue)							
135 litre, 3 star	(P.C. $ 1270.00) ...	No	1850.00	1900.00	1900.00	1900.00	2000.00
170 litre, 3 star	(P.C. $ 1460.00) ...	No	2050.00	2100.00	2100.00	2100.00	2200.00
External							
90 litre, 4 star	(P.C. $ 1040.00) ...	No	1350.00	1365.00	1365.00	1365.00	1440.00
135 litre, 4 star	(P.C. $ 1040.00) ...	No	1365.00	1380.00	1395.00	1395.00	1470.00
170 litre, 4 star	(P.C. $ 1160.00) ...	No	1500.00	1525.00	1525.00	1525.00	1600.00
265 litre, 5 star, heavy duty	(P.C. $ 5060.00) ...	No	5800.00	5800.00	5800.00	5800.00	5800.00

STORAGE TANKS

Commercial storage tank, mains pressure :							
Internal/External							
325 litre	(P.C. $ 1930.00) ...	No	2350.00	2400.00	2400.00	2400.00	2450.00
410 litre	(P.C. $ 2570.00) ...	No	3100.00	3100.00	3100.00	3100.00	3200.00
ADD extra for :							
The primary heating source	...	Note					

SOLAR

Small-scale Technology Certificates (STC's) may be created for the installation of the solar water heater and traded, or the right to create STC's may be assigned to a Registered Agent in return for a financial benefit. Contact the solar water heater manufacturer or refer to the Clean Energy Regulator	...	Note					
DOMESTIC, Mains pressure type :							
Standard "open circuit" model with copper/aluminium collectors and electric booster -							
Thermosiphon (roof mounted) :							
180 litre with one (2sqm) collector	...	No	4200.00	4300.00	4500.00	4400.00	4300.00
300 litre with two (4sqm) collectors	...	No	5800.00	6000.00	6200.00	6000.00	6000.00
Pumped (ground/floor mounted) :							
325 litre with two (4sqm) collectors	...	No	5800.00	5800.00	6000.00	6000.00	5800.00

SANITARY PLUMBING

HOT WATER UNITS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SOLAR (cont'd)							
Heat exchange model, anti-frost and anti-corrosive with electric booster -							
Thermosiphon (roof mounted) :							
180 litre with one (2sqm) collector	...	No	4800.00	5000.00	5200.00	5200.00	4800.00
300 litre with two (4sqm) collectors	...	No	6600.00	6800.00	7000.00	6800.00	6600.00
Pumped (ground/floor mounted) :							
270 litre with two (4sqm) collectors	...	No	7400.00	7600.00	7800.00	7800.00	7400.00
ADD extra for :							
Additional collector (2sqm) to -							
Open circuit model	...	No	1305.00	1305.00	1305.00	1305.00	1305.00
Heat exchange model	...	No	1320.00	1320.00	1320.00	1320.00	1320.00
Support frame, 15°/20°pitch	...	No	1080.00	1080.00	1080.00	1080.00	1080.00
Anti-cyclone frame	...	No	930.00	930.00	930.00	930.00	930.00
COMMERCIAL, INSTITUTION type :							
Closed circuit centralised solar preheat system, with anti-frost and high efficiency collectors. Requires commercial type gas, electric or heat pump hot water units for boosting -							
Average daily output :							
1500 litres having 20sqm surface collector	...	No	49000.00	51000.00	50000.00	49000.00	51000.00
2200 litres having 32sqm surface collector	...	No	68000.00	70000.00	69000.00	67000.00	70000.00
3500 litres having 48sqm surface collector	...	No	93000.00	96000.00	94000.00	91000.00	95000.00
5000 litres having 64sqm surface collector	...	No	125000.00	130000.00	127500.00	122500.00	127500.00
ADD extra for :							
Commercial type gas, electric or heat-pump water units for boosting	...	Note					
Flat roof stands, per sqm of panel (average)	...	sqm	130.00	130.00	130.00	130.00	130.00
Pitch collector rails per sqm of panel (average)	...	sqm	18.00	18.00	18.00	18.00	18.00
BOILING WATER UNITS							
Electric boiling water unit :							
3.0 litre, 18 cup (P.C. $ 695.00)	...	No	1005.00	1020.00	1035.00	1035.00	1110.00
5.0 litre, 30 cup (P.C. $ 990.00)	...	No	1320.00	1335.00	1335.00	1350.00	1425.00
7.5 litre, 45 cup (P.C. $1100.00)	...	No	1450.00	1450.00	1475.00	1475.00	1550.00
15.0 litre, 100 cup (P.C. $1300.00)	...	No	1650.00	1675.00	1675.00	1675.00	1750.00
25.0 litre, 150 cup (P.C. $1370.00)	...	No	1725.00	1750.00	1750.00	1750.00	1825.00
ADD extra for :							
Water filter	...	No	500.00	500.00	500.00	500.00	510.00
MECHANICAL SERVICES							
Refer page 617 for hot water systems and boilers	...	Note					

SANITARY PLUMBING

TAPS AND OUTLETS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for taps and outlets include for joining to pipework	...	Note					
Bath :							
Combination set	(P.C. $ 335.00) ...	No	450.00	450.00	450.00	460.00	480.00
Mixer set	(P.C. $ 405.00) ...	No	520.00	530.00	530.00	530.00	550.00
Mixer set with diverter	(P.C. $ 435.00) ...	No	600.00	600.00	610.00	610.00	650.00
Freestanding bath filler with mixer and handshower	(P.C. $ 735.00) ...	No	920.00	920.00	930.00	930.00	960.00
Basin :							
Combination set	(P.C. $ 345.00) ...	No	430.00	430.00	430.00	430.00	440.00
Mixer set	(P.C. $ 515.00) ...	No	610.00	610.00	610.00	610.00	620.00
Sensor, hands free	(P.C. $ 660.00) ...	No	800.00	810.00	810.00	810.00	830.00
Hospital lever combination	(P.C. $ 810.00) ...	No	940.00	950.00	950.00	940.00	960.00
Bib Tap :							
Standard handle	(P.C. $ 165.00) ...	No	200.00	200.00	200.00	200.00	205.00
Lever handle	(P.C. $ 330.00) ...	No	375.00	375.00	375.00	375.00	380.00
Push, timed flow	(P.C. $ 320.00) ...	No	375.00	375.00	375.00	375.00	385.00
Wave sensor, with ss wall plate	(P.C. $ 770.00) ...	No	975.00	975.00	975.00	990.00	1020.00
Laundry :							
Mixer trough set	(P.C. $ 340.00) ...	No	450.00	460.00	460.00	460.00	490.00
Mixer trough set with pull down spray	(P.C. $ 580.00) ...	No	710.00	710.00	720.00	720.00	740.00
Lever handle trough set	(P.C. $ 395.00) ...	No	510.00	520.00	520.00	520.00	540.00
Washing machine set (hot and cold)	(P.C. $ 350.00) ...	No	420.00	420.00	420.00	420.00	440.00
Pillar tap :							
Handle	(P.C. $ 155.00) ...	No	195.00	195.00	195.00	195.00	200.00
Lever handle	(P.C. $ 335.00) ...	No	385.00	385.00	385.00	385.00	390.00
Shower :							
Combination set	(P.C. $ 325.00) ...	No	480.00	490.00	490.00	490.00	530.00
Mixer set	(P.C. $ 405.00) ...	No	560.00	570.00	570.00	580.00	610.00
Arm and rose combination set	(P.C. $ 180.00) ...	No	335.00	340.00	345.00	350.00	385.00
Hand shower on rail	(P.C. $ 425.00) ...	No	590.00	600.00	600.00	610.00	650.00
Sink :							
Combination set	(P.C. $ 585.00) ...	No	720.00	730.00	730.00	730.00	750.00
Mixer set	(P.C. $ 340.00) ...	No	460.00	460.00	470.00	470.00	490.00
Mixer set & vege spray	(P.C. $ 655.00) ...	No	790.00	800.00	800.00	800.00	820.00
Laboratory Set :							
One Way	(P.C. $ 455.00) ...	No	540.00	550.00	550.00	550.00	560.00
Two Way	(P.C. $ 550.00) ...	No	690.00	690.00	690.00	690.00	720.00
Three Way	(P.C. $1230.00) ...	No	1410.00	1410.00	1410.00	1410.00	1440.00
Hot, cold & mixed	(P.C. $1320.00) ...	No	1500.00	1500.00	1500.00	1500.00	1525.00
Special function :							
Sensor, hands free	(P.C. $ 800.00) ...	No	1005.00	1005.00	1005.00	1005.00	1050.00
Knee operated valve	(P.C. $ 435.00) ...	No	690.00	700.00	700.00	710.00	760.00
Foot operated valve	(P.C. $ 155.00) ...	No	360.00	370.00	370.00	380.00	430.00
Outlet only :							
Fixed / swivel	(P.C. $ 250.00) ...	No	305.00	305.00	305.00	305.00	310.00
Spa spout	(P.C. $ 555.00) ...	No	630.00	630.00	630.00	630.00	630.00

SANITARY PLUMBING

THERMOSTATIC MIXING VALVES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for jointing to pipework	...	Note					
Recessed single point T.M.V. with adjustable flow and temperature, 29 L/m @ 300kPa and polished c.p. finish	...	No	580.00	590.00	590.00	590.00	620.00
Multipoint, T.M.V. with service isolation valves, strainers, non return and flat faced screw connections :							
38 L/m @ 300kPa tamperproof adjustment, 15mm in / 20mm out and polished s.s. and c.p. finish (NSW Health approved)	...	No	590.00	610.00	610.00	610.00	660.00
75 L/m @ 300kPa, 20mm in / 25mm out and polished s.s. and c.p. finish	...	No	1065.00	1080.00	1080.00	1080.00	1125.00
39 L/m @ 300kPa, 15mm in / 20mm out, n.p. brass, with recessed s.s. enclosure and secure door	...	No	1110.00	1125.00	1140.00	1140.00	1215.00
Multipoint, T.M.V. with service isolation valves, strainers, non return,flat faced screw connections and tamperproof adjustment (NSW Health approved) :							
42 L/m @ 300kPa, 15mm in / 20mm out n.p. brass, with cold water bypass (4 pipe), surface s.s. enclosure	...	No	1215.00	1230.00	1230.00	1230.00	1290.00
42 L/m @ 300kPa, 15mm in / 20mm out n.p. brass, with cold water bypass (4 pipe), recessed s.s. enclosure and secure door	...	No	1320.00	1335.00	1335.00	1335.00	1410.00

VALVES AND CONTROLS

Hose cock (tap) brass, with vacuum breaker :							
15mm (with 20mm BSP outlet)	...	No	48.00	50.00	50.00	51.00	57.00
20mm (with 25mm BSP outlet)	...	No	90.00	92.00	92.00	92.00	98.00
Hose cock (tap) brass, with removable key control handle and vacuum breaker :							
15mm (with 20mm BSP outlet)	...	No	59.00	61.00	61.00	62.00	68.00
Hose cock (tap) chromium plated brass, with removable key control handle and vacuum beaker :							
15mm (with 20mm BSP outlet)	...	No	106.00	108.00	108.00	108.00	114.00
Brass T-head (gland) stop cock :							
15mm	...	No	41.00	42.00	43.00	43.00	49.00
20mm	...	No	59.00	60.00	61.00	61.00	67.00
25mm	...	No	112.00	112.00	112.00	114.00	120.00
32mm	...	No	124.00	126.00	126.00	126.00	134.00
40mm	...	No	315.00	315.00	315.00	315.00	325.00
50mm	...	No	360.00	360.00	360.00	360.00	375.00
Cistern stop, chromium plated brass, right angle :							
15mm, mini	...	No	34.00	36.00	36.00	37.00	43.00
15mm	...	No	42.00	43.00	44.00	44.00	50.00
Chromium plated brass recess valve, with removable key control handle :							
20mm	...	No	170.00	172.00	172.00	172.00	178.00

SANITARY PLUMBING

VALVES AND CONTROLS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Brass full flow gate/ball valve :							
Screwed -							
15mm	...	No	52.00	54.00	54.00	55.00	61.00
20mm	...	No	62.00	63.00	63.00	64.00	70.00
25mm	...	No	88.00	90.00	90.00	90.00	96.00
32mm	...	No	138.00	140.00	140.00	140.00	148.00
40mm	...	No	195.00	200.00	200.00	200.00	210.00
50mm	...	No	275.00	280.00	280.00	280.00	295.00
65mm	...	No	590.00	600.00	600.00	600.00	620.00
80mm	...	No	850.00	860.00	860.00	860.00	880.00
Cast iron gate valve :							
Flanged -							
80mm	...	No	560.00	570.00	570.00	580.00	630.00
100mm	...	No	710.00	720.00	720.00	730.00	790.00
150mm	...	No	2400.00	2400.00	2400.00	2400.00	2500.00
Brass swing check or non return valve :							
Screwed -							
15mm	...	No	74.00	74.00	74.00	76.00	82.00
20mm	...	No	92.00	94.00	94.00	94.00	100.00
25mm	...	No	134.00	136.00	136.00	136.00	142.00
32mm	...	No	200.00	200.00	200.00	200.00	210.00
40mm	...	No	285.00	290.00	290.00	290.00	300.00
50mm	...	No	380.00	380.00	380.00	380.00	395.00
65mm	...	No	770.00	780.00	780.00	780.00	800.00
80mm	...	No	1125.00	1140.00	1140.00	1140.00	1155.00
Cast iron swing check or non return valve :							
Flanged :							
80mm	...	No	620.00	630.00	640.00	640.00	690.00
100mm	...	No	660.00	670.00	680.00	680.00	740.00

FLUSHING CONTROLS

Electronic sensor controlled flushing system including wall or ceiling mounted sensor, control equipment, solenoid valve, filter, isolating valve and vacuum breaker, to suit :							
W.C.	...	No	2350.00	2350.00	2350.00	2350.00	2400.00
Urinal, single stall	...	No	1410.00	1425.00	1425.00	1425.00	1455.00
Urinal, twin stall	...	No	1525.00	1550.00	1550.00	1550.00	1575.00
ADD extra for :							
Power outlet	...	Note					
Sanitary fixture	...	Note					
"Sloan" push button and remote pressure valve	...	No	770.00	770.00	770.00	770.00	800.00

PRESSURE LIMITING VALVES

PLV/PRV CW pressure limiting valve, brass :							
15mm	...	No	92.00	94.00	94.00	96.00	106.00
20mm	...	No	106.00	108.00	108.00	110.00	120.00
25mm	...	No	176.00	178.00	178.00	180.00	190.00
32mm	...	No	520.00	520.00	520.00	520.00	530.00
40mm	...	No	890.00	890.00	890.00	890.00	900.00
50mm	...	No	1020.00	1035.00	1035.00	1020.00	1035.00

SANITARY PLUMBING

VALVES AND CONTROLS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
HWU TPR/PTR VALVES							
TPR/PTR temperature and pressure relief valve, brass :							
15mm	…	No	104.00	106.00	106.00	108.00	118.00
20mm	…	No	210.00	215.00	215.00	215.00	225.00
HWU ECV VALVES							
ECV expansion control valve, brass :							
15mm	…	No	78.00	80.00	80.00	82.00	90.00
20mm	…	No	126.00	128.00	128.00	128.00	138.00
TEMPERATURE LIMITING DEVICES							
Refer to Thermostatic Mixing Valves, page 471	…	Note					
Hot water tempering valve, brass :							
15mm	...	No	148.00	152.00	152.00	154.00	168.00
20mm	...	No	176.00	180.00	180.00	182.00	196.00
BACKFLOW PROTECTION							
Dual mini check valve (low hazard) :							
Chromium plated brass -							
15mm	...	No	132.00	140.00	140.00	144.00	174.00
Brass -							
20mm	...	No	76.00	78.00	78.00	78.00	84.00
25mm	...	No	96.00	96.00	96.00	98.00	104.00
40mm	...	No	430.00	430.00	430.00	430.00	440.00
Double check valve assembly (medium hazard) :							
20mm	...	No	370.00	370.00	370.00	370.00	375.00
25mm	...	No	420.00	420.00	420.00	420.00	420.00
40mm	...	No	880.00	880.00	880.00	880.00	890.00
Reduced pressure zone device and backflow preventer, with isolation valves (high hazard) :							
15mm	…	No	450.00	460.00	460.00	460.00	490.00
20mm	…	No	460.00	470.00	470.00	470.00	500.00
25mm	…	No	570.00	580.00	580.00	580.00	610.00
32mm	…	No	860.00	870.00	870.00	880.00	910.00
40mm	…	No	1035.00	1050.00	1050.00	1050.00	1095.00
50mm	…	No	1125.00	1125.00	1125.00	1125.00	1155.00
FIXTURE CONNECTORS							
Flexible fixture connector, rubber and braided stainless steel :							
8mm hose x 450 long	...	No	16.50	17.50	17.50	18.00	21.50
8mm hose x 600 long	...	No	18.50	19.50	19.50	20.00	23.50
10mm hose x 1000 long	...	No	33.00	34.50	34.50	35.00	39.50
20mm hose x 450 long	...	No	62.00	63.00	63.00	64.00	68.00

TRAPS

Plastic P or S trap and connector:							
40mm	...	No	29.50	30.50	31.00	31.50	35.50
50mm	...	No	32.50	33.50	33.50	34.00	38.50
ADD extra for :							
50mm double bowl connector	...	No	32.50	33.50	33.50	34.00	38.50
Copper P trap with c.p. finish							
40mm	...	No	118.00	120.00	120.00	120.00	126.00
50mm	...	No	170.00	172.00	172.00	172.00	180.00

SANITARY PLUMBING

TRAPS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Copper union S trap with c.p. finish							
40mm	...	No	130.00	130.00	132.00	132.00	138.00
50mm	...	No	178.00	180.00	180.00	180.00	188.00
Copper union S trap with drop waste and c.p. finish						\	
40mm	...	No	195.00	200.00	200.00	200.00	210.00
50mm	...	No	245.00	245.00	245.00	245.00	255.00
ADD extra for :							
50mm c.p. double bowl connector	...	No	158.00	158.00	160.00	160.00	166.00

UNTRAPPED FLOOR WASTES AND GRATING

FLOOR WASTES, TUNDISHES

UPVC untrapped floor waste and grating :							
80mm dia. grate x 50/80mm dia.	...	No	32.00	34.00	34.00	35.00	41.00
100mm dia. grate x 100mm dia.	...	No	29.50	31.00	31.00	32.00	38.00
ADD extra for :							
UPVC gully "P" trap - 80 x 50/65/80mm dia.	...	No	44.00	46.00	46.00	47.00	55.00
UPVC gully "P" trap - 100 x 65mm dia.	...	No	52.00	55.00	55.00	56.00	66.00
UPVC 80mm dia. four way riser junction (2x40/2x50mm dia.)	...	No	66.00	68.00	68.00	69.00	77.00
UPVC 100mm riser junction	...	No	40.00	42.00	42.00	43.00	51.00
Chrome plated brass untrapped floor waste and grating :							
50mm dia. grate x 50mm dia.	...	No	27.00	28.50	29.00	29.50	35.50
80mm dia. grate x 80mm dia.	...	No	43.00	45.00	45.00	46.00	53.00
90mm x 90mm grate x 80mm dia.	...	No	44.00	46.00	46.00	47.00	54.00
110 x 110mm grate x 100mm dia.	...	No	56.00	58.00	59.00	60.00	70.00
Stainless steel floor waste :							
100mm dia. x 50mm dia.	...	No	144.00	148.00	148.00	150.00	164.00
100mm dia. x 80mm dia.	...	No	152.00	156.00	156.00	158.00	172.00
150mm x 150mm x 100mm dia.	...	No	230.00	235.00	235.00	235.00	255.00
150mm dia. x 100mm dia.	...	No	215.00	220.00	220.00	220.00	235.00
Commercial chrome plated floor waste and grating :							
100mm dia. grate x 80mm dia. with c.p. finish	...	No	205.00	210.00	210.00	210.00	225.00
150mm dia. grate x 100mm dia. with c.p. finish	...	No	295.00	300.00	300.00	300.00	320.00
ADD extra for :							
Cast iron vertical assembly -							
80mm dia.	...	No	195.00	195.00	200.00	200.00	210.00
100mm dia.	...	No	138.00	142.00	142.00	144.00	160.00
150mm dia.	...	No	180.00	185.00	185.00	190.00	210.00
Cast iron bend assembly -							
50mm dia.	...	No	350.00	350.00	355.00	355.00	365.00
80mm dia.	...	No	380.00	380.00	380.00	380.00	400.00
100mm dia.	...	No	250.00	255.00	255.00	255.00	275.00
ADD extra for :							
Traps - Refer to Waste pipework, pages 476, 479, 480	...	Note					
Riser extension	...	Note					
Safe tray / leak control flange	...	Note					

SANITARY PLUMBING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
UNTRAPPED FLOOR WASTES AND GRATING (cont'd)							
FLOOR WASTES, TUNDISHES (cont'd)							
Tundish:							
Plastic 80 dia. tapered with - 40 dia. outlet	...	No	26.00	27.50	27.50	28.50	34.50
Stainless steel :							
100 dia. to 38 dia. with 300mm long tail	...	No	295.00	300.00	300.00	300.00	315.00
190mm wide x 130mm high x 68mm deep in-wall with window and 50 dia. threaded outlet	...	No	315.00	325.00	325.00	325.00	350.00
150 dia. to 95 dia. conical 95 dia. outlet	...	No	310.00	310.00	310.00	315.00	325.00
Copper :							
100 dia. tapered with 20 dia. outlet	...	No	132.00	136.00	136.00	136.00	150.00
100 dia. tapered with 50 dia. outlet	...	No	142.00	144.00	144.00	146.00	158.00
100 dia. tapered flat back with 40 dia. outlet	...	No	142.00	146.00	146.00	148.00	160.00
ADD extra for :							
Chrome plating	...	No	112.00	112.00	112.00	112.00	114.00
SAFETY UNITS AND SHOWERS							
Prices include for installation, fixings and jointing to pipes but do not include pipework, traps or alarms	...	Note					
Laboratory spray, aerated eye wash unit, hand held, bench mounted (SE 927) (P.C. $ 430.00)	...	No	550.00	550.00	560.00	560.00	580.00
Eye wash, aerated, free standing, hand operated (SE 545) (P.C. $ 760.00)	...	No	1020.00	1020.00	1035.00	1035.00	1080.00
Eye and face wash, aerated, free standing, hand operated with spray (SE 420) (P.C. $ 950.00)	...	No	1215.00	1230.00	1230.00	1230.00	1290.00
Deluge shower, ceiling mounted (SE 220) (P.C. $ 230.00)	...	No	390.00	400.00	400.00	400.00	440.00
Decontamination station, free standing, platform operated (SE 663) (P.C. $3785.00)	...	No	4300.00	4400.00	4400.00	4400.00	4400.00
HEAT LOSS TRACE SYSTEM							
Refer Mechanical Services page 635	...	Note					

SANITARY PLUMBING

PIPEWORK

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for pipework include for all jointing and fixings when applicable	...	Note					
ADD extra for chasing, refer page 209	...	Note					
Prices for pipework of less than 32mm dia. include for all fittings such as reducers, bends, tees, junctions and inspection openings; prices for pipework of 32mm dia. and over are given as excluding all fittings	...	Note					

SOIL WASTE AND VENT

LOW NOISE EPOXY LINED CAST IRON

Ensign low noise cast iron epoxy lined plain end (double spigot) pipe with s/s rapid W2 couplings, fixed to wall or soffit :							
50mm dia.	...	m	88.00	92.00	90.00	94.00	104.00
70mm dia.	...	m	110.00	114.00	114.00	118.00	134.00
100mm dia.	...	m	138.00	142.00	142.00	196.00	166.00
Ensign low noise cast iron epoxy lined plain end (double spigot) pipe with s/s rapid W2 couplings, fixed to wall or soffit (cont'd) :							
150mm dia.	...	m	200.00	210.00	210.00	215.00	240.00
200mm dia.	...	m	300.00	310.00	305.00	320.00	350.00
250mm dia.	...	m	430.00	440.00	430.00	450.00	490.00
300mm dia.	...	m	500.00	520.00	510.00	530.00	570.00
Short piece, with access door :							
100mm dia.	...	No	150.00	152.00	148.00	154.00	156.00
150mm dia.	...	No	305.00	305.00	300.00	310.00	310.00
Bend, plain (45°) :							
50mm dia.	...	No	57.00	58.00	58.00	59.00	64.00
70mm dia.	...	No	69.00	71.00	70.00	72.00	78.00
100mm dia.	...	No	78.00	78.00	78.00	80.00	88.00
150mm dia.	...	No	128.00	130.00	130.00	134.00	142.00
200mm dia.	...	No	245.00	245.00	245.00	250.00	265.00
250mm dia	...	No	560.00	560.00	560.00	570.00	590.00
300mm dia.	...	No	890.00	890.00	880.00	910.00	910.00
Bend, plain (87°) :							
50mm dia.	...	No	57.00	58.00	58.00	59.00	64.00
70mm dia.	...	No	69.00	71.00	70.00	72.00	78.00
100mm dia.	...	No	78.00	80.00	80.00	82.00	88.00
150mm dia.	...	No	140.00	142.00	142.00	146.00	154.00
200mm dia.	...	No	255.00	255.00	255.00	260.00	275.00
250mm dia	...	No	860.00	860.00	840.00	870.00	870.00
300mm dia.	...	No	1170.00	1170.00	1140.00	1185.00	1185.00
Bend, with access door (87°) :							
100mm dia.	...	No	118.00	118.00	118.00	122.00	126.00
150mm dia.	...	No	166.00	168.00	166.00	172.00	178.00
Junction, plain (87°) :							
70mm dia.	...	No	82.00	84.00	84.00	86.00	90.00
100mm dia.	...	No	114.00	116.00	116.00	120.00	126.00
150mm dia.	...	No	220.00	225.00	220.00	230.00	235.00
200mm dia.	...	No	690.00	690.00	680.00	710.00	710.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

LOW NOISE EPOXY LINED CAST IRON (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Junction, with access door (87°) :							
100mm dia.	...	No	160.00	162.00	158.00	164.00	170.00
Reducer :							
100 x 50mm	...	No	94.00	96.00	94.00	98.00	104.00
150 x 100mm	...	No	122.00	126.00	124.00	128.00	138.00
200 x 100mm	...	No	225.00	230.00	225.00	235.00	250.00
250 x 150mm	...	No	510.00	510.00	510.00	520.00	540.00
300 x 200mm	...	No	620.00	620.00	620.00	630.00	650.00
Traps :							
100mm 'P' trap plain	...	No	120.00	122.00	120.00	126.00	132.00
100mm 'P' trap, with access door	...	No	162.00	164.00	160.00	168.00	172.00
150mm 'P' trap, with access door	...	No	275.00	280.00	275.00	285.00	290.00

COPPER

Copper pipe to A.S. 1432-1996, Type B, with soldered joints, fixed to wall or soffit :							
32mm dia. x 1.22mm	...	m	57.00	60.00	60.00	58.00	70.00
40mm dia. x 1.22mm	...	m	64.00	68.00	67.00	65.00	78.00
50mm dia. x 1.22mm	...	m	80.00	84.00	84.00	80.00	96.00
65mm dia. x 1.22mm	...	m	96.00	100.00	100.00	96.00	114.00
80mm dia. x 1.63mm	...	m	136.00	144.00	142.00	134.00	160.00
100mm dia. x 1.63mm	...	m	180.00	190.00	185.00	175.00	210.00
150mm dia. x 2.03mm	...	m	300.00	315.00	305.00	285.00	335.00
Bend, plain :							
32mm x 45°, 85°	...	No	20.00	21.50	21.00	21.50	25.50
40mm x 45°	...	No	23.50	25.00	25.00	25.00	30.00
x 85°	...	No	29.00	30.50	30.00	30.00	35.00
50mm x 45°	...	No	33.00	35.00	35.00	35.00	42.00
x 85°	...	No	34.00	36.00	36.00	36.00	43.00
65mm x 45°	...	No	42.00	44.00	44.00	44.00	52.00
x 85°	...	No	45.00	48.00	47.00	47.00	55.00
80mm x 45°	...	No	55.00	59.00	58.00	57.00	67.00
x 85°	...	No	61.00	64.00	63.00	62.00	73.00
100mm x 45°	...	No	74.00	78.00	78.00	76.00	90.00
x 85°	...	No	88.00	94.00	92.00	90.00	104.00
150mm x 45°	...	No	195.00	205.00	200.00	195.00	220.00
x 85°	...	No	176.00	186.00	182.00	176.00	198.00
Bend, I.O. :							
32mm x 45°, 85°	...	No	33.00	34.50	34.00	33.00	38.00
40mm x 45°	...	No	46.00	48.00	47.00	46.00	52.00
x 85°	...	No	46.00	48.00	47.00	46.00	52.00
50mm x 45°	...	No	52.00	54.00	53.00	52.00	60.00
x 85°	...	No	57.00	60.00	59.00	57.00	66.00
65mm x 45°	...	No	59.00	62.00	61.00	60.00	69.00
x 85°	...	No	67.00	70.00	69.00	67.00	77.00
80mm x 45°	...	No	60.00	63.00	62.00	62.00	72.00
x 85°	...	No	76.00	80.00	78.00	76.00	88.00
100mm x 45°	...	No	100.00	106.00	102.00	100.00	114.00
x 85°	...	No	124.00	132.00	128.00	124.00	140.00
Junction, single oblique :							
32mm	...	No	62.00	66.00	66.00	66.00	80.00
40mm	...	No	78.00	82.00	82.00	84.00	102.00
50mm	...	No	90.00	96.00	96.00	98.00	120.00
65mm	...	No	130.00	138.00	136.00	136.00	164.00
80mm	...	No	142.00	150.00	150.00	150.00	180.00
100mm	...	No	185.00	195.00	195.00	195.00	235.00
150mm	...	No	380.00	400.00	390.00	390.00	450.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

COPPER (cont'd)

Copper pipe to A.S. 1432-1996, Type B, with soldered joints, fixed to wall or soffit (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Junction, double oblique :							
32mm	...	No	88.00	94.00	94.00	94.00	112.00
40mm	...	No	104.00	112.00	110.00	112.00	134.00
50mm	...	No	118.00	126.00	126.00	126.00	152.00
65mm	...	No	158.00	168.00	166.00	166.00	198.00
80mm	...	No	195.00	210.00	205.00	205.00	240.00
100mm	...	No	270.00	285.00	285.00	280.00	330.00
150mm	...	No	500.00	530.00	520.00	510.00	710.00
Reducer :							
40 x 32mm	...	No	21.50	23.00	23.00	23.50	28.50
50 x 40mm	...	No	28.00	30.00	30.00	30.50	36.50
65 x 50mm	...	No	35.00	37.00	37.00	38.00	45.00
80 x 65mm	...	No	44.00	47.00	47.00	47.00	56.00
100 x 80mm	...	No	56.00	60.00	60.00	60.00	72.00
150 x 100mm	...	No	94.00	100.00	98.00	98.00	116.00
Expansion joint :							
40mm	...	No	78.00	82.00	80.00	76.00	86.00
50mm	...	No	86.00	90.00	88.00	84.00	94.00
65mm	...	No	100.00	106.00	102.00	98.00	110.00
80mm	...	No	112.00	118.00	114.00	110.00	124.00
100mm	...	No	158.00	166.00	160.00	154.00	172.00
150mm	...	No	340.00	355.00	345.00	325.00	360.00
Pan adaptor, with rubber and clip :							
100 x 150mm	...	No	74.00	78.00	78.00	78.00	92.00
100 x 225mm	...	No	82.00	86.00	86.00	86.00	98.00
100 x 300mm	...	No	92.00	98.00	96.00	96.00	110.00
100 x 350mm	...	No	92.00	98.00	96.00	96.00	110.00
100 x 450mm	...	No	102.00	106.00	104.00	104.00	118.00
100 x 600mm	...	No	114.00	120.00	118.00	116.00	132.00
Vent cowl :							
50mm	...	No	56.00	59.00	57.00	54.00	60.00
70mm	...	No	56.00	59.00	57.00	54.00	60.00
100mm	...	No	56.00	59.00	57.00	54.00	60.00

BRASS

Brass pipe 70/30 to A.S.1572-1998 (Type B) with soldered joints, fixed to wall or soffit :

50mm dia. x 1.2mm	...	m	225.00	240.00	230.00	260.00	240.00
65mm dia. x 1.2mm	...	m	280.00	295.00	285.00	320.00	300.00
Bend, plain :							
65mm x 45°	...	No	46.00	48.00	48.00	51.00	56.00
x 85°	...	No	58.00	61.00	60.00	65.00	68.00
Bend, I.O. :							
65mm x 45°	...	No	46.00	48.00	48.00	51.00	56.00
x 85°	...	No	58.00	61.00	60.00	65.00	68.00
Junction, plain :							
50mm	...	No	63.00	66.00	65.00	71.00	73.00
65mm	...	No	62.00	65.00	64.00	69.00	74.00
Reducer :							
65 x 50mm	...	No	51.00	54.00	68.00	61.00	62.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

STAINLESS STEEL

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Push to join, 316 stainless steel, pipe (EuroPipe) to AS/NZS 3500, with socketed EPDM ring sealed joints, fixed to wall or soffit :							
50mm dia.	...	m	82.00	86.00	84.00	86.00	96.00
75mm dia.	...	m	120.00	126.00	126.00	130.00	142.00
110mm dia.	...	m	140.00	148.00	148.00	150.00	168.00
160mm dia.	...	m	235.00	245.00	245.00	250.00	275.00
Bend, plain :							
50mm x 45°	...	No	40.00	43.00	43.00	44.00	49.00
x 90°	...	No	49.00	51.00	51.00	52.00	58.00
75mm x 45°	...	No	57.00	60.00	60.00	62.00	70.00
x 90°	...	No	57.00	60.00	60.00	62.00	70.00
110mm x 45°	...	No	80.00	86.00	84.00	88.00	98.00
x 90°	...	No	94.00	100.00	98.00	102.00	112.00
160mm x 45°	...	No	140.00	148.00	146.00	150.00	162.00
x 90°	...	No	172.00	182.00	180.00	184.00	196.00
Bend, sweep :							
50mm x 90°	...	No	63.00	66.00	68.00	67.00	73.00
75mm x 90°	...	No	86.00	90.00	92.00	92.00	100.00
110mm x 90°	...	No	146.00	154.00	158.00	156.00	166.00
160mm x 90°	...	No	270.00	280.00	285.00	285.00	295.00
Junction, single oblique :							
50mm x 45°	...	No	68.00	71.00	71.00	72.00	77.00
75mm x 45°	...	No	92.00	96.00	94.00	96.00	104.00
110mm x 45°	...	No	128.00	134.00	134.00	136.00	146.00
160mm x 45°	...	No	250.00	265.00	260.00	265.00	275.00
Junction, double oblique :							
50mm x 45°	...	No	152.00	160.00	156.00	160.00	164.00
75mm x 45°	...	No	230.00	240.00	235.00	240.00	245.00
110mm x 45°	...	No	310.00	325.00	320.00	325.00	330.00
160mm x 45°	...	No	450.00	470.00	460.00	470.00	480.00
Pipe, access -							
110mm	...	No	124.00	132.00	130.00	132.00	144.00
160mm	...	No	285.00	300.00	295.00	300.00	310.00
Eccentric increaser -							
50 x 75mm	...	No	78.00	82.00	82.00	82.00	88.00
50 x 110mm	...	No	106.00	112.00	110.00	112.00	120.00
75 x 110mm	...	No	116.00	122.00	122.00	124.00	136.00
110 x 160mm	...	No	190.00	200.00	195.00	200.00	210.00
Expansion socket -							
110mm	...	No	86.00	90.00	90.00	92.00	104.00
160mm	...	No	128.00	134.00	132.00	136.00	150.00
Traps :							
50mm 'P' trap plain	...	No	108.00	114.00	112.00	116.00	122.00
75mm 'P' trap plain	...	No	140.00	146.00	144.00	148.00	154.00
110mm 'P' trap plain	...	No	200.00	210.00	210.00	210.00	220.00
160mm 'P' trap plain	...	No	510.00	540.00	530.00	540.00	540.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

HIGH DENSITY POLYETHYLENE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
H.D.P.E. pipes to ISO 770/8772, with electrofusion joints, fixed to wall or soffit :							
50mm dia.	...	m	37.00	38.00	39.00	39.00	49.00
75mm dia.	...	m	49.00	50.00	52.00	52.00	64.00
90mm dia.	...	m	60.00	62.00	64.00	64.00	80.00
110mm dia.	...	m	104.00	110.00	112.00	112.00	136.00
160mm dia.	...	m	155.00	160.00	170.00	170.00	205.00
250mm dia.	...	m	280.00	280.00	295.00	285.00	340.00
Bend, 90° -							
50mm dia.	...	No	56.00	59.00	60.00	61.00	76.00
75mm dia.	...	No	66.00	68.00	70.00	70.00	88.00
90mm dia.	...	No	78.00	80.00	84.00	84.00	104.00
110mm dia.	...	No	88.00	92.00	94.00	96.00	118.00
160mm dia.	...	No	160.00	160.00	170.00	165.00	205.00
250mm dia.	...	No	590.00	560.00	610.00	570.00	680.00
Reducer -							
75 x 50mm	...	No	76.00	78.00	82.00	82.00	102.00
90 x 50mm	...	No	76.00	80.00	82.00	84.00	106.00
90 x 75mm	...	No	78.00	82.00	84.00	86.00	106.00
110 x 75mm	...	No	86.00	92.00	94.00	96.00	120.00
110 x 90mm	...	No	90.00	96.00	98.00	100.00	126.00
160 x 110mm	...	No	128.00	134.00	138.00	140.00	176.00
Reducing junction -							
75 x 50mm	...	No	74.00	78.00	80.00	80.00	100.00
90 x 75mm	...	No	88.00	90.00	94.00	94.00	116.00
110 x 50mm	...	No	96.00	100.00	104.00	104.00	128.00
110 x 75mm	...	No	98.00	102.00	106.00	106.00	130.00
110 x 90mm	...	No	118.00	122.00	126.00	128.00	158.00
160 x 110mm	...	No	185.00	185.00	195.00	190.00	230.00
250 x 160mm	...	No	450.00	420.00	470.00	430.00	510.00
Trap :							
110mm dia. adjustable disconnector trap	...	No	100.00	95.00	100.00	95.00	115.00

POLYPROPYLENE

Polypropylene (PP-MD) SN16 pipe, with clamped push-to-join socketed SBR ring-sealed joints, fixed to wall or soffit :							
110mm dia.	...	m	112.00	116.00	108.00	108.00	138.00
160mm dia.	...	m	165.00	170.00	165.00	165.00	205.00
200mm dia.	...	m	170.00	180.00	235.00	230.00	220.00
250mm dia.	...	m	280.00	290.00	280.00	275.00	340.00
Bend, 45° :							
110mm dia.	...	m	82.00	82.00	88.00	82.00	94.00
160mm dia.	...	m	154.00	156.00	168.00	158.00	178.00
200mm dia.	...	m	200.00	200.00	215.00	205.00	230.00
250mm dia.	...	m	260.00	260.00	280.00	265.00	305.00
Eccentric increaser :							
110 x 160mm	...	m	124.00	126.00	136.00	128.00	146.00
160 x 200mm	...	m	170.00	172.00	186.00	172.00	196.00
200 x 250mm	...	m	210.00	215.00	230.00	215.00	245.00
Reducing junction :							
160 x 110mm	...	m	230.00	235.00	250.00	235.00	270.00
250 x 160mm	...	m	340.00	350.00	370.00	350.00	410.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

POLYPROPYLENE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Polypropylene (PP-MD) SN16 pipe, with clamped push-to-join socketed SBR ring-sealed joints, fixed to wall or soffit (cont'd) :							
Pipe access :							
110mm dia.	...	m	160.00	162.00	176.00	162.00	182.00
160mm dia.	...	m	245.00	250.00	270.00	250.00	280.00
200mm dia.	...	m	280.00	285.00	310.00	285.00	320.00
Polypropylene pipe with Vulcathene mechanical system joints, fixed to wall or soffit :							
38mm dia.	...	m	45.00	47.00	48.00	49.00	56.00
51mm dia.	...	m	61.00	64.00	64.00	66.00	75.00
76mm dia.	...	m	100.00	104.00	104.00	108.00	118.00
102mm dia.	...	m	124.00	128.00	130.00	134.00	146.00
Bend, plain -							
38mm x 45°, 85°	...	No	27.00	28.00	28.00	29.00	30.50
51mm x 45°, 85°	...	No	36.00	37.00	37.00	39.00	40.00
76mm x 45°, 85°	...	No	92.00	94.00	94.00	98.00	98.00
102mm x 45°, 85°	...	No	110.00	112.00	112.00	116.00	116.00
Junction, single 'Y', 45°, 85° -							
38mm	...	No	35.00	35.50	36.00	37.50	39.00
51mm	...	No	47.00	47.00	48.00	50.00	51.00
76mm	...	No	100.00	102.00	102.00	106.00	106.00
102mm	...	No	134.00	136.00	136.00	142.00	142.00
Expansion unit -							
38mm	...	No	90.00	92.00	92.00	96.00	94.00
51mm	...	No	142.00	142.00	142.00	148.00	146.00
76mm	...	No	245.00	245.00	245.00	255.00	250.00
102mm	...	No	295.00	295.00	295.00	310.00	300.00

PVC

PVC pipes (SH) to A.S./N.Z.S. 1260-2002 with solvent welded joints, fixed to wall or soffit :							
40mm dia.	...	m	32.00	34.00	34.00	35.00	43.00
50mm dia.	...	m	39.00	41.00	42.00	43.00	53.00
65mm dia.	...	m	47.00	50.00	50.00	52.00	63.00
80mm dia.	...	m	54.00	57.00	57.00	59.00	72.00
100mm dia.	...	m	62.00	66.00	66.00	68.00	84.00
150mm dia.	...	m	96.00	100.00	102.00	104.00	126.00
Bend, plain -							
40mm x 45°, 85°	...	No	16.50	17.50	18.00	18.50	23.00
50mm x 45°	...	No	20.50	22.00	22.00	23.00	28.50
x 85°	...	No	20.50	22.00	22.00	23.00	28.50
65mm x 45°	...	No	27.50	29.50	29.50	30.50	37.00
x 85°	...	No	28.00	29.50	30.00	30.50	37.50
80mm x 45°	...	No	38.00	40.00	40.00	41.00	49.00
x 85°	...	No	38.00	40.00	40.00	41.00	49.00
100mm x 45°	...	No	39.00	42.00	42.00	43.00	54.00
x 88°	...	No	40.00	43.00	43.00	44.00	55.00
150mm x 45°	...	No	68.00	72.00	72.00	74.00	86.00
x 88°	...	No	68.00	72.00	72.00	74.00	86.00

SANITARY PLUMBING

PIPEWORK (cont'd)

SOIL WASTE AND VENT (cont'd)

PVC (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Bend, I.O. -							
40mm x 45°	...	No	22.00	23.00	23.50	24.00	28.00
x 85°	...	No	18.50	19.50	20.00	20.50	24.50
50mm x 45°	...	No	26.00	27.00	27.50	28.00	33.50
x 85°	...	No	22.50	24.00	24.00	25.00	30.50
65mm x 45°	...	No	33.00	34.00	35.00	35.00	42.00
x 85°	...	No	32.00	34.00	34.00	35.00	41.00
80mm x 45°	...	No	54.00	55.00	56.00	57.00	65.00
x 85°	...	No	56.00	57.00	58.00	59.00	67.00
100mm x 45°	...	No	68.00	72.00	72.00	74.00	84.00
x 85°	...	No	51.00	54.00	54.00	56.00	66.00
Junction, plain -							
40 x 40mm x 45°, 85°	...	No	22.50	23.50	23.50	24.50	29.00
50 x 50mm x 45°	...	No	25.50	27.00	27.50	28.00	34.00
x 85°	...	No	26.50	28.00	28.00	29.00	34.50
65 x 65mm x 45°	...	No	33.00	35.00	35.00	36.00	43.00
x 85°	...	No	36.00	37.00	38.00	39.00	46.00
80 x 80mm x 45°, 85°	...	No	52.00	54.00	55.00	56.00	64.00
100 x 100mm x 45°	...	No	44.00	47.00	47.00	49.00	60.00
x 85°	...	No	45.00	48.00	48.00	50.00	60.00
150 x 150mm x 45°	...	No	82.00	86.00	86.00	88.00	100.00
x 85°	...	No	98.00	100.00	102.00	102.00	116.00
Junction, I.O. -							
40 x 40mm x 45°	...	No	80.00	82.00	82.00	82.00	88.00
50 x 50mm x 45°	...	No	36.00	37.00	38.00	38.00	44.00
65 x 65mm x 45°	...	No	72.00	72.00	74.00	74.00	82.00
80 x 80mm x 45°	...	No	148.00	150.00	150.00	150.00	160.00
100 x 100mm x 45°	...	No	68.00	72.00	72.00	74.00	84.00
Level invert taper -							
65 x 50mm	...	No	28.50	30.50	30.50	31.50	39.00
80 x 65mm	...	No	34.00	36.00	36.00	38.00	46.00
100 x 50mm	...	No	42.00	44.00	45.00	46.00	57.00
100 x 65mm	...	No	42.00	45.00	45.00	47.00	57.00
100 x 80mm	...	No	50.00	53.00	54.00	55.00	66.00
Socket reducer -							
50 x 40mm	...	No	24.00	25.50	25.50	26.50	32.50
65 x 50mm	...	No	28.50	30.50	31.00	31.50	39.00
80 x 50mm	...	No	39.00	41.00	41.00	42.00	51.00
80 x 65mm	...	No	34.00	36.00	36.00	38.00	46.00
100 x 50mm	...	No	41.00	44.00	45.00	46.00	57.00
100 x 80mm	...	No	50.00	52.00	53.00	54.00	65.00
Adaptor, PVC to iron -							
40 x 40mm	...	No	15.00	16.00	16.50	17.00	20.50
50 x 50mm	...	No	21.00	22.00	22.50	23.00	28.00
Vent cowl, plain -							
50mm	...	No	15.50	16.50	17.00	17.00	21.00
80mm	...	No	35.00	36.00	36.00	37.00	42.00
100mm (with wire)	...	No	47.00	48.00	48.00	49.00	55.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER

COPPER

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling when applicable, refer pages 489 and 490 ...	Note					
Copper pipe - Type A with soldered/press fitted joints :						
Laid in trench -						
15mm dia. x 1.02mm (including fittings)...	m	21.00	22.00	22.00	21.50	26.50
20mm dia. x 1.42mm (including fittings)...	m	32.50	34.00	33.50	31.50	37.50
25mm dia. x 1.63mm (including fittings)...	m	48.00	51.00	50.00	46.00	54.00
32mm dia. x 1.63mm ...	m	56.00	59.00	57.00	52.00	61.00
40mm dia. x 1.63mm ...	m	66.00	69.00	67.00	61.00	72.00
50mm dia. x 1.63mm ...	m	90.00	94.00	90.00	82.00	96.00
Fixed to wall or soffit -						
15mm dia. x 1.02mm (including fittings)...	m	43.00	46.00	46.00	47.00	58.00
20mm dia. x 1.42mm (including fittings)...	m	55.00	38.00	38.00	39.00	67.00
25mm dia. x 1.63mm (including fittings)...	m	74.00	78.00	76.00	72.00	86.00
32mm dia. x 1.63mm ...	m	74.00	78.00	76.00	72.00	84.00
40mm dia. x 1.63mm ...	m	88.00	92.00	90.00	84.00	98.00
50mm dia. x 1.63mm ...	m	112.00	118.00	116.00	108.00	126.00
Copper pipe - Type B with soldered/press fitted joints :						
Laid in trench -						
15mm dia. x 0.91mm (including fittings)...	m	20.00	21.50	21.00	21.00	25.50
20mm dia. x 1.02mm (including fittings)...	m	23.50	25.00	24.50	23.50	28.50
25mm dia. x 1.22mm (including fittings)...	m	33.00	35.00	34.00	32.50	39.00
32mm dia. x 1.22mm ...	m	36.00	38.00	37.00	35.00	41.00
40mm dia. x 1.22mm ...	m	43.00	45.00	44.00	42.00	49.00
50mm dia. x 1.22mm ...	m	55.00	58.00	57.00	53.00	63.00
65mm dia. x 1.22mm ...	m	68.00	72.00	70.00	65.00	77.00
80mm dia. x 1.63mm ...	m	98.00	104.00	100.00	92.00	108.00
100mm dia. x 1.63mm ...	m	122.00	128.00	124.00	114.00	134.00
150mm dia. x 2.03mm ...	m	235.00	245.00	235.00	215.00	250.00
Fixed to wall or soffit -						
15mm dia. x 0.91mm (including fittings)...	m	42.00	45.00	45.00	46.00	57.00
20mm dia. x 1.02mm (including fittings)...	m	45.00	48.00	48.00	47.00	57.00
25mm dia. x 1.22mm (including fittings)...	m	57.00	60.00	59.00	58.00	70.00
32mm dia. x 1.22mm ...	m	54.00	57.00	56.00	55.00	65.00
40mm dia. x 1.22mm ...	m	64.00	67.00	66.00	64.00	76.00
50mm dia. x 1.22mm ...	m	78.00	82.00	82.00	78.00	92.00
65mm dia. x 1.22mm ...	m	96.00	100.00	98.00	94.00	112.00
80mm dia. x 1.63mm ...	m	132.00	140.00	136.00	128.00	150.00
100mm dia. x 1.63mm ...	m	164.00	172.00	168.00	158.00	186.00
150mm dia. x 2.03mm ...	m	280.00	295.00	285.00	265.00	310.00
Bend - soldered -						
32mm ...	No	20.50	22.00	22.00	23.00	26.00
40mm ...	No	27.50	29.00	29.00	29.00	34.00
50mm ...	No	33.00	35.00	35.00	35.00	42.00
65mm ...	No	42.00	44.00	44.00	44.00	52.00
80mm ...	No	57.00	60.00	59.00	59.00	69.00
100mm ...	No	80.00	84.00	84.00	82.00	96.00
150mm ...	No	230.00	240.00	235.00	225.00	250.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

COPPER (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Copper pipe - Type B with soldered/press fitted joints joints (cont'd) :							
Tee - soldered -							
32mm	...	No	25.00	26.00	26.00	26.00	31.00
40mm	...	No	29.50	31.00	31.00	31.00	36.50
50mm	...	No	45.00	47.00	47.00	47.00	54.00
65mm	...	No	49.00	52.00	51.00	51.00	60.00
80mm	...	No	63.00	67.00	66.00	65.00	76.00
100mm	...	No	90.00	94.00	94.00	92.00	106.00
150mm	...	No	240.00	250.00	245.00	235.00	265.00
Reducer - soldered -							
40 x 32mm	...	No	21.00	22.50	22.50	23.00	28.00
50 x 40mm	...	No	27.50	29.50	29.50	30.00	36.50
65 x 50mm	...	No	34.00	36.00	36.00	37.00	44.00
80 x 65mm	...	No	43.00	45.00	45.00	46.00	55.00
100 x 80mm	...	No	54.00	57.00	57.00	58.00	69.00
150 x 100mm	...	No	86.00	90.00	90.00	90.00	108.00
Bend - press fitted -							
32mm	...	No	20.50	26.50	26.00	26.50	29.50
40mm	...	No	27.00	34.00	33.50	32.50	37.50
50mm	...	No	39.00	49.00	48.00	47.00	54.00
65mm	...	No	90.00	104.00	100.00	96.00	108.00
80mm	...	No	120.00	138.00	132.00	126.00	140.00
100mm	...	No	210.00	235.00	225.00	215.00	235.00
Tee - press fitted -							
32mm	...	No	29.00	36.00	35.00	34.00	39.00
40mm	...	No	43.00	52.00	50.00	48.00	55.00
50mm	...	No	57.00	69.00	67.00	64.00	73.00
65mm	...	No	112.00	130.00	124.00	118.00	132.00
80mm	...	No	132.00	152.00	146.00	138.00	154.00
100mm	...	No	235.00	260.00	250.00	235.00	260.00
Reducer - press fitted -							
40 x 32mm	...	No	20.50	28.00	27.50	27.50	32.00
50 x 40mm	...	No	30.00	40.00	39.00	38.00	45.00
65 x 50mm	...	No	66.00	78.00	76.00	74.00	84.00
80 x 65mm	...	No	70.00	84.00	82.00	78.00	90.00
100 x 80mm	...	No	136.00	156.00	152.00	144.00	162.00
Prelagged/pre-insulated Type B copper pipe with soldered/press fitted joints fixed to wall or soffit :							
15mm dia. x 0.91mm (including fittings)	...	m	44.00	47.00	46.00	46.00	55.00
20mm dia. x 1.02mm (including fittings)	...	m	57.00	60.00	59.00	57.00	68.00
25mm dia. x 1.22mm (including fittings)	...	m	84.00	88.00	86.00	82.00	96.00
32mm dia. x 1.22mm	...	m	112.00	118.00	114.00	106.00	124.00
40mm dia. x 1.22mm	...	m	142.00	150.00	146.00	134.00	158.00
50mm dia. x 1.22mm	...	m	215.00	230.00	220.00	200.00	235.00
Chromium plated copper pipe, fixed to wall or soffit :							
15mm dia. x 0.91mm (including fittings)	...	m	65.00	68.00	67.00	65.00	76.00
20mm dia. x 1.02mm (including fittings)	...	m	88.00	94.00	92.00	86.00	102.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

ABS THERMOPLASTIC

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
ABS thermoplastic pressure pipe to AS/NZS 3518 "DURAFLO" with solvent joints, laid in trench :							
100mm dia. PN 12 (12 bar)	...	m	118.00	120.00	116.00	122.00	130.00
150mm dia. PN 12 (12 bar)	...	m	235.00	240.00	235.00	240.00	255.00
200mm dia. PN 9 (9 bar)	...	m	290.00	295.00	290.00	300.00	320.00
250mm dia. PN 9 (9 bar)	...	m	355.00	365.00	360.00	370.00	395.00
300mm dia. PN 9 (9 bar)	...	m	550.00	560.00	560.00	570.00	590.00
400mm dia. PN 9 (9 bar)	...	m	950.00	960.00	950.00	960.00	990.00
Elbow, 45° -							
100mm dia.	...	No	138.00	142.00	128.00	146.00	150.00
150mm dia.	...	No	290.00	295.00	280.00	300.00	305.00
200mm dia.	...	No	620.00	630.00	610.00	630.00	640.00
Elbow, 90° -							
100mm dia.	...	No	128.00	132.00	120.00	138.00	142.00
150mm dia.	...	No	410.00	420.00	400.00	420.00	430.00
200mm dia.	...	No	670.00	680.00	660.00	680.00	690.00
Tee -							
100mm dia.	...	No	148.00	152.00	138.00	158.00	162.00
150mm dia.	...	No	440.00	450.00	430.00	460.00	460.00
200mm dia.	...	No	750.00	760.00	740.00	780.00	780.00

GALVANISED STEEL

Galvanised steel pipe with screwed joints, fixed to wall or soffit :							
15mm dia. (including fittings)	...	m	38.00	41.00	40.00	42.00	48.00
20mm dia. (including fittings)	...	m	43.00	46.00	46.00	47.00	54.00
25mm dia. (including fittings)	...	m	47.00	50.00	49.00	51.00	58.00
Galvanised steel pipe with roll grooved joints, fixed to wall or soffit :							
40mm dia.	...	m	55.00	58.00	57.00	59.00	67.00
50mm dia.	...	m	68.00	72.00	72.00	74.00	84.00
65mm dia.	...	m	90.00	96.00	94.00	98.00	110.00
80mm dia.	...	m	106.00	112.00	110.00	114.00	126.00
100mm dia.	...	m	124.00	132.00	130.00	134.00	148.00
150mm dia.	...	m	180.00	195.00	190.00	195.00	220.00
Elbow -							
40mm dia.	...	No	45.00	47.00	46.00	48.00	52.00
50mm dia.	...	No	48.00	51.00	49.00	51.00	55.00
65mm dia.	...	No	60.00	63.00	62.00	64.00	69.00
80mm dia.	...	No	74.00	78.00	76.00	80.00	86.00
100mm dia.	...	No	104.00	110.00	108.00	112.00	118.00
150mm dia.	...	No	156.00	166.00	160.00	168.00	176.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

GALVANISED STEEL (cont'd)

Galvanised steel pipe with roll grooved joints, fixed to wall or soffit (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Tee, plain -							
40mm dia.	...	No	61.00	64.00	63.00	65.00	69.00
50mm dia.	...	No	70.00	73.00	71.00	74.00	78.00
65mm dia.	...	No	86.00	92.00	88.00	92.00	98.00
80mm dia.	...	No	66.00	70.00	70.00	72.00	138.00
100mm dia.	...	No	172.00	180.00	174.00	182.00	188.00
150mm dia.	...	No	190.00	200.00	195.00	205.00	215.00
Tee, reducer :							
65 x 50 mm dia.	...	No	100.00	104.00	102.00	106.00	110.00
80 x 50 mm dia.	...	No	68.00	72.00	70.00	74.00	138.00
80 x 65 mm dia.	...	No	68.00	72.00	70.00	74.00	138.00
100 x 25 mm dia.	...	No	195.00	205.00	200.00	210.00	210.00
100 x 50 mm dia.	...	No	170.00	178.00	172.00	180.00	186.00
100 x 65 mm dia.	...	No	170.00	178.00	172.00	180.00	186.00
100 x 80 mm dia.	...	No	158.00	166.00	160.00	168.00	174.00
Tee, plain :							
150 x 50 mm dia.	...	No	255.00	270.00	260.00	270.00	280.00
150 x 80 mm dia.	...	No	255.00	270.00	260.00	270.00	280.00
150 x 100 mm dia.	...	No	220.00	235.00	225.00	235.00	245.00
Reducer :							
50 x 40 mm dia.	...	No	48.00	51.00	49.00	51.00	55.00
65 x 50 mm dia.	...	No	60.00	63.00	62.00	64.00	69.00
80 x 50 mm dia.	...	No	72.00	76.00	74.00	78.00	84.00
80 x 65 mm dia.	...	No	74.00	78.00	76.00	80.00	86.00
100 x 65 mm dia.	...	No	90.00	96.00	94.00	98.00	104.00
100 x 80 mm dia.	...	No	90.00	96.00	94.00	98.00	104.00
150 x 80 mm dia.	...	No	144.00	152.00	148.00	154.00	164.00
150 x 100 mm dia.	...	No	144.00	152.00	148.00	154.00	164.00
Flange adaptor :							
50mm dia.	...	No	60.00	63.00	61.00	64.00	66.00
65mm dia.	...	No	74.00	78.00	76.00	78.00	82.00
80mm dia.	...	No	86.00	92.00	88.00	92.00	98.00
100mm dia.	...	No	106.00	112.00	108.00	112.00	118.00
150mm dia.	...	No	164.00	174.00	168.00	176.00	184.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

GLASS FILAMENT REINFORCED THERMOSETTING PLASTIC (GRP)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
Glass reinforced plastic (GRP) pressure pipe to AS 3571 with rubber ring coupling joints (Class PN16 (16 bar) SN10000), laid in trench :							
300mm dia.	...	m	355.00	360.00	360.00	370.00	375.00
375mm dia.	...	m	430.00	440.00	440.00	450.00	450.00
450mm dia.	...	m	510.00	520.00	520.00	530.00	540.00
525mm dia.	...	m	590.00	600.00	600.00	610.00	620.00
600mm dia.	...	m	700.00	710.00	710.00	720.00	740.00
675mm dia.	...	m	790.00	800.00	800.00	810.00	820.00
750mm dia.	...	m	1095.00	1110.00	1110.00	1125.00	1140.00
Bend, 45° :							
300mm	...	No	1440.00	1455.00	1455.00	1455.00	1470.00
375mm	...	No	2000.00	2000.00	2000.00	2000.00	2000.00
450mm	...	No	2500.00	2500.00	2500.00	2500.00	2500.00
525mm	...	No	3400.00	3450.00	3450.00	3450.00	3450.00
600mm	...	No	3900.00	3900.00	3900.00	3900.00	3900.00
675mm	...	No	4800.00	4800.00	4800.00	4800.00	4900.00
750mm	...	No	6000.00	6000.00	6000.00	6000.00	6200.00
Bend, 90° :							
300mm	...	No	2100.00	2100.00	2100.00	2100.00	2100.00
375mm	...	No	2850.00	2900.00	2900.00	2900.00	2900.00
450mm	...	No	3700.00	3700.00	3700.00	3700.00	3700.00
525mm	...	No	5000.00	5000.00	5000.00	5000.00	5000.00
600mm	...	No	5800.00	5800.00	5800.00	5800.00	5800.00
675mm	...	No	7200.00	7200.00	7200.00	7200.00	7200.00
750mm	...	No	9200.00	9200.00	9200.00	9200.00	9200.00
Air Valve Tee, 100mm :							
300mm	...	No	1525.00	1550.00	1550.00	1550.00	1575.00
375mm	...	No	2100.00	2100.00	2100.00	2100.00	2100.00
450mm	...	No	2550.00	2600.00	2600.00	2600.00	2650.00
525mm	...	No	3050.00	3050.00	3050.00	3050.00	3100.00
600mm	...	No	3400.00	3400.00	3400.00	3500.00	3500.00
675mm	...	No	4100.00	4200.00	4200.00	4200.00	4200.00
750mm	...	No	5600.00	5600.00	5600.00	5600.00	5600.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

DUCTILE IRON, CEMENT LINED

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
Ductile iron cement lined pipe (PN35) to A.S./NZ 2280 with rubber ring joints, laid in trench :							
100mm dia.	...	m	106.00	114.00	108.00	120.00	116.00
150mm dia.	...	m	170.00	178.00	172.00	186.00	182.00
200mm dia.	...	m	230.00	240.00	230.00	250.00	250.00
225mm dia.	...	m	260.00	275.00	265.00	285.00	280.00
250mm dia.	...	m	295.00	310.00	300.00	320.00	320.00
300mm dia.	...	m	370.00	390.00	380.00	400.00	400.00
375mm dia.	...	m	540.00	560.00	550.00	570.00	580.00
450mm dia.	...	m	730.00	760.00	750.00	780.00	810.00
Supply costs of fittings vary considerably depending upon quantity. The prices given hereafter are based on quantities of 10 to 15	...	Note					
Bend, 90° :							
100mm	...	No	275.00	275.00	275.00	280.00	260.00
150mm	...	No	440.00	450.00	440.00	460.00	430.00
200mm	...	No	1005.00	1020.00	1005.00	1035.00	990.00
225mm	...	No	1230.00	1245.00	1230.00	1260.00	1215.00
250mm	...	No	1305.00	1320.00	1290.00	1335.00	1275.00
300mm	...	No	1925.00	1950.00	1900.00	1975.00	1875.00
375mm	...	No	3500.00	3500.00	3500.00	3600.00	3400.00
450mm	...	No	4800.00	5000.00	4800.00	5000.00	4800.00
Tee :							
100mm	...	No	355.00	360.00	355.00	365.00	340.00
150mm	...	No	560.00	570.00	560.00	580.00	540.00
200mm	...	No	1185.00	1200.00	1185.00	1215.00	1155.00
225mm	...	No	1410.00	1425.00	1395.00	1440.00	1365.00
250mm	...	No	1750.00	1775.00	1725.00	1800.00	1700.00
300mm	...	No	2350.00	2400.00	2300.00	2400.00	2300.00
375mm	...	No	3050.00	3100.00	3000.00	3150.00	2950.00
450mm	...	No	3900.00	4000.00	3800.00	4000.00	3800.00

POLYETHYLENE

ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
Polyethylene pipe to A.S./N.Z.S. 4130/ 4131-2003 with mechanical compression joints :							
Laid in trench -							
PN16/SDR9 (Class 15) :							
20mm dia. (including fittings)	...	m	12.50	13.25	13.25	13.75	16.50
25mm dia. (including fittings)	...	m	14.50	15.25	15.50	16.00	19.25
32mm dia.	...	m	13.25	14.00	14.00	14.50	17.25

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

POLYETHYLENE (cont'd)

Polyethylene pipe to A.S./N.Z.S. 4130/
4131-2003 with mechanical compression joints (cont'd) :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Laid in trench (cont'd) -							
PN12.5/SDR11 (Class 12) :							
40mm dia.	...	m	15.50	16.25	16.25	16.75	19.75
50mm dia.	...	m	19.50	20.00	20.50	21.00	24.50
63mm dia.	...	m	27.00	28.00	28.00	28.50	33.00
75mm dia.	...	m	28.00	29.50	29.50	30.50	35.50
90mm dia.	...	m	41.00	43.00	43.00	44.00	50.00
110mm dia.	...	m	57.00	59.00	60.00	60.00	68.00
Fixed to wall or soffit -							
PN16/SDR9 (Class 15) :							
20mm dia. (including fittings)	...	m	29.50	31.50	31.50	32.50	39.50
25mm dia. (including fittings)	...	m	35.00	37.00	37.00	38.00	46.00
32mm dia.	...	m	30.00	32.00	32.00	33.00	40.00
PN12.5/SDR11 (Class 12) :							
40mm dia.	...	m	35.00	37.00	38.00	39.00	47.00
50mm dia.	...	m	43.00	45.00	46.00	47.00	56.00
63mm dia.	...	m	48.00	57.00	58.00	59.00	70.00
75mm dia.	...	m	47.00	61.00	62.00	64.00	76.00
90mm dia.	...	m	58.00	84.00	84.00	86.00	102.00
110mm dia.	...	m	70.00	104.00	104.00	108.00	126.00
Elbow -							
32mm	...	No	24.00	24.50	25.00	25.00	27.50
40mm	...	No	35.50	36.50	36.50	37.00	39.50
50mm	...	No	45.00	46.00	46.00	46.00	50.00
63mm	...	No	56.00	57.00	57.00	58.00	62.00
75mm	...	No	82.00	84.00	84.00	84.00	88.00
90mm	...	No	114.00	116.00	116.00	116.00	122.00
110mm	...	No	172.00	174.00	174.00	174.00	180.00
Tee -							
32mm	...	No	35.50	36.00	36.00	36.50	39.50
40mm	...	No	48.00	49.00	49.00	49.00	52.00
50mm	...	No	63.00	64.00	64.00	64.00	68.00
63mm	...	No	78.00	78.00	78.00	80.00	84.00
75mm	...	No	114.00	116.00	116.00	116.00	122.00
90mm	...	No	162.00	164.00	164.00	164.00	172.00
110mm	...	No	280.00	280.00	280.00	280.00	290.00
Reducer -							
40 x 32mm	...	No	27.00	27.50	28.00	28.00	31.00
50 x 40mm	...	No	36.00	37.00	37.00	37.00	41.00
63 x 50mm	...	No	41.00	41.00	42.00	42.00	46.00
75 x 63mm	...	No	78.00	80.00	80.00	80.00	84.00
90 x 75mm	...	No	110.00	112.00	112.00	112.00	118.00
110 x 90mm	...	No	172.00	174.00	174.00	174.00	180.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

PE-XA HIGH DENSITY POLYETHYLENE

Cross-linked high density polyethylene pipe, Class PN-20, for hot and cold water applications with polymer fittings and compressed sleeved joints :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fixed to wall or soffit -							
16mm dia. x 2.22mm (including fittings)	...	m	30.00	32.00	32.00	33.00	41.00
20mm dia. x 2.8mm (including fittings)	...	m	34.00	36.00	37.00	38.00	46.00
25mm dia. x 3.5mm	...	m	33.00	34.00	35.00	36.00	42.00
32mm dia. x 4.4mm	...	m	41.00	43.00	43.00	44.00	51.00
40mm dia. x 5.5mm	...	m	51.00	53.00	53.00	54.00	63.00
50mm dia. x 6.9mm	...	m	62.00	65.00	65.00	66.00	76.00
63mm dia. x 8.6mm	...	m	72.00	74.00	74.00	76.00	86.00
Pre-fitted conduit -							
16mm dia. x 2.2mm (including fittings)	...	m	38.00	40.00	40.00	41.00	49.00
20mm dia. x 2.8mm (including fittings)	...	m	44.00	46.00	47.00	48.00	57.00
Straight MI threaded connector -							
25mm	...	No	23.50	24.00	24.00	24.50	27.00
32mm	...	No	32.50	33.00	33.50	33.50	37.50
40mm	...	No	55.00	56.00	56.00	56.00	61.00
50mm	...	No	76.00	78.00	78.00	78.00	84.00
63mm	...	No	104.00	106.00	106.00	106.00	112.00
Elbow -							
25mm	...	No	23.50	24.00	24.00	24.50	27.00
32mm	...	No	44.00	45.00	45.00	46.00	49.00
40mm	...	No	73.00	74.00	74.00	75.00	79.00
50mm	...	No	126.00	128.00	128.00	128.00	134.00
63mm	...	No	188.00	190.00	190.00	190.00	196.00
Tee -							
25mm	...	No	29.00	30.00	30.00	30.50	34.00
32mm	...	No	52.00	53.00	53.00	54.00	58.00
40mm	...	No	112.00	114.00	114.00	114.00	120.00
50mm	...	No	174.00	176.00	176.00	176.00	182.00
63mm	...	No	230.00	235.00	235.00	235.00	240.00

PE-XA HIGH DENSITY POLYETHYLENE FOR RECYCLED WATER OR RAIN WATER

Cross-linked high density polyethylene pipe, Class PN-20, for recycled water (lilac) or rain water (green) applications with polymer fittings and compressed sleeved joints :

16 dia x 2.2mm (including fittings)	...	m	33.00	35.00	35.00	36.00	43.00
20 dia x 2.8mm (including fittings)	...	m	38.00	40.00	40.00	41.00	50.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

PVC

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
PVC pipe to A.S./N.Z.S. 1477-1999 with solvent joints :							
Laid in trench -							
PN/Class 18 :							
15mm dia. (including fittings)	...	m	17.50	19.00	19.00	19.50	24.00
20mm dia. (including fittings)	...	m	18.50	19.50	19.50	20.00	24.50
25mm dia. (including fittings)	...	m	22.00	23.50	23.50	24.50	29.50
PN/Class 9 :							
32mm dia.	...	m	19.00	20.00	20.50	21.00	25.50
40mm dia.	...	m	20.00	21.00	21.50	22.00	27.00
50mm dia.	...	m	24.00	25.50	26.00	26.50	32.00
65mm dia.	...	m	32.00	33.00	33.00	34.00	41.00
PN/Class 12 :							
80mm dia.	...	m	36.00	38.00	38.00	39.00	46.00
100mm dia.	...	m	49.00	51.00	51.00	52.00	60.00
150mm dia.	...	m	80.00	84.00	84.00	86.00	98.00
200mm dia.	...	m	122.00	126.00	128.00	130.00	146.00
225mm dia	...	m	156.00	160.00	162.00	164.00	182.00
Fixed to wall or soffit -							
PN/Class 18 :							
15mm dia. (including fittings)	...	m	28.00	29.50	30.00	31.00	38.00
20mm dia. (including fittings)	...	m	29.50	31.00	31.50	32.50	39.50
25mm dia. (including fittings)	...	m	35.00	37.00	37.00	38.00	46.00
PN/Class 9 :							
32mm dia.	...	m	30.00	32.00	33.00	34.00	41.00
40mm dia.	...	m	34.00	36.00	36.00	38.00	46.00
50mm dia.	...	m	41.00	44.00	44.00	45.00	55.00
65mm dia.	...	m	52.00	55.00	55.00	57.00	68.00
PN/Class 12 :							
80mm dia.	...	m	58.00	62.00	62.00	64.00	77.00
100mm dia.	...	m	82.00	86.00	86.00	88.00	104.00
150mm dia.	...	m	136.00	142.00	144.00	146.00	174.00
200mm dia.	...	m	175.00	185.00	185.00	190.00	220.00
225mm dia.	...	m	215.00	220.00	220.00	225.00	255.00
Elbow -							
32mm	...	No	14.00	14.75	14.75	15.25	18.00
40mm	...	No	15.00	15.75	16.00	16.50	19.75
50mm	...	No	19.50	20.50	20.50	21.00	25.00
65mm	...	No	38.00	39.00	39.00	40.00	45.00
80mm	...	No	41.00	43.00	43.00	44.00	50.00
100mm	...	No	68.00	70.00	72.00	72.00	80.00
150mm	...	No	130.00	134.00	134.00	134.00	144.00
Tee -							
32mm	...	No	15.50	16.00	16.50	17.00	20.00
40mm	...	No	17.00	17.50	18.00	18.50	22.50
50mm	...	No	22.00	23.00	23.50	24.00	28.50
65mm	...	No	55.00	57.00	57.00	57.00	63.00
80mm	...	No	46.00	48.00	48.00	49.00	55.00
100mm	...	No	78.00	80.00	82.00	82.00	90.00
150mm	...	No	215.00	215.00	215.00	220.00	230.00

SANITARY PLUMBING

PIPEWORK (cont'd)

WATER (cont'd)

PVC (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PVC pipe to A.S./N.Z.S. 1477 (Series 2) with rubber ring joints and compatible with cast iron outside diameters :							
Laid in trench -							
PN/Class 12 :							
100mm dia.	...	m	76.00	80.00	82.00	84.00	100.00
150mm dia.	...	m	114.00	120.00	120.00	122.00	142.00
200mm dia.	...	m	170.00	180.00	180.00	185.00	210.00
PN/Class 20 :							
100mm dia.	...	m	86.00	90.00	90.00	92.00	108.00
150mm dia.	...	m	140.00	146.00	146.00	148.00	168.00
200mm dia.	...	m	215.00	220.00	220.00	225.00	255.00
Bend socketed (ductile iron, blue) -							
100mm x 45°	...	No	185.00	190.00	190.00	190.00	200.00
150mm x 45°	...	No	265.00	270.00	270.00	270.00	285.00
100mm x 90°	...	No	205.00	210.00	210.00	210.00	220.00
150mm x 90°	...	No	335.00	335.00	335.00	340.00	350.00
Tee socketed (ductile iron, blue) -							
100mm	...	No	220.00	225.00	225.00	225.00	240.00
150mm	...	No	440.00	440.00	450.00	450.00	460.00
Flange adaptor socketed (ductile iron, blue) -							
100mm	...	No	190.00	190.00	190.00	190.00	200.00
150mm	...	No	255.00	255.00	255.00	255.00	270.00

GAS

POLYETHYLENE

ADD for trench excavation and backfilling when applicable, refer pages 489 and 490	...	Note					
Polyethylene pipe to A.S./N.Z.S. 4130-1997 Series 2 with electrofusion joints, laid in trench :							
20mm dia.	...	m	15.50	16.50	16.50	17.00	20.50
25mm dia.	...	m	24.00	25.00	25.50	26.00	30.00
40mm dia.	...	m	22.50	23.50	24.00	24.50	29.50
63mm dia.	...	m	32.00	33.00	33.00	34.00	41.00
110mm dia.	...	m	49.00	51.00	51.00	52.00	61.00
Elbow -							
25mm	...	No	34.00	35.00	35.00	36.00	42.00
40mm	...	No	42.00	43.00	44.00	44.00	50.00
63mm	...	No	52.00	54.00	54.00	55.00	62.00
110mm	...	No	104.00	108.00	108.00	110.00	126.00
Tee -							
25mm	...	No	48.00	50.00	50.00	51.00	58.00
40mm	...	No	64.00	66.00	67.00	67.00	75.00
63mm	...	No	64.00	66.00	66.00	67.00	76.00
110mm	...	No	170.00	180.00	180.00	180.00	205.00

SANITARY PLUMBING

PIPEWORK (cont'd)

GAS (cont'd)

PE-XA HIGH DENSITY POLYETHYLENE

Cross-linked high density polyethylene pipe, for gas applications with brass fittings and compressed sleeved joints :

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fixed to wall or soffit -							
16mm dia. x 2.6mm (including fittings)	...	m	32.00	34.00	35.00	36.00	43.00
20mm dia. x 2.9mm (including fittings)	...	m	37.00	40.00	40.00	41.00	49.00
25mm dia. x 3.7mm	...	m	33.00	34.00	35.00	35.00	42.00
32mm dia. x 4.7mm	...	m	40.00	42.00	42.00	43.00	51.00
Protective conduit, flexible zinc plated -							
to suit 16/20mm dia. pipe	...	m	18.00	23.00	23.50	24.00	28.00
to suit 25mm dia. pipe	...	m	21.50	27.50	28.00	28.50	33.00
to suit 32mm dia. pipe	...	m	22.50	29.00	29.50	30.00	34.50
Straight MI threaded connector -							
25mm	...	No	23.00	23.50	24.00	24.00	27.00
32mm	...	No	38.00	39.00	39.00	39.00	43.00
Elbow -							
25mm	...	No	30.50	31.50	31.50	31.50	34.50
32mm	...	No	55.00	56.00	56.00	57.00	60.00
Tee -							
25mm	...	No	42.00	43.00	43.00	43.00	47.00
32mm	...	No	76.00	76.00	76.00	76.00	82.00

PIPEWORK, MECHANICAL

Refer to Mechanical Services Pipework pages 624 to 633 for other piping systems	...	Note					

PIPE INSULATION

20mm thick tube insulation to pipe :							
15mm dia.	...	m	11.25	11.50	12.00	12.00	14.50
20mm dia.	...	m	12.75	13.50	13.50	13.75	16.00
25mm dia.	...	m	17.00	17.50	18.50	18.00	21.50
32mm dia.	...	m	18.50	19.00	19.50	19.50	23.50
40mm dia.	...	m	23.00	23.50	24.00	24.50	28.00
50mm dia.	...	m	36.00	37.00	38.00	38.00	45.00
25mm thick 4 zero pipe insulation to pipe :							
25mm dia.	...	m	28.00	28.50	30.00	29.00	33.50
32mm dia.	...	m	34.00	34.00	36.00	35.00	41.00
40mm dia.	...	m	40.00	41.00	42.00	42.00	46.00
50mm dia.	...	m	45.00	46.00	48.00	48.00	56.00
All purpose petrolatum tape Denso 600 wrapped with 55% overlap and layer of overwrapping to pipe :							
15mm dia.	...	m	34.00	35.00	36.00	36.00	43.00
20mm dia.	...	m	40.00	41.00	43.00	43.00	50.00
25mm dia.	...	m	47.00	48.00	50.00	50.00	59.00
32mm dia.	...	m	58.00	59.00	62.00	61.00	72.00
40mm dia.	...	m	68.00	68.00	72.00	70.00	84.00
50mm dia.	...	m	82.00	82.00	86.00	86.00	100.00
65mm dia.	...	m	100.00	100.00	106.00	104.00	120.00
80mm dia.	...	m	118.00	118.00	126.00	122.00	142.00
100mm dia.	...	m	158.00	160.00	168.00	166.00	192.00
Refer to page 634 for Acoustic Insulation and Mechanical Services Piping Insulation	...	Note					

SANITARY PLUMBING

WATER TANKS

DOMESTIC/SMALL COMMERCIAL

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Polyethylene lined water tank, fabricated from ribbed zincalume or colorbond sheet steel, with domed roof and zinc anode; capacity :							
NOMINAL 2.18m HIGH -							
27000 litres, 4.0m dia.	...	Sum	5450.00	5825.00	5450.00	4625.00	5450.00
48000 litres, 5.4m dia.	...	Sum	6450.00	6850.00	6450.00	5600.00	6450.00
75000 litres, 6.7m dia.	...	Sum	8450.00	8875.00	8450.00	7200.00	8450.00
NOMINAL 2.18m HIGH -							
91000 litres, 7.4m dia.	...	Sum	9400.00	9400.00	9400.00	8200.00	9400.00
127000 litres, 8.7m dia.	...	Sum	11700.00	12600.00	11700.00	10500.00	11700.00
148000 litres, 9.3m dia.	...	Sum	13400.00	13600.00	13400.00	11400.00	13400.00
198000 litres, 10.7m dia.	...	Sum	16300.00	17100.00	16300.00	14600.00	16300.00
244000 litres, 12.0m dia.	...	Sum	19500.00	19700.00	19500.00	16900.00	19500.00
ADD extra for :							
High wind rated conical roof	...	+12%					
External access ladder, valves, pumps, pipework and overflows, electrical installation and level controls	...	Note					
Foundation base (stable compacted sand)	...	Note					

FIRE PROTECTION

Fire Protection water tanks for fire hydrants, fire sprinklers and fire hose reels to AS 2304/AS 2419 are project specific - size, standard and price varies between Domestic/Small Commercial tanks and Heavy Commercial tanks	...	Note					

HEAVY COMMERCIAL

0.60mm thick PVC lined tank, fabricated from ribbed zincalume or colorbond sheet steel, with ladder and roof; capacity :							
200000 litres, nominal 2.18m high	...	Sum	63300.00	63300.00	63300.00	63300.00	63300.00
300000 litres, nominal 3.23m high	...	Sum	76500.00	76500.00	76500.00	76500.00	76500.00
500000 litres, nominal 4.37m high	...	Sum	110900.00	110900.00	110900.00	110900.00	110900.00
1000000 litres, nominal 5.31m high	...	Sum	179300.00	179300.00	179300.00	179300.00	179300.00
ADD extra for :							
Cyclone rating	...	+10%					
Electronic cathodic protection system	...	Note					
Valves, controls, pipework and overflows	...	Note					
Fall and safety equipment system	...	Note					
Reinforced concrete slab or ring beam and compacted sand base	...	Note					

HEAVY COMMERCIAL - SEVERE ENVIRONMENTS
(for severe coastal and industrial environments)

0.60mm thick PVC lined tank, fabricated from flat bolted modular colorbond ultra coated steel, with ladder and roof; capacity :							
200000 litres, nominal 2.18m high	...	Sum	67300.00	67300.00	67300.00	67300.00	67300.00
300000 litres, nominal 3.23m high	...	Sum	82300.00	82300.00	82300.00	82300.00	82300.00
500000 litres, nominal 4.37m high	...	Sum	118300.00	118300.00	118300.00	118300.00	118300.00
1000000 litres, nominal 5.31m high	...	Sum	193300.00	193300.00	193300.00	193300.00	193300.00
ADD extra for :							
Reinforced concrete slab, valves, hydraulic and electrical services, overflows, pumps and fittings	...	Note					
Fuel tanks, refer page 702	...	Note					

DRAINAGE

TRENCH EXCAVATION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices for trench and pits include for :							
Maintaining sides of excavation, ne 1.0m deep	...	Note					
Backfilling with excavated material	...	Note					
ADD extra for :							
Disposal of surplus excavated material	...	Note					
Reinstating disturbed surfaces	...	Note					
TRENCHES (x 100 to 150mm wide)							
Excavate trench 100 to 150mm wide by machine (trench digger) :							
500mm total depth in -							
Sand	...	m	4.70	2.35	3.10	1.85	3.25
Light soil	...	m	4.75	2.75	4.00	2.25	3.25
Clay	...	m	5.25	4.25	4.25	3.00	4.00
1000mm total depth in -							
Sand	...	m	9.50	4.75	6.25	3.75	6.50
Light soil	...	m	9.50	5.50	8.00	4.00	6.50
Clay	...	m	10.75	8.75	8.50	5.75	7.75
Excavate trench 100 to 150mm wide by hand :							
500mm total depth in -							
Sand	...	m	8.25	5.50	10.75	4.25	11.75
Light soil	...	m	8.25	6.00	10.75	5.25	11.75
Clay	...	m	9.75	8.00	11.75	7.00	13.75
TRENCHES (x 300mm wide)							
Excavate trench 300mm wide by machine (backhoe, small bucket) :							
500mm total depth in -							
Sand	...	m	11.25	5.75	7.50	4.75	8.75
Light soil	...	m	11.25	7.00	9.75	5.00	8.75
Clay	...	m	12.75	10.25	10.25	7.25	10.75
1000mm total depth in -							
Sand	...	m	22.50	11.50	15.00	9.50	17.50
Light soil	...	m	22.50	14.00	19.50	10.00	17.50
Clay	...	m	25.50	20.50	20.50	14.50	21.50
Excavate trench 300mm wide by hand :							
500mm total depth in -							
Sand	...	m	19.75	13.00	25.50	10.00	28.00
Light soil	...	m	19.50	14.50	25.50	12.50	28.00
Clay	...	m	23.50	19.00	28.50	17.00	32.50
1000mm total depth in -							
Sand	...	m	39.50	26.00	51.00	20.00	56.00
Light soil	...	m	39.00	29.00	51.00	25.00	56.00
Clay	...	m	47.00	38.00	57.00	34.00	65.00

DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
TRENCH EXCAVATION (cont'd)							
TRENCHES (exceeding 300mm wide)							
Excavate trenches or pits by machine :							
Not exceeding 1000mm total depth in -							
Sand	...	cum	75.00	37.00	49.00	30.00	52.00
Light soil	...	cum	75.00	45.00	64.00	32.00	52.00
Clay	...	cum	86.00	68.00	68.00	46.00	62.00
Soft rock	...	cum	96.00	90.00	102.00	94.00	222.00
Hard rock	...	cum	210.00	140.00	325.00	210.00	335.00
1000/2000mm total depth in -							
Sand	...	cum	75.00	39.00	51.00	31.00	59.00
Light soil	...	cum	75.00	46.00	65.00	34.00	59.00
Clay	...	cum	86.00	68.00	68.00	48.00	72.00
Soft rock	...	cum	86.00	90.00	108.00	98.00	224.00
Hard rock	...	cum	210.00	140.00	330.00	220.00	335.00
2000/3000mm total depth in -							
Sand	...	cum	75.00	39.00	51.00	31.00	59.00
Light soil	...	cum	76.00	46.00	66.00	34.00	60.00
Clay	...	cum	86.00	68.00	68.00	48.00	72.00
Soft rock	...	cum	86.00	90.00	110.00	98.00	224.00
Hard rock	...	cum	210.00	140.00	330.00	220.00	335.00
ADD extra for :							
Double cutting of trenches where required	...	Note					
Excavate trenches or pits by hand :							
Not exceeding 1000mm total depth in -							
Sand	...	cum	132.00	86.00	170.00	68.00	188.00
Light soil	...	cum	132.00	98.00	170.00	84.00	188.00
Clay	...	cum	156.00	128.00	190.00	112.00	218.00
BACKFILLING							
Backfilling with excavated material is included in the foregoing excavation prices	...	Note					
Backfilling trenches with :							
Clean sand	...	cum	69.00	66.00	52.00	20.00	66.00
20mm crushed rock	...	cum	108.00	86.00	86.00	68.00	76.00
Flowable (self levelling) material	...	cum	124.00-166.00	124.00-166.00	124.00-166.00	124.00-166.00	124.00-166.00
SEWER DRAINS							
Prices for pipework include for laying and jointing in trench but exclude excavation and backfilling	...	Note					
CAST IRON							
ADD for trench excavation and backfilling	...	Note					
Cast iron pipe with mechanical joints :							
100mm dia	...	m	124.00	126.00	128.00	132.00	148.00
150mm dia	...	m	174.25	175.00	176.00	183.25	200.25
Bend :							
100mm, plain	...	No	152.00	152.00	152.00	158.00	172.00
150mm, plain	...	No	205.00	205.00	205.00	215.00	230.00
100mm, I.O.	...	No	195.00	190.00	195.00	200.00	215.00

DRAINAGE

SEWER DRAINS (cont'd)

CAST IRON (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Access door / short pipe :							
100mm dia	...	No	260.00	255.00	255.00	270.00	275.00
150mm dia	...	No	440.00	430.00	430.00	450.00	460.00
Junction, plain :							
100 x 100mm - 45°	...	No	240.00	245.00	245.00	255.00	280.00
150 x 100mm - 45°	...	No	285.00	285.00	285.00	300.00	330.00
100 x 100mm - 85°	...	No	235.00	235.00	235.00	245.00	270.00
150 x 100mm - 85°	...	No	275.00	280.00	280.00	290.00	320.00
Joint to 100 / 100mm copper or pvc pipe	...	No	80.00	82.00	82.00	86.00	94.00

CONCRETE

ADD for trench excavation and backfilling	...	Note					
Concrete pipe to AS 4058-1992 with rubber ring joints :							
Class 2 -							
300mm dia.	...	m	140.00	164.00	138.00	132.00	172.00
375mm dia.	...	m	185.00	210.00	165.00	160.00	205.00
450mm dia.	...	m	245.00	280.00	215.00	215.00	260.00
525mm dia.	...	m	290.00	335.00	245.00	250.00	295.00
600mm dia.	...	m	340.00	400.00	290.00	290.00	340.00
750mm dia.	...	m	490.00	570.00	410.00	410.00	480.00
900mm dia.	...	m	700.00	870.00	560.00	550.00	630.00
1200mm dia.	...	m	1125.00	1410.00	870.00	885.00	1005.00
1500mm dia.	...	m	1550.00	1975.00	1225.00	1525.00	1375.00
Class 3 -							
300mm dia.	...	m	144.00	182.00	146.00	144.00	176.00
375mm dia.	...	m	160.00	235.00	180.00	180.00	210.00
450mm dia.	...	m	220.00	320.00	230.00	235.00	265.00
525mm dia.	...	m	280.00	385.00	270.00	275.00	305.00
600mm dia.	...	m	340.00	460.00	330.00	330.00	350.00
750mm dia.	...	m	390.00	700.00	460.00	450.00	500.00
900mm dia.	...	m	600.00	1080.00	600.00	615.00	660.00
1200mm dia.	...	m	900.00	1750.00	975.00	1050.00	1100.00
1500mm dia.	...	m	-	2450.00	1300.00	1850.00	1500.00
Class 4 -							
300mm dia.	...	m	165.00	205.00	160.00	155.00	185.00
375mm dia.	...	m	225.00	265.00	190.00	180.00	220.00
450mm dia.	...	m	305.00	370.00	240.00	260.00	275.00
525mm dia.	...	m	360.00	450.00	290.00	310.00	320.00
600mm dia.	...	m	-	540.00	350.00	360.00	370.00
750mm dia.	...	m	-	780.00	480.00	510.00	530.00
900mm dia.	...	m	-	1215.00	645.00	690.00	705.00
1200mm dia.	...	m	-	1975.00	1075.00	1075.00	1175.00
1500mm dia.	...	m	-	2750.00	1450.00	1950.00	1650.00
Bends, junctions, etc. :							
Special fittings can be manufactured to order, alternatively manholes can be used at change of direction	...	Note					

DRAINAGE

SEWER DRAINS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GLASS FILAMENT REINFORCED THERMOSETTING PLASTIC (GRP)							
ADD for trench excavation and backfilling	...	Note					
Glass reinforced plastic (GRP) pipe to AS 3571 with rubber ring coupling joints (Class PN1 non pressure SN10000) :							
300mm dia.	...	m	290.00	295.00	295.00	305.00	315.00
375mm dia.	...	m	360.00	365.00	370.00	380.00	390.00
450mm dia.	...	m	430.00	430.00	440.00	450.00	460.00
525mm dia.	...	m	490.00	500.00	500.00	510.00	530.00
600mm dia.	...	m	590.00	600.00	600.00	610.00	640.00
675mm dia.	...	m	670.00	680.00	680.00	690.00	720.00
750mm dia.	...	m	920.00	930.00	930.00	940.00	970.00
Bend, 45° -							
300mm dia.	...	No	1170.00	1185.00	1185.00	1200.00	1200.00
375mm dia.	...	No	1525.00	1525.00	1525.00	1525.00	1550.00
450mm dia.	...	No	1725.00	1750.00	1750.00	1750.00	1775.00
525mm dia.	...	No	2200.00	2200.00	2200.00	2200.00	2250.00
600mm dia.	...	No	2600.00	2600.00	2600.00	2600.00	2650.00
675mm dia.	...	No	3050.00	3150.00	3050.00	3050.00	3100.00
750mm dia.	...	No	3700.00	3700.00	3700.00	3700.00	3700.00
Bend, 90° -							
300mm dia.	...	No	1700.00	1700.00	1700.00	1725.00	1725.00
375mm dia.	...	No	2200.00	2200.00	2200.00	2200.00	2200.00
450mm dia.	...	No	2550.00	2550.00	2550.00	2550.00	2550.00
525mm dia.	...	No	3200.00	3200.00	3200.00	3200.00	3250.00
600mm dia.	...	No	3800.00	3800.00	3800.00	3800.00	3900.00
675mm dia.	...	No	4500.00	4500.00	4500.00	4500.00	4600.00
750mm dia.	...	No	5400.00	5400.00	5400.00	5400.00	5600.00
POLYPROPYLENE							
ADD for trench excavation and backfilling	...	Note					
Extruded polypropylene non pressure corrugated pipe with smooth internal wall and rubber ring joints ("SewerPRO", "Sewer Max", SN10, ASI 5065) :							
225mm dia.	...	m	114.00	112.00	116.00	124.00	128.00
300mm dia.	...	m	166.00	166.00	170.00	178.00	188.00
375mm dia.	...	m	220.00	220.00	225.00	235.00	245.00
450mm dia.	...	m	285.00	285.00	290.00	295.00	315.00
525mm dia.	...	m	350.00	350.00	360.00	365.00	390.00
600mm dia.	...	m	440.00	440.00	450.00	450.00	480.00
Bend, 45°, GRP -							
225mm dia.	...	No	590.00	590.00	590.00	600.00	610.00
300mm dia.	...	No	780.00	780.00	780.00	790.00	800.00
375mm dia.	...	No	950.00	950.00	960.00	970.00	980.00
450mm dia.	...	No	1155.00	1155.00	1155.00	1170.00	1185.00
525mm dia.	...	No	1335.00	1350.00	1350.00	1365.00	1380.00
600mm dia.	...	No	1550.00	1550.00	1550.00	1550.00	1600.00

DRAINAGE

SEWER DRAINS (cont'd)

POLYPROPYLENE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Extruded polypropylene non pressure corrugated pipe with smooth internal wall and rubber ring joints ("SewerPRO", "Sewer Max", SN10, ASI 5065) (cont'd) :							
Bend, 90°, GRP -							
225mm dia.	...	No	740.00	750.00	750.00	760.00	770.00
300mm dia.	...	No	990.00	990.00	990.00	990.00	1005.00
375mm dia.	...	No	1215.00	1215.00	1215.00	1230.00	1245.00
450mm dia.	...	No	1475.00	1475.00	1475.00	1475.00	1500.00
525mm dia.	...	No	1700.00	1700.00	1700.00	1725.00	1750.00
600mm dia.	...	No	1950.00	1950.00	1950.00	1950.00	2000.00
Equal tee, GRP -							
225mm dia.	...	No	1020.00	1020.00	1020.00	1035.00	1065.00
300mm dia.	...	No	1290.00	1305.00	1305.00	1305.00	1335.00
375mm dia.	...	No	1600.00	1600.00	1600.00	1600.00	1625.00
450mm dia.	...	No	1925.00	1925.00	1925.00	1950.00	1975.00
525mm dia.	...	No	2250.00	2250.00	2250.00	2250.00	2300.00
600mm dia.	...	No	2550.00	2600.00	2600.00	2600.00	2650.00
Manhole connector, GRP -							
225mm dia.	...	No	285.00	280.00	285.00	295.00	290.00
300mm dia.	...	No	350.00	350.00	350.00	360.00	360.00
375mm dia.	...	No	410.00	410.00	410.00	420.00	420.00
450mm dia.	...	No	480.00	480.00	480.00	490.00	490.00
525mm dia.	...	No	550.00	550.00	550.00	560.00	560.00
600mm dia.	...	No	660.00	660.00	660.00	670.00	680.00

PVC

ADD for trench excavation and backfilling	...	Note					
P.V.C. DWV pipe (formally class SH/SEH) to A.S./N.Z.S. 1260-2017 **with solvent welded joints** :							
100mm dia. (SN 6)	...	m	25.50	27.00	27.50	28.00	34.00
150mm dia. (SN 4)	...	m	40.00	41.00	42.00	43.00	50.00
150mm dia. (SN 8)	...	m	44.00	46.00	47.00	47.00	55.00
225mm dia. (SN 4)	...	m	80.00	82.00	84.00	84.00	94.00
225mm dia. (SN 8)	...	m	94.00	98.00	98.00	98.00	108.00
300mm dia. (SN 4)	...	m	122.00	126.00	128.00	128.00	144.00
300mm dia. (SN 8)	...	m	162.00	166.00	166.00	168.00	182.00
Bend, 42.5°, 45° :							
100mm plain (x 42.5°)	...	No	18.50	20.00	20.00	20.50	25.50
100mm access (x 42.5°)	...	No	22.00	23.00	23.50	24.00	29.00
150mm plain (x 42.5°)	...	No	36.00	38.00	38.00	39.00	45.00
300mm plain (x 45°)	...	No	240.00	245.00	245.00	245.00	260.00
Bend, 88° :							
100mm plain	...	No	18.50	20.00	20.00	20.50	25.50
100mm access	...	No	29.50	30.50	31.00	31.50	36.50
150mm plain	...	No	40.00	41.00	42.00	42.00	48.00

DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SEWER DRAINS (cont'd)							
PVC (cont'd)							
Junction 45° :							
100 x 100mm plain	...	No	25.50	27.00	27.00	28.00	34.00
100 x 100mm access	...	No	40.00	42.00	42.00	43.00	49.00
150 x 100mm plain	...	No	54.00	56.00	57.00	57.00	65.00
150 x 150mm plain	...	No	51.00	53.00	54.00	54.00	62.00
225 x 100mm plain	...	No	104.00	108.00	108.00	110.00	122.00
225 x 225mm plain	...	No	150.00	154.00	154.00	156.00	168.00
300 x 150mm plain	...	No	275.00	280.00	280.00	280.00	300.00
300 x 225mm plain	...	No	330.00	335.00	335.00	340.00	355.00
300 x 300mm plain	...	No	450.00	460.00	460.00	460.00	480.00
Test / inspection / access opening with cap :							
100mm	...	No	27.00	28.50	28.50	29.50	35.50
150mm	...	No	68.00	70.00	70.00	71.00	78.00
225 x 150mm inspection	...	No	240.00	240.00	240.00	240.00	250.00
300 x 225mm inspection	...	No	470.00	470.00	470.00	470.00	480.00
Level invert taper :							
150 x 100mm	...	No	36.00	37.00	37.00	38.00	44.00
225 x 150mm	...	No	104.00	106.00	106.00	108.00	118.00
300 x 225mm	...	No	166.00	170.00	170.00	172.00	186.00
Trap, disconnector :							
100mm	...	No	90.00	96.00	98.00	102.00	126.00
P.V.C. DWV pipe (formally class SH/SEH) to A.S./N.Z.S. 1260-2017 **with rubber ring joints** :							
100mm dia. (SN 10)	...	m	30.50	32.00	32.50	33.50	39.50
150mm dia. (SN 8)	...	m	45.00	47.00	47.00	48.00	55.00
225mm dia. (SN 8)	...	m	84.00	86.00	86.00	88.00	98.00
300mm dia. (SN 8)	...	m	130.00	134.00	134.00	136.00	150.00
Bend, plain :							
100mm, 45°	...	No	38.00	39.00	39.00	40.00	45.00
150mm, 45°	...	No	54.00	55.00	56.00	56.00	61.00
100mm, 90°	...	No	47.00	49.00	49.00	50.00	56.00
150mm, 90°	...	No	78.00	80.00	80.00	80.00	86.00
Junction, plain 45° :							
100 x 100mm	...	No	54.00	55.00	55.00	56.00	62.00
150 x 100mm	...	No	68.00	70.00	70.00	71.00	78.00
150 x 150mm	...	No	74.00	76.00	76.00	78.00	86.00
Level invert taper :							
150 x 100mm	...	No	49.00	50.00	51.00	52.00	59.00

STORMWATER DRAINS

Prices for pipework include for laying and jointing in trench but exclude excavation and backfilling ... Note

CONCRETE

Refer page 491 ... Note

DRAINAGE

STORMWATER DRAINS (cont'd)

FIBRE REINFORCED CEMENT

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling	...	Note					
Perth	...	P.O.A.					
F.R.C. non pressure pipe (Class 2) to AS 4139-2003 with collar and dual rubber ring joints (FRC Pipe) :							
225mm dia.	...	m	92.00	88.00	96.00	106.00	110.00
300mm dia.	...	m	120.00	114.00	124.00	-	146.00
375mm dia.	...	m	146.00	136.00	152.00	-	166.00
450mm dia.	...	m	175.00	155.00	180.00	-	190.00
525mm dia.	...	m	215.00	190.00	220.00	-	225.00
600mm dia.	...	m	255.00	220.00	260.00	-	260.00
675mm dia.	...	m	315.00	280.00	320.00	-	325.00
750mm dia.	...	m	360.00	325.00	365.00	-	395.00
F.R.C. non pressure pipe (Class 4) to AS 4139-2003 with collar and dual rubber ring joints (FRC Pipe+) :							
225mm dia.	...	m	-	108.00	-	126.00	136.00
300mm dia.	...	m	-	130.00	-	-	170.00
375mm dia.	...	m	-	155.00	-	-	195.00
450mm dia.	...	m	-	180.00	-	-	230.00
525mm dia.	...	m	-	210.00	-	-	270.00
600mm dia.	...	m	-	260.00	-	-	310.00
675mm dia.	...	m	-	320.00	-	-	390.00
750mm dia.	...	m	-	370.00	-	-	460.00
F.R.C. non pressure pipe (Class 4) to AS 4139-2003 with collar and dual rubber ring joints (FRC PipeR+) :							
225mm dia.	...	m	112.00	110.00	116.00	-	-
300mm dia.	...	m	144.00	140.00	148.00	-	-
375mm dia.	...	m	190.00	170.00	190.00	-	-
450mm dia.	...	m	220.00	200.00	225.00	-	-
525mm dia.	...	m	280.00	250.00	280.00	-	-
600mm dia.	...	m	375.00	310.00	370.00	-	-
675mm dia. (Class 3)	...	m	390.00	340.00	390.00	-	-
750mm dia. (Class 2)	...	m	390.00	350.00	390.00	-	-
Bend -							
225mm x 45º	...	No	360.00	300.00	350.00	-	335.00
225mm x 88º	...	No	485.00	400.00	470.00	-	445.00
300mm x 45º	...	No	455.00	390.00	450.00	-	450.00
300mm x 88º	...	No	730.00	600.00	710.00	-	690.00
375mm x 45º	...	No	550.00	470.00	540.00	-	540.00
375mm x 88º	...	No	980.00	810.00	950.00	-	910.00
450mm x 45º	...	No	760.00	640.00	740.00	-	750.00
450mm x 88º	...	No	1380.00	1140.00	1335.00	-	1290.00
525mm x 45º	...	No	870.00	735.00	855.00	-	855.00
600mm x 45º	...	No	945.00	795.00	930.00	-	915.00
675mm x 45º	...	No	975.00	825.00	960.00	-	945.00
750mm x 45º	...	No	1155.00	975.00	1125.00	-	1110.00

DRAINAGE

STORMWATER DRAINS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PVC							
ADD for trench excavation and backfilling	...	Note					
PVC pipe to AS 1254-2010 with solvent welded joints :							
90mm dia.	...	m	20.00	27.00	25.00	24.50	32.50
150mm dia.	...	m	38.00	40.00	41.00	42.00	49.00
225mm dia.	...	m	76.00	78.00	80.00	80.00	92.00
300mm dia.	...	m	126.00	130.00	132.00	134.00	152.00
375mm dia.	...	m	154.00	160.00	160.00	164.00	184.00
Bend :							
90mm	...	No	19.50	21.00	21.00	21.50	26.50
150mm	...	No	32.00	33.50	34.00	34.50	40.00
225mm	...	No	80.00	82.00	82.00	82.00	90.00
300mm	...	No	166.00	168.00	170.00	170.00	176.00
Junction -							
90 x 90mm	...	No	20.50	21.50	22.00	22.50	27.50
150 x 90mm	...	No	49.00	51.00	51.00	52.00	58.00
150 x 150mm	...	No	46.00	48.00	48.00	49.00	56.00
300 x 150mm	...	No	186.00	188.00	188.00	188.00	196.00
300 x 225mm	...	No	275.00	275.00	275.00	275.00	285.00
300 x 300mm	...	No	290.00	295.00	295.00	295.00	305.00
375 x 150mm	...	No	440.00	450.00	450.00	450.00	470.00
375 x 375mm	...	No	640.00	640.00	650.00	650.00	670.00
Level invert taper -							
225 x 150mm	...	No	106.00	108.00	108.00	108.00	116.00
300 x 150mm	...	No	160.00	164.00	164.00	166.00	178.00
300 x 250mm	...	No	180.00	185.00	185.00	185.00	200.00
Downpipe adaptor -							
100 x 50 x 90mm	...	No	14.50	15.50	15.75	16.25	19.75
100 x 100 x 100mm	...	No	29.50	30.50	30.50	31.00	35.50
150 x 160 x 150mm	...	No	46.00	48.00	48.00	48.00	53.00
POLYETHYLENE - HDPE							
ADD for trench excavation and backfilling	...	Note					
STEEL REINFORCED POLYETHYLENE (SRP) SN8 pipe with rubber ring gasket joints :							
300mm dia.	...	m	98.00	102.00	102.00	104.00	116.00
375mm dia.	...	m	110.00	116.00	116.00	118.00	132.00
450mm dia.	...	m	160.00	166.00	168.00	170.00	190.00
525mm dia.	...	m	210.00	220.00	220.00	220.00	245.00
600mm dia.	...	m	245.00	255.00	255.00	255.00	285.00
750mm dia.	...	m	300.00	310.00	310.00	315.00	345.00
Bends and junctions :							
Special fittings can be manufactured to order	...	Note					

DRAINAGE

STORMWATER DRAINS (cont'd)

POLYPROPYLENE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling	...	Note					
Extruded polypropylene non pressure corrugated pipe with smooth internal wall and rubber ring joints ("BlackMax"/"StormPRO", SN8, ASI 5065) :							
225mm dia.	...	m	82.00	82.00	84.00	96.00	98.00
300mm dia.	...	m	104.00	104.00	106.00	116.00	120.00
375mm dia.	...	m	122.00	124.00	126.00	136.00	142.00
450mm dia.	...	m	154.00	156.00	158.00	168.00	176.00
525mm dia.	...	m	200.00	200.00	205.00	215.00	225.00
600mm dia.	...	m	255.00	255.00	260.00	270.00	285.00
Bend, 45° -							
225mm dia.	...	No	310.00	310.00	310.00	320.00	325.00
300mm dia.	...	No	440.00	440.00	450.00	460.00	460.00
375mm dia.	...	No	540.00	540.00	540.00	550.00	560.00
450mm dia.	...	No	820.00	830.00	830.00	840.00	850.00
525mm dia.	...	No	1170.00	1170.00	1170.00	1185.00	1215.00
600mm dia.	...	No	1475.00	1475.00	1475.00	1500.00	1525.00
Bend, 90° -							
225mm dia.	...	No	390.00	390.00	390.00	400.00	400.00
300mm dia.	...	No	560.00	560.00	560.00	570.00	580.00
375mm dia.	...	No	680.00	680.00	690.00	690.00	700.00
450mm dia.	...	No	1050.00	1050.00	1050.00	1065.00	1080.00
525mm dia.	...	No	1475.00	1500.00	1500.00	1500.00	1525.00
600mm dia.	...	No	1875.00	1900.00	1900.00	1900.00	1925.00
Equal Tee -							
225mm dia.	...	No	570.00	570.00	570.00	580.00	590.00
300mm dia.	...	No	690.00	690.00	700.00	700.00	720.00
375mm dia.	...	No	975.00	975.00	990.00	990.00	1005.00
450mm dia.	...	No	1475.00	1475.00	1475.00	1500.00	1525.00
525mm dia.	...	No	2050.00	2100.00	2100.00	2100.00	2100.00
600mm dia.	...	No	2950.00	2950.00	2950.00	2950.00	3000.00
Equal Junction -							
225mm dia.	...	No	620.00	630.00	630.00	640.00	650.00
300mm dia.	...	No	760.00	760.00	770.00	770.00	790.00
375mm dia.	...	No	1080.00	1080.00	1080.00	1095.00	1110.00
450mm dia.	...	No	1625.00	1625.00	1625.00	1650.00	1675.00
525mm dia.	...	No	2300.00	2300.00	2300.00	2300.00	2350.00
600mm dia.	...	No	3250.00	3250.00	3250.00	3250.00	3300.00

DRAINAGE

SUBSOIL DRAINS

FIBRE REINFORCED CEMENT

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD for trench excavation and backfilling	...	Note					
Perth	...	P.O.A.					
F.R.C. slotted drain pipe to AS 4139-2003 with rubber ring joints (Aquifer) :							
225mm dia. class 4	...	m	106.00	102.00	114.00	-	124.00
300mm dia. class 4	...	m	142.00	134.00	150.00	-	166.00
375mm dia. class 4	...	m	175.00	165.00	185.00	-	200.00
450mm dia. class 4	...	m	185.00	175.00	200.00	-	215.00
525mm dia. class 4	...	m	225.00	215.00	240.00	-	260.00
600mm dia. class 4	...	m	280.00	260.00	300.00	-	315.00
675mm dia. class 4	...	m	320.00	300.00	340.00	-	360.00
750mm dia. class 4	...	m	390.00	360.00	410.00	-	430.00

POLYETHYLENE

ADD for trench excavation and backfilling	...	Note					
FLEXIBLE COIL							
Slotted polyethylene flexible coil pipe, Class 400 with filter sock :							
50mm dia.	...	m	12.00	12.50	12.75	12.75	15.25
65mm dia.	...	m	10.75	11.25	11.25	11.50	13.75
80mm dia.	...	m	15.25	15.75	16.00	16.00	18.50
100mm dia.	...	m	13.50	14.00	14.25	14.50	16.75
160mm dia.	...	m	21.00	21.50	21.50	22.00	24.50
Bends and junctions :							
Special fittings can be manufactured to order	...	Note					
FLEXIBLE PANEL							
Perforated recycled HDPE flexible panel (Megaflo) :							
170mm high panel, unwrapped	...	m	13.25	13.75	14.00	14.00	16.50
300mm high panel, unwrapped	...	m	17.50	18.00	18.00	18.50	20.50
450mm high panel, unwrapped	...	m	22.00	22.50	23.00	23.00	25.50
170mm high panel, A24 Bidum wrapped	...	m	18.00	18.50	19.00	19.00	21.50
300mm high panel, A24 Bidum wrapped	...	m	17.50	18.00	18.50	18.50	20.50
450mm high panel, A24 Bidum wrapped	...	m	22.50	23.00	23.00	23.50	25.50
ADD extra for :							
170mm high panel, end out / bend	...	Each	28.00	28.50	29.00	29.50	33.50
300mm high panel, end out / bend	...	Each	35.00	36.00	37.00	37.00	42.00
450mm high panel, end out / bend	...	Each	39.00	40.00	41.00	41.00	46.00
170mm high panel, side out / junction	...	Each	35.00	36.00	37.00	37.00	42.00
300mm high panel, side out / junction	...	Each	40.00	41.00	41.00	42.00	48.00
450mm high panel, side out / junction	...	Each	49.00	51.00	51.00	52.00	58.00

POLYPROPYLENE

ADD for trench excavation and backfilling when required	...	Note					
Geotextile placed in position - general use (2m wide x 50m roll)	...	sqm	3.20	3.30	3.35	3.45	4.00
Modular Flo-Cell of polypropylene and recycled materials, placed horizontally on sub-grade or slab :							
30mm drainage cell 30x500x600mm	...	sqm	21.00	21.00	21.00	22.00	24.00
50mm drainage cell 50x500x600mm	...	sqm	32.50	32.00	32.00	33.50	34.50

DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SUBSOIL DRAINS (cont'd)							
THERMOPLASTIC CORED							
ADD for trench excavation and backfilling when required	...	Note					
Thermoplastic cored sheet drainage faced one side with geotextile filter fabric, placed :							
Horizontally on sub-grade or slab :							
12mm sheet (Cordrain)	...	sqm	24.00	25.00	25.00	25.00	27.50
18mm sheet (standard)	...	sqm	28.00	29.00	29.00	29.00	31.50
20mm sheet (Plazadeck)	...	sqm	58.00	59.00	60.00	60.00	65.00
Vertically against wall face including adhesive daubs :							
12mm sheet	...	sqm	32.00	33.00	33.00	33.50	38.00
18mm sheet (standard)	...	sqm	37.00	38.00	38.00	39.00	43.00
Thermoplastic cored 40mm thick strip drain wrapped with geotextile filter fabric and placed vertically in trench, strip drain width :							
100mm	...	m	9.00	9.25	9.25	9.50	10.75
200mm	...	m	12.00	12.25	12.25	12.50	13.75
300mm	...	m	15.25	15.75	15.75	15.75	17.00
400mm	...	m	19.00	19.50	19.50	20.00	21.00
500mm	...	m	21.50	22.00	22.00	22.00	23.50
900mm	...	m	33.00	33.50	33.50	33.50	35.50
ADD extra for :							
Outlet, end or side (average)	...	No	41.00	42.00	42.00	42.00	45.00
ACO SURFACE DRAINAGE SYSTEMS							
Prices include excavation in typical material	...	Note					
LIGHT DUTY CHANNELS - KLASSIKDRAIN (AS 3996 Class A-B)							
100mm wide to 300mm deep channel with locking grate :							
Galvanised steel slotted	...	m	160.00	154.00	158.00	166.00	170.00
Plastic intercept heelsafe anti-slip	...	m	178.00	172.00	176.00	186.00	188.00
Stainless steel 5 star heelsafe anti-slip	...	m	520.00	490.00	500.00	540.00	510.00
Stainless steel wedgewire heelsafe anti-slip	...	m	520.00	490.00	500.00	540.00	510.00
Stainless steel wedgewire heelguard	...	m	520.00	490.00	500.00	540.00	510.00
Stainless steel slotted	...	m	295.00	285.00	285.00	305.00	300.00
200mm wide to 400mm deep channel with locking grate:							
Galvanised steel transverse	...	m	360.00	345.00	350.00	375.00	360.00
Stainless steel 5 star heelsafe anti-slip	...	m	870.00	830.00	830.00	900.00	840.00
Stainless steel wedgewire heelsafe anti-slip		m	870.00	830.00	830.00	900.00	840.00
300mm wide to 500mm deep channel with locking grate:							
Galvanised steel transverse	...	m	490.00	460.00	470.00	500.00	480.00
S.S. 5 star heelsafe anti-slip	...	m	1155.00	1095.00	1095.00	1200.00	1110.00
S.S. wedgewire heelsafe anti-slip	...	m	1155.00	1095.00	1095.00	1200.00	1110.00

DRAINAGE

ACO SURFACE DRAINAGE SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
MEDIUM DUTY CHANNELS - KLASSIKDRAIN (AS 3996 Class C-D)							
100mm wide to 300mm deep channel with locking grate :							
Ductile iron wave heelsafe anti-slip	...	m	235.00	225.00	225.00	240.00	240.00
Ductile iron decorative heelsafe anti-slip		m	235.00	225.00	225.00	240.00	240.00
Ductile iron slotted	...	m	215.00	205.00	210.00	220.00	220.00
Ductile iron intercept heelsafe anti-slip	...	m	215.00	205.00	210.00	220.00	220.00
Ductile iron galvanised intercept heelsafe anti-slip	...	m	235.00	225.00	225.00	240.00	240.00
Galvanised steel 10mm wide slot BRICKSLOT (for paving)	...	m	235.00	225.00	225.00	240.00	240.00
Galvanised steel mesh	...	m	176.00	170.00	172.00	182.00	184.00
Galvanised steel slotted	...	m	200.00	190.00	195.00	205.00	205.00
Resin composite slotted	...	m	235.00	230.00	230.00	245.00	240.00
Stainless steel 10mm wide BRICKSLOT (for paving)	...	m	470.00	450.00	460.00	490.00	460.00
Stainless steel mesh	...	m	540.00	520.00	520.00	560.00	530.00
Stainless steel slotted	...	m	430.00	410.00	420.00	450.00	430.00
ADD extra for :							
200mm wide up to 400mm deep channel	...	m	430.00-820.00	410.00-780.00	420.00-790.00	450.00-850.00	430.00-800.00
300mm wide up to 500mm deep channel	...	m	590.00-1425.00	570.00-1350.00	570.00-1365.00	610.00-1470.00	580.00-1365.00
HEAVY DUTY CHANNELS - POWERDRAIN (AS 3996 Class E-G)							
100mm wide to 300mm deep channel with locking grate:							
Ductile iron slotted	...	m	305.00	295.00	300.00	320.00	315.00
Ductile iron intercept heelsafe anti-slip	...	m	335.00	320.00	325.00	345.00	340.00
200mm wide to 400mm deep channel with locking grate:							
Ductile iron hiflo	...	m	480.00	460.00	460.00	490.00	480.00
Ductile iron slotted	...	m	500.00	480.00	480.00	510.00	500.00
Ductile iron intercept heelsafe anti-slip	...	m	540.00	520.00	520.00	560.00	540.00
300mm wide to 500mm deep channel with locking grate:							
Ductile iron hiflo	...	m	620.00	590.00	600.00	640.00	610.00
Ductile iron slotted	...	m	640.00	610.00	620.00	660.00	640.00
Ductile iron intercept heelsafe anti-slip	...	m	690.00	660.00	670.00	720.00	690.00
SUMPS							
Extra Light Duty (330kg) - Stainless steel heelguard antislip grated polymer concrete sump, sump size :							
300 x 300 x 450mm deep	...	No	540.00	510.00	520.00	560.00	530.00
450 x 450 x 600mm deep	...	No	910.00	860.00	880.00	940.00	900.00
600 x 600 x 600mm deep	...	No	1305.00	1245.00	1260.00	1350.00	1290.00
600 x 600 x 900mm deep	...	No	1410.00	1350.00	1365.00	1455.00	1410.00
900 x 500 x 600mm deep	...	No	1365.00	1305.00	1320.00	1410.00	1350.00
900 x 600 x 600mm deep	...	No	1775.00	1675.00	1700.00	1825.00	1725.00
900 x 900 x 500mm deep	...	No	2450.00	2350.00	2350.00	2550.00	2400.00

DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ACO SURFACE DRAINAGE SYSTEMS (cont'd)							
SUMPS (cont'd)							
Light Duty (2670kg) - Galvanised steel transverse grated polymer concrete sump, sump size :							
300 x 300 x 450mm deep	...	No	225.00	215.00	220.00	230.00	235.00
450 x 450 x 600mm deep	...	No	430.00	410.00	420.00	440.00	450.00
600 x 600 x 600mm deep	...	No	650.00	620.00	630.00	660.00	660.00
600 x 600 x 900mm deep	...	No	760.00	720.00	750.00	770.00	780.00
900 x 500 x 600mm deep	...	No	730.00	700.00	720.00	740.00	740.00
900 x 600 x 600mm deep	...	No	870.00	830.00	850.00	890.00	880.00
900 x 900 x 500mm deep	...	No	1365.00	1305.00	1335.00	1395.00	1380.00
Medium Duty (5000kg) - Galvanised steel transverse grated polymer concrete sump, sump size :							
450 x 450 x 600mm deep	...	No	480.00	460.00	470.00	490.00	490.00
600 x 600 x 600mm deep	...	No	760.00	720.00	740.00	780.00	770.00
600 x 600 x 900mm deep	...	No	870.00	830.00	850.00	880.00	890.00
900 x 500 x 600mm deep	...	No	830.00	790.00	810.00	840.00	840.00
900 x 600 x 600mm deep	...	No	1275.00	1215.00	1230.00	1305.00	1260.00
900 x 900 x 500mm deep	...	No	1575.00	1500.00	1525.00	1625.00	1575.00
ADD Extra for :							
Riser to foregoing, sump size -							
300 x 300 x 300mm deep	...	No	112.00	104.00	110.00	106.00	118.00
450 x 450 x 300mm deep	...	No	182.00	168.00	184.00	170.00	198.00
600 x 600 x 300mm deep	...	No	230.00	215.00	230.00	220.00	245.00
900 x 500 x 300mm deep	...	No	250.00	235.00	250.00	240.00	265.00
900 x 600 x 300mm deep	...	No	275.00	260.00	275.00	270.00	290.00
900 x 900 x 300mm deep	...	No	355.00	335.00	355.00	350.00	375.00
BEDS AND SURROUNDS							
Concrete surround 150mm thick including all additional excavation to :							
100mm pipe (0.149 cum/m)	...	m	86.00	86.00	90.00	70.00	92.00
150mm pipe (0.175 cum/m)	...	m	102.00	100.00	106.00	82.00	108.00
225mm pipe (0.223 cum/m)	...	m	130.00	128.00	136.00	106.00	138.00
TRAPS							
Prices for traps include for connection to pipes	...	Note					
Cylindrical bucket trap with removable grate and fine mesh bucket (for kitchen / prep areas)	...	No	2700.00	2750.00	2750.00	2750.00	2850.00
Cylindrical plaster trap with removable grate and settling basket (for art rooms / laboratories)	...	No	2650.00	2700.00	2700.00	2700.00	2800.00
Flushing rim / flushing floor waste (add extra for flush valve / cistern)	...	No	2400.00	2450.00	2450.00	2450.00	2600.00
225mm dia. boundary trap, complete with inlet riser, cast iron box and concrete	...	No	2050.00	2050.00	2100.00	2100.00	2250.00
150mm dia. boundary trap with inspection gate on outlet and riser approximately 750mm long incorporating junctions and with blank flange to top of riser	...	No	1475.00-2100.00	1500.00-2150.00	1525.00-2200.00	1500.00-2150.00	1675.00-2400.00

DRAINAGE

MANHOLES, SOAKWELLS, SUMPS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices include for excavation in material other than rock, backfilling, benching, channels, step irons and connection to pipes	...	Note					
All dimensions are internal dimensions	...	Note					
MANHOLES							
Manhole 900 x 900mm internally with 150mm concrete base, 230mm brick sides rendered internally and light duty cover and frame :							
1000mm deep	...	No	-	2550.00	-	2750.00	-
2000mm deep	...	No	-	3700.00	-	4200.00	-
3000mm deep	...	No	-	5000.00	-	5600.00	-
Manhole 600 x 600mm internally with 150mm concrete base, precast concrete wall sections and cover slab :							
600mm deep	...	No	1260.00	-	1410.00	-	1335.00
900mm deep	...	No	1575.00	-	1825.00	-	1675.00
1200mm deep	...	No	1950.00	-	2250.00	-	2050.00
ADD extra for :							
Cast iron light weight cover	...	No	500.00	-	520.00	-	660.00
Manhole 900 x 900mm internally with 150mm concrete base, precast concrete wall sections and cover slab :							
600mm deep	...	No	1650.00	-	2200.00	-	2350.00
900mm deep	...	No	2150.00	-	2750.00	-	2900.00
1200mm deep	...	No	2700.00	-	3350.00	-	3450.00
ADD extra for :							
Cast iron light weight cover	...	No	930.00	-	945.00	-	1095.00
Manhole 1000 to 1100mm dia. with concrete base and precast concrete wall sections :							
1000mm deep	...	No	1800.00	1600.00	1475.00	1875.00	1525.00
2000mm deep	...	No	2250.00	2200.00	2400.00	2300.00	2700.00
ADD extra for :							
Light slab cover	...	No	765.00	735.00	945.00	840.00	1215.00
Heavy slab cover	...	No	855.00	825.00	1035.00	930.00	1350.00
Light tapered cover	...	No	1025.00	1000.00	1300.00	-	1650.00
Heavy tapered cover	...	No	1175.00	1100.00	1400.00	-	1800.00
SOAKWELLS							
150L domestic 600 x 600 x 250mm deep Reln pvc soakwell with sand sock precast concrete lid 3m x 90mm pvc pipe and downpipe adaptor	...	No	-	-	-	230.00	-
Soakwell with slotted precast concrete liner, geofabric wrap, 200 deep blue metal base and 2-part trafficable concrete cover :							
1200mm I.D. x 1200mm deep	...	No	-	-	-	1135.00	-
1800mm I.D. x 2400mm deep	...	No	-	-	-	2200.00	-

DRAINAGE

MANHOLES, SOAKWELLS, SUMPS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SUMPS AND PITS							
Covers are priced separately	...	Note					
BRICK							
600 x 600 x 600mm deep with 150mm concrete base and 230mm brick walls rendered internally	...	No	-	760.00	-	800.00	-
ADD extra for :							
Each additional 100mm in depth	...	No	-	94.00	-	102.00	-
900 x 900 x 900mm deep with 150mm concrete base and 230mm brick walls rendered internally	...	No	-	1475.00	-	1550.00	-
ADD extra for :							
Each additional 100mm in depth	...	No	-	130.00	-	142.00	-
IN SITU CONCRETE							
600 x 600 x 600mm deep with 150mm concrete base and walls	...	No	1080.00	975.00	1125.00	1080.00	1200.00
ADD extra for :							
Each additional 100mm in depth	...	No	158.00	134.00	158.00	148.00	164.00
900 x 900 x 900mm deep with 150mm concrete base and walls	...	No	2150.00	1900.00	2200.00	2100.00	2350.00
ADD extra for :							
Each additional 100mm in depth	...	No	200.00	180.00	210.00	195.00	220.00
Covers are priced separately	...	Note					
PRECAST CONCRETE							
900mm dia. x 900mm deep with precast base and walls	...	No	1110.00	1080.00	1170.00	1110.00	1395.00
ADD extra for :							
Each additional 100mm in depth	...	No	78.00	72.00	80.00	58.00	84.00
PRECAST CONCRETE COVERS							
Precast concrete cover :							
Standard to suit pit size -							
600 x 600mm	...	No	66.00	64.00	74.00	70.00	86.00
900 x 900mm	...	No	175.00	165.00	190.00	180.00	225.00
900mm dia.	...	No	175.00	165.00	190.00	180.00	225.00
Trafficable -							
600 x 600mm	...	No	114.00	114.00	132.00	124.00	154.00
900 x 900mm	...	No	275.00	270.00	305.00	285.00	355.00
900mm dia.	...	No	265.00	265.00	295.00	275.00	340.00
METAL SUMP AND GULLY GRATINGS							
Refer page 362	...	Note					

DRAINAGE

LARGE UNDERGROUND STORMWATER SYSTEMS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices generally include installation and connection to pipes (but exclude earthworks including dewatering, soil retention, excavation, consolidation and backfilling where required)	...	Note					
HYDRODYNAMIC SEPARATORS							
"HumeCeptor" system, ex works, (holding capacity) :							
STC2 (1.7kl)	...	No	12500.00	14250.00	13000.00	12750.00	13000.00
STC3 (3.4kl)	...	No	18750.00	21500.00	19500.00	18000.00	19500.00
STC9 (9.1kl)	...	No	35500.00	41000.00	37000.00	32500.00	36500.00
STC18 (18.0kl)	...	No	48000.00	56000.00	51000.00	52000.00	50000.00
STC27 (27.3kl)	...	No	84000.00	97000.00	88000.00	69000.00	88000.00
STC40 DuoCeptor (43.4kl)	...	No	125000.00	145000.00	130000.00	120000.00	130000.00
STC50 DuoCeptor (50.5kl)	...	No	145000.00	165000.00	155000.00	-	155000.00
STC60 DuoCeptor (60.25kl)	...	No	180000.00	205000.00	190000.00	-	190000.00
ADD extra for :							
Transportation (special escort may be required, subject to road authority requirements), installation and connection to pipes	...	Note					
GROSS POLLUTANT TRAPS							
"HumeGard" inline system (max in/out pipe size and P.C. amount, ex works) :							
HG12 (300dia. P.C. $13500)	...	No	19500.00	18750.00	20250.00	20000.00	20750.00
HG15 (450dia. P.C. $15750)	...	No	21750.00	21000.00	22500.00	20500.00	23000.00
HG18 (600dia. P.C. $19250)	...	No	26000.00	25000.00	26500.00	29500.00	27500.00
HG27 (750dia. P.C. $32500)	...	No	41500.00	39500.00	42500.00	37000.00	42500.00
HG30A (900dia. P.C. $43000)	...	No	54000.00	52000.00	55000.00	49000.00	56000.00
HG35A (1050dia. P.C. $51500)	...	No	64000.00	61000.00	65000.00	53000.00	65000.00
HG40A (900dia. P.C. $59750)	...	No	74000.00	70000.00	74000.00	57000.00	75000.00
HG40B (1200dia. P.C. $69750)	...	No	85000.00	81000.00	86000.00	66000.00	86000.00
HG45A (1350dia. P.C. $76000)	...	No	93000.00	88000.00	94000.00	78000.00	94000.00
STORMWATER HARVESTING AND REUSE SYSTEMS							
"RainVault" modular system - captures, treats and stores rainwater for non-potable use - includes filters and pump kits (model and P.C. amount, ex works) :							
250cum -							
1200dia. (RV1200, P.C. $680/cum)	...	No	200000.00	200000.00	195000.00	185000.00	215000.00
1950dia. (RV1950, P.C. $575/cum)	...	No	170000.00	170000.00	160000.00	155000.00	180000.00
2100dia. (RV2100, P.C. $525/cum)	...	No	150000.00	155000.00	145000.00	140000.00	165000.00
2250dia. (RV2250, P.C. $680/cum)	...	No	200000.00	200000.00	195000.00	185000.00	215000.00
2400dia. (RV2400, P.C. $575/cum)	...	No	170000.00	170000.00	160000.00	155000.00	180000.00
3000dia. (RV2700, P.C. $595/cum)	...	No	175000.00	175000.00	170000.00	160000.00	190000.00

DRAINAGE

LARGE UNDERGROUND STORMWATER SYSTEMS (cont'd)

STORMWATER HARVESTING AND REUSE SYSTEMS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
"RainVault" modular system - captures, treats and stores rainwater for non-potable use - includes filters and pump kits (model and P.C. amount, ex works) (cont'd) :							
500cum -							
1200dia. (RV1200, P.C. $680/cum)	...	No	400000.00	400000.00	390000.00	370000.00	430000.00
1950dia. (RV1950, P.C. $525/cum)	...	No	300000.00	310000.00	290000.00	280000.00	330000.00
2100dia. (RV2100, P.C. $455/cum)	...	No	270000.00	270000.00	260000.00	250000.00	290000.00
2250dia. (RV2250, P.C. $680/cum)	...	No	400000.00	400000.00	390000.00	370000.00	430000.00
2400dia. (RV2400, P.C. $575/cum)	...	No	340000.00	340000.00	320000.00	310000.00	360000.00
3000dia. (RV2700, P.C. $595/cum)	...	No	350000.00	350000.00	340000.00	320000.00	380000.00
1000cum -							
1200dia. (RV1200, P.C. $690/cum)	...	No	800000.00	810000.00	770000.00	740000.00	860000.00
1950dia. (RV1950, P.C. $515/cum)	...	No	610000.00	610000.00	590000.00	560000.00	660000.00
2100dia. (RV2100, P.C. $460/cum)	...	No	540000.00	550000.00	530000.00	500000.00	590000.00
2250dia. (RV2250, P.C. $690/cum)	...	No	800000.00	810000.00	770000.00	740000.00	860000.00
2400dia. (RV2400, P.C. $575/cum)	...	No	670000.00	680000.00	650000.00	620000.00	720000.00
3000dia. (RV2700, P.C. $595/cum)	...	No	700000.00	700000.00	670000.00	640000.00	750000.00

LARGE VOLUME TRAFFICABLE STORMWATER SYSTEMS

"StormTrap" modular detention system - in various heights (P.C. amount, ex works) :							
SingleTrap -							
100cum :							
600mm high (P.C. $640/cum)	...	No	73000.00	74000.00	71000.00	68000.00	79000.00
1500mm high (P.C. $345/cum)	...	No	40500.00	41000.00	39500.00	37500.00	44000.00
500cum :							
600mm high (P.C. $485/cum)	...	No	280000.00	280000.00	270000.00	260000.00	300000.00
1500mm high (P.C. $330/cum)	...	No	190000.00	190000.00	185000.00	175000.00	205000.00
1000cum :							
600mm high (P.C. $455/cum)	...	No	520000.00	520000.00	500000.00	480000.00	560000.00
1500mm high (P.C. $320/cum)	...	No	370000.00	370000.00	350000.00	340000.00	390000.00
DoubleTrap -							
100cum :							
1200mm high (P.C. $640/cum)	...	No	74000.00	75000.00	72000.00	68000.00	80000.00
3000mm high (P.C. $350/cum)	...	No	41000.00	42000.00	40000.00	38000.00	45000.00
500cum :							
1200mm high (P.C. $500/cum)	...	No	280000.00	290000.00	270000.00	260000.00	310000.00
3000mm high (P.C. $340/cum)	...	No	195000.00	200000.00	190000.00	180000.00	210000.00
1000cum :							
1200mm high (P.C. $465/cum)	...	No	530000.00	540000.00	510000.00	490000.00	570000.00
3000mm high (P.C. $320/cum)	...	No	380000.00	380000.00	370000.00	350000.00	410000.00

DRAINAGE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SEPTIC SYSTEMS							
Prices include for excavation in material other than rock, backfilling, benching, step irons and connection to pipes	...	Note					
STANDARD SYSTEMS							
Septic tank comprising one 1800mm dia. x 1500mm deep and one 1500mm dia. x 1500mm deep precast concrete tanks together with 100mm concrete bases, precast concrete lids and all internal pipework	...	No	4000.00	4300.00	4700.00	4000.00	4600.00
Leach drain comprising 350/600mm dia. plastic sections and blue metal filling	...	m	185.00-195.00	185.00-200.00	185.00-205.00	145.00-160.00	200.00-210.00
ADD extra for :							
Removing top soil	...	Note					
Filling with light soil or sand	...	Note					
Leach drain, 600 x 600mm internal with blue metal filling in :							
Precast concrete, “U” shaped with lid	...	m	-	-	-	235.00	-
Precast blocks with lid	...	m	-	-	-	184.00	-
ADD extra for :							
Additional excavation or sand filling	...	Note					
ENVIRONMENTAL SYSTEMS							
Domestic systems :							
One tank system	...	System	16250.00	17000.00	16500.00	15500.00	16750.00
Two tank system	...	System	16250.00	17000.00	16500.00	15500.00	16750.00
ADD extra for :							
Irrigation system to dispose of treated water	...	System	Incl.	Incl.	Incl.	Incl.	Incl.
Maintenance agreement	...	Per annum	610.00	610.00	610.00	610.00	610.00
Commercial or public building systems :							
Capable of handling per day -							
3060 litres (or 17 persons)	...	System	24000.00	24250.00	24000.00	23500.00	24000.00
3600 litres (or 20 persons)	...	System	26750.00	27500.00	26500.00	26000.00	27000.00
5400 litres (or 30 persons)	...	System	42000.00	43000.00	42000.00	41000.00	42500.00
7200 litres (or 40 persons)	...	System	55500.00	56000.00	55000.00	54000.00	56000.00
9000 litres (or 50 persons)	...	System	68750.00	70000.00	69000.00	67000.00	69000.00
10800 litres (or 60 persons)	...	System	80250.00	82000.00	80000.00	79000.00	81000.00
ADD extra for :							
Irrigation system to dispose of treated water. is subject to local authority requirements	...	Note					
Maintenance agreement - to be negotiated	...	Note					
SEWAGE TREATMENT PLANT							
Package sewage treatment plant 30000 litres capacity complete with steel tank, aeration chamber, pumps, control panel and discharge	...	No	225000.00-245000.00	225000.00-245000.00	225000.00-245000.00	225000.00-245000.00	225000.00-245000.00
PUMPING STATIONS							
Pumps and pumping stations, refer pages 621 and 622	...	Note					

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

PRICE PER POINT

DOMESTIC LIGHTING

Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and luminaires	...	Note	
Circuits generally are TPS cabled	...	Note	
Single light point :			
With 1-one way switch	...	Point	75.00-96.00
With 2-two way switches, spaced 10m apart	...	Point	158.00-200.00
With 2-two way switches and 1-intermediate Switch (6m between switches)	...	Point	240.00-305.00
ADD extra for :			
Additional light point (commonly switched)	...	Point	21.00-30.00
Dimmer (for suitable LED/incandescent light) in lieu of switch	...	Each	35.00-43.00
LUMINAIRES (Lighting fittings) :			
Refer to the following typical luminaires; alternatively refer to LUMINAIRES pages 588 to 598	...	Note	
Domestic Luminaires -			
Average cost of luminaire installed excluding circuitry :			
600mm long batten with prismatic lens	...	Each	55.00-120.00
1200mm long batten with prismatic lens	...	Each	55.00-120.00
Multi light pendant	...	Each	150.00-305.00
Traditional pendant	...	Each	165.00-370.00
Fancy pendant	...	Each	165.00-1245.00
Ceiling mounted fitting	...	Each	35.00-200.00
Downlight	...	Each	30.00-125.00
“Do it yourself” lampholder type glass bowl diffuser	...	Each	15.00-50.00
Exterior wall bracket	...	Each	45.00-235.00
Exterior floodlight	...	Each	30.00-250.00

COMMERCIAL LIGHTING

Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and luminaires	...	Note	
Circuits are TPS cabled where viable, otherwise circuits are PVC conduited.	...	Note	
Single light point :			
With 1-one way light switch -			
In service area, toilet, tea room, store room	...	Point	98.00-126.00
In display area, public area, lobby	...	Point	128.00-178.00
In plant room, plumbing duct	...	Point	142.00-188.00
In undercover car parking (PVC conduited)	...	Point	176.00-245.00

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

COMMERCIAL LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Single light point :			
With 2-two way light switches -			
In room with switches spaced 15m apart	...	Point	210.00-270.00
In corridor with switches spaced 30m apart	...	Point	245.00-330.00
In stairway with switches on each level (PVC conduited)	...	Point	310.00-430.00
Single light point :			
With 2-two way switches and 1-intermediate switch -			
In room with each switch spaced 15m apart	...	Point	355.00-460.00
In corridor with each switch spaced 30m apart	...	Point	410.00-550.00
In stairway with a switch on each level (PVC conduited)	...	Point	390.00-540.00
Security light points -			
Spaced 20m apart	...	Point	73.00-104.00
Spaced 30m apart	...	Point	122.00-176.00
Self contained emergency light point	...	Point	40.00-58.00
Self contained exit sign point	...	Point	50.00-72.00
ADD extra for :			
Up to 5 additional light points (commonly switched) -			
In corridor, passage	...	Point	18.00-35.00
In service area, toilet, etc.	...	Point	10.00-23.00
In display area, public area, lobby	...	Point	16.00-35.00
In plant room	...	Point	39.00-67.00
In undercover car park (PVC conduited)	...	Point	77.00-110.00
5 - 15 additional light points (commonly switched) -			
In corridor, passage	...	Point	18.00-35.00
In display area, public area, lobby	...	Point	16.00-50.00
In open planned office	...	Point	12.00-51.00
In shopping market	...	Point	15.00-27.00
In shopping mall	...	Point	20.00-58.00
In undercover car park (PVC conduited)	...	Point	77.00-110.00
In stairway (PVC conduited)	...	Point	25.00-45.00
Additional intermediate light switch -			
In room with each switch spaced 15m apart	...	Point	138.00-182.00
In corridor with each switch spaced 30m apart	...	Point	194.00-260.00
In stairway with a switch on each level	...	Point	86.00-112.00
Standby alternator power supplied circuit	...	Plus	8.00-35.00
ADD extra per point :			
Fully PVC conduited circuit (in lieu of TPS) - approximate guide			
Cast into concrete	...	+65% to 75%	
Installed in roof space	...	+105% to 300%	
LUMINAIRES (LIGHTING FITTINGS)			
Refer to LUMINAIRES pages 588 to 598 for individual prices of luminaires; alternatively refer to averaged prices for luminaires at the end of this section	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
INDUSTRIAL LIGHTING			
Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and luminaires	...	Note	
Circuits generally are TPS cabled.	...	Note	
Single light point :			
With 1-one way light switch -			
In factory/workshop with lights 4m above floor level	...	Point	178.00-240.00
In factory/workshop with lights 6m above floor level	...	Point	230.00-340.00
Security light -			
Spaced 20m apart	...	Point	128.00-188.00
Spaced 40m apart	...	Point	176.00-265.00
Up to 5 additional light points (commonly switched) :			
In factory/workshop 4m above floor level (for LED/fluorescent lights)	...	Point	56.00-86.00
In factory/workshop 6m above floor level (for LED/high intensity discharge (H.I.D.) lights)	...	Point	56.00-122.00
ADD extra for :			
LUMINAIRES (LIGHTING FITTINGS)			
Refer to LUMINAIRES pages 588 to 598 for individual prices of luminaires; alternatively refer to the following averaged prices for luminaires	...	Note	
LUMINAIRES			
Averaged overall cost of luminaires installed excluding circuitry :			
Home unit	...	Each	76.00-192.00
Shopping centre	...	Each	112.00-265.00
Office building	...	Each	160.00-280.00
Hospital	...	Each	210.00-440.00
Nursing home	...	Each	100.00-250.00
School	...	Each	148.00-265.00
University	...	Each	235.00-460.00
Industrial	...	Each	220.00-880.00
EXTERIOR AND SPORTS LIGHTING			
Prices include columns, luminaires, accessories and wiring from nearby switchboard but EXCLUDE switchboard and RCD+MCB/RCB.	...	Note	
Sports lighting is sub-telecast standard. Ground conditions (e.g. location of existing structures) may affect design/pricing	...	Note	
EXTERIOR LIGHTING			
Bollard type path lighting -			
Domestic	...	Point	390.00-1,005.00
Commercial	...	Point	1,050.00-2,650.00
Small car park pole lighting-			
Commercial	...	Point	2,800.00-4,300.00
Column mounted floodlighting -			
Commercial	...	Point	4,800.00-8,600.00

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

INDUSTRIAL LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
SPORTS LIGHTING			
Tennis -			
Training, single court, 4 poles, 8 lights, 200lx	...	Court	14,750.00-26,000.00
Training, twin court, 4 poles, 12 lights, 200lx	...	Twin Court	25,500.00-31,500.00
Competition, one court, 6 poles, 18 lights, 300lx	...	Court	38,000.00-46,000.00
Netball / basketball -			
Mid grade, twin court, 6 poles, 12 lights, 120lx	...	Twin Court	40,000.00-49,000.00
Lawn bowls -			
Medium competition, 4 poles, 8 lights, 75lx	...	Green	38,500.00-49,000.00
Top competition, 4 poles, 8 lights, 140lx	...	Green	44,000.00-55,000.00
Hockey/rugby/soccer -			
Training, 2 poles, 8 lights, 50lx	...	Field	39,500.00-50,000.00
Club, medium/amateur, 4 poles, 16 lights, 115lx	...	Field	83,000.00-105,000.00
Australian Rules -			
Training, 2 poles, 6 lights, 50lx	...	Oval	43,000.00-57,000.00
Club, amateur, 4 poles, 12 lights, 50lx	...	Oval	96,000.00-122,500.00
Club, semi-prof/amateur, 6 poles, 36 lights, 150lx	...	Oval	187,500.00-237,500.00

DOMESTIC POWER

Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and appliances.	...	Note	
Circuits generally are TPS cabled.	...	Note	
Power outlet -			
Single	...	Point	76.00-96.00
Double	...	Point	82.00-102.00
15Amp	...	Point	110.00-132.00
Double with RCD/safety switch	...	Point	200.00-250.00
10A load circuit, with isolating switch -			
Single phase	...	Point	110.00-192.00
Three phase	...	Point	160.00-245.00
Kitchen appliance circuit -			
Single phase, 5kW oven	...	Point	116.00-225.00
Single phase, 6kW cooking top	...	Point	132.00-285.00
Single phase,13kW stove	...	Point	230.00-375.00
Ceiling exhaust fan outlet with remote switch	...	Point	90.00-100.00
Ceiling sweep fan and remote controller circuit (excluding fan and controller)	...	Point	122.00-144.00
ADD extra for :			
Appliances -			
250mm ceiling exhaust fan	...	Each	78.00-94.00
1200mm ceiling sweep fan and speed controller	...	Each	198.00-240.00
1300mm decorative ceiling sweep fan and speed controller	...	Each	225.00-275.00
Electronic exhaust fan run-on timer (in lieu of foregoing fan switch)	...	Each	130.00-148.00

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

COMMERCIAL POWER

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and appliances.	...	Note	
Circuits are TPS cabled where viable, otherwise circuits are conduited.	...	Note	
Single power outlet :			
In service area, tea room, corridor -			
Air-conditioned area	...	Point	124.00-180.00
Non air-conditioned area	...	Point	130.00-188.00
In display area, public area, lobby -			
Air-conditioned area	...	Point	152.00-240.00
Non air-conditioned area	...	Point	156.00-245.00
In skirting duct (excluding skirting duct), In air-conditioned area	...	Point	82.00-134.00
In open planned office, spaced 15m apart, In air-conditioned area -			
Floor mounted	...	Point	164.00-345.00
Column mounted	...	Point	110.00-285.00
Demountable partition mounted	...	Point	72.00-138.00
In plant room (protected type)	...	Point	126.00-260.00
In undercover car park, spaced 30m apart	...	Point	370.00-520.00
Double power outlet :			
In service area, tea room, corridor -			
Air-conditioned area	...	Point	134.00-194.00
Non air-conditioned area	...	Point	144.00-205.00
In display area, public area, lobby -			
Air-conditioned area	...	Point	162.00-250.00
Non air-conditioned area	...	Point	170.00-265.00
In skirting duct (excluding skirting duct), In air-conditioned area	...	Point	92.00-148.00
In open planned office, spaced 15m apart -			
Floor mounted	...	Point	174.00-360.00
Column mounted	...	Point	122.00-305.00
Demountable partition mounted	...	Point	82.00-150.00
In plant room (protected type)	...	Point	136.00-430.00
In undercover car park, spaced 30m apart	...	Point	370.00-520.00
Floor box, flush into floor, with two 10A double power outlets :			
On two shared circuits (at 15m centres)	...	Point	730.00-1,035.00
Each on dedicated circuit (40m from switchboard)	...	Point	1,035.00-1,485.00
ADD extra for :			
Data outlets and cabling, etc if required	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

COMMERCIAL POWER (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Single 15A single phase power outlet :			
In general office/display area -			
25m from switchboard	...	Point	156.00-265.00
40m from switchboard	...	Point	235.00-380.00
60m from switchboard	...	Point	355.00-550.00
In open planned office, 40m from switchboard -			
Floor mounted	...	Point	530.00-780.00
Column mounted	...	Point	215.00-360.00
Demountable partition mounted	...	Point	182.00-315.00
In plant room (protected type), 40m from switchboard	...	Point	240.00-395.00
Appliance circuit :			
30m from switchboard, with isolating switch -			
Single phase 3kW load	...	Point	150.00-205.00
Three phase 7.2kW load	...	Point	325.00-450.00
Three phase 14kW load	...	Point	440.00-610.00
Stove circuit :			
30m from switchboard -			
Two phase 14kW	...	Point	350.00-470.00
Three phase 25kW	...	Point	440.00-630.00
ADD extra for :			
Stove isolator / plug and socket - refer pages 568 to 571	...	Note	
Fully PVC conduited circuit (in lieu of TPS) - approximate guide	...	+75% to 425%	

INDUSTRIAL POWER

Prices include accessories and wiring from switchboard but EXCLUDE switchboard, RCD+MCB/RCBO and appliances.	...	Note	
Circuits are TPS cabled where viable, otherwise circuits are conduited.	...	Note	
Single power outlet, protected type weatherproof, (1 point per circuit) :			
20m from switchboard	...	Point	172.00-325.00
40m from switchboard	...	Point	275.00-450.00
ADD extra for :			
Double power outlet in lieu of single	...	Point	14.00-178.00
Additional single power outlets on common circuit (max. 7 single outlets/circuit) -			
Max. 10m apart	...	Point	164.00-295.00
Max. 20m apart	...	Point	205.00-375.00
Additional double power outlets on common circuit (max. 3 double outlets/circuit) -			
Max. 10m apart	...	Point	178.00-470.00
Max. 20m apart	...	Point	220.00-550.00
Single 15A single phase power outlet, protected type (weatherproof) -			
20m from switchboard	...	Point	178.00-335.00
40m from switchboard	...	Point	280.00-460.00
Additional single 15A single phase outlet on common circuit (max. 2 outlets/circuit) :			
Max. 10m apart	...	Point	220.00-365.00
Max. 30m apart	...	Point	265.00-440.00

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
INDUSTRIAL POWER (cont'd)			
Double 15A single phase power outlet, protected type (weatherproof) :			
20m from switchboard	...	Point	265.00-470.00
40m from switchboard	...	Point	350.00-600.00
20A three phase 4 pin outlet (1 point per circuit) :			
20m from switchboard - Protected type (weatherproof)	...	Point	430.00-580.00
40m from switchboard - Protected type (weatherproof)	...	Point	570.00-790.00
32A three phase 4 pin outlet (1 point per circuit) :			
20m from switchboard - Protected type (weatherproof)	...	Point	540.00-710.00
- Decontactor type	...	Point	600.00-760.00
40m from switchboard - Protected type (weatherproof)	...	Point	710.00-970.00
- Decontactor type	...	Point	770.00-980.00
63A three phase 4 pin outlet (1 point per circuit) :			
20m from switchboard - Protected type (weatherproof)	...	Point	1,950.00-2,050.00
- Decontactor type	...	Point	1,200.00-1,500.00
40m from switchboard - Protected type (weatherproof)	...	Point	2,400.00-2,700.00
- Decontactor type	...	Point	1,625.00-2,150.00
Three phase appliance circuit (5 wire R,W,B,N,E) e.g. power hacksaw, lathe, grinder (max. 4kW) EXCLUDING motor starter :			
20m from switchboard	...	Point	325.00-510.00
40m from switchboard	...	Point	430.00-660.00
FLOOR HEATING			
Prices include accessories but excluding wiring from switchboard	...	Note	
Storage floor heating cast in slab - Domestic	...	sqm	96.00-148.00
- Commercial	...	sqm	96.00-148.00
CEILING HEATING			
Prices include accessories and wiring from switchboard but EXCLUDE switchboard and ceiling insulation	...	Note	
Radiant insulated strip ceiling heating (120W-175W - e.g. nursing homes, housing etc.) - heated area	...	sqm	96.00-148.00
ADD extra for :			
Glasswool batt insulation - refer Carpentry pages 336 and 337	...	Note	
DOMESTIC FIRE ALARMS (NON MONITORED DETECTORS)			
Prices include accessories and wiring from switchboard but EXCLUDE switchboard and RCD+MCB/RCBO.	...	Note	
Residential type smoke alarms (combined detector and alarm) :			
Basic (non-circuit) type, with 9V battery	...	Point	27.00-41.00
Interconnected type on 240V supply and 9V battery (backup) -			
Surface - Ionisation type	...	Point	102.00-120.00
- Photoelectric type	...	Point	114.00-134.00
- Heat alarm type	...	Point	132.00-154.00
Flush - Ionisation type	...	Point	134.00-156.00
- Photoelectric type	...	Point	150.00-174.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PRICE PER POINT (cont'd)			
TELEPHONE/DATA PRE-CONDUITING			
Prices include conduit, draw wires, accessories, IDF and FDP cabinets but EXCLUDE telephone instruments, handsets, cabling, cable troughs, etc.	...	Note	
Commercial data/telephone point	...	Point	140.00-305.00
CATEGORY 5 / 6 DATA OR TELEPHONE POINTS			
Prices include typical open plan office wiring and RJ45 accessories and central patch panel but EXCLUDE skirting duct, telephone instruments and work beyond the patch panel	...	Note	
Telephone or data outlet :			
CAT 5 system	...	Point	220.00-340.00
CAT 6 system	...	Point	280.00-410.00
DATA/TELEPHONE CABINETS AND COMPONENTS			
Prices include for installation	...	Note	
Telecommunications utilities box - 600 x 600 x 200 (empty)	...	Each	315.00-410.00
Complete assemblies :			
20 pair FDP complete with plastic cover and termination modules	...	Each	135.00-195.00
Distribution enclosures and frames :			
Plastic type -			
30 pair, without lock	...	Each	245.00-350.00
30 pair, with lock	...	Each	320.00-430.00
100 pair, without lock	...	Each	365.00-490.00
External, type -			
30 pair	...	Each	215.00-320.00
50 pair	...	Each	235.00-340.00
100 pair	...	Each	295.00-410.00
MDF frame on wall, including jumper guides :			
500 pair	...	Each	285.00-400.00
600 pair	...	Each	285.00-410.00
Telephone cable termination frame (patch panel type) -			
150 pair	...	Each	290.00-410.00
ADD extra for :			
Krone connect module (1 per 10 pair)	...	Each	25.00-35.00
Krone disconnect module (1 per 10 pair)	...	Each	30.00-40.00
Earth module (1 per enclosure, if required)	...	Each	90.00-140.00
Surge arrestor magazine with gas arrestors (1 magazine per 10 pair)	...	Each	150.00-170.00
Standard/surge protection earthing	...	Note	
Non standard enclosure colour	...	Note	
Wall openings, supports and trimmers	...	Note	
Conduit links to skirting duct/ceiling space	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

TELEPHONE EQUIPMENT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include installation but EXCLUDE pre-conduiting, cabling, accessories, maintenance and phone line	...	Note	
Telephone instruments :			
Basic instrument with LCD display, 6 one touch memories, hands free and 70 phonebook memory	...	Each	77.00-116.00
Executive instrument with backlit display, 500 phonebook records, full duplex hands free and headset port	...	Each	275.00-420.00
Conference Phone with backlit display -			
Medium room	...	Each	1,005.00-1,185.00
Corporate (large room)	...	Each	4,000.00-4,500.00
Cordless phone - base and handset - range of single and multi pack	...	Range	116.00-345.00
ADD extra for :			
Noise cancelling head set -			
Corded, basic/top function	...	Range	128.00-330.00
Behind ear cordless, basic/top function	...	Range	315.00-680.00
Emergency or tough conditions - telephone instruments :			
IP65 weatherproof coloured housing (without door), hotline hands free (no handset), one button call, 21 day standby battery, keypad and time out	...	Each	1,005.00-1,125.00
Weatherproof ss housing (without door), vandal resistant handset with ss armoured cable, noise cancelling, keypad and time out	...	Each	840.00-960.00
Weatherproof coloured lockable cabinet with spring loaded door, vandal resistant handset, noise cancelling, keypad and time out	...	Each	820.00-930.00
IP66 weatherproof coloured lockable cabinet with spring loaded door, vandal resistant handset with ss armoured cable, noise cancelling, LCD display, keypad and time out	...	Each	820.00-930.00
Assistance for impaired - telephone instruments :			
Big button phone -			
Basic	...	Each	118.00-178.00
With loud speaking and battery backup	...	Each	305.00-380.00
With loud speaking and battery backup and separate 130mm function switch	...	Each	420.00-510.00
Cordless phone -			
Hands free, hearing aid t-coil, extra loud visual ring indicator :			
With base and handset	...	Each	250.00-400.00
With base, handset and splashproof emergency alert pendant having call button and handsfree function	...	Each	330.00-500.00
Long range cordless instruments :			
EnGenius type phone/radio (coverage- 1200ha open land / 12 floors / 2.3ha) -			
Base and handset	...	Each	1,215.00-1,335.00
Base and twin handsets	...	Each	1,850.00-2,050.00
Industrial - base and handset	...	Each	1,110.00-1,215.00
Industrial - base and four handsets	...	Each	3,600.00-4,100.00
External antenna, installed	...	Each	410.00-1,065.00
Coin or card operated telephones :			
Coin operated, for supervised location, Solitaire 6000	...	Each	1,900.00-2,350.00
For semi-supervised location, Solitaire 6000HS	...	Each	3,050.00-3,500.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

PRICE PER POINT (cont'd)

KEY OPERATED TELEPHONE SYSTEMS

Combined telephone plus intercom system (NEC Small Business Systems)	...	Note	
Prices include supply and installation of standard unit, hand sets and cabling of extensions but EXCLUDE pre-conduiting (refer to TELEPHONE PRE-CONDUITING for costs of pre-conduiting per point)	...	Note	
Systems are available by outright purchase, lease or rental (plus installation charges). They can be provided in various configurations, to suit specific needs, which together with other factors (such as the availability of stock and the relative market competition) can influence prices. A wide range of equipment options is available	...	Note	

TYPICAL SYSTEMS (* Installation is often included)	PURCHASE*	60 MTHS RENTAL*	INSTALLATION*
	$/SUM	$/ANNUM	$/SUM
Up to 4 (SIP) lines with			
4 handsets	3,550-4,250	775-900	1,050-1,400
8 handsets	7,100-8,450	1,575-1,825	1,100-1,500
Up to 8 (SIP) lines with			
16 handsets	13,550-15,900	2,975-3,425	1,200-1,700
24 handsets	19,850-23,200	4,375-5,000	1,300-1,875
Up to 12 (SIP) lines with			
16 handsets	13,550-15,900	2,975-3,425	1,200-1,700
24 handsets	19,850-23,200	4,375-5,000	1,300-1,875
Up to 24 (SIP) lines with			
24 handsets	19,850-23,200	4,375-5,000	1,300-1,875
40 handsets	31,700-36,550	6,975-7,850	1,475-2,225

ADD extra for :
Handsets, remote IP phones, networked systems, call-centre applications.

Note :
Modern IP telephony systems are expandable to the customer's desired quantity of (SIP) lines and extensions.

Second-hand telephone systems are also available at approximately 50% of new costs.

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

PBX (PABX) & IP TELEPHONY SYSTEMS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Standard PBX (PABX) system using IP telephony technology with voice telephone calls digitised and sent over a common IP network, combined with general data.	...	Note	
Typical IP VOIP solutions enable a single data socket to connect a VOIP telephone and a personal computer	...	Note	
Prices include supply and installation of PBX unit and operator training.	...	Note	
Allow lead time for procurement and programming – varies with system (eg minimum of 3 months)	...	Note	
PBX units are available by outright purchase, lease, rental or fully managed services with nominal installation charge	...	Note	
Medium to large standard VOIP PBX system with programming and unified comms built in (excluding installation of extensions) with :			
40 extensions	...	Sum	20,000.00-35,000.00
60 extensions	...	Sum	40,000.00-65,000.00
144 extensions	...	Sum	80,000.00-100,000.00
Annual Maintenance - ADD to standard PBX system (including at initial purchase)	...	Per Annum	10%
Unified messaging compatibility (to provide voice mail, email and presence) - ADD to standard PBX system, per extension (varies)	...	Each Extn	150.00
ADD extra for :			
Integral handsets (per extension) -			
Basic analogue handset	...	Each	75.00-105.00
IP or SIP suitable handset (range of features)	...	Each	100.00-450.00
Basic digital handsets	...	Each	120.00-220.00
Advanced digital handsets	...	Each	180.00-350.00
Pre-conduiting - refer to TELEPHONE PRE-CONDUITING for cost of pre-conduiting per extension point	...	Note	
Facility and backbone cabling	...	Note	
Number, Patch, Test (per extension point)	...	Each	40.00-43.00
TIMS (Telephone Information Management System), on accounting and telephone use facility -			
Software only	...	Sum	1,000.00-2,000.00
Software and hardware for 100 extensions (and below)	...	Sum	4,000.00-6,000.00
Software and hardware for 800 extensions (and below)	...	Sum	15,000.00-20,000.00
Operator's console with directory package (provides operator with VDU displayed information, e.g. extension number, name, status, title, section, etc.) with single or multiple screens as required	...	Sum	2,000.00-4,000.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PRICE PER POINT (cont'd)			
INTERCOM			
Prices include accessories, wiring and power supply but EXCLUDE conduiting (refer to TELEPHONE PRE-CONDUITING for costs of pre-conduiting per point)	...	Note	
Loud speaking type :			
2 station	...	Station	255.00-365.00
10 station	...	Station	500.00-660.00
ACCOMMODATING LOCAL AREA NETWORKS (LAN/WLAN/WI-FI)			
It is common to hard-wire data outlets with a minimum of 2 points / work space and 4+ points to other activity areas such as Meeting Rooms. Building systems (eg access control, security, CCTV, fire, inter-comm, etc) may require additional cabled infrastructure to non-work areas, such as entrances, loading bays, etc. Commonly data cabling is via service risers, ducting and ceiling voids.	...	Note	
Indoor Wi-Fi is commonly distributed via Access Points (AP) which typically provides 50m coverage around each Point. Systems with more than 2-3 APs require a WLAN Wireless Control System. Data cabled equipment such as APs, IP cameras and VoIP phones may be powered via Power-over-Ethernet (POE), rather than being powered from power outlets.	...	Note	
The growth of WLAN/Wi-Fi systems is demanding higher capacities (eg for voice / video calls, media streaming and frequent file storage). Wireless connectivity for personal devices has expanded to include equipment location tracking and security and data transfer with a requirement for improved band width and low latency and packet loss.	...	Note	
ICN (OPTICOMM)			
Integrated Communications Networks provide Gigabit-capable Passive Optical Networks (GPON) using point to multi-point fibre optics and network integration.	...	Note	
The ICN supports building and telecommunication systems including phone, broadband, building management, security, access control, video intercom, cctv and pay and free television services.	...	Note	
For large to medium residential, commercial and institutional developments.	...	Note	
MIXED USE MULTI-DWELLING-UNIT (MDU), residential apartment complex :			
Typical MDU installation comprising 300 residential apartments and 20 retail shops - example	...	Range Sum	84,500.00-180,000.00
ADD extra for :			
Main fibre optic cable pathways including tray and conduit	...	Note	
Cat 5e (or better) between ICN and data outlet / point of use connection	...	Note	
General 'Builder's Work'	...	Note	
End user contracts and agreements with service providers	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ICN (OPTICOMM)			
SPECIFIC ICN ELEMENTS - examples :			
Headworks and Backhaul (eg small sub-networks) - including fibre to the premises (FTTP) networks and base networks capability for superfast broadband, telephone, pay TV and free to air TV services	…	Range Sum	35,000.00-48,000.00
Apartment Network Termination Device (NTD) - to convert optical (light) signals to electrical signals for connection to copper networks and equipment	…	Range Sum	150.00-400.00
Retail fibre provision (provision for retail fitout)	…	Range Sum	150.00-400.00
Building Management System (BMS) Ethernet Port (eg BMS equipment interface)	…	Range Sum	199.00-257.00
Systems Integration (eg selection and testing of building services such as IP intercom, smart metering and CCTV systems)	…	Range Sum	15,500.00-31,000.00
CLOSED CIRCUIT TELEVISION			
Mid to high end commercial colour CCTV surveillance system including accessories wiring and equipment	...	Note	
High performance IP system with intelligent monitoring and potential threat alerts	...	Note	
Prices include accessories, wiring and equipment	...	Note	
COMPLETE CCTV SYSTEM - with 16 cameras and digital video recording	...	System	35,000.00-45,500.00
INDIVIDUAL MAIN COMPONENTS (included in the foregoing system) :			
Cameras -			
Dummy camera, with fixed mount	...	Each	172.00-230.00
Camera :			
Basic	...	Point	710.00-1,095.00
High end	...	Point	1,700.00-2,050.00
Motorised pan, tilt, zoom, auto iris and focus	...	Point	5,800.00-7,400.00
Dome Camera -			
Mid range, with lens	...	Each	520.00-780.00
With lens and IR illuminator	...	Point	1,275.00-5,400.00
With lens and IR illuminator and auto zoom	...	Point	1,775.00-2,350.00
Camera enclosure (add for camera) -			
Rugged weatherproof housing	...	Each	114.00-520.00
Vandal resistant housing	...	Each	1,260.00-1,700.00
Monitors and Switchers :			
Monitor, with video input -			
17" / 43cm TFT-LCD	...	Point	570.00-880.00
19" / 48cm TFT-LCD	...	Point	730.00-1,095.00
Digital video recorder (DVR) -			
4 Channel	...	Sum	1,140.00-1,850.00
8 Channel	...	Sum	5,600.00-6,800.00
16 Channel	...	Sum	7,600.00-12,750.00
Remote Keyboard	...	Sum	2,400.00-4,900.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

PRICE PER POINT (cont'd)

CLOSED CIRCUIT TELEVISION (cont'd)

IP CCTV ENTRY SYSTEMS

Multi-dwelling unit (MDU) Intercom Entry System :			
Main door digital alpha numeric display and keyboard -			
Per entry :			
B&W	...	Each	1,700.00-2,100.00
Colour	...	Each	2,000.00-2,500.00
Main backbone, power supply, floor distributors per apartment	...	Point	1,000.00-1,245.00
Apartment handset -			
B&W	...	Point	285.00-345.00
Colour	...	Point	460.00-570.00
1 to 1 kit type CCTV (household type), single point with indoor monitor and main door camera unit - range available	...	Sum	285.00-460.00

INTERNET PROTOCOL TELEVISION SYSTEMS

IPTV offers high performance TV reception over a LAN network.	...	Note	
Free-to-air, pay-TV, in-house broadcasts, Server recorded playback and other generated sources can be viewed on PC's and TV's including corporate content and digital signage. IPTV is networked via an IPTV system LAN or a building's general LAN (whilst enabling access to internet data without adversely affecting internet bandwidth) .	...	Note	
Individual or grouped TV's / PC's can be centrally controlled. Options include control of content, sound, channel change and power on/off.	...	Note	
Data points can service telephone, data and IPTV to save cabling.	...	Note	

TELEVISION ANTENNA

Prices include accessories, wiring and antenna	...	Note	
Household antenna in prime signal area :			
First outlet	...	Point	340.00-480.00
Additional outlets	...	Point	124.00-170.00
Master antenna (MATV) :			
Home units (10 to 20 outlets)	...	Point	154.00-215.00
Home units (20 to 50 outlets)	...	Point	220.00-295.00
Hospital bed (100 to 200 outlets)	...	Point	225.00-370.00
Pay-TV MDU point with 5 wire backbone, 2 x RG6 laterals and phone point at each Pay-TV location :			
Backbone cabling, per apartment	...	Point	1,290.00-1,440.00
Pay-TV 5 wire outlet	...	Point	205.00-320.00
ADD extra for :			
Pre-conduiting and power outlets	...	Note	
Coaxial cable may be installed as part of the general electrical installation	...	Note	
Second MATV system for cable TV, may be required	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

TELEVISION ANTENNA (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Satellite TV, KU Band system :			
Satellite decoder -			
Single channel	...	Sum	700.00-820.00
Multi channel	...	Sum	990.00
Satellite dish -			
65cm to 1.2m	...	Range Sum	66.00-370.00
1.8m to 3.2m	...	Range Sum	520.00-2,350.00
ADD extra for :			
Dish mount	...	Range Sum	57.00-1,850.00
LNB	...	Range Sum	77.00-500.00
Installation - varies with mount type, council requirements, building application costs and drawing requirements	...	Range Sum	265.00-1,750.00

SECURITY SYSTEMS

Local alarm system or central station monitored alarm system linked to central security monitoring station by direct phone or adsl line. Prices include circuits, accessories and equipment.	...	Note	
LOCAL ALARM SYSTEM :			
Control unit - 8 sector with remote arming	...	Sum	1,035.00-1,365.00
Sirens - standard or satellite	...	Sum	215.00-290.00
DIRECT CENTRAL STATION SYSTEM			
IP System, Security Service Monitoring Central station system fee (for medium project in city)	...	Week	8.00-27.00
ADD extra for :			
Access to the building's ADSL installation	...	Note	
AUTOMATIC DIALLER CENTRAL STATION SYSTEM :			
Automatic digital dialler control unit	...	Sum	1,200.00-1,575.00
Secondary monitoring using MobileNet	...	Sum	1,950.00-2,600.00
Security service monitoring fee	...	Week	8.00-10.00
Telephone/landline rental and connection fee is excluded as it is covered under conventional telephone fees and charges	...	Note	
DETECTION EQUIPMENT :			
Motion detector - for use with preceding systems -			
Wide angle infra red (covers approx. 60sqm, 60° angle)	...	Point	215.00-290.00
Dual technology detectors	...	Point	380.00-500.00
Control unit (8 - 24 sectors)	...	Point	5,400.00-7,200.00
Magnetic reed door contact	...	Point	104.00-136.00
ELECTRIFIED FENCE :			
Electric security to fence, 24 strands	...	m	190.00-250.00
ADD extra for :			
Chain mesh security fence - refer to EXTERNAL WORKS, pages 226 and 227			
For surveillance monitoring and apartment combined video and intercom systems refer to section on CLOSED CIRCUIT TELEVISION, page 519 and 520	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ENERGY MANAGEMENT SYSTEM			
Microprocessor based building energy controller, used to minimise peak energy demands and to maintain optimum, building environment and efficiency	...	Note	
Prices include accessories, control wiring and equipment	...	Note	
MAIN CONTROL UNIT :			
Controlling 1 to 8 items of plant	...	Sum	5,400.00-7,200.00
Controlling 9 to 20 items of plant	...	Sum	8,400.00-11,250.00
Controlling 21 to 40 items of plant	...	Sum	16,250.00-21,500.00
ADD extra for :			
Remote control via powerline using signal injection -			
Transmitter (up to 65 channels)	...	Sum	2,600.00-3,500.00
Remote receiver (switching up to 2 items of equipment)	...	Each	190.00-250.00
Field sensors -			
Temperature	...	Each	190.00-250.00
Relative humidity	...	Each	710.00-930.00
PC interrogation software	...	Sum	3,350.00-4,500.00
ACCESS CONTROL			
Prices include accessories, control wiring and equipment	...	Note	
SINGLE CARD ACCESS CONTROL			
Complete system including card key reader, electrically controlled lock and strike	...	Point	1,950.00-2,600.00
MULTIPLE ACCESS AND CONTROL SYSTEM			
Typical system comprising central processing unit and access control to 10 doors (including card key readers, electrically controlled locks and strikes)	...	Sum	54,000.00-72,000.00
ADD extra for :			
Lift monitoring (nominally 3 lifts)	...	Sum	38,000.00-50,000.00
Card keys	...	Each	12.00-16.00
Electric supply circuits to doors	...	Note	
Refer also to PARKING EQUIPMENT page 699	...	Note	
CALL SYSTEM			
Basic hard wired call system including accessories, wiring and equipment	...	Note	
Ceiling dome indicator light	...	Point	148.00-178.00
Ceiling dome indicator light and buzzer	...	Point	158.00-190.00
Wall mounted call button	...	Point	166.00-198.00
Pendant cord call button and wall receptacle	...	Point	215.00-260.00
LITHIUM ION BATTERY RACK MONITORING SYSTEM (LI-ION TAMER)			
The Li-ion Tamer device monitors lithium-ion battery off-gas events, early in the failure mode of lithium-ion batteries - to avoid catastrophic thermal runaway failure.	...	Note	
The Li-ion Tamer uses plug-and-play off-gas sensors and logic controller with Digital and Modbus serial outputs - to trigger electrical isolation of batteries, activate alarms, and signal a ventilation system. Sensors monitor ambient environment and air inlets - to cancel common mode signals.	...	Note	
Battery Room, with one controller, for			
Up to 2.1 sqm with 1 sensor and 1 reference sensor		Sum	10,000.00-11,000.00
Up to 20 sqm with 6 sensors and 3 reference sensors		Sum	21,000.00-24,000.00
Up to 20 sqm with 12 sensors and 3 reference sensors		Sum	32,000.00-36,000.00
Up to 40 sqm with 15 sensors and 6 reference sensors		Sum	48,000.00-53,000.00
Up to 40 sqm with 30 sensors and 6 reference sensors		Sum	79,000.00-88,000.00

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
VESDA® LASER & VESDA E SERIES® ASPIRATING SMOKE DETECTION SYSTEM			
The early warning of fire is provided through a highly sensitive laser based air sampling smoke detection system with highly efficient filtering and air pump systems. Smoke levels are continually monitored over a broad range (from 0.005%/m to 20%/m) allowing up to 4 levels of alarm, a staged response and an appropriate operation of suppression equipment	...	Note	
Sampling point, detector, accessories, circuits and equipment :			
Aircraft Hanger -			
Ceiling protection using 1 x VESDA VEP or VEU	...	Sum	10,000.00-13,000.00
Cable Tunnel -			
300m length ceiling sampling, using 1 x VESDA Laser Industrial VLI	...	Sum	8,000.00-10,000.00
Harsh Application -			
Up to 2000sqm , sampling on ceiling (or capillary tubes) 1 x VESDA VLI High Temperature Sampling Pipe (HFT)	...	Sum	10,000.00-12,000.00
Clean Room -			
Return air sampling, using 1 x VESDA Laser Focus VLF-250	...	Sum	3,000.00-5,000.00
General Areas -			
Up to 250sqm , sampling on ceiling (or capillary tubes) 1 x VESDA Laser Focus VLF-250	...	Sum	3,000.00-5,000.00
Up to 500sqm , sampling on ceiling (or capillary tubes) 1 x VESDA Laser Focus VLF-500	...	Sum	5,000.00-7,000.00
Up to 1000sqm , sampling on ceiling (or capillary tubes) 1 x VESDA VEP-1P (one pipe)	...	Sum	6,000.00-8,000.00
Up to 2000sqm , sampling on ceiling (or capillary tubes) 1 x VESDA VEP-4P (one zone)	...	Sum	8,000.00-10,000.00
Up to 2000sqm , sampling on ceiling (or capillary tubes) 1 x VESDA VES-4P (four zone)	...	Sum	10,000.00-12,000.00
Up to 6500sqm , sampling on ceiling (or capillary tubes) 1 x VESDA VEU-4P (one zone)	...	Sum	10,000.00-13,000.00
Conveyor Protection -			
300m length ceiling sampling, using 1 x VESDA Laser Industrial VLI	...	Sum	8,000.00-10,000.00
Coal Bunker Protection (retrofit) -			
2000 sqm ceiling sampling, using 1 x VESDA Laser Industrial VLI	...	Sum	8,000.00-10,000.00
Freezer / Cold Store -			
2000 sqm ceiling sampling, using 1 x VESDA VEP or VESDA Laser Industrial VLI with Refrigerated Sample Points	...	Sum	12,000.00-15,000.00
Office -			
up to 2000sqm - ceiling sampling, concealed, using 1 x VESDA VEP or VEU	...	Sum	10,000.00-13,000.00
Prison / Correction Facility, with on-ceiling tamper-proof sampling points -			
4 zone VESDA VES Scanner for 4 prison cells, each with 1 sampling point	...	Sum	10,000.00-12,000.00
1 zone VESDA Laser Focus VLF-250 for 1 prison cell, with 1 sampling point	...	Sum	3,000.00-5,000.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PRICE PER POINT (cont'd)			
VESDA® LASER & VESDA E SERIES® ASPIRATING SMOKE DETECTION SYSTEM (cont'd)			
Typical VESDA Aspirating Smoke Detection installations, including sampling point, detector, accessories, circuits and equipment :			
Psychiatric Facility, with on-ceiling tamper-proof sampling points -			
4 zone VESDA VES Scanner	...	Sum	10,000.00-12,000.00
1 zone LaserFocus for 1 or more rooms, each with 1 sampling point	...	Sum	3,000.00-5,000.00
Warehouse (retrofit) -			
per 2000sqm or less, using 1 x VESDA VEP or VESDA Laser Industrial VLI	...	Sum	8,000.00-10,000.00
ADD extra for :			
Automatic Fire Alarm FIP alarm group (if required)	...	Note	
Fire suppression systems, refer to FIRE PROTECTION, page 668	...	Note	
VESDA Laser Filter Technology maintenance :			
1 x non reusable filter	...	Each	100.00-200.00
AUTOMATIC FIRE ALARM			
Prices include accessories, control wiring and equipment.	...	Note	
Fire indicator panel to suit :			
2 - 10 alarm groups	...	Sum	4,300.00-6,400.00
12 - 20 alarm groups	...	Sum	5,800.00-8,800.00
22 - 30 alarm groups	...	Sum	6,600.00-9,400.00
32 - 50 alarm groups	...	Sum	9,200.00-12,750.00
Micro processor based fire indicator panel with facility for programmable addressable detectors - range of product varies with size and complexity	...	Range	23,250.00-87,000.00
Typical circuit and accessory :			
Thermal detector, ceiling mounted	...	Point	186.00-225.00
Smoke detector -			
Combustion type, ceiling mounted	...	Point	305.00-410.00
Photo optic type, ceiling mounted	...	Point	305.00-410.00
Photo optic type, mounted in air duct	...	Point	840.00-1,080.00
Manual break glass alarm	...	Point	200.00-240.00
Magnetic door holders :			
Door closer release type (including closer) -			
Double leaf door	...	Point	1,155.00-1,975.00
Wall or floor mounted electromagnetic hold open type -			
Single leaf door	...	Point	530.00-880.00
Double leaf door	...	Point	680.00-1,065.00
Internal flashing warning light	...	Point	290.00-400.00
External flashing warning light	...	Point	305.00-540.00
200 dia. fire bell	...	Point	230.00-255.00
ADD extra for :			
Conduiting and ducting to circuits if required	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
EMERGENCY WARNING AND INTERCOMMUNICATION SYSTEM			
Composite price per floor - typical mutli-storey office including equipment, cabling and installation	...	Floor	7,600.00-9,400.00
Individual prices :			
Master control panel to AS2220 for up to 12 loud speaker zones and 48 fire telephones circuits	...	Sum	27,500.00-38,000.00
Master control panel expansion -			
Additional loud speaker zone	...	Zone	2,450.00-3,000.00
Additional 6 fire telephones	...	Set of 6	610.00-760.00
Warden (fire) telephone with warning label	...	Each	245.00-310.00
Loud speaker with nominal 10m length of local circuit cabling -			
200mm ceiling recessed speaker	...	Each	84.00-130.00
5W horn speaker	...	Each	285.00-365.00
ADD extra for :			
Fire rated cabling - allow a separate 2 core 1.5sqmm high fire resistant or MIMS cable to each speaker zone (floor) and to each fire telephone (refer to pages 562 to 564 for cable prices)	...	Note	
Cable tray where required	...	Note	
BACKGROUND MUSIC			
Various (post construction) systems are available.	...	Note	
Categories of music include			
Over 30 different Channels available	...	Note	
Performance rights payment :			
Systems may be installed and maintained by rental agreement in lieu of outright purchase	...	Note	
Performance rights payment - One Music (formerly APRA & PPCA) fees vary with the size, venue and licence required - quoted on a case by case basis.	...	Note	
Music systems, examples :			
storePlay - https://storeplay.com.au/			
Music	...	Month (From)	35.00
Establishment	...	Established (From)	200.00
Soundtrack Your Brand -			
https://soundtrack.marketingmelodies.com.au/			
Music	...	Month (From)	45.00
Establishment	...	Established (From)	300.00
ADD extra for :			
Installation of the background music audio distribution system including :			
Amplifier,			
Rising and main feeders,			
Floor/zone control equipment,			
Speakers and cabling (and cable ducting, trays and conduits if required)			
Connecting system to PBX for music on hold telephone facility	...	Note	

ELECTRICAL SERVICES

PRICE PER POINT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
PUBLIC ADDRESS AND BACKGROUND MUSIC			
General PA system (not for emergency warning) including accessories and wiring	...	Note	
Performance rights licencing may be required for background music; consult One Music APRA.	...	Note	
Professional/commercial quality equipment :			
Amplifier (120 W rms)	...	Each	500.00-900.00
Media player (playback from variety of media formats) with AM/FM tuner - background music source		Each	550.00-650.00
Local ceiling loudspeaker and circuit	...	Each	125.00-175.00
Local column array loudspeaker and circuit	...	Each	190.00-290.00
Local horn loudspeaker and circuit	...	Each	130.00-230.00
Local volume control and circuit	...	Each	145.00-230.00
Local wireless microphone and receiver	...	Each	650.00-1500.00
ADD extra for :			
Remote aerial for AM and FM/digital reception if required	...	Note	
Primary audio distribution installation including panels, main cabling and installation, ambient noise monitors and remote microphones circuitry	...	Note	
CLOCKS			
240V Mains Clocks :			
HMS Digital Clock – 240V - KT160S	...	Each	840.00-970.00
Time and Date Digital Clock – 240V - KT160C	...	Each	1,080.00-1,245.00
ADD extra for :			
Electrical supply circuit to mains clocks	...	Point	104.00-146.00
PoE Power over Ethernet Clock			
HMS Digital Clock – PoE - KT160SP	...	Each	930.00-1,065.00
Time and Date Digital Clock – PoE - KT160CP	...	Each	1,215.00-1,380.00
Dual Theatre Timer Clock with Controller – PoE – KT-DT-POE	...	Each	2,650.00-3,100.00
ADD extra for :			
Data point circuit to PoE clock	...	Point	126.00-190.00
Master Clocks :			
Network Clock Controller – KT2050	...	Each	920.00-1,125.00
GPS SNTP Timeserver – 240V – KT2000	...	Each	1,050.00-1,275.00
GPS SNTP Timeserver – PoE – KT2002	...	Each	1,125.00-1,350.00
Master Clock with TCXO – KT3000	...	Each	1,230.00-1,470.00
Master Clock – GPS Reference – KT3000GPS	...	Each	1,500.00-1,775.00
Network Master Clock – KT4000	...	Each	2,150.00-2,500.00
ADD extra for :			
Power/data/control circuit to Master Clocks	...	Note	
Slave Clocks :			
Syncroline Analogue Slave Clock – 300mm – SL300	...	Each	305.00-365.00
Syncroline HMS Digital Slave Clock – SL160S	...	Each	710.00-820.00
Syncroline Time and Date Digital Slave Clock – SL160C	...	Each	920.00-1,065.00
Dual Theatre Timer Clock with Controller – Syncroline – KT-DT-SL	...	Each	2,400.00-2,850.00
ADD extra for :			
Electrical control circuit to slave clocks	...	Point	126.00-190.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PRICE PER POINT (cont'd)			
LIGHTNING PROTECTION SYSTEM			
Prices are based on typical commercial high rise installation including accessories but EXCLUDING any special building facilities such as rising shaft or moisture proof penetrations	...	Note	
The terminal is usually mounted 5 metres above the highest point on building/structure. Typically one terminal and down conductor can serve a 70 metre radius (max.) of building/structure	...	Note	
Lightning terminal :			
Dynasphere proprietary triggering terminal	...	Each	3,650.00-5,000.00
Down conductor fixed to concrete :			
ERICORE proprietary insulated and shielded cable (can be installed within building)	...	m	58.00-78.00
ADD extra for :			
Support mast -			
5 metre mast bolted to structure and guyed	...	Each	1,150.00-1,450.00
15 metre free standing mast remote from structure (where required) - refer to lighting columns pages 599 and 600	...	Note	
Earth system (10 Ohm static) -			
1800mm copper-clad steel rod electrode and inspection pit	...	Each	1,400.00-2,300.00
Multiple electrodes, linked with copper tape or wire - the quantity of the electrodes and the spacing varies with earth resistance	...	Note	
Buried mesh, bare copper wire or tape electrode (where required)	...	Note	
Termination and bonding to concrete reinforcement/ structural steel (where required)	...	Note	
SURGE PROTECTION AND POWER CONDITIONING SYSTEMS			
• Prices include accessories, control wiring and equipment	...	Note	
LIGHTNING AND SURGE REDUCTION FILTERS			
(Transient and surge protection device, usually mounted within or close to the main switchboard, in series with circuit) - various options available across the range :			
Single phase, 240V -			
10A	...	Each	220.00-340.00
20A	...	Each	340.00-440.00
63A	...	Each	2,100.00-2,650.00
125A	...	Each	3,150.00-3,700.00
Three phase, 240V/415V -			
63A/phase	...	Each	3,500.00-5,200.00
125A/phase	...	Each	4,900.00-6,200.00
250A/phase	...	Each	6,400.00-9,600.00
500A/phase	...	Each	10,000.00-16,500.00
800A/phase	...	Each	14,100.00-20,500.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PRICE PER POINT (cont'd)			
SURGE PROTECTION AND POWER CONDITIONING SYSTEMS (cont'd)			
SURGE DIVERTERS OR SURGE PROTECTION DEVICES (Shunt connected device, switchboard mounted)			
Three phase, 240V/415V :			
40kA surge rating per phase	...	Each	550.00-820.00
100kA surge rating per phase	...	Each	1,275.00-2,050.00
200kA surge rating per phase	...	Each	1,750.00-2,450.00
POWER LINE FILTERS			
Plug in at equipment for differential and surge protection :			
Single phase -			
10A, 240V	...	Each	210.00-320.00
TRANSIENT SAFETY BARRIERS			
Various degrees of protection available, hard wired and plug-in :			
Data signalling and telephone line protection -			
Multi-stage UTB, for PLC and control input and output	...	Each	162.00-220.00
COMPUTER/DATA LINE PROTECTORS			
LN RJ45-Cat 6 surge protector	...	Each	182.00-220.00
TELEPHONE PROTECTORS			
For MDF or twisted pair computer wiring :			
Line protector for up to 10 pair (Krone compatible) -			
Single stage	...	Each	116.00-156.00
Multi-stage	...	Each	255.00-305.00

HIGH VOLTAGE CABLE

Installed in position complete	...	Note	
COPPER STEEL WIRE ARMOURED (CU XLPE SWA PVC) cable drawn into underground conduit :			
Three core 11kV -			
50 sqmm	...	m	166.00-214.00
95 sqmm	...	m	210.00-265.00
120 sqmm	...	m	235.00-300.00
Three core 22kV -			
50 sqmm	...	m	182.00-234.00
95 sqmm	...	m	225.00-290.00
120 sqmm	...	m	260.00-330.00
MINIMUM ORDER QUANTITIES MAY APPLY :			
Nominally 500m	...	Note	
ADD extra for :			
Excavation - refer to DRAINAGE, pages 489 and 490	...	Note	
Conduit - refer to CABLE CARRIERS, pages 541 to 550	...	Note	

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
HIGH VOLTAGE CABLE (cont'd)			
COPPER STEEL WIRE ARMOURED (CU XLPE SWA PVC) cable drawn into underground conduit (cont'd) :			
1300 x 900 x 1000mm deep brick cable pit including heavy duty gatic cover, excavation, etc., complete	...	Each	4,400.00-7,200.00
2000 x 1500 x 1000mm deep concrete cable pit including heavy duty gatic cover, excavation, etc., complete	...	Each	8,600.00-15,250.00
COPPER XLPE INSULATED and heat resistant PVC sheathed UNARMOURED cables fixed to and including cable ladder :			
Three core 11kV -			
50 sqmm	...	m	225.00-280.00
95 sqmm	...	m	260.00-330.00
120 sqmm	...	m	275.00-350.00
Three core 22kV -			
50 sqmm	...	m	230.00-285.00
95 sqmm	...	m	280.00-360.00
120 sqmm	...	m	305.00-400.00
MINIMUM ORDER QUANTITIES MAY APPLY :			
Nominally 500m	...	Note	
11KV HIGH VOLTAGE TRANSFORMERS			
Installed in position complete :			
Pole mounted type - 200kVA	...	Sum	7,400.00-16,750.00
Cable box type free standing on rollers -			
500kVA	...	Sum	22,250.00-44,000.00
1000kVA	...	Sum	31,500.00-67,000.00
Pad-mount type with metal enclosure HV and LV end cubicles and concrete base -			
500kVA	...	Sum	70,000.00-120,000.00
Indoor dry type Resibloc (encapsulated) with IP21 enclosure -			
1000kVA	...	Sum	53,000.00-105,000.00
11KV HIGH VOLTAGE SWITCHGEAR			
The electrical supply authority may provide some high voltage services for which charges may apply	...	Note	
630A/1,250A, fixed/disconnectable circuit breaker complete	...	Sum	32,000.00-46,500.00
630A/1,250A, air insulated withdrawable SF6/ vacuum circuit breaker complete	...	Sum	40,000.00-59,000.00
ADD extra for :			
Earthing and substation/transformer switchroom or fenced compound where applicable	...	Note	

ELECTRICAL SERVICES

EMERGENCY POWER SUPPLY

		UNIT	Adel. Bris. Melb. Per. Syd. $
Complete system including control switchboard but EXCLUDING distribution cabling and switchboards, etc.	...	Note	
48V DC power supply system with a high power density and operating efficiency rectifier with a high degree of scalability for future growth.			
Batteries are valve regulated maintenance free lead acid (FAA and IATA approved as non-hazardous).			
Includes 600 x 600 x 2100 cabinet, system controller, power rectifier, battery pack and distribution panel (unpopulated) :			
48VDC, 2 hour run time -			
1800W, with 4 x 100AH battery pack	...	Sum	8,000.00-10,250.00
3600W, with 4 x 150AH battery pack	...	Sum	9,250.00-11,750.00
5000W, with 8 x 150AH battery pack	...	Sum	12,000.00-15,000.00
Diesel standby alternator with automatic mains failure start system and residential silencer :			
20kVA	...	Sum	32,000.00-44,500.00
35kVA	...	Sum	35,000.00-55,000.00
50kVA	...	Sum	45,000.00-67,000.00
80kVA	...	Sum	62,000.00-89,000.00
100kVA	...	Sum	70,000.00-99,000.00
125kVA	...	Sum	70,000.00-99,000.00
150kVA	...	Sum	76,000.00-99,000.00
225kVA	...	Sum	90,000.00-120,000.00
250kVA	...	Sum	105,000.00-140,000.00
300kVA	...	Sum	115,000.00-150,000.00
350kVA	...	Sum	120,000.00-165,000.00
450kVA	...	Sum	145,000.00-180,000.00
500kVA	...	Sum	175,000.00-215,000.00
600kVA	...	Sum	220,000.00-280,000.00
750kVA	...	Sum	310,000.00-380,000.00
900kVA	...	Sum	340,000.00-430,000.00
1000kVA	...	Sum	370,000.00-450,000.00
1200kVA	...	Sum	430,000.00-525,000.00
ADD extra for :			
Fuel tanks (storage and day tanks), fuel line, acoustic treatment, ventilation, attenuation and fuel to tank	...	Note	
Sound attenuated acoustic enclosure for residential operation -			
17kVA to 100kVA	...	Range	15,800.00-18,800.00
150kVA to 250 kVA	...	Range	21,700.00-28,000.00

ELECTRICAL SERVICES

UNINTERRUPTABLE POWER SUPPLY

		UNIT	Adel. Bris. Melb. Per. Syd. $
Complete system with control switchboard and with input and outputs wired but EXCLUDING distribution cabling and switchboards, etc. (may be supplied by a standby alternator).	...	Note	
No-break computer room or plant room UPS providing a maximum load and battery support time (BST) :			
1 phase output (with 1 phase input) -			
3kVA / 2.1kW - 15 Minute B S T	...	sum	6,600.00-7,400.00
- 30 Minute B S T	...	sum	8,200.00-9,200.00
- 60 Minute B S T	...	sum	10,000.00-11,250.00
6kVA / 4.2kW - 15 Minute B S T	...	sum	8,800.00-9,800.00
- 30 Minute B S T	...	sum	12,500.00-14,000.00
- 60 Minute B S T	...	sum	20,000.00-22,250.00
1 phase output (with 3 phase input) -			
7.5kVA / 5.25kW - 15 Minute B S T	...	sum	18,500.00-20,750.00
- 30 Minute B S T	...	sum	22,000.00-24,750.00
- 60 Minute B S T	...	sum	29,500.00-33,000.00
10kVA / 7kW - 10 Minute B S T	...	sum	19,250.00-21,500.00
- 30 Minute B S T	...	sum	22,750.00-25,500.00
- 60 Minute B S T	...	sum	30,000.00-33,500.00
15kVA / 10.5kW - 15 Minute B S T	...	sum	22,000.00-24,750.00
- 30 Minute B S T	...	sum	30,500.00-34,500.00
- 60 Minute B S T	...	sum	36,500.00-41,000.00
20kVA / 14kW - 10 Minute B S T	...	sum	25,000.00-28,000.00
- 30 Minute B S T	...	sum	39,000.00-43,500.00
- 60 Minute B S T	...	sum	43,500.00-48,500.00
3 phase output (with 3 phase input) -			
10kVA / 8kW - 15 Minute B S T	...	sum	19,750.00-22,250.00
- 30 Minute B S T	...	sum	23,750.00-26,500.00
- 60 Minute B S T	...	sum	31,500.00-35,000.00
15kVA / 12kW - 15 Minute B S T	...	sum	23,000.00-26,000.00
- 30 Minute B S T	...	sum	32,500.00-36,000.00
- 60 Minute B S T	...	sum	38,500.00-43,000.00
20kVA / 16kW - 15 Minute B S T	...	sum	26,500.00-29,500.00
- 30 Minute B S T	...	sum	41,000.00-45,500.00
- 60 Minute B S T	...	sum	45,500.00-51,000.00
30kVA / 24kW - 15 Minute B S T	...	sum	36,500.00-40,500.00
- 30 Minute B S T	...	sum	43,500.00-48,500.00
- 60 Minute B S T	...	sum	51,000.00-57,000.00
40kVA / 32kW - 15 Minute B S T	...	sum	45,000.00-50,000.00
- 30 Minute B S T	...	sum	51,000.00-57,000.00
- 60 Minute B S T	...	sum	62,000.00-69,000.00
60kVA / 48kW - 15 Minute B S T	...	sum	63,000.00-71,000.00
- 30 Minute B S T	...	sum	73,000.00-82,000.00
- 60 Minute B S T	...	sum	93,000.00-105,000.00
80kVA / 64kW - 15 Minute B S T	...	sum	80,000.00-90,000.00
- 30 Minute B S T	...	sum	93,000.00-105,000.00
- 60 Minute B S T	...	sum	125,000.00-140,000.00

ELECTRICAL SERVICES

UNINTERRUPTABLE POWER SUPPLY (cont'd)

No-break computer or plant room UPS providing a maximum load and battery support time (BST) : (cont'd)

				UNIT	Adel. Bris. Melb. Per. Syd. $
120kVA /	96kW	- 15 Minute B S T	...	sum	100,000.00-110,000.00
		- 30 Minute B S T	...	sum	120,000.00-130,000.00
		- 60 Minute B S T	...	sum	170,000.00-190,000.00
200kVA /	160kW	- 10 Minute B S T	...	sum	170,000.00-190,000.00
		- 15 Minute B S T	...	sum	175,000.00-195,000.00
		- 30 Minute B S T	...	sum	210,000.00-240,000.00
300kVA /	240kW	- 10 Minute B S T	...	sum	220,000.00-245,000.00
		- 15 Minute B S T	...	sum	240,000.00-270,000.00
		- 30 Minute B S T	...	sum	280,000.00-320,000.00
400kVA /	320kW	- 10 Minute B S T	...	sum	290,000.00-330,000.00
400kVA /	320kW	- 30 Minute B S T	...	sum	350,000.00-400,000.00
500kVA /	400kW	- 10 Minute B S T	...	sum	350,000.00-390,000.00
600kVA /	480kW	- 10 Minute B S T	...	sum	470,000.00-525,000.00
700kVA /	560kW	- 10 Minute B S T	...	sum	550,000.00-600,000.00
800kVA /	640kW	- 10 Minute B S T	...	sum	600,000.00-675,000.00

ELECTRICAL SERVICES

MAINS/SUBMAINS INSTALLATIONS

Prices are for typical 3 phase installations and include for trenches, cable protection slabs, conduit, cable tray, mounting struts and tee-off boxes where applicable.

Restrictions may apply to the types of consumer's mains that may be connected to supply authority owned substation equipment compared to privately owned equipment, e.g. MIMS-ESR system may not be acceptable for consumers' mains.

Consumer's mains and relevant submains may be required to be fire rated e.g. power supplies to lifts, fire pumps, smoke exhaust, stair pressurisation and emergency power supplies. This may require specific installation methods; fire rated cable may be used, or alternatively fire rated elements of building construction could be used.

TABLE 1 - TYPICAL MAINS/SUBMAINS SIZES (MAXIMUM DEMANDS)

TYPE OF INSTALLATION	MAINS/SUBMAINS SERVING :				
	SOLE SWITCHBOARD A/phase		MULTIPLE SWITCHBOARDS A/phase		
	LIGHT LOAD	HEAVY LOAD	LIGHT LOAD	MEDIUM LOAD	HEAVY LOAD
Home units	45A	200A	100A	150A	450A
Factory (medium)	45A	200A	150A	200A	500A
Shopping centre	45A	200A	150A	1000A	2500A
Shop	45A	100A	75A	100A	160A
Office building	75A	150A	100A	500A	2500A
Hospital	100A	300A	200A	800A	3500A
Nursing home	75A	150A	150A	250A	450A
Primary school	45A	100A	100A	175A	200A
High school	75A	200A	150A	350A	800A
Technical college	75A	200A	150A	500A	1000A
University	75A	200A	150A	500A	3500A

NOTE : Refer also to Electrical Services Design Data sections

		UNIT	Adel. Bris. Melb. Per. Syd. $
CONSUMER'S MAINS			
Allow for costs and charges associated with the provision of Supply Authorities equipment e.g. cabinets, link frames, pole fuses, transformer compounds, substation construction, cable ducts, trenches, incoming cables, conduits, and draw-in pits etc., as required. Liaise with local electricity supply authority.	...	Note	
MAINS POLE			
Prices are for poles fabricated from GWP, installed in position including concrete base.			
100 dia. HDG steel domestic intermediate pole	...	Each	690.00-840.00
125 dia. HDG steel commercial point of attachment pole with goose neck and cross arm	...	Each	1,410.00-1,700.00

ELECTRICAL SERVICES

MAINS/SUBMAINS INSTALLATIONS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
RAISER BRACKET			
Raiser bracket, domestic type, range of sizes and types available :			
Single phase type	...	Each	158.00-230.00
Three phase type	...	Each	210.00-285.00
HOUSE MAINS			
63A single phase -			
Underground	...	m	30.50-40.00
For overhead supply	...	m	14.25-29.50
32A three phase -			
Underground	...	m	34.00-49.00
For overhead supply	...	m	13.50-36.00

MAINS/SUBMAINS — **CURRENT RATINGS (M.D.)**

50m Nominal Route Length	**50A $/m**	**100A $/m**	**200A $/m**	**400A $/m**	**500A $/m**	**800A $/m**	**1000A $/m**
PVC/XLPE cable installed UNDERGROUND :							
Buried direct (incl. mech. protection) -							
Aluminium conductors	-	-	58-85	110-161	156-223	**207-297**	**296-418**
Copper conductors	27-44	43-69	65-103	156-245	238-367	453-696	**482-740**
In HD PVC conduit -							
Aluminium conductors	-	-	80-132	183-291	208-324	**331-499**	**376-559**
Copper conductors	43-83	60-109	110-182	181-289	327-509	507-794	**649-1014**
PVC cable installed in conduit :							
Fixed to structure -							
Copper conductors	30-65	48-93	117-171	326-506	386-600	**560-872**	**777-1208**
PVC/XLPE cable installed :							
On cable tray -							
Aluminium conductors	-	-	130-177	210-287	242-329	**372-506**	**435-590**
Copper conductors	79-110	104-145	166-235	310-451	427-620	**575-842**	**820-1192**
2 HOUR FIRE RATED cable on cable ladder :							
ESR-MIMS cable	190-254	199-266	220-293	314-413	338-446	496-648	574-744
Double insulated plastic fire resistant cable	183-246	190-256	240-321	321-427	393-523	559-739	692-908
MIMS cable	209-278	220-294	241-321	343-450	372-490	556-725	642-832
BUS DUCT with 3 (nominal) bends :							
Copper conductors	-	-	-	-	-	524-567	537-582
RISING MAIN with 10 unfused tee-offs :							
PVC/XLPE cable on tray -							
Aluminium conductors	-	-	-	422-530	454-573	**585-749**	**648-833**
Copper conductors	-	-	-	522-694	639-863	**788-1085**	**1033-1435**
ESR-MIMS cable on struts	-	-	-	361-430	385-463	550-664	628-760
Double insulated plastic fire resistant cable on struts	-	-	-	414-504	455-559	632-788	714-891
MIMS cable on struts	-	-	-	355-436	385-476	565-701	651-808
Copper bus duct	-	-	-	-	-	584-642	597-656

NOTE : Figures given in **bold** type are based on TWO parallel cables.
Figures apply to installation methods not requiring derating factors (as AS 3008.1).

ELECTRICAL SERVICES

Adel. Bris. Melb.
Per. Syd.
$

MAINS/SUBMAINS INSTALLATIONS (cont'd)

MAINS/SUBMAINS (cont'd) — **CURRENT RATINGS (M.D.)**

100m Nominal Route Length	**50A $/m**	**100A $/m**	**200A $/m**	**400A $/m**	**500A $/m**	**800A $/m**	**1000A $/m**
PVC/XLPE cable installed UNDERGROUND :							
Buried direct (incl. mech. protection) -							
Aluminium conductors	-	-	69-102	136-197	158-226	**207-297**	**296-418**
Copper conductors	39-64	47-75	71-112	156-245	248-383	453-696	**482-740**
In HD PVC conduit -							
Aluminium conductors	-	-	87-137	182-282	198-301	**318-473**	**364-533**
Copper conductors	50-91	60-105	107-174	220-343	321-495	497-772	**637-988**
PVC cable installed in conduit :							
Fixed to structure -							
Copper conductors	39-79	49-92	116-193	324-503	384-596	**555-865**	**772-1201**
PVC/XLPE cable installed :							
On cable tray -							
Aluminium conductors	-	-	130-177	216-296	271-369	**372-506**	-
Copper conductors	99-139	112-157	170-240	310-451	499-728	**428-632**	**820-1192**
2 HOUR FIRE RATED cable on cable ladder :							
ESR-MIMS cable	199-266	220-293	260-344	338-446	430-564	**676-892**	**820-1072**
Double insulated plastic heat resistant cable	188-254	218-292	267-358	393-523	473-630	**776-1032**	**895-1186**
MIMS cable	217-290	241-321	289-381	380-500	479-627	**744-980**	**912-1190**
BUS DUCT with 5 (nominal) bends :							
- Copper conductors	-	-	-	-	-	513-556	527-571

200m Nominal Route Length	**50A $/m**	**100A $/m**	**200A $/m**	**400A $/m**	**500A $/m**	**800A $/m**	**1000A $/m**
PVC/XLPE cable installed UNDERGROUND :							
Buried direct (incl. mech. protection) -							
Aluminium conductors	-	69-102	113-165	**207-297**	**296-418**	-	-
Copper conductors	47-75	71-112	156-245	**307-475**	**389-597**	-	-
In HD PVC conduit							
Aluminium conductors	-	85-134	131-196	**251-378**	**360-525**	-	-
Copper conductors	58-102	83-140	174-274	**349-553**	**430-672**	-	-
PVC cable installed in conduit :							
Fixed to structure							
Copper conductors	44-86	75-132	169-272	**356-568**	**437-687**	-	-
PVC/XLPE cable installed :							
On cable tray							
Aluminium conductors	-	130-177	185-254	**323-443**	**435-590**	-	-
Copper conductors	112-157	149-210	246-356	**463-674**	**575-842**	-	-
2 HOUR FIRE RATED cable on cable ladder :							
ESR-MIMS cable	220-293	260-344	338-446	**676-892**	**820-1072**	-	-
Double insulated plastic fire resistant cable	218-292	267-358	393-523	**776-1032**	**895-1186**	-	-
MIMS	241-321	289-381	380-500	**744-980**	**912-1190**	-	-
BUS DUCT with 6 (nominal) bends :							
Copper conductors	-	-	-	-	-	504-546	515-557

NOTE : Figures given in **bold** type are based on TWO parallel cables.
Figures apply to installation methods not requiring derating factors (as AS 3008.1).

ELECTRICAL SERVICES

Adel. Bris. Melb.
Per. Syd.
$

MAINS/SUBMAINS INSTALLATIONS (cont'd)

MAINS/SUBMAINS (cont'd)

300m Nominal Route Length	CURRENT RATINGS (M.D.) 50A $/m	100A $/m	200A $/m	400A $/m	500A $/m	800A $/m	1000A $/m
PVC/XLPE cable installed UNDERGROUND :							
Buried direct (incl. mech. protection)							
Aluminium conductors	58-85	82-123	**155-223**	-	-	-	-
Copper conductors	53-86	111-176	293-452	**575-882**	**917-1397**	-	-
In HD PVC conduit							
Aluminium conductors	73-116	103-161	**215-312**	-	-	-	-
Copper conductors	66-114	129-206	318-490	**631-977**	**967-1481**	-	-
PVC cable installed in conduit :							
Fixed to structure							
Copper conductors	70-123	120-198	323-502	**645-1003**	**981-1507**	-	-
PVC/XLPE cable installed :							
On cable tray							
Aluminium conductors	117-159	151-206	**271-369**	-	-	-	-
Copper conductors	126-177	197-283	427-620	**810-1179**	**1212-1773**	-	-
2 HOUR FIRE RATED cable on cable ladder :							
ESR-MIMS cable	236-314	314-413	496-648	**952-1240**	-	-	-
Double insulated plastic heat resistant cable	267-358	321-427	641-842	**1231-1606**	-	-	-
MIMS cable	261-347	348-457	556-725	**1066-1386**	-	-	-
BUS DUCT with 10 (nominal) bends :							
Copper conductors	-	-	-	-	-	505-547	518-561

Heavy Current Bus Duct With Copper Conductors	CURRENT RATINGS (M.D.) 1600A $/m	2000A $/m	2500A $/m	3150A $/m	4000A $/m	5600A $/m	7000A $/m
50m Route Length, with 3 (nominal) bends	685-745	1055-1140	1335-1445	1420-1540	1955-2115	2670-2890	4055-4390
100m Route Length, with 5 (nominal) bends	815-885	1040-1125	1325-1435	1685-1825	1940-2100	2630-2845	4005-4335
200m Route Length, with 6 (nominal) bends	800-870	1030-1115	1295-1400	1655-1795	1910-2065	2585-2800	3960-4290
300m Route Length, with 10 (nominal) bends	800-870	1030-1115	1295-1400	1660-1795	1910-2070	2590-2805	3965-4290

NOTE : Figures given in **bold** type are based on TWO parallel cables.
Figures apply to installation methods not requiring derating factors (as AS 3008.1).

ELECTRICAL SERVICES

MAIN SWITCHBOARDS, CUSTOM BUILT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Custom built cabinet by switchboard manufacturer including submains distribution equipment	...	Note	
FACTORY :			
Light load	...	Sum	2,050.00-5,800.00
Medium load	...	Sum	6,000.00-13,750.00
Heavy load	...	Sum	20,500.00-42,000.00
HOME UNITS:			
3 to 6 units	...	Sum	1,215.00-2,950.00
6 to 15 units	...	Sum	2,250.00-4,300.00
15 to 25 units	...	Sum	3,300.00-7,000.00
25 to 50 units	...	Sum	5,400.00-17,750.00
SHOPPING CENTRE :			
Small	...	Sum	7,600.00-15,750.00
Medium	...	Sum	12,000.00-23,500.00
Large	...	Sum	23,500.00-69,000.00
OFFICE BUILDING :			
Small	...	Sum	8,200.00-15,750.00
Medium	...	Sum	12,000.00-21,500.00
Large	...	Sum	20,500.00-53,000.00
HOSPITAL:			
Small	...	Sum	45,500.00-120,000.00
Medium	...	Sum	91,000.00-155,000.00
Large	...	Sum	210,000.00-390,000.00
NURSING HOME :			
Small	...	Sum	15,000.00-23,500.00
Medium	...	Sum	21,250.00-29,500.00
Large	...	Sum	24,250.00-43,000.00
PRIMARY SCHOOL :			
Small	...	Sum	6,000.00-11,750.00
Medium	...	Sum	9,000.00-15,750.00
Large	...	Sum	13,500.00-23,500.00
HIGH SCHOOL :			
Small	...	Sum	15,000.00-21,500.00
Medium	...	Sum	18,250.00-31,500.00
Large	...	Sum	27,000.00-51,000.00
TECHNICAL COLLEGE :			
Small	...	Sum	18,250.00-29,500.00
Medium	...	Sum	24,250.00-43,000.00
Large	...	Sum	60,000.00-105,000.00
UNIVERSITY :			
Small	...	Sum	21,250.00-31,500.00
Medium	...	Sum	30,000.00-59,000.00
Large	...	Sum	60,000.00-235,000.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

DISTRIBUTION SWITCHBOARDS (SUB-BOARDS), CUSTOM BUILT

Custom built cabinet by switchboard manufacturer including final sub-circuit distribution equipment	...	Note	
FACTORY :			
Light load	...	Sum	2,100.00-8,600.00
Medium load	...	Sum	9,000.00-18,500.00
Heavy load	...	Sum	20,500.00-40,000.00
SHOPPING CENTRE :			
Light load	...	Sum	3,300.00-6,600.00
Medium load	...	Sum	6,000.00-9,600.00
Heavy load	...	Sum	9,000.00-12,750.00
OFFICE BUILDING :			
Small	...	Sum	3,300.00-6,400.00
Medium	...	Sum	6,000.00-8,000.00
Large	...	Sum	7,600.00-9,600.00
HOSPITAL :			
Small	...	Sum	4,800.00-6,400.00
Medium	...	Sum	10,500.00-17,500.00
Large	...	Sum	21,250.00-35,000.00
NURSING HOME :			
Small	...	Sum	4,800.00-6,400.00
Medium	...	Sum	9,000.00-11,250.00
Large	...	Sum	12,000.00-16,000.00
PRIMARY SCHOOL :			
Small	...	Sum	1,065.00-2,250.00
Medium	...	Sum	6,000.00-8,000.00
Large	...	Sum	7,600.00-9,600.00
HIGH SCHOOL :			
Small	...	Sum	3,900.00-5,400.00
Medium	...	Sum	6,000.00-8,000.00
Large	...	Sum	9,000.00-14,250.00
TECHNICAL COLLEGE :			
Small	...	Sum	4,500.00-9,600.00
Medium	...	Sum	10,500.00-12,750.00
Large	...	Sum	12,000.00-22,250.00
UNIVERSITY :			
Small	...	Sum	4,500.00-9,000.00
Medium	...	Sum	8,400.00-11,250.00
Large	...	Sum	12,000.00-22,250.00

ELECTRICAL SERVICES

Adel. Bris. Melb. Per. Syd.

PROPRIETARY SWITCHBOARDS

Installed, including connections to circuits	...	Note	
DOMESTIC STYLE SHEET METAL CABINET (METERBOX TYPE)			
Built-in recessed sheet metal box	...	Note	
ADD extra for :			
Circuit breakers (mcb's) - listed separately	...	Note	
Utility and consumer sections can vary with the project's requirements. The following are examples :			
Box with meter / consumer sections and capacity for -			
18 mcb modules only (no space for meters)	...	Sum	130.00-194.00
Two meters only (consumer section is elsewhere)	...	Sum	184.00-270.00
One meter and 24 mcb modules	...	Sum	172.00-255.00
Two meters and 24 mcb modules	...	Sum	245.00-355.00
Two meters and 48 mcb modules	...	Sum	295.00-430.00
Four meters and 36 mcb modules	...	Sum	410.00-600.00
Two meters & 24 mcb modules section and gas section		Sum	500.00-720.00
ADD extra for :			
Circuit breakers (mcb's) - listed separately	...	Note	
Painting, if required	...	Note	
Check with Utility for meter and connection costs	...	Note	
STANDARD PATTERN PROPRIETARY SWITCHBOARDS			
Compact "off the shelf" self-contained DIN rail mcb switchboards with hinged lid/door	...	Note	
ADD extra for :			
Circuit breakers (mcb's) - listed separately	...	Note	
PLASTIC ENCLOSURE, DOMESTIC TYPE			
Surface enclosure with capacity for :			
6 mcb modules	...	Sum	82.00-98.00
11 mcb modules	...	Sum	124.00-146.00
17 mcb modules	...	Sum	148.00-174.00
24 mcb modules	...	Sum	168.00-198.00
30 mcb modules	...	Sum	194.00-230.00
45 mcb modules	...	Sum	235.00-275.00
Recessed, built-in enclosure with capacity for :			
11 mcb modules	...	Sum	152.00-178.00
24 mcb modules	...	Sum	205.00-245.00
36 mcb modules	...	Sum	260.00-305.00
45 mcb modules	...	Sum	285.00-335.00
Surface IP56 weatherproof enclosure with capacity for :			
11 mcb modules	...	Sum	188.00-220.00
24 mcb modules	...	Sum	310.00-365.00
36 mcb modules	...	Sum	440.00-520.00

ELECTRICAL SERVICES

PROPRIETARY SWITCHBOARDS (cont'd)

DOMESTIC STYLE CIRCUIT BREAKERS (MCB'S/RCBOS)

		UNIT	Adel. Bris. Melb. Per. Syd. $
6kA rated, DIN rail mounted, including circuit connections :			
Main switch -			
1 ph 80A mcb (1 module)	...	Sum	22.00-26.00
3 ph 80A mcb (3 modules)	...	Sum	61.00-72.00
1 ph 6A to 63A mcb (1 module)	...	Sum	22.50-32.00
3 ph 6A to 63A mcb (3 modules)	...	Sum	88.00-130.00
RCD's / ELCB's / Safety Switches / RCBO's and other components - refer to page 576	...	Note	

METAL CABINET, COMMERCIAL / INDUSTRIAL TYPE

Surface single phase load centre with in-line 63A main switch :			
12 mcb module capacity	...	Sum	235.00-325.00
16 mcb module capacity	...	Sum	245.00-335.00
32 mcb module capacity	...	Sum	345.00-500.00
48 mcb module capacity	...	Sum	490.00-610.00
Three phase mcb distribution board with 160A main switch :			
18 mcb module capacity	...	Sum	1,010.00-1,260.00
24 mcb module capacity	...	Sum	1,185.00-1,500.00
36 mcb module capacity	...	Sum	1,525.00-1,925.00
48 mcb module capacity	...	Sum	1,650.00-2,100.00
60 mcb module capacity	...	Sum	1,850.00-2,350.00
Three phase mcb distribution board with 250A main switch :			
36 mcb module capacity	...	Sum	2,150.00-2,700.00
48 mcb module capacity	...	Sum	2,350.00-2,950.00
60 mcb module capacity	...	Sum	2,550.00-3,200.00
72 mcb module capacity	...	Sum	2,800.00-3,500.00
Three phase IP66 weatherproof mcb distribution board with 160A main switch :			
24 mcb module capacity	...	Sum	1,950.00-2,400.00
36 mcb module capacity	...	Sum	2,350.00-2,950.00
48 mcb module capacity	...	Sum	2,550.00-3,250.00
60 mcb module capacity	...	Sum	2,850.00-3,600.00
Three phase IP66 weatherproof mcb distribution board with 250A main switch:			
36 mcb module capacity	...	Sum	1,875.00-2,400.00
48 mcb module capacity	...	Sum	2,050.00-2,650.00
60 mcb module capacity	...	Sum	2,300.00-2,950.00
72 mcb module capacity	...	Sum	2,650.00-3,400.00
ADD extra for :			
1 ph 6A to 63A mcb (1 module) -			
6kA	...	Each	20.50-26.00
10kA	...	Each	33.00-41.00
3 ph 6A to 63A mcb (3 modules) -			
6kA	...	Each	52.00-66.00
10kA	...	Each	75.00-92.00
Recess mounting of distribution board into prepared opening, including surround	...	Sum	275.00-345.00
RCD's / ELCB's / Safety Switches / RCBO's and other components - refer to page 576	...	Note	

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

CABLE CARRIERS

CABLE TRAY

Including trapesium brackets	...	Note	
Tray - Admiralty pattern, galvabond :			
100mm	...	m	45.00-61.00
150mm	...	m	50.00-68.00
225mm	...	m	57.00-77.00
300mm	...	m	67.00-90.00
450mm	...	m	92.00-124.00
600mm	...	m	122.00-164.00
ADD extra for :			
90° bend -			
100mm	...	Each	64.00-84.00
150mm	...	Each	67.00-88.00
225mm	...	Each	73.00-96.00
300mm	...	Each	92.00-120.00
450mm	...	Each	152.00-198.00
600mm	...	Each	176.00-230.00
Tee -			
100mm	...	Each	80.00-106.00
150mm	...	Each	84.00-110.00
225mm	...	Each	92.00-120.00
300mm	...	Each	110.00-146.00
450mm	...	Each	176.00-230.00
600mm	...	Each	245.00-320.00
Riser -			
100mm	...	Each	73.00-96.00
150mm	...	Each	76.00-100.00
225mm	...	Each	82.00-108.00
300mm	...	Each	98.00-130.00
450mm	...	Each	148.00-196.00
600mm	...	Each	180.00-240.00

MESH CABLE TRAY

Including trapesium supports and NEMA VE2 compliance	...	Note	
Wire Mesh Tray - Electro zinc plated x 50mm high			
150mm	...	m	51.00-68.00
300mm	...	m	67.00-90.00
450mm	...	m	77.00-102.00
500mm	...	m	90.00-120.00
600mm	...	m	108.00-144.00
ADD extra for :			
Site made horizontal bend -			
150mm	...	Each	77.00-104.00
300mm	...	Each	118.00-158.00
450mm	...	Each	162.00-215.00
500mm	...	Each	198.00-260.00
600mm	...	Each	250.00-330.00
Site made horizontal tee -			
150mm	...	Each	88.00-118.00
300mm	...	Each	134.00-176.00
450mm	...	Each	180.00-235.00
500mm	...	Each	215.00-285.00
600mm	...	Each	270.00-355.00
Site made riser -			
150mm	...	Each	86.00-114.00
300mm	...	Each	130.00-172.00
450mm	...	Each	176.00-230.00
500mm	...	Each	215.00-280.00
600mm	...	Each	270.00-350.00

ELECTRICAL SERVICES

CABLE CARRIERS (cont'd)

CABLE LADDER (BURNDY CSS)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Including basic support brackets	...	Note	
Laddertray - LT3 medium duty :			
LT3H hot dipped galvanised -			
150mm	...	m	134.00-182.00
300mm	...	m	154.00-210.00
450mm	...	m	192.00-260.00
600mm	...	m	245.00-335.00
LT3G galvabond finish – as for galvanised less approximately	...	less/m 6%-13%	
ADD extra for :			
Site made fittings, including the required tray -			
Horizontal bend :			
150mm	...	Each	176.00-240.00
300mm	...	Each	235.00-315.00
450mm	...	Each	350.00-460.00
600mm	...	Each	510.00-670.00
Horizontal tee :			
150mm	...	Each	245.00-325.00
300mm	...	Each	325.00-430.00
450mm	...	Each	470.00-620.00
600mm	...	Each	690.00-910.00
Riser :			
150mm	...	Each	188.00-255.00
300mm	...	Each	255.00-345.00
450mm	...	Each	380.00-510.00
600mm	...	Each	570.00-750.00
Laddertray - LT5 medium duty :			
LT5H hot dipped galvanised -			
150mm	...	m	192.00-265.00
300mm	...	m	220.00-300.00
450mm	...	m	255.00-345.00
600mm	...	m	300.00-400.00
LT5G galvabond finish – as for galvanised less approximately	...	less/m 2%-8%	
ADD extra for :			
Site made fittings, including the required tray -			
Horizontal bend :			
150mm	...	Each	250.00-340.00
300mm	...	Each	325.00-440.00
450mm	...	Each	450.00-600.00
600mm	...	Each	600.00-800.00
Horizontal tee :			
150mm	...	Each	340.00-460.00
300mm	...	Each	440.00-590.00
450mm	...	Each	610.00-810.00
600mm	...	Each	820.00-1,080.00
Riser :			
150mm	...	Each	270.00-365.00
300mm	...	Each	355.00-480.00
450mm	...	Each	500.00-670.00
600mm	...	Each	680.00-900.00

ELECTRICAL SERVICES

CABLE CARRIERS (cont'd)

LADDERTRAY COVERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Cover - flat galvanised to suit Type HLT3 or HLT5 Laddertray :			
150mm	...	m	22.50-29.00
300mm	...	m	28.00-36.50
450mm	...	m	39.00-51.00
600mm	...	m	52.00-67.00
ADD extra for :			
Horizontal bend -			
150mm	...	Each	28.00-37.00
300mm	...	Each	51.00-66.00
450mm	...	Each	98.00-126.00
600mm	...	Each	162.00-210.00
Horizontal tee -			
150mm	...	Each	27.50-36.50
300mm	...	Each	60.00-78.00
450mm	...	Each	120.00-156.00
600mm	...	Each	210.00-270.00
Riser -			
150mm	...	Each	23.50-31.00
300mm	...	Each	51.00-66.00
450mm	...	Each	104.00-132.00
600mm	...	Each	180.00-230.00
NEMA Cable Ladder - medium duty :			
NEMA 1 hot dipped galvanised -			
150mm	...	m	192.00-260.00
300mm	...	m	215.00-290.00
450mm	...	m	235.00-320.00
600mm	...	m	275.00-375.00
ADD extra for :			
Manufactured fittings -			
Horizontal bend :			
150mm	...	Each	285.00-375.00
300mm	...	Each	320.00-420.00
450mm	...	Each	360.00-480.00
600mm	...	Each	400.00-530.00
Horizontal tee :			
150mm	...	Each	470.00-620.00
300mm	...	Each	510.00-670.00
450mm	...	Each	570.00-750.00
600mm	...	Each	620.00-810.00
Riser :			
150mm	...	Each	460.00-600.00
300mm	...	Each	490.00-650.00
450mm	...	Each	550.00-730.00
600mm	...	Each	600.00-790.00

ELECTRICAL SERVICES

CABLE CARRIERS (cont'd)

NEMA CABLE LADDER COVERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Cover - flat, galvanised to suit Type NEMA 1 cable tray :			
150mm	...	m	26.00-34.00
300mm	...	m	37.00-48.00
450mm	...	m	42.00-55.00
600mm	...	m	59.00-77.00
ADD extra for :			
Flat cover to -			
Horizontal bend :			
150mm	...	Each	114.00-148.00
300mm	...	Each	156.00-200.00
450mm	...	Each	162.00-210.00
600mm	...	Each	225.00-290.00
Horizontal tee :			
150mm	...	Each	166.00-215.00
300mm	...	Each	176.00-225.00
450mm	...	Each	215.00-275.00
600mm	...	Each	275.00-350.00
Riser :			
150mm	...	Each	76.00-98.00
300mm	...	Each	90.00-114.00
450mm	...	Each	102.00-132.00
600mm	...	Each	130.00-168.00

CABLE TROUGHING (CABLE DUCT)

Including lid and basic support brackets and nominal number of fittings :			
50 x 50mm	...	m	39.50-51.00
100 x 75mm	...	m	45.00-58.00
150 x 100mm	...	m	60.00-79.00
300 x 75mm	...	m	94.00-124.00

SKIRTING DUCT

Extruded aluminium skirting duct, with powdercoat finish :			
Power and data divided, with drop in front cover -			
100x35mm	...	m	37.00-52.00
125x40mm	...	m	41.00-56.00
150x35mm	...	m	44.00-63.00
150x40mm	...	m	45.00-65.00
150x50mm	...	m	47.00-67.00
Power and data divided, with clip in front cover -			
100x35mm	...	m	42.00-57.00
125x40mm	...	m	41.00-57.00
150x35mm	...	m	47.00-66.00
150x40mm	...	m	48.00-67.00
150x50mm	...	m	50.00-70.00
ADD extra for :			
Cover and shroud section, for outlets (excluding outlet) -			
Single / double power outlet	...	Each	47.00-75.00
Data / telephone outlet plate	...	Each	47.00-75.00
Bend / stop end	...	Each	18.50-35.00
Additional divider	...	m	9.50-15.25

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
CABLE CARRIERS (cont'd)			
SKIRTING DUCT (cont'd)			
Rigid plastic skirting duct / bench duct (in black or white) :			
Power and data divided -			
125x35mm	...	m	47.00-68.00
ADD extra for :			
Cover and shroud section, for outlets (excluding outlet) -			
Single / double power outlet	...	Each	36.00-51.00
Data / telephone outlet plate	...	Each	36.00-51.00
Bend / stop end	...	Each	12.25-17.75
WALL / BENCH DUCT			
Natural anodised or powder coated aluminium body with snap in cover :			
Power or data -			
85 x 50mm	...	m	35.50-52.00
Power and data divided -			
85 x 50mm	...	m	45.00-64.00
ADD extra for :			
Cover and shroud section, for outlets (excluding outlet) -	...	Each	41.00-66.00
Single / double power outlet	...	Each	41.00-66.00
Data / telephone outlet plate	...	Each	41.00-66.00
Bend or End	...	Each	35.50-60.00
SERVICE ACCESS COLUMN			
Nominal 3 metre long natural anodised aluminium column installed floor to ceiling :			
50 x 50 with 1 division	...	Each	250.00-330.00
50 x 50 with 2 divisions	...	Each	280.00-365.00
75 x 50 with 1 division	...	Each	260.00-340.00
75 x 50 with 2 divisions	...	Each	260.00-340.00
75 x 50 with 3 divisions	...	Each	270.00-350.00
80 x 80 with 2 divisions	...	Each	260.00-340.00
100 x 80 with 2 divisions	...	Each	315.00-410.00
ADD extra for :			
Additional length (1m or less)	...	Each	104.00-146.00
Power/lighting accessory mount (excluding accessory)	...	Each	54.00-84.00
Telephone or data accessory mount (excluding accessory)	...	Each	54.00-84.00
Powder coat finish	...	Note	
PEDESTAL POLES			
Short floor/bench mounted post with accessory mounting and cabling access (excluding accessories) :			
390mm high, to suit 4 x standard data/power accessories	...	Each	184.00-220.00
460mm high, to suit 8 x standard data/power accessories	...	Each	205.00-240.00
500mm high, galvanised, to suit 1 x 56 series accessory	...	Each	295.00-400.00
500mm high, galvanised, to suit 2 x 56 series accessories	...	Each	320.00-440.00
500mm high, galvanised, to suit 4 x 56 series accessories	...	Each	390.00-520.00
ADD extra for :			
Under bench service termination box, where required	...	Note	

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

CABLE CARRIERS (cont'd)

INFLOOR TRENCH DUCTING

Three channel flush floor type ducting with removable access lid to suit vinyl tiled floors :			
110mm wide	...	m	90.00-128.00
225mm wide	...	m	230.00-305.00
300mm wide	...	m	240.00-315.00
ADD extra for :			
Elbow -			
110mm wide	...	m	120.00-182.00
225mm wide	...	m	265.00-360.00
300mm wide	...	m	285.00-395.00
Tee -			
110mm wide	...	m	198.00-310.00
225mm wide	...	m	340.00-480.00
300mm wide	...	m	390.00-570.00
Cross -			
110mm wide	...	m	225.00-360.00
225mm wide	...	m	430.00-600.00
300mm wide	...	m	450.00-670.00
Floor outlet boxes (range)	...	Note	

UNDERFLOOR TROUGHING

Flush carpeted floor type fixed in floor recess with grano screeding :			
1 channel	...	m	54.00-82.00
2 channel	...	m	71.00-104.00
3 channel	...	m	92.00-130.00
ADD extra for :			
Junction box (nominal 3 way) -			
1 channel	...	Each	118.00-210.00
2 channel	...	Each	164.00-270.00
3 channel	...	Each	205.00-325.00

FIRE STOPPING

PVC conduit, cable tray and cable troughing may require fire stopping and smoke sealing where penetrating fire walls/ceilings and floor slabs. Refer to FIREPROOFING SECTION pages 673 and 674 for prices	...	Note	

MINI PVC TRUNKING

Including nominal number of fittings :			
16 x 10mm	...	m	11.75-16.00
16 x 16mm	...	m	12.75-17.00
25 x 16mm	...	m	15.50-21.00
25 x 25mm	...	m	16.75-22.00
25 x 50mm	...	m	26.00-34.00
40 x 16mm	...	m	24.00-32.00
40 x 25mm	...	m	27.50-36.50
50 x 25mm	...	m	31.50-42.00

ELECTRICAL SERVICES

CABLE CARRIERS (cont'd)

CONDUITS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Including nominal number of fittings :	...	Note	
MEDIUM DUTY GREY UPVC (formerly called LD)			
MD UPVC fixed to masonry :			
20mm dia.	...	m	11.50-16.00
25mm dia.	...	m	14.00-19.25
32mm dia.	...	m	16.50-23.00
40mm dia.	...	m	18.50-25.50
50mm dia.	...	m	21.00-29.50
MD UPVC cast into concrete :			
20mm dia.	...	m	4.95-6.75
25mm dia.	...	m	6.25-8.75
32mm dia.	...	m	9.00-12.50
MD UPVC installed in roof space :			
20mm dia.	...	m	8.25-32.00
25mm dia.	...	m	10.25-33.50
32mm dia.	...	m	12.50-38.50
40mm dia.	...	m	14.25-42.00
50mm dia.	...	m	15.50-43.00
FLEXIBLE CORRUGATED - MEDIUM DUTY			
Corrugated MD UPVC installed in roof space :			
20mm dia.	...	m	8.25-12.00
25mm dia.	...	m	10.25-14.50
32mm dia.	...	m	12.50-17.75
Corrugated MD UPVC cast into concrete :			
20mm dia.	...	m	5.00-7.50
25mm dia.	...	m	6.50-9.25
32mm dia.	...	m	9.25-13.25
FLEXIBLE CORRUGATED - HEAVY DUTY			
Corrugated HD UPVC installed in roof space :			
20mm dia.	...	m	9.00-12.75
25mm dia.	...	m	11.00-15.75
32mm dia.	...	m	13.50-19.00
Corrugated HD UPVC cast into concrete :			
20mm dia.	...	m	5.75-8.25
25mm dia.	...	m	7.50-10.50
32mm dia.	...	m	10.25-14.25
Corrugated HD UPVC buried underground excluding trenching :			
20mm dia.	...	m	4.95-7.25
25mm dia.	...	m	6.00-8.50
32mm dia.	...	m	7.25-10.25
HFT (Halogen free, fire resistant, temperature stable) rigid black non-metallic conduit fixed to masonry :			
20mm dia.	...	m	21.00-27.50
25mm dia.	...	m	24.50-32.00
32mm dia.	...	m	33.50-42.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
CABLE CARRIERS (cont'd)			
CONDUITS (cont'd)			
GALVANISED STEEL			
Fixed to masonry :			
20mm dia.	...	m	19.75-27.00
25mm dia.	...	m	27.50-37.50
32mm dia.	...	m	39.00-53.00
40mm dia.	...	m	48.00-65.00
50mm dia.	...	m	57.00-75.00
Cast into concrete :			
20mm dia.	...	m	14.00-18.75
25mm dia.	...	m	19.25-26.00
32mm dia.	...	m	28.00-37.50
Installed in roof space :			
20mm dia.	...	m	16.50-40.00
25mm dia.	...	m	22.50-44.00
32mm dia.	...	m	32.50-55.00
40mm dia.	...	m	41.00-67.00
50mm dia.	...	m	48.00-74.00
HEAVY DUTY (AS 2053-2)			
HD orange UPVC buried underground excluding trenching and bends :			
20mm dia. (O.D.)	...	m	3.45-4.65
25mm dia. (O.D.)	...	m	3.70-4.95
32mm dia. (O.D.)	...	m	5.00-6.75
40mm dia. (O.D.)	...	m	5.75-7.75
50mm dia. (O.D.)	...	m	7.50-10.00
63mm dia. (O.D.)	...	m	9.25-12.25
80mm dia. (I.D.)	...	m	12.50-16.25
100mm dia. (I.D.)	...	m	19.50-25.00
125mm dia. (I.D.)	...	m	29.50-37.50
150mm dia. (I.D.)	...	m	29.50-38.00
22.5° sweep bend to HD orange UPVC conduit :			
80mm dia. (O.D.) - radius 584mm	...	Each	21.00-32.00
100mm dia. (I.D.) - radius 584mm	...	Each	30.00-46.00
100mm dia. (I.D.) - radius 1200mm	...	Each	39.50-61.00
150mm dia. (I.D.) - radius 635mm	...	Each	70.00-110.00
45° sweep bend to HD orange UPVC conduit :			
20mm dia. (O.D.)	...	Each	6.00-8.75
25mm dia. (O.D.)	...	Each	7.50-11.00
32mm dia. (O.D.)	...	Each	9.50-14.00
40mm dia. (O.D.)	...	Each	11.00-16.25
50mm dia. (O.D.)	...	Each	12.75-18.75
63mm dia. (O.D.)	...	Each	19.50-29.00
80mm dia. (I.D.) - radius 356mm	...	Each	23.00-34.50
80mm dia. (I.D.) - radius 550mm	...	Each	25.00-38.00
100mm dia. (I.D.) - radius 600mm	...	Each	32.00-49.00
100mm dia. (I.D.) - radius 1200mm	...	Each	48.00-73.00
125mm dia. (I.D.) - radius 600mm	...	Each	68.00-106.00
150mm dia. (I.D.) - radius 635mm	...	Each	88.00-136.00

ELECTRICAL SERVICES

CABLE CARRIERS (cont'd)

CONDUITS (cont'd)

HEAVY DUTY (AS 2053-2) (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
90° sweep bend to HD orange UPVC conduit :			
20mm dia. (O.D.)	...	Each	7.00-10.25
25mm dia. (O.D.)	...	Each	7.75-11.50
32mm dia. (O.D.)	...	Each	10.25-15.00
40mm dia. (O.D.)	...	Each	12.25-18.25
50mm dia. (O.D.)	...	Each	14.25-21.50
63mm dia. (O.D.)	...	Each	21.50-32.50
80mm dia. (I.D.) - radius 356mm	...	Each	26.00-39.50
80mm dia. (I.D.) - radius 550mm	...	Each	28.50-43.00
100mm dia. (I.D.) - radius 600mm	...	Each	45.00-69.00
100mm dia. (I.D.) - radius 1200mm	...	Each	66.00-102.00
125mm dia. (I.D.) - radius 635mm	...	Each	82.00-128.00
150mm dia. (I.D.) - radius 760mm	...	Each	106.00-166.00
150mm dia. (I.D.) - radius 1200mm	...	Each	225.00-355.00
HD Corflo orange UPVC buried underground excluding trenching and bends :			
100mm dia. (I.D.)	...	m	14.75-22.00
125mm dia. (I.D.)	...	m	18.25-27.50
150mm dia. (I.D.)	...	m	22.50-34.50
ADD extra for :			
Trenching and clean fill - refer to Drainage pages 489 and 490	...	Note	

LIGHT DUTY ORANGE CONDUIT (previously known as ducting)

UPVC electrical orange conduit (not heavy duty) buried underground excluding trenching and bends :			
80mm dia. (I.D.)	...	m	9.50-14.00
100mm dia. (I.D.)	...	m	11.75-17.50
125mm dia. (I.D.)	...	m	16.25-24.50
150mm dia. (I.D.)	...	m	20.50-31.00

COMMUNICATIONS CONDUIT (AS/NZ 2053) (AS/ACIF S008)

Approved white, branded, telephone and communications conduit buried underground excluding trench :			
MD UPVC conduit -			
20mm dia. (O.D.)	...	m	3.40-4.80
25mm dia. (O.D.)	...	m	3.60-5.00
32mm dia. (O.D.)	...	m	5.00-7.00
50mm dia. (O.D.)	...	m	7.75-11.00
80mm dia. (I.D.)	...	m	12.50-17.75
100mm dia. (I.D.)	...	m	20.50-29.00
Corflo UPVC conduit -			
100mm dia. (I.D.)	...	m	14.75-22.00
125mm dia. (I.D.)	...	m	18.25-27.50
150mm dia. (I.D.)	...	m	22.50-34.50
ADD extra for :			
Trenching and clean fill - refer to Drainage pages 489 and 490	...	Note	

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

CABLE CARRIERS (cont'd)

CONDUITS (cont'd)

COMMUNICATIONS CONDUIT (AS/NZ 2053) (AS/ACIF S008) (cont'd)

45° sweep bend to white communications conduit :			
20mm dia. (O.D.)	...	Each	8.50-12.00
25mm dia. (O.D.)	...	Each	8.50-12.25
32mm dia. (O.D.)	...	Each	12.00-17.25
50mm dia. (O.D.)	...	Each	14.25-20.50
80mm dia. (I.D.)	...	Each	41.00-60.00
100mm dia. (I.D.)	...	Each	69.00-102.00
90° sweep bend to white communications conduit :			
20mm dia. (O.D.)	...	Each	8.50-12.25
25mm dia. (O.D.)	...	Each	8.75-12.50
32mm dia. (O.D.)	...	Each	12.75-18.50
50mm dia. (O.D.)	...	Each	16.25-23.50
80mm dia. (I.D.)	...	Each	43.00-64.00
100mm dia. (I.D.) - radius 760mm	...	Each	84.00-124.00

WHITE COMMUNICATIONS PIPE (AS 1477)

Class 9 white, branded, UPVC pressure pipe for telephone and communications conduit buried underground excluding trenching and bends :			
20mm dia. (I.D.)	...	m	4.05-6.00
25mm dia. (I.D.)	...	m	4.30-6.50
32mm dia. (I.D.)	...	m	7.00-10.75
40mm dia. (I.D.)	...	m	6.25-9.75
50mm dia. (I.D.)	...	m	8.75-13.25
65mm dia. (I.D.)	...	m	19.25-31.00
80mm dia. (I.D.)	...	m	15.50-25.00
100mm dia. (I.D.)	...	m	20.00-31.50
125mm dia. (I.D.)	...	m	30.50-49.00
150mm dia. (I.D.)	...	m	39.50-63.00
ADD extra for :			
Trenching and clean fill - refer to DRAINAGE pages 489 and 490	...	Note	
45° sweep bend to white UPVC pressure pipe :			
20mm dia. (I.D.)	...	Each	17.75-24.00
25mm dia. (I.D.)	...	Each	22.50-31.00
32mm dia. (I.D.)	...	Each	26.50-36.00
50mm dia. (I.D.)	...	Each	30.00-41.00
80mm dia. (I.D.)	...	Each	78.00-108.00
100mm dia. (I.D.) - radius 584mm	...	Each	108.00-148.00
150mm dia. (I.D.) - radius 635mm	...	Each	300.00-410.00
90° sweep bend to white UPVC pressure pipe :			
20mm dia. (I.D.)	...	Each	16.75-23.00
25mm dia. (I.D.)	...	Each	50.00-69.00
32mm dia. (I.D.)	...	Each	58.00-79.00
40mm dia. (I.D.)	...	Each	31.50-43.00
50mm dia. (I.D.)	...	Each	37.00-51.00
80mm dia. (I.D.)	...	Each	82.00-114.00
100mm dia. (I.D.) - radius 457mm	...	Each	158.00-215.00
100mm dia. (I.D.) - radius 584mm	...	Each	142.00-194.00
150mm dia. (I.D.) - radius 635mm	...	Each	430.00-590.00

ELECTRICAL SERVICES

CABLE PITS

		UNIT	Adel. Bris. Melb. Per. Syd. $
ACO CABLEMATE - Plastic PE pit with (Class A) concrete lid :			
Type 1 - 430 x 150 x 355mm deep	...	Each	148.00-192.00
Type 2 - 570 x 200 x 520mm deep	...	Each	182.00-235.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	98.00-166.00
Steel Urbanfil/Pavermate recessed sealed cover	...	Each	680.00-950.00
Type 53 - 460 x 215 x 430mm deep	...	Each	152.00-196.00
Type 43 - 340 dia. x 545mm deep	...	Each	184.00-240.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	124.00-215.00
Type 55 - 450 x 450 x 450mm deep	...	Each	275.00-350.00
Type 77 - 600 x 600 x 605mm deep	...	Each	375.00-480.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	182.00-490.00
Steel PowerLok lid	...	Each	380.00-560.00
Steel Urbanfil/Pavermate recessed sealed cover	...	Each	820.00-1,395.00
Type 3 - 455 x 225 x 475mm deep	...	Each	215.00-275.00
Type 4 - 605 x 300 x 745mm deep	...	Each	235.00-300.00
Type 5 - 655 x 355 x 580mm deep	...	Each	240.00-305.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	152.00-220.00
Steel PowerLok lid	...	Each	300.00-385.00
Steel Urbanfil/Pavermate recessed sealed cover	...	Each	730.00-1,110.00
Type 6 - 1270 x 455 x 615mm deep	...	Each	800.00-1,010.00
Type 7 - 910 x 455 x 1090 deep	...	Each	740.00-930.00
Type 8 - 1270 x 455 x 835mm deep	...	Each	880.00-1,110.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	192.00-365.00
Steel PowerLok lid	...	Each	530.00-680.00
Steel Urbanfil/Pavermate recessed sealed cover	...	Each	1,335.00-2,250.00
Type 9 - 1925 x 455 x 835mm deep	...	Each	2,250.00-2,800.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	...	Each	850.00-1,050.00
Steel PowerLok lid	...	Each	1,170.00-1,455.00
Steel Urbanfil/Pavermate recessed sealed cover	...	Each	1,800.00-2,850.00
ADD extra for :			
Concrete collars and surround to pits where required and paving filling to Urbanfil/Pavermate recessed covers	...	Note	

ELECTRICAL SERVICES

CABLE PITS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ACO CABLEMATE - Polymer Concrete pit with polymer (Class A) concrete lid :			
Type 33 - 280 x 280 x 405mm deep	…	Each	220.00-280.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	…	Each	116.00-152.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	630.00-820.00
Type 52 - 430 x 145 x 395mm deep	…	Each	235.00-305.00
Type 63 - 585 x 235 x 690mm deep	…	Each	380.00-480.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	…	Each	142.00-220.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	680.00-930.00
Type 45 - 435 x 435 x 530mm deep	…	Each	365.00-460.00
Type 66 - 590 x 590 x 550mm deep	…	Each	550.00-690.00
Type 66H - 580 x 580 x 840mm deep	…	Each	670.00-850.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	…	Each	200.00-275.00
Steel PowerLok lid	…	Each	400.00-620.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	630.00-1,065.00
Type 95 - 895 x 435 x 520mm deep	…	Each	580.00-730.00
Type 96 - 900 x 600 x 550mm deep	…	Each	710.00-890.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	…	Each	280.00-365.00
Steel PowerLok lid	…	Each	530.00-710.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	920.00-1,245.00
Type 99 - 875 x 875 x 510mm deep	…	Each	990.00-1,260.00
ADD extra for :			
Pit with AS 3996 Class B lid :			
Steel lid	…	Each	500.00-620.00
Steel PowerLok lid	…	Each	720.00-900.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	960.00-1,320.00
Type 8 - 1265 x 455 x 840mm deep	…	Each	950.00-1,215.00
ADD extra for :			
Pit with AS 3996 Class B lid -			
Steel lid	…	Each	290.00-365.00
Steel PowerLok lid	…	Each	540.00-680.00
Steel Urbanfil/Pavermate recessed sealed cover	…	Each	1,335.00-2,000.00
ADD extra for :			
Concrete collars and surround to pits where required and paving filling to Urbanfil/Pavermate recessed covers	…	Note	

ELECTRICAL SERVICES

CABLE PITS (cont'd)

			Adel. Bris. Melb. Per. Syd.
Fibre cement cable pit with concrete lid :			
450 x 300 x 400mm deep (Main Earth pit)	...	Each	146.00-194.00
560 x 330 x 525mm deep	...	Each	160.00-210.00
715 x 400 x 820mm deep	...	Each	230.00-295.00
760 x 700 x 825mm deep	...	Each	395.00-500.00
1350 x 575 x 900mm deep	...	Each	560.00-690.00
900 x 900 x 1000mm deep	...	Each	550.00-690.00
1280 x 900 x 1000mm deep	...	Each	800.00-980.00
760 x 700 x 825mm deep (Medium Duty 4000kg wheel load)	...	Each	1,305.00-1,575.00
900 x 900 x 1000mm deep (Medium Duty 4000kg wheel load)	...	Each	1,425.00-1,700.00
1500 x 1500 x 1000mm deep (Medium Duty 4000kg wheel load)	...	Each	4,700.00-5,600.00
1280 x 900 x 1000mm deep (Heavy Duty 9000kg wheel load)	...	Each	3,000.00-3,600.00
ADD extra for :			
Concrete collars and surround to pits where required	...	Note	

CABLE PROTECTION COVERS (SLABS)

CATEGORY B SYSTEM INSTALLATION

Cable covers laid over cables excluding trenching :			
Polyethylene slabs, 3mm thick x 2m long -			
150mm wide	...	m	13.25-18.00
300mm wide	...	m	25.00-34.00
Polyethylene rolls, 5mm thick x 25m long -			
150mm wide	...	m	8.25-11.25
200mm wide	...	m	10.00-13.50
300mm wide	...	m	13.75-19.00
ADD extra for :			
Trenching and clean fill - refer to Drainage pages 489 and 490	...	Note	

CABLE PROTECTORS (ON FLOOR)

Cable protector (to reduce trip hazard and for temporary protection) :			
Plastic, laid on floor -			
One channel, 3m run	...	Each	57.00-69.00
Two channel, 3m run	...	Each	77.00-90.00

ELECTRICAL SERVICES

BUSWAY (Busbar Trunking Systems)

			UNIT	Adel. Bris. Melb. Per. Syd. $
Low impedance copper Electrobar totally enclosed within powder coated sheet metal ducting, fixed to building structure, including brackets :				
Feeder busway -				
800A	- busway section	...	m	480.00-640.00
	- straight flanged end	...	Each	395.00-530.00
	- elbow	...	Each	460.00-620.00
1000A	- busway section	...	m	490.00-660.00
	- straight flanged end	...	Each	395.00-530.00
	- elbow	...	Each	460.00-620.00
1250A	- busway section	...	m	530.00-700.00
	- straight flanged end	...	Each	395.00-530.00
	- elbow	...	Each	460.00-620.00
1600A	- busway section	...	m	630.00-840.00
	- straight flanged end	...	Each	395.00-530.00
	- elbow	...	Each	460.00-620.00
2000A	- busway section	...	m	810.00-1,095.00
	- straight flanged end	...	Each	430.00-560.00
	- elbow	...	Each	520.00-660.00
2500A	- busway section	...	m	1,010.00-1,350.00
	- straight flanged end	...	Each	770.00-1,020.00
	- elbow	...	Each	890.00-1,185.00
3150A	- busway section	...	m	1,305.00-1,750.00
	- straight flanged end	...	Each	840.00-1,125.00
	- elbow	...	Each	960.00-1,290.00
4000A	- busway section	...	m	1,500.00-2,000.00
	- straight flanged end	...	Each	970.00-1,305.00
	- elbow	...	Each	1,065.00-1,425.00
5600A	- busway section	...	m	2,050.00-2,600.00
	- straight flanged end	...	Each	1,080.00-1,600.00
	- elbow	...	Each	1,410.00-1,900.00
7000A	- busway section	...	m	3,150.00-4,200.00
	- straight flanged end	...	Each	1,320.00-1,775.00
	- elbow	...	Each	1,525.00-2,050.00
ADD extra for :				
Waterproof type		...	Plus/m 10%	

ELECTRICAL SERVICES

BUSWAY (Busbar Trunking Systems) (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Low impedance copper Electrobar totally enclosed within powder coated sheet metal ducting, fixed to building structure, including brackets (cont'd) :			
Plug-in busway -			
Busbar section :			
80A	...	m	104.00-130.00
100A	...	m	128.00-154.00
125A	...	m	140.00-166.00
160A	...	m	156.00-186.00
210A	...	m	210.00-245.00
315A	...	m	235.00-270.00
440A	...	m	315.00-375.00
540A	...	m	440.00-510.00
640A	...	m	540.00-630.00
700A	...	m	680.00-820.00
ADD extra for :			
Plug-in take-off box with MCB/RCBO -			
Single phase :			
10A to 32A	...	Each	144.00-180.00
Three phase :			
10A to 32A	...	Each	210.00-255.00
40A to 63A	...	Each	320.00-380.00
80A to 100A	...	Each	540.00-670.00

ELECTRICAL SERVICES

			UNIT	Adel. Bris. Melb. Per. Syd. $

CABLES

Prices are for cable installed in position including fixings but EXCLUDING cable carriers (e.g. excluding conduit, cable tray or struts) ... Note

ALUMINIUM

One x single core XLPE/PVC cable (single double insulated / SDI) drawn into conduit :

95sqmm	(37/1.78)	...	m	10.50-14.00
120sqmm	(37/2.03)	...	m	13.50-18.25
150sqmm	(37/2.25)	...	m	14.00-18.50
185sqmm	(37/2.52)	...	m	16.00-21.50
240sqmm	(61/2.25)	...	m	20.00-27.00
300sqmm	(61/2.52)	...	m	22.50-30.00
400sqmm	(61/2.85)	...	m	28.50-38.50
500sqmm	(61/3.20)	...	m	35.00-47.00
630sqmm	(127/2.52)	...	m	40.00-53.00

One x single core XLPE/PVC (SDI) cable installed on cable tray :

95sqmm	(37/1.78)	...	m	17.00-23.00
120sqmm	(37/2.03)	...	m	20.50-28.00
150sqmm	(37/2.25)	...	m	22.50-30.00
185sqmm	(37/2.52)	...	m	25.50-34.50
240sqmm	(61/2.25)	...	m	30.50-41.00
300sqmm	(61/2.52)	...	m	33.00-45.00
400sqmm	(61/2.85)	...	m	40.00-54.00
500sqmm	(61/3.20)	...	m	49.00-66.00
630sqmm	(127/2.52)	...	m	57.00-77.00

COPPER

One x single core TPI cable (single insulated PVC cable / building wire / earth wire) drawn into conduit :

1.5sqmm	(7/0.50)	...	m	0.90-1.30
2.5sqmm	(7/0.67)	...	m	1.10-1.60
4sqmm	(7/0.85)	...	m	1.65-2.40
6sqmm	(7/1.04)	...	m	1.95-2.85
10sqmm	(7/1.35)	...	m	2.75-4.05
16sqmm	(7/1.70)	...	m	3.65-5.50
25sqmm	(19/1.35)	...	m	6.25-9.25
35sqmm	(19/1.53)	...	m	7.50-11.00
50sqmm	(19/1.78)	...	m	11.25-16.75
70sqmm	(19/2.14)	...	m	15.00-22.50
95sqmm	(37/1.78)	...	m	19.00-28.50
120sqmm	(37/2.03)	...	m	25.00-37.50
150sqmm	(37/2.25)	...	m	31.50-47.00

One x single core PVC/PVC cable (single double insulated / SDI) drawn into conduit :

1.5sqmm	(7/0.50)	...	m	1.25-1.80
2.5sqmm	(7/0.67)	...	m	1.50-2.15
4sqmm	(7/0.85)	...	m	2.05-3.00
6sqmm	(7/1.04)	...	m	2.60-3.85
10sqmm	(7/1.35)	...	m	3.40-5.00
16sqmm	(7/1.70)	...	m	4.20-6.25

ELECTRICAL SERVICES

			UNIT	Adel. Bris. Melb. Per. Syd. $
CABLES (cont'd)				
COPPER (cont'd)				
One x single core XLPE/PVC (SDI) cable drawn into conduit :				
25sqmm	(19/1.35)	...	m	7.25-10.75
35sqmm	(19/1.53)	...	m	9.50-14.00
50sqmm	(19/1.78)	...	m	10.25-15.50
70sqmm	(19/2.14)	...	m	14.25-21.50
95sqmm	(37/1.78)	...	m	20.50-30.50
120sqmm	(37/2.03)	...	m	25.50-38.50
150sqmm	(37/2.25)	...	m	29.50-44.00
185sqmm	(37/2.52)	...	m	36.50-55.00
240sqmm	(61/2.25)	...	m	48.00-72.00
300sqmm	(61/2.52)	...	m	61.00-92.00
400sqmm	(61/2.85)	...	m	74.00-112.00
500sqmm	(61/3.20)	...	m	92.00-140.00
630sqmm	(127/2.52)	...	m	122.00-184.00
One x single core XLPE/PVC (SDI) cable fixed to cable tray :				
50sqmm	(19/1.78)	...	m	21.00-30.00
70sqmm	(19/2.14)	...	m	26.50-38.00
95sqmm	(37/1.78)	...	m	33.50-48.00
120sqmm	(37/2.03)	...	m	39.50-58.00
150sqmm	(37/2.25)	...	m	45.00-66.00
185sqmm	(37/2.52)	...	m	53.00-78.00
240sqmm	(61/2.25)	...	m	69.00-102.00
300sqmm	(61/2.52)	...	m	86.00-126.00
400sqmm	(61/2.85)	...	m	104.00-152.00
500sqmm	(61/3.20)	...	m	128.00-188.00
630sqmm	(127/2.52)	...	m	160.00-235.00
Single core TPS cable installed in roof space :				
1sqmm	(1/1.13)	...	m	1.00-1.55
1.5sqmm	(7/0.50)	...	m	1.10-1.70
2.5sqmm	(7/0.67)	...	m	1.30-2.00
4sqmm	(7/0.85)	...	m	1.70-2.55
6sqmm	(7/1.04)	...	m	2.30-3.50
10sqmm	(7/1.35)	...	m	3.15-4.80
Two core flat TPS cable installed in roof space :				
1sqmm	(1/1.13)	...	m	1.35-2.05
1.5sqmm	(7/0.50)	...	m	1.50-2.30
2.5sqmm	(7/0.67)	...	m	1.95-2.95
4sqmm	(7/0.85)	...	m	2.65-4.00
6sqmm	(7/1.04)	...	m	3.55-5.50
10sqmm	(7/1.35)	...	m	6.75-10.25
16sqmm	(7/1.70)	...	m	10.00-15.25
Two core + E flat TPS cable installed in roof space :				
1sqmm	(1/1.13)	...	m	1.85-2.65
1.5sqmm	(7/0.50)	...	m	2.00-2.90
2.5sqmm	(1/1.78)	...	m	2.90-4.20
2.5sqmm	(7/0.67)	...	m	2.65-3.90
4sqmm	(7/0.85)	...	m	3.50-5.25
6sqmm	(7/1.04)	...	m	4.25-6.25
10sqmm	(7/1.35)	...	m	8.75-13.00
16sqmm	(7/1.70)	...	m	10.00-15.00
Three core + E circular PVC/PVC cable installed in roof space :				
1.5sqmm	(7/0.50)	...	m	3.45-5.25
2.5sqmm	(7/0.67)	...	m	4.95-7.50
4sqmm	(7/0.85)	...	m	6.75-10.25
6sqmm	(7/1.04)	...	m	7.25-11.00
10sqmm	(7/1.35)	...	m	10.75-16.25

ELECTRICAL SERVICES

			UNIT	Adel. Bris. Melb. Per. Syd. $
CABLES (cont'd)				
COPPER (cont'd)				
Four core + E circular PVC/PVC cable installed in roof space :				
1.5sqmm	(7/0.50)	...	m	3.90-5.75
2.5sqmm	(7/0.76)	...	m	5.25-7.50
4sqmm	(7/0.85)	...	m	7.25-10.50
6sqmm	(7/1.04)	...	m	9.25-13.75
10sqmm	(7/1.35)	...	m	13.50-19.75
16sqmm	(7/1.70)	...	m	19.00-28.00
Drawn into conduit -				
1.5sqmm	(7/0.50)	...	m	3.90-5.75
2.5sqmm	(7/0.76)	...	m	4.80-7.00
4sqmm	(7/0.85)	...	m	6.75-10.00
6sqmm	(7/1.04)	...	m	8.75-13.00
10sqmm	(7/1.35)	...	m	12.25-18.25
16sqmm	(7/1.70)	...	m	18.00-27.00
Four core + E circular XLPE/PVC cable installed in roof space :				
25sqmm	(19/1.35)	...	m	29.50-44.00
35sqmm	(19/1.53)	...	m	39.00-59.00
50sqmm	(19/1.78)	...	m	57.00-86.00
70sqmm	(19/2.14)	...	m	82.00-122.00
Drawn into conduit -				
25sqmm	(19/1.35)	...	m	28.00-42.00
35sqmm	(19/1.53)	...	m	38.00-57.00
50sqmm	(19/1.78)	...	m	55.00-82.00
70sqmm	(19/2.14)	...	m	78.00-118.00
CIRCULAR MULTICORE, COPPER				
1.5sqmm PVC/PVC cable, fixed to cable tray :				
6 core +E		...	m	10.50-15.00
8 core +E		...	m	14.00-19.75
10 core +E		...	m	14.75-21.00
12 core +E		...	m	16.75-24.00
15 core +E		...	m	18.50-26.50
20 core +E		...	m	21.00-30.50
25 core +E		...	m	25.00-36.50
2.5sqmm PVC/PVC cable, fixed to cable tray :				
6 core +E		...	m	14.50-21.00
8 core +E		...	m	17.50-25.50
10 core +E		...	m	20.50-30.00
12 core +E		...	m	23.00-32.50
15 core +E		...	m	24.50-35.50
20 core +E		...	m	30.50-44.00
25 core +E		...	m	34.50-50.00
STEEL WIRED ARMOURED, COPPER				
Two core + E PVC insulated SWA cable, fixed to cable tray :				
1.5sqmm	(7/0.50)	...	m	12.50-17.75
2.5sqmm	(1/1.78)	...	m	14.50-20.50
4sqmm	(7/0.85)	...	m	16.00-23.00
6sqmm	(7/1.04)	...	m	19.75-28.00
10sqmm	(7/1.35)	...	m	25.00-35.50
16sqmm	(7/1.70)	...	m	29.50-43.00

ELECTRICAL SERVICES

			UNIT	Adel. Bris. Melb. Per. Syd. $

CABLES (cont'd)

STEEL WIRED ARMOURED, COPPER (cont'd)

Three core + E PVC insulated SWA cable, fixed to cable tray :

1.5sqmm	(7/0.50)	...	m	14.00-19.75
2.5sqmm	(7/0.76)	...	m	16.00-22.50
4sqmm	(7/0.85)	...	m	20.00-28.50
6sqmm	(7/1.04)	...	m	21.50-31.00
10sqmm	(7/1.35)	...	m	28.50-41.00
16sqmm	(7/1.70)	...	m	35.50-52.00
25sqmm	(19/1.35; XLPE/PVC)	...	m	47.00-69.00
35sqmm	(19/1.53; XLPE/PVC)	...	m	59.00-86.00
50sqmm	(19/1.78; XLPE/PVC)	...	m	77.00-114.00

Four core + E PVC insulated SWA cable, fixed to cable tray :

1.5sqmm	(7/0.50)	...	m	15.25-22.00
2.5sqmm	(7/0.76)	...	m	18.75-27.00
4sqmm	(7/0.85)	...	m	20.50-29.50
6sqmm	(7/1.04)	...	m	25.50-37.00
10sqmm	(7/1.35)	...	m	32.50-47.00
16sqmm	(7/1.70)	...	m	42.00-61.00
25sqmm	(19/1.35; XLPE/PVC)	...	m	58.00-86.00
35sqmm	(19/1.53; XLPE/PVC)	...	m	73.00-106.00
50sqmm	(19/1.78; XLPE/PVC)	...	m	96.00-140.00

HIGH FIRE RESISTANT

(2 hour fire and mechanically rated cables tested to AS 3013-2005)
RADOX FR-3013 flame retardant, halogen free, low smoke,
125°C rated flexible cables with copper conductors

FR telephone and data cable (Krone compatible) :
0.37mm indoor unscreened type, fixed to cable tray -

2 pair	...	m	10.00-13.00
3 pair	...	m	12.25-16.00
4 pair	...	m	15.25-19.75
5 pair	...	m	18.00-23.50
6 pair	...	m	20.50-26.50
10 pair	...	m	31.00-40.00
15 pair	...	m	43.00-55.00
20 pair	...	m	54.00-70.00
25 pair	...	m	65.00-84.00
35 pair	...	m	88.00-112.00
50 pair	...	m	124.00-158.00
100 pair	...	m	235.00-295.00

Communications 300V/50V double insulated circular cable :
One pair FR cable, fixed to cable tray -

0.50sqmm	(7/0.30)	...	m	8.50-11.25
0.75sqmm	(7/0.37)	...	m	9.00-12.00
1.00sqmm	(7/0.42)	...	m	10.50-14.00
1.50sqmm	(7/0.53)	...	m	12.25-16.25

ELECTRICAL SERVICES

CABLES (cont'd)

HIGH FIRE RESISTANT (cont'd)

			UNIT	Adel. Bris. Melb. Per. Syd. $
RADOX FR - 3013 flame retardant, etc., cables (cont'd)				
Communications 300V/50V double insulated circular cable (cont'd) :				
Two pair FR cable, fixed to cable tray -				
0.50sqmm	(7/0.30)	...	m	11.00-14.25
0.75sqmm	(7/0.37)	...	m	11.50-15.00
1.0 sqmm	(7/0.42)	...	m	11.00-14.50
1.5 sqmm	(7/0.53)	...	m	13.25-17.75
Three pair FR cable, fixed to cable tray -				
0.50sqmm	(7/0.30)	...	m	11.75-15.25
0.75sqmm	(7/0.37)	...	m	12.50-16.50
1.0 sqmm	(7/0.42)	...	m	13.00-17.00
1.5 sqmm	(7/0.53)	...	m	15.75-20.50
Five pair FR cable, fixed to cable tray -				
0.50sqmm	(7/0.30)	...	m	17.75-23.00
0.75sqmm	(7/0.37)	...	m	19.00-25.00
1.0 sqmm	(7/0.42)	...	m	22.50-29.50
1.5 sqmm	(7/0.53)	...	m	26.50-34.50
Seven pair FR cable, fixed to cable tray -				
0.50sqmm	(7/0.30)	...	m	21.00-27.00
0.75sqmm	(7/0.37)	...	m	24.00-31.50
1.0 sqmm	(7/0.42)	...	m	28.50-37.00
1.5 sqmm	(7/0.53)	...	m	36.00-46.00
Ten pair FR cable, fixed to cable tray -				
0.50sqmm	(7/0.30)	...	m	29.00-37.50
0.75sqmm	(7/0.37)	...	m	33.00-43.00
1.0 sqmm	(7/0.42)	...	m	39.00-51.00
1.5 sqmm	(7/0.53)	...	m	48.00-62.00
Power and control 600V/1000V double insulated circular cable :				
One x single core FR cable, fixed to cable tray -				
2.5sqmm	(7/0.68)	...	m	7.00-9.25
4sqmm	(7/0.85)	...	m	8.25-11.00
6sqmm	(7/1.04)	...	m	9.25-12.25
10sqmm	(7/1.35)	...	m	11.75-15.50
16sqmm	(7/1.70)	...	m	13.50-17.75
25sqmm	(7/2.14)	...	m	17.50-23.00
35sqmm	(7/2.52)	...	m	22.50-29.50
50sqmm	(19/1.80)	...	m	29.50-38.50
70sqmm	(19/2.14)	...	m	36.50-48.00
95sqmm	(19/2.52)	...	m	42.00-55.00
120sqmm	(37/2.05)	...	m	50.00-65.00
150sqmm	(37/2.25)	...	m	60.00-78.00
185sqmm	(37/2.52)	...	m	69.00-90.00
240sqmm	(61/2.25)	...	m	86.00-112.00
300sqmm	(61/2.52)	...	m	110.00-142.00
400sqmm	(61/2.85)	...	m	134.00-172.00
500sqmm	(61/3.20)	...	m	148.00-190.00

ELECTRICAL SERVICES

CABLES (cont'd)

HIGH FIRE RESISTANT (cont'd)

RADOX FR - 3013 flame retardant, etc., cables (cont'd)

Power and control 600V/1000V double insulated circular cable (cont'd) :

			UNIT	Adel. Bris. Melb. Per. Syd. $
Two core FR cable, installed in roof space -				
0.75sqmm	(7/0.37)	...	m	8.00-11.00
1.0sqmm	(7/0.42)	...	m	9.75-13.00
1.5sqmm	(7/0.53)	...	m	10.00-13.50
2.5sqmm	(7/0.68)	...	m	11.50-15.50
4sqmm	(7/0.85)	...	m	15.50-20.50
6sqmm	(7/1.04)	...	m	18.25-24.00
Two core FR cable, fixed to cable tray -				
0.75sqmm	(7/0.37)	...	m	7.50-10.00
1.0sqmm	(7/0.42)	...	m	8.50-11.25
1.5sqmm	(7/0.53)	...	m	9.50-12.75
2.5sqmm	(7/0.68)	...	m	11.25-14.75
4sqmm	(7/0.85)	...	m	15.50-20.50
6sqmm	(7/1.04)	...	m	18.75-24.50
Two core and earth FR cable, installed in roof space -				
1.5sqmm	(7/0.37)	...	m	10.50-14.00
2.5sqmm	(7/0.42)	...	m	12.50-16.50
4sqmm	(7/0.53)	...	m	17.75-23.00
6sqmm	(7/1.04)	...	m	22.50-29.50
Two core and earth FR cable, fixed to cable tray -				
1.5sqmm	(7/0.37)	...	m	11.00-14.50
2.5sqmm	(7/0.42)	...	m	13.75-18.00
4sqmm	(7/0.53)	...	m	19.00-24.50
6sqmm	(7/1.04)	...	m	23.50-31.00
Three core FR cable, fixed to cable tray -				
0.75sqmm	(7/0.37)	...	m	8.50-11.00
1.0sqmm	(7/0.42)	...	m	9.00-11.75
1.5sqmm	(7/0.53)	...	m	10.50-13.75
2.5sqmm	(7/0.68)	...	m	13.50-17.75
4sqmm	(7/0.85)	...	m	18.75-24.50
Three core and earth FR cable, fixed to cable tray -				
1.5sqmm	(7/0.53)	...	m	12.25-16.00
2.5sqmm	(7/0.68)	...	m	15.50-20.50
4sqmm	(7/0.85)	...	m	22.00-28.50
6sqmm	(7/1.04)	...	m	28.50-37.00
10sqmm	(7/1.35)	...	m	35.00-45.00
16sqmm	(7/1.70)	...	m	46.00-59.00
25sqmm	(7/2.14)	...	m	59.00-76.00
35sqmm	(7/2.52)	...	m	73.00-94.00
Four core FR cable, fixed to cable tray -				
0.75sqmm	(7/0.37)	...	m	9.75-12.75
1.0sqmm	(7/0.42)	...	m	10.50-13.75
1.5sqmm	(7/0.53)	...	m	12.25-16.00
2.5sqmm	(7/0.68)	...	m	15.50-20.50
Four core and earth FR cable, installed in roof space -				
1.5sqmm	(7/0.53)	...	m	14.50-19.00
2.5sqmm	(7/0.68)	...	m	17.25-22.50
4sqmm	(7/0.85)	...	m	24.50-31.50
6sqmm	(7/1.04)	...	m	30.00-39.00
10sqmm	(7/1.35)	...	m	37.50-48.00

ELECTRICAL SERVICES

			UNIT	Adel. Bris. Melb. Per. Syd. $
CABLES (cont'd)				
HIGH FIRE RESISTANT (cont'd)				
RADOX FR - 3013 flame retardant, etc., cables (cont'd)				
Power and control 600V/1000V double insulated circular cable (cont'd) :				
Four core and earth FR cable, installed in roof space (cont'd) -				
16sqmm	(7/1.70)	...	m	53.00-68.00
25sqmm	(7/2.14)	...	m	69.00-88.00
35sqmm	(7/2.52)	...	m	78.00-100.00
50sqmm	(19/1.80)	...	m	104.00-132.00
Four core and earth FR cable, fixed to cable tray -				
1.5sqmm	(7/0.53)	...	m	14.00-18.25
2.5sqmm	(7/0.68)	...	m	18.00-23.50
4sqmm	(7/0.85)	...	m	27.00-35.00
6sqmm	(7/1.04)	...	m	33.50-43.00
10sqmm	(7/1.35)	...	m	43.00-56.00
16sqmm	(7/1.70)	...	m	60.00-78.00
25sqmm	(7/2.14)	...	m	78.00-100.00
35sqmm	(7/2.52)	...	m	88.00-114.00
50sqmm	(19/1.80)	...	m	120.00-156.00
Multi-core FR cable, fixed to cable tray -				
1.5sqmm	(7/0.53) 7 core	...	m	17.50-23.00
1.5sqmm	(7/0.53) 10 core	...	m	25.00-33.00
1.5sqmm	(7/0.53) 11 core and earth	...	m	29.00-37.50
1.5sqmm	(7/0.53) 16 core	...	m	36.00-47.00
1.5sqmm	(7/0.53) 20 core and earth	...	m	48.00-63.00
1.5sqmm	(7/0.53) 40 core and earth	...	m	80.00-104.00
2.5sqmm	(7/0.68) 6 core and earth	...	m	22.00-28.50
2.5sqmm	(7/0.68) 10 core and earth	...	m	39.00-51.00
2.5sqmm	(7/0.68) 20 core and earth	...	m	76.00-98.00
MINERAL INSULATED METAL SHEATHED (MIMS, PYROTENAX - Pentair)				
Light duty, 600V/600V cable, fixed to cable tray :				
Two core, bare sheath -				
1sqmm		...	m	11.00-14.25
1.5sqmm		...	m	11.75-15.50
Two core, served sheath -				
1sqmm		...	m	11.75-15.00
1.5sqmm		...	m	12.50-16.50
Three core, bare sheath -				
1sqmm		...	m	13.00-16.50
1.5sqmm		...	m	14.00-18.00
2.5sqmm		...	m	17.25-22.00
Three core, served sheath -				
1sqmm		...	m	13.75-17.50
1.5sqmm		...	m	14.75-19.00
2.5sqmm		...	m	18.00-23.00

ELECTRICAL SERVICES

CABLES (cont'd)

MINERAL INSULATED METAL SHEATHED
(MIMS, PYROTENAX - Pentair) (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Light duty 600V/600V, cable fixed to cable tray (cont'd) :			
Four core, bare sheath -			
1sqmm	...	m	14.00-17.75
1.5sqmm	...	m	15.25-19.75
2.5sqmm	...	m	19.50-25.00
Four core, served sheath -			
1sqmm	...	m	14.25-18.25
1.5sqmm	...	m	15.75-20.50
2.5sqmm	...	m	20.00-25.50
Heavy duty, 1000V/1000V cable, fixed to struts :			
One x Single core, bare sheath -			
10sqmm	...	m	18.75-24.00
16sqmm	...	m	21.50-28.00
25sqmm	...	m	25.00-32.50
35sqmm	...	m	28.50-37.00
50sqmm	...	m	34.00-44.00
70sqmm	...	m	42.00-54.00
95sqmm	...	m	49.00-63.00
120sqmm	...	m	60.00-77.00
150sqmm	...	m	68.00-88.00
185sqmm	...	m	80.00-104.00
240sqmm	...	m	92.00-118.00
300sqmm	...	m	114.00-146.00
400sqmm	...	m	140.00-178.00
One x Single core, served sheath -			
10sqmm	...	m	19.50-25.00
16sqmm	...	m	22.50-28.50
25sqmm	...	m	26.00-33.50
35sqmm	...	m	29.50-37.50
50sqmm	...	m	35.50-45.00
70sqmm	...	m	43.00-56.00
95sqmm	...	m	51.00-65.00
120sqmm	...	m	62.00-78.00
150sqmm	...	m	69.00-88.00
185sqmm	...	m	82.00-106.00
240sqmm	...	m	94.00-120.00
300sqmm	...	m	116.00-148.00
400sqmm	...	m	144.00-186.00
Two core, bare sheath -			
1.5sqmm	...	m	19.00-24.00
2.5sqmm	...	m	22.50-28.50
4sqmm	...	m	27.50-35.00
6sqmm	...	m	34.50-44.00
10sqmm	...	m	44.00-55.00
Two core, served sheath -			
1.5sqmm	...	m	21.00-27.00
2.5sqmm	...	m	24.50-31.00
4sqmm	...	m	29.50-37.50
6sqmm	...	m	37.00-47.00

ELECTRICAL SERVICES

CABLES (cont'd)

MINERAL INSULATED METAL SHEATHED
(MIMS, PYROTENAX - Pentair) (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Heavy duty 1000V/1000V, fixed to struts (cont'd) :			
Three core, bare sheath -			
1.5sqmm	...	m	21.00-26.50
2.5sqmm	...	m	25.50-32.50
4sqmm	...	m	31.00-39.00
6sqmm	...	m	38.50-49.00
10sqmm	...	m	52.00-66.00
16sqmm	...	m	70.00-88.00
Three core, served sheath -			
1.5sqmm	...	m	23.00-29.50
2.5sqmm	...	m	27.50-35.00
4sqmm	...	m	33.50-43.00
6sqmm	...	m	41.00-52.00
10sqmm	...	m	55.00-69.00
16sqmm	...	m	73.00-92.00
25sqmm	...	m	106.00-130.00
Four core, bare sheath -			
1.5sqmm	...	m	24.50-31.50
2.5sqmm	...	m	30.00-38.00
4sqmm	...	m	36.50-46.00
6sqmm	...	m	46.00-58.00
10sqmm	...	m	62.00-79.00
16sqmm	...	m	86.00-108.00
25sqmm	...	m	120.00-150.00
Four core, served sheath -			
1.5sqmm	...	m	27.00-34.00
2.5sqmm	...	m	32.50-41.00
4sqmm	...	m	39.00-50.00
6sqmm	...	m	50.00-62.00
10sqmm	...	m	66.00-84.00
16sqmm	...	m	90.00-114.00
25sqmm	...	m	126.00-158.00
Seven core, bare sheath -			
1.5sqmm	...	m	33.50-42.00
2.5sqmm	...	m	39.00-50.00
Seven core, served sheath -			
1.5sqmm	...	m	34.00-43.00
2.5sqmm	...	m	40.00-51.00
Twelve core, bare sheath -			
1.5sqmm	...	m	48.00-61.00
2.5sqmm	...	m	58.00-74.00
Twelve core, served sheath -			
1.5sqmm	...	m	50.00-64.00
2.5sqmm	...	m	62.00-78.00
Nineteen core, bare sheath -			
1.5sqmm	...	m	70.00-88.00
Nineteen core, served sheath -			
1.5sqmm	...	m	71.00-90.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
CABLES (cont'd)			
TELEVISION ANTENNA CABLE			
75 ohm coaxial cable drawn into conduit or fixed in roof space	...	m	4.50-6.25
TELEPHONE CABLE			
Indoor unscreened type 0.5mm drawn into conduit :			
1 quad	...	m	2.65-3.65
3 pair	...	m	2.80-3.85
10 pair	...	m	3.70-5.00
20 pair	...	m	4.65-6.25
25 pair	...	m	5.00-6.75
50 pair	...	m	8.75-11.50
100 pair	...	m	14.00-18.75
Indoor unscreened type 0.5mm fixed to cable tray :			
25 pair	...	m	10.75-14.75
50 pair	...	m	13.75-18.50
100 pair	...	m	20.00-27.00
Underground 0.64mm jelly filled, polyethylene insulated and nylon jacketed drawn into conduit :			
10 pair	...	m	6.25-8.50
30 pair	...	m	11.50-15.25
50 pair	...	m	17.25-23.00
100 pair	...	m	30.50-40.00
DATA CABLE			
Data cable and local area network cable installed within skirting duct :			
Coaxial cable -			
RG58	...	m	6.25-8.75
RG59	...	m	6.25-8.75
RG62	...	m	6.50-9.00
Instrolex (Dekoron) digital cable installed within cable troughing :			
Overall screened 7/0.30 cable -			
2 pair	...	m	8.25-12.00
4 pair	...	m	9.25-13.50
6 pair	...	m	11.25-16.00
8 pair	...	m	13.00-18.50
10 pair	...	m	14.75-21.00
12 pair	...	m	17.50-25.00
16 pair	...	m	22.50-31.50
20 pair	...	m	28.00-39.00
24 pair	...	m	33.50-47.00
36 pair	...	m	54.00-76.00
50 pair	...	m	64.00-88.00
AUDIO AND SPEAKER CABLE			
Microphone cable installed in roof space :			
2 core -			
7/0.20 cable	...	m	4.80-6.75
14/0.20 cable	...	m	4.95-6.75
24/0.20 cable	...	m	5.00-7.00
Figure 8, single insulated, loud speaker cable installed in roof space :			
2 core -			
24/0.20 (0.75sqmm, 250V/250V) cable	...	m	4.35-6.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

CAT 5 / CAT 6 DATA AND TELEPHONE INSTALLATION

STRUCTURED CABLING SYSTEMS

Distribution system providing integrated backbone cabling for telephone (voice) and computer (data) use. Each voice or data socket is individually (radially) wired from a centrally located patch panel. The patch panel enables facile alteration of sockets and cabling configurations to suit changing telephone and facsimile machine extensions and a variety of computer networks - all fully tested to certify a Category 5/6 system to AS/NZS 3080 ... Note

MUTOs (multi user telecommunication outlets and CPs (consolidation points) are a concept in open office environments where moveable partitions, desks and communications sockets are required. MUTOs and CPs provide better flexibility for office furniture and staff changes. MUTOs are a group of RJ45 sockets in a common location with flexible cable connecting to telephones, PCs etc. CPs are similar in concept to FDPs in traditional telephone installations, installed to provide a closer link for the cabling of RJ45 sockets ... Note

Cat 5 includes the Cat 5 enhanced (Cat 5e) system and is also known as Class D (Cat 4 or Cat 3 have been overtaken by Cat 5 systems) ... Note

CABLING - DATA & TELEPHONE :

4 pair Cat 5 UTP (unshielded twisted pair) -			
Cable :			
Drawn into conduit	...	m	4.75-8.25
Fixed to cable tray	...	m	7.50-12.25
Tied to catenary wire in ceiling space	...	m	8.75-15.00
Underground cable, drawn into conduit	...	m	11.00-19.00
OM1 optical Fibre 62.5/125µM fixed to tray -			
4 core	...	m	25.00-34.00
6 core	...	m	25.00-34.00
12 core	...	m	25.00-34.00
OM3 optical Fibre 50/125µM fixed to tray -			
4 core	...	m	39.00-47.00
6 core	...	m	39.00-47.00
12 core	...	m	39.00-47.00
ADD extra for :			
Conduit, catenary wire and cable tray	...	Note	

FIBRE OPTIC CONNECTIONS :

Z Type splice holder (un-enclosed) -			
6 fibre	...	Each	160.00-225.00
12 fibre	...	Each	225.00-315.00
Splice holder and 215 x 215 x 75mm connection box -			
6 fibre	...	Each	205.00-290.00
Splice holder and 335 x 215 x 75mm connection box -			
12 fibre	...	Each	285.00-390.00
Fibre connector for use with splice holders - per fibre	...	Each	175.00-275.00
ADD extra for :			
Patch Cord and pigtails, duplex (supplied loose) -			
1m long	...	Each	36.00-45.00
2m long	...	Each	37.00-47.00
3m long	...	Each	38.50-48.00

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

CAT 5 / CAT 6 DATA AND TELEPHONE INSTALLATION (cont'd)

STRUCTURED CABLING SYSTEMS (cont'd)

DATA & TELEPHONE SOCKETS (Cat 5e compliant) :			
RJ45 socket modular socket clipped into prepared plate	...	Each	26.50-34.00
Single RJ45 socket and flush face plate	...	Each	36.50-45.00
Twin RJ45 socket and flush face plate	...	Each	56.00-69.00
Triple RJ45 socket and flush face plate	...	Each	76.00-94.00
Quad RJ45 socket and flush face plate	...	Each	96.00-118.00
ADD extra for :			
Skirting duct fittings and floor or bench boxes/enclosures - refer to pages 544 to 546, 582 and 583	...	Note	
DATA & TELEPHONE CABINETS (for patch panels) :			
19" rack type cabinet with lockable hinged door and to suit patch panels -			
400mm deep x 325mm high - 6 rack units (wall mounted)	...	Each	460.00-580.00
700mm deep x 1000mm high - 18 rack units	...	Each	1,500.00-1,775.00
700mm deep x 1300mm high - 27 rack units	...	Each	2,050.00-2,450.00
700mm deep x 1900mm high - 39 rack units	...	Each	2,600.00-3,050.00
850mm deep x 2200mm high - 45 rack units	...	Each	3,000.00-3,500.00
ADD extra for :			
Cable tray -			
150mm wide	...	Each	106.00-126.00
225mm wide	...	Each	136.00-162.00
300mm wide	...	Each	156.00-184.00
Shelf -			
Fixed	...	Each	146.00-172.00
Telescopic	...	Each	305.00-350.00
240V 10A power rail -			
10 socket-outlet	...	Each	335.00-380.00
16 socket-outlet	...	Each	580.00-660.00
20 socket-outlet	...	Each	600.00-680.00
2 roof mounted fans	...	Each	280.00-325.00
4 roof mounted fans	...	Each	510.00-580.00
240V supply and data/telephone patch panels and cords	...	Note	
RCD protection	...	Note	
PBX and computer equipment	...	Note	
DATA & TELEPHONE PATCH PANELS (Cat 5/5e compliant) :			
RJ45 Patch Panel (for 19" rack) -			
24 port (1 rack unit)	...	Each	295.00-395.00
32 port (2 rack units)	...	Each	520.00-710.00
50 port, for voice applications only (1 rack unit)	...	Each	640.00-880.00
Patch Cord and 2 x RJ45 plugs (supplied loose) -			
1m long	...	Each	3.70-4.80
2m long	...	Each	4.40-5.75
3m long	...	Each	5.00-6.50
PABX and data equipment and interconnection	...	Note	

CAT 6 AND CAT 6A or AUGMENTED DATA AND TELEPHONE INSTALLATION STRUCTURED CABLING SYSTEMS

The cost of cabling and equipment for Cat 6, Cat 6A or augmented systems is more expensive than for Cat 5. These systems provide for a greater volume and speed of data transmission with increased distances over unshielded twisted pair (UTP) cabling. These systems require higher standards of material, installation and testing methods.

ADD to foregoing Cat 5 installation for :			
Cat 6 system	...	Range	10% to 20%
Cat 6A Augmented system	...	Range	15% to 30%

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

POWER AND LIGHTING EQUIPMENT

Prices may vary significantly and can be affected by the size/quantity of less common items in an order

SWITCHES

Lighting and small power switches (10A, 1-way / 2-way) in standard size flush plate, plastic :			
1 switch	...	Each	24.50-31.50
2 switches	...	Each	28.50-36.50
3 switches	...	Each	36.50-46.00
4 switches	...	Each	47.00-57.00
5 switches	...	Each	62.00-75.00
6 switches	...	Each	69.00-82.00
ADD extra for :			
Duo mechanism, two x 1way switches in space of one	...	Each	11.50-14.75
Intermediate mechanism	...	Each	25.50-30.50
16A mechanism	...	Each	8.50-10.75
20A mechanism	...	Each	11.75-14.50
35A mechanism	...	Each	16.75-21.50
Double pole mechanism	...	Each	14.25-17.25
Rotary mechanism	...	Each	7.00-9.00
Press button mechanism	...	Each	4.25-6.00
Press button mechanism with LED indicator	...	Each	6.50-9.00
Factory marked 'FAN', 'HEAT', etc.	...	Each	6.75-8.75
Neon indicator ...		Each	28.00-33.00
Anchor grommet, for flexible cable entering plate	...	Each	9.75-14.75
Plate and surround options, refer to Surrounds and Boxes page 568	...	Note	
Engraving, refer to Engraving page 574	...	Note	
Surrounds and Boxes :			
Excel or 2000 Series surround options, (in lieu of standard plastic) -			
Large plate size (2 gang)	...	Each	1.90-5.25
Coloured (non surcharged) plastic flush plate and switch mechanism (in lieu of plain white)	...	Each	1.90-5.25
Excel or 2000 Series flush plate (in lieu of plain white plate) -			
Surcharged colour (red, grey, sand, etc)	...	Each	4.85-8.75
Stainless steel	...	Each	13.75-19.00
Polished brass	...	Each	26.50-34.00
Excel or 2000 Series surround only -			
In different colour to plate	...	Each	4.00-7.75
In metal finish	...	Each	3.85-7.50
In different colour to plate	...	Each	4.00-7.75
Ganging of standard plates - Excel or 2000 Series ganged together with common surround -			
2 plates ganged	...	Each	6.50-10.50
3 plates ganged	...	Each	8.75-13.25
4 plates ganged	...	Each	12.00-16.75
Weather protected covering to cover standard plastic switch / socket-outlet GPO (ADD extra for outlet / switch) -			
Semi-flush clear flap type cover, plastic	...	Each	28.50-35.50
Surface box and clear flap type cover, plastic	...	Each	33.50-41.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SWITCHES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Large-toggle lighting switches, prestige / elderly / disabled, plastic :			
1 switch, 10A, 115 x 70mm flush plate	...	Each	32.50-41.00
2 switches, 10A, 115 x 70mm flush plate	...	Each	39.50-49.00
3 switches, 10A, 115 x 115mm flush plate	...	Each	56.00-68.00
4 switches, 10A, 115 x 115mm flush plate	...	Each	66.00-80.00
ADD extra for :			
Metallic cover - aluminium, chrome, brass, gold	...	Each	7.00-11.00
Lighting switches, Heritage style, surface polished brass with porcelain base and fancy timber block :			
Single 10A switch	...	Each	75.00-88.00
Intermediate 10A switch	...	Each	98.00-114.00
Ceiling pull cord, 10A polished brass covered switch	...	Each	88.00-106.00
Multi-switch stainless steel assemblies, 10A :			
plain face flush plate type -			
119 x 120mm plate with 4 switches (2 rows of 2)	...	Each	180.00-210.00
211 x 120mm plate with 8 switches (2 rows of 4)	...	Each	335.00-390.00
395 x 120mm plate with 16 switches (2 rows of 8)	...	Each	690.00-820.00
257 x 310mm plate with 30 switches (6 rows of 5)	...	Each	1,050.00-1,260.00
label face flush plate with provision with traffolyte strips -			
119 x 120mm plate with 4 switches (2 rows of 2)	...	Each	290.00-340.00
211 x 120mm plate with 8 switches (2 rows of 4)	...	Each	480.00-560.00
395 x 120mm plate with 16 switches (2 rows of 8)	...	Each	910.00-1,065.00
257 x 310mm plate with 30 switches (6 rows of 5)	...	Each	1,380.00-1,650.00
ADD extra for :			
Engraving, refer to Engraving page 574	...	Note	
Cord pull ceiling switch, 10A plastic	...	Each	37.50-46.00
Key operated, standard size flush plastic switch, 240V 16A	...	Each	86.00-104.00
Keytag switch, standard size flush plastic with neon, to switch hotel room controlling 3 or 4 x 20A circuits and including keytag	...	Each	178.00-210.00
ADD extra for :			
Extra neon and switch	...	Each	15.50-19.75
Replacement keytag only	...	Each	16.50-20.50
Electric Range Power switches, standard size flush, plastic :			
45A 240V, 1-pole type	...	Each	34.50-43.00
40A 415V, 2-pole type	...	Each	46.00-56.00
ADD extra for neon indicator	...	Each	21.50-26.50
Three phase power switches, standard size flush, plastic :			
35A 415V, 3-pole	...	Each	96.00-116.00
35A 415V, 3-pole with neon	...	Each	120.00-142.00
Electric Water Heater isolating switch, standard size flush plastic with flexible conduit and cable connection to heater - 20A 240V	...	Each	142.00-170.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SWITCHES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Weatherproof IP66 (bathroom / laundry / wet area), standard size plastic, flush 240V lighting / power switches :			
1 switch, 10A / 20A	...	Each	40.00-51.00
2 switches, 10A / 20A	...	Each	47.00-59.00
3 switches, 10A / 20A	...	Each	55.00-68.00
4 switches, 10A / 20A	...	Each	61.00-75.00
ADD extra for :			
Plate and surround options, refer to Surrounds and Boxes page 568	...	Note	
Weatherproof IP56/IP66 (general external), standard size plastic, 240V lighting / power switches :			
Surface type -			
1 switch, 16A	...	Each	42.00-53.00
2 switch, 16A 2-pole	...	Each	56.00-69.00
2 switches, 16A	...	Each	52.00-65.00
1 switch, 20A	...	Each	58.00-71.00
Semi-flush type -			
1 switch, 16A	...	Each	48.00-61.00
2 switches, 16A	...	Each	57.00-71.00
3 switches, 16A	...	Each	66.00-80.00
ADD extra for :			
Neon indicator, per indicator	...	Each	33.00-41.00
Weatherproof IP56/IP66 surface, padlockable plastic switches, with mounting enclosure :			
Power / isolating switches -			
20A (AC21) 250V, 1-pole (AC23 16A M220)	...	Each	63.00-77.00
20A (AC21) 250V/500V, 2-pole (AC23 16A M180)	...	Each	65.00-79.00
40A (AC21) 250V, 1-pole (AC23 16A M220)	...	Each	75.00-92.00
40A (AC21) 250V/500V, 2-pole (AC23 16A M180)	...	Each	82.00-98.00
20A (AC21) 500V, 3-pole (AC23 16A M180)	...	Each	69.00-84.00
40A (AC21) 500V, 3-pole (AC23 16A M180)	...	Each	90.00-106.00
40A (AC21) 500V, 4-pole (AC23 20A M100)	...	Each	104.00-124.00
55A (AC21) 500V, 3-pole (AC23 20A M160)	...	Each	102.00-122.00
63A (AC21) 500V, 3-pole (AC23 40A M330)	...	Each	136.00-162.00
63A (AC21) 500V, 4-pole (AC23 40A M330)	...	Each	158.00-186.00
80A (AC21) 500V, 3-pole (AC23 55A M330)	...	Each	154.00-182.00
80A (AC21) 500V, 4-pole (AC23 55A M330)	...	Each	182.00-215.00
ADD extra for :			
Neon indicator (plus extra cabling), per indicator	...	Each	27.00-32.50

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SWITCHES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Weatherproof IP56/IP66 surface, padlockable plastic switches, with mounting enclosure (cont'd) :			
Industrial modular lighting / power / isolating switches -			
10 A (AC21) 250V, 1-pole (AC23 8A M80)	...	Each	72.00-86.00
10 A (AC21) 250V, 2-way	...	Each	88.00-106.00
10 A (AC21) 250V, intermediate	...	Each	114.00-136.00
15 A (AC21) 250V, 1-pole (AC23 8A M80)	...	Each	77.00-92.00
15 A (AC21) 250V, 1-pole (AC23 8A M80), key operated	...	Each	150.00-176.00
20 A (AC21) 250V, 1-pole (AC23 20A M260)	...	Each	79.00-96.00
20 A (AC21) 250V, 1-pole (AC23 20A M260), key operated		Each	154.00-182.00
32 A (AC21) 250V, 1-pole (AC23 M260)	...	Each	96.00-114.00
50 A (AC21) 250V, 1-pole (AC23 25A M250)	...	Each	108.00-130.00
20 A (AC21) 500V, 3-pole (AC23 20A M100)	...	Each	112.00-134.00
20 A (AC21) 500V, 3-pole (AC23 20A M100), key operated		Each	220.00-255.00
32 A (AC21) 500V, 3-pole (AC23 20A M100)	...	Each	150.00-176.00
32 A (AC21) 500V, 4-pole (AC23 20A M100)	...	Each	200.00-235.00
50 A (AC21) 500V, 3-pole (AC23 25A M200)	...	Each	176.00-205.00
50 A (AC21) 500V, 3-pole (AC23 25A M200), key operated		Each	255.00-300.00
63 A (AC21) 500V, 3-pole (AC23 25A M200)	...	Each	198.00-230.00
ADD extra for :			
Flush surround, for switch	...	Each	47.00-58.00
Neon indicator (plus extra cabling), per indicator	...	Each	25.00-30.00
High pressure gasket (1600psi)	...	Each	20.50-25.00
Double - use the cost of the switches, plus	...	Each	32.50-37.00
Cast aluminium type instead of plastic	...	Each	71.00-136.00
Heavy duty industrial modular motor rated power switches -			
25A (AC21) 690V, 7.5kW, 3-pole (AC23 25A)	...	Each	162.00-190.00
25A (AC21) 690V, 7.5kW, 4-pole (AC23 25A)	...	Each	174.00-205.00
40A (AC21) 690V, 11kW, 3-pole (AC23 25A)	...	Each	184.00-215.00
40A (AC21) 690V, 11kW, 4-pole (AC23 25A)	...	Each	198.00-235.00
63A (AC21) 690V, 25kW, 3-pole (AC23 63A)	...	Each	210.00-245.00
63A (AC21) 690V, 25kW, 4-pole (AC23 40A)	...	Each	220.00-260.00
80A (AC21) 690V, 3-pole (AC23 63A)	...	Each	220.00-255.00
ADD extra for :			
Stainless steel enclosure, instead of plastic	...	Each	420.00-480.00
Industrial push button station switches -			
Single button, Start or Stop	...	Each	136.00-160.00
Single button, Emergency Stop (latching mushroom)	...	Each	186.00-220.00
Two buttons, Start and Stop	...	Each	210.00-250.00
Two buttons, Start & Emergency Stop (latching mushroom)		Each	265.00-305.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

DIMMERS AND FAN CONTROLLERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Lighting dimmers are designed to suit the luminaire type. Leading Edge dimmers suit incandescent lights (without transformers) and lights with wire wound transformers. Trailing Edge dimmers suit incandescent lights (without transformers) and lights with electronic transformers. Universal dimmers suit incandescent lights and both transformer types.	...	Note	
DALI and DSI type dimmers suit digitally controllable electronic ballasts (eg when fitted to fluorescent lights). A digital signal to each ballast (via a control cable) is used - the ballast does the dimming.	...	Note	
1V to 10V DC signalled analogue electronic ballasts can also be used. The controller sends 1V for minimum light level through to 10V for maximum level.	...	Note	
Some dimmers can be designed to universally control all of the foregoing light types and some can dim fluorescent lights without a DSI or DALI operation. Interface units to allow a mix of systems are also available.	...	Note	
Fluorescent lights can also be dimmed by fitting switch dimmable electronic ballasts – a push button is used for the on-off switch. A long push operates the dimming function. No dimmer is required, but an extra cable may be needed.	...	Note	
LED lights are dimmable if the light and dimming equipment are compatible	...	Note	
Fan speed controller, electronic type (matched to fan), 240V with switch and standard size flush plate, to suit :			
Exhaust fan, 60VA to 300VA	...	Each	48.00-59.00
Ceiling fan, 100VA, 3 speed de-hummer	...	Each	57.00-69.00
Ceiling/exhaust fan -			
40VA to 500VA	...	Each	65.00-78.00
60VA to 650VA, with 2 hour timer	...	Each	96.00-114.00
Lighting dimmer, 250V with switch and standard size flush plate :			
Domestic / light duty -			
Leading edge, 450VA	...	Each	60.00-73.00
Trailing edge, 450VA	...	Each	68.00-82.00
Universal, 450VA	...	Each	77.00-92.00
Leading edge, 1kVA	...	Each	132.00-156.00
Trailing edge, 1kVA	...	Each	124.00-146.00
Commercial / heavy duty, leading edge type, with separate remote controller :			
1kVA (including controller)	...	Each	205.00-250.00
2kVA (including controller)	...	Each	260.00-315.00
2.5kVA (ADD for preset pushbutton)	...	Each	580.00-670.00
5kVA (ADD for preset pushbutton)	...	Each	790.00-930.00
Pushbutton control (or stop) -			
1 preset pushbutton	...	Each	154.00-180.00
2 preset pushbuttons	...	Each	170.00-200.00
3 preset pushbuttons	...	Each	186.00-220.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

DIMMERS AND FAN CONTROLLERS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Commercial / heavy duty, leading edge type, with separate remote controller (cont'd) :			
8 channels, each 1kVA (ADD for keypad)	...	Each	1,900.00-2,200.00
4 channels, each 2.4kVA (ADD for keypad)	...	Each	1,500.00-1,775.00
Keypad for 4 or 8 channel dimmer -			
1 button	...	Each	80.00-96.00
2 button	...	Each	88.00-108.00
4 button	...	Each	102.00-124.00
Commercial / architectural C-Bus controlled :			
1 phase, 4 x 2A channels	...	Each	580.00-690.00
1 phase, 4 x 5A channels	...	Each	1,005.00-1,185.00
1 phase, 2 x 10A channels	...	Each	1,005.00-1,185.00
1 phase, 1 x 20A channel	...	Each	1,005.00-1,185.00
Commercial / architectural C-Bus controlled, universal type :			
1 phase, 4 x 2.5A channels	...	Each	630.00-780.00
Commercial / heavy duty C-Bus controlled :			
3 phase, 12 x 10A channels	...	Each	5,600.00-6,400.00
3 phase, 12 x 12A channels	...	Each	5,800.00-6,800.00
3 phase, 12 x 16A channels	...	Each	6,400.00-7,400.00
3 phase, 12 x 20A channels	...	Each	6,800.00-7,800.00
Commercial 100 ballast DSI type	...	Each	1,080.00-1,275.00
1V to 10V dimmer controller for analogue electronic dimmable ballasted lights - 550VA, up to 30 ballasts (ADD for ballasts)	...	Each	106.00-132.00
Domestic / light duty DSI type with standard size flush plate :			
30 ballast capacity (ADD for ballasts)	...	Each	365.00-430.00
Commercial / heavy duty C-Bus, DSI type :			
8 x 100 ballast capacity channels (ADD for ballasts)	...	Each	1,080.00-1,290.00
Commercial / heavy duty C-Bus, DALI type :			
2 x 64 ballast capacity channels (ADD for ballasts)	...	Each	1,080.00-1,290.00
ADD extra for :			
C-Bus switch, plastic with standard flush plate -			
I button	...	Each	90.00-108.00
2 buttons	...	Each	100.00-120.00
4 buttons	...	Each	114.00-138.00
C-Bus interface (if required to be left on site)	...	Each	480.00-560.00
Dimmable fluorescent electronic ballasts, refer page 572	...	Note	
Circuitry including extra low voltage cabling	...	Note	

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

ENGRAVING

		UNIT	Adel. Bris. Melb. Per. Syd. $
Engraving - laser cut engraving of electrical accessories :			
Standard switch plate or power outlet -			
On plastic :			
Black characters	...	Each line	7.25-9.75
Coloured characters	...	Each line	10.75-14.00
On stainless steel :			
Black characters	...	Each line	10.75-14.00
Coloured characters	...	Each line	14.00-18.25
Multi switch assemblies and complex work - by quotation	...	Note	
Laminated colour Traffolyte / Romark, screw fixed label -			
Up to 50 x 25mm, 2 lines of text	...	Each	11.25-15.25
Up to 100 x 25mm, 2 lines of text	...	Each	16.00-21.50
Up to 200 x 60mm, 3 lines of text	...	Each	28.00-36.50
Stainless steel, screw fixed label -			
Up to 50 x 25mm :			
Natural	...	Each	17.50-23.00
Paint filled characters	...	Each	21.50-28.00
Up to 100x25mm :			
Natural	...	Each	26.00-34.00
Paint filled characters	...	Each	35.50-46.00
ADD extra for :			
Hand, machine cut, engraving (to foregoing)	...	+ 200%	
Transport to and from engraver	...	Note	

AUTOMATIC OR REMOTE CIRCUIT CONTROL

Lighting / small power circuit controllers, 240V :			
P.E. Cell (Photo Electric) 10A daylight switch -			
With standard plastic flush plate and off timer	...	Each	134.00-158.00
With surface NEMA base	...	Each	114.00-136.00
With surface IP56 heavy duty enclosure	...	Each	122.00-146.00
With surface IP56 heavy duty enclosure and off timer	...	Each	168.00-200.00
Electronic 16A light intensity switch (DIN) and remote sensor		Each	265.00-310.00
Timer, 10A, with standard plastic flush plate (installer sets time period) -			
Pneumatic	...	Each	74.00-90.00
Pneumatic with extra switch	...	Each	90.00-110.00
Electronic, with operation push button	...	Each	162.00-190.00
Electronic, for fan run-on timing with light switch	...	Each	200.00-235.00
Time switch / timer, analogue 15A AC1, adjustable multi-cycle, with deep standard plastic flush plate and 2 control switches -			
1 hour timer, multiple on/off	...	Each	200.00-235.00
24 hour timer, multiple on/off	...	Each	176.00-205.00
24 hour timer, multiple on/off and 15A 3 pin socket-outlet	...	Each	245.00-290.00
7 day time switch	...	Each	190.00-220.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

AUTOMATIC OR REMOTE CIRCUIT CONTROL (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Lighting / small power circuit controllers, 240V (cont'd) :			
IP56 time switch / timer, 15A AC1 adjustable multi-cycle, plastic with battery reserve and surface enclosure -			
2 hour process timer, double pole with neon (no reserve)	...	Each	275.00-320.00
24 hour timer, synchronous	...	Each	280.00-330.00
24 hour timer, quartz	...	Each	320.00-370.00
24 hour timer, sync with 2 switches and 15A 3 pin outlet	...	Each	420.00-490.00
7 day time switch, synchronous	...	Each	345.00-400.00
7 day time switch, digital	...	Each	400.00-470.00
DIN mounted (switchboard) time switch / timer, 16A AC1 adjustable multi-cycle, 1 module wide, plastic -			
Interval timer, 24 hour, 30 min interval, mechanical	...	Each	112.00-134.00
Interval timer, 10 hour, 1s interval, digital	...	Each	164.00-192.00
Interval timer, 10 hour, 1s interval, digital multi-function	...	Each	205.00-240.00
7 day time switch, electro-mechanical	...	Each	100.00-120.00
7 day time switch, electro-mechanical, 100hr reserve	...	Each	142.00-168.00
7 day time switch, digital, 6 year reserve - 1 channel	...	Each	144.00-172.00
7 day time switch, digital, 6 year reserve - 2 channel	...	Each	168.00-198.00
Sunrise/sunset time switch, digital, 6 yr reserve - 2 channel	...	Each	560.00-680.00
Annual time switch, digital, 6 year reserve	...	Each	730.00-880.00
Contactor, DIN rail mount (for switchboard) -			
240V: 20A AC1, 1.3kW AC3 :			
1 NO, 1 module wide	...	Each	68.00-82.00
1 NC, 1 module wide	...	Each	77.00-92.00
2 NO, 1 module wide	...	Each	77.00-92.00
2 NC, 2 NO, 2 module wide	...	Each	104.00-122.00
4 NO, 2 module wide	...	Each	100.00-118.00
415V: 25A AC1, 4kW AC3, 4 NC, 2 module wide	...	Each	134.00-158.00
Contactor (without an enclosure) -			
240V: 32A AC1 :			
3kW 12A AC3, 4 NC	...	Each	220.00-245.00
5.5kW 20A AC3, 4 NC	...	Each	255.00-285.00
240V: 65A AC1, 10kW 30A AC3, 4 NC	...	Each	365.00-400.00
ADD extra for :			
Latching function	...	Each	196.00-215.00
Enclosure for up IP54 plastic to 3 contactors	...	Each	124.00-146.00
Occupancy detector, passive infra red (PIR) and ultrasonic motion detector and timer -			
Ceiling mounted, 360 degree detection	...	Each	365.00-410.00
Wall mounted, 180 degree detection	...	Each	400.00-440.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SAFETY SWITCH, RCD (RESIDUAL CURRENT DEVICE / ELCB)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Lighting / small power circuit controllers, 240V (cont'd) :			
Motion detector, passive infra red (PIR), with time delay -			
Indoor, dome (6 x 6m detection) :			
2A, 2 wire	...	Each	134.00-158.00
10A, 3 wire	...	Each	126.00-148.00
Indoor, flush (360deg x 6m detection) :			
2A, 2 wire	...	Each	168.00-198.00
10A, 2 wire	...	Each	180.00-210.00
Outdoor IP56 (18m x 110deg detection) :			
5A, 2 wire	...	Each	170.00-205.00
10A, 3 wire	...	Each	180.00-215.00
10A, 3 wire with 30 or 60 minute timer	...	Each	225.00-265.00
Switch Multiplier (16A) relay and special switches to convert a 2 wire switch circuit without rewiring to -			
2 x 1-way circuit	...	Each	162.00-192.00
2 x 2-way circuit	...	Each	176.00-210.00
Water heater element relay switch (off when hot) 4.8kW 20A	...	Each	144.00-170.00
Motor starters, including protection and enclosure -			
D.O.L. type, 5.5kW (7HP)	...	Each	325.00-365.00
Automatic star delta, 11kW (15HP)	...	Each	1,500.00-1,650.00
Auto transformer, 15kW (20HP)	...	Each	3,300.00-3,600.00
Soft starter :			
15kW (20HP)	...	Each	1,825.00-2,000.00
30kW (40HP)	...	Each	3,700.00-4,100.00
Wall type with standard plastic flush plate :			
2 pole 20A -			
30mA	...	Each	156.00-184.00
10mA	...	Each	178.00-210.00
Power outlet / switched socket outlet with RCD / safety switch - refer to SOCKETS, OUTLETS AND PLUGS, page 578	...	Note	
DIN rail mounted, direct connect RCD (RCCB) :			
30mA 1 phase, 6kA rated (domestic/small switchboard) - Type AC -			
2 pole 40A	...	Each	78.00-100.00
2 pole 63A	...	Each	82.00-104.00
2 pole 80A	...	Each	106.00-132.00
1 pole combined RCD+MCB / RCBO - 10A to 32A	...	Each	82.00-104.00
30mA 3 phase, 6kA rated (domestic/small switchboard) - Type AC -			
4 pole 40A	...	Each	100.00-126.00
4 pole 80A	...	Each	104.00-130.00
30mA 1 phase, 10kA rated (commercial/large switchboard) - Type AC -			
2 pole 25A	...	Each	144.00-176.00
2 pole 63A	...	Each	152.00-184.00
2 pole 80A	...	Each	156.00-190.00
2 pole combined RCD+MCB / RCBO :			
10A to 16A	...	Each	118.00-156.00
20A to 40A	...	Each	128.00-166.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SAFETY SWITCH, RCD (RESIDUAL CURRENT DEVICE / ELCB) (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
DIN rail mounted, direct connect RCD (RCCB) :			
30mA 3 phase, 10kA rated (commercial/large switchboard)			
Type AC -			
4 pole 25A	...	Each	162.00-196.00
4 pole 63A	...	Each	182.00-220.00
4 pole 80A	...	Each	230.00-275.00
30mA 1 phase, 10kA rated (commercial/large switchboard)			
Type A -			
2 pole 25A	...	Each	166.00-200.00
2 pole 63A	...	Each	182.00-220.00
30mA 3 phase, 10kA rated (commercial/large switchboard)			
Type A -			
4 pole 25A	...	Each	182.00-220.00
4 pole 63A	...	Each	215.00-255.00
10mA 1 phase, 10kA rated (hospital switchboard) -			
2 pole combined RCD+MCB :			
RCBO - 10A to 20A	...	Each	168.00-205.00
RCBO - 32A to 40A	...	Each	180.00-215.00
E01 style - 10A to 32A	...	Each	210.00-250.00
100mA, 10kA rated (surge protected, special application) -			
1 phase, 2 pole - 40A Type G	...	Each	220.00-260.00
3 phase, 4 pole - 63A Type A	...	Each	390.00-460.00
3 phase, 4 pole - 63A Type G	...	Each	250.00-300.00
Hospital relay type, switchboard mount, with shunt tripped ACB :			
10mA -			
2 pole 4A to 63A	...	Each	540.00-650.00
4 pole 8A to 63A	...	Each	840.00-990.00
4 pole 80A to 100A (non auto)	...	Each	910.00-1,065.00
Hospital type, wall recessed, with stainless steel flush front :			
Body Protected Area EO1 style - 10mA, 2 pole 10A to 40A	...	Each	620.00-780.00
Cardiac Protected Area EO2 style - 10mA, 2 pole 20A with alarm	...	Each	810.00-990.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SOCKETS, OUTLETS AND PLUGS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Standard lampholders, plastic :			
Flexible cord type -			
B.C.	...	Each	18.00-24.00
E.S.	...	Each	27.00-34.50
Battenholder type -			
B.C.	...	Each	19.00-25.00
E.S.	...	Each	29.50-37.50
Clock socket-outlets and plugs, 5A 240V recessed plastic :			
3 pin socket-outlet -	...	Each	30.50-38.00
4 pin socket-outlet	...	Each	35.00-44.00
3 pin plug	...	Each	21.00-26.50
4 pin plug	...	Each	24.50-30.50
Surface sockets-outlets (plug-bases) without switch, plastic :			
Single -			
10A, 240V 2 parallel pins	...	Each	18.00-23.00
15A, 32V 2 pin polarised	...	Each	22.50-28.50
10A, 110V 2 round pins	...	Each	28.00-34.50
10A, 240V 3 pin, with round earth pin	...	Each	25.50-31.50
10A, 240V 3 pin	...	Each	16.75-21.50
Quick connect 10A, 240V 3 pin	...	Each	12.75-16.00
10A, 240/500V 4 pin	...	Each	22.00-27.50
Double -			
10A, 240V 3 pin	...	Each	26.00-32.00
10A, 240/500V 4 pin	...	Each	35.50-43.00
General power outlets (240V GPO), standard size plate, plastic :			
Socket-outlets, with single pole switches -			
Single :			
10A	...	Each	26.50-34.00
10A, with exchangeable extra switch	...	Each	44.00-55.00
10A, with 2 exchangeable extra switches	...	Each	61.00-76.00
10A, with 2 USB 5VDC charging sockets (no data)	...	Each	80.00-96.00
10A, with safety shutter	...	Each	38.00-47.00
10A, with round earth pin	...	Each	42.00-51.00
15A, with shutter and neon indicator	...	Each	61.00-73.00
20A, with neon indicator	...	Each	70.00-86.00
Double :			
10A	...	Each	32.50-40.00
10A, with exchangeable extra switch	...	Each	52.00-65.00
10A, with safety shutters	...	Each	50.00-60.00
10A, with shutter and neon indicator	...	Each	68.00-82.00
10A, with power available neon, 'RCD Protected' label	...	Each	66.00-79.00
10A, with 2 USB 5VDC charging sockets (no data)	...	Each	79.00-94.00
10A, with round earth pins	...	Each	65.00-78.00
10A, with surge protection	...	Each	110.00-130.00
10A, with RCD safety switch (30mA)	...	Each	150.00-178.00
10A, with RCD safety switch (10mA medical areas)	...	Each	168.00-200.00
15A, with shutter and neon indicator	...	Each	94.00-110.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SOCKETS, OUTLETS AND PLUGS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
General power outlets (240V GPO), standard size plate, plastic (cont'd) :			
Socket-outlets, with automatic (single pole) switches :			
Single -			
10A	...	Each	37.50-47.00
15A, with shutter and neon indicator	...	Each	64.00-77.00
20A	...	Each	54.00-66.00
Double -			
10A	...	Each	59.00-71.00
Socket-outlets, with double pole switches :			
Single -			
10A	...	Each	39.50-49.00
20A	...	Each	74.00-88.00
20A, with neon indicator	...	Each	92.00-110.00
Double -			
10A	...	Each	52.00-63.00
10A, with safety shutter	...	Each	64.00-77.00
10A, with shutter and neon indicator	...	Each	88.00-104.00
15A	...	Each	90.00-106.00
Four socket outlets (200 x 80mm) -			
10A	...	Each	74.00-90.00
10A, with safety shutter	...	Each	120.00-142.00
REFER to Surrounds and Boxes, page 568 for :			
Various colours / material options	...	Note	
Ganging of multiple socket-outlets / switches	...	Note	
REFER to Switches, page 568 for switch and neon options	...	Note	
Large-toggle socket-outlet GPO's, 3 pin 240V with single pole switch - prestige / elderly / disabled, plastic :			
Single -			
10A, standard size (116 x 70mm)	...	Each	34.00-43.00
10A, standard size (116 x 70mm), with shutter	...	Each	34.00-43.00
10A, 2 gang size (116 x 116mm), with neon	...	Each	46.00-57.00
Double -			
10A, 2 gang size (116 x 116mm)	...	Each	54.00-65.00
10A, 2 gang size, with neon	...	Each	86.00-104.00
Plugs, rewireable, connected to multicore cable (on site) :			
Standard -			
3 flat pin 10A, 240V	...	Each	17.50-23.50
2 flat pin, round earth pin 10A, 240V	...	Each	19.75-26.00
3 flat pin 15A, 240V	...	Each	21.00-28.00
3 flat pin 20A, 240V	...	Each	27.50-35.50
4 flat pin 10A, 240V/500V	...	Each	24.00-31.50
Electric Range socket and plug, standard size plate, plastic :			
32A 240V unswitched socket	...	Each	58.00-71.00
32A 240V plug, 1.8m 4sqmm 3 core flex and connection	...	Each	86.00-104.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SOCKETS, OUTLETS AND PLUGS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Universal shaver socket-outlets - 110V and 240V :			
For double insulated appliances only, no earth	...	Each	140.00-164.00
Plus standard outlet	...	Each	200.00-235.00
Plus standard outlet and RCD protection	...	Each	355.00-420.00
Plus standard outlet and RCD protection, stainless steel	...	Each	580.00-670.00
Suspension / pendant type 2-pole switched socket-outlet, plastic :			
10A 3 flat pin, weather protected type	...	Each	65.00-79.00
15A 3 flat pin, with neon, weather protected type	...	Each	98.00-118.00
10A 3 round pin, IP65/IP66 weatherproof type	...	Each	116.00-136.00
15A 3 round pin, IP65/IP66 weatherproof type	...	Each	128.00-150.00
ADD extra for :			
Ceiling Rose	...	Each	28.00-36.50
1m drop of 3 core 1.5 sqmm flexible cable	...	Each	19.00-26.00
1m drop of galvanised suspension wire	...	Each	19.25-25.50
Weather protected covering to cover standard size plastic socket-outlet GPO or switch (ADD extra for outlet or switch) :			
Semi-flush clear flap type cover, plastic	...	Each	28.50-35.50
Surface box and clear flap type cover, plastic	...	Each	39.00-48.00
Weather protected IP53/IP54 (domestic / small commercial) standard plate size, 240V surface switched socket-outlets :			
10A, single socket-outlet	...	Each	53.00-65.00
15A, single socket-outlet	...	Each	58.00-71.00
10A, double socket-outlet	...	Each	68.00-82.00
10A, double socket-outlet with additional switch	...	Each	90.00-110.00
ADD extra for :			
Neon indicator, per indicator	...	Each	33.00-41.00
Industrial / Weatherproof modular IP56 power outlet, plastic surface pad-lockable switched socket-outlets :			
240V single phase, 3 flat pin -			
10A, single	...	Each	100.00-118.00
10A, single, keylock switch	...	Each	176.00-205.00
15A, single	...	Each	112.00-134.00
15, single, keylock switch	...	Each	184.00-215.00
10A, single, RCD protected	...	Each	240.00-280.00
415V three phase, 4 round pin, single -			
10A	...	Each	194.00-230.00
20A	...	Each	205.00-245.00
20A, RCD protected	...	Each	510.00-590.00
32A	...	Each	315.00-370.00
40A	...	Each	510.00-600.00
50A	...	Each	610.00-700.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

SOCKETS, OUTLETS AND PLUGS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Industrial / Weatherproof modular IP56 power outlet, plastic surface pad-lockable switched socket-outlets (cont'd) :			
415V three phase, 5 round pin, single :			
10A	...	Each	230.00-270.00
10A, single, keylock switch	...	Each	325.00-380.00
20A	...	Each	240.00-285.00
20A, RCD protected	...	Each	550.00-640.00
32A	...	Each	330.00-385.00
40A	...	Each	530.00-620.00
50A	...	Each	620.00-720.00
ADD extra for :			
Flush surround, for wall recessed outlet	...	Each	53.00-65.00
Neon indicator to power outlet	...	Each	11.75-15.00
High pressure gasket (1600psi)	...	Each	20.50-25.00
Interlock mechanism (so switch must be off to plug in)	...	Each	29.00-35.00
Double - use the cost of two single outlets, plus	...	Each	52.00-60.00
Appliance Inlets - Industrial / Weatherproof IP56, modular, plastic :			
240V single phase, 3 flat pin - 15A	...	Each	134.00-158.00
415V three phase, 5 round pin -			
20A	...	Each	210.00-245.00
50A	...	Each	355.00-420.00
Plugs - Industrial / Weatherproof IP56, rewireable, connected to multicore cable :			
3 flat pin -			
10A, 240V	...	Each	40.00-50.00
15A, 240V	...	Each	50.00-61.00
20A, 240V	...	Each	61.00-74.00
3 round pin -			
20A, 240V	...	Each	104.00-124.00
32A, 240V	...	Each	152.00-178.00
4 round pin -			
10A, 415V	...	Each	86.00-104.00
20A, 415V	...	Each	96.00-114.00
32A, 415V	...	Each	132.00-158.00
40A, 415V	...	Each	190.00-220.00
50A, 415V	...	Each	250.00-295.00
ADD extra for :			
5 pin instead of 4 pin type	...	Each	13.00-15.75

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

BENCH/FLOOR OUTLET BOXES

		UNIT	Adel. Bris. Melb. Per. Syd. $
Floor or bench mounted enclosure in plastic, stainless steel or painted metal, for standard size plate or outlet, surface mounted (power and data socket-outlets excluded) :			
Plastic, for -			
One data or power plate/outlet	...	Each	68.00-80.00
Two plates/outlets with separator	...	Each	76.00-90.00
Four plates/outlets with separator	...	Each	160.00-186.00
Stainless steel or painted, for :			
One data or power plate/outlet	...	Each	53.00-64.00
Two plates/outlets with separator	...	Each	68.00-82.00
Four plates/outlets with separator	...	Each	108.00-130.00
ADD extra for :			
Power / data outlets	...	Note	
Blanking plates, where required	...	Each	9.75-12.75
Bench recessed box, with flush hinged lid, automatic socket-outlets, soft-wired lead and plate for data (set in prepared opening, sizes nominal) :	...	Note	
Pop Up style 120 x 120mm box with 1 x double power outlet	...	Each	250.00-340.00
250 x 120mm box with 4xpower outlets and space for 4 x data sockets	...	Each	170.00-245.00
250 x 160mm box with dual lids, 4 x power outlets and space for 1 x VGA & 4 x data sockets	...	Each	270.00-365.00
475 x 130mm box with 4 x power outlets and space for 4 x data sockets	...	Each	205.00-290.00
Desk-top aluminium power and data box with 4xpower outlets, plate for 4 x data sockets and soft-wired lead	...	Each	122.00-164.00

FLUSH FLOOR OUTLET BOXES

Steel box, set flush into prepared floor opening, for electrical socket-outlets, etc with flush, or floor covering rebated, hinged lid (lid frame may extend beyond stated box size)	...	Note	
Pop Up style 120x120x80mm deep floor/desk box with 1 x double auto power outlet or data socket plate	...	Each	340.00-500.00
Shallow compact 65mm deep floor box with flush stainless steel lid :			
135 x 135 with 1 x double auto power outlet	...	Each	490.00-680.00
Compact 100mm deep floor box with flush stainless steel lid :			
135 x 130 with 1 x double power outlet	...	Each	500.00-700.00
135 x 130 with 1 x double power outlet and plate for 3 x data sockets	...	Each	500.00-700.00
150 x 150 with 20mm floor covering recess lid, 1 x double power outlet and plate for 3 x data sockets	...	Each	610.00-830.00
ADD extra for :			
Data sockets, refer pages 567	...	Note	

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

FLUSH FLOOR OUTLET BOXES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Round compact 110mm deep floor box with flush stainless steel lid :			
125 x 125 with 165dia round lid, 1 x double power outlet and plate for 3 x data sockets (fits 165dia core hole)	...	Each	470.00-660.00
ADD extra for :			
Data sockets, refer pages 567	...	Note	
Shallow 70mm deep floor box with floor covering rebated plastic lid :			
240 x 150mm with 2 x auto power outlets and plate for 3 x data sockets	...	Each	395.00-570.00
240 x 240mm with 4 x auto power outlets and plate for 6 x data sockets	...	Each	470.00-660.00
360 x 240mm with 8 x auto power outlets and plate for 8 x data sockets	...	Each	600.00-820.00
ADD extra for :			
Data sockets, refer pages 567	...	Note	
110mm floor box with floor covering rebated plastic lid and space for standard size power and data plates :			
240 x 150mm to accept 1 x power & 1 x data plates	...	Each	315.00-460.00
240 x 240mm to accept 2 x power & 2 x data plates	...	Each	330.00-480.00
360 x 240mm to accept 3 x power & 3 x data plates	...	Each	490.00-680.00
ADD extra for :			
Data / power sockets, refer pages 567 and 578	...	Note	
110mm floor box with floor covering rebated stainless steel lid and space for standard size power and data plates :			
240 x 150mm to accept 1 x power & 1 x data plates	...	Each	460.00-650.00
240 x 240mm to accept 2 x power & 2 x data plates	...	Each	600.00-820.00
360 x 240mm to accept 3 x power & 3 x data plates	...	Each	770.00-1,020.00
ADD extra for:			
Data / power sockets, refer pages 567 and 578	...	Note	
Circular water resistant floor box with flat cast brass gunmetal framed lid and 55mm deep box for IP56/IP67 56-Series electrical accessories :			
170mm diameter, to suit one 107x101 accessory	...	Each	1,155.00-1,485.00
260mm diameter, to suit two 107x101 accessories	...	Each	1,675.00-2,150.00
ADD extra for :			
56-Series 107x101 accessory with -			
IP56 automatic 3 flat pin 10A socket	...	Each	82.00-94.00
IP56 twin RJ45 data/voice sockets	...	Each	80.00-92.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

HEATERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Radiant (strip) heater - wall mounted infra red :			
Plug in type -			
Halogen IR lamp type :			
1000W	...	Each	315.00-355.00
1500W	...	Each	325.00-365.00
Weatherproof (IP56), with bracket / chain :			
900W	...	Each	320.00-360.00
1800W	...	Each	440.00-490.00
Direct connect, quartz element type -			
Pull cord switch type (domestic style) :			
1000W	...	Each	98.00-122.00
Weatherproof (IPX4) stainless steel :			
2500W	...	Each	810.00-910.00
3000W	...	Each	920.00-1,035.00
4000W	...	Each	1,020.00-1,155.00
6000W	...	Each	1,230.00-1,380.00
8000W	...	Each	1,320.00-1,485.00
Weatherproof (IPX4) s/s, with gold coated elements :			
2500W	...	Each	960.00-1,080.00
3000W	...	Each	1,065.00-1,200.00
4000W	...	Each	1,170.00-1,305.00
6000W	...	Each	1,320.00-1,485.00
ADD extra for :			
Flush mounting (if needed)	...	Each	670.00-790.00
1, 2 or 3 switches and circuit (if needed)	...	Note	
Heat lamp heater - ceiling mounted/recessed :			
275W heat lamp type, plus R80 100W lighting lamp :			
2 heat lamps	...	Each	152.00-184.00
2 heat lamps and exhaust fan	...	Each	205.00-255.00
4 heat lamps and exhaust fan	...	Each	290.00-345.00
4 heat lamps and exhaust fan with 3m of flexible ducting	...	Each	410.00-500.00
4 heat lamps and exhaust fan and automatic operation	...	Each	270.00-325.00
375W heat lamp type, plus R80 100W lighting lamp :			
2 heat lamps and exhaust fan	...	Each	245.00-295.00
4 heat lamps and exhaust fan	...	Each	350.00-420.00
Halogen infra red, large area / industrial, heater - surface :			
One 1300W lamp	...	Each	315.00-390.00
3000W (two 1500W lamps)	...	Each	750.00-910.00
ADD extra for :			
Safety grill, if required	...	Note	
Baseboard / skirting heater - wall mounted :			
2000W plug in type, with built in thermostat	...	Each	470.00-550.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

HEATERS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Panel heater (convection and radiant), wall mounted, with thermostat :			
750W	...	Each	290.00-340.00
1000W	...	Each	305.00-355.00
1200W	...	Each	325.00-380.00
1500W	...	Each	335.00-395.00
2000W	...	Each	370.00-430.00
2400W	...	Each	400.00-460.00
Panel heater (convection and radiant), wall mounted, with thermostat and timer :			
750W	...	Each	340.00-400.00
1000W	...	Each	355.00-410.00
1200W	...	Each	375.00-430.00
1500W	...	Each	385.00-450.00
2000W	...	Each	420.00-490.00
2400W	...	Each	450.00-520.00
ADD extra for :			
Smart-phone/manual energy control system capability	...	Each	102.00-132.00
Energy Control System wireless router hub interface	...	Each	220.00-260.00
Fan heater, surface with cord switch and thermostat :			
1100W/2200W, automatic setting -			
White finish	...	Each	235.00-295.00
Stainless steel	...	Each	320.00-420.00
1100W/2200W, adjustable setting -			
White	...	Each	305.00-385.00
Stainless steel	...	Each	410.00-520.00
Fan heater, industrial wall mounted Xpelair :			
Metal case -			
3000W	...	Each	560.00-680.00
6000W	...	Each	850.00-1,035.00
ADD extra for :			
Industrial Xpelair type for thermostatic controller	...	Each	225.00-280.00
Towel rail heater - wall mounted :			
80W, 4 bar, painted	...	Each	172.00-210.00
65W, 8 bar, polished stainless steel	...	Each	295.00-355.00
130W, 10 bar, polished stainless steel	...	Each	510.00-600.00
175W, 5 looped bars, chrome finish, with switch	...	Each	335.00-410.00
ADD extra for :			
Socket outlet / power switch	...	Note	
Timer, refer to page 574	...	Note	
Oil filled column heater :			
5 fin, 1200W	...	Each	76.00-90.00
9 fin, 2000W	...	Each	114.00-134.00
11 fin, 2400W with timer	...	Each	146.00-170.00
Logfire heater - range of designs and options, 420W to 2400W	...	Range	400.00-3,050.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

FANS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Oscillating fan, table/wall mount type - 400mm	...	Each	82.00-104.00
ADD extra for shelf	...	Note	
Ceiling sweep fan and controller :			
3 metal blades -			
900mm	...	Each	215.00-275.00
1200mm	...	Each	210.00-270.00
1400mm	...	Each	225.00-295.00
4 rattan blades -			
1200mm	...	Each	235.00-305.00
Combined light, exhaust fan and heater - refer to Heaters, page 584			

EXHAUST FANS

Ceiling (into void) :			
200mm dia. -			
Basic	...	Each	82.00-106.00
With light	...	Each	200.00-250.00
Metal	...	Each	166.00-210.00
250mm dia. -			
Basic	...	Each	82.00-106.00
Retracting	...	Each	144.00-182.00
Decorative	...	Each	148.00-184.00
Metal	...	Each	174.00-220.00
300mm dia. -			
Basic	...	Each	210.00-260.00
Metal	...	Each	196.00-245.00
Pull cord operated :			
150mm dia. -			
Wall	...	Each	240.00-305.00
WIndow	...	Each	220.00-275.00
200mm dia. -			
Wall	...	Each	275.00-345.00
WIndow	...	Each	255.00-315.00
250mm dia. -			
Wall/window	...	Each	510.00-630.00
Automatic shutter :			
150mm dia. -			
Wall	...	Each	225.00-285.00
Window	...	Each	205.00-260.00
200mm dia. -			
Wall	...	Each	240.00-305.00
Window	...	Each	240.00-300.00
250mm dia. -			
Wall/window	...	Each	560.00-700.00
Window/wall & SELV control	...	Each	700.00-870.00
300mm dia. -			
Wall/window	...	Each	720.00-890.00
Window/wall & SELV control	...	Each	820.00-1,020.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

EXHAUST FANS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Continuous duty, gravity shutter, wall :			
300mm dia.	...	Each	800.00-990.00
400mm dia.	...	Each	1,095.00-1,350.00
In-line fan and ducted kit with flex duct, inlet and outlet :			
100mm dia axial, 3m duct 80cum/hr fan	...	Each	220.00-295.00
100mm dia centrifugal, 6m duct 270cum/hr fan	...	Each	450.00-580.00
150mm dia axial, 6m duct 245cum/hr fan	...	Each	355.00-460.00
150mm dia centrifugal, 6m duct 500cum/hr fan	...	Each	520.00-660.00
200mm dia axial, 6m duct 520cum/hr fan	...	Each	440.00-570.00
200mm dia centrifugal, 6m duct 900cum/hr fan	...	Each	720.00-900.00
250mm dia axial, 6m duct 875cum/hr fan	...	Each	530.00-680.00
250mm dia centrifugal, 6m duct 1200cum/hr fan	...	Each	900.00-1,125.00
ADD extra for :			
Gravity shutter / draught preventer, to ceiling fan	...	Each	46.00-59.00
Gravity shutter, to in-line fan (across size range)	...	Each	106.00-192.00
Cowling and roof flue, to ceiling fan	...	Note	
Opening in window glazing, refer page 447	...	Note	
Double glazing type, to 250 window fan	...	Each	260.00-335.00
Double glazing type, to 300 window fan	...	Each	300.00-385.00
Speed and direction controller, refer page 572	...	Note	
Thermostat / humidistat / direction controls	...	Note	

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES

Luminaires are offered in a wide variety of styles, standards and lighting sources and in a broad range of prices.	...	Note	
Prices are for EMC (electromagnetic compatibility) compliant luminaires complying to AU/NZ standards.	...	Note	
Low energy lighting can provide significant energy and maintenance savings and payback periods can be attractive. LED lighting is particularly popular. Non LED fittings can often accept LED lamps.	...	Note	
Energy savings are achieved with designs that automatically adjust lighting levels. Artificial lighting is altered to complement natural lighting. Full lighting is varied to suit occupancy and task requirements. Automation includes sensors (PIR/IR/MW) and run-on timers integrated into luminaires or for banks of luminaires. Wireless, IP, WiFi and controllers provide ready management.	...	Note	
Less popular fittings are anticipated to be offered at reduced prices as stocks are run down.	...	Note	
When estimating allow for non-standard job specific costs such as component assembly system, access requirements and facilitating future maintenance including cleaning of diffuser/surfaces.	...	Note	
Light sources are typically ordered in a colour temperature (Warm-White: 2700K-3000K, Natural: 4000K-4500K and Cool-White: 5000K-6000K). Some LED light sources are colour changeable, often at an additional cost including for the control method. Controllers are sized to suit the load (number of luminaires) and often include fading, sequencing, scenes, dimming and remote control functions in a range from basic single light upwards.	...	Note	
Colour correction (eg for medical, food, catering, photography, butcher's shop, panel shop, etc.) may require additional cost depending on the luminaire quality, function and light source.	...	Note	
Some LED/fluorescent luminaires are not dimming compatible and additional cost may be required depending on the light source, make model and manufacturer.	...	Note	
The following light fittings are based on popular manufacturer's good-quality fittings and include lamps, fixings and installation.	...	Note	

CEILING MOUNTED LIGHTING

LED Panel luminaire, fixed to plasterboard :			
600mm x 300mm -			
1450 lumen	...	Each	110.00-360.00
1600 lumen	...	Each	110.00-360.00
600mm x 600mm -			
2700 lumen	...	Each	114.00-365.00
3400 lumen	...	Each	114.00-365.00
1200mm x 300mm -			
2770 lumen	...	Each	102.00-315.00
3400 lumen	...	Each	102.00-315.00
1200mm x 600mm -			
5000 lumen	...	Each	230.00-780.00
5700 lumen	...	Each	250.00-850.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

CEILING MOUNTED LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Tube style batten luminaire, with LED lamps, fixed to plasterboard :			
Single tube -			
600mm long, 1 x 9W	...	Each	42.00-90.00
1200mm long, 1 x 18W	...	Each	46.00-94.00
1500mm long, 1 x 25W	...	Each	59.00-130.00
Twin tube -			
600mm long, 2 x 9W	...	Each	56.00-118.00
1200mm long, 2 x 18W	...	Each	73.00-146.00
1500mm long, 2 x 25W	...	Each	88.00-192.00
Tube style batten luminaire, with T5 (T16) / T8 (T26) fluorescent lamps, fixed to plasterboard :			
Single tube -			
600mm long, 1 x 14W/18W	...	Each	46.00-86.00
1200mm long, 1 x 28W/36W	...	Each	47.00-88.00
1500mm long, 1 x 58W	...	Each	69.00-126.00
Twin tube -			
600mm long, 2 x 14W/18W	...	Each	59.00-110.00
1200mm long, 2 x 28W/36W	...	Each	72.00-132.00
1500mm long, 2 x 58W	...	Each	102.00-188.00
Tube style moulded diffuser luminaire, with LED lamps, fixed to plasterboard :			
Single tube -			
600mm long, 1 x 9W	...	Each	62.00-220.00
1200mm long, 1 x 18W	...	Each	74.00-265.00
1500mm long, 1 x 25W	...	Each	96.00-330.00
Twin tube -			
600mm long, 2 x 9W	...	Each	90.00-265.00
1200mm long, 2 x 18W	...	Each	118.00-360.00
1500mm long, 2 x 25W	...	Each	156.00-430.00
Tube style moulded diffuser luminaire, with T5 (T16) / T8 (T26) fluorescent lamps, fixed to plasterboard :			
Single tube -			
600mm long, 1 x 14W/18W	...	Each	58.00-215.00
1200mm long, 1 x 28W/36W	...	Each	67.00-255.00
1500mm long, 1 x 58W	...	Each	92.00-325.00
Twin tube -			
600mm long, 2 x 14W/18W	...	Each	84.00-260.00
1200mm long, 2 x 28W/36W	...	Each	104.00-345.00
1500mm long, 2 x 58W	...	Each	152.00-420.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

CEILING MOUNTED LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Tube style tee-bar ceiling recessed luminaire, with LED lamps, including diffuser :			
Twin tube -			
600mm long (300mm wide tile), 2 x 9W	...	Each	86.00-245.00
1200mm long (300mm wide tile), 2 x 18W	...	Each	96.00-270.00
1200mm long (600mm wide tile), 2 x 18W	...	Each	172.00-500.00
Triple tube -			
1200mm long (300mm wide tile), 3 x 18W	...	Each	148.00-420.00
1200mm long (600mm wide tile), 3 x 18W	...	Each	198.00-560.00
Four tube -			
600mm long (600mm wide tile), 4 x 9W LED	...	Each	120.00-335.00
Tube style tee-bar ceiling recessed luminaire, with T5 (T16) / T8 (T26) fluorescent lamps, including diffuser :			
Twin tube -			
600mm long (300mm wide tile), 2 x 14W/18W	...	Each	78.00-240.00
1200mm long (300mm wide tile), 2 x 28W/36W	...	Each	82.00-255.00
1200mm long (600mm wide tile), 2 x 28W/36W	...	Each	158.00-480.00
Triple tube -			
1200mm long (300mm wide tile), 3 x 28W/36W	...	Each	128.00-395.00
1200mm long (600mm wide tile), 3 x 28W/36W	...	Each	176.00-540.00
Four tube -			
600mm long (600mm wide tile), 4 x 14W/18W	...	Each	106.00-320.00
ADD extra for :			
Wire guard to batten type luminaire, 600mm to 1500mm	...	Each	27.50-61.00
Industrial reflector to batten luminaire, 600mm to 1500mm	...	Each	27.00-65.00

LED TUBE SUBSTITUTION

Re-lamp existing luminaire with LED substitute lamp in exchange for a fluorescent tube :			
Direct replacement (no wiring necessary, but higher running costs) -			
10W, 600mm long (in lieu of 18W)	...	Each	22.50-28.50
15W, 900mm long (in lieu of 30W)	...	Each	29.00-36.50
19W, 1200mm long (in lieu of 36W)	...	Each	25.50-32.50
With luminaire components rewired -			
10W, 600mm long (in lieu of 18W)	...	Each	26.50-34.00
15W, 900mm long (in lieu of 30W)	...	Each	33.00-42.00
19W, 1200mm long (in lieu of 36W)	...	Each	29.50-38.00

STEEL CHANNEL SUSPENSION SYSTEM

Unistrut P1000 channel suspended with 1500mm long rod supports (LUMINAIRE suspension)	...	m	31.50-46.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ARCHITECTURAL LIGHTING TRACK			
Aluminium lighting track fixed to flat surface :			
Single circuit, rectangular section	...	m	62.00-114.00
Multi circuit, rectangular section	...	m	98.00-164.00
ADD extra for :			
Spotlight/floodlight	...	Note	
ARCHITECTURAL TUBE LIGHTING SYSTEMS			
Modular extruded aluminium tubular lighting system hung from ceiling on wire suspensions :			
Lighting track module, multi circuit	...	m	210.00-440.00
ADD extra for :			
Rigid bend	...	Each	210.00-470.00
Flexible bend	...	Each	225.00-420.00
Spotlight/floodlight	...	Note	
INTERIOR SPOTLIGHTS AND FLOODLIGHTS			
Surface or track mounted architectural spotlight fitting:			
Basic	...	Each	45.00-150.00
Full feature, adjustable	...	Each	170.00-500.00
DOWNLIGHTS			
Downlight in a wide range of designs including fixed / tilt / gimbal base - range (sample) :			
90mm dia.	...	Each	57.00-225.00
100mm dia. 120 minute fire rated	...	Each	150.00-275.00
100mm dia. x 75mm surface mounted	...	Each	98.00-230.00
100mm x 100mm square	...	Each	67.00-205.00
170mm dia. shop display light (hinged/adjustable)	...	Each	255.00-630.00
200mm dia.	...	Each	205.00-700.00
225mm dia.	...	Each	225.00-820.00
240mm x 145mm shop display light (hinged/adjustable)	...	Each	280.00-910.00
240mm dia.	...	Each	355.00-880.00
ADD extra for :			
Weatherproof design (where applicable)	...	Each	15.00-155.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

EMERGENCY LIGHTING

		UNIT	Adel. Bris. Melb. Per. Syd. $
Terminology :			
Maintained: lamp with emergency supply used for normal lighting;			
Non-maintained (formally Sustained): lamp with emergency supply that only illuminates for emergency purposes;			
Combined non-maintained: 2 lamps, one on standard supply and, one non-maintained lamp.	...	Note	
Self contained Single Point Unit (SPU) with LED light source and minimum 2 hour battery (classifications are for Class C90) :			
Disk style -			
Recessed (in-ceiling battery pack) :			
2W (D40)	...	Each	118.00-144.00
3W (D50)	...	Each	118.00-144.00
Surface (enclosed battery pack) :			
2W (D40)	...	Each	160.00-196.00
3W (D50)	...	Each	160.00-196.00
Weatherproof/vandal resistant, surface (enclosed battery pack) :			
2W (D40)	...	Each	400.00-490.00
3W (D50)	...	Each	400.00-490.00
Batten style -			
600mm long bare - 8W (D63)	...	Each	245.00-300.00
600mm long bare - 2 x 8W (D63)	...	Each	270.00-335.00
1200mm long bare - 16W (D63)	...	Each	275.00-335.00
1200mm long bare - 2 x 16W (D63)	...	Each	305.00-375.00
600mm long diffused - 8W (D40)	...	Each	255.00-310.00
600mm long diffused - 2 x 8W (D40)	...	Each	295.00-360.00
1200mm long diffused - 16W (D40)	...	Each	300.00-365.00
1200mm long diffused - 2 x 16W (D40)	...	Each	335.00-410.00
600mm long weatherproof - 8W (D50)	...	Each	310.00-380.00
600mm long weatherproof - 2 x 8W (D50)	...	Each	340.00-420.00
1200mm long weatherproof - 16W (D50)	...	Each	340.00-420.00
1200mm long weatherproof - 2 x 16W (D50)	...	Each	380.00-460.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

EMERGENCY LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Self contained Single Point Unit (SPU) with LED light source and minimum 2 hour battery (classifications are for Class C90) (cont'd) :			
Batten style, with in-built microwave motion detection -			
600mm long diffused - 8W (D40)	...	Each	335.00-410.00
600mm long diffused - 2 x 8W (D40)	...	Each	375.00-460.00
1200mm long diffused - 16W (D40)	...	Each	380.00-460.00
1200mm long diffused - 2 x 16W (D40)	...	Each	420.00-510.00
600mm long weatherproof - 8W (D40)	...	Each	420.00-510.00
600mm long weatherproof - 2 x 8W (D50)	...	Each	450.00-550.00
1200mm long weatherproof - 16W (D50)	...	Each	450.00-550.00
1200mm long weatherproof - 2 x 16W (D50)	...	Each	490.00-600.00
Oyster style - 350mm dia. - 20W (D80)	...	Each	230.00-285.00
Twin spot lights with surface IP65 box - 2x10W (indirect)	...	Each	335.00-410.00
ADD extra for :			
5 year warranty (longer battery life)	...	Note	
Emergency lighting monitoring system (for large numbers of SPU's)	...	Note	
Switchboard mounted emergency lighting (timed) testing control (for small quantity of SPU's)	...	Note	
Convert normal fluorescent luminaire to include 4W - 80W emergency lamp, factory fitted (requires lamp and space for lamp)	...	Each	166.00-205.00
EXIT SIGN self contained Single Point Unit (SPU) with LED light source and minimum 2 hour battery (classifications are for Class C90, with Maximum Viewing Distance) :			
Ceiling recessed box with slide-in sign sheet, 24m (4W)	...	Each	340.00-410.00
Surface box with slide-in sign sheet, 24m (4W)	...	Each	390.00-470.00
Surface weather proof IP67 box, 24m (4W)	...	Each	350.00-430.00
Large 620mm x 345mm surface box, 32m	...	Each	560.00-680.00
Large 620mm x 417mm surface box, 48m	...	Each	570.00-700.00
Large 770mm x 490mm surface box, 60m	...	Each	700.00-850.00

HIGH BAY LIGHTING

High bay luminaire with integral control gear (suitable for mounting above 5m high) :			
Induction light source -			
120W (in-lieu of 250W metal halide)	...	Each	450.00-560.00
150W	...	Each	510.00-620.00
200W (in-lieu of 400W metal halide)	...	Each	520.00-630.00
250W	...	Each	640.00-790.00
LED light source -			
100W (in-lieu of 250W metal halide)	...	Each	250.00-310.00
160W (in-lieu of 400W metal halide)	...	Each	345.00-420.00
200W (brighter than 400W metal halide)	...	Each	410.00-500.00
250W	...	Each	490.00-600.00
300W	...	Each	470.00-570.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

HIGH BAY LIGHTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
High bay luminaire with integral control gear (suitable for mounting above 5m high) (cont'd) :			
Mercury discharge -			
250W	...	Each	225.00-500.00
400W	...	Each	215.00-440.00
Metal halide discharge -			
250W	...	Each	420.00-780.00
400W	...	Each	420.00-820.00
Sodium discharge -			
250W	...	Each	460.00-620.00
400W	...	Each	520.00-670.00
ADD extra for :			
Wire guard	...	Each	55.00-79.00
Dust proof construction	...	Each	305.00-460.00

LOW BAY LIGHTING

LOW bay luminaire with integral control gear (for a mounting between 3m and 5m high) :			
LED light source -			
100W (in-lieu of 250W metal halide)	...	Each	250.00-380.00
160W (in-lieu of 400W metal halide)	...	Each	245.00-370.00
Mercury discharge -			
400W	...	Each	260.00-390.00
Metal halide discharge -			
250W	...	Each	310.00-470.00
400W	...	Each	340.00-510.00
Sodium discharge -			
250W	...	Each	450.00-830.00
400W	...	Each	450.00-870.00

WEATHERPROOF (IP44 or better) SECURITY AND AREA LIGHTING (high quality luminaire with integral control gear)

Vandal-proof wall or soffit mounted luminaire :			
LED light source -			
20W	...	Each	94.00-245.00
50W	...	Each	108.00-290.00
100W	...	Each	192.00-540.00
150W	...	Each	395.00-590.00
200W	...	Each	490.00-730.00
300W	...	Each	590.00-900.00
400W	...	Each	460.00-690.00
Compact single lamp fluorescent -			
11W	...	Each	118.00-192.00
Standard single lamp 600mm LED -			
12W	...	Each	60.00-355.00
20W	...	Each	42.00-470.00
Compact twin lamp fluorescent -			
9W	...	Each	160.00-260.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
WEATHERPROOF (IP44 or better) SECURITY AND AREA LIGHTING (high quality luminaire with integral control gear) (cont'd)			
Vandal-proof wall or soffit mounted luminaire (cont'd) :			
Standard twin lamp 1200mm LED -			
10W	...	Each	62.00-385.00
24W	...	Each	78.00-540.00
Mercury discharge -			
50W	...	Each	250.00-410.00
80W	...	Each	250.00-410.00
250W	...	Each	500.00-820.00
Sodium discharge -			
18W	...	Each	290.00-480.00
70W	...	Each	275.00-460.00
150W	...	Each	550.00-890.00
RECESSED WALL/STEP LIGHTING			
LED weatherproof light recessed into prepared opening :			
160mm X 60mm, 2.5W	...	Each	73.00-450.00
240mm X 100mm, 2.5W	...	Each	73.00-450.00
125mm circular, 1.5W	...	Each	73.00-450.00
IN-GROUND, IN-FLOOR, IN-WALL LIGHTING			
In-Ground / In-Floor / In-Step / In-Wall light (in a wide range of designs and light sources), in prepared opening or built in - range (sample) :			
Deck Mounted	...	Each	46.00-720.00
In-Wall - small, compact fluorescent	...	Each	73.00-450.00
In-Wall - large	...	Each	98.00-570.00
Walk Over	...	Each	146.00-1,080.00
In-Ground - small, compact fluorescent	...	Each	170.00-750.00
In-Ground - large	...	Each	305.00-1,950.00
Drive Over - small	...	Each	245.00-3,450.00
Drive Over - large	...	Each	305.00-3,500.00
ADD extra for :			
Artistic feature / special lens / material	...	Each	50.00-500.00
BOLLARD LIGHTING			
Bollard (in a wide range of designs and sizes) with post nominally 500mm high, compact fluorescent - range (sample) :			
75mm dia. post and concrete footing	...	Each	740.00-1,200.00
75mm dia. post and baseplate	...	Each	650.00-1,050.00
100mm dia. post and concrete footing	...	Each	820.00-1,320.00
100mm dia. post and baseplate	...	Each	760.00-1,305.00
150mm dia. post and concrete footing	...	Each	1,200.00-2,000.00
150mm dia. post and baseplate	...	Each	1,185.00-2,000.00
200mm dia. post and concrete footing	...	Each	1,200.00-2,100.00
200mm dia. post and baseplate	...	Each	1,065.00-2,050.00
ADD extra for :			
1000mm high post (in lieu of 500mm)	...	Each	77.00-128.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

DECORATIVE POLE-MOUNTED LIGHTING

		UNIT	Adel. Bris. Melb. Per. Syd. $
Decorative Post-Top Lantern (in a wide range of designs and sizes) with integral control gear :			
Traditional / basic	...	Each	100.00-830.00
Mid range	...	Each	205.00-1,020.00
Avant-Garde / advanced	...	Each	330.00-1,395.00
ADD extra for mounting options :			
Wall bracket (range of styles), fixed to wall	...	Each	53.00-400.00
Pole, refer to LIGHTING COLUMNS pages 599 and 600	...	Note	
Opal Sphere (in a wide range of designs, light sources and sizes) with integral "in pipe" control gear :			
300mm diameter	...	Each	178.00-530.00
400mm diameter	...	Each	295.00-1,080.00
450mm diameter	...	Each	330.00-1,200.00
500mm diameter	...	Each	430.00-1,550.00
ADD extra for :			
Poles, refer to LIGHTING COLUMNS pages 599 and 600	...	Note	

FLOODLIGHTING - POLE OR WALL BRACKET MOUNTED (weatherproof - IP44 or better - floodlight with mounting bracket)

General purpose floodlight with integral control gear :			
LED light source -			
10W	...	Each	148.00-320.00
50W	...	Each	162.00-365.00
100W	...	Each	295.00-680.00
200W	...	Each	510.00-760.00
400W	...	Each	600.00-900.00
Mercury discharge -			
250W	...	Each	800.00-1,575.00
400W	...	Each	970.00-1,900.00
1000W	...	Each	910.00-2,550.00
Metal halide discharge -			
250W	...	Each	640.00-1,975.00
400W	...	Each	670.00-2,150.00
1000W	...	Each	1,125.00-2,900.00
1500W	...	Each	2,000.00-3,600.00
Sodium discharge -			
150W	...	Each	650.00-1,675.00
250W	...	Each	650.00-1,875.00
400W	...	Each	1,065.00-2,250.00
1000W	...	Each	1,260.00-2,950.00
General purpose floodlight with separate interior control gear located in pole compartment :			
Mercury discharge -			
400W	...	Each	660.00-1,320.00
1000W	...	Each	990.00-2,150.00
Metal halide discharge -			
400W	...	Each	760.00-1,500.00
1000W	...	Each	1,260.00-2,600.00
1500W	...	Each	1,410.00-2,850.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
FLOODLIGHTING - POLE OR WALL BRACKET MOUNTED (weatherproof - IP44 or better - floodlight with mounting bracket) (cont'd)			
General purpose floodlight with separate interior control gear located in pole compartment (cont'd) :			
Sodium discharge -			
400W	...	Each	790.00-1,550.00
1000W	...	Each	1,290.00-2,800.00
General purpose floodlight with separate weatherproof control gear :			
Mercury discharge -			
400W	...	Each	760.00-1,485.00
1000W	...	Each	1,125.00-2,400.00
Metal halide discharge -			
400W	...	Each	870.00-1,700.00
1000W	...	Each	1,525.00-3,000.00
1500W	...	Each	1,800.00-3,150.00
2000W	...	Each	1,800.00-3,350.00
Sodium discharge -			
400W	...	Each	910.00-1,725.00
1000W	...	Each	1,440.00-3,000.00
STREET LIGHTING (weatherproof lantern suitable for outreach arm mounting) :			
Accessway and security lantern - twin 9W LED	...	Each	200.00-300.00
Minor street lantern with integral control gear :			
LED light source -			
30W	...	Each	530.00-840.00
50W	...	Each	550.00-870.00
70W	...	Each	640.00-1,020.00
Mercury discharge -			
50W	...	Each	210.00-395.00
80W	...	Each	210.00-420.00
Sodium discharge -			
18W	...	Each	300.00-440.00
50W	...	Each	245.00-355.00
70W	...	Each	245.00-650.00
Main road lantern with integral control gear :			
Induction -			
150W	...	Each	650.00-980.00
LED -			
90W	...	Each	760.00-1,200.00
120W	...	Each	790.00-1,260.00
150W	...	Each	830.00-1,335.00
180W	...	Each	890.00-1,410.00
Mercury discharge -			
250W	...	Each	420.00-690.00
400W	...	Each	470.00-760.00
Sodium discharge -			
150W	...	Each	420.00-830.00
250W	...	Each	430.00-850.00
400W	...	Each	520.00-850.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LUMINAIRES (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
STREET LIGHTING (weatherproof lantern suitable for outreach arm mounting) (cont'd) :			
Highway and industrial lantern with separately mounted weatherproof control gear :			
Mercury discharge -			
250W	...	Each	540.00-900.00
400W	...	Each	620.00-1,010.00
Sodium discharge -			
150W	...	Each	870.00-1,440.00
250W	...	Each	710.00-1,575.00
400W	...	Each	790.00-1,305.00
ADD extra for :			
900mm outreach arm fixed to wall	...	Each	98.00-162.00
Pole and outreach arm, refer to LIGHTING COLUMNS page 600	...	Note	
12V / 24V LED AREA AND SOLAR POWERED LIGHTING			
Solar powered LED street light and powdercoat paint finished steel pole, with top mounted solar panel, 2m outreach arm at 6m AGL, HD pole mounted weatherproof battery/equipment enclosure, MMPT charge control, base plate, reinforced concrete base and erection	...	Note	
For 7 hours operation (and 5 day reserve) :			
62W 6400lm - approx 35m x 20m lighting area	...	Sum	6,400.00-7,400.00
For all night operation (and 5 day reserve) :			
30W 2800lm - approx 25m x 15m lighting area	...	Sum	6,400.00-7,200.00
62W 6400lm - approx 35m x 20m lighting area	...	Sum	7,400.00-8,400.00
Fork lift relocatable solar powered LED street light and hot dip galvanised finished steel pole, with top mounted solar panel, 2m outreach arm at 6m AGL, HD base plate mounted weatherproof battery/equipment enclosure, MMPT charge control and installation	...	Note	
For 7 hours operation (and 5 day reserve) :			
62W 6400lm - approx 35m x 20m lighting area	...	Sum	8,600.00-9,600.00
For all night operation (and 5 day reserve) :			
30W 2800lm - approx 25m x 15m lighting area	...	Sum	8,400.00-9,600.00
62W 6400lm - approx 35m x 20m lighting area	...	Sum	9,400.00-10,500.00

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LIGHTING COLUMNS

Tapered octagonal hot dip galvanised steel column (to suit localities with up to 45m/s wind speed) with cable termination compartment, spigotted top, rag bolts, concrete base and erection

GENERAL LIGHTING COLUMNS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Standard type -			
5.5m high	...	Sum	940.00-1,200.00
8m high	...	Sum	1,675.00-2,100.00
10m high	...	Sum	2,550.00-3,200.00
12m high	...	Sum	2,950.00-3,700.00
12m high (for cyclonic winds duty)	...	Sum	4,300.00-5,400.00
15m high	...	Sum	4,000.00-5,000.00
18m high	...	Sum	7,200.00-9,000.00
20m high	...	Sum	8,200.00-10,500.00
20m high (for cyclonic winds duty)	...	Sum	11,500.00-14,500.00
Hinged see-saw or break back type -			
8m high	...	Sum	2,500.00-3,150.00
10m high	...	Sum	3,700.00-4,600.00
12m high	...	Sum	3,900.00-4,800.00
15m high	...	Sum	5,400.00-6,600.00
18m high	...	Sum	9,400.00-11,750.00
20m high	...	Sum	9,400.00-11,750.00

Tapered octagonal hot dip galvanised steel column (to suit localities with up to 45m/s wind speed) with cable termination compartment, spigotted top, rag bolts, concrete base and erection

GENERAL LIGHTING COLUMNS

Hinged hydraulic type -			
8m high	...	Sum	5,400.00-6,400.00
10m high	...	Sum	6,000.00-7,400.00
12m high	...	Sum	6,200.00-7,600.00
15m high	...	Sum	13,000.00-15,500.00
25m high	...	Sum	32,000.00-39,000.00
30m high	...	Sum	34,000.00-41,500.00
35m high	...	Sum	61,000.00-73,000.00
40m high	...	Sum	62,000.00-75,000.00
ADD extra for :			
Powdercoat painting, in standard colours	...	m	25.50-30.00
Cross arm	...	Each	142.00-166.00
Climbing rungs (per metre of pole serviced)	...	m	48.00-58.00
Portable electric hydraulic power unit -			
To suit 8m to 15m high hinged columns	...	Each	8,200.00-9,800.00
To suit 25m to 30m high hinged columns	...	Each	19,500.00-22,750.00
To suit 35m to 40m high hinged columns	...	Each	36,000.00-42,000.00
Luminaires, electrical circuitry, banner arms and painting	...	Note	

ELECTRICAL SERVICES

POWER AND LIGHTING EQUIPMENT (cont'd)

LIGHTING COLUMNS (cont'd)

STREET LIGHTING COLUMNS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Standard column with curved 1.5m outreach arm -			
4.5m high	...	Sum	880.00-1,095.00
5.5m high	...	Sum	1,000.00-1,245.00
6m high	...	Sum	1,230.00-1,525.00
8m high	...	Sum	1,575.00-1,975.00
8m high (for cyclonic winds duty)	...	Sum	1,900.00-2,350.00
10m high	...	Sum	2,350.00-2,950.00
12m high	...	Sum	2,600.00-3,300.00
ADD extra for :			
Luminaires, electrical circuitry and painting	...	Note	

OUTDOOR, FULL COLOUR SIGNS (SIGNTRONICS.COM.AU)

Prices include equipment, controls and installation.	...	Note	
The following are examples of the wide variety of options, sizes and formats available.	...	Note	
Outdoor, full colour sign - mountable in any orientation - with interlinked 10mm or 6.67mm pixel pitch 320mm x 320mm LED display modules, examples :			
Single sided -			
640mm x 640mm	...	Each	1,825.00-2,400.00
640mm x 960mm	...	Each	2,650.00-3,500.00
640mm x 1280mm	...	Each	3,450.00-4,600.00
960mm x 1600mm	...	Each	6,200.00-8,400.00
960mm x 1920mm	...	Each	7,600.00-10,000.00
1280mm x 2240mm	...	Each	11,500.00-15,250.00
1600mm x 2240mm	...	Each	14,500.00-19,000.00
1920mm x 2880mm	...	Each	22,000.00-29,000.00
Double sided -			
640mm x 640mm	...	Each	3,450.00-4,600.00
640mm x 960mm	...	Each	5,000.00-6,600.00
640mm x 1280mm	...	Each	6,600.00-8,800.00
960mm x 1600mm	...	Each	12,250.00-16,250.00
960mm x 1920mm	...	Each	14,750.00-19,500.00
1280mm x 2240mm	...	Each	22,750.00-30,000.00
1600mm x 2240mm	...	Each	28,500.00-37,500.00
1920mm x 2880mm	...	Each	43,500.00-57,500.00
ADD extra for :			
Electrical supply circuit, refer ELECTRICAL SERVICES, pages 507 to 601	...	Note	
Wi-Fi/WLAN installation	...	Sum	124.00-700.00
Support columns / mounting framing	...	Range	620.00-4,800.00
Larger and more complex project-specific LED signs, screens and displays are available including high quality high-resolution video-walls, creative LED screen entrances, auditorium viewing screens and artistic video structures.			

ELECTRICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
POWER AND LIGHTING EQUIPMENT (cont'd)			
ELECTRIC SIGNS			
LED Internally Illuminated Light Box Signs :			
Single faced 2.4m x 0.4m	...	Each	1,875.00-1,975.00
Double faced 2.4m x 0.4m	...	Each	2,150.00-2,250.00
Single faced 1.8m x 1.2m	...	Each	2,700.00-2,800.00
Double faced 1.8m x 1.2m	...	Each	2,850.00-3,000.00
Single faced 2.4m x 1.2m	...	Each	3,300.00-3,400.00
Double faced 2.4m x 1.2m	...	Each	3,500.00-3,700.00
Single faced 3.6m x 1.8m	...	Each	6,400.00-7,000.00
Double faced 3.6m x 1.8m	...	Each	6,800.00-7,400.00

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING

ESTIMATING - THREE PHASE SUBMAIN OR SUBCIRCUIT

To forecast the size/cost of a circuit, establish:

1. **Circuit load** or maximum demand of the circuit - if unknown some examples are included in Table 1 (page 533) and Table 2. Circuit design may also require spare load capacity (e.g. 20% added to the calculated Total Load).
2. **Cable type** and installation method - e.g. refer to the examples in the Cable Selection Tables 5, 6, 7 or 8.
3. **Route length** of the cable – e.g. the actual distance the cable measures between switchboards.

Method – select Cable Selection Table 5, 6, 7 or 8:

1. Refer to directions provided for Table Selection Tables, page 603 and plot the load, route length, and determine the cable size.
2. Once the cable size is known, the conduit size, if required, can be found from Table 9 (page 608).
3. If single core cables are to be used, and an Earth cable is required, the Earth cable can be sized from Table 3.
4. Price the circuit by referring to the respective Detailed Prices section's cost data (page pages 541-564). Where necessary add the cost of cable carriers (conduit, trench, pits, cable tray, cable ladder, etc).
5. If single core cables are chosen, calculate a composite $/m rate for the circuit - remember to multiply the cost of the Active cable by four and, if required, to add the Earth cable cost.

For examples of composite Mains/Submains prices – refer to pages 534-536.

TABLE 2 – MAXIMUM DEMAND EXAMPLE, Domestic Installations

** Residential unit refers to single residence, block of home units or flats, etc.

Number of residential units (flats)**:	1	2	5	10	15	20	25	30	40	50
Lighting, 21 to 40 per unit	5	6	6	8	9	10	13	15	20	25
10A socket outlets, 21 to 40 per unit	15	15	15	45	49	53	75	79	88	98
3.60kW storage hot water unit	16	12	30	60	90	120	120	124	132	140
5.5kW electric cook top	12	15	15	28	42	56	70	84	112	140
7.5kW electric oven	16	15	15	28	42	56	70	84	112	140
14kW (h/c capacity) air-conditioning	13	26	65	130	195	260	325	390	520	650
3.2kW electric spa	10	23	62	127	192	257	322	387	517	647
TOTAL LOAD (A):	**87**	**112**	**208**	**426**	**619**	**812**	**995**	**1163**	**1501**	**1840**
Communal lighting and power										
TOTAL LOAD (A):										

NOTE: (If a 3 phase supply divide Total Load by 3)

- Mains/Submains to each unit are subject to minimum sizes - refer to supply authority.
- Items are to be deleted or modified to match designs e.g. delete electric cook tops if gas is used.
- Allow for spare capacity (e.g. add 25% to the calculated Total Load).
- Communal lighting and power is to be added to the Total Load.
- Sizes and quantities are examples.
- A double socket outlet is counted as 2 socket outlets.

TABLE 3 – EARTH SELECTION TABLE,
for size of Coper Earth in sqmm for selected Circuit (Active/Neutral).

Size of Active (sqmm):	2.5	4	6	10	16	25	35	50	70	95	120	150	185	240	300	400	500	630
for Single Al Actives	—	—	—	—	4	6	6	10	10	16	25	25	35	50	70	≥95†	≥95†	≥120†
for Single Cu Actives	2.5	2.5	2.5	4	6	6	10	16	25	25	35	50	70	95	120	≥120†	≥120†	≥120†
for 2 Parallel Al Actives	—	—	—	6	6	10	10	25	25	50	50	70	≥95†	≥95†	≥120†	≥120†	≥120†	≥120†
for 2 Parallel Cu Actives	—	4	6	6	10	16	25	35	50	95	95	120	≥120†	≥120†	≥120†	≥120†	≥120†	≥120†

† A larger earthing conductor may be required (refer to AS/NZS 3000), which will increase the cost.
• A submain to an outbuilding of an electrical installation forming a separate MEN installation will not require an Earth cable.
• This table gives typical sizes of Copper Earth cable for Active conductors (refer to AS/NZS 3000 for complete details).
• Larger Earth cables, than shown, may be required (e.g. for long cable runs).
• For prices of Earth cable use the same prices as given for TPI (single insulated PVC) cable.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

TABLE 4 – VOLTAGE DROP

§§ Tables 5, 6, 7 and 8 are based on a 2.5% Voltage Drop (VD). For another VD apply a correction factor.

REQUIRED VD	CORRECTION §§		REQUIRED VD	CORRECTION §§
5.00%	(2.00)		4.50%	(1.80)
4.00%	(1.60)		3.50%	(1.40)
3.00%	(1.20)		2.00%	(0.80)
1.50%	(0.60)		1.00%	(0.40)

Example: a 100A circuit @ 2.5% VD and 100m route length from the Cable Selection Table – 50sqmm cable.
The same circuit @ 2% VD requires 100m/0.8 correction = 125m = 70sqmm cable.

ENERGY COST OF CABLE SELECTION

Every cable will have less than 100% efficiency due to the power lost transferring the energy from source to load. A cable sized to carry close to its maximum capacity will be less efficient than a cable that is larger than maximum capacity - an initial extra outlay for a larger cable may provide a payback in a few years – and savings in the running cost will be on going. However, increasing the size of a cable that carries an occasional load is less likely to provide an attractive payback period.

For example: A cable is to operate at 75% of its rated maximum current, with a circuit load that runs for close to 24 hours/day 365 days/year. If the design increases the cable by 2 sizes the power loss in the cable will reduce accordingly. An Operations Benefit / Cost Analysis may show the initial extra outlay, saving in energy cost, payback period for the extra outlay and the resulting ongoing reduction in energy cost.

CABLE SELECTION, TABLES 5, 6, 7 AND 8

The tables are for estimating purposes, based on a maximum voltage drop of 2.5% of 400V/230V circuits and installed in conventional manner and excluding all extraneous circumstances. Before using a table, consider if correction factors should be applied. The cable selection tables are a prelude to established information, which when available, should be used. Procedure :

1. **Decide on the circuit's current load (A) and route length (m).**
2. **Decide on the type of cable and method of installation.**
3. **Plot the circuit load (select the next highest figure when an exact match is not available).**
4. **Choose a cable length that equals or exceeds the route length. Check the method of installation is acceptable for that size cable or choose a larger cable.**

Example: A 4 core sheathed subcircuit, 85m long, installed in conduit in the ground, has a 63A load. From Table 7 the cable size is 25sqmm.

Notes :

- Tables 6, 7 & 8 assume a balanced three phase circuit. The given cable size may need to be increased if a phase is significantly out of balance (e.g. the circuit could increase one or two sizes). The size of the Neutral cable can be different from the Active cables for some loads and circuit configurations (refer to AS/NZS 3000 for complete details).

- MIMS cables utilise their metal sheath as an Earth conductor and therefore do not require a separate Earth cable.

Cable selection is influenced by many factors including circuit current, circuit length, size and type of circuit protection, size of earth cable, type of cable and the method of installation. These can affect the circuit design and therefore the cost. Unforeseen changes in cost can occur with long cable runs and include enhanced discounts, costs for non-standard stock runs, long cable installation difficulties, housing of cable joints, mitigating thermal shock and thermal expansion, issues with fault loop and 3rd harmonic design requirements. Multi-core cable can include a standard Earth within the cable (which is included in the rate). The maximum voltage drop, from the point of attachment by the supply authority to the furthest point, motor, light, etc., is normally not to exceed 5.0% of the mains voltage.

Values used in the tables assume the following typical criteria :

- Ambient = 40°C, Soil temperature = 25°C, Thermal resistivity of soil = 1.2°C.m/W.
- Depth of laying in ground = 500mm
- The space between multiple circuits is not less than the minimum spacings allowed before derating factors apply.
- Circuit is protected by Type C circuit breaker.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

SINGLE PHASE CABLE SELECTION

Table 5 provides 230V single phase two core cable sizes. Cable selection tables 6 and 8 can be adapted to find a 230V single phase cable size - multiply the route length (m) by two prior to applying the product to the table to find the required cable size of an equivalent single phase circuit. Example: Table 6 shows 10sqmm for 55m @ 32A 3 phase load. For single phase use 2x55m=110m = 16sqmm).

CABLE SELECTION - TABLE 5 - refer to pages 602 and 603 for explanatory notes.

230V, TWO CORE INSULATED AND SHEATHED COPPER CONDUCTORS - maximum route length at stated current

Cable Size sqmm	10A	15A	20A	25A	32A	40A	50A
4	51m #	34m #	26m f	21m f	16m abgh	-	-
6	77m #	51m #	38m #	31m f	24m f	19m abgh	15m gh
10	129m #	86m #	65m #	52m #	40m f	32m f	26m cef
16	205m #	137m #	102m #	82m #	64m #	51m #	41m f
25	264m #	176m #	132m #	106m #	82m #	66m #	53m #
35	424m #	283m #	212m #	170m #	133m #	106m #	85m #

Cable Size sqmm	63A	80A	100A	125A	150A	175A	300A
6	-	-	-	-	-	-	-
10	20m agh	-	-	-	-	-	-
16	33m cf	26m abgh	20m g	-	-	-	-
25	42m f	33m cf	26m abgh	21m g	-	-	-
35	67m #	53m f	42m cf	34m cde	28m g	24m g	-

Approved installation methods for this cable, of selected size and current:

a Spaced from Surface, or on Ladder/Tray/Catenary	g Buried Direct in Ground
b Touching Wall/Ceiling/Floor	h Underground in Single Enclosure

EXCLUDED installation methods for this cable, of selected size and current:

c Exposed to Sun	e Partially Surrounded in Thermal Insulation
d Wiring Enclosure in Air	f Completely Surrounded in Thermal Insulation

m # :Any installation method (a to f)

m f :Any installation method except for f (Completely Surrounded in Thermal Insulation)

-- :Not recommended

Note:

- If more than one installation method is required – size the cable to suit the largest size e.g. a cable in conduit runs 10m in air and 15m in ground: size as if 25m in conduit in air.
- Refer to AS/NZS 3000 and AS/NZS 3008.1 for complete cable sizing requirements. Circuits are based on 0.8pf, 75C, 2.5% VD with stranded Copper conductors with or without earth core, armoured or non-armoured.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

CABLE SELECTION - TABLE 6 - refer to pages 602 and 603 for explanatory notes.

400V, SINGLE CORE INSULATED AND SHEATHED CU CONDUCTORS LAID IN TREFOIL - maximum route length at stated current

Cable Size

sqmm	10A	15A	20A	25A	32A	40A	50A	63A	80A
4	103m #	69m #	51m cf	41m cef	32m abgh	--	--	--	--
6	154m #	103m #	77m #	62m cf	48m cef	39m abgh	--	--	--
10		173m #	130m #	104m #	81m cf	65m cef	52m abgh	--	--
16			206m #	165m #	129m #	103m cf	82m cf	65m abgh	--
25					203m #	162m #	130m f	103m cf	81m cef
35						223m #	179m #	142m cf	112m cf
50							240m f	190m f	150m cf
70									212m f
95									

Cable Size

sqmm	100A	125A	150A	175A	200A	250A	300A	400A	500A
16	--	--	--	--	--	--	--	--	--
25	65m agh	--	--	--	--	--	--	--	--
35	89m cef	70m a	--	--	--	--	--	--	--
50	120m cef	96m abgh	80m a	--	--	--		--	--
70	170m cf	136m cef	113m cef	97m abgh	--	--	--	--	--
95	228m f	182m cf	152m cef	130m cef	114m abgh	--	--	--	--
120		223m cf	186m cf	159m cef	139m cef	111m abgh	--	--	--
150			219m cf	187m cf	164m cef	131m abgh	109m ab	--	--
185				219m cf	192m cf	153m cef	128m abgh	--	--
240					227m cf	182m cf	152m cef	114m ab	--
300						205m cf	171m cf	128m abgh	103m a
400						231m cf	193m cf	145m cef	116m ab
500							214m cf	160m cf	128m cef

Approved installation methods for this cable, of selected size and current:

a Spaced from Surface, or on Ladder/Tray/Catenary	g Buried Direct in Ground
b Touching Wall/Ceiling/Floor	h Underground in Single Enclosure

EXCLUDED installation methods for this cable, of selected size and current:

c Exposed to Sun	e Partially Surrounded in Thermal Insulation
d Wiring Enclosure in Air	f Completely Surrounded in Thermal Insulation

m #	:Any installation method (a to f)	m f	:Any installation method except for f (Completely Surrounded in Thermal Insulation)
--	:Not recommended		:Outside of the limits of this table - consult the Standards

Note:

- If more than one installation method is required – size the cable to suit the largest size e.g. a cable in conduit runs 25m in air and 75m in ground: size as if 100m in conduit in air.
- Refer to AS/NZS 3000 and AS/NZS 3008.1 for complete cable sizing requirements. Circuits are based on 0.8pf, 75C, 2.5% VD with stranded Copper conductors, non-armoured sheathed or unsheathed.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

CABLE SELECTION - TABLE 7 - refer to pages 602 and 603 for explanatory notes.

400V, THREE CORE AND FOUR CORE INSULATED AND SHEATHED CU CONDUCTORS - maximum route length at stated current

Cable Size

sqmm	10A	15A	20A	25A	32A	40A	50A	63A	80A
4	103m #	69m f	51m f	41m cef	32m gh	--	--	--	--
6	154m #	103m #	77m f	62m f	48m cf	39m agh	--	--	--
10		173m #	130m #	104m f	81m f	65m cef	52m abgh	--	--
16			206m #	165m #	129m f	103m f	82m ef	65m abgh	51m g
25					203m #	162m f	130m f	103m ef	81m agh
35						225m #	180m f	143m f	113m cef
50							241m f	191m f	151m cef
70									214m f

Cable Size

sqmm	100A	125A	150A	175A	200A	250A	300A	400A	500A
16	--	--	--	--	--	--	--	--	--
25	65m g	52m g	--	--	--	--	--	--	--
35	90m abgh	72m g	60m g	--	--	--	--	--	--
50	121m cef	97m abgh	80m g	69m g	--	--		--	--
70	172m f	137m cef	114m abgh	98m ag	86m g	--	--	--	--
95	232m f	186m f	155m ef	133m cef	116m abgh	93m g	--	--	--
120		228m f	190m f	163m cef	142m cef	114m ag	95m g	--	--

Approved installation methods for this cable, of selected size and current:

a Spaced from Surface, or on Ladder/Tray/Catenary	g Buried Direct in Ground
b Touching Wall/Ceiling/Floor	h Underground in Single Enclosure

EXCLUDED installation methods for this cable, of selected size and current:

c Exposed to Sun	e Partially Surrounded in Thermal Insulation
d Wiring Enclosure in Air	f Completely Surrounded in Thermal Insulation

m # :Any installation method (a to f)

m f :Any installation method except for f (Completely Surrounded in Thermal Insulation)

-- :Not recommended

( ) :Outside of the limits of this table - consult the Standards

Note:

- If more than one installation method is required – size the cable to suit the largest size e.g. a cable runs 25m in Sun and 75m on Tray: size as if 100m in Sun.
- Refer to AS/NZS 3000 and AS/NZS 3008.1 for complete cable sizing requirements. Circuits are based on 0.8pf, 75C, 2.5% VD with stranded Copper conductors with or without earth core, armoured or non-armoured.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

CABLE SELECTION - TABLE 8 - refer to pages 602 and 603 for explanatory notes.

400V, 1/1kV SINGLE CORE AND MULTICORE MIMS CABLES WITH CU CONDUCTORS - maximum route length at stated current

Cable Size

sqmm	10A	15A	20A	25A	32A	40A	50A	63A	80A
4	110m #	73m #	55m #	44m #	34m #	28m abcd	-	-	-
6	165m #	110m #	83m #	66m #	52m #	41m #	33m abcd	-	-
10		183m #	137m #	110m #	86m #	68m #	55m #	43m #	34m a
16			219m #	175m #	137m #	110m #	88m #	70m #	55m #
25					214m #	171m #	137m #	109m #	86m #
35						238m *	190m *	151m *	119m *
50								214m *	169m *
70									232m *

Cable Size

sqmm	100A	125A	150A	175A	200A	250A	300A	400A	500A
16	44m ab	-	-	-	-	-	-	-	-
25	68m #	55m abcd	-	-	-	-	-	-	-
35	95m *	76m *	63m *	-	-	-	-	-	-
50	135m *	108m *	90m *	77m *	67m ab	-	-	-	-
70	186m *	148m *	124m *	106m *	93m *	74m ab	-	-	-
95	244m *	195m *	163m *	139m *	122m *	98m *	81m ab	-	-
120		240m *	200m *	172m *	150m *	120m *	100m *	75m a	-
150			238m *	204m *	179m *	143m *	119m *	89m ab	-
185				235m *	206m *	165m *	137m *	103m *	82m ab
240					243m *	194m *	162m *	121m *	97m *
300						220m *	183m *	137m *	110m *
400						248m *	207m *	155m *	124m *

Approved installation method for Single Core cable, of selected size and current:

a Flat Horizontal - Spaced from Wall
b Trefoil - Spaced from Wall
c Flat Vertical - Clipped to Wall

Approved installation method for 3 or 4 Core cable, of selected size and current:

d Spaced from Wall
e Clipped to Wall

m # :Any cable / installation method (a to e)

m * :Any single core installation method a to c

-- :Not recommended

:Outside of the limits of this table - consult the Standards

Note:

- If more than one installation method is required – size the cable to suit the largest size e.g. a cable runs with 25m clipped to wall and 25m spaced from wall: size as if 50m clipped to wall.
- Refer to AS/NZS 3000 and AS/NZS 3008.1 for complete cable sizing requirements. Circuits are based on 0.8pf, 75C, 2.5% VD bare MIMS Copper conductors with single core laid in trefoil and multicore cable.

ELECTRICAL SERVICES

DESIGN DATA, FOR ESTIMATING (cont'd)

CONDUIT SELECTION
These tables are a rule-of-thumb guide to the capacity of a range of conduits and cables.
Product dimensions can vary between manufacturers. Determine actual conduit capacity, in accordance with the standards and rules, from manufacturer's cable and conduit sizes.

CONDUIT SELECTION TABLE 9 - for electrical circuits/cables (pick a larger conduit for long or complicated conduit runs).

Electrical Conduit Size and Type:		25 MD	32 MD	40 MD	50 MD	25 HD	32 HD	40 HD	50 HD	63 HD	80 HD	100 HD	125 HD	150 HD
Single Insulated Cable														
Set of 2 single-core	sqmm:	16	35	70	120	16	35	50	95	185	400	630	630	630
Set of 3 single-core	sqmm:	16	25	50	95	10	25	50	70	150	300	500	630	630
Set of 4 single-core	sqmm:	10	16	35	70	6	16	35	50	95	240	400	630	630
Set of 8 single-core	sqmm:	2.5	6	16	25	2.5	6	16	25	50	120	185	300	400
Double Insulated Cable														
Set of 2 single-core	sqmm:	10	16	50	70	6	16	35	70	150	300	630	630	630
Set of 3 single-core	sqmm:	6	16	35	70	6	16	25	50	95	240	500	630	630
Set of 4 single-core	sqmm:	4	10	16	50	4	10	16	35	70	185	300	500	630
Set of 8 single-core	sqmm:	1.5	2.5	6	16	-	2.5	6	16	25	70	150	240	300
Multi Core Cable														
2-core and earth	sqmm:	6	16	50	70	4	16	35	70	120	>120	>120	>120	>120
Set of two 2-core and earth	sqmm:	-	1.5	4	10	-	1.5	4	10	35	95	120	120	>120
Set of four 2-core and earth	sqmm:	-	-	1.5	4	-	-	1.5	4	10	50	95	120	120
3-core and earth	sqmm:	4	10	25	50	2.5	10	25	50	95	185	300	300	>300
4-core and earth	sqmm:	2.5	6	16	35	2.5	6	16	35	70	150	240	300	300
Set of two 4-core and earth	sqmm:	-	-	2.5	6	-	-	2.5	6	10	50	70	120	185
Set of four 4-core and earth	sqmm:	-	-	-	2.5	-	-	-	1.5	6	25	50	70	95

CONDUIT SELECTION TABLE 10 - for telecommunications circuits/cables (pick larger conduit for long or complicated conduit runs).

Telecommunications Conduit Size and Type:		20	25	32	50	80	100	P20	P50	P100	50	80	100	150
				Austel / S009				PN12	PN9	PN9		Heavy Duty		
Data Cable														
4 pair - Cat 5e Solid	Max Qty:	2	6	10	>20	>20	>20	7	>20	>20	>20	>20	>20	>20
4 pair - Cat 6 Underground	Max Qty:	2	4	7	19	>20	>20	5	>20	>20	17	>20	>20	>20
Telephone Cable														
2 pair - 0.5mm Internal	Max Qty:	5	9	16	>20	>20	>20	12	>20	>20	>20	>20	>20	>20
3 pair - 0.5mm Internal	Max Qty:	4	8	13	>20	>20	>20	10	>20	>20	>20	>20	>20	>20
10 pair - 0.5mm Internal	Max Qty:	1	2	4	10	>20	>20	2	16	>20	10	>20	>20	>20
25 pair - 0.5mm Internal	Max Qty:	0	1	1	4	16	>20	1	7	>20	4	15	>20	>20
50 pair - 0.5mm Internal	Max Qty:	0	0	1	2	9	15	0	4	15	2	8	14	>20
10 pair - 0.64mm External	Max Qty:	0	1	1	3	13	>20	1	5	>20	3	12	>20	>20
30 pair - 0.64mm External	Max Qty:	0	0	0	1	5	9	0	2	9	1	5	9	18
100 pair - 0.64mm External	Max Qty:	0	0	0	0	1	2	0	1	2	0	1	2	6

MECHANICAL SERVICES

ROOM AIR-CONDITIONING UNITS

The performance of air-conditioners and heat pumps is required to be tested and rated to AS/NZS 3823. Minimum Energy Performance Standards (MEPS) with capacity output in kW for cooling and heating are provided by equipment manufacturers. The kW capacity refers to the amount of heat removed / added (not the electrical load). A star rating system also applies.

TYPICAL BUILDING COOLING LOADS PER SQUARE METRE :

Building Type	Typical Cooling loads
Domestic :	125 to 155W per sqm
Office :	135 to 160W per sqm
Shop :	150 to 190W per sqm

EMPIRICAL COSTS - BY CAPACITY		UNIT	Adel. Bris. Melb. Per. Syd. $
Split type, reverse cycle, room air-conditioner with basic installation :			
2kW to 3kW cooling	...	kW	520.00-990.00
3kW to 5kW cooling	...	kW	450.00-780.00
5kW to 8kW cooling	...	kW	420.00-600.00
Installation only including electrical	...	Each	710.00-1,155.00

WALL OR WINDOW MOUNTED AIR-CONDITIONERS

Prices include standard typical installation	...	Note	
Cooling only :			
2.05kW cooling	...	Each	580.00-880.00
2.60kW cooling	...	Each	630.00-940.00
3.60kW cooling	...	Each	800.00-1,140.00
4.70kW cooling	...	Each	1,020.00-1,525.00
5.33kW cooling	...	Each	1,125.00-1,650.00
Heat / Cool :			
2.70kW cooling / 2.6kW heating	...	Each	730.00-1,065.00
3.45kW cooling / 3.2kW heating	...	Each	850.00-1,215.00
4.30kW cooling / 3.9kW heating	...	Each	1,080.00-1,600.00
ADD extra for :			
Openings in existing walls, refer to DEMOLITION Openings, page 207	...	Note	
10A/15A single phase power outlet, refer to PRICE PER POINT, pages 510 and 511	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
SPLIT TYPE ROOM AIR-CONDITIONERS			
Prices include standard typical installation	...	Note	
Cool only inverter :			
2.5kW cooling	...	Each	1,625.00-2,250.00
3.5kW cooling	...	Each	1,850.00-2,500.00
5.0kW cooling	...	Each	2,200.00-3,150.00
7.1kW cooling	...	Each	2,350.00-3,350.00
8.0kW cooling	...	Each	2,950.00-4,100.00
Heat/Cool inverter :			
2.5kW cooling / 3.2kW heating	...	Each	1,600.00-2,350.00
3.5kW cooling / 4.3kW heating	...	Each	1,775.00-2,600.00
4.2kW cooling / 4.1kW heating	...	Each	1,925.00-2,800.00
5.0kW cooling / 6.0kW heating	...	Each	2,350.00-3,600.00
6.0kW cooling / 7.2kW heating	...	Each	2,500.00-3,800.00
7.1kW cooling / 8.0kW heating	...	Each	2,700.00-4,100.00
ADD extra for :			
Additional piping, insulation, refrigerant gas, etc., where the condenser and evaporator (i.e. inside and outside cabinets) are greater than approximately 7m apart	...	Note	
Concrete cutting for pipework, refer to CONCRETE CUTTING AND DRILLING, page 243	...	Note	
10A/15A single phase power outlet, refer to PRICE PER POINT, pages 510 and 511	...	Note	
EVAPORATIVE DOMESTIC SYSTEMS			
Roof mounted evaporative air-conditioning including installation, ducting, electrical and plumbing. Cost to air-condition building with a recommended 30 air changes per hour and :			
Nominal 2.6m ceiling height	...	sqm	45.00-54.00
Ceiling height exceeding 2.6m	...	cum	18.25-23.00
DUCTED DOMESTIC SYSTEMS			
Ducted refrigerated domestic air-conditioning including installation, ducting and electrical. Cost to air-condition building using an alternative day or night zoning (i.e. cool bedrooms at night, living area during day) :			
Nominal 2.6m ceiling height	...	sqm	100.00-118.00
Ceiling height exceeding 2.6m	...	cum	40.00-50.00
SOLID FUEL HEATERS			
Prices include standard installation and flue	...	Note	
Log burning slow combustion stove type heater with large heat resistant glass panel door :			
Free standing type to suit single room	...	Sum	1,625.00-3,300.00
Fireplace type to suit single room (excluding brick hearth)	...	Sum	2,300.00-4,900.00

MECHANICAL SERVICES

GAS HEATERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include standard installation and flue where required	...	Note	
Log / embers fire place with heat resistant viewing glass door			
Indoor (30MJ/8.3kWh or 28000 BTU/hr)	...	sum	2,850.00-6,000.00
Outdoor 1100mm x 500mm (47MJ/13kWh or 45000 BTU/hr)	...	sum	4,200.00-8,000.00
Overhead radiant heater, mounted 2.5metres (minimum) above floor :			
12 MJ (3.4kW or 11,000 BTU/hr) -			
Manual, cord operated	...	Sum	770.00-840.00
Automatic, 240V operated	...	Sum	1,000.00-1,080.00
24 MJ (6.7kW or 23,000 BTU/hr) -			
Manual, cord operated	...	Sum	780.00-860.00
Automatic, 240V operated	...	Sum	1,125.00-1,230.00
40 MJ (11.1kW or 38,000 BTU/hr) -			
Manual, cord operated	...	Sum	960.00-1,065.00
Automatic, 240V operated	...	Sum	1,305.00-1,425.00
48 MJ (13.3kW or 46,000 BTU/hr) -			
Manual, cord operated	...	Sum	1,260.00-1,380.00
Automatic, 240V operated	...	Sum	1,440.00-1,575.00
ADD extra for :			
Durable stainless steel outer case	...	sum	198.00-230.00
Floor mounted radiant heater with fan assistance :			
25 MJ (6.9kW or 24,000 BTU/hr) -			
Unflued, console type	...	Sum	1,095.00-1,200.00
Flued, inbuilt type	...	Sum	1,575.00-1,775.00
33 MJ (9.2kW or 31,000 BTU/hr) -			
Flued, console type	...	Sum	1,750.00-1,975.00
Wall furnace convection heater with fan assistance, time switch and thermostatic control - surface mounted console type with concealed flue :			
26 MJ (7.2kW or 25,000 BTU/hr) -			
Manual	...	Sum	1,950.00-2,200.00
Automatic	...	Sum	2,050.00-2,350.00
40 MJ (11.1kW or 38,000 BTU/hr) -			
Manual	...	Sum	2,100.00-2,400.00
Automatic	...	Sum	2,250.00-2,550.00
Portable radiant convector type heater with flexible hose and bayonet connector :			
15 MJ (4.2kW or 14,000 BTU/hr)	...	Each	870.00-1,035.00
18 MJ (5kW or 17,000 BTU/hr)	...	Each	900.00-1,080.00
21 MJ (5.8kW or 20,000 BTU/hr)	...	Each	910.00-1,095.00
25 MJ (6.9kW or 24,000 BTU/hr)	...	Each	940.00-1,125.00
Outdoor heater (4.5m radius) including gas bottle	...	Each	540.00-750.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

EVAPORATIVE COOLING UNITS

Prices include standard installation	...	Note	
Single phase :			
6000 cum/hr (or 1700 L/s)	...	Sum	2,700.00-2,800.00
8000 cum/hr (or 2200 L/s)	...	Sum	3,000.00-3,200.00
10000 cum/hr (or 2800 L/s)	...	Sum	3,700.00-3,900.00
12000 cum/hr (or 3300 L/s)	...	Sum	4,300.00-4,500.00
Three phase :			
15000 cum/hr (or 4200 L/s)	...	Sum	5,600.00-6,000.00
17000 cum/hr (or 4700 L/s)	...	Sum	6,200.00-6,600.00
21000 cum/hr (or 5800 L/s)	...	Sum	7,000.00-7,400.00
26000 cum/hr (or 7200 L/s)	...	Sum	7,800.00-8,200.00
31000 cum/hr (or 8600 L/s)	...	Sum	8,800.00-9,400.00
43000 cum/hr (or 12000 L/s)	...	Sum	11,000.00-11,500.00
51000 cum/hr (or 14000 L/s)	...	Sum	12,000.00-12,750.00
ADD extra for :			
Electrical service (nom. 30m run) -			
Single phase	...	Sum	240.00-325.00
Three phase up to 31000 cum/hr	...	Sum	530.00-1,000.00
31000 cum/hr and over	...	Sum	800.00-2,800.00
Water service (nom. 20m run)	...	Sum	570.00-775.00
Drain (PVC pipe, nom. 20m run)	...	Sum	166.00-275.00
Plenum diffuser, 4 way -			
9000 cum/hr	...	Each	470.00-500.00
15000 cum/hr	...	Each	530.00-575.00
20000 cum/hr	...	Each	890.00-975.00
30000 cum/hr	...	Each	1,365.00-1,525.00
Ducting, duct fittings and air diffusers	...	Note	
Insulation	...	Note	
Relief air system	...	Note	
Automatic controls	...	Note	
Roof reinforcement where required	...	Note	
Painting	...	Note	

DUCTED GAS HEATERS

Prices include standard installation.			
Compact furnace, high efficiency heater, including standard installationstandard installation :			
Internal -			
16.5kW	...	Sum	2,550.00-3,050.00
19.5kW	...	Sum	2,750.00-3,250.00
26.5kW	...	Sum	3,050.00-3,550.00
31.5kW	...	Sum	3,250.00-3,800.00
External -			
19.5kW	...	Sum	2,350.00-2,850.00
26.5kW	...	Sum	2,900.00-3,450.00
31.5kW	...	Sum	3,150.00-3,750.00

MECHANICAL SERVICES

PACKAGED AIR-CONDITIONING UNITS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Complete installed system. Capacities given represent net cooling capacities to ARI Standard (35°C. D.B. outdoor air, 27°C. D.B. and 19.5°C. W.B. air entering evaporator)			
FULLY SELF CONTAINED roof top type with net cooling capacity of :			
Reverse cycle type -			
14 kW (4 ton), 850 L/s @ 120 Pa resistance	...	Sum	9,600.00-10,600.00
20 kW (6 ton), 1400 L/s @ 150 Pa resistance	...	Sum	11,000.00-12,250.00
30 kW (8 ton), 1630 L/s @ 150 Pa resistance	...	Sum	18,000.00-19,500.00
40 kW (13 ton), 2300 L/s @ 200 Pa resistance	...	Sum	21,250.00-23,250.00
60 kW (17 ton), 3000 L/s @ 250 Pa resistance	...	Sum	31,500.00-34,500.00
87 kW (26 ton), 4300 L/s @ 400 Pa resistance	...	Sum	44,500.00-48,500.00
SMALL SPLIT SYSTEM, with net cooling capacity of :			
Under ceiling type -			
Reverse cycle :			
7.5kW, 350 L/s	...	Sum	5,200.00-6,000.00
11kW, 500 L/s	...	Sum	6,000.00-7,000.00
13kW, 573 L/s	...	Sum	7,600.00-8,800.00
Cassette type -			
Reverse cycle :			
7.5kW, 350 L/s	...	Sum	3,900.00-4,600.00
11kW, 500 L/s	...	Sum	6,400.00-7,400.00
13kW, 573 L/s	...	Sum	7,800.00-9,200.00
PACKAGED (Plant room) TYPE with remote air cooled condenser and net cooling capacity of :			
Reverse cycle type -			
14 kW (4 ton), 850 L/s @ 120 Pa resistance	...	Sum	9,000.00-10,000.00
20 kW (6 ton), 1400 L/s @ 150 Pa resistance	...	Sum	10,500.00-11,750.00
30 kW (8 ton), 1630 L/s @ 150 Pa resistance	...	Sum	17,000.00-18,750.00
40 kW (13 ton), 2300 L/s @ 200 Pa resistance	...	Sum	22,750.00-24,750.00
60 kW (17 ton), 3000 L/s @ 250 Pa resistance	...	Sum	35,000.00-38,000.00
70 kW (20 ton), 4300 L/s @ 400 Pa resistance	...	Sum	37,500.00-41,000.00
ADD extra for :			
Ductwork	...	Note	
Duct insulation	...	Note	
Diffusers	...	Note	
Filters	...	Note	
Terminal reheating	...	Note	
Thermostats	...	Note	
Reduced voltage starters	...	Note	
Hoisting	...	Note	
Roof bracing and supports	...	Note	
Piping	...	Note	
Pipe insulation	...	Note	
Electrical supply circuitry	...	Note	
Refrigerant charging	...	Note	
Balancing and commissioning	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

PROCESS COOLING UNITS

Complete installed system. Capacities given represent nominal gross capacities. Units suit down-flow and up-flow air or duct connection and two levels of micro processor control modes.

COMPUTER ROOM ENVIRONMENT Air-Conditioning Units giving heating, cooling, humidifying, de-humidification and 95% efficiency filtration (room condition 22°C. D.B. and 50% R.H. and condensing temperature 45°C.) :

Direct expansion with remote air cooled condenser -			
8 kW	...	sum	22,500.00-25,500.00
12 kW	...	sum	24,750.00-27,500.00
18 kW	...	sum	30,500.00-33,000.00
23 kW	...	sum	34,500.00-37,500.00
27 kW	...	sum	39,500.00-45,000.00
38 kW	...	sum	53,000.00-60,000.00
48 kW	...	sum	57,000.00-62,000.00
53 kW	...	sum	61,000.00-67,000.00
63 kW	...	sum	67,000.00-72,000.00
68 kW	...	sum	70,000.00-76,000.00
78 kW	...	sum	80,000.00-85,000.00
88 kW	...	sum	84,000.00-89,000.00
Direct expansion, water cooled -			
8 kW	...	sum	22,750.00-25,500.00
12 kW	...	sum	25,000.00-27,500.00
18 kW	...	sum	29,500.00-32,000.00
23 kW	...	sum	37,000.00-37,500.00
27 kW	...	sum	38,500.00-44,000.00
38 kW	...	sum	49,500.00-55,000.00
48 kW	...	sum	53,000.00-55,000.00
53 kW	...	sum	58,000.00-63,000.00
63 kW	...	sum	61,000.00-63,000.00
68 kW	...	sum	65,000.00-70,000.00
78 kW	...	sum	71,000.00-77,000.00
88 kW	...	sum	75,000.00-80,000.00
Chilled water only -			
12 kW	...	sum	13,000.00-17,000.00
23 kW	...	sum	18,750.00-22,000.00
27 kW	...	sum	20,000.00-23,750.00
38 kW	...	sum	21,500.00-24,500.00
48 kW	...	sum	24,750.00-27,500.00
53 kW	...	sum	30,000.00-32,500.00
63 kW	...	sum	31,500.00-34,000.00
68 kW	...	sum	37,500.00-43,000.00
78 kW	...	sum	40,500.00-43,000.00
88 kW	...	sum	37,500.00-40,000.00
103 kW	...	sum	39,000.00-41,500.00
ADD extra for :			
Ductwork and diffusers	...	Note	
Duct insulation	...	Note	
Thermostats	...	Note	
Hoisting, roof bracing and supports	...	Note	
Additional piping and valves	...	Note	
Piping insulation	...	Note	
Cooling towers	...	Note	
Electrical supply circuitry	...	Note	
Balancing and commissioning	...	Note	

MECHANICAL SERVICES

AIR-CONDITIONING EQUIPMENT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include for installation but EXCLUDE hoisting, electrical circuitry, piping, ducting, controls, refrigerant, hot water, steam (unless typically included as normal part of packaged equipment).	...	Note	
WATER CHILLERS			
Centrifugal Compressor :			
Water cooled hermetic type (R514A, R1233, R513A or R134A) -			
1000 kW (285 ton)	...	Each	290,000.00-330,000.00
1400 kW (400 ton)	...	Each	330,000.00-380,000.00
2100 kW (630 ton)	...	Each	400,000.00-450,000.00
3500 kW (1000 ton)	...	Each	550,000.00-600,000.00
Helical Rotary (Screw or Scroll Compressor) :			
Air cooled, self contained packaged type -			
200 kW (60 ton)	...	Each	105,000.00-120,000.00
300 kW (85 ton)	...	Each	115,000.00-130,000.00
400 kW (115 ton)	...	Each	135,000.00-150,000.00
630 kW (170 ton)	...	Each	180,000.00-230,000.00
800 kW (230 ton)	...	Each	230,000.00-300,000.00
1000 kW (285 ton)	...	Each	310,000.00-350,000.00
Water cooled (EXCLUDING cooling tower) -			
200 kW (60 ton)	...	Each	88,000.00-100,000.00
300 kW (85 ton)	...	Each	96,000.00-110,000.00
400 kW (115 ton)	...	Each	105,000.00-120,000.00
630 kW (170 ton)	...	Each	125,000.00-150,000.00
800 kW (230 ton)	...	Each	185,000.00-210,000.00
1000 kW (285 ton)	...	Each	205,000.00-230,000.00
Absorption :			
Steam or hot water operated, water cooled (EXCLUDING cooling tower) packaged type - single stage -			
350 kW (100 ton)	...	Each	370,000.00-430,000.00
1050 kW (300 ton)	...	Each	490,000.00-550,000.00
2100 kW (630 ton)	...	Each	675,000.00-725,000.00
3500 kW (1000 ton)	...	Each	1,100,000.00-1,225,000.00
COOLING TOWERS			
Prices include assembly and typical installation. Performance duty : inlet water = 35°C, outlet water = 29.5°C, air entering = 23°C, wet bulb.	...	Note	
Standard square type (induced draft counter flow) with full load cooling capacity of :			
2.0 L/s (approx. 36kW or 10 ton refrig.)	...	Each	6,000.00-7,800.00
4.8 L/s (approx. 88kW or 24 ton refrig.)	...	Each	7,200.00-9,200.00
13.0 L/s (approx. 238kW or 66 ton refrig.)	...	Each	14,500.00-18,500.00
32.0 L/s (approx. 587kW or 163 ton refrig.)	...	Each	26,000.00-33,000.00
66.0 L/s (approx. 1210kW or 336 ton refrig.)	...	Each	43,500.00-58,000.00
Standard square type (forced draft) with full load cooling capacity of :			
66.0 L/s (approx. 1210kW or 336 ton refrig.)	...	Each	47,000.00-61,000.00
ADD extra for :			
Electrical supply and control circuitry	...	Note	
Pumps, piping and valves	...	Note	
Mountings and foundations	...	Note	
Water treatments	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
AIR-CONDITIONING EQUIPMENT (cont'd)			
FAN COIL UNITS			
Central station air handling :			
Low/medium pressure -			
950 L/s (2000 cfm)	...	Each	6,600.00-8,000.00
2800 L/s (6300 cfm)	...	Each	11,500.00-13,750.00
4700 L/s (10000 cfm)	...	Each	15,500.00-19,000.00
8500 L/s (18000 cfm)	...	Each	25,500.00-34,000.00
14 cum/s (30000 cfm)	...	Each	42,000.00-49,000.00
Individual room, wall cabinet type :			
90 L/s (200 cfm)	...	Each	2,600.00-3,100.00
280 L/s (630 cfm)	...	Each	3,250.00-3,900.00
570 L/s (1200 cfm)	...	Each	5,200.00-6,400.00
COOLING AND HEATING COILS			
Fin-tube, drainable :			
Chilled water type -			
457 x 3050mm (6 row)	...	Each	4,700.00-5,600.00
838 x 3050mm (6 row)	...	Each	7,400.00-8,200.00
1220 x 3050mm (6 row)	...	Each	11,000.00-11,750.00
Refrigerant type -			
457 x 3050mm (6 row)	...	Each	5,400.00-6,400.00
838 x 3050mm (6 row)	...	Each	7,600.00-9,000.00
1220 x 3050mm (6 row)	...	Each	10,500.00-12,500.00
Hot water type -			
457 x 1676mm (1 row)	...	Each	1,300.00-1,500.00
610 x 1676mm (1 row)	...	Each	1,575.00-1,900.00
762 x 1676mm (1 row)	...	Each	1,850.00-2,200.00
Steam type -			
457 x 1676mm (1 row)	...	Each	1,375.00-1,600.00
610 x 1676mm (1 row)	...	Each	1,675.00-2,000.00
762 x 1676mm (1 row)	...	Each	1,950.00-2,300.00
VARIABLE AIR VOLUME TERMINALS			
Single duct unit, 600 to 900L/s :			
Cooling only	...	Each	2,250.00-2,800.00
Cooling and electrical reheat	...	Each	3,800.00-4,500.00
Cooling only, low profile -			
With parallel fan	...	Each	3,500.00-4,300.00
With series fan	...	Each	4,000.00-4,800.00
MOTORISED AIR DAMPERS			
99.9% airtight by design	...	Note	
800mm x 300mm	...	Each	355.00-510.00
1600mm x 400mm	...	Each	790.00-1,170.00
1800mm x 600mm	...	Each	1,200.00-1,700.00
2000mm x 800mm	...	Each	1,600.00-2,350.00
ADD extra for :			
Actuator	...	Each	295.00-540.00

MECHANICAL SERVICES

WATER HEATING

		UNIT	Adel. Bris. Melb. Per. Syd. $
GAS FIRED MAINS PRESSURE STORAGE HOT WATER SYSTEM including flue pipe, with 60°C outlet temperature :			
Direct storage, commercial instantaneous -			
700L/1st hour with 1,280L/hr recovery	...	Sum	9,800.00-10,800.00
850L/1st hour with 1,529L/hr recovery	...	Sum	10,000.00-11,000.00
1,050/1st hour with 1,529L/hr recovery	...	Sum	13,250.00-14,500.00
1900L/1st hour with, 2,311L/hour recovery	...	Sum	15,750.00-17,250.00
2050L/1st hour with, 2,311L/hour recovery	...	Sum	13,500.00-15,000.00
2900L/1st hour with, 3,591L/hour recovery	...	Sum	24,250.00-26,500.00
3380L/1st hour with, 3,591L/hour recovery	...	Sum	24,250.00-26,750.00
Indirect, with copper heat exchange coils and integrated package burner -			
1900L/1st hour with, 2,546L/hour recovery	...	Sum	22,500.00-25,000.00
2050L/1st hour with, 2,546L/hour recovery	...	Sum	22,500.00-26,250.00
2900L/1st hour with, 5,246L/hour recovery	...	Sum	32,000.00-35,000.00
3380L/1st hour with, 5,246L/hour recovery	...	Sum	32,000.00-35,500.00
ELECTRIC, THREE PHASE, MAINS PRESSURE STORAGE HOT WATER SYSTEM including weather proof isolating switch with 60°C outlet temperature :			
700L storage, 1000L/1st hour (requires two 20A circuit)	...	Sum	15,750.00-17,250.00
700L storage, 1300L/1st hour (requires two 35A circuit)	...	Sum	16,500.00-18,000.00
1500L storage, 2000L/1st hour (requires four 20A circuit)	...	Sum	31,000.00-33,500.00
1500L storage, 2630L/1st hour (requires four 35A circuit)	...	Sum	32,500.00-35,500.00
2000L storage, 3000L/1st hour (requires six 20A circuit)	...	Sum	P.O.A.
2000L storage, 4000L/1st hour (requires six 35A circuit)	...	Sum	P.O.A.
Hot water units are also given in SANITARY PLUMBING SECTION pages 462 and 463	...	Note	
SIMONS GAS FIRED BOILERS			
High efficiency hot water heater with modulating gas burner :			
175kW	...	Sum	35,000.00-42,500.00
300kW	...	Sum	36,500.00-44,500.00
500kW	...	Sum	46,000.00-55,000.00
620kW	...	Sum	47,500.00-57,000.00
800kW	...	Sum	56,000.00-67,000.00
1000kW	...	Sum	58,000.00-69,000.00
1500kW	...	Sum	75,000.00-89,000.00
1800kW	...	Sum	76,000.00-89,000.00
2000kW	...	Sum	84,000.00-99,000.00
Add extra for :			
Flueing	...	Note	
Drainage and plumbing supply installation	...	Note	
Gas supply installation	...	Note	
Valves, pumps and vents	...	Note	
Electrical supply installation	...	Note	
Heat exchanges	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

STEAM BOILERS

Simons commercial gas fuelled unattended steam boiler

Variety of options and installation requirements are common and usually necessitate project specific installation methods.

Example:			
100kw	…	Sum	60,000.00-74,000.00
150kW	…	Sum	68,000.00-83,000.00
200kW	…	Sum	72,000.00-87,000.00
300kW	…	Sum	88,000.00-105,000.00
400kW	…	Sum	110,000.00-130,000.00
500kW	…	Sum	120,000.00-145,000.00
600kw	…	Sum	130,000.00-155,000.00
Add extra for :			
Flueing	…	Note	
Drainage and plumbing supply installation	…	Note	
Gas supply installation	…	Note	
Valves, pumps and vents	…	Note	
Electrical supply installation	…	Note	
Heat exchanges	…	Note	

HEAT EXCHANGERS

PLATE TYPE

Plate type heat exchanger with STEAM TO WATER HEAT transfer. 316 grade stainless steel plate material with M6-FM epoxy painted ms frame and EPDM gasket material. BSP steam connections and triclamp or BSM connections :

20 plate, 63kW, 700mm height x 300mm wide frame	...	Sum	5,200.00-6,000.00
40 plate, 166kW, 1000mm height x 450mm wide frame	...	Sum	10,500.00-11,750.00

MECHANICAL SERVICES

PUMPS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include installation but EXCLUDE electrical circuitry, piping and flexible pipe joints	...	Note	
GENERAL PURPOSE PUMPS			
CONDENSER WATER, CHILLED WATER, HEATING WATER PUMPS			
LONG COUPLED ISO PUMP SET			
ISO back pull-out pump (to ISO2858 international standard with cast iron casing, bronze impeller and stainless steel shaft fitted with mechanical seal or packed gland), spacer coupling and guard, T.E.F.C. 1440 r/min electric motor, and mild steel baseplate :			
Up to -			
5 L/s @ 10m head (1.1kW)	...	Each	3,850.00-4,800.00
10 L/s @ 9m head (1.5kW)	...	Each	6,200.00-7,700.00
8 L/s @ 14m head (2.2kW)	...	Each	6,500.00-8,300.00
10 L/s @ 22m head (4.0kW)	...	Each	8,800.00-9,000.00
Up to -			
11 L/s @ 25m head (5.5kW)	...	Each	7,200.00-9,400.00
20 L/s @ 22m head (7.5kW)	...	Each	8,000.00-10,000.00
20 L/s @ 32m head (11.0kW)	...	Each	10,200.00-12,800.00
40 L/s @ 22m head (15.0kW)	...	Each	10,600.00-13,200.00
35 L/s @ 35m head (22.0kW)	...	Each	14,500.00-18,500.00
COMPACT (CLOSE COUPLED) ISO PUMP SET			
ISO pump (to ISO2858 international standard with cast iron casing, bronze impeller and stainless steel shaft fitted with a mechanical seal), close coupled to a T.E.F.C. 1440 r/min electric motor via stub shaft :			
Up to -			
5 L/s @ 10m head (1.5kW)	...	Each	4,050.00-5,150.00
10 L/s @ 9m head (1.5kW)	...	Each	4,400.00-5,700.00
8 L/s @ 14m head (3.0kW)	...	Each	4,450.00-5,850.00
10 L/s @ 22m head (5.5kW)	...	Each	5,800.00-7,500.00
Up to -			
11 L/s @ 25m head (5.5kW)	...	Each	5,800.00-7,500.00
20 L/s @ 22m head (7.5kW)	...	Each	6,500.00-8,100.00
20 L/s @ 32m head (11.0kW)	...	Each	7,400.00-9,400.00
40 L/s @ 22m head (18.0kW)	...	Each	7,800.00-10,000.00
35 L/s @ 35m head (22.0kW)	...	Each	10,200.00-12,800.00
HOT WATER IN LINE CIRCULATING PUMP			
Multi speed single phase circulation pump :			
0.3 L/s @ 5m head (110W)	...	Each	650.00-910.00
SUBMERSIBLE SUMP PUMP			
Single phase electric motor and pump including automatic float switch control :			
3.2 L/s @ 12m head (750W)	...	Each	2,325.00-2,975.00
ADD extra for :			
Control panel for three phase operation including duty alternator, alarm level, pump failure alarm	...	Each	5,100.00-6,500.00
Set of two alarm level float switches	...	Set	1,300.00-1,670.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PUMPS (cont'd)			
PACKAGED COMPACT CLOSE COUPLED PRESSURE SYSTEMS			
Set of two (three phase electric motor driven) pumps with 2 x 30L draw off pressure tanks, control panel and interconnecting pipework :			
5.3 L/s @ 290kPa (2.2kW)	...	Set	25,500.00-32,750.00
4.5 L/s @ 585kPa (5.5kW)	...	Set	30,750.00-42,750.00
22 L/s @ 345kPa (11.0kW)	...	Set	36,000.00-50,000.00
ADD extra for :			
Additional 30L draw off pressure tank	...	Each	2,025.00-2,865.00
FIRE HYDRANT / SPRINKLER PUMPS to AS 2419/AS 2118			
Comprises complete unit including drive, base and control panel all to AS 2419/AS 2118			
ELECTRIC DRIVEN FIRE BOOSTER PUMP SET :			
Three phase electrically driven -			
10 L/s @ 35m head (7.5 kW)	...	Each	10,400.00-13,200.00
16 L/s @ 30m head (11.0kW)	...	Each	11,800.00-19,600.00
DIESEL DRIVEN FIRE BOOSTER PUMP SET :			
Standby diesel driven booster pump unit including fuel tank and batteries -			
10 L/s @ 35m head	...	Each	40,500.00-49,500.00
16 L/s @ 30m head	...	Each	40,500.00-49,500.00
FIRE JACKING PUMP :			
Including three phase electric motor complete with hydropneumatic accumulator -			
0.3 L/s @ 800 kPa (1.5kW)	...	Each	5,750.00-8,050.00
DIESEL FUEL PUMP			
Long coupled positive displacement pump with cast iron casing, cast iron rotor and stainless steel shaft. Either mechanical seal or packed gland, with coupling, guard, 1440 r/min, three phase, Class 1, Zone 1 electric motor on cast iron baseplate :			
Up to -			
0.165 L/s @ 400 kPa (550W)	...	Each	4,400.00-4,800.00
0.4 L/s @ 500 kPa (1.5 kW) (geared motor 496 r/min)	...	Each	7,200.00-9,600.00
2.47 L/s @ 970 kPa (4.0 kW) (geared motor 379 r/min)	...	Each	9,400.00-10,800.00
SUBMERSIBLE PUMPS FOR DIRTY WATER			
SUBMERSIBLE DRAINAGE PUMPS with multi-vane open or semi-shrouded impellers for dirty water containing abrasive particles :			
Up to -			
2L/s @ 8m head (0.4kW)	...	Each	1,305.00-1,575.00
5L/s @ 10m head (1.0kW)	...	Each	3,450.00-4,150.00
8L/s @ 14m head (2.2kW)	...	Each	4,600.00-5,500.00
20L/s @ 16m head (2.2kW)	...	Each	7,800.00-9,400.00
8L/s @ 50m head (8.0kW)	...	Each	12,750.00-15,250.00
30L/s @ 45m head (20.0kW)	...	Each	19,000.00-22,750.00
Available up to 280L/s, @ 380m head, (200kW)	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PUMPS (cont'd)			
SUBMERSIBLE PUMPS FOR DIRTY WATER (cont'd)			
SUBMERSIBLE WASTEWATER PUMPS with shrouded channel impellers for clean and wastewater containing solids and fibres :			
Up to -			
3L/s @ 6m head (0.6kW)	...	Each	2,375.00-2,850.00
5L/s @ 10m head (1.1kW)	...	Each	2,550.00-3,050.00
7L/s @ 10m head (1.2kW)	...	Each	2,850.00-3,450.00
8L/s @ 4m head (1.3kW)	...	Each	3,950.00-4,750.00
20L/s @ 6m head (2.0kW)	...	Each	4,800.00-5,800.00
10L/s @ 13m head (2.4kW)	...	Each	4,800.00-5,800.00
25L/s @ 9m head (3.1kW)	...	Each	6,400.00-7,800.00
18L/s @ 14m head (4.4kW)	...	Each	6,200.00-7,400.00
30L/s @ 12m head (5.9kW)	...	Each	8,800.00-10,600.00
20L/s @ 14m head (5.9kW)	...	Each	7,400.00-9,000.00
10L/s @ 30m head (7.4kW)	...	Each	8,000.00-9,800.00
50L/s @ 11m head (9.0kW)	...	Each	14,250.00-17,000.00
35L/s @ 16m head (9.0kW)	...	Each	11,200.00-13,400.00
50L/s @ 15m head (13.5kW)	...	Each	17,250.00-20,750.00
40L/s @ 20m head (13.5kW)	...	Each	15,000.00-18,000.00
10L/s @ 48m head (15.0kW)	...	Each	14,500.00-17,500.00
Available up to 1800L/s, @ 75m head, (520kW)	...	Note	
SUBMERSIBLE GRINDER PUMPS for pressure sewage system with grinder cutter and semi-shrouded impeller for wastewater containing fibres and solids :			
Up to -			
2L/s @ 11m head (1.2kW)	...	Each	3,950.00-4,700.00
2L/s @ 28m head (2.6kW)	...	Each	4,700.00-5,700.00
7L/s @ 17m head (4.4kW)	...	Each	6,800.00-8,200.00
4L/s @ 30m head (4.4kW)	...	Each	6,800.00-8,200.00
7L/s @ 23m head (7.4kW)	...	Each	8,600.00-10,400.00
4L/s @ 40m head (7.4kW)	...	Each	8,600.00-10,400.00
PACKAGED PUMPING STATIONS - FIBREGLASS			
SINGLE PUMP DOMESTIC PUMPING STATION - Fibreglass tank with gas tight gatic type lid, internal pipework, non-return valve, and knife-gate valve (to Water Authority requirements) :			
Up to -			
200 litres (630mm dia. x 930mm)	...	Each	3,950.00-4,900.00
900 litres (1000mm dia. x 1350mm)	...	Each	8,000.00-9,800.00
ADD extra for :			
Excavation - refer to Drainage pages 489 and 490	...	Note	
Wastewater pump (typically 5L/s @ 10m head) or grinder pump (typically 2L/s @ 11m head)	...	Note	
Electrical installation	...	Note	
Pump controller - single phase	...	Each	2,025.00-2,425.00
Three level float switches	...	Note	

MECHANICAL SERVICES

PUMPS (cont'd)

PACKAGED PUMPING STATIONS - FIBREGLASS (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
DUAL PUMP DOMESTIC AND COMMERCIAL PUMPING STATIONS - Fibreglass tank with gas tight gatic type lid, all internal pipework, fittings, ball check valve and knife-gate valve (to MWS and DB - W10-8(h), AS1170-2002 Part 1, AS1546-2001, AS2634-1983 and BS4994-1987, Water Authority and Government requirements) :			
1.0m diameter -			
1600m deep	...	Each	10,200.00-12,200.00
1.4m to 1.7m diameter -			
2000mm deep	...	Each	30,000.00-36,000.00
3000mm deep	...	Each	34,500.00-41,500.00
4000mm deep	...	Each	41,500.00-50,000.00
5000mm deep	...	Each	49,000.00-58,500.00
6000mm deep	...	Each	41,000.00-49,000.00
7000mm deep	...	Each	67,000.00-81,000.00
1.9m to 2.2m diameter -			
2000mm deep	...	Each	40,500.00-49,000.00
3000mm deep	...	Each	46,000.00-55,000.00
4000mm deep	...	Each	56,500.00-68,000.00
5000mm deep	...	Each	68,000.00-81,000.00
6000mm deep	...	Each	80,000.00-96,000.00
7000mm deep	...	Each	99,000.00-119,000.00
ADD extra for :			
Excavation - Refer to DRAINAGE pages 489 and 490	...	Note	
Two waste water pumps (typically 10L/s @ 13m head)			
Two grinder pumps (typically 7L/s @ 17m head)	...	Note	
Electrical installation `	...	Note	
Dual pump control cabinet - locked weatherproof type with audible and visual alarms	...	Each	4,550.00-5,600.00
Four level float switches	...	Note	
LEVEL FLOAT SWITCHES, fixed within tank, cable drawn into conduit :			
Plastic tilt switch (13 metre cable)	...	Each	355.00-385.00
Plastic tilt switch (20 metre cable)	...	Each	385.00-420.00
ADD extra for :			
Conduit	...	Note	
Support bracket	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $

VALVES

INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS

GATE VALVES

Stainless steel, with threaded ends :

15mm	...	Each	68.00-82.00
20mm	...	Each	78.00-94.00
25mm	...	Each	98.00-116.00
32mm	...	Each	134.00-158.00
40mm	...	Each	166.00-194.00
50mm	...	Each	225.00-260.00

Cast iron, resilient seated, with flanged ends :

65mm	...	Each	345.00-410.00
80mm	...	Each	430.00-510.00
100mm	...	Each	790.00-980.00
150mm	...	Each	1,215.00-1,500.00
200mm	...	Each	1,675.00-2,050.00

GLOBE VALVES

Stainless steel, with threaded ends :

15mm	...	Each	68.00-82.00
20mm	...	Each	78.00-94.00
25mm	...	Each	92.00-110.00
32mm	...	Each	140.00-164.00
40mm	...	Each	172.00-200.00
50mm	...	Each	220.00-255.00

Cast steel, with flanged ends :

50mm	...	Each	440.00-520.00
65mm	...	Each	550.00-640.00
80mm	...	Each	700.00-810.00
100mm	...	Each	1,170.00-1,395.00
150mm	...	Each	2,100.00-2,500.00
200mm	...	Each	2,950.00-3,400.00

BALL VALVES

Stainless steel, two piece, with threaded ends :

15mm	...	Each	56.00-68.00
20mm	...	Each	69.00-84.00
25mm	...	Each	84.00-100.00
32mm	...	Each	114.00-136.00
40mm	...	Each	160.00-188.00
50mm	...	Each	230.00-265.00

Stainless steel, fire safe, with flanged ends :

50mm	...	Each	560.00-630.00
65mm	...	Each	1,020.00-1,185.00
80mm	...	Each	1,260.00-1,440.00
100mm	...	Each	1,950.00-2,200.00
150mm	...	Each	3,900.00-4,500.00

Stainless steel, 3 way, with threaded ends :

15mm	...	Each	98.00-116.00
20mm	...	Each	102.00-122.00
25mm	...	Each	154.00-178.00
32mm	...	Each	265.00-305.00
40mm	...	Each	340.00-385.00
50mm	...	Each	475.00-500.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
VALVES (cont'd)			
INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)			
BUTTERFLY VALVES			
Cast iron, with stainless steel disc and lugged body :			
50mm	...	Each	230.00-285.00
80mm	...	Each	420.00-540.00
100mm	...	Each	620.00-780.00
150mm	...	Each	940.00-1,200.00
200mm	...	Each	1,290.00-1,625.00
ADD extra for :			
Flanges, gaskets and bolts	...	Note	
Insulation	...	Note	

PIPEWORK

INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (NOT PETROCHEMICAL INDUSTRY)			
For petrochemical installations refer page 633	...	Note	
Refer pages 803 to 807 for typical labour constants of industrial/commercial pipework and increased loading for petrochemical installations	...	Note	
INDUSTRIAL/COMMERCIAL INSTALLATIONS			
FIREPIPE, GALVANISED STEEL			
Installed complete but excluding supports	...	Note	
Approved for fire sprinkler systems (AS2118 through to AS4118 Part 2.1)	...	Note	
Hot dip galvanised sprinkler and hydrant AS2419 and AS4118 steel piping with VICTAULIC roll grooved coupling joints, diameter :			
65mm	...	m	97.00-119.00
80mm	...	m	108.00-132.00
100mm	...	m	118.00-146.00
150mm	...	m	180.00-222.00
200mm	...	m	256.00-315.00
ADD extra for :			
Pipe supports, refer page 633			
Insulation, refer page 634	...	Note	
Painting, refer page 449			
Victaulic fittings -			
Painted, refer page 626			
Galvanised, refer page 632	...	Note	
PRESSFITTED, STAINLESS STEEL			
High pressure 316 stainless steel piping with pressfitting joints, diameter :			
20mm (1.2mm wall x 22mm)	...	m	40.00-50.00
25mm (1.2mm wall x 28mm)	...	m	49.00-60.00
32mm (1.5mm wall x 35mm)	...	m	62.00-76.00
40mm (1.5mm wall x 42mm)	...	m	74.00-90.00
50mm (1.5mm wall x 54mm)	...	m	88.00-108.00
75mm (2.0mm wall x 76.1mm)	...	m	130.00-152.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PIPEWORK (cont'd)			
INDUSTRIAL/COMMERCIAL INSTALLATIONS (cont'd)			
PRESSFITTED, STAINLESS STEEL (cont'd)			
Standard coupling (stainless steel) - pressfitting type :			
20mm	...	Each	16.75-20.00
25mm	...	Each	21.00-25.50
32mm	...	Each	29.50-36.50
40mm	...	Each	40.00-49.00
50mm	...	Each	55.00-69.00
75mm	...	Each	114.00-132.00
45° bend (stainless steel) - pressfitting type :			
20mm	...	Each	24.00-29.00
25mm	...	Each	30.00-36.50
32mm	...	Each	45.00-55.00
40mm	...	Each	60.00-72.00
50mm	...	Each	82.00-100.00
75mm	...	Each	162.00-186.00
90° bend (stainless steel) - pressfitting type :			
20mm	...	Each	23.50-28.50
25mm	...	Each	30.00-36.50
32mm	...	Each	47.00-56.00
40mm	...	Each	65.00-77.00
50mm	...	Each	90.00-108.00
75mm	...	Each	196.00-220.00
Tee (stainless steel) - pressfitting to pressfitting type :			
20mm	...	Each	32.00-38.00
25mm	...	Each	43.00-51.00
32mm	...	Each	58.00-70.00
40mm	...	Each	75.00-90.00
50mm	...	Each	106.00-128.00
75mm	...	Each	250.00-285.00
Reducing tee (stainless steel) - pressfitting to pressfitting type :			
25mm - 20mm	...	Each	42.00-50.00
32mm - 20mm	...	Each	55.00-67.00
32mm - 25mm	...	Each	61.00-75.00
40mm - 20mm	...	Each	75.00-90.00
40mm - 25mm	...	Each	76.00-92.00
50mm - 20mm	...	Each	104.00-128.00
50mm - 25mm	...	Each	104.00-128.00
50mm - 40mm	...	Each	108.00-132.00
75mm - 40mm	...	Each	205.00-235.00
75mm - 50mm	...	Each	205.00-240.00
Reducing tee (stainless steel) - pressfitting type with BSP thread tee :			
20mm - 15mm	...	Each	44.00-53.00
25mm - 20mm	...	Each	57.00-69.00
32mm - 15mm	...	Each	68.00-84.00
32mm - 20mm	...	Each	70.00-84.00
40mm - 15mm	...	Each	86.00-104.00
40mm - 20mm	...	Each	86.00-106.00
50mm - 15mm	...	Each	120.00-148.00
50mm - 20mm	...	Each	120.00-148.00
50mm - 25mm	...	Each	120.00-148.00
75mm - 50mm	...	Each	295.00-335.00

MECHANICAL SERVICES

PIPEWORK (cont'd)

INDUSTRIAL/COMMERCIAL INSTALLATIONS (cont'd)

PRESSFITTED, STAINLESS STEEL (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Reducer (stainless steel, M-F) - pressfitting to pressfitting type :			
25mm - 20mm	...	Each	21.00-25.50
32mm - 20mm	...	Each	30.50-37.00
32mm - 25mm	...	Each	31.50-38.50
40mm - 20mm	...	Each	42.00-51.00
40mm - 25mm	...	Each	41.00-50.00
50mm - 20mm	...	Each	64.00-78.00
50mm - 25mm	...	Each	60.00-73.00
75mm - 50mm	...	Each	110.00-130.00
Adapter / reducer (stainless steel) - pressfitting type with BSP thread type :			
20mm - 15mm	...	Each	26.50-31.50
20mm - 20mm	...	Each	28.00-33.00
20mm - 25mm	...	Each	30.50-35.50
25mm - 20mm	...	Each	34.50-41.00
25mm - 25mm	...	Each	38.00-45.00
32mm - 25mm	...	Each	50.00-59.00
40mm - 40mm	...	Each	72.00-84.00
50mm - 50mm	...	Each	98.00-116.00
Flange adaptor (stainless steel) - pressfitting type :			
20mm	...	Each	126.00-136.00
25mm	...	Each	116.00-126.00
32mm	...	Each	118.00-130.00
40mm	...	Each	154.00-170.00
50mm	...	Each	210.00-235.00
75mm	...	Each	305.00-345.00

PAINTED CARBON STEEL

Medium duty piping to AS-1074 installed complete but excluding supports	...	Note	
Red oxide painted carbon steel piping with screwed joints, diameter :			
25mm	...	m	41.00-53.00
32mm	...	m	48.00-63.00
40mm	...	m	56.00-73.00
50mm	...	m	70.00-92.00
Red oxide painted carbon steel piping with VICTAULIC joints, diameter :			
65mm	...	m	79.00-102.00
80mm	...	m	92.00-118.00
100mm	...	m	102.00-132.00
150mm	...	m	170.00-215.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Painting, refer page 449	...	Note	
Galvanised VICTAULIC fittings (in lieu of painted), refer page 632	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PIPEWORK (cont'd)			
INDUSTRIAL/COMMERCIAL INSTALLATIONS (cont'd)			
PAINTED CARBON STEEL (cont'd)			
ADD extra for (cont'd) :			
Bend, with screwed joints :			
25mm	...	Each	55.00-82.00
32mm	...	Each	70.00-102.00
40mm	...	Each	86.00-124.00
50mm	...	Each	104.00-148.00
Elbow, ductile iron, painted, with VICTAULIC joints :			
65mm	...	Each	126.00-152.00
80mm	...	Each	154.00-184.00
100mm	...	Each	190.00-230.00
150mm	...	Each	310.00-375.00
Tee, with screwed joints :			
25mm	...	Each	71.00-106.00
32mm	...	Each	92.00-136.00
40mm	...	Each	112.00-160.00
50mm	...	Each	132.00-190.00
Tee, ductile iron, painted, with VICTAULIC joints :			
65mm	...	Each	170.00-200.00
80mm	...	Each	205.00-240.00
100mm	...	Each	250.00-295.00
150mm	...	Each	395.00-470.00
Single flange VICTAULIC adaptor with gasket and bolts - Table E :			
80mm	...	Each	255.00-305.00
100mm	...	Each	340.00-410.00
150mm	...	Each	490.00-580.00
STANDARD FIRE PROTECTION SPRINKLER HEADS			
Standard sprinkler head connected to piping :			
Flush mounting	...	Each	46.00-51.00
Concealed or exposed	...	Each	38.50-43.00
Automatic on-off sprinkler head connected to piping :			
Pendant	...	Each	255.00-285.00
Fusible link head :			
Pendant	...	Each	39.50-44.00
ADD extra for :			
Dropper, up to 1000mm long for flush head	...	Each	72.00-80.00

MECHANICAL SERVICES

PIPEWORK (cont'd)

COPPER

		UNIT	Adel. Bris. Melb. Per. Syd. $
TYPE B piping to AS-1432 installed complete including standard supports but excluding insulation	...	Note	
Copper piping with silver soldered or press-fit joints, diameter :			
15mm	...	m	30.00-40.00
20mm	...	m	36.50-48.00
25mm	...	m	46.00-59.00
32mm	...	m	55.00-69.00
40mm	...	m	64.00-80.00
50mm	...	m	78.00-98.00
65mm	...	m	98.00-122.00
80mm	...	m	136.00-168.00
100mm	...	m	174.00-215.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Insulation, refer page 634	...	Note	
Bend (soldered) :			
15mm	...	Each	18.25-22.50
20mm	...	Each	26.50-31.50
25mm	...	Each	31.00-36.50
32mm	...	Each	38.00-45.00
40mm	...	Each	47.00-56.00
50mm	...	Each	60.00-72.00
Elbow (press-fit) :			
50mm	...	Each	79.00-90.00
65mm	...	Each	108.00-122.00
80mm	...	Each	148.00-168.00
100mm	...	Each	295.00-325.00
Tee (soldered) :			
32mm	...	Each	33.50-41.00
40mm	...	Each	42.00-52.00
50mm	...	Each	51.00-63.00
Tee (press-fit) :			
50mm	...	Each	72.00-84.00
65mm	...	Each	138.00-158.00
80mm	...	Each	168.00-194.00
100mm	...	Each	265.00-300.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PIPEWORK (cont'd)			
INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)			
COPPER (cont'd)			
TYPE B piping to AS-1432 installed complete including standard supports but excluding insulation (cont'd)			
Copper piping with silver soldered or press-fit joints, diameter (cont'd) :			
Flange adaptor, with gasket and bolts (soldered) :			
20mm	...	Each	50.00-56.00
25mm	...	Each	54.00-61.00
32mm	...	Each	60.00-68.00
40mm	...	Each	68.00-78.00
50mm	...	Each	84.00-98.00
Flange adaptor, with gasket and bolts (press-fit) :			
65mm	...	Each	144.00-162.00
80mm	...	Each	200.00-225.00
100mm	...	Each	270.00-300.00
GALVANISED CARBON STEEL			
Heavy duty piping to AS-1074 installed complete but excluding insulation and supports	...	Note	
Galvanised carbon steel piping with screwed joints, diameter :			
20mm (medium duty)	...	m	39.00-51.00
25mm	...	m	49.00-63.00
32mm	...	m	59.00-76.00
40mm	...	m	69.00-88.00
50mm	...	m	86.00-110.00
65mm	...	m	110.00-142.00
80mm	...	m	138.00-176.00
100mm	...	m	172.00-220.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Insulation, refer page 634	...	Note	
Bend :			
20mm	...	Each	47.00-71.00
25mm	...	Each	53.00-78.00
32mm	...	Each	63.00-94.00
40mm	...	Each	74.00-108.00
50mm	...	Each	92.00-134.00
65mm	...	Each	140.00-198.00
80mm	...	Each	174.00-245.00
100mm	...	Each	290.00-395.00
Tee :			
20mm	...	Each	57.00-86.00
25mm	...	Each	66.00-98.00
32mm	...	Each	78.00-118.00
40mm	...	Each	88.00-130.00
50mm	...	Each	106.00-156.00
65mm	...	Each	142.00-205.00
80mm	...	Each	178.00-260.00
100mm	...	Each	280.00-400.00
Single flange - with gasket and bolts - Table D :			
50mm	...	Each	106.00-148.00
65mm	...	Each	134.00-186.00
80mm	...	Each	160.00-225.00
100mm	...	Each	215.00-305.00

MECHANICAL SERVICES

PIPEWORK (cont'd)

INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)

BLACK CARBON STEEL

		UNIT	Adel. Bris. Melb. Per. Syd. $
Schedule 40 piping installed complete but excluding insulation and supports	...	Note	
Black carbon steel piping with butt welded joints, diameter :			
20mm	...	m	46.00-60.00
25mm	...	m	54.00-70.00
32mm	...	m	61.00-79.00
40mm	...	m	72.00-92.00
50mm	...	m	88.00-114.00
65mm	...	m	116.00-150.00
80mm	...	m	142.00-182.00
100mm	...	m	178.00-225.00
150mm	...	m	250.00-315.00
200mm	...	m	380.00-480.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Insulation, refer page 634	...	Note	
Long radius elbow :			
20mm	...	Each	88.00-116.00
25mm	...	Each	96.00-126.00
32mm	...	Each	110.00-144.00
40mm	...	Each	128.00-168.00
50mm	...	Each	150.00-198.00
65mm	...	Each	182.00-240.00
80mm	...	Each	220.00-290.00
100mm	...	Each	295.00-385.00
150mm	...	Each	440.00-570.00
200mm	...	Each	670.00-860.00
Tee :			
20mm	...	Each	134.00-178.00
25mm	...	Each	144.00-188.00
32mm	...	Each	162.00-215.00
40mm	...	Each	182.00-240.00
50mm	...	Each	210.00-275.00
65mm	...	Each	265.00-345.00
80mm	...	Each	330.00-440.00
100mm	...	Each	410.00-540.00
150mm	...	Each	580.00-750.00
200mm	...	Each	800.00-1,035.00
Single flange - with gasket and bolts - 300LB :			
50mm	...	Each	205.00-260.00
65mm	...	Each	275.00-345.00
80mm	...	Each	325.00-410.00
100mm	...	Each	390.00-490.00
150mm	...	Each	550.00-690.00
200mm	...	Each	730.00-930.00

MECHANICAL SERVICES

PIPEWORK (cont'd)

INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)

STAINLESS STEEL

		UNIT	Adel. Bris. Melb. Per. Syd. $
Schedule 40 piping installed complete but excluding insulation and supports	...	Note	
Stainless steel (Grade 316) piping with butt welded joints, diameter :			
20mm	...	Each	73.00-94.00
25mm	...	Each	84.00-106.00
32mm	...	Each	98.00-122.00
40mm	...	Each	112.00-140.00
50mm	...	Each	136.00-168.00
65mm	...	Each	178.00-215.00
80mm	...	Each	210.00-250.00
100mm	...	Each	265.00-315.00
150mm	...	Each	410.00-480.00
200mm	...	Each	580.00-670.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Insulation, refer page 634	...	Note	
Bend :			
20mm	...	Each	112.00-148.00
25mm	...	Each	126.00-166.00
32mm	...	Each	140.00-184.00
40mm	...	Each	160.00-210.00
50mm	...	Each	180.00-235.00
65mm	...	Each	240.00-310.00
80mm	...	Each	305.00-395.00
100mm	...	Each	420.00-540.00
150mm	...	Each	640.00-790.00
200mm	...	Each	1,005.00-1,215.00
Tee :			
20mm	...	Each	174.00-230.00
25mm	...	Each	194.00-255.00
32mm	...	Each	215.00-280.00
40mm	...	Each	240.00-310.00
50mm	...	Each	265.00-345.00
80mm	...	Each	390.00-500.00
100mm	...	Each	470.00-600.00
150mm	...	Each	670.00-840.00
200mm	...	Each	1,095.00-1,305.00
Single flange - with gasket and bolts - 300LB :			
50mm	...	Each	290.00-390.00
80mm	...	Each	450.00-590.00
100mm	...	Each	600.00-760.00
150mm	...	Each	900.00-1,110.00
Stainless steel (Grade 304) piping with butt welded joints, diameter :			
As a guide use Grade 316 stainless steel prices -			
Less 10% to less 16% for largest size piping	...	Note	
Less 2% to less 10% for largest size, bend, tee, etc.	...	Note	

MECHANICAL SERVICES

PIPEWORK (cont'd)

INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)

GALVANISED MILD STEEL

		UNIT	Adel. Bris. Melb. Per. Syd. $
Heavy duty piping to AS-1074 installed complete but excluding insulation and supports	...	Note	
Galvanised mild steel piping with VICTAULIC joints, diameter :			
50mm	...	m	80.00-102.00
65mm	...	m	112.00-142.00
80mm	...	m	132.00-166.00
100mm	...	m	170.00-215.00
150mm	...	m	260.00-330.00
ADD extra for :			
Pipe supports, refer page 633	...	Note	
Insulation, refer page 634	...	Note	
Painted VICTAULIC fittings (in lieu of galvanised), refer page 626	...	Note	
Coupling :			
50mm	...	Each	106.00-120.00
80mm	...	Each	142.00-166.00
100mm	...	Each	220.00-250.00
150mm	...	Each	260.00-305.00
Elbow :			
50mm	...	Each	116.00-134.00
80mm	...	Each	186.00-215.00
100mm	...	Each	275.00-310.00
150mm	...	Each	345.00-410.00
45° elbow :			
80mm	...	Each	220.00-260.00
100mm	...	Each	275.00-315.00
150mm	...	Each	330.00-390.00
Reducer :			
80mm - 50mm	...	Each	156.00-184.00
100mm - 50mm	...	Each	172.00-205.00
100mm - 80mm	...	Each	182.00-215.00
150mm - 100mm	...	Each	290.00-340.00
Tee :			
50mm	...	Each	160.00-182.00
80mm	...	Each	260.00-300.00
100mm	...	Each	370.00-420.00
150mm	...	Each	480.00-550.00
Blank end :			
80mm	...	Each	90.00-106.00
100mm	...	Each	138.00-158.00
150mm	...	Each	166.00-198.00
Male threaded adaptor :			
50mm	...	Each	82.00-98.00
100mm	...	Each	200.00-235.00
Single flange with gasket and bolts - Table E :			
100mm	...	Each	380.00-450.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PIPEWORK (cont'd)			
INDUSTRIAL/HEAVY COMMERCIAL INSTALLATIONS (cont'd)			
PIPE SUPPORTS			
Hot dipped galvanised heavy duty pipe supports installed complete	...	Note	
Rod hanger fixed to concrete (nominal 1500mm drop), per metre of pipe run, diameter :			
20mm	...	m	27.00-34.00
25mm	...	m	27.00-34.00
32mm	...	m	27.00-34.00
40mm	...	m	27.50-34.50
50mm	...	m	34.50-43.00
65mm	...	m	34.50-43.00
80mm	...	m	38.00-48.00
100mm	...	m	38.50-49.00
150mm	...	m	48.00-60.00
Suspended pipe roller assembly fixed to concrete (nominal 1500mm drop), per metre of pipe run, diameter :			
20mm	...	m	49.00-62.00
25mm	...	m	49.00-62.00
32mm	...	m	49.00-62.00
40mm	...	m	51.00-63.00
50mm	...	m	56.00-70.00
65mm	...	m	56.00-70.00
80mm	...	m	66.00-82.00
100mm	...	m	71.00-86.00
150mm	...	m	82.00-100.00
Rubber lined single boss swing clip assembly (nominal 300mm drop with M10 rod), per metre of pipe run, diameter :			
12 - 15mm (83kg safe working load)	...	m	22.50-29.00
15 - 20mm (83kg safe working load)	...	m	22.50-29.50
20 - 30mm (83kg safe working load)	...	m	22.50-29.50
33 - 37mm (83kg safe working load)	...	m	22.50-29.50
36 - 46mm (83kg safe working load)	...	m	29.50-38.50
47 - 52mm (83kg safe working load)	...	m	30.00-39.00
54 - 63mm (116kg safe working load)	...	m	32.50-42.00
68 - 77mm (133kg safe working load)	...	m	33.00-43.00
82 - 103mm (133kg safe working load)	...	m	34.00-44.00
110 - 114mm (133kg safe working load)	...	m	34.50-44.00
124 - 142mm (316kg safe working load)	...	m	39.50-51.00
150 - 168mm (316kg safe working load)	...	m	45.00-58.00
PETROCHEMICAL INDUSTRY INSTALLATIONS (WITH QUALITY ASSURANCE)			
The standard required for petrochemical installations may increase the foregoing pipework	...	Note	
Labour constants for pipework are given on pages 803 to 807 and may assist in general adjustment of the foregoing figures	...	Note	

MECHANICAL SERVICES

PIPING INSULATION

		UNIT	Adel. Bris. Melb. Per. Syd. $
Insulate concealed pipework to commercial standard :			
Heating water pipe -			
Up to 50 NB	...	m	30.50-43.00
100 NB	...	m	51.00-77.00
150 NB	...	m	88.00-104.00
Heating water pipe bend -			
Up to 50 NB	...	Each	38.00-50.00
100 NB	...	Each	73.00-90.00
150 NB	...	Each	132.00-154.00
Heating water pipe flanged valve -			
Up to 50 NB	...	Each	80.00-94.00
100 NB	...	Each	112.00-154.00
150 NB	...	Each	182.00-225.00
Chilled water pipe -			
Up to 50 NB	...	m	58.00-140.00
100 NB	...	m	132.00-182.00
Chilled water pipe bend -			
Up to 50 NB	...	Each	110.00-168.00
100 NB	...	Each	174.00-225.00
Chilled water pipe flanged valve -			
Up to 50 NB	...	Each	220.00-300.00
100 NB	...	Each	300.00-370.00
Insulate exposed pipework to commercial standard :			
Heating water pipe -			
Up to 50 NB	...	m	62.00-82.00
100 NB	...	m	86.00-104.00
150 NB	...	m	108.00-126.00
Heating water pipe bend -			
Up to 50 NB	...	Each	70.00-90.00
100 NB	...	Each	108.00-126.00
150 NB	...	Each	156.00-194.00
Heating water pipe flanged valve -			
Up to 50 NB	...	Each	140.00-170.00
100 NB	...	Each	178.00-215.00
150 NB	...	Each	210.00-275.00
Chilled water pipe -			
Up to 50 NB	...	m	104.00-182.00
100 NB	...	m	174.00-240.00
Chilled water pipe bend -			
Up to 50 NB	...	Each	144.00-220.00
100 NB	...	Each	230.00-295.00
Chilled water pipe flanged valve -			
Up to 50 NB	...	Each	290.00-400.00
100 NB	...	Each	395.00-490.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
PIPING INSULATION (cont'd)			
Acoustically insulate pipework with Quadzero Wavebar 6kg/25mm foam barrier (or Deci-Bar ARF) faced with 4 Zero foil :			
Copper piping, diameter -			
40mm	...	m	57.00-62.00
65mm	...	m	72.00-75.00
100mm	...	m	104.00-106.00
ADD extra for :			
Bend -			
40mm	...	Each	30.00-31.50
65mm	...	Each	37.00-39.50
100mm	...	Each	62.00-66.00
Tee/junction -			
40mm	...	Each	27.00-31.50
65mm	...	Each	33.00-35.50
100mm	...	Each	54.00-55.00
Floor Waste Gully -			
100mm	...	Each	142.00-156.00
Refer to SANITARY PLUMBING page 486 for basic/domestic insulation types	...	Note	
HEAT TRACE CABLE INSTALLATION			
Hot water mains heat trace cable installation - plastic insulated cable installed along one pipe to maintain the water temperature when flow is static, including related accessories - HWAT Red (60°C) system - range	...	m	26.00-41.00
ADD extra for :			
Pipe installation	...	Note	
Electrical supply circuit to each run of heat trace cable	...	Note	
COMMERCIAL DUCTWORK			
FLEXIBLE DUCTING			
Prices include for installation	...	Note	
COMPRESSIBLE FABRIC FLEXIBLE DUCTING :			
150mm dia. -			
Bare	...	m	19.00-23.50
Rm 1.0 insulated	...	m	23.50-29.00
Rm 1.6 insulated	...	m	28.50-36.00
Rm 2.0 insulated	...	m	34.00-43.00
200mm dia. -			
Bare	...	m	18.50-26.50
Rm 1.0 insulated	...	m	25.00-29.50
Rm 1.6 insulated	...	m	31.50-39.00
Rm 2.0 insulated	...	m	37.50-48.00
300mm dia. -			
Bare	...	m	19.00-29.50
Rm 1.0 insulated	...	m	33.00-45.00
Rm 1.6 insulated	...	m	41.00-54.00
Rm 2.0 insulated	...	m	50.00-64.00

MECHANICAL SERVICES

COMMERCIAL DUCTWORK (cont'd)

FLEXIBLE DUCTING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
COMPRESSIBLE FABRIC FLEXIBLE DUCTING (cont'd) :			
400mm dia. -			
Bare	...	m	25.50-41.00
Rm 1.0 insulated	...	m	35.50-47.00
Rm 1.6 insulated	...	m	46.00-61.00
Rm 2.0 insulated	...	m	57.00-75.00
500mm dia. -			
Bare	...	m	31.50-56.00
Rm 1.0 insulated	...	m	43.00-66.00
Rm 1.6 insulated	...	m	55.00-86.00
Rm 2.0 insulated	...	m	66.00-106.00
SEMI RIGID FLEXIBLE ALUMINIUM DUCTING :			
150mm dia. -			
Bare	...	m	32.50-40.00
Rm 1.0 insulated	...	m	37.50-46.00
Rm 1.6 insulated	...	m	44.00-55.00
Rm 2.0 insulated	...	m	50.00-63.00
200mm dia. -			
Bare	...	m	33.00-47.00
Rm 1.0 insulated	...	m	42.00-49.00
Rm 1.6 insulated	...	m	50.00-62.00
Rm 2.0 insulated	...	m	58.00-75.00
300mm dia. -			
Bare	...	m	39.50-61.00
Rm 1.0 insulated	...	m	47.00-64.00
Rm 1.6 insulated	...	m	71.00-94.00
Rm 2.0 insulated	...	m	96.00-124.00
400mm dia. -			
Bare	...	m	51.00-84.00
Rm 1.0 insulated	...	m	62.00-84.00
Rm 1.6 insulated	...	m	86.00-116.00
Rm 2.0 insulated	...	m	110.00-148.00
500mm dia. -			
Bare	...	m	64.00-114.00
Rm 1.0 insulated	...	m	84.00-128.00
Rm 1.6 insulated	...	m	98.00-154.00
Rm 2.0 insulated	...	m	114.00-182.00
ADD extra for :			
Branch take off (Rm 0.4 insulated) -			
300mm dia. (light commercial type)	...	No	94.00-130.00
400mm dia. (light commercial type)	...	No	132.00-170.00
500mm dia. (light commercial type)	...	No	154.00-178.00
Reducer (Rm 0.4 insulated) -			
300mm dia. (light commercial type)	...	No	37.50-61.00
400mm dia. (light commercial type)	...	No	50.00-73.00
500mm dia. (light commercial type)	...	No	66.00-98.00

MECHANICAL SERVICES

COMMERCIAL DUCTWORK (cont'd)

RIGID DUCTING

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include for installation and hangers but EXCLUDE insulation. Ducting is to SMACNA specifications, for low velocity installations.	...	Note	
Measure ducting over all bends, branches, etc. (and then add extra for those fittings).	...	Note	
For ductwork manufactured in stainless steel multiply prices by 3.5	...	Note	
For ductwork manufactured in 316 Grade L low carbon stainless steel multiply prices by 5.5	...	Note	
RECTANGULAR SHEETMETAL DUCT of thickness to suit girth :			
0.5m girth	...	m	96.00-126.00
1.0m girth	...	m	118.00-142.00
1.5m girth	...	m	160.00-174.00
2.0m girth	...	m	190.00-210.00
2.5m girth	...	m	235.00-260.00
3.0m girth	...	m	280.00-300.00
3.5m girth	...	m	330.00-350.00
4.0m girth	...	m	375.00-400.00
ADD extra for :			
Bend with turning vanes, in duct -			
0.5m girth	...	No	64.00-94.00
1.0m girth	...	No	75.00-110.00
1.5m girth	...	No	142.00-220.00
2.0m girth	...	No	220.00-330.00
2.5m girth	...	No	390.00-450.00
3.0m girth	...	No	420.00-500.00
3.5m girth	...	No	490.00-610.00
4.0m girth	...	No	600.00-770.00
Branch spigot with splitter, in duct -			
0.5m girth	...	No	51.00-68.00
1.0m girth	...	No	69.00-110.00
1.5m girth	...	No	126.00-150.00
2.0m girth	...	No	158.00-194.00
2.5m girth	...	No	225.00-270.00
3.0m girth	...	No	250.00-335.00
3.5m girth	...	No	345.00-440.00
4.0m girth	...	No	450.00-580.00
Stop end, in duct -			
0.5m girth	...	No	21.50-28.00
1.0m girth	...	No	28.00-32.50
1.5m girth	...	No	39.50-48.00
2.0m girth	...	No	59.00-69.00
2.5m girth	...	No	71.00-80.00
3.0m girth	...	No	74.00-82.00
3.5m girth	...	No	84.00-92.00
4.0m girth	...	No	92.00-100.00
Transition, in duct (largest size) -			
0.5m girth	...	No	72.00-92.00
1.0m girth	...	No	96.00-106.00
1.5m girth	...	No	118.00-144.00
2.0m girth	...	No	146.00-164.00
2.5m girth	...	No	176.00-210.00
3.0m girth	...	No	196.00-235.00
3.5m girth	...	No	280.00-340.00
4.0m girth	...	No	335.00-385.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
COMMERCIAL DUCTWORK (cont'd)			
RIGID DUCTING (cont'd)			
SPIRAL SHEETMETAL DUCT (of thickness to suit diameter) with slip joints and internal joining collars :			
200mm dia	...	m	118.00-152.00
300mm dia	...	m	138.00-205.00
400mm dia	...	m	215.00-275.00
500mm dia	...	m	220.00-285.00
600mm dia	...	m	310.00-360.00
700mm dia	...	m	360.00-420.00
ADD extra for :			
Painting -			
200mm dia	...	m	14.75-20.00
300mm dia	...	m	18.75-23.50
400mm dia	...	m	23.50-29.00
500mm dia	...	m	29.00-36.00
600mm dia	...	m	33.00-40.00
700mm dia	...	m	38.50-44.00
Bend -			
200mm dia	...	No	196.00-245.00
300mm dia	...	No	360.00-410.00
400mm dia	...	No	460.00-520.00
500mm dia	...	No	770.00-870.00
600mm dia	...	No	830.00-910.00
700mm dia	...	No	990.00-1,125.00
Transition in duct (largest size) -			
300mm dia	...	No	250.00-290.00
400mm dia	...	No	295.00-310.00
500mm dia	...	No	320.00-395.00
600mm dia	...	No	410.00-480.00
700mm dia	...	No	430.00-500.00
SPIRAL SHEETMETAL DUCT (of thickness to suit diameter) with angle flanged joints :			
ADD 12% to 15% to foregoing slip jointed spiral ducting figures	...	Note	
RIGID PHENOLIC FOAM DUCT (KoolDuct)			
Rigid light-weight ducting manufactured from self insulating, closed cell phenolic foam board			
Non-toxic, non-flammable and faced internally and externally with washable aluminium film			
For general use and where fibreless insulation and cleanable surfaces are required. Suitable for alteration work on site	...	Note	
Use RECTANGULAR INTERNALLY INSULATED SHEETMETAL DUCT prices less 10%	...	Note	

DUCT INSULATION

Acoustic (internal) insulation with capping and corner angles :			
Rm 1.2 (50mm thick) insulated	...	sqm	68.00-104.00
Rm 2.0 (75mm thick) insulated	...	sqm	82.00-126.00
Rm 3.0 (100mm thick) insulated	...	sqm	96.00-144.00
External insulation :			
R1.2 (50mm thick, site applied)	...	sqm	40.00-64.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
FIRE DAMPERS			
Prices include installation	...	Note	
Two Hour fire rated metal air duct fire damper :			
800 x 300mm	...	No	620.00-740.00
1600 x 400mm	...	No	1,305.00-1,550.00
1800 x 600mm	...	No	1,800.00-2,200.00
2000 x 800mm	...	No	2,300.00-2,800.00
Two Hour smoke-fire rated metal air duct fire damper :			
800 x 300mm	...	No	1,010.00-1,155.00
1600 x 400mm	...	No	2,150.00-2,500.00
1800 x 600mm	...	No	3,250.00-3,800.00
2000 x 800mm	...	No	4,000.00-4,700.00
ADD extra for :			
Spring return actuator	...	No	390.00-460.00
TWO HOUR FIRE RATED DUCTING			
Prices are for surface area of duct wall and include installation	...	Note	
Fire rated coating, applied to rigid sheetmetal ducting :			
Plain coating	...	sqm	142.00-178.00
Plain coating with 50mm thick high temperature mineral wool insulation, applied externally	...	sqm	275.00-350.00
ADD extra for :			
Rigid sheetmetal ducting	...	Note	
Fire resistant building board	...	sqm	325.00-410.00
Durasteel metal	...	sqm	530.00-620.00
AIR FILTERS			
Standard size fixed panel modules available in a range of medias. Prices include for media, frame (except where given separately), and standard installation. Capacities and filter ratings are based on nominal clean resistance and efficiency generally applied to AS1324.2.4-5 - 2003 : Air Filter Performance. Filter banks are formed to obtain the required air flow ability by fitting together groups of filter modules.			
ELECTROSTATIC FILTERS			
Ionitron agglomerator (washable type) :			
80% to 92% (No 1 dust to AS1324) efficient -			
940 L/s	...	Each	14,500.00-16,500.00
7550 L/s	...	Each	37,000.00-44,000.00
21200 L/s	...	Each	79,000.00-95,000.00
ADD extra for :			
Electrical circuitry	...	Note	
METAL VISCOUS (Grease) FILTERS			
400 x 500mm (450 L/s)	...	Each	116.00-128.00
630 x 630mm (830 L/s)	...	Each	158.00-174.00

MECHANICAL SERVICES

AIR FILTERS (cont'd)

DRY ARRESTANCE FILTERS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Replacement medias to disposable filters form approximately 50% of the initial filter cost	...	Note	
Reloadable frames with replacement media. pads/service pleats :			
G4 Rating, up to 0.36sqm face area -			
50mm service pleat	...	Each	36.00-41.00
10-15mm flat panel dry media	...	Each	53.00-63.00
Disposable Panel, (cardboard frame) type :			
G4 Rating, 25mm thick -			
500mm x 500mm (230 L/s @ 45 Pa)	...	Each	42.00-49.00
600mm x 600mm (330 L/s @ 45 Pa)	...	Each	42.00-49.00
G4 & F5 Ratings, 50mm thick -			
500 x 500mm (460 L/s @ 40 Pa)	...	Each	60.00-68.00
630 x 630mm (670 L/s @ 50 Pa)	...	Each	60.00-68.00
G4 & F5 Ratings, 100mm thick -			
500 x 500mm (460 L/s @ 40 Pa)	...	Each	72.00-80.00
630 x 630mm (670 L/s @ 40 Pa)	...	Each	72.00-80.00
Meshed Panel (woven media) type :			
G2 & G3 Ratings, 25mm thick -			
400 x 500mm (290 L/s @ 30 Pa)	...	Each	100.00-114.00
630 x 630mm (560 L/s @ 30 Pa)	...	Each	146.00-166.00
Layer Vee form (woven media) type :			
G3 & G4 Ratings, 50mm thick -			
400 x 500mm (290 L/s @ 54 Pa)	...	Each	92.00-114.00
630 x 630mm (560 L/s @ 54 Pa)	...	Each	100.00-114.00
G3 & G4 Ratings, 100mm thick -			
500 x 500mm (380 L/s @ 58 Pa)	...	Each	112.00-118.00
630 x 630mm (560 L/s @ 58 Pa)	...	Each	120.00-134.00
Duo Vee 2 & 3-pocket deep bed type (economy deep bed option) :			
G4 & F5 Ratings, 380mm deep -			
600mm x 300mm (500 L/s @ 27 Pa - 62 Pa)	...	Each	132.00-146.00
600mm x 600mm (940 L/s @ 27 Pa - 62 Pa)	...	Each	156.00-172.00
Multi Peak 3-pocket deep bed type :			
G4 & F5 Ratings, 570mm deep -			
630 x 300mm (500 L/s @ 27 Pa - 62 Pa)	...	Each	164.00-225.00
630 x 630mm (940 L/s @ 27 Pa - 62 Pa)	...	Each	178.00-250.00
Four Peak deep bed type :			
G4 & F5 Ratings, 381mm deep -			
630 x 300mm (500 L/s @ 40 Pa - 65 Pa)	...	Each	186.00-225.00
630 x 630mm (940 L/s @ 40 Pa - 65 Pa)	...	Each	198.00-250.00
Pyramid indented cube deep bed type :			
G4 & F5 Ratings, 660mm deep -			
630 x 300mm (470 L/s @ 22 Pa - 52 Pa)	...	Each	205.00-245.00
630 x 630mm (850 L/s @ 22 Pa - 52 Pa)	...	Each	265.00-310.00
Interpocket/Univee 6 & 8 pocket type :			
F6 to F9 Ratings, 560 - 762mm depths -			
630 x 300mm (470 L/s @ 22 Pa - 68 Pa)	...	Each	245.00-350.00
630 x 630mm (944 L/s @ 22 Pa - 68 Pa)	...	Each	280.00-410.00
Interfirm rigid deep bed filters, cardboard frame :			
F6 to F9 Ratings, 150 - 300mm depths -			
600 x 600mm x 150mm (944 L/s @ 86 Pa - F6 rating)	...	Each	205.00-350.00
600 x 600mm x 300mm (944 L/s @ 72 Pa - F6 rating)	...	Each	225.00-410.00

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
AIR FILTERS (cont'd)			
DRY ARRESTANCE FILTERS (cont'd)			
NanoVee rigid deep bed filters with Nano fibre media, polypropylene frame 3-Vee construction (also available in 50mm Flat-Panel type filters):			
F5 to F9 Ratings, 300mm depths -			
600mm x 600mm x 300mm (944 L/s @ 47 Pa - F6 rating)	...	Each	225.00-410.00
HIGH EFFICIENCY AIR FILTERS			
High-performance air filters and housing systems to suit critical and semi-critical applications.			
Prices include for filter, frame (except where given separately), and standard installation.			
HEPA Filters individually tested to verify conformance to AS 4260.			
ULTRA CLEAN VENTILATORS (UCV)			
Operating theatre module (for new or existing theatres) handling 100 room air changes per hour/500 clean-zone changes per hour - 3800 x 3800mm (overall), S.A. @ 2.98cum/s and average down-flow @ 0.38sqm/s)	...	Sum	65,000.00-90,000.00
UCV alternative - budget arrangement - 4 x HEPA filter modules (either 900mm x 600mm or 1200mm x 600mm), fan assisted or supply from plantroom	...	Sum	24,500.00-28,500.00
ADD extra for :			
Light fittings, etc usually in a "all in one" UCV unit	...	Note	
Econocell/Bio-HEPA type :			
95% efficient -			
150mm deep x 590 x 290mm			
(290 L/s - 470 L/s @ 190 Pa - 320 Pa)	...	Each	500.00-590.00
290mm deep x 590 x 590mm			
(560 L/s - 940 L/s @ 280 Pa - 320 Pa)	...	Each	810.00-900.00
Mounting frame and clamps	...	Each	395.00-410.00
Absolute HEPA filter, concertina glass paper cell (disposable) type :			
99.99% & 99.995% efficient,150mm deep -			
EN1822 H13 and H14 ratings :			
610 x 610mm (360 L/s @ 250 Pa)	...	Each	750.00-850.00
610 x 1200mm (440 L/s @ 250 Pa)	...	Each	960.00-1,080.00
Cold DOP (factory) test certification	...	Each	88.00-108.00
Site test HEPA filter to AS 1806/7 (per filter)	...	Each	82.00-124.00
Site test HEPA filter to AS 1806/7 (per room)	...	Each	820.00-3,600.00
ADD extra for :			
Supply in fan-assisted or fanless ceiling module	...	Note	
Typical fan-assisted ceiling module arrangement (per filter)	...	Each	2,050.00-4,100.00
Fanless HEPA module (per filter)	...	Each	1,550.00-3,100.00
Absolute "Minipleat" HEPA filter, concertina glass paper cell (disposable) type :			
99.99 & 99.995% efficient, 70mm deep -			
EN1822 H13 and H14 Ratings :			
610mm x 610mm (360 L/s @ 250 Pa)	...	Each	350.00-600.00
610mm x 1200mm (440 L/s @ 250 Pa)	...	Each	450.00-700.00
Cold DOP (factory) test certification	...	Each	88.00-108.00
Site test HEPA filter to AS 1806/7 (per filter)	...	Each	82.00-124.00
Site test HEPA filter to AS 1806/7 (per room)	...	Each	820.00-3,600.00
Fanless Minipleat HEPA module (per filter)	...	Each	700.00-900.00
Also referred to as 'TFP' or disposable HEPA modules			
ADD extra for :			
Support frame to banks of filters	...	Note	

MECHANICAL SERVICES

		UNIT	Adel. Bris. Melb. Per. Syd. $
AIR DIFFUSION EQUIPMENT			
Double deflection register with turning vane volume control :			
150mm x 150mm	...	Each	84.00-106.00
200mm x 100mm	...	Each	90.00-114.00
400mm x 200mm	...	Each	104.00-136.00
Louvre face diffuser with OBD :			
300 x 300mm	...	Each	98.00-128.00
630 x 630mm	...	Each	140.00-174.00
4000mm long - 3 slot linear diffuser with boot, spigot and volume control damper	...	m	200.00-235.00
AIR-CONDITIONING BALANCING AND COMMISSIONING			
Prices include for all miscellaneous work, etc., to fully balance and commission a medium sized air-conditioning installation	...	Note	
Air diffuser equipment (register, grille, etc.)	...	Each	63.00-88.00
Air handler :			
Air portion	...	Each	710.00-850.00
Refrigerant portion	...	Each	235.00-310.00
Main item of plant, condenser/chiller	...	Each	690.00-890.00
NATURAL VENTILATION			
Prices include for prepared opening and upstand base but EXCLUDE pneumatic line or electrical supply (refer to Electrical Services from page 510 for 240V single phase outlet and control switch; cost varies with installation specification and route length and BMS interface, if required)	...	Note	
Refer to Mechanical Services Design Data page 648 for information on natural ventilator selection	...	Note	
The capacity of each ventilator is dependent upon its aerodynamic throat area. All areas given in natural ventilator descriptions refer to aerodynamic throat area.	...	Note	
NATURAL VENTILATOR, vertical discharge heavy duty continuous ridge or slope roof, all weather type with optional variable damper :			
300mm throat, 0.18sqm/m clear opening size ventilator -			
Zincalume	...	m	320.00-390.00
Colorbond	...	m	360.00-430.00
Colorbond Ultra	...	m	450.00-540.00
ADD extra for :			
Option (where required) -			
Variable twin damper with cord operation	...	m	255.00-320.00
Variable twin damper with electrical operation	...	m	335.00-400.00
600mm throat, 0.36sqm/m clear opening size ventilator :			
Zincalume	...	m	510.00-550.00
Colorbond	...	m	540.00-600.00
Colorbond Ultra	...	m	750.00-850.00
ADD extra for :			
Option (where required) -			
Variable twin damper with cord operation	...	m	255.00-320.00
Variable twin damper with electrical operation	...	m	390.00-470.00

MECHANICAL SERVICES

NATURAL VENTILATION (cont'd)

NATURAL VENTILATOR (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
1000mm throat, 0.62sqm/m clear opening size ventilator :			
Zincalume	…	m	1,275.00-1,525.00
Colorbond	…	m	1,305.00-1,550.00
Colorbond Ultra	…	m	1,725.00-2,100.00
ADD extra for :			
Option (where required) -			
Variable twin damper with cord operation	…	m	255.00-320.00
Variable twin damper with electrical operation	…	m	390.00-470.00

OPERABLE ROOF AND WALL NATURAL/FIRE VENTILATOR

1000 x 1200mm, 0.773 sqm/m clear opening size unit :			
Zincalume	…	Each	1,215.00-1,455.00
Colorbond	…	Each	1,290.00-1,525.00
Aluminium	…	Each	1,395.00-1,625.00
ADD extra for option :			
Pneumatic operation	...	Each	335.00-420.00
Electrical operation	...	Each	770.00-830.00
1000 x 1800mm, 1.227 sqm/m clear opening size unit :			
Zincalume	…	Each	1,380.00-1,625.00
Colorbond	…	Each	1,500.00-1,750.00
Aluminium	…	Each	1,600.00-1,850.00
ADD extra for option :			
Pneumatic operation	...	Each	335.00-420.00
Electrical operation	...	Each	770.00-830.00
1000 x 2400mm, 1.681 sqm/m clear opening size unit :			
Zincalume	…	Each	1,575.00-1,800.00
Colorbond	…	Each	1,700.00-1,950.00
Aluminium	…	Each	1,850.00-2,100.00
ADD extra for option :			
Pneumatic operation	...	Each	335.00-420.00
Electrical operation	...	Each	770.00-830.00
1400 x 2400mm, 2.432 sqm/m clear opening size unit :			
Zincalume	…	Each	1,625.00-1,875.00
Colorbond	…	Each	1,775.00-2,050.00
Aluminium	…	Each	1,925.00-2,200.00
ADD extra for option :			
Pneumatic operation	...	Each	335.00-420.00
Electrical operation	...	Each	770.00-830.00
1400 x 3000mm, 3.089 sqm/m clear opening size unit :			
Zincalume	…	Each	1,825.00-2,100.00
Colorbond	…	Each	2,000.00-2,300.00
Aluminium	…	Each	2,150.00-2,450.00
ADD extra for option :			
Pneumatic operation	...	Each	335.00-420.00
Electrical operation	...	Each	770.00-830.00

MECHANICAL SERVICES

NATURAL VENTILATION (cont'd)

OPERABLE ROOF AND WALL NATURAL/FIRE VENTILATOR (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ADD extra for associated work (examples) :			
Modification of ventilators used for automatic fire smoke and heat ventilation louvres in open position when air is off (fail safe type)	...	Each	100.00-142.00
Automatic control system to suit multi function ventilators consisting of pneumatic controls, rain sensor, fire control and compressed air supply - typical for warehouse and large commercial type heat/fire/smoke venting	...	Sum	10,500.00-16,250.00
Automatic control system to suit open louvre type ventilators consisting of pneumatic controls, rain sensor, fire control and compressed air supply - typical for supermarket/shopping centre type heat/fire/smoke venting	...	Sum	11,500.00-20,000.00
Air compressor for pneumatic control system (required where a compressed air supply is not readily available)	...	Note	
Pneumatic line from compressed air supply for pneumatic control system	...	m	65.00-82.00
Electric circuit to automatic control centre from power supply - nominal 30m	...	Sum	410.00-530.00
Smoke detectors where required	...	Note	
Fire rated ducting between ceiling and ventilator where required	...	Note	
Manually operated mechanical cord controls - typical for warehouse/factory -			
Single ventilator control	...	Sum	67.00-142.00
Four ventilator control	...	Sum	200.00-400.00
Inlet louvres where required	...	Note	

INLET LOUVRES

Refer to Mechanical Services Design Data page 662 for information on Inlet Louvre Selection	...	Note	
Adjustable WALL LOUVRE installed complete :			
Galvabond steel	...	sqm	770.00-870.00
Colorbond steel	...	sqm	870.00-960.00
Aluminium	...	sqm	1,020.00-1,125.00
Non-Adjustable WALL LOUVRE installed complete :			
Galvabond steel	...	sqm	385.00-470.00
Colorbond steel	...	sqm	490.00-550.00
Aluminium	...	sqm	710.00-800.00
ADD extra for :			
Wall penetration and structural supports	...	Note	

MECHANICAL SERVICES

POWERED VENTILATION

		UNIT	Adel. Bris. Melb. Per. Syd. $
Refer to Mechanical Services Design Data page 663 for information on power ventilator selection	...	Note	
ROOF MOUNTED EXHAUST VENTILATORS (non fire rated type) - e.g. light factory, shop, warehouse, sports hall :			
Single speed single phase low profile fan with free inlet capacity of -			
0.40 cum/s	...	Each	720.00-940.00
0.70 cum/s	...	Each	1,320.00-1,455.00
1.40 cum/s	...	Each	1,725.00-1,975.00
Single speed three phase low profile fan with free inlet capacity of -			
0.40 cum/s	...	Each	990.00-1,140.00
0.70 cum/s	...	Each	1,230.00-1,410.00
1.40 cum/s	...	Each	1,650.00-1,875.00
Two speed three phase low profile fan with free inlet capacity of -			
0.40 cum/s	...	Each	990.00-1,140.00
0.70 cum/s	...	Each	1,230.00-1,410.00
1.40 cum/s	...	Each	1,650.00-1,875.00
Single speed single phase high capacity vertical discharge fan with free inlet capacity of -			
0.75 cum/s	...	Each	1,305.00-1,485.00
1.45 cum/s	...	Each	1,775.00-1,975.00
2.45 cum/s	...	Each	1,975.00-2,400.00
Two speed single phase high capacity vertical discharge fan with free inlet capacity of -			
0.75 cum/s	...	Each	1,365.00-1,525.00
1.45 cum/s	...	Each	1,825.00-2,050.00
2.45 cum/s	...	Each	2,300.00-2,800.00
Single speed three phase high capacity vertical discharge fan with free inlet capacity of -			
0.75 cum/s	...	Each	1,215.00-1,380.00
1.45 cum/s	...	Each	1,650.00-1,800.00
2.45 cum/s	...	Each	1,875.00-2,400.00
3.75 cum/s	...	Each	2,450.00-3,300.00
4.85 cum/s	...	Each	2,750.00-3,400.00
ROOF MOUNTED EXHAUST VENTILATORS (non fire rated type) - e.g. light factory, shop, warehouse, sports hall (cont'd) :			
Two speed three phase high capacity vertical discharge fan with free inlet capacity of :			
0.75 cum/s	...	Each	1,215.00-1,380.00
1.45 cum/s	...	Each	1,650.00-1,800.00
2.45 cum/s	...	Each	1,875.00-2,400.00
3.75 cum/s	...	Each	2,400.00-3,500.00
4.85 cum/s	...	Each	2,750.00-3,400.00
ADD extra for :			
Roof opening and prepared upstand	...	Each	255.00-460.00
Typical sports hall electrical installation from power supply to a single fan via a control switch and a separate motor isolating switch with a nominal circuit length of 35m -			
Single phase single speed	...	Each	250.00-355.00
Single phase two speed	...	Each	320.00-490.00
Three phase single speed	...	Each	410.00-590.00
Three phase two speed	...	Each	890.00-1,200.00
Thermal overload protection to fan motors where required	...	Note	

MECHANICAL SERVICES

POWERED VENTILATION (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ROOF MOUNTED VERTICAL DISCHARGE EXHAUST FAN with fibreglass housing, venturi type inlet and outlet and backwardly inclined centrifugal impellor with nominal system resistance of 60 Pa and sized to suit areas with quiet noise levels (e.g. offices, hotels, residential, schools) :			
Three phase, single speed, direct drive -			
0.05 to 0.1 cum/s (240 V, 1 phase)	...	Each	500.00-720.00
0.15 to 0.5 cum/s	...	Each	1,410.00-1,725.00
0.6 to 0.8 cum/s	...	Each	1,410.00-1,725.00
0.9 to 2.0 cum/s	...	Each	2,350.00-2,900.00
2.1 to 4.0 cum/s	...	Each	4,400.00-5,200.00
Three phase, single speed, belt drive -			
0.4 to 0.6 cum/s	...	Each	1,800.00-2,450.00
0.7 to 1.1 cum/s	...	Each	1,800.00-3,000.00
1.3 to 1.7 cum/s	...	Each	2,450.00-4,100.00
2.0 to 2.6 cum/s	...	Each	3,900.00-5,000.00
3.0 to 3.6 cum/s	...	Each	4,600.00-5,600.00
ADD extra for :			
Single phase motor	...	Sum	92.00-192.00
Three phase, two speed motor	...	Sum	610.00-790.00
Single phase, dual motor, direct drive up to 1.7 cum/s	...	Sum	770.00-1,125.00
Three phase, dual motor, direct drive up to 1.7 cum/s	...	Sum	610.00-960.00
Upward dispersion design suitable for use with chemical, grease and destructive fumes	...	Note	
Typical office type electrical installation including separate control cabinet, motor overload protection, fan isolating switch and a nominal circuit length of 30m -			
Single speed fan with manual "on/off" control :			
Single phase	...	Each	365.00-490.00
Three phase	...	Each	490.00-670.00
Two speed fan with manual "high/low/off" control -			
Single phase	...	Each	430.00-580.00
Three phase	...	Each	980.00-1,395.00
Dual motor fan with manual fan "1/fan 2/off" control :			
Single phase	...	Each	420.00-570.00
Three phase	...	Each	960.00-1,185.00
Dual motor fan with automatic fan change over and time switch control :			
Single phase	...	Each	1,260.00-1,410.00
Three phase	...	Each	1,290.00-1,550.00
Roof opening and prepared upstand for fan :			
up to 0.8 cum/s	...	Each	170.00-194.00
0.8 to 3.6 cum/s	...	Each	190.00-490.00
Sound attenuators, ducting, grilles, etc.	...	Note	

MECHANICAL SERVICES

POWERED VENTILATION (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
ROOF MOUNTED HIGH VELOCITY VERTICAL DISCHARGE EXHAUST VENTILATOR of three phase single speed industrial type design with wind baffles, butterfly dampers and aerofoil blade with nominal system resistance of 60 Pa and sized to suit areas where higher noise levels are acceptable (e.g. commercial, industrial) :			
3.5 cum/s	...	Each	3,300.00-4,300.00
5.0 cum/s	...	Each	4,100.00-5,400.00
6.5 cum/s	...	Each	5,000.00-6,400.00
9.5 cum/s	...	Each	5,800.00-7,400.00
12.0 cum/s	...	Each	6,800.00-8,600.00
15.0 cum/s	...	Each	7,800.00-10,250.00
17.0 cum/s	...	Each	9,000.00-12,000.00
23.0 cum/s	...	Each	11,000.00-13,750.00
ADD extra for :			
Roof opening and prepared upstand for fan -			
up to 5.0 cum/s fan	...	Each	265.00-470.00
5.0 to 9.5 cum/s fan	...	Each	370.00-580.00
9.5 to 23.0 cum/s fan	...	Each	480.00-650.00
Cost of electrical installation of aforementioned fans - refer to ELECTRICAL SERVICES section for prices	...	Note	

Electrical installation* guide -

Up to 6.5 cum/s capacity fan allow for :
4 kW DOL starter
Circuit cabled in 3 x 2.5 sqmm + earth

Up to 17 cum/s capacity fan allow for :
11 kW star delta starter
Circuit cabled in 6 x 2.5 sqmm + earth

Up to 23 cum/s capacity fan allow for :
18.5 kW auto transformer starter
Circuit cabled in 3 x 6 sqmm + earth

* Based on conventional installation methods with 50m maximum circuit route length and nominal 2.5% volt drop

MECHANICAL SERVICES

DESIGN DATA

TYPICAL BUILDING AIR-CONDITIONING COOLING LOADS

The following forecast figures are indicative and do not purport to represent actual project values. Specific loads, special features and abnormalities in design and function must be included for.

A guide to cooling loads for comparative cost estimating is given for each capital city in the cooling load guide tables following this section

			Adel.	Bris.	Mel.	Per.	Syd.
BANK	W/sqm	...	95-215	120-265	95-210	115-260	105-240
	ton (ref)/1000 sqm	...	27- 61	34- 75	27- 60	33- 74	30- 68
	sqm/ton (ref)	...	37- 16	29- 13	37- 17	31- 14	33- 15
CLUBHOUSE	W/sqm	...	130-190	155-220	130-190	155-220	145-205
	ton (ref)/1000sqm	...	37-54	44-63	37-54	44-63	41-58
	sqm/ton (ref)	...	27-19	23-16	27-19	23-16	24-17
DEPARTMENT STORE - STREET LEVEL	W/sqm	...	130-175	175-235	130-175	165-220	150-200
	ton (ref)/1000 sqm	...	37- 50	50- 67	37- 50	47- 63	43- 57
	sqm/ton (ref)	...	27- 20	20- 15	27- 20	21- 16	23- 18
DEPARTMENT STORE - OTHER LEVELS	W/sqm	...	85-105	110-140	85-105	100-130	100-125
	ton (ref)/1000 sqm	...	24- 30	31- 40	24- 30	28- 37	28- 36
	sqm/ton (ref)	...	41- 33	32- 25	41- 33	35- 27	35- 28
HOSPITAL	W/sqm	...	125-165	150-195	125-165	145-195	135-180
	ton (ref)/1000 sqm	...	36- 47	43- 55	36- 47	41- 55	38- 51
	sqm/ton (ref)	...	28- 18	23- 18	28- 21	24- 18	26- 20
HOTEL ACCOMODATION	W/sqm	...	65-100	65-105	65-100	70-110	65-100
	ton (ref)/1000 sqm	...	18- 28	18- 30	18- 28	20- 31	18- 28
	sqm/ton (ref)	...	54- 35	54- 33	54- 35	50- 32	54- 35
HOTEL BAR	W/sqm	...	205-240	255-300	205-240	250-295	230-270
	ton (ref)/1000 sqm	...	58- 68	73- 85	58- 68	71- 84	65- 77
	sqm/ton (ref)	...	17- 15	14- 12	17- 15	14- 12	15- 13
HOTEL FOYER	W/sqm	...	110-170	110-170	105-165	115-175	105-165
	ton (ref)/1000 sqm	...	31- 48	31- 48	30- 47	33- 50	30- 47
	sqm/ton (ref)	...	32- 21	32- 21	33- 21	31- 20	33- 21
LIBRARY	W/sqm	...	85-115	100-130	85-115	100-125	95-120
	ton (ref)/1000 sqm	...	64- 78	80- 98	64- 78	78- 97	73- 88
	sqm/ton (ref)	...	16- 13	13- 1	16- 13	13- 10	14- 11
OFFICE	W/sqm	...	90-145	100-150	85-140	100-150	85-145
	ton (ref)/1000 sqm	...	26- 41	28- 43	24- 40	28- 43	24- 41
	sqm/ton (ref)	...	39- 24	35- 23	41- 25	35- 23	41- 24
RESTAURANT	W/sqm	...	225-275	280-345	225-275	275-340	255-310
	ton (ref)/1000 sqm	...	64- 78	80- 98	64- 78	78- 97	73- 88
	sqm/ton (ref)	...	16- 13	13- 10	16- 13	13- 10	14- 11
SHOPPING ARCADE	W/sqm	...	180-210	240-280	180-210	225-265	215-250
	ton (ref)/1000 sqm	...	51- 60	68- 80	51- 60	64- 75	61- 71
	sqm/ton (ref)	...	20- 17	15- 13	20- 17	15- 13	16- 14
SUPERMARKET	W/sqm	...	120-185	155-245	120-185	150-235	140-215
	ton (ref)/1000 sqm	...	34- 53	44- 70	34- 53	43- 67	40- 61
	sqm/ton (ref)	...	29- 19	23- 14	29- 19	23- 15	25- 16
TAVERN	W/sqm	...	165-210	195-245	165-210	195-245	180-225
	ton (ref)/1000 sqm	...	47- 60	55- 70	47- 60	55- 70	51- 64
	sqm/ton (ref)	...	21- 17	18- 14	21- 17	18- 14	20-169

CONVERSIONS : ton (ref) = (W ÷ 3517) • Btu/hour = (W x 3.412) • kW = (W ÷ 1000)

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING COOLING LOAD GUIDE FOR ADELAIDE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

WINDOWS		FACTOR*							
NO external shading :		N	NE	E	SE	S	SW	W	NW
Clear glass	...	195	239	255	191	131	274	406	389
Medium performance tinted glass	...	172	204	216	168	124	247	355	309
Clear glass with Verosol 816 pleated shades	...	89	95	100	86	71	181	259	188
High performance tinted glass	...	117	133	140	114	92	160	219	192
Mirror film tinted clear glass	...	114	128	134	112	92	145	193	174
High performance tinted double glazing	...	74	86	91	72	55	107	151	130

CORRECTIONS :

- For windows in shade, select the factor listed for south facing windows
- Allow for some opening of shades by occupants, e.g. if 50% of shades open to windows (with sun off) then take 50% clear glass factor plus 50% shade factor
- Allow to increase selected factors by + 15% to + 20% for small to medium size systems, (i.e. where related zones aren't controlled individually)

WALLS (excluding area of windows) :		FACTOR* N	E	S	W
100mm concrete	...	78	47	45	95
Concrete with 80mm thick insulated plasterboard	...	10	6	6	14
190mm hollow block	...	52	31	30	63
240mm plasterboard/brick veneer	...	37	22	21	45
270mm (cavity) clay brick	...	30	24	15	28
125mm metal/75mm insulation/plasterboard	...	10	6	6	14

CORRECTION : For walls in shade select the factor listed for south facing walls

110mm clay brick partition bordering unconditioned space		14
100mm plasterboard partition with 50mm insulation bordering unconditioned space		3
Party-wall air-conditioned both sides		NIL
FLOORING		
150mm concrete, carpet :	... Over unconditioned room	8
	... Slab on ground or sealed crawl space	NIL
ROOFING		
Metal decking with suspended ceiling :	... Without insulation	64
	With 100mm insulation	8
Clay or concrete tile with suspended ceiling :	... Without insulation	64
	With 100mm insulation	8
150mm concrete with suspended ceiling :	... Without insulation	36
	With 100mm insulation	5

CORRECTION : Multiply foregoing roofing cooling factors by 0.37 if roofing is shaded — Note

FUNCTIONAL LOADINGS		
Hotel air-conditioned area :	... Heavy population, e.g. cafe, restaurant, bar	280
	... Medium population, e.g. foyer, reception	58
	... Light population, e.g. accommodation	27
	... Transient population, e.g. corridors	12
Library air-conditioned area :	... Reading zones	96
	... Storage zones	13
Office air-conditioned area :	...	53
Shopping air-conditioned area :	... Street level and basement level	156
	All other levels	92
Showroom air-conditioned area :	...	73

Area sqm	Factor xW/sqm	= Load Watts
N	*	
NE	*	
E	*	
SE	*	
S	*	
SW	*	
W	*	
NW	*	
	Note	
N	*	
E	*	
S	*	
W	*	
	Note	

EVAPORATIVE COOLING SIZING:
Total Air Flow (L/s) = cooling load (W) x Indicative Evaporative Cooler Efficiency (range):

Adel.	Bris.	Mel.	Per.	Syd.
0.21 - 0.40	0.56 - 1.05	0.19 - 0.30	0.23 - 0.48	0.27 - 0.42

TOTAL COOLING LOAD (Watts) = ___

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING COOLING LOAD GUIDE FOR BRISBANE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

WINDOWS		FACTOR*							
NO external shading :		N	NE	E	SE	S	SW	W	NW
Clear glass	...	97	158	204	150	99	226	338	261
Medium performance tinted glass	...	88	133	168	128	91	200	292	206
Clear glass with Verosol 816 pleated shades	...	46	61	76	62	52	151	217	126
High performance tinted glass	...	61	85	103	82	63	124	174	125
Mirror film tinted clear glass	...	60	81	97	79	61	109	149	112
High performance tinted double glazing	...	38	56	70	54	40	86	123	86

CORRECTIONS :

- For windows in shade, select the factor listed for south facing windows
- Allow for some opening of shades by occupants, e.g. If 50% of shades open to windows (with sun off) then take 50% clear glass factor plus 50%shade factor
- Allow to increase selected factors by + 15% to + 20% for small to medium size systems, (i.e. where related zones aren't controlled individually)

WALLS (excluding area of windows) :		FACTOR* N	E	S	W
100mm concrete	...	61	30	27	78
Concrete with 80mm thick insulated plasterboard	...	10	6	6	14
190mm hollow block	...	40	20	18	52
240mm plasterboard/brick veneer	...	29	14	13	37
270mm (cavity) clay brick	...	22	16	7	20
125mm metal/75mm insulation/plasterboard	...	8	4	4	12

CORRECTION : For walls in shade select the factor listed for south facing walls

110mm clay brick partition bordering unconditioned space		8
100mm plasterboard partition with 50mm insulation bordering unconditioned space		2
Party-wall air-conditioned both sides		NIL
FLOORING		
150mm concrete, carpet :	... Over unconditioned rooms	5
	... Slab on ground or sealed crawl space	NIL
ROOFING		
Metal decking with suspended ceiling :	... Without insulation	55
	... With 100mm insulation	7
Clay or concrete tile with suspended ceiling :	... Without insulation	55
	... With 100mm insulation	6
150mm concrete with suspended ceiling :	... Without insulation	29
	With 100mm insulation	4

CORRECTION : Multiply foregoing roofing cooling factors by 0.37 if roofing is shaded

FUNCTIONAL LOADINGS		
Hotel air-conditioned area :	... Heavy population, e.g. cafe, restaurant, bar	373
	... Medium population, e.g. foyer, reception	67
	... Light population, e.g. accommodation	33
	... Transient population, e.g. corridors	15
Library air-conditioned area :	... Reading zones	143
	... Storage zones	16
Office air-conditioned area :	...	62
Shopping air-conditioned area :	... Street level and basement level	218
	... All other levels	123
Showroom air-conditioned area :	...	92

Area sqm	Factor xW/sqm	= Load Watts
N	*	
NE	*	
E	*	
SE	*	
S	*	
SW	*	
W	*	
NW	*	
	Note	
N	*	
E	*	
S	*	
W	*	
	Note	

EVAPORATIVE COOLING SIZING:
Total Air Flow (L/s) = cooling load (W) x Indicative Evaporative Cooler Efficiency (range):

Adel.	Bris.	Mel.	Per.	Syd.
0.21 - 0.40	0.56 - 1.05	0.19 - 0.30	0.23 - 0.48	0.27 - 0.42

TOTAL COOLING LOAD (Watts) = ___

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING COOLING LOAD GUIDE FOR MELBOURNE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

										Area	Factor	Load
WINDOWS					FACTOR*					sqm	xW/sqm	= Watts
NO external shading :		N	NE	E	SE	S	SW	W	NW			
Clear glass	...	204	235	244	180	120	261	393	389	N	*	
Medium performance tinted glass	...	176	197	204	156	113	234	342	305	NE	*	
Clear glass with Verosol 816 pleated shades	...	91	91	94	78	65	174	252	188	E	*	
High performance tinted glass	...	116	127	131	106	83	150	209	187	SE	*	
Mirror film tinted clear glass	...	111	121	124	102	82	135	182	168	S	*	
High performance tinted double glazing	...	75	83	85	67	50	101	145	128	SW	*	
										W	*	
										NW	*	

CORRECTIONS :

- For windows in shade, select the factor listed for south facing windows
- Allow for some opening of shades by occupants, e.g. if 50% of shades open to windows (with sun off) then take 50% clear glass factor plus 50% shade factor
- Allow to increase selected factors by + 15% to + 20% for small to medium size systems, (i.e. where related zones aren't controlled individually)

				FACTOR*					
WALLS (excluding area of windows) :		N		E	S	W			Note
100mm concrete	...	71		40	37	88			*
Concrete with 80mm thick insulated plasterboard	...	10		6	6	14	N		*
190mm hollow block	...	47		27	25	58	E		*
240mm plasterboard/brick veneer	...	33		19	18	41	S		*
270mm (cavity) clay brick	...	27		20	12	25	W		
125mm metal/75mm insulation/plasterboard	...	9		5	5	13			Note
CORRECTION : For walls in shade select the factor listed for south facing walls									12
110mm clay brick partition bordering unconditioned space									3
100mm plasterboard partition with 50mm insulation bordering unconditioned space									NIL
Party-wall air-conditioned both sides	...								
FLOORING									
150mm concrete, carpet :	...	Over unconditioned rooms							7
	...	Slab on ground or sealed crawl space							NIL
ROOFING									
Metal decking with suspended ceiling :	...	Without insulation							60
	...	With 100mm insulation							7
Clay or concrete tile with suspended ceiling :	...	Without insulation							60
	...	With 100mm insulation							7
150mm concrete with suspended ceiling :	...	Without insulation							33
	...	With 100mm insulation							5
CORRECTION : Multiply foregoing roofing cooling factors by 0.37 if roofing is shaded									Note
FUNCTIONAL LOADINGS									
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant, bar							281
	...	Medium population, e.g. foyer, reception							58
	...	Light population, e.g. accommodation							27
	...	Transient population, e.g. corridors							12
Library air-conditioned area :	...	Reading zones							97
	...	Storage zones							13
Office air-conditioned area :	...								53
Shopping air-conditioned area :	...	Street level and basement level							157
	...	All other levels							92
Showroom air-conditioned area :	...								74

EVAPORATIVE COOLING SIZING:
Total Air Flow (L/s) = cooling load (W) x Indicative Evaporative Cooler Efficiency (range):

Adel.	Bris.	Mel.	Per.	Syd.
0.21 - 0.40	0.56 - 1.05	0.19 - 0.30	0.23 - 0.48	0.27 - 0.42

TOTAL COOLING LOAD (Watts) = ___

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING COOLING LOAD GUIDE FOR PERTH

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

WINDOW		FACTOR*							
NO external shading :		N	NE	E	SE	S	SW	W	NW
Clear glass	...	159	218	243	183	128	261	384	349
Medium performance tinted glass	...	145	188	206	162	122	236	336	279
Clear glass with Verosol 816 pleated shades	...	76	88	96	81	70	172	245	170
High performance tinted glass	...	102	125	135	112	90	154	209	176
Mirror film tinted clear glass	...	102	121	130	110	91	140	184	161
High performance tinted double glazing	...	63	80	87	70	54	102	144	118

CORRECTIONS :

- For windows in shade, select the factor listed for south facing windows
- Allow for some opening of shades by occupants, e.g. if 50% of shades open to windows (with sun off) then take 50% clear glass factor plus 50% shade factor
- Allow to increase selected factors by + 15% to + 20% for small to medium size systems, (i.e. where related zones aren't controlled individually)

WALLS (excluding area of windows) :		FACTOR* N	E	S	W
100mm concrete	...	76	45	42	93
Concrete with 80mm thick insulated plasterboard	...	10	6	6	14
190mm hollow block	...	50	30	28	62
240mm plasterboard/brick veneer	...	36	21	20	44
270mm (cavity) clay brick	...	29	23	14	27
125mm metal/75mm insulation/plasterboard	...	9	6	6	14

CORRECTION : For walls in shade select the factor listed for south facing walls

110mm clay brick partition bordering unconditioned space		3
100mm plasterboard partition with 50mm insulation bordering unconditioned space		NIL
Party-wall air-conditioned both sides		

FLOORING			
150mm concrete, carpet :	...	Over unconditioned rooms	8
	...	Slab on ground or sealed crawl space	NIL

ROOFING			
Metal decking with suspended ceiling :	...	Without insulation	63
	...	With 100mm insulation	8
Clay or concrete tile with suspended ceiling :	...	Without insulation	63
	...	With 100mm insulation	7
150mm concrete with suspended ceiling :	...	Without insulation	35
	...	With 100mm insulation	5

CORRECTION : Multiply foregoing roofing cooling factors by 0.37 if roofing is shaded — Note

FUNCTIONAL LOADINGS			
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant, bar	347
	...	Medium population, e.g. foyer, reception	64
	...	Light population, e.g. accommodation	32
	...	Transient population, e.g. corridors	14
Library air-conditioned area :	...	Reading zones	130
		Storage zones	15
Office air-conditioned area :	...		59
Shopping air-conditioned area :	...	Street level and basement level	201
	...	All other levels	114
Showroom air-conditioned area :	...		87

Area sqm	Factor xW/sqm	= Load Watts
N	*	
NE	*	
E	*	
SE	*	
S	*	
SW	*	
W	*	
NW	*	
	Note	
N	*	
E	*	
S	*	
W	*	
	Note	

EVAPORATIVE COOLING SIZING:
Total Air Flow (L/s) = cooling load (W) x Indicative Evaporative Cooler Efficiency (range):

Adel.	Bris.	Mel.	Per.	Syd.
0.21 - 0.40	0.56 - 1.05	0.19 - 0.30	0.23 - 0.48	0.27 - 0.42

TOTAL COOLING LOAD (Watts) = ___

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING COOLING LOAD GUIDE FOR SYDNEY

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

WINDOWS		FACTOR*							
NO external shading :		N	NE	E	SE	S	SW	W	NW
Clear glass	...	143	190	208	148	93	225	347	325
Medium performance tinted glass	...	123	156	170	126	86	199	298	251
Clear glass with Verosol 816 pleated shades	...	62	70	76	60	49	150	223	154
High performance tinted glass	...	79	97	104	81	60	123	177	149
Mirror film tinted clear glass	...	74	90	96	76	57	106	150	131
High performance tinted double glazing	...	51	64	70	53	37	84	125	104

CORRECTIONS :

- For windows in shade, select the factor listed for south facing windows
- Allow for some opening of shades by occupants, e.g. if 50% of shades open to windows (with sun off) then take 50% clear glass factor plus 50% shade factor
- Allow to increase selected factors by + 15% to + 20% for small to medium size systems, (i.e. where related zones aren't controlled individually)

WALLS (excluding area of windows) :		FACTOR* N	E	S	W
100mm concrete	...	67	36	34	84
Concrete with 80mm thick insulated plasterboard	...	10	6	6	14
190mm hollow block	...	45	24	23	56
240mm plasterboard/brick veneer	...	32	17	16	40
270mm (cavity) clay brick	...	25	19	10	23
125mm metal/75mm insulation/plasterboard	...	8	5	5	13

CORRECTION : For walls in shade select the factor listed for south facing walls

110mm clay brick partition bordering unconditioned space		7
100mm plasterboard partition with 50mm insulation bordering unconditioned space		2
Party-wall air-conditioned both sides		NIL
FLOORING		
150mm concrete, carpet :	... Over unconditioned rooms	4
	... Slab on ground or sealed crawl space	NIL
ROOFING		
Metal decking with suspended ceiling :	... Without insulation	58
	... With 100mm insulation	7
Clay or concrete tile with suspended ceiling :	... Without insulation	58
	... With 100mm insulation	7
150mm concrete with suspended ceiling :	... Without insulation	32
	... With 100mm insulation	5

CORRECTION : Multiply foregoing roofing cooling factors by 0.37 if roofing is shaded

FUNCTIONAL LOADINGS		
Hotel air-conditioned area :	... Heavy population, e.g. cafe, restaurant, bar	328
	... Medium population, e.g. foyer, reception	62
	... Light population, e.g. accommodation	30
	... Transient population, e.g. corridors	13
Library air-conditioned area :	... Reading zones	120
	... Storage zones	14
Office air-conditioned area :	...	57
Shopping air-conditioned area :	... Street level and basement level	188
	... All other levels	108
Showroom air-conditioned area :	...	83

Area sqm	Factor xW/sqm	= Load Watts
N	*	
NE	*	
E	*	
SE	*	
S	*	
SW	*	
W	*	
NW	*	
	Note	
N	*	
E	*	
S	*	
W	*	

EVAPORATIVE COOLING SIZING:
Total Air Flow (L/s) = cooling load (W) x Indicative Evaporative Cooler Efficiency (range):

Adel.	Bris.	Mel.	Per.	Syd.
0.21 - 0.40	0.56 - 1.05	0.19 - 0.30	0.23 - 0.48	0.27 - 0.42

TOTAL COOLING LOAD (Watts) = ___

MECHANICAL SERVICES

DESIGN DATA

TYPICAL BUILDING AIR-CONDITIONING HEATING LOADS

The following forecast figures are indicative and do not purport to represent actual project values. Specific loads, special features and abnormalities in design and function must be included for.

A guide to heating loads, for comparative cost estimating, is given for each capital city in the heating load guide tables, following this section.

Where heating is intermittent or a fast pick-up is desired, oversizing of heating capacity may be necessary, e.g. an allowance of up to 60% additional capacity may be required.

			Adel.	Bris.	Mel.	Per.	Syd.
BANK	W/sqm	...	60-150	40-105	70-175	45-120	55-140
	Btu/hr/sqm	...	205-512	136-358	239-597	154-409	188-478
	sqm/kW	...	17- 7	25- 10	14- 6	11- 8	18- 7
CLUBHOUSE	W/sqm	...	70-105	50-70	85-120	60-85	70-100
	Btu/hr/sqm	...	239-358	171-239	290-409	205-290	239-341
	sqm/kW	...	14-10	20-14	12-8	17-12	14-10
DEPARTMENT STORE - STREET LEVEL	W/sqm	...	80-100	55- 70	95-120	65- 80	75- 95
	Btu/hr/sqm	...	273-341	188-239	324-409	222-273	256-324
	sqm/kW	...	13- 10	18- 14	11- 8	15- 13	13- 11
DEPARTMENT STORE - OTHER LEVELS	W/sqm	...	40- 55	30- 35	50- 60	35- 40	40- 50
	Btu/hr/sqm	...	136-188	102-119	171-205	119-136	136-171
	sqm/kW	...	25- 18	33- 29	20- 17	29- 25	25- 20
HOSPITAL	W/sqm	...	70- 95	50- 65	85-110	55- 75	70- 90
	Btu/hr/sqm	...	239-324	171-222	290-375	188-256	239-307
	sqm/kW	...	14- 11	20- 15	12- 9	18- 13	14- 11
HOTEL ACCOMODATION	W/sqm	...	35-65	25-40	40-70	25-50	35-55
	Btu/hr/sqm	...	119-222	85-136	136-239	85-171	119-188
	sqm/	...	29-15	40-25	25-14	40-20	29-18
HOTEL BAR	W/sqm	...	110-140	80- 95	130-165	90-110	105-130
	Btu/hr/sqm	...	375-478	273-324	444-563	307-375	358-444
	sqm/kW	...	9- 7	13- 11	8- 6	11- 9	10- 8
HOTEL FOYER	W/sqm	...	60- 85	40- 60	70-100	50- 65	60- 80
	Btu/hr/sqm	...	205-290	136-205	239-341	171-222	205-273
	sqm/kW	...	17- 12	25- 17	14- 10	20- 15	17- 13
LIBRARY	W/sqm	...	50-55	35-40	55-70	35-45	50-55
	Btu/hr/sqm	...	171-188	119-136	188-239	119-154	171-188
	sqm/kW	...	20-18	29-25	18-14	29-22	20-18
OFFICE	W/sqm	...	40-60	30-40	50-70	35-50	40-60
	Btu/hr/sqm	...	136-205	102-136	171-239	119-171	136-205
	sqm/kW	...	25-17	33-25	20-14	29-20	25-17
RESTAURANT	W/sqm	...	120-160	90-110	145-190	100-125	115-150
	Btu/hr/sqm	...	409-546	307-375	495-648	341-427	392-512
	sqm/kW	...	8-6	11-9	7-5	10-8	9-7
SHOPPING ARCADE	W/sqm	...	80-100	55- 70	95-120	65- 80	75-100
	Btu/hr/sqm	...	273-341	188-239	324-409	222-273	256-341
	sqm/kW	...	13- 10	18- 14	11- 8	15- 13	13- 10
SUPERMARKET	W/sqm	...	85-110	60- 75	100-130	70- 85	85-150
	Btu/hr/sqm	...	290-375	205-256	341-444	239-290	358-512
	sqm/kW	...	12- 9	17- 13	10- 8	14- 12	12- 10
TAVERN	W/sqm	...	90-115	65- 80	105-135	75- 95	85-110
	Btu/hr/sqm	...	307-392	222-273	358-461	256-324	290-375
	sqm/kW	...	11- 9	15- 13	10- 7	13- 11	12- 9

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING HEATING LOAD GUIDE FOR ADELAIDE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*.

			Area sqm	Factor xW/sqm	= Load Watts
WINDOWS					
Clear glass or mirror film tinted glass	...			106	
Clear glass with fully closed heavy curtains	...			80	
Clear glass with Verosol 816 pleated shades	...			62	
High performance tinted double glazing	...			50	
CORRECTION : Allow for some opening of curtains or shades by occupants, e.g. where 80% of curtains are open take 80% clear glass factor plus 20% curtain or shade factor				Note	
WALLS					
100mm concrete	...			76	
Concrete with 80mm thick insulated plasterboard	...			6	
190mm hollow block	...			49	
240mm plasterboard/brick veneer	...			35	
270mm (cavity) clay brick	...			33	
125mm metal/75mm insulation/plasterboard	...			9	
110mm clay brick partition bordering unconditioned space				19	
100mm plasterboard partition with 50mm insulation bordering unconditioned space				5	
Party-wall air-conditioned both sides	...			NIL	
FLOORING					
150mm concrete, carpet :	...	Over unconditioned rooms		10	
	...	Over sealed crawl space		8	
	...	Slab on ground		3	
ROOFING					
Metal decking with suspended ceiling :	...	Without insulation		38	
	...	With 100mm insulation		6	
Clay or concrete tile with suspended ceiling :	...	Without insulation		39	
	...	With 100mm insulation		6	
150mm concrete with suspended ceiling :	...	Without insulation		28	
	...	With 100mm insulation		5	
FUNCTIONAL LOADINGS					
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant and bar		99	
	...	Medium population e.g. foyer, reception		10	
	...	Light population e.g. accommodation		7	
	...	Transient population e.g. corridors		3	
Library air-conditioned area :	...	Reading zones		50	
	...	Storage zones		3	
Office air-conditioned area :	...			10	
Shopping air-conditioned area :	...	Street level and basement level		66	
	...	All other levels		33	
Showroom air-conditioned area :	...			20	

ADDITONAL CAPACITY ALLOWANCE :

When heating is intermittent, or a fast pick-up is required, extra heating capacity will be required. It varies with application and is influenced by the temperature difference and the period of heating provided prior to occupation.

e.g. 1.1 to 1.25 for large systems
1.25 to 1.6 for small systems
1.6 to 2.0 for rapid pick-up applications

HEATING LOAD (Watts) =
ADDITIONAL CAPACITY x ____
TOTAL HEATING LOAD (Watts) =

CONVERSIONS : Btu/hour	=	W x 3.412
kW	=	W ÷ 1000

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING HEATING LOAD GUIDE FOR BRISBANE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*

			Area sqm	Factor xW/sqm	= Load Watts
WINDOWS					
Clear glass or mirror film tinted glass	...			74	
Clear glass with fully closed heavy curtains	...			56	
Clear glass with Verosol 816 pleated shades	...			44	
High performance tinted double glazing	...			35	
CORRECTION : Allow for some opening of curtains or shades by occupants, e.g. where 80% of curtains are open take 80% clear glass factor plus 20% curtain or shade factor				Note	
WALLS					
100mm concrete	...			53	
Concrete with 80mm thick insulated plasterboard	...			6	
190mm hollow block	...			34	
240mm plasterboard/brick veneer	...			24	
270mm (cavity) clay brick	...			23	
125mm metal/75mm insulation/plasterboard	...			6	
110mm clay brick partition bordering unconditioned space				13	
100mm plasterboard partition with 50mm insulation bordering unconditioned space				3	
Party-wall air-conditioned both sides	...			NIL	
FLOORING					
150mm concrete, carpet :	...	Over unconditioned rooms		7	
	...	Over sealed crawl space		6	
	...	Slab on ground		3	
ROOFING					
Metal decking with suspended ceiling :	...	Without insulation		27	
	...	With 100mm insulation		4	
Clay or concrete tile with suspended ceiling :	...	Without insulation		27	
	...	With 100mm insulation		4	
150mm concrete with suspended ceiling :	...	Without insulation		19	
	...	With 100mm insulation		4	
FUNCTIONAL LOADINGS					
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant and bar		69	
	...	Medium population e.g. foyer, reception		7	
	...	Light population e.g. accommodation		5	
	...	Transient population e.g. corridors		2	
Library air-conditioned area :	...	Reading zones		35	
	...	Storage zones		2	
Office air-conditioned area :	...			7	
Shopping air-conditioned area :	...	Street level and basement level		46	
	...	All other levels		23	
Showroom air-conditioned area :	...			14	

ADDITONAL CAPACITY ALLOWANCE :

When heating is intermittent, or a fast pick-up is required, extra heating capacity will be required. It varies with application and is influenced by the temperature difference and the period of heating provided prior to occupation.

e.g. 1.1 to 1.25 for large systems
1.25 to 1.6 for small systems
1.6 to 2.0 for rapid pick-up applications

CONVERSIONS : Btu/hour	=	W x 3.412
kW	=	W ÷ 1000

HEATING LOAD (Watts) =
ADDITIONAL CAPACITY x____
TOTAL HEATING LOAD (Watts) =

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING HEATING LOAD GUIDE FOR MELBOURNE

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*

			Area sqm	Factor xW/sqm	= Load Watts
WINDOWS					
Clear glass or mirror film tinted glass	...			125	
Clear glass with fully closed heavy curtains	...			95	
Clear glass with Verosol 816 pleated shades	...			74	
High performance tinted double glazing	...			58	
CORRECTION : Allow for some opening of curtains or shades by occupants, e.g. where 80% of curtains are open take 80% clear glass factor plus 20% curtain or shade factor				Note	
WALLS					
100mm concrete	...			89	
Concrete with 80mm thick insulated plasterboard	...			6	
190mm hollow block	...			58	
240mm plasterboard/brick veneer	...			41	
270mm (cavity) clay brick	...			38	
125mm metal/75mm insulation/plasterboard	...			11	
110mm clay brick partition bordering unconditioned space				22	
100mm plasterboard partition with 50mm insulation bordering unconditioned space				5	
Party-wall air-conditioned both sides	...			NIL	
FLOORING					
150mm concrete, carpet :	...	Over unconditioned rooms		11	
	...	Over sealed crawl space		10	
	...	Slab on ground		4	
ROOFING					
Metal decking with suspended ceiling :	...	Without insulation		45	
	...	With 100mm insulation		7	
Clay or concrete tile with suspended ceiling :	...	Without insulation		46	
	...	With 100mm insulation		7	
150mm concrete with suspended ceiling :	...	Without insulation		33	
	...	With 100mm insulation		6	
FUNCTIONAL LOADINGS					
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant and bar		117	
	...	Medium population e.g. foyer, reception		12	
	...	Light population e.g. accommodation		8	
	...	Transient population e.g. corridors		4	
Library air-conditioned area :	...	Reading zones		58	
	...	Storage zones		4	
Office air-conditioned area :	...			12	
Shopping air-conditioned area :	...	Street level and basement level		78	
	...	All other levels		39	
Showroom air-conditioned area :	...			23	

ADDITONAL CAPACITY ALLOWANCE :

When heating is intermittent, or a fast pick-up is required, extra heating capacity will be required. It varies with application and is influenced by the temperature difference and the period of heating provided prior to occupation.

e.g. 1.1 to 1.25 for large systems
1.25 to 1.6 for small systems
1.6 to 2.0 for rapid pick-up applications

CONVERSIONS : Btu/hour	=	W x 3.412
kW	=	W ÷ 1000

HEATING LOAD (Watts) =
ADDITIONAL CAPACITY x ____
TOTAL HEATING LOAD (Watts) =

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING HEATING LOAD GUIDE FOR PERTH

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*

			Area sqm	Factor xW/sqm	= Load Watts
WINDOWS					
Clear glass or mirror film tinted glass	...			85	
Clear glass with fully closed heavy curtains	...			64	
Clear glass with Verosol 816 pleated shades	...			50	
High performance tinted double glazing	...			40	
CORRECTION : Allow for some opening of curtains or shades by occupants, e.g. where 80% of curtains are open take 80% clear glass factor plus 20% curtain or shade factor				Note	
WALLS					
100mm concrete	...			61	
Concrete with 80mm thick insulated plasterboard	...			6	
190mm hollow block	...			39	
240mm plasterboard/brick veneer	...			28	
270mm (cavity) clay brick	...			26	
125mm metal/75mm insulation/plasterboard	...			7	
110mm clay brick partition bordering unconditioned space				15	
100mm plasterboard partition with 50mm insulation bordering unconditioned space				4	
Party-wall air-conditioned both sides	...			NIL	
FLOORING					
150mm concrete, carpet :	...	Over unconditioned rooms		8	
	...	Over sealed crawl space		7	
	...	Slab on ground		3	
ROOFING					
Metal decking with suspended ceiling :	...	Without insulation		31	
	...	With 100mm insulation		5	
Clay or concrete tile with suspended ceiling :	...	Without insulation		31	
	...	With 100mm insulation		5	
150mm concrete with suspended ceiling :	...	Without insulation		22	
	...	With 100mm insulation		4	
FUNCTIONAL LOADINGS					
Hotel air-conditioned area :	...	Heavy population, e.g. cafe, restaurant and bar		79	
	...	Medium population e.g. foyer, reception		8	
	...	Light population e.g. accommodation		5	
	...	Transient population e.g. corridors		3	
Library air-conditioned area :	...	Reading zones		40	
	...	Storage zones		3	
Office air-conditioned area :	...			8	
Shopping air-conditioned area :	...	Street level and basement level		53	
	...	All other levels		26	
Showroom air-conditioned area :	...			16	

ADDITONAL CAPACITY ALLOWANCE :

When heating is intermittent, or a fast pick-up is required, extra heating capacity will be required. It varies with application and is influenced by the temperature difference and the period of heating provided prior to occupation.

e.g. 1.1 to 1.25 for large systems
1.25 to 1.6 for small systems
1.6 to 2.0 for rapid pick-up applications

CONVERSIONS : Btu/hour = W x 3.412
kW = W ÷ 1000

HEATING LOAD (Watts) =
ADDITIONAL CAPACITY x ____
TOTAL HEATING LOAD (Watts) =

MECHANICAL SERVICES

DESIGN DATA

AIR-CONDITIONING HEATING LOAD GUIDE FOR SYDNEY

This abridged, comparative cost estimating method applies the area of the perimeter windows, walls, and so on, bordering the air-conditioned space, multiplied by the corresponding indicative load factor*

			Area sqm	Factor xW/sqm	= Load Watts
WINDOWS					
Clear glass or mirror film tinted glass	...			102	
Clear glass with fully closed heavy curtains	...			77	
Clear glass with Verosol 816 pleated shades	...			60	
High performance tinted double glazing	...			47	
CORRECTION : Allow for some opening of curtains or shades by occupants, e.g. where 80% of curtains are open take 80% clear glass factor plus 20% curtain or shade factor				Note	
WALLS					
100mm concrete	...			73	
Concrete with 80mm thick insulated plasterboard	...			6	
190mm hollow block	...			47	
240mm plasterboard/brick veneer	...			33	
270mm (cavity) clay brick	...			31	
125mm metal/75mm insulation/plasterboard	...			9	
110mm clay brick partition bordering unconditioned space				18	
100mm plasterboard partition with 50mm insulation bordering unconditioned space				4	
Party-wall air-conditioned both sides	...			NIL	
FLOORING					
150mm concrete, carpet :	...	Over unconditioned rooms		9	
	...	Over sealed crawl space		8	
	...	Slab on ground		3	
ROOFING					
Metal decking with suspended ceiling :	...	Without insulation		37	
	...	With 100mm insulation		6	
Clay or concrete tile with suspended ceiling :	...	Without insulation		38	
	...	With 100mm insulation		6	
150mm concrete with suspended ceiling :	...	Without insulation		26	
	...	With 100mm insulation		5	
FUNCTIONAL LOADINGS					
Hotel air-conditioned area :	...	Heavy population e.g. cafe, restaurant and bar		95	
	...	Medium population e.g. foyer, reception		9	
	...	Light population e.g. accommodation		6	
	...	Transient population e.g. corridors		3	
Library air-conditioned area :	...	Reading zones		47	
	...	Storage zones		3	
Office air-conditioned area :	...			9	
Shopping air-conditioned area :	...	Street level and basement level		63	
	...	All other levels		32	
Showroom air-conditioned area :	...			19	

ADDITONAL CAPACITY ALLOWANCE :

When heating is intermittent, or a fast pick-up is required, extra heating capacity will be required. It varies with application and is influenced by the temperature difference and the period of heating provided prior to occupation.

e.g. 1.1 to 1.25 for large systems
1.25 to 1.6 for small systems
1.6 to 2.0 for rapid pick-up applications

CONVERSIONS : Btu/hour = W x 3.412
kW = W ÷ 1000

HEATING LOAD (Watts) =
ADDITIONAL CAPACITY x ___
TOTAL HEATING LOAD (Watts) =

MECHANICAL SERVICES

DESIGN DATA

NATURAL VENTILATOR SELECTION

The following is a cost estimating method of Natural Ventilator Selection :

Step 1 Determine the heat load of the building in kW by using Table 1 and the formula :

$$\frac{\text{Floor Area x Value from Table 1 + any additional heat sources}}{1000} = \text{kW}$$

Step 2 Decide on an acceptable indoor to outdoor temperature difference, generally around 14°C (K)

Step 3 Find the ventilators air velocity by applying temperature rise (from Step 2) and ventilators height above floor to table 2.

Step 4 Apply data gathered from steps 1, 2 and 3 to the ventilator formula :

$$\frac{\text{Heat Load (Step 1)}}{\text{1.23 (constant) x temp. difference (Step 2) x air velocity (Step 3)}} = \text{Total aerodynamic throat area of ventilators}$$

Step 5 Decide on the number and size of ventilators required based on the total area resulting from Step 4 (by dividing the Step 4 result by the size of ventilator chosen, the number required can be found).

Step 6 Refer to Inlet . Selection.

EXAMPLE What natural ventilation is required for a 1000 sqm uninsulated metal shed if the ventilators are to be mounted 8m above the floor if a temperature rise of 14°C is preferred and wind velocity is 6 km/hour.

Solution, following steps as before :

Step 1 Heat load of building is $\frac{\text{1000 sqm x 250 W/sqm}}{1000}$ = 250kW

Step 2 Indoor temperature difference chosen is 14°C (K)

Step 3 Enter the temperature difference (from Step 2) of 14°C (K) and the given ventilator mounting height of 8m onto Table 2 to find a velocity of 1.16 m/s. To this velocity add 0.81 m/s which is the increase in extract velocity due to the increase in wind velocity (refer to note 4 - e.g. wind velocity is 3 km/hr greater, therefore 3 x 0.27m/s = 0.81 m/s increase), i.e. 1.16 + 0.81 = 1.97 m/s.

Step 4 Apply the ventilator formula : $\frac{\text{250 kW}}{\text{1.23 x 14K x 1.97 m/s}}$ = 7.37 sqm total

Step 5 Choose say, a ventilator with 0.38 sqm (aerodynamic throat) area; then from Step 4 :

$$\frac{\text{7.37 sqm}}{\text{0.38 sqm}} = \text{20m of ventilator length required}$$

Step 6 Refer to Inlet Louvre Selection.

MECHANICAL SERVICES

DESIGN DATA

NATURAL VENTILATOR SELECTION (cont'd)

TABLE 1 - TYPICAL HEAT LOADS

For Natural Ventilation Calculations

WALL TYPE	Metal or Fibre Cement Sheeted	
	UNINSULATED ROOF	INSULATED ROOF
	Watts per sqm of Floor Area	
Metal or fibre cement sheeted wall - with 25mm thick insulation	165 to 190	45 to 70
Clay brick cavity wall	175 to 200	55 to 80
Concrete hollow block wall	185 to 215	65 to 95
Concrete wall 100mm thick	195 to 230	80 to 110
Metal or fibre cement sheeted wall - unlined and uninsulated	210 to 250	95 to 130

Notes :1. Values given in Table 1 have been calculated and based on Typical Factory/Warehouse design with 35 degree pitched roof and 5m high walls.
2. Insulation is mineral wool batts with foil lined building paper, 50mm thick to roofing and 25mm thick to wall sheeting.
3. Data is based on design indoor temperature being 14°C(K) lower than the outdoor temperature. Pro-rata values for alternative temperature differences.
4. *Table 2 includes an allowance for extract flow due to wind velocity (approximately 3 km/hour). The size of ventilators can be reduced when greater wind velocities may be relied on (approximate increase in extract velocity is 0.27 m/s per 1 km/hour increase in wind velocity).
5. The range in heat loads given corresponds to the range in anticipated wall/floor ratios, and solar aspects and window sizes. Heat loads will be lower where a passive building design is adopted.
6. An allowance for heat gain due to glazing has been included. Glazing is assumed to be untreated and spread uniformaly around building with the area of glass not exceeding 20% of wall area.
7. ADD to total for internal heat loads e.g. boilers, machinery, plant, etc.
8. ADD to total for roof lights (use 115W/sqm of glazing for shaded south facing lights).

TABLE 2 - CHOOSING EXTRACT VELOCITIES FOR NATURAL VENTILATION CALCULATIONS*

FOR INDOOR TO OUTDOOR TEMPERATURE DIFFERENCE IN °C (K)

FOR HEIGHT IN METRES	4	5	6	7	8	9	10	12	14	16	18	20
2	0.59	0.67	0.71	0.77	0.81	0.84	0.87	0.90	0.97	1.00	1.02	1.05
3	0.65	0.71	0.77	0.81	0.84	0.87	0.90	0.97	1.00	1.02	1.05	1.16
4	0.67	0.77	0.81	0.84	0.87	0.90	0.97	1.00	1.02	1.05	1.10	1.14
5	0.71	0.81	0.84	0.87	0.90	0.97	1.00	1.02	1.05	1.10	1.14	1.16
6	0.77	0.84	0.87	0.90	0.97	1.00	1.02	1.05	1.10	1.14	1.16	1.18
7	0.81	0.87	0.90	0.97	1.00	1.02	1.05	1.10	1.14	1.16	1.18	1.20
8	0.84	0.90	0.97	1.00	1.02	1.05	1.10	1.14	1.16	1.18	1.20	1.22
9	0.90	0.97	1.00	1.02	1.05	1.10	1.14	1.16	1.18	1.20	1.22	1.26
10	0.97	1.00	1.02	1.05	1.10	1.14	1.16	1.18	1.20	1.22	1.26	1.30
11	1.00	1.02	1.05	1.10	1.14	1.16	1.18	1.20	1.22	1.26	1.30	1.34
12	1.02	1.05	1.10	1.14	1.16	1.18	1.20	1.22	1.26	1.30	1.34	1.38

NATURAL VENTILATION EXTRACT VELOCITIES IN m/s

MECHANICAL SERVICES

DESIGN DATA

AUTOMATIC SMOKE AND HEAT VENTILATOR SELECTION

Automatic smoke and heat ventilation is often used as an alternative to electrically powered smoke and heat exhaust systems. Capacity and type of system is dependent upon building design and local authorities' requirements and may vary from project to project. As an aid to forecasting a building's requirements for cost estimating/feasibility purposes prior to detailed design information, the presumption that the aggregate airway (i.e. total aerodynamic throat area) of ventilators in each building compartment be usually required to be not less than 3% of the compartment's area, may be made. Example : An 800 sqm supermarket may require 24 sqm of vent opening/aerodynamic area approximately equivalent to 40.8 sqm net throat area of ventilator.

The Building Code of Australia (eg parts E2 and F4) has specific requirements for Automatic Smoke and Heat Ventilation and Natural Ventilation.

INLET LOUVRE SELECTION

To allow unrestricted natural ventilation the total area of inlet louvre is generally two to three times the total aerodynamic throat area of the natural ventilators.

Any increase in the ventilator (i.e. outlet) size over the inlet (effective opening area) size can increase the air flow.

Adjust ventilator sizes for cost estimating purposes in line with the following guide :

INCREASE In Either Inlet or Outlet Area	Approximate Resulting Increase In Air Flow
+ 50%	+ 17%
+ 100%	+ 27%
+ 200%	+ 34%
+ 630%	+ 37%

EXAMPLE Increase the area of the inlet louvre to three times the aerodynamic throat area of outlet ventilator in the previous Natural Ventilator Selection example.

Step 1 Obtain the approximate increase in air flow from selection guide, (i.e. 3 times = + 200%, then resulting increase in air flow is + 34%).

Step 2 Apply the 34% increase to the previously calculated area of 7.37 sqm :

$$\frac{7.37 \text{ sqm} \times 100\%}{100\% + 34\% \text{ increase}} = 5.5 \text{ sqm total (with an inlet area of 3 x outlet area)}$$

Step 3 Similar to step 5 in previous example :

$$\frac{5.50 \text{ sqm}}{0.46 \text{ sqm}} = 12 \text{ ventilators required}$$

MECHANICAL SERVICES

DESIGN DATA

POWERED ROOF VENTILATOR SELECTION

The following is a cost estimating method of powered ventilator selection :

Steps 1 & 2 — Determine the anticipated building heat load and temperature difference in the same manner as described for Natural Roof Ventilator Selection steps 1 and 2.

Step 3 — Apply the data from Steps 1 and 2 to the powered ventilator formula :

$$\frac{\text{Heat Load (Step 1)}}{\text{1.23 (constant) x temp. difference}} = \text{Total free inlet capacity of powered ventilators required (i.e. total ventilation rate) in cum/s}$$

Step 4 — Decide on the number and size of powered ventilators required (by dividing the result from Step 3 by the size of powered ventilator chosen, the number required can be found).

Step 5 — Take system resistance into consideration - refer to section on System Resistance.

EXAMPLE What would be the requirements for power ventilating the shed given in the previous Natural Ventilation example :

Steps 1 & 2 — Follow Steps 1 and 2 in the same manner as given in the previous Natural Ventilation example.

Step 3 — Apply the powered ventilator formula : $\frac{250\text{ kW}}{1.23 \times 14\text{K}} = 14.52$ cum/s total

Step 4 — Choose, say, a powered ventilator with 2.46 cum/s capacity, then :

from Step 3 : $\frac{14.52\text{ cum/s}}{2.46\text{ cum/s}} = 6$ ventilators

Step 5 — System resistance can be ignored if ducting is not required and prevailing wind does not affect performance.

Note : 1000 L/s = 1 cum/s and 1 cum/s = 2119 c.f.m.

SYSTEM RESISTANCE

Capacities given in fan items are nominal and assume either a free inlet (i.e. negligible resistance) or a typical system resistance (given in pascals).

Each fan's performance can differ from installation to installation and is affected by the system's resistance. System resistance is influenced by factors such as length size and type of ducting, duct fittings, filters, grilles, etc. and by adverse pressure differences created by prevailing winds.

As the system's resistance is increased, the efficiency of the fan is reduced and a fan with a larger free inlet capacity is required. The reduction in a fan's handling capacity due to the introduction of system resistance may, as a cost estimating guide, be anticipated to range between -5% and -15% at 60 Pa resistance, -15% and -30% at 120 Pa resistance and -25% and -45% at 185 Pa resistance.

Example 1- 2.4 cum/s is required to be extracted from an installation with a long duct run creating 180 Pa resistance. The size of the fan chosen to handle 2.4 cum/s at 180 Pa resistance would have a capacity when connected as a free inlet ranging approximately between :

$\frac{2.4\text{ cum/s} \times 100\%}{100\% - 25\%\text{ reduction}} = 3.2$ cum/s and $\frac{2.4\text{ cum/s} \times 100\%}{100\% - 45\%\text{ reduction}} = 4.36$ cum/s

Example 2- If the resistance in Example 1 was 60 Pa then the fan would have a capacity when connected as a free inlet ranging approximately between :

$\frac{2.4\text{ cum/s} \times 100\%}{100\% - 5\%} = 2.53$ cum/s and $\frac{2.4\text{ cum/s} \times 100\%}{100\% - 15\%} = 2.82$ cum/s

TRANSPORTATION SYSTEMS

The purpose of the following tables is to facilitate the preparation of estimates. ... Note

Prices given hereafter apply to ADELAIDE, BRISBANE, MELBOURNE, PERTH and SYDNEY and represent the indicative cost of installation complete of “Custom” equipment by a specialist company. Lift cars are based on quality design, entrance frames and door panel finishes are stainless steel. To the figures :

ADD extra for :
Lift enclosure walls, lift pit, plant room, Builder’s profit and attendance ... Note

PASSENGER LIFTS

Building	Type	Floors Served	Speed M/Sec	No of Passengers	Installation Cost $	Additional Floor Cost $
Hospitals	Hydraulic	3	0.50	20	150,000	11,500
	Hydraulic	3	0.50	25	155,000	11,500
	Hydraulic	3	0.50	30	165,000	11,500
	Geared, variable speed	3	1.00	20	210,000	11,500
	Hydraulic	5	0.65	20	170,000	11,500
	Hydraulic	5	0.65	25	175,000	11,500
	Geared, variable speed	6	1.00	20	235,000	11,500
	Geared, variable speed	6	1.00	25	240,000	11,500
	Geared, variable speed	8	1.50	25	265,000	11,500
Offices, Hotels						
Low Rise (to 10 floors)	Hydraulic	3	0.50	8/12	105,000	10,000
	Hydraulic	3	0.50	14/17	125,000	10,000
	Geared, variable speed	4	1.00	14/17	205,000	10,000
	Hydraulic	5	0.75	8/12	135,000	10,000
	Hydraulic	5	0.75	14/17	145,000	10,000
	Geared, variable speed	6	1.00	14/17	230,000	10,500
	Geared, variable speed	6	1.75	14/17	240,000	10,500
	Geared, variable speed	10	1.75	20	305,000	10,500
High Rise (10 floors plus)	Geared, variable speed	10*	1.75	16	300,000	10,600
	Geared, variable speed	10*	1.75	23	315,000	10,600
	Gearless, variable speed	15*	2.50	20	455,000	11,000
	Gearless, variable speed	15*	3.50	20	470,000	11,000
	Gearless, variable speed	15*	5.00	20	500,000	11,000
	Gearless, variable speed	15*	6.00	20	540,000	11,000
	Gearless, variable speed	20*	3.50	23	530,000	11,000
	Gearless, variable speed	20*	6.00	23	590,000	11,000
	Gearless, variable speed	20*	7.00	23	650,000	11,000
	Gearless, variable speed	20*	8.00	23	700,000	11,000
Residential	Hydraulic	3	0.50	8/12	95,000	10,000
	Hydraulic	3	0.50	14/17	105,000	10,000
	Hydraulic	5	0.75	8/12	130,000	10,000
	Hydraulic	5	0.75	14/17	145,000	10,000
	Geared, variable speed	5	1.00	11/12	210,000	9,600
	Geared, variable speed	6	1.75	11/12	230,000	9,600
	Geared, variable speed	10*	1.75	14/17	285,000	9,600
	Gearless, variable speed	15*	3.00	14/17	470,000	11,000
Retail (refer page 665)						

* Excludes express zones by-passed

ADD extra for :
Express zones, floors by-passed (average) Per Car/Floor $6,500.00

TRANSPORTATION SYSTEMS

PASSENGER LIFTS (cont'd)

Building	Type	Floors Served	Speed M/Sec	No of Passengers	Installation Cost $	Additional Floor Cost $
Retail	Hydraulic	3	0.5	8/12	100,000	9,700
	Hydraulic	3	0.5	20	150,000	11,400
	Hydraulic	3	0.5	26	155,000	11,400
	Geared, variable speed	3	1.0	20	210,000	9,500
	Geared, variable speed	6	1.75	23	260,000	11,000

GLASS LIFTS, Refer page 666

GOODS LIFTS

Building	Type	Floors Served	Speed M/Sec	Load kg	Installation Cost $	Additional Floor Cost $
General	Hydraulic	3	0.45	3000	185,000	12,600
Industrial Buildings, Hospitals	Geared	5	0.50	3000	190,000	12,600
Offices, Hotels	Gearless	10	2.50	1800	405,000	12,600

SERVICE LIFTS, DUMBWAITERS

Building	Type	Floors Served	Speed M/Sec	Load kg	Installation Cost $	Additional Floor Cost $
All buildings	Standard	2	0.25	100	29,000	4,600
	Standard	2	0.20	450	50,000	8,400
	Standard	6	0.50	100	42,500	8,400

All above service lifts have single sliding door and automatic call system

ESCALATORS

	Characteristics	Installation Cost, One Way $	Additional Cost Per M/Rise, One Way (up to 6.0m overall height) $
Solid balustrade	3.0 metres rise		
	x 600mm carriage way	135,000	11,500
	x 1000mm carriage way	140,000	11,500
Glass balustrade	3.0 metres rise		
	x 600mm carriage way	140,000	15,000
	x 1000mm carriage way	145,000	15,000

TRANSPORTATION SYSTEMS

MOVING WALKS

Characteristics		Installation Cost $	Additional Costs $
			Per additional metre
Horizontal	30.0 metre unit x 1000mm carriage way		up to 60.0 metres
Solid balustrade		280,000	13,000
Glass balustrade		300,000	14,000
			Per additional metre
Inclined at maximum 12°	3.0 metre rise x 1000mm carriage way		rise up to 5.0 metres
Solid balustrade		160,000	35,000
Glass balustrade		165,000	35,000
ADD extra for :			
Weatherproofing if required		30,000-35,000	
Balustrade to well opening		Note	

GLASS LIFTS

Price should be based on the total installation cost of “PASSENGER LIFTS” on page 664 with the addition of + 50% to 70% to cover glass car and upgrading of finishes and working parts. $

ADD extra for :		
Glazed enclosure walling to form shaft	...	Note

WHEELCHAIR LIFTS

Platform lift serving two levels :		
Straight	...	38,000
With bend	...	51,000
Stair lift serving two levels :		
Straight	...	15,500
Curved/spiral	...	22,500

HANDICAPPED PERSON LIFTS

Lift serving two levels, push button operation	...	51,000

DOCUMENT TRANSPORT SYSTEMS

Refer page 704	...	Note

FIRE PROTECTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
EXTINGUISHERS							
Wall mounted extinguisher and bracket :							
Dry chemical, capacity -							
1.00 kg	...	No	85.00	85.00	85.00	85.00	85.00
1.50 kg	...	No	90.00	90.00	90.00	90.00	90.00
2.00 kg	...	No	95.00	95.00	95.00	95.00	95.00
2.50 kg	...	No	110.00	110.00	110.00	110.00	110.00
4.50 kg	...	No	140.00	140.00	140.00	140.00	140.00
9.00 kg	...	No	160.00	160.00	160.00	160.00	160.00
Wet chemical, capacity -							
2.00 kg	...	No	155.00	155.00	155.00	155.00	155.00
7.00 kg	...	No	300.00	300.00	300.00	300.00	300.00
CO_2, capacity -							
2.00 kg	...	No	230.00	230.00	230.00	230.00	230.00
3.50 kg	...	No	245.00	245.00	245.00	245.00	245.00
5.00 kg	...	No	255.00	255.00	255.00	255.00	255.00
Foam, 9 litre	...	No	245.00	245.00	245.00	245.00	245.00
Water stored pressure type -							
Standard :							
9 litres	...	No	225.00	225.00	225.00	225.00	225.00
Mobile extinguisher :							
Dry chemical, capacity -							
25.00kg	...	No	1125.00	1125.00	1125.00	1125.00	1125.00
50.00kg	...	No	1440.00	1440.00	1440.00	1440.00	1440.00
CO_2 , capacity -							
32.00kg	...	No	1435.00	1435.00	1435.00	1435.00	1435.00
45.00kg	...	No	1715.00	1715.00	1715.00	1715.00	1715.00
Water air foam, AFFF -							
45 litres	...	No	1690.00	1690.00	1690.00	1690.00	1690.00
75 litres	...	No	1875.00	1875.00	1875.00	1875.00	1875.00
120 litres	...	No	4995.00	4995.00	4995.00	4995.00	4995.00
HYDRANT POINTS, HOSE REELS, CABINETS							
HYDRANT POINT							
65mm dia. valve including average cost of supply piping in :							
Single storey	...	No	1160.00	1180.00	1210.00	1165.00	1240.00
2/15 storey	...	No	1705.00	1600.00	1715.00	1690.00	1755.00
16/35 storey	...	No	1805.00	1775.00	1860.00	1780.00	1885.00
ADD extra for :							
Tank when required in multi-storey buildings, refer page 488	...	Note					
Standard spring fire hydrant, 80mm dia. connected to 100mm dia. main, including tee riser not exceeding 600mm long and concrete thrust block	...	No	1240.00	1250.00	1285.00	1230.00	1320.00
ADD extra for :							
Cast iron box and cover, surrounded in concrete	...	No	335.00	325.00	345.00	335.00	350.00
Posts, signs, etc.	...	Sum	60.00	60.00	60.00	60.00	60.00

FIRE PROTECTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
HYDRANT POINTS, HOSE REELS, CABINETS (cont'd)							
BOOSTER VALVE ASSEMBLY							
To suit 65mm dia. piping	...	Note					
Mains booster valve assembly (average)	...	No	2850.00	2800.00	2850.00	2800.00	2900.00
Tank booster valve assembly (average)	...	No	1450.00	1450.00	1450.00	1400.00	1500.00
HOSES, HOSE REELS							
Wall mounted and connection to hydrant point :							
Hose reel -							
19mm dia. x 27m long	...	No	755.00	665.00	805.00	790.00	830.00
19mm dia. x 36m long	...	No	790.00	700.00	840.00	820.00	865.00
Hose -							
65mm dia. x 27m long	...	No	555.00	615.00	590.00	545.00	610.00
Recess or cabinet mounted with swinging bracket and connection to hydrant point :							
Hose reel -							
19mm dia. x 27m long	...	No	840.00	755.00	900.00	815.00	940.00
19mm dia. x 36m long	...	No	880.00	795.00	925.00	880.00	975.00
Hose -							
65mm dia. x 27m long	...	No	555.00	615.00	590.00	545.00	610.00
CABINETS							
Cabinet with flush door and baked enamel finish to exposed models :							
Fire reel cabinet -							
Wall mounted	...	No	545.00	565.00	555.00	530.00	580.00
Recessed	...	No	480.00	470.00	490.00	460.00	510.00
Fire reel and hydrant cabinet -							
Wall mounted	...	No	670.00	655.00	695.00	635.00	725.00
Recessed	...	No	655.00	650.00	685.00	630.00	710.00
AUTOMATIC EXTINGUISHING SYSTEMS							
STANDARD WATER SPRINKLER SYSTEM							
COST PER SQUARE METRE							
Average cost per sqm of sprinkler area with allowance for service mains but excluding storage tank, pumps or booster :							
Extra light hazard, exposed -							
Residential, public halls, etc.	...	sqm	39.70	40.70	39.70	44.70	41.60
Ordinary hazard, exposed -							
Offices, shops, car parks, factories, etc.	...	sqm	43.20	43.40	42.60	51.50	44.50

FIRE PROTECTION

AUTOMATIC EXTINGUISHING SYSTEMS (cont'd)

STANDARD WATER SPRINKLER SYSTEM (cont'd)

COST PER SQUARE METRE (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Extra high hazard, exposed -							
At roof level only	...	sqm	66.50	65.60	63.10	84.20	65.10
At rack levels (allowance per metre length per level)	...	m	181.00	184.00	181.00	185.00	189.00
ADD extra for :							
Allowance for tanks, pumps, boosters, etc. to complete installation :							
Ordinary hazard	...	sqm	10.35-17.45	10.55-17.55	10.20-17.10	11.05-19.55	10.90-17.70
Extra high hazard	...	sqm	38.30-44.40	38.40-44.40	37.20-43.20	38.50-45.30	38.40-44.60
Automatic on-off sprinkler heads (average)	...	sqm	18.80	18.70	18.30	19.60	18.75
Additional system to concealed ceiling space	...	sqm	21.40	21.10	22.30	23.80	23.00

COST PER SPRINKLER HEAD

Prices exclude service mains, storage tank, pumps or booster	...	Note					
Sprinkler head including 3 metres x 25mm dia. straight run pipe and supports :							
Standard flush mounted	...	No	335.00	327.00	328.00	327.00	339.00
Standard concealed or exposed	...	No	345.00	337.00	338.00	348.00	349.00
SPRINKLER HEADS							
Standard sprinkler head connected to piping :							
Flush mounted	...	No	36.40	34.90	35.30	43.30	36.60
Concealed or exposed	...	No	31.30	30.40	30.30	38.40	31.90
Automatic on-off sprinkler head connected to piping :							
Pendant	...	No	205.00	195.00	197.00	207.00	203.00
Fusible link head :							
Pendant	...	No	29.80	29.60	29.40	38.40	30.70
ADD extra for :							
Dropper, 1000mm maximum long for flush head	...	No	53.10	51.10	51.30	54.70	54.30
PIPEWORK							
Refer Mechanical Services pages 624 to 633	...	Note					
PUMPS, FIRE INDICATOR BOARDS, ETC.							
As these items vary depending on the size of the project, it is advisable to contact a consultant or fire sprinkler installation firm for price guide lines	...	Note					

FIRE PROTECTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
AUTOMATIC EXTINGUISHING SYSTEMS (cont'd)							
STANDARD DRY PIPE WATER SPRINKLER SYSTEM							
Average cost per sqm of sprinkled area with allowance for service mains and connection to mains water supply, excluding storage tank :							
Ordinary hazard -							
Standard installation, shops, offices, etc.	...	sqm	79.00	80.00	79.00	82.00	82.00
Extra high hazard -							
Cold stores, high temperature areas	...	sqm	105.00	106.00	105.00	109.00	107.00
CARBON DIOXIDE SYSTEM							
Cost range :							
50 - 100 cum	...	cum	80.00-110.00	80.00-110.00	80.00-110.00	80.00-110.00	80.00-110.00
INERGEN SYSTEM							
Inergen, a mixture of nitrogen, argon and carbon dioxide gases and a replacement for Halon systems	...	cum	250.00	250.00	250.00	250.00	250.00
ADD extra for :							
Fire alarms	...	cum	100.00	100.00	100.00	100.00	100.00
DETECTOR AND ALARM SYSTEMS							
THERMAL DETECTOR SYSTEM							
Refer page 523	...	Note					
SMOKE, COMBUSTION DETECTOR SYSTEM							
Refer page 523	...	Note					
AUTOMATIC FIRE ALARM							
Refer page 524	...	Note					
EMERGENCY WARNING SYSTEM							
Refer page 525	...	Note					

FIREPROOFING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $

FIREPROOFING APPLICATIONS, SYSTEMS, ETC.

- Prices are based on 1000 sqm and include all labours and narrow widths ... Note

The Hp/A ratio given hereafter is calculated as follows :

$$\text{Hp/A} = \frac{\text{heated perimeter of exposed steelwork in m}}{\text{cross-section area of steel member in sqm}}$$

DRY BOARD SYSTEMS

Rigid exfoliated vermiculite board box encasing (Vermiculux) :							
Internally -							
Structural steel columns :							
One hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.50	89.00	84.50	90.50
61 - 150 Hp/A	...	sqm	86.00	88.50	90.00	85.50	91.50
151 - 215 Hp/A	...	sqm	87.00	90.00	91.00	86.50	92.50
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.50	89.00	84.50	90.50
61 - 150 Hp/A	...	sqm	86.00	88.50	90.00	85.50	91.50
151 - 215 Hp/A	...	sqm	87.00	90.00	91.00	86.50	92.50
Two hour rating -							
0 - 60 Hp/A	...	sqm	104.00	107.00	110.00	105.00	110.50
61 - 150 Hp/A	...	sqm	105.00	108.00	111.00	106.00	111.50
151 - 215 Hp/A	...	sqm	106.50	109.00	112.00	107.00	113.00
Structural steel beams :							
One hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.50	89.00	84.50	90.50
61 - 150 Hp/A	...	sqm	86.00	88.50	90.00	85.50	91.50
151 - 215 Hp/A	...	sqm	87.00	90.00	91.00	86.50	92.50
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.50	89.00	84.50	90.50
61 - 150 Hp/A	...	sqm	86.00	88.50	90.00	85.50	91.50
151 - 215 Hp/A	...	sqm	87.00	90.00	91.00	86.50	92.50
Two hour rating -							
0 - 60 Hp/A	...	sqm	104.00	107.00	110.00	105.00	110.50
61 - 150 Hp/A	...	sqm	105.00	108.00	111.00	106.00	111.50
151 - 215 Hp/A	...	sqm	106.50	109.00	112.00	107.00	113.00
Rigid silicate bonded exfoliated vermiculite board box encasing (Vicuclad) :							
Internally -							
Flat metal soffit :							
One hour rating	...	sqm	96.00	88.50	88.00	84.00	91.00
Two hour rating	...	sqm	103.00	107.00	109.00	102.00	110.50
Three hour rating	...	sqm	169.00	172.50	176.50	167.50	178.50
Four hour rating	...	sqm	223.50	232.50	236.50	223.50	242.50
Structural steel columns :							
One hour rating -							
0 - 60 Hp/A	...	sqm	85.00	88.50	88.00	84.00	91.00
61 - 150 Hp/A	...	sqm	85.00	88.50	88.00	84.00	91.00
151 - 215 Hp/A	...	sqm	85.00	88.50	88.00	84.00	91.00

FIREPROOFING

FIREPROOFING APPLICATIONS, SYSTEMS, ETC. (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DRY BOARD SYSTEMS (cont'd)							
Rigid silicate bonded exfoliated vermiculite board box encasing (Vicuclad) (cont'd) :							
Internally (cont'd) -							
Structural steel columns (cont'd) :							
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	85.00	88.50	88.00	84.00	91.00
61 - 150 Hp/A	...	sqm	92.00	95.50	96.50	91.00	100.00
151 - 215 Hp/A	...	sqm	108.50	112.00	113.50	107.50	118.00
Two hour rating -							
0 - 60 Hp/A	...	sqm	91.00	95.50	96.50	91.00	100.00
61 - 150 Hp/A	...	sqm	111.50	115.50	118.00	110.50	123.00
151 - 215 Hp/A	...	sqm	130.50	133.50	137.00	129.50	143.00
Structural steel beams :							
One hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.00	88.50	84.00	92.00
61 - 150 Hp/A	...	sqm	85.00	87.00	88.50	84.00	92.00
151 - 215 Hp/A	...	sqm	85.00	87.00	88.50	84.00	92.00
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	85.00	87.00	88.50	84.00	92.00
61 - 150 Hp/A	...	sqm	85.00	87.00	88.50	84.00	92.00
151 - 215 Hp/A	...	sqm	92.00	95.50	96.50	91.00	99.00
Two hour rating -							
0 - 60 Hp/A	...	sqm	92.00	95.50	96.50	91.00	99.00
61 - 150 Hp/A	...	sqm	92.00	95.50	96.50	91.00	99.00
151 - 215 Hp/A	...	sqm	111.50	115.50	118.00	110.50	122.00
ADD extra for :							
0.7mm galvanised steel cladding, preformed with interlocking corners	...	sqm	61.00	61.50	63.50	62.00	64.50
Render or plaster application	...	Note					
Fire rated plasterboard encasing comprising two skins each of two layers finishing 32mm thick with 30mm furring channels between, corner angles and corner beads to structural steel columns and beams :							
Two hour rating	...	sqm	242.00	227.00	208.00	224.00	244.00
INTUMESCENT COATINGS							
Sprayed composite coating of solvent based intumescent and top sealer (Nullifire) :							
Internally -							
Flat metal soffit :							
Sixty minute rating	...	sqm	178.00	176.00	178.00	177.00	180.00
Structural steel columns and beams :							
Sixty minute rating	...	sqm	198.00	194.00	195.00	197.00	199.00
ADD extra for :							
Small section members, i.e. angles and tubing	...	+ 30%					

FIREPROOFING

FIREPROOFING APPLICATIONS, SYSTEMS, ETC. (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SPRAYED VERMICULITE APPLICATIONS							
ADD extra to the following treatments for :							
One coat silicone treatment	...	sqm	9.20	8.90	9.10	9.50	8.90
Sprayed pre-mixed vermiculite and cement (Mandolite P20, Monokote, Roberts VP30) :							
Internally -							
Structural steel columns :							
One hour rating -							
0 - 60 Hp/A	...	sqm	62.00	62.70	60.70	62.90	63.00
61 - 150 Hp/A	...	sqm	110.00	110.00	107.00	110.50	110.50
151 - 215 Hp/A	...	sqm	139.00	140.50	136.00	141.00	141.50
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	47.30	47.90	46.40	48.10	48.10
61 - 150 Hp/A	...	sqm	75.30	76.10	73.70	76.50	76.40
151 - 215 Hp/A	...	sqm	97.90	98.80	96.00	99.20	99.50
Two hour rating -							
0 - 60 Hp/A	...	sqm	70.60	71.40	69.40	71.60	71.60
61 - 150 Hp/A	...	sqm	100.50	102.00	98.80	102.00	102.50
151 - 215 Hp/A	...	sqm	124.00	125.50	121.50	126.00	125.50
Structural steel beams :							
One hour rating -							
0 - 60 Hp/A	...	sqm	43.20	43.60	39.20	43.70	43.80
61 - 150 Hp/A	...	sqm	71.80	72.80	70.60	73.00	83.60
151 - 215 Hp/A	...	sqm	93.80	94.70	91.80	95.40	95.00
One and a half hour rating -							
0 - 60 Hp/A	...	sqm	65.10	65.80	63.80	66.20	66.00
61 - 150 Hp/A	...	sqm	71.80	72.70	70.50	73.00	83.60
151 - 215 Hp/A	...	sqm	86.40	87.40	84.60	87.80	87.70
Two hour rating -							
0 - 60 Hp/A	...	sqm	54.80	55.40	53.80	55.60	55.60
61 - 150 Hp/A	...	sqm	77.50	78.30	76.10	78.70	78.60
151 - 215 Hp/A	...	sqm	96.90	97.60	94.50	98.20	98.00

FIRE SEPARATION

Fire protection sleeve through two hour fire rated wall/floor, to suit PVC or polyethylene pipe size :							
50mm dia.	...	No	142.00	146.00	147.00	142.00	147.00
65mm dia.	...	No	152.00	159.00	160.00	152.00	160.00
80mm dia.	...	No	183.00	191.00	191.00	183.00	191.00
100mm dia.	...	No	229.00	237.00	237.00	229.00	237.00
125mm dia.	...	No	270.00	280.00	280.00	270.00	280.00
150mm dia.	...	No	525.00	540.00	540.00	525.00	540.00
Fire stop collar through floor to suit PVC or polyethylene pipe size :							
40mm dia.	...	No	49.90	51.80	51.80	49.90	51.80
50mm dia.	...	No	53.10	54.80	54.80	53.10	54.80
65mm dia.	...	No	64.70	67.20	67.70	64.70	67.70
80mm dia.	...	No	81.10	84.00	83.50	81.10	83.60
100mm dia.	...	No	109.50	113.00	113.50	109.50	113.50

FIREPROOFING

FIRE SEPARATION (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Fire stop collar through wall to suit PVC or polyethylene pipe size :							
40mm dia.	...	No	56.10	58.00	58.50	56.10	58.50
50mm dia.	...	No	61.20	63.50	63.50	61.20	63.50
65mm dia.	...	No	73.00	75.30	75.30	73.00	75.30
80mm dia.	...	No	102.00	105.50	106.50	102.00	106.50
100mm dia.	...	No	131.50	136.00	136.50	131.50	136.50
Fire stop spring choker to PVC or polyethylene pipe size :							
40mm dia.	...	No	12.70	13.00	13.30	12.70	13.30
50mm dia.	...	No	12.70	13.00	13.30	12.70	13.30
65mm dia.	...	No	14.90	15.50	15.80	14.90	15.80
80mm dia.	...	No	18.90	19.60	19.70	18.90	19.70
100mm dia.	...	No	24.40	25.10	26.00	24.40	26.00
Fire stop to wall and floor through penetrations with expanding pillows (costs based on open area and simple installation conditions) :							
Four hour, remountable	...	sqm	4000.00	4100.00	4050.00	4000.00	4050.00
Ditto but with mortar or panel seals :							
Two/three hour, fixed	...	sqm	1710.00	1765.00	1740.00	1710.00	1740.00
Remountable barrier and duct casing in galvanised sheet steel clad panel with core of fibre cement sheet (for areas subject to mechanical impact and/or exterior weather conditions) :							
Two hour	...	sqm	228.50	240.50	240.50	228.50	240.50
Four hour	...	sqm	265.00	276.00	276.00	265.00	276.00

FIRE STOPPING OF PENETRATIONS

Reinstatement of penetrations through fire resistant elements of construction using Fire Research systems :							
This cost can vary greatly depending on the nature of the building separation element and the complexity of service penetrations (cost range)	...	sqm	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00	1000.00-6000.00

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $

CATERING EQUIPMENT

The purpose of the following tables is to facilitate the preparation of estimates.	...	Note	
Prices given are for the installation complete by a specialist company.	...	Note	
ADD extra to all figures for :			
Building works, enclosure walls, etc.	...	Note	
Plumbing and drainage services	...	Note	
Electrical services	...	Note	
Air-conditioning, mechanical ventilation and special services	...	Note	
Tables, chairs, crockery, glasses, utensils, cash registers, etc.	...	Note	
Prices include for cooking equipment, dish and glass washers, exhaust hoods, benching, sinks, racks, counters, tray races, display cases, shelving, coolrooms, refrigerators, freezers and where applicable all fixed in position and connected to services provided elsewhere.	...	Note	
Refer Commercial Kitchen Equipment on pages 676 to 679 for prices of specific items	...	Note	
COFFEE SHOP			
Serving per hour :			
50 persons	...	Sum	95,000.00
100 persons	...	Sum	145,000.00
SNACK BAR			
Serving per hour :			
100 persons	...	Sum	68,000.00
CAFETERIA			
Serving per sitting (usually 2 sittings per meal) :			
50 persons	...	Sum	290,000.00
100 persons	...	Sum	320,000.00
250 persons	...	Sum	480,000.00
500 persons	...	Sum	715,000.00
TABLE D'HOTE CATERING			
Serving per single sitting :			
50 persons	...	Sum	145,000.00
100 persons	...	Sum	200,000.00
200 persons	...	Sum	289,000.00
A LA CARTE CATERING			
Serving per single sitting :			
50 persons	...	Sum	170,000.00
100 persons	...	Sum	270,000.00
200 persons	...	Sum	355,000.00
FUNCTION CENTRES			
Serving per single sitting :			
100 persons	...	Sum	220,000.00
250 persons	...	Sum	310,000.00
500 persons	...	Sum	500,000.00
1000 persons	...	Sum	680,000.00

MISCELLANEOUS

COMMERCIAL KITCHEN EQUIPMENT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices are for the installation complete by a specialist company	...	Note	
ADD extra for :			
Plumbing and drainage services	...	Note	
Electrical services	...	Note	
SHELVING AND BENCHWORK			
Benches, stainless steel, 600mm wide :			
Open under	...	m	1,155.00
One shelf under	...	m	1,300.00
Two shelves under	...	m	1,600.00
ADD extra for :			
Two tier shelves over	...	m	600.00
Sink -			
Small	...	No	630.00
Large	...	No	1,315.00
Central plating system (range)	...	Sum	80,000.00-960,000.00
Counter, cafeteria service with tray race, stainless steel, with :			
Heated and cold display	...	m	18,600.00
Drinks and beverages	...	m	24,300.00
Shelving, cool rooms, electro galvanised :			
4 tier, 1800mm high x 450mm wide	...	m	875.00
5 tier, ditto	...	m	1,220.00
Shelving, display, stainless steel :			
3 tier, 600mm wide	...	m	1,490.00
4 tier, 600mm wide	...	m	2,070.00
5 tier, 600mm wide	...	m	2,420.00
Shelving, general storage, 1800mm high :			
Electro galvanised	...	m	875.00
Shelving, mobile, stainless steel :			
3 tier, 900 x 600mm	...	Each	1,390.00
4 tier, 1500 x 600mm	...	Each	1,650.00
Tables with shelf under, stainless steel :			
900 x 600 x 900mm high	...	Each	725.00
1800 x 600 x 900mm high	...	Each	1,630.00
PREPARATION			
Chipper :			
Hand	...	Each	450.00
Automatic	...	Each	11,200.00
Cutters :			
457mm	...	Each	8,200.00
Drink mixing machine :			
Single	...	Each	1,150.00
Double	...	Each	2,300.00
Ice cube and flake machine :			
25kg	...	Each	2,100.00
100kg	...	Each	5,100.00

MISCELLANEOUS

COMMERCIAL KITCHEN EQUIPMENT (cont'd)

PREPARATION (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Juice extractor	...	Each	2,400.00
Mincer :			
136kg/hour	...	Each	1,200.00
432kg/hour	...	Each	2,800.00
Mixer :			
20 litre	...	Each	2,900.00
40 litre	...	Each	4,950.00
Peeler :			
6.3kg	...	Each	6,850.00
13.5kg	...	Each	10,300.00
Processor :			
Bench, 3 litre	...	Each	3,500.00
Floor, 36 litre	...	Each	25,000.00
Scales, bench type :			
3kg	...	Each	250.00
20kg	...	Each	300.00
Slicer :			
Manual carriage	...	Each	1,250.00
Automatic carriage	...	Each	8,650.00

COOKING

Bain marie :			
4 containers	...	Each	4,570.00
6 containers	...	Each	6,500.00
Hot press	...	Each	9,150.00
6 containers and warming oven	...	Each	17,350.00
Barbeque, 900mm	...	Each	4,000.00
Boiling table :			
900mm	...	Each	5,200.00
1800mm	...	Each	10,400.00
Boiling water units, refer page 463	...	Note	
Coffee making machines :			
Automatic	...	Each	5,200.00
Espresso	...	Each	10,500.00
Food warmer	...	Each	7,000.00
Fryer and stand :			
Single tank	...	Each	3,600.00
Double tank	...	Each	4,900.00
Grill on stand :			
600mm	...	Each	3,800.00
900mm	...	Each	5,100.00

MISCELLANEOUS

COMMERCIAL KITCHEN EQUIPMENT (cont'd)

COOKING (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
Kettles :			
Jacketed	...	Each	21,000.00
Tilting	...	Each	33,200.00
Milkshake maker	...	Each	350.00
Stock pot :			
90 litre	...	Each	16,300.00
135 litre	...	Each	17,750.00
Toasted sandwich maker	...	Each	1,200.00
Toaster :			
Conveyor	...	Each	950.00
Pop-up	...	Each	500.00
Rotary	...	Each	9,450.00
Urns, refer page 463	...	Note	

DISPLAY AND SERVING

Beverage dispensers :			
Single	...	Each	1,350.00
Double	...	Each	2,850.00
Triple	...	Each	4,000.00
Cafeteria service counter, refer page 676	...	Note	
Food display cabinet, counter type refrigerated	...	Each	6,100.00
Food conveyor (Mobile cabinet)	...	Each	7,900.00
Heated food cabinet, vertical :			
One door	...	Each	4,500.00
Two doors	...	Each	10,000.00
Refrigerated food cabinet :			
600 litre	...	Each	2,800.00
1000 litre	...	Each	3,200.00
1500 litre	...	Each	5,000.00
Shelving, display, refer page 676	...	Note	

WASH-UP

Utencil washer :			
Standard	...	Each	11,800.00
Large	...	Each	22,200.00
Dishwasher :			
Small	...	Each	5,000.00
Medium	...	Each	6,500.00-12,000.00
Large	...	Each	23,000.00-52,000.00
Glass washer :			
Freestanding	...	Each	4,500.00
Underbar	...	Each	2,200.00-3,900.00
Sinks, refer Sanitary Plumbing	...	Note	

MISCELLANEOUS

COMMERCIAL KITCHEN EQUIPMENT (cont'd)

SUNDRIES

		UNIT	Adel. Bris. Melb. Per. Syd. $
Cool rooms, refer page 704	...	Note	
Insect control unit :			
Single sided	...	Each	355.00
Double sided	...	Each	555.00
Lowerator, plate dispenser	...	Each	5,250.00
Multipots :			
10 litre	...	Each	355.00
45 litre	...	Each	610.00
Oven :			
Convection	...	Each	4,000.00-14,000.00
Pizza, single	...	Each	2,800.00-115,000.00
Microwave	...	Each	1,500.00-31,000.00
Pan, self generating, steam jacketed :			
90 litre	...	Each	31,100.00
180 litre	...	Each	33,300.00
275 litre	...	Each	40,200.00
Pie warmer	...	Each	980.00
Range, electric :			
900mm	...	Each	11,300.00
Double oven	...	Each	22,600.00
Range, gas :			
600mm	...	Each	5,800.00
900mm	...	Each	7,200.00
1200mm	...	Each	8,900.00
Double oven	...	Each	12,100.00
Salamander equipment :			
900mm	...	Each	3,150.00
Steam equipment :			
Bain marie (average)	...	Each	18,400.00
Jacketed stockpot (average)	...	Each	27,500.00
Pressure steamer	...	Each	45,200.00
Steam generator, 48kW	...	Each	26,300.00
Steamers :			
Atmospheric	...	Each	10,500.00
Combi	...	Each	15,000.00-32,600.00

MISCELLANEOUS

LAUNDRY EQUIPMENT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include fixing in position and connecting to services provided elsewhere	...	Note	
WASHING MACHINES			
Fully automatic, electric, coin operated, capacity :			
6kg	...	No	1,800.00
8kg	...	No	2,600.00
Fully automatic, electric, non-coin operated, capacity :			
16kg, freestanding, high speed	...	No	7,700.00
25kg, freestanding, high speed	...	No	13,200.00
35kg, freestanding, high speed	...	No	15,400.00
TUMBLE DRYERS			
Coin operated, capacity :			
9kg -			
Gas	...	No	1,800.00
Electric	...	No	1,650.00
13.5kg -			
Gas	...	No	5,000.00
Non-coin operated, capacity :			
9kg -			
Gas	...	No	1,400.00
Electric	...	No	1,200.00
22.5kg -			
Gas	...	No	4,000.00
34.0kg -			
Gas	...	No	6,600.00
54.5kg -			
Gas	...	No	14,500.00
IRONING CENTRE			
Folding wall unit and cabinet	...	No	450.00-500.00
COMMERCIAL ROTARY IRONER (De luxe)			
Range, small to very large heavy duty	...	No	2,000.00-20,000.00
Electric roller ironers -			
1.2m wide	...	No	4,400.00
1.4m wide	...	No	5,500.00
1.6m wide	...	No	7,700.00
SUNDRIES			
Third arm sheet folder, manual	...	No	750.00
Mobile sheet folding table	...	No	900.00

MISCELLANEOUS

DOMESTIC APPLIANCES

			UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include for fixing in position and connecting to services provided elsewhere		...	Note	
COOKING				
STOVE, electric or gas, free standing, four plates, grill, lower oven	(Average)	...	No	750.00-1,400.00
STOVE, electric or gas, free standing, four plates, grill, lower and top ovens	(Average)	...	No	1,400.00-2,100.00
COOKING TOP				
Electric	(Average)	...	No	750.00-1,400.00
Gas hob (5 burner)	(De luxe)	...	No	2,600.00
Ceramic hob	(De luxe)	...	No	3,200.00
Induction	(De luxe)	...	No	4,500.00
WALL OVEN, electric, single, including grilling compartment	(Average)	...	No	1,000.00-1,400.00
	(De luxe)	...	No	8,000.00
WALL OVEN, electric, double, including grill in top oven	(Average)	...	No	1,250.00-2,500.00
	(De luxe)	...	No	12,000.00
UNDERBENCH OVEN	(De luxe)	...	No	1,800.00-4,000.00
OVEN, microwave	(Average)	...	No	350.00-1,000.00
	(De luxe)	...	No	2,700.00
OVEN/MICROWAVE combination	(De luxe)	...	No	3,500.00-6,400.00
REFRIGERATORS, FREEZERS				
REFRIGERATOR, free standing, capacity :				
150 litre	(Average)	...	No	400.00-650.00
300 litre	(Average)	...	No	750.00-1,100.00
REFRIGERATOR, free standing, incorporating deep freezer, capacity :				
250 litre (refrigerator)				
50 litre (freezer)	(Average)	...	No	775.00-1,300.00
300 litre (refrigerator)				
100 litre (freezer)	(Average)	...	No	900.00-1,800.00
FREEZER, free standing, capacity :				
150 litre	(Average)	...	No	550.00-650.00
200 litre	(Average)	...	No	600.00-850.00
300 litre	(Average)	...	No	1,000.00-1,300.00
WASHING				
DISHWASHER, automatic :				
Free standing	(Average)	...	No	800.00-1,400.00
	(De luxe)	...	No	1,800.00
Built-in	(Average)	...	No	900.00-1,600.00
	(De luxe)	...	No	5,600.00
Mobile	(Average)	...	No	500.00-850.00

MISCELLANEOUS

DOMESTIC APPLIANCES (cont'd)

WASHING (cont'd)

			UNIT	Adel. Bris. Melb. Per. Syd. $
WASHING MACHINE, automatic,				
7.5kg capacity	(Average)	...	No	800.00-1,500.00
	(De luxe)	...	No	1,400.00-4,500.00
DRYERS, tumble, 3.5 to 5kg capacity	(Average)	...	No	370.00-700.00
7kg capacity	(De luxe)	...	No	1,600.00-4,500.00

SUNDRIES

EXHAUST FAN, fitted into :				
Window 200mm dia.	(Average)	...	No	160.00-200.00
Wall 250mm dia.	(Average)	...	No	245.00-310.00
Ceiling 250mm dia.	(Average)	...	No	75.00-100.00
FOOD WASTE disposer, sink type	(Minimum)	...	No	350.00
	(Maximum)	...	No	1,000.00
STOVE HOOD, two speed vented,				
600 to 900mm wide	(Minimum)	...	No	500.00
	(Maximum)	...	No	7,500.00
WATER HEATERS, refer pages 461 to 463		...	Note	

BANK FIT-OUT

The following prices are applicable to fit-out of branch premises		...	Note	
Plain counter	(Minimum)	...	m	1,500.00
	(Maximum)	...	m	5,000.00
Bullet resistant counter				
prepared for rising screens	(Average)	...	m	5,500.00
Bulkhead, ditto	(Average)	...	m	2,000.00
ADD extra for :				
Tellers box including fit-out -				
Single teller	(Average)	...	No	4,000.00
Dual tellers	(Average)	...	No	7,250.00
Enquiries screens	(Average)	...	m	2,000.00
Bullet resistant glass screen, 37mm thick x 1200mm high, RISI rated		...	m	4,000.00
Bullet resistant rising screen, 1000mm high above counter and 1050mm below, covering 2000mm overall high below pelmet -				
1500mm wide (standard size)		...	No	12,500.00
2100mm wide		...	No	17,000.00
Bullet resistant partition or wall x 2700mm high with :				
10mm Bissalloy and plasterboard		...	m	2,500.00
15-35mm aluminium and 6mm plywood		...	m	3,900.00
Bullet resistant door	(Average)	...	No	6,000.00

MISCELLANEOUS

BANK FIT-OUT (cont'd)

			UNIT	Adel. Bris. Melb. Per. Syd. $
Writing slope on :				
Brackets		...	m	120.00-200.00
Island stand		...	m	300.00-400.00
Night depository		...	Each	500.00
Automatic teller		...	Each	150,000.00-180,000.00
Strong room doors, refer page 701		...	Note	
Modular vaults, refer page 701		...	Note	
Partitions to interview offices, etc., refer pages 393 to 395		...	Note	

DORMITORY FIT-OUT

Overall cost for furniture, per student :				
	(Minimum)	...	Student	1,500.00
	(Maximum)	...	Student	5,000.00
Bed, free standing, excluding mattress	(Average)	...	Each	375.00
Chair	(Average)	...	Each	100.00
Drawer unit	(Average)	...	Each	600.00
Desk top	(Average)	...	m	250.00
Wardrobe	(Average)	...	Each	800.00
Cubicle	(Average)	...	Each	100.00
Trundle	(Average)	...	Each	350.00
Mattress	(Average)	...	Each	200.00
LED lighting to cubicle	(Average)	...	Each	200.00
Modular wiring	(Average)	...	Each	280.00
Pinboard	(Average)	...	Each	250.00

ECCLESIASTICAL FIT-OUT

Altar :				
Plain	(Average)	...	Each	2,800.00
Deluxe	(Average)	...	Each	17,500.00
Lectern :				
Plain	(Average)	...	Each	800.00
Deluxe	(Average)	...	Each	3,000.00
Pulpit :				
Plain	(Average)	...	Each	2,000.00
Deluxe	(Average)	...	Each	8,000.00
Pews :				
Hardwood bench	(Average)	...	m	675.00
ADD extra for :				
Padded seat/cushion	(Average)	...	m	80.00
Kneeler	(Average)	...	m	150.00
Seating, individual :				
Stack type -				
Standard		...	No	45.00
Deluxe		...	No	90.00

MISCELLANEOUS

HOSPITAL FIT-OUT

		UNIT	Adel. Bris. Melb. Per. Syd. $
• Prices include for fixing in position and connecting to services provided elsewhere	...	Note	
AUTOPSY TABLE, 2440 x 790mm stainless steel, built in	...	No	20,000.00
ADD extra for :			
Medical vacuum	...	Note	
Exhaust air duct to suit ventilated type	...	Note	
Special drainage requirements	...	Note	
BED PAN FLUSHER / SANITISER, single phase powered unit, cold water supply :			
Free standing	...	No	10,200.00
Built-in	...	No	10,500.00
BLOOD BANK REFRIGERATOR, freestanding :			
Two drawer, 60 unit	...	No	16,000.00
Five drawer, 250 unit	...	No	20,000.00
FUME CUPBOARDS, refer page 692	...	Note	
HOODS :			
Biohazard)			
Laminar Flow) refer page 692	...	Note	
HYDRO-POOL HOIST			
Manual	...	No	20,000.00
Electric	...	No	30,000.00-35,000.00
ADD extra for :			
Equipment (stretcher, commode, trolley, pool lift transporter, swim chair)	...	set	17,000.00
VACCINE REFRIGERATOR			
220 litre	...	No	5,350.00
400 litre	...	No	6,000.00
MORTUARY CABINET, refrigerated, side entry :			
Single bier	...	No	13,800.00
Double bier	...	No	20,000.00
WASHER, DISINFECTOR, computerised controls, integral data logging, web interface for remote viewing and fault diagnosis, building management systems compatibility, automatic door, 305 litre :			
Single door	...	No	48,000.00
Double door	...	No	52,500.00

MISCELLANEOUS

HOTEL FIT-OUT

OVERALL

			UNIT	Adel. Bris. Melb. Per. Syd. $
FITTING OUT of total building including reception, bars, restaurants, bedrooms, together with all furniture, furnishings, crockery, linen, etc. :				
City Hotel -				
Basic standard	(Minimum)	...	Bedroom	23,000.00
	(Maximum)	...	Bedroom	36,000.00
Intermediate standard	(Minimum)	...	Bedroom	40,000.00
	(Maximum)	...	Bedroom	50,000.00
High standard	(Minimum)	...	Bedroom	65,000.00
	(Maximum)	...	Bedroom	78,000.00

BARS, RESTAURANTS

BAR COUNTERS

Prices are for counters only. Under-bar equipment, piping, refrigerated cabinets, etc., are included in BAR EQUIPMENT	...	Note	
Public Bar :			
Front counter -			
Basic standard	...	m	2,000.00
Medium standard	...	m	2,500.00
Back counter/display -			
Basic standard	...	m	2,800.00
Medium standard	...	m	3,800.00
Lounge Bar :			
Front counter -			
Medium standard	...	m	2,500.00
High standard	...	m	3,300.00
Back counter/display -			
Medium standard	...	m	3,700.00
High standard	...	m	5,000.00
Cocktail Bar :			
Front counter -			
Medium standard	...	m	2,900.00
High standard	...	m	3,400.00
Back counter/display -			
Medium standard	...	m	3,800.00
High standard	...	m	5,600.00

MISCELLANEOUS

HOTEL FIT-OUT (cont'd)

BARS, RESTAURANTS (cont'd)

BAR EQUIPMENT

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include for under-bar equipment, refrigeration together with refrigerated storage cabinets and keg storage, insulation, bar plumbing, beer and cool drink supply lines, glass washers, glass racks and trays, ice making machines, all fixed in position and connected to services provided elsewhere	...	Note	
Prices are per metre of front counter or back counter/display	...	Note	
Public/Lounge Bar :			
Basic standard	...	m	3,800.00
Medium standard	...	m	5,300.00
Cocktail Bar :			
Medium standard	...	m	5,200.00
High standard	...	m	6,000.00

SEATING

Chairs :			
Stack type	...	No	50.00
Upright, moulded plastic	...	No	130.00
Upright, upholstered, fabric covered	...	No	250.00
Lounge, upholstered, fabric covered	...	No	1,500.00
Bar Stools :			
Vinyl covered	...	No	350.00
Fabric covered	...	No	550.00
Booth (logue) unit to seat four persons :			
Moulded plastic	...	Unit	2,700.00
Upholstered, vinyl covered	...	Unit	6,600.00
Upholstered, fabric covered	...	Unit	7,000.00
Semi-circular booth unit to seat four persons :			
Upholstered, vinyl covered	...	Unit	5,600.00
Upholstered, fabric covered	...	Unit	6,500.00
Bench seating :			
Upholstered, fabric covered	...	m	1,400.00

TABLES

To seat four persons :			
Basic standard	...	No	150.00
Medium standard	...	No	300.00
High standard	...	No	1,000.00
To seat six persons :			
Basic standard	...	No	230.00
Medium standard	...	No	450.00
High standard	...	No	1,100.00
Occasional :			
Medium standard	...	No	660.00
High standard	...	No	1,200.00

MISCELLANEOUS

HOTEL FIT-OUT (cont'd)

BARS, RESTAURANTS (cont'd)

BULKHEADS

			UNIT	Adel. Bris. Melb. Per. Syd. $
Dropped bulkhead over bar/servery :				
Medium standard		...	sqm	650.00
High standard		...	sqm	1,200.00
ADD extra for :				
Floor, wall and ceiling finishes upgraded as necessary		...	Note	
Curtains and drapes		...	Note	
Paintings, murals		...	Note	
Lighting and air-conditioning, upgraded as necessary		...	Note	
Dumb waiter, if required		...	Note	
Keg hoist, if required		...	No	90,000.00

BEDROOMS

Wardrobe, fully fitted	(Minimum)	...	m	850.00
	(Maximum)	...	m	1,600.00
Case and tray rack	(Minimum)	...	m	220.00
	(Maximum)	...	m	450.00
Writing desk/bench	(Minimum)	...	m	650.00
	(Maximum)	...	m	1,600.00
Bed, including bed head :				
Single	(Minimum)	...	No	850.00
	(Maximum)	...	No	2,000.00
Double	(Minimum)	...	No	1,500.00
	(Maximum)	...	No	3,500.00
ADD extra for :				
Tables and chairs		...	Note	
Refrigerator		...	Note	
Radio/TV		...	Note	
Call and communication system		...	Note	

KITCHEN EQUIPMENT refer pages 676 to 679 ... Note

LAUNDRY EQUIPMENT refer page 680 ... Note

DOMESTIC APPLIANCES refer pages 681 to 682 ... Note

MISCELLANEOUS

LABORATORY FIT-OUT

		UNIT	Adel. Bris. Melb. Per. Syd. $
BENCHING			
Fixed perimeter wall benching on steel frames with stainless steel levelling feet :			
Fixed bench 750mm deep -			
Laminate top	...	m	500.00
Solid surface (12mm)	...	m	720.00
Trespa	...	m	740.00
Add extra for :			
150mm high splash back to wall	...	m	200.00
Fixed bench 750mm deep with 300mm splash back to wall -			
Stainless steel (316) with stainless steel legs	...	m	1,600.00
Fixed bench 1500mm deep -			
Laminate top	...	m	960.00
Solid surface (12mm)	...	m	1,500.00
Trespa	...	m	1,580.00
SERVICE RETICULATION SPINES			
Service spine 300mm high including service 'bollards' and two tier shelving :			
Laminate top	...	m	1,560.00
Solid surface (12mm)	...	m	1,700.00
Trespa	...	m	1,740.00
Compact service spine 300mm high including 'bollards' and two tier shelving :			
Laminate top	...	m	2,500.00
Solid surface (12mm)	...	m	2,700.00
Trespa	...	m	2,740.00
SERVICE DROPPERS			
Aluminium powder coated suspended laboratory spine including two tier shelving	...	m	3,500.00
UNDER BENCH STORAGE			
Under bench mobile storage :			
Cupboard unit -			
600 x 600 x 840mm high :			
Pre-finished	...	No	480.00
Polyurethane	...	No	640.00
900 x 600 x 840mm high :			
Pre-finished	...	No	540.00
Polyurethane	...	No	860.00
Drawer unit -			
600 x 600 x 840mm high :			
Pre-finished	...	No	640.00
Polyurethane	...	No	840.00
900 x 600 x 840mm high :			
Pre-finished	...	No	800.00
Polyurethane	...	No	980.00
Under bench fixed storage :			
Cabinet -			
1000 x 600 x 865mm high :			
Pre-finished	...	No	960.00
Polyurethane	...	No	1,260.00



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MISCELLANEOUS

LABORATORY FIT-OUT (cont'd)

WALL STORAGE UNITS

		UNIT	Adel. Bris. Melb. Per. Syd. $
Wall storage unit including doors :			
1000 x 300 x 2100mm high -			
Pre-finished	...	No	1,100.00
Polyurethane	...	No	1,620.00
1000 x 600 x 2100mm high -			
Pre-finished	...	No	1,460.00
Polyurethane	...	No	2,060.00
Wall storage unit including glass doors :			
1000 x 300 x 2100mm high -			
Pre-finished	...	No	1,420.00
Polyurethane	...	No	2,100.00
1000 x 600 x 2100mm high -			
Pre-finished	...	No	1,780.00
Polyurethane	...	No	2,480.00
Wall storage shelving unit, no door :			
1000 x 300 x 2100mm high -			
Pre-finished	...	No	980.00
Polyurethane	...	No	1,060.00
1000 x 600 x 2100mm high -			
Pre-finished	...	No	1,120.00
Polyurethane	...	No	1,480.00
Wall storage unit including sliding doors :			
450 x 750 x 2100mm high -			
Pre-finished	...	No	2,420.00
Polyurethane	...	No	3,420.00

STAINLESS STEEL SINK BENCHES

Stainless steel sink bench (316) including one bowl on fixed cupboard :			
1200 x 750 x 300mm high	...	No	3,660.00
1500 x 750 x 300mm high	...	No	4,240.00
Stainless steel sink bench (316) including one bowl on stainless steel frame :			
1200 x 750 x 300mm high	...	No	2,560.00
1500 x 750 x 300mm high	...	No	2,820.00
Splashbacks to sink benches :			
Glass 1500 x 800mm high 10mm toughened glass	...	No	900.00

MISCELLANEOUS

LABORATORY FIT-OUT (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
OVER BENCH SHELVING			
Overbench cupboard unit :			
1000 x 300 x 750mm high -			
Pre-finished	...	m	640.00
Polyurethane	...	m	900.00
Overbench cupboard unit including glass doors :			
1000 x 300 x 750mm high, pre-finished	...	m	860.00
Overbench shelf unit :			
1000 x 300 x 750mm high -			
Pre-finished	...	m	460.00
Polyurethane	...	m	600.00
Bulkhead up to 600mm high :			
Pre-finished	...	m	220.00
Polyurethane	...	m	260.00
Metal shelving, 1000mm wide bays including two tier shelves with pre-finished backing board :			
Powdercoated steel	...	m	700.00
Stainless steel	...	m	800.00
BALANCE BENCH			
Bench comprising 12mm solid surface filled with reinforced concrete (balance) :			
1500 x 750 x 900mm high	...	m	2,700.00
SAFETY STATION BACKING PANELS			
Safety station backing panel :			
Laminate	...	sqm	500.00
Polyurethane	...	sqm	550.00
Trespa (10-13mm)	...	sqm	680.00
LIGHTING			
In-line LED "softwired" in series - excluding starter sockets	...	No	55.00
POWER			
In-line auto-switched 240 quadruple power outlets - excluding starter sockets	...	No	80.00
LABORATORY STOOL			
PU seat on castors :			
With back	...	No	265.00
No back	...	No	190.00

MISCELLANEOUS

LABORATORY FIT-OUT (cont'd)

AUTOCLAVES

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices include installation, computerised control, integrated data logging and building management systems compatibility	...	Note	
General purpose, standard range autoclave size 1300mm long x 660mm wide x 660mm high, automatic sliding door, recessed / built in installation :			
Single door	...	No	111,500.00
Single door, electric	...	No	120,000.00
Double door	...	No	144,000.00
Double door, electric	...	No	159,500.00
General purpose, standard range autoclave size 1300mm long x 660mm wide x 900mm high, automatic sliding door :			
Single door	...	No	131,000.00
Single door, electric	...	No	146,000.00
Double door	...	No	171,500.00
Double door, electric	...	No	181,500.00

FUME CUPBOARDS, ETC.

Prices include fixing in position, standard flues and connecting to services provided elsewhere	...	Note	
Fume cupboard 750 x 1200 x 1500mm high, PVC, bench mounted (school type)	...	No	27,500.00
Fume cupboard, PVC, stand mounted (laboratory type) :			
750 x 1200 x 1500mm high	...	No	27,500.00
750 x 1800 x 1500mm high	...	No	33,000.00
Biological safety cabinet, freestanding, 750 deep x 900, 1200 or 1800mm wide, auto electric or fixed stand mounted	...	No	10,800.00-15,500.00
Biological hood, bench mounted, 750 deep x 900, 1200 or 1800mm wide	...	No	10,000.00-14,000.00
Laminar flow hood, freestanding (range) :			
1200mm module	...	No	6,000.00-7,500.00
1800mm module	...	No	7,500.00-8,500.00

MISCELLANEOUS

LIBRARY FIT-OUT

		UNIT	Adel. Bris. Melb. Per. Syd. $
BOOK DETECTION SYSTEM comprising lockable entry and exit gates, book check, desensitiser, fully installed to suit :			
Small library	...	System	10,000.00
ADD extra for :			
Access gates	...	Sum	8,500.00
Tertiary library -			
Single corridor :			
Basic	...	System	10,000.00
Top of range	...	System	15,000.00
Dual corridor :			
Basic	...	System	14,000.00
Top of range	...	System	22,000.00
BOOKSHELF UNITS in sheet steel with powdercoat finish, six intermediate shelves, 250mm deep x 1875mm high :			
End unit, 900mm long -			
Single sided	...	No	500.00
Double sided	...	No	700.00
Mid unit, 900mm long -			
Single sided	...	No	380.00
Double sided	...	No	560.00
ADD extra for :			
Canopy -			
Single sided 900mm long	...	No	45.00
Double sided 900mm long	...	No	80.00
PERIODICAL DISPLAY STAND in sheet metal with powdercoat finish, 1875mm high :			
End unit, 900mm long -			
Single sided	...	No	550.00
Double sided	...	No	810.00
Mid unit, 900mm long -			
Single sided	...	No	430.00
Double sided	...	No	670.00
ADD extra for :			
Canopy -			
Single sided 900mm long	...	No	45.00
Double sided 900mm long	...	No	80.00
CARREL, modular unit, 900/1000mm long with laminate finished top and dividing screens	...	No	650.00
ISSUE AND RETURN DESK, 750mm wide x 1000mm high with laminate finished top and face, open shelves under, bookshelf on top :			
2700mm long	...	No	1,800.00
6000mm long L shaped	...	No	3,000.00
STUDY TABLE, with laminate finished top :			
1200 x 900 x 700mm high	...	No	250.00
1500 x 900 x 700mm high	...	No	280.00
ADD extra for :			
Standard chair	...	No	115.00
OPAQUE TABLE, standard model	...	No	520.00
BROWSER BOX, standard model	...	No	485.00

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $

RETAIL FIT-OUT

SHOWCASES

Floor model, full display, 6mm float glass and anodised aluminium frame, with sliding doors and two adjustable shelves, laminate finished base on legs, 450mm wide x 900mm high x :			
1200mm long	...	No	3,540.00
1800mm long	...	No	4,365.00
Floor model, half display, 6mm float glass and anodised aluminium frame, laminate finished ends, doors, storage space, base on legs, 450mm wide x 900mm high x :			
1200mm long	...	No	4,010.00
1800mm long	...	No	5,015.00
Wall model, full display, 6mm float glass and anodised aluminium frame, with sliding doors and adjustable shelves, laminate finished wall panel, base and top, on plinth, 600mm wide x 2100mm high x :			
1200mm long	...	No	6,370.00
1800mm long	...	No	8,025.00

GONDOLAS

Gondola unit, sheet metal with powder coat finish including base shelf, intermediate division, 1200mm high :			
End unit, 1200mm long -			
Single sided	...	No	2,880.00
Double sided	...	No	3,480.00
Mid unit, 1200mm long -			
Single sided	...	No	2,460.00
Double sided	...	No	2,820.00
ADD extra for :			
Laminated particle board shelves			
1200mm long x 300mm wide, including brackets	...	No	106.00

WALL SHELVING

Steel wall stripping, powder coat finish (average) :			
Single slot	...	m	46.00
Double slot	...	m	60.00
Steel tube, slot, powder coat finish (average) :			
Single slot	...	m	106.00
Double slot	...	m	113.00
Shelves, 300mm wide including brackets :			
6mm float glass	...	m	133.00
13mm laminated particle board	...	m	106.00

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $
RETAIL FIT-OUT (cont'd)			
LATERAL MERCHANDISING SYSTEM			
Eyeline system comprising backing of aluminium interlocking members with fixing grooves at 100mm or 150mm centres, fixed to face of wall, all powder coat finished (average) :	...	sqm	290.00
ADD extra for :			
Accessories to suit particular merchandising application (approx cost range)	...	sqm	67.00-103.00
VISUAL MERCHANDISING SYSTEM			
Slatwall system comprising laminate faced fibreboard backing board in 1200mm high modules and having grooves at 100mm to 200mm centres vertically each with aluminium fixing extrusion insert, all fixed to face of wall (average) :	...	sqm	210.00
ADD extra for :			
Accessories to suit particular merchandising application (approx cost range)	...	sqm	65.00-100.00
CLOTHING RACKS, chromium plated finish (cost range)	...	No	800.00-945.00
CASH REGISTER STAND, laminate finished, 600mm long x 600mm wide x 900mm high	...	No	1,890.00
CHECKOUTS, laminate finished :			
2000mm long x 900mm wide x 900mm high -			
Standard model	...	No	5,900.00
De luxe model	...	No	9,910.00
3000mm long x 1150mm wide x 900mm high, scanning model	...	No	16,700.00
TURNSTILE, chromium plated, three arm with speed adjustment	...	No	1,380.00
TROLLEY RAIL, 25mm dia. x 2400mm long chromium plated tube in top rail and with posts at 1000mm centres	...	No	1,030.00
SPACEFRAME, (average cost)			
PVC ultralight	...	sqm	255.00
Aluminium	...	sqm	875.00-1,105.00
SHOPFRONTS			
Refer page 367	...	Note	
STORAGE SYSTEMS			
Refer pages 311 to 313	...	Note	

MISCELLANEOUS

RETAIL FIT-OUT (cont'd)

REFRIGERATED FOOD CASES

		UNIT	Adel. Bris. Melb. Per. Syd. $
Generally in modules of 2440 or 3650mm	...	Note	
Dairy, multi deck	...	m	1,280.00
Meat case, single deck	...	m	9,060.00
Produce, single deck	...	m	10,140.00
Service delicatessen, 3650mm long single deck	...	No	20,400.00
Self contained, 1800mm high :			
Two door	...	No	5,220.00
Three door	...	No	7,500.00
Four door	...	No	9,840.00
ADD extra for :			
Remote refrigerator motor	...	No	1,380.00

FREEZER / FROZEN FOOD CASES

Frozen food, island case (in modules of 2440 or 3650mm)	...	m	13,560.00
Self contained, 1800mm high :			
Two door	...	No	10,680.00
Three door	...	No	17,520.00
Four door	...	No	19,800.00
ADD extra for :			
Remote freezer motor	...	No	1,860.00

OFFICE FURNITURE

BOARDROOM

Tables :			
Medium range, metal base frame, basic veneer top	...	Per Head	600.00
High standard, timber base frame, detailed veneer top	...	Per Head	800.00
Prestige, stone / timber / lacquer base frame and top	...	Per Head	1,200.00
Chairs :			
Medium, Chrome base and arms, fabric seat and back	...	No	600.00
High Standard, chrome base, leather seat, back, arms	...	No	1,800.00
Credenza / sideboard 2700mm high x 600mm deep, with doors :			
Medium range, basic veneer with doors and drawers	...	m	1,250.00
High standard, detailed veneer with doors and drawers	...	m	2,500.00

MISCELLANEOUS

OFFICE FURNITURE (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
CONFERENCE ROOMS			
Tables, chrome or powder coated steel base frames :			
Medium range, ABS edged 25mm melamine board top	...	Per Head	250.00
High standard, basic veneer top	...	Per Head	600.00
Chairs :			
Medium range	...	No	400.00
High standard	...	No	1,200.00
Credenza / sideboard 900 to 1200mm high x 600mm deep, with doors :			
Medium range, melamine board with doors	...	m	500.00
High standard, polished timber veneer with doors	...	m	1,250.00
TRAINING ROOMS			
Tables, 1600 X 800 X 725mm high, edged 25mm melamine board top :			
Flip-top frame on castors	...	Per Head	700.00
Simple powder coated metal frame	...	Per Head	350.00
Chairs, all medium range :			
Metal frame, stackable upholstered seat only	...	No	180.00
Metal frame, stackable upholstered seat and back	...	No	250.00
Credenza, medium range, 900 to 1200mm high x 600mm deep, with doors	...	m	500.00
BREAKOUT AREA			
Tables to seat 4, ABS edged 25mm melamine board top	...	No	300.00
Chairs, metal frame, stackable, upholstered seat only	...	No	160.00
MANAGING DIRECTOR/DIRECTOR OFFICES			
Setting, desk with drawers, return and storage cabinets :			
Medium range	...	No	6,000.00
Prestige	...	No	10,000.00
Desk chair, fully adjustable with arms and castors :			
Medium range	...	No	600.00
Prestige	...	No	2,500.00
Desk visitors chairs :			
Medium range	...	No	300.00
Prestige	...	No	1,500.00
Visitors armchair / settee setting with low table :			
Medium range	...	Per Head	500.00
Prestige	...	Per Head	1,350.00

MISCELLANEOUS

OFFICE FURNITURE (cont'd)

		UNIT	Adel. Bris. Melb. Per. Syd. $
MIDDLE MANAGEMENT OFFICES			
Setting, desk with drawers, return and storage cabinets :			
Medium range with 3 legged sit-stand system	...	No	4,000.00
Medium range, static	...	No	3,000.00
Prestige with 3 legged sit-stand system	...	No	6,000.00
Prestige, static	...	No	5,000.00
Desk chair, fully adjustable with arms and castors :			
Medium range	...	No	500.00
Prestige	...	No	900.00
Desk visitors chairs :			
Medium range	...	No	300.00
Prestige	...	No	600.00
GENERAL STAFF			
Desk, L shaped, 1800 x 1800mm including mobile drawers :			
Medium range with 3 legged sit-stand system	...	No	3,000.00
Medium range static	...	No	2,200.00
Desk, rectangular 1800 x 800mm including mobile drawers :			
Medium range with 2 legged sit-stand system	...	No	2,000.00
Medium range static	...	No	1,350.00
Desk chair, fully adjustable with castors and no arms	...	No	400.00
RECEPTION			
Counter, split level, 1100mm visitor side, 725mm receptionist :			
Medium range	...	m	1,200.00-3,000.00
High standard	...	m	3,000.00-5,000.00
Prestige	...	m	5,000.00-10,000.00
SOFT WIRING			
Soft wiring excluding electrician provided starter sockets autoswitched four GPO outlet with 'in desktop' box and cable tray	...	No	200.00
DOCUMENT STORAGE CABINETS			
Document storage cabinets with doors and adjustable height shelving :			
900 x 900 x 450 mm, melamine board, 2 x 25mm shelves	...	No	400.00
1800 x 900 x 450 mm, melamine board, 2 x 25mm shelves	...	No	550.00
2700 x 900 x 450 mm, melamine board, 2 x 25mm shelves	...	No	650.00
SCREENS			
Powdercoated aluminium frames, frabic or melamine panels :			
Medium range	...	sqm	130.00
High standard	...	sqm	250.00
ACOUSTIC			
Acoustic fabric 3mm thick	...	sqm	75.00
Acoustic self-adhesive tile 6-10mm thick	...	sqm	125.00
Acoustic ceiling sheet 50mm thick	...	sqm	150.00
Suspended ceiling "Lattice Clouds" 2400 x 1200 x 300mm high	...	No	1,250.00

MISCELLANEOUS

SEATING

		UNIT	Adel. Bris. Melb. Per. Syd. $
Prices are based on 100+ units	...	Note	
AUDITORIUM			
Cinema seating and theatre seating (live theatre)	...	No	335.00-440.00
Concert seating (performing arts complexes, etc.)	...	No	335.00-790.00
Lecture hall seating	...	No	170.00-330.00
ADD extra for :			
Seat numbering	...	No	3.00
Aisle lighting	...	Point	45.00
Writing tablet	...	No	125.00-170.00
LECTURE THEATRES			
Fixed continuous seat and writing tablet in moulded polypropylene and steel framing :			
Upholstered and covered with vinyl fabric (basic)	...	Per Seat	215.00-325.00
Fully upholstered and covered with wool fabric (De luxe)	...	Per Seat	495.00
STADIA			
In moulded polypropylene :			
Fixed one piece shell with back	...	No	35.00-55.00
Fixed seat with back (average)	...	No	55.00-65.00
Tip-up seat with fixed back (average)	...	No	75.00-150.00
In anodised aluminium :			
Fixed continuous bench seat	...	m	95.00
Ditto with backrest	...	m	130.00

PARKING EQUIPMENT

Boom gate, single lane, two way	...	Each	3,500.00
Control systems :			
Single car reader	...	Each	380.00
PC based reader system	...	Each	4,500.00
Ticket dispenser	...	Each	8,000.00
Fee terminal	...	Each	8,000.00
Bar code car park control systems :			
Ticket dispenser	...	Each	7,000.00
Fee terminal	...	Each	8,000.00
Magnetic stripe car park control systems :			
Ticket dispenser/permanents reader	...	Each	8,000.00
Fee terminal	...	Each	8,000.00
Management computer	...	Each	35,000.00
Pay and display ticket machines	...	Each	11,000.00
Traffic infringement enforcement, handheld computers	...	Each	3,000.00
Automatic payment station	...	Each	36,000.00

MISCELLANEOUS

CAR PARKING SYSTEMS

		UNIT	Adel. Bris. Melb. Per. Syd. $
ADD extra to all figures for :			
Building works, i.e. slabs, enclosure walls, roof, etc.	...	Note	
Plumbing and drainage services	...	Note	
Electrical services	...	Note	
LATERAL SYSTEM			
Automatic lateral sliding platforms	...	Note	
Spaces provided :			
Ten	...	Space	11,000.00
Twenty and over	...	Space	10,500.00
LONGITUDINAL SYSTEM			
Automatic longitudinal sliding platforms (to augment fixed parking)	...	Note	
Spaces provided :			
Up to ten single units (each providing one space)	...	Space	1,750.00
Up to five double units (each providing two spaces)	...	Space	12,000.00
LIFTING SYSTEM			
Automatic lifting system	...	Note	
Dependant lifting, i.e. move one to access the other, single phase power :			
Single unit providing two spaces out of one on plan area (one above the other) units provided -			
One to nine	...	Unit	9,750.00
Ten +	...	Unit	8,600.00
Independent lifting, three phase power :			
Single unit providing two spaces out of one on plan area (one above the other) units provided -			
One to nine	...	Unit	6,000.00-14,000.00
Ten +	...	Unit	6,000.00-14,000.00
ADD extra for :			
Sliding or roller doors to each unit	...	Note	
VERTICAL STACK SYSTEM			
Automatic vertical parking system (cost range)	...	Space	38,000.00-80,000.00
ADD extra for :			
Builders work in pits, electrical services and drainage	...	Note	
TURNTABLES			
Refer page 702	...	Note	

PARKING SPACE PROTECTORS

Parking space protector :			
Manual type	...	Each	285.00
Automatic type	...	Each	830.00

MISCELLANEOUS

THEATRE, CINEMAS

			UNIT	Adel. Bris. Melb. Per. Syd. $
Prices are for a 100/400 seat cinema		...	Note	
Front end including curtains	(Average)	...	Item	30,000.00
Projection equipment	(Average)	...	Item	300,000.00
Sound equipment	(Average)	...	Item	300,000.00
Seats, upholstered	(Minimum)	...	Each	400.00
	(Maximum)	...	Each	6,000.00

SECURITY

DATA CABINETS

Standard data cabinet :			
Cube safe, 32 litre	...	Each	750.00
Dataguard NT, 48 litre	...	Each	1,990.00
Dataguard NT, 89 litre	...	Each	2,650.00
Data plus series, 111 litre	...	Each	8,100.00
Data plus series, 221 litre	...	Each	10,800.00
Data plus series, 556 litre	...	Each	18,000.00

SAFES

HOTEL safe :			
AIR Hotel safe -			
Volume 24 litre, including anchoring kit and securely fixed to prevent removal	...	Each	320.00
STANDING			
OMNI safe :			
Volume 15 litre	...	Each	1,150.00
Volume 43 litre	...	Each	1,700.00
Volume 103 litre	...	Each	2,150.00
Volume 236 litre	...	Each	2,700.00
OXLEY MARK 3 safe :			
Volume 60 litre	...	Each	3,700.00
Volume 179 litre	...	Each	4,000.00
Volume 345 litre	...	Each	5,900.00
EUROPA GRADE 3 safe :			
Volume 276 litre	...	Each	6,700.00
Volume 503 litre	...	Each	9,000.00
Volume 634 litre	...	Each	10,200.00

STRONGROOM DOORS

Prices by special quote only	...	Note	
Lightweight door and frame built into wall	...	Allow	5,500.00-7,000.00
Commercial standard door and frame built into wall	...	Allow	11,000.00-19,000.00
Security Doors, Refer page 381	...	Note	

MODULAR VAULTS

Prices by special quote only	...	Note	
High security modular vault comprising precast security concrete floor, wall and roof panels, size :			
2500 x 1650 x 2400mm high	...	Allow	34,000.00-43,000.00
2500 x 2075 x 2400mm high	...	Allow	38,000.00-48,000.00
2500 x 2500 x 2500mm high	...	Allow	42,000.00-55,000.00

MISCELLANEOUS

TURNTABLES

		UNIT	Adel. Bris. Melb. Per. Syd. $
Motorised turntable installed in position :			
Light duty (cars) -			
4000mm dia., 4t capacity	...	Each	18,000.00
4500mm dia., 4t capacity	...	Each	18,750.00
5000mm dia., 4t capacity	...	Each	22,000.00
Heavy duty (trucks) -			
6000mm dia., 12t capacity	...	Each	34,250.00
7800mm dia., 20t capacity	...	Each	88,750.00
10000mm dia., 30t capacity	...	Each	110,750.00
ADD extra for :			
Builders work in pits, electrical service, drainage	...	Note	

INDUSTRIAL

DOCK BUMPERS

Resilient laminated rubber dock bumper including steel end angles and fixing to concrete :			
Horizontal -			
300mm wide x 250mm high x 115mm deep	...	No	365.00
600mm wide x 250mm high x 115mm deep	...	No	440.00
600mm wide x 250mm high x 230mm deep (heavy duty)	...	No	960.00
Vertical -			
280mm wide x 500mm high x 115mm deep	...	No	510.00
ADD extra for :			
Galvanising	...	+ 15%	

DOCK LEVELLERS

Hydraulic dock leveller, 12,500kg capacity, 2820mm long x 2110mm wide, complete with angle kit to edge of concrete, dock bumper, all necessary accessories and fully installed	...	No	9,000.00
Alternative installation :			
Self contained frame cradling leveller	...	No	3,050.00

FUEL TANKS

Steel fuel tank, fully installed including excavation, sheet piling, backfilling with sand and pipework to pumps, capacity :			
27500 litre	...	No	25,000.00-35,000.00
50000 litre	...	No	42,000.00-50,000.00
Water tanks, refer pages 487 and 488	...	Note	

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $

CLEANING, WASTE DISPOSAL

DUCTED VACUUM CLEANING SYSTEM

Domestic Type System			
Ducted system, single operator :			
3 outlets	...	System	2,000.00
10 outlets	...	System	2,900.00
15 outlets	...	System	3,500.00
Commercial Type System			
Ducted system to suit up to three operators and in excess of twenty outlets	...	Outlet	1,800.00

GARBAGE COMPACTOR

Bin carousel for 240 and 360L bins			
Without compaction	...	Each	24,250.00
With compaction	...	Each	26,500.00
ADD extra for :			
Bins	...	Note	

INCINERATORS

Electrically powered central plant incinerator, for general combustible waste, fully installed, and including immediate enclosure costs, capacity :			
250kg/hour	...	Each	4,500,000.00
ADD extra for :			
Each additional 50kg/hour	...	Each	1,250,000.00

REFUSE DISPOSAL CHUTE SYSTEMS

Aluminised steel chute, 610mm dia., including sprinkler head and fire rated intake door size 457 x 457mm to each floor level (based on 3.00m floor/floor height)	...	Floor	3,350.00
Plastic LLPD chute, 550mm dia., including sprinkler head and fire rated intake size 457 x 457mm to each floor level (based on 3.00m floor/floor height)	...	Floor	2,900.00

BAGGAGE HANDLING SYSTEMS

Baggage make-up or claim conveyors :			
Make-up conveyor, sheet steel clad, painted or galvanised	...	m	3,600.00
Passenger claim conveyor, stainless steel clad	...	m	4,200.00
Ticket counter check-in :			
Stainless steel clad baggage check-in weigh scale x 1500mm long and induction belt conveyor x 1200mm long	...	Each	35,000.00
Behind ticket counter exposed to public conveyors :			
Baggage take away straight belt conveyor, stainless steel clad	...	m	5,500.00
Ditto 90° flat belt curve	...	Each	38,000.00
Transportation conveyors, behind draught curtain and not exposed to public :			
Sheet steel clad with painted or galvanised finish -			
Straight belt, level	...	m	3,900.00
Straight belt, inclined	...	m	4,100.00
90° flat belt, curve	...	Each	38,000.00

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $
DOCUMENT TRANSPORT SYSTEMS			
DOCUMENT LIFT			
Document lift with 10kg payload, serving two stations 4.50m apart	...	Sum	12,000.00
PNEUMATIC TUBE SYSTEM			
Single tube point to point system, 85mm diameter for small documents, serving two stations not exceeding 40.0 metres apart, fully installed, (Average)	...	Sum	16,000.00
Automatic system, 85mm diameter tube for small documents, 40.0m overall length and serving three stations, fully installed	...	Sum	47,000.00
COOL ROOMS			
Cool room with walls and roof constructed of foam insulation clad both sides with Colorbond® Steel sheet, insulated floor and complete with door, cooling equipment, internal size :			
3000 x 3000 x 2400mm high, (Average)	...	Each	17,000.00
3000 x 5000 x 2400mm high, (Average)	...	Each	27,000.00
KILN			
Kiln, on and including steel frame, 3 phase power, ambient to 1280°C with fully programmable digital controller :			
130 litre capacity	...	Each	8,500.00
220 litre capacity	...	Each	9,500.00
340 litre capacity	...	Each	11,000.00
Electrical connection by others	...	Note	
SAUNAS			
Sauna bath including heater and controls :			
Domestic freestanding -			
1800 x 1200 x 1900mm high, 4 person, infrared (Average)	...	Each	3,000.00
Built-in -			
1800 x 1500 x 2000mm high (4.4kW heater) (Average)	...	Each	8,000.00
2000 x 2000 x 2000mm high (8.0kW heater) (Average)	...	Each	11,000.00
2400 x 1800 x 2100mm high (8.75kW heater) (Average)	...	Each	12,000.00
Enclosure walls by others	...	Note	
SPA POOLS			
Spa pool including installation, plumbing, electrical connection, air supply, pumps, heating and filters :			
1700 dia. x 700mm deep pool (Average)	...	Each	10,000.00
2600 x 2300 x 870mm deep pool (Average)	...	Each	12,500.00
Enclosure walls by others	...	Note	
Spa baths, refer page 458	...	Note	

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $
WAVE POOLS			
Pneumatic system for wave generation to suit pool :			
10.0/15.0m wide, 100kW, 2 fan	...	Each	320,000.00-350,000.00
16.0/21.0m wide, 165kW, 3 fan	...	Each	370,000.00-400,000.00
22.0/25.0m wide, 220kW, 4 fan	...	Each	420,000.00-450,000.00
ADD extra for :			
Pool construction and all associated works, filtration, etc.	...	Note	
DAM AND POND LINERS			
Synthetic rubber liner laid on prepared surface including site vulcanised or heat welded joints :			
EPDM liner 1.14mm thick in panels up to 30 x 15 metres	...	sqm	25.00
Butylclad liner in panels up to 30 x 30 metres -			
1.00mm thick	...	sqm	31.50
1.50mm thick	...	sqm	35.50
ADD extra for :			
Excavation and preparation of surface to receive liner	...	Note	
All other siteworks, containment walls, landscaping, etc.	...	Note	
GOLF COURSES			
Prices are for an 18 hole course and include : Light to moderate earthworks as described; Shaping and trimming; Construction of fairways, tees, greens and bunkers; Grassing, fertilising, watering and maintaining for twelve months up to the course opening; Post construction subsoil drainage, survey fees and construction management of the works	...	Note	
Land costs not included	...	Note	
Good quality club course on lightly contoured land, light to moderate earthworks (area required 50-60 hectares) (Cost range)	...	per course	4,700,000.00-8,200,000.00
Championship course on lightly contoured land, light to moderate earthworks (area required 60-75 hectares) :			
Average standard, (Cost range)	...	per course	10,700,000.00-15,500,000.00
Premium standard, (Cost range)	...	per course	14,000,000.00-25,000,000.00
ADD extra for :			
Irrigation, underground, fully automatic -			
Good quality club course (Cost range)	...	per course	850,000.00-1,450,000.00
Championship course average standard (Cost range)	...	per course	1,400,000.00-1,650,000.00
Championship course premium standard (Cost range)	...	per course	1,550,000.00-2,850,000.00
Synthetic turf greens, tees and walkways refer page 234	...	Note	
Course landscaping -			
Good quality course (Cost range)	...	per course	320,000.00-575,000.00
Championship course average standard (Cost range)	...	per course	750,000.00-1,300,000.00
Championship course premium standard (Cost range)	...	per course	800,000.00-2,050,000.00
Greenkeeping plant and equipment, course furniture, maintenance compound etc. -			
Good quality course (Cost range)	...	per course	540,000.00-780,000.00
Championship course average standard (Cost range)	...	per course	700,000.00-1,000,000.00
Championship course premium standard (Cost range)	...	per course	1,100,000.00-1,500,000.00

MISCELLANEOUS

		UNIT	Adel. Bris. Melb. Per. Syd. $
GOLF COURSES (cont'd)			
ADD extra for (cont'd) :			
Location factor where applicable	...	Note	
Clubhouse, ancillary buildings, plant maintenance buildings and surrounding areas	...	Note	
Parking areas	...	Note	
Services and access roads to the site of the course	...	Note	
Major earthworks and imported topsoil	...	Note	
OVALS			
Sporting oval on lightly contoured land, light to moderate earthworks, grassing, fertilising, water and maintenance for twelve months, irrigation and floodlighting	...	Each	1,050,000.00-1,500,000.00
Clubhouse, ancillary buildings, plant maintenance buildings and surrounding areas	...	Note	
Parking areas	...	Note	
IRRIGATION AND RETICULATION			
RETICULATION			
The following prices include only for reticulation piping and sprinkler heads in position in the ground, bores and pumps are excluded :	...	Note	
Playing Fields (Average)	...	Ha	11,750.00-14,250.00
Golf Courses-Refer page 705	...	Note	
Residential (Average)	...	1000sqm	2,500.00-4,200.00
PUMPS AND BORES			
The price of pumps and bores varies according to the volume of water required, depth of water table, type of soil. For instance the price of pumps and bores for a playing field may vary between $10,000 and $60,000.	...	Note	
DRILLING			
The following prices include for drilling in sand, clay or gravel but exclude drilling in rock.	...	Note	
Drilling, including installing PVC casing :			
100mm dia. -			
0.00/45.00m	...	m	68.00
Over 45.00m	...	m	77.50
125mm dia. -			
0.00/45.00m	...	m	84.00
Over 45.00m	...	m	97.00
3000mm stainless steel screen at bottom of bore, 100mm or 125mm dia.	...	No	435.00
Bore development :			
Allow at $50.00/hour (usually 3 to 4 hours, minimum charge $200.00)	...	Note	

MISCELLANEOUS

TRANSPORTABLE BUILDINGS

		UNIT	Melb. Syd. $	Adel. Bris. Per. $
Prices are for buildings constructed generally of steel framing and supported on stumps, external walls insulated and clad with painted Hardiplank or Zincalume® Steel decking, aluminium sliding windows; low pitch roof lined with zinc-alumed decking and insulated; finishes comprising floors of particle board panels and finished with tiles in wet areas, walls and ceilings of painted plasterboard sheeting; hollow core doors in metal frames; electric light and power installation of no more than average standard and, where applicable, sanitary plumbing and fixtures together with hot water system.	...	Note		
Prices include for erection on a flat site with reasonable ground conditions, NO allowance is included for transportation costs.	...	Note		
Refer to the end of this section for ADDITIONAL costs in respect of erection and other works	...	Note		

EDUCATIONAL

Prices include for construction to government specifications, lockable store or similar to each classroom, carpet and vinyl floor finishes, chalk and pin-up boards, store shelving :				
No allowance for furniture	...	Note		
Classroom, approximately 9.60 x 7.20m	...	sqm	1,210.00- 1,330.00	1,145.00- 1,255.00
Home economics, approximately 13.00 x 7.20m including benches, sinks and stoves	...	sqm	1,825.00- 1,975.00	1,705.00- 1,850.00
Manual arts including work benches	...	sqm	1,285.00- 1,420.00	1,225.00- 1,350.00
Pre-primary, approximately 14.00 x 7.20m including w.c.'s and basins	...	sqm	1,580.00- 1,740.00	1,480.00- 1,695.00
Science laboratory 13.00 x 7.20m including work benches and sinks	...	sqm	1,680.00- 1,820.00	1,590.00- 1,725.00
Toilet block approximately 10.00 x 4.20m with six w.c.'s and six basins	...	sqm	2,590.00- 2,840.00	2,440.00- 2,660.00

MOTEL UNITS

14.40 x 4.20m section comprising three units each of 19sqm and containing twin bedroom and bathroom, including carpet, bench cupboard unit, case rack, and air conditioner. No allowance for refrigerator or furniture and fit-out	...	sqm	1,565.00- 1,715.00	1,450.00- 1,600.00
	...	Unit	32,900.00- 36,000.00	30,500.00- 33,600.00

MISCELLANEOUS

TRANSPORTABLE BUILDINGS (cont'd)

		UNIT	Melb. Syd. $	Adel. Bris. Per. $
OFFICES				
14.00 x 3.60m section subdivided into four rooms. No allowance for floor coverings, plumbing or air conditioning	...	sqm	825.00-925.00	770.00-865.00
RESIDENTIAL				
Basic standard house :				
70-100 sqm	...	sqm	985.00-1,090.00	900.00-1,000.00
110-130 sqm	...	sqm	970.00-1,065.00	880.00-980.00
Medium standard house :				
70-100 sqm	...	sqm	1,100.00-1,450.00	1,050.00-1,350.00
110-130 sqm	...	sqm	1,090.00-1,430.00	1,030.00-1,320.00
ADD extra for :				
Log Cabin style	...	sqm	120.00-135.00	120.00-135.00
ADDITIONAL COSTS				
ADD extra for :				
Cartage from capital city including -				
Transporation	...	Note		
Pilot car	...	Note		
Police escort	...	Note		
Overtime	...	Note		
Permits	...	Note		
Special allowances	...	Note		
Travelling time	...	Note		
The foregoing costs can vary appreciably depending upon location and particular state requirements, obtain advice from a transporter in respect of permissible loads and approximate costs of cartage	...	Note		
Site clearance and levelling	...	Note		
Raising to 2.40m above ground	...	sqm	52.00-62.00	50.50-60.50
Reinforced and prestressed concrete slab, excluding cartage	...	sqm	90.00-100.00	80.00-90.00
Rat walls	...	Note		
Prepainted finish to metal wall and roof cladding (sqm/surface area)	...	sqm	12.50	12.50
Stairs, external x 1.00m wide, steel framed with balustrade	...	M/rise	1,330.00	1,300.00
Cyclone conditions, strengthening of all framing and lining walls with plywood in addition (plan area)	...	sqm	90.00-110.00	89.00-109.00
Balconies/verandahs including site erection of component parts -				
Deck only	...	sqm	165.00-210.00	145.00-195.00
Deck with roof over	...	sqm	450.00-540.00	400.00-530.00
ADD extra for :				
Transportation of foregoing	...	Note		

MISCELLANEOUS

TRANSPORTABLE BUILDINGS (cont'd)

ADDITIONAL COSTS (cont'd)

		UNIT	Melb. Syd. $	Adel. Bris. Per. $
ADD extra for (cont'd) :				
External services outside building, refer costs given elsewhere, ADD country loading -				
Water	...	Note		
Septic	...	Note		
Stormwater	...	Note		
Electrical	...	Note		
Floor finishes other than to wet areas and unless stated as included -				
Vinyl tiles	...	sqm	22.00-35.00	22.00-33.00
Carpet (average)	...	sqm	30.00-56.00	30.00-55.00
Ceramic tiles	...	sqm	105.00-135.00	105.00-140.00
Air-conditioning, refer pages 609 and 610	...	Note		
Ceiling fans, refer page 586	...	Note		
Curtains, refer pages 384 to 386	...	Note		
Domestic appliances, refer pages 681 and 682	...	Note		
Furniture and furnishings	...	Note		
Building to government specification, approximate allowance to be added to all costs other than Educational	...	+10-15%		

CIVIL ENGINEERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are based on large quantities for work in typical conditions of the respective cities and within 100 kilometres	...	Note					
Prices do not include allowance for Preliminaries	...	Note					

SITE PREPARATION

Refer page 211	...	Note					

EXCAVATION

TOP SOIL

Excavate to remove top soil average 150mm deep and deposit in spoil heaps within 500m, in :							
Sand	...	cum	4.45	5.30	4.70	4.85	5.20
Light soil	...	cum	4.70	5.60	5.00	5.50	5.75
Clay	...	cum	6.80	8.55	7.20	8.10	9.15
Excavate top soil from spoil heap, cart not exceeding 500m and spread and level average 150mm thick :							
Level areas in -							
Sand	...	cum	4.45	5.80	5.15	5.35	5.60
Light soil	...	cum	5.05	6.40	6.25	5.85	6.15
Clay	...	cum	7.40	9.75	9.25	8.70	9.45
Battered areas in -							
Sand	...	cum	5.45	6.55	6.25	6.00	6.15
Light soil	...	cum	6.30	7.40	6.90	6.75	7.10
Clay	...	cum	8.90	10.80	10.55	10.00	10.50

BULK

Excavate to reduce levels and deposit in spoil heaps within 1km, in :							
Sand	...	cum	4.45	4.65	4.20	4.80	5.45
Light soil	...	cum	5.45	5.30	5.00	5.30	5.60
Clay	...	cum	5.20	8.20	7.25	7.95	8.30
ADD extra for :							
Cartage, per additional km	...	cum	0.58	0.55	0.63	0.53	0.56
Excavate to reduce levels and deposit, spread and level within 1km, in :							
Sand	...	cum	6.45	7.40	7.00	7.20	8.10
Light soil	...	cum	7.50	8.25	8.40	7.70	9.00
Clay	...	cum	8.95	10.45	10.75	10.55	11.40
ADD extra for :							
Cartage, per additional km	...	cum	0.58	0.55	0.63	0.53	0.56
Compaction to 90%	...	cum	3.85	4.45	4.00	4.40	4.45
Excavate from borrow pit and deposit as fill within 1km including compaction to 90% in :							
Sand	...	cum	7.80	9.75	9.05	9.45	9.55
Light soil	...	cum	8.80	10.35	10.15	10.35	10.45
Clay	...	cum	11.20	13.75	13.35	13.50	13.95

CIVIL ENGINEERING

EXCAVATION (cont'd)

FOUNDATIONS

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Excavate at river bed level for abutment, retaining wall footing or the like, including maintaining sides, working space allowance, keeping free of water :							
N/E 1.50m deep -							
Sand or silt	...	cum	69.00	75.70	72.40	70.60	78.80
Clay	...	cum	91.60	103.50	95.80	98.80	100.70
ADD extra for :							
Excavation around driven piles	...	cum	46.90	49.30	48.80	49.40	51.00
Excavate for abutment, retaining wall footing or the like, including maintaining sides, working space allowance :							
N/E 1.5m deep in -							
Sand	...	cum	28.40	28.10	29.10	28.80	32.20
Light soil	...	cum	29.50	38.00	30.30	31.90	33.40
Clay	...	cum	39.90	43.60	43.00	43.10	47.20
1.5/3.0m deep in -							
Sand	...	cum	31.40	32.80	34.10	31.90	34.90
Light soil	...	cum	32.70	34.40	35.80	33.40	36.10
Clay	...	cum	43.60	50.80	47.80	46.90	50.00

ROCK

ADD extra to foregoing excavation prices for :							
Ripping out rock in -							
Reduced level excavation	...	cum	17.30	17.70	22.50	20.60	22.50
Footing excavation	...	cum	27.00	26.70	31.40	30.40	30.70
Blasting out rock in -							
Reduced level excavation	...	cum	60.50	52.20	56.00	45.20	61.90
Footing excavation	...	cum	96.10	104.00	139.90	46.60	131.90

DISPOSAL

ADD extra to the foregoing prices for :							
Tip fees, refer page 213 for a guide to tip fees	...	Note					
Cartage to tip, per km	...	cum	0.58	0.55	0.63	0.53	0.56

CIVIL ENGINEERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DREDGING							
Prices are for dredging to form marinas (NOT DEEP WATER HARBOURS) and assume quantities in the vicinity of 75,000/100,000 cum	...	Note					
Dredge in sand or silt and deposit on adjoining land (base cost includes transporting dredge to works not exceeding 100km, setting up, dismantling and removal)	...	Base Cost	25000.00-57500.00	25000.00-57500.00	25000.00-57500.00	25000.00-57500.00	25000.00-57500.00
(@ 350 cum/hr)	...	+/cum	3.30-4.95	3.40-5.10	3.35-5.00	3.30-4.95	3.50-5.15
Excavate dredged material stockpile, cart not exceeding 500m and spread and level on site including compaction	...	cum	2.85	3.60	3.10	2.95	3.15
FOUNDATION IMPROVEMENT							
Excavate below footing level and stockpile on site within 500m, in :							
Sand	...	cum	29.20	23.70	31.10	29.10	32.50
Light soil	...	cum	30.40	25.00	32.30	32.20	34.30
Clay	...	cum	42.10	33.10	45.80	44.20	46.90
Excavate from stockpile within 500m, backfill and compact to 90% in layers not exceeding 200mm below footings, in :							
Sand	...	cum	10.35	11.10	11.65	11.40	11.15
Light soil	...	cum	11.00	12.20	12.40	11.55	11.90
Clay	...	cum	15.60	16.50	15.90	16.35	17.35
SURFACE TREATMENTS							
Level and grade subgrade to receive sub-base	...	sqm	1.55	2.15	2.75	2.00	2.75
Trim excavation to batter	...	sqm	2.00	2.15	3.45	2.00	3.05
Trim filling to batter	...	sqm	2.00	2.15	3.45	2.00	3.05
FILLING							
Imported sand filling, carted not exceeding 10km, deposited, spread and levelled and compacted to 90%	...	cum	59.80	59.00	50.00	22.50	63.00
Decomposed granite filling, ditto	...	cum	-	73.00	71.00	-	-
ADD extra for :							
Cartage, per additional 5km	...	cum	3.15	3.00	3.50	2.95	3.10
Filling from borrow pits refer page 710	...	Note					
PITCHING							
Pitching to embankments 150mm to 200mm thick, embedded in mortar :							
Granite	...	sqm	76.30	78.40	82.30	74.50	82.20
Limestone	...	sqm	-	-	-	95.00	-

CIVIL ENGINEERING

EROSION AND SEDIMENT CONTROL

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Silt fence comprising geotextile fabric 500mm high above ground and 300mm girth in trench, fixed to and including steel pickets at 3000mm centres, excavation and backfilling of trench, removal and reinstatement on completion	…	m	15.00-17.00	15.00-17.00	15.00-17.00	15.00-17.00	15.00-17.00
Jute mesh comprising open-weave biodegradable blanket made of jute. Temporary erosion control where flows are not concentrated	…	sqm	0.55-0.85	0.55-0.85	0.55-0.85	0.55-0.85	0.55-0.85
Jute mat comprising densely packed jute fibres forming a biodegradable erosion control mat suitable for use in areas of concentrated water flow	…	sqm	1.15-4.10	1.15-4.10	1.15-4.10	1.15-4.10	1.15-4.10
Erosion control log constructed from compressed, durable, organic materials such as straw, coir and jute. Up to 300mm in diameter, they can be used as soil stabilisers through placement along the contour to interrupt surface water flow	…	m	14.50-31.00	14.50-31.00	14.50-31.00	14.50-31.00	14.50-31.00
Silt sausage comprising tubes of geotextile fabric designed to be filled with sand to form a mouldable barrier suitable for use as check dams or in front of drop inlet structures	…	m	3.50-4.30	3.50-4.30	3.50-4.30	3.50-4.30	3.50-4.30
Cellular confinement sytem comprising interconnected cells designed to prevent soil movement on steep slopes and batters, whilst allowing free growth of vegetation	…	sqm	7.00-11.30	7.00-11.30	7.00-11.30	7.00-11.30	7.00-11.30
Floating silt curtain comprising geotextile fabric attached to floats and weights and extending from the floor to the surface of a water body, used to trap sediments	…	m	50.00-70.00	50.00-70.00	50.00-70.00	50.00-70.00	50.00-70.00

REVETMENT WALLS

Revetment walls to foreshore embedded in mortar, average 450mm thick laid dry :							
Granite	...	sqm	111.50	120.50	125.50	123.50	125.50
Limestone	...	sqm	-	-	-	152.50	-

CIVIL ENGINEERING

RETAINING WALLS ETC.

REINFORCED EARTH STRUCTURES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ADD extra to each of the following for :							
Selected backfill compacted in maximum 375mm lifts, pro-rata to filling on pages 710 and 712	...	Note					
Reinforced earth structure with the facing element as described hereafter, including soil, reinforcing strips and all necessary labour, accessories, hoisting, bracing and propping :							
Concrete masonry blocks with split face finish -							
Up to 3.00m high	...	sqm	490.00	475.00	510.00	460.00	475.00
3.00 to 6.00m high	...	sqm	515.00	500.00	540.00	490.00	500.00
Exceeding 6.00m high	...	sqm	535.00	515.00	555.00	510.00	515.00
ADD extra for :							
Backfill as above	...	cum	53.50	51.50	49.50	50.00	55.00
Drainage zone of 10mm aggregate 300mm deep behind blocks	...	Note					
Special surface finish to blocks	...	Note					
Precast concrete wall panels with a smooth off-form finish -							
Up to 3.00m high	...	sqm	485.00	480.00	515.00	495.00	535.00
3.00 to 6.00m high	...	sqm	530.00	520.00	555.00	535.00	585.00
Exceeding 6.00m high	...	sqm	560.00	555.00	590.00	575.00	630.00
ADD extra for :							
Backfill as above	...	cum	53.50	51.50	49.50	50.00	55.00
Special surface finish to panels	...	Note					
Prefabricated steel facing panels -							
Up to 3.00m high	...	sqm	535.00	550.00	585.00	565.00	545.00
3.00 to 6.00m high	...	sqm	585.00	605.00	640.00	620.00	595.00
Exceeding 6.00m high	...	sqm	630.00	645.00	685.00	665.00	640.00
ADD extra for :							
Backfill as above	...	cum	53.50	51.50	49.50	50.00	55.00
Hot dipped galvanised finish	...	Note					
Prefabricated steel wire (Terratrel) facing panels with hot dipped galvanised finish -							
Up to 3.00m high	...	sqm	450.00	475.00	470.00	465.00	470.00
3.00 to 6.00m high	...	sqm	475.00	505.00	495.00	495.00	495.00
Exceeding 6.00m high	...	sqm	525.00	555.00	545.00	545.00	545.00
ADD extra for :							
Backfill as above	...	cum	53.50	51.50	49.50	50.00	55.00
Gabion stone to a nominal depth of 600mm behind facing panel	...	Note					
Reduce price for temporary structures (less than 5 years) by using ungalvanised wire and reinforcing strips and no gabion stone	...	Note					

DIAPHRAGM WALLS

Refer page 216	...	Note					

CIVIL ENGINEERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
RETAINING WALLS ETC. (cont'd)							
SECANT PILE WALLS							
Refer page 216	...	Note					
RETAINING WALLS GENERALLY							
Brick, block, etc., refer pages 168 and 169	...	Note					
ROOT BARRIER							
PVC root barrier placed vertically in trench, barrier width :							
600mm	...	m	11.75	12.65	12.75	12.60	14.25
900mm	...	m	15.70	16.70	16.85	16.65	18.50
1200mm	...	m	19.65	20.80	21.00	20.70	22.80
ADD extra for :							
Trench excavation and backfilling, refer pages 489 and 490	...	Note					
STABILISATION							
Filter fabric, non-woven geotextile fabric with overlapped joints laid on sub-grade/sub-base :							
Polypropylene/Polyethylene -							
1.5mm (115g/sqm)	...	sqm	3.30	3.65	3.70	3.65	4.25
1.9mm (190g/sqm)	...	sqm	3.65	4.00	4.05	4.00	4.60
2.3mm (270g/sqm)	...	sqm	4.90	5.35	5.45	5.35	6.15
2.8mm (320g/sqm)	...	sqm	5.70	6.15	6.20	6.15	6.95
4.4mm (510g/sqm)	...	sqm	7.70	8.30	8.35	8.25	9.30
General purpose mat, high density polyethylene, laid on embankment, mesh size :							
6 x 6mm	...	sqm	14.95	15.55	15.60	15.50	16.50
12 x 12mm	...	sqm	12.70	13.30	13.35	13.25	14.30
20 x 20mm	...	sqm	13.90	14.50	14.55	14.45	15.50
26 x 26mm	...	sqm	13.30	13.85	13.95	13.85	14.85
40 x 60mm	...	sqm	9.80	10.35	10.45	10.30	11.35
Geotextile envelope comprising high strength rigid mattress with pumped grout fill, laid on embankment :							
Filter point fabric -							
125mm F.P. (centres)	...	sqm	58.40	58.40	58.40	58.40	58.40
200mm F.P. (centres)	...	sqm	66.00	66.00	66.00	66.00	66.00
Mat comprising layer of coir fibre between two layers of polymer mesh :							
8 to 10mm thick (medium)	...	sqm	6.50	7.10	7.15	7.05	8.10
15 to 20mm thick (heavy)	...	sqm	7.20	7.80	7.85	7.75	8.80
ADD extra for :							
Anchor trench 300mm wide x 250mm deep, including excavation, backfilling and pinning edge of matting	...	m	7.60	5.20	6.80	4.20	5.75
Uniform cross section impermeable mat	...	sqm	70.90	70.90	70.90	70.90	70.90
Hydro mulch, sprayed grass seed compound	...	Ha	3350.00	3350.00	3400.00	3400.00	3350.00

CIVIL ENGINEERING

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SOILING AND GRASSING							
Refer pages 228	...	Note					
ROADWORKS							
BASE COURSE							
Prices include grading, rolling and compaction	...	Note					
Yellow sand :							
150mm thick	...	sqm	-	-	9.40	5.70	-
300mm thick	...	sqm	-	-	16.70	8.50	-
450mm thick	...	sqm	-	-	24.40	9.85	-
Crushed limestone :							
150mm thick	...	sqm	-	-	-	8.35	-
300mm thick	...	sqm	-	-	-	18.95	-
Crushed rock/blue metal :							
100mm thick	...	sqm	8.85	7.35	10.25	8.10	12.05
200mm thick	...	sqm	20.40	13.35	17.55	13.45	20.10
300mm thick	...	sqm	29.70	20.20	26.40	19.75	29.30
Bitumen stabilised crushed limestone :							
75mm thick	...	sqm	-	-	-	9.80	-
100mm thick	...	sqm	-	-	-	11.95	-
Lean concrete (18:1) :							
100mm thick	...	sqm	22.40	16.85	15.20	-	14.55
150mm thick	...	sqm	32.80	24.80	22.90	-	24.80
Local laterite (country locations) :							
175mm thick	...	sqm	-	-	-	19.50	22.10
PAVING ETC.							
Prime and sprayed bitumen sealing :							
Two coats	...	sqm	15.20	9.10	8.50	14.50	7.90
Three coats	...	sqm	17.20	13.70	12.85	18.20	12.10
Hot bituminous concrete including tack coat :							
25mm thick	...	sqm	19.70	17.25	15.00	11.65	16.60
40mm thick	...	sqm	23.00	24.40	18.50	18.00	23.50
50mm thick	...	sqm	23.00	29.30	23.80	25.30	27.00
ADD extra for :							
Prime coat of emulsion and crushed rock/blue metal chips	...	sqm	5.10	5.30	4.15	3.70	4.30
SALVIACIM oil and chemical resistant wearing course 40mm thick :							
In areas not less than 1000sqm	...	sqm	-	-	-	46.00	-
In areas not less than 4000sqm	...	sqm	-	-	-	41.20	-
ADD extra for :							
Smaller quantities	...	Note					
Work outside metropolitan area	...	Note					

CIVIL ENGINEERING

ROADWORKS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
PAVING ETC. (cont'd)							
Concrete (32MPa), unreinforced :							
150mm thick	...	sqm	40.10	35.50	45.90	42.50	54.50
200mm thick	...	sqm	53.50	47.30	61.20	56.70	72.70
ADD extra for :							
Fabric reinforcement, refer page 249	...	Note					
Keyed and filled joint to 150mm thick concrete	...	m	30.90	34.70	31.20	33.30	31.70
Dowelled expansion joint to 150mm thick concrete	...	m	50.10	49.40	53.10	51.40	54.20
Kerbs - refer page 224	...	Note					
Marking - refer page 224	...	Note					
REPROFILING							
Reprofiling top only of bituminous concrete paving including taking up and putting back x 30mm thick including top up (based on 2000sqm)	...	sqm	22.50-25.50	23.00-26.00	24.00-29.00	20.00-22.00	27.50-31.50
ADD extra for :							
Smaller quantities	...	+25/50%					
Night work	...	+25/30%					
CRASH AND GUARD RAILS							
Refer page 304	...	Note					
NOISE BARRIERS							
Prices are an approximate indication only, factors that can have an appreciable effect on the price include; performance criteria, height of barrier, topography, location, etc. :	...	Note					
Aerated concrete 100mm thick panels up to 4.0m long, planks laid horizontally between UB columns -							
125mm thick	...	sqm	324.00	332.00	324.00	319.00	293.00
150mm thick	...	sqm	351.00	358.00	351.00	345.00	317.00
ADD extra to aerated panels for :							
Supporting steelwork	...	Note					
Panel finishes	...	Note					
PROTECTION SAFETY BARRIERS							
Precast concrete shaped barrier placed in position :							
770mm high	...	m	300.00	335.00	345.00	330.00	350.00
970mm high	...	m	355.00	385.00	395.00	380.00	395.00
Wire rope safety barrier fully installed	...	m	50.00-100.00	50.00-100.00	50.00-100.00	50.00-100.00	50.00-100.00

CIVIL ENGINEERING

ROADWORKS (cont'd)

CULVERTS, PRECAST CONCRETE

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Add for trench excavation and backfilling	...	Note					
Smaller sized culverts, refer page 224	...	Note					
Precast concrete box culvert laid on ground including minimal excavation size :							
1200 x 900mm	...	m	1030.00	1555.00	900.00	1120.00	1385.00
1200 x 1200mm	...	m	1095.00	1905.00	1015.00	1490.00	1575.00
Precast concrete box culvert laid on ground excluding excavation :							
4000 x 3115mm	...	m	6420.00	9910.00	6250.00	6800.00	7590.00
Precast concrete pipe Class 2, butt and interlock joint in culvert laid on ground including minimal excavation, pipe size :							
300mm dia.	...	m	120.00	-	124.50	142.50	162.50
450mm dia.	...	m	186.50	238.00	175.00	225.00	253.00
600mm dia.	...	m	242.00	310.00	212.00	300.00	331.00
750mm dia.	...	m	405.00	515.00	361.00	490.00	549.00
1200mm dia.	...	m	830.00	1120.00	680.00	974.00	1080.00
1500mm dia.	...	m	1175.00	1610.00	957.00	1560.00	1725.00
2100mm dia.	...	m	-	-	-	-	-
Precast concrete headwall laid on ground and including additional excavation for toe, to suit culvert size :							
300mm dia.	...	No	615.00	478.00	507.00	626.00	790.00
450mm dia.	...	No	615.00	572.00	519.00	626.00	790.00
600mm dia.	...	No	826.00	776.00	686.00	834.00	1055.00
750mm dia.	...	No	1190.00	1195.00	868.00	1120.00	1410.00
Headwall to suit culvert, in reinforced concrete :							
Foundation including excavation	...	cum	548.00	460.00	497.00	447.00	573.00
200mm thick wall including formwork	...	sqm	426.00	409.00	449.00	355.00	476.00

CULVERTS, PIPE ARCHES, UNDERPASSES - STEEL

CULVERTS

ADD for trench excavation and backfilling	...	Note					
Prices for helically lock seamed pipes include average delivery cost from either Perth, Sydney or Townsville, prices given can vary depending upon the mix of sizes ordered	...	Note					
Corrugated galvanised steel pipes, helically lock seamed and laid on ground, pipe size :							
300mm dia. x 15 kg/m	...	m	83.70	86.30	86.90	80.70	87.80
375mm dia. x 19 kg/m	...	m	109.50	114.50	113.00	105.50	115.50
450mm dia. x 22 kg/m	...	m	127.50	132.50	131.50	122.00	133.50
600mm dia. x 30 kg/m	...	m	165.00	170.50	169.00	157.00	170.00
750mm dia. x 37 kg/m	...	m	207.00	214.00	212.00	197.00	214.00
900mm dia. x 44 kg/m	...	m	267.00	280.00	278.00	260.00	287.00
1050mm dia. x 51 kg/m	...	m	317.00	329.00	327.00	305.00	333.00
1200mm dia. x 58 kg/m	...	m	357.00	370.00	368.00	342.00	372.00
1350mm dia. x 80 kg/m	...	m	450.00	463.00	460.00	427.00	460.00
1500mm dia. x 89 kg/m	...	m	500.00	514.00	511.00	474.00	510.00

CIVIL ENGINEERING

ROADWORKS (cont'd)

CULVERTS, PIPE ARCHES, UNDERPASSES - STEEL (cont'd)

CULVERTS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Corrugated galvanised steel pipes, helically lock seamed and laid on ground, pipe size (cont'd) :							
1650mm dia. x 106 kg/m	...	m	584.00	599.00	596.00	552.00	592.00
1800mm dia. x 115 kg/m	...	m	636.00	652.00	649.00	601.00	646.00
2100mm dia. x 134 kg/m	...	m	789.00	810.00	806.00	747.00	804.00
3000mm dia. x 357 kg/m	...	m	1985.00	2005.00	2000.00	1840.00	1935.00
Corrugated aluminium pipes, helically lock seamed and laid on ground, pipe size :							
300mm dia. x 5 kg/m	...	m	72.70	76.00	76.50	74.10	80.60
450mm dia. x 7 kg/m	...	m	110.50	116.50	115.50	112.00	122.00
600mm dia. x 10 kg/m	...	m	149.00	155.50	154.50	149.50	161.50
750mm dia. x 12 kg/m	...	m	189.50	198.50	197.00	191.00	207.00
900mm dia. x 14 kg/m	...	m	247.00	261.00	260.00	252.00	277.00
1050mm dia. x 17 kg/m	...	m	292.00	307.00	305.00	296.00	322.00
1200mm dia. x 23 kg/m	...	m	387.00	402.00	400.00	388.00	416.00
1350mm dia. x 29 kg/m	...	m	420.00	436.00	434.00	421.00	450.00
1500mm dia. x 32 kg/m	...	m	467.00	486.00	483.00	468.00	501.00
1650mm dia. x 35 kg/m	...	m	540.00	561.00	558.00	541.00	577.00
1800mm dia. x 38 kg/m	...	m	593.00	616.00	613.00	594.00	635.00
2100mm dia. x 57 kg/m	...	m	899.00	927.00	923.00	894.00	946.00
2250mm dia. x 62 kg/m	...	m	943.00	975.00	971.00	940.00	998.00
2400mm dia. x 91 kg/m	...	m	1275.00	1310.00	1305.00	1260.00	1325.00
Multi-plate corrugated galvanised steel pipe laid on ground, pipe size :							
1500mm dia. x 174 kg/m	...	m	1115.00	1145.00	1125.00	1150.00	1105.00
4050mm dia. x 730 kg/m	...	m	4965.00	5085.00	5005.00	5120.00	4890.00
6000mm dia. x 1093 kg/m	...	m	7475.00	7655.00	7535.00	7705.00	7360.00
8550mm dia. x 2116 kg/m	...	m	14270.00	14625.00	14395.00	14715.00	14065.00

PIPE ARCHES

Add for trench excavation and backfilling	...	Note					
Multi-plate corrugated galvanised steel arch, laid on ground, arch size :							
2000 x 890mm rise x 169 kg/m	...	m	1250.00	1275.00	1255.00	1285.00	1220.00
4000 x 1970mm rise x 361 kg/m	...	m	2535.00	2595.00	2555.00	2615.00	2495.00
6000 x 2430mm rise x 475 kg/m	...	m	3370.00	3450.00	3395.00	3475.00	3310.00
8500 x 4210mm rise x 1043 kg/m	...	m	7620.00	7780.00	7665.00	7850.00	7465.00
11660 x 7290mm rise x 1760 kg/m	...	m	12525.00	12810.00	12615.00	12915.00	12300.00

UNDERPASSES

Add for trench excavation, backfilling and headwalls	...	Note					
Multi-plate corrugated galvanised steel pipe underpass, laid on ground, size :							
3220 x 2780mm rise x 562 kg/m	...	m	3920.00	4015.00	3950.00	4045.00	3855.00
4080 x 3350mm rise x 691 kg/m	...	m	5020.00	5130.00	5050.00	5175.00	4920.00
4960 x 4000mm rise x 834 kg/m	...	m	6065.00	6195.00	6100.00	6250.00	5945.00
6010 x 4680mm rise x 997 kg/m	...	m	7220.00	7380.00	7270.00	7445.00	7080.00

CIVIL ENGINEERING

ROADWORKS (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
DRAINAGE							
Stormwater drainage, refer pages 494 to 497	...	Note					
Subsoil drainage, refer pages 497 and 499	...	Note					
TRAFFIC SIGNALS							
Set of programmed signals including fully activated detectors, electrical services between signals, control box, pedestrian push buttons and indication signs to intersection of :							
Access road with dual carriageway	...	Set	1115000.00-145000.00	115000.00-145000.00	115000.00-145000.00	115000.00-145000.00	125000.00-155000.00
ADD extra for :							
Electrical service to site of signals	...	Note					
ROAD SIGNS							
Non flexible powder coated steel sign post and reflective sign to Australian Standard 1743 and Local Road Authority Specification including concrete base :							
Sign 450 x 450mm	...	No	390.00-410.00	390.00-410.00	390.00-410.00	390.00-410.00	390.00-410.00
Sign 450 x 600mm	...	No	390.00-410.00	390.00-410.00	390.00-410.00	390.00-410.00	390.00-410.00
Sign 600 x 600mm	...	No	425.00-445.00	425.00-445.00	425.00-445.00	425.00-445.00	425.00-445.00
Sign 600mm dia.	...	No	425.00-445.00	425.00-445.00	425.00-445.00	425.00-445.00	425.00-445.00
Sign 600 x 750mm	...	No	450.00-470.00	450.00-470.00	450.00-470.00	450.00-470.00	450.00-470.00
Sign 900 x 900mm	...	No	500.00-520.00	500.00-520.00	500.00-520.00	500.00-520.00	500.00-520.00
COMPOSITE PRICES							
Prices include minimal cut and fill but EXCLUDE lighting and drainage	...	Note					
Suburban road with in situ concrete kerbs :							
6.0m wide	...	m	355.00-395.00	310.00-350.00	310.00-350.00	250.00-290.00	355.00-395.00
8.0m wide	...	m	445.00-485.00	400.00-440.00	390.00-430.00	285.00-335.00	435.00-535.00
Country highway with shoulders :							
Two lane	...	m	385.00-435.00	415.00-475.00	595.00-655.00	385.00-435.00	595.00-645.00
Three lane	...	m	485.00-555.00	615.00-700.00	880.00-950.00	550.00-620.00	870.00-940.00

CIVIL ENGINEERING

ROADWORKS (cont'd)

COMPOSITE PRICES (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
City highway/freeway with median strip and emergency lanes :							
Duplicate two lane	...	m	2150.00-2430.00	2700.00-3000.00	2350.00-2650.00	2120.00-2420.00	2510.00-2810.00
Duplicate three lane	...	m	2650.00-2950.00	2990.00-3290.00	3060.00-3360.00	2440.00-2740.00	3090.00-3490.00
ADD extra for :							
Paved footpath -							
1200mm wide	...	m	78.00	80.00	74.40	62.90	76.20
1500mm wide	...	m	90.70	93.50	86.40	75.30	89.00

BRIDGEWORKS

APPROACH WORKS

Excavation, filling, refer aforementioned prices	...	Note					
Concrete (32MPa) in footings, abutments, retaining walls and approach slabs	...	cum	286.00	263.00	329.00	307.00	394.00
Class F1 formwork to :							
Sides of footings	...	sqm	102.00	90.00	80.00	95.00	101.00
Sides of walls	...	sqm	165.00	171.00	155.00	140.00	191.00
Class F2 formwork to :							
Sides of walls	...	sqm	175.00	190.00	172.00	168.00	190.00
Edges of slabs	...	sqm	160.00	173.00	120.00	159.00	195.00
Class F3 formwork to :							
Sides of columns	...	sqm	224.00	210.00	210.00	172.00	230.00
Reinforcement :							
Deformed bar reinforcement	...	t	2690.00	2510.00	2500.00	2470.00	2490.00

BRIDGE DECKS (CONVENTIONAL)

Concrete (32MPa) in :							
Bridge deck, thickenings and beams	...	cum	286.00	263.00	329.00	307.00	394.00
Upstand beams, walls	...	cum	299.00	277.00	345.00	322.00	412.00
Class F2 formwork to :							
Soffit of slabs (excluding falsework)	...	sqm	157.00	138.00	185.00	126.00	189.00
Edge of slabs	...	sqm	179.00	199.00	172.00	189.00	205.00
Sides of walls	...	sqm	178.00	193.00	175.00	186.00	216.00
Sides and soffit of beams and thickenings	...	sqm	219.00	181.00	195.00	186.00	238.00
ADD extra for :							
Falsework to support soffit formwork (range)	...	sqm	63.00-118.00	64.00-118.00	58.00-108.00	56.00-106.00	67.00-123.00
Falsework to support soffit formwork over active thoroughfare	...	Note					

CIVIL ENGINEERING

BRIDGEWORKS (cont'd)

BRIDGE DECKS (CONVENTIONAL) (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Reinforcement :							
Deformed bar reinforcement	...	t	2690.00	2510.00	2500.00	2470.00	2490.00
Prestressing including all sundries	...	t	6200.00-7600.00	6300.00-7700.00	6200.00-7600.00	6200.00-7600.00	6500.00-7900.00
ADD extra for :							
Waterproof membrane (average)	...	sqm	34.20	33.50	32.60	25.80	32.10
Surfacing, refer page 716	...	Note					
Guardrails, refer page 304 (average)	...	m	211.00	243.00	247.00	236.00	260.00
Balustrade (average)	...	m	295.00	290.00	300.00	295.00	290.00
Lighting of bridge and piers	...	Note					
Sewerage, gas and water mains	...	Note					
Power supply	...	Note					
Strengthening Bridge Decks							
Carbon fibre reinforcement polymer strengthening strips, refer page 257	...	Note					

BRIDGE DECKS (INCREMENTALLY LAUNCHED)

Casting bay (plan area measurement)	...	sqm	3500.00	3600.00	3350.00	3450.00	3450.00
Concrete (40MPa)	...	cum	329.00	311.00	378.00	355.00	453.00
Reinforcement	...	t	2690.00	2510.00	2500.00	2470.00	2490.00
Prestressing :							
Concentric	...	t	8400.00	8500.00	8400.00	8400.00	8600.00
Continuity	...	t	6200.00	6300.00	6200.00	6200.00	6400.00
Launching	...	cum	220.00-320.00	220.00-320.00	220.00-320.00	220.00-320.00	220.00-320.00
ADD extra for :							
Waterproof membrane (average)	...	sqm	34.20	33.50	32.60	25.80	32.10
Surfacing, refer page 716	...	Note					
Guardrails refer page 304 (average)	...	m	211.00	243.00	247.00	236.00	260.00
Balustrade (average)	...	m	295.00	290.00	300.00	295.00	290.00
Lighting of bridge and piers	...	Note					
Sewerage, gas and water mains	...	Note					
Power supply	...	Note					

CIVIL ENGINEERING

BRIDGEWORKS (cont'd)

COMPOSITE PRICES

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Prices are for reinforced concrete single span bridges including safety rails, balustrades but excluding approach works, abutments and piling	...	Note					
Footbridge :							
Conventional	...	sqm	755.00-905.00	790.00-930.00	770.00-930.00	765.00-915.00	785.00-935.00
Single lane overpass 7m wide :							
Conventional	...	sqm	1760.00-1860.00	1825.00-1925.00	1705.00-1805.00	1595.00-1695.00	1860.00-1970.00
Incremental	...	sqm	-	-	-	1290.00-1470.00	-
Two lane 11m wide :							
Conventional	...	sqm	1700.00-1800.00	1785.00-1905.00	1700.00-1780.00	1520.00-1640.00	1780.00-1880.00
Incremental	...	sqm	-	-	-	1265.00-1465.00	-
Four lane 20m wide :							
Conventional	...	sqm	1675.00-1795.00	1785.00-1905.00	1660.00-1780.00	1530.00-1655.00	1750.00-1870.00
Incremental	...	sqm	-	-	-	1215.00-1415.00	-
Six lane 27m widec :							
Conventional	...	sqm	1640.00-1770.00	1745.00-1865.00	1580.00-1700.00	1500.00-1625.00	1730.00-1850.00
Incremental	...	sqm	-	-	-	1195.00-1395.00	-

DOMESTIC CONSTRUCTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
The prices given hereafter are indicative average prices and apply to either domestic projects or similarly constructed commercial/industrial projects in the $100,000 to $1,000,000 range, and having average site conditions.	...	Note					
An allowance of 7.5% has been included for Builder's overheads and profit.	...	Note					
Prices exclude the Goods and Services Tax (G.S.T.).	...	Note					

PRELIMINARIES

The prices given hereafter do not include any allowance for preliminaries. Allow separately for all usual applicable Builder's preliminaries	...	Note					

EXCAVATION

EXCAVATION

Excavate over site to reduce levels (x 500 cum) :							
In sand or light soil	...	cum	21.30	16.65	18.85	15.60	24.00
In clay .	..	cum	23.40	17.85	21.80	23.50	35.90
In hard rock	...	cum	112.00	103.50	136.50	126.50	130.00
Excavate trench not exceeding 1.00m deep including retaining sides :							
In sand or light soil	...	cum	61.30	37.80	54.50	29.90	51.90
In clay	...	cum	69.00	72.70	57.20	42.80	62.90
In hard rock	...	cum	171.50	151.50	271.00	194.00	340.00

FILLING

Excavated material as filling (on site)	...	cum	24.50	15.00	17.90	7.75	9.85
Clean sand filling	...	cum	72.30	65.00	58.00	26.50	71.00
Crushed rock filling	...	cum	114.00	85.00	92.00	74.00	85.00

EXTERNAL WORKS

PAVING, ETC.

Crushed rock/blue metal base course 150mm thick including grading, rolling and consolidating to receive paving :							
100 to 200 sqm	...	sqm	17.70	14.45	18.25	14.55	18.90
500 to 1000 sqm	...	sqm	16.15	13.75	17.10	13.65	17.95
Prime and two coats sprayed bitumen seal :							
100 to 200 sqm	...	sqm	23.10	29.60	11.60	38.20	11.20
500 to 1000 sqm	...	sqm	20.40	26.70	10.20	19.85	9.45
Hot bituminous concrete 25mm thick :							
100 to 200 sqm	...	sqm	28.20	30.00	26.10	35.40	22.90
500 to 1000 sqm	...	sqm	23.70	24.20	20.50	17.50	20.60
Clay brick paving laid on prepared foundation :							
230 x 110/115 x 50mm	...	sqm	72.20	77.60	67.60	49.40	68.90
230 x 110/150 x 65mm	...	sqm	-	77.70	68.70	-	76.40
230 x 150 x 50mm	...	sqm	-	73.70	-	-	67.90
230 x 230 x 50/60mm	...	sqm	-	83.80	-	61.30	-

DOMESTIC CONSTRUCTION

EXTERNAL WORKS (cont'd)

PAVING ETC. (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Interlocking 60mm thick concrete block paving laid on prepared foundation	...	sqm	65.00	76.00	75.40	49.80	80.30
For additional paving brick types refer page 223	...	Note					
ADD extra for :							
75mm thick compacted sand bed	...	sqm	6.50	6.65	7.30	5.55	6.65
Black polythene underlay	...	sqm	2.30	2.35	2.40	2.55	2.85
25mm mortar bed	...	sqm	8.30	6.55	6.70	6.05	6.65

FENCING

Standard butt jointed paling fence 1500mm high (or butt jointed picket fence in Perth)	...	m	96.50	85.50	73.50	99.00	78.00
Standard brush fence on steel framing with capping and set in concrete, 1500mm high	...	m	213.00	-	208.00	230.00	247.00
Additional fence types, refer pages 225 to 228	...	Note					

CONCRETE WORK

REINFORCED CONCRETE 20MPa

Strip footing, pier foundation	...	cum	264.00	240.00	286.00	263.00	311.00
Slabs and thickening on fill	...	cum	257.00	217.00	305.00	283.00	332.00
Suspended slabs	...	cum	257.00	214.00	301.00	280.00	328.00
Attached beams	...	cum	306.00	214.00	300.00	276.00	327.00

FORMWORK

Prices are based on strutting of 2.50 to 3.00 metres	...	Note					
Standard Formwork (Class 3, limited imperfections) :							
Soffit of suspended slab up to 250mm thick	...	sqm	120.00	111.00	129.00	89.00	133.00
Sides and soffit of attached beams	...	sqm	137.00	128.00	140.00	127.00	159.00
Edges, sides or risers -							
N.E. 100mm high	...	m	21.50	26.50	55.00	18.00	35.00
100/200mm high	...	m	26.00	31.00	37.00	20.00	47.00
200/300mm high	...	m	32.00	37.50	42.00	26.00	59.00

REINFORCEMENT

Tempcore, deformed bar reinforcement (Y16)	...	t	2925.00	2280.00	2720.00	2660.00	2955.00
SL72 fabric	...	sqm	11.65	9.10	12.65	10.60	12.55
SL62 fabric	...	sqm	9.60	7.55	10.55	8.80	10.35

DOMESTIC CONSTRUCTION

BRICKWORK

		Adel. $	Bris. $	Mel. $	Per. $	Syd. $
Supply costs of bricks are those of cheapest brick of the particular type ... Note						
BASE PRICES						
Bricks delivered to site :						
Bricks, standard clay common (48.5/sqm) ...	per 1000	824.00	705.00	750.00	871.00	830.00
Facing bricks clay, standard (48.5/sqm) ...	per 1000	1187.00	1025.00	900.00	963.00	1090.00
Cement mortar per 1000 standard bricks ...	per 1000	95.00	92.50	107.50	92.50	152.50
Labour - sub-contract rate (excluding mortar) for :						
Standard bricks ...	per 1000	1060.00	1330.00	1500.00	1250.00	1710.00
BRICKWORK						
Standard clay common brickwork :						
Reduced brickwork ...	per 1000	2150.00	2305.00	2555.00	2405.00	2915.00
110mm wall, leaf of wall ...	sqm	104.50	112.00	124.00	116.50	141.50
Standard clay face brickwork, including fair face and pointing one side :						
Reduced brickwork ...	per 1000	3010.00	2775.00	2935.00	2610.00	3405.00
110mm wall, leaf of wall ...	sqm	146.00	134.50	142.50	126.50	165.50
ADD or DEDUCT extra for :						
Each variation of $5.00 per thousand in supply cost of facing bricks in stretcher bond ...	sqm	0.26	0.26	0.26	0.26	0.26
ADD extra for :						
Cleaning down face brickwork ...	sqm	3.25	3.35	3.30	3.25	3.40

CARPENTRY

			Treated Pine MGP10	Loc Hwd F14	Hwd F17	Loc Hwd F8	Nth Hwd F11/14
FLOOR FRAMING							
100 x 25mm plate ...	m		9.35	10.45	7.00	8.65	9.55
100 x 38mm ditto ...	m		12.95	13.55	9.50	12.05	13.60
100 x 75mm bearer ...	m		26.00	32.40	19.20	24.10	30.40
125 x 75mm ditto ...	m		31.80	39.80	22.90	28.60	38.60
							Pine Treated F7
100 x 50mm joist ...	m		16.85	18.05	12.65	17.80	16.10
150 x 50mm ditto ...	m		25.60	28.20	18.15	26.30	26.90
WALL FRAMING			Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10	Pine MGP10
75 x 38mm plate ...	m		8.90	9.85	8.55	9.60	9.95
100 x 38mm ditto ...	m		12.05	13.25	10.85	12.50	13.15
75 x 38mm stud ...	m		8.20	9.15	7.90	8.80	9.25
100 x 38mm ditto ...	m		10.65	11.80	9.55	10.90	11.75
75 x 38mm nogging ...	m		8.90	9.85	8.55	9.60	9.95
100 x 38mm ditto ...	m		11.35	12.55	10.20	11.70	12.45

DOMESTIC CONSTRUCTION

CARPENTRY (cont'd)

ROOF AND CEILING FRAMING		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
			Pine MGP10	Loc Hwd F14	Pine MGP10	Loc Hwd F8	Pine MGP10
100 x 50mm rafter	...	m	12.60	19.50	10.85	19.40	11.95
150 x 50mm ditto	...	m	19.25	30.40	18.35	28.70	18.15
150 x 50mm purlin	...	m	19.90	31.10	19.00	29.50	18.85
150 x 75mm ditto	...	m	28.30	48.70	25.10	37.40	32.10
200 x 38mm ridge, hip, valley board	...	m	-	34.80	20.50	29.30	20.20
						Pine MGP10	
100 x 38mm ceiling joist	...	m	10.35	16.05	9.25	10.50	11.40
100 x 50mm ditto	...	m	12.60	19.50	10.85	13.45	11.95
200 x 50mm hanger	...	m	25.10	41.50	23.80	25.70	22.30
FLOORING							
Compressed fibre cement sheets fixed to timber :							
15mm thick	...	sqm	96.60	97.70	97.90	100.00	99.30
18mm thick	...	sqm	110.00	111.50	112.00	114.00	113.50
Particle board panels with tongued and grooved edges to timber :							
19mm thick standard (yellow)	...	sqm	27.00	27.70	28.00	29.20	28.80
22mm thick mid-range (red)	...	sqm	33.00	33.70	33.90	35.20	34.80
Tongued and grooved, (standard grade), 25mm thick in 100mm widths, fixed to timber :							
Radiata pine	...	sqm	64.60	66.10	67.80	70.40	65.30
Mixed hardwood or jarrah	...	sqm	125.50	93.60	-	96.10	124.00
Tasmanian oak	...	sqm	97.40	89.00	82.50	118.50	136.50
WALL LININGS							
Fibre cement flexible sheeting fixed to timber framing :							
4.5mm thick (exterior)	...	sqm	30.40	31.40	31.80	33.60	33.10
6.0mm thick (exterior)	...	sqm	38.70	39.80	40.00	42.20	41.40
Fibre cement decorative sheeting fixed to timber framing :							
6.0mm thick Stucco	...	sqm	53.70	55.10	55.60	58.00	57.30
Fibre cement decorative planks fixed to timber framing :							
230mm x 7.5mm thick smooth finish	...	sqm	51.80	53.30	54.20	56.50	56.10
300mm x 9.0mm thick Weatherboard	...	sqm	68.10	69.20	70.00	71.70	71.50
Wallboard fibre cement sheeting flush jointed and fixed to timber framing :							
6.0mm thick	...	sqm	67.70	69.30	70.10	72.60	72.10
Interior standard grade plywood for paint finish fixed to timber framing :							
4.0/5.0mm thick	...	sqm	29.20	29.90	30.10	33.20	31.50

DOMESTIC CONSTRUCTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
JOINERY							
SKIRTINGS							
Rounded or splayed fixed to timber :							
100 x 25mm -							
Oregon	...	m	-	22.20	18.60	22.40	15.35
Local hardwood	...	m	24.90	20.80	13.30	19.00	21.40
Tasmanian oak	...	m	26.30	25.80	20.10	21.90	23.20
Pacific maple	...	m	13.25	-	13.30	16.45	13.45
Western red cedar	...	m	27.60	19.95	23.30	24.70	28.60
MD Fibreboard	...	m	10.50	11.00	11.80	11.90	11.90
CUPBOARDS AND FITTINGS							
All purpose made	...	Note					
Refer pages 354 to 358	...	Note					
WINDOWS AND DOORS							
WINDOWS							
Refer pages 363 and 366	...	Note					
DOOR FRAMES							
Standard zincanneal door frame to suit door size 2040 x 820mm, built in :							
Knock down frame	...	No	140.00	142.50	141.50	157.00	141.50
Fully welded domestic frame	...	No	103.00	105.50	104.50	116.50	104.50
Timber linings and frames fixed to brickwork :							
150 x 38mm plain lining -							
Oregon	...	m	-	48.80	36.70	48.90	-
Local hardwood	...	m	54.40	45.60	37.30	45.30	46.80
Pacific maple	...	m	36.20	-	37.30	35.50	38.20
Western red cedar	...	m	57.40	43.80	51.10	54.10	62.80
100 x 75mm frame, twice rebated -							
Oregon	...	m	-	66.30	46.80	66.70	48.40
Local hardwood	...	m	73.00	62.20	54.60	62.00	86.00
Pacific maple	...	m	56.30	-	54.60	49.00	61.40
Western red cedar	...	m	82.20	59.80	71.30	73.50	85.00
DOORS							
Standard hollow core door, 2040 x 820 x 35mm thick (excluding edge strips, hardware and painting), faced both sides with :							
Prime coated hardboard	...	No	132.00	140.50	141.00	153.50	147.50
Sliced Pacific maple veneer	...	No	163.00	171.50	170.00	217.00	195.00
Standard solid core door, 2040 x 820 x 40mm thick (excluding edge strips, hardware and painting), faced both sides with :							
Prime coated hardboard	...	No	192.50	179.50	198.50	209.00	195.00
Sliced Pacific maple veneer	...	No	244.00	223.00	228.00	255.00	245.00
ADD extra for :							
Edge strips to four edges of 2040 x 820mm -							
Hollow core door	...	No	82.30	115.60	109.40	107.60	113.90
Solid core door	...	No	78.10	115.60	109.40	107.60	113.90

DOMESTIC CONSTRUCTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
ROOFING							
STEEL							
Prices are based on areas of 250 to 500sqm	...	Note					
Corrugated, Zincalume® Steel :							
0.42mm	...	sqm	39.00	40.60	43.50	39.50	49.20
Ribbed, zincalume coated :							
0.42mm (Klip-lok®, Speedek)	...	sqm	50.30	51.90	54.80	50.80	60.50
0.48mm (Klip-lok®, Speedek)	...	sqm	52.00	53.60	56.50	52.50	62.20
ADD extra for :							
Colour coating -							
Corrugated	...	sqm	8.30	8.30	8.30	8.30	8.30
Ribbed.	..	sqm	10.85	10.85	10.85	10.85	10.85
Cappings (average)	...	m	37.60	38.50	38.00	40.80	38.50
Stop ending	...	m	10.00	10.50	10.25	11.80	10.50
Closing ends and lipping	...	m	13.35	14.00	13.65	15.75	14.00
TILE							
Prices include 38 x 35mm battens and for laying to slope not exceeding 30°	...	Note					
Concrete tiles, standard pattern and colour	...	sqm	36.00	43.60	52.50	45.20	65.90
Terracotta tiles, French pattern, standard colour	...	sqm	52.00	80.00	60.00-70.00	52.00-56.00	96.70
ADD extra for :							
Ridge and hip tiles (average)	...	m	37.50	40.80	42.80	23.60	45.10
PLASTERING							
Cement render, two coats 16mm thick wood float finish on brick walls including narrow widths and labours (openings deducted)	...	sqm	47.90	43.20	46.50	41.30	44.90
Hardwall plaster, two coats 16mm thick on brick walls including narrow widths and labours (openings deducted)	...	sqm	54.70	43.60	49.90	30.90	49.80
Plasterboard, flush finished :							
1 x 10mm thick to -							
Timber wall framing	...	sqm	34.40	36.00	29.50	28.00	30.40
Steel wall framing	...	sqm	34.90	36.00	29.50	28.00	30.80
Timber ceiling framing	...	sqm	36.90	37.40	30.50	29.50	30.80
Wall framing of cold formed steel sections in top and bottom plates, studs at 600mm centres and one row of noggings, including angles and intersections, in :							
51mm framing	...	sqm	35.70	38.40	33.00	31.30	37.40
76mm framing	...	sqm	36.70	40.00	35.00	33.00	40.60
PAVIOR AND TILER							
Power float monolithic finish	...	sqm	7.35	5.35	7.15	5.70	6.60
Tiling, Non-Resilient and Resilient Finishes refer pages 432 to 442	...	Note					

DOMESTIC CONSTRUCTION

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
GLAZING							
Clear float glass and glazing to metal :							
3mm	...	sqm	96.00	95.00	97.00	98.00	100.00
4mm	...	sqm	106.00	105.00	105.00	110.00	109.00
5mm	...	sqm	119.00	116.00	121.00	122.00	122.00
6mm	...	sqm	124.00	121.00	125.00	129.00	125.00
PAINTING							
On general surfaces :							
Woodwork -							
Prime, one sealer undercoat and two coats gloss acrylic	...	sqm	12.35	11.85	14.75	18.20	16.30
Prime, one undercoat and two coats enamel	...	sqm	17.05	15.40	14.70	19.10	16.30
Two coats clear polyurethane	...	sqm	8.80	11.05	8.05	11.00	8.35
Metalwork -							
One coat zinc phosphate primer	...	sqm	4.55	3.85	4.20	4.95	5.15
Touch up primer, one undercoat and two coats enamel	...	sqm	14.90	13.95	12.55	17.00	15.35
Walls and ceilings, smooth surface -							
Seal and two coats low gloss acrylic	...	sqm	12.20	10.60	11.25	11.95	11.15
One coat sealer, one coat undercoat and one coat enamel	...	sqm	12.85	12.25	11.95	12.55	11.05
Seal and two coats acrylic latex	...	sqm	11.70	10.25	10.30	11.10	11.05
PLUMBING							
Prices for the following sanitary fixtures include for installation including jointing to pipes but do not include for pipework. PC Amounts show the included costs of fixtures including tapware.	...	Note					
Prices for pipework include for fixing in position.	...	Note					
Refer page 159 for a guide to composite prices including pipework to a main, riser or to the building.	...	Note					
INSIDE BUILDING							
BASINS							
Vitreous china basin:							
Wall, 10L (P.C. $ 420.00)	...	No	620.00	620.00	620.00	640.00	650.00
Vanity, 12L (P.C. $ 355.00)	...	No	450.00	450.00	450.00	460.00	460.00
BATHS							
Porcelain enamelled steel bath :							
1675 x 760mm (P.C. $ 355.00)	...	No	970.00	970.00	980.00	990.00	1000.00
W.C. SUITES							
Vitreous china pan with integrated plastic cistern, lid and seat (P.C. $ 435.00)	...	No	690.00	690.00	710.00	730.00	740.00
Vitreous china cistern and pan with plastic lid and seat (P.C. $ 900.00)	...	No	1200.00	1200.00	1215.00	1230.00	1245.00
Vitreous china wall hung pan with plastic lid and seat, inwall cistern, fascia and buttons (P.C. $ 1050.00)	...	No	1380.00	1380.00	1395.00	1425.00	1425.00
SHOWER BASES							
Porcelain enamelled steel, 900 x 900mm (P.C. $ 540.00)	...	No	750.00	750.00	760.00	780.00	780.00
SHOWER BATH							
Reinforced polyester, 900 x 765 x 1875mm high (P.C. $ 1250.00)	...	No	1750.00	1750.00	1775.00	1825.00	1825.00

DOMESTIC CONSTRUCTION

PLUMBING (cont'd)

INSIDE BUILDING (cont'd)

		UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
SINKS							
Kitchen Sink :							
Stainless steel, standard grade -							
Under mount -							
Single bowl (P.C. $ 340.00)	...	No	430.00	430.00	440.00	440.00	440.00
Double bowl (P.C. $ 620.00)	...	No	740.00	740.00	740.00	750.00	750.00
Single bowl, with drainer -							
1234mm long (P.C. $ 540.00)	...	No	650.00	650.00	650.00	660.00	660.00
Double bowl, with drainer -							
1538mm long (P.C. $ 925.00)	...	No	1070.00	1070.00	1070.00	1080.00	1080.00
LAUNDRY TROUGHS							
Stainless steel :							
Single compartment 45L laundry trough -							
Tub, flush into benchtop (P.C. $425.00)	...	No	530.00	530.00	530.00	540.00	540.00
Single compartment 70L laundry trough -							
Tub, flush into benchtop (P.C. $685.00)	...	No	810.00	810.00	810.00	820.00	820.00
HOT WATER UNITS							
50L storage electric hot water heater	...	No	610.00	610.00	610.00	620.00	630.00
20L/min continuous flow 6 star external gas hot water heater	...	No	930.00	945.00	945.00	960.00	1020.00
WATER PIPEWORK							
16mm dia. PE-Xa pipe (including fittings)	...	m	28.00	28.50	29.50	31.50	32.00
20mm dia. PE-Xa pipe (including fittings)	...	m	31.50	32.50	33.50	36.00	36.50
20mm thick tube insulation, for 16mm pipe	...	m	10.75	10.75	11.25	11.75	12.25
20mm thick tube insulation, for 20mm pipe	...	m	12.25	12.50	12.75	13.25	13.50
PVC DRAINAGE WASTE AND VENT PIPE							
40mm pipe, with solvent welded joints	...	m	26.50	27.50	28.00	30.00	30.50
65mm pipe, with solvent welded joints	...	m	39.00	40.00	41.00	44.00	45.00
100mm pipe, with solvent welded joints	...	m	52.00	53.00	55.00	59.00	60.00
40mm pipe bend	...	No	20.00	20.50	21.00	22.00	22.00
65mm pipe bend	...	No	29.00	29.50	30.00	31.50	32.00
100mm pipe bend	...	No	60.00	61.00	61.00	63.00	63.00
100mm pipe junction	...	No	62.00	63.00	64.00	67.00	67.00

ELECTRICAL

Individual house, medium standard, excluding external service	...	sqm	68.00	70.00	71.00	71.00	79.00
Power point :							
Single	...	Point	82.00	80.00	80.00	84.00	86.00
Double	...	Point	88.00	86.00	86.00	90.00	92.00
Lighting (excluding luminaires), one switch	...	Point	82.00	80.00	80.00	84.00	88.00
For additional costs refer pages 507 to 601	...	Note					

WAGE RATES

Based on typical rates as at December 1, 2020 and on a 38 hour week (36 hour week for Melbourne).

Rates are for metropolitan areas only and, whilst they include for site allowance at the average rate per hour, they do not include any other special allowances such as structural and height allowances, etc.

The Contract Charge-Out rate assumes a negotiated rate, this should not be confused with the usually much higher rate charged for non-contract works.

ADELAIDE		Av. Tender (Costing) Rate Incl. Overheads and Profit	Av. Contract Charge-Out Rate Incl. Overheads and Profit
BUILDING TRADES		$/Hour	$/Hour
Bricklayer		65.00-68.75	74.75-82.00
Carpenter/Joiner		64.50-68.25	74.25-82.00
Painter/Glazier		64.50-68.25	74.25-82.00
Plasterer		64.50-68.25	74.25-82.00
Plumber/Drainer		77.00-81.50	88.50-97.25
Labourer	- Group 1	64.00-67.75	73.75-81.50
	- Group 2	63.25-67.25	73.75-81.50
	- Group 3	62.75-66.75	72.50-80.25
	- Group 4	62.25-66.00	71.50-78.75
MECHANICAL SERVICES			
Refrigeration Fitter		64.50-68.25	74.25-82.00
Sheet Metalworker	- Class 1	64.50-68.25	74.25-82.00
	- Class 2	65.50-69.25	75.50-82.00
Lagger		65.50-69.25	75.50-82.00
Electrician		77.00-83.50	88.50-97.25
Plumber		74.25-79.75	85.75-93.75
ELECTRICAL SERVICES			
Electrical Installer		80.25-86.25	92.25-100.50

BRISBANE		Av. Tender (Costing) Rate Incl. Overheads and Profit	Av. Contract Charge-Out Rate Incl. Overheads and Profit
BUILDING TRADES		$/Hour	$/Hour
Bricklayer		77.00-81.00	87.00-91.00
Carpenter/Joiner		77.00-81.00	87.00-91.00
Painter/Glazier		75.00-79.00	85.00-89.00
Plasterer		77.00-81.00	87.00-91.00
Plumber/Drainer		82.00-86.00	92.00-96.00
Labourer	- Group 1	72.00-76.00	82.00-86.00
	- Group 2	71.00-75.00	81.00-85.00
	- Group 3	70.00-74.00	80.00-84.00
	- Group 4	69.00-73.00	79.00-83.00
MECHANICAL SERVICES			
Refrigeration Mechanic		79.00-83.00	82.00-93.00
Sheet Metalworker	- Class 1	76.00-80.00	81.00-90.00
	- Class 2	70.00-74.00	80.00-84.00
Lagger		70.00-74.00	79.00-84.00
Electrician		88.00-92.00	98.00-102.00
Plumber		82.00-86.00	82.00-96.00
ELECTRICAL SERVICES			
Electrical Installer		88.00-92.00	98.00-102.00

NOTE - BRISBANE RATES ARE ‘EBA/MARKET’ RATES FOR A COMMERCIAL PROJECT

WAGE RATES

MELBOURNE		Av. Tender (Costing) Rate Incl. Overheads and Profit	Av. Contract Charge-Out Rate Incl. Overheads and Profit
BUILDING TRADES		$/Hour	$/Hour
Bricklayer		78.00-85.00	90.00-97.00
Carpenter/Joiner		78.00-85.00	90.00-97.00
Painter/Glazier		77.00-83.00	89.00-96.00
Plasterer		78.00-85.00	90.00-97.00
Plumber/Drainer		83.00-91.00	90.00-97.00
Labourer	- Group 1	78.00-85.00	90.00-97.00
	- Group 2	77.00-83.00	89.00-96.00
	- Group 3	75.00-81.00	87.00-94.00
	- Group 4	-	-
MECHANICAL SERVICES			
Refrigeration Fitter		79.00-86.00	91.00-98.00
Sheet Metalworker	- Class 1	77.00-83.00	91.00-87.00
	- Class 2	75.00-79.00	84.00-93.00
Lagger		79.00-86.00	91.00-98.00
Electrician		82.00-90.00	93.00-98.00
Plumber		84.00-92.00	91.00-98.00
ELECTRICAL SERVICES			
Electrical Installer		84.00-92.00	90.00-98.00

PERTH		Av. Tender (Costing) Rate Incl. Overheads and Profit	Av. Contract Charge-Out Rate Incl. Overheads and Profit
BUILDING TRADES		$/Hour	$/Hour
Bricklayer		78.50-81.50	85.00-90.00
Carpenter/Joiner		79.00-82.00	85.50-90.50
Painter/Glazier		78.50-81.50	85.00-90.00
Plasterer		78.50-81.50	85.00-90.00
Plumber/Drainer		91.75-93.75	99.25-104.25
Labourer	- Group 1	71.50-74.50	80.50-85.50
	- Group 2	71.50-74.00	79.50-84.50
	- Group 3	71.00-73.50	79.00-84.00
	- Group 4	70.00-73.00	78.50-83.50
MECHANICAL SERVICES			
Refrigeration Fitter		86.75-89.75	93.25-98.25
Sheet Metalworker	- Class 1	85.50-89.50	93.00-98.00
	- Class 2	85.25-88.25	91.75-96.75
Lagger		84.25-87.25	90.75-95.75
Electrician		87.25-90.25	93.75-98.75
Plumber		91.25-93.75	99.25-104.25
ELECTRICAL SERVICES			
Electrical Installer		89.25-99.25	95.25-99.25

WAGE RATES

SYDNEY		Av. Tender (Costing) Rate Incl. Overheads and Profit	Av. Contract Charge-Out Rate Incl. Overheads and Profit
BUILDING TRADES		$/Hour	$/Hour
Bricklayer		65.00-90.00	83.00-107.00
Carpenter/Joiner		70.00-95.00	90.00-113.00
Painter/Glazier		60.00-90.00	77.00-107.00
Plasterer		65.00-90.00	83.00-107.00
Plumber/Drainer		85.00-115.00	109.00-137.00
Labourer	- Group 1	55.00-80.00	71.00-95.00
	- Group 2	50.00-75.00	64.00-89.00
	- Group 3	47.50-72.00	61.00-86.00
	- Group 4	45.00-70.00	58.00-83.00
MECHANICAL SERVICES			
Refrigeration Fitter		80.00-110.00	103.00-131.00
Sheet Metalworker	- Class 1	75.00-95.00	96.00-113.00
	- Class 2	70.00-90.00	90.00-107.00
Lagger		70.00-90.00	90.00-107.00
Electrician		90.00-120.00	116.00-143.00
Plumber		90.00-120.00	116.00-143.00
ELECTRICAL SERVICES			
Electrical Installer		85.00-115.00	109.00-137.00

WAGE RATES

ON-SITE LABOUR COST RATES

The following is a **guide only** for calculation of hourly on-site labour cost rates.

Adjust the respective factors to accord with local practice.

CALCULATION OF NET NUMBER OF HOURS PER ANNUM

		Days
Number of ordinary days per annum		261
LESS : Annual leave	- 20 days	
Statutory holidays*	- 11 days	
Sick leave	- 10 days	41
		220

	Hours
Number of hours worked per annum - 220 x 7.6	1672.00
LESS : 41 days x 0.4 hours per day accrued	16.40
	1655.60
Allowance for inclement weather** 5%	82.78
	1572.82
Rest hours*** -	
Bricklayer @ 30 minutes per day $\frac{1572.82 \times 30 \text{ minutes}}{7.6}$	103.48
Other trades @ 20 minutes per day $\frac{1572.82 \times 20 \text{ minutes}}{7.6}$	68.98
Prescribed hours per annum -	
Bricklayer (1572.82 - 103.48)	1469.34
Other trades (1572.82 - 68.98)	1503.84

*** Rest hours may vary from state to state and award to award.

** This percentage will vary depending upon the Contractor's past experience.

* The number of days will vary from state to state.

WAGE RATES

ON-SITE LABOUR COST RATES (cont'd)

CALCULATION OF HOURLY RATE

GROSS WEEKLY WAGE $

- Base Rate -
- Industry Allowance -
- Tool Allowance -
- Lost Time Allowance, Follow Job -
- Special Allowance -
- Additional Payments -
- Registration Allowance -
- Plumbing or Other Trade Allowance-_____

GROSS WEEKLY WAGE =
GROSS HOURLY WAGE (38) = _____

COST PER ANNUM $

- Gross Hourly Rate x Ordinary Paid Hours Per Annum (261 x 7.6 = 1983.6 Hours) -
- Award Travel Allowance (Based on 220 Days) 220 x 7.6 / 38 x $_____ -
- Annual Leave Loading 4 x (Weekly Award + Travel) x .175 -
- Long Service Leave -
- Training Levy -
- Superannuation -
- Severance Pay -_____

=

- Payroll Tax @ ' '% -_____

COST PER ANNUM = _____ - excluding Workers' Compensation, Workcare, etc. Costs

ADD WORKERS' COMPENSATION, WORKCARE, ETC.	Up to 2 Storey $	Above 2 Storey $	
1.0 Cost per annum from above	-		
2.0 Workers' Compensation*, average rate (' '% and ' '%)	-		
3.0 Accident Insurance average rate (' '%)	-		
4.0 Government Surcharge, at ' '% of '2.0'	-		
5.0 Stamp Duty at ' '% of '2.0' and '3.0' -_____			
TOTAL COST PER ANNUM for ordinary Paid Days =	_____		- excluding site and multi-storey allowances
COST PER HOUR Bricklayers ÷ 1469.36 =			- excluding site and multi-storey allowances
Others ÷ 1503.84	_____		

ADD for SITE AND MULTI-STOREY ALLOWANCES

- Cost Per Hour from Above -
- Site Allowance) -
- Multi-storey Allowance) plus Payroll Tax -
- Any Other Allowance) -
- Workers' Compensation and Other charges as '2.0' to '5.0' above - _____

COST PER HOUR including all usual allowances
Bricklayers ÷ 1469.36 =
Others ÷ 1503.84 = _____

ADD to COST PER HOUR for :

- Other charges not included above
- Time off for - Picnic days, funerals and other days
- Overheads
- Profit

PLANT HIRE RATES

Refer Preliminaries for :

Cranes -
- Mobile - page 173
- Tower - page 173

Hoists -
- Barrow - page 174
- Platform - page 174
- Man and material - page 174

Scaffolding, etc. -
- Tubular - page 174
- Mobile - page 174
- Scissor platform - page 174
- Cherry pickers - page 174

Offices and sheds - page 176

EARTHMOVING AND ROADMAKING PLANT

The rates given indicate the average range of hire rates on a working hourly rate basis and are typical for the mainland capital cities. It should be appreciated that the rates can vary appreciably according to the market conditions at a particular time.

Unless otherwise indicated, the rates given include allowance for operator and fuel costs.

Whilst hourly rates are indicated, minimum hire times of four to eight hours would apply to most equipment.

Hire rates for extended periods could be reduced through negotiation and also depend upon the market conditions at the time of hire.

Where classifications are given they are ERG classification.

The rates exclude Goods and Services Tax (G.S.T.).

ADD to the following rates for :
- Supervision.
- General labour.
- Penalty rates and overtime to the foregoing and operators.
- Transportation.
- Pilot car.
- Travelling time.
- Special allowances.
- Permits and licence fees.
- Fuel, power, etc., costs where not included.

BACKHOE/LOADER		$/hour
Typical Model :		
JCB 2CX	- Digging depth to 3.5m	90-110
JCB 3CX	- Digging depth to 5.0m	115-135
Cat 444F	- Digging depth to 6.0m	120-140

BULLDOZERS - SWAMP	Mass (kg)	$/hour
Typical Model :		
Cat D3K LGP	8093	130-150
Cat D6K2 LGP	13950	145-165
Komatsu D61PXi-23	18770	165-185
Cat D6T LPG	22450	170-190
Komatsu D85PX-15E0	25150	230-260
Cat D8T LGP	37750	260-290

BULLDOZERS	Mass (kg)	$/hour
Typical Model :		
Cat D3K XL	7670	130-150
Komatsu D51EX-22	12710	145-165
Cat D6N XL	16510	170-200
Komatsu D65WX-16	20360	180-210
Cat D7E	26055	220-250
Cat D8T	35585	260-290
Cat D9T	43820	290-320
Cat D10T	63850	500-540

PLANT HIRE RATES

EARTHMOVING AND ROADMAKING PLANT (cont'd)

EXCAVATORS, HYDRAULIC (TRACKED)	Mass (kg)	$/hour
Typical Model :		
Takeuchi TB108	855	80-95
JCB 8014CTS	1634	85-100
Cat 8025ZTS	2685	90-105
Komatsu PC35MR-3	3575	95-110
Cat 304E CR	4009	100-115
Cat 305.5E CR	5415	105-125
Komatsu PC88MR-8	8395	110-130
Cat 312D2L	13900	115-135
Komatsu PC160LC-6	16680	120-140
Komatsu PC220LC-8	24610	125-145
Cat 324E L	26307	130-150
Komatsu PC300-8	33010	150-180
Cat 336D2 L	37100	160-200
Komatsu PC450LC-8	44320	175-215
Cat 349E L	53300	190-230
Komatsu PC600LC-8E0	61000	225-275
Cat 390F L	75200	265-315

GRADERS		
Typical Model :	Mass (kg)	$/hour
Champion C80C	6800	120-150
Cat 120M	16875	155-185
Cat 14M	23900	190-230
Komatsu GD825A-2	26350	215-255

LOADERS, WHEELED		
Typical Model :	Mass (kg)	$/hour
Cat 906H2	5630	105-125
Hitachi	6890	110-120
Komatsu WA150-5	7845	120-140
Cat 930H HL	13322	130-160
Volvo L60F	11600	140-170
Komatsu WA250PZ-6	15445	145-175
Cat 972H	25510	165-205
Cat 980H	30925	175-215

ROLLERS, VIBRATORY	
Smooth drum rollers :	$/hour
- Up to 5.0t (various)	90-110
- 5.0t-7.5t (various)	100-120
- 7.5t-10.0t (various)	110-130
- 10.0t-12.5t (various)	120-140
- 12.5t-15.0t (various)	120-150
- 15.0t-17.5t (various)	130-160
- 17.5t-20.0t (various)	135-170
Padfoot rollers :	
- Up to 5.0t (various)	80-100
- 5.0t-7.5t (various)	105-125
- 7.5t-10.0t (various)	110-130
- 10.0t-12.5t (various)	130-155
- 12.5t-15.0t (various)	130-160
- 15.0t-17.5t (various)	135-165
- 17.5t-20.0t (various)	140-170
- 20.0t-25.0t (various)	150-180

DUMP TRUCKS, RIGID FRAME		
	Heaped Capacity (cum)	$/hour
30t to 40t (various)	24.0-25.0	255-295
40t to 60t (various)	27.0-36.0	340-380
60t to 90t (various)	40.0-42.0	415-455

DUMP TRUCKS, ARTICULATED		
	Heaped Capacity (cum)	$/hour
20t to 25t (various)	15.0-15.5	190-220
25t to 30t (various)	16.5-18.0	200-230
30t to 35t (various)	19.7-20.5	215-245
35t to 40t (various)	23.1-24.0	260-290
Over 40t (various)	24.0-28.8	265-295

PLANT HIRE RATES

GENERAL AND SMALL PLANT

- Hire rates for extended periods would generally be reduced.
- **The rates exclude Goods and Services Tax (G.S.T.).**

ADD extra for :
Delivery and return.
Erection and dismantling where applicable.
Operators.
Fuel, power and other consumables.

	$/day
ACROW PROPS	
1.6m to 3.3m	5.00
AIR COMPRESSORS	
Portable :	
4 litre/second, petrol or electric	50-60
6 litre/second, petrol or electric	55-65
Silenced :	
47 litre/second, diesel	115-155
61 litre/second, diesel	115-155
83 litre/second, diesel	135-175
189 litre/second, diesel	200-250
425 litre/second, diesel	250-350
AIR HOSES	
19mm x 18m	13-17
50mm x 18m	45-55
AIR TOOLS	
Breaker :	
Light, 18 litre/second	35-45
Medium, 26 litre/second	50-60
Heavy, 28 litre/second	60-75
Chipping hammer, 12 litre/second	40-55
Spade bits	10-12
Drills :	
6mm/13mm, 8 litre/second	35-45
19mm/32mm, 32 litre/second	55-65
Grinders, 9 litre/second	40-50
Impact Wrench :	
13mm, 11 litre/second	40-50
19mm, 14 litre/second	45-55
25mm, 24 litre/second	50-60
38mm, 47 litre/second	80-95
Orbital Sander	35-45
Rock Drill :	
Small, 12 litre/second	35-45
Medium, 21 litre/second	50-65
Drill Steels -	
Up to 1.8m	15-20
1.8 to 3.0m	15-20
Sander, 18 litre/second	28-38

AIR TOOLS (cont'd)	$/day
Scabbler :	
Single held, 4 litre/second	60-75
Triple head, wall, 11 litre/second	75-90
Multiple, 5 head, floor, 94 litre/second	270-310
CONCRETE AND MASONRY EQUIPMENT	
Brick/Block splitter	35-45
Concrete mixer (petrol) 3ft^3	35-50
Saws (excluding blades) :	
Brick (electric, petrol)	130-150
Concrete	105-135
Diamond blade 305mm	40-50
Trowelling machine	75-85
Vibrators :	
Drive unit (petrol)	65-75
Immersion vibe shaft and head	70-85
Shaft extension	40-50
CHAIN SAW	60-80
COMPACTION EQUIPMENT	
Plate compactor (Refer also Earthmoving page 737) :	
Light (petrol)	70-85
Medium/heavy (diesel)	120-160
DEHUMIDIFIER	50-60
ELECTRIC TOOLS AND EQUIPMENT	
Drills (electric)	35-45
Kango hammer	60-70
Extension leads, 30m :	
Single phase	7-11
Three phase	30-40
Sanders :	
100mm belt	25-35
175mm disc	25-35
200mm floor	60-80
Orbital	20-30
FLOOR GRINDER (petrol, electric)	70-85
FLOOR SANDER	60-80
FLOOR TILE STRIPPER	80-100

PLANT HIRE RATES

GENERAL AND SMALL PLANT (cont'd)

	$/day
GENERATORS AND LIGHTING	
Generators, petrol :	
Unsilenced, petrol -	
2 KVA, 1 phase, 8 amp	40-50
3 KVA, 1 phase, 12 amp	50-60
5 KVA, 1 phase, 20 amp	70-80
8 KVA, 1 phase, 20 amp	105-135
Silenced, diesel -	
6 KVA, 1 phase, 24 amp	70-80
8 KVA, 1 phase, 32 amp	70-85
15 KVA, 3 phase, 20 amp	80-95
20 KVA, 3 phase, 27 amp	100-115
35 KVA, 3 phase, 47 amp	130-145
50 KVA, 3 phase, 69 amp	145-170
60 KVA, 3 phase, 83 amp	180-205
80 KVA, 3 phase, 111 amp	200-230
100 KVA, 3 phase, 139 amp	230-250
200 KVA, 3 phase, 278 amp	370-420
350 KVA, 3 phase, 435 amp	480-550
Silenced, biodiesel -	
12 KVA	85-110
20 KVA	95-135
37 KVA	135-160
50 KVA	165-200
80 KVA	220-260
120 KVA	310-365
300 KVA	395-450
365 KVA	450-530
550 KVA	670-800
Floodlights :	
On stand	15-20
Portable	20-35
Lighting towers, mobile, hydraulic	100-340
HANDLING, HOISTING AND LIFTING EQUIPMENT	
Belt conveyor :	
10.0m	145-170
Chain block :	
1 tonne	45-50
2 tonne	65-75
3 tonne	75-85
5 tonne	100-130
Hoists and hoisting-refer Preliminaries	
Jacks, hydraulic :	
11 tonne	35-45
20 tonne	50-60
50 tonne	70-85

	$/day
PUMPS	
Centrifugal pumps :	
50mm, petrol, 525 litres/minute	60-70
50mm, diesel, 525 litres/minute	65-75
Centrifugal trash pumps :	
50mm, petrol, 525 litres/minute	65-70
75mm, petrol, 825 litres/minute	70-80
100mm, diesel, 1900 litres/minute	85-100
150mm, diesel, 5400 litres/minute	160-200
Diaphragm pumps :	
50mm, petrol, 150 litres/minute	80-90
75mm, diesel, 300 litres/minute	90-110
Hoses :	
Delivery -	
50mm x 15m	11-13
75mm x 15m	15-20
100mm x 15m	20-25
150mm x 15m	25-35
Suction -	
75mm x 6m	7-10
100mm x 6m	10-14
150mm x 3m	6-9
ROLLERS	
Pedestrian roller, 700mm, petrol	110-135
Single drum padfoot }	
Single drum smooth } Refer to "Earthmoving" plant	
Tandem drum smooth } on page 737	
SKID STEER LOADERS	
Medium	200-250
Large	250-300
Extra large	275-320
Rubber tracked	275-340
Medium with trailer	280-320
WATER TESTING	
Portable tubidity meter	70-80
WELDING EQUIPMENT	
Welders, diesel :	
300 amp	65-75
400 amp	95-115
Welders, electric	
140 amp, single phase	45-55
200 amp, three phase	50-60
300 amp, three phase	70-85
400 amp, three phase	95-110
Welding leads	7-13

PLANT HIRE RATES

GENERAL AND SMALL PLANT (cont'd)

	$/day
MISCELLANEOUS	
Barrows	15-17
Battery charger	20-25
Bitumen sprays :	
Hand	70-85
Power	90-115
Cleaning charge, additional per hour	35-40
Bolt cutters	21-23
Drain rods, 18m	24-26
Dumpy level	55-65
Earth auger :	
Hand	13-16
Petrol	70-95
Earth leakage unit	21-23
Fans, exhaust	50-70
Flashing lights (per week)	8-10
Ladders :	
3.0m/4.8m	19-21
4.2m/7.2m	25-30
5.4m/10.0m	33-40
Nail gun	50-60
Paint burner including gas	25-30
Penetrometer	17-19
Pipe bender, (hydraulic) up to 50mm	60-70
Pipe cutter, manual	35-40
Pipe threaders, electric :	
Up to 50mm	110-135
Up to 100mm	125-160
Pipe vice and stand	19-23
Planks (aluminium)	10-14
Ramset gun	55-70
Road warning signs	4.00
Space heater, 180000 BTU	65-80
Tarpaulin, 6m x 5.5m	21-24

	$/day
MISCELLANEOUS (cont'd)	
Tension wrenches :	
12mm	19-23
19mm	25-28
25mm	35-40
Tile cutter, manual, ceramic	26-30
Tile saw, electric	75-90
Trestles :	
1.8m	10-15
2.4m	10-15
3.0m	15-20
3.6m	15-20
4.2m	15-20
Tube bender	20-25
Vacuum cleaner	50-70
Wallpaper stripper, electric	35-45
Water tank, 1500 litres	35-45
Witches hat (per week)	15.00

HAULAGE RATES

Prices are average as ruling at December 1, 2020.	...	Note
Within metropolitan areas allow minimum one hour travelling time	...	Note
ADD extra for unloading - Refer CRANES-MOBILE page 173	...	Note
The rates exclude Goods and Services Tax (G.S.T.)	...	Note

NEW SOUTH WALES

SYDNEY		COUNTRY	
COST PER VEHICLE CAPACITY	COST PER HOUR	VEHICLE CAPACITY	TONNE PER 200 KM
5t	$59.50-$67.00	5t	)
8t	$62.50-$73.00	8t	)
12t	$72.00-$85.50	12t	) $81.50 (average)
16t	$83.00-$106.00	16t	)
20t	$91.00-$113.00	20t	)

QUEENSLAND

BRISBANE		COUNTRY	
COST PER VEHICLE CAPACITY	COST PER HOUR	VEHICLE CAPACITY	TONNE PER 200 KM
5t	$60.50	5t	)
8t	$68.50	8t	) $60.50 for the first 200km
12t	$82.50	12t	)
16t	$91.50	16t	) $64.00 thereafter
20t	$118.00	20t	)

HAULAGE RATES

SOUTH AUSTRALIA

ADELAIDE		COUNTRY	
COST PER VEHICLE CAPACITY	COST PER HOUR	VEHICLE CAPACITY	TONNE PER 200 KM
5t	$48.50-$56.50	5t	)
8t	$52.50-$61.50	8t	)
12t	$59.50-$72.50	12t	) $66.00 (average)
16t	$60.50-$86.00	16t	)
20t	$76.00-$98.00	20t	)

VICTORIA

MELBOURNE		COUNTRY	
COST PER VEHICLE CAPACITY	COST PER HOUR	VEHICLE CAPACITY	TONNE PER 200 KM
5t	$59.00-$67.00	4 - 8t	)
8t	$64.00-$76.00		)
12t	$72.00-$84.00	12 - 16t	) $78.50 (average)
16t	$74.00-$87.00		)
20t	$91.50-$114.00	20 - 24t	)

WESTERN AUSTRALIA

PERTH		COUNTRY	
COST PER VEHICLE CAPACITY	COST PER HOUR	VEHICLE CAPACITY	TONNE PER 200 KM
5t	$61.00-$70.00	4 - 8t	)
8t	$68.00-$75.00		)
12t	$74.00-$88.50	12 - 16t	) $85.50 (average)
16t	$86.50-$108.00		)
20t	$98.50-$118.00	20 - 24t	)

REFURBISHING AND RECYCLING BUILDINGS

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REFURBISHING AND RECYCLING BUILDINGS

MEANING

- The terms refurbishing, recycling and regeneration are used to describe the building operations involved in the upgrading or conversion of an existing building, the extent of the works varying from little more than a superficial treatment of the building to total reconstruction.
- Whilst opinions on terminology vary somewhat the following is a reasonably representative definition of the scope of each category.

Minor Refurbish : The minimum work necessary, little more than a superficial treatment of the building including making good faults and enabling it to operate without obvious deficiencies, would include :

- Cleaning and checking watertightness of facade.
- Remodelling of main entry and lobby.
- Upgrading of finishes in lettable space.
- Refurbishing of toilets.
- Minor improvements and repairs to air-conditioning and electrical services.
- Refurbishing of lifts.

Medium Refurbish : Sufficient work to restore the building and its functions to its original design standards with limited improvements only, would include :

- Partial upgrading, cleaning and checking watertightness of facade.
- Remodelling of main entry and lobby.
- Upgrading of finishes in lettable space.
- Refurbishing of toilets including new tiling and fixtures.
- Upgrading of staircases and core components.
- Upgrading of air-conditioning and electrical services, ensuring that they are energy efficient.
- Refurbishing of lifts.

Major Refurbish : The upgrading of the building to a standard of appearance and functionality that will meet both current and future client expectations, would include :

- New facade to visible elevations.
- Remodelling of main entry and lobby.
- Total upgrading of lettable space.
- Re-planning and upgrading of core and its components.
- Fully upgraded state of the art and energy efficient services.
- Modernised and/or new lifts.

Recycle, regenerate : The upgrading of the building to provide virtually a new building, one whose efficiency and image will be equal to a new building, would include :

- Total re-design and replacement of facade with likely only the basic structure retained.
- Re-planning and upgrading of ground floor and entry and of lettable floors together with service core and its components.
- New state of the art and energy efficient services.
- New lift installation.

REFURBISHING AND RECYCLING BUILDINGS

FACTORS FOR AND AGAINST

The ever increasing interest in refurbishment by building owners can be attributed to many factors, amongst them being :

- The need to maintain the value of the investment.
- The likelihood of capital appreciation.
- Cost effective means of restoring an asset to maximum income earning potential and rejuvenation of investment growth.
- Refurbished stock can compete with new, but with lower rents.
- Improved appearance and functions of the building will hold or attract new tenants, if nothing done tenants leave for a better building.
- Possibility of maintaining some cash flow from existing tenants during currency of the works.
- Prime established city locations made more available.
- Current regulations and plot or floor space ratios could disadvantage redevelopment.
- Likely cost saving when compared with demolition and re-building.
- Faster building completion.
- Less risk of time delays caused by the likes of disputes and wet weather, therefore less consequential costs and more accurate budget feasibilities possible.
- Improved maintenance and running costs for the refurbished building.
- If a historic building, the concern for the preservation of our past heritage.

The negative aspects of refurbishing are limited to a few factors such as :

- Commitment to the possible shortcomings of an existing building in under development of available plot ratio.
- The inability, due to structural limitations, to replace the service core.
- Possible higher maintenance and running costs.
- Commitment to the possible outdated facade of an existing building and lack of choice in creating a corporate image.

However in respect of the last two negative aspects there is unlikely to be such a problem when opting for a full recycle or regeneration.

REFURBISHING AND RECYCLING BUILDINGS

DEVELOPMENT PROCESS

As soon as the need to refurbish the properties within an investment portfolio is recognised it is then necessary to follow an orderly approach through the various stages :

PORTFOLIO ANALYSIS

The various buildings within a particular investment portfolio would be examined to determine the priorities applicable to each in respect of refurbishment.

PROPERTY ANALYSIS

An analysis which investigates the various alternatives for the particular building on the current and future market place.

BUILDING ANALYSIS

A detailed audit and analysis of the fabric, finishes, services and operating costs of the building.

DESIGN AND DOCUMENTATION

At this stage the design team should be appointed, the team should include real estate and client representatives. The refurbishment development process can now proceed, this consists of :

- From the preceding building analysis it can be established what state the building is in and what the appropriate action should be. A decision then has to be made as to the level of refurbishing required :
  Minor Refurbishment,
  Medium Refurbishment,
  Major Refurbishment,
  Recycle, Regenerate.
- Preparation of a detailed programme of the various procedures including construction.
- Determine the market to be targeted.
- Presentation and assessment of design alternatives.
- Defining the standards necessary to satisfy the specific market needs.
- Cost evaluation of design alternatives, cost plan and continuing cost control with remedial action as necessary to maintain overall budget.
- Documentation and selection of contractual system.

IMPLEMENTATION AND CONSTRUCTION

Ensure that the construction programme reflects the most efficient and cost effective means of carrying out the works, for example it may well be practical to start from the top floor and work down, thus premium floors are on the market first.

Ensure that minimum inconvenience is caused to existing tenants. The degree of disruption to tenants can affect rental payments.

Essential that the cost effect of unforeseen difficulties be minimised.

MAINTENANCE MANUAL

Prepare a comprehensive handover and maintenance manual, the object being to ensure that both the property manager and the building engineer operate the refurbished building in the most efficient manner possible. The manual should also include recommendations and likely costs in respect of any ongoing maintenance inspection programmes and necessary remedial work.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE COSTS PER SQUARE METRE

- Costs given hereafter should be used for initial feasibility studies only.
- Costs given are average prices for typical buildings at December 31, 2020 within the metropolitan areas including allowance for Preliminaries and Builders profit and overheads. ADD additional allowances for increased costs (refer Anticipated Building Price Adjustment page 15).
- Costs are based on the total floor area of all levels measured between the inner faces of external walls.
- Costs EXCLUDE basements, external services outside 3.0m from the outside face of the building, external works other than those immediately adjacent to the buildings, loose or special equipment, furniture, furnishings, legal and professional fees.
- For a more detailed breakdown and description of the works, and to assist in the compilation of a more accurate estimate refer to the Elemental Costs given on pages 750 to 773, the page references being given immediately after the scope of the works.
- Costs given are for Finished Floor Lettable buildings as described on page 32.
- **Costs exclude the Goods and Services Tax (G.S.T.)**.

	UNIT	Adel. $	Bris. $	Mel. $	Per. $	Syd. $
HOTEL - MEDIUM TO HIGH RISE - THREE STAR						
Medium refurbish (refer pages 750 and 751) ...	sqm	1715.00-1865.00	1605.00-1755.00	1530.00-1680.00	1795.00-1945.00	1830.00-1980.00
HOTEL - HIGH RISE - FIVE STAR						
Medium refurbish (refer pages 752 and 753) ...	sqm	2385.00-2535.00	2345.00-2495.00	2005.00-2155.00	2330.00-2480.00	2650.00-2800.00
OFFICE BUILDING - 3 STOREY - standard finishes, air-conditioned, lift :						
Minor refurbish (refer pages 754 and 755) ...	sqm	480.00-540.00	415.00-475.00	435.00-495.00	505.00-565.00	510.00-570.00
Medium refurbish (refer pages 756 and 757) ...	sqm	935.00-1085.00	820.00-970.00	880.00-1030.00	980.00-1130.00	985.00-1135.00
OFFICE BUILDING - 7 to 20 STOREY - standard finishes and facilities, air-conditioning, medium speed lifts, sprinklers :						
Minor refurbish (refer pages 758 and 759) ...	sqm	670.00-730.00	625.00-685.00	595.00-655.00	660.00-720.00	665.00-725.00
Medium refurbish (refer pages 760 and 761) ...	sqm	1435.00-1585.00	1360.00-1510.00	1320.00-1470.00	1420.00-1570.00	1445.00-1595.00
Major refurbish (refer pages 762 and 763) ...	sqm	3155.00-3405.00	3010.00-3260.00	2955.00-3205.00	3150.00-3400.00	3285.00-3535.00
Recycle, regenerate (refer pages 764 and 765) ...	sqm	3900.00-4200.00	3645.00-3945.00	3655.00-3955.00	3815.00-4115.00	4120.00-4420.00
OFFICE STOREY - 21 to 35 STOREY - standard finishes and facilities, air-conditioned, multiple high speed lifts, sprinklers :						
Medium refurbish (refer pages 766 and 767) ...	sqm	1660.00-1810.00	1595.00-1745.00	1475.00-1625.00	1660.00-1810.00	1940.00-2090.00
Major refurbish (refer pages 768 and 769) ...	sqm	4210.00-4460.00	3980.00-4230.00	3795.00-4045.00	4265.00-4515.00	5065.00-5315.00
Recycle, regenerate (refer pages 770 and 771) ...	sqm	5205.00-5505.00	4855.00-5155.00	4675.00-4975.00	5225.00-5525.00	6335.00-6635.00
OFFICE BUILDING CLASSIFIED - 5 STOREY - standard finishes and facilities, air-conditioned, slow lifts, no sprinklers :						
Recycle, regenerate (refer pages 772 and 773) ...	sqm	2935.00-3185.00	2595.00-2845.00	2825.00-3075.00	3035.00-3285.00	3145.00-3395.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

HOTEL - MEDIUM TO HIGH RISE - THREE STAR

SCOPE - **MEDIUM REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 12 storey
- All facades visible
- Fully serviced
- 3.30m floor/floor. Wall/floor ratio 0.50
- Fully enclosed covered area 8500sqm

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	360.25-390.25	375.25	278.75	341.75	246.75	353.25
SUBSTRUCTURE - Upgrade entrance steps	...	0.75-1.75	1.25	1.25	1.25	1.25	1.25
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Upgrading of finishes, redecoration of balustrades, soffits and strings	...	13.50-16.50	15.00	14.50	14.75	14.25	17.50
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	3.00-5.00	4.00	4.25	4.00	4.75	3.50
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	75.50-95.50	85.50	88.25	80.25	78.75	85.25
External Doors - Replace entry doors, being part of remodelling of main entry and foyer	...	13.25-15.25	14.25	14.75	11.00	12.00	14.25
Internal Walls - Remodelling of ground floor foyer/reception	...	9.75-12.75	11.25	11.50	9.75	10.00	13.00
Internal Screens - New shower and toilet screens	...	10.25-12.25	11.25	12.25	9.75	9.75	11.25
Internal Doors - Replace hardware, redecoration of existing doors, some new doors	...	56.50-66.50	61.50	58.75	53.50	54.25	61.75
FINISHES							
Wall - New tiling in bathrooms, redecoration of existing painted surfaces, replacement of finishes to lift lobbies, new finishes to main entry and foyer	...	79.75-89.75	84.75	87.00	84.75	77.50	84.75
Floor - New carpet throughout, re-tiling of bathroom/toilets	...	128.25-138.25	133.25	133.50	117.75	117.50	131.75
Ceiling - New suspended ceilings to foyer, reception and lift lobbies on all floors, elsewhere redecoration	...	48.25-58.25	53.25	55.25	52.75	48.00	57.75
		c/f	850.50	760.00	781.25	674.75	835.25

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

HOTEL - MEDIUM TO HIGH RISE - THREE STAR

SCOPE - **MEDIUM REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			850.50	760.00	781.25	674.75	835.25
FITTINGS							
Fitments - Take out built-in fitments and replace with new, renew toilet accessories	...	154.50-174.50	164.50	163.25	149.25	147.00	174.50
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Replacement of sanitary fixtures, alterations to and upgrading of pipework	...	169.00-179.00	174.00	177.00	151.25	160.00	176.00
Mechanical - Upgrading of air-conditioning and improved mechanical ventilation of bathrooms	...	193.50-243.50	218.50	196.75	166.50	177.50	192.50
Fire - Upgrading of existing equipment	...	48.25-58.25	53.25	50.25	44.25	50.50	52.75
Electrical - Upgrading of electrical services including new light fittings	...	180.50-200.50	190.50	187.75	164.25	170.25	185.50
Transportation - Existing installation fully upgraded, partially replaced	...	126.00-166.00	146.00	153.25	130.25	133.50	149.50
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Some demolitions in service core, all other demolitions included in respective elements	...	4.50-6.50	5.50	5.75	4.00	5.25	4.75
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	5.00-9.00	7.00	6.50	5.00	6.00	5.75
CONTINGENCY - Allowance	...	-	95.25	89.50	84.00	80.25	93.50
TOTAL			1905.00	1790.00	1680.00	1605.00	1870.00

ADD extra for : Asbestos removal.
Basements including tanking.
Canopies.
Lifts, replace with new.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling
Sun protection, adequacy.
Fitting out, total building, - to be assessed as a percentage of the cost of similar new hotel.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

HOTEL - HIGH RISE - FIVE STAR

SCOPE - **MEDIUM REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 5 storey and plant/lift motor room
- All facades visible
- Fully serviced
- 3.50m floor/floor. Wall/floor ratio 0.50
- Fully enclosed covered area 21000sqm

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY		Adel.	Bris.	Mel.	Per.
		Cost Range	Mean Cost	Mean Cost	Mean Cost	Mean Cost	Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	549.25-584.25	566.75	429.00	483.75	325.00	475.25
SUBSTRUCTURE - Upgrade entrance steps	...	1.25-2.25	1.75	1.75	3.25	1.75	2.25
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Upgrading of finishes, redecoration of balustrades, soffits and strings	...	15.75-18.75	17.25	16.25	17.75	14.25	17.75
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	4.75-6.75	5.75	4.00	5.00	4.25	4.00
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	128.75-148.75	138.75	137.50	134.00	108.75	125.00
External Doors - Replace entry doors, being part of remodelling of main entry and foyer	...	17.25-19.25	18.25	17.25	16.50	14.00	17.00
Internal Walls - Remodelling of ground floor foyer/reception	...	9.50-12.50	11.00	13.00	11.75	9.25	13.00
Internal Screens - New shower and toilet screens	...	8.50-10.50	9.50	9.00	8.50	8.50	8.50
Internal Doors - Replace hardware, redecoration of existing doors, some new doors	...	65.50-75.50	70.50	73.25	70.25	62.00	73.50
FINISHES							
Wall - New tiling in bathrooms, redecoration of existing painted surfaces, replacement of finishes to lift lobbies, new finishes to main entry and foyer	...	119.25-134.25	126.75	124.00	133.25	108.00	117.50
Floor - New carpet throughout, re-tiling bathroom/toilets	...	167.00-182.00	174.50	180.50	168.25	153.50	167.00
Ceiling - New suspended ceilings to foyer, reception and lift lobbies on all floors, elsewhere redecoration	...	65.75-75.75	70.75	70.00	71.75	60.50	72.25
		c/f	1211.50	1075.50	1124.00	869.75	1093.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

HOTEL - HIGH RISE - FIVE STAR

SCOPE - **MEDIUM REFURBISH**

ELEMENT		SYDNEY		Adel.	Bris.	Mel.	Per.
		Cost Range	Mean Cost	Mean Cost	Mean Cost	Mean Cost	Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			1211.50	1075.50	1124.00	869.75	1093.00
FITTINGS							
Fitments - Take out built-in fitments and replace with new, renew toilet accessories	...	237.00-257.00	247.00	246.25	234.75	211.00	236.25
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Replacement of sanitary fixtures, alterations to and upgrading of pipework	...	232.75-252.75	242.75	237.75	222.25	210.25	222.25
Mechanical - Upgrading of air-conditioning and improved mechanical ventilation of bathrooms	...	350.25-410.25	380.25	308.50	276.25	264.50	289.00
Fire - Upgrading of existing equipment	...	59.75-69.75	64.75	55.25	54.25	54.50	53.50
Electrical - Upgrading of electrical services including new light fittings	...	244.50-264.50	254.50	227.00	214.25	199.75	216.50
Transportation - Existing installation fully upgraded, and partially replaced	...	152.25-192.25	172.25	173.00	160.75	155.25	162.75
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Some demolitions in service core, all other demolitions included in respective elements	...	6.00-8.00	7.00	5.75	5.50	5.00	4.75
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	6.75-10.75	8.75	8.00	7.00	6.00	6.75
CONTINGENCY - Allowance	...	-	136.25	123.00	121.00	104.00	120.25
TOTAL			2725.00	2460.00	2420.00	2080.00	2405.00

ADD extra for : Asbestos removal.
Basements including tanking.
Canopies.
Lifts, replace with new.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.
Fitting out, total building, - to be assessed as a percentage of the cost of similar new hotel.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 3 STOREY

SCOPE - **MINOR REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- All facades visible.
- Fully serviced.
- 3.50m floor/floor. Wall/floor ratio 0.60.
- Fully enclosed covered area 2000sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	68.25-78.25	73.25	61.50	58.50	55.25	83.50
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	8.00-13.00	10.50	10.00	9.50	8.50	13.25
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Make good and/or replace walking surfaces, redecoration of balustrades, soffits and strings	...	6.00-8.00	7.00	7.25	7.25	7.50	7.00
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	5.00-7.00	6.00	6.00	5.50	5.75	5.00
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, redecoration of painted surfaces	...	70.50-90.50	80.50	75.75	65.50	65.00	76.25
External Doors - Replace entry doors, being part of remodelling of main entry	...	9.75-12.75	11.25	10.00	9.00	10.25	11.00
Internal Walls - Remodelling ground floor lobby only	...	2.75-4.75	3.75	4.50	4.50	2.50	3.00
Internal Screens - New toilet screens	...	7.50-9.50	8.50	9.50	7.25	8.25	8.00
Internal Doors - Replace hardware, redecoration of existing	...	6.00-10.00	8.00	6.75	6.25	7.50	6.50
FINISHES							
Wall - New tiling in toilets, redecoration of all plastered surfaces, new finishes to main entry and lobby	...	20.75-30.75	25.75	24.75	23.00	23.25	24.50
Floor - New carpet to lettable, new tile finish to main entry and lobby, new tile finish to toilets	...	94.75-104.75	99.75	100.00	84.50	91.75	98.25
Ceiling - Make good and redecorate all lettable and service core area ceilings, new ceilings to main entry and lobby	...	15.50-25.50	20.50	19.25	19.25	18.00	20.75
		c/f	354.75	335.25	300.00	303.50	357.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 3 STOREY

SCOPE - **MINOR REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			354.75	335.25	300.00	303.50	357.00
FITTINGS							
Fitments - Make good and redecorate fitments, renew all toilet accessories	...	5.00-8.00	6.50	5.00	5.25	5.75	4.25
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Partial replacement of sanitary fixtures, minor alteration and pipework	...	29.50-39.50	34.50	32.50	27.00	29.25	33.00
Mechanical - Minor improvements, replacements and repairs	...	38.50-48.50	43.50	40.00	32.25	37.75	42.00
Fire - Check for and make good any defects	...	3.50-4.50	4.00	2.50	2.25	3.00	2.75
Electrical - Minor improvements and repairs	...	37.00-47.00	42.00	40.25	31.50	38.00	41.00
Transportation - Refurbish lift cars and entrance doors	...	20.75-30.75	25.75	26.75	21.75	22.75	26.00
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Minor only, included in respective elements	...	-	-	-	-	-	-
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	1.00-3.00	2.00	2.25	2.75	1.75	2.50
CONTINGENCY - Allowance	...	-	27.00	25.50	22.25	23.25	26.50
TOTAL			540.00	510.00	445.00	465.00	535.00

ADD extra for : Access flooring.
Asbestos removal.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 3 STOREY

SCOPE - **MEDIUM REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- All facades visible
- Fully serviced
- 3.50m floor/floor. Wall/floor ratio 0.60
- Fully enclosed covered area 2000sqm

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	127.00-142.00	134.50	112.25	120.00	113.50	131.50
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	8.25-13.25	10.75	10.25	9.50	10.00	12.50
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Upgrading of finishes, redecoration of balustrades, soffits and strings	...	7.75-10.75	9.25	9.50	9.00	8.50	9.50
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	5.25-7.25	6.25	6.00	5.50	5.50	5.50
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	84.00-104.00	94.00	93.75	84.00	82.50	94.25
External Doors - Replace entry doors, being part of remodelling of main entry	...	10.75-13.75	12.25	13.00	11.00	11.25	12.75
Internal Walls - Remodelling of ground floor lobby, limited remodelling of service core	...	7.00-10.00	8.50	8.75	6.75	6.50	7.00
Internal Screens - New toilet screens	...	8.25-10.25	9.25	9.50	7.75	8.50	8.50
Internal Doors - Replace hardware, redecoration of existing, some new doors	...	10.50-16.50	13.50	13.50	11.00	12.25	12.75
FINISHES							
Wall - New tiling in toilets and tea rooms, redecoration of existing painted surfaces, upgrading of finishes to lift lobbies, new finishes to main entry and lobby	...	24.50-34.50	29.50	28.75	28.25	28.50	30.25
Floor - New carpet to lettable, new tile finish to main entry and lobby, new tile finish to toilets	...	92.75-102.75	97.75	103.25	84.00	89.25	99.25
Ceiling - Make good and redecorate ceilings in service core areas; new suspended ceilings to lettable areas, main entry and lobby and lift lobbies on upper floors	...	75.75-85.75	80.75	79.50	71.25	72.75	89.75
		c/f	506.25	488.00	448.00	449.00	513.50

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 3 STOREY

SCOPE - **MEDIUM REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			506.25	488.00	448.00	449.00	513.50
FITTINGS							
Fitments - Take out fitments and replace with new, renew all toilet accessories	...	21.50-23.50	22.50	21.00	18.50	20.50	21.75
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Replacement of sanitary fixtures, alterations to and upgrading of pipework	...	58.50-68.50	63.50	62.75	52.25	60.50	65.25
Mechanical - Upgrading of air-conditioning and improved mechanical ventilation of toilets	...	179.25-229.25	204.25	184.25	162.00	182.75	193.75
Fire - Upgrading of existing equipment	...	6.00-8.00	7.00	6.50	5.50	7.50	6.00
Electrical - Upgrading of electrical services including new light fittings	...	161.75-181.75	171.75	163.25	135.75	155.75	168.50
Transportation - Refurbish lift cars and entrance doors	...	24.25-29.25	26.75	28.25	22.00	25.75	27.25
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Some demolitions in service core, all other demolitions included in respective elements	...	1.50-3.50	2.50	2.75	3.00	2.50	3.25
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	1.00-4.00	2.50	2.75	3.25	2.75	3.00
CONTINGENCY - Allowance	...	-	53.00	50.50	44.75	47.75	52.75
TOTAL			1060.00	1010.00	895.00	955.00	1055.00

ADD extra for : Access flooring.
Asbestos removal.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MINOR REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 13 storey and plant/lift motor room.
- All facades visible.
- Fully serviced.
- 3.30m floor/floor. Wall/floor ratio 0.50.
- Fully enclosed covered area 13500sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY		Adel.	Bris.	Mel.	Per.
		Cost Range	Mean Cost	Mean Cost	Mean Cost	Mean Cost	Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	111.25-126.25	118.75	100.00	115.75	97.50	120.50
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	5.25-7.25	6.25	6.50	6.50	5.75	5.75
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Make good and/or replace walking surfaces, redecoration of balustrades, soffits and strings	...	8.75-10.75	9.75	8.50	8.25	8.25	9.25
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	0.75-2.75	1.75	2.00	2.75	1.75	2.50
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	50.00-65.00	57.50	61.25	56.75	47.50	58.00
External Doors - Replace entry doors, being part of remodelling of main entry	...	9.75-14.75	12.25	13.75	12.25	12.25	12.00
Internal Walls - Remodelling of ground floor lobby only	...	3.50-5.50	4.50	5.50	4.75	3.75	3.50
Internal Screens - New toilet screens	...	7.25-9.25	8.25	9.00	8.25	8.00	8.00
Internal Doors - Replace hardware, redecoration of existing	...	9.75-11.75	10.75	11.00	9.25	9.00	8.75
FINISHES							
Wall - New tiling in toilets and tea rooms, redecoration of all plastered surfaces, new finishes to main entry and lobby	...	24.50-34.50	29.50	30.00	33.25	29.50	31.75
Floor - New carpet to lettable, new granite or marble finish to main entry and lobby, new tile finish to toilets	...	122.50-132.50	127.50	128.75	120.00	117.00	126.00
Ceiling - Make good and redecorate all lettable and service core area ceilings; new suspended ceiling to main entry and lobby	...	23.00-33.00	28.00	28.00	26.75	25.75	29.00
		c/f	414.75	404.25	404.50	366.00	415.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MINOR REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			414.75	404.25	404.50	366.00	415.00
FITTINGS							
Fitments - Make good and redecorate fitments, renew all toilet accessories	...	10.00-13.00	11.50	12.25	11.25	10.25	10.75
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Partial replacement of sanitary fixtures, minor alterations to pipework	...	35.25-45.25	40.25	44.75	35.75	36.25	36.25
Mechanical - Minor improvements, replacements and repairs	...	59.25-69.25	64.25	63.75	52.50	59.25	62.75
Fire - Check for and make good any defects	...	24.50-29.50	27.00	26.75	24.00	25.75	25.50
Electrical - Minor improvements and repairs	...	43.00-58.00	50.50	55.25	44.75	49.00	52.25
Transportation - Refurbish lift cars and entrance doors	...	43.25-53.25	48.25	53.00	44.25	44.00	48.75
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Minor only, included in respective element	...	-	-	-	-	-	-
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	1.75-5.75	3.75	5.00	5.25	3.25	4.25
CONTINGENCY - Allowance	...	-	34.75	35.00	32.75	31.25	34.50
TOTAL			695.00	700.00	655.00	625.00	690.00

ADD extra for : Access flooring.
Asbestos removal.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MEDIUM REFURBISH**

DESCRIPTION OF EXISTING BUILDING
- 13 storey and plant/lift motor room
- All facades visible
- Fully serviced
- 3.30m floor/floor. Wall/floor ratio 0.50
- Fully enclosed covered area 13500sqm

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	259.00-279.00	269.00	235.25	258.50	227.50	267.00
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	6.25-8.25	7.25	6.75	6.50	6.50	6.25
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Upgrading of finishes, redecoration of balustrades, soffits and strings	...	9.50-12.50	11.00	10.50	11.00	8.75	11.25
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	1.25-3.25	2.25	2.00	3.50	2.25	2.75
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	79.00-99.00	89.00	96.00	87.50	81.25	86.00
External Doors - Replace entry doors, being part of remodelling of main entry	...	12.75-15.75	14.25	15.75	13.75	13.75	15.00
Internal Walls - Remodelling of ground floor lobby, limited remodelling of service core	...	17.75-25.75	21.75	25.50	22.50	21.75	27.00
Internal Screens - New toilet screens	...	7.50-9.50	8.50	8.75	7.75	8.25	8.00
Internal Doors - Replace hardware, redecoration of existing, some new doors	...	15.25-22.25	18.75	20.00	17.25	17.00	18.00
FINISHES							
Wall - New tiling in toilets and tea rooms, redecoration of existing painted surfaces, upgrading of finishes to lift lobbies, new finishes to main entry and lobby	...	31.75-41.75	36.75	39.00	39.25	36.50	39.75
Floor - New carpet to lettable, new tile finish to main entry and lobby, new tile finish to toilets	...	119.75-129.75	124.75	131.50	119.25	117.75	123.00
Ceiling - Make good and redecorate ceilings in service core areas; new suspended ceilings to lettable areas, main entry and lobby and lift lobbies on upper floors	...	94.75-104.75	99.75	103.75	100.25	94.75	111.50
		c/f	703.00	694.75	687.00	636.00	715.50

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MEDIUM REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			703.00	694.75	687.00	636.00	715.50
FITTINGS							
Fitments - Take out fitments and replace with new, renew all toilet accessories	...	31.75-35.75	33.75	35.50	31.75	31.25	36.25
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Replacement of sanitary fixtures, alterations to and upgrading of pipework	...	71.75-81.75	76.75	81.25	72.75	75.50	74.50
Mechanical - Upgrading of air-conditioning and improved mechanical ventilation of toilets	...	269.75-319.75	294.75	281.50	264.50	266.75	266.75
Fire - Upgrading of existing equipment	...	49.25-59.25	54.25	51.25	48.75	50.00	49.00
Electrical - Upgrading of electrical services including new light fittings	...	212.75-227.75	220.25	227.25	204.25	208.75	220.00
Transportation - Refurbish lift cars and entrance doors	...	40.50-55.50	48.00	51.25	44.00	45.50	48.00
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Some demolitions in service core, all other demolitions included in respective elements	...	6.50-8.50	7.50	7.00	5.50	6.25	4.50
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	3.75-7.75	5.75	4.75	4.75	5.25	5.75
CONTINGENCY - Allowance	...	-	76.00	75.50	71.75	69.75	74.75
TOTAL			1520.00	1510.00	1435.00	1395.00	1495.00

ADD extra for : Access flooring.
Asbestos removal.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MAJOR REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 13 storey and plant/lift motor room.
- All facades visible.
- Fully serviced.
- 3.50m floor/floor. Wall/floor ratio 0.50.
- Fully enclosed covered area 13500sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	679.50-779.50	729.50	625.50	695.25	643.00	718.25
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	6.25-8.25	7.25	6.75	7.50	6.25	6.00
SUPERSTRUCTURE							
Columns - Work associated with remodelling at ground, mezzanine and first floor levels	...	8.00-11.00	9.50	8.50	7.50	7.50	8.25
Upper Floors - Work associated with remodelling at ground, mezzanine and first floor levels, minor work in service core at all levels	...	18.75-28.75	23.75	25.25	22.00	22.75	28.25
Staircases - Make good and/or replace walking surfaces, redecoration of balustrades, soffits and strings at all levels	...	8.75-11.75	10.25	9.25	11.25	9.00	11.25
Roof - Partial re-roofing including new to plant room, remedial work as required and redecoration of painted surfaces	...	25.75-35.75	30.75	27.25	31.75	27.50	30.50
External Walls, Windows - Reclad lettable area facades with new windows/curtain walling and metal or other to solid surfaces; new textured finish to service core walls; remodelling of ground floor and main entry	...	534.25-604.25	569.25	589.50	582.50	531.25	542.00
External Doors - New doors to ground floor entries	...	14.00-16.00	15.00	15.25	14.25	14.25	15.50
Internal Walls - Remodelling of service core areas, approx. 35% of existing walls replaced/upgraded	...	45.75-55.75	50.75	58.50	47.00	45.75	57.75
Internal Screens - New toilet screens	...	7.00-9.00	8.00	8.50	7.75	9.25	7.75
Internal Doors - New doors and fire doors throughout	...	35.50-39.50	37.50	40.50	34.50	35.25	36.50
FINISHES							
Wall - New finishes to service core, foyers and lettable areas, redecoration of staircase walls	...	76.75-86.75	81.75	83.25	83.75	77.50	86.50
Floor - New floor finishes throughout	...	124.75-134.75	129.75	136.00	121.75	123.25	126.50
Ceiling - New suspended ceilings throughout	...	105.25-115.25	110.25	116.25	116.75	104.50	127.00
		c/f	1813.25	1750.25	1783.50	1657.00	1802.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **MAJOR REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			1813.25	1750.25	1783.50	1657.00	1802.00
FITTINGS							
Fitments - New fitments and toilet accessories throughout	...	32.50-37.50	35.00	37.50	33.00	33.75	36.50
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - New sanitary fixtures, upgrading and replacement of most pipework	...	138.50-150.50	144.50	152.00	130.25	138.75	137.25
Mechanical - Total upgrading of air-conditioning and mechanical ventilation system	...	569.50-629.50	599.50	531.50	466.00	513.50	517.75
Fire - Replacement of virtually all fire service	...	136.25-146.25	141.25	131.75	116.75	125.50	124.50
Electrical - Total upgrading of electrical services, new light and power installations throughout	...	271.50-281.50	276.50	258.00	222.00	238.50	249.25
Transportation - Existing installation fully upgraded/partially replaced	...	147.75-197.75	172.75	192.00	167.25	164.75	181.50
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Take down facade cladding and windows, demolish 35% of service core walls, strip off existing finishes as necessary	...	77.75-87.75	82.75	87.00	83.00	77.75	88.00
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	7.00-9.00	8.00	8.75	7.75	7.25	7.25
CONTINGENCY - Allowance	...	-	136.50	131.25	125.50	123.25	131.00
TOTAL			3410.00	3280.00	3135.00	3080.00	3275.00

ADD extra for : Access flooring.
Asbestos removal.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **RECYCLE, REGENERATE**

DESCRIPTION OF EXISTING BUILDING -

- 13 storey and plant/lift motor room.
- All facades visible.
- Fully serviced.
- 3.30m floor/floor. Wall/floor ratio 0.50.
- Fully enclosed covered area 13500sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	986.25-1086.25	1036.25	843.75	887.75	875.00	918.25
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	5.75-7.75	6.75	6.50	7.50	6.25	6.75
SUPERSTRUCTURE							
Columns - Work associated with remodelling at ground, mezzanine and first floor levels	...	10.25-13.25	11.75	11.75	11.00	9.50	11.25
Upper Floors - Work associated with remodelling at ground, mezzanine and first floor levels, alterations and new work in totally re-planned service core	...	45.25-55.25	50.25	51.75	46.50	48.75	53.75
Staircases - New fire escape stairs	...	34.75-39.75	37.25	37.50	39.50	36.25	41.25
Roof - Re-roofing including structural steel framing as required, remedial work to concrete slabs under built-up roofing, new rainwater disposal	...	30.25-40.25	35.25	33.75	35.25	35.00	36.50
External Walls, Windows - Reclad entire facade including remodelling of ground floor and main entry	...	642.25-682.25	662.25	687.75	648.50	585.25	604.75
External Doors - New doors to ground floor entries and to plant rooms	...	14.50-16.50	15.50	15.00	14.00	14.25	15.50
Internal Walls - Remodelling of service core areas, approx. 90% of existing walls replaced/upgraded	...	118.00-128.00	123.00	136.50	116.00	115.50	131.25
Internal Screens - New toilet screens	...	6.75-8.75	7.75	8.75	7.75	8.00	7.50
Internal Doors - New doors and fire doors throughout	...	34.25-38.25	36.25	39.25	33.75	35.00	36.25
FINISHES							
Wall - New finishes throughout	...	81.00-91.00	86.00	89.00	90.00	82.75	91.75
Floor - New floor finishes throughout	...	125.25-135.25	130.25	137.00	124.00	124.25	127.75
Ceiling - New suspended ceilings throughout	...	107.00-117.00	112.00	118.50	117.25	106.75	127.50
		c/f	2350.50	2216.75	2178.75	2082.50	2210.00

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 7 to 20 STOREY

SCOPE - **RECYCLE, REGENERATE**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			2350.50	2216.75	2178.75	2082.50	2210.00
FITTINGS							
Fitments - New fitments and toilet accessories throughout	...	33.50-37.50	35.50	37.75	33.00	35.00	36.00
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - New installation	...	161.00-173.00	167.00	178.25	150.50	163.25	157.25
Mechanical - New installation	...	685.50-725.50	705.50	621.50	549.75	598.25	606.25
Fire - New services	...	148.25-158.25	153.25	139.50	122.00	141.75	133.75
Electrical - New installation	...	315.25-330.25	322.75	302.25	256.00	287.00	289.00
Transportation - New installation	...	239.75-279.75	259.75	281.75	246.75	247.25	266.25
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Take down facade cladding and windows, demolish most internal core walls, strip off existing finishes	...	103.00-123.00	113.00	115.25	111.00	104.50	114.25
EXTERNAL SERVICES - Allowance	...	2.25-4.25	3.25	4.25	4.00	3.50	3.50
EXTERNAL WORKS - Allowance	...	7.50-12.50	10.00	11.00	10.50	8.75	10.00
CONTINGENCY - Allowance	...	-	149.50	141.75	132.75	133.25	138.75
TOTAL			4270.00	4050.00	3795.00	3805.00	3965.00

ADD extra for : Access flooring.
Asbestos removal.
Atriums.
Basements including tanking.
Canopies.
Lifts, replace with new.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **MEDIUM REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 30 storey and plant/lift motor room.
- All facades visible.
- Fully serviced.
- 3.50m floor/floor. Wall/floor ratio 0.50.
- Fully enclosed covered area 39000sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	378.50-408.50	393.50	269.75	344.75	241.50	334.25
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	6.75-8.75	7.75	6.00	7.00	6.25	6.25
SUPERSTRUCTURE							
Columns - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Upper Floors - No remedial or new structural work allowed for	...	-	-	-	-	-	-
Staircases - Upgrading of finishes, redecoration of balustrades, soffits and strings	...	13.00-16.00	14.50	13.75	13.00	11.75	13.75
Roof - Check watertightness and carry out remedial work as required, redecoration of painted surfaces	...	1.00-2.00	1.50	1.50	2.50	1.75	2.50
External Walls, Windows - Clean and check watertightness of facade, remodelling of main entry, applied textured finish to existing brick/concrete surfaces, redecoration of painted surfaces	...	146.25-166.25	156.25	153.75	134.50	118.50	142.25
External Doors - Replace entry doors, being part of remodelling of main entry	...	15.75-18.75	17.25	18.50	15.50	14.50	16.75
Internal Walls - Remodelling of ground floor lobby, limited remodelling of service core	...	28.75-36.75	32.75	34.75	28.00	24.75	33.25
Internal Screens - New toilet screens	...	11.25-13.25	12.25	13.25	11.50	12.50	12.50
Internal Doors - Replace hardware, redecoration of existing doors, some new doors	...	16.75-24.75	20.75	20.50	17.00	17.50	18.00
FINISHES							
Wall - New tiling in toilets and tea rooms, redecoration of existing painted surfaces, upgrading of finishes to lift lobbies, new finishes to main entry and lobby	...	43.75-53.75	48.75	45.75	46.25	40.75	47.25
Floor - New carpet to lettable, new tile finish to main entry and lobby, new tile finish to toilets	...	138.75-148.75	143.75	144.00	127.50	125.25	133.50
Ceilings - Make good and redecorate ceilings in service core areas; new suspended ceilings to lettable areas and lift lobbies on upper floors	...	111.00-121.00	116.00	109.75	104.00	99.25	115.25
		c/f	965.00	831.25	851.50	714.25	875.50

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **MEDIUM REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			965.00	831.25	851.50	714.25	875.50
FITTINGS							
Fitments - Take out fitments and replace with new, renew all toilet accessories	...	34.75-38.75	36.75	36.75	32.25	32.00	36.50
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - Replacement of sanitary fixtures, alterations to and upgrading of pipework	...	95.00-105.00	100.00	92.00	83.75	89.25	84.75
Mechanical - Upgrading of air-conditioning and improved mechanical ventilation of toilets	...	361.25-411.25	386.25	305.00	276.00	288.25	295.25
Fire - Upgrading of existing equipment	...	59.50-69.50	64.50	52.75	48.25	53.75	49.00
Electrical - Upgrading of electrical services including light fittings	...	280.25-305.25	292.75	259.00	230.50	235.25	244.00
Transportation - Refurbish lift and entrance doors	...	50.00-65.00	57.50	59.50	51.50	51.75	54.00
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Some demolition in service core, all others included in respective elements	...	7.50-9.50	8.50	8.25	8.50	5.75	6.00
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	1.50-4.50	3.00	3.75	4.25	2.25	3.25
CONTINGENCY - Allowance	...	-	100.75	86.75	83.50	77.50	86.75
TOTAL			2015.00	1735.00	1670.00	1550.00	1735.00

ADD extra for : Access flooring. Asbestos removal. Basements including tanking. Canopies. Lifts, replace with new. Levelling screeds for floors. Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling. Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **MAJOR REFURBISH**

DESCRIPTION OF EXISTING BUILDING -

- 30 storey and plant/lift motor room
- Fully serviced
- All facades visible
- 3.50m floor/floor. Wall/floor ratio 0.50
- Fully enclosed covered area 39000sqm

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	1252.25-1377.25	1314.75	898.50	1002.25	875.25	1055.00
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	7.50-9.50	8.50	6.50	7.25	6.50	6.75
SUPERSTRUCTURE							
Columns - Work associated with remodelling at ground, mezzanine and first floor levels	...	6.75-8.75	7.75	6.50	6.50	6.50	5.00
Upper Floors - Work associated with remodelling at ground, mezzanine and first floor levels, minor work with service core at all levels	...	26.25-36.25	31.25	29.75	24.25	27.50	31.75
Staircases - Make good and/or replace walking surfaces, redecoration of balustrades, soffits and strings	...	14.25-17.25	15.75	14.25	15.75	14.00	15.00
Roof - Partial re-roofing including new to plant room, remedial work as required and redecoration of painted surfaces	...	23.75-31.75	27.75	23.75	24.25	22.25	25.75
External Walls, Windows - Reclad lettable area facades with new windows/curtain walling and metal or other solid surfaces; new textured finish to service core walls; remodelling of ground floor and main entry	...	1063.75-1123.75	1093.75	1015.25	943.25	855.75	968.50
External Doors - New doors to ground floor entries	...	29.50-31.50	30.50	16.75	14.50	14.25	16.00
Internal Walls - Remodelling of service core areas, approx. 35% of existing walls replaced/upgraded	...	84.00-99.00	91.50	91.00	71.25	71.25	88.00
Internal Screens - New toilet screens	...	11.00-14.00	12.50	13.25	12.25	11.75	12.75
Internal Doors - New doors and fire doors throughout	...	40.75-44.75	42.75	42.50	37.00	35.75	38.25
FINISHES							
Wall - New finishes to service core, foyers and lettable areas, redecoration of staircase walls	...	104.25-114.25	109.25	100.00	101.25	89.50	101.25
Floor - New floor finishes throughout	...	147.50-157.50	152.50	144.50	133.75	129.00	139.00
Ceiling - New suspended ceilings throughout	...	126.00-136.00	131.00	125.50	119.50	107.00	134.25
		c/f	3069.50	2528.00	2513.00	2266.25	2637.25

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **MAJOR REFURBISH**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			3069.50	2528.00	2513.00	2266.25	2637.25
FITTINGS							
Fitments - New fitments and toilet accessories throughout	...	40.50-44.50	42.50	39.50	33.50	37.50	38.50
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - New sanitary fixtures, upgrading and replacement of most pipework	...	183.75-198.75	191.25	180.00	158.25	165.75	170.75
Mechanical - Total upgrading of air-conditioning and mechanical ventilation system	...	756.00-796.00	776.00	597.75	516.25	560.50	587.00
Fire - Replacement of virtually all fire service	...	166.50-176.50	171.50	143.00	126.75	141.50	141.00
Electrical - Total upgrading of electrical services, new light fittings and power installations throughout	...	359.75-374.75	367.25	304.75	268.00	278.75	296.25
Transportation - Existing installation fully upgraded/partially replaced	...	249.50-299.50	274.50	282.00	247.75	242.50	262.25
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Take down facade and windows, demolish 35% of service core walls, strip off existing finishes as necessary	...	99.25-115.25	107.25	99.75	90.25	83.25	95.50
EXTERNAL SERVICES - Not applicable	...	-	-	-	-	-	-
EXTERNAL WORKS - Minor only	...	7.50-9.50	8.50	8.50	7.50	6.75	7.75
CONTINGENCY - Allowance	...	-	181.75	151.75	143.75	137.25	153.75
TOTAL			5190.00	4335.00	4105.00	3920.00	4390.00

ADD extra for : Access flooring.
Asbestos removal.
Atriums.
Basements including tanking.
Canopies.
Lifts, replace with new.
Levelling screeds for floors.
Repairs to concrete i.e. : exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **RECYCLE, REGENERATE**

DESCRIPTION OF EXISTING BUILDING -

- 30 storey and plant/lift motor room.
- Fully serviced.
- All facades visible.
- 3.50m floor/floor. Wall/floor ratio 0.50.
- Fully enclosed covered area 39000sqm.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	1772.75 - 1902.75	1837.75	1196.50	1313.00	1168.25	1422.50
SUBSTRUCTURE - Upgrade entrance steps, new ramp for disabled	...	8.00- 10.00	9.00	6.50	7.25	6.25	7.50
SUPERSTRUCTURE							
Columns - Work associated with remodelling at ground, mezzanine and first floor levels	...	13.00- 16.00	14.50	12.00	11.50	11.25	12.50
Upper Floors - Work associated with remodelling at ground, mezzanine and first floor levels, alterations and new work in totally re-planned service core	...	56.25- 66.25	61.25	58.25	48.00	51.50	55.50
Staircases - New fire escape stairs	...	37.50- 42.50	40.00	37.75	38.75	34.00	39.75
Roof - Re-roofing including structural steel framing as required, remedial work to concrete slabs under built-up roofing, new rainwater disposal	...	41.25- 51.25	46.25	42.25	43.00	36.25	44.00
External Walls, Windows - Reclad entire facade including remodelling of ground floor and main entry	...	1162.25- 1222.25	1192.25	1108.50	1003.25	946.50	1007.25
External Doors - New doors to ground floor entries and to plant rooms	...	15.25- 17.25	16.25	15.50	14.50	13.75	16.00
Internal Walls - Remodelling of service core areas, approx. 90% of existing walls replaced/upgraded	...	206.75- 226.75	216.75	222.25	173.00	168.50	214.75
Internal Screens - New toilet screens	...	11.00- 14.00	12.50	13.25	12.25	11.50	12.50
Internal Doors - New doors and fire doors throughout	...	41.00- 45.00	43.00	42.25	36.25	35.00	38.25
FINISHES							
Wall - New wall finishes throughout	...	113.50- 123.50	118.50	110.75	110.75	96.25	109.25
Floor - New floor finishes throughout	...	146.25- 156.25	151.25	149.00	134.75	129.25	139.75
Ceiling - New suspended ceilings throughout	...	128.00- 138.00	133.00	126.75	120.00	108.25	133.25
		c/f	3892.25	3141.50	3066.25	2816.50	3252.75

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING - 21 to 35 STOREY

SCOPE - **RECYCLE, REGENERATE**

ELEMENT		SYDNEY		Adel.	Bris.	Mel.	Per.
		Cost Range	Mean Cost	Mean Cost	Mean Cost	Mean Cost	Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			3892.25	3141.50	3066.25	2816.50	3252.75
FITTINGS							
Fitments - New fitments and toilet accessories throughout	...	43.00-47.00	45.00	40.75	34.25	39.50	39.00
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - New installation	...	207.00-222.00	214.50	199.00	174.75	184.75	188.00
Mechanical - New installation	...	894.00-914.00	904.00	700.00	602.25	650.75	681.25
Fire - New services	...	181.00-191.00	186.00	152.75	131.00	150.75	140.75
Electrical - New installation	...	437.00-452.00	444.50	360.25	315.00	334.75	346.50
Transportation - New installation	...	391.50-411.50	401.50	417.50	361.25	350.75	387.50
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Take down facade cladding and windows, demolish most internal core walls, strip off existing finishes	...	146.00-161.00	153.50	138.75	129.25	114.00	135.75
EXTERNAL SERVICES - Allowance	...	5.00-7.00	6.00	5.50	5.25	5.25	4.75
EXTERNAL WORKS - Allowance	...	8.25-13.25	10.75	11.50	10.50	9.00	10.75
CONTINGENCY - Allowance	...	-	227.00	187.50	175.25	169.00	188.00
TOTAL			6485.00	5355.00	5005.00	4825.00	5375.00

ADD extra for : Access flooring.
Asbestos removal.
Atriums.
Basements including tanking.
Canopies.
Levelling screeds for floors.
Repairs to concrete ie: exposed reinforcement, lack of cover, spalling.
Sun protection, adequacy.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING WITH CLASSIFIED FACADE

SCOPE - **RECYCLE, REGENERATE**

DESCRIPTION OF EXISTING BUILDING -

- 5 storey and plant/lift motor room.
- Existing facades to be retained, two with street frontage, the other two adjoining adjacent buildings.
- 4.00m floor/floor. Wall/floor ratio 0.75.
- Fully enclosed covered area of new building 3300sqm.
- New building within existing facades.

Note : Refer page 749 for notes on what the costs given include and exclude.

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
PRELIMINARIES - Builders site establishment, administration, plant, scaffolding, protection, insurances, fees, etc.	...	681.50-781.50	731.50	584.50	608.75	624.00	656.50
SUBSTRUCTURE - Reinforced concrete footings to columns and internal walls, new reinforced concrete ground slab	...	56.50-61.50	59.00	53.50	51.00	53.50	55.75
SUPERSTRUCTURE							
Columns - Reinforced concrete columns to all suspended slabs and to underside of roof	...	65.25-70.25	67.75	67.50	54.25	66.00	68.50
Upper Floors - Reinforced concrete suspended slabs to all levels	...	290.50-305.50	298.00	303.25	239.00	280.50	299.25
Staircases - One new reinforced concrete fire staircase	...	25.50-30.50	28.00	29.00	29.50	25.50	30.00
Roof - New roof comprising structural steel framing, purlins, insulation, steel decking, rainwater disposal	...	70.00-80.00	75.00	71.00	70.75	68.50	78.25
External Walls, Windows - The two visible masonry facades to be steam cleaned, checked for watertightness, made good and pointed or painted, new windows to both facades, the two rear facades to be checked for watertightness and made good/rendered where visible	...	207.25-247.25	227.25	228.00	211.50	207.50	208.75
External Doors - New main entry doors to accord with classified facade	...	12.75-14.75	13.75	14.00	11.75	13.00	14.50
Internal Walls - Reinforced concrete and brick or block enclosure walls to service core and shafts	...	116.50-128.50	122.50	134.75	104.50	106.25	135.75
Internal Screens - Toilet screens	...	9.25-11.25	10.25	11.00	9.00	9.50	11.25
Internal Doors - Doors and service core and fire staircase	...	32.75-37.75	35.25	35.50	29.50	31.25	34.75
FINISHES							
Wall - Standard wall finishes throughout	...	71.75-81.75	76.75	75.75	72.50	71.25	79.75
Floor - Standard floor finishes throughout	...	93.75-103.75	98.75	99.50	84.00	90.75	96.25
Ceiling - Suspended ceilings throughout	...	79.75-89.75	84.75	84.00	79.50	78.75	96.25
		c/f	1928.50	1791.25	1655.50	1726.25	1865.50

REFURBISHING AND RECYCLING BUILDINGS

INDICATIVE ELEMENTAL COST

OFFICE BUILDING WITH CLASSIFIED FACADE

SCOPE - **RECYCLE, REGENERATE**

ELEMENT		SYDNEY Cost Range	SYDNEY Mean Cost	Adel. Mean Cost	Bris. Mean Cost	Mel. Mean Cost	Per. Mean Cost
		$/sqm	$/sqm	$/sqm	$/sqm	$/sqm	$/sqm
TOTAL (MEAN) COST brought forward			1928.50	1791.25	1655.50	1726.25	1865.50
FITTINGS							
Fitments - Cupboards and fitments to service core areas, toilet accessories	...	27.25-30.25	28.75	26.50	23.00	27.00	28.00
Special - Not applicable	...	-	-	-	-	-	-
SERVICES							
Plumbing - New installation	...	137.75-152.75	145.25	148.75	122.00	135.75	146.00
Mechanical - New installation	...	431.00-451.00	441.00	399.50	333.00	397.25	407.25
Fire - New services	...	23.00-26.00	24.50	22.50	18.00	23.00	21.25
Electrical - New installation	...	208.00-218.00	213.00	209.00	173.50	198.50	217.00
Transportation - New installation	...	195.25-215.25	205.25	203.75	165.50	190.50	199.50
Special - Not applicable	...	-	-	-	-	-	-
DEMOLITIONS - Demolition of existing building apart from facade, temporary supports to facade	...	151.25-171.25	161.25	144.75	127.50	141.25	157.50
EXTERNAL SERVICES - Allowance	...	5.25-7.25	6.25	5.25	5.00	5.25	5.00
EXTERNAL WORKS - Allowance	...	1.25-2.25	1.75	1.75	1.75	2.00	2.50
CONTINGENCY - Allowance	...	-	114.50	107.00	95.25	103.25	110.50
TOTAL			3270.00	3060.00	2720.00	2950.00	3160.00

ADD extra for : Access flooring.
Asbestos removal.
Basement including tanking.
Canopies.

Check : All construction to accord with current regulations i.e. : fire rating of walls, ceilings, doors, panels, etc.; fire sprinklers; fireproofing of structural steelwork; glazing; staircases; plumbing, etc.

REFURBISHING AND RECYCLING BUILDINGS

DETAILED PRICES

In Refurbish and Recycle projects the wide variety of factors affecting cost make it impractical to give typical Detailed Prices. However, to give some indication of cost, a suitable method would be to take the Detailed Prices (pages 171 to 709) and apply the aggregate adjustment for the factors relevant to the particular project, these adjustment percentages could be in the order of :

		Structural, etc. Trades • Demolitions & Alterations • Excavation • Underpinning • Concrete • Brickwork & Blockwork • Masonry • Structural Steelwork	**General Trades** • Metalwork • Carpentry • Joinery • Windows & Doors • Hardware • Partitions • Roofing & Roof Plumbing • Miscellaneous	**Finishing Trades** • Plastering • Suspended Ceilings • Pavior & Terrazzo • Tiling - Non-Resilient • Resilient Finishes • Glazing • Painting	**Services** • Sanitary Plumbing • Drainage • Electrical • Mechanical • Transportation • Fire Protection
		Add	Add	Add	Add
Plant and equipment -					
Restrictions on use i.e. access and nuisance to occupiers	...	5%-7.5%	2.5%-5%	0%-5%	0%-2.5%
Inability to use, allowance for alternatives	...	5%-7.5%	2.5%-5%	0%-2.5%	0%-2.5%
Materials -					
Handling problems	...	1%-2.5%	1.5%-2.5%	1.5%-2.5%	1.5%-2.5%
Storage restraints	...	3%-5%	0%-2.5%	0%-2.5%	0%-2.5%
Labour -					
Loss of productivity on account of -					
Working in partially occupied buildings i.e. other floors occupied	...	5%-10%	1.5%-5%	2.5%-5%	2.5%-5%
Access and travelling within existing building	...	2%-3%	1%-2%	1%-2%	1%-2%
Work outside normal hours	...	20%-30%	10%-15%	10%-15%	10%-15%
Protection of existing building	...	2.5%-5%	0%-1.5%	0%-2.5%	0%-1.5%
Dust protection -					
Partially occupied building i.e. other than floor being worked on	...	1.5%-2.5%	0%-1%	1.5%-2.5%	0%-1%
Isolating areas on same floor	...	5%-7.5%	1%-2.5%	2.5%-5%	1.5%-2.5%
Matching existing construction, additional cost in cutting to fit and forming neat junction	...	5%-10%	5%-10%	5%-10%	2.5%-5%
Locality factor i.e. CBD loadings	...	10%-20%	5%-10%	2.5%-10%	2.5%-7.5%
Temporary shoring and supports as required - evaluate for particular project	...	Note			
Security if more than usual - evaluate for particular project	...	Note			
Heritage factor, could be significant - examine and evaluate for any specific problems	...	Note			
Preliminaries - evaluate for particular project	...	Note			
Safety, health and welfare regulations and requirements - evaluate for particular project	...	Note			

LABOUR AND PLANT CONSTANTS

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The labour constants given are for the major items of on-site works for which prices are given in the Detailed Prices section.

An examination of the Detailed Prices indicates that, in certain instances, considerable variation can occur in the constants for the respective cities. This can sometimes be due to the differing output of Sub-Contractor's labour compared with the labour employed directly by the Contractor or to the established labour factors of a particular city.

In using the constants, it must be appreciated that each Contractor will have his own opinion as to what the constant should be on variables such as the type of job, site conditions, location and scope for mechanical plant use. For these reasons, the following constants are intended as an average guide only.

EXCAVATION

		UNIT	Tradesman Hours	Labourer Hours
EXCAVATION AND BACKFILLING				
Hand excavation in :				
Trenches N/E 1.00m deep -				
in sand	...	cum		1.50
in medium soil	...	cum		2.50
in heavy soil	...	cum		3.50
Loading and removal :				
in sand	...	cum		0.60
in medium soil	...	cum		0.80
in heavy soil	...	cum		1.00
Wheel N/E 50.00m, spread and level	...	cum		1.00
Backfilling and compaction	...	cum		0.60
Machine excavation and backfilling in :				
Trenches N/E 1.00m deep in medium soil	...	cum		*0.20
*ADD :				
Backhoe + operator - 0.20 hours/cum				
F/E loader + operator - 0.08 hours/cum				
Tipper + operator - 0.10 hours/cum				
Vibrating plate - 0.10 hours/cum				
SURFACE TREATMENTS				
Compaction of :				
Trenches, pits and the like	...	sqm		0.08
Ground under slabs	...	sqm		0.05
Level and grade surface	...	sqm		0.05
MEMBRANES				
Plastic film underlay	...	sqm		0.06

CONCRETE WORK

		UNIT	Tradesman Hours	Labourer Hours
CONCRETE				
Unreinforced readymixed concrete placed in :				
Blinding layer 50mm thick	...	cum		2.00
Column and pier foundations	...	cum		0.90
Strip footings	...	cum		0.90
Machine bases	...	cum		1.70
Plinths, kerbs and steps	...	cum		1.70
Cavity wall filling	...	cum		2.70
Reinforced readymixed concrete placed :				
Raft foundations	...	cum		0.80
Column and pier foundations	...	cum		0.90
Strip footings	...	cum		0.90
Slabs and thickenings on fill	...	cum		1.25
Suspended slabs 150/300mm thick	...	cum		1.25
Walls 150/300mm thick	...	cum		1.30
Columns	...	cum		1.80
Attached beams	...	cum		1.25
Isolated beams	...	cum		2.00
Stairs and landings	...	cum		1.80
Lintels	...	cum		2.80
Casing :				
Columns	...	cum		2.40
Beams	...	cum		1.85
Lintels	...	cum		3.45
Filling :				
Piers	...	cum		5.50
Cores of hollow blocks	...	cum		6.50
Lintel blocks	...	cum		5.75
JOINTS				
Keyed construction joint in :				
150mm slab	...	m	0.30	
300mm slab	...	m	0.35	
Expansion joint in :				
150mm slab	...	m	0.50	
300mm slab	...	m	0.60	
150mm wall	...	m	0.60	
300mm wall	...	m	0.70	
WATERSTOPS				
Double bulb waterstop, cast in, excluding angles and intersections :				
150mm wide	...	m	0.250	
200mm wide	...	m	0.275	

CONCRETE WORK

		UNIT	Tradesman Hours	Labourer Hours
SURFACE FINISHES				
Broomed finish	...	sqm		0.12
Monolithic finish :				
Hand trowel	...	sqm		0.15
Machine trowel	...	sqm		0.08
FORMWORK				
• Constants assume five uses, where the number of uses is less, the constants should be adjusted by :				
Four uses + 2.5%				
Three uses + 6.0%				
Two uses + 12.5%				
One use + 20.0%	...	Note		
Sides of column or pier foundation :				
Class 5	...	sqm	0.65	0.22
Sides of strip footing :				
Class 5	...	sqm	0.65	0.22
Soffit of suspended slab :				
Class 2	...	sqm	1.10	0.35
Class 3	...	sqm	0.95	0.33
Class 4	...	sqm	0.90	0.30
Sides of wall :				
Class 2	...	sqm	1.15	0.38
Class 3	...	sqm	1.05	0.35
Class 4	...	sqm	0.95	0.33
Sides of column :				
Class 2	...	sqm	1.55	0.52
Class 3	...	sqm	1.40	0.47
Class 4	...	sqm	1.35	0.45
Sides of attached beam :				
Class 2	...	sqm	1.15	0.38
Class 3	...	sqm	1.05	0.35
Class 4	...	sqm	0.95	0.33
Soffits of stairs and landings :				
Class 2	...	sqm	1.60	0.54
Class 3	...	sqm	1.50	0.50
Class 4	...	sqm	1.40	0.47
Edges, sides or risers, Class 3 :				
N/E 100mm high	...	100m	16.00	5.35
100/200mm high	...	100m	24.00	8.00
200/300mm high	...	100m	32.00	10.70
Form groove, 25 x 25mm	...	m	0.13	0.04
Form rebate, 100 x 50mm	...	m	0.20	0.07
Form hole/opening in concrete :				
100/200mm thick, size -				
100 x 100mm	...	No	0.24	0.08
300 x 300mm	...	No	0.40	0.14
500 x 500mm	...	No	0.60	0.20

CONCRETE WORK

		UNIT	Tradesman Hours	Labourer Hours
FORMWORK (cont'd)				
Form hole/opening in concrete (cont'd) :				
200/300mm thick, size -				
100 x 100mm	...	No	0.30	0.10
300 x 300mm	...	No	0.50	0.17
500 x 500mm	...	No	0.65	0.22
Coffer pans (handling, placing, fixing, stripping and cleaning), size :				
500 x 500 x 200mm deep	...	No	0.20	0.07
1000 x 1000 x 250mm deep	...	No	0.40	0.13
Tube formers :				
Fibre tube, including stripping -				
300 I.D.	...	m	0.93	0.31
500 I.D.	...	m	1.16	0.39
F.R.C. left in -				
200mm dia.	...	m	0.82	0.27
345mm dia.	...	m	1.00	0.33
585mm dia.	...	m	1.20	0.40
Spirally wound plastic tube, including stripping -				
300 I.D.	...	m	0.79	0.26
500 I.D.	...	m	0.98	0.33
REINFORCEMENT				
Bar reinforcement, placing and fixing :				
6 - 12mm ligatures	...	t	17.00	
16 - 33mm reinforcement in -				
Slabs	...	t	9.00	
Walls	...	t	11.00	
Columns and beams	...	t	11.00	
Mesh reinforcement, placing and fixing :				
Medium weight mesh in -				
Slabs	...	sqm	0.08	
Walls	...	sqm	0.10	
PRECAST CONCRETE				
Erection and fixing in position (excluding craneage and operators) :				
Wall panel -				
100mm thick	...	sqm	0.70-1.10	
150mm thick	...	sqm	0.80-1.20	
Hollow slab units -				
150mm thick	...	sqm	0.25-0.40	
200mm thick	...	sqm	0.30-0.50	

BRICKWORK AND BLOCKWORK

		UNIT	Tradesman Hours	Labourer Hours
BRICKWORK				
Stretcher bond brickwork :				
Mass brickwork -				
Standard	...	1000 bks	13.00	6.50
Modular	...	1000 bks	17.00	8.50
General walls -				
Standard	...	1000 bks	14.00	7.00
Modular	...	1000 bks	18.00	9.00
Facing, pointing and cleaning down -				
Standard	...	100 sqm	15.00	5.00
Modular	...	100 sqm	15.00	5.00
BRICKWORK LABOURS ETC.				
Cutting etc. :				
Rough raking cutting (per 110mm thickness)	...	m	0.18	0.18
Rough circular cutting (per 110mm thickness)	...	m	0.23	0.23
Rough chase, 50 x 25mm deep	...	m	0.23	0.23
Fair raking cutting (per 110mm thickness)	...	m	0.21	0.21
Fair circular cutting (per 110mm thickness)	...	m	0.28	0.28
Cut, tooth and bond new work to old (per 110mm thickness)	...	m	0.40	0.40
Level and prepare old work for raising :				
110mm wide	...	m	0.11	0.11
230mm wide	...	m	0.16	0.16
Pin up new work to old soffit (per 110mm thickness)	...	m	0.21	0.21
Lintels :				
Building in of -				
Steel flat arch bar 76 x 10mm	...	m	0.03	0.03
Steel angle :				
76 x 76 x 8mm	...	m	0.04	0.04
152 x 89 x 10mm	...	m	0.04	0.04
Sills, Copings :				
Brick on edge, sill set sloping	...	m	0.45	0.15
Brick on edge coping	...	m	0.36	0.12
Tile sill in 150 x 150mm tiles	...	m	0.32	0.11
BLOCKWORK				
Walls of :				
Hollow blocks -				
400 x 200 x 100mm	...	100 sqm	43.00	21.50
400 x 200 x 150mm	...	100 sqm	50.00	25.00
400 x 200 x 200mm	...	100 sqm	56.00	28.00
Solid blocks -				
400 x 200 x 100mm	...	100 sqm	47.00	23.50
400 x 200 x 150mm	...	100 sqm	57.00	28.50
400 x 200 x 200mm	...	100 sqm	65.00	32.50
400 x 200 x 200mm hollow block threaded over bar reinforcement	...	100 sqm	96.00	28.00
Pointing and cleaning down	...	100 sqm	9.00	4.00

BRICKWORK AND BLOCKWORK

		UNIT	Tradesman Hours	Labourer Hours
BLOCKWORK LABOURS ETC.				
Cutting, etc. :				
Fair raking cutting -				
100mm hollow block	...	m	0.18	0.18
200mm hollow block	...	m	0.28	0.28
Concrete filling to cores of hollow blocks :				
150mm blocks	...	sqm		0.45
200mm blocks	...	sqm		0.55
Lintel block including filling :				
200mm wide x 200mm high x 200mm long	...	m	0.40	0.14
200mm wide x 400mm high x 200mm long	...	m	0.54	0.18
BRICKWORK AND BLOCKWORK SUNDRIES				
Damp-proof courses, building in of bitumen felt or metal cored material :				
100/110mm wide	...	m	0.005	
230mm wide	...	m	0.008	
Joints and sealants :				
9mm wide control joint in 110mm wall including pointing one face	...	m	0.20	
6mm wide mastic pointing between metal window and wall	...	m	0.10	

STRUCTURAL STEELWORK

STRUCTURAL STEELWORK				
Framing and bracing including connections :				
Heavy section (UB, UC, RSJ and PFC) -				
Fabrication	...	t	23.00	
Erection	...	t	7.00	
Light section (angles etc.) -				
Fabrication	...	t	40.00	
Erection	...	t	5.00	
Hollow sections and connections -				
Large sections :				
Fabrication	...	t	30.00	
Erection	...	t	6.00	
Small sections :				
Fabrication	...	t	35.00	
Erection	...	t	5.00	
Trusses and connections -				
Up to 10m span :				
Fabrication	...	t	50.00	
Erection	...	t	5.00	
Over 10m span :				
Fabrication	...	t	35.00	
Erection	...	t	8.00	

STRUCTURAL STEELWORK

		UNIT	Tradesman Hours	Labourer Hours
PURLINS AND GIRTS				
Proprietary purlins and girts including laps, bracing, etc. :				
Up to 150mm high	...	m	0.085	
Over 150mm high	...	m	0.12	
SUNDRIES				
Site fillet weld :				
6mm	...	m	0.30	
12mm	...	m	0.70	
Fixings including drilling concrete :				
Sleeve anchor, 16mm dia. x 110mm	...	No	0.14	
Stud anchor, 16mm dia. x 150mm	...	No	0.14	
SURFACE TREATMENT				
Sand blast and paint :				
Light sections	...	t	8.00	
Heavy sections	...	t	5.00	

METALWORK

Air relief grille, take delivery and fix only to timber :				
600 x 150mm	...	Each	0.50	
600 x 300mm	...	Each	0.50	
600 x 600mm	...	Each	0.55	
Angle trim, aluminium, screwed to backing surface :				
12 x 12mm angle	...	m	0.10	
25 x 25mm angle	...	m	0.10	
45 x 25mm angle	...	m	0.11	
Bench supports, tubular framing with welded joints :				
Steel	...	t	200.00-300.00	
Stainless steel	...	t	250.00-375.00	
Brackets, small section steel angle or flat	...	t	200.00-250.00	
Cladding, sheet material fixed to framing :				
Aluminium, 1.6mm	...	sqm	2.00	
Galvanised or Zincalume® steel, 0.9/1.0mm	...	sqm	1.45	
Stainless steel, 0.9mm	...	sqm	2.40	
Covers and gratings, take delivery, build in frame and place in position :				
Cast iron manhole covers, including concrete filling -				
Light duty :				
450 x 450mm	...	Each	0.70	0.70
600 x 600mm	...	Each	0.85	0.85
900 x 600mm	...	Each	1.00	1.00
Medium duty :				
450 x 450mm	...	Each	0.70	0.70
600 x 600mm	...	Each	0.90	0.90
900 x 600mm	...	Each	1.10	1.10
Heavy duty :				
450 x 450mm	...	Each	0.75	0.75
600 x 600mm	...	Each	0.95	0.95
900 x 600mm	...	Each	1.15	1.15

METALWORK

		UNIT	Tradesman Hours	Labourer Hours
Covers and gratings, take delivery, build in frame and place in position :				
Cast iron trench grating -				
Light service :				
225mm wide	...	m	0.35	0.35
450mm wide	...	m	0.60	0.60
Heavy service :				
225mm wide	...	m	0.50	0.50
450mm wide	...	m	0.75	0.75
Ladder, 450mm wide fully welded steel and shop primed	...	t	100.00	
Staircases, balustrades, fully welded steel, shop primed :				
Staircase including treads	...	t	90.00-120.00	
Balustrade 900mm high with balusters at 150mm centres	...	t	100.00	
Toilet accessories fixed to brick/blockwork :				
Small	...	Each	0.30-0.50	
Large	...	Each	1.00-2.00	

CARPENTRY

- Constants based on softwood. For hardwood ADD 5%-12.5% ... Note
- All sawn timber unless otherwise described ... Note

FLOOR FRAMING

Ground Floor :			
100 x 25mm plate	...	100m	6.00
100 x 75mm bearer	...	100m	15.00
150 x 75mm bearer	...	100m	19.50
100 x 50mm joist	...	100m	10.00
150 x 50mm joist	...	100m	14.50
200 x 50mm joist	...	100m	18.50
250 x 50mm joist	...	100m	22.50
Upper Floor :			
150 x 50mm joist	...	100m	15.50
200 x 50mm joist	...	100m	20.50
250 x 50mm joist	...	100m	24.50
150 x 50mm blocking	...	100m	17.50
200 x 50mm blocking	...	100m	22.50
250 x 50mm blocking	...	100m	26.50

WALL FRAMING

75 x 38mm plate	...	100m	10.50
75 x 50mm plate	...	100m	14.50
100 x 50mm plate	...	100m	17.50
75 x 38mm stud	...	100m	9.50
100 x 38mm stud	...	100m	11.50
100 x 50mm stud	...	100m	13.50
100 x 100mm stud	...	100m	21.00
75 x 38mm nogging	...	100m	10.50
100 x 38mm nogging	...	100m	14.00
100 x 50mm nogging	...	100m	15.50
150 x 50mm lintel	...	100m	25.00
200 x 50mm lintel	...	100m	31.50

CARPENTRY

		UNIT	Tradesman Hours	Labourer Hours
ROOF AND CEILING FRAMING				
100 x 38mm plate	...	100m	11.00	
100 x 50mm plate	...	100m	13.50	
100 x 38mm rafter	...	100m	9.50	
150 x 38mm rafter	...	100m	13.00	
100 x 50mm rafter	...	100m	12.50	
150 x 50mm rafter	...	100m	18.00	
200 x 50mm rafter	...	100m	23.00	
100 x 50mm purlin	...	100m	14.00	
150 x 50mm purlin	...	100m	19.00	
200 x 50mm purlin	...	100m	21.00	
150 x 25mm ridge	...	100m	11.50	
200 x 25mm ridge	...	100m	14.00	
200 x 25mm hip, valley board	...	100m	14.00	
200 x 38mm hip, valley board	...	100m	18.50	
100 x 50mm ceiling joist	...	100m	12.50	
150 x 50mm ceiling joist	...	100m	17.00	
200 x 50mm ceiling joist	...	100m	22.00	
200 x 50mm hanger	...	100m	22.00	
250 x 50mm hanger	...	100m	26.50	
75 x 50mm struts, ties	...	100m	11.50	
100 x 50mm struts, ties	...	100m	15.00	
Fabricated roof trusses, erection :				
6000mm span	...	Each	1.25	
9000mm span	...	Each	1.65	
12000mm span	...	Each	2.10	
Grounds, battens :				
25 x 19mm	...	100m	2.75	
38 x 19mm	...	100m	2.75	
38 x 25mm	...	100m	2.75	
50 x 25mm	...	100m	3.25	
50 x 38mm	...	100m	3.25	
75 x 38mm	...	100m	4.25	
Fascia, barges, etc. :				
175 x 31mm	...	100m	21.00	
200 x 31mm	...	100m	21.50	
250 x 31mm	...	100m	23.00	
BEAMS				
Beam :				
150 x 50mm	...	100m	21.00	
200 x 50mm	...	100m	28.00	
250 x 50mm	...	100m	35.00	
300 x 75mm	...	100m	56.00	
Laminated timber beams, erection :				
190 x 80mm	...	100m	25.00	
270 x 80mm	...	100m	30.00	
380 x 80mm	...	100m	40.00	
460 x 130mm	...	100m	60.00	
POSTS				
100 x 100mm post	...	100m	28.50	

CARPENTRY

		UNIT	Tradesman Hours	Labourer Hours
INSULATION				
Aluminium foil insulation laid over steel or timber framing	...	100sqm	5.25	
Batt insulation :				
Laid between steel or timber framing	...	100sqm	5.75	
Fixed vertically to framing	...	100sqm	12.75	
Blanket insulation :				
Laid over purlin	...	100sqm	5.25	
FLOORING				
• All fixed to timber framing	...	Note		
Compressed fibre cement sheets :				
12mm thick	...	100sqm	30.50	
18mm thick	...	100sqm	39.00	
Hardboard sheets 4.8mm thick	...	100sqm	19.00	
Particle board panels :				
19mm thick	...	100sqm	21.00	
22mm thick	...	100sqm	22.00	
Parquetry block/mosaic block including bedding and polishing	...	100sqm	80.00	
Plywood, tongued and grooved :				
15mm thick	...	100sqm	19.00	
19mm thick	...	100sqm	21.00	
Timber :				
Decking -				
75 x 25mm thick	...	100sqm	52.50	
100 x 25mm thick	...	100sqm	42.00	
Tongued and grooved -				
100 x 25mm thick	...	100sqm	42.50	
WALL LININGS				
• All fixed to timber framing	...	Note		
ADD extra for :				
Fixing to metal framing	...	100sqm	3.00	
Caneite sheet, 12.7mm thick	...	100sqm	26.50	
Fibreboard :				
Plain sheet -				
6mm thick	...	100sqm	34.00	
9mm thick	...	100sqm	36.00	
12mm thick	...	100sqm	38.00	
18mm thick	...	100sqm	42.50	
25mm thick	...	100sqm	47.50	
ADD extra for :				
Veneer faced	...	100sqm	10.00	
Melamine faced	...	100sqm	10.00	

CARPENTRY

		UNIT	Tradesman Hours	Labourer Hours
WALL LININGS (cont'd)				
Fibre cement sheet :				
Flexible -				
4.5mm thick	...	100sqm	32.00	
6.0mm thick	...	100sqm	35.00	
Decorative, planks, 7.5mm thick -				
205mm wide	...	100sqm	52.50	
230mm wide	...	100sqm	47.50	
Compressed -				
6mm thick	...	100sqm	47.50	
9mm thick	...	100sqm	52.50	
12mm thick	...	100sqm	62.50	
15mm thick	...	100sqm	68.75	
18mm thick	...	100sqm	73.75	
Villaboard -				
6mm thick	...	100sqm	50.00	
9mm thick	...	100sqm	57.50	
12mm thick	...	100sqm	65.00	
Hardboard :				
Standard, 3.2mm thick -	...	100sqm	28.50	
4.8mm thick	...	100sqm	29.50	
6.4mm thick	...	100sqm	34.00	
Pegboard, 4.8mm thick	...	100sqm	31.50	
Tempered, 4.8mm thick	...	100sqm	29.50	
Planks, 9.5mm thick -				
200mm wide	...	100sqm	47.50	
300mm wide	...	100sqm	42.00	
Particle board :				
Plain sheet -				
9mm thick	...	100sqm	36.00	
12mm thick	...	100sqm	38.00	
18mm thick	...	100sqm	42.50	
25mm thick	...	100sqm	47.50	
ADD extra for :				
Veneer faced	...	100sqm	10.00	
Melamine faced	...	100sqm	10.00	
Plywood :				
Plain sheet -				
4mm thick	...	100sqm	34.50	
6mm thick	...	100sqm	37.00	
9mm thick	...	100sqm	41.00	
12mm thick	...	100sqm	45.00	
15mm thick	...	100sqm	49.50	
18mm thick	...	100sqm	52.50	
25mm thick	...	100sqm	55.50	
ADD extra for :				
Veneer faced	...	100sqm	10.00	
Timber boarding :				
Weatherboarding, 25mm thick	...	100sqm	47.50	
Shiplap panelling, 25mm thick	...	100sqm	60.00	
Tongued and grooved, 15/19mm thick	...	100sqm	57.50	

CARPENTRY

		UNIT	Tradesman Hours	Labourer Hours
CEILING AND EAVES LININGS				
Cellulose fibre cement, 4.5mm thick fixed to timber framing in :				
Ceiling linings	...	100sqm	50.50	
Eaves linings -				
N/E 2000mm wide	...	100sqm	56.00	
2000/4000mm wide	...	100sqm	53.00	
Exceeding 4000mm wide	...	100sqm	50.00	
FIXINGS				
Drill brick wall for 12mm dia. masonry anchor	...	Each	0.10	
Drill concrete wall for 12mm dia. masonry anchor	...	Each	0.13	
Drill timber for 6/12mm dia. bolt	...	Each	0.05	
Drill steel angle for 6/12mm dia. bolt	...	Each	0.13	

JOINERY

COVER MOULDS				
19mm quadrant fillet fixed to timber	...	100m	8.00	
38mm sector fixed to timber	...	100m	9.00	
DOOR FRAMES, ARCHITRAVES				
• Refer WINDOWS AND DOORS	...	Note		
HAT AND COAT RAIL				
100 x 25mm rail plugged to brick	...	100m	19.00	
PELMETS				
150 x 25mm fascia or top	...	100m	17.00	
Returned end	...	Each	0.23	
SKIRTINGS				
Skirting fixed to timber :				
75 x 25mm	...	100m	12.00	
150 x 25mm	...	100m	15.00	
150 x 25mm (moulded)	...	100m	16.00	
ADD extra for :				
Plugging to brick or concrete	...	100m	9.50	
SEAT BATTEN				
75 x 38mm seat batten fixed to metal	...	100m	14.00	
STAIRCASES				
Staircases comprising strings, treads, landing supports and balustrades	...	M/Rise	10.00	
Handrail, 150 x 50mm fixed to steel including ramps and wreaths	...	100m	55.00	

JOINERY

		UNIT	Tradesman Hours	Labourer Hours
SHELVING				
50 x 25mm ladder framing	...	100m	21.00	
50 x 25mm slat shelving with battens at 75mm centres	...	sqm	1.15	
Shelving :				
150 x 25mm	...	100m	20.00	
200 x 25mm	...	100m	22.50	
16mm particle board shelving :				
450mm wide	...	100m	27.50	
600mm wide	...	100m	31.00	
CUPBOARDS AND FITTINGS				
Fabricate and fix :				
Bench cupboard unit 450mm wide x 900mm high, hinged doors, laminate top -				
Finished for painting	...	m	6.00-8.00	
Laminate finish	...	m	7.00-9.00	
ADD extra for :				
Drawer	...	No	1.25	
Wall cupboard 300mm wide x 600mm high, sliding doors -				
Finished for painting	...	m	4.00-5.00	
Laminate finish	...	m	4.50-6.00	
Vanity bench top 450mm wide x 200mm high, laminate finished and prepared to receive basin	...	m	3.50-4.50	
NOTICE BOARDS, CHALKBOARDS				
Pinboard, welded vinyl on Caneite and particle board backing and with angle surround	...	sqm	1.50	
Pegboard, 5mm pegboard on 19mm timber framing and with aluminium angle surround	...	sqm	1.40	
Chalkboard, 4.8mm Timsonplate on 19mm timber framing and aluminium angle surround :				
Fixed boards	...	sqm	1.40	
Pivoting or sliding boards	...	sqm	2.00	
White board, 0.5mm vitreous porcelain finished steel laminated to styrene foam with particle board backing	...	sqm	1.40	
LETTERING, FIX ONLY				
Metal lettering plugged and screwed to brickwork :				
50mm high	...	Each	0.25	
100mm high	...	Each	0.35	
200mm high	...	Each	0.45	

WINDOWS AND DOORS

		UNIT	Tradesman Hours	Labourer Hours
WINDOWS, FIX ONLY				
Aluminium :				
600 x 1200mm wide	...	Each	0.45	
1200 x 1200mm wide	...	Each	0.50	
1200 x 2400mm wide	...	Each	0.70	
2000 x 1200mm wide	...	Each	0.70	
2000 x 2400mm wide	...	Each	0.90	
Timber :				
600 x 1200mm wide	...	Each	0.50	
1200 x 1200mm wide	...	Each	0.60	
1200 x 2400mm wide	...	Each	0.80	
2000 x 1200mm wide	...	Each	0.80	
2000 x 2400mm wide	...	Each	1.00	
METAL DOOR FRAMES, FIX ONLY				
Metal frame to suit door size :				
2040 x 820mm in 110 brick wall	...	Each	0.60	
2640 x 820mm in 110 brick wall	...	Each	0.80	
TIMBER WINDOW AND DOOR FRAMES				
Fabricate and fix :				
Plain lining -				
100 x 25mm	...	100m	23.00	
100 x 38mm	...	100m	23.50	
150 x 25mm	...	100m	23.50	
150 x 38mm	...	100m	24.00	
Frame -				
100 x 50mm	...	100m	25.50	
100 x 75mm	...	100m	27.00	
150 x 50mm	...	100m	27.00	
150 x 75mm	...	100m	31.50	
ADD extra for :				
Rebate	...	100m	3.75	
Throat	...	100m	3.75	
Weather	...	100m	4.75	
TIMBER DOORS, FIX ONLY				
Hollow core flush door, size :				
2040 x 820mm on two hinges	...	Each	1.30	
2640 x 820mm on three hinges	...	Each	1.50	
Solid core flush door, size :				
2040 x 820mm on three hinges	...	Each	1.50	
2640 x 820mm on four hinges	...	Each	1.70	
Panelled, framed and glazed door, size :				
2040 x 820mm on two hinges	...	Each	1.50	
2640 x 820mm on three hinges	...	Each	1.70	
GLAZING BEADS, STOPS, FILLETS				
Glazing bead -				
19 x 13mm	...	100m	8.50	
25 x 13mm	...	100m	9.50	
Stop, fillet -				
38 x 19mm	...	100m	9.00	
50 x 25mm	...	100m	9.50	

WINDOWS AND DOORS

		UNIT	Tradesman Hours	Labourer Hours
ARCHITRAVES				
Architrave to timber door frame :				
50 x 25mm	...	100m	8.00	
75 x 25mm	...	100m	8.50	
100 x 25mm	...	100m	9.00	
ROLLER SHUTTERS, FIX ONLY				
Domestic shutter, 2100 x 2400mm wide	...	Each	4.00	
Industrial shutter, 3000 x 3000mm wide	...	Each	6.00	

HARDWARE AND DOOR SUNDRIES

FIX ONLY				
Hinges - included with hanging door	...	Note		
Lockset and furniture	...	Set	0.85	
Closers :				
Surface mounted	...	Each	1.25	
Concealed	...	Each	1.65	
Floor spring, light duty	...	Each	2.50	
Bolt :				
Barrel, 200mm	...	Each	0.29	
Flush, 200mm	...	Each	0.85	
Panic bolt :				
Vertical rod and crossbar to single door	...	Set	1.10	
Vertical rod and (2) crossbars to double door	...	Set	2.40	
Terrazzo set	...	Set	1.55	
Push plate, 300 x 75mm, metal	...	Each	0.13	
Kick plate, 820 x 225mm, metal	...	Each	0.21	
Sliding track including fittings (heavy duty) to :				
820mm wide door	...	Set	1.35	
1200mm wide door	...	Set	1.60	

ROOFING AND ROOF PLUMBING

TILE ROOFING				
Standard pattern with 38 x 25mm battens	...	100sqm	17.00	8.50
Ridge, hip, verge tiles	...	m	0.24	0.12
FIBRE CEMENT ROOFING				
Corrugated roofing with safety mesh, fixed to timber	...	100sqm	14.00	7.00
Ridge, hip, barge moulds	...	m	0.14	0.07
FIBREGLASS ROOFING				
Corrugated translucent fibreglass fixed to timber	...	100sqm	21.00	

ROOFING AND ROOF PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
POLYCARBONATE ROOFING				
Corrugated/ribbed polycarbonate, 0.8mm thick, fixed to timber	...	100sqm	22.50	
METAL ROOFING				
Corrugated steel roofing on timber purlins	...	100sqm	20.00	
Ribbed steel/aluminium roofing on timber purlins	...	100sqm	22.00	
ADD extra for :				
Fixing to steel	...	100sqm	4.50	
INSULATION				
Refer CARPENTRY	...	Note		
ROOF PENETRATIONS				
Tiled roof :				
Penetration for 100mm dia. pipe and 600 x 600mm tray	...	Each	1.00	
Metal roof :				
Penetration for 100mm dia. pipe, 600 x 600mm tray and seal	...	Each	1.00	
FLASHINGS				
Copper flashing :				
Straight -				
250mm girth	...	m	0.425	
500mm girth	...	m	0.525	
750mm girth	...	m	0.625	
Stepped -				
250mm girth	...	m	0.925	
500mm girth	...	m	1.050	
750mm girth	...	m	1.150	
Zincalume steel flashing :				
Straight -				
250mm girth	...	m	0.375	
500mm girth	...	m	0.475	
750mm girth	...	m	0.575	
Stepped -				
250mm girth	...	m	0.875	
500mm girth	...	m	1.000	
750mm girth	...	m	1.100	
Stainless steel flashing :				
Straight -				
250mm girth	...	m	0.625	
500mm girth	...	m	0.725	
750mm girth	...	m	0.825	
Stepped -				
250mm girth	...	m	1.150	
500mm girth	...	m	1.250	
750mm girth	...	m	1.350	

ROOFING AND ROOF PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
DOWNPIPES				
Downpipe with brackets at 900mm centres :				
Copper, 100mm dia.	...	m	0.335	
Zincalume® steel, 100mm dia.	...	m	0.335	
Stainless steel, 100mm dia.	...	m	0.465	
PVC, 100 x 50mm	...	m	0.290	
ADD extra for :				
Bend, elbow or shoe -				
Copper, 100mm dia.	...	Each	0.950	
Zincalume® steel, 100mm dia.	...	Each	0.575	
Stainless steel, 100mm dia.	...	Each	1.175	
PVC, 100 x 50mm	...	Each	0.365	
EAVES GUTTERS				
Eaves gutters with brackets/straps at 900mm centres :				
Copper	...	m	0.390	
Zincalume® steel	...	m	0.340	
Stainless steel	...	m	0.440	
PVC	...	m	0.290	
ADD extra for :				
Mitred angle, stopped end, outlet and joint -				
Copper	...	Each	0.640	
Zincalume® steel	...	Each	0.520	
Stainless steel	...	Each	0.725	
PVC	...	Each	0.490	
Expansion joint -				
Copper	...	Each	1.325	
Zincalume® steel	...	Each	1.225	
Stainless steel	...	Each	1.525	
BOX GUTTERS				
Box gutter 1000mm girth with metal straps/brackets at 900mm centres :				
Copper	...	m	0.750	
Zincalume® steel	...	m	0.650	
Stainless steel	...	m	0.800	
ADD extra for :				
Mitred angle -				
Copper	...	Each	0.875	
Zincalume® steel	...	Each	0.625	
Stainless steel	...	Each	0.925	
Stopped end -				
Copper	...	Each	1.075	
Zincalume® steel	...	Each	0.775	
Stainless steel	...	Each	1.175	
Expansion joint -				
Copper	...	Each	1.925	
Zincalume® steel	...	Each	1.825	
Stainless steel	...	Each	2.850	
Outlet and joint -				
Copper	...	Each	1.075	
Zincalume® steel	...	Each	0.775	
Stainless steel	...	Each	1.175	

ROOFING AND ROOF PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
WALL CLADDING				
Wall cladding to steel girts :				
Corrugated fibreglass	...	100sqm	31.00	
Corrugated steel	...	100sqm	29.00	
Ribbed steel/aluminium	...	100sqm	31.00	

PLASTERING

Render for wall tiling, cement and sand in one coat 13mm thick	...	sqm	0.20	0.10
Cement render, one coat 13mm thick on :				
Walls	...	sqm	0.25	0.11
Soffits	...	sqm	0.28	0.11
Columns	...	sqm	0.42	0.18
Beams	...	sqm	0.42	0.18
Cement render, two coats 19mm thick on :				
Walls	...	sqm	0.40	0.17
Soffits	...	sqm	0.45	0.19
Columns	...	sqm	0.50	0.20
Beams	...	sqm	0.50	0.20
Hardwall plaster, two coats 16mm thick on :				
Walls	...	sqm	0.42	0.18
Soffits	...	sqm	0.50	0.20
Columns	...	sqm	0.55	0.25
Beams	...	sqm	0.55	0.25
Skim coat hardwall plaster on soffits	...	sqm	0.22	0.08
Expanded metal lath fixed to :				
Wall studs -				
Timber	...	sqm	0.125	
Metal	...	sqm	0.140	
Soffit of timber joists	...	sqm	0.155	
Metal corner bead	...	100m	10.00	
Plasterboard sheeting 10mm thick and fixing to timber framing in :				
Walls	...	sqm	0.30	
Ceilings	...	sqm	0.36	
ADD extra for :				
Fixing to steel framing	...	sqm	0.04	

SUSPENDED CEILINGS

SUSPENSION SYSTEMS				
Fully concealed suspension system suspended 600mm from concrete soffit or steel purlins	...	sqm	0.375	
Exposed grid system suspended 600mm from concrete soffit or steel purlins	...	sqm	0.325	

SUSPENDED CEILINGS

		UNIT	Tradesman Hours	Labourer Hours
CEILINGS				
• ADD for suspension system as foregoing	...	Note		
Plasterboard 10mm thick or veneered plywood fixed to concealed suspension system	...	sqm	0.40	
Plasterboard tiles set into exposed grid system	...	sqm	0.20	
Veneered plywood panels set into exposed grid system	...	sqm	0.20	

PAVIOR

Granolithic 19mm thick in :				
Paving to -				
Floors	...	sqm	0.34	0.11
Surfaces of ducts, pits, etc.	...	sqm	0.43	0.12
Skirting x 100mm high	...	m	0.25	
Stair tread x 300mm wide	...	m	0.25	
Stair riser x 150mm high	...	m	0.25	
Division strip, brass angle or flat, FIX ONLY	...	m	0.10	
Matwell frame, 1000 x 700mm in brass angle, FIX ONLY	...	Each	0.60	

TILING - NON RESILIENT

Wall tiles, ceramic or mosaic fixed with adhesive and including pointing and cleaning	...	sqm	1.00	
Floor tiles, ceramic or mosaic including bedding, pointing and cleaning	...	sqm	1.00	

RESILIENT FINISHES

Carpet laying :				
Directly fixed carpet	...	sqm	0.150	
Carpet and rubber underlay	...	sqm	0.175	
Cork tiles and clear plastic finish	...	sqm	0.60	
Linoleum or vinyl fixed with adhesive :				
300 x 300 x 2.5mm thick tiles	...	sqm	0.25	
2.5mm thick sheet	...	sqm	0.25	
Foam backed sheet	...	sqm	0.30	

GLAZING

Glazing in regular panes approximately 1000 x 1000mm to :				
Timber with putty	...	sqm	0.90	
Timber with beads	...	sqm	0.90	
Metal with putty/mastic and/or neoprene strips	...	sqm	1.00	

PAINTING

		UNIT	Tradesman Hours	Labourer Hours
PAINTING				
Acrylic, low sheen finish, prepare and apply two coats on smooth finished :				
Walls	...	sqm	0.15	
Ceilings and soffits	...	sqm	0.17	
Acrylic, latex, flat finish, prepare and apply two coats on smooth finished :				
Walls	...	sqm	0.13	
Ceilings and soffits	...	sqm	0.15	
Acrylic, semi gloss finish, prepare, prime and apply one coat sealer undercoat and two coats finish on general timber/metal surfaces	...	sqm	0.30	
Enamel, prepare and apply one coat sealer and two coats finish on smooth finished :				
Walls	...	sqm	0.24	
Ceilings and soffits	...	sqm	0.27	
Enamel, gloss finish, prepare, prime and apply one undercoat and two coats finish on general timber/metal surfaces	...	sqm	0.34	
Polyurethane, clear finish, prepare and apply two coats on general timber surfaces	...	sqm	0.16	
Stain, prepare and apply two coats on general timber surfaces	...	sqm	0.14	
PAPER HANGING				
Prepare and hang on :				
Walls	...	sqm	0.25	
Ceilings	...	sqm	0.27	

SANITARY PLUMBING

SANITARY FIXTURES				
Fix only including brackets, fixings, etc :				
Vanity basin	...	Each	0.75	
Wall basin	...	Each	2.00	
Bath	...	Each	2.00	
Bidet (excluding heater)	...	Each	2.75	
W.C. suite	...	Each	2.75	
W.C. pan -				
floor mounted	...	Each	1.50	
wall mounted	...	Each	3.00	
Cistern, wall type	...	Each	1.00	
Shower base	...	Each	2.00	
Shower bath	...	Each	3.00	
Shower enclosure	...	Each	3.00	
Cleaner's sink	...	Each	2.00	
Kitchen sink	...	Each	0.75	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
SANITARY FIXTURES (cont'd)				
Spa bath	...	Each	3.00-4.00	
Laundry trough	...	Each	0.75	
Slop hopper	...	Each	2.75	
Urinal -				
Single stall	...	Each	2.00	
1800mm long	...	Each	4.00	
3000mm long	...	Each	6.00	
HOT WATER UNITS				
Fix only including connections, to piping :				
Instantaneous multi-point	...	Each	2.00	
160 litre mains pressure, glass lined	...	Each	4.00	
TAPS AND OUTLETS				
Fix only including connections to piping :				
Basin combination set	...	Set	0.50	
Bath combination set	...	Set	1.00	
Laundry combination set	...	Set	1.00	
Pillar tap	...	Each	0.25	
Shower combination set	...	Set	1.50	
Sink combination set	...	Set	1.00	
Washing machine cock	...	Each	0.25	
Laboratory set, two way	...	Set	1.00	
TRAPS				
Fix only including connections :				
Copper or brass S or P trap -				
40mm dia.	...	Each	0.25	
50mm dia.	...	Each	0.33	
COCKS AND VALVES				
Fix only including connections :				
Stop cocks, gate valves -				
15 to 25mm dia.	...	Each	0.25	
32mm dia.	...	Each	0.30	
40mm dia.	...	Each	0.40	
50mm dia.	...	Each	0.50	
65mm dia.	...	Each	0.75	
80mm dia.	...	Each	1.00	
SOIL, WASTE AND VENT PIPES				
• Constants include fixing to walls and testing	...	Note		
• Pipe constants do not include for bends, junctions, etc., to include in metre constant allow for one fitting to every 2.0 to 3.0m of pipe.	...	Note		
Cast iron pipe :				
100mm dia.	...	m	1.15	
150mm dia.	...	m	1.35	
225mm dia.	...	m	2.00	
Bend -				
100mm	...	Each	0.40	
150mm	...	Each	0.50	
225mm	...	Each	1.25	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
SOIL, WASTE AND VENT PIPES (cont'd)				
Cast iron pipe (cont'd) :				
Junction -				
100mm	...	Each	0.45	
150mm	...	Each	0.60	
225mm	...	Each	1.35	
Copper pipe :				
32mm dia.	...	m	0.375	
40mm dia.	...	m	0.400	
50mm dia.	...	m	0.470	
65mm dia.	...	m	0.530	
80mm dia.	...	m	0.680	
100mm dia.	...	m	0.900	
150mm dia.	...	m	1.150	
Bend -				
32mm	...	Each	0.15	
40mm	...	Each	0.18	
50mm	...	Each	0.25	
65mm	...	Each	0.30	
80mm	...	Each	0.35	
100mm	...	Each	0.45	
150mm	...	Each	0.65	
Junction -				
32mm	...	Each	0.20	
40mm	...	Each	0.25	
50mm	...	Each	0.30	
65mm	...	Each	0.35	
80mm	...	Each	0.40	
100mm	...	Each	0.50	
150mm	...	Each	0.75	
PVC pipe :				
32mm dia.	...	m	0.30	
40mm dia.	...	m	0.32	
50mm dia.	...	m	0.40	
65mm dia.	...	m	0.47	
80mm dia.	...	m	0.52	
100mm dia.	...	m	0.65	
Bend -				
32mm	...	Each	0.16	
40mm	...	Each	0.18	
50mm	...	Each	0.22	
65mm	...	Each	0.27	
80mm	...	Each	0.33	
100mm	...	Each	0.42	
150mm	...	Each	0.50	
Junction -				
32mm	...	Each	0.18	
40mm	...	Each	0.20	
50mm	...	Each	0.24	
65mm	...	Each	0.30	
80mm	...	Each	0.35	
100mm	...	Each	0.45	
150mm	...	Each	0.53	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
WATER PIPES				
• Pipe constants do not include for bends, tees, etc.,	...	Note		
Copper pipe :				
Laid in trench -				
15mm dia.	...	m	0.11	
20mm dia.	...	m	0.11	
25mm dia.	...	m	0.13	
32mm dia.	...	m	0.16	
Laid in trench -				
40mm dia.	...	m	0.19	
50mm dia.	...	m	0.23	
65mm dia.	...	m	0.26	
80mm dia.	...	m	0.30	
100mm dia.	...	m	0.33	
150mm dia.	...	m	0.50	
Fixed to wall or soffit -				
15mm dia.	...	m	0.22	
20mm dia.	...	m	0.24	
25mm dia.	...	m	0.26	
32mm dia.	...	m	0.30	
40mm dia.	...	m	0.35	
50mm dia.	...	m	0.40	
65mm dia.	...	m	0.47	
80mm dia.	...	m	0.55	
100mm dia.	...	m	0.65	
150mm dia.	...	m	0.85	
• Bends, tees, refer SOIL, WASTE AND VENT PIPES	...	Note		
Ductile iron cement lined pipe with rubber ring joints laid in trench :				
100mm dia.	...	m	0.40	0.13
150mm dia.	...	m	0.53	0.18
200mm dia.	...	m	0.75	0.25
225mm dia.	...	m	0.80	0.27
250mm dia.	...	m	0.90	0.30
300mm dia.	...	m	1.15	0.38
375mm dia.	...	m	1.33	0.45
450mm dia.	...	m	2.25	0.75
Bend -				
100mm	...	Each	0.45	
150mm	...	Each	0.55	
200mm	...	Each	0.70	
225mm	...	Each	0.75	
250mm	...	Each	1.00	
300mm	...	Each	1.25	
375mm	...	Each	1.50	
450mm	...	Each	1.75	
Tee -				
100mm	...	Each	0.50	
150mm	...	Each	0.60	
200mm	...	Each	0.75	
225mm	...	Each	0.80	
250mm	...	Each	1.05	
300mm	...	Each	1.35	
375mm	...	Each	1.60	
450mm	...	Each	1.90	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
WATER PIPES (cont'd)				
Mild steel cement lined pipe with rubber ring joints laid in trench :				
100mm dia.	...	m	0.40	0.13
150mm dia.	...	m	0.53	0.18
200mm dia.	...	m	0.75	0.25
250mm dia.	...	m	0.90	0.30
300mm dia.	...	m	1.15	0.38
400mm dia.	...	m	1.50	0.50
Bend -				
100mm	...	Each	0.45	
150mm	...	Each	0.55	
200mm	...	Each	0.70	
Tee -				
100mm	...	Each	0.50	
150mm	...	Each	0.60	
200mm	...	Each	0.75	
Polyethylene pipe with compression joints laid in trench :				
16mm dia.	...	m	0.09	
20mm dia.	...	m	0.10	
25mm dia.	...	m	0.11	
32mm dia.	...	m	0.12	
40mm dia.	...	m	0.13	
50mm dia.	...	m	0.15	
63mm dia.	...	m	0.18	
75mm dia.	...	m	0.21	
90mm dia.	...	m	0.25	
110mm dia.	...	m	0.30	
Elbow -				
16mm	...	Each	0.08	
20mm	...	Each	0.08	
25mm	...	Each	0.08	
32mm	...	Each	0.10	
40mm	...	Each	0.12	
50mm	...	Each	0.14	
63mm	...	Each	0.16	
75mm	...	Each	0.20	
90mm	...	Each	0.24	
110mm	...	Each	0.28	
Tee -				
16mm	...	Each	0.10	
20mm	...	Each	0.10	
25mm	...	Each	0.10	
32mm	...	Each	0.12	
40mm	...	Each	0.14	
50mm	...	Each	0.16	
63mm	...	Each	0.18	
75mm	...	Each	0.22	
90mm	...	Each	0.26	
110mm	...	Each	0.28	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
WATER PIPES (cont'd)				
Polyethylene pipe with mechanical joints, laid in trench :				
125mm dia.	...	m	0.35	
140mm dia.	...	m	0.40	
160mm dia.	...	m	0.45	
200mm dia.	...	m	0.55	
225mm dia.	...	m	0.60	
250mm dia.	...	m	0.70	
315mm dia.	...	m	0.85	
355mm dia.	...	m	1.00	
Bend -				
140mm	...	Each	0.55	
160mm	...	Each	0.60	
200mm	...	Each	0.70	
225mm	...	Each	0.75	
250mm	...	Each	1.00	
315mm	...	Each	1.30	
355mm	...	Each	1.40	
Tee -				
140mm	...	Each	0.60	
160mm	...	Each	0.65	
200mm	...	Each	0.75	
250mm	...	Each	1.05	
315mm	...	Each	1.40	
355mm	...	Each	1.50	
PVC pipe :				
Laid in trench -				
15mm dia.	...	m	0.13	
20mm dia.	...	m	0.14	
25mm dia.	...	m	0.16	
32mm dia.	...	m	0.19	
40mm dia.	...	m	0.20	
50mm dia.	...	m	0.23	
65mm dia.	...	m	0.26	
80mm dia.	...	m	0.30	
100mm dia.	...	m	0.33	
150mm dia.	...	m	0.55	
200mm dia.	...	m	0.70	
300mm dia.	...	m	1.10	
Fixed to wall or soffit -				
15mm dia.	...	m	0.20	
20mm dia.	...	m	0.20	
25mm dia.	...	m	0.23	
32mm dia.	...	m	0.30	
40mm dia.	...	m	0.34	
50mm dia.	...	m	0.40	
65mm dia.	...	m	0.47	
80mm dia.	...	m	0.53	
100mm dia.	...	m	0.67	

SANITARY PLUMBING

		UNIT	Tradesman Hours	Labourer Hours
WATER PIPES (cont'd)				
Polyethylene pipe with mechanical joints, laid in trench (cont'd) :				
Elbow -				
15mm	...	Each	0.10	
20mm	...	Each	0.10	
25mm	...	Each	0.10	
32mm	...	Each	0.12	
40mm	...	Each	0.14	
50mm	...	Each	0.17	
65mm	...	Each	0.20	
80mm	...	Each	0.25	
100mm	...	Each	0.33	
150mm	...	Each	0.42	
Tee -				
15mm	...	Each	0.125	
20mm	...	Each	0.125	
25mm	...	Each	0.125	
32mm	...	Each	0.14	
40mm	...	Each	0.16	
50mm	...	Each	0.19	
65mm	...	Each	0.22	
80mm	...	Each	0.27	
100mm	...	Each	0.35	

DRAINAGE

SEWER AND STORMWATER DRAINS

• Constants include laying, jointing and testing but EXCLUDE excavation and backfilling	...	Note		
Concrete pipe with rubber ring joints :				
150mm dia.	...	m	0.425	
225mm dia.	...	m	0.50	0.17
300mm dia.	...	m	0.65	0.22
375mm dia.	...	m	0.75	0.25
450mm dia.	...	m	1.00	0.33
Fibre reinforced cement pipe with rubber ring joints :				
225mm dia.	...	m	0.45	0.15
300mm dia.	...	m	0.60	0.20
375mm dia.	...	m	0.675	0.225
450mm dia.	...	m	0.72	0.24
Glass reinforced plastic pipe with rubber ring joints :				
300mm dia.	...	m	0.45	0.15
375mm dia.	...	m	0.50	0.17
450mm dia.	...	m	0.66	0.22
Polyethylene pipe with rubber ring joints :				
375mm dia.	...	m	0.50	0.17
450mm dia.	...	m	0.66	0.22
PVC pipe with solvent welded joints :				
100mm dia.	...	m	0.25	
150mm dia.	...	m	0.30	
225mm dia.	...	m	0.40	
300mm dia.	...	m	0.60	

DRAINAGE

		UNIT	Tradesman Hours	Labourer Hours
SEWER AND STORMWATER DRAINS				
Bends -				
100mm dia.	...	Each	0.20	
150mm dia.	...	Each	0.25	
225mm dia.	...	Each	0.35	
300mm dia.	...	Each	0.60	
Junctions -				
100mm dia.	...	Each	0.25	
150mm dia.	...	Each	0.30	
225mm dia.	...	Each	0.50	
300mm dia.	...	Each	0.75	

ELECTRICAL SERVICES

CONDUIT

- Constants include for installing a nominal quantity of bends and conduit fittings and saddling to SAA wiring rules requirements where required. ... Note

UPVC conduit :				
Fixed to masonry maximum 4.0m above floor -				
20mm dia.	...	m	0.15	
32mm dia.	...	m	0.18	
50mm dia.	...	m	0.22	
Cast into suspended concrete slab -				
20mm dia.	...	m	0.05	
25mm dia.	...	m	0.06	
32mm dia.	...	m	0.08	
Fixed within suspended ceiling space -				
20mm dia.	...	m	0.11	
32mm dia.	...	m	0.12	
50mm dia.	...	m	0.15	

CABLE

- Constants include for drawing cabling in sets of four cables plus approved size earth cable into adequately sized conduits with minimal long sweep bends and for fixing cabling to top of horizontal single side supported trays. ... Note

TPI cable in conduit :				
2.5 sqmm	...	m	0.03	
6 sqmm	...	m	0.04	
25 sqmm	...	m	0.05	
35 sqmm	...	m	0.06	
95 sqmm	...	m	0.09	
Four core and earth PVC/PVC circular cable on tray :				
2.5 sqmm	...	m	0.08	
6 sqmm	...	m	0.11	
16 sqmm	...	m	0.17	
35 sqmm	...	m	0.23	
95 sqmm	...	m	0.42	
150 sqmm	...	m	0.59	

ELECTRICAL SERVICES

		UNIT	Tradesman Hours	Labourer Hours
CABLE				
MIMS/PVC cable on tray :				
Single core -				
6 sqmm	...	m	0.11	
10 sqmm	...	m	0.13	
25 sqmm	...	m	0.17	
50 sqmm	...	m	0.23	
70 sqmm	...	m	0.25	
120 sqmm	...	m	0.32	
Four core -				
4 sqmm	...	m	0.19	
6 sqmm	...	m	0.21	
10 sqmm	...	m	0.25	
16 sqmm	...	m	0.29	
25 sqmm	...	m	0.35	

MECHANICAL SERVICES

		UNIT	Tradesman Hours
PIPEWORK			**Industrial / Heavy Commercial**
• Constants do not include bends, junctions, tees and flanges	...	Note	
COPPER			
Copper piping with silver soldered joints, installed complete including standard supports but EXCLUDING insulation, diameter :			
15mm	...	m	0.25
20mm	...	m	0.28
25mm	...	m	0.32
32mm	...	m	0.37
40mm	...	m	0.43
50mm	...	m	0.50
65mm	...	m	0.62
80mm	...	m	0.75
100mm	...	m	0.95
150mm	...	m	1.50
Long Sweep Bend -			
15mm	...	Each	0.16
20mm	...	Each	0.16
25mm	...	Each	0.18
32mm	...	Each	0.23
40mm	...	Each	0.30
50mm	...	Each	0.37
65mm	...	Each	0.50
80mm	...	Each	0.70
100mm	...	Each	0.85
150mm	...	Each	1.00
Junction -			
15mm	...	Each	0.20
20mm	...	Each	0.20
25mm	...	Each	0.22
32mm	...	Each	0.27
40mm	...	Each	0.33
50mm	...	Each	0.43
65mm	...	Each	0.65
80mm	...	Each	0.85
100mm	...	Each	1.00
150mm	...	Each	1.30

MECHANICAL SERVICES

		UNIT	Tradesman Hours	Tradesman Hours
PIPEWORK (cont'd)			**Industrial / Heavy Commercial**	**Petrochemical**
COPPER (cont'd)				
Single Bronze flange - with gasket and bolts - Table E :				
15mm	...	Each	0.16	
20mm	...	Each	0.16	
25mm	...	Each	0.18	
32mm	...	Each	0.23	
40mm	...	Each	0.30	
50mm	...	Each	0.40	
65mm	...	Each	0.56	
80mm	...	Each	0.76	
100mm	...	Each	0.93	
150mm	...	Each	1.15	
BLACK CARBON STEEL				
Schedule 40 piping with butt welded joints, installed complete but EXCLUDING insulation and supports, diameter :				
20mm	...	m	0.46	1.80
25mm	...	m	0.50	2.05
32mm	...	m	0.57	2.05
40mm	...	m	0.67	2.10
50mm	...	m	0.81	2.25
65mm	...	m	1.00	2.50
80mm	...	m	1.80	2.60
100mm	...	m	1.38	2.70
150mm	...	m	1.66	3.00
Bend -				
20mm	...	Each	1.00	2.70
25mm	...	Each	1.10	2.70
32mm	...	Each	1.25	2.80
40mm	...	Each	1.45	3.00
50mm	...	Each	1.70	3.60
65mm	...	Each	2.00	4.15
80mm	...	Each	2.40	4.70
100mm	...	Each	3.00	5.60
150mm	...	Each	4.10	7.70
Tee -				
20mm	...	Each	1.50	4.00
25mm	...	Each	1.60	4.00
32mm	...	Each	1.80	4.25
40mm	...	Each	2.00	4.45
50mm	...	Each	2.30	5.35
65mm	...	Each	2.80	6.20
80mm	...	Each	3.50	7.10
100mm	...	Each	4.20	8.40
150mm	...	Each	5.20	11.60
Single flange - with gasket and bolts - 300LB -				
20mm	...	Each	1.15	2.40
25mm	...	Each	1.25	2.40
32mm	...	Each	1.40	2.45
40mm	...	Each	1.60	2.50
50mm	...	Each	1.80	2.80
65mm	...	Each	2.20	3.10
80mm	...	Each	2.60	3.55
100mm	...	Each	3.40	4.60
150mm	...	Each	4.60	6.10

MECHANICAL SERVICES

PIPEWORK (cont'd)

		UNIT	Tradesman Hours **Industrial / Heavy Commercial**	Tradesman Hours **Petrochemical**
GALVANISED CARBON STEEL				
Heavy duty piping to AS-1074 with screwed joints, installed complete but EXCLUDING insulation and supports, diameter :				
20mm	...	m	0.38	1.40
25mm	...	m	0.44	1.60
32mm	...	m	0.52	1.60
40mm	...	m	0.61	1.60
50mm	...	m	0.73	1.70
65mm	...	m	0.91	1.80
80mm	...	m	1.12	2.00
100mm	...	m	1.31	2.10
150mm	...	m	2.12	2.45
Bend -				
20mm	...	Each	0.60	1.60
25mm	...	Each	0.65	1.60
32mm	...	Each	0.75	1.70
40mm	...	Each	0.85	1.75
50mm	...	Each	1.00	2.10
65mm	...	Each	1.25	2.60
80mm	...	Each	1.50	3.00
100mm	...	Each	2.10	4.00
150mm	...	Each	3.80	7.10
Tee -				
20mm	...	Each	0.75	2.00
25mm	...	Each	0.85	2.15
32mm	...	Each	1.00	2.40
40mm	...	Each	1.10	2.45
50mm	...	Each	1.30	3.00
65mm	...	Each	1.60	3.60
80mm	...	Each	2.00	4.00
100mm	...	Each	3.00	6.00
150mm	...	Each	5.00	11.10
Single flange - with gasket and bolts - Table D -				
20mm	...	Each	0.60	1.25
25mm	...	Each	0.65	1.25
32mm	...	Each	0.75	1.30
40mm	...	Each	0.85	1.35
50mm	...	Each	1.00	1.60
65mm	...	Each	1.25	1.80
80mm	...	Each	1.54	2.10
100mm	...	Each	2.10	2.80
150mm	...	Each	3.80	5.00
STAINLESS STEEL				
Schedule 40 piping with welded joints, installed complete but EXCLUDING insulation and supports, diameter :				
20mm	...	m	0.70	2.75
25mm	...	m	0.75	3.10
32mm	...	m	0.82	3.10
40mm	...	m	0.92	3.15
50mm	...	m	1.06	3.20
65mm	...	m	1.22	3.40
80mm	...	m	1.32	3.60
100mm	...	m	1.44	3.80
150mm	...	m	1.91	4.40

MECHANICAL SERVICES

		UNIT	Tradesman Hours	Tradesman Hours
PIPEWORK (cont'd)			**Industrial / Heavy Commercial**	**Petrochemical**
STAINLESS STEEL (cont'd)				
Schedule 40 piping with welded joints, installed complete but EXCLUDING insulation and supports, diameter (cont'd) :				
Bend :				
20mm	...	Each	1.30	3.80
25mm	...	Each	1.45	3.80
32mm	...	Each	1.60	4.00
40mm	...	Each	1.80	4.25
50mm	...	Each	2.00	5.10
65mm	...	Each	2.50	6.00
80mm	...	Each	3.15	7.00
100mm	...	Each	4.00	8.70
150mm	...	Each	4.90	12.50
Tee :				
20mm	...	Each	2.00	5.70
25mm	...	Each	2.20	5.70
32mm	...	Each	2.40	6.00
40mm	...	Each	2.65	6.40
50mm	...	Each	2.90	7.70
65mm	...	Each	3.30	9.10
80mm	...	Each	4.00	10.50
100mm	...	Each	4.60	13.00
150mm	...	Each	5.45	18.75
Single flange - with gasket and bolts - 300LB :				
20mm	...	Each	1.50	2.90
25mm	...	Each	1.60	2.90
32mm	...	Each	1.80	3.00
40mm	...	Each	2.00	3.15
50mm	...	Each	2.30	3.60
65mm	...	Each	2.80	4.10
80mm	...	Each	3.50	4.70
100mm	...	Each	4.20	6.10
150mm	...	Each	5.20	8.50
MILD STEEL				
Heavy duty piping with VICTAULIC joints, installed complete but EXCLUDING insulation and supports, diameter :				
50mm	...	m	0.56	
80mm	...	m	0.86	
100mm	...	m	0.95	
150mm	...	m	1.50	
Bend -				
50mm	...	Each	0.37	
80mm	...	Each	0.60	
100mm	...	Each	0.75	
150mm	...	Each	1.10	
Tee -				
50mm	...	Each	0.47	
80mm	...	Each	0.70	
100mm	...	Each	0.85	
150mm	...	Each	1.30	

MECHANICAL SERVICES

PIPEWORK (cont'd)		UNIT	Tradesman Hours **Industrial / Heavy Commercial**	Tradesman Hours **Petrochemical**
MILD STEEL				
Heavy duty piping with VICTAULIC joints, installed complete but EXCLUDING insulation and supports, diameter (cont'd):				
Single flange adaptor - with gasket and bolts - 300LB -				
50mm	...	Each	0.41	
80mm	...	Each	0.65	
100mm	...	Each	0.80	
150mm	...	Each	1.20	
PIPE SUPPORTS				
• Typical fixing centres are given in parenthesis	...	Note		
Rod hanger fixed to concrete, diameter :				
20mm (@ 2m centres)	...	Each	0.45	
25mm (@ 2m centres)	...	Each	0.45	
32mm (@ 2m centres)	...	Each	0.45	
40mm (@ 2m centres)	...	Each	0.45	
50mm (@ 2.4m - 3m centres)	...	Each	0.60	
65mm (@ 2.4m - 3m centres)	...	Each	0.60	
80mm (@ 2.4m - 3m centres)	...	Each	0.65	
100mm (@ 3m - 4m centres)	...	Each	0.65	
150mm (@ 3m - 5m centres)	...	Each	0.80	
Suspended pipe roller assembly fixed to concrete, per metre run, diameter :				
20mm (@ 2m centres)	...	Each	0.80	
25mm (@ 2m centres)	...	Each	0.80	
32mm (@ 2m centres)	...	Each	0.80	
40mm (@ 2m centres)	...	Each	0.80	
50mm (@ 2.4m - 3m centres)	...	Each	0.90	
65mm (@ 2.4m - 3m centres)	...	Each	0.90	
80mm (@ 2.4m - 3m centres)	...	Each	1.00	
100mm (@ 3m - 4m centres)	...	Each	1.00	
150mm (@ 3m - 5m centres)	...	Each	1.10	

CIVIL ENGINEERING - EARTHWORKS

EARTHMOVING PLANT, PRODUCTION ETC.

- For information in respect to the cycle times and the various production factors of equipment such as bulldozers, excavators, loaders, etc., refer to the reference handbook issued by the Caterpillar Tractor Co.

DENSITY AND LOAD FACTORS

- The following is based on information given in the reference handbook issued by the Caterpillar Tractor Co.
- Whilst measured quantities are generally in situ or in place quantities, the actual quantity to be removed and/or spread and levelled is, in practical terms, the loose volume.

  The following table indicates the approximate density (weight) and load factor of common materials :

APPROXIMATE DENSITY AND LOAD FACTOR OF COMMON MATERIALS

Material		Density - In situ kg/cum	Load Factor	Density - Loose kg/cum
Clay	- Dry excavated	1840	0.81	1480
	- Wet excavated	2080	0.80	1660
Clay and Gravel	- Dry	1660	0.85	1420
	- Wet	1840	0.85	1540
Decomposed Rock	- 75% Rock 25% Earth	2790	0.70	1960
	- 50% Rock 50% Earth	2280	0.75	1720
	- 25% Rock 75% Earth	1960	0.80	1570
Earth	- Wet excavated	2020	0.79	1600
Gravel	- Pitrun	2170	0.89	1930
	- Dry	1690	0.89	1510
Sand	- Dry, loose	1600	0.89	1420
	- Wet	2080	0.89	1840
Sand and Gravel	- Dry	1930	0.89	1720
	- Wet	2230	0.91	2020
Stone	- Crushed	2670	0.60	1600
Top Soil		1370	0.70	950

CIVIL ENGINEERING - EARTHWORKS

TRUCKS AND HAULAGE

TRUCK CAPACITY

- The volumes given hereafter are arrived at by dividing the rated capacity of the truck by the loose or in situ densities of the various materials as listed in the Density and Load Factor table.

		VEHICLE CAPACITY							
		8t		12t		16t		20t	
		Cum In situ	Cum Loose	Cum In situ	Cum Loose	Cum In situ	Cum Loose	Cum In situ	Cum Loose
Clay	- Dry excavated	4.35	5.41	6.52	8.11	8.70	10.81	10.87	13.51
	- Wet excavated	3.85	4.82	5.77	7.23	7.69	9.64	9.62	12.05
Clay and Gravel	- Dry	4.82	5.63	7.73	8.45	9.64	11.26	12.05	14.08
	- Wet	4.35	5.19	6.52	7.79	8.70	10.39	10.87	12.99
Decomposed Rock	- 75% Rock 25% Earth	2.87	4.08	4.30	6.12	5.73	8.16	7.17	10.20
	- 50% Rock 50% Earth	3.51	4.65	5.26	6.98	7.02	9.30	8.77	11.63
	- 25% Rock 75% Earth	4.08	5.10	6.12	7.64	8.16	10.19	10.20	12.74
Earth	- Wet excavated	3.96	5.00	5.94	7.50	7.92	10.00	9.90	12.50
Gravel	- Pitrun	3.69	4.15	5.53	6.22	7.37	8.29	9.21	10.36
	- Dry	4.73	5.30	7.10	7.95	9.47	10.59	11.83	13.24
Sand	- Dry, loose	5.00	5.63	7.50	8.45	10.00	11.26	12.50	14.08
	- Wet	3.85	4.35	5.77	6.52	7.69	8.69	9.62	10.87
Sand & Gravel	- Dry	4.15	4.65	6.22	6.98	8.29	9.30	10.36	11.63
	- Wet	3.59	3.96	5.38	5.94	7.17	7.92	8.97	9.90
Stone	- Crushed	3.00	5.00	4.49	7.50	5.99	10.00	7.49	12.50
Top Soil		5.84	8.42	8.76	12.63	11.68	16.84	14.59	21.05

HAULING TIMES

- Based on light traffic and travelling at an average 35km/hour.
- Includes loading time of two minutes, travel to, dump time of four minutes, and return time.

LOADS PER HOUR

DISTANCE TRAVELLED TO TIP							
1km	2km	3km	4km	5km	10km	20km	30km
6.63	4.67	3.67	3.04	2.59	1.49	0.80	0.55

ADJUST for :

Medium traffic + 20% on travelling time.
Heavy traffic + 40% to 50% on travelling time.
Waiting time at loading point.

NOTE :

The above times are a very approximate guide only. Location, route, and time of day can have an appreciable effect on figures given.

TRADE RATIOS

The following schedule of trade ratios is intended as an approximate guide to the proportion of labour, material, and plant that can be anticipated on an average project. Proportions exclude off site overheads and profit components.

TRADE	PROPORTION OF TOTAL			
	LABOUR OFF SITE	ON SITE	MATERIAL	PLANT
Preliminaries ...		55	5	40
Demolitions ...		75	15	10
Excavations - bulk ...		5	10	85
Excavations - detail ...		20	15	65
Piling ...		20	50	30
Concrete ...		15	75	10
Formwork ...		70	30	
Reinforcement ...		20	80	
Precast concrete ...	30	10	50	10
Brick/block ...		45	55	
Masonry ...	25	10	65	
Structural steel ...	35	10	45	10
Metalwork ...	10	15	75	
Carpentry ...		55	45	
Joinery ...	10	30	60	
Aluminium windows/doors ...	35	15	50	
Hardware ...		15	85	
Roofing - membrane ...		30	70	
Roofing - sheet ...		40	60	
Plastering ...		80	20	
Plasterboard - linings ...		45	55	
Suspended ceilings ...		40	60	
Paving - in situ ...		75	25	
Paving - units ...		50	50	
Tiling ...		55	45	
Resilient finishes ...		20	80	
Painting ...		80	20	
Paper hanging ...		70	30	
Plumbing - pipework ...		55	45	
Plumbing - fittings ...		20	80	
Drainage ...		50	40	10
Electrical ...		40	60	
Air-conditioning ...	35	10	55	
Lifts ...	30	10	60	
Roadworks ...		20	50	30
Fencing ...		60	40	
Landscaping ...		65	35	

BUILDING INDICES

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PRICE INDICES

This information is based on historical data prepared over the years by contributing associated offices and is intended as a guide to the effect on building costs brought about by periodic variations in the rates of labour and materials as well as reflecting the cost effect of building activity and resource availability (i.e. market competition) at any time.

The indices are not valid for housing, small projects, remote country work or work not exceeding $250,000.00.

The indices do not reflect the relative cost differential of the various cities.

PERIOD YEAR	BUILDING PRICE INDEX							
	ADEL	BRIS	CAN	DWN	HOB	MEL	PER	SYD
JUNE 30:								
1979	21.46	18.50	17.67	20.02	18.67	16.42	16.42	16.77
1980	23.05	20.74	19.95	22.93	20.98	18.75	18.10	19.51
1981	25.80	24.16	22.54	25.74	23.30	21.98	19.88	23.08
1982	29.64	27.38	28.69	29.00	27.15	26.58	22.41	25.97
1983	30.48	27.08	30.85	30.61	28.54	24.97	23.84	27.28
1984	32.32	28.18	34.04	31.95	30.39	27.37	25.24	28.85
1985	36.76	30.06	38.66	33.78	34.40	31.87	28.74	31.52
1986	39.25	32.38	42.77	36.22	41.50	37.22	32.39	35.90
1987	42.31	34.00	45.90	38.28	45.75	42.41	34.25	39.31
1988	45.69	36.40	47.76	40.39	48.30	46.44	36.50	42.59
1989	49.81	39.69	50.84	41.96	51.92	50.46	41.22	46.41
1990	51.23	41.85	52.65	44.22	53.29	53.76	43.67	49.34
1991	48.38	42.06	55.60	46.21	54.35	49.52	45.18	47.45
1992	48.87	40.58	55.55	48.51	54.40	46.65	45.18	45.07
1993	48.87	41.18	55.55	49.24	53.00	46.05	45.18	45.36
1994	49.92	43.46	56.95	50.72	53.61	47.43	45.18	46.66
1995	51.92	47.82	58.67	53.01	55.47	48.63	45.86	48.06
1996	52.70	47.82	60.56	55.93	56.14	50.95	47.47	50.71
1997	53.76	48.77	61.92	57.85	58.63	52.15	48.42	54.67
1998	53.89	50.10	62.38	58.35	59.70	54.77	49.14	58.29
1999	54.57	51.11	63.16	59.00	60.03	56.95	49.88	62.43
2000	57.46	52.64	64.11	59.89	60.87	60.98	52.14	65.24
2001	58.30	51.36	65.07	59.89	61.75	63.41	52.65	64.72
2002	59.93	53.43	66.37	61.69	62.97	66.85	53.71	66.38
2003	63.37	60.93	71.77	64.16	64.32	70.87	56.71	69.89
2004	67.27	77.36	75.67	66.74	75.55	73.50	62.73	73.21
2005	72.61	84.20	80.63	71.06	78.35	78.01	69.82	76.49
2006	76.97	89.18	85.05	74.96	81.07	81.90	76.88	81.49
2007	81.19	95.42	88.86	80.23	83.88	85.83	85.30	87.17
2008	88.58	102.10	92.42	87.48	88.11	90.66	92.09	91.53
2009	93.35	100.00	93.33	92.64	93.87	93.34	95.63	93.34
2010	97.09	100.00	96.15	96.34	98.07	94.74	99.01	97.09
2011	99.03	100.00	98.79	99.02	100.00	98.55	99.26	99.03
2012	100.00	100.50	100.50	100.90	96.00	101.00	100.00	101.00
2013	100.50	101.32	100.49	103.02	96.00	103.02	101.00	103.02
2014	101.00	102.78	102.52	106.11	97.45	105.08	102.51	106.12
2015	102.78	107.44	105.09	108.38	99.40	107.45	105.08	109.31
2016	105.35	111.72	107.98	110.06	101.39	109.49	107.17	113.11
2017	107.99	116.19	112.06	110.39	103.42	112.78	109.10	116.85
2018	111.50	118.78	116.54	110.50	104.97	116.16	110.74	120.00
2019	114.56	122.95	120.61	111.27	108.12	120.23	112.40	122.39
2020	117.43	124.47	123.91	111.60	110.81	125.03	113.75	127.68

PRICE INDICES - ADELAIDE

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	21.46	23.05	25.80	29.64	30.48	32.32	36.76	39.25	42.31	45.69	49.81	51.23	48.38	48.87
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	48.87	49.92	51.92	52.70	53.76	53.89	54.57	57.46	58.30	59.93	63.37	67.27	72.61	76.97
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	80.25	86.27	92.44	96.14	98.05	100.00	100.50	100.75	102.27	104.33	106.93	110.41	113.72	116.56
June 30	81.19	88.58	93.35	97.09	99.03	100.00	100.50	101.00	102.78	105.35	107.99	111.50	114.56	117.43
Sept 30	82.78	90.26	94.72	97.56	99.51	100.50	100.75	101.51	103.29	105.87	109.05	112.33	115.13	117.43
Dec 31	83.96	91.52	95.18	97.56	100.00	100.50	100.75	101.76	103.80	106.39	109.59	112.87	115.69	118.00(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 105.35

June 2013 = 100.50

4.85

$$\frac{4.85 \times 100.00}{100.50} = 4.83\%$$

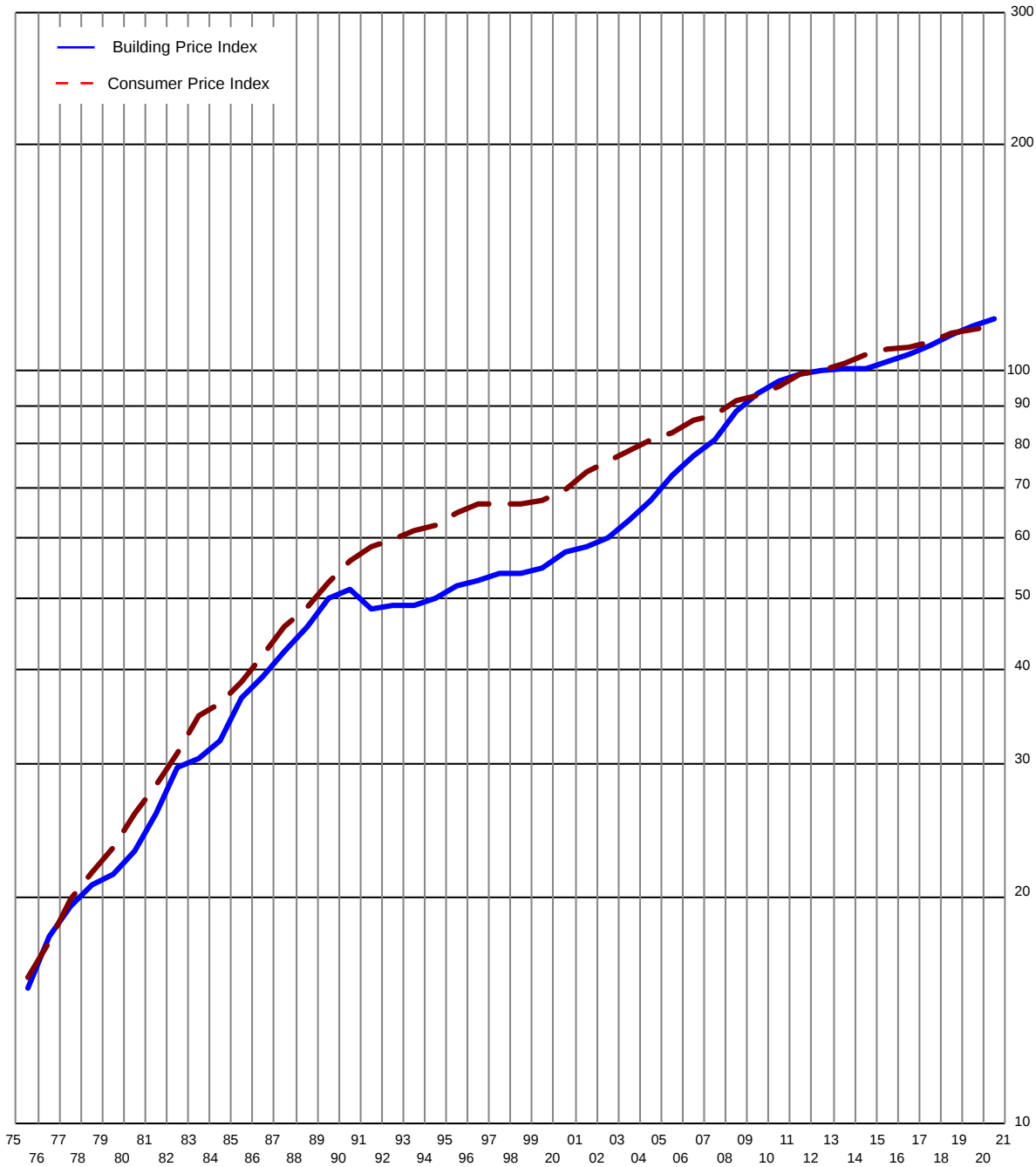
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	23.3	25.8	28.1	31.0	34.8	36.1	38.7	41.8	45.6	48.5	52.3	57.2	58.4	59.6
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	61.2	62.3	64.7	66.5	66.4	66.7	67.3	69.5	73.6	75.7	78.6	80.9	82.7	85.8
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.3	90.1	92.2	94.6	98.1	99.9	102.1	105.1	106.3	107.0	109.1	111.6	113.1	115.8
June 30	87.3	91.3	92.7	95.3	99.0	100.2	102.3	105.5	106.8	107.5	109.2	112.1	113.7	114.6
Sept 30	88.0	92.5	93.7	96.2	100.0	101.7	103.7	105.9	107.1	108.4	110.4	112.4	114.5	115.7
Dec 31	88.9	92.2	94.1	96.5	100.0	102.1	104.4	106.2	107.3	108.7	111.2	113.0	115.4	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - ADELAIDE

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - BRISBANE

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	18.50	20.74	24.16	27.38	27.08	28.18	30.06	32.38	34.00	36.40	39.69	41.85	42.06	40.58
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	41.18	43.46	47.82	47.82	48.77	50.10	51.11	52.64	51.36	53.43	60.93	77.36	84.20	89.18
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	94.49	101.11	103.09	100.00	100.00	100.00	101.25	102.27	105.88	110.35	115.05	117.90	122.34	124.47
June 30	95.42	102.10	100.00	100.00	100.00	100.50	101.32	102.78	107.44	111.72	116.19	118.78	122.95	124.47
Sept 30	97.27	103.10	100.00	100.00	100.00	101.00	101.76	103.54	108.48	113.09	117.04	119.66	123.56	125.09
Dec 31	99.13	103.09	100.00	100.00	100.00	101.00	102.01	104.81	109.53	113.91	117.90	121.43	124.47	126.03(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 111.72

June 2013 = 101.32 / 10.40

$$\frac{10.40 \times 100.00}{101.32} = 10.27\%$$

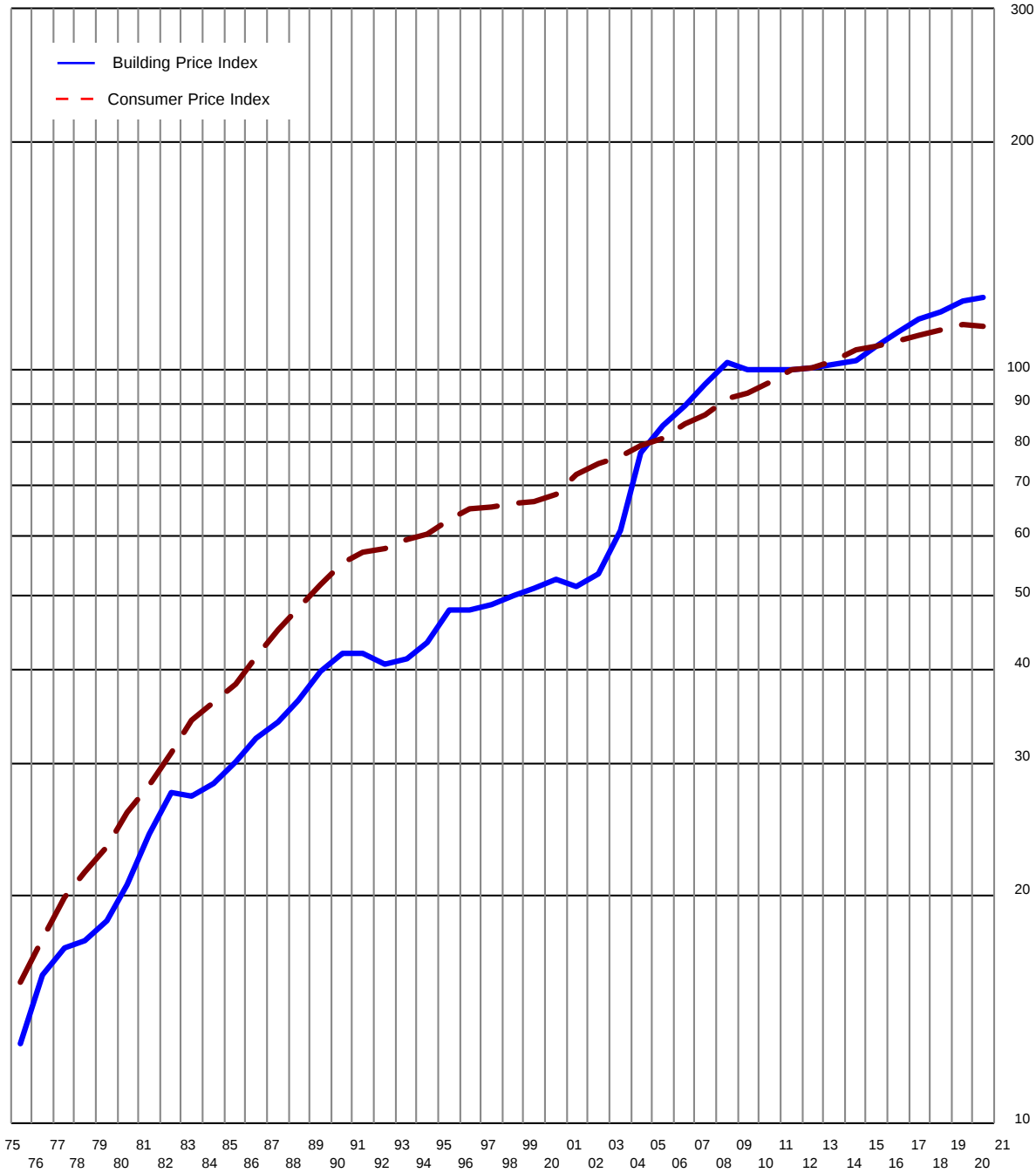
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	23.3	25.8	28.1	31.0	34.3	36.2	38.3	41.4	45.1	48.3	51.8	55.3	57.2	57.9
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	59.3	60.3	63.2	65.1	65.5	66.2	66.6	68.3	72.5	74.7	76.7	79.1	81.1	84.5
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	85.5	89.6	92.4	95.2	98.6	99.9	102.0	105.2	106.7	108.5	110.5	112.4	114.1	116.2
June 30	86.7	91.1	92.9	95.9	99.6	100.5	102.5	105.8	107.4	109.0	111.0	112.9	114.8	113.6
Sept 30	87.5	92.4	94.2	96.9	99.9	101.6	103.8	106.5	108.1	109.7	111.4	113.4	115.5	116.2
Dec 31	88.4	92.2	94.5	97.4	99.7	101.9	104.6	106.7	108.5	110.2	112.3	114.0	116.3	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - BRISBANE

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - CANBERRA

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	17.67	19.95	22.54	28.69	30.85	34.04	38.66	42.77	45.90	47.76	50.84	52.65	55.60	55.55
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	55.55	56.95	58.67	60.56	61.92	62.38	63.16	64.11	65.07	66.37	71.77	75.67	80.63	85.05
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	88.00	91.51	92.87	95.20	97.82	100.00	100.49	102.52	104.57	107.45	109.32	115.41	118.85	122.99
June 30	88.86	92.42	93.33	96.15	98.79	100.50	100.49	102.52	105.09	107.98	112.06	116.54	120.61	123.91
Sept 30	89.73	92.87	93.79	96.62	99.52	101.00	101.50	103.54	105.61	108.52	112.60	117.36	121.20	124.52
Dec 31	90.16	92.87	94.26	97.09	100.00	101.50	102.01	104.05	106.65	109.32	113.14	117.67	122.38	125.44(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 107.98

June 2013 = 100.49

7.50

$$\frac{7.50 \times 100.00}{100.49} = 7.46\%$$

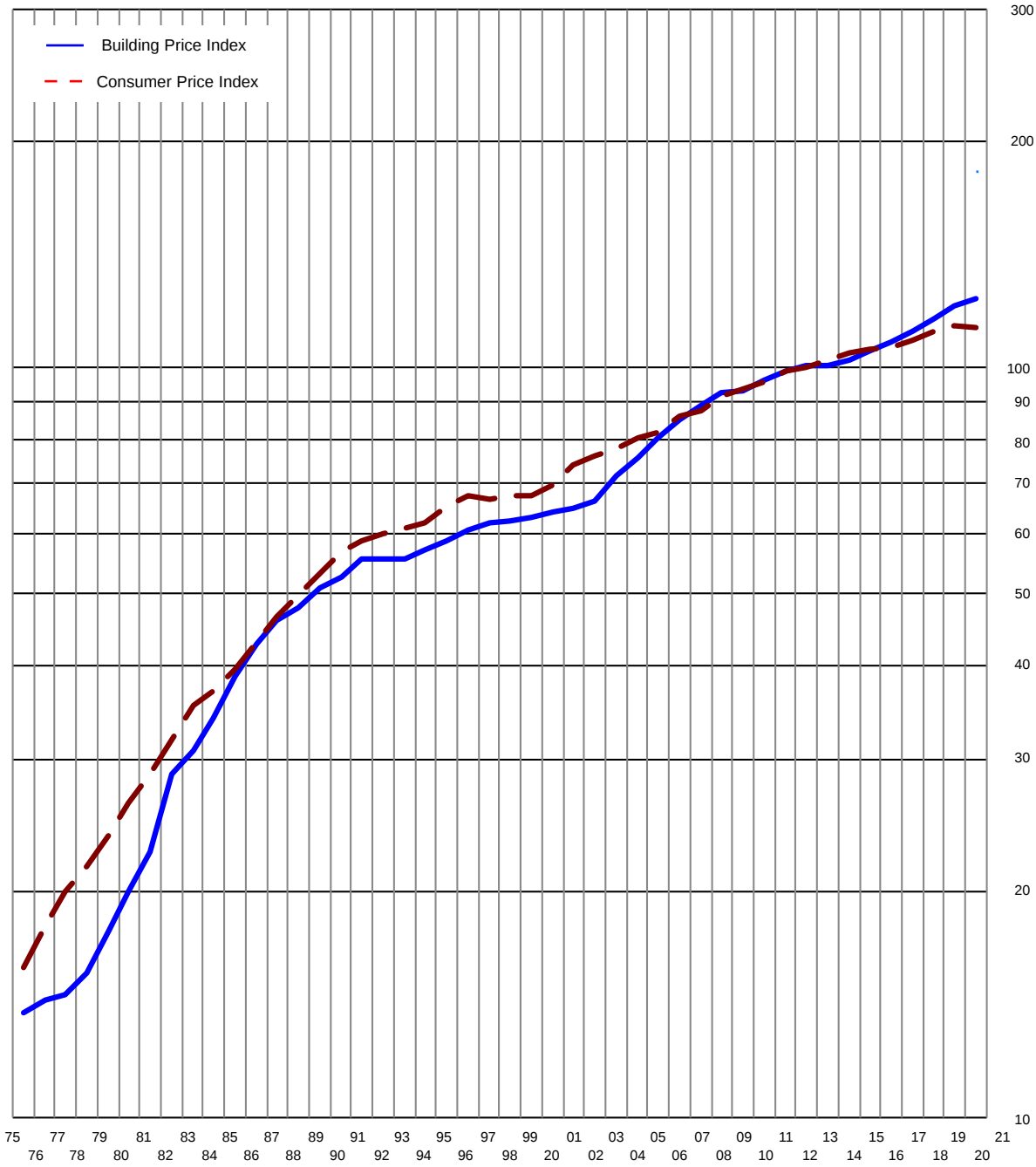
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	23.7	26.3	28.7	31.9	35.5	37.0	39.6	43.0	46.5	49.8	53.1	56.8	58.6	59.9
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	61.2	62.2	65.3	67.4	66.8	67.3	67.4	69.9	74.0	76.2	78.1	80.4	82.1	86.0
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.5	90.5	92.9	95.3	98.1	99.7	101.9	104.6	105.2	106.2	108.6	111.2	113.2	115.5
June 30	87.7	91.6	93.5	95.6	99.2	100.3	102.5	104.8	105.6	106.4	108.6	111.6	113.5	112.8
Sept 30	88.4	93.0	94.3	96.3	99.8	101.4	103.1	105.2	105.8	107.3	109.6	112.3	114.3	115.4
Dec 31	89.2	92.6	94.7	96.7	100.1	101.8	104.1	105.3	106.0	107.9	110.3	113.1	115.0	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - CANBERRA

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - DARWIN

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	20.02	22.93	25.74	29.00	30.61	31.95	33.78	36.22	38.28	40.39	41.96	44.22	46.21	48.51
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	49.24	50.72	53.01	55.93	57.85	58.35	59.00	59.89	59.89	61.69	64.16	66.74	71.06	74.96
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	78.69	85.82	92.18	95.40	98.52	100.30	102.00	105.06	108.21	109.84	110.28	110.50	111.05	111.27
June 30	80.23	87.48	92.64	96.34	99.02	100.90	103.02	106.11	108.38	110.06	110.39	110.50	111.27	111.60
Sept 30	81.00	89.15	93.55	97.29	99.51	101.40	104.04	107.16	108.54	110.06	110.39	110.83	111.27	111.60
Dec 31	83.32	90.82	94.45	98.23	100.00	102.00	105.06	108.21	109.84	110.06	110.50	111.05	111.27	111.60(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 110.06

June 2013 = 103.02

$$\frac{7.04}{103.02} \times \frac{100.00}{1} = 6.83\%$$

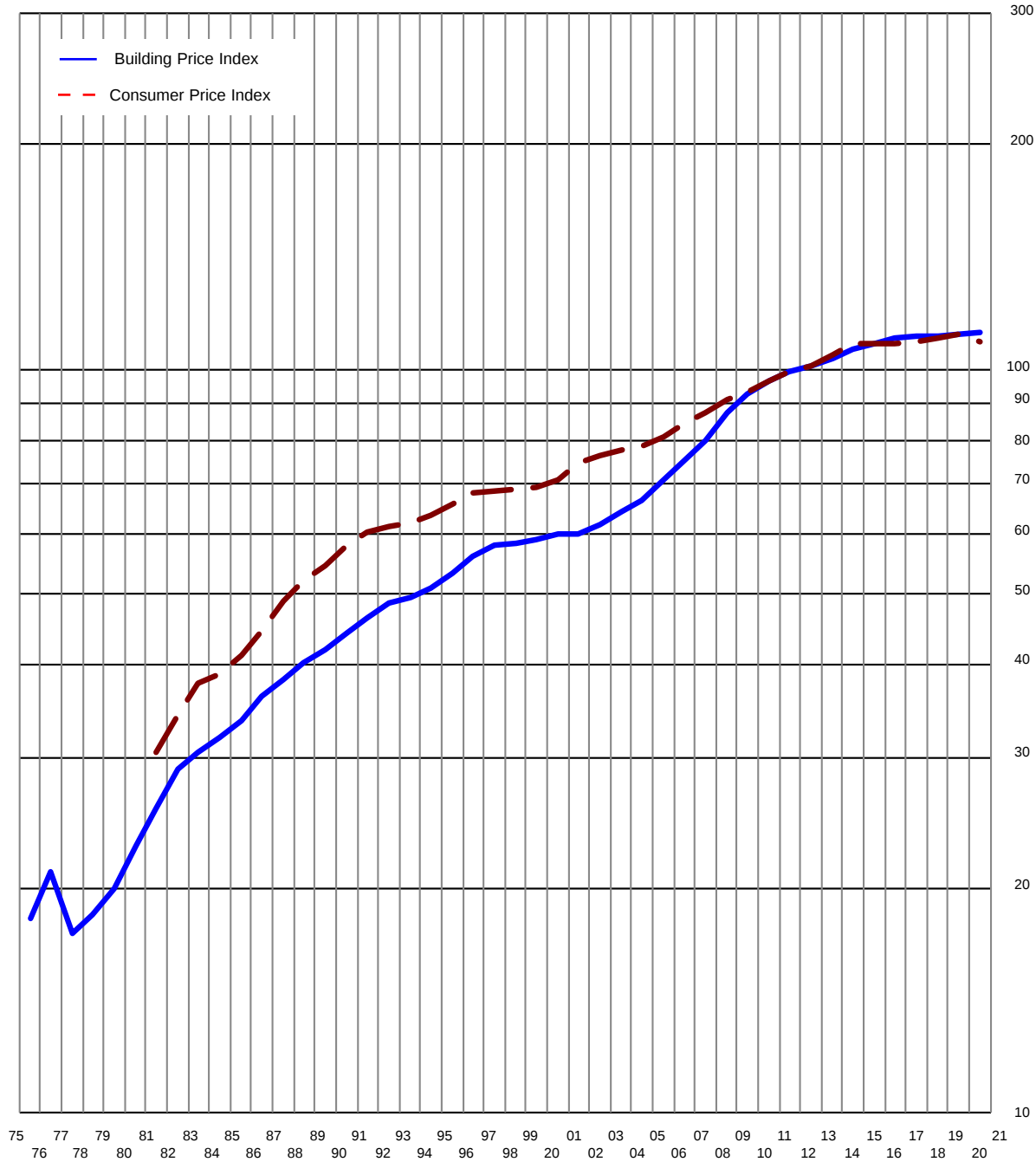
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	0.0	0.0	30.6	34.2	37.8	39.0	41.2	44.4	48.8	51.9	54.4	57.9	60.3	61.3
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	62.2	63.5	66.0	68.3	68.7	68.9	69.4	71.1	74.7	76.3	78.0	78.9	80.9	84.4
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.3	89.6	92.2	95.4	98.2	99.9	103.7	107.4	108.3	108.0	108.5	109.7	110.1	111.8
June 30	87.5	90.9	93.2	96.2	99.2	100.7	104.6	108.1	108.3	108.3	108.8	110.1	111.0	109.0
Sept 30	88.5	92.5	95.0	97.2	99.9	102.0	105.5	108.3	108.7	108.7	109.4	110.8	111.3	110.8
Dec 31	88.8	92.1	94.9	97.1	99.5	102.0	106.5	108.5	109.0	108.6	109.7	111.0	111.5	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - DARWIN

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - HOBART

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	18.67	20.98	23.30	27.15	28.54	30.39	34.40	41.50	45.75	48.30	51.92	53.29	54.35	54.40
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	53.00	53.61	55.47	56.14	58.63	59.70	60.03	60.87	61.75	62.97	64.32	75.55	78.35	81.07
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	83.15	87.20	91.39	96.88	99.50	98.00	96.00	96.96	98.90	100.64	102.40	104.44	106.80	110.26
June 30	83.88	88.11	93.87	98.07	100.00	96.00	96.00	97.45	99.40	101.39	103.42	104.97	108.12	110.81
Sept 30	84.97	89.06	94.78	99.27	100.00	95.00	96.48	97.93	99.89	101.89	103.93	105.48	109.18	111.36
Dec 31	86.13	90.26	95.68	99.50	100.00	96.00	96.48	98.41	99.89	101.89	103.93	106.00	109.71	112.45(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 101.39
June 2013 = 96.00

$$\frac{101.39}{96.00} = \frac{5.39 \times 100.00}{96.00} = 5.61\%$$

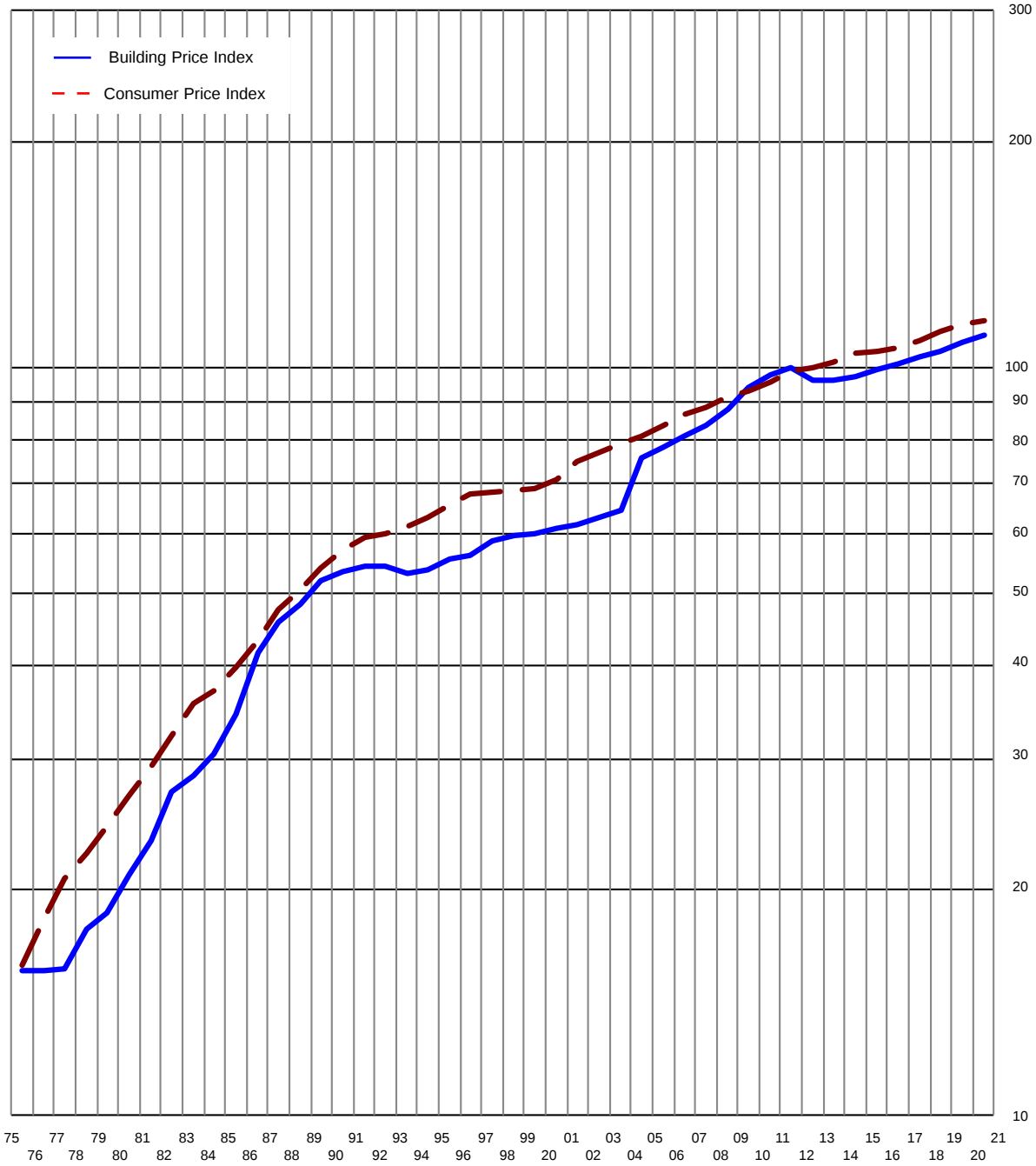
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	24.4	26.8	29.2	32.1	35.6	37.0	39.7	43.1	47.4	50.5	53.9	57.2	59.4	60.1
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	61.4	63.1	65.7	67.7	68.1	68.5	68.8	71.0	74.9	76.9	79.1	81.0	83.5	86.4
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	87.3	90.5	92.5	95.4	98.2	100.3	101.3	104.1	105.0	106.4	108.9	111.1	113.4	117.2
June 30	88.4	91.5	93.0	95.8	99.1	99.9	101.7	104.5	105.1	106.4	108.9	111.5	114.1	115.6
Sept 30	88.6	92.5	94.1	96.8	99.9	100.6	102.6	104.6	105.7	107.1	109.2	112.2	114.7	116.7
Dec 31	89.4	92.3	94.7	96.9	100.0	101.0	103.6	104.7	106.6	108.0	110.3	113.6	116.7	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - HOBART

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - MELBOURNE

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	16.42	18.75	21.98	26.58	24.97	27.37	31.87	37.22	42.41	46.44	50.46	53.76	49.52	46.65
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	46.05	47.43	48.63	50.95	52.15	54.77	56.95	60.98	63.41	66.85	70.87	73.50	78.01	81.90
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	84.91	89.12	92.42	94.27	97.58	100.50	102.00	104.30	106.92	108.95	110.57	115.02	119.05	124.72
June 30	85.83	90.66	93.34	94.74	98.55	101.00	103.02	105.08	107.45	109.49	112.78	116.16	120.23	125.03
Sept 30	86.92	91.53	93.57	95.68	100.00	101.50	103.53	105.60	107.66	110.03	113.33	116.73	121.70	125.34
Dec 31	88.02	92.42	93.80	96.62	100.00	102.00	104.04	105.86	107.87	110.57	113.88	117.87	122.88	125.34(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 109.49

June 2013 = 103.02

$$\frac{6.47}{103.02} \times 100.00 = 6.28\%$$

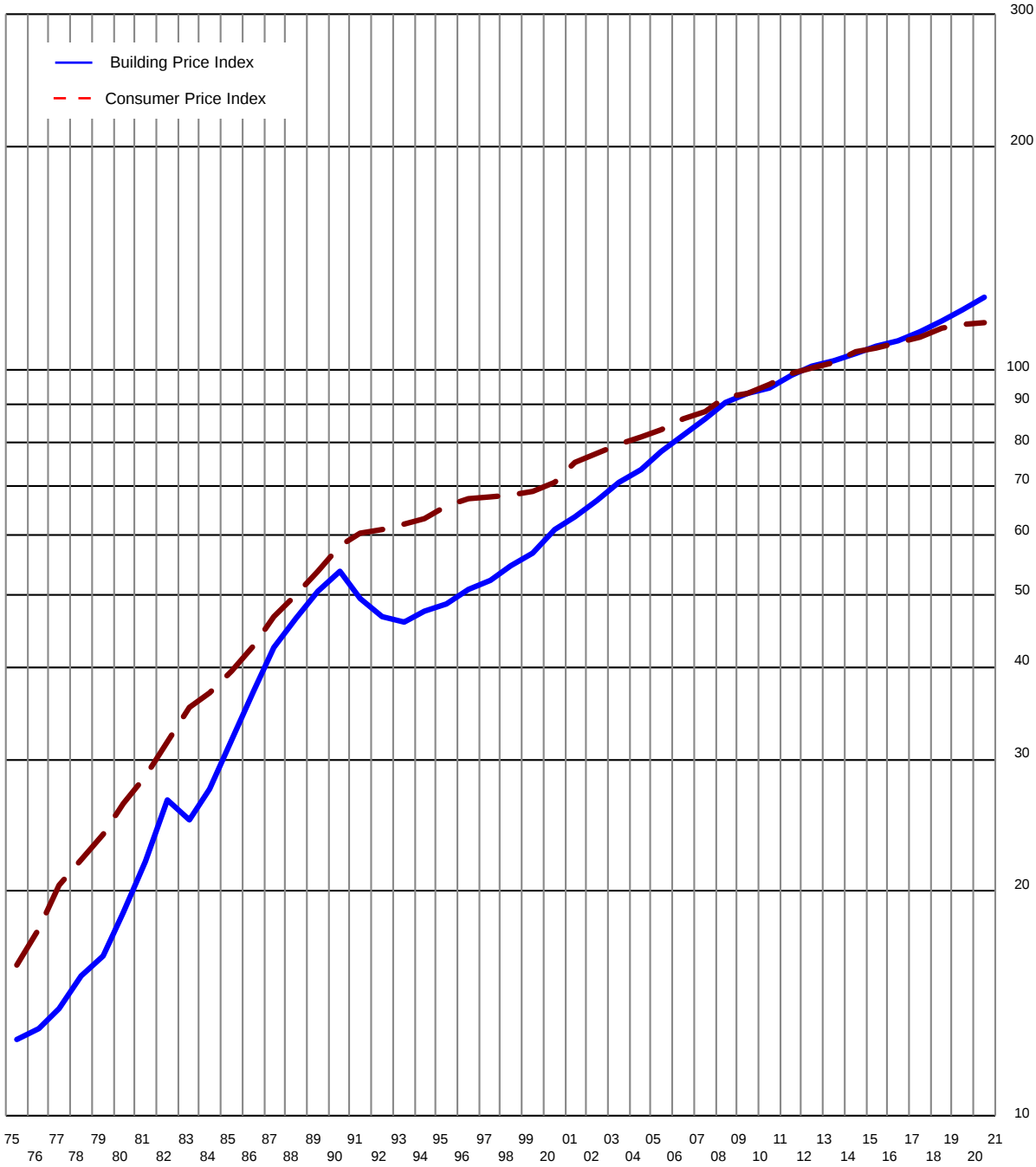
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	23.8	26.3	28.6	31.7	35.4	36.9	39.5	42.8	46.8	50.1	53.8	58.0	60.3	61.1
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	62.2	63.3	65.6	67.3	67.7	68.0	68.6	70.9	75.1	77.3	79.6	81.3	83.0	86.2
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.9	90.7	92.6	95.2	98.5	99.9	102.4	105.3	106.4	108.2	110.9	113.3	114.7	117.8
June 30	87.9	91.8	92.9	95.8	99.2	100.4	102.6	105.9	107.1	108.6	111.0	113.8	115.3	115.7
Sept 30	88.6	92.9	93.4	96.3	99.8	101.6	104.0	106.1	107.6	109.1	111.5	114.0	115.9	116.7
Dec 31	89.5	92.3	94.0	96.9	99.9	102.0	104.8	106.3	108.3	109.9	112.3	114.6	116.9	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - MELBOURNE

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - PERTH

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	16.42	18.10	19.88	22.41	23.84	25.24	28.74	32.39	34.25	36.50	41.22	43.67	45.18	45.18
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	45.18	45.18	45.86	47.47	48.42	49.14	49.88	52.14	52.65	53.71	56.71	62.73	69.82	76.88
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	82.86	90.32	95.63	97.55	99.01	100.00	100.75	102.01	104.56	106.65	108.24	110.30	111.96	113.46
June 30	85.30	92.09	95.63	99.01	99.26	100.00	101.00	102.51	105.08	107.17	109.10	110.74	112.40	113.75
Sept 30	86.92	94.75	96.59	99.01	99.51	100.50	101.25	103.02	105.60	107.71	109.54	111.18	112.85	114.31
Dec 31	88.55	95.63	97.55	99.01	100.00	100.50	101.50	103.53	106.11	108.24	109.86	111.51	113.18	114.59(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 107.17
June 2013 = 101.00

$$\frac{6.17}{101.00} \times 100.00 = 6.11\%$$

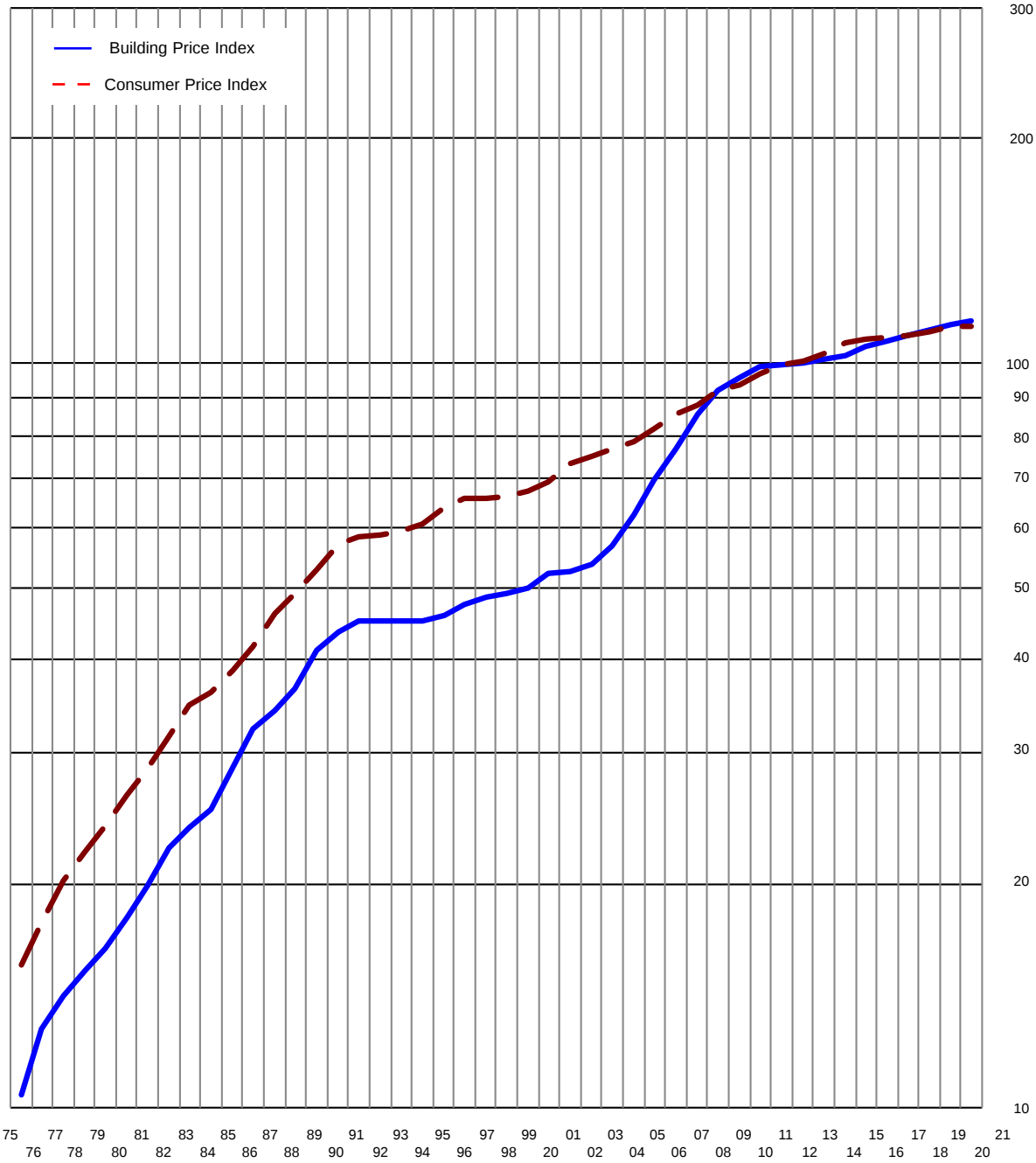
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	24.0	26.3	28.5	31.6	34.8	36.2	38.7	41.7	46.0	49.1	52.8	57.3	58.6	58.8
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	59.5	60.8	64.0	65.7	65.8	66.2	67.3	69.1	73.2	75.0	76.5	78.6	81.5	85.4
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.8	90.5	92.5	95.6	98.1	100.0	102.4	105.6	107.1	107.9	109.0	110.0	111.2	113.5
June 30	88.0	92.0	93.3	96.5	99.4	100.5	103.0	106.4	107.7	108.2	109.0	110.2	112.0	112.1
Sept 30	88.6	92.9	94.0	96.9	99.6	101.6	104.2	106.9	108.1	108.6	109.5	110.8	112.6	114.1
Dec 31	89.3	92.6	94.5	97.0	99.8	101.9	104.9	107.0	108.6	109.0	109.9	111.3	113.1	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - PERTH

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



Building Price Index and Consumer Price Index are base dated to 2011-2012

PRICE INDICES - SYDNEY

BUILDING PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	16.77	19.51	23.08	25.97	27.28	28.85	31.52	35.90	39.31	42.59	46.41	49.34	47.45	45.07
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	45.36	46.66	48.06	50.71	54.67	58.29	62.43	65.24	64.72	66.38	69.89	73.21	76.49	81.49
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	85.49	89.77	92.41	97.09	99.03	101.00	102.51	105.60	107.71	113.11	115.70	119.40	121.18	126.43
June 30	87.17	91.53	93.34	97.09	99.03	101.00	103.02	106.12	109.31	113.11	116.85	120.00	122.39	127.68
Sept 30	87.59	92.41	94.26	97.09	100.00	101.50	103.53	106.12	109.83	114.22	118.22	120.59	123.60	128.43
Dec 31	88.01	92.41	95.18	97.09	100.00	102.00	104.04	106.12	110.89	114.55	118.22	121.18	125.30	128.81(F)

(P) PROVISIONAL (F) FORECAST (R) REVISED

Example :

To assess the percentage variation in cost between June 2013 and June 2016 :

June 2016 = 113.11

June 2013 = $\frac{103.02}{10.09}$ $\frac{10.09 \times 100.00}{103.02}$ = 9.79%

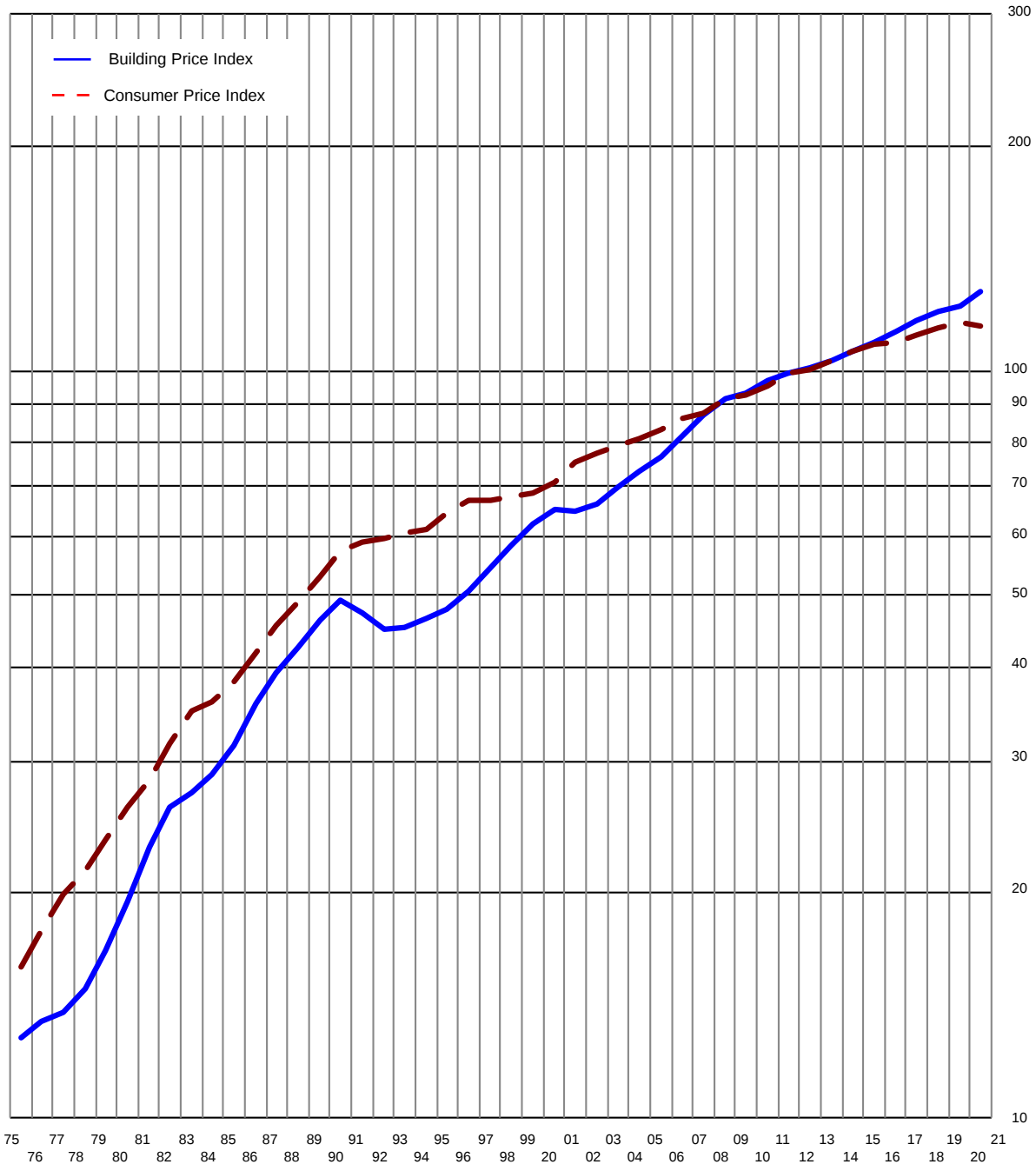
CONSUMER PRICE INDEX

	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992
June 30	23.5	26.1	28.4	31.6	35.1	36.1	38.4	41.8	45.7	49.0	52.9	57.2	58.9	59.5
	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006
June 30	60.5	61.4	64.4	67.0	67.1	67.8	68.7	70.9	75.4	77.5	79.4	81.2	83.2	86.4
	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
March 31	86.9	90.3	92.5	95.2	98.2	99.9	102.7	105.6	107.3	108.7	111.3	113.6	115.1	117.4
June 30	87.9	91.7	92.9	95.6	99.2	100.5	103.1	106.0	108.3	109.3	111.7	114.0	115.9	114.7
Sept 30	88.3	92.7	93.9	96.3	99.9	102.2	104.3	106.6	108.6	110.4	112.5	114.7	116.5	116.8
Dec 31	89.1	92.4	94.4	96.7	99.8	102.3	105.0	106.8	108.9	110.9	113.3	115.2	117.1	

Consumer Price Index : All Group Index Numbers - as issued by the Australian Bureau of Statistics

PRICE INDICES - SYDNEY

COMPARISON OF CONSUMER PRICE INDEX AND RAWLINSONS BUILDING PRICE INDEX



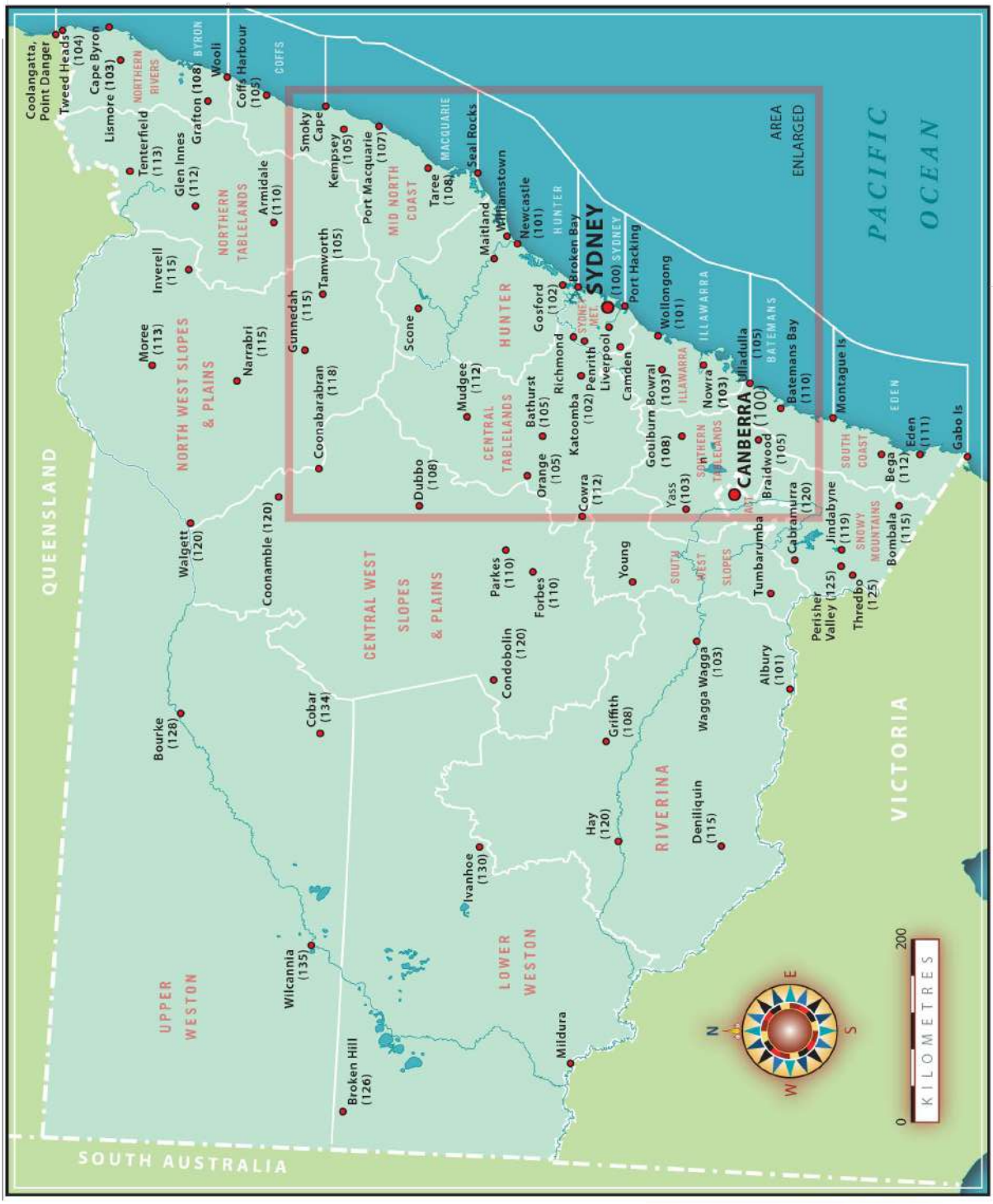
Building Price Index and Consumer Price Index are base dated to 2011-2012

REGIONAL INDICES

Pages

- NEW SOUTH WALES 834-835
- QUEENSLAND. 836-837
- SOUTH AUSTRALIA 838-839
- VICTORIA. 840-841
- WESTERN AUSTRALIA. 842-843
- NORTHERN TERRITORY AND TASMANIA 844

Base **- SYDNEY 100**



NOTE :

The above index is a broad indication only of cost variation within New South Wales and is applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

REGIONAL INDICES NEW SOUTH WALES

Base **- SYDNEY 100**



Note: Sydney CBD (103)

REGIONAL INDICES QUEENSLAND

Base **- BRISBANE 100**



NOTE :

The above index is a broad indication only of cost variation within Queensland and is applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

REGIONAL INDICES QUEENSLAND

Base **- BRISBANE 100**



Note: Brisbane CBD (102)

REGIONAL INDICES SOUTH AUSTRALIA

Base **- ADELAIDE 100**

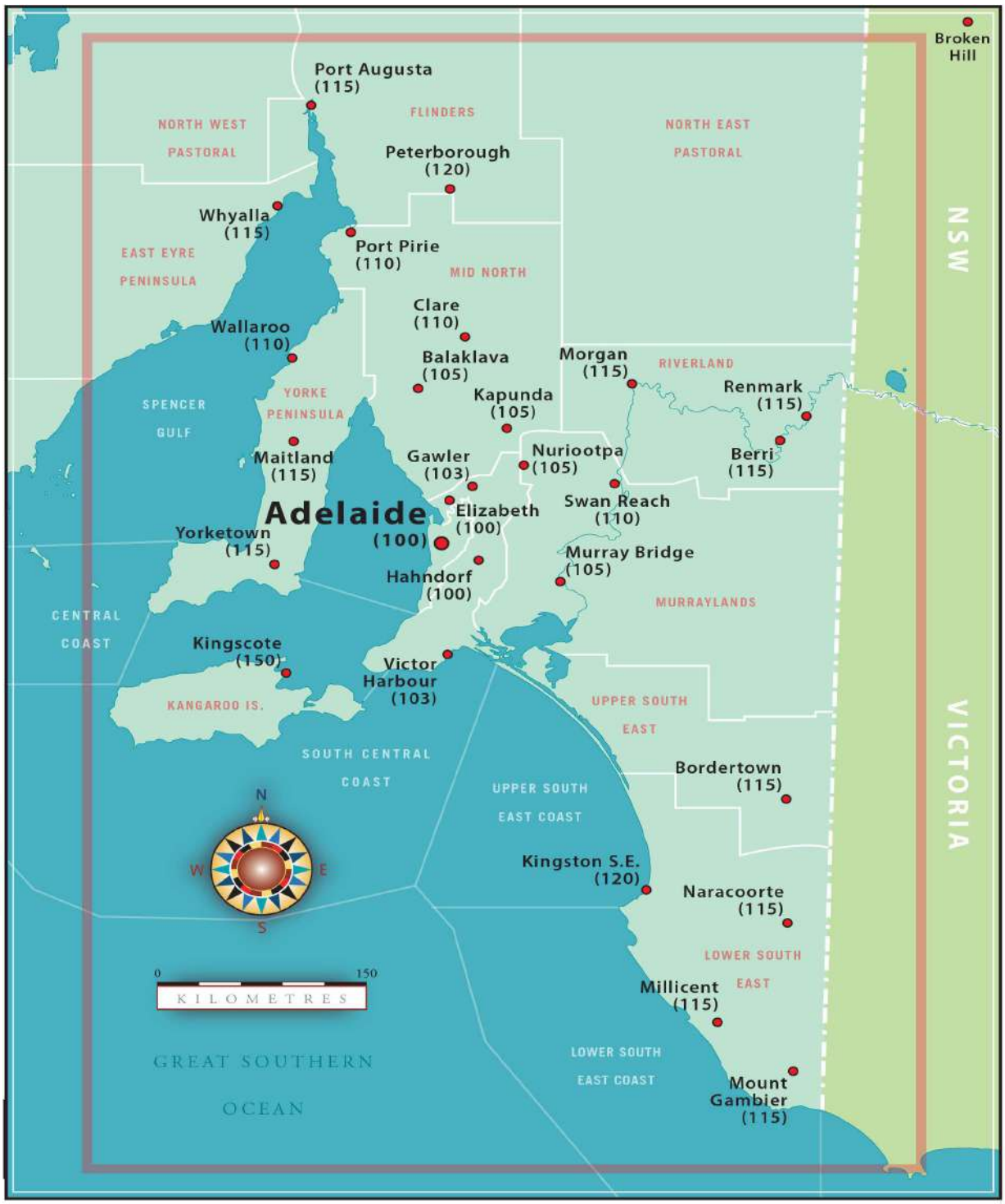


NOTE :

The above index is a broad indication only of cost variation within South Australia and is applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

REGIONAL INDICES SOUTH AUSTRALIA

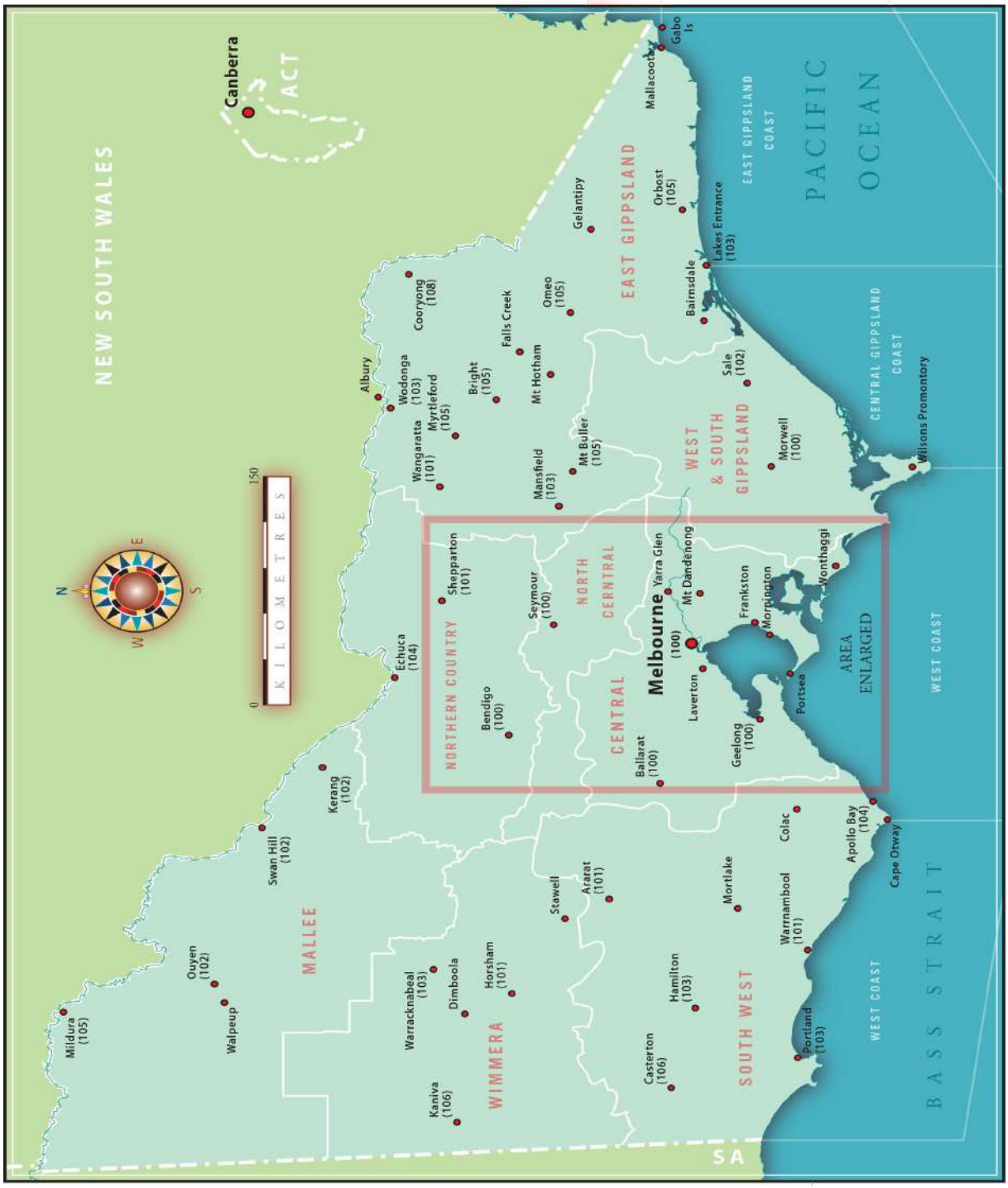
Base **- ADELAIDE 100**



Note: Adelaide CBD (103)

REGIONAL INDICES VICTORIA

Base - **MELBOURNE 100**



NOTE :

The above index is a broad indication only of cost variation within Victoria and is applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

REGIONAL INDICES VICTORIA

Base **- MELBOURNE 100**



Note: Melbourne CBD (103)

REGIONAL INDICES WESTERN AUSTRALIA

Base **- PERTH 100**



NOTE :

The above index is a broad indication only of cost variation within Western Australia and is applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

REGIONAL INDICES WESTERN AUSTRALIA

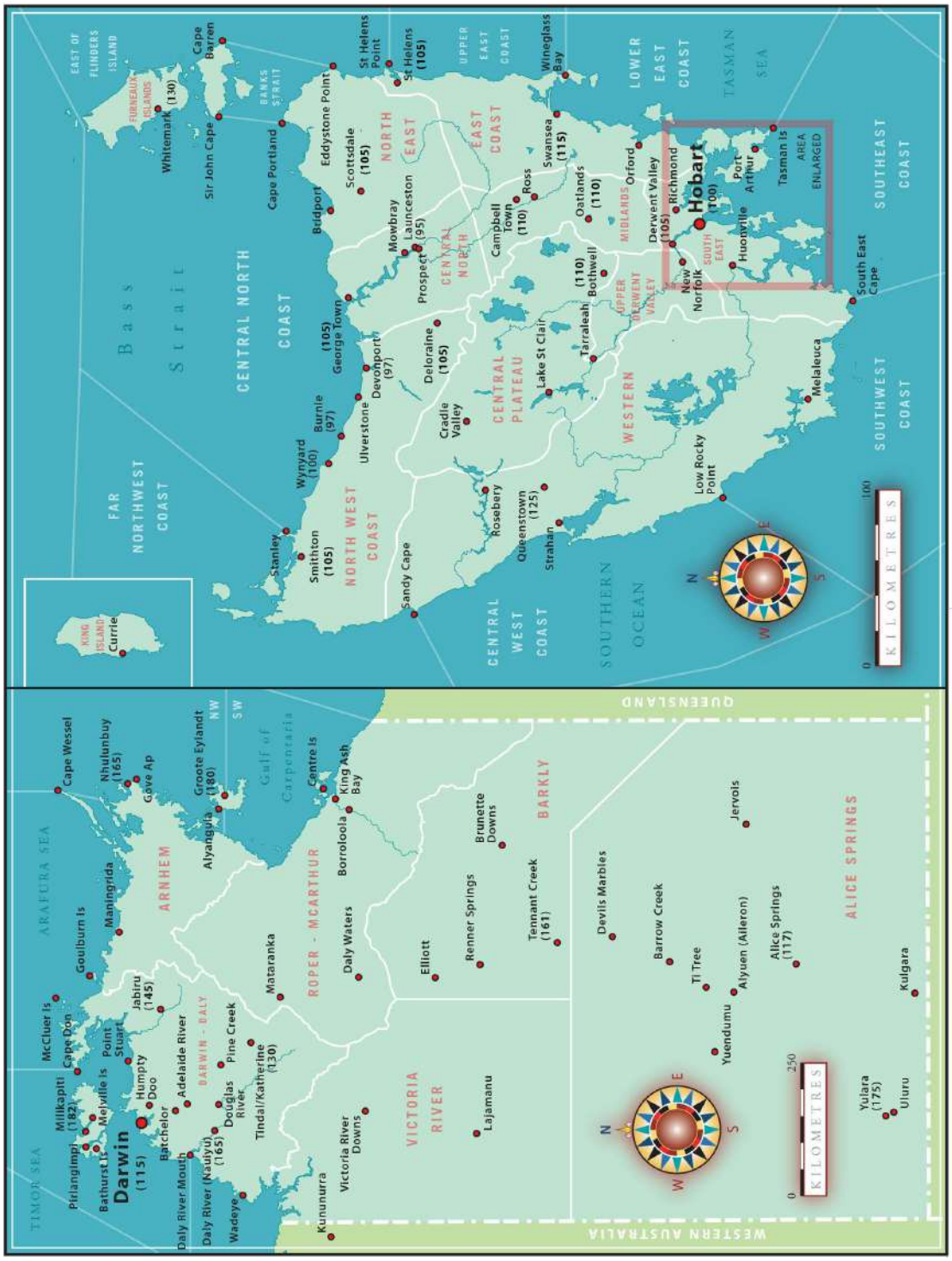
Base **- PERTH 100**



Note: Perth CBD (103)

REGIONAL INDICES NORTHERN TERRITORY AND TASMANIA

Base **- HOBART 100**



Base **- ADELAIDE 100**

DEVELOPMENT, PLANNING AND MANAGEMENT

CONTENTS

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  - Background - State Planning Policy 3.6. 846-849
  - Cost Guidelines for Items Suitable For the Development Contribution Plan. 850-856

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

BACKGROUND - STATE PLANNING POLICY 3.6

Developer contributions for infrastructure have been part of the process of development for many decades. Planning Bulletin 18 was developed in 1997 to address concerns that some local governments were levying contributions from developers for services and infrastructure beyond the scope of the policies of the Western Australian Planning Commission (the Commission). There were further concerns that developer contributions were being levied as a condition or pre-requisite of rezoning without any head of power or proper legislative basis for such charges.

At that time the power to require developer contributions derived from the Town Planning and Development Act 1928 (as amended) (now the Planning and Development Act 2005) which is legislation of the State Government administered by the Commission. Local governments do not have the authority under planning policy to require developer contributions except where expressly provided for in local planning schemes (formerly "town planning schemes") which have been recommended by the Commission and approved by the Minister.

Developers routinely provide the onsite physical infrastructure associated with water supply, sewerage, drainage, roads and power in the course of creating lots. Since 1956 developers have also been required to give up the equivalent of 10 percent of the gross subdivisible area for Public Open Space (Development Control Policy 2.3). Land developers are also required to contribute to other local infrastructure such as primary school sites, regional roads and, in some cases, public transport reserves.

In some areas where there are multiple landowners, private land pooling schemes were introduced which, among other things, set out scheme works and cost contributions to be shared between the owners. In some areas these schemes were the only way the land could be developed as they provided an equitable way to share the costs of many of the basic infrastructure items with credits for those developers who completed the work themselves.

Through the early cost sharing schemes, some key principles had been established to guide the application of the requirements for contributions including:

- The subject subdivision must create or contribute to the need for the particular infrastructure or facility for which the contribution is being sought.
- The contribution must be fair and reasonable and reflect the true costs of the infrastructure or facility.
- The contribution should be fairly apportioned between multiple landowners proportional to the share of the need created by each landowner's subdivision.
- A financial contribution must be spent within a reasonable period of time.
- There should be accountability in the manner in which contributions are determined and expended.

During the upswing in the land development sector during the mid-2000's, industry became concerned about the increasing use of legal agreements which created an avenue for the delivery of developer funded infrastructure which sat outside of the WAPC policies and the intent of Planning Bulletin 18. The process was uncontrolled, was often not founded on the key principles of need, nexus, equity and fairness, and was having an impact on housing affordability.

There was acknowledgement by the industry that there were impacts on services outside of the boundaries of the property, including demand for community centres and libraries. Many local governments in the growth corridors were struggling to meet the escalation in services and it was agreed that the delivery of community infrastructure needed to be discussed.

This led to a joint project with the Western Australian Local Government Association (WALGA) and the Urban Development Institute of Australia, with guidance from the Department of Planning. That project work underpinned the development of State Planning Policy 3.6 Development Contributions for Infrastructure (SPP 3.6) which superseded the existing policy settings when gazetted in November 2009. Planning Bulletin 100 was also developed to provide further implementation guidance.

SPP 3.6 sought to replace the existing framework (which was at mainly designed to fund 'hard' infrastructure, such as roads, sewer, water and power services) and the ad hoc approach to delivering 'soft' infrastructure, such as community centres and libraries, with a controlled process and a scope of inclusions reflective of the needs of the population in the new areas, even though the services may be located outside of the precinct in some cases.

The objectives of SPP 3.6 are:

- to promote the efficient and effective provision of public infrastructure and facilities to meet the demands arising from new growth and development;
- ensure that development contributions are necessary and relevant to the proposed development and are charged equitably between those benefiting from the infrastructure and facilities provided;
- to ensure consistency and transparency in the system for apportioning, collecting and spending development contributions.
- It should be noted that at the time of writing, a draft revision of SPP 3.6 had been released for public consultation by the Department of Planning (2016). As this draft revision is not yet operational, references herein pertain to the 2009 SPP 3.6, unless otherwise specified. Changes associated with the current draft revision, largely seek to provide further guidance on the policies operation for clarity and consistency. They are not anticipated to substantially vary the processes outlined in this part.

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

BACKGROUND – STATE PLANNING POLICY 3.6 (cont'd)

Developer contributions for infrastructure are not intended to replace normal government expenditure and are intended to supplement services provided through rates and government grants. At all times the balance between housing affordability and liveability in new areas are to be maintained with the need for infrastructure included in the development contribution plan which needs to be clearly demonstrated and the connection between the development and the demand created be clearly established.

It is also critical that both the method for calculating the development contribution and the manner in which it is applied should be clear, transparent and simple to understand and administer. Each developer contribution plan must set out in detail the calculation of the cost. This is where the work of Rawlinsons Pty Ltd becomes invaluable in providing guidance on standard infrastructure costs.

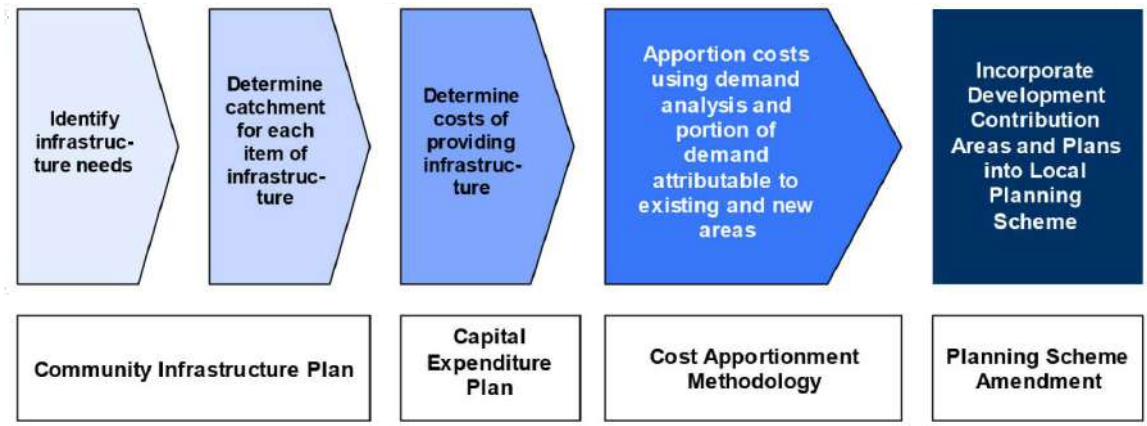
Types of Development Contribution Schemes :

Contribution schemes can serve a variety of purposes. One of the most common purposes is to assist in the equal sharing of the costs of development of land with fragmented ownership. These schemes typically cover the provision of the standard infrastructure of water, waste water, power and roads, as well as parks and other essential services.

The other common purpose of a traditional contribution scheme, relates to services outside the boundary of the development area and can include upgrades to roads and other services, where those upgrades are required to cater for demands generated by new communities, along with agreed community infrastructure.

Contribution schemes are not to be used for infrastructure which is not relevant to the development, or steps beyond the provision of basic liveability items.

A Simple Guide to a Developer Contribution Scheme



Step 1 – Identify Infrastructure Needs

The starting point is to define the catchment area for the contribution plan. In some instances there will be areas which require more services than others so the defined catchment can be broken down into precincts or cells which have local infrastructure items identified, along with those at district scale.

The lifespan of the contribution plan then needs to be agreed; this is the time the identified infrastructure should be constructed and completed. The lifespan should not be so extensive that contributions collected may not be expended for many years after their contribution. The population projections, along with the demographic profile of expected residents, needs to be established. This helps to identify the scope of the infrastructure that needs to be provided, as well as underpinning an apportionment schedule. This is a key piece of work as underestimating population growth can lead to overdemand for available services. Underestimation of the infrastructure and services required can put a local authority at risk of a funding shortfall.

An audit should then be undertaken to identify gaps between the existing infrastructure, the infrastructure that was already planned for delivery for current residents and the infrastructure which will be required for liveability for new residents. In some instances the gap will be insufficient to warrant the time and administrative costs of developing and maintaining a development contribution plan, in other areas the gap may be significant.

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

BACKGROUND – STATE PLANNING POLICY 3.6 (cont'd)

It is important to confirm the infrastructure needs in relation to the existing community. Whilst these can be identified within a plan, they will not be included in the cost apportionment. This could include instances where a community hall is included to cater for all local residents, however only that proportion of the cost relating the residents in new areas is included in the cost apportionment per lot/dwelling.

Step 2 – Determine the catchment for the Infrastructure

The Development Contribution Area (DCA) is the area to which the Development Contribution Plan (DCP) will apply. This may cover all or part of a local government area.

Defining catchment areas for items of infrastructure can be quite complex. Whilst there are indicative figures for the number of persons per facility, patterns of usage may vary on how far people are likely to travel to use a facility.

In addition, major barriers to travelling to a facility (main roads, freeways, rivers etc) can also help to define a catchment area.

As a further complication, most community infrastructure standards are defined in terms of ratios per person rather than per dwelling. An assumption will need to be made about the occupancy rates for dwellings to convert population data to dwelling unit equivalents.

Catchments should be identified physically and described in terms of the number of demand units. Demand units for community facilities will usually relate to a lot or dwelling unit.

Step 3 – Determine the cost of providing the infrastructure

Once the scope of the infrastructure to be included in the plan has been identified, the construction and fit out standard needs to be determined. This is often one of the more complex issues to work though, however the standard should meet the basic and reasonable community needs and expectations. From there costings can be developed and a capital expenditure plan developed.

The capital infrastructure plan should include all of the capital costs of providing the required infrastructure, including design, land and construction costs as well as the cost of financing the infrastructure when it is required in advance of revenue collection.

Once the capital infrastructure plan has been established, the sources of funding should be identified. All potential sources should be explored as it is not the intent that contributions should replace other sources of capital as housing affordability must remain a priority.

The residual funding required after the identification of all other sources then needs to be allocated. The need generated by the existing community or the residents forecast to join the community outside of the timeframe of the developer contribution plan should be identified and deducted from the cost of each relevant item prior to the calculation of the apportionment.

Step 4 – Apportion Costs

Whilst equity is one of the guiding principles of developing a contribution scheme, that does not mean that every contribution is the same. Some cells within the broader catchment may have more infrastructure requirements than others creating a variation in the resulting contribution required.

Step 5 – Give Effect to the Plan

DCPs are subject to a statutory public consultation process when it is included in a local planning scheme, but it is strongly recommended that consultation also occurs during the preparation, to allow the community to analyse the proposed scope of works and raise any issues or concerns before the content of the DCP is included in a statutory scheme amendment document, subject to formal consideration by Local Government, the WAPC and the Minister for Planning.

A DCP does not have effect until it is incorporated into a local planning scheme, either as part of a new scheme or through an amendment to an existing scheme. It is a requirement that within 90 days of a DCP coming into effect, the Local Government is to adopt and make available a DCP report and cost apportionment schedule to all affected owners.

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

BACKGROUND – STATE PLANNING POLICY 3.6 (cont'd)

Step 6 - Payment of Contributions

Under SPP 3.6 there are three stages to the calculation and payment of individual contributions. There are three stages to levying development contributions:

1. Development Contributions are **formulated and agreed**
   The Development Contribution Plan is used to prescribe the cost contributions for owners in a Development Contribution Area.

2. Development contributions are **calculated and applied**
   Development contributions are generally calculated and applied by way of conditions of subdivision, strata subdivision or development, particularly in greenfield areas. Development contributions may also be sought in infill and redevelopment areas at the time of subdivision, strata subdivision or development.

3. Development contributions become **due and payable**
   Development contributions become due and payable as part of the subdivision clearance process or prior to the commencement of development. Clearance of Deposited Plans, or Strata Plans, as the case may be, to enable the issuance of titles should not occur until full payment, as calculated and applied, has been finalised. Development contributions are only payable on the proportion of land within a Plan being requested for clearance in a development.

Development contributions are not to be imposed as a condition of rezoning.

There may be variations on this where it relates to the ceding of land or other in kind contributions. In the circumstance where the contribution is made ahead of the schedule above, the prefunded item will be used as a credit against any future contributions which may be owed.

Resolution of disputes will be through arbitration in accordance with the Commercial Arbitration Act 1985, with the exception of disputes relating to land value.

Step 7 - Review of the Plan

It is a requirement that the DCP is reviewed five years from the date of gazettal of the new or amended local planning scheme. Any modification will require an amendment to the relevant local planning scheme.

Further detailed information can be found on the Department of Planning's website: www.planning.wa.gov.au.

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN

Costs given hereafter should be used for the Development Contribution Scheme calculation only.

Costs given are the average prices for typical buildings at December 31 2020 within the metropolitan areas. ADD additional allowances for increased costs (refer Anticipated Building Price Adjustment page 15) and when referring to regional areas (refer Regional Indices pages 834-844).

PUBLIC PARKLAND AND RECREATION

Multi-purpose district sport grounds (at district parks) including grassed playing surfaces, maximum four multi-purpose hard surface outdoor courts, lighting, fencing but excluding sports equipment

Total area of approximately 7 hectares	$1,935,000	
consisting of :		
	ESTIMATED PROJECT COST	**NOTES**
4-6 sporting ovals of total area 6 hectares	$885,000 per 4 ha $187,000 per ha thereafter	
Multi-purpose plexi-pave hard surface outdoor courts (includes fencing, spoon drains and site preparation)	$285,000 per 2 sets of two courts	
Lighting to outdoor courts	$95,000 per 2 sets of two courts	**1**
Sports change rooms – total area of 70sqm (One set of male and female changing area with two showers, two toilet pans and two basins in each)	$204,000 $2,915/sqm of additional area	**2**
Public toilets attached – total area of 15sqm (Constructed under the roof of the change rooms)	$48,000	**2**
Public toilets isolated – total area of 15sqm	$44,000	**2**

Excludes
- Car parking, footpaths, perimeter bollards and the like
- Lighting to the sports ovals
- Overall site clearance of vegetation, sloping site or rock encountered and demolitions
- Utility headworks and external services outside 3m from the building
- Loose or special equipment, furniture, furnishings and legal fees
- Goods and Services Tax

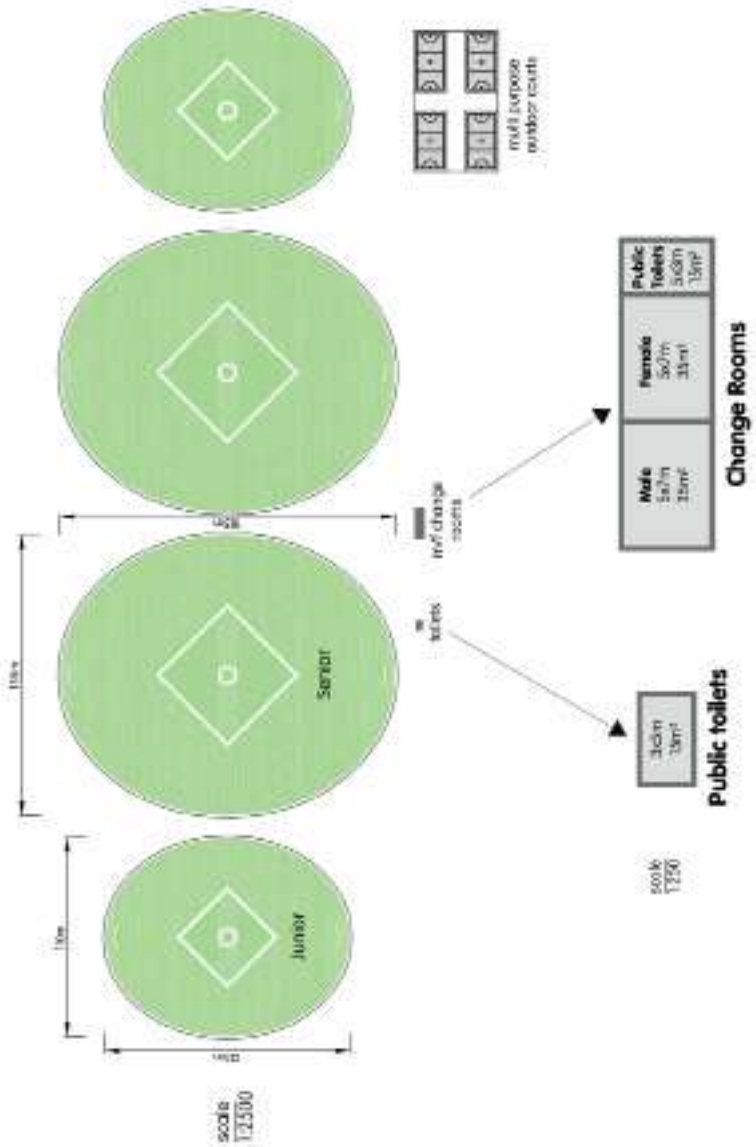
Notes
1. Based on mains power supply connected within 3m of the building
2. Based on mains water and power supply connected within 3m of the building

All the above costs include for main contractors margin and overheads, BCA requirements

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

PUBLIC PARKLAND AND RECREATION (cont'd)



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DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

MULTI-PURPOSE DISTRICT COMMUNITY BUILDING

Multi-purpose district community building and basic facilities (including, for example, meeting rooms, indoor recreation rooms, small scale flexible performance space, kitchen facilities, toilets)

Total area of approximately 730sqm	$1,930,000	
consisting of :		
	ESTIMATED CONSTRUCTION COST	**NOTES**
Indoor recreation rooms - total area 250sqm (removable wall between the rooms, storerooms)	$570,000	**1**
Meeting rooms - total area 30sqm (includes operable walls)	$ 65,000	
Small scale flexible performance space - 120sqm (storeroom and servery, no stage or bio-box)	$215,000	
Change rooms and toilets - 110sqm	$465,000	**2**
Kitchen and servery - 90sqm (full frying facility with cool and dry store)	$350,000	**3**
Lobby, entry and circulation - 90sqm	$190,000	**4**
Storerooms for external access - 40sqm	$ 75,000	

Excludes
- Car parking, footpaths, perimeter bollards and the like
- Overall site clearance of vegetation, sloping site or rock encountered, demolitions
- Landscaping
- Utility headworks and external services outside 3m from the building
- Loose or special equipment, furniture and furnishings
- Design and legal fees
- Goods and Services Tax

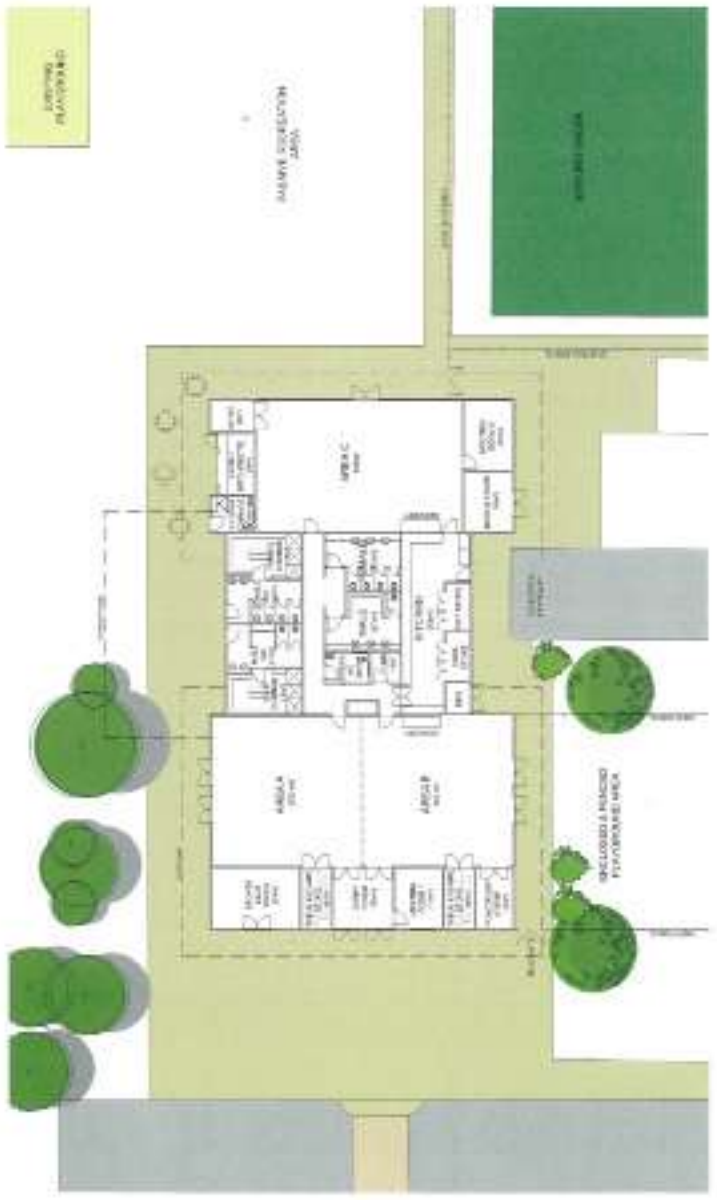
Notes
1. Multi-purpose activity rooms marked for multiple activities and includes storage rooms and circulation area
2. Two blocks – one comprising male and female change rooms, showers and toilets and the other comprising toilets only central to the meeting rooms, performance space and kitchen
3. Excludes all equipment but includes all the services particularly mechanical and hydraulics for a fully operating commercial kitchen
4. Allows for entry statement, open foyer area and circulation

All the above costs include for main contractors margin and overheads, BCA requirements

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

MULTI PURPOSE DISTRICT COMMUNITY BUILDING (cont'd)



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DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

DISTRICT LIBRARY BUILDING

District library buildings and basic facilities (excluding specialist interior fit-out and technology)
Built single storey and air conditioned
Total area of approximately 840sqm $2,095,000

Consisting of :

	Estimated Project Cost	Notes
Open plan reception and books display – 500sqm	$1,320,000	**1**
Study and quiet rooms – 70sqm	$115,000	
Sound proof technology rooms – 60sqm	$165,000	**2**
Administration offices and workrooms - 60sqm	$105,000	
Toilets and auxiliary rooms – 80sqm	$225,000	
Entry statement and foyer	$165,000	

Excludes

- Car parking, footpaths, perimeter bollards and the like
- Overall site clearance of vegetation, sloping site or rock encountered, demolitions
- Landscaping
- Utility headworks and external services outside 3m from the building
- Loose or special equipment, furniture, furnishings and legal fees
- Goods and Services Tax

Notes

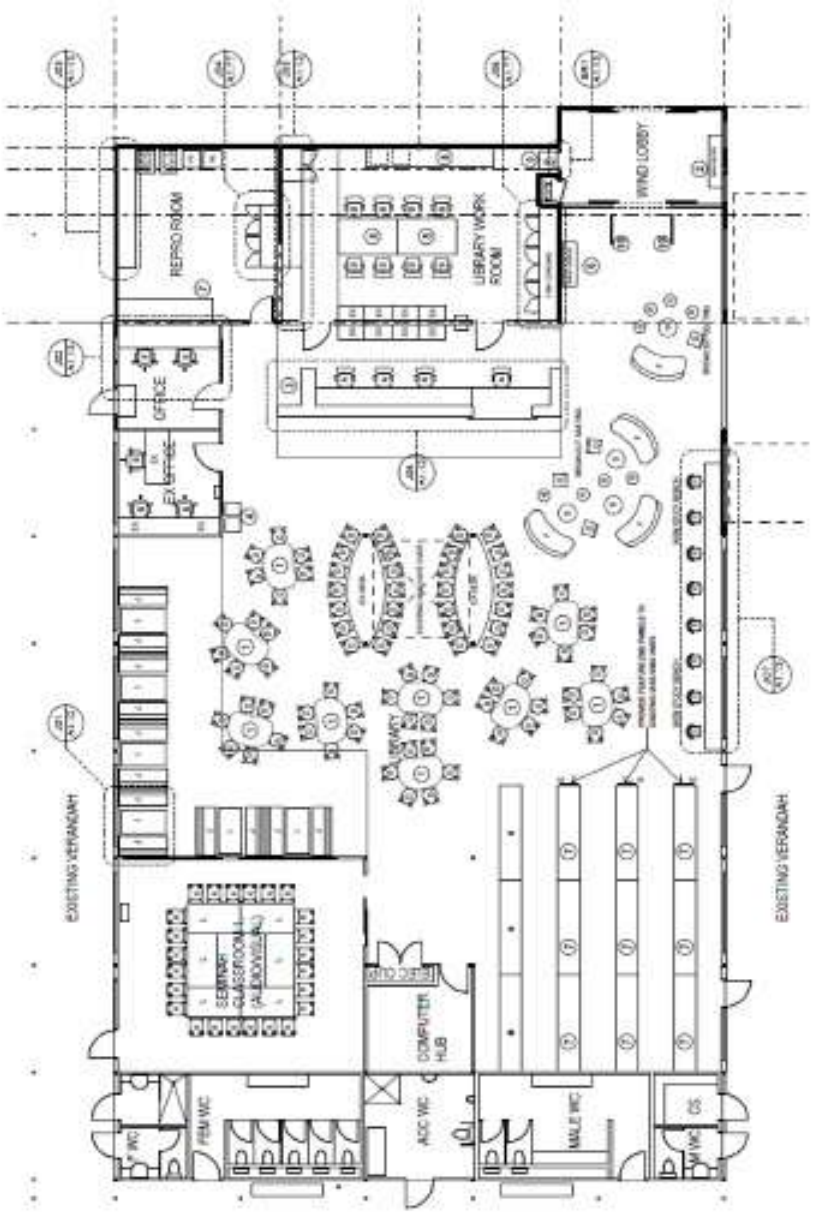
1. No allowance is included for any fixed or loose shelving units or other storage furniture and fitments specific to the library business
2. Allowance has been included for acoustic treatment to the walls, ceilings and doors. Provision for technology has been included but no equipment or dedicated cabling

All the above costs include for main contractors margin and overheads, BCA requirements.

DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

DISTRICT LIBRARY BUILDING (cont'd)



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DEVELOPMENT CONTRIBUTIONS FOR INFRASTRUCTURE

COST GUIDELINES FOR ITEMS SUITABLE FOR THE DEVELOPMENT CONTRIBUTION PLAN (cont'd)

CURRENT URBAN DEVELOPMENT ISSUES

The policy settings for urban development are continuously changing and evolving to meet the needs and expectations of our communities. Many of the detailed policies are invisible to the buyer but are there to ensure that they can live in comfort and safety in the location of their choice. Policies cover how stormwater is managed to avoid flooding, the quality of the roads, connection to power, water and sewer as well as the layout of estates to enable traffic flow and pedestrian safety. Add to that environmental management, which can cover anything from endangered species to mosquito management, consideration of parkland design and controls of the built environment, urban development is a complex process involving many government agencies. At last count there were twenty eight government agencies that could potentially be involved in the approvals process, so getting the policy settings right is an ongoing challenge, particularly as Western Australia is going through a period of significant growth. The below is a list of just some of the substantial reforms progressed during 2016, impacting the urban development industry.

2016 REFORMS

- The Perth and Peel Green Growth Plan for 3.5 million (draft) was released for public consultation during early 2016. The Plan is the outcome of a strategic environmental assessment of the Perth and Peel regions, targeting identification of Matters of National Environmental Significance (MNES), assessment of impacts from a range of actions and management of these impacts. A number of these 'classes of action' relate to activities associated with urbanisation and thus the management of these impacts will have a significant impact on where and how development will take place, as well as associated costs.

- Draft Subregional Planning Frameworks, released in conjunction with the Perth and Peel@3.5 Million suite of documents were refined, with final release anticipated in early 2017. These frameworks are perhaps the most significant planning documents released in recent years, impacting the supply, location and staging of land available for urban uses.

- Banksia Woodlands of the Swan Coastal Plain were listed as a Threatened Ecological Community (TEC) under the Commonwealth Environment Protection and Biodiversity Conservation Act 1999 (EPBC Act). Banksia Woodlands are present across much of WA and thus increased protection of the community is likely to have a significant impact on development activities across the state.

- An omnibus amendment to the Planning and Development (Local Planning Schemes) Regulations was progressed in 2016, affecting processes and timeframes associated with development planning across WA.

- Draft amendments to the Residential Design Codes (State Planning Policy 3.1) have been released, which seek to amend provisions in line with related reforms.

- Reforms in relation to Liveable Neighbourhoods, the operational policy which guides the planning subdivision for greenfield and large brownfield sites in WA, have progressed. These reforms have the potential to substantially alter the design and planning of land development across the state. A revised policy is expected for release in 2017.

- Substantial reform of the Strata Titles Act has progressed during 2016, with significant implications for industry. Once finalised, these reforms will facilitate new built form development product through policy provisions which create new management arrangement for grouped and apartment housing.

- Reform of the Bushfire Risk Management Framework has progressed, with draft revision of the Bushfire Guidelines progressing, as well as further guidance and changes to administrative procedures, aimed at creating greater consistency in application of new policy provisions.

- Design WA was released by the Department of Planning in late 2016, a suite of documents which will guide design and planning of the built environment in WA. The suite includes guidelines for the design of apartment buildings, as well as policy provisions intended to replace r-codes requirements for certain classes of development.

BUILDING PLANNING, ADMINISTRATION AND MANAGEMENT

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GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT

Prepared and Edited by Garry Baverstock AM, Adjunct Professor School of Engineering, Murdoch University, Director of Wise Earth Research Centre and Innovation Australia.

INTRODUCTION

Green Design is rapidly becoming a mainstream approach to developing our built environment. It has taken a great time for this philosophy, principles and techniques to evolve, be accepted and after a long gestation period lead to practical solutions. Sustainable Development has been a term used for the last twenty five years to describe the major challenge for the world, socially, economically and environmentally. Without green design of buildings achievement of true sustainability is impossible.

Given that a large proportion of emissions of CO2 comes from the built environment it really is the main way forward at this time to achieve a low carbon polluting economy and a more stable climate on planet earth. Buildings in Australia use about 50% of the electricity generation. The current focus on direct action for renewable energy and a renewed focus on energy efficiency by the Federal Government will result in an accentuation of built environment advances to reduce carbon pollution.

Federal Governments will continue with stronger compliance levels at building license stage and result in an accentuation of built environment advances to further reduce carbon pollution and meet its International committed obligations.

WHAT IS GREEN DESIGN?

Green design relates to preservation and improvement of our living environment. The natural world is the beneficiary of such a “green” approach that in turns helps to make human existence on earth more sustainable and eventually assured once Climate Change along with global population has been stabilised. Provided this approach is efficiently applied, the massive growth in consumption of materials over the last 50 years and the ever-increasing pollution of air and depletion of water resources can be significantly remediated and eventually a sustainable balance achieved.

INCREASING ACCEPTANCE OF GREEN PRINCIPLES

The recognition of the importance of such green design principles is seen in the steadily increasing energy efficiency requirements in the Building Code of Australia, the various rating options available for domestic dwellings (House Energy Rating Software http://www.nathers.gov.au/software/index.html) and commercial buildings (NABERS, http://www.nabers.com.au/, and Greenstar, http://www.gbca.org.au/green-star/rating-tools/), and the gradual introduction of provisions requiring energy efficiency rating statements to be provided when a building is being sold or leased.

The opportunity to showcase designs using these rating systems are leading to changes in the way buildings are developed. The fact that the NABERS scheme has added a star to the star rating options bringing it to 6 star is an indication that an increasing number of designers are taking on board the possibilities offered by good “design for sustainability” principles. Obviously, the next step is to increase the rating option higher in the future.

Apart from the basics of good passive solar design the main focus needs to be the application of new energy and water technologies. The latest building materials that science has to offer, will become an important driver of change for the rest of the 21st century. Undoubtedly as the ravages of Climate Change become obvious to everyone, compliance and design guidelines will increase in stringency.

WHAT CAN BE ACHIEVED BY GREEN DESIGN?

In most temperate regions of Australia, energy savings of over 50% in commercial and institutional buildings are possible. These savings can be achieved by observing simple design methods and applying rating systems, promoted by the Green Building Council of Australia (GBCA) (http://www.gbca.org.au/), once the basic science and design principles are clearly understood. With residential buildings, passive solar design and the installation of low energy/solar water heating and energy efficient lighting appliances, the savings in energy can easily be as high as 90% when compared with the normal lifestyle prevalent in Australia in the late 20th century.

Significant savings in water use can also be achieved in both the domestic and commercial spheres. The benefits of the use of water saving appliances and fittings can be augmented by grey water recycling and harvesting systems and the planting of water-wise gardens.

THE GLOBAL CHALLENGE

The challenge for the building design professional is to cater for current owner/occupier requirements while adopting an intelligent approach to delivering practical environmental outcomes. Sustainability in construction aims not only in meeting or bettering the currently legislated requirements but also to produce outcomes that address important issues. In this way there will be a meaningful contribution to addressing the world’s very serious pollution and climate change problems.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

WHAT IS GREEN DESIGN? (cont'd)

IMPORTANT ELEMENTS OF GREEN DESIGN

Although buildings have functional, cost, quality, aesthetic, health and welfare constraints, it is imperative that built environment professionals place the following principles and targets high on the agenda during the design process; otherwise very little will be achieved. This will require integration of these constraints with intelligent planning and a systematic selection of materials, systems and technologies at key times during the design process.

1. **Ecologically sustainable design (ESD)** – ESD does not need to have a detrimental impact on the aesthetics of buildings, in fact the contrary. About 50% of greenhouse gasses (GHGs) in Australia is attributed to the built environment (including infrastructure), so the real potential for Green Design and Sustainable Planning cannot be ignored any longer. Use of design tools such as 'ecotect' an early integrated environmental program, now owned and operated by Autodesk, or referral to the library of design tools developed by the GBCA, along with the application of the latest high standards mandated by the rating systems for Australia such as AccuRate and NABERS, is an important arsenal of tools for the contemporary architect or building designer. However there is no substitute for ongoing practical learning online or through higher studies offered by Universities for post graduate studies in the field. Understanding the concepts and how to understand latest innovations in materials, systems and technologies can be a big advantage to architects and designers wishing to stay at the cutting edge (refer to http://solar-e.net/elearning/). There are many academic programs offered by universities and technical colleges through the country and internet resources like the solar-e.net learning that can lead into formal qualifications from a University.

2. **Prudent master planning** – Shaping and orienting buildings on site is a crucial factor in leaving the window of opportunity to maximize passive solar benefits. Planning construction options from before the finalisation concept stage will lead to form and function aligning and with aesthetics and economic efficiency. This is probably the most important measure a designer can initially take.

3. **Operating efficiency** – Energy management control systems are important to enhance the green design strategies. These systems facilitate the establishment of an efficient use culture by the occupant. However, some of the best efficiencies are achievable by simply turning off the air conditioning plant and opening windows to gain natural ventilation. Research has shown a good use culture by occupants can create a 10% reduction in energy use or more.

4. **Innovative construction** – Energy and water efficient technologies should not be an afterthought. The reality is that new technologies are often compatible with innovative construction techniques and therefore suit a contemporary approach to the aesthetics. It is important that the physics, such as thermal performance, is clearly understood. Their selection needs scrutiny for cost effectiveness as well as the physics, if comparable or superior to the conventional approach. The advancement in nanotechnology is starting to provide possibilities that architects can consider and is set to radically change architectural design over the coming decades (https://www.understandingnano.com/nanotechnology-building-materials.html). The innovation process requires early research for the correct design decisions to be established at the concept stage of design. Use of thermal inertia system is now emerging as a new frontier for building science. This may be as simple as building in mass materials and using them more prudently as an energy storage system for night time comfort in buildings. Linkage with PV systems using specially controlled technologies will provide a cheaper option than storing energy in batteries from PV systems.

5. **Resources consumption** – Consider recycling, sourcing local and the use of low embodied energy materials. Where applicable 'Ecospecifier' (http://www.ecospecifier.com.au) can be utilised to review the latest products and materials. It is important to consider rebate schemes to improve the economics for the client or building owner. These rebate schemes mainly applied to solar water heaters and PV roof top systems are rapidly being reduced or eliminated now in Australia as new innovative systems can reduce capital costs and deliver better energy use. The dumping of renewable energy excesses in the grid is an emerging advance, using new control systems approved by power suppliers. Hot water systems connected to grid connected PV may avoid the need for battery storage (which has a Life Cycle Cost of about 33 c/kWh) by acting as a thermal energy battery.

6. **Costs** – Cost neutral designs and materials can provide mutually beneficial outcomes and must be an important objective of the designer. However some extra cost implications may occur with new innovative approaches. This cost must be an investment rather than a wasted cost and sold to the property owner as such. This inevitably leads to the 'light-weight/heavy weight arguments', with lightweight solutions often appearing cheaper. It is not so simple. The longevity of a building becomes a serious issue when writing off the initial capital investment and the life cycle ramifications. Another key issue is the choice of cooling and heating strategy. The life cycle costing of the structure needs to include the thermal inertia component to be able to understand the economics in the longer term. Thermal mass linkage can be more important with certain hybrid AC/ventilation systems. Such systems maximize the advantages of heavy weight construction through the use of night ventilation cooling to reducing summer peak loads. Similarly the use of solar sources of passive and active heating in winter can keep the mass temperature of the building naturally warmer and save on heating bills.

7. **Retrofits** – Retrofitting of existing buildings is important to improve the energy and building efficiency. The majority of environmental problems relate to vast numbers of inefficient existing buildings. Over the next 40 years building recycling or retrofitting will be a realistic path in reducing energy and water use in the built environment. The largest savings in greenhouse gases until 2050 are predicted to occur in the building recycling and retrofitting industries because much of the embodied energy involved in manufacture of new materials is avoided. (Ref. http://www.academia.edu/212378/Using_building_adaptation_to_deliver_sustainability_in_Australia). There are over 7.5 million existing houses in Australia. They cannot all be demolished and replaced. We need to adapt.

8. **Innovative technologies** – new PV driven technologies for water heating are emerging as viable alternatives. The integration into the aesthetics is the main challenge ahead as the economies of mass production are expanding the viable range of choices for architects and builders. We are entering a new period where technological solutions developed in the 20th century will need to be adapted or replaced. One key area will be new ways of using the sun to heat hot water for building occupants.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

CLIMATE CHANGE AND CARBON POLLUTION

An emerging government policy in Australia to address climate change due to increasing carbon pollution on the planet is somehow to place a price on carbon. Modelling the economic realities of carbon pollution is a large challenge in making decisions based on facts and data. It is very complex with many variables in simulating the outcomes of any action. The environmental damage cost needs to be calculated so that economic decisions that affect the built environment results in decisions that are of benefit inter-generationally. This approach is currently in progress in most countries but is in its infancy. The built environment will be a particular target for such policy development as it is one of the major causes of climate change. For some time direct action will be the most likely step forward until more scientific analysis is available to Government to support the case for carbon pricing as a wholesale policy. The CRC on Low Carbon Living currently in progress will undoubtedly provide the basis of new initiatives in the near future. With the program now finished there will be a number of informative videos being produced to educate the public and professions to take advantage of the research completed over the life of the CRC (http://www.lowcarbonlivingcrc.com.au).

THE 'RIGHT' PRICE

It is generally recognized by scientists that the cost of every product or system is directly proportional to the cost of the energy that goes into the manufacture, operation of the system or building and the maintenance and capital replacement regimes, during the life of the built environment or component part of the total system.

For that reason the 'right' price on carbon should be structured to be an investment in an economic future for the built environment. The difficulty has been and will probably continue to be for a number of decades, is how this price is to be applied and the effectiveness in delivering changes that create a low carbon producing but fully functioning economy. In the near future there will be many Internet resources tracking the real price of carbon from the perspective of environmental damage costs (https://carbonpricingdashboard.worldbank.org/what-carbon-pricing for general reference and in Australia specifically https://en.wikipedia.org/wiki/Carbon_pricing_in_Australia).

THE REALITY OF GROWTH IN CARBON POLLUTION

Large factors affecting the amount of pollution emanating from the built environment are population growth as a prime mover of economic development and the growth in the rate of consumption in Australia.

One only needs to refer to the extensive growth in the sizes of houses in Australia since WWII (Fig.1). A green architect therefore must be interested in reducing the overall building footprint not just the carbon footprint or rate of carbon generation from a project.

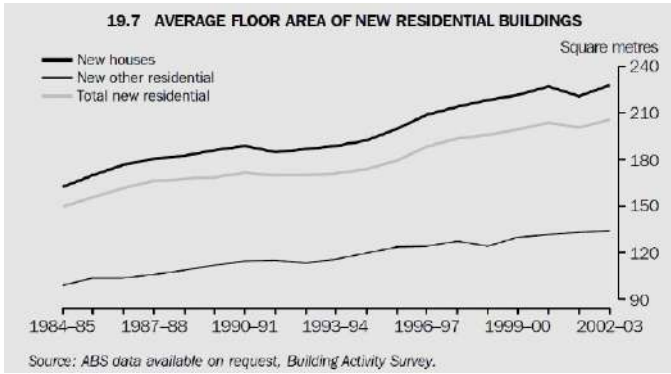


Fig. 1 – Average floor area of new residential buildings
(Source: http://www.abs.gov.au/ausstats, 1301.0 - Year Book Australia, 2005)

Over the next 20 years, Australia as well as the rest of the world needs to move to a higher quality of built environment as a driver of our economy rather than business as usual. The past reliance on population growth and increasing sizes of buildings creating wasteful consumption of building materials, will tend to diminish as energy and material costs rise. This is an underlying reality of carbon pollution. The construction industry and design professions need to adapt current cultures to produce measurable outcomes for a new cleaner, efficient economy.

An underlying reality is the diminishing amount of available energy and food for the growing world's population this century. This will increase the pressure for energy use in making materials and driving construction processes for the built environment. This will put enormous pressure on the next generation of building design professionals to institute Green Design everywhere during their careers. (refer to https://population.org.au).

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

CLIMATE CHANGE AND CARBON POLLUTION (cont'd)

MACRO LEVEL

In the meantime, an important focus will be how we adapt our infrastructure such as transport systems to help create a more efficient macro plan for our cities.

Public transport initiatives (also known as a Green Traffic Plan; incorporates suitable public transport links and aims to remove single occupant vehicles from the road) - will include shower facilities and bicycle lockers. Reductions of 5 to 10% of the carbon pollution are possible with smart transport planning. The potential for building development around such nodes known as Transit Oriented Development (TODs) are becoming increasingly popular and a target for preferred densification of cities in Australia. These projects also have potential to integrate green design in a very beneficial and multi-dimensional manner as the use of share water and energy infrastructure make green design far more economically attractive. (Refer to: www.swanbournevillagetrust.com).

Green energy (energy sourced from renewable sources such as the sun, wind and hydro) - 90% of Australia's energy comes from coal or gas fired generators that produce large amounts of greenhouse gases. Until recently, green energy used around 5 cents per kWh extra, dependent on location (http://www.synergy.net.au). However, the dramatic reductions of PV power and the introduction of batteries the 24 hour power price is heading to 15 c/kWh averaged over 24 hours. Each state has a different policy for offering a Green Power scheme, so it is worth checking each year.

The challenge is integration of green energy into our built and natural environments to be effective economically as well as abating carbon. To stay up to date with progress in green energy systems, the Australia Solar Energy Society has been for the last 50 years a very useful body (https://www.smartenergy.org.au). This non-profit body is related to the world's peak organisation for green energy and has operated nationally and at international conferences over the last 50 years. ISES (the international body) (www.ises.org) has predicted that adoption 25% - 50% renewable energy is possible and practical as industries grow over the next 50 years and has published a number of white papers detailing how this can occur. More technical inputs are available from other organisations such as the ATA and SEN (Sustainable Energy Network). It is worth checking out current active organisations that provide the latest up to date information.

MICRO LEVEL

Low embodied energy material selection (considers the energy required to manufacture each building product) – opt for products with the lowest total energy. Total energy includes resource excavation, extraction from the earth, transportation, energy in production and complexity of manufacture. Usually the more natural and simpler the process, the less harm is done to the environment. (A truly green project should endeavour to utilise materials with small-embodied energy coefficients. For example, structural steel uses in the region of 35 MJ/kg whilst extruded aluminium uses six times as much. (See the Embodied Energy Fact Sheet, produced by the Australian Government, should be available at http://www.yourhome.gov.au).

Local sourcing of materials and labour (to ensure that transportation costs are reduced) - a secondary effect may be the stimulation of the local economy.

Offset carbon trading for embodied energy - negates the release of one tonne of CO_2e (carbon dioxide equivalent) by avoiding the release of or removing from the atmosphere the same amount of CO_2e somewhere else. What may be viewed as a secondary level 'green' item, as it is better not to produce the carbon in the first place. As governments introduce a cost penalty for carbon pollution, there will be more incentives for direct action through green building design. There seems to be a diversion from such schemes by most governments but it may return as a governance mechanism in the future. A factor often ignored with embodied energy is the longevity of the material when built into the building. This factor can make a huge difference to over simplistic off set schemes. Buildings in many loactions in Europe built in mass stone of brick are still in existence 500 to a 1000 years later. In heritage creating architecture analysing the embodied energy is superfluous.

RECOMMENDED DESIGN PROCESS AND PRINCIPLES

Designs should relate to its local climate, soil conditions and availability of cost effective building materials and technologies. Cultural aspects are extremely important in final design decisions. The way people currently use buildings or capacity to learn new user-habits will affect the systems introduced and the control systems employed. Cultural effects must always be considered as related to longevity which relies on how a culture actually admires its buildings.

CLIENT REQUIREMENTS

Early decisions need complete involvement of the client. The decision to have a naturally ventilated building or a solely air conditioned space, or even a hybrid solution of incorporating both using a mixture of natural and conditioned approaches, and even manual and automatic controls, will have a large impact on the final result and the efficiency performance. These decisions require a collegiate approach by the architect, client and building services engineers. Similar considerations to decide on natural and artificial lighting strategies, use-regimes/ protocols and control systems should be an early consideration in the design process. Water use, waste management and landscape approaches also need early consideration at the briefing stage or during the master-planning phase, long before the concept design has been formulated. Client requirements in the future should also comply with generally recommended principles of sustainability. The professional must guide this process to happen.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

RECOMMENDED DESIGN PROCESS AND PRINCIPLES (cont'd)

CLIMATIC INFLUENCE

The basic climatic decisions have the greatest impact on choices for shaping and orienting a building as well as deciding the physical architectural features of the structure. For instance, the decision to choose a heavy thermal mass building with large north facing glazed walls may be ideal for Hobart, whilst lightweight construction with screened porches that suit cool breezes may be ideal for Darwin. Every site, every client's requirements and cost effective construction options are different for every situation. Therefore, the sustainable solution ends up being a function of the process that was undertaken for the particular climate.

WEIGHING UP THE COST EFFECTIVENESS

Understanding the benefits to the environment and the building occupier is an important part of the process and is imperative for true sustainability to be achieved. For example, movement detectors and timers for lights in intermittently used areas could have a low cost and high benefit. A ground source heat pump for 50% heat and hot water could have a high cost but low benefit. If such decisions are left too late, it is easy for a concept design to be left irrelevant or inappropriate due to a poorly managed design process.

FEASIBILITY STUDIES

A successful, professional practice should do numerous generic studies prior to embarking on a green or sustainable design. This will avoid confusion and back tracking, if the viability is proven to be faulty at a later stage. Building types and their particular building services systems can be improved by life cycle costing exercises taking into account running costs, embodied energy, maintenance, capital replacement and carbon abatement performance.

Checking the level of benefit against the cost should always be undertaken before final implementation. The money may be better used on another element, material or system or indicate that a completely different strategy be adopted. It is important to check the cost of maintaining, operating or parts replacement, over its expected life, compared to the initial cost and benefits.

Products or systems with comparatively low energy consumption, but a comparatively short life, may, after a number of replacement cycles, actually be worse for the environment and be poor economic decision making in the long term.

ONGOING PROFESSIONAL DEVELOPMENT

Keeping updated of the latest changes in environmental design methods and processes will be an important habit of the modern architect, builder and developer for most of this century. Formal courses and computer packages will increasingly facilitate the design process. It is important to understand that there are distinct stages of application if the following effective design process is to be applied :

1. **Conceptual** – online education that teaches the principles and basic physics behind effective technologies and techniques;
2. **Design development** – use of computer packages to test the performance before finding a concept;
3. **Technology selection**, cost analysis and progressive engineering input before finalising the design;
4. **Compliance** and rating considerations of the final designs;
5. **Monitoring** and post occupancy for continual improvement and awareness of changes in materials, technologies and systems in the building industry;
6. **Latest research and innovations** that may change future architectural concepts that become mainstream practice.

Such courses are now available from many Australian Universities and TAFE Colleges and the growing number of online resources for self learning.

GREEN MATERIALS, PRODUCTS AND SYSTEMS

Automation, control and operation (the efficient operation of energy consuming devices) – usually addressed by the building services consultants and contractors but should also be coordinated in conjunction with the architect and the client to ensure good standards and sustainable results. Design decisions range from the convenient location of a control switch through to complex fully automated building management systems. This usually includes the energy efficiency of air-conditioning, ventilation, water heating, transportation and lighting systems. Often they encompass individual components and the overall operation of each system and should be integral with the other Green Building initiatives.

Offering systems that can deliver shutdown times and facilitate user participation in increasing the use of natural lighting and ventilation systems is a very worthwhile approach often overlooked in the quest for energy efficiency.

Rain water harvesting (collecting and storing rainwater for use in and around buildings) – uses include irrigation, toilet and urinal flushing and laundry. The water may be used for other purposes, subject to suitable filtering/treatment.

Refer to Sanitary Plumbing for piping and tanks and Mechanical Services for pumps – a small domestic system with 3000 litre tank and irrigation pump costs around $2500.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

RECOMMENDED DESIGN PROCESS AND PRINCIPLES (cont'd)

GREEN MATERIALS, PRODUCTS AND SYSTEMS (cont'd)

Grey water recycling and storage (suitably treated wastewater from baths, showers, clothes washers, and basins) – uses include irrigation, toilet and urinal flushing and laundry (http://www.yourhome.gov.au).

Grey water treatment requires the physical removal, adsorption and microbial induced aerobic degradation of the pathogens from humans and the chemicals and solids from soaps, shampoos and laundry detergents. Small systems totally isolated and connected to a main sewer diversion are very cost effective, costing only about $1000 installed, but only appropriate to a single resident/small household and applicable to laundry water only (only a simple filter is required and needs cleaning once every 2 weeks).

Refer to Sanitary Plumbing for piping and tanks and Mechanical Services for pumps. The demand on potable water in a household can be reduced by 40% by utilising grey water – a small domestic system costs around $12,000.

Black water treatment and recycling (suitably treated wastewater from toilets, kitchen sinks and dishwashers) – uses vary depending on the level of filtration and treatment (http://www.yourhome.gov.au).

Black water treatment requires the higher-grade pathogens, solids etc. to be removed by a triple barrier disinfection system. For domestic purposes a combined Black water and Grey water treatment system, recycling up to 2,200 litres of wastewater per day and producing 99.999% germ free water, costs around $14,000. In environmentally sensitive areas usually close to watercourses and sub soil chemistry that is conducive for the build up of harmful pathogens, these systems are particularly relevant. For the average consumer with single residential plot the costs are probative for use in most projects and economically and environmentally not easily justified.

Solar water heaters (using solar energy to heat water) – water is circulated through solar collectors or heat exchangers linked to solar collectors and stored in an insulated storage tank. Alternative heating sources (boosters) may also be necessary (e.g. an electric element in the storage tank). There are Government subsidies still available for these products. Up to 30% GHGs can be reduced in a home with a solar water heater (http://www.cleanenergyregulator.gov.au/Pages/default.aspx) .

For domestic purposes solar water heaters can reduce hot water bills between 50% and 90%. In most Australian warm temperate and tropical climates flat plate technology is all that is needed and is most efficient and long lasting. In areas where solar radiation is scarcer, evacuated tube collectors are more effective. Throughout Australia and New Zealand a new range of solar water heaters operating from PV are emerging which are showing excellent potential for further reduction of emissions. On going innovation is now happening at a fast rate.
Large commercial and industrial systems are available and have very short pay back periods (often 3 – 5 years).

Refer to Sanitary Plumbing, page 462 – a commercial grade 2,500 litre daily output costs around $65,000.

Solar protection of glazed facades (blocking solar radiation prior to or at external glazed surfaces) – includes eaves, overhangs, sunscreens and louvers prior to glazing and solar control glass and solar control film at the glazing. Eaves and louvers must be appropriately designed; the harsh summer sun should be blocked whilst the solar gain from the winter sun is not obstructed. Solar glass and solar control film can reduce the sought after winter solar heat gain.

Extra over costs from standard 6mm clear float glass :

- Solar control, heat absorbing $19/sqm
- Solar control, heat absorbing, reflective $76/sqm
- Double glazing 6mm / 6mm $155/sqm
- Solar control film, high performance costs $77/sqm

External shading (such as with overhangs, shutters and awnings) is always the most effective way of controlling direct radiation. Most glass films can reduce a substantial amount of the radiation heat component but are now most effective in reducing radiant temperature effects in using low–e films. Of course a major issue with glass is the high heat transmission compared with solid walls. Films can help, and when combined with double-glazing or laminated glass, very good efficiencies can be attained. With new nanotechnologies such as 'aerogel' reaching the R&D stage, exciting advances are set to change the way we apply glazing systems (https://en.wikipedia.org/wiki/Aerogel).

For further cost information - refer to Windows and Doors, pages 363 to 387 and for differences in glass costs refer to Glazing, from page 443. For solar control film refer to page 447.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

RECOMMENDED DESIGN PROCESS AND PRINCIPLES (cont'd)

GREEN MATERIALS, PRODUCTS AND SYSTEMS (cont'd)

Photovoltaic systems – an array of photovoltaic modules which convert sunlight directly into DC electricity, which is used in conjunction with an inverter and control system to provide grid-quality AC power. Such systems can also incorporate battery storage. DC systems are more applicable to remote loations.

Payback periods vary considerably with the extremely variable and seemingly constantly changing feed-in tariff arrangements around the country. Consider mounting space and access requirements when considering the inclusion of such systems. Mass production from countries such as China is changing the cost ramifications of PV systems, but the major challenge over the next decade will be quality control and longevity of the systems as the market grows enormously.

A wide variety of sizes and configurations are available. For domestic purposes, a 1500W solar panel with mains inverter can cost as little as $4000 installed, when the Small Technology Certificates produced under the Small Generator Unit provisions of the expanded Renewable Energy Target scheme (http://www.cleanenergyregulator.gov.au/RET/Scheme-participants-and-industry/Agents-and-installers/Small-scale-technology-certificates) are taken over by the supplier. Quality of the panels, inverters and controls is a major factor as well as the cost. With the expected long life of most buildings, capital cost replacement and maintenance cost issues need to be considered for such systems. With prices falling rapidly to $2 - $3 per peak Watt in 2011, it's advisable to check all prices with actual quotes. Mass production in Germany, particularly in China and now in the USA is driving costs down at a high rate. For houses in most temperate climates a 3 – 5 kW system is the most economic for the consumer. Most States now limit the size of a grid connection system to 6 kW. This is to restrict the disruption of the quality of grid power used by an over supply of renewable energy when the power is not needed or used.

The price of PV has reduced dramatically in recent years. Supplied at 30 to 50 c/peak watt for the supply of the panels, when installed and maintained over the life cycle of the system, the actual cost is 10 – 15c/kWh. For daytime use direct or storing into batteries power should average about 24 c/kWh which is cheaper than peak power rates charged by the utilities, but running hot water systems or air conditioners off batteries simply is not cost effective. For power and lights it does make economic sense in many locations in Australia where the smart power option is selected for billing.

Natural ventilation / cross ventilation – (passive supply and removal of air by natural means, for ventilation and cooling) – uses cross ventilation and stack ventilation. Cross ventilation is achieved through ambient wind patterns and resultant pressure differentials. Stack ventilation (also referred to as the chimney effect) generally involves airflow in the vertical direction caused by temperature and pressure differences (https://www.tlj-eng.com/news/53-stack-effect-or-chimney-effect-2.html).

Natural ventilation may replace some mechanical ventilation / cooling systems and reduce capital expenditure and operating costs. In some circumstances the designs can virtually eliminate heating and cooling requirements in several of Australia's climate zones, particularly for residential buildings.

Natural light harvesting (the maximization of natural light through a building) – for example the use of light wells, roof lights, light shelves and interior colours to allow natural light to flow into the centre of buildings.
Used in conjunction with electrical light dimming to reduce power consumption when natural light is available. Fluorescent, HID and lights with transformers need to suit dimming systems. Modern design has increased the quantity of light fittings, which increases the demand for energy; excessive light fittings increase building heat load. The main challenge for a Green Architect is to integrate lighting systems into a daylighting strategy to create a natural ambience and minimise energy use.

Refer to Electric Services, Dimmers pages 572 and 573 and Luminaires from page 588.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

RECOMMENDED DESIGN PROCESS AND PRINCIPLES (cont'd)

GREEN MATERIALS, PRODUCTS AND SYSTEMS (cont'd)

Recycling and re-use of building components and construction waste (utilising material that otherwise would go to landfill and the consequential saving of energy to manufacture and construct them) - 40% of Australia's waste results from construction and demolition activities with up to 80% from discarded materials ideal for reuse or recycling (https://www.environment.gov.au/system/files/resources/b0ac5ce4-4253-4d2b-b001-0becf84b52b8/files/case-studies.pdf)**.**

Material	Reuse and recycling
Masonry, concrete, construction aggregate	Pavements, roads, car parks, driveways, pipe bedding, shoulder aggregate, erosion control, landscaping.
Bricks, tiles	Salvaged and resold.
Timber	Floor boards, reconstituted panelboards, doors, furniture, fencing, mulch.
Glass	Crushed glass reused as compaction fill.
Insulation material	Housing insulation.
Metals	Remelted and recycled as secondary materials for structural steel, roofing, piping etc.
Trees, organic matter	Compost.
Plastics	Recycled as secondary materials for products e.g. playgrounds, park benches, tables.
Paper, cardboard	Mulch, compost, animal bedding, newsprint, egg cartons, packaging material.
Fittings (bath, toilets, sinks)	Salvaged and resold.

Source: KESAB environmental solutions

Use of sustainable timber resources (defined as a harvested tree will be replaced by another tree) - trees are nature's air-conditioner, they are fuel, they are low embodied energy construction materials; the tree is a good 'all-rounder'.

Processed timber can be treated with toxic chemicals that can affect air quality within the building after construction. Low toxicity treatments should be favoured.

Cost Implications of Green Design

The cost analysis of green buildings is complex and must include embodied energy involved in the manufacture of products and technology as well as the energy used in the operation of a building and the maintenance as well as capital replacement impacts over the full life of the product, system or material used in a building. Demolition of the building should be included in a cradle to grave analysis. E-Tool is a multidisciplinary consultancy that specializes in such modelling and policy advice.

The purpose of a price on carbon is to stimulate innovation and a more focused industry action in order to generate cleaner and greener forms of energy which was a phase of investigation by Government. Politics prevented such an implementation by way of a carbon tax, or a market mechanism via a Carbon Pollution Reduction Scheme (CPRS). Other alternatives available to government by direct action of installing renewable or clean energy systems by direct funding or tax incentives has been preferred to date and will be likely for some time. By selecting clean energy, innovative construction techniques and products for rebates or government direct subsidy payment in the way of renewable energy certificates or grants seems the only way, progress will be made into the near future. This system has been in operation in Australia for way over a decade and more. However, a wholesale price on environmental damage is not possible using this technique and if applied to all areas of feasible ways of reducing the build up carbon in the atmosphere the cost of governance would be very expensive.

No matter what socio-economic situation is eventually attained to deal with environmental matters, it will always be relevant in the 21st century for green architects and engineers to be aware of the carbon consequences of design decisions and the whole of life economics of their decision-making.

The fundamental influence on design choices for construction materials and energy and water systems integrated into the building is the planned longevity or obsolescence of the building development.

GREEN DESIGN, CONSTRUCTION AND DEVELOPMENT (cont'd)

RECOMMENDED DESIGN PROCESS AND PRINCIPLES (cont'd)

Embodied Carbon Emissions of Different Wall Types Including Recurring Impacts, Measured in kgCO2e/m2

The life cycle implications for various materials need a great deal more work than has been taken to date. The basic considerations are worth noting (https://en.wikipedia.org/wiki/Life-cycle_assessment).

It is generally understood that the longevity of solid mass materials will last longer that framed or lightweight panelised construction and be more durable. History is full of such heritage-listed buildings. The challenge for the Green Architect is to choose a construction system suitable for the predicted duration of the land use.

It is not just the cost of capital and maintenance replacement over the life of the building but to quantify the energy saving benefits of the material over its life.

The architectural quality and aesthetics is also a big factor in how long a building, will be allowed to stand before demolition occurs, and cannot be ill considered in a proper sustainability analysis.

Therefore it is not possible to generalise on the sustainability of various construction systems and use of materials as there are so many variables in the analysis for each building type. Construction systems are also changing all the time as new innovations occur.

SUMMARY

Green design is a very broad subject and reliant on a complex relationship of environmental, social economic issues that need to be seen in a holistic manner. The two main challenges for the 21st century will be the serious threats of Climate Change and the diminishing availability of fossil fuels, water, food and many finite resources such as timber products and many minerals as well as the crucial impacts on the natural world.

How this is handled by ‘green’ architects, designers, builders, and developers and of course the regulatory framework will define eventually what it means to achieve ‘green design’ in our built environments. The way we look at costs and investments, in a new contemporary economic construct, will determine how successful the current and emerging generation of professionals will be in achieving success and ultimately true sustainability for future generations.

RESOURCE MANAGEMENT AND AUDITING

Resource Management is a method of monitoring and reducing the overall use of energy and water. Energy and Water use is one of the factors of production and building operating costs which can be controlled.

RESOURCE MANAGEMENT REQUIRES

Resource management requirements are:

- a systematic collection of information,
- a reporting system, and
- a well defined responsibility in the implementation of corrective action.

A resource management program is a means of reducing operating costs. Commitment from senior management is essential to the success of a resource management program.

A RESOURCE AUDIT WILL DETERMINE

- the extent and types of resources used (i.e. electricity, gas, fuel, water),
- the quantities and areas of consumption,
- potential areas to improve efficiency and reduce consumption and costs.

SAVINGS IN RESOURCE COSTS

Resource management is a complex area, but with the right audit and engineering solutions, there can be substantial savings for tenants and building owners. Savings in resource costs of between 10% and 30% can often be achieved.

The cost of an energy audit is usually offset by the resultant reduction in operating costs. Besides allowing for savings any financial budget should include the borrowing costs for capital works and differences resulting from the tax deductibility of capital costs versus running costs.

RESOURCE SUPPLY COSTS

It is well understood by most in the industry that the cost of energy and water is likely to continue to rise over the coming years. Issues such as any carbon pollution reduction schemes and higher energy demands are likely to drive costs of utilities higher. For this reason, the effective management of resource use will become beneficial and cost effective.

A RESOURCE AUDIT COMPRISES

Audit of Historical Data

- Collection and analysis of historical records of resource use
- Review of the costs and quantities of energy and water used
- Determination of annual and seasonal trends
- Comparison of resource use with other similar organisations.

Screening Survey

Visit building and review existing drawings and documentation to :

- Indicate major consuming plant
- Locate obvious waste and inefficiencies
- Interview appropriate staff and establish present management practices
- Locate gaps in metering and recording.

Analysis

- Investigate supply and tariff options
- Investigate the various saving options identified in detail
- Perform cost benefit analysis.

Report

- Review of the audit document and discussion of recommendations
- Set priorities
- Discussion of resource awareness and staff attitudes
- Recommend management procedures.

ENERGY MANAGEMENT AND ENERGY AUDITING (cont'd)

IMPLEMENTATION COMPRISES

- Preparation of documents for each cost saving solution
- Calling of tenders
- Supervision of the installation and commissioning of each solution
- Education of staff for various levels and functions
- Introducing an on-going energy management program.

Example

The following example is given in a simplistic form. A financial analysis of any energy management proposal would also include factors such as taxation, borrowing costs, projected energy costs, depreciation, etc.

Potential Saving

Say a 5000sqm office building consumed 600,000 units (i.e. 600,000 kWh) of electricity in one year.

Then	600,000 kWh ÷ 5000m^2	=	120 kWh/m^2/P.A.
or	(600,000 kWh x 3.6 MJ/kWh) ÷ 5000m^2	=	35MJ/m^2/P.A.

Say the electricity cost (excluding any fixed charges) was $150,000 P.A. (i.e. $0.25/kWh) then, if a 20% saving was achieved after the implementation of energy management, the electricity saving is **$30,000 P.A**.

Implementation

Referring to the above example the following figures may apply :

	$
• Audit costs	7,000
• Capital cost to implement audit recommendations	40,000
• Professional fees to implement audit recommendations	5,000
	52,000

SUSTAINABILITY ASSESSMENT

NATIONAL AUSTRALIAN BUILT ENVIRONMENT RATING SYSTEM

The National Australian Built Environment Rating System (NABERS) is a performance-based environmental rating tool used to assess office, hotel, retail, data centres and multi-unit residential buildings. Tools are currently being developed for other asset types as well. The environmental categories which can be assessed under this scheme include Energy/Greenhouse Gas emissions, Water, Waste (limited building types) and Indoor Environment Quality (limited building types). The performance of a building is represented on a scale of one (worst) to six (best) stars based on benchmarks established by the current stock of Australian buildings.

The NABERS-Energy tool is the most widely used component and assesses the Greenhouse Gas emissions resulting from the operation of a building. The NABERS-Energy rating is both an important energy management tool and a well-recognised method of promoting an organisation's sustainability performance. Buildings can either be self-rated (not formally recognised) or rated using the services of an Accredited Assessor.

A self rating can be performed by downloading the rating software from the website. The software requires the input of various building performance parameters. The output of the software is normalised to the particular usage profiles and then compared to a representative sample of surveyed buildings within each state to determine the star rating. A network of Accredited Assessors is available within each state to carry out certified ratings of buildings. An accredited rating allows for the open promotion of the building or tenancy rating using the NABERS logo.

PREDICTED RATING

During the design phase of a new building, a predicted rating can be obtained with the use of a computer simulation package. Strict protocols are provided for the use of such simulations and should be closely scrutinised given the idealised nature of modelling. The main advantage of a predicted rating is the minimisation of risk associated with achieving a desired rating. It can be a valuable tool in the assessment and quantification of various design solutions.

Utilising a predicted rating, a Commitment Agreement can be sought to allow the marketing and promotion of a predicted NABERS rating prior to 12 months of building operational data being available.

For more information please visit the NABERS website (www.nabers.com.au).

ENERGY MANAGEMENT AND ENERGY AUDITING (cont'd)

SUSTAINABILITY ASSESSMENT (cont'd)

COMMERCIAL BUILDING DISCLOSURE

The Commercial Building Disclosure (CBD) Program requires a building owner to include energy efficiency information at the point of advertising the space for lease or sale. This requirement applies to any commercial office building with a total space greater than 1,000sqm per building advertised at any time.

The building owner must obtain a Building Energy Efficiency Certificate (BEEC) which consists of 3 components :

- A NABERS Energy Star rating
- A tenancy lighting assessment
- A general energy efficiency guide

The BEEC is valid for a period of one year, must be registered on the publicly accessible Building Energy Efficiency Register and a valid BEEC must be provided to potential buyers of lessors at the time of sale, lease or sublease. In addition, the NABERS energy rating for the space must be included on any advertising material in a font not smaller than the standard font used on the advertising material.

For more information please visit the CBD website (www.cbd.gov.au)

NATIONWIDE HOUSE ENERGY RATING SCHEME

The Nationwide House Energy Rating Scheme (NatHERS) is a star rating system which benchmarks the energy efficiency of a dwelling or a group of dwellings (class 1 or 2). The energy required to maintain comfort conditions (theoretical heating and cooling) is compared to benchmarks established for 69 climate zones within Australia. The benchmarks relate to a star rating between 0 (worst) and 10 (best) stars.

Only the energy efficiency of the building envelope is considered, that is, the building fabric, external glazing, building sealing and air movement. The efficiency of lighting, AC or appliances are not assessed under this scheme.

The NatHERS program can be used to demonstrate compliance to the energy efficiency requirements (Section J) of the National Construction Code (NCC). The compliance requirement for a class 1 and Class 2 building have been adopted in various forms in each state. Please refer to the state specific requirement of the latest version of the NCC for specific targets. Heating and cooling load limits appropriate for each climate zone have been introduced for both individual and average star ratings as part of NCC 2019 changes.

A number of approved software packages are available for the assessment of the NatHERS performance. They are all based on the Chenath calculation engine, developed by the CSIRO. The calculation engine processes project specific information relating to climate, building form and materials. An accredited assessor must be used to obtain a formal rating.

Refer to NatHERS website (www.nathers.gov.au) for further details and a list of approved software.

GREEN STAR

Buildings have a significant impact on the environment and consume approximately one third of the world's resources. This includes approximately 12% of its water and up to 40% of its energy. In turn, they also produce 40% of the waste going to landfill and 40% of air emissions. Green Star is a national voluntary rating scheme operated by the Green Building Council of Australia that evaluates the environmental performance of buildings based on a number of criteria :

- Management
- Indoor environment quality
- Energy
- Transport
- Water
- Materials
- Land use, site selection and ecology
- Emissions
- Innovation

The Green Star environmental rating system was created to :

- Establish a common language
- Set a standard of measurement for green buildings
- Promote integrated, whole building design
- Recognise environmental leadership
- Identify building life-cycle impacts
- Raise awareness of green building benefits
- Reduce the environmental impact of development

Green Star accredited assessors are trained in the application of the assessment criteria and are able to provide a preliminary evaluation of a building. Each of the nine main categories for evaluation has multiple criteria that are individually assessed to determine the category score. The credits available correlate with the environmental impact of the item considered.
Only buildings achieving a rating of four, five or six stars will receive official certification from the Green Building Council of Australia, unless the Performance tool is used.

There are 4 rating tools that cover the design, construction and operation phases of all buildings, fitouts and communities. The 4 rating tools are:

- Green Star Design and As Built
- Green Star Communities
- Green Star Performance
- Green Star Interiors

Refer to GBCA website (www.new.gbca.org.au) for further information.

Stantec



Building our communities through design

Design with community in mind
stantec.com/en/offices/australia-locations-hub

ENERGY MANAGEMENT AND ENERGY AUDITING (cont'd)

SUSTAINABILITY ASSESSMENT (cont'd)

WELL BUILDING STANDARD

The WELL Building Standard is a holistic standard and certification tool which is aimed at promoting health and wellness of building users. The standard promotes building design features and operational procedures that will have a positive effect on the mental, physical and psychological health of occupants. The standard was designed by a team of physicians and architects over seven years, aimed at putting the user's health in the centre of the design. Given staff salary costs can contribute 90% of the operational cost of an office-based company, features that improve the health and wellness of the building occupants can have large returns on investment.

The WELL Building Standard considers over 100 features within eight factors:

- Air
- Light
- Mind
- Water
- Fitness
- Innovation
- Nourishment
- Comfort

The standard can be applied to many building types, and the International Well Building Institute (IWBI) has developed individual pathways within the tool for specific applications, such as healthcare, hospitality and retail. The standard is broken into the following typologies with different Features appropriate for each typology: :

- New and Existing Buildings – applied to fit-out only projects and whole buildings i.e. both base build with fit-out projects
- Core and Shell

Refer to the WELL Building Standard website (www.wellcertified.com) for more information.

CARBON NEUTRAL

Carbon neutral buildings achieve net zero overall carbon emissions. Buildings are required to reduce emissions where possible and compensate for the remainder by investing in projects that generate carbon credits or "offsets". Offsets are generated from an activity that prevents, reduces or removes greenhouse gas emissions from being released into the atmosphere.

The Climate Active Carbon Neutral standard is Australia's first national standard which defines the process and certification is licensed by the Federal Government. Offset units' credibility is dictated by international standards and vetted by the Department of Industry, Science and Agriculture to ensure validity. Offsets also often provide additional social, environmental and economic benefits that align with organisation's corporate social responsibility endeavours.

Annual carbon emissions are calculated from the operational data of the building and offset units are purchased equal to the carbon footprint. The annual emissions data must come from either a NABERS Energy Assessment or a Green Star Performance rating. Accredited consultants lodge the submission on behalf of the client which is reviewed prior to the certification being issued. An ongoing commitment to continually reduces emissions is required and the certification is renewed annually.

Further information can be found at https://www.industry.gov.au/data-and-publications/climate-active-carbon-neutral-standard-for-buildings

ONE PLANET LIVING

Globally, consumption of naturally renewing resources and pollution are at a level forty per cent higher than the earth can sustain. If everyone consumed as much as the average Australian, 3.7 planets worth of resources would be needed.

One Planet Living takes ecological and carbon footprinting to create a framework to assist companies, councils and communities in operating within the natural constraints of the environment. In the context of communities, the tool operates at single building and precinct level scales.

The framework provides a set of guidelines under the following categories:

- Health and Happiness
- Land Use and Wildlife
- Sustainable Materials
- Zero Carbon
- Equity and Local Economy
- Sustainable Water
- Sustainable Transport
- Culture and Community
- Local and Sustainable Food
- Zero Waste

The One Planet Living framework provides detailed goals and guidance but allows for complete flexibility in responding to each category. This allows for the most suitable features to be implemented within the project's local context instead of meeting a set of prescriptive requirements.

Certification is provided by Bioregional and consists of a review and audit of the project's One Planet Living Action Plan. The Action Plan details the project's response to each of the 10 categories and outlines commitments for ongoing monitoring and improvement. Judgement is then made on the overall credibility of the chosen features in responding to each category aim. Certification is then provided, if the Action Plan sufficiently meets the aims of each category and requires continuing implementation of the Action Plan and reporting annually to Bioregional in order to retain certification. Refer to the Bioregional Australia website (https://bioregional.com.au) for further information.

ENERGY MANAGEMENT AND ENERGY AUDITING (cont'd)

SUSTAINABILITY ASSESSMENT (cont'd)

ENVIRODEVELOPMENT

In order to increase the uptake of sustainability in all development projects, the Urban Development Institute of Australia (UDIA) has created a development wide environmental branding system – EnviroDevelopment.

EnviroDevelopment is a scientifically based branding system based on the recognition of high achievement across a wide range of development types and situations including residential, retail, commercial and industrial as well as large scale mixed-use community developments.

This tool offers independent certification of the sustainability credentials of a development in six key areas of environmental and community sustainability :

- Ecosystems
- Water
- Materials
- Energy
- Waste
- Community

Unlike other rating tools which focus on a particular building type, this tool rates the development in its entirety, from a single building scale, to large scale proposed new towns and communities. It includes considerations of the physical environment around the development, energy and water efficiency, sustainability of materials used throughout the development area, as well as sustainable community objectives.

Certification is approved by the EnviroDevelopment Board of Management once the requirements of the EnviroDevelopment standards have been met and the appropriate documentation has been submitted by the developer.

Certified developments will have been designed to use resources responsibly and protect the environment. These projects are entitled then to advertise their achievements by displaying the icons. Developments can apply to be rated for any one of the sustainability indexes above or all six of them. This rating is designed to make it easier for purchasers to recognise and thereby select more environmentally sustainable homes and lifestyles.

Refer to EnviroDevelopment website (www.envirodevelopment.com.au) for further information.

LEED

The Leadership in Energy and Environmental Design (LEED) tool provides a framework for virtually all building, community and home project types to create healthy, efficient and cost-saving green buildings. LEED is delivered by the United States Green Building Council (USGBC) but is applicable globally and is increasingly being applied in Australia as organisations seek to benchmark buildings against a globally uniform standard. LEED operates in over 167 countries and have certified more than 96,000 projects.

Applicable building project types are:

- Building Design and Construction
- Neighbourhood Development
- Interior Design and Construction
- Homes
- Building Operations and Maintenance
- Cities and Communities

Each of the above rating types provide a list of relevant credits that can be targeted under the following categories:

- Location and Transportation
- Energy and Atmosphere
- Innovation
- Sustainable Sites
- Material and Resources
- Regional Priority
- Water Efficiency
- Indoor Environment Quality

Various rating levels are available and follow the following scale :

Rating Level	Min. Score Required
Certified	40
Silver	50
Gold	60
Platinum	80

Formal certification requires the involvement of a LEED Accredited Professional who will manage the process of coordinating and documenting targeted credit requirements. The certification process involves the submission of a comprehensive set of documents, drawings and reports to the Green Business Certification Inc. – an independent auditing body that provides certification and verification for several sustainability rating schemes related to the built environment. Each credit will be assessed based on the relevant criteria and certification provided according to the final score achieved.

Refer to the LEED website (https://new.usgbc.org) for more information.

ENVIRONMENTAL MANAGEMENT SYSTEM

An Environmental Management System (EMS) is a tool for managing the current and future impacts of a construction contractor's activities on the environment. It provides a structured approach to planning and implementing environment protection measures. An EMS monitors environmental performance and integrates environmental management into construction activities. It can assist construction contractors in minimising environmental risk, maximising the efficient use of resources, reduce waste, demonstrate a good corporate image, build awareness of environmental concerns, improve environmental performance and efficiency and increase profits.

Many Australian State and Territory government entities require construction contractors to have an acceptable corporate EMS in order to be engaged for work on major projects. Such an EMS must comply with the documentation requirements of AS/NZS 14001:2004 Environmental Management Systems and include the following components:

- Environmental policy
- Environmental impact identification
- Objectives and targets
- Operational and emergency procedures
- Documentation and records
- Defined roles and responsibilities
- Environmental training
- Internal auditing procedures

More information can be found on the Department of the Environment's website: http://www.environment.gov.au/node/20494

BEST PRACTICE ENVIRONMENTAL MANAGEMENT IN CONSTRUCTION

Best practice environmental management in construction can vary between different locations and project situations. Listed below are several methods which can be used to ascertain best practice :

- Comparisons of current, previously established and external environmental management plans
- Costs associated with environmental conformance
- Formally or informally prepared performance indicators against environmental management objectives
- Performance measurements against industry, local authority, regulatory or world standards
- Specific environmental targets, including emissions, noise, dust, waste disposal, storm and ground water disposal, waste segregation and hazardous materials
- Statistical record keeping, using at start, ongoing and at conclusion measurements
- Strategic comparisons of resources in and residuals out

CONSTRUCTION ENVIRONMENTAL MANAGEMENT PLAN

A Construction Environmental Management Plan (CEMP) is a site or project specific plan developed to ensure that appropriate environmental management practices are followed during a project's construction. CEMP's are often required as part of tendering for projects and are essential for some types of government projects. An effective CEMP will help construction contractors to :

- Apply best practice environmental management to construction
- Implement measures or conditions outlined in a project's environmental assessment documentation
- Comply with environmental legislation
- Properly manage environmental risks associated with construction

The following elements form a CEMP :

- Introduction and background to the project
- Environmental objectives and targets
- Description of project's environmental risks
- Plans incorporating controls and measures to manage environmental risks
- Environmental incident and emergency procedures
- Procedures for monitoring, auditing and reporting
- Description of roles and responsibilities relating to the project
- Forms and registers for tracking compliance with the CEMP

ENVIRONMENTAL MANAGEMENT SYSTEM (cont'd)

PROTECTING THE ENVIRONMENT DURING CONSTRUCTION

MELBOURNE

Civil Contractors Federation Environmental Guidelines for Civil Construction
http://www.civilcontractors.com/Uploads/files/Victoria/Environment/Environmental%20Guidelines%20for%20Civil%20Construction%20CCF.pdf

PROTECTING THE ENVIRONMENT DURING CONSTRUCTION IN OR NEAR WATERWAYS

ADELAIDE
Protecting Waterways Manual
http://www.dpti.sa.gov.au/__data/assets/pdf_file/0005/35717/DOCS_AND_FILES-3476572-v2-Environment_-_Water_-_Protecting_Waterways_Manual_-_Chapters_1-10_.pdf

BRISBANE
Minor waterway barrier works (various guidelines)
http://www.daff.qld.gov.au/fisheries/habitats/fisheries-development/self-assessable-codes

MELBOURNE
Constructing waterway crossings
http://www.melbournewater.com.au/Planning-and-building/Forms-guidelines-and-standard-drawings/Documents/Constructing-waterway-crossings-guidelines.pdf

SYDNEY
Guidelines for controlled activities on waterfront land
http://www.water.nsw.gov.au/Water-licensing/Approvals/Controlled-activities/Controlled-activities

ENVIRONMENTAL PROFESSIONALS

The Certified Environmental Practitioner (CEnvP) Scheme, initiated by the Environment Institute of Australia and New Zealand (EIANZ) is a licencing scheme exclusively for environmental practitioners. CEnvP accreditation provides a level of assurance to the construction sector of the credentials of an environmental professional who are often engaged to assist with the application of environmental best practice to a construction project.

A directory of CEnvP accredited professionals is available on the CEnvP website:

http://www.cenvp.org/pages/b_directory.php

STATE GOVERNMENTAL EPA DEPARTMENTS

Environment Protection Authority South Australia
http://www.epa.sa.gov.au

Department of Environment and Resource Management
http://www.derm.qld.gov.au

Department of the Environment, Climate Change, Energy and Water
http://www.environment.act.gov.au

Environment Protection Authority Northern Territory
http://www.epa.nt.gov.au

Environment Protection Authority Tasmania
http://www.epa.tas.gov.au

Environment Protection Authority Victoria
http://www.epa.vic.gov.au

Environment Protection Authority Western Australia
http://www.epa.wa.gov.au

Office of Environment and Heritage
http://www.environment.nsw.gov.au

ENVIRONMENTAL MANAGEMENT SYSTEM (cont'd)

MINIMISING NOISE CONSTRUCTION

Adelaide
Construction noise
http://www.epa.sa.gov.au/xstd_files/Noise/Information%20sheet/info_noise_construction.pdf

Brisbane
Noise control measures
http://www.deir.qld.gov.au/workplace/business/construction/safety/index.htm

Canberra
Noise environment protection policy
http://www.environment.act.gov.au/__data/assets/pdf_file/0003/198525/Noise_EPP_web.pdf

Melbourne
Noise control guidelines
http://epanote2.epa.vic.gov.au/EPA/Publications.nsf/2f1c2625731746aa4a256ce90001cbb5/7f958a3b5e6a087eca2574dd00090419/$FILE/1254.pdf

Perth
Guidance for the assessment of environmental factors – Environmental noise No. 8
http://www.epa.wa.gov.au/EPADocLib/2495_GS8.pdf

Sydney
Construction noise guideline
http://www.environment.nsw.gov.au/resources/noise/09265cng.pdf

PREVENTING STORMWATER POLLUTION FROM CONSTRUCTION ACTIVITIES

Adelaide
Stormwater pollution prevention code of practice for the building and construction industry
http://www.epa.sa.gov.au/xstd_files/Water/Code%20of%20practice/bccop1.pdf

Brisbane
EPA Best practice urban stormwater management – Erosion and sediment control
http://www.derm.qld.gov.au/register/p02301aa.pdf

Canberra
Preventing pollution from concreting operations – Business and industry information sheet
http://www.environment.act.gov.au/__data/assets/pdf_file/0018/146223/PreventingPollutionfromConcretingOperationsJuly2010.pdf

Melbourne
Reducing stormwater pollution from construction sites
http://epanote2.epa.vic.gov.au/EPA/Publications.NSF/2f1c2625731746aa4a256ce90001cbb5/13b115278e9b7b95ca257000002675bf/$FILE/981.pdf

Perth
Stormwater management manual for West Australia
http://www.water.wa.gov.au/PublicationStore/first/84954.pdf

Sydney
Stormwater pollution from building sites
http://www.environment.nsw.gov.au/water/stormwater.htm

MEASUREMENT OF BUILDING AREAS

This method has been adopted for use by N.P.W.C. members, the Australian Institute of Quantity Surveyors and the Royal Australian Institute of Architects.

It should be noted that local authorities may adopt other methods for the likes of plot ratio calculations and others.

GROSS FLOOR AREA (G.F.A.)

The sum of the Fully Enclosed Covered Area and Unenclosed Covered Area as defined.
Unit of Measurement - square metres (sqm)

FULLY ENCLOSED COVERED AREA (F.E.C.A.)

The sum of all such areas at all building floor levels, including basements (except unexcavated portions), floored roof spaces and attics, garages, penthouses, enclosed porches and attached enclosed covered ways alongside buildings, equipment rooms, lift shafts, vertical ducts, staircases and any other fully enclosed spaces and usable areas of the building, computed by measuring from the normal inside face of exterior walls but ignoring any projections such as plinths, columns, piers, and the like which project from the normal inside face of exterior walls. It shall not include open courts, light wells, connecting or isolated covered ways and net open areas of upper portions of rooms, lobbies, halls, interstitial spaces and the like, which extend through the storey being computed.

Unit of Measurement - square metres (sqm)

UNENCLOSED COVERED AREA (U.C.A.)

The sum of all such areas at all building floor levels, including roofed balconies, open verandahs, porches and porticos, attached open covered ways alongside buildings, undercrofts and usable space under buildings, unenclosed access galleries (including ground floor) and any other trafficable covered areas of the building which are not totally enclosed by full height walls, computed by measuring the area between the enclosing walls or balustrade (i.e. from the inside face of the U.C.A. excluding the wall or balustrade thickness). When the covering element (i.e. roof or upper floor) is supported by columns, is cantilevered or is suspended, or any combination of these, the measurements shall be taken to the edge of the paving or to the edge of the cover, whichever is the lesser. U.C.A. shall not include eaves, overhangs, sun shading, awnings and the like where these do not relate to clearly defined trafficable covered areas, nor shall it include connecting or isolated covered ways.

Unit of Measurement - square metres (sqm)

BUILDING AREA (B.A.)

The total enclosed and unenclosed area of the building at all building floor levels measured between the normal outside face of any enclosing walls, balustrades and supports.

Unit of Measurement - square metres (sqm)

USABLE FLOOR AREA (U.F.A.)

The sum of the floor areas measured at floor level from the general inside face of walls of all interior spaces related to the primary function of the building. This will normally be computed by calculating the Fully Enclosed Covered Area (F.E.C.A.) and deducting all of the following areas supplementary to the primary function of the building.

DEDUCTIONS FROM (U.F.A.)

(a) Common Use Areas :
All floored areas in the building for circulation and standard facilities provided for the common use of occupiers, tenants and/or the public such as lobbies and foyers to entrances, lifts, stairways, landings and fire escapes, verandahs and balconies, corridors and passages, toilet and rest room areas, cloak and locker areas, cleaners rooms including stores and cupboards, tea making and similar amenities areas.

MEASUREMENT OF BUILDING AREAS (cont'd)

USABLE FLOOR AREA (U.F.A.) (cont'd)

(b) Service Areas :
All areas set aside for building plant supplying services and facilities common to the building for the use of occupants, tenants and/or public, such as mechanical plant and equipment rooms, electrical equipment and switchrooms, tank rooms, lift motor rooms, meter cupboards, telecommunication switchrooms, refuse collection areas, loading bays and all car parks including access ways thereto.

(c) Non-Habitable Areas :
All non-habitable building space such as that occupied by internal columns and other structural supports, internal walls and permanent partitions, lift shafts, service ducts and the like.

Unit of Measurement - square metres (sqm)

TREATED AREA (T.A.)

The sum of all areas at all building levels to which a particular engineering service is provided. The area is computed by measuring from the normal inside face of exterior walls, but ignoring any projections such as plinths, columns, piers and the like which project from the normal inside face of exterior walls, to the centre line of internal walls, as applicable, which enclose the area treated.

Notes :
(i) A Treated Area may need to be computed separately for each engineering service.
(ii) The Treated Area for such services as space heating, ventilation, evaporative cooling and air-conditioning should include all areas to which treated air is supplied.
(iii) The Treated Areas for fire protection comprises the sum of protected floor space, the area of protected ceiling spaces and the area of protected under-floor spaces.

Unit of Measurement - square metres (sqm)

NET LETTABLE AREA, OFFICE BUILDINGS (N.L.A.)

The sum of all lettable areas within a commercial type office building, measured from the internal finished surfaces of permanent walls and from the internal finished surfaces of dominant portions of the permanent outer building walls, and including the area occupied by structural columns and engaged perimeter columns, all in accordance with the Method for the Measurement for Lettable Areas issued by the Property Council of Australia.

DEDUCTIONS FROM N.L.A. -

(a) All stairs, toilets, cleaners cupboards, lift shafts, escalators and tea rooms where provided as standard facilities in the building.

(b) Lobbies between lifts facing other lifts serving the same floor.

(c) Areas set aside as public space or thoroughfares and not used exclusively by occupiers of the building.

(Note : excludes any additional common areas resulting from sub-division of a whole floor to accommodate more than one tenant).

(d) Areas set aside as plant and lift motor rooms or for the provision of facilities or services to the building and not for the exclusive use of an occupier or occupiers of the building.

(e) Areas set aside for use by service vehicles and for delivery of goods and access ways thereto.

(f) Areas set aside for car parking and access ways thereto.

(g) Areas where clear height is less than 1.50 metres.

Unit of Measurement - square metres (sqm)

MEASUREMENT OF BUILDING AREAS (cont'd)

ASSOCIATED DEFINITIONS

STOREYS :

The number of storeys in a building is the number of main floors above ground level including the ground floor but excluding penthouses or machine or plant rooms above the main roof level. Basements below ground level should be stated separately (e.g. 2B + 12 storeys).

BUILDING HEIGHT :

The distance measured between the main ground floor level and the upper limit of the structure of the building including parapet walls but excluding guard rails and the like.

INTERSTITIAL SPACE :

The intervening space between floors providing occasional access for the installation and maintenance of building, engineering and other equipment.

WALL / FLOOR AREA RATIO :

A measure of plan efficiency, computed by dividing the face area of the external surfaces of the building* by the Fully Enclosed Covered Area (F.E.C.A.) and expressing the result as a ratio to 1.

* Excludes area of gable walls, parapet walls and walls below lowest finished floor level.

P.O.P. RATIO :

A measure of plan compactness, computed by relating the perimeter of a building at a typical floor level to that of a circle enclosing the same area and expressing the result as a percentage.

The formula for P.O.P. Ratio is :

$$\text{P.O.P.} = \frac{2\sqrt{\pi A}}{P} \times 100 \text{ where}$$

A = The Fully Enclosed Covered Area (F.E.C.A.) of a typical floor level.

P = The perimeter enclosing that area, measured internally.

(N.B. : The P.O.P. Ratio for a circle = 100%
The P.O.P. Ratio for a square = 89%)

AREA EFFICIENCY :

A measure of design efficiency, computed by dividing the Usable Floor Area* (U.F.A.) by the Gross Floor Area (G.F.A.) and expressing the result as a percentage.

* Net Rentable Area (N.R.A.) may be substituted for Usable Floor Area (U.F.A.) in commercial type premises.

AREA PERCENTAGE :

The percentage that the Fully Enclosed Covered Area (F.E.C.A.) and the Unenclosed Covered Area (U.C.A.) each bear to the Gross Floor Area (G.F.A.).

NOTE : In measuring building areas from drawings, figured dimensions should be used in preference to scaled dimensions.

FEASIBILITY STUDIES

The purpose of feasibility studies is to calculate the return which will be derived from a particular project.

The return can be expressed as an annual percentage return or as a terminal percentage return. An annual percentage return will be used where the project will be generating rent for the owner, while a terminal return is used when the project is to be sold.

The calculated returns are important only for comparisons, i.e. to compare one project with another or to compare one project's return with the return that would be achieved by investing elsewhere.

The following proforma represents a suggested set out to calculate the return. Item 3.0 is shown in alternative forms. The first alternative is applicable to an annual return and the second to a terminal return.

1.0 PRIMARY INFORMATION

Site Area -	sqm	
Frontages -	m	
Gross Floor Area -	sqm	
Net Rentable Area -	sqm	
Parking Provision -	No	(Cars)
Land Purchase Price -	$	
Building Cost -	$	
Vacant Possession Costs -	$	
Preliminary Sketch Plans -	Months	
Development Approvals -	Months	
(Design Development Stage to calling of Tenders and including Bill of Quantities) -	Months	
Calling Tenders and Awarding Contract -	Months	
Construction Time -	Months	

2.0 CAPITAL EXPENDITURE

LAND COSTS	$		$
Purchase Price -			
Stamp Duty @ % on first $ -			
Plus % on remainder -			
Legal Costs -			
Vacant Possession Costs -			
Soil Tests -			
Land Surveyor's Fees -	_____	=	_____
BUILDING COSTS			
Demolitions -			
Contract Price -			
Architects, Engineers and Consultants Fees -			
Project Management Fees -			
Quantity Surveyors Fees -			
Local Council and Planning Authority Fees -			
Environmental Fees -	_____	=	_____

C/f'wd

FEASIBILITY STUDIES (cont'd)

2.0 CAPITAL EXPENDITURE (cont'd)

						$ Per Annum		$ B/fwd
		RATES AND TAXES						
Council Rates @	in $	on the relevant value -						
Land Tax @	in $	on the relevant value -						
		Water -						
		Sewerage -						
		Drainage -				_____	=	
		(+ 12)	=	$			per month	
		Planning and Construction Time						
		( ) months	=	x		_____	months	

ADD :

Interest (@	% per month simple)							
	On Land Cost -							
	On Rates and Charges -							
	On Building Costs -					_____	=	
	SUNDRY COSTS							
	Owner's Moving Expenses -							
	Furnishing -							
	Market Studies -							
	Advertising and Signs -							
	Premises Department's Costs -							
	Owner's Overhead Expenses -							
	Letting Agents Fees @ % -							
	Tenant Inducements -							
	Contingency -					_____	=	_____
	TOTAL CAPITAL EXPENDITURE						$	*
	Allow for G.S.T. on applicable items of the foregoing @ 10% -							
	G.S.T. refund for applicable clients/owners as defined in TAXABLE on page 923 -							
								_____
	TOTAL EXPENDITURE INCLUDING G.S.T.						$	_____

* NO ALLOWANCE FOR LIFE CYCLE COSTING

ANNUAL INCOME AND EXPENDITURE

FEASIBILITY STUDIES (cont'd)

3.0 ANNUAL INCOME AND EXPENDITURE

INCOME	$		$
Car Parking cars @ $ -			
Ground Floor sqm @ $ -			
Upper Floors sqm @ $ -	_____	=	_____
TOTAL ANNUAL INCOME			$
ANNUAL EXPENDITURE			
Council Rates -			
Land Tax -			
Water & Sewerage Rates -			
Insurance combined @ % -			
Electricity to Public Areas -			
Garbage Removal -			
Caretaker and Cleaning @ per sqm -			
Window Cleaning -			
Security Service -			
Fire Alarm and Sprinkler Service -			
Lift Maintenance and Operation -			
A.C. Maintenance and Operation -			
Management Fees at Scale (say 4%)			
of Total Letting -	_____	=	
PROVISIONS			
Building Maintenance -			
Building Depreciation years @ % -			
Plant Depreciation years @ % -			
Vacancies - say % -	_____	=	_____
TOTAL ANNUAL EXPENDITURE			$
			_____
NET ANNUAL INCOME			$
			_____

ANNUAL RETURN = Net Annual Income $ ___ x 100 / Total Capital Expenditure $ 1 = %

ALTERNATIVE

3.0 TERMINAL INCOME AND EXPENDITURE

	$		$
SALE PRICE	_____	=	
SELLING EXPENDITURE			
Council Rates -			
Land Tax -			
Water and Sewerage Rates -			
Insurance -			
Security Service -			
Agents Fees -			
Interest -	_____	=	_____
NET SALE PRICE			$
			_____

TERMINAL RETURN = Net Sale Price - Total Capital Expenditure / Total Capital Expenditure x 100 / 1 = %

COST PLANNING PROCEDURES

Cost planning is quite simply the process of designing to or within a pre-calculated cost, determined by the finances available, so as to obtain optimum value for available money whereby a building owner obtains a building tailored to his budget.

Most building owners today are compelled to adopt tight financial control on all costs used in feasibility studies if the intended revenue obtained from buildings is to be achieved. Because building costs are a substantial part of any feasibility, it follows that a logical process of establishing a target cost at the outset of any project, and the realisation of that same cost at completion, is essential if credibility is to be maintained.

Cost planning should not be seen negatively as a restrictive influence on the work of the designer, but more of a positive, pre-determined, disciplinary process, where money is consciously allocated to the various building elements or features, so as to obtain a building conforming to budget but possessing pre-planned embellishments in certain areas, if so intended.

Two great personalities, born 350 years apart, had this to say on the subject :

Sir Thomas Bennett, FRIBA -

> ".....the client should be able to set out to spend a given sum of money and finish with a building that has cost precisely this sum".

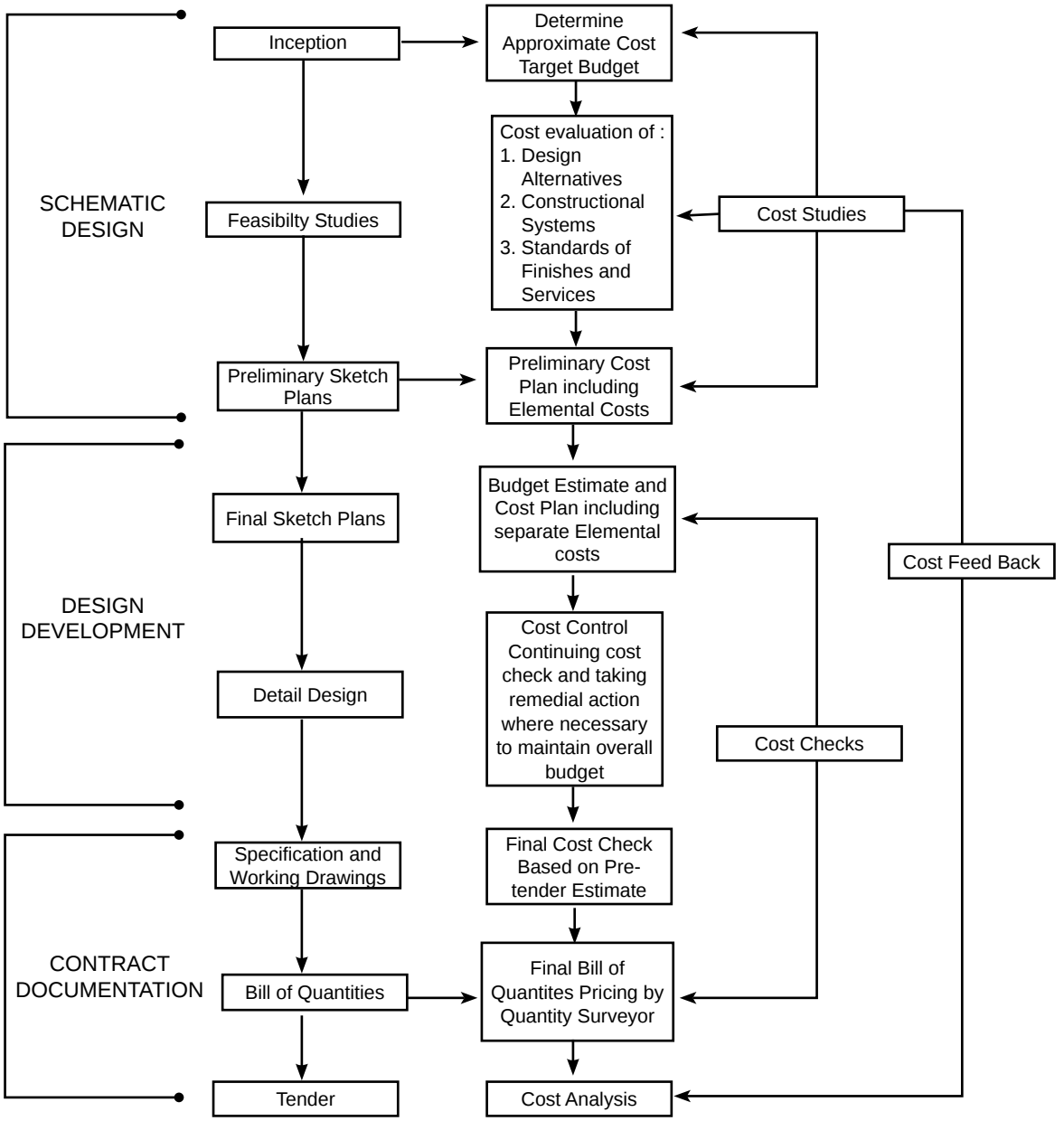
William Shakespeare

> "When we mean to build,
> We first survey the plot, then draw the model;
> And when we see the figure of the house,
> Then we must rate the cost of the erection;
> Which, if we find outweighs ability,
> What do we then, but draw anew the model
> In fewer offices; or at least desist to build at all."
>
> - Henry IV

The system flow chart shown, indicates the cost planning processes related to the architectural stages of a typical project from inception through to tender.

COST PLANNING PROCEDURES (cont'd)

SYSTEM FLOW CHART



COST PLANNING PROCEDURES (cont'd)

Architectural - Schematic Design Stage :

The cost planning processes are carried out by a quantity surveyor in close liaison with the architect and include the establishment of a cost limit or budget at inception, obtained from empirical or recorded costs of similar past projects, usually on a cost per sqm basis. This process proceeds on to cost evaluation of design alternatives, constructional systems and standards of finish and services pertinent to the particular project, which is then included in a preliminary cost plan at the end of the schematic design stage.

Architectural - Design Development Stage :

As final sketch plans are completed, the preliminary cost plan is expanded into a budget estimate and cost plan, containing separate budget costs for all the building elements (e.g. substructure, columns, upper floors, external walls, etc.), reflecting the design input from architect, structural engineer, mechanical and electrical consultants, etc. As detail design proceeds, these cost elements are periodically monitored against the detail design and, where an element appears to be exceeding the budget, recommended remedial action is submitted to the architect by the quantity surveyor, with the object of staying within the overall set budget.

Architectural - Contract Documentation Stage :

Usually the point of no return - the project cost is committed at this stage and can only accept major alterations at the expense of disruption to the documentation activities, and resultant delays.

This stage sees the final cost check in the form of a detailed pre-tender elemental estimate, followed by quantity surveying pricing of the bill of quantities, if commissioned.

If the various stages of the cost planning activities are efficiently carried out, and close liaison and co-operation maintained with the design team, then the final outcome of tenders should conform to the established budget and the building owner can then proceed immediately to signing a contract with the builder, satisfied that all reasonable steps have been taken to ensure a good budgetary and design conclusion.

CONTRACTUAL SYSTEMS

This section is a brief commentary on the various contractual arrangements commonly available to prospective building owners who are considering some form of property development. Various adaptations and intermixes of these basic forms evolve from time to time but for the sake of simplicity this section is confined to :

- Traditional Tender
- Competitive Negotiation
- Cost Plus
- Private-Public Partnership (PPP) Alliance
- Design and Construct
- Design Build Finance and Manage
- Construction Management

Refer page 887 for definition of the terms Construction Management and Project Management.

TRADITIONAL TENDER SYSTEM

Tenders may be called for the construction on either :

- A selected basis - where tenderers are restricted in number by a process of selection according to their qualification and experience in the type of project in question.
- An open or public basis - where no such restriction exists and public accountability and total market exposure is paramount, sometimes at the expense of suitability and selective expertise.

Resumé

The building owner appoints a project design team comprising an architect, structural engineer, quantity surveyor, mechanical and electrical engineer and other specialist consultants as required, depending on the nature and complexity of the project.

The building owner commissions the full design documentation of the project within the cost parameters set by the quantity surveyor's cost plan.

Tenders, usually lump sum, are then called from building contractors to ascertain the price of the works, prior to the final decision to proceed.

On awarding the contract to the successful tenderer, the site is handed over to the building contractor and the contract administered by the architect on behalf of the building owner in accordance with the contract documents.

The construction work is carried out by the building contractor often using sub-contract trades.

The design performance obligations rest with the design team and any risks are invariably underwritten by the individual team members' professional indemnity insurance. The construction risks (contractual risks) rest with the builder contractor. The building owner is therefore reasonably well protected from risks inherent in design and construction.

Salient Factors

- Price is known before building owner is committed to construction - allowing remedial action to be taken before the event.
- Price is the true, competitive market price.
- Building owner is insulated, for the most part, from risks, or at least has contractual recourse.
- Design and tender documentation is complete before proceeding to tender obviating the incidence of major variation costs.

CONTRACTUAL SYSTEMS (cont'd)

COMPETITIVE NEGOTIATION

Resumé

The building owner appoints a consultant team to prepare schematic design drawings up to preliminary working drawings stage, outlining specification including a schedule of construction and finishes, a provisional bill of quantities and a form of building contract.

Tenders are called from a selected list of building contractors, for the following elements :

(a) Preliminaries - i.e. builders price for site mobilisation, day to day running and final demobilisation, construction plant including cranes, scaffold, etc., builders insurance, temporary on-site services, water, phones, electricity, periodic and final clean-up and builders site administration including foremen.

(b) A tendered percentage for margins to be based on the value of work when known.

(c) A tendered percentage for off-site overheads.

(d) A tendered percentage for attendance on sub-trades.

(e) Statement of the time required to complete the project accompanied by the builder's bar chart or other programme scheduling.

Tenders, submitted in accordance with the above requirements, are evaluated by the consultant team and a recommendation is made to the building owner.

On a recommendation in favour of one of the building contractors being accepted, that builder then joins the project team as a building consultant and his practical building expertise is then used in final design documentation before he proceeds to perform the building works.

The appointed builder prices documentation as it becomes available for final acceptance by the building owner. This is usually done by the builder calling competitive bids from say three sub-contractors for each trade package.

Salient Factors

- Is essentially a fast-tracking system of procuring a contract, the main object being to install a selected builder on site and have him working before formal design documentation is complete resulting, hopefully, in an earlier completion and all the advantages stemming therefrom.
- The selected builder becomes a member of the team and is available to add his expertise to advantage the project.
- Sub-contract prices are tendered just prior to when needed thereby obtaining current market prices and possibly avoiding formula-based Rise and Fall.

COST PLUS CONTRACT

The building owner selects a building contractor who contracts to perform the building works in accordance with the contract documents at cost plus a percentage of the cost. These documents can be based on any one of those mentioned in Contract Conditions. In this arrangement it is extremely important to define cost. The plus is easily defined. The cost usually includes all on site activities whilst the plus covers head office overheads and profit. The plus percentage can vary from 1% to 10% depending on the particular project. Alternatively the plus can be an agreed fixed fee.

PRIVATE-PUBLIC PARTNERSHIPS (PPP)

Broadly the use of PPP contracts transfers the responsibility and risks from the public sector to the private. The design, finance, construction and often the maintenance and operation of a facility is undertaken by a private party for an agreed amount which is of course paid for by the public sector. Unfortunately if the enterprise fails to return the profit expected by the private sector partner, the public sector partner may be left to carry future costs.

CONTRACTUAL SYSTEMS (cont'd)

ALLIANCE

An alliance contract is an agreement between two or more entities that undertake to work cooperatively, reaching decisions jointly by consensus and using intensive relationship facilitation. The entities work together to achieve agreed outcomes and share project risks and rewards, relying on good faith and trust.

An alliance contract would usually require early identification of a Target Cost for the whole of the project, with actual project costs to be within the Target Cost and losses and gains, risks and rewards to be shared.

The success of an alliance contract is entirely dependant on trust between participants and a realistic if not generous target cost.

The alliance contract option is used mainly by government authorities whilst the private sector has a very small participation percentage.

DESIGN AND CONSTRUCT CONTRACT

Resumé

This system is more commonly known as the Package Deal and usually commences when a client approaches a Package Dealer with a brief for a particular project. Alternatively the client may be approached by a Package Dealer with a development proposal for his site.

The Package Dealer submits a preliminary proposal incorporating outline aspects for the intended design and construction. This proposal would include estimates of time and cost to complete the project. For the Investment client the proposal may also incorporate projected return figures for the project.

If the preliminary proposal is accepted by the building owner, the Package Dealer would work up and submit a final development proposal, which would incorporate in most instances a guarantee of maximum price for the project and offering the building owner a share in any savings achieved in such maximum price. The final development proposal would comprise schematic design drawings to a reasonably advanced stage, and an outline specification incorporating a schedule of construction and finishes.

The system may be either with or without a savings participation clause.

DESIGN BUILD FINANCE AND MANAGE

Resumé

In this arrangement the Design and Construct system is extended to include the financing of the project and the operation of the completed facility for a fixed period. The contract period covered therefore could last for 25 years. These contracts are usually between government authorities and Contractor consortia where both parties have proven records of long lasting stability.

CONSTRUCTION MANAGEMENT CONTRACT

Resumé

When embarking on this form of contractual arrangement, the building owner would usually appoint a project manager and the other members of the consultant team.

Once the initial schematic design is formulated a construction manager is appointed to the team to assist in design considerations and to provide practical building expertise, procedures, etc., to the project team.

Construction activities are sub-let to firms or companies specialising in the various trade work required. These trades are selected on a fully competitive delayed letting basis, and enter into direct contract agreements with the client.

A general foreman supervises all on site activities; a cost clerk and a limited number of carpenters and labourers are also engaged to attend on other trades and execute minor sundry works.

Costs are controlled by the quantity surveyors continuous audit of actual costs incurred.

Payments are made to trade contractors, suppliers and on-site employees by the client.

CONTRACTUAL SYSTEMS (cont'd)

DEFINITIONS

CONSTRUCTION MANAGEMENT

The construction manager, who may be either an individual, consultancy firm, or a builder, is selected and appointed by the proprietor or by the project manager on the proprietor's behalf, and carries out the work normally performed by a builder in a traditional system. He organises, manages and co-ordinates the construction of the project on a consultancy basis for a fee. This means that the construction manager does not have any direct commercial interest in the work carried out by sub-contractors as all sub-contracts are between proprietor and sub-contractor. Thus he neither benefits nor suffers from the contractual risks attendant upon traditional contract.

PROJECT MANAGEMENT

A project manager is an agent of the proprietor and assumes the role of the proprietor in a construction project where the proprietor lacks the technical and organisational skills for the task. As such, the project manager becomes responsible for preparing a brief, and appointing and co-ordinating the consultants. The proprietor's requirements are conveyed to the project manager who ensures that they are acted upon by the consultant and contractor. He may, if engaged for a total service, be responsible for such matters as site selection, feasibility studies, preparation of consultants briefs, selection of consultancy team together with negotiation and documentation of conditions of engagement, evaluation of design profiles, supervision of documentation, implementation of programme, selection of contractual system, contract financial control, supervision of letting as well as many other services. It should be noted that the engagement of a project manager is not affected by the contractual arrangement adopted. He can serve the client in virtually the same capacity whether it be a traditional tender or a construction management contract.

CONTRACT CONDITIONS

A number of contract agreements exist to cater for a variety of building and civil engineering projects. It is not the purpose of this book to compare, in any detail, the differences in the agreements but to briefly examine the type of project for which the particular contract is best suited.

It is also the intent to concentrate on the clauses of the contracts having cost implications.

AUSTRALIAN BUILDING INDUSTRY CONTRACT - ABIC MW-1 - Major works contract

This contract has been developed jointly by the Royal Australian Institute of Architects and the Master Builders Australia Inc. and supersedes the JCC contracts that are now withdrawn.

It claims to use plain english and to involve less risk to architects than the ABS contract.

It is recommended for use on building works of a major nature ($250,000 to $25million) where an architect is engaged to administer the contract. In its present form it does not appear to have any advantages over the ABS contracts.

AUSTRALIAN BUILDING INDUSTRY CONTRACT - SW-1

This contract is similar to MW-1 but is intended for use on building works from $50,000 to $3million.

GENERAL CONDITIONS OF CONTRACT - NATIONAL PUBLIC WORKS CONFERENCE EDITION 3

Issued by the Department of Administrative Services - Australian Construction Services, these conditions are still in use by some of the Public Works Departments in all states.

AUSTRALIAN STANDARD GENERAL CONDITIONS OF CONTRACT AS2124-1992

This Australian standard prepared by a number of scientific, industrial and governmental organisations is still the preferred document for many government departments. It provides for either a lump sum or a schedule of rates type of contract and the bill of quantities may or may not form part of the contract or it may form part of the specification.

AUSTRALIAN STANDARD GENERAL CONDITIONS OF CONTRACT AS4000

This document updates AS2124-1992 to cater for changes in construction practice and law and the AS4000 series of documents are probably the most widely used in major works.

AUSTRALIAN STANDARD GENERAL CONDITIONS OF CONTRACT AS4905

The provisions of this document are similar to AS4000 but are for minor works.

PROJECT CONTRACT PC-1 1998

Produced by the Property Council of Australia, this document can be seen as an alternative to ABIC MW1 and AS contracts. It is suitable for Design and Construct projects, as well as Construct only contracts.

COST PLUS CONTRACT FOR BUILDING WORKS - FF/C

This form of contract is intended for building works (including alterations) where the contract is to be administered by an architect and where payment to the builder is to be on the basis of the actual cost of the works plus a fee. This fee may be either a lump sum or a percentage of the cost of the works.

GENERAL CONDITIONS OF CONTRACT FOR DESIGN AND CONSTRUCTION (AS4902)

AS4902 provides choices for the project procurement :

Design and construct - the Principal provides the Principal's project requirements and does not normally provide a detailed preliminary design and does not require novation of any sub-contractors (including consultants);

Design development and construct - the Principal provides the Principal's project requirements together with a preliminary design;

Design, Novate and Construct - the Principal provides the Principal's project requirements together with a preliminary design and requires selected sub-contractors to be novated to the Contractor.

CONTRACT CONDITIONS (cont'd)

CONSTRUCTION MANAGEMENT AGREEMENT

This agreement is for use where no head contractor exists and the owner pays to the manager a fee for his services. The agreement is issued by the Royal Australian Institute of Architects (Victorian Chapter).

The clauses with cost implications are :

VARIATIONS : Where the value of variations cannot be made by agreement then :

ABIC MW-1 - 2003 — A maximum percentage for the Contractor's overheads and profit is nominated in Schedule 1 of the contract and is applied to the value of variations (additions and omissions).

N.P.W.C.3 — Where the variation is valued by applying bill or schedule rates, no percentage is added. Where daywork rates are applied an agreed charge to cover overheads, administrative costs, site supervision, establishment costs, attendance and profit is made.

AS2124 — As for N.P.W.C.3

AS4000 AS4902 AS4905 — Variations are valued by the superintendant using rates or prices in the contract. Where the bill of quantities or schedule of rates is not a contract document the rates shall still apply. No percentage is added to or deducted from the rates, but variations of omission include profit but not overheads. As overheads are not defined in the conditions it is common to accept the preliminaries on the overheads.

P.C.1 1998 — Variations are valued using rates or prices which appear in the Cost Schedule or the priced Bill of Quantities where applicable. The percentages stated in the Contract Particulars are added to the value calculated. The percentages are also to be applied where a reasonable amount is agreed or determined. In agreeing or determining the reasonable amount, no mark-ups are to be included.

S.B.W.2. — Variations are valued in accordance with rates and prices nominated in a schedule, and these amounts should be inclusive of all profit, overheads, etc. Where the variation results in an addition the percentage rate nominated in the Appendix is added to the variation total.

TIME EXTENSION COSTS :

Refer TIME EXTENSION COSTS, pages 896 and 897.

ADJUSTMENT OF PROVISIONAL SUMS :

ABIC - MW-1 - 2003 — A percentage nominated in Schedule 1 of the contract is applied to the difference between the provisional sum and the cost of performance of the work.

In the case where the performance of the work cost is more than the provisional sum, a percentage nominated in Schedule 1 of the contract is added to the difference and the resultant is added to the contract sum. Where the performance of the work cost is less than the provisional, the difference is deducted from the contract sum.

N.P.W.C.3 — No allowance is made on account of profit to or attendance on the adjustment of provisional sums.

AS2124 — A percentage nominated in the Annexure is applied to the amount actually paid for each provisional sum and can therefore result in an increase or decrease in the contract sum depending on whether the actual amount was more or less than the provisional allowance.

AS4000 AS4902 AS4905 — The provisions are the same as for AS2124.

P.C.-1 1998 — No percentage adjustment or mark-up is required until the increase of the let amount of provisional sum work exceeds 20% of the allowed amount. Thereafter, the difference is increased by the percentage stated in the Contract Particulars.

CONTRACT CONDITIONS (cont'd)

The clauses with cost implications are (cont'd) :

FAILURE TO PAY :

ABIC - MW-1	The rate of interest to be applied to contractually outstanding payments is nominated in Schedule 1.
N.P.W.C.3	No mention of an interest rate is made.
AS2124	The rate of interest is stated in the Annexure on contractually overdue payments.
AS4000	The rate of interest is stated in the Annexure.
P.C.-1 1998	The interest rate is stated in the Contract Particulars on contractually outstanding amounts.
S.B.W.2	The rate of interest is stated in the Appendix.
AS4902	The rate of interest is stated in the Annexure.

LIQUIDATED AND ASCERTAINED DAMAGES

Each set of contract conditions contains in its Annexure of Schedule a space for inserting a rate for liquidated and ascertained damages. This rate is a genuine pre-estimate of damages that the owner will incur if the project is not completed by the authorised date for completion. Once contractually accepted the rate will apply whether the actual damages are higher or lower than the pre-estimate. Liquidated and ascertained damages are always calculated on calendar days.

CONSTRUCTION CONTRACTS ACT 2004

An act -

- to prohibit or modify certain provisions in construction contracts;
- to imply provisions in construction contracts about certain matters if there are no written provisions about the matters in the contracts;
- to provide a means for adjudicating payment disputes arising under construction contracts, and for related purposes.

The adjudication of payment disputes means that a claim made may be determined under the Act and payment made without reference to the contact conditions.

The payment however "is to be taken to be an advance towards the total amount payable under the contract".

CONSTRUCTION PERIODS

The time required for the construction of a building is so closely linked to the time required for design and documentation that project period is probably a better term than construction period. There is, however, an important reason why the two are separated.

During design and documentation the capital expenditure on a project is far less than during the construction phase, and for this reason in commercial building undertakings, it is important that the construction phase is kept as short as possible.

An unrealistically short documentation period invariably results in an extended construction period and attendant extra costs for variations, time extensions and finance holding costs. It is sometimes argued that a short documentation period will result in savings because of the reduced period in which inflation can occur. This is not so, as inflation can only affect committed funds, and the documentation period does not attract much in the way of committed funds.

The most effective financial situation for a commercial building would be sufficient documentation period to enable an optimum construction period.

In certain cases, however, the advantages and savings occasioned by sufficient documentation and optimum construction period are outweighed by commercial consideration. Such an example would be a shopping centre which, regardless of cost, must open to take advantage of the Christmas buying season.

Unfortunately, even when the condition just described does not occur, the building owner usually wants his building immediately. This leads to rushed documentation and the hazards already explained.

The relationship between project costs and project period is not mathematical because of the numerous factors including complexity, material choice and values, critical activities and techniques. The relationship does, nevertheless, when based on historical data, provide one of the most reliable methods of establishing project and construction periods. For example, if records show that high rise multi-storey office blocks during construction period average monthly payments of $3,000,000, then the total estimated cost of a proposed building divided by the monthly amount will give a construction period.

To establish the documentation period is considerably more difficult, as the design process is less mechanical and therefore less predictable than construction. Records concerning documentation periods are also less precise and less plentiful than those concerning construction.

In the following guide the documentation periods shown do not include for delays occasioned by client or committee approvals, rejections and modifications. The times stated assume a minimum of abortive time but do not include for delays caused by industrial action, inclement weather, etc.

	Value $m	Documentation Period (Weeks)	Construction Period (Weeks)	Project Period (Weeks)
Simple projects e.g. warehouses, etc.	2.0 - 4.5	4 - 9	17 - 26	21 - 34
	4.5 - 7.5	7 - 10	26 - 43	32 - 53
Average projects e.g. multi-storey offices, units, etc.	15.0 - 25.0	26 - 34	43 - 60	68 - 94
	40.0 - 70.0	34 - 43	85 - 128	119 - 170
	150.0 - 200.0	51 - 77	111 - 136	162 - 221
Complex projects	5.0	12	26	37
e.g. concert halls, hospitals, art galleries, etc.	6.0 - 12.0	15 - 21	34 - 51	49 - 72
	25.0 - 40.0	43 - 51	51 - 68	94 - 119
	75.0 - 150.0	68 - 85	128 - 170	196 - 255

CASH FLOW

Cash Flow prediction is the forecasting of cash needed to be provided by the building owner at pre-determined dates in a building construction programme for the purpose of making progress payments to the builder for work executed. The extremely high interest charges on money today make it imperative on any construction project for reasonably accurate estimates of such costs to be made.

Such estimates are usually prepared before the builder commences work and are obtained by preparing a programme of construction activities for the project, costing these activities and obtaining anticipated expenditure on a month to month basis.

The example given is a simplified cash flow for a part single, part two storey 40 bed aged persons home costing $4,800,000 with a 10 month construction period. The example assumes a firm price contract with no Rise and Fall adjustments. It also assumes that a bank guarantee will be operating with no retention held.

It should also be noted that a delay of 14 or so days often elapses between the end of a construction month and the actual payment for the work done in that period, and this delay should be allowed for in any cash flow.

The cash flow example given is for construction work only and excludes other periodic payments to which the building owner will be committed, e.g. architect's and other consultant's fees, land costs, interest and holding cost, furniture, etc.

CASH FLOW (cont'd)

EXAMPLE

	CONSTRUCTION PERIOD IN MONTHS										
	1	2	3	4	5	6	7	8	9	10	TOTAL
ACTIVITY	$	$	$	$	$	$	$	$	$	$	$
Site establishment)											
Site clearance)											
Bulk excavation and fill)											
Building set-out)	32,000										32,000
Excavation and pour footings	62,500	78,500									141,000
Columns -											
Concrete ground - first		8,000									8,000
Steel ground - roof											
First - roof		10,500	16,000								26,500
Suspended floor slab)											
Stairs and beams)		32,000	82,500	129,500	20,000						264,000
Roof steelwork -											
Single storey			4,000	11,000	1,000						16,000
Two storey					9,500	3,500					13,000
Roof framing, purlins, insulation,											
Roofing R.W. goods -											
Single storey				80,000	96,000	16,000					192,000
Two storey					139,000	85,00					224,000
Services under ground slab				13,000							13,000
Ground slab on fill			32,000	64,000	13,000						109,000
Lift		10,500				48,000	80,000	21,500			160,000
Mechanical					136,000	136,000	160,000	144,000	24,000	8,000	608,000
Brickwork, windows, door frames -											
Ground - first)		51,000	128,000	144,000	144,000	133,000					600,000
First - roof)											
Services -											
Rough plumbing						54,000	50,000	40,000)			
Electrical						56,000	80,000	32,000)			312,000
Wall plaster/screeds							16,000	16,000			32,000
Glazing							6,500				6,500
Ceilings							80,000	80,000			160,000
Hot water boiler, etc.						3,000	35,500				38,500
Electrical/plumber fit-off						24,000	64,000	139,000			227,000
Wall and floor tiles fit-off							48,000	93,000			141,000
Carpentry and joinery fit-off							38,500	112,000	57,500		208,000
Cupboards, fitments, white goods								9,500	9,500	9,500	28,500
Kitchen/laundry equipment									96,000	96,000	192,000
Painting									40,000	56,000	96,000
Carpet/vinyl									25,000	94,000	119,000
Clean-up/commission										16,000	16,000
External Works -											
Bitumen car park, roads and kerbs								4,000	11,500	5,500)	
Brick paving									75,000	16,000)	
Fencing/crossovers								1,500	13,000	8,00)	
Planting/pergolas, etc.									41,500	32,000)	208,000
External Services -											
Stormwater drains	17,500	17,500	17,500		13,000	13,000				)	
Sewer drains	8,000	8,000	8,000		6,500	6,500				)	160,000
Water/fire/gas	9,500	9,500	9,500		8,000	8,000				)	
Electrical								13,000			13,000
Clean-up externally										4,000	4,000
Preliminaries	86,000	38,000	38,000	38,000	38,000	38,000	38,000	38,000	38,000	42,000	432,000
TOTALS FOR PAYMENT NUMBER	215,500	263,500	335,500	479,500	624,000	624,000	696.500	743,500	431,000	387,000	
CUMULATIVE TOTALS	215,500	479,000	814,500	1,294,000	1,918,000	2,542,000	3,238,500	3,982,000	4,413,000	4,800,000	4,800,000

CASH FLOW (cont'd)

The calculation of the cash flow shown can be quite laborious and, for a guide only, many projects have been analysed and a typical cash flow curve developed.

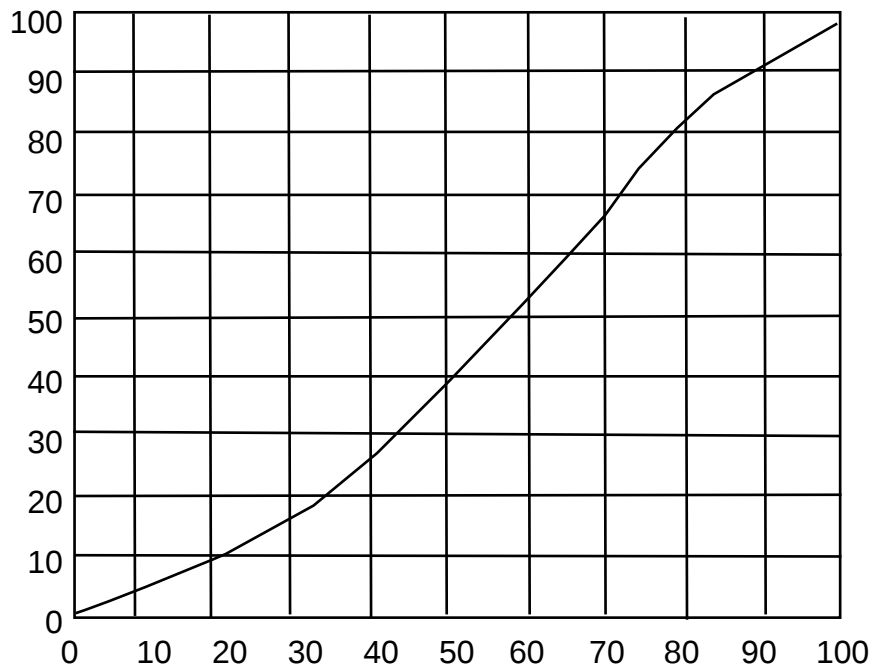
It shows expenditure as following a lazy S curve for construction purposes.

It is stressed that this is intended as a guide only and variations will apply according to the total cost of the work and whether or not lifts, sprinklers, air-conditioning and other specialist services are involved.

After reading off the theoretical amount expended against the time scale, additions or adjustments should be made to accommodate unusual or extraordinary items such as :

- Interest
- Professional Fees
- Pre-ordering

CONSTRUCTION COST %



CONTRACT CONSTRUCTION PERIOD %

From the graph it can be seen that 50% of the contract value is expended at 60% of the contract period.

TIME EXTENSION COSTS

Reimbursement as a result of a delay in the progress of the works is treated differently in the various contracts.

ABIC MW-1	Allows reimbursement for causes of delay listed in clause L1.1 clause 1 to 13. This provision therefore provides reimbursement to the builder even though the cause of delay was not a breach of contract on the part of the proprietor. Where the delay is caused by breach on the part of the proprietor, the builder is entitled to recover damages sustained and incurred.
NPWC 3	No reimbursement is allowed unless the cause of the delay was due to any breach of the provisions of the contract by or any other act or omission on the part of the Principal, etc.
AS2124-1992	Reimbursement is limited to the reasons agreed and stated in the Annexure to the contract, where however the extension is a result of a delay caused by the Principal, the Superintendent or the Principal's employees, consultants, etc., then the reimbursement becomes automatic without reference to the Annexure. Proof of extra costs for reimbursement is required.
AS4000	Reimbursement applies to compensable causes which are stated and can be expanded by completing the appropriate item in the Appendix.
P.C.-1 1998	Where a time extension is due to a breach of the contract by the Owner, the Contractor is entitled to reimbursement at the rate stated in the Contract Particulars. If an extension of time is granted as a result of a variation, then reasonable costs and expenses incurred are to be added to the value of the variation.
Lump Sum Contract for Simple Building Works - SBW2	Reimbursement is allowable only for certain causes of delay and for example inclement weather, although a cause for time extension, is not a cause which attracts time extension costs.
AS4902	Reimbursement applies to time extensions resulting from delay or disruption caused by Principal, Superintendent and their employees etc. Any other events for which costs for delay or disruption are payable must be shown in the Annexure.
AS4905	As AS4902 but the Annexure does not have the provision for additional compensation clauses.

The quantifying of the reimbursable amount has been, and probably will always be, a point of contention between the parties to the contract.

The traditional means of arriving at the amount has been for the builder to give written details substantiating any such damages, costs and expenses and the actual or closely estimated amounts thereof. It should be remembered that the reimbursement is meant to cover actual loss to the builder only. It is not intended that any profit should be returned to the builder.

Certain formulae have been applied in an attempt to establish a reimbursement cost before the actual event. Like any average formula, the results are average and do not reflect the particular circumstances of any one project.

One such formula is :

$$\frac{\text{Contract Sum}}{\text{Contract period (in working days)}} \times 4.5\% = \text{Reimbursement per day}$$

Another means of assessing reimbursement is by allocating those items of a builder's preliminaries which are time affected, to the contract period and thus establishing a cost per day for running the job. To this must be added head office overheads and profit, which is very difficult to establish.

A formula used for this purpose is :

$$\text{H.O./Profit Percentage} \times \frac{\text{Contract Sum}}{\text{Contract period (in weeks)}} \times \text{Period of Delay (in weeks)}$$

TIME EXTENSION COSTS (cont'd)

By far the most satisfactory means of establishing a reimbursement amount is to afford the builder an opportunity to tender the amount. The amount tendered is then used in the evaluation of the tender price and during the contract when a time extension reimbursement is required.

In a contract running over a long period and in which the items of plant, etc. will vary greatly, it is advisable to divide the contract into periods (i.e. first quarter, second quarter, etc.). The builder is then invited to tender different rates for the various periods of the contract.

Where a contract is subject to Rise and Fall the tendered reimbursement rates should be subject to the same provisions. The loss or expense occasioned by delay is affected by the size of the project, its complexity and plant involvement.

The following table has been compiled from historical data obtained over a wide range of contracts. It is stressed that the information is indicative only.

It should be noted that time extension costs are intended to reimburse the builder for loss and expense suffered as a result of being delayed. The reimbursement should not include any profit. Where the cause of the delay is brought about by the Principal it is usual that reimbursement to the builder will follow. Where however the cause of the delay is brought about as a result of something or an event beyond the control of both builder and Principal then each party bears the cost of the delay, the builder by not being reimbursed for loss and the principal by not being able to recover liquidated and ascertained damages.

Project	Value $m	PERIOD OF DELAY		
		First Third	Second Third	Final Third
		$/Day	$/Day	$/Day
Administrative Office 1-3 Storey	6.0	625	625	500
Civic Centre 500-700 Capacity	11.0	775	775	625
Secondary School Classrooms and Laboratories	5.5	425	425	425
University Residential College	5.5	425	425	425
Cinema Complex 2000 Capacity	25.0	2000	2100	1700
General Hospital 200 Bed	115.0	9000	9000	8250
City Hotel	65.0	7250	7250	6250
Factory/Warehouse	12.0	950	950	950
High Rise Office Block	195.0	10500	12500	10500
Multi Level Car Park	40.0	5500	5500	5500
Regional Shopping Centre	65.0	7750	7750	7750

TAX DEPRECIATION OF BUILDINGS

Capital depreciation of property is covered by the Income Tax Assessment Act 1936 and subsequent amendments.

The rates and applications of depreciation have changed frequently and are summarised :

1. Building Depreciation Rates (Pre 1979) - A building (i.e. the part not listed in the Tax Order 1217) was not eligible for depreciation before August 1979.

2. Building Depreciation Rates (1979 - 1984) - Certain income producing buildings (those used for short term traveller accommodation) constructed after August 1979, attracted 2.5% per annum allowance.

   All income producing buildings constructed after June 19, 1982 attracted a 2.5% per annum allowance.

3. Building Depreciation Rate (1984 - 1987) - Income producing buildings constructed after October 19, 1984 had the depreciation rate raised to 4% per annum.

   As a result of the 1987/88 budget the 4% allowance would apply only to buildings where :

   (a) Construction commenced before September 15, 1987.
   (b) A contract for the construction was entered into before September 1987.
   (c) Finance for the construction of the building was borrowed under a contract entered into, before September 15, 1987 and the land on which the building was to be constructed was held before September 17, 1987, or an interest in that land had been acquired under a contract entered into before September 15, 1987.

4. Building Depreciation Rates (1987 Onward) - Where construction commenced, or contracts were entered into, after September 15, 1987, the depreciation allowance on income producing buildings is 2.5% per annum. In the case of buildings used to provide short term accommodation for travellers the rate is 4% per annum.

5. Depreciation : New Rules From July 1,1991 to February 26, 1992 - The March 1991 Industry Statement changes designed to simplify the operations of the depreciation provisions of the income tax law and to allow plant and equipment write off rates for taxation purposes to be set by reference to economic factors such as predictable obsolescence, limitations on use caused by project life, or other special factors. In comparing the existing rates with the broadbanded rates, it is important to remember that no allowance has been made for the possible effects of the proposed change to the way in which life of plant acquired on or after 1 July 1991 may be determined. That is, the schedules show how broadbanding would affect existing rates, on an assumption that the effective life of the particular item remains the same.

   From 1 July 2000, an optional low-value pooling arrangement for plant has been introduced. It will apply to plant costing less than $1000 or having an undeducted cost of less than $1000 at the start of the income year provided it has been depreciated using the diminishing value method.

   There are two statutory depreciation rates that must be used together to calculate the total depreciation deduction for the low-value pool in a year of income.

   - For low-cost plant that has been allocated to the pool for the first time in that income year - 18.75% (which is 50% of the pool depreciation rate) of the total cost of that plant allocated, regardless of when during that year the item(s) was actually acquired. This eliminates the need to calculate a pro-rata depreciation deduction for each item of plant, based on the actual date it was pooled. It is consistent with the simplification aim of the low value pooling measure and;
   - For all other plant in the pool (that is, plant that has been allocated to the pool in previous year and low undeducted-cost plant allocated to the pool in the year of income) - 37.5% of the sum of :
     - The pool closing balance from the previous year and
     - The total undeducted cost of any low undeducted-cost plant allocated to the pool in the year of income.

6. Depreciation : From February 26, 1992 - An option of self estimation of effective life of items was introduced or the choice of IT2685 (the Commissioners Determination) was given.

7. Taxation Ruling IT2685 has been withdrawn as of 1 January 2001 and replaced by Taxation Ruling TR2000/18 and 18C7. These taxation rulings list the Effective Lives' changes between 1 January 2001 to 1 July 2004. A brief summary of other significant changes is :

   a) Simplified depreciation rules for those who qualify for and elect to use the Simplified Tax System after 1 July 2001. **This does not apply to taxpayers owning rental premises.**

   b) Removal of accelerated depreciation for plant acquired or commenced to construct after 21 September 1999.

TAX DEPRECIATION OF BUILDINGS (cont'd)

7. c) The effective life of plant is given under Industry categories (Table A) for the stated industries and Asset Categories (Table B) for generic assets. Table A must be used by members of the industries stated and Table B for others. Where an asset is not listed in Table A (extract), then Table B (extract) should be used and where the asset is not in either, the effective life must be worked out by the taxpayer.

d) Employee Amenties and plant used for Scientific Research retain the same rates as the previous taxation ruling IT2583.

For those taxpayers using the Simplified Tax system the rules for tangible depreciable assets are :

- Immediate write-off of items costing less than $1,000.
- Pooling of all assets with an effective life of less than 25 years at 30% rate using the diminishing value basis and immediate write off for a pool balance of less than $1,000.

For taxpayers NOT using the simplified tax system, depreciation should be calculated using Table A and/or Table B (shown on pages 910 to 916) with the following formulae.

1a) Diminishing Value Method - Pre May 10, 2006

$$\frac{\text{Base Value} \times \text{Days Held} \times 150\%}{365 \times \text{Asset's Effective Life}}$$

Diminishing Value Method - Post May 10, 2006

$$\frac{\text{Base Value} \times \text{Days Held} \times 200\%}{365 \times \text{Asset's Effective Life}}$$

1b) Prime Cost Method

$$\frac{\text{Asset's Cost} \times \text{Days Held} \times 100\%}{365 \times \text{Asset's Effective Life}}$$

TAX DEPRECIATION OF BUILDINGS (cont'd)

THE SCHEDULE applying to items 5 and 6 on page 898

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Air-conditioning plant (see also Ventilating plant)						
Central type, including ducting and vents	9	13.5	13	20		
Structural alterations and additions associated with the installation of such plant which forms an integral part of the plant	3	4.5	7	10		
Room units	12	18	17	25		
Solar energy powered	9	13.5	13	20		
Alarms	6	9	13	20		
Amenities provided for employees -						
Fittings and Fixtures						
General	33⅓	50	33	50		
Plumbing -						
(a) forming part of plant or equipment of cafeteria, kitchen, dining, mess, recreation or rest rooms, etc.	33⅓	50	33	50		
(b) sanitary ware, etc., forming part of toilet accommodation or washing facilites	6	9	13	20		
Plant and equipment used in providing meals or facilities for meals for employees	33⅓	50	33	50		
Automatic fire sprinklers	6	9	13	20		
Bedding, linen, crockery, etc. (Hospitals, hotels and boarding houses)	Replacement†		Replacement†			
Boilers (See also salvage machinery)	6	9	13	20		
Boom gates	12	18	17	25		
Building and Construction Industry -						
Earth moving plant and heavy equipment -						
Bulldozers, ditchers, excavators, graders, loaders, rippers, rollers, rooters, scrapers, shovels, tractors and similar associated machines and attachments	18	27	20	30		
(Note : Rates of 20% [prime cost] and 30% (diminishing value) may be allowed if claimed and supported)						
General Plant -						
Bending machines (bar, angle or rod)	12 or Replacement†	18 or	18 or Replacement†	18 or		
Brick elevators (portable)	24	36	27	40		
Chain blocks, rod shears, jacks, etc.	9 or Replacement†	13.5 or	13 or Replacement†	20 or		
Compressors	12	18	17	25		
Concreting plant -						
Batching plant -						
Static	9	13.5	13	20		
Portable and demountable	18	27	20	30		
Concrete buggies or dumpers (motorised)	24	36	27	40		

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

Refer Pages 910-916

TAX DEPRECIATION OF BUILDINGS (cont`d)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Building and Construction Industry - (cont'd)						
Concrete immersion vibrators	40	60	40	60		
Concrete mixers	12	18	17	25		
Concrete pumps	12	18	17	25		
Concrete transit mixers - mixing bowl, separate motor and drive mechanism	18	27	20	30		
Concrete vibrating screeders	40	60	40	60		
Hoppers, skips and hoist buckets	12 or Replacement†	18 or Replacement†	17 or Replacement†	25 or Replacement†		
Mobile concrete pumping units	18	27	20	30		
Monorails	24	36	27	40		
Rickshaws or dump carts (hand operated)	12 or Replacement†	18 or Replacement†	17 or Replacement†	25 or Replacement†		
Steel formwork, beams and props	12	18	17	25		
Trowelling machines	40	60	40	60		
Cranes -						
Mobile -						
Light and medium, up to 15 tons (15.24 tonnes) lift	18	27	20	30		
Heavy, over 15 tons (15.24 tonnes) lift	12	18	17	25		
Tower	12	18	17	25		
Derricks	9	13.5	13	20		
Grinding and milling machines	40	60	40	60		
Levels, dumpy, etc.	9 or Replacement†	13.5 or Replacement†	13 or Replacement†	20 or Replacement†		
Lift slab equipment	24	36	27	40		
Loose tools and equipment	24 or Replacement†	36 or Replacement†	27 or Replacement†	40 or Replacement†		
Portable sheds	12	18	17	25		
Power tools, hand operated -						
Electric, pneumatic or power	24 or Replacement†	36 or Replacement†	27 or Replacement†	40 or Replacement†		
Pumps	12	18	17	25		
Saw benches (portable)	9	13.5	13	20		
Scaffolding	12	18	17	25		
Sheds, portable	12	18	17	25		
Tower cranes and hoists	12	18	17	25		
Vehicles -						
Cars and utilities	15	22.5	15	22.5		
Trailers	12	18	17	25		
Trucks -						
Designed to carry 1 tonne or more	18	27	20	30		
Designed to carry less than 1 tonne	15	22.5	15	22.5		
Heavy haulage (if claimed and supported)	24	36	27	40		
Welding units -						
Light type	18	27	20	30		
Medium and other types	12	18	17	25		
Wheelbarrows	12	18	17	25		
Winches	9	13.5	13	20		
Buildings -						
(1) Except in special industries (as mentioned below) -						
(a) to the extent that they form an integral part of plant and machinery -						
Brick, stone or concrete structures	3	4.5	7	10		
Other structures	6	9	7	10		
Gantries	6	9	7	10		

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

Refer Pages 910-916

TAX DEPRECIATION OF BUILDINGS (cont`d)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Buildings (cont'd) -						
(b) which do not form integral parts of plant and machinery (including magazines for storing explosives)	Nil	Nil	Nil	Nil		
(NOTE - In case of buildings used solely for purposes of scientific research related to taxpayers business, a deduction may be allowed under section 73A of the Act. Special deductions may also be available in respect of - buildings used in the Mining and Petroleum industries, for Traveller Accommodation and Non-residential Income producing buildings. See Part 2).						
(2) Special industries - In the case of certain industries, special rates have been fixed in respect of factory buildings, to apply to those buildings to the extent only to which they form integral parts of plant and machinery. The special rates are listed hereunder :						
Class of Building	Proportion of the total cost of building normally regarded as comprising an integral part of plant and machinery					
Bacon Factory -	40 per cent					
Brick, stone or concrete structure	3	4.5	7	10		
Wooden structure	6	9	13	20		
Butter Factory -	66⅔ per cent					
Brick, stone or concrete structure	3	4.5	7	10		
Wooden structure	6	9	13	20		
Wharf structure	3	4.5	7	10		
Meat Processors - 1975 ruling applicable to modern abbatoir complexes including slaughter houses, chillers, freezing rooms, blast tunnels, boning and packing rooms -	66⅔ per cent					
Brick, stone and concrete structures	3	4.5	7	10		
Wooden structures	6	9	13	20		
Stock yards, pens and lairages, both timber and steel, but excluding concrete stockyard floors	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Buildings - (cont'd)						
Freezer Works -						
Brick, stone or concrete structures	3	4.5	7	10		
Wholly wooden structure	6	9	13	20		
Other materials - such rate as may be fixed in the particular case						
Cork board used in insulating cold storage chambers	6	9	13	20		
Primary production, forestry and pearling industries -						
On buildings of the following descriptions, not being structures used for the taxpayers own domestic or residential purposes or structures the costs of which are allowable under section 75 (see Part 2)						
Humidification cotton sheds	6	9	13	20		
Woolsheds -						
With brick, stone or concrete walls	3	4.5	7	10		
With wood and/or iron walls	3	4.5	7	10		
Other -						
With brick, stone or concrete walls	3	4.5	7	10		
With wood and/or iron walls (NOTE - See also explanatory notes re Primary production, forestry, fishing and pearling industries)	6	9	7	10		
Bullet resistant screens (not forming part of the building)	6	9	13	20		
Burglar alarms	6	9	13	20		
Car Parking - Hydraulic elevated platforms and hoists (including control equipment)						
Carpets -						
In business places, hotels, etc.	24	36	27	40		
In houses let furnished	12	18	17	25		
In picture theatres	24	36	27	40		
In professional chambers	12	18	17	25		
In ten-pin bowling centres	40	60	40	60		
Chimney stacks and flues (see also brickmaking plant)						
Integral part of building	Nil	Nil	Nil	Nil		
Concrete stacks in heavy industry qualifying as plant	3	4.5	7	10		
Cold Rooms - prefabricated, with stressed skin panels	9	13.5	13	20		
Compressors (see also building and construction industry and road-making plant)						
Air and oxygen types	6	9	13	20		
Ammonia -						
Vertical	9	13.5	13	20		
Horizontal	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Concrete mixers	12	18	17	25		
Concrete transit mixers - mixing bowl, separate motor and drive mechanism	18	27	20	30		
Concreting plant (see building and construction industry)						
Conveyors (see also materials handling plant and equipment)						
Cranes (see also building and construction industry) -						
Electrical or otherwise	6	9	13	20		
Gantries	6	9	7	10		
Curtains and drapes	18	27	20	30		
Docks, floating	6	9	13	20		
Electrical Machinery and Equipment -						
Accumulators and storage batteries	9	13.5	13	20		
Alternators, dynamos, motor generators, rotary converters	6	9	13	20		
Batteries, storage	9	13.5	13	20		
Boilers	6	9	13	20		
Broadcasting equipment -						
Computer Automated	12	18	17	25		
Buildings - to the extent to which the building forms an integral part of plant and machinery -						
Brick, stone or concrete stuctures	3	4.5	7	10		
Other structures	6	9	7	10		
Cables and wires -						
Overhead -						
Insulated	6	9	13	20		
Bare	3	4.5	7	10		
Underground	3	4.5	7	10		
Chimney flues	Nil	Nil	Nil	Nil		
Choke coils	3	4.5	7	10		
Condensers	6	9	13	20		
Distributing centres - switchgear	6	9	13	20		
Dynamos	6	9	13	20		
Electric transmission lines (NOTE - In addition, the cost of replacements of poles, cross arms, etc., is allowable)	3	4.5	7	10		
Engines, condensers, pumps	6	9	13	20		
Generators, motor						
Hand tools and loose plants	Replacement†		Replacement†			
House installations (owned by electricity suppliers)	6	9	13	20		
Indicators - fixed and portable	9	13.5	13	20		
Instruments, testing apparatus, meters, indicators - fixed and portable	9	13.5	13	20		
Lamps, arc	12	18	17	25		
Lamps,incandescent	Replacement†		Replacement†			
Lighting plant (overall rate)	6	9	13	20		
Lighting units, fluorescent	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Electrical Machinery and Equipment (cont'd) -						
Lightning arresters	3	4.5	7	10		
Loose plant	Replacement†		Replacement†			
Machinery not otherwise specified	6	9	13	20		
Meters	9	13.5	13	20		
Motor generators	6	9	13	20		
Power tools, hand operated	24 or Replacement†	36 or	27 or Replacement†	40 or		
Power factor control	6	9	13	20		
Power station plant	6	9	13	20		
Pumps	6	9	13	20		
Rotary convertors	6	9	13	20		
Standards, iron or steel (including brackets and switch arms)	3	4.5	7	10		
Starting gear, including compensators, switches, etc.	6	9	13	20		
Storage batteries	9	13.5	13	20		
Switchboards	6	9	13	20		
Testing apparatus	9	13.5	13	20		
Transformer boxes	3	4.5	7	10		
Transformers (static)	3	4.5	7	10		
Electric railway						
Bridge works - brick, stone and concrete	3	4.5	7	10		
Other	6	9	7	10		
Electric signs (see also neon signs)	Replacement†		Replacement†			
Engines	6	9	13	20		
Escalators (machinery and other moving parts)	9	13.5	13	20		
Fences -						
Electric	6	9	13	20		
Generally (but see Primary industry)	Nil	Nil	Nil	Nil		
Wire mesh - demountable used for partitioning purposes	6	9	13	20		
Fire control and alarm systems -						
Alarm, hoses and nozzle	6	9	13	20		
Water services	3	4.5	7	10		
Fire sprinklers, automatic	6	9	13	20		
Fire exinguishers - (refills and maintenance s.51)	9	13.5	13	20		
Fittings and fixtures in cafeteria, rest, recreation and locker rooms provided for employees (see also Amenities)	33⅓	50	33	50		
Fittings in shops (see also shops)	6	9	13	20		
Floating docks	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Floating marinas	6	9	13	20		
Floor installations - free access floors in computer rooms	3	4.5	7	10		
Fluorescent lighting units	6	9	13	20		
Foundation for Stacks	Replacement†		Replacement†			
Foundation of plant and machinery forming integral part of such plant and machinery	3	4.5	7	10		
Furniture and fittings (see also Amenities)						
In business places, boarding houses, hotels, motels, houses let furnished, picture theatres and professional offices including those of dentists and medical practitioners	9	13.5	13	20		
Galvanised iron for roofing stacks	Replacement†		Replacement †			
Galvanised iron tanks -						
Borewater	12	18	17	25		
Rainwater	6	9	13	20		
Garbage bins	18	27	20	30		
Hand dryers - electrically operated	12	18	17	25		
Heating units, electronic	12	18	17	25		
Hoardings	6	9	13	20		
Hospitals (See also medical practitioners plant) -						
Beds	9	13.5	13	20		
Bedding, linen, crockery, etc.	Replacement†		Replacement†			
Furniture	6	9	13	20		
Hot water installations -						
On whole installation including boilers and, where installed, pumps	6	9	13	20		
Hotel, motel, boarding house and restaurant plant and equipment -						
Bedding	Replacement†		Replacement†			
Carpets	24	36	27	40		
Crockery, cutlery, glassware and cooking utensils	Replacement†		Replacement†			
Furniture and fittings	9	13.5	13	20		
Linen	Replacement†		Replacement†			
Linoleum and similar floor coverings	12	18	17	25		
Soda water fountains	9	13.5	13	20		
Houses and flats let furnished -						
Bedding	Replacement†		Replacement†			
Blinds, venetian	6	9	13	20		

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

Refer Pages 910-916

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Houses and flats let furnished -						
Carpets	12	18	17	25		
Chain saw	40	60	40	60		
Crockery, cutlery, glassware and cooking utensils	Replacement†		Replacement†			
Curtains and drapes	18	27	20	30		
Electric bed	9	13.5	13	20		
Electric clock	9	13.5	13	20		
Electric heater	12	18	17	25		
Furniture and fittings	9	13.5	13	20		
Garbage units, compacting	18	27	20	30		
Gas coppers	6	9	13	20		
Hot water services	6	9	13	20		
Lawn mowers - motor	18	27	20	30		
- Self propelled	24	36	27	40		
Linen	Replacement†		Replacement†			
Linoleum and similar floor coverings	12	18	17	25		
Microwave ovens	18	27	20	30		
Radios	12	18	17	25		
Refrigerators	9	13.5	13	20		
Solarhart	6	9	13	20		
Stoves	6	9	13	20		
Sun louvres	Nil	Nil	Nil	Nil		
Television sets	12	18	17	25		
Vacuum cleaners	12	18	17	25		
Washing machines	18	27	20	30		
Incinerettes (gas or electrically fired) (see also Refuse Destructors)	6	9	13	20		
Jetties, boat shed	3	4.5	7	10		
Laboratory equipment	9	13.5	13	20		
Letter Boxes - private, polycarbonate	9	13.5	13	20		
- aluminium, nylon, brass	3	4.5	7	10		
Lift - Boom	40	60	40	60		
- Scissor	40	60	40	60		
Lift slab equipment	24	36	27	40		
Lifts and elevators -						
Electric	9	13.5	13	20		
Hydraulic	6	9	13	20		
Lift well and other structural alterations	Nil	Nil	Nil	Nil		
Lighting control system (microprocessor based)	24	36	27	40		
Lighting plant (Electric)	6	9	13	20		
Lighting system, fluorescent	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Linoleum, vinyl, and similar floor coverings	12	18	17	25		
Materials handling plant and equipment :						
Container port loading facilities -						
Portainer cranes	6	9	13	20		
Straddle carriers	24	36	27	40		
Wharves	3	4.5	7	10		
Containers (transportable) -						
Generally	18	27	20	30		
On hire or pool units	24	36	27	40		
Security type (heavy steel)	12	18	17	25		
Conveyors (production or freight handling) -						
Belts (rubber or vinyl)	18	27	20	30		
Overhead production lines	12	18	17	25		
Rollers (static or movable)	12	18	17	25		
Forklifters, including automatic loaders, transporters and front end loaders	18	27	20	30		
Pallets	24	36	27	40		
Racks, stillages, trollies and baskets	12	18	17	25		
Refrigeration equipment -						
Clip-on, integrally mounted or static	12	18	17	25		
Slings (Rope or steel wire)	Replacement†		Replacement†			
Neon Signs - owned by taxpayer	6	9	13	20		
PABX	6	9	13	20		
Paging and music while you work systems. (Up to 15% [prime cost] or 22.5% [diminishing value] may be fixed in special circumstances)	12	18	17	25		
Partitions - demountable	6	9	13	20		
Portable sheds - nomadic type industry	12	18	17	25		
Portable toilet	12	18	17	25		
Refrigerating plant and machinery (see also ice making plant)						
Buildings (see building)						
Cold rooms - prefabricated with stressed skin panels	9	13.5	13	20		
Compressors, ammonia -						
Vertical	9	13.5	13	20		
Horizontal	6	9	13	20		
Condenser pipes	9	13.5	13	20		
Cork board for insulating cold storage chambers	6	9	13	20		
Expansion pipes	3	4.5	7	10		
General machinery	9	13.5	13	20		
Refrigeration (freezing) units, including compressors for shops	12	18	17	25		
Refrigerators (gas, electric or fuel burning)	6	9	13	20		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

ITEM	Acquired Pre 27.2.92		Acquired between 26.2.92 and 1.1.01		Acquired Post 1.1.2001	
	Prime Cost Rate Rate	Diminishing Value	Prime Cost Rate Rate	Diminishing Value		
	%	%	%	%	%	%
Safes, bank -						
Portable	3	4.5	7	10		
Other	Nil	Nil	Nil	Nil		
Scaffolding	12	18	17	25		
Screens - bullet resistant (not forming part of the building)	6	9	13	20		
Security systems -						
Bullet resistant screens (not forming part of the building)	6	9	13	20		
Burglar alarms	18	27	20	30		
Camera scanning - of type used in large retail establishment	18	27	20	30		
Electronic tags, releases - retail stores	18	27	20	30		
Sewage treatment plant	6	9	13	20		
Signs	6	9	13	20		
Sprinklers, automatic fire	6	9	13	20		
Strongrooms - demountable	3	4.5	7	10		
Supa grass - synthetic tennis court surface	12	18	17	25		
Swimming pools -						
Above ground	12	18	17	25		
Concrete	3	4.5	7	10		
Fibreglass	6	9	13	20		
Filtration equipment	9	13.5	13	20		
Other equipment	9	13.5	13	20		
Tanks (see also Oil companies plant and machinery) -						
Galvanised iron -						
Rain water	6	9	13	20		
Bore water	12	18	17	25		
Earth	Nil	Nil	Nil	Nil		
Reinforced concrete or masonry	3	4.5	7	10		
Telephone installations owned by taxpayers -						
Answering machines - car phone if purchased prior to delivery	18	27	20	30		
Car phone if purchased after delivery	18	27	20	30		
Cellular mobile	18	27	20	30		
Complete telephone system (comprising switchboards, instruments, cables, etc.)	6	9	13	20		
Computerised PABX equipment	6	9	13	20		
Public telephones	12	18	17	25		
Reservation system (data print)	6	9	13	20		
Turnstiles	6	9	13	20		
Vinyl, linoleum and similar floor coverings	12	18	17	25		
Water mains	3	4.5	7	10		

Refer Pages 910-916

† Immediate deduction where plant costing $300 or less acquired after 30.6.91 * Effective life not provided for pre -13.3.91 plant.

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE A - EFFECTIVE LIFE TABLES AS AT 1 JULY 2020 (INDUSTRY CATEGORY) applying to item 7 on pages 898 and 899

ELECTRICITY, GAS, WATER AND WASTE SERVICES (26110 to 29220)

ASSET	Life (Years)	Date of Application
Electricity supply (26110 to 26400) :		
Electricity distribution :		
Control, monitoring, communications and protection systems	10	1 Jan 2002
Customer meters (incorporating load and time switches if fitted)	25	1 Jan 2002
Customer service mains or cable, above ground	40	1 Jan 2002
Customer service mains or cable, underground	50	1 Jan 2002
Distribution lines :		
Above ground (incorporating conductors; cross arms, insulators and fittings; poles-concrete, wood, steel or stobie; and transformers - pole or ground pad mounted)	45	1 Jan 2002
Combination of above ground and underground	47½	1 Jan 2002
Underground (incorporating cables, fittings and ground pad mounted transformers)	50	1 Jan 2002
Distribution substations/transformers, pole or ground pad mounted	40	1 Jan 2002
Distribution zone substations (excluding control, monitoring, communications and protection systems)	40	1 Jan 2002
Nightwatchman's lights	15	1 Jan 2002
Street lights	15	1 Jan 2002
Electricity generation :		
Gas turbine:		
Control and monitoring system	15	1 Jan 2002
Emergency power supply assets (including batteries and uninterruptible power supply assets)	15	1 Jan 2002
Gas turbine generators	30	1 Jan 2002
Generator transformer and unit transformer in sub-tropical area	30	1 Jan 2002
Generator transformer and unit transporment in tropical area	25	1 Jan 2002
Miscellaneous assets	30	1 Jan 2002
On-site switchyard with conventional outdoor switchgear	30	1 Jan 2002
On-site switchyard with gas insulated switchgear	30	1 Jan 2002
Station and auxiliary electrical systems within the power station	30	1 Jan 2002
Hydro-electric :		
Control and monitoring system	15	1 Jan 2002
Emergency power supply assets (including batteries and uninterruptible power supply assets)	15	1 Jan 2002
Generator transformer and unit transformer in sub-tropical area	30	1 Jan 2002

ELECTRICITY, GAS, WATER AND WASTE SERVICES (26110 to 29220) (cont'd)

ASSET	Life (Years)	Date of Application
Electricity supply (26110 to 26400) (cont'd) :		
Hydro-electric (cont'd):		
Generator transformer and unit transformer in tropical area	25	1 Jan 2002
Hydro turbines and generators	40	1 Jan 2002
Miscellaneous assets	40	1 Jan 2002
On-site switchyard with conventional outdoor switchgear	40	1 Jan 2002
On-site switchyard with gas insulated switchgear	35	1 Jan 2002
Station and auxiliary electrical systems within the power station	40	1 Jan 2002
Solar :		
Photovoltaic electricity generating system assets (incorporating photovoltaic panels, mounting frames and inverters)	20	1 Jul 2011
Thermal :		
Condensing and feed heating assets	30	1 Jan 2002
Control and monitoring system	15	1 Jan 2002
Emergency power supply assets (including batteries and uninterruptible power supply assets)	15	1 Jan 2002
Generator transformer and unit transformer in sub-tropical area	30	1 Jan 2002
Generator transformer and unit transformer in tropical area	25	1 Jan 2002
Miscellaneous assets	30	1 Jan 2002
On-site switchyard with conventional outdoor switchgear	30	1 Jan 2002
On-site switchyard with gas insulated switchgear	30	1 Jan 2002
Primary dust collection system (incorporating electrostatic precipitators or baghouse filters)	30	1 Jan 2002
Solid fuel preparation assets (including fuel feeders and milling assets)	30	1 Jan 2002
Station and auxiliary electrical systems within the power station	30	1 Jan 2002
Steam generator	30	1 Jan 2002
Steam turbine generator	30	1 Jan 2002
Wind :		
Generator transformer and unit transformer in sub-tropical area	30	1 Jan 2002
Generator transformer and unit transformer in tropical area	25	1 Jan 2002
Wind turbine	20	1 Jan 2002

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE A - EFFECTIVE LIFE TABLES AS AT 1 JULY 2020 (INDUSTRY CATEGORY) applying to item 7 on pages 898 and 899

ELECTRICITY, GAS, WATER AND WASTE SERVICES (26110 to 29220) (cont'd)

ASSET	Life (Years)	Date of Application
Electricity supply (26110 to 26400) (cont'd) :		
Power station civil and structural works :		
Chimney stack :		
Concrete surround	30	1 Jan 2002
Steel flues	20	1 Jan 2002
Cooling tower (concrete or timber)	30	1 Jan 2002
Cooling water system (excluding cooling towers and condensing assets)	30	1 Jan 2002
Power station buildings, to the extent that they form an integral part of plant	30	1 Jan 2002
Workshop machinery and tools	20	1 Jan 2002
Gas supply (27000) :		
Gas distribution :		
Control systems (excluding computers)	10	1 Jul 2002
Gas meter	15	1 Jul 2002
Low pressure (LP) gas storage holder	40	1 Jul 2002
Pigging device	5	1 Jul 2002
Pipeline (including high, medium or low pressure trunk, primary or secondary mains or services) :		
Generally	50	1 Jul 2002
PVC pipeline	30	1 Jul 2002
Regulators (including gate stations, subgate stations, block valve stations, pressure regulating stations and district regulating stations)	40	1 Jul 2002
Gas transmission :		
Compressor gas turbine (GT) driver	20	1 Jul 2002
Compressor station assets	30	1 Jul 2002
Control systems (excluding computers)	10	1 Jul 2002
Gas meter	15	1 Jul 2002
Gas pipeline LNG station assets	30	1 Jul 2002
Pigging device	5	1 Jul 2002
Pipeline - transmission, spur or lateral	50	1 Jul 2002
Regulators (including gate stations, subgate stations, block valve stations, pressure regulating stations and district regulating stations)	40	1 Jul 2002
Underground gas storage asset	40	1 Jul 2002

ACCOMMODATION AND FOOD SERVICES (44000 to 45302)

Accommodation (44000) :		
Carpets	7	1 Jul 2005
Door control and motor drive system for automatic sliding doors and revolving doors (incorporating chains, controls, motors and sensors), excluding doors	10	1 Jul 2005
Furniture, freestanding:		
Generally (including guestrooms)	7	1 Jul 2005
Outdoor	5	1 Jul 2005
Garage doors, electric (excluding doors) :		
Controls and motors	5	1 Jul 2005
Gates, electric (excluding gates) :		
Controls and motors	5	1 Jul 2005
Guestroom assets :		
Bathroom assets:		
Heated towel rails, electric	5	1 Jul 2005
Spa bath pumps	7	1 Jul 2005
Kitchen assets:		
Bar refrigerators	10	1 Jul 2005
Microwave ovens	5	1 Jul 2005
Laundry assets in guestrooms:		
Clothes dryers	7	1 Jul 2005
Washing machines	7	1 Jul 2005
Hot water systems (excluding commercial boilers and piping)	10	1 Jul 2005

TRANSPORT, POSTAL AND WAREHOUSING (46100 to 53090)

ASSET	Life (Years)	Date of Application
Other warehousing and storage services (53090) :		
Warehouse and distribution centre equipment and machines :		
Automated storage and retrieval machines	20	1 Jul 2011
Balers	15	1 Jul 2011
Battery assets for warehouse vehicles (including pallet trucks and forklifts) :		
Batteries (detachable for re-charging)	5	1 Jul 2011
Battery chargers :		
Forklifts	11	1 Jul 2011
Other	10	1 Jul 2011
Handling assets :		
Battery tuggers	10	1 Jul 2011
Racking roller beds	15	1 Jul 2011
Transfer carts	10	1 Jul 2011
Washers	10	1 Jul 2011
Carts/buggies	15	1 Jul 2011
Conveyors	15	1 Jul 2011
Door control and motor drive systems (incorporating chains, controls, motor and sensors), excluding doors :		
External	20	1 Jul 2011
Internal	10	1 Jul 2011
Floor sweepers/scrubbers	10	1 Jul 2011

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE A - EFFECTIVE LIFE TABLES AS AT 1 JULY 2020 (INDUSTRY CATEGORY) applying to item 7 on pages 898 and 899

TRANSPORT, POSTAL AND WAREHOUSING (46100 to 53090) (cont'd)

ASSET	Life (Years)	Date of Application
Other warehousing and storage services (53090) (cont'd) :		
Forklift attachments :		
Cages	10	1 Jul 2011
Push pull units	11	1 Jul 2011
Inflatable dock bags/seals/shelters	10	1 Jul 2011
Pallet assets :		
Dispensers	15	1 Jul 2011
Lift tables	10	1 Jul 2011
Racks	20	1 Jul 2011
Radio frequency terminal assets :		
Barcode readers/scanners	5	1 Jul 2011
Portable/handheld and vehicle mounted terminal devices	4	1 Jul 2011
Roll cages	10	1 Jul 2011
Trolleys	10	1 Jul 2011
Voice picking assets :		
Battery charges	4	1 Jul 2011
Headsets	4	1 Jul 2011
Terminals (on person)	4	1 Jul 2011
Waste compactors (used for cardboard and plastic) :		
Electric	15	1 Jul 2011
Hydraulic	20	1 Jul 2011

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) :		
Assets generally:		
Air-conditioning assets (excluding ducting, pipes and vents):		
Air handling units	20	1 Jul 2003
Chillers :		
Absorption	25	1 Jul 2003
Centrifugal	20	1 Jul 2003
Volumetrics (including reciprocating, rotary, screw, scroll) :		
Air-cooled	15	1 Jul 2003
Water-cooled	20	1 Jul 2003
Condensing sets	15	1 Jul 2003
Cooling towers	15	1 Jul 2003
Damper motors (including variable air volume box controller)	10	1 Jul 2003
Fan coil units (connected to condensing set)	15	1 Jul 2003
Mini split systems up to 20kW (including ceiling, floor and high wall split system)	10	1 Jul 2003
Packaged air-conditioning units	15	1 Jul 2003
Pumps	20	1 Jul 2003
Room units	10	1 Jul 2003

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200) (cont'd)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) :		
Assets generally:		
Ceiling fans	5	1 Jul 2004
Clocks, electric	10	1 Jul 2004
Digital peepholes	10	1 Jul 2019
Digital video display (DVD) players	5	1 Jul 2004
Door closers	10	1 Jul 2004
Door stops, freestanding	10	1 Jul 2004
Escalators (machinery and moving parts)	20	1 Jan 2003
Evaporative coolers :		
Fixed (excluding ducting and vents)	15	1 Jul 2019
Portable	10	1 Jul 2005
Floor coverings (removable without damage) :		
Carpet	8	1 Jul 2019
Floating timber	15	1 Jul 2004
Linoleum	10	1 Jan 2001
Vinyl	10	1 Jan 2001
Furniture, freestanding	13⅓	1 Jan 2001
Garbage bins	10	1 Jul 2004
Garbage compacting systems (excluding chutes)	6⅔	1 Jan 2001
Generators	20	1 Jan 2001
Gym assets :		
Cardiovascular	5	1 Jul 2004
Resistance	10	1 Jul 2004
Hand dryers, electrical	10	1 Jan 2001
Heaters :		
Fixed :		
Electric	15	1 Jul 2004
Gas :		
Ducted central heating unit	20	1 Jul 2004
Other	15	1 Jul 2004
Freestanding	15	1 Jul 2004
Home automation control assets	10	1 Jul 2019
Hot water systems (excluding piping) :		
Electric	12	1 Jul 2004
Gas	12	1 Jul 2004
Solar	15	1 Jul 2004
Hypronic assets :		
Controls	10	1 Jul 2019
Water heaters	15	1 Jul 2019
Intercom system assets	10	1 Jul 2004
Lifts (including hydraulic and traction lifts)	30	1 Jan 2003
Lights :		
Fittings (excluding hardwired)	5	1 Jul 2004
Freestanding	5	1 Jul 2004
Shades, removable	5	1 Jul 2004
Linen	5	1 Jul 2004
Master antenna television (MATV) assets :		
Amplifiers	10	1 Jul 2004
Modulators	10	1 Jul 2004
Power sources	10	1 Jul 2004

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE A - EFFECTIVE LIFE TABLES AS AT 1 JULY 2020 (INDUSTRY CATEGORY) applying to item 7 on pages 898 and 899

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200) (cont'd)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) (cont'd):		
Mirrors, freestanding	15	1 Jul 2004
Radios	10	1 Jan 2001
Rugs	7	1 Jul 2004
Skylights :		
Controls	10	1 Jul 2019
Motors	10	1 Jul 2019
Solar power generating system (incorporating batteries, inverters, solar panels, regulators)	20	1 Jul 2004
Stereo systems (incorporating amplifiers, cassette players, CD players, radios and speakers)	7	1 Jul 2004
Surround sound systems (incorporating audio-video receivers and speakers)	10	1 Jul 2004
Telecommunications assets :		
Cordless phones	4	1 Jul 2004
Telephone hand sets	6	1 Jul 2019
Television antennas, freestanding	5	1 Jul 2004
Television sets	8	1 Jul 2019
Vacuum cleaners :		
Ducted :		
Hoses	10	1 Jul 2004
Motors	10	1 Jul 2004
Wands	10	1 Jul 2004
Portable	10	1 Jan 2001
Ventilation fans	20	1 Jul 2004
Video cassette recorder systems (VCR)	5	1 Jul 2004
Water pumps :		
Multi storey	20	1 Jul 2019
Rainwater tanks	5	1 Jul 2019
Single residence pressure pumps	8	1 Jul 2019
Window blinds, internal	10	1 Jul 2004
Window curtains	6	1 Jul 2004
Window shutters, automatic :		
Controls	10	1 Jul 2004
Motors	10	1 Jul 2004
Bathroom assets :		
Accessories, freestanding (including shower caddies, soap holders, toilet brushes)	3	1 Jul 2019
Exhaust fans (including light/heating)	10	1 Jul 2004
Heated towel rails, electric	10	1 Jul 2004
Shower curtains (excluding curtain rods and screens)	2	1 Jul 2004
Spa bath pumps	10	1 Jul 2019
Fire control assets :		
Alarms :		
Heat	6	1 Jul 2004
Smoke	6	1 Jul 2004
Detection and alarm systems :		
Alarm bells	12	1 Jul 2004
Detectors (including addressable manual call points, heat, multi type and smoke)	20	1 Jul 2004
Fire indicator panels	12	1 Jul 2004

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200) (cont'd)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) (cont'd) :		
Fire control assets (cont'd) :		
Emergency warning and intercommunication systems (EWIS) :		
Master emergency control panels	12	1 Jul 2004
Speakers	12	1 Jul 2004
Strobe lights	12	1 Jul 2004
Warden intercom phone	12	1 Jul 2004
Extinguishers	15	1 Jul 2004
Hoses and nozzles	10	1 Jul 2004
Pumps (including diesel and electric)	25	1 Jul 2004
Stair pressurisation assets :		
AC variable speed drives	10	1 Jul 2004
Pressurisation and extraction fans	25	1 Jul 2004
Sensors	10	1 Jul 2004
Kitchen assets :		
Cook tops	12	1 Jul 2004
Crockery	5	1 Jul 2004
Cutlery	5	1 Jul 2004
Dishwashers	8	1 Jul 2019
Freezers	12	1 Jul 2004
Garbage disposal units	10	1 Jul 2004
Microwave ovens	8	1 Jul 2019
Ovens	12	1 Jul 2004
Range hoods	12	1 Jul 2004
Refrigerators	12	1 Jul 2004
Stoves	12	1 Jul 2004
Water filters, electrical	15	1 Jul 2004
Laundry assets :		
Clothes dryers	7	1 Jul 2019
Ironing boards, freestanding	5	1 Jul 2019
Irons	5	1 Jul 2004
Washing machines	8	1 Jul 2019
Outdoor assets :		
Automatic garage doors :		
Controls	5	1 Jul 2004
Motors	10	1 Jul 2004
Barbecue assets :		
Fixed barbecue assets :		
Sliding trays and cookers	10	1 Jul 2004
Freestanding barbecues	5	1 Jul 2004
Floor carpet (including artificial grass and matting)	5	1 Jul 2004
Folding arm awnings :		
Controls	10	1 Jul 2019
Motors	10	1 Jul 2019
Furniture, freestanding	5	1 Jul 2004
Gardening watering installations :		
Control panels	5	1 Jul 2004
Pumps	5	1 Jul 2004
Timing devices	5	1 Jul 2004

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE A - EFFECTIVE LIFE TABLES AS AT 1 JULY 2020 (INDUSTRY CATEGORY) applying to item 7 on pages 898 and 899

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200) (cont'd)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) (cont'd) :		
Garden lights, solar	5	1 Jul 2019
Garden sheds, freestanding	15	1 Jul 2004
Gates, electrical :		
Controls	5	1 Jul 2004
Motors	10	1 Jul 2004
Operable pergola louvres :		
Controls	15	1 Jul 2004
Motors	15	1 Jul 2004
Rainwater tanks :		
Galvanised steel	25	1 Jul 2019
Polyethylene	15	1 Jul 2019
Roller blinds :		
Controls	10	1 Jul 2019
Motors	10	1 Jul 2019
Sauna heating assets	15	1 Jul 2004
Sewage treatment assets :		
Controls	8	1 Jul 2004
Motors	8	1 Jul 2004
Spas :		
Fixed spa assets :		
Chlorinators	10	1 Jul 2019
Filtration assets (including pumps)	10	1 Jul 2019
Heaters (electric or gas)	15	1 Jul 2004
Freestanding spas (incorporating blowers, controls, filters, heaters and pumps)	17	1 Jul 2004
Swimming pool assets :		
Chlorinators	10	1 Jul 2019
Cleaning assets	7	1 Jul 2004
Covers (including blankets)	8	1 Jul 2019
Filtration assets (including pumps)	10	1 Jul 2019
Heaters :		
Electric	15	1 Jul 2004
Gas	15	1 Jul 2004
Solar	20	1 Jul 2004
Tennis court assets :		
Cleaners	3	1 Jul 2004
Drag brooms	3	1 Jul 2004
Nets	5	1 Jul 2004
Rollers	3	1 Jul 2004
Umpire chairs	15	1 Jul 2004
Security and monitoring assets :		
Access control systems :		
Code pads	5	1 Jul 2004
Door controllers	5	1 Jul 2004
Readers :		
Proximity	7	1 Jul 2004
Swipe card	3	1 Jul 2004

RENTAL, HIRING AND REAL ESTATE SERVICES (66110 to 67200) (cont'd)

ASSET	Life (Years)	Date of Application
Residential property operators (67110) (cont'd) :		
Security and monitoring assets (cont'd) :		
Closed circuit television systems :		
Cameras	4	1 Jul 2004
Monitors	4	1 Jul 2004
Recorders :		
Digital	4	1 Jul 2004
Time lapse	2	1 Jul 2004
Switching units (including multiplexes)	5	1 Jul 2004
Security systems :		
Code pads	5	1 Jul 2004
Control panels	5	1 Jul 2004
Detectors (including passive infra-red, photo sensors and vibration)	5	1 Jul 2004
Global System for Mobiles (GSM) Units	5	1 Jul 2004
Noise makers (including bells and sirens)	5	1 Jul 2004
Non-residential property operators (67120)		
Commercial office building assets :		
Boilers	20	1 Jul 2005
Boiler pumps	5	1 Jul 2005
Carpets	8	1 Jul 2005
Door control and motor drive system for automatic sliding doors (incorporating chains, controls, motors and sensors, excluding doors)	15	1 Jul 2005
Hot water installations (excluding commercial boilers and piping)	15	1 Jul 2005
Power supply assets :		
Emergency or standby :		
Generator assets :		
Acoustic hoods and canopies	20	1 Jul 2005
Generators (incorporating attached engine management and generator monitoring instruments)	25	1 Jul 2005
Power management units	15	1 Jul 2005
Lighting control systems (microprocessor based)	5	1 Jul 2014
Uninterruptible power supply (UPS) systems :		
Line interactive types	5	1 Jul 2013
On line double conversion types	10	1 Jul 2013
Window blinds	20	1 Jul 2005

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE B - EFFECTIVE LIVES (ASSET CATEGORIES) AS AT 1 JULY 2020

ASSET	Life (Years)	Date of Application
Air compression assets:		
Air compression assets generally (including air compressors, air dryers and air receivers)	15	1 Jul 2014
Air compressors (portable):		
Compressors - reciprocating	7	1 Jul 2014
Compressors - rotary screw	10	1 Jul 2014
Air-conditioning assets (excluding pipes, duct work and vents) :		
Air handling units	20	1 Jul 2003
Cooling towers	15	1 Jul 2003
Condensing sets	15	1 Jul 2003
Chillers :		
Absorption	25	1 Jul 2003
Centrifugal	20	1 Jul 2003
Volumetrics (including reciprocating, rotary, screw, scroll) :		
Air-cooled	15	1 Jul 2003
Water-cooled	20	1 Jul 2003
Damper motors (including variable air volume box controller)	10	1 Jul 2003
Fan coil units (connected to a condensing set)	15	1 Jul 2003
Humidifiers (steam generator)	10	1 Jul 2003
Mini split system up to 20kW (including ceiling, floor and high wall split system)	10	1 Jul 2003
Packaged air conditioning units	15	1 Jul 2003
Pumps	20	1 Jul 2003
Room units	10	1 Jul 2003
Airless sprayers (used in painting, epoxy and polyurethane coating, priming, etc.) :		
Electrically and petrol-driven units	7	1 Jul 2013
Pneumatically-driven units	15	1 Jul 2013
Artworks qualifying as depreciating assets (restricted to works of art and reproductions of artwork that are tangible in nature, such as paintings, sculptures, drawings, engravings and photographs, that are displayed in open viewing areas in premises used for	100	1 Jul 2013
Automatic external defibrillators (AEDs)	8	1 Jul 2020
Automatic teller machine (ATMs)	7	1 Jul 2019
Bending machines (bar, angle and rod)	10	1 Jul 2008
Block and brick elevators, portable	10	1 Jul 2008
Boilers	20	1 Jul 2005
Boom gates :		
Electro Mechanically operated	7	1 Jul 2013
Hydraulically operated	10	1 Jul 2013

ASSET	Life (Years)	Date of Application
Car park assets :		
Automatic and semi-automatic parking systems (including platforms, stackers, turntables, etc.)	20	1 Jul 2016
Automatic payment machines/stations	10	1 Jul 2016
Car park floodlighting	20	1 Jul 2016
Car park signage	7	1 Jul 2016
Entry and exit stations/columns (including RFID card readers, ticket dispensing machines, ticket readers, etc.)	10	1 Jul 2016
Licence plate recognition cameras and systems	5	1 Jul 2016
Parking guidance assets :		
Automatic vehicle counter systems (incorporating counter modules and signage)	7	1 Jul 2016
Parking guidance systems (incorporating controllers, detectors, dynamic signage, lighting, etc.)	7	1 Jul 2016
Pay and display machines/Pay and go machines	8	1 Jul 2016
Security and monitoring assets:		
Intercommunication (intercom) system assets	7	1 Jul 2016
Concreting assets :		
Brick and paving saws	5	1 Jul 2008
Concrete demolition saws	3	1 Jul 2008
Concrete kibble buckets	15	1 Jul 2008
Concrete mixers	4	1 Jul 2008
Concrete preparation assets (including floor grinders)	5	1 Jul 2008
Concrete trowels :		
Walk behind	5	1 Jul 2008
Ride on	7	1 Jul 2008
Concrete vibrating screeders	5	1 Jul 2008
Concrete vibrators :		
Bushcutter style	5	1 Jul 2008
Drive units	6	1 Jul 2008
Flexible shaft units	6	1 Jul 2008
Vibrating shaft	3	1 Jul 2008
Concrete wheeled saws	6	1 Jul 2008
Cranes (gantry and overhead)	25	1 Jul 2014
Curtains and drapes	6	1 Jul 2004
Escalators (machinery and their moving parts)	20	1 Jan 2003
Fences :		
Portable electric fences	20	1 Jul 2014
Wire mesh (demountable used for partitioning purposes, including portable electric fences)	20	1 Jul 2014

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE B - EFFECTIVE LIVES (ASSET CATEGORIES) AS AT 1 JULY 2020

ASSET	Life (Years)	Date of Application
Fire control and alarm assets :		
Alarms :		
Heat	6	1 Jul 2004
Smoke	6	1 Jul 2004
Detection and alarm systems :		
Alarm bells	12	1 Jul 2004
Detectors :		
Aspirated smoke	12	1 Jul 2004
Heat	20	1 Jul 2004
Manual call point (addressable type only)	20	1 Jul 2004
Multi type	20	1 Jul 2004
Smoke	20	1 Jul 2004
Fire indicator panels	12	1 Jul 2004
Gas suppression cylinders	25	1 Jul 2004
Emergency warning and intercommunication systems :		
Master emergency control panels	12	1 Jul 2004
Speakers	12	1 Jul 2004
Strobe lights	12	1 Jul 2004
Warden intercom phones	12	1 Jul 2004
Extinguishers	15	1 Jul 2004
Hoses and nozzles	10	1 Jul 2004
Pumps (including diesel and electric)	25	1 Jul 2004
Stair pressurisation assets :		
A C variable speed drive	10	1 Jul 2004
Pressurisation and extraction fans	25	1 Jul 2004
Sensors	10	1 Jul 2004
Floor coverings (removable without damage) :		
Carpets (excluding fixed carpet tiles)	8	1 Jul 2016
Floating timber	15	1 Jul 2016
Linoleum and vinyl	10	1 Jul 2016
Foundations for plant and machinery (integral to the operation of such plant and machinery, but not incorporated into the plant and machinery itself)	40	1 Jul 2014
Generators (see in Table B, Power supply assets)		
Hand dryers (electrical)	10	1 Jul 2013
Labratory assets used in quality control, sample checking, etc. :		
Automated and electronically based laboratory assets (including analysers, refractometers), etc.	7	1 Jul 2014
Other laboratory assets (including autoclaves, centrifuges, microscopes, ovens etc)	10	1 Jul 2014
Lifts (including dumbwaiters, hydraulic lifts and traction lifts)	30	1 Jan 2003
Loading bay assets :		
Dock levellers	20	1 Jul 2005
Pallet jacks and pallet trucks	10	1 Jul 2005
Scissor lifts	15	1 Jul 2005
Lockers, freestanding	20	1 Jul 2019
Moving walks	20	1 Jan 2003
Office furniture, freestanding :		
Bookcases :		
Timber	15	1 Jul 2005
Metal	20	1 Jul 2005
Cabinets (including credenzas, cupboards, filing, mapping, mobile, stationery and storage type) :		
Timber/laminated	15	1 Jul 2005
Metal	20	1 Jul 2005
Chairs	10	1 Jul 2005
Desks	20	1 Jul 2005
Mobile storage units (compactus type)	25	1 Jul 2005
Reception assets (including lobby chairs, desks, lounges, sofas and tables)	10	1 Jul 2005
Screens	20	1 Jul 2005
Tables :		
Boardroom	20	1 Jul 2005
General	10	1 Jul 2005
Workstations (including desks and partitions)	20	1 Jul 2005
Partitions (demountable)	20	1 Jul 2005
Pneumatic air tube systems	10	1 Jul 2013
Power supply assets :		
Emergency or standby :		
Generator assets :		
Acoustic hoods and canopies	20	1 Jul 2005
Generators (incorporating attached engine management and generator monitoring instruments)	25	1 Jul 2005
Power management units	15	1 Jul 2005
Uninterruptible power supply (UPS) system	15	1 July 2013
Generators, portable (incorporating attached engine management and generator monitoring instruments) :		
Diesel	10	1 Jul 2008
Petrol	5	1 Jul 2008
Public address and paging system assets (including amplifiers, audio speakers and microphones)	12	1 Jul 2005
Rainwater tanks :		
Galvanised steel	25	1 Jul 2019
Polyethylene	15	1 jul 2019
Refrigeration assets :		
Compressors, condensers, evaporators, etc.	15	1 Jul 2014
Insulation panels used in cool or freezer rooms	40	1 Jul 2014

TAX DEPRECIATION OF BUILDINGS (cont'd)

EXTRACTS FROM TABLE B - EFFECTIVE LIVES (ASSET CATEGORIES) AS AT 1 JULY 2020

ASSET	Life (Years)	Date of Application
Security and monitoring assets :		
Access control systems :		
Code pads	5	1 Jul 2004
Door controllers	5	1 Jul 2004
Readers :		
Proximity	7	1 Jul 2004
Swipe card	3	1 Jul 2004
Ballistic and blast resistant screens and barriers (including fixed and rising hypes) not forming part of the building	20	1 Jul 2014
Silos :		
Bulk handling :		
Ancillary mechanical equipment (including augers, bucket excavators, conveyors, etc.)	15	1 Jul 2014
Concrete construction	50	1 Jan 2004
Galvanised construction	30	1 Jan 2004
Steel construction	40	1 Jan 2004
Solar photovoltaic electricity generation system assets	20	1 Jul 2011
Spas used as plant in a business (incorporating blowers, controls, filters, heaters and pumps)	17	1 Jul 2014
Swimming pool assets :		
Chlorinators	12	1 Jul 2004
Cleaning assets	7	1 Jul 2004
Covers (including blankets)	8	1 Jul 2019
Filtration assets (including pumps)	12	1 Jul 2004
Heaters :		
Electric	15	1 Jul 2004
Gas	15	1 Jul 2004
Solar	20	1 Jul 2004

ASSET	Life (Years)	Date of Application
Swimming pool assets (cont'd) :		
Swimming pools (used as plant in a business) :		
Above-ground	10	1 Jul 2005
Concrete	50	1 Jul 2005
Covers (including blankets)	8	1 Jul 2019
Fibreglass	20	1 Jul 2005
Switchboards	20	1 Jul 2014
Synthetic lawn surface	5	1 Jul 2014
Telephony :		
Mobile phones	3	1 Jul 2014
Telephone systems (including analogue and digital telephone systems, PABX/PBX systems, key/commander systems, VoIP systems and hybrid telephone systems such as IP-PBX systems, etc.)	7	1 Jul 2015
Ventilation fans (excluding ducting, piping and vents)	20	1 Jan 2005
Waste storage and disposal bins (including crane bins, hooklift bins, skip bins and other industrial use bins)	10	1 Jul 2016
Weighbridges	20	1 Jul 2014
Window blinds	15	1 Jul 2016

LOADING ON DEPRECIATION RATES

The depreciation rates published before 1991 appeared in Income Tax Order No. 1217 and certain loadings apply to the rates :

Items acquired under a contract or commenced to be constructed after August 19, 1980 and before May 1, 1981	20%
Items acquired under a contract or commenced after April 30, 1981	18%
Items acquired under a contract or commenced after May 25, 1988	20%

ACCELERATED DEPRECIATION

The accelerated depreciation rates of 20 percent and 33⅓ percent which were available from July 19, 1982 on items of plant have now been eliminated as from May 25, 1988. Plant acquired under contracts entered into (and construction had commenced) before May 25, 1988 will retain the accelerated depreciation provided it is used or is ready for use before July 1, 1991.

PROPERTY INSURANCE

Past years of inflationary trends in the cost of building have caused enlightened and prudent building owners to look more closely at their insurance cover and to adopt more frequent reviews and adjustments in their insured building value than hitherto.

Normally a building can be insured in one of two ways - either for Indemnity Value or for Replacement Value.

i) Indemnity Value is the poor relation - a building so insured provides cover for the value of the building as it stands, warts and all i.e. likely depreciated, outdated and not conforming to design regulations. If total loss occurs, the insurer would pay the actual value of the property destroyed. If the loss is a partial one, the insured may be required to contribute to the cost of repairs if the remaining old construction is replaced with new. The policy will only place the insured in the position he was in prior to the loss - no better, no worse.

ii) Replacement Value - an old building might well have an indemnity value of $200,000 and a replacement value of $1,000,000. Replacement involves the construction of a building having the same functional use and of the same usable area as the old building, but built in the modern idiom and, of necessity, conforming to regulations and by-laws which may have been adopted since the original building.

An existing building, of whatever age, adequately insured in this category, would absolve the owner from expense in the event of building loss. An existing building constructed of say pipe columns and metal cladding, if destroyed could well be required to be replaced with a building of reinforced concrete and brick construction. If the insurance premiums covered this extra cost element, in other words was adequately insured, then the owner would be subjected to no expense if loss occurred.

Fire policies have, of late, become subject to averaging and a co-insurance clause has been introduced, the result being that the insured is compelled to maintain full insurance, whereas in the past many building owners have insured on a maximum probable loss basis, producing a sum insured substantially less than either indemnity or replacement value, e.g. a building with a true replacement value of $1,000,000, insured on a replacement basis for $600,000, would result in the insured paying 40% of any claim. The importance of being adequately insured is obvious.

VALUING BUILDINGS FOR INSURANCE PURPOSES

The calculated value of a building comprises several elements :

- Present rebuilding cost
- Allowance for cost escalation during the lead time of planning, calling tenders, construction and fit-out
- Professional fees
- Removal of debris
- Cost escalation in the likely lapse time between the insurance anniversary date and the date of any happening.

Some or all of these cost factors could be automatically included in the premium quoted by the insurer, but it is extremely important to establish the extent of such inclusions.

PROPERTY INSURANCE (cont'd)

VALUING BUILDINGS FOR INSURANCE PURPOSES (cont'd)

Example :

Typical insurance valuation for a building (on a replacement basis) would follow the following general lines :

• Estimated/calculated building value on acquired data at rates current at the time of estimate			$1,000,000
• Cost escalation for period -			
Design and documentation		3 months	
Calling tenders and appraisals		1 month	
Construction period and fit-out (mean cash point)		7 months	
	Total	11 months	
11 months escalation at say 0.5% per month = 5.5%			55,000
			1,055,000
• Professional fees - say 11.5%			121,000
• Cost of removal of debris			10,000
			1,186,000
• Cost escalation in the lapse period between policy renewal dates (worst possible case assumed) 12 months @ say 0.5% = 6%			71,000
		Insured Value	$1,257,000

Such value represents building costs only and excludes cost factors such as loss of revenue in the period between building loss and reinstatement, cost of leasing alternative premises in the same period, removal and upset expenses, etc.

The principal cost of any valuation, and one which affects all other on-costs, is the building cost, and it is advisable to obtain professional advice, through a quantity surveyor or the like, before the insured commits himself to the payment of insurance premiums. Above all it is important that building owners protect their investments by arranging periodic cost reviews of their buildings in terms of inflation and the cost effect of the introduction of statutory and other building regulations, by-laws, etc.

OPERATING COSTS OF BUILDINGS

The function of a building determines its operating cost, and operating costs must be comparative. The operating costs of a cinema will obviously differ from those of an office block.

The methods of calculating operating costs can, however, be applied to any type of building.

The following main cost categories should be included :

1. Rates and Taxes	- Land tax - calculated on the unimproved land value - Water Rates - calculated on the unimproved land value with additional charges for excess water - Council rates - calculated on the gross rental value
2. Insurance	- Fire Insurance - Workers' Compensation - Machinery Breakdown - Public Liability - Burglary/Theft - Plate Glass
3. Air-conditioning	- Maintenance and running costs excluding energy costs
4. Lifts	- Maintenance and running costs excluding energy costs
5. Fire Protection	- Sundry fees, repair and maintenance, etc.
6. Energy	- Replacement costs - Energy costs
7. Cleaning	- Contract cleaning - Toilet consumables, etc.
8. Buildings Staff	- Caretaker - Engineer - Commissionaire
9. Security	- Patrol service - Night watchman
10. Repairs and Maintenance	- Redecorating - Carpet Replacement, etc.
11. Management	- Rent collection - Supervision, etc.
12. Sundries	- Sundry minor costs
13. Void Allowance and Contingency	- To cover running costs brought about by vacancies

Great variation in the cost and categories occurs not only between capital cities, but between councils and local authorities within capital cities, and therefore each project must be individually calculated.

OPERATING COSTS OF BUILDINGS (cont'd)

The following table should be regarded as an **approximate guide only** :

- **These guide costs exclude Goods and Services Tax (G.S.T.).**

COST CATEGORY	BUILDING TYPE							
	LOW RISE OFFICE BUILDING		HIGH RISE OFFICE BUILDING		SUBURBAN SHOPPING CENTRE		REGIONAL SHOPPING CENTRE	
	$/sqm Net Lettable Area/Annum	% Total	$/sqm Net Lettable Area/Annum	% Total	$/sqm Net Lettable Area/Annum	% Total	$/sqm Net Lettable Area/Annum	% Total
1. Rates and Taxes	44.30	29.5	54.70	32.2	37.80	28.0	47.10	32.5
2. Insurances	8.10	5.4	9.20	5.4	6.90	5.1	7.60	5.3
3. Air-Conditioning	8.90	5.9	10.00	5.9	6.50	4.8	7.00	4.8
4. Lifts	7.20	4.8	8.10	4.8	1.30	0.9	1.40	1.0
5. Fire Protection	1.50	1.0	1.60	1.0	-	-	-	-
6. Energy	27.80	18.5	28.10	16.5	20.10	14.9	18.80	13.0
7. Cleaning	15.90	10.7	18.00	10.6	15.70	11.6	14.10	9.7
8. Building Staff	7.40	4.9	7.10	4.2	7.00	5.2	10.30	7.1
9. Security	3.00	2.0	5.10	3.0	2.80	2.1	4.40	3.0
10. Repairs and Maintenance	6.60	4.4	7.60	4.4	7.40	5.5	9.80	6.7
11. Management	11.80	7.9	11.90	7.0	17.40	12.9	14.40	9.9
12. Sundries	4.60	3.1	5.30	3.1	12.10	9.0	10.10	7.0
13. Void Allowance	2.90	1.9	3.30	1.9	-	-	-	-
TOTAL	150.00	100.0	170.00	100.0	135.00	100.0	145.00	100.0

PROFESSIONAL FEES

Over the past few years the character of Professional Fees has undergone great change. For many years the various professional institutes set scales of minimum fees for defined services and terms of engagement. To charge less than the prescribed minimum left the consultant open to action by his Institute.

The various codes of ethics contained a clause which held the under-cutting of fees to be unethical and unprofessional and disciplinary action against transgressors was available to the Institute.

The Trades Practices Commission appears to hold the view that the setting of a minimum fee by an institution is a restrictive trade practice, and it has become common practice for the institutes to publish Guideline Fee Scales only. The guidelines fee scales allow negotiation on terms or fees on any basis whatsoever which need have no relationship with published guidelines.

Copies of relevant fee guides are available from :

The Royal Australian Institute of Architects
Association of Consulting Engineers, Australia
Australian Institute of Quantity Surveyors
Environmental Institute of Australia and New Zealand

For budgeting purposes professional fees should be assessed on the particular type of building as the variation can be significant. The following table is indicative of the likely range of professional fees for various typical projects, the figures being based on the Guideline Fee Scales mentioned above. However, currently many fees are substantially less than the figures given hereafter and discounts of 25% are common with the level of discounting relating to the service provided.

BUILDING TYPE	VALUE $M	CONSULTANTS	FEE %
1-3 Storey Administration	6,000,000	A, SE, EE, HE, ME, LA, QS, ES, BS	11.5 - 14.0
Civic Centre, 500-700 Capacity	11,000,000	A, SE, EE, HE, ME, SC, LA, QS, ES, BS	12.0 - 14.5
Secondary School, Laboratory Building	5,500,000	A, SE, EE, HE, QS, ES, BS	10.5 - 13.5
Multi Storey General Hospital	115,000,000	A, SE, EE, ME, HE, SC, QS, ES, BS	11.5 - 14.0
Warehouse	12,000,000	A, SE, EE, HE, QS, ES, BS	6.5 - 9.5
Office Building, 20 Storey	195,000,000	A, SE, EE, ME, HE, QS, ES, BS	11.0 - 13.5
Parking Station, Multi-Storey	40,000,000	A, SE, EE, ME, HE, QS, ES, BS	10.0 - 12.5
Residential - Individual House	450,000	A, SE, EE, HE, BS	14.0 - 17.0

A = Architect
SE = Structural Engineer
EE = Electrical Engineer
ME = Mechanical and Lift Engineer
ES = Environmental Scientist
HE = Hydraulic Engineer
LA = Landscape Architect
QS = Quantity Surveyor
SC = Specialist Consultant
BS = Building Surveyor

GOODS AND SERVICES TAX (G.S.T.)

These notes are intended not to be a precis of the entire G.S.T. system and its application and accounting, but rather how it applies generally to the Construction Industry and in particular to the rates and prices contained in the Handbook. Some of the terminology used is not directly from the New Tax System (Goods and Services Tax) Act 1999, but merely an attempt to simplify the explanation of the flow of G.S.T. from the supplier to the Tax Office.

GENERALLY

G.S.T. is a tax on consumption, NOT on business. What this means in broad terms is that the end user of goods and services pays the final bill for the G.S.T. and not the suppliers, manufacturers and producers down the chain of manufacture and production. A Builder registered for G.S.T. may claim a refund from the Tax Office equivalent to his input tax. Input tax is the G.S.T. paid on the purchase of most materials and subcontract services in operating a construction business. Consequently, because input tax credits are claimed, G.S.T. is not a cost factor to the Builder in carrying out construction activity, that is, apart from any possible additional administration costs and interest for the period between the G.S.T. remittance* and receipt of the progress claim payment (*unless a ruling is given that G.S.T. does not have to be remitted until after issuing the certificate authorising the progress claim). Therefore, in building up the total cost of a construction project, the Builder will initially add up all the NET (excluding G.S.T.) costs of materials, plant and labour to which will be added overheads, margins and finally G.S.T. (see example below). The G.S.T. added by the Builder onto the net cost of the tender/quotation is known as output tax. Output tax is G.S.T. added to the net cost of most goods and services flowing out of a business to clients/customers. This output tax, when paid by the client/customer, is paid to the Tax Office at the end of every G.S.T. tax period.

	Quotation $	**G.S.T. (Input Tax) $**	**Tender $**
• Materials	110,000	10,000	100,000
• Plant - Hired	11,000	1,000	10,000
• Plant - Own	15,000		15,000
• Labour	120,000		120,000
• Subcontractors - non-registered	75,000		75,000
• Subcontractors - registered	165,000	15,000	150,000
• Overheads	25,000		25,000
• Margin	25,000		25,000
• Total G.S.T. (input tax) to be paid by the Builder to suppliers included in the Tender		26,000	520,000
• G.S.T. (output tax) to be paid by the Client and passed on to the Tax Office by the Builder			52,000
• Total Amount of Tender (including allowance for G.S.T.)			572,000

Assuming the contract was completed in one G.S.T. accounting period, the Builder would submit the following to the Tax Office for this particular contract :

• Output tax paid by Client to the Builder	52,000
• Less : input tax paid to Suppliers by the Builder	26,000
• Balance of G.S.T. remitted to the Tax Office by the Builder	26,000

It can be seen from this example that the Builder's payments and receipts cancel each other out and consequently there is no G.S.T. effect to the Builder.

Having established an estimated cost of a project and derived a cost inclusive of G.S.T., one must examine the Client for whom the work is to be undertaken to establish how the G.S.T. will affect them.

GOODS AND SERVICES TAX (G.S.T.) (cont'd)

GENERALLY (cont'd)

There are two broad categories into which property/building transaction Clients may fall, namely :

- Taxable & Tax-free - this is where the Client is registered for G.S.T. purposes and may claim input credits and would therefore, referring to the foregoing example, claim an input credit for the $52,000 G.S.T. paid to the Builder. Consequently, G.S.T. has NO EFFECT on the cost of the project, with the exception of possible additional administration costs and interest for the period between G.S.T. remittance and refund;

- Input Taxed - this is where a Client is registered for G.S.T. but provides a service which is input taxed. This would preclude the Client from claiming input credits for G.S.T. paid on supplies to create the input taxed service. Consequently, in this situation the G.S.T. would have A DEFINITE IMPACT on the cost of the project, namely $52,000 shown in the above example. This situation will apply mainly to buildings used by the providers of financial services and premises for residential purposes.

  Within the category of Residential Premises, not all are input taxed, there are Commercial Residential premises which are required to have G.S.T. added to their rental or room rates, i.e. :

  - Hotels, motels, inns, hostels and boarding houses.

  - Caravan parks and camping grounds.

RENTAL REVIEW

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- **KNIGHT FRANK**

Note :

This section is intended as a guide only to the various rental values so that, by selection or interpolation, an approximate value can be arrived at, to assist in the preparation of initial feasibility studies and the like.

The information should not be used for valuation or lease negotiation purposes.

To obtain more specific information it is recommended that the advice of a real estate consultant be sought.

ADELAIDE

Source - Colliers International Research

COMMENT

OFFICE

The outbreak of the COVID-19 pandemic at the beginning of the second quarter and the resulting shutdown saw sales and leasing activity ground to a halt, with most companies holding off an any decisions while they understood the impact that this health shock will have on business revenue. We saw many businesses adapt rapidly with many office-based businesses moving to remote working.

The health response to the COVID-19 in South Australia has been swift and effective, with limited community transmission. The level of restrictions put in place has been significantly less restrictive and the reopening of the economy has been faster than in some of the larger states. While the pandemic currently remains contained within South Australia, until there is a vaccine it is unlikely international borders will open and there may still be state border restrictions to contain outbreaks, as seen with the Victorian outbreak.

The ABS has moved up the release of some of their key economic data and have added new data collections which give a faster insight as to the impact the pandemic is having on the economy. One of these additions is the analysis of the ATO payroll data. This gives a snapshot of the industries which have seen the largest job losses and impacts on wages. South Australia has seen a recovery in jobs since the pandemic commenced in March, and occupations which contribute to white collar demand have rebounded faster than accommodation, food services and arts and recreation sectors. South Australia has also seen less losses in wages than other states which will support household expenditure a key component of GDP.

Activity in the leasing markets has been subdued during the second quarter, with a little improvement in the third quarter however, we have seen a significant improvement in enquiry with September seeing record levels of enquiry which points to a more positive fourth quarter. The State Government have committed to ten levels at 83 Pirie Street, for the Department of Transport (DTI). Cbus Property is the developer for this seventeen level 30,000 sqm project which has commenced demolition of the site in mid-September with completion expected in second quarter of 2023.

The Federal Government has committed to leasing 29,000 sqm of a new development at the 60 King William Street which is expected to complete in the fourth quarter 2023 in line with their lease requirements. This building is being developed by Charter Hall and will have a further 10,000 sqm to be leased as the project commences construction. There is only one building currently under construction which is 108 Wakefield Street, due to complete in the second half of 2020. At time of writing there are no pre-commitments on completion.

Vacancy has increased marginally to 14.2 per cent in July 2020 up from the fourteen per cent in January. With significant new supply due to occur in 2023, we are expecting that vacancy will increase further when this supply is delivered and backfill space is released to the market. At this stage there are four projects which are possible in 2023, with two of these expected to commence construction with firm pre-commitments. Vacancy for new generation A Grade space remains low at around four per cent.

We have completed a comprehensive tenant analysis which shows fifty-six per cent of tenants are unlikely to see space requirements impacted as a result of the pandemic and therefore being classed as “resilient tenants”. We have identified around nine pe rcent as being at risk of downsizing. We also found that seventy-seven per cent of tenants occupy less than 2,000 sqm. This means we are at much lower risk of seeing sublease coming to the market which will increase vacancy significantly in the short term.

Our forecasts from the first half now seem a little pessimistic, given all the government stimulus which has been put in place to support the economy. We expect that the worst case is now around ten per cent unemployment rate. Our best case is we expect a peak unemployment rate at nine per cent and base case at nine and a half per cent. At this stage we expect that best case is the most probable. This would see vacancy increase to around fifteen per cent at the beginning of 2021. We expect that vacancy is likely to be ‘hidden’ in smaller tenancies, so we have therefore discounted net absorption over the next two years to account for this hidden vacancy component.

There are several significant unknowns which could create a drag on the office market. The forced adoption of work from home may lead to some companies requiring less space, but the implementation of social distancing and the abandoning of hot desking may lead to some companies needing more space. Both factors could impact future net absorption. At this stage we are not forecasting any rental growth for the next two years and we expect that incentives will increase from current levels by five per cent across the board.

Transaction volumes are expected to pick up over the next quarter with an increasingly large weight of capital seeking investment grade assets. At this stage we are not forecasting changes to yields for prime grade assets other than possibly long WALE assets with blue chip tenants. There is a case for yield compression here given the defensive nature of this asset class. We do expect that there will be some softening for secondary grade assets.

Grade	Precinct	Average Gross Face Rents ($ / sqm per annum)	
		Low	High
Premium	CBD	490	580
A Grade	CBD	424	580
B Grade	CBD	380	480

ADELAIDE

COMMENT (cont'd)

INDUSTRIAL

Despite broader economic headwinds, the Adelaide industrial market has remained stable in recent months, buoyed by a large pipeline of infrastructure investment and continued growth of e-commerce. Over the past quarter, there has been a strong rise in sentiment within both occupier and investment markets with enquiry levels reverting to pre COVID-19 levels, albeit lease deals are taking longer to transact.

The recent 2020-21 Federal Budget was positive for the Adelaide industrial market with increasing levels of funding announced for infrastructure and manufacturing while defence spending has been brought forward. Coupled with growing demand from transport, e-commerce and food logistics occupiers, tenant demand is expected to pick up over the next twelve months.

Infrastructure investment remains key for growth in the Adelaide industrial market with both state and federal governments boosting spending on infrastructure and fast-tracking developments to support the construction industry and jobs. Colliers Research is currently tracking over $8 billion in transport and manufacturing projects across South Australia which will have a direct positive impact on local industrial markets. In addition to this, the completion of the $1.2 billion of defence infrastructure project at Osbourne will support the frigates and submarine projects due to commence in 2020 and 2023 respectively. With the Federal Government investing $270 billion in its national defence capability, Adelaide is well placed to capture a large share of this given the city's strong ties to the sector.

While not specific to Adelaide, online retail sales have grown by almost ninety per cent in the twelve months to August 2020 which is the highest level on record and well above the 15.1 per cent recorded in January prior to the COVID-19 outbreak. Long term, a permanent structural shift to online shopping is anticipated and will underpin industrial demand going forward.

Manufacturing which has seen a decline over the last decade due to cheaper offshore alternatives are now coming into focus as key in managing crisis situations and building onshore capabilities. The recent announcement of a $1.5 billion manufacturing funding program highlights this and will provide funding support for six subsectors, mining, food and drink, medical products, recycling and clean energy, defence and space. Adelaide, with a strong manufacturing base, is well positioned to capitalise on increased levels of demand.

New supply in 2020 is forecast to reach just over 300,000 sqm which is the highest level ever recorded within the Adelaide market and over triple the ten-year average. The Inner North submarket is expected to capture eighty-two per cent of supply additions in 2020 and will stem from major facilities including the Woolworths expansion (94,000 sqm - Gepps Cross) and Metcash's new distribution centre (73,223 sqm – Gepps Cross). Elsewhere, new additions to supply includes the OI Glass facility (37,525 sqm – Kilkenny), Pacific Marine Battery (11,805 sqm – Osborne) and United Steel (10,400 sqm - Salisbury South). Unlike the East Coast capital cities where there has been a steady supply of speculative facilities, recent completions in Adelaide have stemmed from pre-commitments or owner occupier facilities. Looking ahead, supply levels are expected to fall to approximately 70,000 sqm in 2021. However, given the anticipated growth in certain manufacturing subsectors and increased requirements from occupiers in the defence and space industries, it is likely that additional projects will progress in 2021 as pre-commitments are sourced.

The industrial leasing market has been subdued across Adelaide over the past six months as occupiers delay their expansion and relocation decisions. Increased demand from owner occupiers given stamp duty exemptions and low interest rates has also impacted the Adelaide leasing market as occupiers increasingly looking to purchase rather than rent. The Outer North submarket has been the major beneficiary of this with the take-up of land in Direk and Edinburgh being strong and there is now limited serviced land available.

The flight to quality remains an ongoing theme with tenant activity focused on the prime market while activity within the secondary market has been patchy as occupiers take advantage of rising incentives in some selected submarkets. Similarly, recent demand has largely stemmed from the smaller end of the market (sub 2,000 sqm).

Our latest industrial vacancy rate estimates for Adelaide show a 6.9 per cent vacancy rate for buildings sized above 2,000 sqm with just over 450,000 sqm available for lease. By grade, the vacancy rate in the secondary market is more elevated at seven per cent and compares to 6.6 per cent within the prime market. The Outer North had the highest vacancy rate at 7.7 per cent while it's lowest in the Inner North at six and a half per cent.

While subdued leasing demand has not impacted rents, there has been a modest rise in incentive levels, increasing from 11.6 per cent in June 2020 to 13.4 per cent in September 2020 in the prime market and 14.1 per cent to 15.6 per cent in the secondary market over the same period. The largest rise in incentives has occurred in the Outer North submarket and is the result of the more elevated vacancy rate and higher proportion of institutional ownership who have become more aggressive in their approach to secure tenants. Alternatively, markets dominated by private investors are yet to see the same level of increase to incentives.

The outlook for the Adelaide industrial market remains positive. The Adelaide market has strong ties to the manufacturing sector and the recent Government funding to support growth in the sector bodes well for local leasing demand. In addition, with a significant investment in space and defence programs, these industries are expected to grow post pandemic. E-commerce and food logistics are also expected to drive tenant demand going forward as the shift to online purchases becomes permanent.

Although we expect rental growth to remain subdued and incentives to increase in the short-term, we are more positive about the medium and longer terms outlook for the industrial property market. Industries which were traditionally viewed as being in decline are expected to have a resurgence which places Adelaide as a key beneficiary of this growth.

ADELAIDE

COMMENT (cont'd)

RETAIL

	Prime				Secondary			
Market	Net Rent ($/sqm		Average Incentive		Net Rent ($/sqm		Average Incentive	
	Low	High	Low	High	Low	High	Low	High
Outer North	$70	$90	15%	25%	$35	$50	15%	20%
Inner North	$90	$125	5%	18%	$50	$80	10%	20%
West	$125	$170	5%	15%	$75	$100	10%	15%
Inner South	$95	$135	5%	15%	$70	$85	10%	15%
Outer South	$60	$85	10%	15%	$50	$60	15%	20%

The retail sector is one of the largest effected sectors as a result of the restrictions put in place to manage the COVID-19 pandemic. While we are starting to see some normalisation of spending, there is a high degree of uncertainty and several categories such as international tourism which is unable to contribute to the retail sector. Outbound international tourism has accounted for $61 billion last year. With international borders expected to remain closed through into 2021, we expect that some of the cash that has been spent on travel will be spent on other retail categories.

Retail sales have seen dramatic swings and changes in the categories of spending over the last six months, food delivery, alcohol and tobacco, furniture and homewares have all been strong category performers. Gyms and fitness, travel and transport have all seen substantial falls over the pandemic period. Strong growth has been reported in online sales with the market penetration increasing significantly during the pandemic period. Non-Food online retailing nationally has jumped to over seventeen per cent of total sales in August and reached a peak of over twenty per cent during the pandemic.

New supply in the Adelaide market has been mostly limited to the construction of new Aldi Stores which have opened in Salisbury, Mount Gambier, Findon, Norwood and Plympton. Port Adelaide Plaza has also completed a significant upgrade this year. The extension of Burnside Village has been delayed due to the pandemic with no firm construction timeline in place. The redevelopment of the Coles at Norwood is expected to start construction next year.

Rents have fallen for most retail categories in Adelaide over the last twelve months. Incentives have also increased. This trend was underway in the retail sector prior to the pandemic; however, this has been accelerated as a result of the pandemic. There are further revisions expected for rents over the next twelve months with the most likely timing when federal stimulus is withdrawn. We are expecting in the June 2021 quarter is the most likely timing with the current settings.

While there is more certainty than in the June quarter there is still some risks around the economic recovery. As we have seen an outbreak in Melbourne has seen stage four restrictions in place for over twelve weeks with only a short reprieve since the initial restrictions in March. The budget release in October showed a more optimistic outlook for the Australian economy than we have seen from other government bodies such as the Reserve Bank. This however is dependent on the containment of new outbreaks and the wide availability of a vaccine by the end of 2021. There are however some positives for the retail sector. Superannuation withdrawals, extension of job keeper and record federal stimulus for job creation are all expected to be positive for the retail sector. Also, there are some categories such as overseas travel which are unable to be spent due to international border restrictions. This is likely to be the case into 2021. With $61 billion spent on international travel during 2019, it is expected that some of this spending will be reallocated to other categories. It is also worth noting that savings rates have increased substantially in the June quarter up to almost twenty per cent up from six per cent in the March quarter. Once consumers become more confident it is expected that some of this money will be spent in the retail sector. All these factors support a recovery in household consumption over the next couple of quarters.

Adelaide Retail Market Indicators

Precinct	Average Gross Face Rent ($ / sqm per annum)			Per annum growth (per cent)	Average Incentive	
	Low	High	Average		Low	High
Regional Centres	$1,000	$1,750	$1,375	-3.5%	20%	25%
Sub Regional Centre	$500	$880	$650	-5.5%	25%	30%
Neighbourhood Centres	$350	$650	$500	-4.8%	20%	30%
Retail CBD - Rundle Mall	$2,000	$3500	$2,650	-7.8%	25%	30%

BRISBANE

Source - Jones Lang LaSalle

COMMENT

OFFICES

MARKET OVERVIEW

- COVID-19 continued to weigh on demand, with negative 15,728 sqm of net absorption recorded in the second quarter.
- Headline vacancy increased by 0.7 per cent to 12.8 per cent over the quarter, largely attributed to small tenant moves and space hand backs by large occupiers.
- Three projects remain under construction, totalling 117,700 sqm.
- Prime gross effective rents (-0.2 per cent) and secondary gross effective rents (-0.1 per cent) decreased marginally.
- Prime and secondary yields remained stable, off the back of limited transactional evidence.

RENTS

Transactional evidence has been limited due to the slowdown in leasing activity. Nonetheless, prime gross effective rents (PGER) decreased over the second quarter, by -0.2 per cent to $417 / sqm per annum. Albeit marginal, this is the first time since fourth quarter 2017 that rents have fallen. Secondary gross effective rents also fell slightly (-0.1 per cent) to 310 / sqm per annum. Landlords continue to offer high incentives to remain competitive, choosing to re-coupe costs via face rents rather than lower incentives. However, face rents are expected to come under pressure noting the lack of demand. Flight to quality remains a consistent theme in the Brisbane market. The spread between PGER and SGER is $107 / sqm per annum. This is a far tighter gap than in the Sydney CBD, where the prime to secondary rental differential is $256 / sqm per annum. A tighter spread in Brisbane is helping drive demand towards prime grade space, which is demonstrated in the vacancy and net absorption figures.

YIELDS & SALES

Economic uncertainty and a lack of stock on the market has led to no transactions over the quarter. Investment volumes have not only fallen in Brisbane, but nationally, with all major CBD markets recording below average volumes. The prime yield range remained stable for the fifth consecutive quarter at 5.0 per cent to 6.25 per cent noting a lack of transactions. Yields are asset specific with focus on the tenancy and cashflow profile. The mid-point of this range remains the tightest held midpoint in ten-years.

OUTLOOK

The impact of COVID-19 is materialising across the economy. The focus for most occupiers in the short term remains on their operations and getting staff back into the office. As a result, we expect a large number of occupiers to put relocation plans on hold. Over the near term, occupiers will likely reassess and recalibrate their future use of workspace. An extensive work-from-home period may well accelerate the changes that were already taking shape before the crisis. Whether that translates to less shared workspaces (such as hot-desking) or an increase in flexible work arrangements remains to be seen. Government support programs should help insulate landlords from short term vacancy risk, however demand is likely to be softer in a weaker economy. This will drive a correction in rents. Under the Code of Conduct issued by the National Cabinet, landlords will be required to provide rental relief equivalent to lost revenue for qualifying tenants, which could see incentive levels rise from their current level. Transactional activity is expected to remain subdued over the near term. However, demand will persist for selective office assets. As capital in search of yield is drawn to Brisbane, there will still be groups actively looking to purchase assets with sound covenants and value-add opportunities. We expect the upper end of the yield range will remain stable in the short-term, while the lower end will decompress.

INDUSTRIAL

MARKET OVERVIEW

Supply was more than double the five-year quarterly average, totalling 135,012 sqm. The majority of this (eight per cent) is pre-committed. Gross take-up (≥ 3,000 sqm) was relatively subdued (94,860 sqm), in line with economic slowdown. Prime and secondary rents remained stable over the quarter. Despite economic uncertainty, transaction volumes increased over the quarter, totalling $246.0 million. Investment sale evidence resulted in the prime yield range tightening on the upper end of the range this quarter, to 5.00 per cent to 6.25 per cent.

YIELDS

Despite considerable yield tightening in 2019, the lack of sale evidence has caused prime yields to stabilise over the first quarter 2020. The prime yield range for Brisbane remained stable at 5.25 per cent to 6.25 per cent, with a midpoint of 5.69 per cent. This remains the tightest recorded in Brisbane since JLL began recording the market in 2002. The prime yield gap between Brisbane (Southern Precinct) and Sydney (South Precinct) and Melbourne (South East Precinct) is currently 119 basis points and 69 basis points respectively. The secondary yield range tightened by twenty-five basis points on the lower end, bringing the range to 7.00 per cent to 8.25 per cent. The spread between prime and secondary yields remains elevated at 188 basis points, highlighting the increased vacancy risk associated with secondary assets.

BRISBANE

COMMENT (cont'd)

INDUSTRIAL (cont'd)

DEMAND

Take-up in the Northern Precinct accounted for three per cent (3,120 sqm) of gross leasing volumes this quarter. While gross leasing volumes (≥ 3,000 sqm) in the North made up a small proportion of total take up, it is worth noting that a number sub-3,000 sqm deals occurred in the Precinct over the quarter. Monsar Smash Repairs leased 3,120 sqm at 78 Pritchard Road, Virginia for $125 / sqm. The Trade Coast witnessed 27,485 sqm of take-up over the quarter, across four deals. This is a significant increase from last quarter (10,714 sqm), despite being below the five-year quarterly average of 36,310 sqm. Occupier demand in the Trade Coast accounted for twenty-two per cent of total take-up this quarter. ABC Manufacturing leased 10,300 sqm within 157 Holt Street, Pinkenba. Occupier demand in the Southern precinct decreased over the quarter with approximately 43,650 sqm of industrial space leased. Strong levels of demand within the Southern Precinct continued over the first quarter 2020, with the precinct accounting for seventy-five per cent (93,566 sqm) of total take-up. This is twenty-three per cent higher than the five-year quarterly average (72,255 sqm), following a strong twelve months of take up (396,246 sqm), with demand recorded above the precincts' ten-year annual average of 282,765 sqm. Woolworth's 42,000 sqm pre-commitment to a new temperature control facility, located in Heathwood.

SUPPLY

There is currently 14,903 sqm of stock in the pipeline with plans approved and 3,499 sqm with plans submitted. The pipeline is likely to remain muted over the coming twelve months. This is mainly given land availability and locational factors. Over the past five years, annual average completions have been significantly lower in the North than in the Southern and Trade Cost precincts, down eight-six per cent and sixty-nine per cent respectively. There was one completion within the Trade Coast this quarter, totalling 3,000 sqm, developed by Brisbane Airport Corporation a 2,500 sqm warehouse and 500 sqm ancillary office. The project was one hundred per cent pre-committed upon completion. While completions were below average this quarter (19,572 sqm), the forward pipeline looks steady. There is currently 50,690 sqm of stock under construction and 37,020 sqm with plans approved. Currently, eight per cent of the stock expected to complete in 2020 (50,690 sqm) is pre-committed. Developers remain reluctant to build speculatively due to lower price expectations and vacancy risk. Three projects reached completion during the quarter, adding 51,282 sqm of stock to the market. The current supply pipeline looks strong, with 361,275 sqm of industrial space under construction and a further 155,220 sqm with plans approved. Currently, sixty-five per cent of this stock is expected to complete over the remainder of 2020. It is worth noting that the majority of this stock is pre-committed.

ASSET PERFORMANCE

Over the past twelve months, 2,000 sqm lots increased by 5.6 per cent to $475 / sqm, while 1 hectare and 2-5 hectare lots held steady at $350 / sqm and $300 / sqm respectively. No assets transacted in the Northern Precinct this quarter, off the back of record volumes recorded last quarter ($245.7 million). While transaction volumes are typically lower in the first quarter of each year (there were no sales recorded in the Northern Precinct in first quarter 2019), the reduction in investment activity over fourth quarter 2019 to first quarter 2020 has been attributed to the uncertainty surrounding COVID-19. Following subdued rental growth over the last few quarters, prime and secondary net rents broadly appreciated over first quarter 2020. Prime rents increased by 1.4 per cent to $120 / sqm per annum., while secondary rents increased by 2.5 per cent to $94 / sqm per annum. On annual terms, prime and secondary rents have grown by 2.1 per cent and 3.5 per cent respectively. Average prime incentives remained stable at thirteen per cent. Average land values appreciated in the Trade Coast this quarter. While 2,000 sqm lots remained stable, 1 hectare lots increased by 2.8 per cent in Lytton ($370 / sqm) and by 0.9 per cent in Eagle Farm ($565 / sqm). Similarly, 2-5 hectare lots in Eagle Farm increased by 2.4 per cent ($420 / sqm). The value between the two tracked suburbs remains wide, reflective of the divergence in supply and demand. Two transactions occurred in the Trade Coast this quarter, totalling $43.7 million. This is eighteen per cent below the five-year quarterly average ($53.3 million). The largest transaction to occur was 731 Curtin Avenue, Pinkenba. Sydney-based property group, City Freeholds purchased the 4.5 hectare site from fund manager, Arcadia for $34.7 million. Following steady rental growth throughout 2019, rents remained relatively stable over the quarter. Prime existing net rents increased marginally (0.2 per cent) to $110 / sqm per annum. Similarly, secondary rents grew by 0.6 per cent to $89 / sqm per annum. Despite there being limited rental growth over the quarter, rents have continued to grow on annual terms (prime by 2.1 per cent and secondary by 3.4 per cent).

Prime existing net rents within the Southern Precinct continue to remain low relative to the Northern Precinct and Trade Coast. This is due to several locational reasons, the first being the Trade Coast's proximity to the ports, meaning that occupiers have paid a locational premium to benefit from supply chain savings, through decreased road transport times. Equally, rents within the Southern Precinct are lower, given the relative distance from the ports. Institutional developers continue to offer high incentives in the Southern Precinct as they continue to compete aggressively for leasing commitments and a high return on investment. The average prime incentive remained stable at twenty per cent, for the sixth consecutive quarter. Incentives are expected to remain at this level in the near term, given the broader economic climate. Land values for all lot sizes in Yatala and Acacia Ridge remained stable over the quarter. In annual terms, 2,000 sqm lots appreciated by 5.3 per cent (year-over-year) in Acacia Ridge, while Yatala witnessed 3.1 per cent year-over-year growth. Yatala remains the most affordable suburb in the Brisbane industrial market, with 2,000 sqm lots fifty-one per cent lower than in Eagle Farm ($650), which is considered a prime location.

BRISBANE

COMMENT (cont'd)

RETAIL MARKET

OVERVIEW

Queensland's retail turnover growth is currently 5.1 per cent (year-over-year) (as of May), outpacing the national average (2.4 per cent). However, market sentiment remains subdued. Softer leasing demand continues to drive a correction in rents, most notably in the CBD (-10.0 per cent quarter-over-quarter) and regional (-4.0 per cent quarter-over-quarter) centres. Vacancy rates have increased across all sub-sectors since the acceleration of COVID19. Transaction volumes were below average over the second quarter of 2020, totalling $320.0 million. Three out of four assets to transact were neighbourhood centres. Median yields remained stable over the quarter across all sub-sectors, except for the CBD (+25 basis points).

CONSUMER SENTIMENT AND RETAIL TURNOVER

As at May 2020, Queensland's retail turnover growth totalled 5.1 per cent year-over-year. While growth fell in April, the easing of restrictions in Queensland helped strengthen retail trade in May. It remains well above the national average (2.4 per cent year-over-year) and is the highest of all states. It is worth noting that Queensland's recovery is coming off a low base relative to other states and had been assisted (pre-COVID-19) by a recovering housing market and improving unemployment rate. More recently, retail turnover in Queensland has been bolstered by spend in the 'other' (including pharmaceuticals) and food retailing category.

Food retailing (7.6 per cent) was the strongest growth category, reflecting the increase in spending across the supermarket and liquor categories, as people continue to pantry load and are more inclined to eat at home. Household good spending was also strong (six per cent), in part owing to increased working from home purchases such as furniture and electronics. These growth categories are mirrored at a national level, demonstrating the broad trends in consumer behaviour amid the pandemic. Cafes, restaurants and takeaway food (-4.9 per cent) and clothing, footwear and accessories (-1.9 per cent) were the weakest categories, the result of weakened discretionary spending.
The Westpac-Melbourne Institute Consumer Sentiment Index fell 6.1 per cent to 87.9 in July from 93.7 in June. While sentiment was building from April's extreme low of 75, it has fallen again with the resurgence in Coronavirus cases over the last month. While this resurgence has been almost exclusive to Melbourne, it has caused concerns about the virus spreading interstate and the subsequent effects on the wider economy.

RENTS

Softer leasing demand continues to drive a correction in rents. CBD prime (-10.0 per cent) and super-prime (-10.0 per cent) rents declined substantially over the quarter. Tenant demand for retail space in the CBD has been softening over the past number of years and the Queens Wharf (2022) development is beginning to add additional leasing pressure. Regional (-4.0 per cent) and sub-regional (-2.0 per cent) rents also declined over the second quarter of 2020. Their discretionary nature has made them even more vulnerable to store closures, and more recently, social distancing restrictions, circumstances that have had a damaging impact on leasing demand. Neighbourhood and large format centre rents have shown more resilience, remaining stable over the quarter.

Many tenants continue to seek rent relief from landlords, under the Commercial Tenancy Code. In addition to this, the Australian Competition and Consumer Commission (ACCC) announced in April that they are allowing retailers to collectively bargain with landlords for rent relief. This should help tenants better negotiate terms in the wake of COVID-19 and is expected to stand until September 2021, unless otherwise revoked.

INVESTMENT MARKET

Investment volumes (≥ $5.0 million) were subdued over the quarter, totalling $320.0 million. This is well below the five-year quarterly average ($502.6 million). There were four assets to transact, three of which were neighbourhood centres and the other was a sub-regional centre. Neighbourhood assets are likely to remain in demand over the near term, given their non-discretionary base and underlying land value.
The seventy-five per cent stake in Mt Ommaney Shopping Centre was the largest transaction over the quarter, both in S.E. Queensland and nationally. YFG Shopping Centres purchased the stake for $285.0 million after purchasing the other twenty-five per cent in late 2019. It is worth noting that the pricing of the asset had been settled pre-COVID.

Median yields remained stable across all sub-sectors, except for the CBD, which softened by twenty-five basis points. CBD yields softened on the back of income and vacancy pressure. Despite this, the average weighted yield remained stable at 6.13 per cent. Over the past twelve months, regional (twenty-five basis points) and CBD (twenty-five basis points) yields have softened, while sub-regional, neighbourhood and large format yields have remained stable. A large spread between upper and lower yields still persists, reflecting the divergence in asset quality and performance.

CANBERRA

Source - Knight Frank

RENTAL VALUES

MARKET INDICATORS

Canberra's office vacancy rate continues to tighten, decreasing from 11.1 per cent to 10.1 per cent in the twelve months to July 2020. A-grade vacancy has remained unchanged at 6.9 per cent for the last twelve months, whilst secondary vacancy has tightened from sixteen per cent to 14.1 per cent over the same period. Civic A-grade vacancy has increased marginally, from 6.6 per cent in July 2019 to 6.8 per cent in July 2020. Overall, there was little movement in the A-grade market over the past twelve months with the Parliamentary Precinct recording the largest variance, contracting from 4.1 per cent to 2.3 per cent. Supply additions totalled 59,581 sqm in the twelve months to July 2020, dominated by the completion of Civic Quarter in late 2019 (15,934 sqm), as well as several refurbishments. Withdrawals totalled 34,520 sqm over the past twelve months. Supply additions in the second half of 2020 are forecast to comprise new buildings (Constitution Place – 31,685 sqm and DKSN – 13,000 sqm) and refurbishments (15,801 sqm). Canberra office market indicators as at July 2020 are as follows :

Grade	Total Stock (sqm)	Vacancy rate (%)	Annual Net Absorption (sqm)	Annual Net Additions (sqm)	Average Civic-Gross Face Rent ($ / sqm)	Average Incentives (%)
Prime						
A-Grade	1,252,233 (55%)	6.9	41,311	43,744	488	19.5
Secondary	1,031,766 (45%)	14.1	3,946	15,847	424	27.0
Total Market	2,283,999	10,1	45,257	59,581		

Vacancy Rate (%)	A		B		C		D		Total		10 Yr
By Locality/ Grade	**Jul-20**	**Jul-19**	**Jul-20**	**Jul-19**	**Jul-20**	**Jul-19**	**Jul-20**	**Jul-19**	**Jul-20**	**Jul-19**	**Average**
Civic	6.8	6.6	22.1	18.1	18.5	23.8	12.4	15.3	12.3	12.0	**11.4**
CBD Fringe	0.0	0.0	6.5	8.1	11.7	15.6	27.4	24.5	11.1	14.2	**13.2**
Parliamentary Precinct	2.3	4.1	8.8	19.0	4.4	36.7	0.0	0.0	4.0	7.6	**8.2**
Belconnen	38.0	36.2	0.0	0.0	20.5	25.8	0.0	11.2	31.2	31.4	**11.0**
Woden	0.0	0.0	0.0	0.0	62.8	58.2	39.1	31.0	18.5	16.9	**16.9**
Tuggeranong	0.0	0.0	13.3	14.7	28.2	33.1	0.0	0.0	6.6	7.4	**9.2**
Airport	6.9	6.9	12.1	14.1	0.0	0.0	0.0	0.0	10.1	11.1	**33.0**
Total	**6.9**	**6.9**	**12.1**	**14.1**	**15.8**	**17.9**	**14.4**	**14.9**	**10.1**	**11.1**	**12.4**
10 Yr Average	**10.7**	**11.1**	**10.2**	**9.7**	**15.9**	**15.5**	**22.1**	**23.0**	**12.4**	**12.4**	

Source : Source: Knight Frank Research / PCA

Tenant Demand

Recent tenant demand has remained positive, with the Canberra office market recording positive net absorption over the past six months (17,055 sqm) and twelve months (45,257 sqm). The above illustrates the longer term, two-tier nature of the leasing market as tenants continue to pursue a 'flight to quality'. Positive long-term net absorption was evident in key markets such as Civic (13,789 sqm), Belconnen (12,115 sqm), Barton (5,120 sqm) and the Airport (4,353 sqm. However negative demand was recorded in Bruce and Woden in particular. Tenants continue to prioritise buildings offering enhanced amenity, with high quality end of trip facilities becoming vital to attract and retain quality corporate tenants. Landlord funded speculative fit outs continue to drive tenant demand and result in rent premiums, with tenants increasingly seeking to offset costs associated with establishing their new corporate tenancy.

CANBERRA

RENTAL VALUES (cont'd)

Supply and Development

Stock withdrawals totalled 34,520 sqm in the twelve months to July 2020, well down on the ten-year average of 45,601 sqm and the lowest level since January 2018. Ongoing record stock withdrawals had been particularly evident since 2017, which had assisted with maintaining vacancy levels amongst secondary stock in particular. Short term supply additions are dominated by DKSN (13,000 sqm) which is fully pre-committed to the ACT Government (completion mid 2020) and Constitution Place (31,685 sqm total across two buildings, both of which are now pre-committed), which is due for completion in December 2020. The Department of Agriculture, Water and the Environment have committed to Civic Quarter Stage 2, for which a development application was lodged in late 2019. Completion is expected in late 2023. This major Commonwealth pre-commitment will see 35,000 sqm across the whole of 18 Marcus Clarke Street and part of 7 London Circuit vacated, with ISPT to undertake a major refurbishment program of these A-grade assets in an effort to reposition the buildings in the market place. Outside of Civic, Cromwell has proposed a new 18,000 sqm building at 19 National Circuit which remains subject to approval (as at mid-2020). Existing approvals at the Airport include 25 Catalina Drive (up to 25,000 sqm) and 6 Brindabella Circuit (21,000 sqm), while concept plans have been prepared for a further 20,000 sqm at 9-11 Molonglo Drive. The TGA and Department of Health will occupy a total of 13,700 sqm of specialised laboratory space and office accommodation across two new buildings from mid-2022. Several major accommodation briefs remain active as at mid-2020, including the Australian Taxation Office, Department of Defence, and Services Australia. These large Commonwealth briefs have the potential to re-shape the Canberra office market and development pipeline, with a number of agencies appearing to pursue new build options. With COVID-19 impacting the Commonwealth's budgetary position, some requirements appear to have stalled or been deferred. While a number of buildings are expected to be withdrawn in the short to medium term for refurbishment due to tenant displacement, the impact of COVID-19 is expected to stem withdrawals for redevelopment given hotel was often a preferred use targeted by developers.

Vacancy

The overall vacancy rate contracted to 10.1 per cent, from 11.1 per cent in the twelve months to July 2020, two hundred and thirty basis points lower than the ten-year average of 12.4 per cent. A-grade vacancy has remained unchanged since July 2019, at 6.9 per cent. A-grade vacancy excluding the Airport increased from 5.8 per cent to 6.2 per cent in the twelve months to July 2020. Secondary stock tightened from 16.0 per cent to 14.1 per cent in the twelve months to July 2020, with all lower grades tightening in this period. The Civic A-grade market softened slightly, with vacancies increasing from 6.6 per cent to 6.8 per cent in the twelve months to July 2020, with new space within Civic Quarter still yet to be fully absorbed. Overall, total vacancy in Civic softened slightly from 12.0 per cent to 12.3 per cent in the twelve months to July 2020. The Parliamentary Precinct A-grade market remained tightly held with vacancy levels declining from 4.1 per cent to 2.3 per cent over the past twelve months. B-grade vacancies have tightened considerably, declining from 19.0 per cent to 8.8 per cent in the twelve months to July 2020. Overall, total vacancy in the Parliamentary Precinct contracted to four per cent, from 7.6 per cent twelve months prior. Canberra's Town Centre markets continue to record mixed results. Vacancy in Belconnen tightened notionally from 31.4 per cent to 31.2 per cent, with A-grade vacancy at a record high at thirty-eight per cent due largely to stock additions of 17,300 sqm (refurbishments). Woden vacancy rates have increased from 16.9 per cent to 18.5 per cent in July 2020, with C and D-grade vacancy at 62.8 per cent and 39.1 per cent respectively, both up on twelve months prior. The Tuggeranong market has tightened to 6.6 per cent, down from 7.4 per cent in July 2019 with B and C grade vacancy levels contracting over the past twelve months. Overall, the disparity in office vacancy between prime and secondary stock is expected to continue. This will be further exacerbated if the Commonwealth trigger a new supply wave between 2023 and 2025.

Rents and Incentives

Civic and Parliamentary Precinct A-Grade rents have increased over the twelve months to July 2020 to $485/sqm (gross face), showing a 2.2 per cent increase year-over-year. Prime incentives are holding at 18.7 per cent, with incentives in Town Centres and the Airport also remaining steady. Secondary Civic and Parliamentary Precinct gross face rents increased by 1.4 per cent to $424/sqm; and 1.2 per cent to $421/sqm respectively in the twelve months to July 2020. Slight growth was recorded in Canberra's prime and secondary Town Centre markets, with gross face rents increasing by 1.7 per cent in the prime market to $412/sqm and in the secondary market rents increasing by 1.3 per cent to $336/sqm over the twelve months to January 2020. Prime and secondary incentives within the Town Centres remain steady, measuring 24.5 per cent and 29.6 per cent respectively. While leasing evidence during the COVID-19 period generally suggests incentives have so far remain unchanged, they are expected to soften over the short to medium term, particularly in the secondary market as landlord's seek to maximise occupancy in a climate characterised by reduced levels of enquiry and demand. Combined with a lack of short-term face rental growth, this will likely see a softening in effective rents until market conditions stabilise.

Forecasts

New supply in Civic and Dickson totalling approximately 45,000 sqm will represent the tail end of the current supply wave. Tenant demand and market acceptance of premium rentals associated with Civic Quarter and Constitution Place appears to be favourable, however market depth will be tested with sentiment and enquiry levels expected to remain flat given the effect of COVID-19 on the private sector leasing market. Future stock withdrawals include Finlay Crisp Offices in mid to late 2020, with Home Affairs to formally vacate around 21,000 sqm which is expected to lead to a major refurbishment. Belconnen's vacancy rate is expected to tighten following ABS's commitment to expand into 40 Cameron Avenue from mid-2020 (9,300 sqm). Conversely, the Department of Agriculture, Water and the Environment will vacate 18 Marcus Clarke Street and part of 7 London Circuit in 2022 which will see these A-grade assets repositioned in the market place. Incentive levels are unlikely to contract in the short term, particularly in the secondary market, as landlords seek to maximise occupancy in the current COVID-19 environment which is characterised by significantly reduced levels of enquiry and demand. We expect secondary stock and fringe markets to be most affected. The outcome of current Commonwealth accommodation briefs will also have a significant impact on future vacancy and potential for rental growth. Several current briefs imply a heightened level of obsolescence in relation to existing Commonwealth leased office stock, with vacancy likely to spike if agencies opt for new accommodation options.

CANBERRA

RENTAL VALUES (cont'd)

Investment Market Overview

Prime and Secondary yields have traded below their ten-year averages since 2015. A-grade yields tightened eight basis points to 6.4 per cent in the twelve months to July 2020. This continued tightening reflects increased buyer depth for prime assets across the broader Canberra market. Average A-grade yields in the Civic and Parliamentary Precinct markets contracted to 5.88 per cent (minus twelve basis points) and 5.88 per cent (minus twelve basis points) respectively in the twelve months to July 2020. Whilst transactional volumes outside these core markets remains limited and have historically illustrated a risk premium; A-grade Town Centre yields have contracted five basis points in the twelve months to July 2020 to average 7.4 per cent. Secondary yields in Civic and the Parliamentary Precinct average 6.86 per cent and 6.53 per cent respectively, demonstrating a tightening of thirty-four basis points and two basis points respectively in the twelve months to July 2020. Average secondary yields in Town Centre markets remain unchanged at 9.88 per cent over the past twelve months. 2019 transactions indicated increased activity from listed funds compared to historic trends. Major sales included Centuria's acquisition of Nishi in December 2019 ($255.75 million), Elanor's purchase of Garema Court ($71.5 million), Charter Hall's acquisition of 16-18 Mort Street, Canberra ($108.5 million) and Centuria's purchase of 10 Moore Street ($35 million). Additional off market deals included a 50% share in 121 Marcus Clarke Street ($102.9 million) and Finlay Crisp Offices ($62 million, subject to a delayed settlement). 2020 financial year transaction volumes ($10 million plus) totalled $476.6, down from $634.2 million recorded in 2019 financial year. Activity during the first half of 2020 has been limited to two transactions (over $10 million). 5 Tennant Street, Fyshwick sold to an offshore investor for $15.3 million, while 7 National Circuit, Barton was acquired by ISPT off market for $14.5 million. 221 London Circuit, Canberra was offered to the market publicly in mid-2020, however failed to sell despite receiving multiple offers. A lack of stock nationally has seen buyer depth and demand strengthen over the past six-months. Increased volumes in the sub $10 million market demonstrate strengthening market conditions and buyer depth in particular. Indeed assets in non-core locations continue to attract strong demand from interstate and even offshore groups in market sectors historically dominated by local investors.

Sales Evidence

5 Tennant Street, Fyshwick transacted during the COVID-19 pandemic period after a high exposure marketing campaign generated considerable interest. The property is a modern commercial asset, with income heavily weighted toward Commonwealth tenants, however represents a secondary location within an industrial suburb. Our enquiries indicate pricing and demand remained firm, despite the broader economic conditions and sentiment, with the property acquired by a Singaporean based investor. 1 Thynne Street and Garema Court exchanged in early 2020 and late 2019 respectively. Thynne Street was the subject of an off-market transaction, with Garema Court selling via and expressions of interest campaign which closed 14 November 2019. Nishi transacted in late 2019 through a closed expressions of interest campaign with selected investors invited to submit offers to purchase. The asset represents a high profile and unique investment opportunity with the office component anchored by a long-term lease to the Commonwealth. Cinema and Hotel components were also subject to long term leases, however also represented specialised elements of the property. The property generated strong interest from local and offshore groups. The sale indicates the strongest investment return in the Canberra market, and is expected to somewhat 'reset' market expectations around prime grade yields locally. 205 Anketell Street sold to a local investor/ developer with pricing and analysis metrics reflective of the risk premium associated with Canberra's suburban markets due to limited tenant depth and demand. The building had been refurbished in recent years and is well located opposite South.Point Shopping Centre. 40 Allara Street was acquired off market by Meriton with settlement delayed until expiry of all leases (mid 2020). Meriton have announced the building is to be demolished with a five-star, 230 room hotel to be developed on the site.

COVID-19 Market Comment

Real gross domestic product contracted by seven per cent in the June quarter, weaker than economist expectations of a six per cent fall, and the largest fall since quarterly records began in 1959. Over the year to June 2020, output declined by 6.3 per cent, the largest contraction since World War II, and down from 1.6 per cent growth over the year to March. The coronavirus outbreak has clearly slowed momentum and activity within the private sector leasing market throughout the first half of 2020. While the impact is yet to play out, we expect the market to become more competitive going forward, with pressure on incentives and effective rentals anticipated in the short term at least. Demand from government is more difficult to confirm, as these larger requirements are subject to protracted lead in periods and often protracted negotiations. The Commonwealth's intentions regarding several significant accommodation briefs remain unclear as at mid to late 2020; however, some requirements are certainly expected to stall or be deferred which may see agencies roll onto short term options rather than follow through with a competitive tender process. With the investment market starved of stock, and record low interest rates expected to remain for some time, a degree of pent up demand is apparent for well leased investment stock. Transactional activity above $10 million in the current COVID-19 climate is extremely limited, however lower grade stock, particularly sub $5 million has been largely unaffected to date. Overall, a market correction stemming from the impact of COVID-19 on the leasing market and broader economy is yet to materialise within Canberra. With the initial stage of the JobKeeper stimulus package finishing on 30 September 2020, we expect further market corrections may become apparent in the fourth quarter 2020 and into the first quarter 2021. We highlight however, that the impact of the COVID-19 pandemic continues to play out across all facets of the Australian economy and various property markets. Heightened levels of market uncertainty and volatility are expected, and this will become more evident as transaction volumes occur and provide a degree of understanding as to current sentiment. Transaction volumes within both the sales and leasing sectors of the ACT office market remain extremely limited to date, exacerbating the uncertainty within the market.

DARWIN

Source - Colliers International Research

COMMENT

DARWIN OFFICE LEASING MARKET

The Property Council of Australia commenced publishing of the Darwin CBD within its national office market report in January 2010, providing statistics such as total building stock, vacancy rates, supply additions, withdrawals, and net absorption. This information is further broken down by building grade.

We summarise the level of stock within each building grade in the Darwin CBD, from January 2019 to January 2020 below inclusive of the approximate rental range for each classification :

Darwin CBD	Jan-20	Jan-19	Market Rental Rates Per sqm
Total Vacancy Rate	16.8%	17.20%	
Net Absorption (sqm) 12 Months	6,475	4,533	
Premium			
A Grade – Total NLA	142,489	135,876	$375 - $550
A Grade - Vacancy	7.1%	8.1%	
B Grade – Total NLA	61,845	61,778	$250 - $380
B Grade - Vacancy	28.2%	27.4%	
C Grade – Total NLA	16,042	16,112	$180 - $250
C Grade - Vacancy	58.7%	54.8%	
	220,376	213,766	

Source: PCA

It can be seen that approximately sixty-five per cent of the total stock within Darwin is classified A grade, with approximately twenty-eight per cent B grade, and seven per cent C grade. The Property Council of Australia has not classified any buildings within Darwin as Premium, despite several buildings providing premium finishes and a high environmental rating, they remain outside the rigorous national requirements (such as a minimum of 20,000 sqm net lettable area) to receive the premium classification. Also, there are no longer any buildings in the Darwin CBD market classified as D grade.

The overall level of vacancy decreased in the year to January 2020, due to a net absorption of 6,475 sqm, across all building categories. The net absorption was principally due to the demolition of 38 Cavenagh Street in the Darwin CBD which had a net lettable area of 6,613 sqm. 2021 will see a supply addition of approximately 10,000 sqm of A grade office when Manunda Place is completed. There has been a nominal change in vacancy between January 2019 and 2020.

'A' grade vacancy has declined nominally over the past year from 8.1 per cent to 7.1 per cent however based on our own research it is probable that the 2018 figure of 14.6 per cent was incorrect and was actually much closer to eight per cent, therefore here has not been much change in this category. The 'B' grade vacancy has experienced a slight increase in vacancy, increasing from 27.4 per cent at January 2019 to 28.2 per cent in January 2020. We have seen a continual increase in 'C' grade vacancy increasing from 54.8 per cent to 58.7 per cent over the year.

DARWIN INDUSTRIAL LEASING MARKET

The industrial market has been subdued over the past five years with limited demand by both the leasing market and owner occupiers and this may likely to be further subdued in view of the COVID-19 outbreak. Sentiment for premium industrial properties, those of a high quality or well located, is currently subdued and this is likely to continue now that the Inpex Development has reached the end of its construction phase. There continues to be very limited demand to lease older style office/warehouse properties which are smaller in scale.

Let up periods for the premium market are in excess of six months. Secondary industrial properties on the market available for lease can take in excess of twelve months to lease and with limited demand for secondary properties, these can languish in the market for extended periods.

DARWIN

COMMENT (cont'd)

DARWIN INDUSTRIAL LEASING MARKET (cont'd)

The industrial market is characterised by two distinct sub-markets. Firstly large, modern industrial space with good clearance and accessibility is experiencing softening demand from both purchasers and tenants. The largest component of the industrial market are the older style properties which are typically lower clearance, in average condition and have limited access or lay down area. There is significantly less demand for this style of property with sale and letting up periods long and vacancy rates relatively high. The approximate rental range for each classification is shown on the table below :

Industrial Rental Rates

Property Type	Market Rental Rates Per sqm
Prime Industrial	$100 - $140
Secondary Industrial	$60 - $100
Strata Units	$90 - $200

DARWIN RETAIL LEASING MARKET

The Darwin Retail Market is primarily divided up into three main areas, these being the Darwin CBD, Casuarina Square and Palmerston. Scattered further throughout Darwin are a number of strip shops and smaller retail centres.

There has been an extraordinary amount of retail development in Palmerston and inner rural area over the previous four years and it is now becoming very difficult to find tenants for retail space due to the high amount of vacant space and a limited number of retail tenants. Compounding this is a high failure rate of existing retail tenants, with a number of prominent retails closing their doors over the previous eighteen months, in virtually all of the main retail centres. The result of this turmoil is that letting up periods have extended; incentives have come into play and market rentals have declined. The approximate rental range for each classification is shown on the table below:

Retail Rental Rates

Property Type	Market Rental Rates Per sqm
Prime CBD Retail	$300 - $600
Secondary CBD Retail	$175 - $300
Suburban Neighbourhood Centres	$200 - $500

HOBART

Source - Knight Frank, Tasmania

COMMENT

OVERVIEW

Economic contraction stemming from Covid-19 will continue to influence the property market into 2021. Projected impacts are not likely to be uniform and will depend on location, product and the stance of different counter-parties. Retail tenant demand has deteriorated, the office and residential are holding steady and the industrial sector is resilient.

OFFICE

The Hobart office market is dominated by State and Australian Government commitments which we approximate represents up to seventy per-cent of office space in Hobart. The February 2019 A grade vacancy rate was 4.9 per cent representing a reduction in vacant stock. The reduction in the overall vacancy was not as significant in the B grade sector that stood at 9.1 per cent although the C and D grade sectors were both around six per-cent and 6.1 per-cent vacancy respectively for the period to February 2019. The 2020 PCA office market report reported further vacancy reductions in the rate overall to 4.1 per cent (national average 8.3 per cent) with the A grade vacancy to 3.2 per cent.

Current gross face rents per square metre have remained static and without growth over the last twenty-four months and are as follows :-

A-Grade	$330 - $370 per sqm gross
B-Grade	$240 - $280 per sqm gross
C-Grade	$180 - $220 per sqm gross
Suburban	$180 - $250 per sqm gross
Suburban (Call Centre)	$280 - $320 per sqm gross

There have been no major office property transactions in the Hobart market over the course of 2020. This is not so much to do with the Covid-19 restrictions but more so the reasonable market churn in the prior year. Ownership has bedded down and existing owners continue to hold existing real estate investments. The market is tightly held. Investment offerings backed by government tenants have emerged as strongly sought-after investments. .

INDUSTRIAL

Industrial growth in the Tasmanian/Hobart region economy is low volume/scale relative to mainland city markets given the smaller size of the freight and manufacturing sectors in Tasmania as a proportion of the national market.

Sales volumes (freehold real estate basis) in the Tasmanian industrial market are low by national standards. Return on investments of 6.5 per cent to 7.25 per cent are current investment rate albeit a lack of recent investment activity underpinning the current level of return. Industrial leasing became more active throughout 2020.

Rental rates for industrial are as follows :-

Prime Modern Industrial	$100 - $120 per sqm gross
Prime Older Style Industrial	$70 - $90 per sqm gross
Secondary Industrial	$45 - $65 per sqm gross

HOBART

COMMENT (cont'd)

RETAIL

The retail sector was impacted by Covid-19 shutdowns. Non-discretionary retailers have remained stable with may experiencing sales growth. Hardware retailing has been strong throughout 2020 on the back of Australian government assistance via the JobKeeper scheme. National retailers continue to make enquiry to source locations in the Hobart retail market.

The emphasis on property investor demand for non-discretionary retail tenanted investment properties is very strong under the current investor market.

The following is a range of rental rates prevailing in the different categories of the retail sector :-

City CBD Prime	$1,300 - $1,750 per sqm gross
City CBD Secondary	$ 500 - $ 800 per sqm gross
Suburban Neighbourhood Centre Prime	$ 500 - $1,000 per sqm gross
Suburban Neighbourhood Centre Secondary	$ 300 - $ 500 per sqm gross
Suburban Strip Prime	$ 350 - $ 500 per sqm gross
Showroom and Bulky Goods Centres	$ 180 - $ 250 per sqm gross

Returns/yields for retail investments do vary according to location, quality and number of tenants, management complexity and weighted average lease expiry profiles. Selling yields in the current economic climate are around five to six per cent for prime CBD strip retail properties. Shopping centre yields securely leased to quality national anchor tenants are six and a half to seven and a half per cent. Properties with a single lease covenant to national retailers continue to exhibit favourable investor sentiment. Hobart area sales have not been evident in the CBD although transactions in suburban strip retail locations such as North Hobart have continued to underpin the pre-Covid environment at around five and a half per cent.

RESIDENTIAL

Residential real estate activity and average prices were increasing from early 2018 through to April 2019 prior to the May 2019 Federal election. At that time, there was some uncertainty caused by Federal Labour Party policy in respect of negative gearing that resulted in a retraction of buyers until the election outcome. A return to government of the sitting Liberal National Coalition did restore confidence and the market began to stabilise at price levels throughout the remainder of 2019. The national economy, and to that extent all areas of the Tasmanian market, were subject to more limited economic and wages growth which conspired to temper future growth in residential real estate activity and prices. This was perhaps a timely position for the market to take stock which appeared to plateau through early 2020.

Since the outbreak of COVID-19, residential real estate markets have experienced different circumstances within markets across Australia. The observable trend now evident post mid-2020 is a more stable price situation on lower volumes of trading activity in regional areas of Tasmania. The residential land market was very strong through the middle of 2020, largely stimulated by the Australian government's HomeBuilder grants encouraging first home buyers into the land and new homes market.

The Real Estate Institute of Tasmania (REIT) moving annual median sale price for houses in Hobart increased eight per cent to $536,421 in the year to June 2020 following on from the fifteen per cent increase in the year to June 2019 $498,248. The median sale price in Launceston increased 6.8 per cent in the twelve months to June 2020 to $362,027 (up from $338,000 in the prior twelve-month period) and in the NW centres Devonport $293,542 (three and a half per cent annual increase) and Burnie $284,729 (14.6 per cent annual increase).

The state-wide rental market vacancy rate remained steady in Hobart at 2.8 per cent (June 2020) after a period of continuing falling vacancy (1.8 per cent in Hobart August 2019). In this time, rents have continued to increase. According to REIT, Hobart area three- bedroom house rents were $445 per week in the twelve months to June 2020 compared to August 2019 at $430 per week. In Launceston, three-bedroom rents rose from $340 to $345 per week over the same period.

MELBOURNE

Source - Charter Keck Cramer

COMMENT

ECONOMY

Victoria's economic conditions were until recently amongst the strongest in the Country, with key cyclical drivers supporting the diversified and largely non-mining economic base. Robust population growth, driven by overseas and interstate migration supported Victorian economic growth, and consequently, employment growth. It also provided for substantial public investment in transport and social infrastructure as the Government looks to alleviate the pressures around such population growth.

However, the short-term outlook to the economy is now being dominated by the effects of locking down the country to prevent the spread of the COVID-19 virus. The social distancing requirements and economic uncertainty has forced some businesses to close and non-essential spending and business investment to fall. Unemployment has risen sharply with workers stood down in response to declining revenues. Victoria has been more challenged than the other states, with a second wave of infections resulting in a more stringent lockdown of the population, which in turn significantly impacted business activity and economic conditions.

The impact is being somewhat cushioned by a substantial and coordinated fiscal and monetary response designed to support businesses and consumers through the economic shock. Financial institutions are assisting by deferring loan repayments to help businesses and households manage cashflows during this period. The Federal Government has also implemented the JobKeeper and JobSeeker subsidies to support incomes, with the JobKeeper payment designed to maintain the connection between businesses and employees until the current restrictions are eased.

With a significant exposure to international markets, Melbourne and by extension Victoria is likely to be affected harder than the other states, with overseas student numbers and international tourism effectively falling to zero, and this in turn has impacted related businesses. Nevertheless, Australia has been one of the more successful countries in containing the spread of the virus, and there has been a steady re-opening of the economy. However, any return to 'normal' will not be immediate, with limits likely to remain in place on large gatherings and travel, particularly international travel, for some time. The pace of recovery will depend on how much household and business confidence returns as restrictions are progressively lifted. Victoria will benefit significantly from the opening of the economy.

Beyond the recovery and over the long term, the fundamentals for Australia remain positive. Notwithstanding a possible structural change to net overseas migration, the country remains an attractive destination for international investment and migration (which may potentially be enhanced due to its management of the COVID-19 crisis) and this will help to underpin domestic demand and economic growth.

OFFICE

Victoria's recent strong economic growth supported employment growth and the robust take up of office space. Several years of solid white-collar employment growth drove significant net absorption across the major office markets. Strong rental growth and diminishing rents have been the result.

The economic and employment shock due to the COVID crisis through 2020 will result in a reduction in tenant demand as businesses seek to cut costs. At the same time, the elevated supply pipeline that is now coming through after the upturn in rents will work with lower tenant demand to result in upward pressure on vacancy rates and downward pressure on rents. Vacancy rates have started to edge upward through the year, with further rises expected in 2021 as new supply comes on stream.

On the positive side, Melbourne's office markets started 2020 from a position of strength, after a period of strong office space demand and relatively low supply saw vacancy rates tighten across all office markets, and in Melbourne CBD fall to historically low levels. Much of the pipeline of new stock has also been substantially pre-committed, which will also help to somewhat mitigate the extent of the rise in vacancy rates and their subsequent impact on rents.

There remain potential impacts from COVID and ongoing demand for office space in the short and medium term. In the short-term social distancing measures have pushed employees to work from home (where able). In the medium term, any cost savings resultant from a more flexible workspace strategy may be more fully implemented, especially as many have moved to off-site work practices through the disruption. However, any reductions to office space demand could be partly offset by increased need for more space per employee to meet social distancing requirements.

Despite the market disruption, the low interest rate environment means that the investment market is strong, with quality investment grade stock sought after by both domestic and foreign investors as they search for yield. The weight of capital in the market for a limited number of core assets suggests that the investment market will remain competitive.

MELBOURNE

COMMENT (cont'd)

INDUSTRIAL

The onset of COVID-19 resulted in Victoria experiencing the harshest lockdowns in the nation. Despite this, Melbourne's industrial market has withstood the effects that were felt across other asset classes and instead has continued to experience an increase in activity during the pandemic.

The unique combination of proximity to a major CBD, together with the benefit of significant amounts of industrial land underpin the strength of the Melbourne industrial market. Large warehouse spaces supported by major transport infrastructure located in close proximity to company offices in, or near to the Melbourne CBD, provide a strong competitive advantage. As a hub for national distribution activity, Melbourne is anticipated to benefit enormously from continued growth in demand.

The industrial property market continues to evolve through increased integration with e-commerce and warehousing and logistics. A key economic driver of demand for industrial space throughout 2020 has been the sharp increase in e-commerce activity. Online retailing spending increased sharply following the onset of COVID-19 which in turn has created demand for warehousing and supporting infrastructure to enable goods being moved quickly and efficiently around the city. Warehouse automation is expected to increase, while multi-level warehousing is now also being considered, particularly in inner-city locations where land is scarce, and costs are higher. The acceleration in online shopping and a more digitally connected consumer is expected to drive increased demand for industrial real estate post COVID-19.

Strong domestic institutional investor demand has continued to support the growth in industrial values. This is reflective of the ongoing national trend towards industrial as an asset class and which is contributing further to the scarcity factor. Significantly, leasing for speculatively built industrial projects has been solid throughout 2020 and vacancy remains tight in most markets.

Post Covid-19 demand in the Melbourne market, combined with already low vacancy rates, will likely give rise to speculative warehouse developments. This rapidly evolving market is expected to generate opportunities for developers to undertake projects that will can capitalise on tight vacancy rates in areas in high-demand areas. Rather, there is a projected chronic shortage of industrial and logistics particularly in Melbourne's southeast in the Dandenong region.

Supported by healthy demand, positive signs exist for industrial market through 2021, with the prospect that the sector will emerge more strongly than it was performing in the pre-COVID period.

RETAIL

The retail property sector through 2020 has been impacted by the COVID-19 lockdown. The requirement for social distancing, shift of much of the working population to be working from home, and restrictions on population movement have had implications for the volume of retail spending overall and across different sectors.

The increased time at home has caused a rise in spending in retail sectors for goods consumed in the home, including food retailing (supermarkets, grocery and liquor), household goods (incorporating home office goods and general home comforts) and 'other' (pharmaceutical, cosmetics and toiletries, and newspapers and books). Spending in more discretionary sectors, including clothing, footwear and personal accessories has been weaker (also impacted by store closures through the lockdown period). The inability to congregate in large numbers has been most negative for spending in the cafes, restaurants and takeaway foods sector. While there has been some rebound as the state economies have gradually reopened, spending in these weaker sectors has not yet fully recovered.

Retail spending in Victoria was set back after the second wave of COVID-19 infections after an emerging improvement through May to July. However, the progressive scaling back of restrictive measures since October 2020 should see Victoria follow the recovery trajectory of the other states.

Online retailing has surged while outside movement has been restricted. After increasing from just over two per cent in 2013 to around six and a half per cent at the start of 2020, the Australian Bureau of Statistics reports that the online share of retail sales has increased to just over ten per cent of sales since March. Online turnover over April to September 2020 is up by over sixty per cent on the same period in 2019. The National Australia Bank Online Retail Sales Index has reported stronger online growth within Victoria, reflecting its extended lockdown, and all retail stectors have enjoyed growth, the strongest being for games and toys, department stores, and homewares and appliances.

While the share of online retailing is expected to pull back as the ability of the population to physically shop and eat and drink out of the home returns, it is not expected to get back to pre-COVID levels and this will have implications for the retail property sector. Neighbourhood shopping centres that benefit from essential spending look to remain in demand by investors, while large format retailing is currently servicing the surge in household goods spending. However, subregional centres and all but the largest regional centres are likely to be less keenly contested unless they can reinvent themselves in a post-COVID world. CBD retail may similarly struggle, particularly while the working population is not fully back at work and tourism numbers remain low. After being unloved for some time, shops in some neighbourhood strips may experience increased demand as their outdoor nature and a potentially higher propensity to be working from home increases the preference to shop locally.

MELBOURNE

COMMENT (cont'd)

COMMENT (cont'd)

RESIDENTIAL

After a downturn over 2017–2019, encouraging signs that the Melbourne residential market was recovering began to emerge over late 2019/early 2020. However, the onset of the COVID-19 pandemic and ensuing lockdown of the economy brought the nascent recovery to a halt. The challenges of economic recession and rising unemployment through the lockdown have been compounded by the closure of international borders. This has halted net overseas migration inflows, which have been a key driver of the Melbourne residential market. Transaction activity slowed considerably and dwelling prices in Melbourne are estimated to have fallen by around 6% between early in the lockdown in April and October 2020.

The shock to the residential market from the lockdown has been somewhat ameliorated by cuts to interest rates by the Reserve Bank at the onset of the COVID-19 crisis and financial market support being offered to the banks. Mortgage repayment relief support have been offered by the banks, which allowed stressed borrowers to suspend payments to the end of September 2020, and potentially longer on a case-by-case basis. This has prevented distressed sales coming onto the market and possibly driving prices down.

New dwelling completions in Melbourne have remained high in 2020 while the slowdown in migration and weak employment market has caused tenant demand to slow. Vacancy rates have risen, rents have decreased, and investors are retreating from the market. However, with interest rates record lows, this has encouraged demand from owner occupiers, particularly first home buyers, who have provided some support to prices. Notably, with restrictions on movement in Melbourne being lifted in October, transaction activity in the Melbourne market has picked up and there is evidence that prices have bottomed out.

Notwithstanding pauses during successive lockdowns, the greenfield land market in Melbourne has generally been performing well, assisted by the Federal Government's HomeBuilder incentive and the strength of first home buyer demand. In contrast, the weakness of investor confidence has flattened off-the-plan sales for new apartments and there has been a significant fall in new projects on the market.

In contrast to the weakness in Melbourne, conditions in regional Victoria have remained robust across both the established and new home markets, buoyed by the out-migration of Melburnians to the regions. Together with the lower rates of COVID infections in the regions, workers have found the increased acceptance of working from home has allowed them to move further away from their place of employment to enjoy the improved affordability and lifestyle aspects of the regions.

Despite the positive signs emerging since Melbourne's lockdown was eased, question marks remain on the sustainability of any recovery, with residential markets facing a number of potential headwinds including mortgage relief being progressively rolled back, the tapering off of the Federal Government's JobKeeper and JobSeeker support packages, rising vacancy rates and falling rents, and continued risks to the economic and employment outlook. On the positive side, a more robust economic recovery and low interest rates could see some price growth re-emerge once the outlook becomes more optimistic, particularly within the owner-occupier segment of the market.

MELBOURNE

RENTAL VALUES

OFFICES

Rents are net face (before incentives) per square metre per annum for fully serviced buildings, excluding all outgoings such as rates, taxes and other management costs.

Prime net face rents include “Premium” and “A” grade buildings in the CBD and “A” grade only in all other markets.

Office Market	Prime net face $ / sqm per annum	Secondary net face $ / sqm per annum
CBD / Docklands	550 – 850	450 – 550
City Fringe	400 – 500	350 – 440
Inner Suburban	390 – 490	325 – 390
Outer Suburban	270 – 350	220 – 260

INDUSTRIAL

Rents are net face (before incentives) per square metre per annum, excluding all outgoings such as rates, taxes and other management costs.

Assume industrial building of circa 3,000sqm and incorporating an office component of approx.ten per cent of total building area

Industrial Market	Prime net face $ / sqm per annum	Secondary net face $ / sqm per annum
City Fringe	125 – 175	80 – 110
East	90 – 115	70 – 90
South East	85 – 110	60 – 75
North	80 – 95	55 – 70
West	80 – 95	55 – 70

RETAIL

Rents are net face and assume a speciality store on the main trading level of approximately 100sqm in area with rates quoted per square metre per annum excluding all outgoings such as rates, taxes and other management costs.

Prime strip locations include Chapel Street, Toorak Road, Church Street, Burke Road and Acland Street.

Retail Market	Net face $ / sqm per annum
CBD – Prime Retail Core (excl. Bourke St Mall)	1,500 – 5,000
Regional shopping centre	1,250 – 1,500
Sub-regional shopping centre	800 – 1,000
Neighbourhood shopping centre	550 – 750
Retail strip – Prime	500 – 900
Large Format	175 – 250

MELBOURNE

RENTAL VALUES

RESIDENTIAL

Median weekly rents for unfurnished premises. All building insurance, rates and taxes paid by landlord. Electricity, gas, excess water, telephone, internet, cable to be paid by tenant.

Residential Dwelling Type	Median Rent $ / week	Annual Change*
Housing –		
3-bedroom	400	0.0%
4-bedroom	430	-2.3%
Units and Apartments –		
1-bedroom	350	-6.7%
2-bedroom	420	-3.4%

Source: DHHS (2020).

PERTH

Source - Jones Lang LaSalle

COMMENT

COMMERCIAL

PROPERTY LAND VALUES

Property values in 2020 have generally demonstrated a stabilisation in recent years with limited transactional activity as a consequence of the fact that most of the land within the CBD and periphery is in the process of being developed or been developed with few land sales to point to the current market for land values. The approximate range of property values in the city are :

	$ / sqm per annum
Adelaide Terrace (east to west)	2,000 - 3,000
St Georges Terrace (east to west)	5,000 - 10,000
Wellington, Murray, Hay Streets between Hill and Barrack Streets (east to west)	1,500 - 2,500
Wellington, Murray, Hay Streets between William Street and the Freeway (west to east) (includes Hay and Murray Street Malls)	2,500 - 8,000

The approximate ranges of property values outside the CBD are :

	$ / sqm per annum
West Perth	2,000 - 3,000
Applecross	2,500 - 3,500
Nedlands, Subiaco, Claremont, Cottesloe	2,000 - 3,000
Stirling Highway	1,500 - 2,500
Herdsman	700 - 900
Northern prime (Balcatta)	450 - 550
Welshpool	300 - 400

PERTH CBD INVESTMENT DEMAND

- The Perth CBD vacancy increased by 0.6 percentage points over the second quarter of 2020 to 20.1 per cent.
- Net absorption over the last twelve months totalled approximately -4,800 sqm.
- No new commercial projects reached completion in the second quarter of 2020.
- No CBD office sales (≥ $5.0 million) were recorded over the second quarter of 2020.
- Average prime gross effective rents were stable over the second quarter of 2020, at $456 per sqm per annum.

The latest economic growth figures for WA do not capture the impact of the COVID-19 pandemic. However, Deloitte Access Economics (DAE) estimates show that WA's Gross State Product will contract by three per cent in 2020 before growing by an average of 2.4 per cent per annum over the following five years (2021 to 2025), second highest amongst all states. The contraction in economic growth in 2020 is expected to be felt across all states.

There were no new major completions over the June quarter of 2020, with a low future supply pipeline in the short to medium term. Construction works commenced on the corner of Spring Street and Mounts Bay Road for Capital Square Tower 2. The 25,200 sqm office tower is expected to complete by the end of 2023. This is in addition to another project also currently under construction; Chevron's new headquarters at 1 The Esplanade (54,000 sqm), due for completion at the end of 2023. Speculative construction in the Perth market is highly unlikely given the current level of vacancy. A number of new office developments have development approval, or are proposed, but limited information is known on the timeline of these projects.

Stock withdrawals are likely to increase as older stock is removed for conversion or redevelopment. Competition from newer, more affordable, and higher quality buildings makes the leasing environment even more challenging for these assets. However, the level of withdrawals remains considerably lower than seen in previous cycles, and no major buildings in the CBD boundaries have been confirmed for withdrawal. The availability of land for development in the CBD has meant we have yet to witness the need for developers to consider conversions or redevelopments of existing stock through the current cycle.

PERTH

COMMENT (cont'd)

OFFICES

The Perth CBD vacancy rate increased by 0.6 percentage points over the June quarter, to 20.1 per cent. This marks the second quarterly increase in the overall vacancy rate since the third quarter of 2018. The Perth CBD continues to have the highest CBD vacancy rate in Australia. Vacancy in Perth's prime grade buildings increased by 0.9 percentage points over the quarter to 15.3 per cent, while secondary grade vacancy recorded a marginal increase, up 0.1 percentage points to twenty-seven and a half per cent. On the other hand, the vacancy rate in premium-grade buildings declined by 0.2 percentage points to 6.7 per cent, the lowest level since the third quarter of 2017.

Sub-lease vacancy increased marginally over the quarter to 1.9 per cent of total stock, but is still trending close to its lowest level in over five years. Sub-lease vacancy has generally been declining since its most recent peak (5.1 per cent), as multiple occupiers either reabsorbed or removed sub-lease offerings from the market. These areas of sub-lease space were largely offered by mining and engineering companies.

Net absorption totalled -10,500 sqm in the Perth CBD during the second quarter of 2020, with a limited number of deals being concluded. Prior to the last two quarters, the Perth CBD leasing market experienced a tangible improvement in leasing conditions, recording two years of positive net absorption totalling 79,600 sqm. A decrease in occupier demand was recorded nationally with net absorption in the Sydney CBD (-66,600 sqm) and Melbourne CBD (-42,300 sqm) also negative.

Prior to the COVID-19 pandemic, demand for CBD office space was robust with favourable leasing conditions for occupiers. The centralisation of tenants into the CBD from suburban locations dominated leasing deals. Leasing demand was also supported by new entrants setting up operations in the Perth CBD (for example, co-working groups) and existing CBD tenants expanding their operations (predominantly mining services).

However, in 2020 so far, leasing activity has slowed significantly with a reduced number of leasing deals. The COVID-19 pandemic has affected major CBD office markets globally and in Australia, with the Perth CBD also facing office demand headwinds as a result. Prime gross effective rents remained stable in the second quarter of 2020 at $456 per sqm per annum. Prime incentives were also unchanged at 47.1 per cent (based on a ten-year lease).

Average incentives had more than doubled over the three-year period 2014 to 2016 exclusive, in line with increasing vacancy. However, given the recent positive leasing demand over the three-year period up to 2019 and subsequent reduction in vacancy, leasing incentives have scaled back, especially in premium grade buildings. The negative impact of COVID-19 on demand for office space in the CBD has so far not resulted in reduced rents and/or higher incentives.

Secondary gross effective rents also remained stable, but remain 0.3 per cent lower over the year. Secondary incentives were unchanged averaging fifty and a half per cent (based on a ten-year lease) while secondary vacancy increased by 0.1 percentage points over the June quarter to twenty-seven and a half per cent.

Average gross face rental growth is expected to trend lower for the remainder of 2020 before recovering in 2021 and beyond. The incentive cycle was previously expected to continue its downward trajectory. However, in light of the outbreak, incentives may increase in the short term as larger discounts are negotiated with tenants due to softer anticipated demand.

There were no major office sales recorded across the Perth CBD (≥ $5.0 million) over the June 2020 quarter. Over the last twelve months, CBD office sales totalled $635.9 million, across four transactions. This figure sits above the ten-year average of $587.6 million. While transaction volumes are typically smaller in the first half of each year, investment activity over the first half of 2020 has been hindered more by the uncertainty surrounding COVID-19.

Average Perth CBD prime-grade equivalent yields were unchanged over the June quarter, ranging between 5.25 to 7.75 per cent. Secondary yields were also stable, ranging between 7.00 to 9.50 per cent. Yields have generally tightened across prime and secondary grade assets over the last two years, despite elevated vacancy. This reflects the number of active investors and counter-cyclical buyers looking to invest in the Perth market.

While we expect investment activity to be subdued in the near term, demand will persist for selective office assets. As capital in search of yield is drawn to Perth, there will still be groups actively looking to purchase premium assets with sound covenants. On the contrary, demand for secondary assets is expected to be subdued in the near term, particularly as uncertainty prevails.

PERTH

COMMENT (cont'd)

OFFICES (cont'd)

Over the short term, it is likely that investment levels will remain subdued in the Perth CBD office market. However, this trend is expected to be felt across all CBD markets in Australia. Amidst the current volatility in economic conditions, investors may take a conservative approach as far as their appetite for risk is concerned, pausing investment decisions in order to fully assess the impact of the global COVID-19 pandemic on the commercial real estate markets. The longer-term prognostics is still looking positive, with the overall vacancy rate expected to trend lower from 2021 onwards. Prior to the outbreak, sectors such as mining, professional services and healthcare were showing signs of expansion due to an improvement in the broader economic environment. Furthermore, new businesses were also increasingly considering Perth in their set-up plans, demonstrated by recent take-up of office space by major co-working groups. It remains to be seen whether these trends will resume, but the situation will be monitored as it evolves.

We have revised our forecasts to show yield softening amid a slowdown in investment demand. Prime capital values are also expected to follow the same progression. Given the uncertainty of the global COVID-19 pandemic, current forecasts are fluid. We will continue to monitor the situation and will provide real-time updates to our clients on new and emerging trends, as this situation evolves.

INDUSTRIAL

As with all economies, the COVID-19 pandemic is expected to create short-term headwinds for the WA economy due to implications of border closures and forced shutdowns impacting consumer spending, investment, tourism and international trade. WA has traditionally bounced back much stronger than other states following a global economic crisis, led by the resources industry.

Leasing activity across the Perth industrial market stalled over second quarter of 2020, with prospective deals placed on hold following the COVID-19 pandemic. Gross take-up for the quarter totalled 10,000 sqm across two occupier moves. In the largest take-up of industrial space over the quarter, Australia Post (Transport, Postal & Logistics) signed a short-term lease for 6,500 sqm at 123-125 Kewdale Road, Welshpool.

Over the twelve months to June 2020, industrial gross take-up totalled 190,900 sqm across twenty-two occupier moves; below the ten-year annual average of 209,100 sqm. Leasing activity over the last twelve months has consisted primarily of smaller leasing deals, with just five deals recorded of greater than 10,000 sqm. The last twelve months has seen 111,800 sqm of new supply added to the Perth industrial market; just below the 10-year average of 120,400 sqm. Construction activity remains concentrated in the East, accounting for 77.5 per cent of new supply in the last year, followed by the South precinct, twelve and a half per cent.

There are six projects (34,100 sqm) currently under construction and expected to complete by the end of the year. The largest project is a 10,000 sqm warehouse located at 26 Nardine Close, Forrestfield. Siemens has pre-leased a 3,500 sqm warehouse at 15 Radius Loop, Bayswater (North). Three speculative developments are underway including a 4,768 sqm prime grade warehouse located at Perth Airport (East), 3,429 sqm warehouse in Bibra Lake (South) and a 3,410 sqm warehouse in Henderson (South). Two further projects totalling 32,600 sqm have either DA approval or plans submitted.

The industrial supply pipeline may increase with the rezoning of land for industrial use, as lots are sold, and development commences. However, speculative development is anticipated to remain limited, with new industrial projects unlikely to commence without pre commitment as has been the trend with projects currently under construction. Speculative development only accounts for 34% of projects currently under construction. Prime and secondary rents were unchanged across all precincts in 2Q20, with recent deals maintaining pre-COVID rates. Over the last 12 months, prime rents have remained stable in the North and East precincts, while the South precinct has declined by 0.8%.

Rental growth is anticipated to be flat for 2020 and 2021, before beginning to recover from 2022 onwards. In the pre-lease market, developers remain competitive in a bid to secure tenants - offering lower warehouse rents, which could flow through to more competitive pre lease rents. Strong competition for tenants is anticipated to remain, with higher incentives expected to play a larger part in future lease negotiations. Three industrial sales (≥ $5.0 million) including a four asset portfolio sale were recorded over the second quarter of 2020, totalling $ 22.2 million. The largest transaction of the quarter was 22 Renewable Chase, Bibra Lake (South) which sold to an undisclosed buyer for $ 9.8 million.

Primewest purchased a portfolio of four assets located on the corner of Wittenberg Drive and Sorbonne Crescent, Canning Vale. 36 Hemisphere Street, Neerabup sold for $ 5.2 million. Over the last twelve months, investment volumes totalled $272.5 million, across eighteen transactions. This remains below the ten-year average of $347.6 million. Sales activity has primarily occurred in the $5.0 – 20.0 million range, with only three sales greater than $20.0 million recorded in the last twelve months. While the current uncertainty has slowed transactional activity, we expect a number of strategic acquisitions may take place in the second half of the year, particularly as industrial assets have outperformed other commercial property markets through the pandemic. Despite low investment volumes over the last twelve months, capital in search for yield is expected to persist in the industrial sector. Over the medium to long term, investor demand for prime assets is expected to continue, particularly if an asset is supported by food or supply-chain related tenants. Investment mandates from local and institutional buyers exist, however they have been hindered by a limited supply of leased, large-scale assets listed on market.

Perth prime industrial mid-point yields are expected to remain stable over 2020 and 2021, before softening over 2022 as the property market enters a new cycle. With Perth yields sitting above the national average, investment in WA continues to offer an attractive spread to east coast markets. This may attract investors in search of yield in Australia. Prime assets in the Perth market remain tightly held, and future transactional evidence will likely remain dependent on the ability to unlock such assets.

PERTH

RENTAL VALUES

RETAIL

- Retail turnover growth continued to accelerate in WA (4.9 per cent year-over-year as at June 2020). However, market sentiment is subdued.
- One project reached completion in the second quart of 2020, adding 5,800 sqm of large format retail space. 32,700 sqm of new retail floorspace has been added over the last twelve months.
- There is currently 78,400 sqm of retail developments under construction across five projects.
- Average rents declined across most sub-sectors as landlords attempt to combat elevated vacancies.
- Regional and sub-regional yields decompressed over the second quarter of 2020 as rents continue to fall.
- Two retail sales (≥ $5.0 million) were recorded over the second quarter of 2020, totalling $64.3 million. Investment volumes over the last twelve months totalled $791.8 million, across ten transactions.

OFFICES

- Rental values stated are average and strictly net, excluding outgoings.
- Rental values stated for the CBD are for premium buildings.
- Rental values in high rise buildings can vary as much as ten to fifteen per cent above the average, depending on the location of accommodation within the building.
- Outgoings include all rates, taxes, power, cleaning, management costs, etc.
- All fully serviced including air conditioning, lifts, etc. where applicable.

	CENTRAL BUSINESS DISTRICT		WEST PERTH		OTHERS	
	PRIME including ST GEORGE'S TERRACE	**PERIPHERAL including HAY & MURRAY STREETS AND ADELAIDE TCE**	**PRIME KINGS PARK ROAD**	**OTHERS**	**SOUTH PERTH**	**FREMANTLE NEDLANDS SUBIACO VIC. PARK**
	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum
Net Rent	500 - 800	250 - 325	350 - 400	200 - 350	250 - 350	250 - 350
Outgoings	170 - 190	120 - 140	170 - 200	110 - 125	100 - 135	100 - 125

PERTH

RENTAL VALUES (cont'd)

INDUSTRIAL

- Rental values stated are average and strictly net, excluding all outgoings.
- Rental values are primarily influenced by supply and demand, land values and building construction costs. The latter two factors are also influenced by interest rate levels for development funds.

GENERAL INDUSTRIAL

- Typical locations - CANNING VALE, KEWDALE, WELSHPOOL, BELMONT, OSBORNE PARK and MALAGA.
- Self contained typical factory/warehouse generally exceeding 1,000 sqm total area including approximately fifteen per cent office space.

	$ / sqm per annum Net Rent
Low	55 - 65
Average	75 - 85
Purpose built - turnkey	90 - 110
High tech industries	150 - 190

LIGHT INDUSTRIAL

- Typical locations - BALCATTA, MORLEY, MYAREE, O'CONNOR, BIBRA LAKE, BASSENDEAN and WILLETTON.
- Smaller factory/warehouse not exceeding 1,000 sqm, average 400 - 600 sqm, either free-standing, self-contained or estate type unit.

	$ / sqm per annum Net Rent
Low	70 - 80
Average	85 - 100

Note :

- Top of the range at $95.00 - $105.00 per sqm per annum - good size, well maintained, free-standing, self-contained property in prime location such as OSBORNE PARK and CANNING VALE.
- Bottom of the range at $65.00 per sqm per annum - small unit with low exposure on an estate in a peripheral location.

HEAVY INDUSTRIAL

- Typical location - KWINANA.
- Generally heavy industrial property is owner-occupied. However, when properties are available for leasing, the type of accommodation offered is usually of average quality and both rental values and outgoings are influenced by many variable factors, such as :
  - Proportion of building area/site area (i.e. not unusual for a building of relatively small area to occupy a very large site).
  - Type of industry in which user is involved.
  - Type of industry carried out on adjacent properties.
  - Condition of site due to activities of previous occupants.
  - Accommodation requirements.
- Rental values stated are therefore as close to ‘typical’ as can be achieved, bearing in mind the influences of the foregoing factors.

	$ / sqm (BUILDING AREA) per annum Net Rent
Average rental levels	55 - 65

PERTH

RENTAL VALUES (cont'd)

RETAIL

- Rental levels stated are strictly net excluding all outgoings.
- Rental levels in suburban shopping centres vary according to tenant's type of business, location in the centre and shop size.

CENTRAL BUSINESS DISTRICT	$ / sqm per annum Net Rent
Prime :	
Hay Street Mall and surrounds including associated well patronised arcades	
Maximum	3,000 - 4,000
Good	1,500 - 2,500
Average	800 - 1,250

CENTRAL BUSINESS DISTRICT	$ / sqm per annum Net Rent
Peripheral :	
Away from Mall	
Average (good location)	1,200 - 2,000
Low (poor location)	350 - 600
ADD extra for :	
Outgoings (prime or peripheral)	125 - 500

SUBURBAN

	REGIONAL CENTRE	**DISTRICT CENTRE**	**NEIGHBOURHOOD CENTRE**
	(exceeding 20,000 sqm)	**(5,000-20,000 sqm)**	**(1,000-5,000 sqm)**
	$ / sqm per annum Net Rent	**$ / sqm per annum Net Rent**	**$ / sqm per annum Net Rent**
Services - travel/bank/building society	900 - 1,500	650 - 950	400 - 500
Clothing	1,500 - 2,750	650 - 1,250	400 - 500
Food - take away/coffee lounge/ butcher/ delicatessen, etc.	1,200 - 2,000	550 - 1,000	450 - 600
Supermarket	175 - 250	160 - 225	150 - 200
Miscellaneous - jewellers and other specialty shops	2,200 - 3,500	850 - 1,500	400 - 600
Department store	150 - 300	-	-
Discount department store	125 - 180	125 - 150	-
ADD extra for :			
Outgoings	160 - 180	120 - 175	110 - 140

PERTH

RENTAL VALUES (cont'd)

RESIDENTIAL

- Property insurance, rates and taxes and management fees paid by owners.
- Electricity, telephone, excess water paid by tenant.
- All unfurnished.

	PRIME including COMO, CRAWLEY, EAST PERTH, FREMANTLE, NEDLANDS, NORTH PERTH, SOUTH PERTH, SUBIACO, WEST PERTH	PERIPHERAL including MAYLANDS, MOSMAN PARK, MT LAWLEY, OSBORNE PARK, SCARBOROUGH, VICTORIA PARK, WEMBLEY
	$ / Week	$ / Week
UNIT OR TOWN HOUSE		
Average - Medium Standard :		
One-bedroom	330 - 370	270 - 310
Two-bedroom	425 - 465	325 - 365
Three-bedroom	515 - 555	425 - 465
High Standard :		
Three-bedroom luxury	900 - 940	765 - 805
LARGE LUXURY APARTMENTS	1230+	900+

SYDNEY

Source - Cushman & Wakefield

COMMENT

Australia experienced its first recession in twenty-eight years during the June quarter. The impact of the COVID pandemic has resulted in a softer economy with unemployment averaging severn per cent over the September quarter, the highest levels since 1998.

Economic growth is expected to accelerate as the pandemic is contained and the economy reopens. The Reserve Bank of Australia (RBA) estimated in their November 2020 Statement on Monetary Policy, real GDP growth of five per cent in calendar 2021 after a four per cent decline in 2020. Inflation is expected to remain weak over the next few years. The RBA estimating CPI to increase by just one per cent over 2021, below the RBA target band of two to three per cent. Employment is forecast to improve in 2021 with the unemployment rate expected to decline to six and a half per cent by December 2021. However, this is above the pre-pandemic rate that averaged 5.2 per cent over 2019.

OFFICES

After several years of relatively strong growth, which pushed vacancy rates to multi-decade lows, the COVID-19 pandemic has resulted in Sydney's office market becoming much more tenant favourable. While the limited pipeline of new office developments will keep supply below average for the next few years, the pandemic has significantly reduced demand for office space.

Social distancing, delayed decision making as well as many tenants seeking to lower costs and adapt to the rise in flexible working has resulted in reduced leasing activity and rising vacancy. According to the Property Council of Australia, in the six-month period to July 2020, vacancy rates in the Sydney CBD recorded sharp increases across all grades, with total market availability increasing from 3.9 per cent to 5.6 per cent. Estimates of sub-lease availability in the CBD tracked by Cushman & Wakefield indicate available space may have risen from around 40,000 sqm to over 160,000 sqm during the pandemic. With employment conditions forecast to remain soft for several more years, CBD vacancy is expected to rise to over ten per cent in 2021.

The lower demand plus rising vacancy has reduced effective rents for CBD offices. Gross incentives for prime grade stock rose from around twenty-one per cent in December 2019 to average about thirty and a half per cent in the third quarter. Gross face rents also began to slip, declining on average by two and a half per cent over the year to the third quarter. As a result of the higher incentives and softer face rents, prime gross effective rents in the Sydney CBD are now 10.4 per cent lower than the third quarter of 2019. Prime rents range from $845 to $1,020 / sqm per annum with the average prime gross effective rent $935 / sqm per annum. Lower grade stock has suffered more with B-grade gross effective rents averaging $710 / sqm per annum, a decrease of 13.6 per cent year-on-year.

With vacancy expected to continue to move higher in 2021, down pressure on rents should continue with the Sydney office market remaining tenant favourable in 2021.

MARKET	Stock (sqm)	Total Vacancy (%)		Supply Additions (sqm)	Net Absorption (%)
	July 20	January 20	July 20	12 months to July 20	12 months to July 20
Sydney CBD	4,977,938	3.9	5.6	70,198	-98,290
North Sydney	824,767	7.6	8.0	4,230	-8,682
Macquarie Park	894,034	4.6	6.8	35,000	18,098
Parramatta	765,763	3.2	4.5	13,786	34,138

Source : Property Council of Australia

Market Average[1]	Sydney CBD	North Sydney	Macquarie Park[2]	Parramatta
Q3 2020	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum
Premium	1,225	735	-	-
A-Grade	885	675	510	670
B-grade	710	615	485	600

Source: Cushman & Wakefield

1) Gross effective rent.
2) Q2 2020

SYDNEY

COMMENT (cont'd)

INDUSTRIAL

The industrial sector has performed relatively well through the COVID-19 pandemic. A boom in online trade due to social distancing restrictions as well as a desire by many retailers to rationalise and modernise supply chains to reduce costs has resulted in relatively strong demand for logistic space. With online trade continuing to grow, it is expected demand for logistic space should remain solid in 2021.

Despite the strong demand, industrial rents in Sydney have been relatively stable. The average prime net face rent for inner industrial markets, as indicated by the South Sydney market, was unchanged over the twelve months to the third quarter of 2020 at $230 / sqm per annum. The average prime net face rent in outer markets rose 1.3 per cent to $146 / sqm per annum, however most of the gain occurred in the first quarter.

Market Average[3]	South	Central West	South West	Outer West	North West
Q3 2020	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum	$ / sqm per annum
Net Face Rent	230	175	138	135	135
Outgoings	55	33	28	30	25

Source: Cushman & Wakefield

3) Prime grade

INTERNATIONAL CONSTRUCTION COSTS

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INTERNATIONAL CONSTRUCTION COSTS

REGIONAL VARIATION INDICES

The prices given are for the base city, the regional indices are :

AUSTRALIA

Base	- SYDNEY	1.00
Others	- Adelaide	0.99
	- Brisbane	0.90
	- Canberra	1.00
	- Darwin	1.14
	- Hobart	0.98
	- Melbourne	0.92
	- Perth	0.97
	- Queensland Islands	1.19

CANADA

Base	- VANCOUVER	1.00
Others	- Montreal	0.88
	- Toronto	0.93
	- Quebec	0.88
	- Edmonton	0.90
	- Winnipeg	0.85
	- Calgary	0.90

FIJI

Base	- SUVA/LAUTOKA	1.00
Others	- Islands	1.35
	- Ba	0.98
	- Sigatoka	0.98
	- Labasa	1.10

HONG KONG

Base	- HONG KONG	1.00
Others	- Outlying islands	1.25

MALAYSIA

Base	- KUALA LUMPUR	1.00
Others	- Penang	1.01
	- Johor Baru	1.03

NEW ZEALAND

Base	- AUCKLAND	1.00
Others	- Christchurch	1.00
	- Dunedin	0.96
	- Wellington	1.00

PAPUA NEW GUINEA

Base	- PORT MORESBY	1.00
Others	- Inland Towns	1.50
	- Other Locations	2.00

SOUTH AFRICA

Base	- CAPE TOWN	1.00
Others	- Bloemfontein	0.96
	- Durban	0.96
	- East London	1.04
	- Johannesburg	0.97
	- Port Elizabeth	1.00
	- Pretoria	0.97

UNITED KINGDOM

Base	- LONDON, outer	1.00
Others	- London, central	1.04
	- Belfast	0.50
	- Birmingham	0.79
	- East Anglia	0.83
	- Edinburgh	0.77
	- Manchester	0.80
	- South East (excl. London)	0.89
	- Wales	0.76

UNITED STATES

Base	- LOS ANGELES	1.00
Others	- Chicago	1.11
	- Miami	0.72
	- New York	1.26
	- San Francisco	1.13
	- Washington	0.84

VANUATU

Base	- PORT VILA	1.00
Others	- Inland Centres	1.15
	- Islands	1.40

INTERNATIONAL CONSTRUCTION COSTS

OCTOBER 1, 2020

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
AVERAGE TENDER WAGE							
Includes all labour related charges and costs, contractor's overheads and profit :							
Tradesmen (Av) per hour - Local Currency	...		81.00	75.00	14.25	141.00	
Labourer (Av) per hour - Local Currency	...		72.00	50.00	7.75	94.00	
For information only, being of little comparative value due to the widely differing skills, efficiency of labour and use, and availability of plant.							
BUILDING PRICE INDEX							
September 30, 2019 to September 30, 2020	...		3.9%	1-2%	2.1%	-3.9%	
Anticipated, September 30, 2020 to September 30, 2021	...		2.8%	(-)2 to (+)2%	-0.5%	-3.7%	
BUILDING COSTS PER SQUARE METRE							
Costs given are average prices for typical buildings, they provide no more than a rough guide to the probable cost.							
Costs are based on total floor area of all levels measured between the inner face of external walls and without deduction for internal walls.							
Costs exclude abnormals, sloping sites, unusual shape, external works, external services, furniture, furnishings and professional fees.							
Australia, Canada, New Zealand and Singapore figures exclude Goods and Services Tax.							
Fiji, PNG, South Africa, United Kingdom and Vanuatu figures exclude Value Added Tax.							
HOTELS, fully serviced, excluding fitting-out,							
- single/two storey, resort	...	sqm	3380-3520	3080-4100	3000-3575	-	
- medium/high rise, three star tourist	...	sqm	4220-4550	2850-3345	3150-3675	28500-30800	
- medium/high rise, five star international			6080-6555	4285-5155	3325-4000	29800-35700	
ADD : Fitting-out, total building							
INDUSTRIAL							
- factories or warehouses for letting	...	sqm	645-695	1225-1490	1500-2050	9100-10100	
- factories or warehouses for owner occupation			815-875	1490-1880	1575-2375	16800-21600	

Malaysia Kuala Lumpur Ringgit	New Zealand Auckland $/NZ	P.N.G. Pt. Moresby/Lae Kina	Singapore $/SG	South Africa Cape Town Rand	U.K. London £/Stg	U.S. Los Angeles $/US	Vanuatu Port Vila Vatu
21.00	62.00	35.00	14.50	143.00	34.60	94.75	740
12.00	49.00	15.00	11.30	62.00	25.50	75.50	375
0.5%	3.6%	20.0%	0.8%	5.4%	-1.8%	0.1%	1.0%
0.5%	2.3%	25.0%	1.0%	4.6%	0.6%	4.0%	-0.5%
2200-2300	3890-4260	8000-10000	3000-3400	14150-15200	1185-2110	3125-3355	143400-196300
2400-2500	4010-4390	9000-12000	2800-3200	15400-16500	1765-2840	2150-2340	150800-201600
3400-3700	5100-5500	9000-12000	3800-4200	17500-18600	2605-3575	3210-3575	175800-209500
1100-1200	910-1070	6500-7500	1100-1500	7500-8050	590-1290	805-815	65400-91700
1150-1250	980-1190	6500-7500	1300-1800	8150-9200	640-1805	910-1070	78500-117900

INTERNATIONAL CONSTRUCTION COSTS

OCTOBER 1, 2020

BUILDING COSTS PER SQUARE METRE

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
OFFICES, standard finish, lettable :							
- one/two storey, air-conditioned, no lift	...	sqm	2085-2245	2555-3040	2375-2850	-	
- three storey, air-conditioned, lift	...	sqm	2370-2555	3015-3495	2725-3450	-	
- four/ten storey, fully serviced	...	sqm	2650-2860	3705-4730	3575-3800	-	
- ten/twenty storey, fully serviced		sqm	3775-4070	3660-4700	3650-4275	21900-25700	
- twenty/thirty five storey, fully serviced		sqm	5605-6040	3500-4600	-	27000-33000	
ADD : Basements and car parking	...	Note					
Fire sprinklers	...	sqm	86.00	42.10	48.00	580.00	
Sub-divisional partitioning etc.	...	Note					
RESIDENTIAL							
- Individual, excluding air-conditioning :							
- medium standard, 150/250 sqm	...	sqm	1965-2120	3000-3700	1650-2625	30200-35300	
- high standard, 200/300 sqm	...	sqm	2745-2960	3500-6500	2850-4275	39900-45300	
- Multi unit, 2/3 bedroom, excluding air-conditioning							
- medium standard	...	sqm	2515-2710	2550-2800	2650-3150	21900-26000	
- high standard	...	sqm	3120-3360	2800-4000	2850-3575	29700-34800	
RETAIL							
- Shell construction for letting	...	sqm	1245-1305	1800-2280	1850-2150	22400-26000	
- Supermarkets, fully serviced, but excluding cool rooms and shop fittings	...	sqm	1760-1895	2100-2350	2000-2750	29300-34800	

Malaysia Kuala Lumpur Ringgit	New Zealand Auckland $/NZ	P.N.G. Pt. Moresby/Lae Kina	Singapore $/SG	South Africa Cape Town Rand	U.K. London £/Stg	U.S. Los Angeles $/US	Vanuatu Port Vila Vatu
1800- 1900	1820- 2080	7000- 9000	1600- 2000	14000- 15100	1490- 1920	1445- 1625	113000- 150800
2000-2 200	2150- 2400	8500- 10500	1700- 2100	14300- 15400	1430- 2230	1500- 1660	125700- 176000
2200- 2400	2740- 2870	8500- 10500	2100- 2500	14500- 15700	1980- 2300	1590- 1750	150800- 201150
2700- 2900	2870- 3120	8500- 10500	2500- 2800	18300- 19400	-	2005- 2150	176000- 217300
2900- 3100	3490- 3870	8500- 10500	2700- 3000	21300- 22400	-	2175- 2430	-
80.00	86.00	800.00	45.00-55.00	215.00	31.00	41.00	2500
2200- 2300	2040- 2340	7500- 8500	2600- 3200	14000- 16000	885-1960	1220- 1330	95300- 149800
2600- 2800	2480- 2780	8500- 10000	3200- 4000	18200- 22300	1540-2775	1545- 1725	161800- 245100
1900- 2000	2230- 2530	7500- 8500	2100- 2600	13500- 15900	645-1235	1220- 1290	134800- 177000
2100- 2300	2460- 2760	8500- 10000	2700- 3500	16430- 18400	1110-1560	1605- 1785	163400- 245100
1700- 1800	1170- 1290	6500- 7500	1900- 2400	8745- 10400	595-2245	950- 1045	81700- 95300
1800- 1900	1390- 1640	7000-8000	2000- 2700	9500- 10300	1090-2095	1165- 1310	94400- 149800

INTERNATIONAL CONSTRUCTION COSTS

OCTOBER 1, 2020

ESTIMATING PRICES

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
The costs are **all-up estimating prices, i.e. they include allowance for labours and minor items**, the costs are indicative average prices for reasonable quantities of work and would apply to commercial projects of 2,500 to 10,000 sqm							
Australia and New Zealand costs exclude GST							
Fiji, PNG, South Africa, United Kingdom and Vanuatu figures exclude Value Added Tax							
Excavate over site to reduce levels in typical ground conditions	...	cum	31.70	35.00	15.00	208.00	
Excavate trench not exceeding 1.00m deep in typical ground conditions including allowance for planking and strutting	...	cum	62.70	75.70	40.00	209.00	
Sheet steel piling including all supports : Temporary, 74kg	...	sqm	530.00	480.00	385.00	1490.00	
Permanent, 74kg	...	sqm	700.00	655.00	500.00	2820.00	
ADD extra for : Walers and ground anchors if required	...	Note					
Crushed rock/blue metal basecourse to receive paving :							
150mm thick	...	sqm	17.45	14.70	19.50	133.00	
250mm thick	...	sqm	29.70	20.40	32.50	189.00	
Hot bituminous concrete, 25mm thick	...	sqm	19.20	11.85	40.00	128.00	
Concrete, 25 MPa (3500 p.s.i.) in :							
Foundations	...	cum	340.00	220.00	521.00	1370.00	
Ground slab	...	cum	367.00	220.00	521.00	1370.00	
Suspended slab and beams	...	cum	363.00	236.50	526.00	1370.00	
Walls	...	cum	376.00	238.00	526.00	1370.00	
Columns	...	cum	427.00	238.00	526.00	1370.00	
ADD extra for:							
35 MPa (5000 p.s.i.)	...	cum	8.60	21.00	98.60	133.00	
50 MPa (7000 p.s.i.)	...	cum	47.40	92.00	191.50	284.00	
Formwork, average standard (close tolerances, surface with limited imperfections, not subject to close scrutiny) :							
Soffit of suspended slab	...	sqm	133.00	185.00	90.00	390.00	
Sides of wall	...	sqm	172.00	182.00	90.00	390.00	
Sides and soffit of attached beams	...	sqm	162.00	195.00	100.00	390.00	
Sides of rectangular columns	...	sqm	162.00	184.40	90.00	390.00	

Malaysia Kuala Lumpur Ringgit	New Zealand Auckland $/NZ	P.N.G. Pt. Moresby/Lae Kina	Singapore $/SG	South Africa Cape Town Rand	U.K. London £/Stg	U.S. Los Angeles $/US	Vanuatu Port Vila Vatu
22.00	54.00	250.00	14.50	58.00	1.60	7.50	850
26.00	118.00	350.00	26.00	180.00	28.80	15.00	1800
270.00	890.00	2000.00	400.00-500.00	1485.00	229.00	336.00	22550
600.00	920.00	2500.00	650.00-740.00	2100.00	681.50	468.00	29075
20.00	24.30	30.00	9.00	127.00	8.30	7.65	1060
28.00	32.40	55.00	15.00	200.00	13.50	14.75	1765
33.00	-	220.00	10.50	265.00	-	6.45	1860
320.00	402.00	1100.00	123.00	2430.00	120.40	222.00	29175
320.00	417.00	1100.00	124.00	2450.00	123.50	213.00	29175
320.00	430.00	1250.00	124.00	2490.00	152.50	260.00	29500
320.00	430.00	1250.00	125.00	4510.00	141.50	260.00	29500
320.00	490.00	1300.00	125.00		176.70	260.00	29500
60.00	66.00	250.00	7.00	110.00	13.50	13.85	4130
80.00	-	300.00	17.00	220.00	54.00	32.30	-
39.00	196.00	250.00	40.50	370.00	42.80	127.00	5100
39.00	181.00	200.00	40.00	460.00	49.00	142.00	5400
39.00	235.00	250.00	40.50	670.00	56.10	149.50	5400
39.00	192.00	220.00	41.50	550.00	63.20	147.50	5100

INTERNATIONAL CONSTRUCTION COSTS

OCTOBER 1, 2020

ESTIMATING PRICES

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
Bar reinforcement, 16/20mm dia.	...	t	2600.00	2835.00	3800.00	9600.00	
Clay brick wall (commons) to receive plaster :							
90/110mm thick	...	sqm	137.50	285.00	-	204.00	
Hollow concrete block wall to receive plaster :							
100mm thick	...	sqm	138.50	205.00	85.00	178.00	
150mm thick	...	sqm	159.50	230.00	108.00	249.00	
200mm thick	...	sqm	184.00	250.00	132.00	313.00	
Hollow plaster block wall :							
100mm thick	...	sqm	-	198.00	-	-	
Steel stud framed wall comprising 75mm framing with 10mm plasterboard each face	...	sqm	124.50	165.00	115.00	1080.00	
Structural steel including shop drawings, fabrication, erection and zinc phosphate treatment :							
Built up plate section columns and beams	...	t	5150.00	5760.00	6375.00	35400.00	
UB, UC columns and beams	...	t	5450.00	5910.00	6275.00	35400.00	
Angle framing in trusses	...	t	6100.00	6680.00	6450.00	39600.00	
Tubular framing in trusses	...	t	6800.00	7415.00	7500.00	39600.00	
Carpentry framing timber :							
100 x 50mm rafter	...	m	17.05	16.25	9.00	169.00	
150 x 50mm purlin	...	m	26.70	18.50	11.60	259.00	
100 x 38mm stud	...	m	16.20	12.50	7.75	120.00	
100 x 50mm stud	...	m	18.05	12.90	8.00	168.00	
Aluminium windows, fully glazed, commercial standard :							
Fixed lights	...	sqm	560.00	745.00	500.00	2790.00	
50% opening lights	...	sqm	655.00	790.00	750.00	2790.00	
Double glazed opening lights	...	sqm	765.00	1020.00	-	-	
Aluminium shop fronts, commercial standard, glazed with :							
6mm clear float glass	...	sqm	-	561.00	650.00	1890.00	
6mm toughened safety glass	...	sqm	725.00	726.00	875.00	2770.00	

Malaysia Kuala Lumpur Ringgit	New Zealand Auckland $/NZ	P.N.G. Pt. Moresby/Lae Kina	Singapore $/SG	South Africa Cape Town Rand	U.K. London £/Stg	U.S. Los Angeles $/US	Vanuatu Port Vila Vatu
3800.00	4500.00	8500.00	1300.00	16500.00	1245.00	3125.00	187900
46.00	292.00	-	33.50	375.00	45.60	244.00	-
33.00	102.00	220.00	29.00	300.00	31.80	160.50	3300
37.00	118.00	250.00	43.00	450.00	41.30	179.00	4000
42.00	128.00	300.00	57.00	550.00	50.90	188.00	5050
-	-	270.00	-	300.00	-	-	-
59.00	118.00	250.00	61.50	450.00	74.20	101.50	3710
5600.00	6200.00	18000.00	4500.00	56500.00	3070.00	5375.00	381500
5600.00	6200.00	18000.00	4400.00	56200.00	3390.00	5375.00	375800
5900.00	8500.00	18000.00	4500.00	55100.00	2940.00	6840.00	387200
5900.00	7000.00	18000.00	4600.00	60000.00	3325.00	6840.00	448380
12.00	16.60	32.00	21.00	120.00	5.20	8.10	550
12.50	21.10	48.00	27.00	170.00	7.30	11.80	835
10.00	13.10	12.00	13.00	100.00	5.20	-	540
11.50	11.90	32.00	17.00	120.00	6.30	20.20	545
420.00	560.00	2000.00	230.00	3600.00	264.00	636.00	22500
470.00	800.00	2000.00	295.00	3700.00	439.00	672.00	38975
590.00	850.00	2500.00	400.00	5850.00	522.00	776.00	-
420.00	450.00	3000.00	230.00	3600.00	1520.00	791.00	23970
490.00	640.00	6000.00	320.00	3900.00	2050.00	878.00	35950

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		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
Curtain walling comprising 50% colour anodised aluminium spandrils and 50% window :							
Windows							
Single glazed, fixed	...	sqm	835.00	750.00	1650.00	6400.00	
Double glazed, fixed	...	sqm	925.00	1050.00	2250.00	7780.00	
Standard internal flush door, 2040 x 820mm, excluding hardware and painting :							
Hollow core	...	No	212.00	218.00	145.00	2250.00	
Solid core	...	No	268.00	357.00	185.00	7070.00	
Door frame to suit 2040 x 820mm door, excluding painting :							
Timber with architraves	...	No	585.00	254.00	235.00	2090.00	
Welded steel	...	No	128.00	351.00	445.00	3410.00	
Fire door, 2040 x 820mm, two hour rated including steel frame, average quality hardware and closer, excluding painting	...	No	1440.00	1885.00	4250.00	23590.00	
Metal roofing fixed to steel purlins :							
Corrugated x 0.53/0.55, Zincalume® Steel	...	sqm	54.60	70.90	40.00	1550.00	
Ribbed x 0.53/0.55 ditto	...	sqm	60.70	75.00	40.00	1550.00	
Ribbed x 0.53/0.55 colour coated one side	...	sqm	74.40	79.50	-	1550.00	
Three layer bituminous felt or butyl rubber roofing finished with quartz or similar chips including upstands, etc.	...	sqm	105.50	64.50	75.00	386.00	
Cement render, two coats on brick or concrete walls, including narrow widths and labours	...	sqm	46.30	79.00	19.00	110.00	
Smooth plaster, two coats ditto	...	sqm	51.30	88.00	20.00	110.00	
10mm plasterboard fixed to :							
Timber or steel wall framing	...	sqm	33.50	69.50	44.00	543.00	
Timber ceiling framing	...	sqm	34.00	77.00	46.00	543.00	
Suspended ceilings :							
Flush plasterboard with concealed suspension system	...	sqm	85.00	71.30	80.00	721.00	
Acoustic tiles in exposed suspension system	...	sqm	78.00	60.70	120.00	524.00	

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800.00	930.00	3500.00	350-450	3800.00	688.00	1060.00	89965
1000.00	1030.00	-	480-530	5400.00	819.00	1175.00	119850
250.00	178.00	950.00	230.00	1250.00	108.50	275.00	9900
470.00	286.00	1200.00	330.00	1840.00	209.00	455.00	16525
230.00	290.00	300.00	230.00	1620.00	111.00	229.00	12750
290.00	-	1000.00	275.00	1430.00	-	392.00	25500
1800.00	1980.00	5000.00	1350.00-1600.00	11200.00	588.00	1845.00	254745
55.00	46.30	85.00	36.00	445.00	20.60	97.80	2285
51.00	78.00	90.00	40.00	455.00	-	97.90	2285
60.00	105.00	120.00	43.00	470.00	58.40	103.50	-
50.00	144.00	450.00	32.00-35.00	550.00	56.60	44.10	4290
23.00	74.00	40.00	15.50	225.00	11.50	77.60	1145
25.00	54.00	40.00	16.80	245.00	16.50	76.00	1200
40.00	38.00	70.00	18.00	165.00	10.40	23.10	2645
45.00	42.50	70.00	19.00	180.00	9.10	25.90	2755
51.00	52.00	160.00	31.00	460.00	57.20	92.40	4230
58.00	54.00	220.00	31.00	400.00	35.80	43.40	7290

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		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
Ceramic wall tiling, standard 150 x 150mm white glazed, fixed with adhesive	...	sqm	108.00	200.00	87.00	557.00	
Floor tiling laid with adhesive (base price tiles) :							
Mosaic, 50 x 50mm in sheets	...	sqm	131.50	221.00	117.00	380.00	
Quarry, 200 x 200mm	...	sqm	131.50	257.00	117.00	432.00	
Carpet, average commercial standard acrylic or nylon including rubber underlay	...	sqm	47.10	56.00	80.00	459.00	
Vinyl sheet, 2.0mm standard flexible on prepared surface	...	sqm	50.10	46.10	75.00	404.00	
Prime, 1 u/c and 2 f/c alkyd enamel on woodwork	...	sqm	18.15	18.70	15.00	86.00	
1 coat sealer and 2 coats acrylic latex/vinyl emulsion paint on plastered walls	...	sqm	12.00	12.75	12.50	85.00	
Sanitary fixtures, fully installed including pipework to outside face of building at ground level :							
W.C. suite, white vitreous china :							
Single or two storey office building	...	No	3900.00	2310.00	1850.00	-	
Ten to thirty storey office building	...	No	4100.00	1965.00	2150.00	7780.00	
Wall basin, white vitreous china :							
Single or two storey office building	...	No	2750.00	1350.00	1500.00	-	
Ten to thirty storey office building	...	No	2700.00	1255.00	1750.00	5510.00	
Sink and drainer, stainless steel single bowl :							
Single or two storey office building	...	No	3000.00	2050.00	1550.00	-	
Ten to thirty storey office building	...	No	3300.00	1865.00	1750.00	7970.00	
ADD extra for :							
Hot water service	...	Note					
External services	...	Note					

Malaysia Kuala Lumpur Ringgit	New Zealand Auckland $/NZ	P.N.G. Pt. Moresby/Lae Kina	Singapore $/SG	South Africa Cape Town Rand	U.K. London £/Stg	U.S. Los Angeles $/US	Vanuatu Port Vila Vatu
70.00	122.00	220.00	(labour cost only) 35.00-45.00	480.00	52.00	175.50	4795
65.00	165.00	330.00	(labour cost only) 45.00-55.00	1000.00	106.00	175.50	6000
90.00	133.00	330.00	40.00-50.00	850.00	78.50	193.00	6000
61.00	72.00	250.00	45.00-70.00	300.00	34.70	58.20	3900
45.00	74.00	220.00	35.00-55.00	340.00	22.70	62.80	3600
12.00	30.90	40.00	9.00	130.00	12.10	13.40	900
7.00	18.80	55.00	4.00	95.00	6.40	11.55	750
1600.00	4800.00	4000.00	2500.00	13250.00	910.00	2955.00	104860
1750.00	5700.00	4000.00	2550.00	15300.00	-	2715.00	116900
1500.00	2800.00	3500.00	1800.00	11200.00	610.00	2215.00	86900
1650.00	2800.00	-	1850.00	14300.00	-	2125.00	104850
1550.00	2800.00	4000.00	1900.00	13200.00	780.00	2920.00	88860
1700.00	3200.00	4000.00	2200.00	14500.00	-	2810.00	99050

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ESTIMATING PRICES

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
Electrical, the following are average costs for typical buildings including fire alarms but EXCLUDING sub-station work, PABX units and any significant site conditions :							
Hotels -							
Single/two storey resort	...	sqm	235.00	295.00	250.00-325.00	-	
Three star, tourist	...	sqm	275.00	225.00	265.00-380.00	2400.00	
Five star, international	...	sqm	370.00	357.00	300.00-440.00	3160.00	
Industrial -							
Factory or warehouse with small office	...	sqm	75-185	95.20	100.00-150.00	1080.00	
Offices, lettable -							
1/ 3 storey	...	sqm	165-220	177.00	180.00-260.00	-	
4/10 storey	...	sqm	170-260	210.00	255.00-360.00	2270.00	
10/20 storey	...	sqm	235-355	245.00	300.00-420.00	2270.00	
20/35 storey	...	sqm	315-395	245.00	-	2270.00	
Residential -							
Individual, medium standard	...	sqm	60-90	200.00	135.00-190.00	1450.00	
Multi unit medium standard	...	sqm	85-135	258.00	190.00-260.00	1450.00	
Retail -							
Supermarket, fully serviced	...	sqm	115-160	159.00	135.00-185.00	2140.00	
Air-conditioning, typical costs per square metre a/c area or per room :							
Hotels -							
Public areas	...	sqm	770.00	355.00	475.00-570.00	3390.00	
Bedrooms	...	Per room	11070-14885	11950-18000	1125.00-1235.00	96600.00	
Offices, lettable -							
1/ 3 storey, medium standard	...	sqm	360-440	255.00	445.00-540.00	-	
4/10 storey, medium standard	...	sqm	435-540	245.00	475.00-565.00	2510.00	
10/20 storey, medium standard	...	sqm	565-705	240.00	515.00-750.00	2570.00	
20/35 storey, medium standard	...	sqm	665-850	275.00	-	2770.00	
Retail -							
Supermarket, medium standard	...	sqm	285-355	162.00	420.00-540.00	2640.00	

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230.00-250.00	248.00	800.00-1200.00	200.00-250.00	690.00	157.00-303.00	251.00	1350-17750
250.00-260.00	260.00	800.00-1200.00	275.00-335.00	785.00	189.00-346.00	197.00	14600-20700
280.00-300.00	272.00	1200.00-1500.00	315.00-375.00	1000.00	254.00-470.00	245.00	16200-24000
140.00-150.00	86.00	800.00	70.00-130.00	500.00	124.00-524.00	98.00	5400-8500
160.00-170.00	136.00	750.00-1000.00	150.00-180.00	625.00 - 645.00	157.00-313.00	178.00	9750-14200
210.00-220.00	181.00	800.00-1100.00	160.00-190.00	635.00 - 655.00	189.00-335.00	158.00	14100-19950
240.00-260.00	197.00	800.00-1100.00	160.00-200.00	650.00 - 670.00	233.00-579.00	155.00	16150-22900
270.00-300.00	203.00	800.00-1100.00	180.00-260.00	720.00 - 745.00	482.00-579.00	155.00	-
95.00-110.00	60.00	600.00	160.00-215.00	585.00	44.00-157.00	117.00	7600-10650
95.00-110.00	133.00	600.00	110.00-145.00	600.00	56.00-167.00	125.00	10500-14250
210.00-220.00	130.00	750.00	170.00-270.00	675.00	114.00-221.00	155.00	7600-10100
270.00-300.00	360.00-520.00	1200.00	250.00-300.00	1625.00	291.00-524.00	197.00	21000-30100
6500.00-6900.00	6900.00-8500.00	5000.00	7000.00-7500.00	38000.00	7730.00-12220.00	-	61500-67850
140.00-150.00	240.00-330.00	1200.00	125.00-180.00	1400-1450	221.00-476.00	222.00	24300-29650
180.00-190.00	270.00-430.00	1200.00	155.00-200.00	1450-1500	281.00-487.00	205.00	25600-32450
230.00-240.00	330.00-510.00	1200.00	170.00-230.00	1450-1500	460.00-589.00	213.00	28350-41450
230.00-250.00	-	1200.00	180.00-250.00	1450-1500	524.00-768.00	224.00	-
160.00-180.00	170-240	1600.00	170.00-220.00	1230-1280	254.00-514.00	112.00	22900-30400

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ESTIMATING PRICES

		UNIT	Australia Sydney $/Aus	Canada Vancouver $/Can	Fiji Suva $/Fiji	Hong Kong $/HK	
Preliminaries, including insurance, plant, supervision, temporary services, cleaning, maintenance, etc., on :							
Single storey project -							
City Centre	...	ADD%	-	-	7%-15%	-	
Suburbs	...	ADD%	12%	12%-20%	6%-12%	11.0%	
2/3 storey project -							
City Centre	...	ADD%	-		8%-15%	-	
Suburbs	...	ADD%	13%	15%-20%	7%-13%	11.0%	
4/10 storey project -							
City Centre	...	ADD%	16%	12%-18%	11%-20%	10-15%	
Suburbs	...	ADD%	13%	10%-15%	9%-18%	10-15%	
10/20 storey project -							
City Centre	...	ADD%	20%	15%-20%	12%-22%	12-17%	
Suburbs	...	ADD%	17%	10%-15%	12%-20%	12-17%	
20/35 storey project -							
City Centre	...	ADD%	22%	10%-20%	-	15-20%	
Suburbs	...	ADD%	19%	10%-15%	-	15-20%	

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7%-8%	-	20%-30%	-.	10%-15%	-	-	7%-16%
5%-6%	8%-13%	35%-45%	10%-15%	9%-14%	10%-18%	16%-20%	6%-14%
7%-8%	10%-15%	20%-30%	12%-16%	9%-14%	-	-	8%-16%
5%-6%	10%-15%	35%-45%	10%-15%	8%-12%	10%-18%	16%-20%	7%-15%
8%-9%	10%-18%	20%-30%	13%-16%	9%-14%	10%-20%	8%-12%	12%-21%
5%-6%	8%-16%	35%-45%	10%-15%	8%-12%	8%-18%	10%-15%	11%-19%
9%-10%	10%-18%	20%-30%	13%-18%	9%-14%	17%-25%	7%-10%	13%-23%
6%-7%		35%-45%	12%-16%	8%-14%	15%-25%	7%-11%	13%-21%
10%-11%	10%-18%	20%-30%	13%-18%	9%-15%	17%-25%	7%-10%	-
6%-7%	-	35%-45%	12%-16%	8%-15%	15%-25%	9%-12%	-

MENSURATION

MENSURATION

Figure	Perimeter	Surface Area	Volume
Triangle (base a, altitude h)	_	½ ah	_
Triangle (sides a, b,c; ½ perimeter s)	a + b + c = 2s	$\sqrt{s}$ (s - a) (s - b) (s - c)	_
Circle (radius r)	$2\pi r$	πr^2	_
Arc (radius r, subtended angle A)	$\pi r \frac{A}{180}$	_	_
Sector (arc a, radius r, subtended angle A)	_	½ ar or $\pi r^2 \frac{A}{360}$	_
Segment (chord a, altitude h)	_	$\frac{h^3}{2a}$ + (⅔ *a x h*)	_
Annulus (outer radius R, inner radius r)	_	π (R - r) (R + r)	_
Ellipse (axes a, b)	½ π (a + b)	π ¼ (ab)	_
Parallelogram (base a, altitude h)	_	ah	_
Rhombus (base a, altitude h)	_	ah	_
Trapezoid (parallel sides a, b; altitude h)	_	½ (a + b) h	_
Regular polygon (inscribed radius r, side a, number of sides n)	na	½ ran	_
Rectangular solid or cuboid (edges a, b, c)	_	2 (ab + bc + ca)	abc
Cylinder/Prism (radius r, height h)	_	2 π rh (curved surface) 2 π r (h + r) (total surface)	$\pi r^2 h$
Cone (radius r, height h, slant height l)	_	πrl (curved surface) πr (l + r) (total surface)	⅓ $\pi r^2 h$
Pyramid, rectangular or triangular (base perimeter a, base area A, height h, slant height l)	_	½ al + A	⅓ Ah
Sphere (radius r)	_	$4\pi r^2$	1⅓ πr^3
Helix (number of turns n, pitch p, radius r)	n x $\sqrt{p^2 + (2\pi r)^2}$	_	_

Note : Due to the nature of some shapes, the formulae can only give an approximation.

MENSURATION

IRREGULAR AREAS AND VOLUMES

Trapezoidal Rule

Divide the figure into any number of parts by parallel lines (or offsets) at equal distances apart.

Area = ½ the distance between any two adjacent offsets x (first offset + last offset + twice the sum of all the other offsets).

Simpson's Rule

Divide the figure into any even number of parts by parallel lines (or offsets) at equal distances apart.

Area = ⅓ the distance between any two adjacent offsets x (first offset + last offset + twice the sum of the other odd offsets + four times the sum of the even offsets).

Prismoidal Formula

$$\text{Volume} = \frac{1}{6}\ (A_1 + 4A_2 + A_3)$$

Where I is the length
A_1 and A_3 are the areas of the ends
A_2 is the area of the middle section

MEASURES

LINEAR

1,000mm	= 1m
100cm	= 1m
1,000m	= 1km

SQUARE

1,000,000 sqmm (mm^2)	= 1 sqm (m^2)
10,000 sqcm (cm^2)	= 1 sqm (m^2)
10,000 sqm (m^2)	= 1 ha
1,000,000 sqm (m^2)	= 1 sqkm (km^2)

CUBIC

1,000,000,000cumm (mm^3)	= 1 cum (m^3)
1,000,000cucm (cm^3)	= 1 cum (m^3)
1kl	= 1 cum (m^3)

WEIGHT

1,000g	= 1kg
1,000kg	= 1t

LIQUID

1,000ml	= 1 litre
1,000litres	= 1 cum (m^3)
1,000litres	= 1kl

NUMERIC NOTATION

Exa	= 10^{18}
Peta	= 10^{15}
Tera	= 10^{12}
Giga	= 10^{9}
Mega	= 10^{6}
Kilo	= 10^{3}
Hecto	= 10^{2}
Deca	= 10^{1}
Deci	= 10^{-1}
Centi	= 10^{-2}
Milli	= 10^{-3}
Micro	= 10^{-6}
Nano	= 10^{-9}
Pico	= 10^{-12}
Femto	= 10^{-15}
Atto	= 10^{-18}

MENSURATION

MASS DENSITIES OF MATERIAL

Mass densities shown in this selective list may vary, depending on material composition. Where calculations need to be precise, exact density factors should be obtained from manufacturers or other appropriate sources.

Material	kg/m^3
Aluminium, rolled	2710
Barium	3780
Bentonite	593
Bitumen, prepared	1362
Bolts and nuts, bags	1201
Brass, cast	8330
Bricks (common burnt clay):	
stacked	1602 - 1920
brickwork	1920
Cement :	
clinker	1290 - 1540
Portland	1506
mortar	2162
slurry	1442
Clay :	
compacted	1746
excavated, dry	1089
excavated, wet	1826
fire	1362
lump, dry	1073
lump, wet	1602
Coke	481 - 561
Concrete :	
asphalt	2243
reinforced, 2 per cent steel	2420
Copper, drawn and sheet	8938
Cyprus wood	593
Douglas fir	529
Earth :	
excavated, dry	1249
excavated, moist	1442
excavated, wet	1602
loose, dry	1280
loose, wet	1440 - 1600
Galvanised steel, sheets, bundles	897
Glass, plate	2787
Granite, solid	2691
Gravel :	
loose, dry	1522
with sand, natural	1922
Gypsum, plaster	737
Iron, cast	7208
Jarrah	835
Lead, cast or solid	11325
Lime, hydrate, bags	513

Material	kg/m^3
Lime mortar :	
dry	1650
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CANBERRA
DARWIN
HOBART

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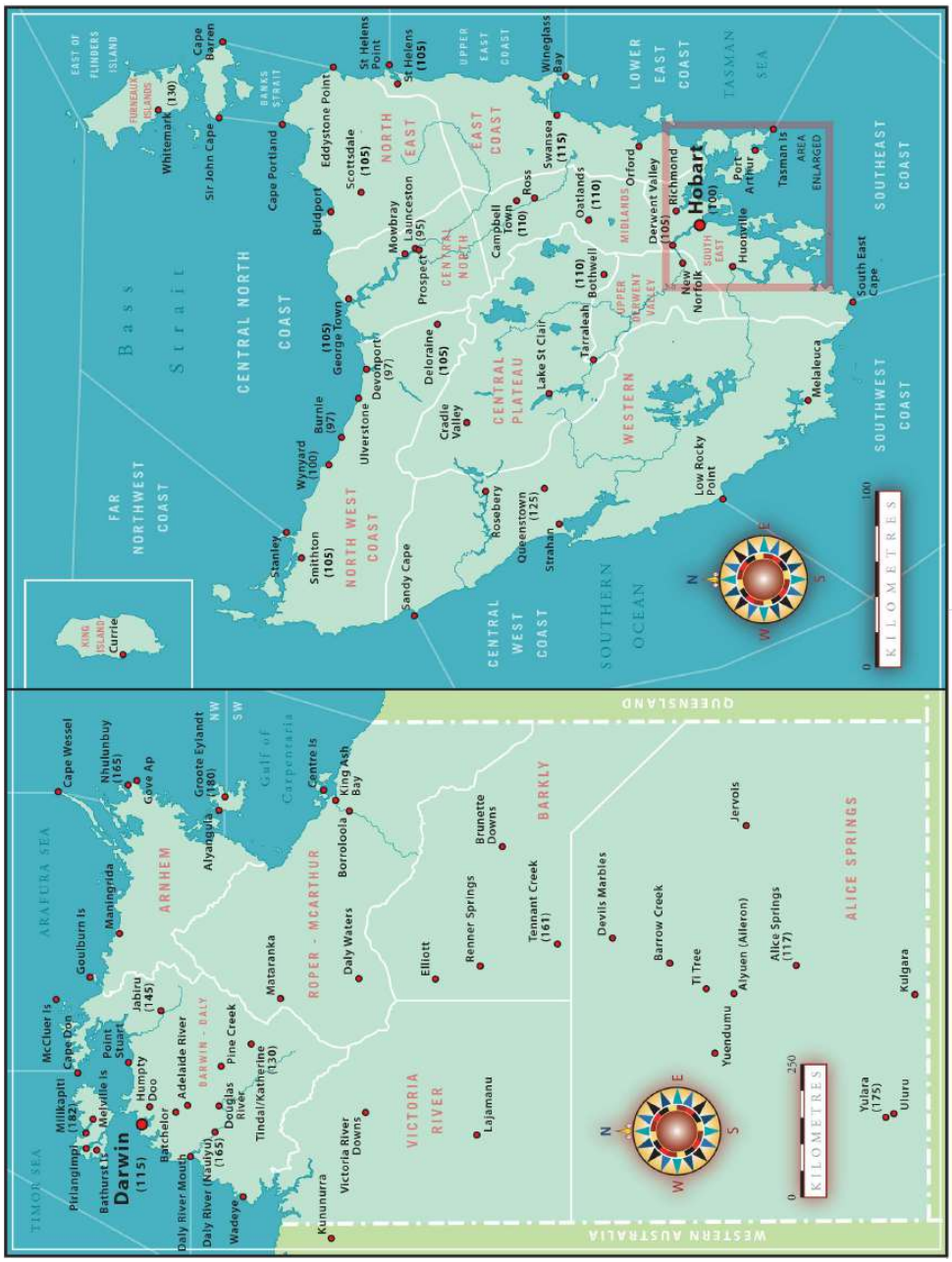
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REGIONAL INDICES NORTHERN TERRITORY

- **Base - ADELAIDE 100**

NOTE :

The above index is a broad indication only of cost variation within the Northern Territory and Tasmania and are applicable to the total project cost. Choice of material, degree of prefabrication and general work load will have a considerable bearing on cost differentials.

ESTIMATING BUILDING COSTS PER SQUARE METRE

CONTENTS

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- Explanatory Notes 1026
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ESTIMATING - BUILDING COSTS PER SQUARE METRE

Costs given hereafter should be used for initial feasibility studies only.

Costs given are average prices for typical buildings at December 31, 2020 within the metropolitan areas, including allowance for Preliminaries and Builders profit and overheads. ADD additional allowances for increased costs (refer Anticipated Building Price Adjustment page 15) and when in country areas (refer Regional Indices page 1024).

Costs given should be adjusted to take account of such factors as sloping sites, foundation problems, high wall to floor ratios, unusual shape and any special design factors where costs can vary considerably from the range given.

Costs are based on the total floor area of all levels measured between the inner faces of external walls.

Costs can provide no more than a rough guide to the probable cost of a building, and whilst in many instances a single rate is sufficient to indicate this ruling average cost, there are many other instances such as banks, function centres, churches, residential buildings, where costs can vary considerably from the range given.

Costs EXCLUDE parking areas. Refer to Sections 9.3 and 10.2 for parking costs.

Costs EXCLUDE land, demolitions, balconies, verandahs, covered ways, external services outside 3.0m from the outside face of the building, external works other than those immediately adjacent to the building, loose or special equipment, furniture, furnishings, legal and professional fees.

For a more detailed breakdown, and to assist in the compilation of a more accurate estimate, refer to pages 49 to 105. For ease of reference identical item numbers have been used.

ADD to the costs given hereafter for : all costs listed above as exclusions, cooling, heating, ventilation, etc., where considered necessary and not included; verandahs and covered ways, particularly for Darwin.

Costs exclude the Goods and Services Tax (G.S.T.).

			UNIT	Can. $	Dwn. $	Hob. $
1.0	**ADMINISTRATION, CIVIC**					
1.1	ADMINISTRATIVE					
1.1.1	ADMINISTRATION OFFICE, SINGLE STOREY - standard finishes, subdivisional partitioning, reception, amenities areas, air-conditioning	...	sqm	2625-2825	2460-2660	2600-2800
1.1.2	ADMINISTRATION OFFICE, 2-3 STOREY - standard finishes, subdivisional partitioning, reception, amenities areas, air-conditioning, lift	...	sqm	2970-3205	2830-3060	2930-3155
1.2	CIVIC					
1.2.3	CIVIC CENTRE, 300-500 CAPACITY - main hall suitable for cabarets, conventions, etc., ante-rooms, small kitchen, standard finishes, no air-conditioning	...	sqm	2605-2805	2400-2580	2205-2375
1.2.5	LIBRARY SUBURBAN - single storey, good standard finishes, air-conditioning, excluding loose fittings	...	sqm	3010-3245	2725-2985	2860-3080
1.2.6	LIBRARY, SUBURBAN - 2-3 storey, good standard finishes, air-conditioned, lift, excluding loose fittings	...	sqm	3065-3305	3075-3320	2985-3220

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGES 52 AND 54 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
1.0	**ADMINISTRATION, CIVIC**					
1.4	DIPLOMATIC					
1.4.1	EMBASSY CHANCERY					
1.4.1.1	Prestige building, high standard finishes and services and including landscaping	...	sqm	3825-4125	-	-
1.4.1.2	Medium standard building with commercial finishes and services	...	sqm	2905-3130	-	-
1.4.2	EMBASSY RESIDENCE					
1.4.2.1	Prestige building, high standard finishes, fully serviced and including landscaping	...	sqm	3870-4175	-	-
1.4.2.2	Medium standard building	...	sqm	2440-2630	-	-
3.0	**BANKS**					
3.1	SUBURBAN, SINGLE STOREY - standard finishes, air-conditioning, bank fittings	...	sqm	3015-3255	3030-3250	3090-3330
3.2	SUBURBAN 2-3 STOREY - standard finishes, air-conditioning, bank fittings, lift	...	sqm	3150-3400	3070-3320	3155-3400
3.4	FIT-OUT ONLY :					
3.4.1	Of lettable space to provide single level suburban bank including standard finishes, air-conditioning, bank fittings	...	sqm	1980-2130	1960-2100	2080-2245
4.0	**EDUCATIONAL**					
4.1	SCHOOLS (TEACHING)					
4.1.1	PRIMARY - single storey, standard finishes, buildings only, no air-conditioning	...	sqm	2200-2375	2385-2550	1855-2000
4.1.2	SECONDARY - maximum three storeys, standard finishes, built-in cupboards and fittings, buildings only, no air-conditioning					
4.1.2.1	Overall	...	sqm	2350-2535	2530-2730	2375-2560
4.1.2.2	Classrooms	...	sqm	2255-2430	2350-2540	2305-2485
4.1.2.3	Laboratories	...	sqm	3100-3340	4400-4750	3025-3260
4.1.2.4	Covered ways and balconies, refer 4.5.5 and 4.5.6	...	Note			
4.1.3	TECHNICAL - maximum two storeys, standard finishes, built-in cupboards and fittings, buildings only, no general air-conditioning					
4.1.3.1	Overall	...	sqm	2435-2620	2750-2960	2490-2685
4.1.3.2	Classrooms	...	sqm	2335-2520	2530-2730	2355-2540
4.1.3.3	Arts and crafts	...	sqm	2310-2490	2530-2730	2375-2560
4.1.3.4	Catering	...	sqm	2585-2785	2740-2960	2570-2770
4.1.3.5	Engineering workshops	...	sqm	2940-3170	2700-2920	2915-3140
4.1.3.6	Covered ways and balconies, refer 4.5.5 and 4.5.6	...	Note			
4.2	SCHOOLS (SUPPORT)					
4.2.1	SECONDARY- maximum three storeys, standard finishes, built-in cupboards and fittings, buildings only, no general air-conditioning					
4.2.1.1	Administration	...	sqm	2255-2430	2430-2640	2175-2345
4.2.1.2	Gymnasium	...	sqm	2175-2340	2630-2840	2220-2390
4.2.1.3	Residential	...	sqm	2375-2565	2425-2650	2385-2570
4.2.1.4	Covered ways and balconies, refer 4.5.5 and 4.5.6	...	Note			

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 57 TO 61 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
4.0	**EDUCATIONAL**					
4.2	SCHOOLS (SUPPORT) (cont'd)					
	ADD extra for :					
4.3	External works and services to secondary schools at approximately 20/30%	...	Note			
4.5.5	BALCONIES - walkway type, reinforced concrete, with balustrade to one side, no roof					
4.5.5.1	Basic standard finish	...	sqm	775-850	480-620	520-665
4.5.5.2	Medium standard finish	...	sqm	850-900	630-980	660-845
4.5.6	COVERED WAYS - open sided					
4.5.6.1	Basic standard finish	...	sqm	500-550	510-590	520-560
4.5.6.2	Medium standard finish	...	sqm	550-650	630-700	610-660
6.0	**HOSPITALS, HEALTH**					
6.1	HOSPITALS					
6.1.1	DISTRICT - single storey, 60 bed, operating theatre, partial air-conditioning	...	sqm	4320-4660	4550-4900	4040-4355
		...	bed	233400-251600	250000-300000	218200-235200
6.1.4	PRIVATE - single storey, 40 bed, operating theatre, partial air-conditioning	...	sqm	3370-3630	4350-4730	3245-3500
		...	bed	144900-156100	200000-280000	139600-150500
6.1.5	NURSING HOME - aged person, refer 13.5.4	...	Note			
6.2	ANCILLARY					
6.2.3	GROUP PRACTICE SURGERY - single storey, standard finishes with consulting rooms, surgery, partial air-conditioning	...	sqm	2200-2375	2520-2730	2220-2390
7.0	**HOTELS, MOTELS, TAVERNS**					
7.1	SUBURBAN					
7.1.1	TAVERN - single storey, standard construction and finishes, partial air-conditioning, excluding loose furniture	...	sqm	3080-3320	3150-3400	3035-3270
7.2	CITY HOTELS					
7.2.1	MEDIUM TO HIGH RISE, INTERMEDIATE STANDARD - three star, medium standard finishes, air-conditioning, excluding fitting out					
7.2.1.1	Overall					
7.2.1.1.1	Hotel with 30.00 to 32.00 sqm total floor area per bedroom	...	sqm	3600-3880	3500-3750	3795-4090
		...	bedroom	111600-120300	110000-130000	117600-126800
7.2.1.1.2	Hotel with 45.00 sqm total floor area per bedroom	...	sqm	3600-3880	3720-4050	3900-4205
		...	bedroom	162000-174500	175000-190000	175500-189000
7.2.1.2	ADD extra for :					
	Fitting out, total building	...	bedroom	41400-44500	45000-50000	39000-42000
7.2.2.6	Balconies, individual, refer 13.3.1.6 to 13.3.1.7	...	Note			

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 64 TO 67 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
7.0	**HOTELS, MOTELS, TAVERNS**					
7.3	COUNTRY MOTELS					
7.3.1	STANDARD - travellers' type, single or two storey, standard finishes, individual facilities, unit air - conditioning, excluding fitting out					
7.3.1.1	Overall	…	sqm	2540-2740	2500-2690	2645-2850
		…	unit	85800	85000	89350
7.3.1.2	Accommodation units	…	sqm	2530-2725	2610-2810	2715-2930
		…	unit	60450	62000	64950
7.3.1.3	Reception, dining, kitchen	…	sqm	2285-2465	2320-2510	2355-2540
		…	unit	22550	23000	23250
7.3.1.4	ADD extra for :					
	Fitting out total building	...	unit	26000-33000	24500-31000	22700-28700
8.0	**INDUSTRIAL**					
8.1	WAREHOUSES					
	Low bay provides 3500/4500mm and high bay 6500/9000mm headroom respectively	...	Note			
8.1.1	SINGLE STOREY for LETTING, standard shell construction, metal roof, roller shutters, electrical service to board, plumbing service only; no ventilation, sprinklers or fit-out :					
	LOW BAY					
8.1.1.1	Brick external walls	...	sqm	775-840	820-880	725-780
8.1.1.2	Precast or tilt-up concrete external walls	...	sqm	830-890	830-890	715-770
	HIGH BAY					
8.1.1.3	Brick external walls	...	sqm	940-1010	930-1010	865-935
8.1.1.4	Precast or tilt-up concrete external walls	...	sqm	925-1000	965-1050	855-925
8.1.2	SINGLE STOREY for OWNER OCCUPATION, metal roof, roller shutters, small office and amenities, all services, no ventilation or sprinklers :					
8.1.2.1	LOW BAY					
	Brick external walls	...	sqm	925-1000	970-1040	865-935
8.1.2.2	Precast or tilt-up concrete external walls	...	sqm	925-1000	1060-1140	865-935
	HIGH BAY					
8.1.2.3	Brick external walls	...	sqm	1025-1105	1070-1170	1025-1105
8.1.2.4	Precast or tilt-up concrete external walls	...	sqm	1025-1105	1150-1250	1000-1075
8.2	FACTORIES					
	Small span is up to 10.0m	...	Note			
8.2.1	SINGLE STOREY for LETTING, standard shell construction, metal roof, roller shutters, electrical service to board, plumbing service only; no ventilation, sprinklers or fit-out :					
8.2.1.1	Framed and metal clad external walls	...	sqm	775-840	800-860	720-775
8.2.1.2	Brick external walls	...	sqm	775-840	800-860	730-785
8.2.1.3	Precast or tilt-up concrete external walls	...	sqm	775-840	820-895	720-775

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 68 TO 71 AND 74 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
8.0	**INDUSTRIAL**					
8.2	FACTORIES (cont'd)					
8.2.2	SINGLE STOREY SMALL SPAN for OWNER OCCUPATION, metal roof, roller shutters, small offices and amenities, all services, no ventilation or sprinklers :					
8.2.2.1	Brick external walls	...	sqm	900-975	980-1060	925-995
8.2.2.2	Precast or tilt-up concrete external walls	...	sqm	925-1000	1075-1160	910-980
9.0	**OFFICES**					
9.1	LETTABLE					
	Costs exclude parking, tenants subdivisional partitions and any other special owner or tenant requirements	...	Note			
	Buildings are all taken as above ground level	...	Note			
	Parking costs, refer 9.3 and 10.2	...	Note			
	Costs given hereafter are for either : FINISHED FLOOR, being where the building is handed over in a fully furnished condition.					
	SHELL AND CORE, being where the typical office floors and lift lobbies of the building are handed over in a bare state whilst the core components are handed over in a finished state - refer page 32 for explanation of SHELL AND CORE					
9.1.1	LOW RISE (BASIC SERVICES)					
	SINGLE STOREY - standard finishes, no air-conditioning -					
9.1.1.1	Finished floor lettable type	...	sqm	1575-1700	1650-1760	1535-1655
9.1.1.2	Shell and core lettable type	...	sqm	1225-1325	1400-1500	1340-1440
	TWO STOREY - standard finishes, no air-conditioning or lifts -					
9.1.1.3	Finished floor lettable type	...	sqm	1765-1905	1780-1920	1765-1900
9.1.1.4	Shell and core lettable type	...	sqm	1605-1735	1680-1810	1595-1720
	LOW RISE (FULLY SERVICED)					
	SINGLE STOREY - standard finishes, air-conditioned -					
9.1.2.1	Finished floor lettable type	...	sqm	1865-2015	2010-2160	1885-2035
9.1.2.2	Shell and core lettable type	...	sqm	1710-1840	1850-1990	1715-1850
	TWO STOREY - standard finishes, air-conditioned, no lift -					
9.1.2.3	Finished floor lettable type	...	sqm	2200-2370	2410-2500	2185-2355
9.1.2.4	Shell and core lettable type	...	sqm	2060-2225	2130-2300	1985-2140
	THREE STOREY - standard finishes, air-conditioned, lift -					
9.1.2.5	Finished floor lettable type	...	sqm	2355-2535	2620-2830	2445-2635
9.1.2.6	Shell and core lettable type	...	sqm	2205-2380	2420-2600	2255-2430
	4 - 7 STOREY - standard finishes and facilities, air-conditioned, slow lifts, no sprinklers -					
9.1.2.7	Finished floor lettable type	...	sqm	2580-2780	2740-2960	2640-2845
9.1.2.8	Shell and core lettable type	...	sqm	2410-2600	2565-2775	2445-2635

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 75 AND 80 TO 83 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
9.0	**OFFICES**					
9.1	LETTABLE (cont'd)					
9.1.3	HIGH RISE (FULLY SERVICED)					
	7 - 20 STOREY - standard finishes and facilities, air-conditioned, medium speed lifts, sprinklers -					
9.1.3.1	Finished floor lettable type	...	sqm	3345-3605	3380-3670	3055-3295
9.1.3.2	Shell and core lettable type	...	sqm	2960-3195	3205-3440	2870-3095
9.1.5	FITTING OUT OFFICES - including partitions, amenities, additional facilities, reception and board room areas, etc., no loose furniture, additional cost :					
9.1.5.1.1	Standard quality fit-out	...	sqm	575-625	620-730	510-545
9.1.5.2.1	Medium quality fit-out	...	sqm	1110-1190	1170-1300	1020-1095
9.1.5.1.2	ADD extra for :					
	Standard work stations (one per 15 sqm)	...	sqm	200-250	180-210	150-170
9.2	ADMINISTRATIVE					
	Refer 1.1	...	Note			
9.3	PARKING					
9.3.1	UNDERCROFT PARKING TO 1 - 3 STOREY OFFICES Open at ground or slightly reduced ground level including additional excavation and structural costs	...	sqm	700-750	870-950	765-825
9.3.2	PARTIALLY UNDERGROUND, one level including additional excavation and structural costs, ramps, sprinklers, minimal mechanical ventilation, NO roof over as within building	...	sqm	840-910	1350-1450	1555-1675
10.0	**PARKING**					
10.2	OPEN PARKING AREAS, including bitumen paving, stormwater drainage, minimal lighting, and some landscaping	...	sqm	100-120	140-170	91-98
		...	car	3600-4320	4900-5900	3275-3530
10.3	PARKING STATIONS					
10.3.1	2 - 3 STOREY - reinforced concrete construction with bitumen paving at ground level, open sides, minimal toilet facilities; no lifts or ventilation					
10.3.1.1	Ground + one Level	...	sqm	625-675	840-900	620-670
		...	car	18100-19600	24300-26200	18000-19400
10.3.1.2	Ground + two Levels	...	sqm	675-725	830-890	785-845
		...	car	18900-20300	23200-25000	22000-23700
	BELOW OFFICES - refer 9.3.1 and 9.3.2					
11.0	**RECREATIONAL**					
11.2	CLUBHOUSES					
11.2.1	CHANGE ROOM/TOILET BUILDING - single storey, standard construction and basic finishes, large change area and adjoining showers/toilets	...	sqm	2660-2870	3100-3350	3100-3340
11.2.2	CLUBHOUSE AND CHANGE ROOMS - single storey, standard construction and finishes, large bar and lounge, small kitchen, dining area, large basic finished change rooms and shower/toilets; no air-conditioning	...	sqm	2630-2835	2900-3130	2835-3055

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 84 TO 86, 88 AND 91 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
11.0	**RECREATIONAL**					
11.5	COMMUNITY RECREATION CENTRE -					
11.5.1	BASIC standard with minimum amenities	...	sqm	1320-1420	1360-1470	1225-1320
11.5.2	MEDIUM standard with moderate amenities	...	sqm	1545-1665	1530-1660	1535-1655
11.6	GYMNASIUM - single storey, standard construction, vinyl sports floor finish; excludes change rooms/toilets (refer separate cost)	...	sqm	2200-2370	2280-2470	2305-2485
11.6.1	ADD extra for : Sprung timber floor	...	sqm	165	155	133
11.7	SQUASH COURTS					
11.7.1	BASIC developer standard including change areas, small toilet, no air-conditioning	...	sqm	1610-1740	1850-2000	1665-1795
		...	court	145100-156400	166800-179700	150000-161700
11.7.2	HIGH standard private club courts including toilets, showers, change area and viewing gallery, no air-conditioning	...	sqm	2130-2295	2330-2520	2170-2340
		...	court	255500-275500	300000-330000	260500-280500
	Add extra for : Glass rear wall	...	court	32000-33000	33000-36000	31000-32000
11.14	CARAVAN PARKS					
11.14.1	BASIC STANDARD, communal facilities, electric power to all bays, small office, roads and parking	...	bay	22100-23900	24200-26100	23000-24800
11.14.2	MEDIUM STANDARD, communal facilities, electric power to all bays, office and retail, laundromat, barbecue areas, small swimming pool, roads and parking	...	bay	31300-33700	35800-38600	31800-34300
12.0	**RELIGIOUS**					
12.1	CHURCH HALL - single storey, standard construction, basic finishes, small kitchen, toilets, no air-conditioning	...	sqm	1495-1615	1550-1710	1275-1375
12.2.2	CHAPEL, CHURCH, SYNAGOGUE - medium standard construction and finishes, including seating	...	sqm	2350-2530	2650-2890	2455-2645
13.0	**RESIDENTIAL**					
	Costs exclude the Goods and Services Tax	...	Note			
13.1	HOUSES - SINGLE UNIT					
	Costs given hereafter are based on the total floor area between inner faces of external walls. To adjust for measurement based on outer face of external walls REDUCE the foregoing Costs Per Square Metre by approximately 10%	...	Note			

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 92 TO 94 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

				UNIT	Can. $	Dwn. $	Hob. $
13.0	**RESIDENTIAL**						
13.1	HOUSES - SINGLE UNIT (cont'd)						
13.1.1	PROJECT HOUSE - tiled roof and built on a flat site :						
	Basic standard finish and fit-out -						
13.1.1.1	90/110sqm - Brick veneer	...		sqm	1395-1505	-	1435-1545
13.1.1.2	- Full brick	...		sqm	1490-1610	1850-1990	1550-1670
13.1.1.3	120/140sqm - Brick veneer	...		sqm	1370-1480	-	1400-1510
13.1.1.4	- Full brick	...		sqm	1395-1505	1920-2070	1500-1615
	Medium standard finish and fit-out -						
13.1.1.5	120/140sqm - Brick veneer	...		sqm	1490-1610	-	1565-1690
13.1.1.6	- Full brick	...		sqm	1590-1710	2050-2220	1690-1820
13.1.1.9	Elevated, concreted lower area, concrete piers, metal clad construction to upper level (upper level measurement)	...		sqm	-	4325-4700	-
	ADD extra for :						
13.1.1.10	Verandah - Single storey	...		sqm	-	630-690	-
13.1.1.11	- Elevated, steel framed, timber deck	...		sqm	-	760-830	-
13.1.2	INDIVIDUAL HOUSE - tiled roof, approximately 150/350 sqm -						
	Medium standard finish						
13.1.2.1	- Framed	...		sqm	1610-1740	2060-2230	1825-1970
13.1.2.2	- Brick veneer	...		sqm	1780-1920	-	1950-2100
13.1.2.3	- Full brick	...		sqm	1805-1945	2050-2220	2085-2250
	High standard finish, no air-conditioning						
13.1.2.4	- Framed	...		sqm	2215-2385	2520-2740	2365-2550
13.1.2.5	- Brick veneer	...		sqm	2360-2540	-	2525-2720
13.1.2.6	- Full brick	...		sqm	2455-2645	2450-2650	2635-2840
13.1.2.9	Elevated type, concreted lower area, metal clad construction to upper level (upper level measurement)	...		sqm	-	4400-4750	-
	ADD extra for :						
	Verandahs refer 13.1.1.10 and 13.1.1.11	...		Note			
13.2	MULTI UNIT - LOW DENSITY						
13.2.1	TOWN HOUSE - two storey, full brick with tiled roof						
13.2.1.1	Medium standard finish	...		sqm	2215-2385	1960-2110	2430-2620
13.2.1.2	High standard finish	...		sqm	2430-2620	2180-2350	2645-2850
13.2.2	APARTMENTS - maximum three stories, one or two bedroom units, excluding balconies, no lift						
13.2.2.1	Basic standard finish	...		sqm	2550-2750	2080-2240	2430-2620
		...		apart	153000-165000	140000-210000	145800-157200
13.2.2.2	Medium standard finish	...		sqm	2840-3060	2150-2320	2550-2750
		...		apart	170400-183600	160000-240000	153000-164900
	ADD extra for :						
	Individual unit balcony, reinforced concrete						
13.3.1.5	Basic standard	...		sqm	750-800	610-660	590-950
13.3.1.6	Medium standard	...		sqm	850-900	665-725	625-985
13.3.1.7	High standard	...		sqm	900-1000	1110-1220	790-1215
	Walkway type balconies, refer 4.5.5	...		Note			

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 95 TO 98 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
13.0	**RESIDENTIAL**					
13.5	AGED PERSONS HOMES					
13.5.1	SINGLE STOREY self care housing units with medium standard finish, partial air-conditioning, communal laundries	...	sqm	2695-2905	2460-2680	2835-3055
		...	unit	161700-174300	210000-230000	170100-183300
	ADD extra for :					
	Support buildings -					
13.5.1.1	Single storey amenities and social centre	...	sqm	2455-2645	2300-2485	2535-2735
13.5.4	NURSING HOME, built as homes, single storey, self care and central care units, medium standard finish, air-conditioning, communal dining, amenities and social areas	...	sqm	2580-2780	2700-2925	2575-2775
		...	unit	161200-173800	220000-240000	160900-173500
	ADD extra for :					
13.5.4.1	High care units	...	sqm	185-200	200-215	158-170
13.7	CARPORTS, GARAGES					
13.7.1	Garage, concrete floor, metal clad walls and roof on timber framing, roller shutter door	...	sqm	675-725	-	905-975
13.7.2	Garage, concrete floor, cavity brick walls, metal deck roof, roller shutter doors	...	sqm	825-890	-	1070-1155
13.7.3	Carport, concrete floor, metal clad roof on steel framing	...	sqm	425-455	610-690	575-620
14.0	**RETAIL**					
14.1	SUBURBAN					
14.1.1	NEIGHBOURHOOD SHOPS - standard shell construction including shop-fronts, plasterboard ceilings, electrical service to board, cold water supply to fixture point only and drainage. No fittings, hot water, air-conditioning, sprinklers or malls					
14.1.1.1	Single storey	...	sqm	1095-1180	1125-1215	970-1045
14.1.1.2	Two storey	...	sqm	1155-1245	1300-1410	1065-1150
	ADD extra for :					
14.1.1.3	FIT OUT shell including finishes, shop fittings and services, but excluding air-conditioning	...	sqm	300	315	228
14.1.1.4	Air-conditioning (Package Unit)	...	sqm	295-315	445-485	330-345
14.1.1.5	Toilets	...	Note			
14.1.1.6	Awnings	...	Note			
14.1.2	SUPERMARKETS - standard construction including shop fronts, standard finishes, all services, air - conditioning and sprinklers, standard amenities, offices, etc.; excluding					
14.1.2.1	cold rooms etc. and shop fittings	...	sqm	1925-2070	1930-2090	1885-2035
14.1.2.2	ADD extra for :					
	Shop fittings and installation of cold and freezer rooms	...	sqm	400-430	410-440	350-380

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGES 101 TO 103 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

			UNIT	Can. $	Dwn. $	Hob. $
14.0	**RETAIL**					
14.1	SUBURBAN (cont'd)					
14.1.3	SHOPPING CENTRE - standard shell construction including shop fronts, plasterboard ceilings, electrical service to board, plumbing service only, air-conditioning, excluding fit-out sprinklers and					
14.1.3.1	mall/arcade areas	...	sqm	1830-1975	1820-1950	1810-1950
	ADD extra for :					
14.1.3.2	Fully enclosed mall/arcade, good standard finishes, air-conditioned and landscaping, etc.	...	sqm	2750-2965	2900-3150	2805-3025
14.1.3.3	Trading area fit-out including wall and floor finishes, additional electrical, plumbing services	...	sqm	675-725	810-870	690-745

NOTE - ADD - INCREASED COSTS
- COUNTRY LOADINGS
- REFER PAGE 1026 FOR NOTES ON USE OF THESE COSTS
- REFER PAGE 103 FOR A DETAILED ELEMENTAL BREAKDOWN OF SYDNEY COSTS

DETAILED PRICES

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The Detailed Prices are indicative average prices for reasonable quantities of work and would apply to projects in the $1,250,000/$10,000,000 range and having average site conditions.

Prices, unless otherwise described, are for the completely installed item or work including sales tax and overheads and profit normally applying at the commencement of 2021.

No allowance has been included in the prices of the respective trade sections for Preliminaries items such as site establishment, supervision, large plant, scaffolding, notices and fees, insurance, etc.; whilst prices include allowance for small tools and hand plant, separate provision must be made for Preliminaries items as described in order to arrive at a realistic total cost.

As prices can differ appreciably because of the nature and specific requirements of each particular contract, it is not recommended that they be used for quotations without checking in detail the requirements of the contract.

Costs excludes the Goods and Services Tax (G.S.T.).

In these Detailed Prices an allowance of 5% has been included for Builder's overheads and profit.

EXCAVATION

		UNIT	Can. $	Dwn. $	Hob. $
EXCAVATION					
Excavate over site to reduce levels :					
In sand	...	cum	-	-	25.30
In clay	...	cum	25.00	37.50	35.20
In hard rock	...	cum	150.00	190.00	162.00
Excavate trench N.E. 1.00m deep :					
In sand	...	cum	-	-	52.60
In clay	...	cum	120.00	90.00	82.10
In hard rock	...	cum	195.00	235.00	243.00
FILLING					
Excavated material as filling	...	cum	18.00	37.50	13.50
Clean sand filling	...	cum	65.00	65.00	81.30
Crushed rock filling	...	cum	75.00	98.00	134.50
SURFACE TREATMENTS					
Trim, grade and compact surfaces of groundworks	...	sqm	5.00	5.00	3.60
Termite treatment	...	sqm	-	17.00	-
PLANKING AND STRUTTING					
Sides of trench excavation N.E. 1.00m deep :					
In sand	...	sqm	Incl.	-	Incl.
In clay	...	sqm	Incl.	Incl.	Incl.
SHEET STEEL PILING					
Sheet steel piling including all supports :					
74kg -					
Temporary	...	sqm	400.00	-	-
Permanent	...	sqm	450.00	-	-
ADD extra for :					
Toe, allow x 1/3 height retained	...	Note			

EXTERNAL WORKS

PAVING, ETC.					
Basecourse including grading, rolling and consolidating to receive paving :					
Crushed rock/blue metal -					
150mm thick	...	sqm	22.00	24.00	21.50
250mm thick	...	sqm	35.00	38.00	36.50
Graded gravel - 150mm thick	...	sqm	22.00	20.00	-
Seal and two coats sprayed bitumen surfacing	...	sqm	24.00	24.00	12.70
Hot bituminous concrete :					
25mm thick	...	sqm	28.00	31.00	28.20
40mm thick	...	sqm	36.00	38.00	31.20
Brick paving laid on prepared foundation :					
200 x 200 x 50mm paving brick	...	sqm	75.00	-	-
200 x 200 x 80mm paving brick	...	sqm	-	-	-
230 x 110 x 65/76mm laid flat	...	sqm	75.00	65.00	80.70
230 x 110 x 76mm laid on edge	...	sqm	90.00	80.00	81.10
225 x 185 x 38mm paving brick	...	sqm	75.00	-	-
230 x 150 x 50mm paving brick	...	sqm	75.00	-	-
400 x 400 x 40mm eurotile	...	sqm	-	-	-

EXTERNAL WORKS

		UNIT	Can. $	Dwn. $	Hob. $
PAVING, ETC. (cont'd)					
Interlocking block paving laid on prepared foundation :					
60mm thick block	...	sqm	75.00	72.00	81.20
80mm thick block	...	sqm	85.00	85.00	87.70
ADD extra for :					
75mm thick compacted sand bed	...	sqm	9.00	8.50	6.70
Black polythene underlay	...	sqm	4.00	5.50	2.45
25mm mortar bed	...	sqm	12.00	10.80	7.50
Cast in situ concrete kerb including reinforcement, formwork and surface finish :					
150 x 150mm	...	m	48.00	48.00	56.90
250 x 150mm	...	m	65.00	62.00	65.20
Extruded in situ concrete kerb :					
300 x 150mm kerb	...	m	65.00	58.00	-
425 x 300mm standard kerb and gutter	...	m	85.00	74.00	-
FENCING					
Standard butt jointed paling fence 1500mm high	...	m	100.00	160.00	70.50
Galvanised chain mesh rail-less fence, 1800mm high	...	m	120.00	95.00	89.80
Security fence of galvanised chain mesh and three rows barbed wire to top, 2100mm high	...	m	150.00	125.00	67.00

CONCRETE WORK

BASE PRICES					
Concrete, delivered to site :					
20 MPa	...	cum	190.00	241.00	190.00
25 MPa	...	cum	205.00	250.00	195.00
32 MPa	...	cum	212.00	262.00	204.00
Bar reinforcement :					
Cut, bent and delivered	...	t	1650.00	1850.00	1700.00
Placing and fixing in position	...	t	850.00	1050.00	1100.00
Fabric reinforcement, SL72 supply only, full sheets	...	sqm	9.50	9.80	6.00
REINFORCED CONCRETE 25 MPa					
Column or pier foundation, strip footing	...	cum	265.00	290.00	276.00
Slabs and thickening on fill :					
Not exceeding 150mm thick	...	cum	296.00	295.00	293.00
150/300mm thick	...	cum	296.00	295.00	293.00
Suspended slabs :					
Not exceeding 150mm thick	...	cum	296.00	310.00	289.00
150/300mm thick	...	cum	296.00	310.00	289.00
Walls :					
Not exceeding 150mm thick	...	cum	305.00	315.00	299.00
150/300mm thick	...	cum	305.00	315.00	299.00

CONCRETE WORK

		UNIT	Can. $	Dwn. $	Hob. $
REINFORCED CONCRETE 25 MPa (cont'd)					
Columns	...	cum	325.00	318.00	326.00
Attached beams	...	cum	290.00	310.00	286.00
ADD extra for :					
White cement	...	cum	95.00	-	132.00
Brightonlite cement	...	cum	75.00	-	115.00
FORMWORK					
Prices assume moderate quantities with five uses and strutting up to 4.50 metres	....	Note			
CLASS 3 FORMWORK					
Soffit of suspended slab :					
100/200mm thick	...	sqm	120.00	110.00	148.50
200/300mm thick	...	sqm	120.00	115.00	151.50
Sides of walls	...	sqm	120.00	115.00	149.50
Sides of isolated square or rectangular columns	...	sqm	120.00	115.00	149.50
Sides and soffit of attached beams	...	sqm	120.00	115.00	156.00
Edges, sides or risers :					
Not exceeding 100mm high	...	m	36.00	26.00	31.80
100/200mm high	...	m	42.00	32.00	50.00
200/300mm high	...	m	48.00	42.00	55.50
REINFORCEMENT					
BAR					
Structural grade round bar reinforcement	...	t	2650.00	2850.00	2940.00
Ditto in ligatures	...	t	2650.00	2850.00	2940.00
Tempcore, deformed bar reinforcement	...	t	2650.00	2850.00	2940.00
FABRIC					
SL92	...	sqm	24.00	24.60	19.55
SL82	...	sqm	18.00	20.00	14.95
SL72	...	sqm	15.00	18.00	14.95
SL62	...	sqm	12.00	14.80	12.45
PRECAST CONCRETE					
Prices include supply, delivery, mobile craneage, erection and fixing in position	...	Note			
WALLING					
Non load bearing wall panel with a smooth off form grey external face and a floated finish internal face :					
150mm thick	...	sqm	250.00	290.00	271.00
175mm thick	...	sqm	275.00	315.00	308.00
200mm thick	...	sqm	300.00	360.00	343.00
ADD extra for :					
Galvanised fixing brackets	...	sqm	25.00	26.00	-
Profiled or fenestrated panels	...	sqm	65.00	66.00	61.50
Surface finish :					
Water washed	...	sqm	40.00	41.00	46.50
Load bearing panels	...	Note			
Tower craneage, additional cost if required	...	Note			

BRICKWORK AND BLOCKWORK

		UNIT	Can. $	Dwn. $	Hob. $
Prices for common brickwork are based on the cheapest general purpose common brick available.	...	Note			
BASE PRICES					
Bricks and blocks delivered to site :					
Bricks, standard common (48.50/sqm)	...	per 1000	600.00	-	956.00
Bricks, modular common (33.50/sqm)	...	per 1000	-	-	-
Facing bricks, standard (48.50/sqm)	...	per 1000	900.00	-	1076.00
Facing bricks, modular (33.50/sqm)	...	per 1000	-	-	-
Blocks, hollow concrete masonry, natural colour :					
100 x 200 x 400mm (12.5/sqm)	...	per 100	280.00	310.00	255.00
200 x 200 x 400mm (12.5/sqm)	...	per 100	350.00	385.00	428.00
Cement mortar per 1000 standard bricks	...	per 1000	100.00	128.00	85.00
Labour - sub-contract rate (excluding mortar) for :					
Standard bricks	...	per 1000	2000.00	-	1100.00
Modular bricks	...	per 1000	-	-	-
Masonry blocks -					
100 x 200 x 400mm, hollow	...	per 100	550.00	390.00	365.00
200 x 200 x 400mm, hollow	...	per 100	650.00	420.00	470.00
BRICKWORK					
Standard common brickwork :					
Reduced brickwork	...	per 1000	2450.00	-	2390.00
110mm wall	...	sqm	130.00	-	116.00
230mm wall	...	sqm	260.00	-	232.00
Face brickwork, extra over base price standard common brickwork for facing with base price facing bricks including pointing and cleaning down, in stretcher bond	...	sqm	20.00	-	22.00
ADD or DEDUCT for each variation of $5.00 per thousand in supply cost of facing bricks in stretcher bond	...	sqm	0.26	-	0.26
BLOCKWORK					
Natural colour hollow concrete blocks, 200mm high coursing in :					
100mm thick wall	...	sqm	85.00	80.00	89.20
150mm ditto	...	sqm	110.00	96.00	108.50
200mm ditto	...	sqm	130.00	120.00	130.50
Natural coloured solid concrete blocks, 200mm high coursing in :					
100mm thick wall	...	sqm	90.00	86.00	119.50
150mm ditto	...	sqm	-	-	149.50

STRUCTURAL STEELWORK

		UNIT	Can. $	Dwn. $	Hob. $
BASE PRICES					
Average supply only price of steelwork (based on 10-20 tonnes) :					
Universal beams and columns	...	t	2425.00	3680.00	1600.00
Taper flange beams and channels	...	t	2560.00	3680.00	2050.00
Angles	...	t	2340.00	3680.00	1800.00
Hollow sections	...	t	3300.00	3680.00	2050.00
Average price of fabrication, delivery to site and erection of steelwork :					
Universal beams and columns	...	t	5000.00	4370.00	2875.00
Taper flange, beams and channels	...	t	5000.00	4370.00	2925.00
Angles	...	t	6000.00	4370.00	3375.00
Hollow sections	...	t	6000.00	4370.00	3125.00
SUPPLY AND ERECT					
Prices include for shop drawings, fabrication, hoisting, erection and fixing complete and are based on a content of at least 10 tonnes	...	Note			
ADD extra for :					
Surface treatment	...	Note			
Universal columns and beams :					
Up to 40 Kg/m	...	t	7500.00	9250.00	5350.00
40 - 100 Kg/m	...	t	5700.00	9250.00	5350.00
Over 100 Kg/m	...	t	5400.00	9250.00	5350.00
Taper flange beams and channels :					
Up to 14 Kg/m	...	t	7500.00	9150.00	6500.00
Over 14 Kg/m	...	t	6500.00	9150.00	6500.00
Hollow sections :					
Up to 15 Kg/m	...	t	7500.00	9500.00	6450.00
Over 15 Kg/m	...	t	6500.00	9500.00	6450.00
Purlins and girts :					
Channels	...	t	8500.00	9250.00	6200.00
Angles and tees	...	t	8500.00	9250.00	6200.00
Plate connections :					
Base and cap plates	...	t	10000.00	11500.00	7150.00
Attached/loose connections	...	t	10000.00	11500.00	7150.00
Proprietary purlins bolted to steel, excluding bolts :					
Z or C 10019	...	m	25.00	22.00	20.00
Z or C 15019	...	m	28.00	26.00	26.80
Z or C 20019	...	m	32.00	36.00	32.10
SURFACE TREATMENT					
Prices given are an indication only and depend on the section and mass of steel to be treated	...	Note			
Lightly sandblast and paint one coat of zinc phosphate primer :					
Columns, beams, channels, etc.	...	t	475.00	580.00	465.00
Small angles, tubular sections, etc.	...	t	600.00	720.00	520.00
Class 3 sandblast and zinc silicate treatment :					
Columns, beams, channels, etc.	...	t	800.00	895.00	800.00
Small angles, tubular sections, etc.	...	t	950.00	1150.00	900.00
Hot dip galvanise :					
Columns, beams, channels, etc.	...	t	1300.00	1550.00	1340.00
Small angles, tubular sections, etc.	...	t	1500.00	1650.00	1500.00

CARPENTRY

BASE MATERIAL PRICES

The following tables indicate the comparative supply prices of commonly used timber sizes in standard lengths and of the stress grade indicated.

CANBERRA

LOCAL HARDWOOD			LOCAL PINE						OREGON		
			Untreated				Pressure Treated				
Sawn Size	Green Sawn F11	Dry Dressed F17	Nom. Green Sawn Size	Dry Dressed Size	Dry Sawn	Dry Dressed	Dry Sawn F7	Dry Dressed	Nom. Size	Merch. Grade	Select Grade Dressed
	$/m	$/m			$/m	$/m	$/m	$/m		$/m	$/m
100 x 25	4.50	9.00	100 x 26	90 x 19	-	6.00	-	-	100 x 25	5.75	7.00
100 x 38	7.00	10.00	100 x 38	90 x 30	-	9.00	-	-	100 x 38	8.60	9.50
100 x 50	9.00	15.00	153 x 38	140 x 30	-	12.00	-	-	150 x 38	13.80	15.00
150 x 50	12.00	20.00	100 x 53	90 x 45	-	12.00	-	-	100 x 50	8.60	10.50
200 x 50	15.00	32.00	206 x 53	190 x 45	-	22.00	-	-	200 x 50	17.80	22.00

DARWIN

LOCAL HARDWOOD			LOCAL PINE						OREGON		
			Untreated				Pressure Treated				
Sawn Size	Green Sawn F11	Dry Dressed	Nom. Green Sawn Size	Dry Dressed Size	Dry Sawn	Dry Dressed	Dry Sawn F7	Dry Dressed	Nom. Size	Merch. Sawn	Select Grade Dressed
	$/m	$/m			$/m	$/m	$/m	$/m		$/m	$/m
100 x 25	5.10	8.05	100 x 26	90 x 19	-	-	-	-	100 x 25	-	-
100 x 38	6.90	12.35	100 x 38	90 x 30	-	-	-	-	100 x 38	-	-
100 x 50	10.20	15.20	153 x 38	140 x 30	-	-	-	-	150 x 38	-	-
150 x 50	10.90	14.10	100 x 53	90 x 45	-	-	-	-	100 x 50	-	-
200 x 50	18.80	27.70	206 x 53	190 x 45	-	-	-	-	200 x 50	-	-

HOBART

LOCAL HARDWOOD			LOCAL PINE						OREGON		
			Untreated				Pressure Treated				
Sawn Size	Green Sawn F11	Dry Dressed	Nom. Green Sawn Size	Dry Dressed Size	Dry Sawn	Dry Dressed	Dry Sawn F7	Dry Dressed	Nom. Size	Merch. Sawn F7	Select Grade Dressed
	$/m	$/m			$/m	$/m	$/m	$/m		$/m	$/m
100 x 25	3.27	4.36	100 x 26	90 x 19	-	2.21	-	2.59	100 x 25	6.06	6.12
100 x 38	6.63	8.84	100 x 38	90 x 30	3.07	3.66	-	4.15	100 x 38	9.69	9.79
100 x 50	11.64	15.52	153 x 38	140 x 30	7.28	3.12	-	6.41	150 x 38	14.53	14.68
150 x 50	9.31	12.42	100 x 53	90 x 45	4.76	4.86	-	5.40	100 x 50	10.01	10.11
200 x 50	24.39	32.53	206 x 53	190 x 45	12.70	12.93	-	11.46	200 x 50	19.98	20.18

CARPENTRY

		UNIT	Can. $	Dwn. $	Hob. $
FLOOR FRAMING			Loc H'wd F17	Loc H'wd F11	Loc H'wd F17
Plates :					
100 x 25mm	...	m	9.50	11.00	9.15
100 x 38mm	...	m	13.00	16.00	15.45
Bearers, beams :					
100 x 75mm	...	m	20.00	27.00	-
125 x 75mm	...	m	28.50	33.00	-
Joists :					
100 x 50mm	...	m	21.00	21.00	21.40
150 x 50mm	...	m	33.50	29.00	34.90
WALL FRAMING			Pine F5		
Plates :					
75 x 38mm	...	m	13.00	-	14.30
75 x 50mm	...	m	17.85	18.70	19.80
100 x 38mm	...	m	21.00	17.60	19.20
90 x 45mm	...	m	21.00	20.10	25.20
Studs :					
75 x 38mm	...	m	-	-	13.55
75 x 50mm	...	m	17.85	18.70	18.30
100 x 38mm	...	m	-	-	17.70
90 x 45mm	...	m	22.00	20.10	22.60
Noggings :					
75 x 38mm	...	m	-	-	14.30
75 x 50mm	...	m	19.00	18.70	19.05
100 x 38mm	...	m	-	-	18.45
90 x 45mm	...	m	21.00	17.60	24.10
ROOF AND CEILING FRAMING					
Rafters :					
100 x 50mm	...	m	22.00	21.00	22.90
150 x 50mm	...	m	34.00	30.00	37.10
Purlins :					
150 x 50mm	...	m	33.60	30.00	37.90
150 x 75mm	...	m	40.00	39.00	-
Ridge, hip valley board :					
200 x 38mm	...	m	37.00	33.00	36.80
Ceiling joists :					
100 x 38mm	...	m	18.00	17.50	18.05
100 x 50mm	...	m	22.00	21.00	22.90
Hangers :					
200 x 50mm	...	m	42.00	38.80	52.80

CARPENTRY

		UNIT	Can. $	Dwn. $	Hob. $
ROOF TRUSSES					
Standard prefabricated roof truss in local sawn softwood or hardwood for tiled roof to 22° pitch and including 750mm overhang each side :					
6000mm span	...	No	260.00	365.00	240.00
8000mm span	...	No	320.00	450.00	305.00
10000mm span	...	No	340.00	550.00	376.00
ADD extra for :					
First truss only	...	No	120.00	150.00-200.00	80.00-90.00
FLOORING					
Compressed fibre cement sheets fixed to timber :					
15mm thick	...	sqm	100.00	145.00	104.50
18mm thick	...	sqm	125.00	175.00	118.50
Particle board panels with tongued and grooved edges to timber :					
19mm thick	...	sqm	32.00	-	29.10
22mm thick	...	sqm	36.00	-	35.20
Tongued and grooved, 25mm thick in 100mm widths, fixed to timber :					
Radiata pine	...	sqm	85.00	110.00	52.90
Local hardwood (standard grade)	...	sqm	140.00	160.00	-
WALL LININGS					
Fibre cement flexible sheeting fixed to timber framing :					
4.5mm thick	...	sqm	26.00	-	33.00
6.0mm thick	...	sqm	42.00	48.00	41.80
Fibre cement decorative sheeting or planks fixed to timber framing :					
6.0mm thick Stucco	...	sqm	65.00	64.00	57.90
Wallboard fibre cement sheeting flush jointed and fixed to timber :					
6.0mm thick	...	sqm	58.00	56.00	45.50
9.0mm thick	...	sqm	82.00	76.00	64.50
Interior standard grade plywood for paint finish fixed to timber framing :					
3.6/4.0mm thick	...	sqm	42.00	-	31.40
6.0mm thick	...	sqm	57.00	-	33.60
Structural grade plywood shearwall lining fixed to timber	...	sqm	65.00	82.00	39.30

JOINERY

		UNIT	Can. $	Dwn. $	Hob. $
BASE MATERIAL PRICES					
The following table indicates the comparative average trade (SUPPLY ONLY) price of timbers in general use.					
All timbers are seasoned, selected grade, dressed all round.					
TIMBER					
Oregon -					
50 x 25mm	...	m	4.50	-*	4.21
100 x 25mm	...	m	9.00	-*	7.07
150 x 38mm	...	m	18.00	-*	15.54
200 x 50mm	...	m	32.00	-*	20.99
Pacific maple -					
50 x 25mm	...	m	5.50	-*	-
100 x 25mm	...	m	11.00	-*	-
150 x 38mm	...	m	18.00	-*	-
200 x 50mm	...	m	38.00	-*	-
Radiata pine -					
50 x 25mm	...	m	4.00	3.50	2.85
100 x 25mm	...	m	5.50	4.80	4.82
150 x 38mm	...	m	12.75	9.70	8.46
200 x 50mm	...	m	19.25	16.30	13.98
Tasmanian oak -					
50 x 25mm	...	m	9.00	7.10	3.46
100 x 25mm	...	m	15.00	12.30	6.84
150 x 38mm	...	m	32.00	27.00	22.49
200 x 50mm	...	m	48.00	40.20	42.69
Western red cedar -					
50 x 25mm	...	m	12.00	-*	10.54
100 x 25mm	...	m	18.20	-*	19.99
150 x 38mm	...	m	38.00	-*	41.07
200 x 50mm	...	m	68.00	-*	65.85
* Prices vary depending upon availability of imported timbers.					
PELMETS					
Fascia or top :					
150 x 25mm -					
Oregon	...	m	42.00	-	-
Local hardwood	...	m	32.00	30.00	25.90
Tasmanian oak	...	m	46.00	38.50	24.30
Pacific maple	...	m	38.00	-	-
ADD extra for :					
Returned end	...	No	25.00	25.00	22.70
Standard bracket (average) fixed to wall	...	No	25.00	25.00	21.20
SKIRTINGS					
Rounded or splayed fixed to timber :					
100 x 25mm -					
Oregon	...	m	15.00	-	-
Local hardwood	...	m	19.00	18.70	19.10
Tasmanian oak	...	m	19.00	-	16.95
Pacific maple	...	m	15.00	-	-

JOINERY

		UNIT	Can. $	Dwn. $	Hob. $
HANDRAILS					
Plain handrail with rounded edges, fixed to balustrade :					
150 x 50mm -					
Oregon	...	m	65.00	-	44.40
Local hardwood	...	m	75.00	85.00	61.70
Tasmanian oak	...	m	85.00	95.00	59.90
Pacific maple	...	m	65.00	-	-
ADD extra for :					
Mitred joint (average)	...	No	65.00	75.00	64.90
CUPBOARDS AND FITTINGS					
All purpose made	...	Note			
COMMERCIAL					
Bench cupboard unit 600mm wide x 900mm high with intermediate shelf and hinged doors, all laminate finished	...	m	920.00	1020.00	960.00
Bench cupboard unit 600mm wide x 900mm high with intermediate shelf and hinged doors and top left open to receive stainless steel sink, all laminate finished (excluding sink)	...	m	960.00	950.00	855.00
ADD extra to the foregoing bench and cupboard units for :					
Drawer.	..	No	160.00	150.00	100.00
Laminate finished end	...	No	160.00	150.00	80.00
Vanity bench top 450mm wide with 200mm high fascia, top prepared to receive basin, all laminate finished and supported on and including metal brackets	...	m	600.00	680.00	525.00

WINDOWS AND DOORS

WINDOWS					
Prices include for stock pattern sections, glazing with clear glass and fixing in position	...	Note			
Clear anodised commercial aluminium window :					
Fixed light	...	sqm	400.00	495.00	487.00
Sliding - 25% opening	...	sqm	440.00	530.00	555.00
Pivoted - 50% opening	...	sqm	600.00	580.00	636.00
Clear anodised commercial double glazed aluminium window :					
Fixed light	...	sqm	600.00	1020.00	658.00
Vertically pivoted	...	sqm	780.00	1100.00	793.00
ADD extra for :					
Integral venetian blind	...	sqm	175.00	195.00	190.00
Powder coated aluminium framed glass louvred window with insect screens	...	sqm	460.00	590.00	-
ADD extra for :					
Amplimesh security screens	...	sqm	500.00	410.00	-

WINDOWS AND DOORS

		UNIT	Can. $	Dwn. $	Hob. $
DOOR FRAMES					
Standard zinc anneal door frame to suit door size 2040 x 820mm, built in :					
Knock down frame	...	No	147.00	220.00	114.00
Fully welded commercial standard frame	...	No	185.00	360.00	117.50
Timber linings and frames fixed to brickwork :					
150 x 38mm plain lining -					
Oregon	...	m	34.00	-	36.60
Local hardwood	...	m	38.00	50.00	44.40
Pacific maple	...	m	44.00	-	-
Western red cedar	...	m	58.00	-	65.40
100 x 75mm frame, once rebated -					
Oregon	...	m	40.00	-	-
Local hardwood	...	m	55.00	74.00	-
Pacific maple	...	m	55.00	-	-
Western red cedar	...	m	74.00	-	-
150 x 75mm frame, twice rebated -					
Oregon	...	m	48.00	-	-
Local hardwood	...	m	72.00	92.00	-
Pacific maple	...	m	80.00	-	-
Western red cedar	...	m	75.00	-	-
DOORS					
Hollow core door, 2040 x 820 x 35mm thick faced both sides with hardboard for paint finish (excluding edge strips, hardware and painting)	...	No	150.00	290.00	151.00
Solid core door, 2040 x 820 x 35mm thick faced both sides with hardboard for paint finish (ditto)	...	No	350.00	365.00	216.00
Industrial roller door, chain operated :					
Continuous curtain slat -					
2700 x 2700mm	...	No	3500.00	3500.00	2935.00
3600 x 3600mm	...	No	5200.00	4900.00	3740.00
ADD extra for :					
Electrically operated opening mechanism	...	No	1500.00	2700.00	1355.00
Fire rated single door 2040 x 820mm and frame comprising door faced both sides with plywood for paint finish, steel frame built in and fitting with butt hinges, surface mounted closer and latchset :					
One hour rated	...	No	1400.00	1650.00	1530.00
Two hour rated	...	No	1500.00	1800.00	1640.00
Three hour rated	...	No	1800.00	2300.00	1760.00

ROOFING

		UNIT	Can. $	Dwn. $	Hob. $
STEEL					
Corrugated, Zincalume® Steel :					
0.42mm	...	sqm	48.00	50.00	39.30
0.60mm	...	sqm	56.00	56.80	58.10
Ribbed, Zincalume® Steel :					
0.42mm (Spandek®)	...	sqm	52.00	50.00	44.60
0.48mm (Spandek®)	...	sqm	56.00	53.60	49.40
0.48mm (Kliplok®)	...	sqm	60.00	59.80	52.40
0.60mm (Kliplok®)	...	sqm	70.00	67.80	64.80
ADD extra for :					
Colour coating one side (average)	...	sqm	15.00	13.00	14.00
Cyclone fixings	...	sqm	-	15.50	-
Cappings	...	m	32.00	43.00	37.60
Stop ending	...	m	13.00	17.00	10.00
Closing ends and lipping	...	m	14.00	18.00	13.35
TILE					
Prices include 38 x 25mm battens and for laying to slope not exceeding 30°	...	Note			
Concrete tiles, standard pattern and colour	...	sqm	52.00	-	48.20
Terracotta tiles, French pattern, standard colour	...	sqm	75.00	-	71.60
ADD extra for :					
Ridge and hip tiles	...	m	Incl.	Incl.	Incl.

PLASTERING

Cement render, wood float finish :					
One coat, 13mm thick on :					
Brick walls	...	sqm	45.00	32.00	33.70
Concrete walls	...	sqm	45.00	32.00	35.30
Two coats, 19mm thick on :					
Brick walls	...	sqm	55.00	40.00	45.60
Concrete walls	...	sqm	55.00	40.00	48.50
Hardwall plaster, two coats 16mm thick on :					
Brick walls	...	sqm	60.00	40.00	50.30
Concrete walls	...	sqm	60.00	40.00	53.20
Vermiculite plaster sprayed finish 10mm thick, including all labours on :					
Concrete soffits	...	sqm	48.00	52.00	31.20
Sides and soffit of beams	...	sqm	48.00	59.00	36.00
Plasterboard, flush finished :					
1 x 10mm thick to -					
Timber wall framing	...	sqm	26.00	34.00	37.80
Steel wall framing	...	sqm	26.00	34.00	37.80
Timber ceiling framing	...	sqm	44.00	44.00	43.50

PLASTERING

		UNIT	Can. $	Dwn. $	Hob. $
Fire rated plasterboard, flush finished :					
1 x 13mm thick to -					
Timber wall framing	...	sqm	36.00	65.00	46.70
Steel wall framing	...	sqm	36.00	65.00	46.70
Timber ceiling framing	...	sqm	50.00	65.00	52.30
2 x 13mm thick to -					
Timber wall framing	...	sqm	62.00	93.00	87.80
Steel wall framing	...	sqm	62.00	93.00	87.80
Timber ceiling framing	...	sqm	85.00	113.00	99.00
Wall framing of cold formed steel sections in top and bottom plates, studs at 600mm centres and one row of noggings, including angles and intersections, in :					
51mm x 0.6mm framing	...	sqm	32.00	-	48.50
76mm x 1.2mm framing	...	sqm	35.00	51.00	51.30
76mm x 1.6mm heavy duty framing	...	sqm	65.00	61.00	-

SUSPENDED CEILINGS

Plasterboard, 10mm thick, flush finish with concealed suspension	...	sqm	70.00	78.00	82.90
Fire rated (one hour) plasterboard, 2 x 16mm layers, flush finish with concealed suspension	...	sqm	130.00	148.00	137.50
Plain plasterboard tiles 600 x 600mm in a pre-finished two-way exposed suspension system	...	sqm	70.00	80.00	82.60

PAVIOR AND TERRAZZO

Power float monolithic finish	...	sqm	9.00	15.00	8.70
Granolithic (1:2:5) with steel trowel finish, in large areas :					
19mm thick	...	sqm	35.00	44.00	29.60
25mm thick	...	sqm	40.00	62.00	39.80
38mm thick	...	sqm	45.00	69.00	41.90

GLAZING

Clear float glass and glazing to metal :					
3mm	...	sqm	120.00	167.00	110.00
4mm	...	sqm	125.00	180.00	117.00
5mm	...	sqm	145.00	198.00	136.00
6mm	...	sqm	155.00	210.00	140.00
8mm	...	sqm	220.00	260.00	187.00
10mm	...	sqm	242.00	360.00	213.00

PAINTING

		UNIT	Can. $	Dwn. $	Hob. $
On general surfaces :					
Woodwork -					
Sealer undercoat and two coats gloss acrylic	...	sqm	20.00	20.00	18.95
Prime, one undercoat and two coats enamel	...	sqm	20.00	20.00	18.95
Two coats clear polyurethane	...	sqm	18.00	18.00	11.65
Metalwork -					
Touch up, prime, one undercoat and two coats enamel	...	sqm	25.00	21.00	16.10
Walls and ceilings, smooth surface -					
One coat sealer and two coats low gloss acrylic	...	sqm	16.00	20.00	15.45
One coat sealer, one coat undercoat and one coat enamel	...	sqm	20.00	20.00	16.70
One coat sealer and two coats acrylic latex	...	sqm	20.00	20.00	14.85

* * * * * *

RENTAL REVIEW

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Rawlinsons

AUSTRALIAN CONSTRUCTION HANDBOOK

QUARTERLY UPDATE
APRIL, 2021

COMMENT

ADELAIDE - South Australia is generally tracking well in most sectors. This has to be underlined with caution with continuing uncertainty impacting on investor and consummer confidence. Low interst rates and the HomeBuilder Scheme are fuelling the residential sector with activity expected to remain strong for the remainder of the year. Current indications note that the strong residential approvals that have been driving the sector are now showing signs of easing. This may be as a result of the HomeBuilder Scheme ending in March 2021 and not being extended. The downsize has been that construction material costs for timber, steel and concrete have increased due to the significant demand. This has impacted on all sectors of the construction industry and is expected to remain so for the ensuing period.

BRISBANE - The construction industry generally has continued to strengthen and is on the upward trend within its cycle. Activity levels in the domestic housing market have remained strong with activity in pre-contract services improving. Significant price increases have been experienced of the past quarter driven by supply and demand and in some cases importation of goods. Further cost escalation is anticipated over the remainder of the year, but the rate is likely to moderate.

CANBERRA - Prices for residential property have increased significantly over the last few months; sales of residential property generally (particularly townhouses) has exceeded all expectations. Overall material costs have increased five to twenty per cent, particularly timber, steel, concrete, tiles, copper and general hardware items due in the main to supply issues. We understand that some subcontractors are holding quotes for a maximum of seven days only. It would appear that the supply issues are COVID-19 related and it is unlikley that any price increase will reduce in the short to medium term. While material price increases have been covered off by the increases in sale prices any increase in interest rates would have a significant impact.

DARWIN - Market is stable with no discernible price rises over this quarter. A number ofdDefence projects are in delivery phase causing the construction industry to remain buoyant and utilising existing spare capacity. Activity in the residential construction is being maintained and housing projects in remote communities are progressing. A number of road infrastructure projects are also being committed. We predict slight price rises over the next quarter as spare capacity reduces in the market.

HOBART - Activity and tendering levels are back to full strength with contractor availability now the major impediment to new building procurement. Selective tendering and opportunistic pricing is now more of a concern than if the industry can rebound from the challenges of 2020. With an under supply of housing to be corrected and infrastructure spending below budget expectations in recent years the forecast is for the current situation to continue into the foreseeable future.

MELBOURNE - The positive signs of activity that showed itself across several industry sectors of the construction industry, in the past quarter, are generally set to continue into the quarter ahead. This is despite there still being, some ongoing softening signs evident, not only in the residential, but also the commercial offices, accommodation and retail sector spaces. The high level of activity in the civil engineering sector, is set to continue to underpin the industry at large, for some time to come.

PERTH - Perth's economy remains resilient against the complexities of COVID-19, the main underpinning being the Mining and Resources boom. The Homebuilder incentive has invigorated the domestic market coupled with the low vacancy rates the market is booming, bricklaying rates have increased significantly. Infrastructure's momentum continues with Metronet. The hotel market is being underpinned by COVID-19 quarantine and state-based recreation; once the domestic boarder fully opens and the internal boarder are relaxed, the occupancy slack will be depleted. All signs are looking towards a strong post-COVID recovery.

SYDNEY - The construction industry in Sydney is showing signs of returning to the levels of activity seen in 2019. The slowdown in 2020 has created a back log of projects which are now starting to come online. The lack of supply of raw materials such as timber, steel and copper are generating increased costs. This is in turn creating an upward trend on tender prices. Overall there is a level of confidence returning to the sector.

To adjust the 2020 Handbook prices from December 31, 2020 to March 31, 2021 :

Adelaide	+0.75%	Brisbane	+1.50%	Canberra	+2.00%	Darwin	0.00%
Hobart	+1.00%	Melbourne	+1.00%	Perth	+0.60%	Sydney	+2.00%

REVISIONS:

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ANTICIPATED BUILDING PRICE ADJUSTMENT, 2021

Prices given in the book are those ruling at the commencement of 2021 with an index of 100.00. To update prices refer to the following, which is a prediction of tender levels, taking into account anticipated market conditions.

	31-Dec-20	31-Mar-21	30-Jun-21	30-Sep-21	31-Dec-21
Adelaide	100.00	100.75	101.00	102.25	102.75
Brisbane	100.00	101.50	102.00	103.00	103.50
Canberra	100.00	102.00	103.00	103.50	104.00
Darwin	100.00	100.00	100.20	100.30	100.40
Hobart	100.00	101.00	102.00	103.00	104.00
Melbourne	100.00	101.00	101.50	102.00	102.50
Perth	100.00	100.60	101.10	101.90	102.60
Sydney	100.00	102.00	102.50	103.00	104.00

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BUILDING PRICE INDEX

	ADEL	BRIS	CAN	DWN	HOB	MEL	PER	SYD
	(page 816)	(page 818)	(page 820)	(page 822)	(page 824)	(page 826)	(page 828)	(page 830)
2020								
June 30	117.43	124.47	123.91	111.60	111.36	125.03	113.75	127.68
Sept 30	117.43	125.09	124.52	111.60	111.36	125.34	114.31	128.43
Dec 31	118.00	126.03	125.44	111.60	112.45	125.34	114.59	128.43
2021								
March 31	118.89(F)	127.92(F)	127.95(F)	111.60(F)	113.57(F)	126.59(F)	115.28(F)	131.00(F)
June 30								
Sept 30								

CONSUMER PRICE INDEX

	ADEL	BRIS	CAN	DWN	HOB	MEL	PER	SYD
	(page 816)	(page 818)	(page 820)	(page 822)	(page 824)	(page 826)	(page 828)	(page 830)
2020								
Sept 30	115.7	116.2	115.4	110.8	116.7	116.7	114.1	116.8
Dec 31	116.5	117.5	115.0	111.5	116.7	116.9	113.1	117.1
2021								
March 31								
June 30								

ESTIMATING – COMPARATIVE COSTS AND DETAILED PRICES

General note:

The overall movement in prices since the start of 2021 is given in the updated 'Anticipated Price Adjustment, 2021. and whilst this adjustment factor already includes due allowance, it should be noted that some trades, when taken in isolation, have varied appreciably from the 'going rate' indicated by the adjustment factor. The following indicates the overall movement for updating the 2021 Handbook prices for the particular trade or category of work where an appreciable variation occurs:

ADELAIDE -	Concrete	- prices for concrete supply have increased by 5-10%
	Timber supply	- prices for timber supply have increased by 10-15%
	Bar reinforcement	- prices for bar reinforcement have increased by 20%
	Reinforcing steel	- prices for reinforcing steel have increased by 10-15%
BRISBANE -	LVL, Laminated Beams	- prices for LVL and laminated beams have increased by 15%
	Timber framing	- prices for timber framing have increased by 15%
	Bar reinforcement	- prices for bar reinforcement supply have increased by $150/t
	Mesh reinforcement	- prices for mesh reinforcement supply have increased by 10%
	Concrete	- prices for concrete supply have increased by $10/cum
	Structural Steel	- prices for structural steel supply have increased by 10%
	Roofing / Purlins	- prices for roofing / purlins have increased by 10%
CANBERRA -	Concrete	- prices for concrete supply have increased by 7%
	Timber supply	- prices for timber supply have increased by 10-20%
	Brick supply	- prices for brick supply have increased by 10%
	Reinforcing steel	- prices for reinforcing steel have increased by 10-15%
PERTH -	Formwork	- prices for formwork have increased by 5-10%
	Brickwork labour	- prices for brickwork labour have increased by 25-45%
	Fabric reinforcement	- prices for fabric reinforcement have increased by 20%

Rawlinsons

AUSTRALIAN CONSTRUCTION HANDBOOK

QUARTERLY UPDATE
JULY, 2021

COMMENT

ADELAIDE - Residential building activity remains strong and will remain so for the remainder of the 2021 period and into first half of 2022. The non-residential sector to remain steady over the next two to three years with tendering to remain very competitive. Infrastructure projects to slow over the ensuing period. Material cost increases are still of a concern, due to the HomeBuilder Scheme, inparticular, timber, concrete, and steel. Timber supply is problematic. The State Government has delivered, on 22nd June 2021 the 21/22 budget with emphasis on major infrastructure projects, in particular the new Women's & Children's Hospital, North/South Corridor extension and Riverbank Arena, including various other works associated with Sport, Education and Health Services. Notwithstanding the current debt level, as a result of the COVID-19 pandemic, this budget was formulated to further stimulate the economy and boost employment and investor confidence. This will be tested over the coming three to four years to determine if this will be a sustainable approach.

BRISBANE - Tender activity continues to remain strong, particularly in the residential sectors of the market. This activity coupled with continued supply and demand volatility for various materials continue to drive cost escalation upward. This volatility is also impacting tender validity periods and contract durations through the incorporation of subcontract lead times. The forecast cost escalation is not anticipated to moderate until 2022.

CANBERRA - Townhouse and apartment construction are dominating the Canberra market. Most mult unit developments are all sold prior to construction commencing. Prices for timber, reinforecment and steel have increased significantly which have been partially off set by higher selling prices. Apart from significant price increases, the other major difficulty for all builders is obtaining framing timbers on a regular basis when required, many developments have been delayed one to three months waiting for deliveries which will increase preliminary costs. Similar to many other States housing stock is low coupled with low interest rates and a strong demand which may taper off later in the year.

DARWIN - With a number of federally funded infrastructure and defence projects coming on board soaking up existing spare capacity and increases in steel prices nationwide, tender prices are starting to trend upwards. Most sectors of the industry are experiencing the start of a growth phase with new optimism in the market. The volume of work coming on the market could well cause a spike in escalation due mainly to potential labour shortages. The market is on a watch list.

HOBART - Contractor availabilty continues to have an impact on pricing levels with activity high across all sectors of the industry. Ongoing material price increases and procurement issues are also impacting tendering and programming with additional increases forecasted into next year. Material shortages and price increases in combination with a buoyant housing sector, private commercial developments and Government stimulus packages are all contributing to higher tendering and activity levels than has been seen in Tasmania in recent years.

MELBOURNE - While there continues to be ongoing softening signs evident within the level of activity across a number of sectors, the overall optimism and health of the construction industry remains reasonably solid. Generally, trends of the previous quarter, are set to continue into the next. The past quarter has witnessed supply issues, as well as significant material supply price increases in some materials, including timber, structural steel and reinforcement. However, the full effect of these delays and price increases, is yet to be fully realised and reflected in tender prices.

PERTH - Western Australia has proved it's resilient economy by being recognised by Standard & Poor as having the best state economy in the world over the past year. The resources market is leading the way with a flow on effect that is stimulating the office market with the vacancy rate contracting. Residentially, a growth cycle is looming as interstate migration to Western Australia has recommenced after a seven-year contraction. The retail market has had minimal disruption over the last twelve months, although being remote, the E-Commerce infrastructure is lacking compared to the Eastern States. Government major projects continue to stimulate the construction industry as a whole. Supply chain and high demand have brought shortages accross a miriad of materials and sub-contract labour.

SYDNEY - The construction market supported by the NSW government is continuing to grow. The government's ability to put in place a comprehensive range of policy measures and keep the spread of Covid-19 at bay, exceeds the forecasted levels. The impact of border closure still has influence on the supply chains and the import time frames with these factors driving price increase and causing delays on project delivery, however, with the ease of restrictions in recent months seeing the gradual opening of interstate borders, the sector's level of confidence is starting to recover its positive trend.

To adjust the 2021 Handbook prices from December 31, 2020 to June 30, 2021 :

Adelaide	+1.25%	Brisbane	+2.50%	Canberra	+7.50%	Darwin	+0.40%
Hobart	+4.00%	Melbourne	+1.50%	Perth	+3.00%	Sydney	+3.00%

REVISIONS:

Page 15

ANTICIPATED BUILDING PRICE ADJUSTMENT, 2021

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Brisbane	100.00	102.00	102.50	103.50	104.50
Canberra	100.00	104.50	107.50	108.00	109.50
Darwin	100.00	100.20	100.40	100.50	100.70
Hobart	100.00	102.00	104.00	105.00	106.00
Melbourne	100.00	101.00	101.50	102.00	102.50
Perth	100.00	101.50	103.00	104.50	106.00
Sydney	100.00	102.00	103.00	103.50	104.00

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BUILDING PRICE INDEX

	ADEL	BRIS	CAN	DWN	HOB	MEL	PER	SYD
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2020								
June 30	117.43	124.47	123.91	111.60	111.36	125.03	113.75	127.68
Sept 30	117.43	125.09	124.52	111.60	111.36	125.34	114.31	128.43
Dec 31	118.00	126.03	125.44	111.60	112.45	125.34	114.59	128.43
2021								
March 31	118.89	128.55	131.08	111.82	114.70	126.59	116.31	131.00
June 30	119.48(F)	129.18(F)	134.85(F)	112.05(F)	116.95(F)	127.22(F)	118.03(F)	132.28(F)
Sept 30								

CONSUMER PRICE INDEX

	ADEL	BRIS	CAN	DWN	HOB	MEL	PER	SYD
	(page 816)	(page 818)	(page 820)	(page 822)	(page 824)	(page 826)	(page 828)	(page 830)
2020								
Sept 30	115.7	116.2	115.4	110.8	116.7	116.7	114.1	116.8
Dec 31	116.5	117.5	116.3	111.5	117.6	118.4	113.0	118.0
2021								
March 31	117.2	118.2	117.3	114.4	118.5	118.8	114.6	118.5
June 30								

REVISIONS (cont'd)

Page 109 - STRIP FOOTINGS
ADD extra for:
Excavation twice depth of footing
Delete Perth 420.00 Insert Perth 83.80

ESTIMATING – COMPARATIVE COSTS AND DETAILED PRICES

General note:

The overall movement in prices since the start of 2021 is given in the updated 'Anticipated Price Adjustment, 2021. and whilst this adjustment factor already includes due allowance, it should be noted that some trades, when taken in isolation, have varied appreciably from the 'going rate' indicated by the adjustment factor. The following indicates the overall movement for updating the 2021 Handbook prices for the particular trade or category of work where an appreciable variation occurs:

ADELAIDE -	Bar reinforcement	- prices for bar reinforcement have increased by 10%
	Concrete	- prices for concrete supply have increased by 5%
	LVL, Laminated Beams	- prices for LVL and laminated beams have increased by 15%
	Plasterboard Accessories	- prices for plasterboard accessories have increased by 6%
	Reinforcing steel	- prices for reinforcing steel have increased by 5-10%
	Timber supply	- prices for timber supply have increased by 25%
BRISBANE -	Concrete	- prices for concrete supply have increased by $10/cum
	LVL, Laminated Beams	- prices for LVL and laminated beams have increased by 15%
	Plasterboard	- prices for plasterboard have increased by 5%
	PVC supply	- prices for PVC supply have increased by 20%
	Reinforcing steel	- prices for reinforcing steel have increased by 10%
	Roofing / Purlins	- prices for roofing / purlins have increased by 10%
	Structural Steel	- prices for structural steel supply have increased by 5%
	Timber framing	- prices for timber framing have increased by 15%
CANBERRA -	Concrete	- prices for concrete supply have increased by 5%
	Timber supply	- prices for timber supply have increased by 45%
	Reinforcing steel	- prices for reinforcing steel have increased by 40%
HOBART -	Timber supply	- prices for timber supply have increased by 15%
	Reinforcing steel	- prices for reinforcing steel have increased by 20%
MELBOURNE -	Concrete	- prices for concrete supply have increased by 1-5%
	Reinforcing steel	- prices for reinforcing steel have increased by 5-10%
	Structural Steel	- prices for structural steel supply have increased by 5-10%
	Timber supply	- prices for timber supply have increased by 10-15%
PERTH -	Brickwork labour	- prices for brickwork labour have increased by 25-45%
	Fabric reinforcement	- prices for fabric reinforcement have increased by 20%
	Formwork	- prices for formwork have increased by 5-10%
	Structural Steel	- prices for structural steel supply have increased by 15%
	Timber supply	- prices for timber supply have increased by 25%